

Customer Interface Specification

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Customer Interface Specification

This document contains technical information and specifications for customers to understand the system functionality and to process transactions through the Mastercard Dual Message Authorization System. System specifications are based on Mastercard's adoption of the International Organization for Standardization (ISO) 8583:1987 message formats.

Audience

This document is intended for parties using the Mastercard Network to process transactions through the Dual Message Authorization System. This includes acquirers, issuers, processors, and Network Enablement Partners.

Product

The Mastercard Dual Message Authorization System is a global real-time message processing system and is a component of the Mastercard Network. The Mastercard Network supports real-time authorization processing followed by batch clearing processing, leading to the settlement of funds between acquirers and issuers.

Table 1: Details

Metadata	Value
Audience	• Acquirer • Issuer • Processor • Network Enablement Partner
Product	Mastercard Network Processing and Operations
Region	Global
Country	Global
Release version	25.Q1
Type	Technical specification guide
Publish date	25 February 2025

Table 2: Summary of changes

Description of Change	Source	Where to look
Added note stating that Mastercard recommends that customers use response code 00 for all Stand-In Advice messages.	System clarification needed	Authorization Advice Response/0130: Issuer-Generated (Responding to a System-Generated 0120 from SAF) Reversal Request/0400 response codes
Expanded DE 48 (Additional Data: Private Use), subelement 22 (Multi-Purpose Merchant Indicator), subfield 04 (Issuer PIN Request in a Single Tap Mode) to include Middle East/Africa, Asia/Pacific, and Latin America/Caribbean regions.	AN 6563 Single Tap and PIN Request Expands to More Regions	Subfield 04 (Issuer PIN Request in a Single Tap Mode)
Updated the Application notes for DE 48, subelement 42, subfield 3 (Reason for UCAF Collection Indicator Downgrade).	System clarification needed	Subfield 3 (Reason for UCAF Collection Indicator Downgrade)
Added application note for European Economic Area (EEA) only in DE 48, subelement 63 (Trace ID).	System clarification needed	Subelement 63 (Trace ID)
Added note to DE 55 stating that subelements not listed as either required or optional must not be included in DE 55.	System clarification needed	DE 55 Subelements
Updated the Application notes for DE 104, subelement 001 (Digital Payment Cryptogram).	System clarification needed	Subelement 001 (Digital Payment Cryptogram)
Corrected name of DE 108 (Additional Transaction Reference Data), subelement 07 (Data Validation Request) and DE 108, subelement 08 (Data Validation Response)	GLB 8429 Introducing On-Behalf Name Match Service and Enhancements to Name Validation Service	Subelement 07 (Data Validation Request) Authorization Request/0100 message edits for DE 108, subelement 07 Subelement 08 (Data Validation Response)

Description of Change	Source	Where to look
Revised the descriptions of Mastercard Digital Enablement Services (MDES) on-behalf services (OBS) as: • OBS 50 = MDES PAN Mapping service • OBS 51 = MDES Pre-Validation Service for Secure Element tokens (ICC or DSRP data) • OBS 52 = MDES Pre-Validation Service for Secure Element tokens (CVC3 data) • OBS 61 = MDES Pre-Validation Service for Cloud and Static tokens (ICC, DSRP, or DTVC data) • OBS 62 = MDES Pre-Validation Service for Cloud tokens (CVC3 data) NOTE: These changes do not indicate a change in functionality.	System clarification needed	Subelement 71 (Valid Subfield 1 and Subfield 2 Value Combinations) Subelement 71 (Valid Subfield 1, Subfield 2, and Subfield 3 value combinations) MDES for Merchants
Corrected note to Payment Transactions in DE 108 (Additional Transaction Reference Data), subelement 02 (Sender Data), subfield 05 (Sender City).	System clarification needed	Subfield 05 (Sender City)
Clarified that DE 122, subelement 001, subfield 010 (Service Location Country Subdivision Code) must be followed by trailing spaces for codes that are less than three positions.	System clarification needed	Subfield 010 (Service Location Country Subdivision Code)
Added DE 124 (Member-Defined Data: Ukraine Domestic Transactions).	EUR 9908.1 Enhancing Domestic Processing in Ukraine	DE 124 (Member-Defined Data: Ukraine Domestic Transactions)
Added information about Post-tokenization Authentication.	System clarification needed	DE 124 Subfields in Authorization Request/0100 for ACN

Description of Change	Source	Where to look
Mastercard enhanced the MDES to include additional authentication methods and issuer notification. This allows the Secure Card on File (SCOF) and Click to Pay programs to provide additional authentication options to customers for tokenized cards and tokenized payment transactions within Token Authentication Framework (TAF). Updated ISO 8583 message specification changes for MDES post-tokenization authentication for SCOF and Click to Pay.	System clarification needed	Authentication Methods Request (AMR) Authorization Request/0100 for AMR DE 124 Subfields in Authorization Request/0100 for AMR Authorization Request Response/0110 for AMR DE 124 Subfields in Authorization Request Response/0110 for AMR
Added: - Activation Method Type values 8 and 9 to the Consumer's Activation Method Preference subfield. - Token Unique Reference subfield. - Values in the Reason Code subfield.	System clarification needed	DE 124 Subfields in Authorization Request/0100 for ACN
Added: - DE 48, subelement 06 (Advanced Digital Transaction Data 1), subfield 1 (Binding ID). - DE 48, subelement 33 (PAN Mapping File Information), subfield 5 (Token Assurance Method). - DE 48, subelement 33 (PAN Mapping File Information), subfield 8 (Storage Technology). - DE 104 (Digital Payment Data), subelement 004 (Digital Service Provider). - DE 104 (Digital Payment Data), subelement 005 (Digital Authentication Data).	System clarification needed	Authorization Request/0100 for TVN

Description of Change	Source	Where to look
Added: • Binding Status subfield. • Values a - Token Binding and b - Account Binding to Tokenization Event Indicator subfield for the value 9=Replacement (token re-digitization).	System clarification needed	DE 124 Subfields in Authorization Request/0100 for TVN
Added: • Token Unique Reference field. • authRequestCorrelationId field. • Activation Method Type values 8 and 9 to the Consumer's Activation Method Preference subfield.	System clarification needed	DE 120 Layout for Administrative Advice/0620 Issuer Token Notification Advice, Activation Code Notification
Added: • Cardholder Verification field. • Authentication Factors field. • Binding Status field. • Binding ID field. • Digital Service Provider ID 1 field. • Digital Service Provider ID 2 field. • Values a - Token Binding and b - Account Binding to Tokenization Event Indicator subfield for the value 9=Replacement (token re-digitization).	System clarification needed	DE 120 Layout for Administrative Advice/0620 Issuer Token Notification Advice, Tokenization Event Notification

Chapter 1 Overview

This section discusses the various conventions that the Mastercard® Authorization Platform has adopted from the International Organization for Standardization (ISO) 8583–1987 message formats. Standard message flows are presented for acknowledgements, advices, and error conditions.

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Customer interface specification format

This document contains the Mastercard implementation of the ISO 8583–1987 international message standard for processing authorization information using the Mastercard Dual Message System, Authorization Platform. It provides Mastercard customers with information necessary for development of an application-level online software interface between customer processing systems (CPSs) and the Mastercard Dual Message System.

Benefits

All programs and services carrying the Mastercard brand use the ISO 8583–1987 message standard.

Benefits of using the ISO 8583–1987 international message standard include:

- **Flexibility:** Customers may use this interface as a "gateway" vehicle to other credit card and debit card networks. These networks include Visa's credit and debit card systems; and all major travel and entertainment (T&E) card authorization services. Customers using these gateway capabilities can eliminate the time and expense involved in developing, operating, and maintaining multiple communication links to various regional, national, and international authorization networks in which they may participate.
- **Capacity:** This interface allows Mastercard and its customers to take full advantage of the Mastercard Network. This network features a fully-distributed network architecture at both the application and data-transport (communication network) levels. It provides direct high-speed "peer-to-peer" transaction routing (for example, acquirer-to-issuer), capable of handling several thousands of transactions per second.
- **Functionality:** Mastercard customers whose proprietary card processing systems support the Authorization Platform standard specified in this document can be assured that their systems will support:
 - All existing Mastercard programs and services, without requiring development of new system interfaces
 - Other national and international networks developed in accordance with ISO 8583–1987 interchange specifications
 - Upgrades related to future revisions of ISO standards

NOTE: Mastercard reserves the right to record, store, and use all data transmitted by the Mastercard Dual Message System in online electronic transactions, subject to Mastercard privacy and security compliance policies and applicable laws and regulations, without further notice.

NOTE: Throughout this document, functional references to "online" indicate an acquirer or issuer that is directly connected to a Mastercard Interface Processor (MIP) and does not imply any web-based or Internet connection.

Issuer Post-on-Authorization

The Authorization Platform and all Mastercard programs and services employ the Post-on-Authorization concept for handling issuer side transaction processing of authorization messages. This concept ensures necessary system integrity and optimizes efficient use of network resources.

The Post-on-Authorization concept is far more efficient than the alternative Post-on-Completion processing method. **It does not require** Completion Confirmation and Completion Response messages in order for the issuer processing system (IPS) to process Authorization Request/0100 messages. In contrast, the Post-on-Completion method **does require** propagation of Completion Confirmation and Completion Response messages between the acquirer and the issuer.

Under the Post-on-Authorization concept, when the Authorization Platform receives an Authorization Request Response/0110 message, it assumes that the issuer's authorization approval affected the cardholder's credit line or open-to-buy limit immediately. The system does not send an Authorization Confirmation message to the issuer. The issuer must assume that the transaction processed normally (**unless advised otherwise** by a Reversal Advice/0420). The Reversal Request/0400 message also may affect Post-on-Authorization processing because it cancels either a part or all of the authorization.

The Authorization Platform sends a response message to the issuer **only** if the Authorization Request Response/0110 message is late. This occurs, for example, when the IPS does not respond within the predetermined time (times-out) and the Authorization Platform forwards the message to the Stand-In System to process the transaction on behalf of the issuer. The Authorization Platform then sends the issuer an Authorization Negative Acknowledgement/0190 message, indicating that the Authorization Request Response/0110 message is unrecognizable.

If the issuer receives an Authorization Negative Acknowledgement/0190 message, the issuer must assume that the Authorization Platform:

- Timed-out the IPS
- Immediately went to the Stand-In System or alternate authorizer processing to service this transaction

If the issuer does not select Stand-In options, the Authorization Platform automatically sends an Authorization Request Response/0110 message to the acquirer with a negative response code (transaction request denied). **The issuer always must reverse any effect upon the cardholder's account.** Later, the issuer receives an Authorization Advice/0120 message indicating the specific action taken by the Stand-In System.

NOTE: The term "post" in Post-on-Authorization does not refer to actual posting of cardholder accounts for billing purposes. Post-on-Authorization refers only to the technique used to maintain accurate settlement reconciliation totals between the Authorization Platform and any attached customer processing system (CPS). The IPS handles actual posting of cardholder account data for cardholder billing purposes. Posting of the cardholder account data is not an Authorization Platform function.

Bit mapped message encoding scheme

A description of the bit map scheme.

All customer interface specification (CIS) messages are variable-length, with a bit map scheme used as the first data element(s) of the message following the Message Type Identifier (MTI) to indicate the presence or absence of additional data elements in the message. Each bit map is a 64-bit string contained within an eight-byte data element. The first bit in each bit map is 1 or 0 to indicate the presence or absence of another (immediately following) bit map data element.

The Authorization Platform uses a maximum of two-bit maps: a "Primary" and a "Secondary" Bit Map. Bits 1 or 0 in the Primary Bit Map indicate the presence or absence of DE 2 (Primary Account Number [PAN]) through DE 64 (Message Authentication Code [MAC]). Bits 1 or 0 in the Secondary Bit Map indicate the presence or absence of DE 66 (Settlement Code) through DE 128 (Message Authentication Code [MAC]).

NOTE: All bit positions are interpreted from left to right within each bit map. For example, within the Primary Bit Map, the leftmost bit is "bit number 1" and the rightmost bit is "bit number 64."

Bit number 1 in the Primary Bit Map and bit number 65 in the Secondary Bit Map (that is, the first bit in each bit map) do not have corresponding data elements. These bits indicate the presence or absence of additional data elements in the message. If bit number 1 is 1, the Secondary Bit Map is present and selected data elements in the range DE 66–DE 128 exist in the Secondary Bit Map of the message. Bit number 65 **must always be 0** because no additional bit maps are defined beyond the Secondary Bit Map.

Each message **must** contain the Primary Bit Map. The Secondary Bit Map must be included only if data elements DE 66–DE 128 are present in the message.

Authorization Platform processing terms and acronyms

Authorization Platform processing terms and acronyms are used in describing the logical flow of an Authorization Platform message from one point to another.

The following Authorization Platform terms or acronyms are used in describing the CIS message format:

- Acquirer processing system (APS)
- Customer processing system (CPS)

- Issuer processing system (IPS)
- Point of Sale (POS)

Customer Interface Specification notations

The Customer Interface Specification notations describe the Customer Interface Specification (CIS) format.

The CIS format is described with the following notations:

- Data length
- Data representation
- Data field
- Data justification
- Date and time
- Entities
- Presence
- Presence requirements
- Program and service category

Data length notations

Data length notations indicate the format of the data length.

Notation	Description
All length fields are encoded as numeric EBCDIC, right-justified with leading zeros. If a customer sends ASCII, it is converted to EBCDIC and back to ASCII again if needed.	
-digit(s)	Fixed length in number of positions. Example: n-3 indicates a three-position numeric data element. Example: an-10 indicates a 10-position alphanumeric data element.
...digit(s)	Variable length, with maximum number of positions specified. Example: n...11 indicates a variable-length numeric data element of 1–11 digits. Example: an...25 indicates a variable-length alphanumeric data element of 1–25 positions.
LLVAR	Present with a variable-length data element attribute, indicates that the data element contains two fields: LL The length field represents the number of positions in the variable-length data field that follows. The length field contains a value in the range 01–99.

Notation	Description
	VAR The variable-length data field Example: "an...25; LLVAR" represents a variable-length alphanumeric data element with a length of 1–25 positions.
LLLVAR	Present with a variable-length data element attribute, indicates that the data element contains two fields: LLL The length field represents the number of positions in the variable-length data field that follows. The length field contains a value in the range 001–999. VAR The variable-length data field Example: "an...500; LLLVAR" indicates a variable-length alphanumeric data element having a length of 1–500 positions.

Data representation notations

Data representation notations indicate how data is represented. All message data elements are aligned on byte boundaries. The following data types are encoded using EBCDIC or ASCII display character representation, except for binary data.

NOTE: Special characters are any non-numeric or non-alphabetic characters (for example, @, #, &, and \$), including spaces.

Notation	Description
a	alphabetic characters A–Z and a–z
n	numeric digits 0–9
an	alphabetic and numeric characters (excluding spaces and special characters)
ans	alphabetic, numeric, space, and special characters
b	space (or blank)
b	binary representation of data in eight-bit bytes All binary data elements are constructed of bit-strings that have lengths that are an integral number of eight-bit bytes. No binary data element has a length of less than eight bits (one byte). b-8 indicates a fixed-length binary field of eight characters (eight bytes, 64 bits)
s	special character Special characters are any non-numeric or non-alphabetic characters (for example, @, #, &, and \$), including spaces. For a complete list of valid character sets, refer to the Character Sets section.

Notation	Description
ns	numeric and special characters

All track 2 or track 3 (attribute "ans") data elements are encoded as EBCDIC representations of the hexadecimal data specified in the ISO 7811 and 7812 specifications. Thus, a hexadecimal D (binary 1101) is encoded as an EBCDIC "D" character, and so forth. The LL or LLL length specification associated with these data elements specifies the data element length in number of **bytes**.

Data field notations

Data field notations indicate where the data exists in the data element.

Notation	Description
Contents of subfields	Subfield number or number range Example: Contents of subfields 1–8
Contents of position(s)	Position number or number range Example: Contents of positions 1–8
N/A	Not applicable

Data justification notations

Data justification indicates the position of the data in the data element.

Notation	Description
Left	Data is left justified
Right	Data is right justified
See subfields	Data justification is defined in the subfield description indicating the subfield justification may vary between subfields
N/A	Not applicable

Date and time notations

Date and time notations indicate the format of the data that represents date and time.

Notation	Description
MM	month (two digits; 01–12)
DD	day (two digits; 01–31)

Notation	Description
YY	year (last two digits of calendar year; 00–99)
hh	hour (two digits; 00–23)
mm	minute (two digits; 00–59)
ss	second (two digits; 00–59)

Entity notations

Entity notations identify who is responsible for a message (acquirer, issuer, processor, or the Authorization Platform) at any given point.

Purpose

Several entities may insert or modify data elements in an Authorization Platform message as it flows from the message origin to the Authorization Platform and from the Authorization Platform to the message destination. These entities typically include the customer or processor at the origin, the Authorization Platform, and the customer or processor at the destination. In the message format layouts, the following three entities provide information to the originator, the Authorization Platform, and destination related to the data element requirements.

Notations

Notation	Description
Org	Originator Requirements. The message originator must satisfy this data element's requirements. Examples of originators include acquirers sending Authorization Request/0100 or Network Management Request/0800 messages.
Sys	Authorization Platform Requirements. The Authorization Platform may insert, correct, modify, or echo this data element while, for example, routing a message from the origin to the destination. The Authorization Platform may overwrite the data element and thereby destroy any previous content.
Dst	Destination Requirements. The message destination must expect this data element (read it) and accept this data element (process it) if the originator requirements are satisfied. Examples of destinations include issuers receiving Authorization Request/0100 or Network Management Request/0800 messages.

Presence notations

Presence notations indicate if and how data is present. These notations appear in the originator (Org), Authorization Platform (Sys), and destination (Dst).

Notation	Description
M	Mandatory. The data element is required in the message.

Notation	Description
C	Conditional. The data element is required in the message if the conditions described in the accompanying text apply.
O	Optional. The data element is not required, but may be included in the message at the message initiator's option.
X	Authorization Platform. The Authorization Platform may (or will) insert or overwrite the data element, depending on specific Authorization Platform support services provided for individual programs and services.
ME	Mandatory Echo. The data element is required in a response message and must contain the same value (echoed) from the original request or advice message.
CE	Conditional Echo. The data element is required in a response message if it was present in the original request or advice message, and it must contain the same value (echoed) from the original message.
XE	Authorization Platform Echo. The data element must contain the value from the original request or advice (echoed), if present.
•	Not Required or Not Applicable. The data element is not required or is not applicable. The transaction originator should not include this data element if this code is present in the Org or Dst column.
P	Sys passes data between Org and Dst.

Program and service category notations

Program and Service Category notations identify the various programs and services.

Notation	Description
MC	All Mastercard activity the Authorization Platform processes.
NP	Network processing services activity (often referred to as private label activity) the Authorization Platform processes that does not fall into one of the following categories: Mastercard (MC), Visa (VI), and Travel and Entertainment (TE). This specification indicates the most general Authorization Platform data element requirements for private label activity. Processors that support all these general data element specifications are able to access or participate in any Authorization Platform credit card or debit card gateway. For detailed information on specific gateways, refer to the appropriate program or service user manual.
VI	Visa activity the Authorization Platform processes. This specification includes Authorization Platform requirements for accepting Visa activity.

Notation	Description
TE	Travel and Entertainment (T&E) card activity the Authorization Platform processes that does not fall into one of the following categories: Mastercard (MC), network processing services activity (NP), and Visa (VI). This specification includes Authorization Platform interface requirements for Travel and Entertainment (T&E) card activity, including American Express, and Diners Club.
MS	Maestro card transactions that the Authorization Platform processes.
CI	Cirrus card transactions that the Authorization Platform processes.

Presence requirement notations

Presence requirement notations describe the possible presence or usage requirements. An entity presence requirement is the three combined values in the originator (Org), Authorization Platform (Sys), and destination (Dst).

Org	Sys	Dst	Description
M	.	M	Mandatory; originator must provide the data element.
M	X	M	Mandatory; originator must provide the data element; Authorization Platform adjusts/appends data.
C	.	C	Conditionally required; conditions are described in Comments column.
C	X	C	Conditionally required; Authorization Platform adjusts/appends data.
.	X	C	Authorization Platform provides the data element conditionally.
.	X	M	Authorization Platform always provides the data element.
O	.	C	Optional; originator may provide the data element.
O	X	C	Optional; originator may provide the data element; Authorization Platform adjusts/appends data.
M	.	.	Mandatory; originator must provide the data element; Authorization Platform does not forward data.
C	.	.	Conditionally required; Authorization Platform does not forward data.
O	.	.	Optional; Authorization Platform does not forward data.
.	X	.	Mastercard use only.

Authorization Platform messages

Authorization Platform messages are those online transaction messages used to transmit authorization, file update, reversal, administrative, and network management data across the Mastercard Network.

The Authorization Platform supports the following online transaction message types:

- Authorization/01xx messages
- Issuer File Update/03xx messages
- Reversal/04xx messages
- Administrative/06xx messages
- Network Management/08xx messages

List of Dual Message Authorization System MTIs

The following tables list the message types the *Dual Message Authorization System* supports and indicates the originating parties for each message type. The *Dual Message Authorization System* may not support all message types for each Mastercard product or service.

Acquirers and issuers are responsible for ensuring correct usage of the message types as described in the table and may be liable for the impacts of transmitting erroneous messages.

Table 3: Authorization/01xx messages

MTI	Description	Acquirer	Issuer	Mastercard Network
0100	Authorization Request	X		
0110	Authorization Request Response		X	X
0120	Authorization Advice: acquirer-generated	X		
0120	Authorization Advice: system-generated			X
0130	Authorization Advice Response: issuer-generated (responding to an acquirer-generated 0120)		X	
0130	Authorization Advice Response: issuer-generated (responding to a system-generated 0120)		X	
0130	Authorization Advice Response: system-generated			X
0180	Authorization Response Acknowledgment	X		
0190	Authorization Response Negative Acknowledgment			X

Table 4: Issuer File Update/03xx messages

MTI	Description	Acquirer	Issuer	Mastercard Network
0302	Issuer File Update Request		X	
0312	Issuer File Update Request Response			X

Table 5: Reversal/04xx messages

MTI	Description	Acquirer	Issuer	Mastercard Network
0400	Reversal Request	X		
0410	Reversal Request Response		X	X
0420	Reversal Advice			X
0430	Reversal Advice Response		X	

Table 6: Administrative/06xx messages

MTI	Description	Acquirer	Issuer	Mastercard Network
0600	Administrative Request	X		
0610	Administrative Request Response		X	X
0620	Administrative Advice	X	X	X
0630	Administrative Advice Response	X	X	X

Table 7: Network Management/08xx messages

MTI	Description	Acquirer	Issuer	Mastercard Network
0800	Network Management Request: sign-on/sign-off Because the <i>Mastercard Network</i> does not track session status information for an <i>acquirer</i> , <i>Dual Message Authorization System</i> sign-on/sign-off for acquirers is not required. However, acquirers may optionally send <i>Dual Message Authorization System</i> sign-on/sign-off messages (for example, if required by the vendor software).	X	X	
0800	Network Management Request: network connection status, member-generated	X	X	
0800	Network Management Request: network connection status, system-generated			X
0800	Network Management Request: host session activation/deactivation	X	X	X
0800	Network Management Request: PEK exchange			X
0800	Network Management Request: PEK exchange, on demand	X	X	
0810	Network Management Request Response: sign-on/sign-off			X
0810	Network Management Request Response: network connection status, customer-generated	X	X	
0810	Network Management Request Response: network connection status, system-generated			X
0810	Network Management Request: host session activation/deactivation	X	X	X
0810	Network Management Request Response: PEK exchange	X	X	
0810	Network Management Request Response: PEK exchange on demand			X
0820	Network Management Advice: PEK exchange			X

Message Type Identifier Presence Requirements by Program and Service

The following table lists the message type indicator (MTI) presence requirements for each authorization message type as it relates to the card program categories.

MTI	Description	MC	NP	VI	TE	MS	CI
0100	Authorization Request	√	√	√	√	√	√
0110	Authorization Request Response	√	√	√	√	√	√
0120	Authorization Advice: acquirer-generated	√	√	•	•	√	√
0120	Authorization Advice: issuer-generated	√	√	√	√	√	√
0120	Authorization Advice: system-generated	√	√	√	√	√	√
0130	Authorization Advice Response: issuer-generated (responding to an Acquirer-generated 0120)	√	√	√	√	√	√
0130	Authorization Advice Response: issuer-generated (responding to a System-generated 0120 from SAF)	√	√	√	√	√	√
0130	Authorization Advice Response: system-generated	√	√	√	√	√	√
0180	Authorization Acknowledgement	X	X	X	X	X	X
0190	Authorization Negative Acknowledgement	√	√	√	√	√	√
0302	Issuer File Update Request	√	√	•	•	√	√
0312	Issuer File Update Request Response	√	√	•	•	√	√
0400	Reversal Request/0400	√	√	√	•	√	√
0410	Reversal Request Response/0410	√	√	√	•	√	√
0420	Reversal Advice	√	√	√	•	√	√
0430	Reversal Advice Response	√	√	√	•	√	√
0600	Administrative Request/0600	√	√	•	•	•	•
0610	Administrative Request Response/0610	√	√	•	•	•	•
0620	Administrative Advice	√	√	√	•	√	√
0630	Administrative Advice Response	√	√	√	•	√	√
0800	Network Management Request	√	√	√	√	√	√
0810	Network Management Request Response	√	√	√	√	√	√
0820	Network Management Advice	√	√	√	√	√	√

Table key

√ = The MTI must be provided for the program or service indicated.

X = Optional support (for example, when individual Mastercard customers may elect to support certain message types at their own discretion).

• = The MTI is unavailable for the program or service.

Character Sets

Character sets define how letters, numbers, symbols, and special characters are displayed in computer text.

What character sets do Mastercard support?

Mastercard supports the following code pages for both standard and extended character sets:

- ASCII (ISO 8859-1)
- EBCDIC (Code Page 1047)

Mastercard supports standard EBCDIC formatted messages and bulk files. In addition to standard EBCDIC, Mastercard also supports the extended EBCDIC format as well as the standard and extended ASCII format for both authorization messages and selected bulk files. Customers using the MIP to transmit bulk files will still need to adhere to the existing Mastercard File Transfer Protocol format—that is, EBCDIC (for Header and Trailer information).

NOTE: Extended characters: Within the full character set, the alphabet is represented as used in the English language (with no character “enhancements”) and also the characters that may be found, for example, in the Spanish alphabet where the tilde is used over the Ñ.

Mastercard uses extended EBCDIC as the default character set type for sending messages unless otherwise specified by the customer.

Customers also have the option to send and receive either standard or extended character sets for both ASCII and EBCDIC formats in specific data elements that contain text. Following are examples of some data elements that may contain standard or extended text characters:

- DE 42 (Acceptor ID)
- DE 43 (Acceptor Name and Location)
- DE 120 (Record Data)
- DE 123 (Receipt Free Text)
- DE 124 (Member-defined Data)

The following bulk files will be supported in ASCII:

- R311—AMS File Updates
- R361—AMS File Updates (TEST)

Customers that want to receive messages in standard EBCDIC or extended or standard ASCII must inform Mastercard.

Character Sets

These tables contains valid ASCII to EBCDIC character sets.

Table 8: ASCII to EBCDIC character set: ASCII hex order

Note: For the purposes of ISO 8583, the < space > character is treated as a special character (attribute "s"). It may, however, also be used in fields designated as alphanumeric (attribute "an") but only in the form of trailing spaces used to pad out significant data to fill a fixed-length field.

Decimal value	ASCII hex	EBCDIC hex	ASCII symbol	Meaning	EBCDIC symbol	Meaning
032	20	40	< space >	Space	DS	digit select
033	21	5A	!	exclamation mark	SOS	start of significance
034	22	7F	"	straight double quotation mark	FS	field separator
035	23	7B	#	number sign	WUS	word underscore
036	24	5B	\$	dollar sign	BYP/INP	bypass/inhibit presentation
037	25	6C	%	percent sign	LF	line feed
038	26	50	&	ampersand	ETB	end of transmission block
039	27	7D	'	apostrophe	ESC	escape
040	28	4D	(	left parenthesis	SA	set attribute
041	29	5D	)	right parenthesis	SFE	
042	2A	5C	*	asterisk	SM/SW	set mode switch
043	2B	4E	+	addition sign	CSP	control sequence prefix
044	2C	6B	,	comma	MFA	modify field attribute

Decimal value	ASCII hex	EBCDIC hex	ASCII symbol	Meaning	EBCDIC symbol	Meaning
045	2D	60	-	subtraction sign	ENQ	enquiry
046	2E	4B	.	period	ACK	acknowledge
047	2F	61	/	right slash	BEL	bell
048	30	F0	0			
049	31	F1	1			
050	32	F2	2		SYN	synchronous idle
051	33	F3	3		IR	index return
052	34	F4	4		PP	presentation position
053	35	F5	5		TRN	
054	36	F6	6		NBS	numeric backspace
055	37	F7	7		EOT	end of transmission
056	38	F8	8		SBS	subscript
057	39	F9	9		IT	indent tab
058	3A	7A	:	colon	RFF	required form feed
059	3B	5E	;	semicolon	CU3	customer use 3
060	3C	4C	<	less than	DC4	device control 4
061	3D	7E	=	equal	NAK	negative acknowledge
062	3E	6E	>	greater than		
063	3F	6F	?	question mark	SUB	substitute
064	40	7C	@	at symbol	SP	space
065	41	C1	A		RSP	
066	42	C2	B		â	
067	43	C3	C		ä	
068	44	C4	D		à	

Decimal value	ASCII hex	EBCDIC hex	ASCII symbol	Meaning	EBCDIC symbol	Meaning
069	45	C5	E		á	
070	46	C6	F		â	
071	47	C7	G		ã	
072	48	C8	H		ç	
073	49	C9	I		ñ	
074	4A	D1	J		¢	cent
075	4B	D2	K		.	period
076	4C	D3	L		<	less than
077	4D	D4	M		(	left parenthesis
078	4E	D5	N		+	addition sign
079	4F	D6	O			logical or
080	50	D7	P		&	ampersand
081	51	D8	Q		é	
082	52	D9	R		ê	
083	53	E2	S		ë	
084	54	E3	T		è	
085	55	E4	U		í	
086	56	E5	V		î	
087	57	E6	W		ï	
088	58	E7	X		ì	
089	59	E8	Y		þ	
090	5A	E9	Z		!	exclamation point
091	5B	AD	[	left bracket	\$	dollar sign
092	5C	E0	\	left slash	*	asterisk
093	5D	BD	]	right bracket	)	right parenthesis
094	5E	5F	^	hat, circumflex	;	semicolon
095	5F	6D	_	underscore	^	hat, circumflex

Decimal value	ASCII hex	EBCDIC hex	ASCII symbol	Meaning	EBCDIC symbol	Meaning
096	60	79	`	grave	-	subtraction sign
097	61	81	a		/	right slash
098	62	82	b		Â	
099	63	83	c		Ã	
100	64	84	d		À	
101	65	85	e		Á	
102	66	86	f		Ä	
103	67	87	g		Å	
104	68	88	h		Ç	
105	69	89	i		Ñ	
106	6A	91	j		¡	split vertical bar
107	6B	92	k		,	comma
108	6C	93	l		%	percent sign
109	6D	94	m		_	underscore
110	6E	95	n		>	greater than
111	6F	96	o		?	question mark
112	70	97	p		Ø	
113	71	98	q		É	
114	72	99	r		Ê	
115	73	A2	s		Ë	
116	74	A3	t		È	
117	75	A4	u		Í	
118	76	A5	v		Î	
119	77	A6	w		Ï	
120	78	A7	x		Ì	
121	79	A8	y		`	grave
122	7A	A9	z		:	colon
123	7B	C0	{	left brace	#	number sign
124	7C	6A		logical or	@	at symbol

Decimal value	ASCII hex	EBCDIC hex	ASCII symbol	Meaning	EBCDIC symbol	Meaning
125	7D	D0	}	right brace	'	apostrophe
126	7E	A1	~	similar, tilde	=	equal

Table 9: ASCII to EBCDIC Character Set—EBCDIC hex order

Note: For the purposes of ISO 8583, the < space > character is treated as a special character (attribute "s"). It may, however, also be used in fields designated as alphanumeric (attribute "an") but only in the form of trailing spaces used to pad out significant data to fill a fixed-length field.

Decimal Value	EBCDIC Hex	ASCII Hex	EBCDIC Symbol	Meaning	ASCII Symbol	Meaning
032	40	20	DS	digit select	< space >	Space
046	4B	2E	ACK	acknowledge	.	period
060	4C	3C	DC4	device control 4	<	less than
040	4D	28	SA	set attribute	(	left parenthesis
043	4E	2B	CSP	control sequence prefix	+	addition sign
038	50	26	ETB	end of transmission block	&	ampersand
033	5A	21	SOS	start of significance	!	exclamation mark
036	5B	24	BYP/INP	bypass/inhibit presentation	\$	dollar sign
042	5C	2A	SM/SW	set mode switch	*	asterisk
041	5D	29	SFE		)	right parenthesis
059	5E	3B	CU3	customer use 3	;	semicolon
094	5F	5E	;	semicolon	^	hat, circumflex
045	60	2D	ENQ	enquiry	-	subtraction sign

Decimal Value	EBCDIC Hex	ASCII Hex	EBCDIC Symbol	Meaning	ASCII Symbol	Meaning
047	61	2F	BEL	bell	/	right slash
124	6A	7C	@	at symbol		logical or
044	6B	2C	MFA	modify field attribute	,	comma
037	6C	25	LF	line feed	%	percent sign
095	6D	5F	^	hat, circumflex	_	underscore
062	6E	3E			>	greater than
063	6F	3F	SUB	substitute	?	question mark
096	79	60	-	subtraction sign	`	grave
058	7A	3A	RFF	required form feed	:	colon
035	7B	23	WUS	word underscore	#	number sign
064	7C	40	SP	space	@	at symbol
039	7D	27	ESC	escape	'	apostrophe
061	7E	3D	NAK	negative acknowledge	=	equal
034	7F	22	FS	field separator	"	straight double quotation mark
097	81	61	/	right slash	a	
098	82	62	Â		b	
099	83	63	Ä		c	
100	84	64	À		d	
101	85	65	Á		e	
102	86	66	Ã		f	
103	87	67	Å		g	
104	88	68	Ç		h	
105	89	69	Ñ		i	
106	91	6A		split vertical bar	j	

Decimal Value	EBCDIC Hex	ASCII Hex	EBCDIC Symbol	Meaning	ASCII Symbol	Meaning
107	92	6B	,	comma	k	
108	93	6C	%	percent sign	l	
109	94	6D	_	underscore	m	
110	95	6E	>	greater than	n	
111	96	6F	?	question mark	o	
112	97	70	Ø		p	
113	98	71	É		q	
114	99	72	Ê		r	
126	A1	7E	=	equal	~	similar, tilde
115	A2	73	Ë		s	
116	A3	74	È		t	
117	A4	75	Í		u	
118	A5	76	Î		v	
119	A6	77	Ï		w	
120	A7	78	Ì		x	
121	A8	79	`	grave	y	
122	A9	7A	:	colon	z	
091	AD	5B	\$	dollar sign	[	left bracket
093	BD	5D	)	right parenthesis	]	right bracket
123	C0	7B	#	number sign	{	left brace
065	C1	41	RSP		A	
066	C2	42	â		B	
067	C3	43	ä		C	
068	C4	44	à		D	
069	C5	45	á		E	
070	C6	46	ã		F	
071	C7	47	å		G	
072	C8	48	ç		H	
073	C9	49	ñ		I	

Decimal Value	EBCDIC Hex	ASCII Hex	EBCDIC Symbol	Meaning	ASCII Symbol	Meaning
125	D0	7D	'	apostrophe	}	right brace
074	D1	4A	□	cent	J	
075	D2	4B	.	period	K	
076	D3	4C	<	less than	L	
077	D4	4D	(	left parenthesis	M	
078	D5	4E	+	addition sign	N	
079	D6	4F		logical or	O	
080	D7	50	&	ampersand	P	
081	D8	51	é		Q	
082	D9	52	ê		R	
092	E0	5C	*	asterisk	\	left slash
083	E2	53	ë		S	
084	E3	54	è		T	
085	E4	55	í		U	
086	E5	56	î		V	
087	E6	57	ï		W	
088	E7	58	ì		X	
089	E8	59	β		Y	
090	E9	5A	!	exclamation point	Z	
048	F0	30			0	
049	F1	31			1	
050	F2	32	SYN	synchronous idle	2	
051	F3	33	IR	index return	3	
052	F4	34	PP	presentation position	4	
053	F5	35	TRN		5	
054	F6	36	NBS	numeric backspace	6	

Decimal Value	EBCDIC Hex	ASCII Hex	EBCDIC Symbol	Meaning	ASCII Symbol	Meaning
055	F7	37	EOT	end of transmission	7	
056	F8	38	SBS	subscript	8	
057	F9	39	IT	indent tab	9	

Extended character sets

These tables contain valid ASCII to EBCDIC extended character sets.

Table 10: Extended Character Set: ASCII Hex order

Decimal value	ASCII hex	EBCDIC hex	ASCII character representation	Meaning	Mapped character	Mapping hex
192	C0	64	À	capital A, grave	A	41
193	C1	65	Á	capital A, acute	A	41
194	C2	62	Â	capital A, circumflex	A	41
195	C3	66	Ã	capital A, tilde	A	41
196	C4	63	Ä	capital A, umlaut	A	41
197	C5	67	Å	capital A, ring	A	41
198	C6	9E	Æ	capital AE, diphthong	E	45
199	C7	68	Ç	capital C, cedilla	C	43
200	C8	74	È	capital E, grave	E	45
201	C9	71	É	capital E, acute	E	45
202	CA	72	Ê	capital E, circumflex	E	45
203	CB	73	Ë	capital E, umlaut	E	45

Decimal value	ASCII hex	EBCDIC hex	ASCII character representation	Meaning	Mapped character	Mapping hex
204	CC	78	Ì	capital I, grave	I	49
205	CE	75	Í	capital I, acute	I	49
206	CD	76	Î	capital I, circumflex	I	49
207	CF	77	Ï	capital I, umlaut	I	49
208	D0	AC	Ð	capital Eth, Icelandic	D	44
209	D1	69	Ñ	capital N, tilde	N	4E
210	D2	ED	Ò	capital O, tilde	O	4F
211	D3	EE	Ó	capital O, acute	O	4F
212	D4	EB	Ô	capital O, circumflex	O	4F
213	D5	EF	Õ	capital O, tilde	O	4F
214	D6	EC	Ö	capital O, umlaut	O	4F
215	D7	BF	×	multiply sign	×	78
216	D8	80	Ø	capital O, slash	O	4F
217	D9	FD	Ù	capital U, grave	U	55
218	DA	FE	Ú	capital U, acute	U	55
219	DB	FB	Û	capital U, circumflex	U	55
220	DC	FC	Ü	capital U, umlaut	U	55
221	DD	BA	Ý	capital Y, acute	Y	59
222	DE	AE	Þ	capital THORN, Icelandic	P	50

Decimal value	ASCII hex	EBCDIC hex	ASCII character representation	Meaning	Mapped character	Mapping hex
223	DF	59	ß	small sap s, German	S	53
224	E0	44	à	small a, grave	a	61
225	E1	45	á	small a, acute	a	61
226	E2	42	â	small a, circumflex	a	61
227	E3	46	ã	small a, tilde	a	61
228	E4	43	ä	small a, umlaut	a	61
229	E5	47	å	small a, ring	a	61
230	E6	9C	æ	small ae diphthong	e	65
231	E7	48	ç	small c, cedilla	c	63
232	E8	54	è	small e, grave	e	65
233	E9	51	é	small e, acute	e	65
234	EA	52	ê	small e, circumflex	e	65
235	EB	53	ë	small e, umlaut	e	65
236	EC	58	ì	small i, grave	i	69
237	ED	55	í	small i, acute	i	69
238	EE	56	î	small i, circumflex	i	69
239	EF	57	ï	small i, umlaut	i	69
240	F0	8C	ð	small eth, Icelandic	d	64
241	F1	49	ñ	small n, tilde	n	6E
242	F2	CD	ò	small o, grave	o	6F
243	F3	CE	ó	small o, acute	o	6F
244	F4	CB	ô	small o, circumflex	o	6F
245	F5	CF	õ	small o, tilde	o	6F

Decimal value	ASCII hex	EBCDIC hex	ASCII character representation	Meaning	Mapped character	Mapping hex
246	F6	CC	ö	small o, umlaut	o	6F
247	F7	E1	÷	division sign	/	2F
248	F8	70	ø	small o, slash	o	6F
249	F9	DD	ù	small u, grave	u	75
250	FA	DE	ú	small u, acute	u	75
251	FB	DB	û	small u, circumflex	u	75
252	FC	DC	ü	small u, umlaut	u	75
253	FD	8D	ý	small y, acute	y	79
254	FE	8E	þ	small thorn, Icelandic	p	70
255	FF	DF	ÿ	small y, umlaut	y	79

Table 11: Extended Character Set: EBCDIC Hex order

Decimal value	EBCDIC hex	ASCII hex	ASCII character representation	Meaning	Mapped character	Mapping hex
226	42	E2	â	small a, circumflex	a	61
228	43	E4	ä	small a, umlaut	a	61
224	44	E0	à	small a, grave	a	61
225	45	E1	á	small a, acute	a	61
227	46	E3	ã	small a, tilde	a	61
229	47	E5	å	small a, ring	a	61
231	48	E7	ç	small c, cedilla	c	63
241	49	F1	ñ	small n, tilde	n	6E

Decimal value	EBCDIC hex	ASCII hex	ASCII character representation	Meaning	Mapped character	Mapping hex
233	51	E9	é	small e, acute	e	65
234	52	EA	ê	small e, circumflex	e	65
235	53	EB	ë	small e, umlaut	e	65
232	54	E8	è	small e, grave	e	65
237	55	ED	í	small i, acute	i	69
238	56	EE	î	small i, circumflex	i	69
239	57	EF	ï	small i, umlaut	i	69
236	58	EC	ì	small i, grave	i	69
223	59	DF	ß	small sap s, German	S	53
194	62	C2	Â	capital A, circumflex	A	41
196	63	C4	Ä	capital A, umlaut	A	41
192	64	C0	À	capital A, grave	A	41
193	65	C1	Á	capital A, acute	A	41
195	66	C3	Ã	capital A, tilde	A	41
197	67	C5	Å	capital A, ring	A	41
199	68	C7	Ç	capital C, cedilla	C	43
209	69	D1	Ñ	capital N, tilde	N	4E
248	70	F8	ø	small o, slash	o	6F
201	71	C9	É	capital E, acute	E	45
202	72	CA	Ê	capital E, circumflex	E	45
203	73	CB	Ë	capital E, umlaut	E	45

Decimal value	EBCDIC hex	ASCII hex	ASCII character representation	Meaning	Mapped character	Mapping hex
200	74	C8	È	capital E, grave	E	45
205	75	CE	Í	capital I, acute	I	49
206	76	CD	Î	capital I, circumflex	I	49
207	77	CF	Ï	capital I, umlaut	I	49
204	78	CC	Ì	capital I, grave	I	49
216	80	D8	Ø	capital O, slash	O	4F
240	8C	F0	ð	small eth, Icelandic	d	64
253	8D	FD	ý	small y, acute	y	79
254	8E	FE	þ	small thorn, Icelandic	p	70
230	9C	E6	æ	small ae diphthong	e	65
198	9E	C6	Æ	capital AE, diphthong	E	45
208	AC	D0	Ð	capital Eth, Icelandic	D	44
222	AE	DE	Þ	capital THORN, Icelandic	P	50
221	BA	DD	Ý	capital Y, acute	Y	59
215	BF	D7	×	multiply sign	×	78
244	CB	F4	ô	small o, circumflex	o	6F
246	CC	F6	ö	small o, umlaut	o	6F
242	CD	F2	ò	small o, grave	o	6F
243	CE	F3	ó	small o, acute	o	6F

Decimal value	EBCDIC hex	ASCII hex	ASCII character representation	Meaning	Mapped character	Mapping hex
245	CF	F5	ô	small o, tilde	o	6F
251	DB	FB	û	small u, circumflex	u	75
252	DC	FC	ü	small u, umlaut	u	75
249	DD	F9	ù	small u, grave	u	75
250	DE	FA	ú	small u, acute	u	75
255	DF	FF	ÿ	small y, umlaut	y	79
247	E1	F7	÷	division sign	/	2F
212	EB	D4	Ô	capital O, circumflex	O	4F
214	EC	D6	Ö	capital O, umlaut	O	4F
210	ED	D2	Ò	capital O, tilde	O	4F
211	EE	D3	Ó	capital O, acute	O	4F
213	EF	D5	Õ	capital O, tilde	O	4F
219	FB	DB	Û	capital U, circumflex	U	55
220	FC	DC	Ü	capital U, umlaut	U	55
217	FD	D9	Ù	capital U, grave	U	55
218	FE	DA	Ú	capital U, acute	U	55

Swedish Domestic Authorization Switching character set

Customers using the Swedish Domestic Authorization Switching Service (SASS) also may support Swedish national characters in EBCDIC as described in the following table.

Table 12: Swedish Domestic Authorization Switching character set

EBCDIC IBM-278	EBCDIC (Hexadecimal)	ASCII (Hexadecimal)	Graphic	Attribute
Å	5B	24	\$	s
Ä	7B	23	#	s
Ö	7C	40	@	s
å	D0	7d	}	s
ä	C0	7b	{	s

Chapter 2 Message Definitions and Flows

This section provides message definitions of all message types the Authorization Platform uses followed by the detailed message flow diagrams that illustrate both the standard and exception (error condition) message processing.

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About Message Definitions

Message definitions describe the general purpose, type, routing, and response information of each Authorization Platform message type.

About Authorization Messages

The authorization message definitions define the authorization request, advice, and response message types.

Authorization/01xx messages transport authorization-related information. "Authorization" is a term applied to transactions for which, by design, there is partial transaction data contained within the individual messages; no actual posting of accounts at the issuer processing system (IPS) occurs during this type of transmission. Consequently, all Authorization/01xx messages and advice transactions assume that follow-up information (for example, paper or electronic clearing transactions) will be used to effect actual settlement, cardholder account posting, and cardholder billing.

Authorization/01xx messages are defined as:

- Authorization Request messages
- Authorization Advice messages
- Authorization Response Acknowledgement messages

Authorization Request/0100

The Authorization Request/0100 message requests approval authorization or guarantee for a transaction to proceed. The Authorization Request/0100 message is not intended to permit the application of this transaction to the cardholder's account for issuing a bill or statement.

Type	Interactive
Routing	• From an acquirer processing system (APS) to the Authorization Platform. • From the Authorization Platform to an IPS.
Response	An Authorization Request Response/0110 message is required.

Authorization Request Response/0110

The Authorization Request Response must be sent in response to an Authorization Request/0100 message; it carries the response information required to service (approve or deny) the Authorization Request/0100 message.

Type	Interactive
Routing	• From an IPS to the Authorization Platform. • From the Authorization Platform to an APS.

Response	The APS may provide an Authorization Acknowledgement/0180 message to the Authorization Platform to acknowledge positive receipt of the Authorization Request Response/0110 message from the Authorization Platform. The Authorization Acknowledgement/0180 message is optional for an APS.
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About Authorization Advice Messages

The Authorization Platform uses guaranteed advice message delivery for all advice messages transmitted using the Authorization Platform.

Guaranteed advice message delivery provides a message routing facility. When an advice message is forwarded to the Authorization Platform, the Authorization Platform:

- Immediately secures the transaction for future delivery
- Responds to the advice message initiator with an advice response message, indicating that it received and secured the message
- Forwards the advice message to the appropriate receiving entity

The Authorization Platform guarantees the delivery of all advice messages routed through the Mastercard Network. Occasionally, the Authorization Platform cannot immediately deliver an advice message (for example, because of a system or communication failure at the destination). In this case, the Authorization Platform stores the message on its store-and-forward (SAF) processing facilities and automatically delivers the message when communication is restored.

The receiver of an advice message must acknowledge receipt with an appropriate advice response message. When the Authorization Platform receives the advice response message, Mastercard considers advice delivery complete and removes the message from SAF processing queues.

The Guaranteed Advice Message Delivery messages include:

- Authorization Advice/0120—Acquirer-generated
- Authorization Advice/0120—System-generated
- Authorization Advice Response/0130—Issuer-generated (Responding to a System-generated 0120 from SAF)
- Authorization Advice Response/0130—Issuer-generated (Responding to an Acquirer-generated 0120)
- Authorization Advice Response/0130—System-generated

Authorization Advice/0120: Acquirer-Generated

The Authorization Advice/0120—Acquirer-generated message advises of an authorization that was carried out on the issuer's behalf. It is not intended to permit the application of this transaction to the cardholder's account for issuing a bill or statement; for example, this is a non-posting advice message.

Type	Noninteractive
Routing	From the APS to an IPS.

Response	An Authorization Advice Response/0130—Issuer-generated message is required.
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Automated Fuel Dispenser

An automated fuel dispenser (AFD) transaction is identified with MCC 5542 and may occur at a cardholder-activated terminal (CAT) or as an electronic commerce transaction. Acquirers of AFD merchants send an Authorization Advice/0120 message containing DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code), value 191 (Acquirer Processing System [APS] Completed Authorization Transaction) to the issuer providing the actual transaction amount for each approved AFD transaction.

Offline Account Status Inquiry for Account Ranges Registered for the Authentication Indicator Service

Acquirers of merchants supporting EMV^{®1} offline account status inquiry transactions send an Authorization Advice/0120 message containing DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code), value 192 (M/Chip Offline Advice to Issuer) to the issuer.

Authorization Advice/0120: Issuer-Generated

Authorization Request/0100 messages may contain Bank Identification Number (BINs) that customers have selected for scoring. For these transactions, issuers create and send Authorization Advice/0120 messages to the risk scoring system. The risk scoring system sends to the issuer an Authorization Advice Response/0130 to acknowledge receipt of the Authorization Advice/0120.

Type	Noninteractive
Routing	From the IPS to the Authorization Platform.
Response	An Authorization Advice Response/0130—System-generated message is required.

Authorization Advice/0120: System-Generated

The Authorization Advice/0120—System-generated message advises of an authorization that was carried out on the issuer's behalf. It is not intended to permit the application of this transaction to the cardholder's account for issuing a bill or statement; for example, this is a non-posting advice message.

Type	Noninteractive
Routing	From the Authorization Platform to an IPS.
Response	An Authorization Advice Response/0130—Issuer-generated message is required.

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Authorization Advice Response/0130: Issuer-Generated (Responding to an Acquirer-Generated 0120)

The Authorization Advice Response/0130—Issuer-generated message indicates positive receipt of an Authorization Advice/0120—Acquirer-generated message.

Type	Noninteractive
Routing	From the IPS to the APS when responding to the Authorization Advice/0120—Acquirer-generated message.
Response	None

Authorization Advice Response/0130: Issuer-Generated (Responding to a System-Generated 0120 from SAF)

The Authorization Advice Response/0130—Issuer-generated message indicates positive receipt of an Authorization Advice/0120—System-generated message from SAF.

NOTE: Mastercard recommends that customers use response code 00 for all Stand-In Advice messages.

Type	Noninteractive
Routing	From an IPS to the Authorization Platform when responding to the Authorization Advice/0120—System-generated message.
Response	None

Authorization Advice Response/0130: System-Generated

The Authorization Advice Response/0130—System-generated message indicates positive receipt of an Authorization Advice/0120 message.

Type	Noninteractive
Routing	From the Authorization Platform to the APS or IPS.
Response	None

About Authorization Response Acknowledgement Messages

For APS customers, the Authorization Platform provides optional interactive response acknowledgement for Authorization/0100 messages. Acknowledgement messages indicate positive receipt of a response or advice message.

Acquirers are not required to support Authorization Response Acknowledgement/0180 messages, but Mastercard encourages acquirer customers to select this option and to support these messages. Customers may select this option at network configuration time.

The Authorization Response Acknowledgement messages include:

- Authorization Response Acknowledgement/0180
- Authorization Response Negative Acknowledgement/0190

Authorization Acknowledgement/0180

The Authorization Advice Acknowledgement/0180 message indicates positive acknowledgement that it received a previous Authorization Request Response/0110 message. The Authorization Acknowledgement/0180 message is optional for an APS.

Type	Interactive
Routing	From an APS to the Authorization Platform
Response	None

Authorization Negative Acknowledgement/0190

The Authorization Platform sends to an IPS an Authorization Negative Acknowledgement/0190 message only in response to late or invalid Authorization Request Response/0110 messages from an IPS. This informs the IPS that the Authorization Platform has timed-out or rejected the Authorization Request Response/0110 message and initiated Stand-In processing or alternate authorization services on the issuer's behalf. The Authorization Platform also may send the Authorization Negative Acknowledgement/0190 in response to a late or invalid Reversal Request Response/0410 message. In this scenario, the Authorization Negative Acknowledgement/0190 message informs the IPS that the Authorization Platform responded to the acquirer.

Type	Interactive
Routing	From the Authorization Platform to an IPS
Response	None

About Issuer File Update Messages

Issuers use Issuer File Update/03xx messages to update individual data files or system parameter tables that the Authorization Platform maintains on their behalf. The Authorization Platform uses these data files to control the operation of standard and optional features that customers may select when they participate in one or more of the Mastercard program and service offerings.

The Issuer File Update messages include:

- Issuer File Update Request/0302
- Issuer File Update Request Response/0312

Issuer File Update Request/0302

The Issuer File Update Request/0302 message requests an Issuer File Update or inquiry action to be initiated on the issuer's behalf.

Type	Interactive
Routing	From an IPS to the Authorization Platform
Response	An Issuer File Update Request Response/0312 message is required

Issuer File Update Request Response/0312

The Issuer File Update Request Response/0312 message must be sent in response to a Issuer File Update Request/0302; it indicates the disposition of the Issuer File Update Request/0302 message.

Type	Interactive
Routing	From the Authorization Platform to an IPS
Response	None

About Reversal Messages

Reversal Request/0400 messages are generated by an acquirer when the acquirer is unable to deliver an issuer's Authorization Request Response/0110 to a merchant. Merchants also may initiate a Reversal Request/0400 message to cancel the full or partial amount of the original authorization amount.

Reversal/04xx messages include:

- Reversal Request/0400
- Reversal Request Response/0410
- Reversal Advice/0420
- Reversal Advice Response/0430

Reversal Request/0400

The Reversal Request/0400 message reverses fully or partially an earlier authorization request.

Type	Interactive
Routing	From acquirer to the issuer or intermediate network facility
Response	A Reversal Request Response/0410 message is required

Reversal Request Response/0410

The Reversal Request Response/0410 message is sent in response to a Reversal Request/0400 message and denotes the disposition of the Reversal Request/0400 message.

Type	Interactive
Routing	From the issuer to the acquirer
Response	None

Reversal Advice/0420

The Reversal Advice/0420 message fully reverses a previous Authorization Request/0100. It advises an issuer of a Reversal Request/0400 responded to by the Authorization Platform.

The Authorization Platform may generate this message on detection of an exception condition while processing a previous Authorization Request Response/0110 message.

The Authorization Platform may generate this message when an issuer is unavailable to respond a Reversal Request/0400 or is delayed in providing a Reversal Request Response/0410 message.

Type	Noninteractive
Routing	From the Authorization Platform to an IPS
Response	A Reversal Advice Response/0430 message is required.

Reversal Advice Response/0430

The Reversal Advice Response/0430 message must be sent in response to a Reversal Advice/0420 message to acknowledge positive receipt of that message.

Type	Noninteractive
Routing	From an IPS to the Authorization Platform.
Response	None

About Administrative Messages

Administrative messages request confirmation of data or transmit data in a format other than identified in this International Standard. Member generated advice messages are used to transmit administrative (free format) textual messages between any two parties participating as customers on the Authorization Platform. Authorization Platform generated advice messages return indecipherable messages to a message originator.

Administrative messages include:

- Administrative Request/0600

- Administrative Request Response/0610
- Administrative Advice/0620—System-generated
- Administrative Advice/0620—Member-generated
- Administrative Advice Response/0630

Administrative Request/0600

The Administrative Request/0600 message contains customer data in DE 113–119 (Reserved for National Use) and the type of usage in DE 60 (Advice Reason Code). Any eligible Administrative Request/0600 message is routed to the issuer associated to the account range within DE 2 (Primary Account Number [PAN]) and the appropriate response timer is set for the issuer Administrative Request Response/0610 message. The response timer is configured by each DE 60 usage type and initially will be set at a default of 30 seconds for each usage type.

Type	Interactive
Routing	From acquirer to issuer
Response	An Administrative Request Response/0610 message is required.

Administrative Request Response/0610

The Authorization Platform forwards from the issuer to the acquirer an Administrative Request Response/0610 message that contains the issuer's response data, if the issuer provided a timely response or DE 39 (Response Code), value 91 (Authorization System or issuer system inoperative), if an issuer response is not provided within applicable time limits.

Type	Interactive
Routing	From issuer to acquirer
Response	none

Administrative Advice/0620

Administrative Advice/06xx messages are administrative messages that two parties participating in a given Mastercard program or service offering may use when using the Authorization Platform.

The Authorization Platform routes messages from an originator to a receiver and, in general, does not distinguish whether the originator or receiver is an issuer or an acquirer.

The Authorization Platform uses only noninteractive Administrative Advice/06xx messages. These messages fall under the general category of advice messages, and as such, are subject to the Authorization Platform Guaranteed Advice Delivery procedures that are standard for all advice messages. If the Authorization Platform cannot immediately deliver these messages to their intended destination, the Authorization Platform automatically assumes the responsibility for storing them in the SAF System. When network delivery-point communication has been reestablished and the receiver requests delivery of the advice messages, the Authorization Platform forwards them to the proper destination.

In all cases, DE 60 (Advice Reason Code) within the Administrative Advice/0620 message determines the specific reason for the advice message.

Following are how the Administrative Advice/06xx messages are used:

- Administrative Advice/0620—System Generated messages are used to:
 - Return indecipherable messages to a message originator with an appropriate error condition code, indicating the point at which the Authorization Platform terminated message parsing or message processing. In general, messages returned in Administrative Advice/0620 messages will have improperly coded or garbled Message Type Indicators (MTIs) or improperly coded bit maps.
- Administrative Advice/0620—Member-generated messages are used to transmit administrative (free format) textual messages between any two parties participating as customers on the Authorization Platform.

Type	Noninteractive
Routing	Between any two parties participating on the Authorization Platform.
Response	An Administrative Advice Response/0630 message is required.

Administrative Advice Response/0630

The Administrative Advice Response/0630 message must be sent in response to an Administrative Advice/0620 message to acknowledge positive receipt of that message.

Type	Noninteractive
Routing	From the receiver to the originator of the related Administrative Advice/0620 message.
Response	None

About Network Management Messages

The Authorization Platform, customer processing systems (CPSs), acquirer processing systems (APSs), issuer processing systems (IPSs), and intermediate network facilities (INFs) use Network Management/08xx messages to coordinate system or network events or tasks. These systems also use these messages to communicate network status conditions.

Network Management/08xx messages include:

- Network Management Request/0800
- Network Management Request Response/0810
- Network Management Advice/0820

Typical uses of Network Management/08xx messages include:

- Sign on to and sign off from the Authorization Platform

- Perform a Network Connection Status to Mastercard
- Perform Host Session Activation or Deactivation
- Perform encryption key management

The Authorization Platform routes Network Management/08xx messages from an originator to a receiver and, in general, does not distinguish whether the originator or receiver is an issuer or an acquirer.

Network Management Request/0800

The Network Management Request/0800 message controls the Mastercard Network by communicating or coordinating system condition or system security. DE 70, a mandatory data element in all Network Management/08xx messages, determines specific Network Management/08xx message functions.

Type	Interactive
Routing	Between the Authorization Platform and any other party (such as CPS, APS, IPS, INF) communicating directly with the Authorization Platform. May be originated by either party.
Response	A Network Management Request Response/0810 message is required.

Network Management Request Response/0810

The Network Management Request Response/0810 message must be sent in response to a Network Management Request/0800 to acknowledge positive receipt of that message.

Type	Interactive
Routing	From "receiver" to "originator" of the related Network Management Request/0800.
Response	None

Network Management Advice/0820

The Network Management Advice/0820 message advises of a previous Network Management Request/0800 message. There is no response to a Network Management Advice/0820 message.

Type	Noninteractive
Routing	From the Authorization Platform to any other party (such as CPS, APS, IPS, INF) communicating directly with the Authorization Platform.
NOTE: May not be originated by any party other than the Authorization Platform.	

Response	None
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About Message Flows

Message flows describe the overall standard and exception process of a message type or message pair in various scenarios. Each message flow provides an illustration of the process with the associated stages that describe each part of the process. All Authorization/01xx message flows are illustrated without showing the optional (acquirer-selected) use of the Authorization Acknowledgement/0180 message.

Authorization Message Routing Timers

Following are the specific Authorization Platform message routing timer values applicable to acquirer-generated 0100, 0120, and 0400 messages. All system-generated Advice/0120, Issuer File Update/03xx, system-generated Reversal/0420, Administrative/06xx, and Network Management/08xx messages remain unchanged and retain their current timer values.

NOTE: Mastercard reserves the right to adjust local market timer values based on specific market conditions and as additional exception countries are added, they will be announced in the *Global Operations Bulletin*.

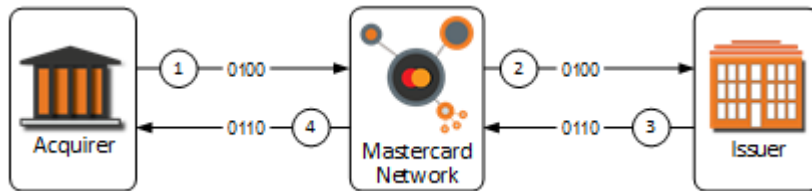
Product and Transaction Type	Issuer Response Time (in seconds)	Alternate Authorization Service Provider (if applicable)	Acquirer Minimum Wait Time (in seconds)
Mastercard Credit—POS	9 ^a	3	12 ^b
Mastercard—POS PIN for Credit	18 ^c	6	24
Mastercard—ATM	12	6	18
Debit Mastercard—POS	Same as Mastercard Credit—POS	Same as Mastercard Credit—POS	Same as Mastercard Credit—POS
Maestro—All	18 ^d	6	24
Cirrus—ATM	18	6	24

- ^a For transactions acquired in the countries of Brazil, Canada, the United Kingdom, and the United States, Mastercard has reduced the timers by 2 additional seconds. In these countries, the maximum time Mastercard will wait before invoking alternate authorization provider processing is reduced to 7 seconds and the minimum time that an acquirer must wait is reduced to 10 seconds.
- ^b Because the Mastercard Network does not track session status information for an acquirer, Authorization Platform sign-on/sign-off for acquirers is not required. However, acquirers optionally may send Authorization Platform sign-on/sign-off messages (for example, if required by vendor software).
- ^c A transaction is considered to be PIN for Credit if it is Dual Message System acquired, but the issuer keys are managed by the Single Message System. If the issuer is Dual Message System managed, the transaction is not flagged as being PIN for Credit and defaults to the standard POS timer of 9 seconds.

Product and Transaction Type	Issuer Response Time (in seconds)	Alternate Authorization Service Provider (if applicable)	Acquirer Minimum Wait Time (in seconds)
Private Label—POS	12	6	18
AMEX—POS	12	6	18
Visa—POS	12	6	18

Authorization Request/0100 and Authorization Request Response/0110

This message flow describes the standard authorization transaction process.

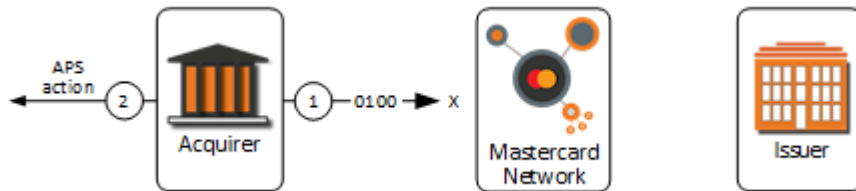


1. The acquirer initiates an Authorization Request/0100 message and sends it to the Authorization Platform.
2. The Authorization Platform forwards the Authorization Request/0100 message to the issuer.
3. The issuer generates an appropriate Authorization Request Response/0110 message and sends it to the Authorization Platform.
4. The Authorization Platform forwards the Authorization Request Response/0110 message to the acquirer.

NOTE: The "issuer" in the illustration may in fact be a Mastercard IPS used for Dual Message System processing. Optional issuer processing services are available to Mastercard customers in support of various programs and services; for example, the Authorization Platform may provide IPS services for customers that elect to use this service.

Authorization Request/0100: Communication Failure at Acquirer

This message flow describes exception processing when communication fails between the APS and the Authorization Platform (resubmit processing is not supported).



^d For Maestro POS transactions acquired in the Netherlands, Mastercard has reduced the timer to 7 seconds, which is the maximum time Mastercard will wait before invoking alternate authorization provider processing. If no response is received within 10 seconds, Mastercard will send a time-out response to the acquirer.

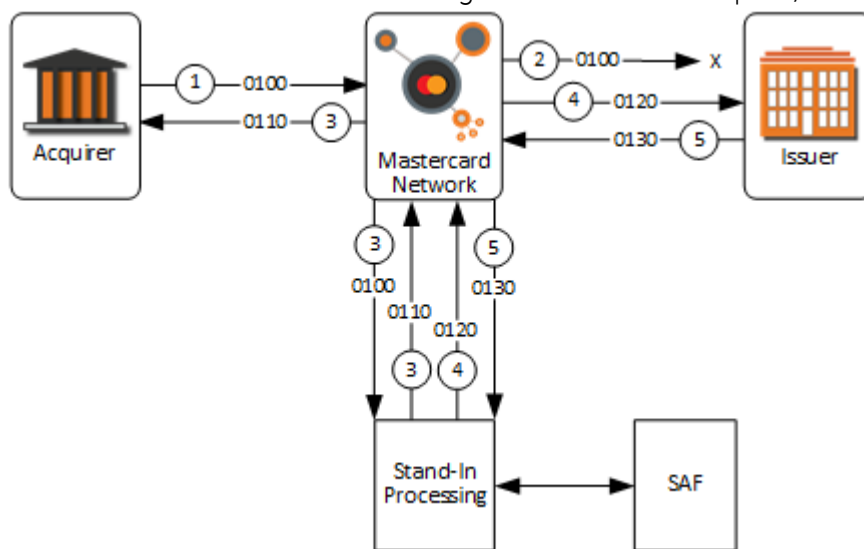
1. The acquirer initiates an Authorization Request/0100 message, but it cannot be delivered to the Authorization Platform because of system failure.
2. The acquirer completes the transaction at the point of interaction. In most cases, the APS simply denies the transaction. However, the APS may elect to approve the transaction for completion for various reasons.

Depending on the type of systems failure and the operating procedures or regulations governing the program or service involved, the acquirer, the merchant, or both may be required to assume full or partial liability for the transaction if it is completed without appropriate issuer authorization.

If the acquirer elects to authorize the transaction at the point of interaction, the acquirer may be required to accept financial liability for the transaction. The acquirer may notify the issuer of the authorization it approved on the issuer's behalf by sending an Authorization Advice/0120 message.

Authorization Request/0100: Communication Failure at Issuer

This message flow describes exception processing when communication fails between the IPS and the Authorization Platform during an Authorization Request/0100 message.



1. The acquirer initiates an Authorization Request/0100 message and sends it to the Authorization platform.
2. The Authorization platform attempts to forward the Authorization Request/0100 message to the issuer but is unable to complete the message transmission because of a communication link failure or IPS failure. As a result, the Authorization platform immediately sends the Authorization Request/0100 message to Stand-In processing for alternate processing functions.
3. After the Authorization platform sends the Authorization Request/0100 message, if the issuer permits Stand-In processing to authorize transactions, Stand-In processing generates an appropriate Authorization Request Response/0110 message on the issuer's behalf and forwards it to the acquirer. Stand-In processing also generates an Authorization Advice/

0120 message and forwards it to the Authorization platform store-and-forward (SAF) process for later transmission to the issuer.

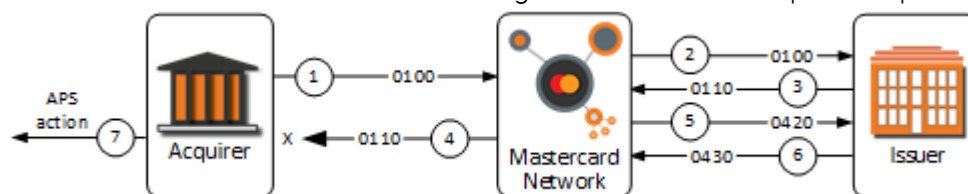
NOTE: Stand-In processing generates and forwards Authorization Advice/0120 messages for "approved," "declined," and "capture card" Authorization Request/0100 messages.

4. When communication re-establishes with the IPS, the Stand-In System SAF process forwards the Authorization Advice/0120 message to the issuer. This action allows the issuer to update its cardholder "open-to-buy" and "velocity" data files in a timely manner.
5. The issuer, upon receiving and securing the Authorization Advice/0120 message, returns an Authorization Advice Response/0130 message to the Authorization Platform to indicate positive receipt of the Authorization Advice/0120 message.

NOTE: Each time the SAF process receives an Authorization Advice Response/0130 message, it sends the next Authorization Advice/0120 message. This process repeats until there are no more Authorization Advice/0120 messages. If the Authorization Platform does not receive an Authorization Advice Response/0130 message, it assumes the issuer did not receive the previous Authorization Advice/0120 message, ceases the SAF session, and places an unqueued Authorization Advice/0120 message into the queue for delivery.

Authorization Request Response/0110: Communication Failure at Acquirer

This message flow describes exception processing when communication fails between the Authorization Platform and the APS during an Authorization Request Response/0110 message.



1. The acquirer initiates an Authorization Request/0100 message and sends it to the Authorization Platform.
2. The Authorization Platform forwards the Authorization Request/0100 message to the issuer.
3. The issuer generates an appropriate Authorization Request Response/0110 message and sends it to the Authorization Platform.
4. The Authorization Platform attempts to forward the Authorization Request Response/0110 message to the acquirer but cannot successfully complete the transmission because of communications link failure or APS failure.
5. The Authorization Platform has determined that the issuer's Authorization Request Response/0110 message is "undeliverable," so the Authorization Platform immediately generates a Reversal Advice/0420 message back to the issuer if the Authorization Request Response/0110 message indicates an approval.

The purpose of this advice is to let the issuer know that its Authorization Request Response/0110 message was undelivered and that the issuer should "reverse" or otherwise adjust its

cardholder authorization files as necessary to reflect that the transaction could not be completed.

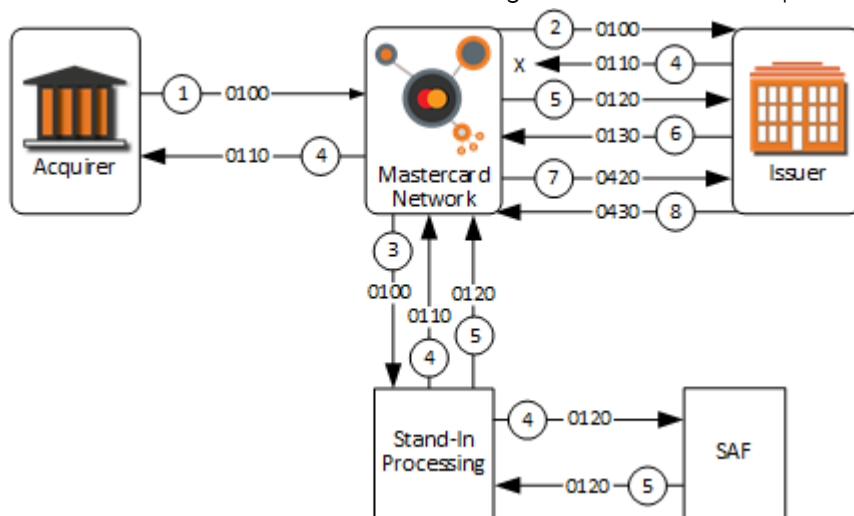
6. The issuer responds back to the Authorization Platform with a Reversal Advice Response/0430 message to acknowledge positive receipt of the Reversal Advice/0420 message.
7. Meanwhile, if the APS is still operating, it detects a time-out condition on the Authorization Request Response/0110 message that it is expecting from the Authorization Platform. When the time-out occurs, the APS must make a decision whether to authorize the transaction at the point of interaction. If the APS denies the transaction, the message flow terminates at this point.

If the acquirer elects to authorize the transaction at the point of interaction, the acquirer may be required to accept financial liability for the transaction. The acquirer may notify the issuer of the authorization it approved on the issuer's behalf by sending an Authorization Advice/0120 message.

NOTE: Does not apply in the case of a Mastercard Hosted Mobile Phone Top-up request. If the issuer approves the Mastercard Hosted Mobile Phone Top-up transaction, the Authorization Platform will complete the Mastercard Hosted Mobile Phone Top-up request. The acquirer should provide the cardholder with notification at the ATM that the top-up request will be complete within 15 minutes.

Authorization Request Response/0110: Communication Failure at Issuer

This message flow describes exception processing when communication fails between the Authorization Platform and the IPS during an Authorization Request Response/0110 message.



1. The acquirer initiates an Authorization Request/0100 message.
2. The Authorization Platform forwards the Authorization Request/0100 message to the issuer.
3. The IPS cannot return the appropriate Authorization Request Response/0110 message because of a communications failure between the IPS and the Authorization Platform. The IPS **must** assume that the Authorization Platform or the acquirer will take appropriate

action at this point and should reverse any effect to its cardholder or velocity files that may have taken place during the authorization process.

If Stand-In processing or an alternate authorization process subsequently approves the transaction, the issuer can receive a later notification by a SAF Authorization Advice/0120 message.

The Authorization Platform detects a time-out condition from the issuer on the Authorization Request Response/0110 message. If the issuer permits Stand-In processing, the Authorization Platform sends an Authorization Request/0100 message to Stand-In processing.

4. Stand-In processing generates an appropriate Authorization Request Response/0110 message on the issuer's behalf and forwards it to the acquirer. It also generates an Authorization Advice/0120 message and forwards it to the Authorization Platform SAF process for later transmission to the issuer.

NOTE: Stand-In processing generates and forwards Authorization Advice/0120 messages for both "approved" and "declined" Authorization Request/0100 messages.

5. When communication re-establishes with the IPS, the Stand-In System SAF process forwards the Authorization Advice/0120 message to the issuer. This action allows the issuer to update its cardholder "open-to-buy" and "velocity" data files in a timely manner.
6. The issuer, on receiving and securing the Authorization Advice/0120 message, returns an Authorization Advice Response/0130 message to the Authorization Platform to indicate positive receipt of the Authorization Advice/0120 message.

NOTE: Each time the SAF process receives an Authorization Advice Response/0130 message, it sends the next Authorization Advice/0120 message. This process repeats until there are no more Authorization Advice/0120 messages. If the Authorization Platform does not receive an Authorization Advice Response/0130 message, it assumes the issuer did not receive the previous Authorization Advice/0120 message and ceases the SAF session.

7. The Authorization Platform sends a Reversal Advice/0420 message containing the following values to the issuer because no issuer response is received:
 - DE 39 = 82 (Time-out at issuer)
 - DE 60, subfield 1 = 402 (Issuer Time-out)
8. When the issuer receives the Reversal Advice/0420 message, it must generate a Reversal Advice Response/0430 message.

Authorization Request Response/0110: Stand-In System Allowed

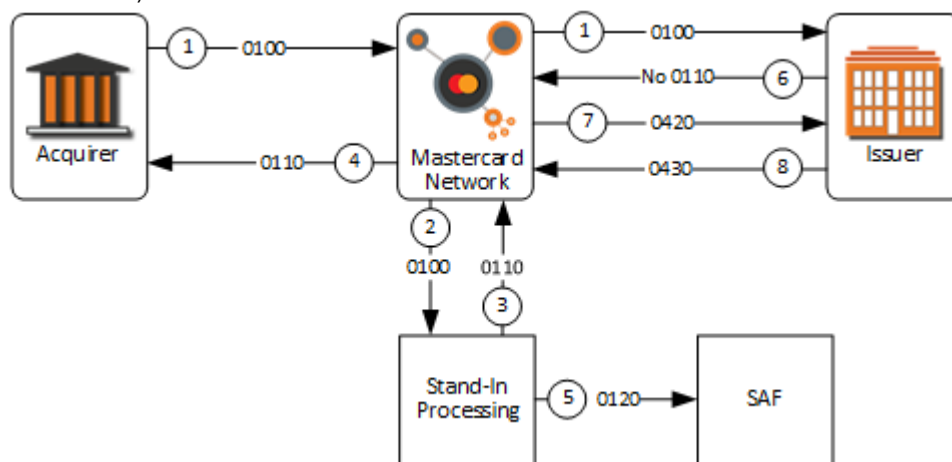
Following are the Authorization Request Response/0110 exception condition message flows that illustrate when Stand-In System processing is allowed on the transaction:

- Authorization Request Response/0110—Not Received within Time Limits
- Authorization Request Response/0110—Received within the Time Limit but after Stand-In System Response
- Authorization Request Response/0110—Received within Time Limit and before Stand-In System Response

NOTE: [Authorization Message Routing Timers](#) provides details regarding specific time limits.

Authorization Request Response/0110: Not Received within the Time Limit

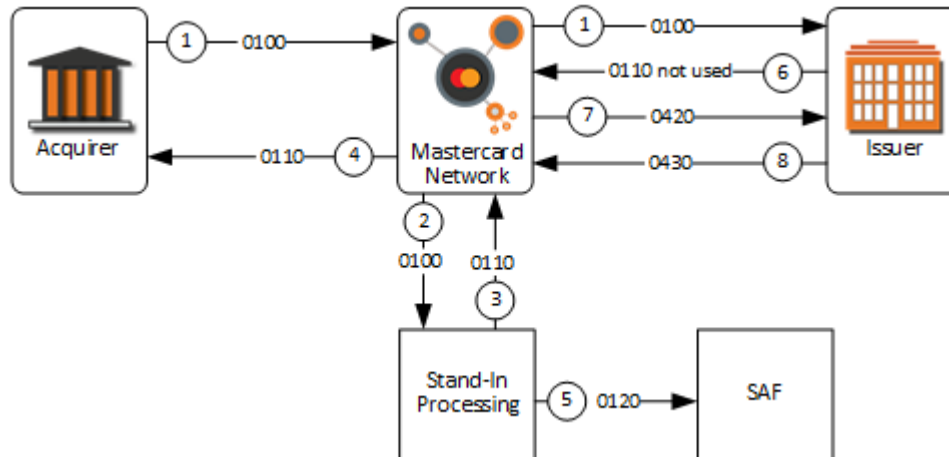
This message flow describes exception processing when no issuer Authorization Request Response/0110 message is received within the defined time limits after receipt of the Authorization Request/0100 message for transactions that are allowed to be processed by the Stand-In System.



1. The acquirer sends the Authorization Request/0100 message to the Authorization Platform, and the Authorization Platform forwards the Authorization Request/0100 message to the issuer.
2. If the Authorization Platform does not receive the Authorization Request Response/0110 message from the issuer within the time limit, the Authorization Platform sends the Authorization Request/0100 message to the Stand-In System.
3. The Stand-In System forwards the Authorization Request Response/0110 message to the Authorization Platform.
4. The Authorization Request Response/0110 message received from the Stand-In System is forwarded by the Authorization Platform to the acquirer.
5. The Stand-In System creates an Authorization Advice/0120 message and stores it in a SAF queue for guaranteed delivery to the issuer.
6. The Authorization Platform does not receive the Authorization Request Response/0110 message from the issuer within the time limit.
7. The Authorization Platform sends a Reversal Advice/0420 message containing the following values to the issuer because no issuer response is received:
 - DE 39 = 82 (Time-out at issuer)
 - DE 60, subfield 1 = 402 (Issuer Time-out)
8. When the issuer receives the Reversal Advice/0420 message, it must generate a Reversal Advice Response/0430 message to acknowledge receipt of the Reversal Advice/0420 message.

Authorization Request Response/0110: Received within the Time Limit but after Stand-In System Response

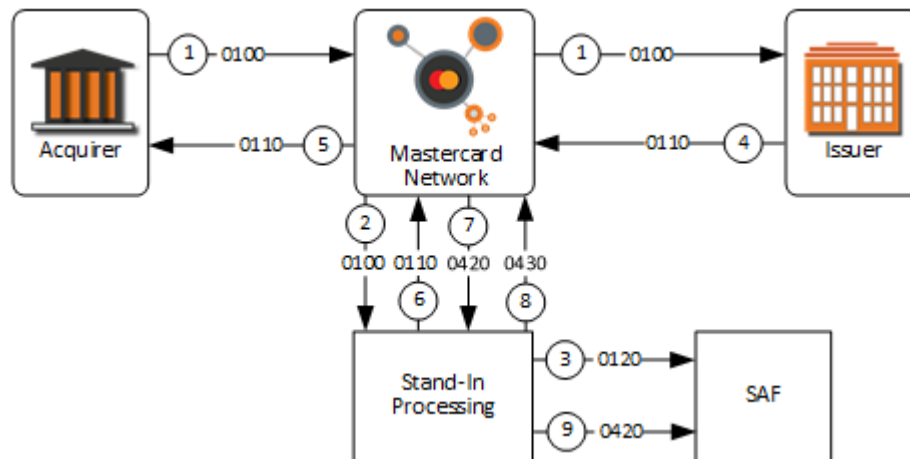
This message flow describes exception processing when the issuer's Authorization Request Response/0110 message is received within the time limit but after the Stand-In System response. Issuers can retrieve SAF messages upon request.



1. The acquirer sends the Authorization Request/0100 message to the Authorization Platform, and the Authorization Platform forwards the Authorization Request/0100 message to the issuer.
2. If the Authorization Platform does not receive the Authorization Request Response/0110 message within the time limit from the issuer, the Authorization Platform sends the Authorization Request/0100 message to the Stand-In System.
3. The Stand-In System forwards the Authorization Request Response/0110 message to the Authorization Platform.
4. The Authorization Platform sends the Authorization Request Response/0110 message to the acquirer.
5. The Stand-In System creates an Authorization Advice/0120 message and stores it in a SAF queue for guaranteed delivery to the issuer.
6. The issuer's Authorization Request Response/0110 message is received within the time limit but the issuer's response is not used because the issuer's response was received after the Authorization Request Response/0110 message from the Stand-In System.
7. The Authorization Platform sends a Reversal Advice/0420 message containing the following values to the issuer to reverse the issuer's response that was not used:
 - DE 39 = the value from the issuer's Authorization Request Response/0110
 - DE 60, subfield 1 = 400 (Banknet advice: APS error; unable to deliver response)
8. When the issuer receives the Reversal Advice/0420 message, it must generate a Reversal Advice Response/0430 message to acknowledge receipt of the Reversal Advice/0420 message.

Authorization Request Response/0110: Received within the Time Limit and before Stand-In System Response

This message flow describes exception processing when the issuer's Authorization Request Response/0110 message is received within the time limit but before the Stand-In System response.



1. The acquirer sends the Authorization Request/0100 message to the Authorization Platform, and the Authorization Platform forwards the Authorization Request/0100 message to the issuer.
2. If the Authorization Platform does not receive the Authorization Request Response/0110 message within the time limit from the issuer, the Authorization Platform sends the Authorization Request/0100 message to the Stand-In System.
3. The Stand-In System creates an Authorization Advice/0120 message and stores it in SAF for guaranteed delivery to the issuer.
4. The issuer sends the Authorization Request Response/0110 message to the Authorization Platform.
5. The Authorization Platform receives the issuer's Authorization Request Response/0110 message within the time limit and before the Stand-In System response, and sends the issuer's Authorization Request Response/0110 message to the acquirer.
6. The Stand-In System forwards the Authorization Request Response/0110 message to the Authorization Platform.
7. The Authorization Platform sends the Reversal Advice/0420 message to the Stand-In System to reverse the Stand-In System's response that was not used.
8. The Stand-In System generates a Reversal Advice Response/0430 message to acknowledge receipt of the Reversal Advice/0420 message.
9. The Stand-In System stores the Reversal Advice/0420 in SAF for guaranteed delivery to the issuer.

Authorization Request Response/0110: Not Eligible for Alternate Processing

These message flows describe authorization message processing when the transaction is not eligible for alternate processing.

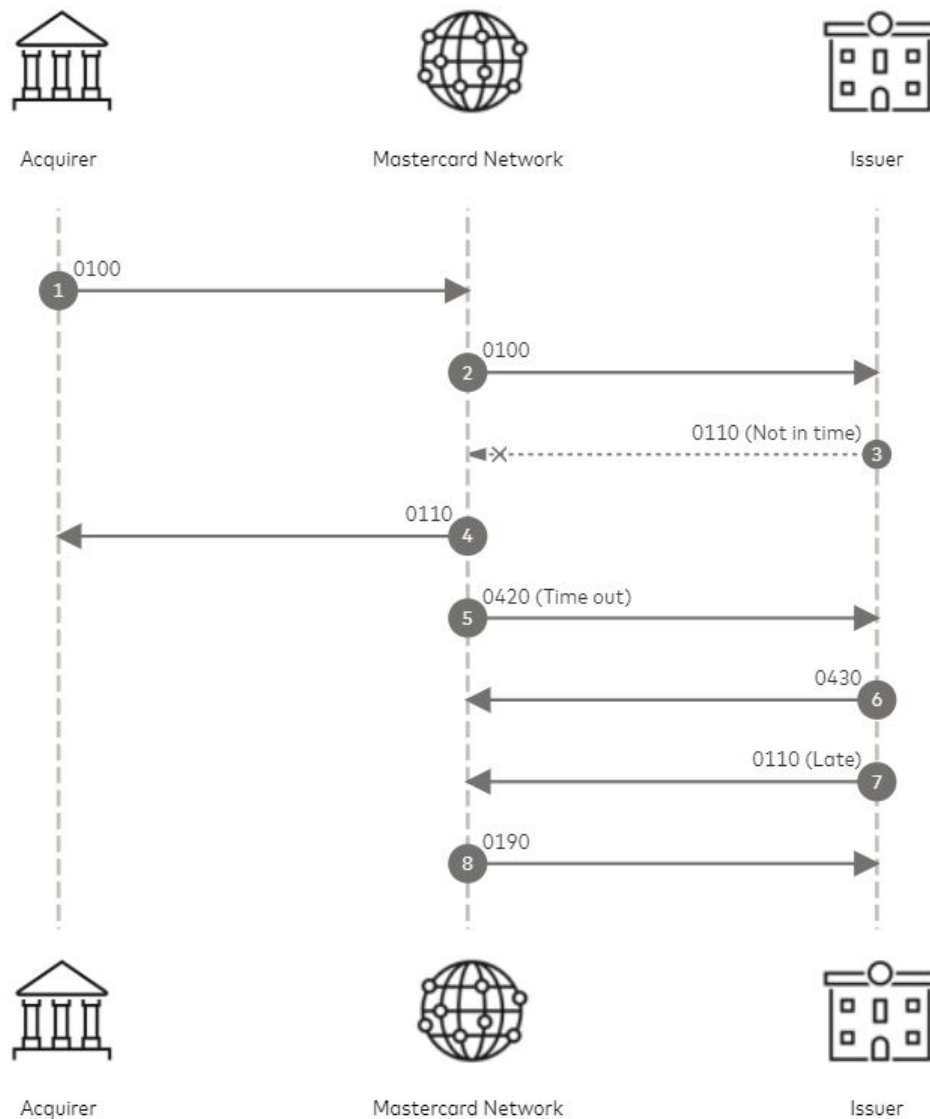
These message flows describe exception processing when:

- Authorization Request Response/0110—No Issuer Response
- Authorization Request Response/0110—Late Issuer Response
- Authorization Request Response/0110—Issuer Edit Error

No alternate processing, issuer late response or not in time limit

This message flow describes *exception processing* when the *issuer's* Authorization Request Response/0110 message is not received within the allowed *issuer* response time limit after receipt of the Authorization Request/0100 message.

Figure 1: No alternate processing, issuer late response or not in time limit flow



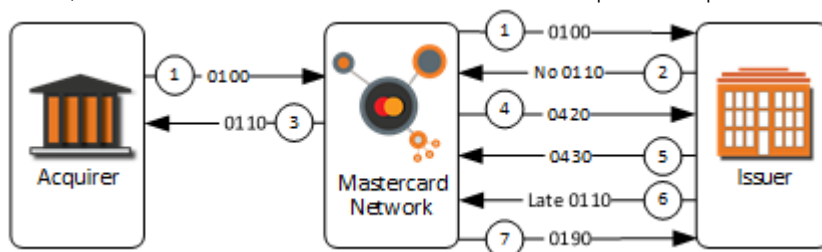
1. The *acquirer* sends the Authorization Request/0100 message to the *Mastercard Network*.
2. The *Mastercard Network* forwards the Authorization Request/0100 message to the *issuer*.

3. The *Mastercard Network* does not receive the *issuer's* Authorization Request Response/0110 within the response time allowed for the *issuer*.
4. The *acquirer* is sent an Authorization Request Response/0110 message indicating that the *issuer* could not be reached, and the transaction is declined.
5. A Reversal Advice/0420 message is sent to the *issuer* to advise the *Mastercard Network* did not receive the *issuer's* Authorization Request Response/0110 message within the allowed response time. Even though no response was received from the *issuer*, this *advice message* indicates to the *issuer* to release any funds they may have held for the transaction.
6. When the *issuer* receives the Reversal Advice/0420 message, it must generate a Reversal Advice Response/0430 message to acknowledge receipt of the Reversal Advice/0420 message.
In case of a late response from the *issuer* the following two steps would occur.
7. The late Authorization Request Response/0110 message is received from the *issuer*.
8. The *Mastercard Network* creates an Authorization Response Negative Acknowledgment/0190 message which is sent to the *issuer*.

NOTE: There is no response necessary from the *issuer* after the *Mastercard Network* sends an Authorization Response Negative Acknowledgment/0190 message.

Authorization Request Response/0110: Late Issuer Response

This message flow describes exception processing when the Authorization Platform detects a time-out condition on an expected Authorization Request Response/0110 message from an issuer, and then receives a late Authorization Request Response/0110 message.



1. The acquirer sends the Authorization Request/0100 message to the Authorization Platform, and the Authorization Platform forwards the Authorization Request/0100 message to the issuer.
2. The Authorization Platform does not receive the Authorization Request Response/0110 message within the time limits from the issuer.
3. The Authorization Platform generates an Authorization Request Response/0110 message where DE 39 (Response Code) = 91 (Authorization System or issuer system inoperative) and forwards it to the acquirer.
4. The Authorization Platform sends a Reversal Advice/0420 message containing the following values to the issuer because no issuer response is received:
 - DE 39 = 82 (Time-out at issuer)
 - DE 60, subfield 1 = 402 (Issuer Time-out)

5. When the issuer receives the Reversal Advice/0420 message, it must generate a Reversal Advice Response/0430 message to acknowledge receipt of the Reversal Advice/0420 message.
6. The Authorization Platform receives a late Authorization Request Response/0110 message from the issuer.
7. The Authorization Platform sends an Authorization Response Negative Acknowledgement/0190 message to the issuer, indicating it has no record of a corresponding Authorization Request/0100 message.

NOTE: There is no response necessary from the issuer after the Authorization Platform sends an Authorization Response Negative Acknowledgement/0190 message.

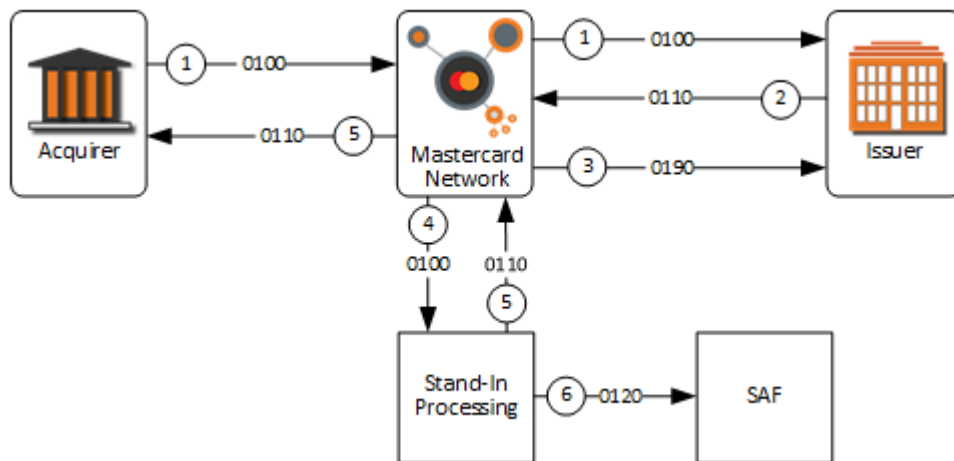
Authorization Request Response/0110: Issuer Edit Error

This message flow describes exception processing when an invalid issuer Authorization Request Response/0110 message is received after receipt of the Authorization Request/0100 message for transactions that are allowed to be processed by the Stand-In System.

Mastercard will provide the issuer's response to the Authorization Request/0110 message whenever possible. If the issuer returns an Authorization Request Response/0110 message, Mastercard evaluates the authorization response to determine if the transaction can be parsed and if data element (DE) 39 (Response Code) can be read. If DE 39 can be read and the issuer has approved the transaction, inclusive of the following response codes, this message flow is used (the request is sent to Stand-In processing):

- 00 Approved or completed successfully
- 10 Partial Approval
- 87 Purchase Amount Only, No Cash Back Allowed
- 91 Authorization System or issuer system inoperative
- 92 Unable to route transaction
- 96 System error (inclusive of Response Codes)

If the response code is any other than those previously listed, Mastercard replaces or removes the data responsible for the format error and forwards the remainder of the message to the acquirer. Mastercard continues to log information relevant to the format error for reference purposes.

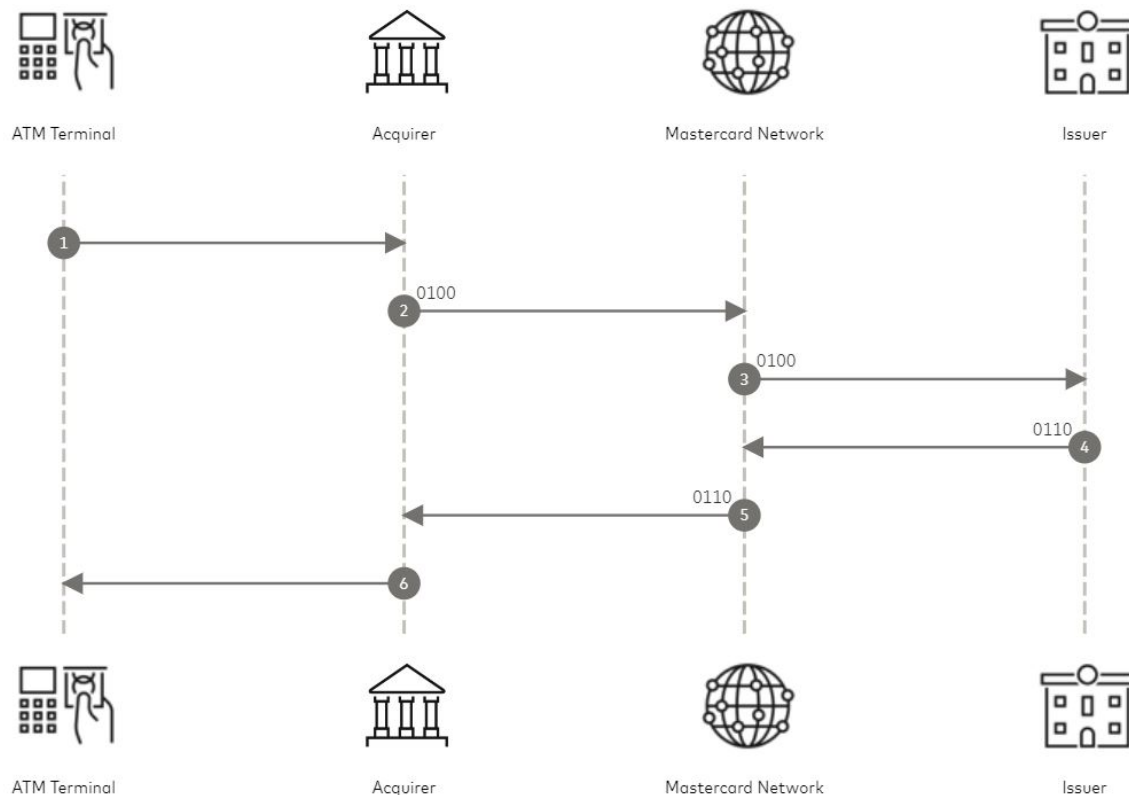


1. The acquirer sends the Authorization Request/0100 message.
2. The issuer generates an invalid or late Authorization Request Response/0110 message.
3. The Authorization Platform sends an Authorization Response Negative Acknowledgement/0190 message to the issuer.
4. The Authorization Platform sends the Authorization Request/0100 message to Stand-In processing.
5. Stand-In processing generates an appropriate Authorization Request Response/0110 message on the issuer's behalf and forwards it to the acquirer.
6. The Stand-In System also generates an Authorization Advice/0120 message and forwards it to the Authorization Platform SAF process for later transmission to the issuer.

Standard Chip-PIN Management request

This message flow describes standard transaction processing when changing or unblocking a PIN on a chip card.

Figure 2: Standard Chip-PIN Management request flow

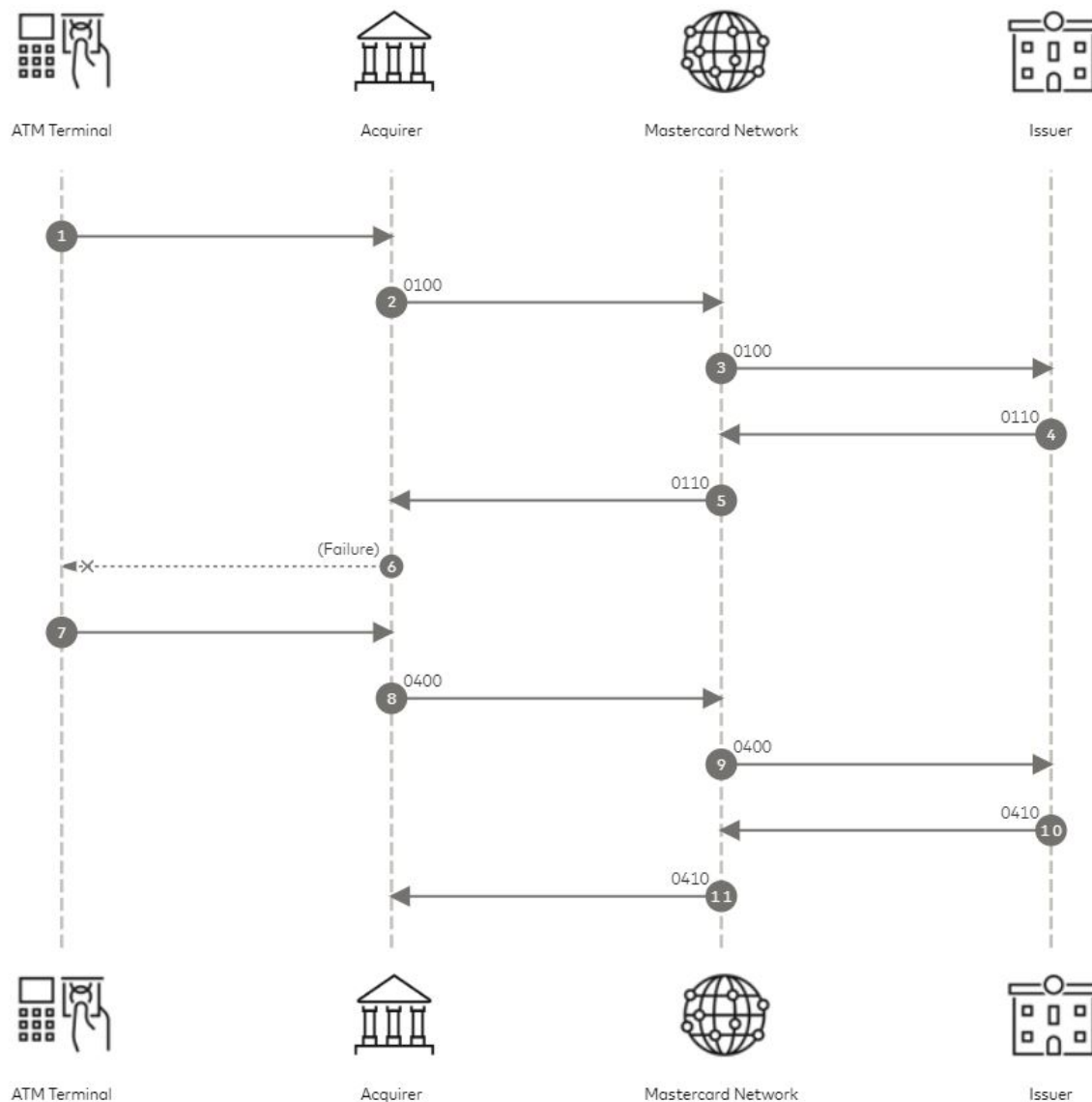


1. The cardholder initiates a PIN management request (PIN change or PIN unblock) at an ATM. The ATM requests data from the chip card and forwards the data in the request to the acquirer.
2. The acquirer sends an Authorization Request/0100 message containing the appropriate values to indicate a Chip PIN Management request to the Mastercard Network.
3. The Mastercard Network forwards the Authorization Request/0100 message to the issuer.
4. The issuer sends an Authorization Request Response/0110 message to the Mastercard Network approving or declining the request. The issuer will make updates within the issuer host system to reflect the PIN change or PIN unblock.
5. The Mastercard Network forwards the Authorization Request Response/0110 message to the acquirer.
6. The acquirer will send the response to the ATM. A script is sent from the ATM to the card to update the chip. After the script is applied, the chip provides the result to the ATM to complete the transaction.

Chip-PIN Management Update cannot be transmitted or applied to chip card

This message flow describes the transaction process when there is a failure in transmitting the script to the chip card, or a failure in applying the script to the chip card.

Figure 3: Chip-PIN Management Update cannot be transmitted or applied to chip card flow



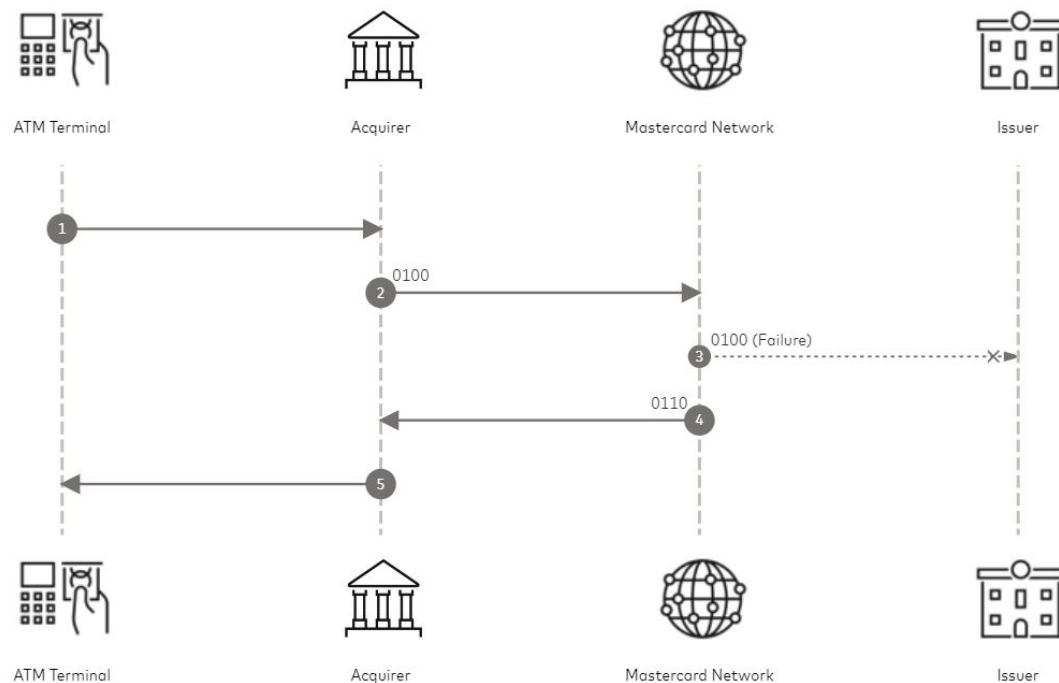
1. The cardholder initiates a Chip PIN management request (PIN change or PIN unblock) at an ATM. The ATM requests data from the chip card and forwards the data in the request to the acquirer.
2. The acquirer sends an Authorization Request/0100 message containing the appropriate values to indicate a Chip PIN Management request to the Mastercard Network.
3. The Mastercard Network forwards the Authorization Request/0100 to the issuer.

4. The issuer sends an Authorization Request Response/0110 message to the Mastercard Network approving or declining the request. The issuer will make updates within the issuer host system to reflect the PIN change or PIN unblock.
5. The Mastercard Network forwards the Authorization Request Response/0110 message to the acquirer.
6. A failure occurs for one of the following reasons:
 - The acquirer cannot send the script in DE 55 to the ATM.
 - The ATM cannot transmit the script to the chip card.
 - The chip card cannot process the script.
7. The ATM sends the processing result to the acquirer indicating the failure.
8. The acquirer generates a Reversal Request/0400 message to the Mastercard Network.
9. The Mastercard Network forwards the Reversal Request/0400 message to the issuer.
10. The issuer sends a Reversal Request Response/0410 message to the Mastercard Network. The issuer must also reverse the PIN change or PIN unblock applied to their system based on the Reversal Request.
11. The Mastercard Network forwards the Reversal Request Response/0410 message to the acquirer.

Chip-PIN Management Request unable to connect to issuer

This message flow describes the transaction process when there is a network failure at the issuer.

Figure 4: Chip-PIN Management Request unable to connect to issuer flow

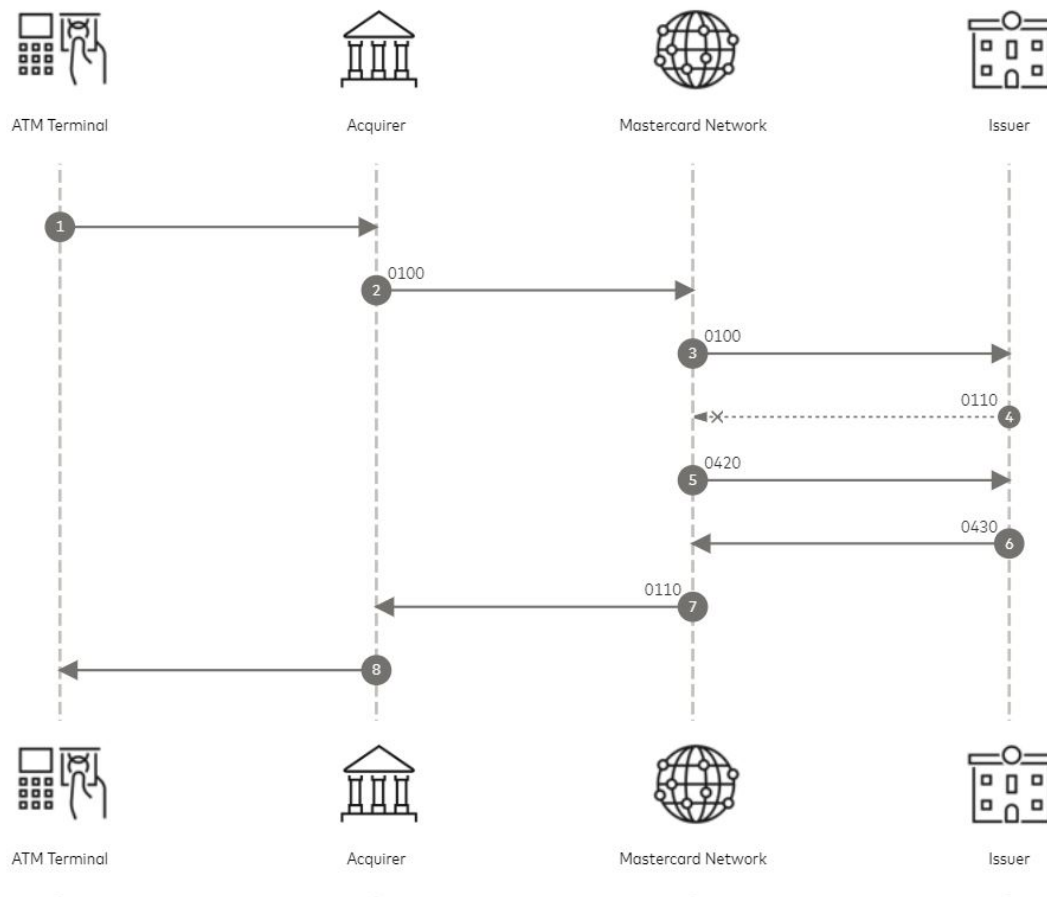


1. The cardholder initiates a Chip PIN management request (PIN change or PIN unblock) at an ATM. The ATM requests data from the chip card and forwards the data in the request to the acquirer.
2. The acquirer sends an Authorization Request/0100 message containing the appropriate values to indicate a Chip PIN Management request to the Mastercard Network.
3. The Mastercard Network attempts to forward the Authorization Request/0100 to the issuer. However the request message does not reach the issuer which results in a time-out for the request.
4. The Mastercard Network sends an Authorization Request Response/0110 message to the acquirer indicating the request could not be delivered.
5. The acquirer sends a message to the ATM advising that the service could not be completed.

PIN-management Response not received from issuer

This message flow describes the transaction process when the issuer response is not delivered to the Mastercard Network or is delivered to late.

Figure 5: PIN-management Response not received from issuer flow



1. The cardholder initiates a PIN management request (PIN change or PIN unblock) at an ATM. The ATM requests data from the chip card and forwards the data in the request to the acquirer.
2. The acquirer sends an Authorization Request/0100 message containing the appropriate values to indicate a Chip PIN Management request to the Mastercard Network.
3. The Mastercard Network forwards the Authorization Request/0100 to the issuer.
4. The issuer fails to send the Authorization Request Response/0110 message to the Mastercard Network in time.
5. The Mastercard Network sends a Reversal Advice/0420 message to the issuer indicating the time-out condition.

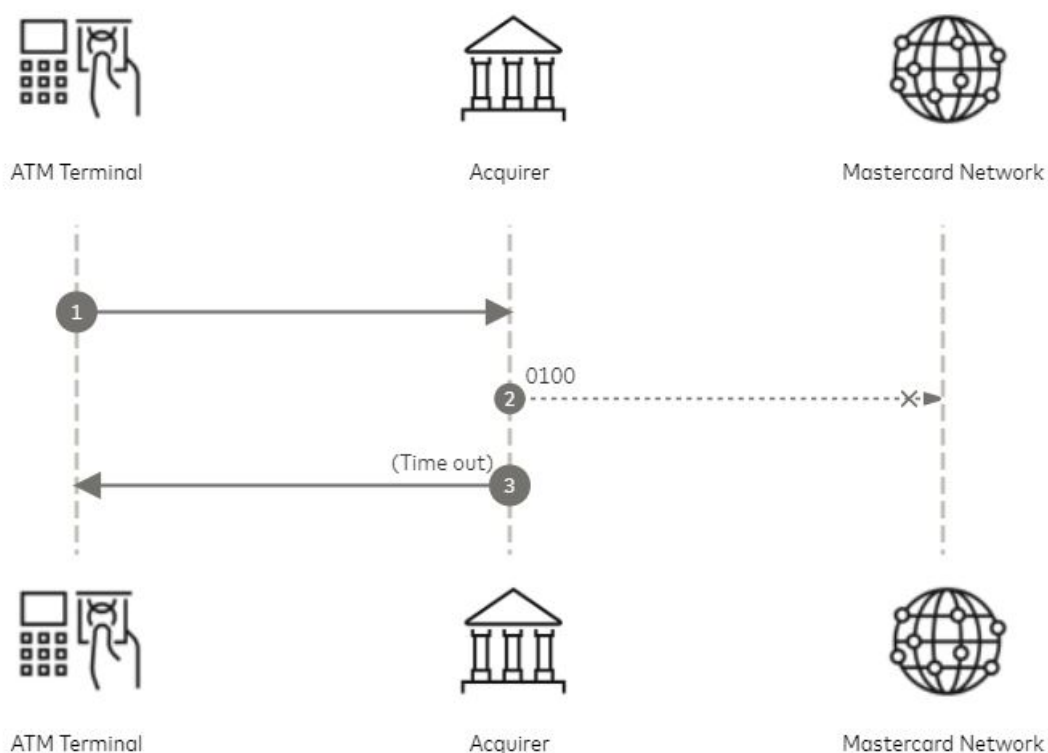
NOTE: If a response is received from the issuer after the allowed response time limit for ATM transactions for Mastercard and Debit Mastercard only, the Mastercard Network will send the issuer an Authorization Response Negative Acknowledgment/0190 message advising the response was late (not illustrated).

6. The issuer sends a Reversal Advice Response/0430 message to the Mastercard Network acknowledging receipt of the Reversal Advice/0420 message. If the original transaction was a PIN change, the issuer restores the old online PIN in their host system.
7. The Mastercard Network sends the acquirer an Authorization Request Response/0110 message indicating the request was not able to be delivered.
8. The acquirer sends a message to the ATM, indicating that the service is unavailable.

Chip-PIN Management Request failure from acquirer

This message flow describes the transaction process when there is a network failure at the acquirer.

Figure 6: Chip-PIN Management Request failure from acquirer flow



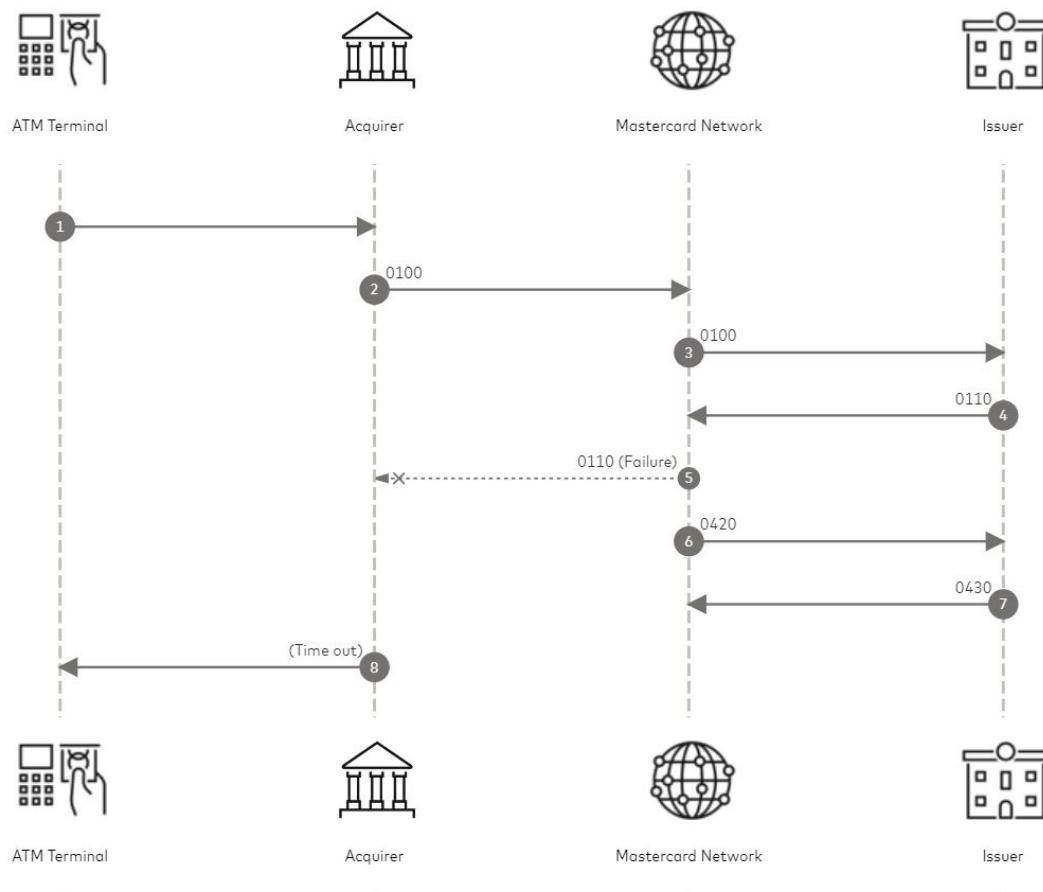
1. The cardholder initiates a PIN management request (PIN change or PIN unblock) at an ATM. The ATM requests data from the chip card and forwards the data in the request to the acquirer.

2. The acquirer sends an Authorization Request/0100 message containing the appropriate values to indicate a Chip PIN Management request to the Mastercard Network. However, the request message does not reach the Mastercard Network.
3. The acquirer sends a message to the ATM advising that the service could not be completed.

Chip-PIN Management Response cannot be delivered to the acquirer

This message flow describes the transaction process when the Mastercard Network is unable to successfully deliver the issuer's response message to the acquirer.

Figure 7: Chip-PIN Management Response cannot be delivered to the acquirer flow



1. The cardholder initiates a PIN management request (PIN change or PIN unblock) at an ATM. The ATM requests data from the chip card and forwards the data in the request to the acquirer.
2. The acquirer sends an Authorization Request/0100 message containing the appropriate values to indicate a Chip PIN Management request to the Mastercard Network.
3. The Mastercard Network forwards the Authorization Request/0100 message to the issuer.

4. The issuer sends an Authorization Request Response/0110 message to the Mastercard Network approving or declining the request. The issuer will make updates within the issuer host system to reflect the PIN change or PIN unblock.
5. The Mastercard Network fails to deliver the Authorization Request Response/0110 message to the acquirer.
6. The Mastercard Network generates a Reversal Advice/0420 message to the issuer.
7. The issuer sends a Reversal Advice Response/0430 message to the Mastercard Network acknowledging receipt of the Reversal Advice/0420 message. The issuer must also reverse the PIN change or PIN unblock applied to their system based on the Reversal Advice.

NOTE: The Mastercard Network will only generate a Reversal Advice/0420 message if the issuer approved the original PIN management request.

8. The request times out at the acquirer and the acquirer sends a message to the ATM advising that the service is unavailable.

NOTE: The issuer may receive two reversals: one from the Mastercard Network to identify the undelivered response and a second from the acquirer as a result of a system timeout.

Guaranteed Advice Message Delivery

The guaranteed advice message delivery process provides a message routing facility. The Authorization Platform uses guaranteed advice message delivery for **all** advice messages transmitted using the Authorization Platform.

The Guaranteed Message Delivery Process

When an advice message is forwarded to the Authorization Platform, the Authorization Platform:

1. Immediately secures the transaction for future delivery
2. Responds to the advice message initiator with an advice response message, indicating that it received and secured the message
3. Forwards the advice message to the appropriate receiving entity

The Authorization Platform guarantees the delivery of all advice messages routed through the Mastercard Network. Occasionally, the Authorization Platform cannot immediately deliver an advice message (for example, because of a system or communication failure at the destination). In this case, the Authorization Platform stores the message on its store-and-forward (SAF) processing facilities and automatically delivers the message when communication is restored.

The receiver of an advice message must acknowledge receipt with an appropriate advice response message. When the Authorization Platform receives the advice response message, Mastercard considers advice delivery complete and removes the message from SAF processing queues.

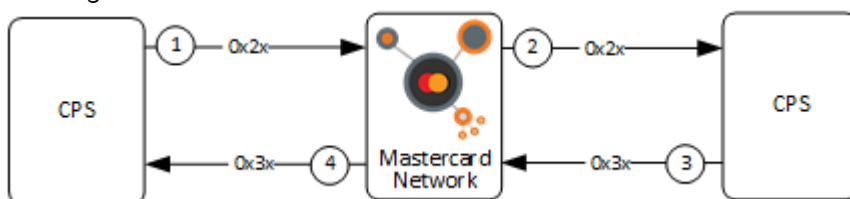
Authorization Advice/0120—Acquirer-generated messages are immediately sent to the issuer. The issuer responds with an Authorization Advice/0130—Issuer-generated message that the Authorization Platform forwards to the acquirer.

The Authorization Platform Guaranteed Advice Message Delivery process processes all Mastercard advice messages including the following:

- Authorization Advice/0120—Acquirer-generated
- Authorization Advice/0120—System-generated
- Reversal Advice/0420
- Administrative Advice/0620

Standard Advice Delivery: All Advice Message Types

This message flow describes the standard transaction processing of all advice messages originating from customer processing systems (CPS) connected to the Authorization Platform. The only exception is the Network Management Advice/0820 message, which has no response message.



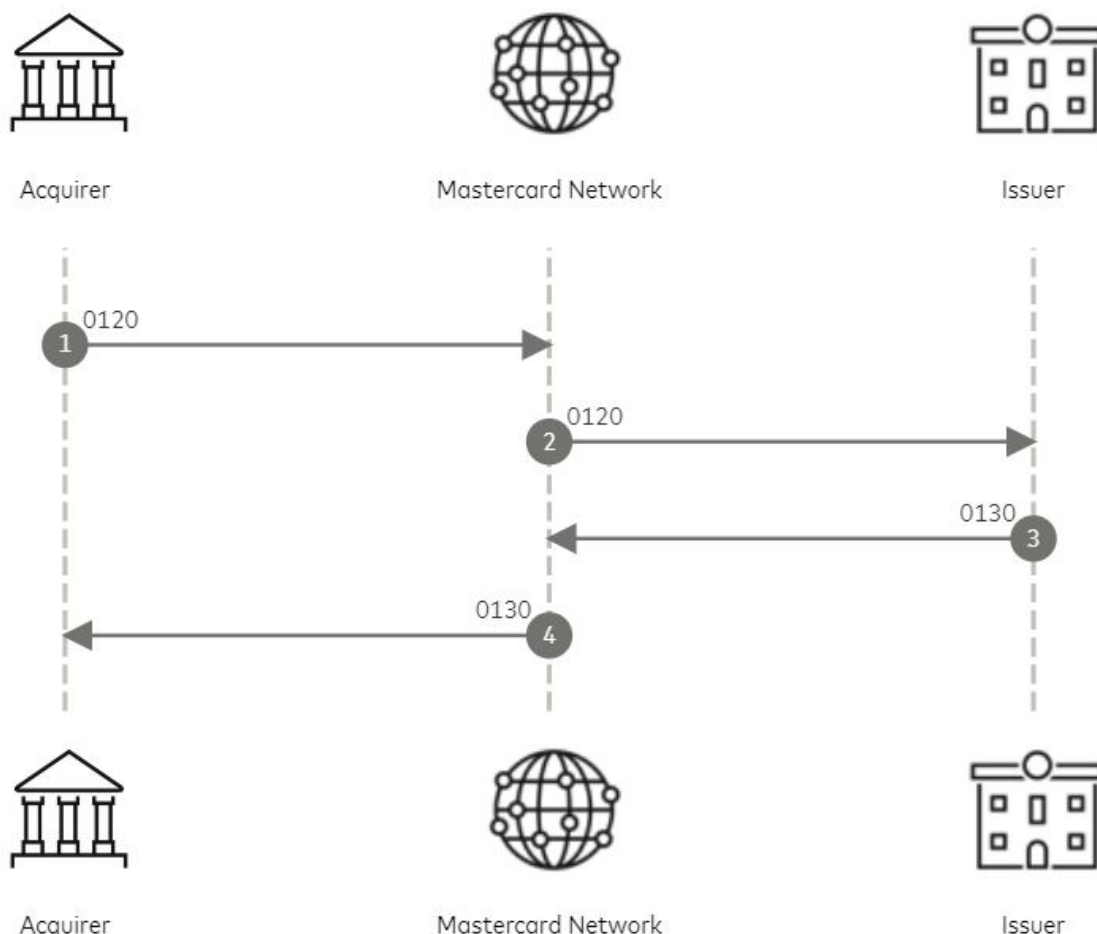
1. The CPS forwards the advice message to the Authorization Platform.
2. The Authorization Platform returns an appropriate advice response message when it secures the advice message.
3. The Authorization Platform forwards the advice message to the receiving entity.
4. The receiving entity returns an appropriate advice response message as positive acknowledgement of receipt when it secures the advice message.

NOTE: Only the Authorization Platform generates Network Management Advice/0820 messages. Network Management Advice/0820 messages do not require any type of response message.

Acquirer-generated Authorization Advice message

This message flow describes the transaction process of the Authorization Advice/0120: acquirer-generated and Authorization Advice Response/0130: issuer-generated message when the *issuer* is available.

Figure 8: Acquirer-generated authorization advice message standard flow

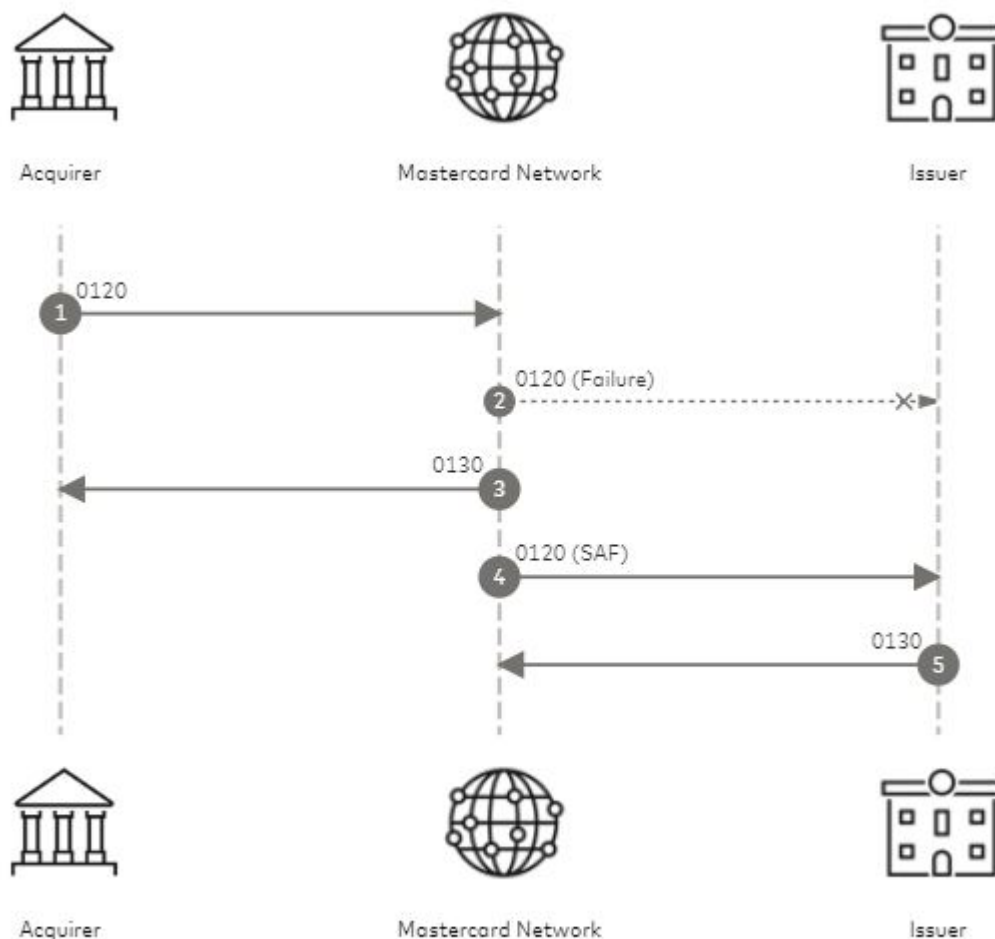


1. The *acquirer* initiates an Authorization Advice/0120: acquirer-generated message and passes the 0120 message to the *Mastercard Network*.
2. The *Mastercard Network* inserts DE 48, subelement 15 (Authorization Platform Advice Date and Time) into the message and forwards the Authorization Advice/0120: acquirer-generated message to the *issuer*.
3. The *issuer* returns an Authorization Advice Response/0130: issuer-generated message to the *Mastercard Network* to indicate positive *receipt* of the Authorization Advice/0120: acquirer-generated message. The *issuer's* advice response message will contain DE 48, subelement 15 (Authorization Platform Advice Date and Time).
4. The *Mastercard Network* forwards the *issuer's* Authorization Advice Response/0130 message to the *acquirer*.

Acquirer-generated Advice message, issuer not available

This message flow describes the transaction process of the Authorization Advice/0120: acquirer-generated and Authorization Advice Response/0130: system-generated message when the issuer is unavailable or there is no issuer response.

Figure 9: Acquirer-generated advice message, issuer not available flow



1. The acquirer initiates an Authorization Advice/0120: acquirer-generated message and forwards the 0120 message to the Mastercard Network.
2. The Mastercard Network determines that the Authorization Advice/0120: acquirer-generated message cannot be delivered to the issuer host. The Mastercard Network routes the Authorization Advice/0120: acquirer-generated message to the Stand-In Authorization to add to the issuer SAF queue.
3. The Stand-In Authorization places the Authorization Advice/0120: acquirer-generated message into the SAF queue for guaranteed delivery to the issuer. The Stand-In

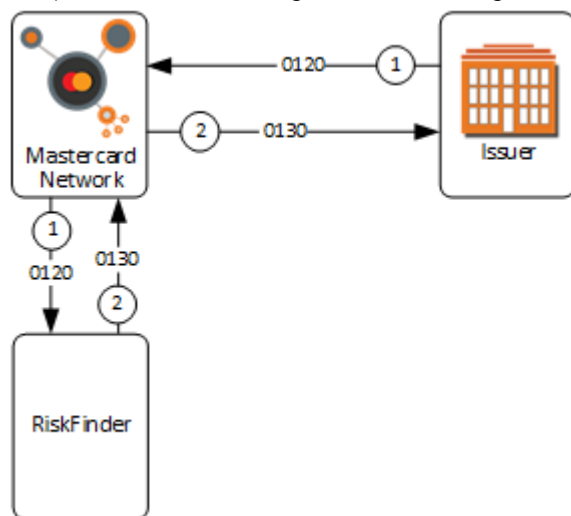
Authorization responds to the acquirer with an Authorization Advice Response/0130: System-generated message containing DE 48, subelement 15 (Authorization Platform Advice Date and Time).

4. The Mastercard Network delivers the Authorization Advice/0120: acquirer-generated message to the issuer once the issuer is available.
5. The issuer will respond with an Authorization Advice Response/0130: issuer-generated message.

NOTE: Customers in the Europe region that route to an alternate issuer host for alternate processing instead of the Stand-In Authorization will still receive an Authorization Advice/0120: acquirer-generated as described here. Alternate issuer host processing does not send Reversal Request/0400 or Authorization Advice/0120 messages to the alternate host.

Authorization Advice/0120: Issuer-Generated

Authorization Request/0100 messages may contain BINs that customers have selected for scoring. For these transactions, issuers create and send Authorization Advice/0120 messages to the risk scoring system. The risk scoring system sends to the issuer an Authorization Advice Response/0130 message to acknowledge receipt of the Authorization Advice/0120 message.

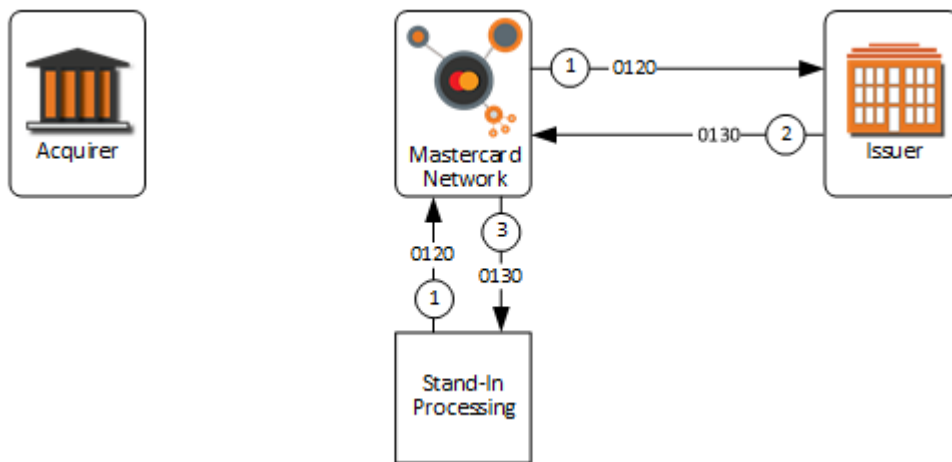


1. The issuer initiates an Authorization Advice/0120—Issuer-generated message and sends it to the Authorization Platform. The Authorization Platform forwards the message to the risk scoring system.
2. The risk scoring system creates an Authorization Advice Response/0130—System-generated message and sends it to the Authorization Platform. The Authorization Platform forwards the Authorization Advice Response/0130—System-generated message to the issuer to indicate positive receipt of the Authorization Advice/0120—Issuer-generated message.

Authorization Advice/0120: System-Generated

When Mastercard responds to an Authorization Request/0100 message on behalf of the issuer, the Authorization Platform generates an authorization response based on the issuer's parameters. The Authorization Platform also generates an Authorization Advice/0120—

System-generated message and stores it in a Store-and-Forward (SAF) queue. The Stand-In System will forward advice messages from the SAF queue when the issuer is available.

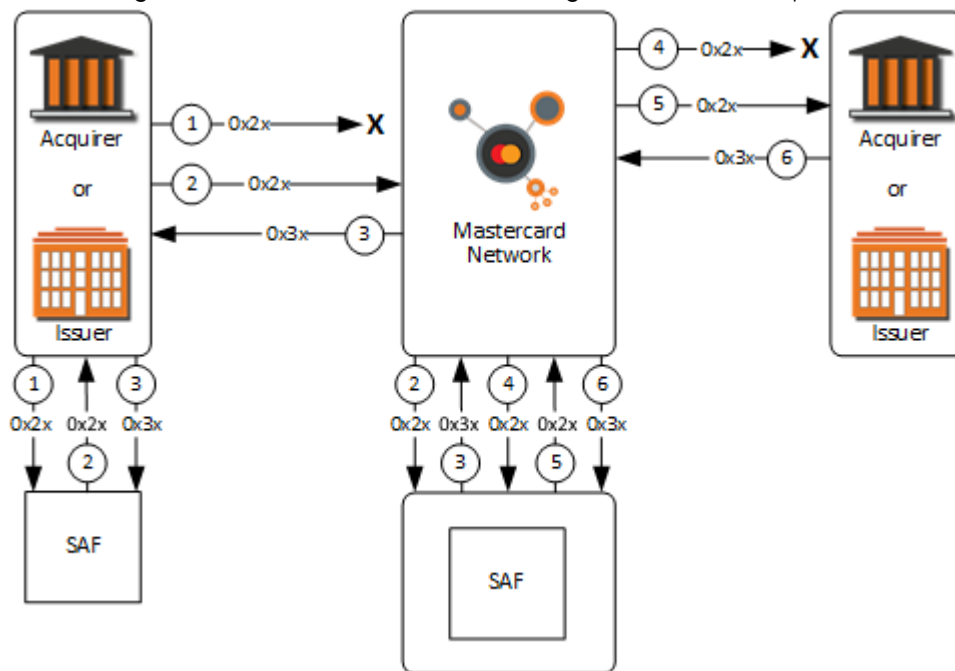


1. Stand-In System processing generates an Authorization Advice/0120—System generated message and sends it to the issuer.
2. The issuer returns an Authorization Advice Response/0130—Issuer-generated message.
3. The Authorization Platform receives the Authorization Advice Response/0130—Issuer-generated to indicate positive receipt of the Authorization Advice/0120—System-generated message.

NOTE: Issuers may differentiate between the acquirer-generated or system-generated Authorization Advice/0120 messages by examining the values in DE 60 (Advice Reason Code).

Advice Message Error Condition

This message flow describes the advice message error-condition process.



1. An issuer or acquirer (any customer processing system [CPS]) generates an advice message. If the message cannot be transmitted immediately, it should be stored by an appropriate Store-and-Forward (SAF) facility on the CPS for later transmission to the Authorization Platform.
2. The CPS forwards the advice message to the Authorization Platform.
3. The Authorization Platform returns an appropriate advice response message when it secures the advice message.
4. The Authorization Platform immediately attempts to deliver the transaction to the receiving entity. If the Authorization Platform cannot deliver the message, it stores it at the Mastercard SAF facility for later delivery.
5. The Authorization Platform forwards the advice message to the receiving entity when communication is restored.
6. The receiving entity returns an appropriate advice response message as positive acknowledgement of receipt when it secures the advice message.

Not all advice messages flow through the Authorization Platform from one CPS to another CPS. For example, the Authorization Platform originates some advice messages and forwards them to a CPS. Similarly, a CPS may initiate some advice messages and forward them to the Authorization Platform as the receiving destination.

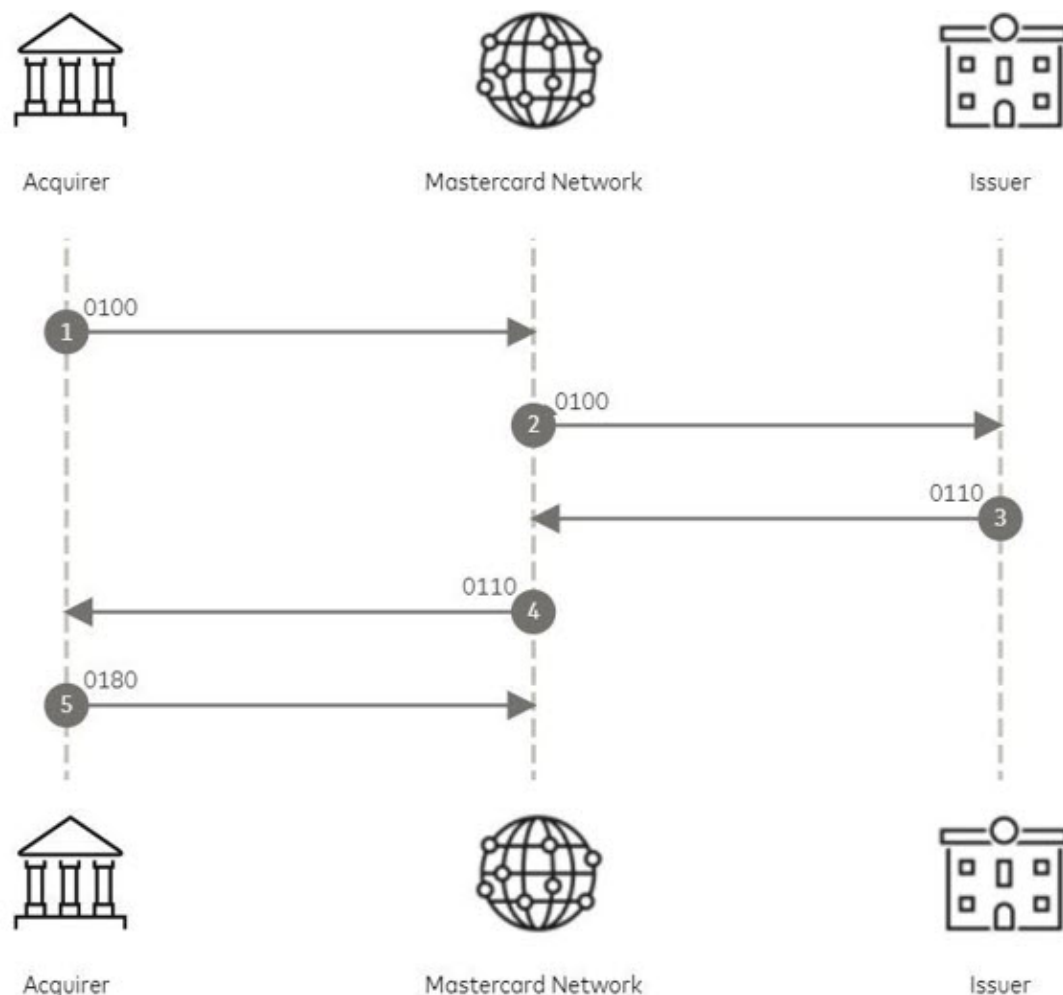
Authorization Response Acknowledgment/0180

The Mastercard Network allows acquirers to optionally acknowledge Authorization Request Response/0110 messages. This process is accomplished using Authorization Response Acknowledgment/0180 messages.

Use of this acknowledgment can provide additional assurance that authorization responses have been delivered to the acquirer or acceptor's device. This may protect the issuer from erroneous debits against cardholder accounts due to incomplete or unsuccessful transactions.

Acquirers are not required to support Authorization Response Acknowledgment/0180, but Mastercard encourages acquirers to select this option and support these messages. Acquirers may select this option at network configuration time.

Figure 10: Acquirer Response Acknowledgment flow



1. The acquirer initiates an Authorization Request/0100 message and sends it to the Mastercard Network.
2. The Mastercard Network forwards the Authorization Request/0100 message to the issuer.
3. The issuer creates the Authorization Request Response/0110 message and sends it to the Mastercard Network.
4. The Mastercard Network forwards the Authorization Request Response/0110 message to the acquirer.
5. The acquirer sends an Authorization Response Acknowledgment/0180 message back to the Mastercard Network, indicating the acquirer successfully received the preceding Authorization Request Response/0110 message at the application level.

In most cases, the acquirer should send the Authorization Response Acknowledgment/0180 message to the Mastercard Network:

- Immediately after it secures, successfully receives, and logs the Authorization Request Response/0110 message
- Before the transaction is completed at the point of interaction

An Authorization Response Acknowledgment/0180 message does not necessarily indicate that the transaction was successfully completed at the point of interaction.

Alternate Issuer Host Processing for Online Europe Region Customers

To support customers in the Europe region that route to an alternate issuer host for alternate processing instead of using the Stand-In System, Mastercard offers alternate issuer host processing.

The system routes transactions to the alternate issuer for those issuers that choose to use alternate issuer host processing as their secondary path.

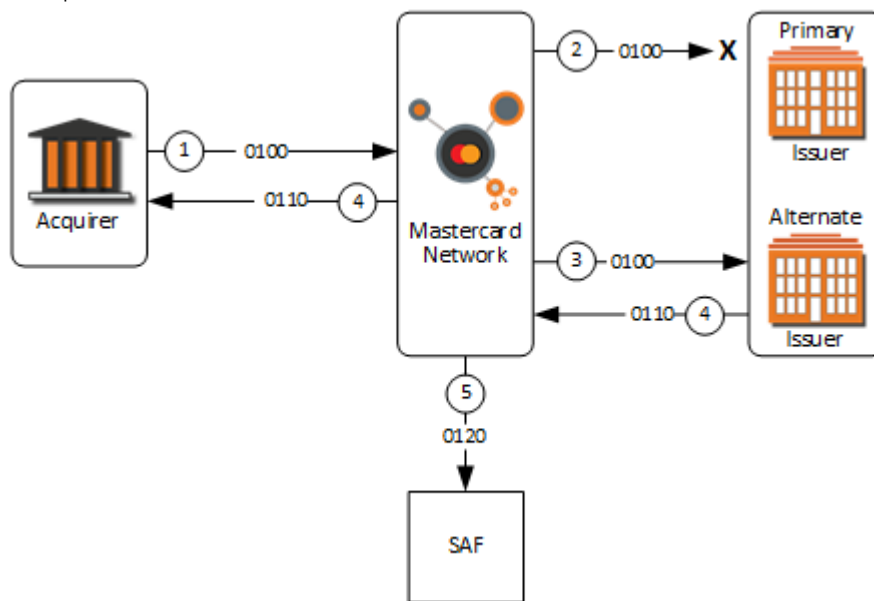
NOTE: [Authorization Message Routing Timers](#) provides details on time limits.

Authorization Request/0100: Communication Failure at Issuer (Issuer is not signed in or transaction cannot be delivered)

This message flow describes exception processing when the primary issuer is not signed in, the transaction could not be delivered to the primary issuer, or when the primary issuer does not

Message Definitions and Flows
Authorization Request/0100: Communication Failure at Issuer (Issuer is not signed in or transaction cannot be delivered)

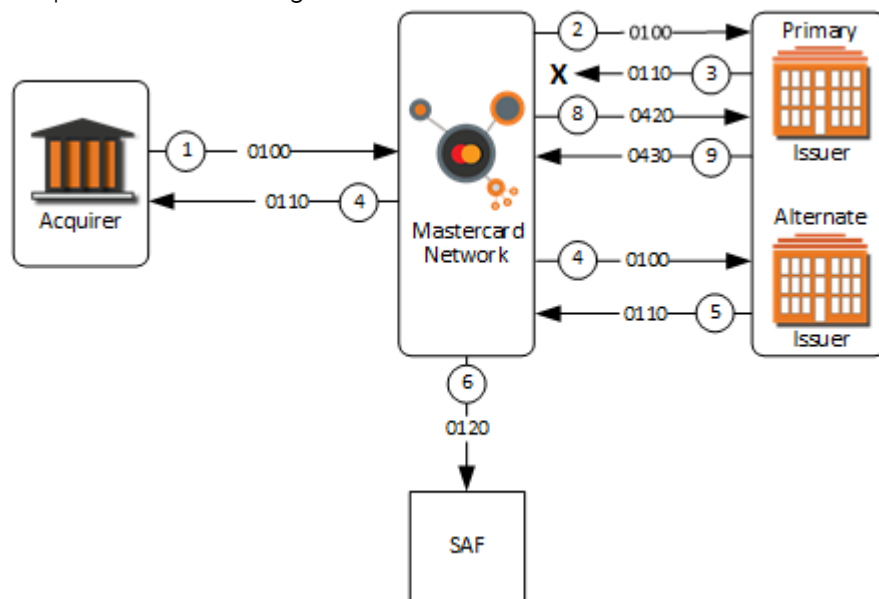
respond with an Authorization Response/0110 message within the time limit defined for the acceptance brand.



1. The acquirer initiates an Authorization Request/0100 message and sends it to the Authorization Platform.
2. The Authorization Platform attempts to forward the Authorization Request/0100 message to the primary issuer but is unable to complete the message transmission because of a communication link failure or IPS failure.
3. The Authorization Platform immediately sends the Authorization Request/0100 message to the alternate issuer.
4. The alternate issuer generates an appropriate Authorization Request Response/0110 message on the primary issuer's behalf and forwards it to the acquirer.
5. The Authorization Platform generates an Authorization Advice/0120 message and forwards it to the Authorization Platform store-and-forward (SAF) process for later transmission to the primary issuer.

Authorization Request Response/0110: Alternate Issuer Allowed

This message flow describes exception processing when communication fails between the Authorization Platform and the issuer processing system (IPS) during an Authorization Request Response/0110 message.

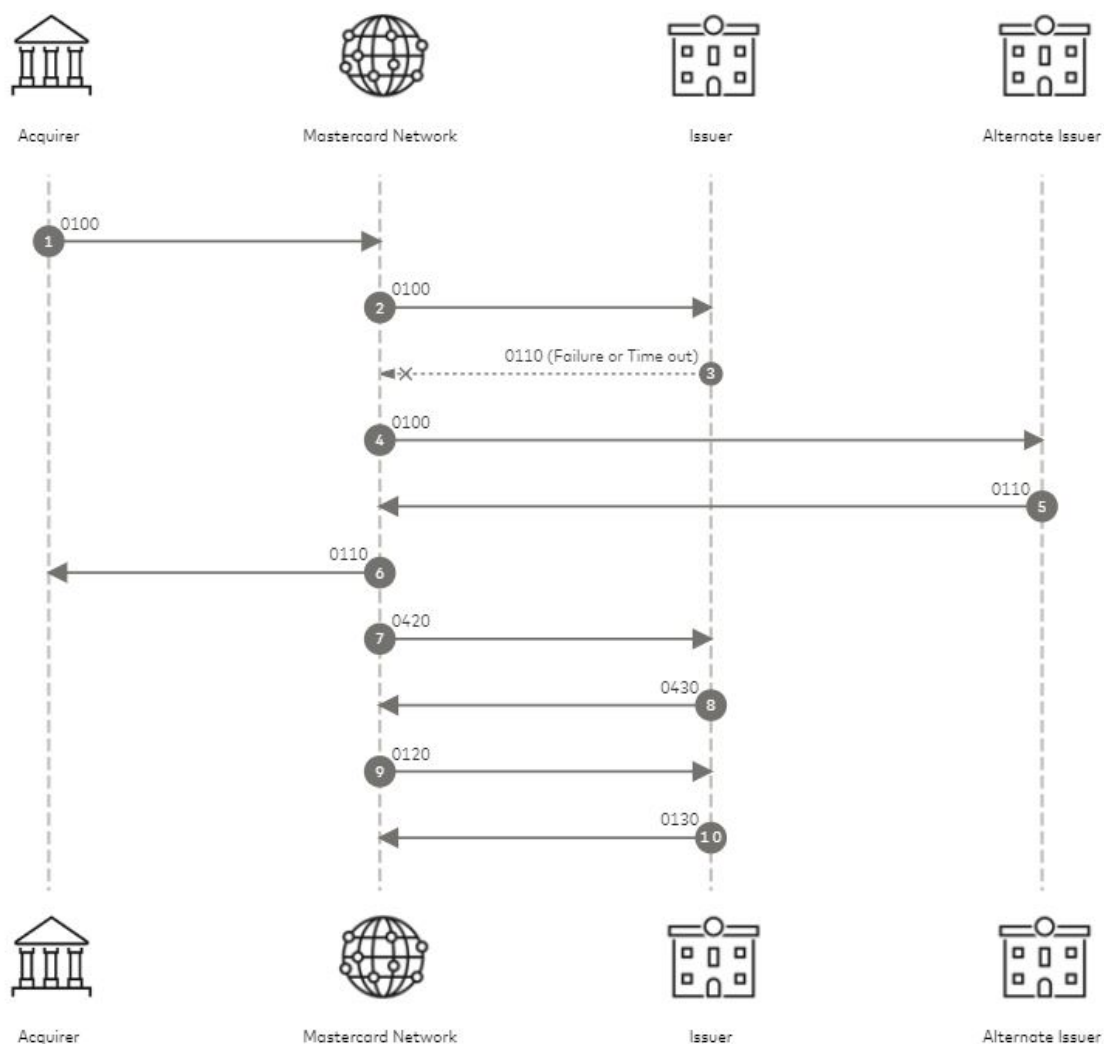


1. The acquirer initiates an Authorization Request/0100 message and sends it to the Authorization Platform.
2. The Authorization Platform forwards the Authorization Request/0100 message to the primary issuer.
3. The IPS cannot return the appropriate Authorization Request Response/0110 message because of a communication failure between the IPS and the Authorization Platform.
4. The Authorization Platform detects an expected time-out condition from the primary issuer on the Authorization Request Response/0110 message. If the primary issuer permits alternate issuer processing, the Authorization Platform sends an Authorization Request/0100 message to the alternate issuer host.
5. The alternate issuer generates an appropriate Authorization Request Response/0110 message on the primary issuer's behalf and forwards it to the acquirer.
6. The Authorization Platform generates an Authorization Advice/0120 message and forwards it to the Authorization Platform SAF process for later transmission to the primary issuer.
7. The Authorization Platform does not receive the Authorization Request Response/0110 message from the primary issuer within the response time limits.
8. The Authorization Platform sends a Reversal Advice/0420 message containing the following values to the primary issuer because no issuer response is received:
 - DE 39 (Response Code) = 82 (Time out at issuer)
 - DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code) = 402 (Issuer Time-out)
9. When the primary issuer receives the Reversal Advice/0420 message, it must generate a Reversal Advice Response/0430 message.

Primary issuer times out

This message flow describes *exception processing* when no *issuer* Authorization Request Response/0110 message is received within the time limit after receipt of the Authorization Request/0100 message for transactions that are allowed to be processed by the *alternate issuer host*.

Figure 11: Primary issuer times out flow



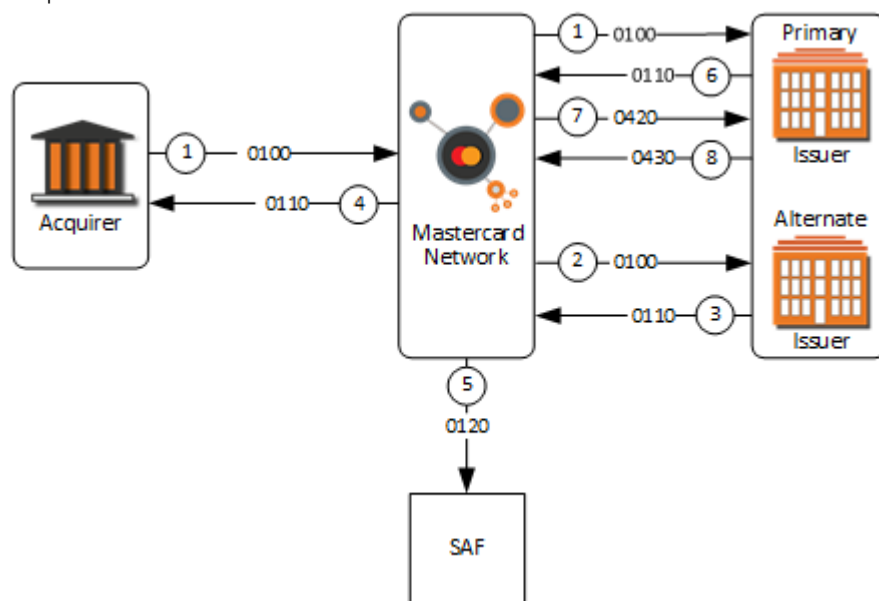
1. The *acquirer* initiates an Authorization Request/0100 message and sends it to the *Mastercard Network*.
2. The *Mastercard Network* forwards the Authorization Request/0100 message to the *primary issuer*.
3. The *Mastercard Network* does not receive the Authorization Request Response/0110 message from the *primary issuer* within the time limit.

Authorization Request Response/0110: Received within the Time Limit but after Alternate Issuer Response

4. The *Mastercard Network* sends the Authorization Request/0100 message to the *alternate issuer*.
5. The *alternate issuer* generates and sends the Authorization Request Response/0110 message to the *Mastercard Network*.
6. The *Mastercard Network* forwards to the *acquirer* the Authorization Request Response/0110 message received from the *alternate issuer*.
7. The *Mastercard Network* creates an Authorization Advice/0120 message and stores it in the SAF queue for guaranteed delivery to the *primary issuer*.
8. Upon receipt, the *issuer* must respond to the SAF advice message with an Authorization Advice Response/0130 message.
9. Since the response was not received within the time limit, the *Mastercard Network* will send an Authorization Advice/0120 message to the *primary issuer* to advise that they timed out on the Authorization Request. Even though no response was received from the *primary issuer*, this *advice message* indicates to the *primary issuer* to release any funds they may have held for the transaction.
10. When the *primary issuer* receives the Authorization Advice/0120 message, it must generate an Authorization Advice Response/0130 message to acknowledge receipt of the Authorization Advice/0120 message.

Authorization Request Response/0110: Received within the Time Limit but after Alternate Issuer Response

This message flow describes exception processing when the issuer's Authorization Request Response/0110 message is received within the time limit but after the alternate issuer response.



1. The acquirer initiates an Authorization Request/0100 message and sends it to the Authorization Platform and then the Authorization Platform forwards the Authorization Request/0100 message to the primary issuer.

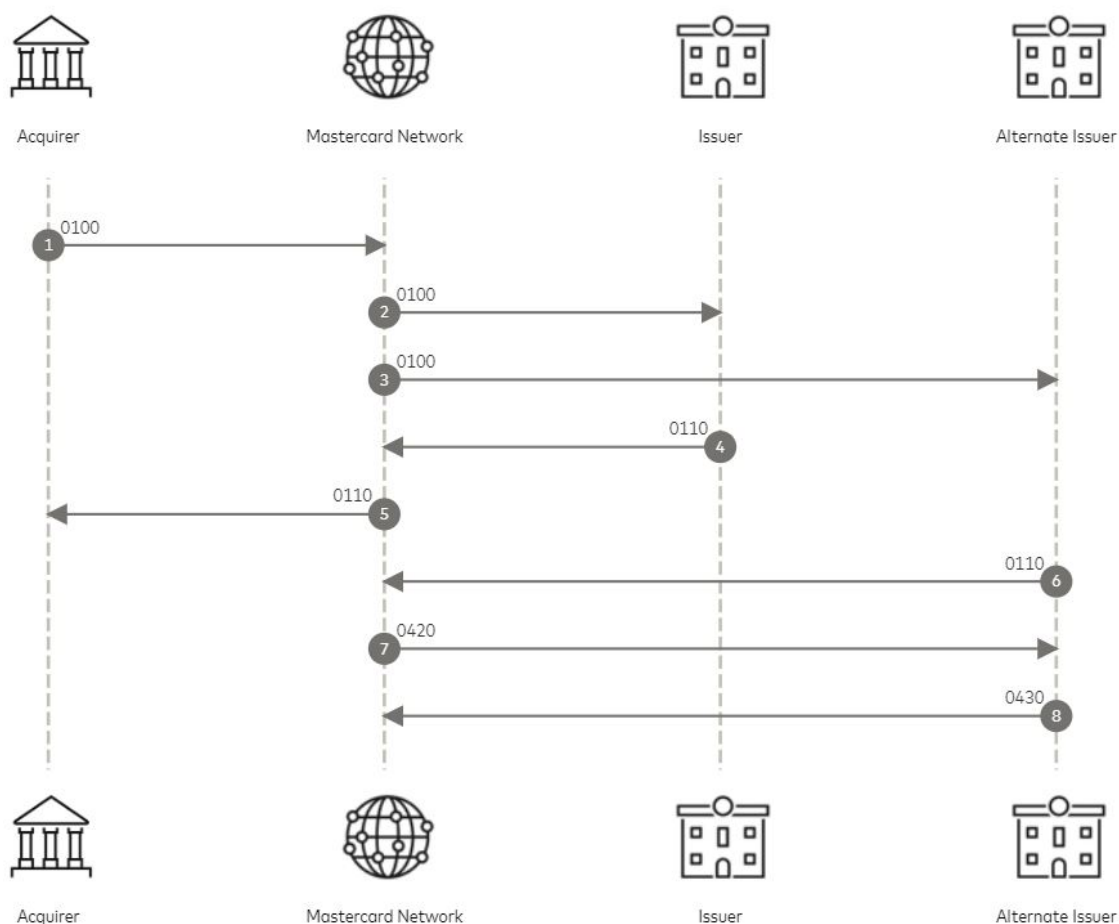
Authorization Request Response/0110: Received within the Time Limit but after Alternate Issuer Response

2. If the Authorization Platform does not receive the Authorization Request Response/0110 message from the primary issuer within the time limit, the Authorization Platform sends the Authorization Request/0100 message to the alternate issuer.
3. The alternate issuer forwards the Authorization Request Response/0110 message to the Authorization Platform.
4. The Authorization Platform sends the alternate issuer Authorization Request Response/0110 message to the acquirer.
5. The Authorization Platform creates an Authorization Advice/0120 message and stores it in a SAF queue for guaranteed delivery to the primary issuer.
6. The primary issuer's Authorization Request Response/0110 message is received within the time limit, but the primary issuer's response is not used because the primary issuer's response was received after the alternate issuer Authorization Request Response/0110 message.
7. The Authorization Platform sends a Reversal Advice/0420 message containing the following values to the primary issuer to reverse the issuer's response that was not used:
 - DE 39 (Response code) = the value from the primary issuer's Authorization Request Response/0110 message
 - DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code) = 400 (Banknet advice: APS error; unable to deliver response)
8. When the primary issuer receives the Reversal Advice/0420 message, it must generate a Reversal Advice Response/0430 message to acknowledge receipt of the Reversal Advice/0420 message.

Primary issuer responds after time limit and primary issuer response received first

This message flow describes *exception processing* when the Authorization Request/0100 message has already been forwarded to the *alternate issuer*, but the *primary issuer's* authorization response is received before the *alternate issuer's* response.

Figure 12: Primary issuer responds after time limit and primary issuer response received first flow

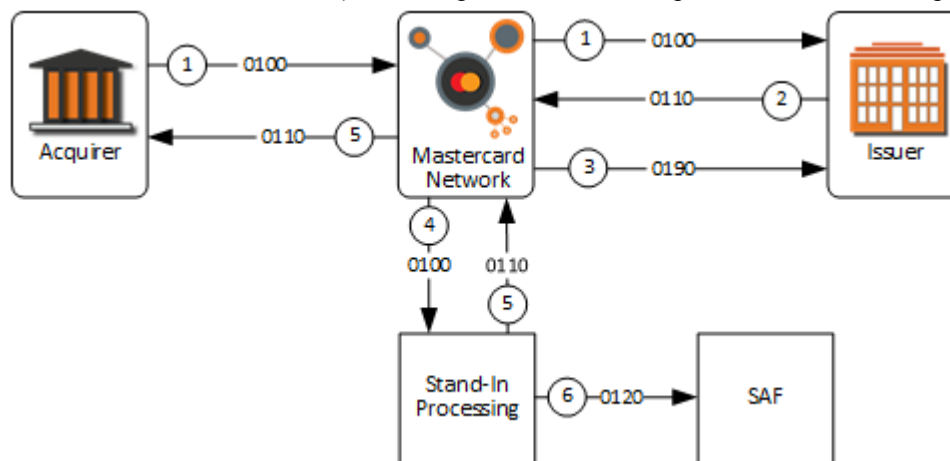


1. The *acquirer* sends the Authorization Request/0100 message to the *Mastercard Network*.
2. The *Mastercard Network* forwards the Authorization Request/0100 message to the *primary issuer*.
3. If the *Mastercard Network* does not receive the Authorization Request Response/0110 message within the time limit from the *primary issuer*, the *Mastercard Network* forwards the Authorization Request/0100 message to the *alternate issuer*.
4. The *primary issuer* sends the Authorization Request Response/0110 message to the *Mastercard Network*.
5. The *Mastercard Network* receives the *primary issuer's* Authorization Request Response/0110 message before the *alternate issuer's* response and sends the *primary issuer's* Authorization Request Response/0110 message to the *acquirer*.

6. The *alternate issuer* sends the Authorization Request Response/0110 message to the *Mastercard Network*.
7. The *Mastercard Network* generates a Reversal Advice/0420 message to the *alternate issuer* to reverse the *alternate issuer's* response that was not used.
8. When the *alternate issuer* receives the Reversal Advice/0420 message, it must generate a Reversal Advice Response/0430 message to acknowledge receipt of the Reversal Advice/0420 message.

Authorization Response Negative Acknowledgement/0190 (Responding to the Authorization Request Response/0110)

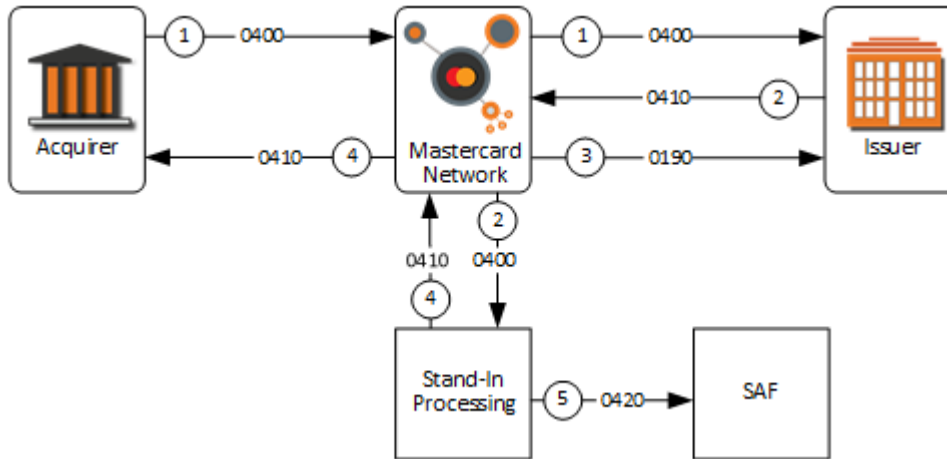
This message flow describes transaction processing when the issuer host generates an invalid or late Authorization Request Response/0110 message, and the Stand-In System processes the transaction and generates a response on behalf of the issuer. The Authorization Platform also sends an Authorization Response Negative Acknowledgement/0190 message to the issuer.



1. The acquirer sends the Authorization Request/0100 message.
2. The issuer generates an invalid or late Authorization Request Response/0110 message.
3. The Authorization Platform sends an Authorization Response Negative Acknowledgement/0190 message to the issuer.
4. The Authorization Platform sends the Authorization Request/0100 message to Stand-In processing.
5. Stand-In processing generates an appropriate Authorization Request Response/0110 message on the issuer's behalf and forwards it to the acquirer.
6. The Stand-In System also generates an Authorization Advice/0120 message and forwards it to the Authorization Platform SAF process for later transmission to the issuer.

Authorization Response Negative Acknowledgement/0190 (Responding to the Reversal Request Response/0410)

This message flow describes transaction processing when the issuer host generates an invalid or late Reversal Request Response/0410 message, and the Authorization Platform processes the transaction and generates a response on behalf of the issuer. The Authorization Platform also sends an Authorization Response Negative Acknowledgement/0190 message to the issuer.



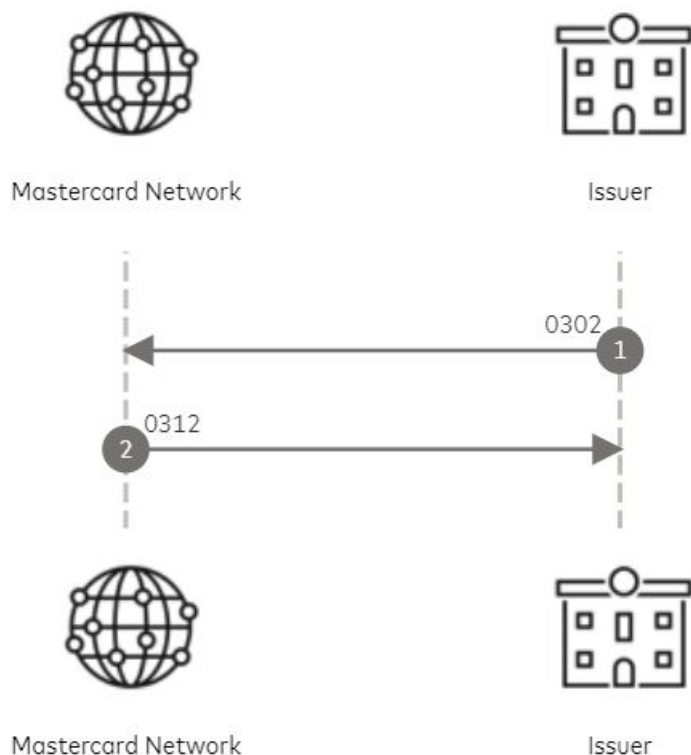
1. The acquirer sends the Reversal Request/0400 message.
2. The issuer generates an invalid or late Reversal Request Response/0410 message and forwards it to the Authorization Platform. The Authorization Platform routes the Reversal Request/0400 to the Stand-In System.
3. The Authorization Platform sends an Authorization Response Negative Acknowledgement/0190 message to the issuer.
4. The Authorization Platform generates the Reversal Request Response/0410 message and sends it to the acquirer.
5. The Stand-In System generates a Reversal Advice/0420 message and forwards it to the Authorization Platform SAF process for later transmission to the issuer.

NOTE: Customers in the Europe region that route to an alternate issuer host for alternate processing instead of the Stand-In System will still receive a Reversal Advice/0420 message as described here. Alternate issuer host processing does not send Reversal Request/0400 or Authorization Advice/0120 messages to the alternate host.

Issuer File Update Requests

This message flow describes transaction processing of the Issuer File Update Request/0302 message.

Figure 13: Issuer file update requests flow



1. Issuer submits the Issuer File Update Request/0302 message to the Mastercard Network. The type of update being requested is specified by the format of the 0302 message. For more details regarding the various request layouts, refer to the [Account Management System User Manual](#).
2. The Mastercard Network performs the requested issuer file update task and responds with an Issuer File Update Request Response/0312 message back to the issuer. The Response Code returned in the Issuer File Update Request Response/0312 message indicates whether the issuer file update was successfully completed.

NOTE: The error-condition message process for Issuer File Update/03xx messages is not illustrated. If an issuer unsuccessfully forwards an Issuer File Update Request/0302 message to the Dual Message Authorization System, the issuer should retransmit the message.

Reversal Messages

Acquirers must send a Reversal Request/0400 message when the acquirer is unable to deliver an issuer's approved Authorization Request Response/0110 to a merchant. Merchants also may request their acquirers to send a Reversal Request/0400 message to cancel the full or partial amount of the original authorization amount.

The Authorization Platform will not attempt to match the contents of the Reversal Request/0400 message with the contents of the original Authorization Request/0100 message.

NOTE:

The acquirer must send the Reversal Request/0400 message immediately, or as soon as possible, after detecting that an approved Authorization Request Response/0110 message cannot be delivered to a merchant. An approved transaction has DE 39 (Response Code) value of 00 (Approved or completed successfully), 08 (Honor with ID), 10 (Partial Approval), or 87 (Purchase amount only, no cash back allowed).

The acquirer must send a Reversal Request/0400 message as a result of an acquirer time-out based on transaction processing rules dictated by the brand. When an acquirer sends a reversal, it is possible that an issuer will receive two reversals for the same authorization request. Issuers must decline subsequent reversal messages using the duplicate transmission response value DE 39 = 94.

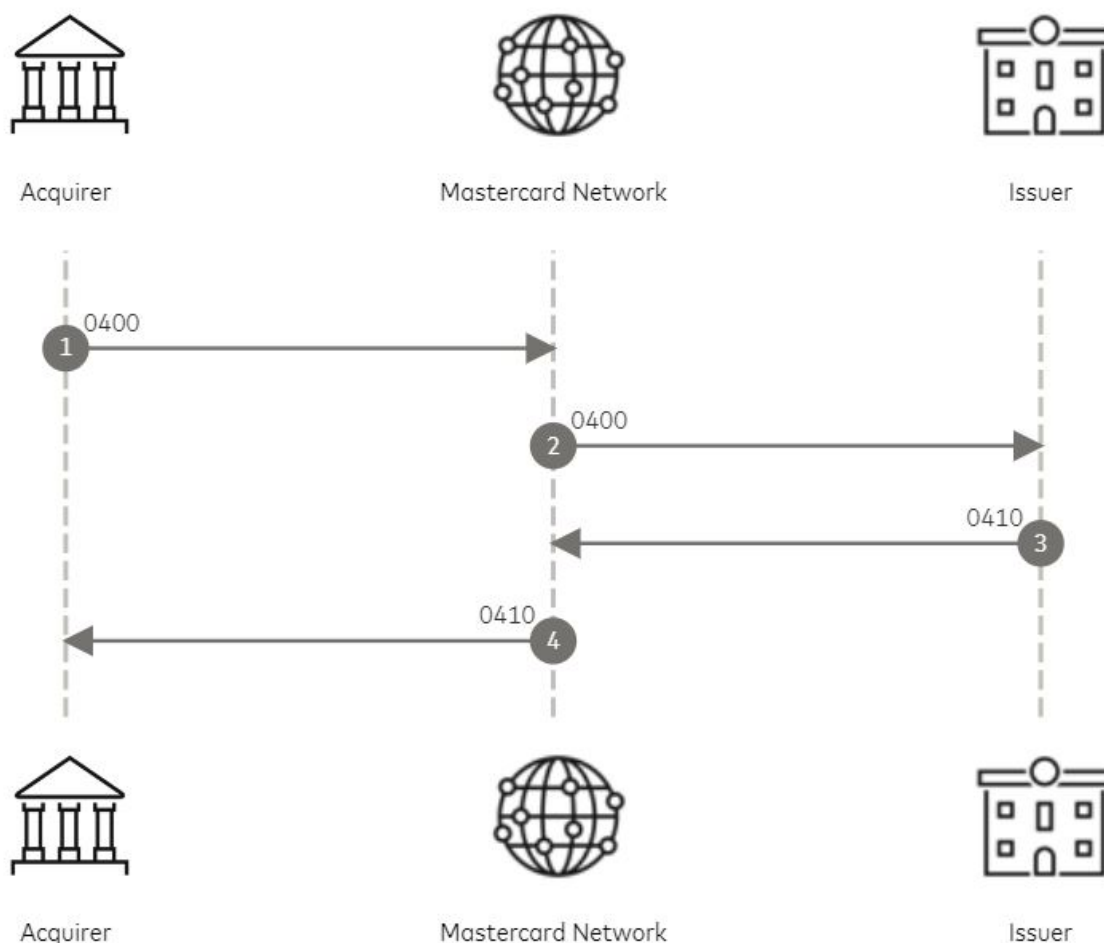
The following message flows describe Stand-In System processing of the Reversal Request/0400 message when:

- The issuer response is received within the time limit
- The issuer response is received after the time limit
- The issuer is signed off
- The issuer response contains errors
- The issuer does not receive the Reversal Request/0400 message

Standard Reversal Request

This message flow describes transaction processing of the Reversal Request/0400 and Reversal Request Response/0410 messages.

Figure 14: Standard Reversal Request flow

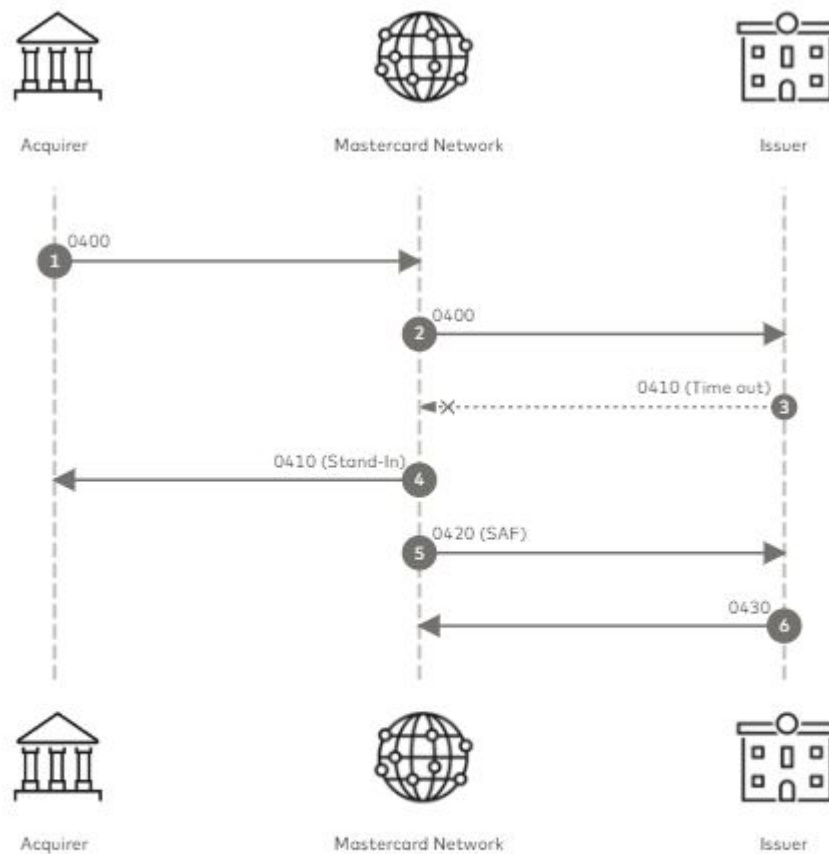


1. The acquirer initiates a Reversal Request/0400 message and sends it to the Mastercard Network.
2. The Mastercard Network forwards the Reversal Request/0400 message to the issuer.
3. The issuer generates a Reversal Request Response/0410 message and sends it to the Mastercard Network.
4. The Mastercard Network forwards the Reversal Request Response/0410 message to the acquirer.

Reversal Request no response from issuer or not within time limit

This message flow describes transaction processing of the Reversal Request/0400 message when the issuer response is not received within the time limit.

Figure 15: Reversal request no response from issuer or not within time limit flow



1. The acquirer sends the Reversal Request/0400 message to the Mastercard Network.
2. The Mastercard Network forwards the Reversal Request/0400 message to the issuer.
3. The Mastercard Network does not receive the issuer's Reversal Request Response/0410 message within the time limit and the Mastercard Network sends the Reversal Request/0400 message to the Stand-In Authorization.
4. The Stand-In Authorization generates the Reversal Request Response/0410 message which is then sent to the acquirer.
5. The Stand-In Authorization creates a Reversal Advice/0420 message and stores it in a SAF queue for guaranteed delivery to the issuer. When the issuer is available to receive the advice

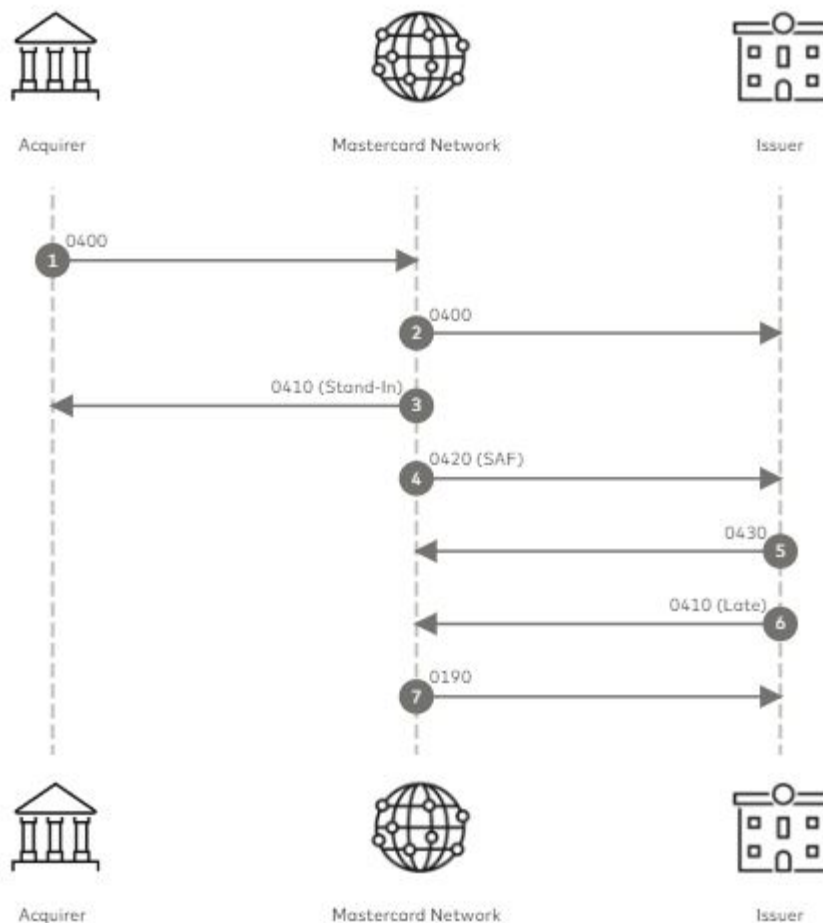
message, the Reversal Advice/0420 message will be delivered to the issuer. The message will indicate that the issuer timed out.

6. The issuer must reply with a Reversal Advice Response/0430 to acknowledge receipt of the advice message.

Reversal Request late response from issuer

This message flow describes transaction processing when the issuer responds to the Reversal Request/0400 message after the time limit.

Figure 16: Reversal request late response from issuer flow



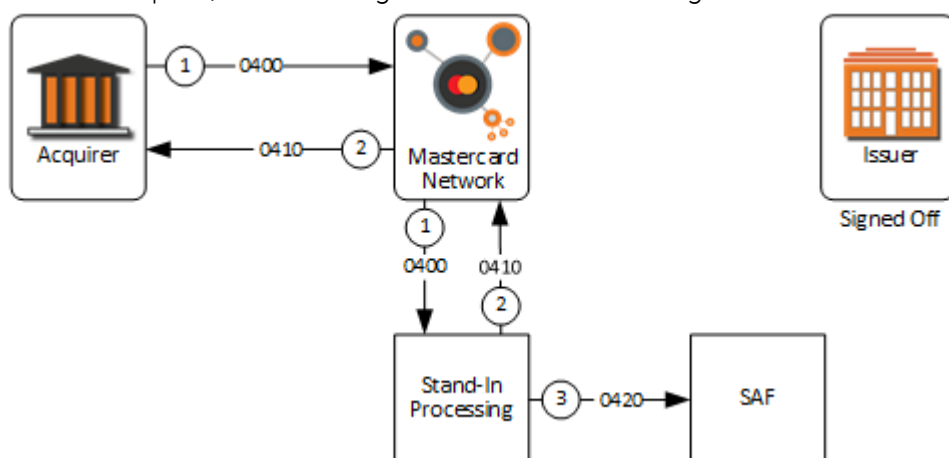
1. The acquirer sends the Reversal Request/0400 message to the Mastercard Network.
2. The Mastercard Network forwards the Reversal Request/0400 message to the issuer.
3. The Mastercard Network does not receive the Reversal Response message from the issuer within the time limit. The Reversal Request/0400 message is sent to the Stand-In Authorization. The Stand-In Authorization will generate the Reversal Request Response/0410 which is then sent to the acquirer.

4. The Stand-In Authorization generates a Reversal Advice/0420 message indicating a time-out condition and places it into the SAF queue for the issuer. When the issuer is available to receive the advice message, the 0420 message will be delivered to the issuer.
5. The issuer must reply with a Reversal Advice Response/0430 message to acknowledge receipt of the advice message.
6. The Mastercard Network receives the late Reversal Request Response/0410 message from the issuer.
7. The Mastercard Network sends the issuer an Authorization Response Negative Acknowledgment/0190 message indicating that the network cannot match the response to a pending Reversal Request message.

NOTE: There is no response necessary from the issuer after the Mastercard Network sends an Authorization Response Negative Acknowledgment/0190 message.

Reversal Request/0400: Issuer Signed Off

This message flow describes transaction processing when the issuer does not respond to the Reversal Request/0400 message because the issuer is signed off.

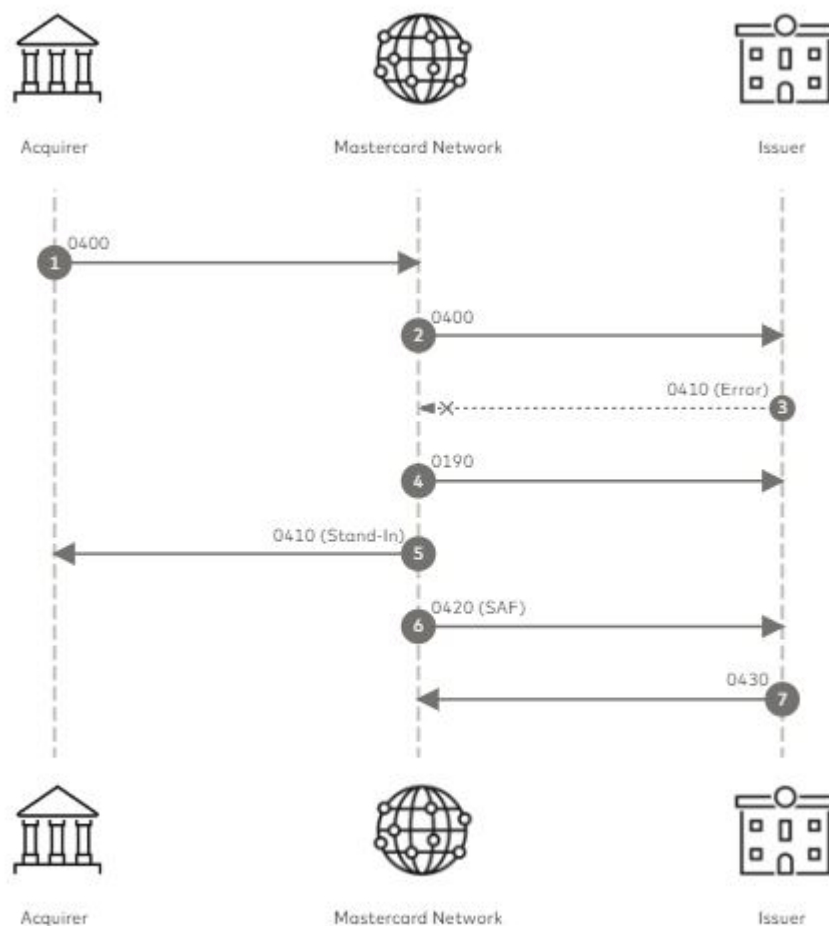


1. The acquirer sends the Reversal Request/0400 message to the Authorization Platform, and then the Authorization Platform forwards the Reversal Request/0400 message to the Stand-In System.
2. The Stand-In System sends the Reversal Request Response/0410 message containing DE 39, value 00 (Approved or completed successfully) to the acquirer.
3. The Stand-In System creates a Reversal Advice/0420 message and stores it in a SAF queue for guaranteed delivery to the issuer where:
 - DE 39 = Response Code value from the original Reversal Request/0400 message
 - DE 60, subfield 1 = 403 (Issuer Sign-out)

Reversal Request response from issuer contains errors

This message flow describes transaction processing when the issuer's response message contains errors.

Figure 17: Reversal request response from issuer contains errors flow



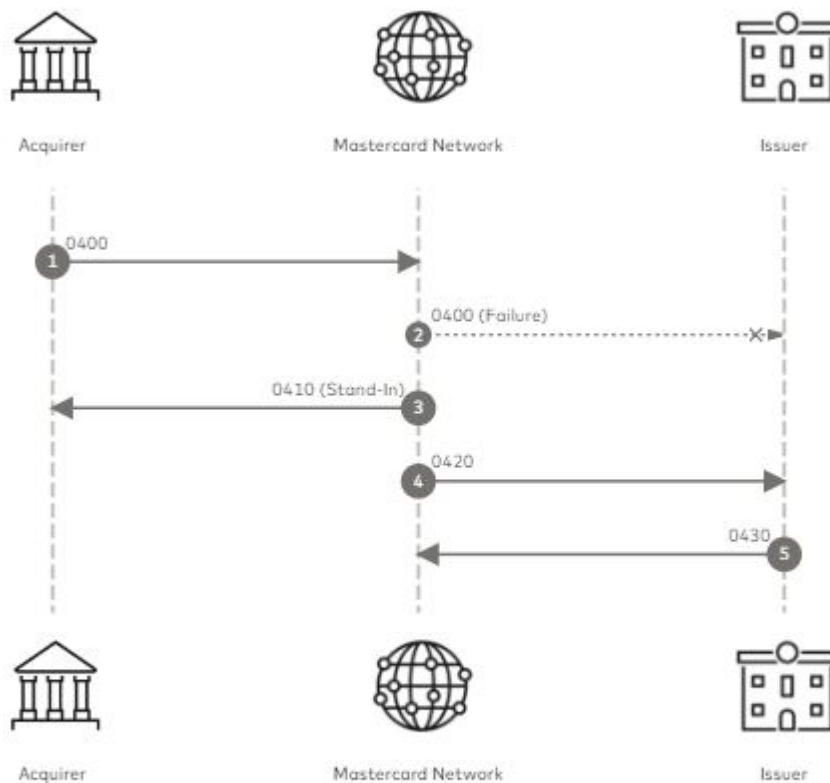
1. The acquirer sends the Reversal Request/0400 message to the Mastercard Network.
2. The Mastercard Network forwards the Reversal Request/0400 message to the issuer.
3. The issuer responds with a Reversal Request Response/0410 message to the Mastercard Network; however, the message contains errors.
4. The Mastercard Network generates an Authorization Response Negative Acknowledgment/0190 message advising the issuer of the error condition in their response.
5. The Mastercard Network will send the Reversal Request/0400 message to the Stand-In Authorization. The Stand-In Authorization will generate a Reversal Request Response/0410 message which is then sent to the acquirer.

6. The Mastercard Network will create a Reversal Advice/0420 message indicating a format error and put it into the SAF queue for the issuer. When the issuer is available to receive the advice message, the advice message will be delivered to the issuer.
7. The issuer must reply with a Reversal Advice Response/0430 message to acknowledge receipt of the advice message.

Reversal Request not delivered to issuer or issuer signed out

This message flow describes transaction processing when the issuer does not receive the Reversal Request/0400 message.

Figure 18: Reversal request not delivered to issuer or issuer signed out flow

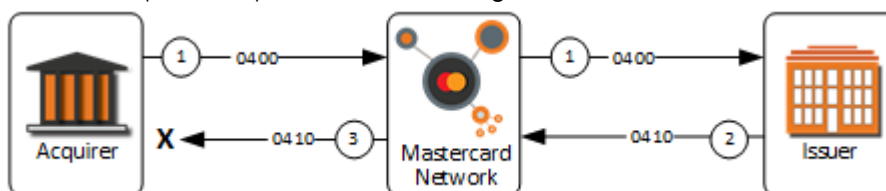


1. The acquirer sends the Reversal Request/0400 message to the Mastercard Network.
2. The Mastercard Network attempts to forward the Reversal Request/0400 message to the issuer. If the issuer is signed out, there is no attempt to route the request to the issuer.
3. The Mastercard Network will send the Reversal Request/0400 message to the Stand-In Authorization. The Stand-In Authorization will generate the Reversal Request Response/0410 message and send it to the acquirer.

4. The Mastercard Network will create a Reversal Advice/0420 message and put it into a SAF queue for guaranteed delivery for the issuer. This advice will indicate if the issuer was signed out or if the request message could not be delivered. When the issuer is available to receive the advice message, the 0420 message will be delivered to the issuer.
5. The issuer must reply with a Reversal Advice Response/0430 message to acknowledge receipt of the advice message.

Reversal Request Response/0410: Not Delivered to Acquirer

This message flow describes transaction processing when the acquirer does not receive the Reversal Request Response/0410 message.

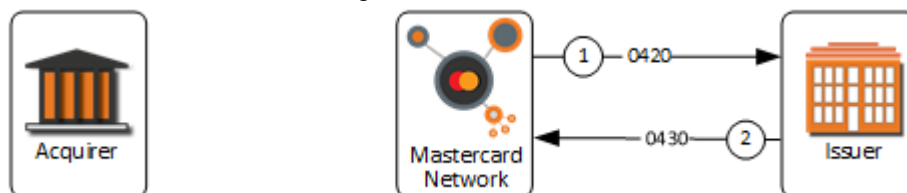


1. The acquirer sends the Reversal Request/0400 message to the Authorization Platform, and then the Authorization Platform forwards the Reversal Request/0400 message to the issuer.
2. The Authorization Platform receives the issuer's Reversal Request Response/0410 message.
3. The Authorization Platform cannot deliver the issuer's Reversal Request Response/0410 message to the acquiring host.
4. The Authorization Platform stores the Reversal Request/0400 and Reversal Request Response/0410 messages and takes no additional action.

NOTE: The acquirer has the responsibility to resend the Reversal Request/0400 message if the acquirer did not receive a response from the Authorization Platform.

Reversal Advice/0420 and Reversal Advice Response/0430

This message flow describes transaction processing of the Reversal Advice/0420 and the Reversal Advice/0430 messages.

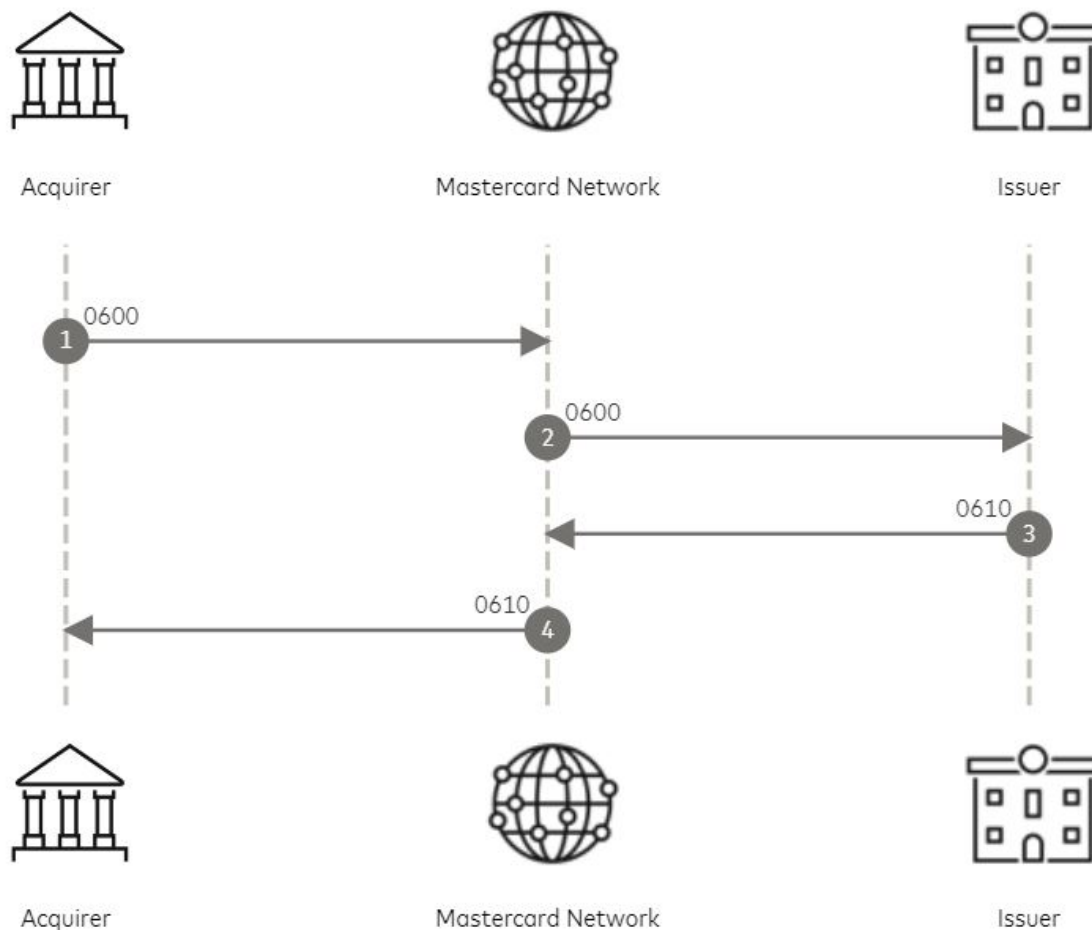


1. The Authorization Platform sends a Reversal Advice/0420 message to the issuer.
2. The issuer responds to the Authorization Platform with a Reversal Advice Response/0430 message to acknowledge positive receipt of the Reversal Advice/0420 message.

Standard Administrative Request

This message flow describes transaction processing of the Administrative Request/0600 and Administrative Request Response/0610 messages.

Figure 19: Standard Administrative request flow

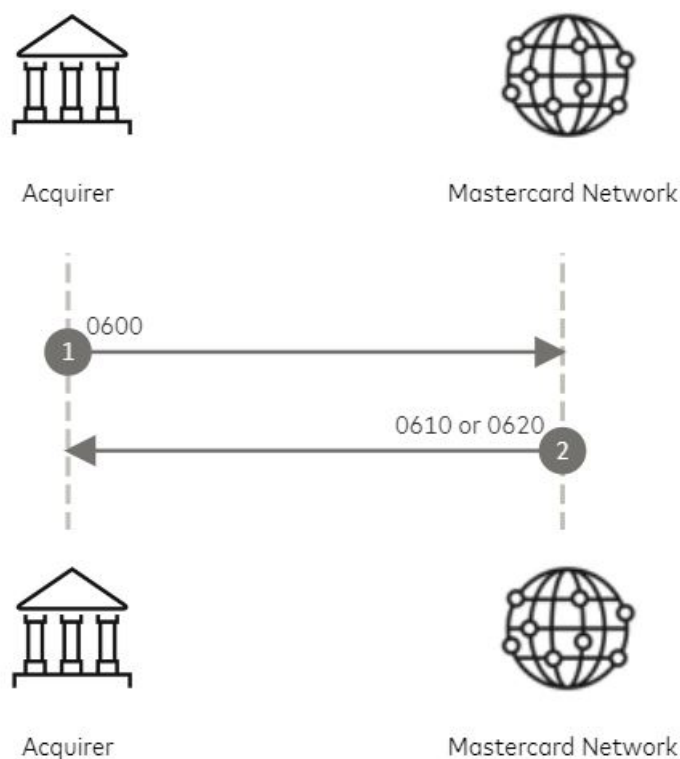


1. The acquirer initiates an Administrative Request/0600 message and sends it to the Mastercard Network.
2. The Mastercard Network forwards the Administrative Request/0600 message to the issuer based on the account range of the Primary Account Number (PAN).
3. The issuer responds with an Administrative Request Response/0610 message to the Mastercard Network.
4. The Mastercard Network forwards the Administrative Request Response/0610 message to the acquirer.

Acquirer-edit Failure

This message flow describes exception processing when the Mastercard Network determines an acquirer edit failure during processing an Administrative Request/0600 message.

Figure 20: Acquirer-edit Failure flow



1. The acquirer initiates an Administrative Request/0600 message and sends it to the Mastercard Network.
2. The Mastercard Network edits the Administrative Request/0600 message and detects an acquirer edit failure. As a result, the Mastercard Network will send the acquirer either an Administrative Request Response/0610 message or an Administrative Advice/0620 message depending on the edit errors.

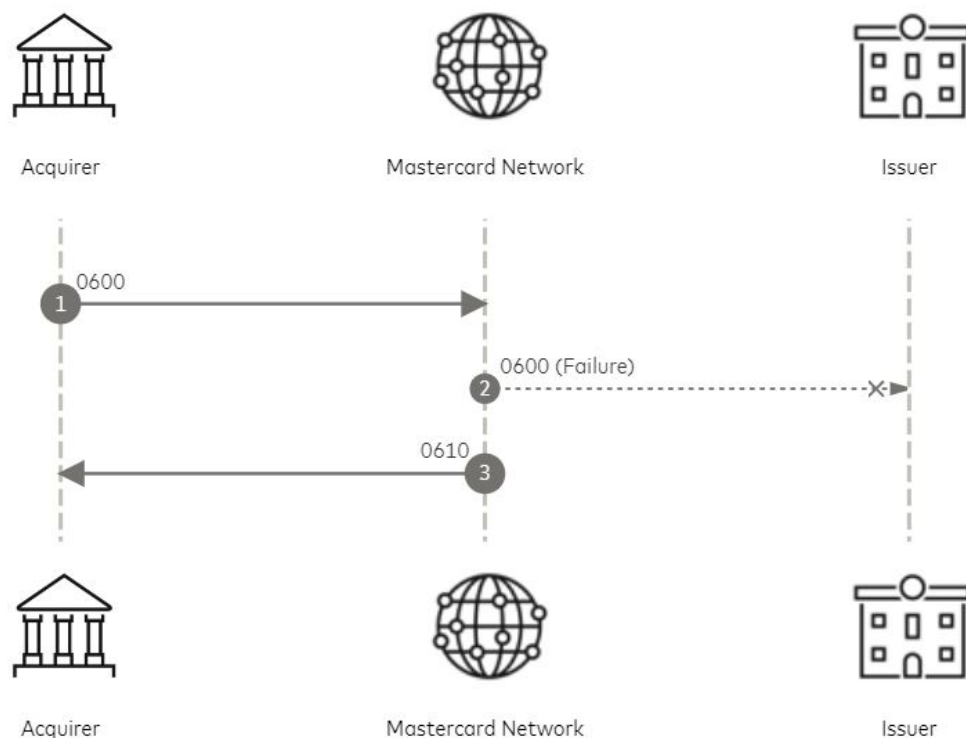
Table 13: Message types and error conditions for response messages

Message type sent	Conditions for the response message
Administrative Request Response/0610	– Is missing mandatory data element – Contains data attribute error or invalid data value – Is from ineligible acquirer/processor – Is sent to ineligible issuer
Administrative Advice/0620	– Cannot be parsed – Exceeds the maximum message length of 8k bytes

Communication Failure at issuer

This message flow describes exception processing when communication fails between the issuer and the Mastercard Network during an Administrative Request/0600 message.

Figure 21: Communication Failure at issuer flow

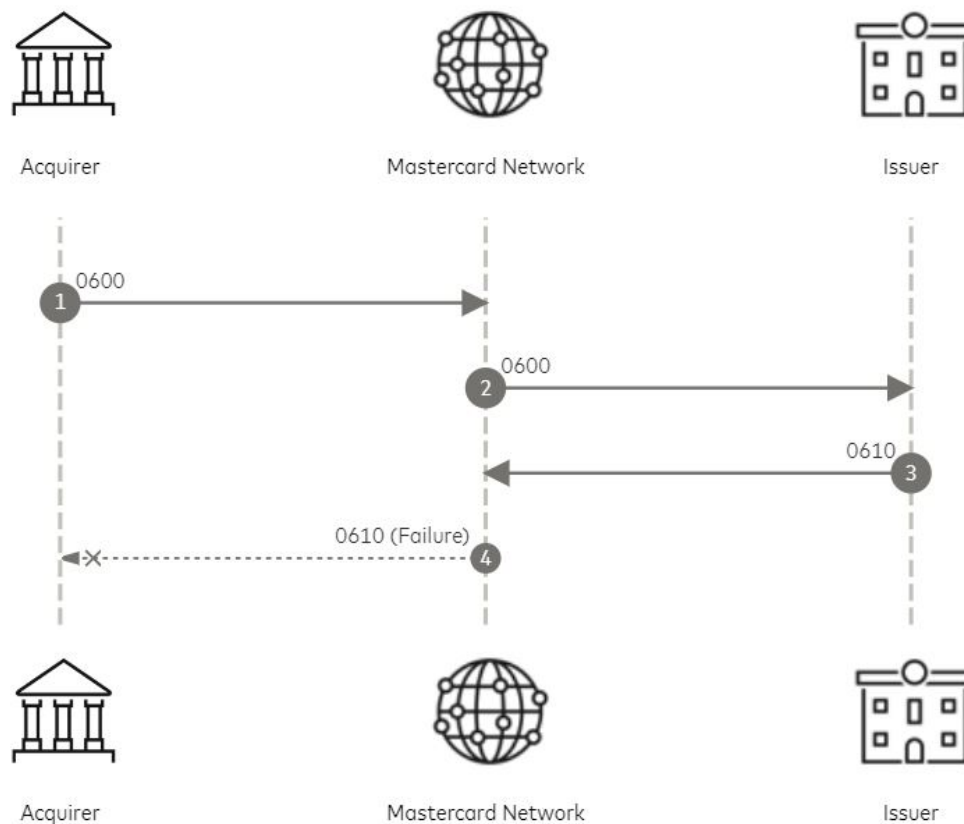


1. The acquirer initiates an Administrative Request/0600 message and sends it to the Mastercard Network.
2. The Mastercard Network attempts to forward the Administrative Request/0600 message to the issuer and is unsuccessful due to a communication failure.
3. The Mastercard Network sends the acquirer an Administrative Request Response/0610 message indicating the request could not be delivered.

Communication Failure at acquirer

This message flow describes exception processing when communication fails between the Mastercard Network and the acquirer during an Administrative Request Response/0610 message.

Figure 22: Communication Failure at acquirer flow

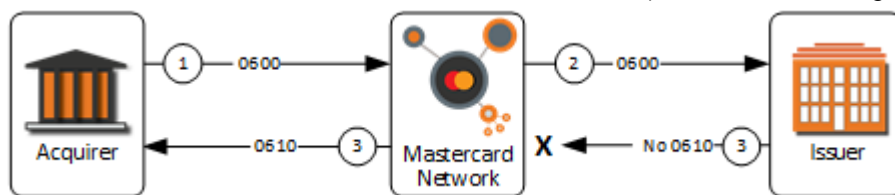


1. The acquirer initiates an Administrative Request/0600 message and sends it to the Mastercard Network.
2. The Mastercard Network forwards the Administrative Request/0600 message to the issuer based upon the account range in the Primary Account Number (PAN).

3. The issuer sends an Administrative Request Response/0610 message to the Mastercard Network.
4. The Mastercard Network attempts to forward the Administrative Request Response/0610 message to the acquirer and is unsuccessful because of a communication link failure or acquirer failure. The Mastercard Network logs the undelivered 0610 message and takes no further action. The acquirer host will time out the 0600 message and resend, if appropriate.

Administrative Request Response/0610: No Issuer Response

This message flow describes exception processing when there is no response from the issuer to the Authorization Platform for an Administrative Request/0600 message.

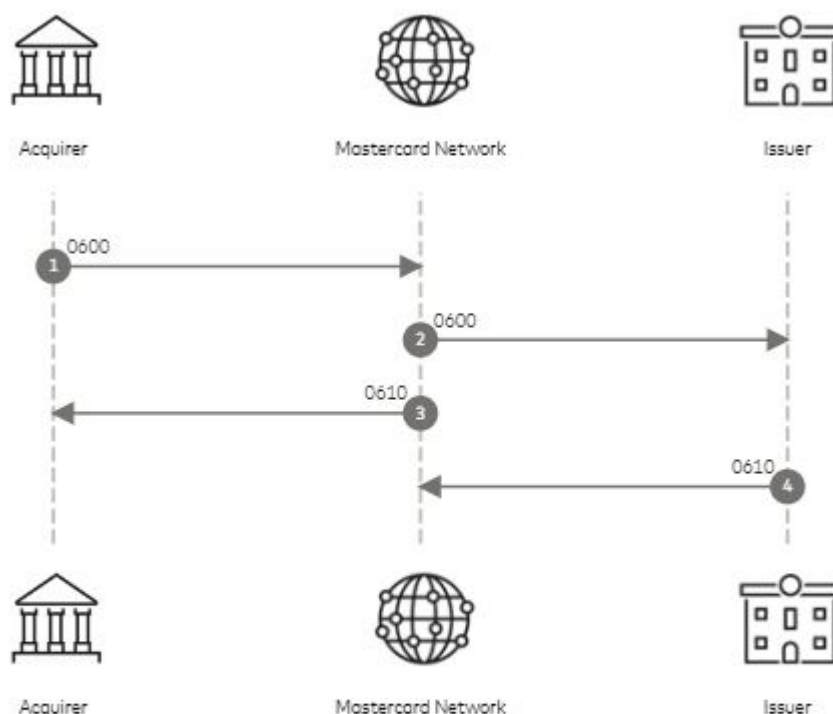


1. The acquirer initiates an Administrative Request/0600 message and sends it to the Authorization Platform.
2. The Authorization Platform forwards the Administrative Request/0600 message to the issuer based upon the account range contained in DE 2 (Primary Account Number [PAN]).
3. If the Authorization Platform receives no response from the issuer before the expiration of the response timer, the Authorization Platform sends the acquirer an Administrative Request Response/0610 message containing DE 39 (Response Code), value 91 (Authorization System or issuer system inoperative).

Late or no issuer response

This message flow describes exception processing when the Mastercard Network either does not receive an Administrative Request Response/0610 message from the issuer or receives a late response.

Figure 23: Late or no issuer response flow

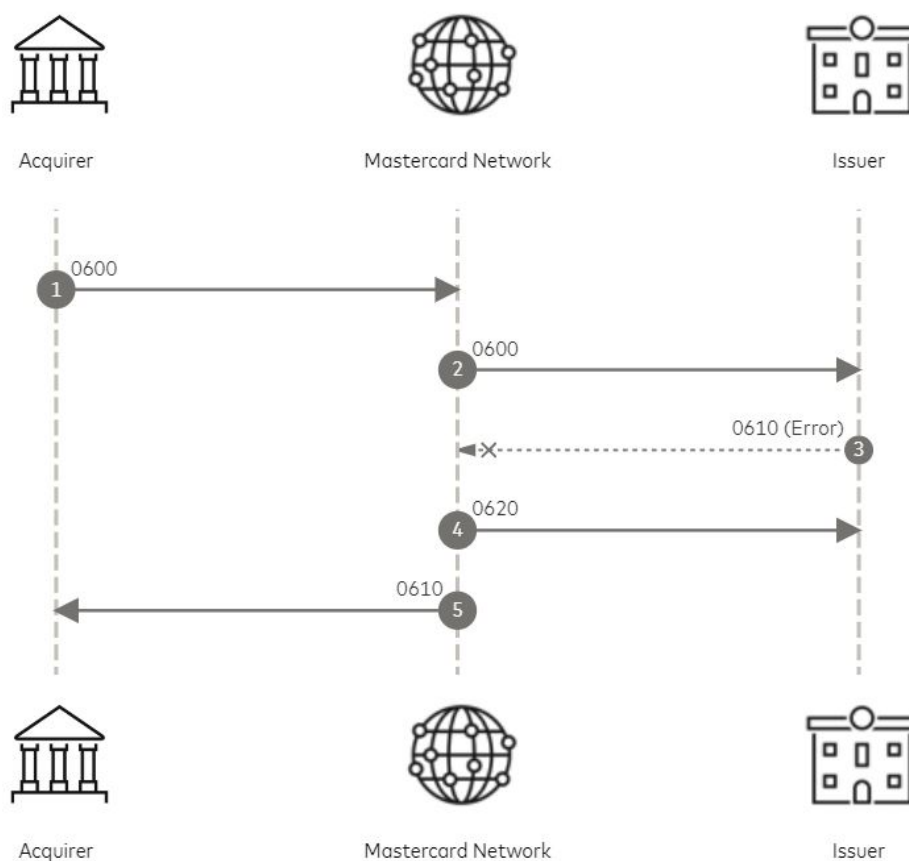


1. The acquirer initiates an Administrative Request/0600 message and sends it to the Mastercard Network.
2. The Mastercard Network forwards the Administrative Request/0600 message to the issuer based upon the account range contained in the PAN.
3. If the Mastercard Network receives no response from the issuer within the time limits, the Mastercard Network sends the acquirer an Administrative Request Response/0610 message indicating the administrative request message could not be delivered.
4. If the Mastercard Network receives the issuer's response after the time limit expired, the Mastercard Network logs the late issuer 0610 message and takes no further action.

Issuer-edit Failure

This message flow describes exception processing when the Mastercard Network receives an Administrative Request Response/0610 message that contains an edit failure from the issuer.

Figure 24: Issuer-edit Failure flow



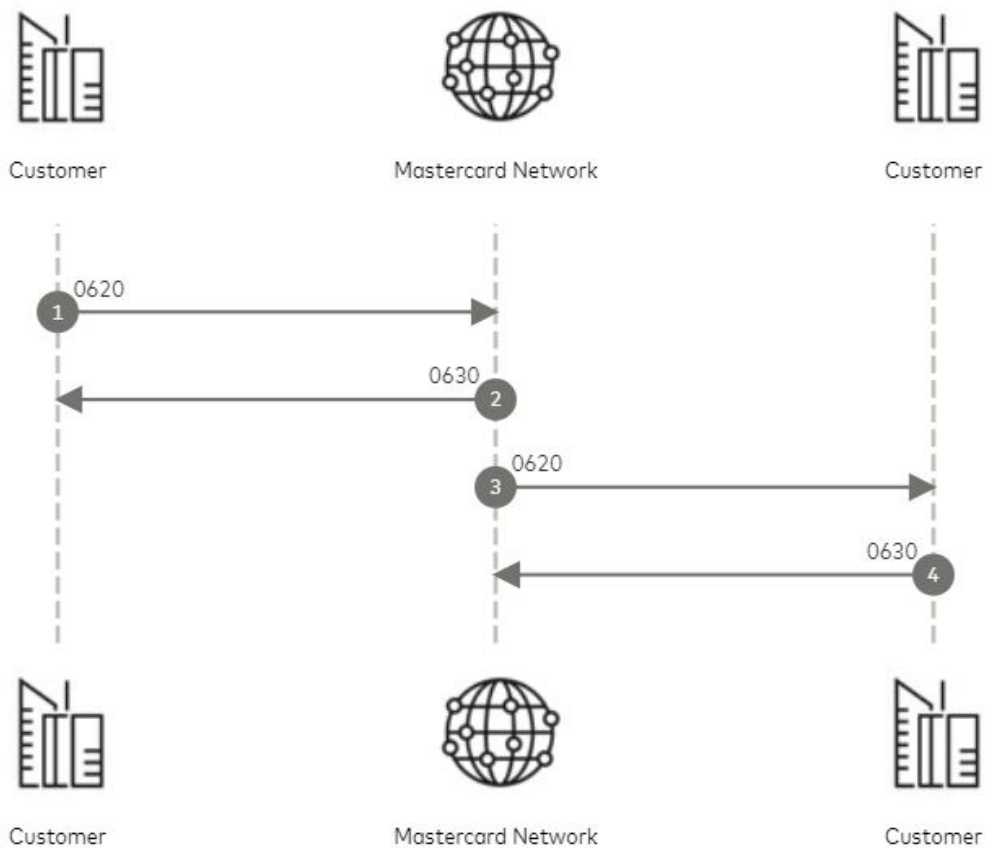
1. The acquirer initiates an Administrative Request/0600 message and sends it to the Mastercard Network.
2. The Mastercard Network forwards the Administrative Request/0600 message to the issuer based upon the account range contained in the PAN.
3. The issuer sends an Administrative Request Response/0610 message to the Mastercard Network. During validation, the Mastercard Network detects edit errors in the response message.
4. The Mastercard Network will generate an Administrative Advice/0620 message to the issuer indicating the following has occurred:
 - Cannot be parsed
 - Exceeds the maximum message length of 8k bytes

- Is missing mandatory data element
 - Contains a data attribute error or invalid data value
5. The Mastercard Network sends the acquirer an Administrative Request Response/0610 message indicating the administrative request was unsuccessful.

Standard Administrative Advice

This message flow describes the standard Administrative Advice/06xx message process between customers. Acquirers or issuers can originate or receive these messages.

Figure 25: Standard Administrative Advice flow



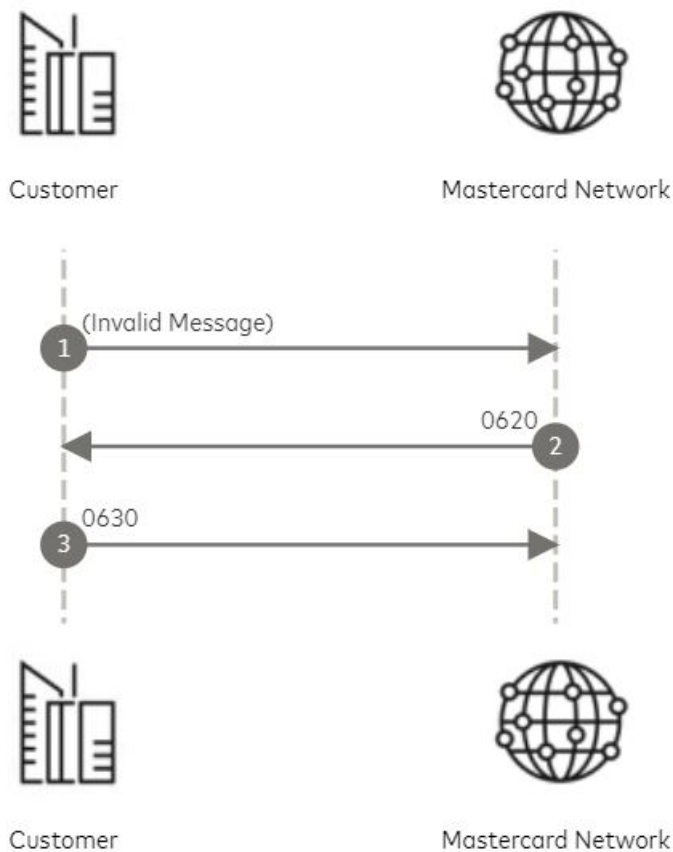
1. The originating customer generates an Administrative Advice/0620 message to the Mastercard Network destined for another customer.
2. The Mastercard Network acknowledges receipt of the Administrative Advice/0620 message by returning an Administrative Advice Response/0630 message to the originating customer.
3. The Mastercard Network forwards the Administrative Advice/0620 message to the intended receiving customer.

4. The receiving customer responds with an Administrative Advice Response/0630 to the Mastercard Network to acknowledge receipt of the advice message.

Failure scenario: Administrative Advice for invalid message

This message flow describes when an invalid message is received and an Administrative Advice/0620: system-generated is sent.

Figure 26: Failure scenario: Administrative Advice for invalid message flow

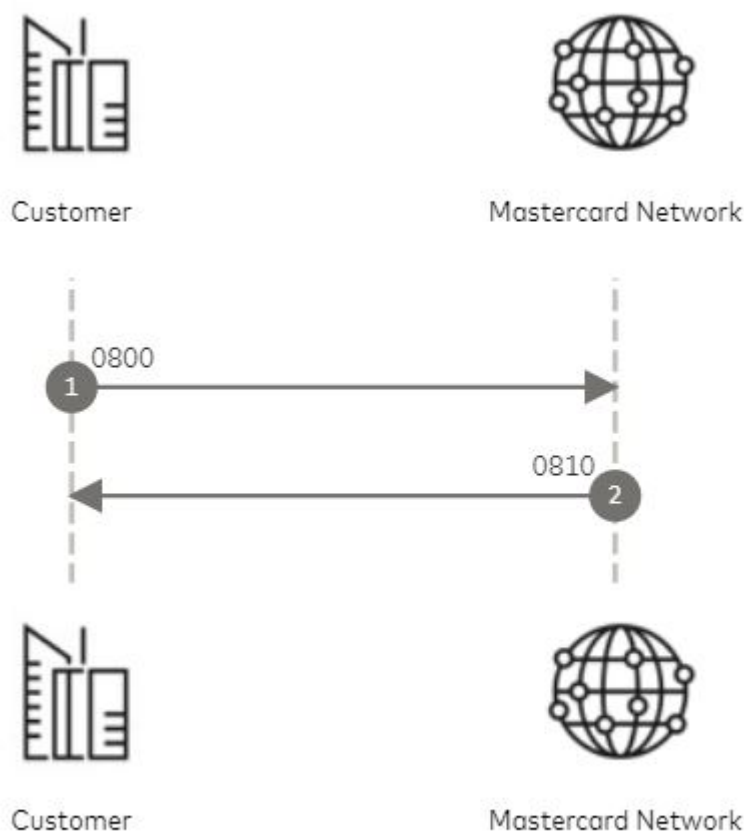


1. The originating customer sends an unidentifiable or invalid message to the Mastercard Network.
2. The Mastercard Network generates an Administrative Advice/0620 message to the originator advising where the failure occurred while parsing or processing the message.
3. The customer acknowledges receipt of the Administrative Advice/0620 message and returns an Administrative Advice Response/0630 message to the Mastercard Network.

Network sign-on or sign-off

This message flow describes the standard Network Management Request/0800 message: sign-on/sign-off message to sign on to or sign off from the Mastercard Network. The originator of these request messages can be either an acquirer or an issuer.

Figure 27: Network sign-on or sign-off flow



1. The customer creates a Network Management Request/0800: sign-on/sign-off message and sends it to the Mastercard Network.
2. After receiving the Network Management Request/0800: sign-on/sign-off message, the Mastercard Network creates a Network Management Request Response/0810: sign-on/sign-off message and sends it to the customer.

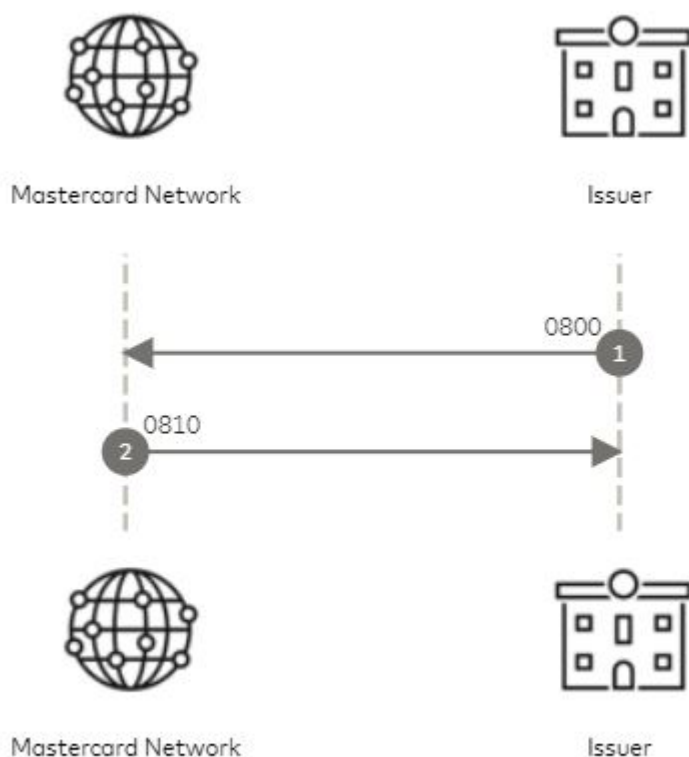
NOTE: Customers are reminded that they should use any one or a combination of the network connectivity management options described in this guide. Mastercard recommends using the varying levels of probes or echo tests to verify host connectivity to the Mastercard Network instead of using the sign-on request message.

Request for issuer-generated or solicited SAF

This message flow describes the issuer-generated (solicited) SAF session using a Network Management Request/0800: SAF Request message.

When an issuer has a large number of SAF records (for example, because of an extended outage), Mastercard can deliver SAF records to the issuer through alternate methods.

Figure 28: Request for issuer-generated or solicited SAF flow

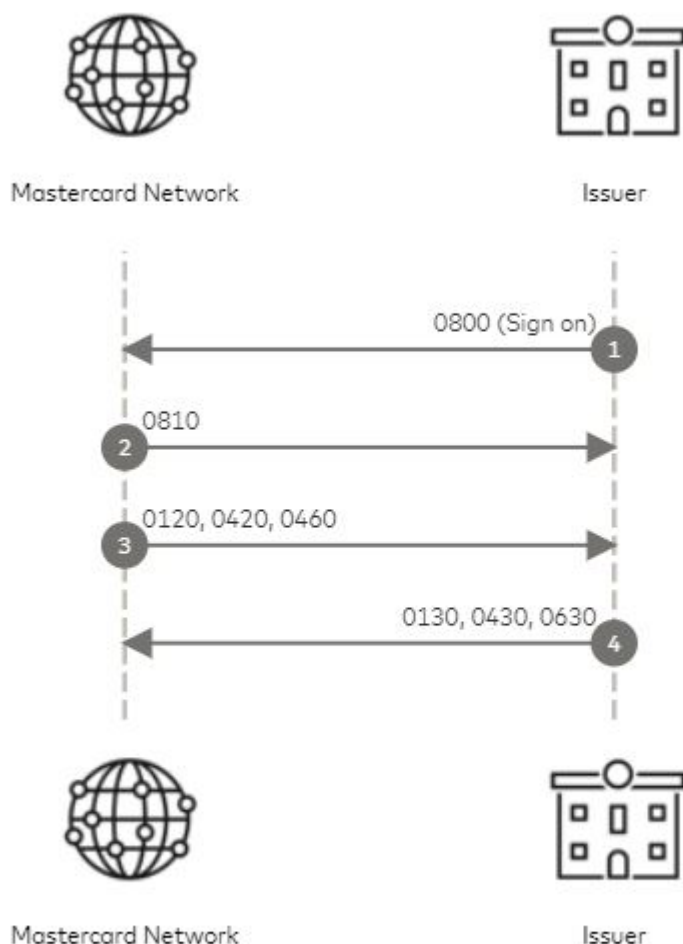


1. The issuer sends a Network Management Request/0800: SAF Request message requesting a SAF session. The request is acknowledged; however, it will not affect SAF processing.
2. The Mastercard Network generates the Network Management Request Response/0810 acknowledgment message. After the Mastercard Network responds to the issuer SAF request message, no further processing of the issuer's request is performed. All SAF messages are processed unsolicited.

Unsolicited SAF after sign-on

This message flow describes the unsolicited SAF session with the issuer after signing on to the Mastercard Network using a Network Management Request/0800: sign-on/sign-off message.

Figure 29: Unsolicited SAF after sign-on flow



1. The issuer sends a Network Management Request/0800: sign-on/sign-off message to sign on to the Mastercard Network.
2. The Mastercard Network sends a Network Management Request Response/0810 message acknowledging the issuer's sign-on request.
3. The Stand-In Authorization checks for SAF messages in the queue and initiates a SAF session. The SAF process forwards an Authorization Advice/0120, Reversal Advice/0420, or Administrative Advice/0620 message to the issuer. The Stand-In Authorization continues to check for SAF messages periodically as long as the issuer is signed in, and forwards messages if there are any available.

4. The issuer generates an Authorization Advice Response/0130, Reversal Advice Response/0430, or Administrative Advice Response/0630 message. These response messages advise the Mastercard Network that the issuer received the previous Authorization Advice/0120, Reversal Advice/0420, or Administrative Advice/0620 message and prompt transmission of the next Authorization Advice/0120, Reversal Advice/0420, or Administrative Advice/0620 message.

NOTE: Issuers receive SAF messages intermixed with authorization and reversal request messages.

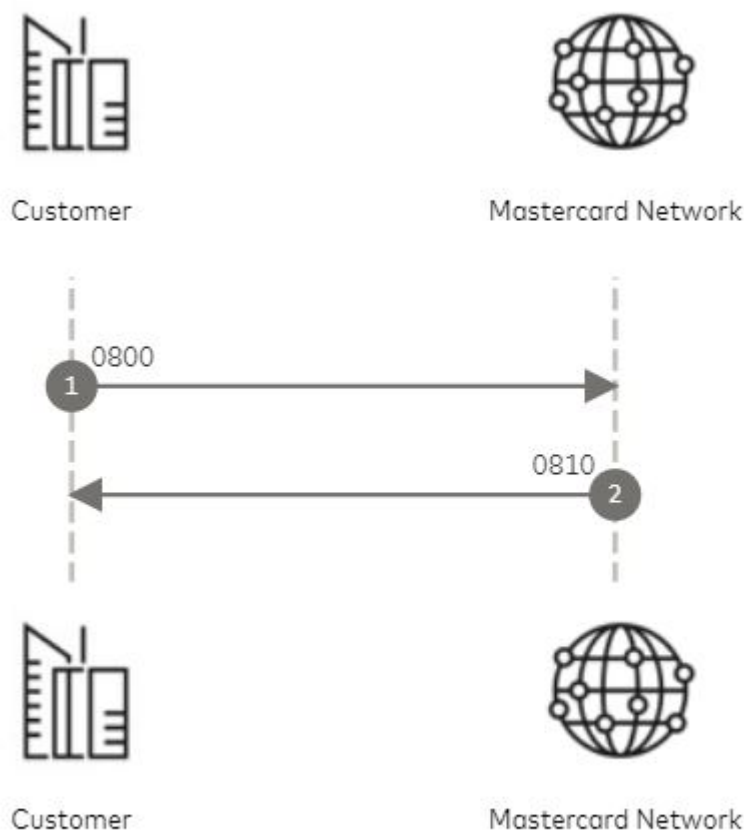
Network-connection status request: customer-generated

This message flow describes transaction processing of the Network Management Request/0800: network connection status: customer-generated message.

Customers should send the Network Management Request Network Connection Status: echo test when they want to verify the status of their connection to the Mastercard Network instead of using a sign-on message.

Customers may not receive a Network Management Request Response/0810 message in the event Mastercard is experiencing an internal network or system issue which impacts the ability to process network management messages. When this occurs, the Mastercard customer must continue to process all financial messages. The customer must report the lack of a Network Management Request Response/0810 message to Mission Control and Operations with details about this scenario and Mastercard will investigate.

Figure 30: Network-connection status request: customer-generated flow



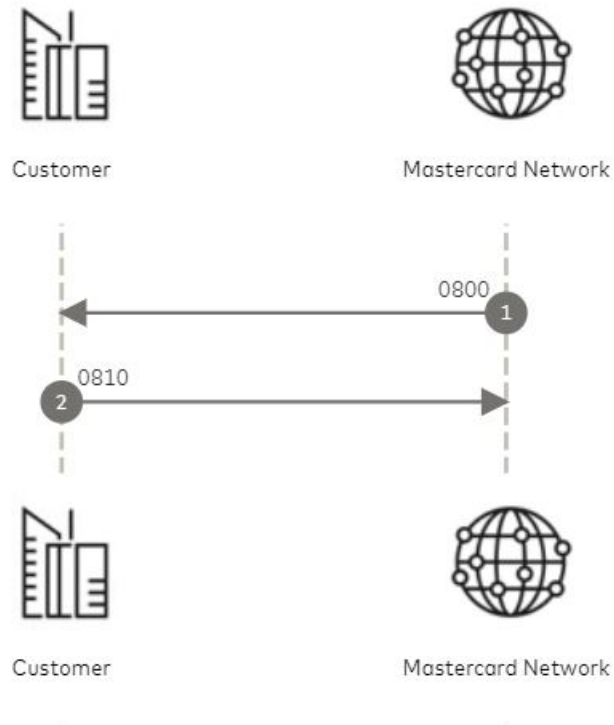
1. The customer sends a Network Management Request/0800: network connection status, customer-generated message indicating an echo test to the Mastercard Network to verify the customer's connection.
2. The Mastercard Network responds with a Network Management Request Response/0810: network connection status, system-generated message to confirm the connection is active.

Network-connection status request: system-generated

This message flow describes transaction processing of the Network Management Request/0800: network connection status: system-generated message.

Customers that request Mastercard to periodically check their connection status to the Mastercard Network will receive the Network Management Request/0800 message from Mastercard.

Figure 31: Network-connection status: system-generated flow

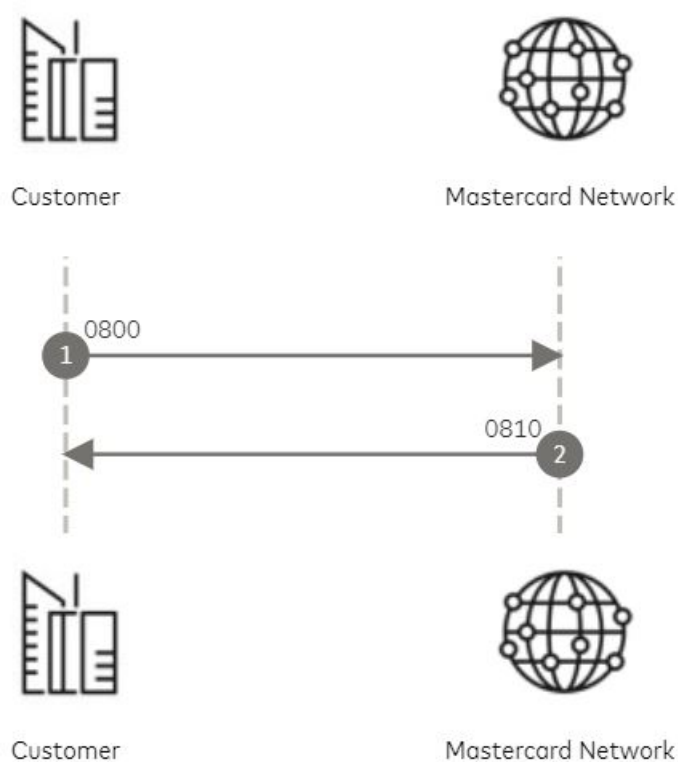


1. The Mastercard Network sends a Network Management Request/0800: network connection status: system-generated message identifying the message as an echo test to the customer to verify the customer's connection.
2. The customer responds with a Network Management Request Response/0810: network connection status: customer-generated message to confirm that the connection is active.

Host activation or deactivation

This message flow describes transaction processing of the Network Management Request/0800: host session activation/deactivation message.

Figure 32: Host activation or deactivation flow

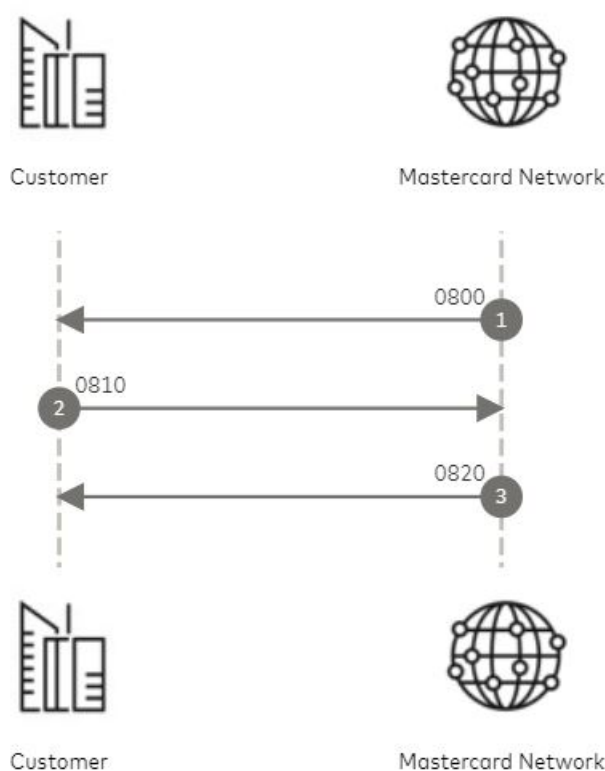


1. The customer sends a Network Management Request/0800: Host Session Activation/Deactivation message to identify session activation or session deactivation, through the value supplied in DE 70, to the Mastercard Network.
2. The Mastercard Network responds with a Network Management Request Response/0810: Host Session Activation/Deactivation message indicating successful completion.

PEK-exchange request: system generated

This message flow describes transaction processing as it exchanges new PEKs with the customer every 24 hours.

Figure 33: PEK exchange request: system-generated flow

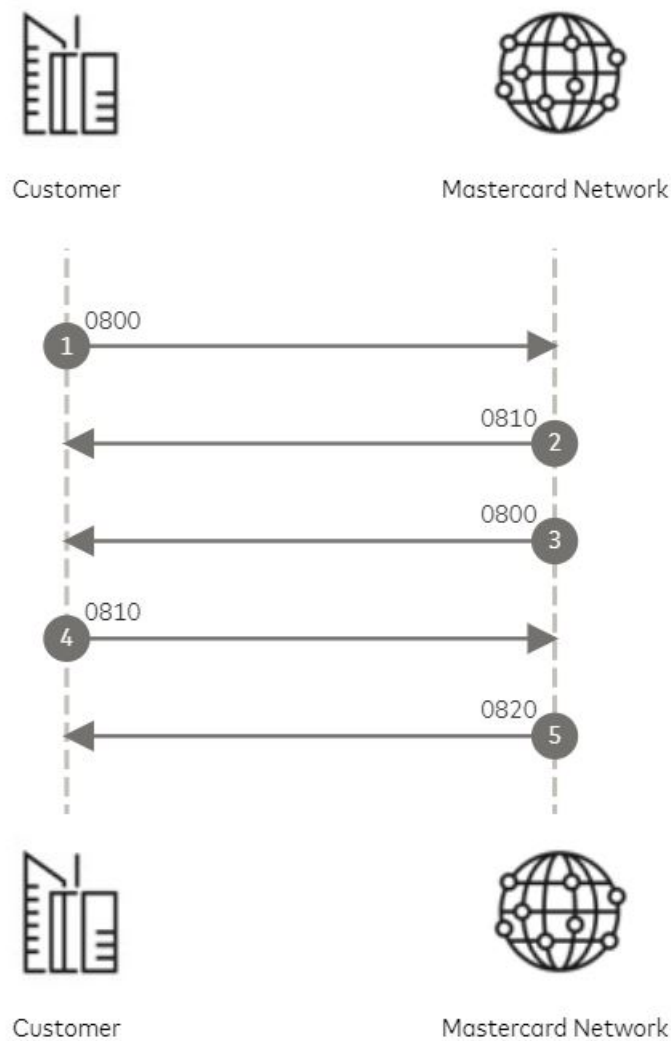


1. The predetermined time has passed for exchanging a new PEK. The Mastercard Network uses the Key Encryption Key (KEK) to encrypt the new PEK. The Mastercard Network sends a Network Management Request/0800: PEK exchange message containing the PEK. The message contains the new PEK (encrypted using the KEK) in *DE 48* or *DE 110* and the customer ID in *DE 2* (*Primary Account Number [PAN]*).
2. The customer stores the new PEK. The customer sends a Network Management Request Response/0810: PEK exchange message to the Mastercard Network acknowledging receipt of the Network Management Request message exchanging the PEK.
3. The Mastercard Network sends a Network Management Advice/0820: PEK exchange message to notify the customer that the new PEK is active and operational or to advise of the failure. If the Network Management Advice/0820: PEK exchange message indicated that the PEK is active and operational, then subsequent Authorization Request/0100 messages must use the new PEK to translate the PIN data.

PIN Encryption Key exchange request: customer generated

Customers must be capable of sending a Network Management Request/0800: PEK exchange customer-generated message to the Mastercard Network requesting a new PEK. The Mastercard Network immediately initiates a PEK exchange with the customer identified in *DE 33 (Forwarding Institution ID Code)* of the Network Management Request/0800: PEK exchange On Demand message. Customers should use this feature when systems problems occur and a re-synchronization of the PEK is necessary. Mastercard exchanges PEKs with customers every 24 hours.

Figure 34: PEK exchange request: customer-generated flow



1. The customer determines there is a problem with its PEK and sends a Network Management Request/0800: PEK exchange On Demand message to the Mastercard Network requesting a new PEK.
2. The Mastercard Network responds with a Network Management Request Response/0810: PEK exchange On Demand message. The Mastercard Network uses the Key Encryption Key (KEK) to encrypt the new PEK.
3. The Mastercard Network sends a Network Management Request/0800: PEK exchange message containing the new PEK. The message contains the new PEK and the associated ID of the customer, identified in *DE 2 (Primary Account Number [PAN])*. The customer stores the new PEK and will use it for PIN encryption in subsequent Authorization Request/0100 messages.
4. The customer sends a Network Management Request Response/0810: PEK exchange message to the Mastercard Network acknowledging receipt of the Network Management Request/0800: PEK exchange message exchanging the PEK.
5. The Mastercard Network sends a Network Management Advice/0820: PEK exchange message to the customer notifying it that the new PEK is active and operational or advising of the failure. If the Network Management Advice/0820: PEK exchange message indicated that the PEK is active and operational, then subsequent Authorization Request/0100 messages must use the new PEK to translate the PIN data.

NOTE: Error-condition message flows for Network Management/08xx messages are not illustrated. If an acquirer or issuer unsuccessfully forwards a Network Management/08xx message to the Mastercard Network, the acquirer or issuer should retransmit the message.

Chapter 3 Message layouts

This section describes the required, conditional, optional, or Authorization Platform–provided data element layouts for all messages that the Authorization Platform supports.

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Authorization Request/0100

This is the message layout for the Authorization Request/0100 message.

Table 14: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	•	M	Constant: 0100 (Authorization Request)
-	Bit Map, Primary	M	P	M	Mandatory for all messages
1	Bit Map, Secondary	C	P	C	Required data element if any DE 65 (Bit Map, Extended) through DE 128 (Message Authentication Code [MAC]) are present in the message
2	Primary Account Number (PAN)	M	X	M	Contains cardholder's Primary Account Number (PAN)
3	Processing Code	M	•	M	Indicates type of transaction and the affected cardholder To or From Account Type
4	Amount, Transaction	M	•	M	Transaction amount in the currency of the acquirer's acceptor. DE 49 (Currency Code, Transaction) also must be present with this data element to identify the transaction currency.
5	Amount, Settlement	•	X	C	Transaction amount in the settlement currency in DE 50 (Currency Code, Settlement). The Dual Message Authorization System provides DE 05 to issuers if they choose to receive settlement amount related data elements.
6	Amount, Cardholder Billing	•	X	M	Transaction amount in the issuer currency. The Dual Message Authorization System provides this data element. DE 51 (Currency Code, Cardholder Billing) will also be present to identify the currency for this amount.
7	Transmission Date and Time	M	•	M	Transaction time stamp; date and time that the originator initiates the message.

DE ID	Data Element Name	Org	Sys	Dst	Comments
9	Conversion Rate, Settlement	•	X	C	The Dual Message Authorization System provides DE 9 if the issuer chooses to receive settlement amount-related data elements.
10	Conversion Rate, Cardholder Billing	•	X	M	The Dual Message Authorization System provides DE 10 with the rate used to convert the transaction currency to the cardholder billing currency.
11	Systems Trace Audit Number (STAN)	M	•	M	Number assigned by the message initiator to uniquely identify a transaction.
12	Time, Local Transaction	C	•	C	Used in all customer-related reports and statements, if available.
13	Date, Local Transaction	C	•	C	Used in all customer-related reports and statements, if available.
14	Date, Expiration	C	•	C	Specifies the year and month a PAN is considered expired by the issuer.
15	Date, Settlement	•	X	M	The acquirer omits this data element, and the Mastercard Network inserts it and forwards it to the issuer.
16	Date, Conversion	•	X	M	Currency conversion rate file effective date. The Mastercard Network provides this data element.
18	Merchant Type	M	•	M	Refer to the Quick Reference Booklet for a listing of MCCs.
19	Acquiring Institution Country Code	•	X	C	Based on Mastercard franchise data information.
20	Primary Account Number (PAN) Country Code	C	•	C	Required for messages when the PAN begins with 59.
22	Point-of-Service (POS) Entry Mode	M	•	M	Used to indicate the method used for PAN data entry to initiate a transaction.
23	Card Sequence Number	C	•	C	Must be present for transactions that include DE 55 (ICC system-related data) and where the ICC provides the application PAN sequence number to the terminal.

DE ID	Data Element Name	Org	Sys	Dst	Comments
26	Point-of-Service (POS) Personal ID Number (PIN) Capture Code	C	•	C	Required when PIN data is present and the terminal PIN capture capability is other than 12 characters.
28	Amount, Transaction Fee	C	•	C	Must contain fee amount, if applied.
32	Acquiring Institution ID Code	M	•	M	Contains acquirer's identification information.
33	Forwarding Institution ID Code	C	•	C	Contains the processor ID number forwarding this message to the Mastercard Network.
35	Track 2 Data	C	X	C	Required if the transaction entry point captured Track 2 data. Required for ATM transactions. For Maestro transactions, such as e-commerce, the Dual Message Authorization System creates Track 2 data if it is not provided by the acquirer.
37	Retrieval Reference Number	C	•	C	Required for ATM, Chip and POS card-read transactions.
41	Acceptor Terminal ID	C	•	C	Required for all ATM and card-read POS transactions.
42	Acceptor ID	C	•	C	Mandatory for POS transaction types containing DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code), values 00 (Purchase), 09 (Purchase with Cash Back), and 28 (Payment Transaction).
43	Acceptor Name and Location	C	•	C	Required for Chip and POS card-read transactions. Required for Mastercard Hosted Mobile Phone Top-up transactions.
45	Track 1 Data	C	•	C	Required if the transaction entry point captured Track 1 data.
48	Additional Data: Private Use	M	X	M	Contains a variety of subelements depending on the card and purpose of the request.
49	Currency Code, Transaction	M	•	M	Identifies the currency code of DE 4 (Amount, Transaction).

DE ID	Data Element Name	Org	Sys	Dst	Comments
50	Currency Code, Settlement	•	X	C	Identifies the currency code of DE 5 (Amount, Settlement). The Dual Message Authorization System provides DE 50 if the issuer chooses to receive settlement amount-related data elements.
51	Currency Code, Cardholder Billing	•	X	M	Defines the currency of DE 6 (Amount, Cardholder Billing), DE 8 (Amount, Cardholder Billing Fee), and DE 95 (Replacement Amounts), subfield 3 (Actual Amount, Cardholder Billing).
52	Personal ID Number (PIN) Data	C	X	C	Required for ATM transactions. NOTE: DE 110 pin data is mutually exclusive of DE 52, DE 53 and DE 125. Either one of them has to be provided.
53	Security-Related Control information	C	X	C	Required for acquirers using the Dual Message Authorization System to perform PIN translation services. Acquirers provide DE 53 to identify the PIN Block Format and key used for PIN encryption. The Dual Message Authorization System provides issuers using the Dual Message Authorization System to perform PIN translation the PIN Block Format and key used by Mastercard to encrypt the PIN before sending the Authorization Request/0100 to the issuer. NOTE: DE 110 pin data is mutually exclusive of DE 52, DE 53 and DE 125. Either one of them has to be provided.
54	Additional Amounts	C	X	C	The Dual Message Authorization System will forward the occurrence of each DE 54 amount type provided by the acquirer in the acquirer's currency and will provide an additional occurrence of each DE 54 amount type in the issuer's currency.
55	Integrated Circuit Card (ICC) System-Related Data	C	•X	C	Conditional data for chip-based transactions.

DE ID	Data Element Name	Org	Sys	Dst	Comments
56	Payment Account Data	•	X	C	When Mastercard is the BIN Controller (as defined by EMVCo) DE 56 will be present and contain the PAR value when one is associated with the PAN.
61	Point-of-Service (POS) Data	M	•	M	Describes the conditions present at the point-of-interaction at the time when the originator initiates the transaction.
62	Intermediate Network Facility (INF) Data	O	•	O	May be used to provide unique acquiring network ID codes.
63	Network Data	•	X	M	Contains the financial network code and reference data for the transaction.
104	Digital Payment Data	C	X	C	Contains applicable subelement data.
105	Multi-Use Transaction Identification Data	C	X	C	
106	Fleet Service Data	O	•	C	
108	Additional Transaction Reference Data	C	•	C	Mandatory for originating institution to submit DE 108 on all Mastercard Send transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions, and Mastercard™ Merchant Presented QR Payment and Funding Transactions). Optional on Mastercard Merchant Presented QR Refund Payment Transactions.
110	Encryption Data	C	X	C	Includes PIN encryption data. NOTE: DE 110 pin data is mutually exclusive of DE 52, DE 53 and DE 125. Either one of them has to be provided.
112	Additional Data (National Use)	C	•	C	Contains applicable subelement data.
120	Record Data	C	X	C	Used to transmit file record data or textual character string data.
122	Additional Record Data	C	•	C	
124	Member-defined Data	C	•	C	May contain customer-defined data.

DE ID	Data Element Name	Org	Sys	Dst	Comments
125	New PIN Data	C	X	C	Must be present for all PIN change transactions; otherwise not present. NOTE: DE 110 pin data is mutually exclusive of DE 52, DE 53 and DE 125. Either one of them has to be provided.
127	Private Data	O	X	.	Private data for message initiator's use.

Authorization Request Response/0110

This is the message layout for the Authorization Request Response/0110 message.

Table 15: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	.	M	Constant: 0110 (Authorization Request Response).
-	Bit Map, Primary	M	P	M	Mandatory for all messages.
1	Bit Map, Secondary	C	P	C	Mandatory if DE 65 (Bit Map, Extended) through DE 128 (Message Authentication Code [MAC]) are present in the message.
2	Primary Account Number (PAN)	ME	X	ME	Must be the same value as in the original Authorization Request/0100.
3	Processing Code	ME	.	ME	Must be the same value as in the original Authorization Request/0100.
4	Amount, Transaction	CE	X	M	Must be the same value as in the original Authorization Request/0100 except when DE 39 (Response Code) contains value 10 (Partial approval) or value 87 (Purchase Amount Only: No Cash Back Allowed).

DE ID	Data Element Name	Org	Sys	Dst	Comments
5	Amount, Settlement	CE	X	C	Must be the same value as in the original Authorization Request/0100, if present, except when DE 39 (Response Code) contains value 10 (Partial approval) or value 87 (Purchase Amount Only: No Cash Back Allowed). The Mastercard Network provides this data element if the acquirer chooses to receive settlement amount-related data elements. This data will be present, as defined, except when the Mastercard Network has declined an Authorization Request/0100 message and was unable to complete currency conversion processing.
6	Amount, Cardholder Billing	C	•	C	Must be the same value as in the original Authorization Request/0100, except when DE 39 (Response Code) contains value 10 (Partial approval) or value 87 (Purchase Amount Only: No Cash Back Allowed). This data will be present, as defined, except when the Mastercard Network has declined an Authorization Request/0100 message and was unable to complete currency conversion processing.
7	Transmission Date and Time	ME	•	ME	Must be the same value as in the original Authorization Request/0100.
9	Conversion Rate, Settlement	CE	X	C	Must be the same value as in the original Authorization Request/0100, if present. The Dual Message Authorization System provides this data element if the acquirer chooses to receive settlement amount-related data elements. This data will be present, as defined, except when the Mastercard Network has declined an Authorization Request/0100 message and was unable to complete currency conversion processing.

DE ID	Data Element Name	Org	Sys	Dst	Comments
10	Conversion Rate, Cardholder Billing	C	•	C	Must be the same value as in the original Authorization Request/0100 message. This data will be present, as defined, except when the Mastercard Network has declined an Authorization Request/0100 message and was unable to complete currency conversion processing.
11	Systems Trace Audit Number (STAN)	ME	•	ME	Must be the same value as in the original Authorization Request/0100 message.
15	Date, Settlement	ME	•	M	Must be the same value as in the original Authorization Request/0100 message.
16	Date, Conversion	C	•	C	Must be the same value as in the original Authorization Request/0100 message. This data will be present, as defined, except when the Mastercard Network has declined an Authorization Request/0100 message and was unable to complete currency conversion processing.
20	Primary Account Number (PAN) Country Code	CE	•	CE	Must be the same value as in the original Authorization Request/0100 message, if present.
23	Card Sequence Number	CE	•	C	Must be the same value as in the original Authorization Request/0100 message, if present.
28	Amount, Transaction Fee	CE	X	CE	Must be the same value as in the Authorization Request/0100 message, if present.
32	Acquiring Institution ID Code	ME	•	ME	Must be the same value as in the original Authorization Request/0100 message.
33	Forwarding Institution ID Code	CE	•	CE	Must be the same value as in the original Authorization Request/0100 message, if present.
37	Retrieval Reference Number	CE	•	CE	Must be the same value as in the original Authorization Request/0100 message, if present.

DE ID	Data Element Name	Org	Sys	Dst	Comments
38	Authorization ID Response	C	•	C	Response ID that the authorizing institution or agent assigned for an approved request.
39	Response Code	M	•	M	Contains message-specific values. Refer to Data element definitions for list of values by message type. Issuers responding with DE 39 (Response Code), value 10 (Partial approval) or value 87 (Purchase amount only, no cash back allowed) will not be required to echo DE 4 (Amount, Transaction) in the Authorization Request Response/0110. Likewise, if DE 5 (Amount, Settlement) was present in the Authorization Request/0100 message to the issuer, the issuer will not be required to echo DE 5 in the Authorization Request Response/0110 message when responding with DE 39, value 10 or value 87. The issuer will provide the partial approval amount in DE 6 (Amount, Cardholder Billing).
41	Acceptor Terminal ID	CE	•	CE	Must be the same value as in the original Authorization Request/0100 message, if present.
44	Additional Response Data	C	•	C	May be used for referral phone numbers in denied transactions, or cardholder ID information in approved transactions.
48	Additional Data: Private Use	C	X	C	Contains applicable subelement data.
49	Currency Code, Transaction	ME	•	ME	Must be the same value as in the original Authorization Request/0100 message.

DE ID	Data Element Name	Org	Sys	Dst	Comments
50	Currency Code, Settlement	CE	X	C	Must be the same value as in the original Authorization Request/0100 message, if present. The Mastercard Network provides this data element if the acquirer chooses to receive settlement amount-related data elements. This data will be present, as defined, except when the Mastercard Network has declined an Authorization Request/0100 message and was unable to complete currency conversion processing.
51	Currency Code, Cardholder Billing	C	•	C	Must be the same value as in the original Authorization Request/0100 message. This data will be present, as defined, except when the Mastercard Network has declined an Authorization Request/0100 message and was unable to complete currency conversion processing.
54	Additional Amounts	C	X	C	The Dual Message Authorization System will forward the occurrence of each DE 54 amount type as provided by the issuer in the issuer's currency and will provide an additional occurrence of each DE 54 amount type in the acquirer's currency.
55	Integrated Circuit Card (ICC) System-Related Data	C	•	C	Conditional data for chip-based transactions.
56	Payment Account Data	C	X	C	When Mastercard is the BIN Controller (as defined by EMVCo), the issuer is not required to provide DE 56 in the response message. If a PAR value is associated with the PAN, Mastercard will insert DE 56 containing the PAR value in the response message before forwarding the message to the acquirer. When the issuer is the BIN Controller and has associated a PAR value with the PAN, the issuer must include DE 56 containing the PAR value in the response message.

DE ID	Data Element Name	Org	Sys	Dst	Comments
62	Intermediate Network Facility (INF) Data	CE	•	CE	Must be the same value as in the original Authorization Request/0100 message, if present.
63	Network Data	ME	•	M	Must be the same value as in the original Authorization Request/0100 message.
102	Account ID-1	C	•	C	May contain cardholder "from" account number.
103	Account ID-2	C	•	C	May contain cardholder "to" account number.
105	Multi-Use Transaction Identification Data	CE	X	C	
106	Fleet Service Data	C	•	C	
108	Additional Transaction Reference Data	C	•	C	Mandatory for originating institution to submit DE 108 on all Mastercard Send transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions, and Mastercard™ Merchant Presented QR Payment and Funding Transactions). Optional on Mastercard Merchant Presented QR Refund Payment Transactions.
112	Additional Data (National Use)	C	•	C	Contains applicable subelement data.
120	Record Data	CE	•	CE	Must be the same value as in the original Authorization Request/0100 message, if present.
121	Authorizing Agent ID Code	C	•	C	Contains the Mastercard customer ID of an alternate authorizer, if authorized by other than the issuer or issuer's primary authorizer.

DE ID	Data Element Name	Org	Sys	Dst	Comments
123	Receipt Free Text	O	X	C	Optional for issuers. Present for acquirers if provided by the issuer. If not supported by the acquirer, the Dual Message Authorization System removes DE 123 from the message. NOTE: Applicable only to Swedish Domestic Authorization Switching Service (SASS) or Peru domestic POS transactions.
124	Member-defined Data	C	•	C	May contain issuer-defined data.
127	Private Data	O	X	CE	Private data for message initiator's use. The Dual Message Authorization System will echo DE 127 from the original Authorization Request/0100 message, if present.

Authorization Advice/0120: acquirer-generated

This is the message layout for the Authorization Advice/0120: acquirer-generated message.

Table 16: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	•	M	Constant: 0120 (Authorization Advice).
-	Bit Map, Primary	M	P	M	
1	Bit Map, Secondary	C	P	C	Mandatory if DE 65 (Bit Map, Extended) through DE 128 (Message Authentication Code [MAC]) are present in the message.
2	Primary Account Number (PAN)	M	X	M	Must be the same value as in the original Authorization Request/0100 message.
3	Processing Code	M	•	M	Must be the same value as in the original Authorization Request/0100 message.

DE ID	Data Element Name	Org	Sys	Dst	Comments
4	Amount, Transaction	M	•	M	Must be the same value as in the original Authorization Request/0100 message unless the message is submitted to complete a preauthorized transaction.
5	Amount, Settlement	•	X	C	The Dual Message Authorization System provides this data element if the issuer chooses to receive settlement amount-related data elements.
6	Amount, Cardholder Billing	•	X	M	The Dual Message Authorization System provides this data element.
7	Transmission Date and Time	M	•	M	Must be the same value as in the original Authorization Request/0100 message.
9	Conversion Rate, Settlement	•	X	C	The Dual Message Authorization System provides this data element if the issuer chooses to receive settlement amount-related data elements.
10	Conversion Rate, Cardholder Billing	•	X	M	The Dual Message Authorization System provides this data element.
11	Systems Trace Audit Number (STAN)	M	•	M	Must be the same value as in the original Authorization Request/0100 message.
12	Time, Local Transaction	C	•	C	Must be the same value as in the original Authorization Request/0100 message, if present.
13	Date, Local Transaction	C	•	C	Must be the same value as in the original Authorization Request/0100 message, if present.
14	Date, Expiration	C	•	C	Must be the same value as in the original Authorization Request/0100 message, if present.
15	Date, Settlement	•	X	M	The Dual Message Authorization System provides this data element.
16	Date, Conversion	•	X	M	The Dual Message Authorization System provides this data element.
18	Merchant Type	M	•	M	Must be the same value as in the original Authorization Request/0100 message.
19	Acquiring Institution Country Code	•	X	C	Based on Mastercard franchise data information.

DE ID	Data Element Name	Org	Sys	Dst	Comments
20	Primary Account Number (PAN) Country Code	C	•	C	Must be the same value as in the original Authorization Request/0100 message, if present.
22	Point-of-Service (POS) Entry Mode	M	•	M	Must be the same value as in the original Authorization Request/0100 message.
23	Card Sequence Number	C	•	C	Must be the same value as in the original Authorization Request/0100 message, if present.
28	Amount, Transaction Fee	C	•	C	Contains an online transaction fee as permitted by the operating rules of a bank card product.
32	Acquiring Institution ID Code	M	•	M	Must be the same value as in the original Authorization Request/0100 message.
33	Forwarding Institution ID Code	C	•	C	Must be the same value as in the original Authorization Request/0100 message, if present.
35	Track 2 Data	O	•	O	Must be the same value as in the original Authorization Request/0100 message, if present.
37	Retrieval Reference Number	C	•	C	Must be the same value as in the original Authorization Request/0100 message, if present.
38	Authorization ID Response	C	•	C	Must be either the same value as in the original Authorization Request Response/0110 message, if present, or a value provided by the acquirer if the acquirer or merchant processed and approved the transaction. Must be the same value as in the original Authorization Request Response/0110 message provided by the acquirer, if present.
39	Response Code	M	•	M	Must be either the same value as in the original Authorization Request Response/0110 message or a value provided by the acquirer if the acquirer or merchant processed and approved the transaction.
41	Acceptor Terminal ID	C	•	C	Must be the same value as in the original Authorization Request/0100 message, if present.

DE ID	Data Element Name	Org	Sys	Dst	Comments
42	Acceptor ID	C	•	C	Must be the same value as in the original Authorization Request/0100 message, if present.
43	Acceptor Name and Location	C	•	C	Must be the same value as in the original Authorization Request/0100 message, if present.
44	Additional Response Data	C	•	C	Must be the same value as in the original Authorization Request Response/0110 message, if present.
45	Track 1 Data	O	•	O	Must be the same value as in the original Authorization Request/0100 message, if present.
48	Additional Data: Private Use	M	X	M	Contains applicable subelement data.
49	Currency Code, Transaction	M	•	M	Must be the same value as in the original Authorization Request/0100 message.
50	Currency Code, Settlement	•	X	C	The Dual Message Authorization System provides this data element if the issuer chooses to receive settlement amount-related data elements.
51	Currency Code, Cardholder Billing	•	X	M	The Dual Message Authorization System provides this data element.
54	Additional Amounts	C	X	C	Must be the same value as in the original Authorization Request/0100 message, if present. The Dual Message Authorization System will forward the occurrence of each DE 54 amount type provided by the acquirer in the acquirer's currency and will provide an additional occurrence of each DE 54 amount type in the issuer's currency.
55	Integrated Circuit Card (ICC) System-Related Data	C	•	C	Must be the same value as in the original Authorization Request/0100 message, if present.
56	Payment Account Data	•	X	C	When Mastercard is the BIN Controller, the PAR value is inserted when available.

DE ID	Data Element Name	Org	Sys	Dst	Comments
60	Advice Reason Code	M	•	M	DE 60, subfield 1 (Advice Reason Code) must contain the value 190 (acquirer approved), value 191 (Acquirer Processing System Completed Authorization Transaction), or value 192 (M/Chip Offline Advice to Issuer).
61	Point-of-Service (POS) Data	M	•	M	Must be the same value as in the original Authorization Request/0100 message.
62	Intermediate Network Facility (INF) Data	O	•	O	Must be the same value as in the original Authorization Request/0100 message, if present.
63	Network Data	•	X	M	The Mastercard Network provides this data element.
102	Account ID-1	C	•	C	Must be the same value as in the original Authorization Request Response/0110 message, if present.
103	Account ID-2	C	•	C	Must be the same value as in the original Authorization Request Response/0110 message, if present.
105	Multi-Use Transaction Identification Data	C	X	C	
106	Fleet Service Data	O	•	C	
112	Additional Data (National Use)	C	•	C	Contains applicable subelement data.
121	Authorizing Agent ID Code	C	•	C	Must be the Mastercard assigned customer ID of the authorizing entity, if present. Must be the same value as in the original Authorization Request Response/0110 message, if present.
122	Additional Record Data	C	•	C	
124	Member-defined Data	C	•	C	May contain acquirer-defined data.
127	Private Data	O	X	•	Private data for message initiator's use.

Authorization Advice/0120: Issuer-Generated

Following is the list of the data elements applicable to this message.

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	M	•	Constant—0120 (Authorization Advice).
-	Bit Map, Primary	M	M	•	
1	Bit Map, Secondary	C	C	•	Mandatory if DE 65–DE 128 are present in the message.
2	Primary Account Number (PAN)	M	M	•	Must be the same value as in the original Authorization Request/0100 message.
3	Processing Code	M	M	•	Must be the same value as in the original Authorization Request/0100 message.
4	Amount, Transaction	M	M	•	Must be the same value as in the original Authorization Request/0100 message or the partial approval amount or purchase only approval amount.
5	Amount, Settlement	M	M	•	Must be the same value as in the original Authorization Request/0100 message or the partial approval amount or purchase only approval amount, if present.
6	Amount, Cardholder Billing	M	M	•	Must be the same value as in the original Authorization Request/0100 message or the partial approval amount or purchase only approval amount.
7	Transmission Date and Time	M	M	•	Must be the same value as in the original Authorization Request/0100 message.
9	Conversion Rate, Settlement	M	M	•	Must be the same value as in the original Authorization Request/0100 message.
10	Conversion Rate, Cardholder Billing	M	M	•	Must be the same value as in the original Authorization Request/0100 message.
11	Systems Trace Audit Number (STAN)	M	M	•	Must be the same value as in the original Authorization Request/0100 message.

DE ID	Data Element Name	Org	Sys	Dst	Comments
12	Time, Local Transaction	C	C	•	Must be the same value as in the original Authorization Request/0100 message, if present.
13	Date, Local Transaction	C	C	•	Must be the same value as in the original Authorization Request/0100 message, if present.
14	Date, Expiration	C	C	•	Must be the same value as in the original Authorization Request/0100 message, if present.
15	Date, Settlement	M	M	•	Must be the same value as in the original Authorization Request/0100 message, if present.
16	Date, Conversion	M	M	•	Must be the same value as in the original Authorization Request/0100 message.
18	Merchant Type	M	M	•	Must be the same value as in the original Authorization Request/0100.
20	Primary Account Number (PAN) Country Code	C	C	•	Must be the same value as in the original Authorization Request/0100, if present.
22	Point-of-Service (POS) Entry Mode	M	M	•	Must be the same value as in the original Authorization Request/0100.
23	Card Sequence Number	C	C	•	Must be the same value as in the original Authorization Request/0100, if present.
26	Point-of-Service (POS) Personal ID Number (PIN) Capture Code	C	C	•	Must be the same value as in the original Authorization Request/0100, if present.
32	Acquiring Institution ID Code	M	M	•	Must be the same value as in the original Authorization Request/0100.
33	Forwarding Institution ID Code	C	C	•	Must be the same value as in the original Authorization Request/0100, if present.
35	Track 2 Data	C	C	•	Must be the same value as in the original Authorization Request/0100, if present.
37	Retrieval Reference Number	C	C	•	Must be the same value as in the original Authorization Request/0100, if present.

DE ID	Data Element Name	Org	Sys	Dst	Comments
38	Authorization ID Response	C	C	•	Must be the same value as in the original Authorization Request Response/0110, if present.
39	Response Code	M	M	•	Must be the same value as in the original Authorization Request Response/0110.
41	Acceptor Terminal ID	C	C	•	Must be the same value as in the original Authorization Request/0100, if present.
42	Acceptor ID	C	C	•	Must be the same value as in the original Authorization Request/0100, if present.
43	Acceptor Name and Location	C	C	•	Must be the same value as in the original Authorization Request/0100, if present.
44	Additional Response Data	C	C	•	Must be the same value as in the original Authorization Request Response/0110, if present.
45	Track 1 Data	C	C	•	Must be the same value as in the original Authorization Request/0100, if present.
48	Additional Data—Private Use	M	M	•	Contains applicable subelement data.
49	Currency Code, Transaction	M	M	•	Must be the same value as in the original Authorization Request/0100.
50	Currency Code, Settlement	M	M	•	Must be the same value as in the original Authorization Request/0100.
51	Currency Code, Cardholder Billing	M	M	•	Must be the same value as in the original Authorization Request/0100.
54	Additional Amounts	C	C	•	Must be the same value as in the original Authorization Request/0100, if present.
55	Integrated Circuit Card (ICC) System-Related Data	C	C	•	Must be the same value as in the original Authorization Request/0100, if present.
60	Advice Reason Code	M	M	•	Specifies reason for this Authorization Advice/0120. 6500030 = for customer-generated transactions

DE ID	Data Element Name	Org	Sys	Dst	Comments
61	Point-of-Service (POS) Data	M	M	•	Must be the same value as in the original Authorization Request/0100.
62	Intermediate Network Facility (INF) Data	C	C	•	Must be the same value as in the original Authorization Request/0100, if present.
63	Network Data	M	M	•	Must be the same value as in the original Authorization Request/0100.
102	Account ID-1	C	C	•	Must be the same value as in the original Authorization Request/0100, if present.
103	Account ID-2	C	C	•	Must be the same value as in the original Authorization Request/0100, if present.
112	Additional Data (National Use)	C	C	•	Contains applicable subelement data.
121	Authorizing Agent ID Code	C	C	•	Must be the issuer customer ID if the issuer authorized the request; otherwise, must be the same value as in the Authorization Advice/0120—System-generated message for Mastercard processed authorizations.

Authorization Advice/0120: system-generated

This is the message layout for the Authorization Advice/0120: system-generated message.

Table 17: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	•	M	M	Constant: 0120 (Authorization Advice).
-	Bit Map, Primary	•	X	M	
1	Bit Map, Secondary	•	X	C	Mandatory if DE 65 (Bit Map, Extended) through DE 128 (Message Authentication Code [MAC]) are present in the message.
2	Primary Account Number (PAN)	•	M	M	Must be the same value as in the original Authorization Request/0100 message.

DE ID	Data Element Name	Org	Sys	Dst	Comments
3	Processing Code	•	M	M	Must be the same value as in the original Authorization Request/0100 message.
4	Amount, Transaction	•	X	M	Must be the same value as in the original Authorization Request/0100 message.
5	Amount, Settlement	•	X	C	The Dual Message Authorization System provides this data element if the issuer chooses to receive settlement amount-related data elements.
6	Amount, Cardholder Billing	•	M	M	The Dual Message Authorization System provides this data element.
7	Transmission Date and Time	•	X	M	Must be the same value as in the original Authorization Request/0100 message.
9	Conversion Rate, Settlement	•	X	C	The Dual Message Authorization System provides this data element if the issuer chooses to receive settlement amount-related data elements.
10	Conversion Rate, Cardholder Billing	•	X	M	The Dual Message Authorization System provides this data element.
11	Systems Trace Audit Number (STAN)	•	X	M	Must be the same value as in the original Authorization Request/0100 message.
12	Time, Local Transaction	•	C	C	Must be the same value as in the original Authorization Request/0100 message, if present.
13	Date, Local Transaction	•	X	C	Must be the same value as in the original Authorization Request/0100 message, if present.
14	Date, Expiration	•	C	C	Must be the same value as in the original Authorization Request/0100 message, if present.
15	Date, Settlement	•	X	M	The Dual Message Authorization System provides this data element.
16	Date, Conversion	•	X	M	The Dual Message Authorization System provides this data element.
18	Merchant Type	•	X	M	Must be the same value as in the original Authorization Request/0100 message.

DE ID	Data Element Name	Org	Sys	Dst	Comments
19	Acquiring Institution Country Code	•	X	C	
20	Primary Account Number (PAN) Country Code	•	X	C	Must be the same value as in the original Authorization Request/0100 message, if present.
22	Point-of-Service (POS) Entry Mode	•	X	M	Must be the same value as in the original Authorization Request/0100 message.
23	Card Sequence Number	•	X	C	Must be the same value as in the original Authorization Request/0100 message, if present.
26	Point-of-Service (POS) Personal ID Number (PIN) Capture Code	•	C	C	Must be the same value as in the original Authorization Request/0100 message, if present.
28	Amount, Transaction Fee	•	X	C	Must contain the same value as the original Authorization Request/0100 message, if present.
32	Acquiring Institution ID Code	•	X	M	Must be the same value as in the original Authorization Request/0100 message.
33	Forwarding Institution ID Code	•	X	C	Must be the same value as in the original Authorization Request/0100 message, if present.
35	Track 2 Data	•	X	C	Must be the same value as in the original Authorization Request/0100 message, if present.
37	Retrieval Reference Number	•	X	C	Must be the same value as in the original Authorization Request/0100 message, if present.
38	Authorization ID Response	•	X	C	Must be the same value as in the original Authorization Request Response/0110 message, if present.
39	Response Code	•	X	M	Must be the same value as in the original Authorization Request Response/0110 message.
41	Acceptor Terminal ID	•	X	C	Must be the same value as in the original Authorization Request/0100 message, if present.
42	Acceptor ID	•	X	C	Must be the same value as in the original Authorization Request/0100 message, if present.

DE ID	Data Element Name	Org	Sys	Dst	Comments
43	Acceptor Name and Location	•	X	C	Must be the same value as in the original Authorization Request/0100 message, if present.
44	Additional Response Data	•	X	C	Must be the same value as in the original Authorization Request Response/0110 message, if present.
45	Track 1 Data	•	X	C	Must be the same value as in the original Authorization Request/0100 message, if present.
48	Additional Data: Private Use	•	X	M	Contains applicable subelement data.
49	Currency Code, Transaction	•	X	M	Must be the same value as in the original Authorization Request/0100 message.
50	Currency Code, Settlement	•	X	C	The Dual Message Authorization System provides this data element if the issuer chooses to receive settlement amount-related data elements.
51	Currency Code, Cardholder Billing	•	X	M	The Dual Message Authorization System provides this data element.
54	Additional Amounts	•	X	C	Must be the same value as in the original Authorization Request/0100 message, if present.
55	Integrated Circuit Card (ICC) System-Related Data	•	X	C	Must be the same value as in the original Authorization Request/0100 message, if present.
56	Payment Account Data	•	X	C	When Mastercard is the BIN Controller, the PAR value is inserted when available.
60	Advice Reason Code	•	X	M	Specifies reason for this Authorization Advice/0120 message.
61	Point-of-Service (POS) Data	•	X	M	Must be the same value as in the original Authorization Request/0100 message.
62	Intermediate Network Facility (INF) Data	•	X	C	Must be the same value as in the original Authorization Request/0100 message, if present.
63	Network Data	•	X	M	Must be the same value as in the original Authorization Request/0100 message. The Mastercard Network provides this data element, if required.

Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)

DE ID	Data Element Name	Org	Sys	Dst	Comments
102	Account ID-1	•	C	C	Must be the same value as in the original Authorization Request Response/0110 message, if present.
103	Account ID-2	•	X	C	Must be the same value as in the original Authorization Request Response/0110 message, if present.
105	Multi-Use Transaction Identification Data	•	X	C	
108	Additional Transaction Reference Data	•	C	C	Must be the same value as in the original Authorization Request/0100 message, if present. Optional for Mastercard Merchant Presented QR Transactions.
112	Additional Data (National Use)	•	C	C	Contains applicable subelement data.
121	Authorizing Agent ID Code	•	C	C	Must be the same value as in the original Authorization Request Response/0110 message, if present.
122	Additional Record Data	•	X	C	
124	Member-defined Data	•	X	C	Must be the same value as in the original Authorization Request/0100 message, if present.

Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)

This is the message layout for the Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120) message.

Table 18: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	M	•	Constant: 0130 (Authorization Advice Response).
-	Bit Map, Primary	M	X	•	
1	Bit Map, Secondary	C	X	•	Must be the same value as in the Authorization Advice/0120 message, if present.

DE ID	Data Element Name	Org	Sys	Dst	Comments
2	Primary Account Number (PAN)	ME	ME	•	Must be the same value as in the Authorization Advice/0120 message.
3	Processing Code	ME	ME	•	Must be the same value as in the Authorization Advice/0120 message.
4	Amount, Transaction	ME	XE	•	Must be the same value as in the Authorization Advice/0120 message.
5	Amount, Settlement	CE	XE	•	Must be the same value as in the Authorization Advice/0120 message, if present.
6	Amount, Cardholder Billing	ME	XE	•	Must be the same value as in the Authorization Advice/0120 message.
7	Transmission Date and Time	ME	XE	•	Must be the same value as in the Authorization Advice/0120 message.
9	Conversion Rate, Settlement	CE	XE	•	Must be the same value as in the Authorization Advice/0120 message, if present.
10	Conversion Rate, Cardholder Billing	ME	XE	•	Must be the same value as in the Authorization Advice/0120 message.
11	Systems Trace Audit Number (STAN)	ME	XE	•	Must be the same value as in the Authorization Advice/0120 message.
15	Date, Settlement	ME	XE	•	Must be the same value as in the Authorization Advice/0120 message.
16	Date, Conversion	ME	XE	•	Must be the same value as in the Authorization Advice/0120 message.
20	Primary Account Number (PAN) Country Code	CE	XE	•	Must be the same value as in the Authorization Advice/0120 message, if present.
23	Card Sequence Number	CE	XE	•	Must be the same value as in the Authorization Advice/0120 message, if present.
28	Amount, Transaction Fee	CE	CE	•	Must be the same value as in the Authorization Advice/0120 message, if present.
32	Acquiring Institution ID Code	ME	XE	•	Must be the same value as in the Authorization Advice/0120 message.
33	Forwarding Institution ID Code	CE	XE	•	Must be the same value as in the Authorization Advice/0120 message, if present.

DE ID	Data Element Name	Org	Sys	Dst	Comments
37	Retrieval Reference Number	CE	XE	•	Must be the same value as in the Authorization Advice/0120 message, if present.
39	Response Code	M	X	•	For more detail, refer to the data element definition for DE 39.
41	Acceptor Terminal ID	CE	XE	•	Must be the same value as in the Authorization Advice/0120 message, if present.
44	Additional Response Data	CE	CE	•	May contain additional response information for certain error conditions in original Authorization Advice/0120 message.
49	Currency Code, Transaction	ME	XE	•	Must be the same value as in the Authorization Advice/0120 message.
50	Currency Code, Settlement	CE	XE	•	Must be the same value as in the Authorization Advice/0120 message, if present.
51	Currency Code, Cardholder Billing	ME	XE	•	Must be the same value as in the Authorization Advice/0120 message.
56	Payment Account Data	C	X	•	When Mastercard is the BIN Controller, issuers are not required to send PAR. Mastercard will insert the PAR in the message before sending to the acquirer, when available. When Mastercard is not the BIN Controller, the issuer may provide the PAR in the response message when available. Mastercard will pass this value to the acquirer.
62	Intermediate Network Facility (INF) Data	CE	XE	•	Must be the same value as in the Authorization Advice/0120 message, if present.
63	Network Data	ME	XE	•	Must be the same value as in the Authorization Advice/0120 message.

Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120 response or an Authorization Advice Request/0120)

This is the message layout for the Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120 response or an Authorization Advice Request/0120) message.

Table 19: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	•	M	Constant: 0130 (Authorization Advice Response).
-	Bit Map, Primary	M	P	M	
1	Bit Map, Secondary	C	P	C	Must be the same value as in the Authorization Advice/0120: acquirer-generated, if present.
2	Primary Account Number (PAN)	ME	X	ME	Must be the same value as in the Authorization Advice/0120: acquirer-generated.
3	Processing Code	ME	•	ME	Must be the same value as in the Authorization Advice/0120: acquirer-generated.
4	Amount, Transaction	ME	•	ME	Must be the same value as in the Authorization Advice/0120: acquirer-generated.
5	Amount, Settlement	CE	•	C	Must be the same value as in the Authorization Advice/0120: acquirer-generated, if present.
6	Amount, Cardholder Billing	CE	•	C	Must be the same value as in the Authorization Advice/0120: acquirer-generated.
7	Transmission Date and Time	ME	•	ME	Must be the same value as in the Authorization Advice/0120: acquirer-generated.
9	Conversion Rate, Settlement	CE	•	C	Must be the same value as in the Authorization Advice/0120: acquirer-generated, if present.
10	Conversion Rate, Cardholder Billing	CE	•	C	Must be the same value as in the Authorization Advice/0120: acquirer-generated.

Message layouts

Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120 response or an Authorization Advice Request/0120)

DE ID	Data Element Name	Org	Sys	Dst	Comments
11	Systems Trace Audit Number (STAN)	ME	•	ME	Must be the same value as in the Authorization Advice/0120: acquirer-generated.
15	Date, Settlement	ME	•	M	Must be the same value as in the Authorization Advice/0120: acquirer-generated.
16	Date, Conversion	CE	•	C	Must be the same value as in the Authorization Advice/0120: acquirer-generated, if present.
20	Primary Account Number (PAN) Country Code	CE	•	CE	Must be the same value as in the Authorization Advice/0120: acquirer-generated, if present.
23	Card Sequence Number	CE	•	CE	Must be the same value as in the Authorization Advice/0120: acquirer-generated, if present.
28	Amount, Transaction Fee	CE	XE	•	Must be the same value as in the Authorization Advice/0120 message, if present.
32	Acquiring Institution ID Code	ME	•	ME	Must be the same value as in the Authorization Advice/0120: acquirer-generated.
33	Forwarding Institution ID Code	CE	•	CE	Must be the same value as in the Authorization Advice/0120: Acquirer-generated, if present.
37	Retrieval Reference Number	CE	•	CE	Must be the same value as in the Authorization Advice/0120: acquirer-generated, if present.
39	Response Code	M	•	M	Response Code for the Authorization Advice Response/0130 message. For values, refer to DE 39.
41	Acceptor Terminal ID	CE	•	CE	Must be the same value as in the Authorization Advice/0120 acquirer-generated, if present.
44	Additional Response Data	CE	•	CE	May contain additional response information for certain error conditions in the original Authorization Advice/0120: acquirer-generated, if present.
48	Additional Data: Private Use	ME	•	ME	Must be the same value as in the Authorization Advice/0120: acquirer-generated.

Message layouts

Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120 response or an Authorization Advice Request/0120)

DE ID	Data Element Name	Org	Sys	Dst	Comments
49	Currency Code, Transaction	ME	•	ME	Must be the same value as in the Authorization Advice/0120: acquirer-generated.
50	Currency Code, Settlement	CE	•	CE	Must be the same value as in the Authorization Advice/0120: acquirer-generated, if present.
51	Currency Code, Cardholder Billing	CE	•	C	Must be the same value as in the Authorization Advice/0120: acquirer-generated.
56	Payment Account Data	C	X	C	When Mastercard is the BIN Controller, issuers are not required to send PAR. Mastercard will insert the PAR in the message before sending to the acquirer, when available. When Mastercard is not the BIN Controller, the issuer may provide the PAR in the response message when available. Mastercard will pass this value to the acquirer.
62	Intermediate Network Facility (INF) Data	CE	•	CE	Must be the same value as in the Authorization Advice/0120: acquirer-generated, if present.
63	Network Data	ME	•	M	Must be the same value as in the Authorization Advice/0120: acquirer-generated.
105	Multi-Use Transaction Identification Data	CE	X	C	
106	Fleet Service Data	C	•	C	
127	Private Data	CE	X	CE	Private data for message originator's use. The Dual Message Authorization System will echo in DE 127 from the original Authorization Advice/0120: acquirer-generated, if present.

Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)

Following is the list of the data elements applicable to this message.

Table 20: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	X	•	Constant: 0130 (Authorization Advice Response).
-	Bit Map, Primary	M	X	•	
1	Bit Map, Secondary	C	X	C	Must be the same value as in the Authorization Advice/0120 message, if present.
2	Primary Account Number (PAN)	ME	ME	•	Must be the same value as in the Authorization Advice/0120 message.
3	Processing Code	ME	ME	•	Must be the same value as in the Authorization Advice/0120 message.
4	Amount, Transaction	ME	ME	•	Must be the same value as in the Authorization Advice/0120 message.
5	Amount, Settlement	CE	XE	•	Must be the same value as in the Authorization Advice/0120 message, if present.
6	Amount, Cardholder Billing	ME	•	ME	Must be the same value as in the Authorization Advice/0120 message.
7	Transmission Date and Time	ME	XE	•	Must be the same value as in the Authorization Advice/0120 message.
9	Conversion Rate, Settlement	CE	XE	•	Must be the same value as in the Authorization Advice/0120 message, if present.
10	Conversion Rate, Cardholder Billing	ME	•	ME	Must be the same value as in the Authorization Advice/0120 message.
11	Systems Trace Audit Number (STAN)	ME	XE	•	Must be the same value as in the Authorization Advice/0120 message.
15	Date, Settlement	•	X	M	Must be the same value as in the Authorization Advice/0120 message.
16	Date, Conversion	ME	XE	•	Must be the same value as in the Authorization Advice/0120 message.

Message layouts

Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)

DE ID	Data Element Name	Org	Sys	Dst	Comments
20	Primary Account Number (PAN) Country Code	CE	XE	•	Must be the same value as in the Authorization Advice/0120 message, if present.
23	Card Sequence Number	CE	XE	•	Must be the same value as in the Authorization Advice/0120 message, if present.
28	Amount, Transaction Fee	CE	XE	•	Must be the same value as in the Authorization Advice/0120 message, if present.
32	Acquiring Institution ID Code	ME	XE	•	Must be the same value as in the Authorization Advice/0120 message.
33	Forwarding Institution ID Code	CE	XE	•	Must be the same value as in the Authorization Advice/0120 message, if present.
37	Retrieval Reference Number	CE	XE	•	Must be the same value as in the Authorization Advice/0120 message, if present.
39	Response Code	M	X	•	For more detail, refer to the data element definition for DE 39.
41	Acceptor Terminal ID	CE	XE	•	Must be the same value as in the Authorization Advice/0120 message, if present.
44	Additional Response Data	C	X	•	May contain additional response information for certain error conditions in original Authorization Advice/0120 message.
48	Additional Data: Private Use	ME	XE	•	Must be the same value as in the Authorization Advice/0120 message.
49	Currency Code, Transaction	•	XE	ME	Must be the same value as in the Authorization Advice/0120 message.
50	Currency Code, Settlement	CE	XE	•	Must be the same value as in the Authorization Advice/0120 message, if present.
51	Currency Code, Cardholder Billing	ME	XE	•	Must be the same value as in the Authorization Advice/0120 message.
56	Payment Account Data	C	X	•	When Mastercard is the BIN Controller, the PAR value is inserted when available.

DE ID	Data Element Name	Org	Sys	Dst	Comments
62	Intermediate Network Facility (INF) Data	CE	XE	•	Must be the same value as in the Authorization Advice/0120 message, if present.
63	Network Data	ME	XE	•	Must be the same value as in the Authorization Advice/0120 message.
105	Multi-Use Transaction Identification Data	•	X	C	
127	Private Data	•	X	CE	Must be the same value as in the Authorization Advice/0120 message.

Authorization Advice Response/0130: System-Generated (Responding to an Issuer-Generated 0120)

Following is the list of the data elements applicable to this message.

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	•	M	M	Constant—0130 (Authorization Advice Response).
-	Bit Map, Primary	•	M	M	
1	Bit Map, Secondary	•	C	C	Must be the same value as in the Authorization Advice/0120—Issuer-generated, if present.
2	Primary Account Number (PAN)	•	ME	ME	Must be the same value as in the Authorization Advice/0120—Issuer-generated.
3	Processing Code	•	ME	ME	Must be the same value as in the Authorization Advice/0120—Issuer-generated.
4	Amount, Transaction	•	ME	ME	Must be the same value as in the Authorization Advice/0120—Issuer-generated.
5	Amount, Settlement	•	ME	ME	Must be the same value as in the Authorization Advice/0120—Issuer-generated.
6	Amount, Cardholder Billing	•	ME	ME	Must be the same value as in the Authorization Advice/0120—Issuer-generated.

DE ID	Data Element Name	Org	Sys	Dst	Comments
7	Transmission Date and Time	•	ME	ME	Must be the same value as in the Authorization Advice/0120—Issuer-generated.
9	Conversion Rate, Settlement	•	ME	ME	Must be the same value as in the Authorization Advice/0120—Issuer-generated.
10	Conversion Rate, Cardholder Billing	•	ME	ME	Must be the same value as in the Authorization Advice/0120—Issuer-generated.
11	Systems Trace Audit Number (STAN)	•	ME	ME	Must be the same value as in the Authorization Advice/0120—Issuer-generated.
15	Date, Settlement	•	ME	ME	Must be the same value as in the Authorization Advice/0120—Issuer-generated.
16	Date, Conversion	•	ME	ME	Must be the same value as in the Authorization Advice/0120—Issuer-generated.
20	Primary Account Number (PAN) Country Code	•	CE	CE	Must be the same value as in the Authorization Advice/0120—Issuer-generated, if present.
23	Card Sequence Number	•	CE	CE	Must be the same value as in the Authorization Advice/0120—Issuer-generated, if present.
32	Acquiring Institution ID Code	•	ME	ME	Must be the same value as in the Authorization Advice/0120—Issuer-generated.
33	Forwarding Institution ID Code	•	CE	CE	Must be the same value as in the Authorization Advice/0120—Issuer-generated, if present.
37	Retrieval Reference Number	•	CE	CE	Must be the same value as in the Authorization Advice/0120—Issuer-generated, if present.
39	Response Code	•	M	M	Response Code for this Authorization Advice Response/0130.
41	Acceptor Terminal ID	•	CE	CE	Must be the same value as in the Authorization Advice/0120—Issuer-generated, if present.

DE ID	Data Element Name	Org	Sys	Dst	Comments
44	Additional Response Data	•	X	C	May contain additional response information for certain error conditions in original Authorization Advice/0120—Issuer-generated.
49	Currency Code, Transaction	•	ME	ME	Must be the same value as in the Authorization Advice/0120—Issuer-generated.
50	Currency Code, Settlement	•	ME	ME	Must be the same value as in the Authorization Advice/0120—Issuer-generated.
51	Currency Code, Cardholder Billing	•	ME	ME	Must be the same value as in the Authorization Advice/0120—Issuer-generated.
62	Intermediate Network Facility (INF) Data	•	CE	CE	Must be the same value as in the Authorization Advice/0120—Issuer-generated, if present.
63	Network Data	•	X	M	The Authorization Platform provides this data element, if required.
105	Multi-Use Transaction Identification Data	•	X	C	

Authorization Response Acknowledgment/0180

Following is the list of the data elements applicable to this message.

Table 21: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	M	•	Constant: 0180 (Authorization Response Acknowledgment).
-	Bit Map, Primary	M	X	•	
1	Bit Map, Secondary	C	X	•	Mandatory if DE 65 (Bit Map, Extended) through DE 128 (Message Authentication Code [MAC]) are present in the message.
7	Transmission Date and Time	ME	XE	•	Must be the same value as in the original Authorization Request Response/0110 message.

DE ID	Data Element Name	Org	Sys	Dst	Comments
11	Systems Trace Audit Number (STAN)	ME	XE	•	Must be the same value as in the original Authorization Request Response/0110 message.
39	Response Code	M	X	•	Must be 00 to indicate positive Response Acknowledgment.
63	Network Data	ME	XE	•	Must be the same value as in the original Authorization Request Response/0110 message.
127	Private Data	CE	•	•	Private data for message initiator's use.

Authorization Response Negative Acknowledgment/0190

Following is the list of the data elements applicable to this message.

Table 22: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	•	M	M	Constant: 0190 (Authorization Response Negative Acknowledgment).
-	Bit Map, Primary	•	X	M	
1	Bit Map, Secondary	•	X	C	Required data element if DE 65 (Bit Map, Extended) through DE 128 are present in the message.
7	Transmission Date and Time	•	XE	ME	Must be the same value as in the original Authorization Request Response/0110 message or Reversal Request Response/0410 message.
11	Systems Trace Audit Number (STAN)	•	XE	ME	Must be the same value as in the original Authorization Request Response/0110 message or Reversal Request Response/0410 message.
39	Response Code	•	X	M	Indicates specific reason for this Negative Response Acknowledgment.
44	Additional Response Data	•	X	C	For a listing of values, refer to Data element definitions .

DE ID	Data Element Name	Org	Sys	Dst	Comments
63	Network Data	•	XE	ME	Must be the same value as in the original Authorization Request Response/0110 message or Reversal Request Response/0410 message.
127	Private Data	•	XE	CE	Must be the same value as in the original Authorization Request Response/0110 message or Reversal Request Response/0410 message.

Issuer File Update Request/0302

Following is the list of the data elements applicable to this message.

Table 23: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	M	•	Constant: 0302 (Issuer File Update Request).
-	Bit Map, Primary	M	X	•	Mandatory.
1	Bit Map, Secondary	M	X	•	Mandatory.
2	Primary Account Number (PAN)	C	C	•	May contain PAN information if required for this Issuer File Update Request/0302 message.
7	Transmission Date and Time	M	X	•	Transaction time stamp; UTC date and time that this transaction was entered into interchange.
11	Systems Trace Audit Number (STAN)	M	X	•	Transaction trace number; must be unique value for transaction initiator within each UTC day.
33	Forwarding Institution ID Code	M	X	•	Mastercard-assigned customer ID code. Identifies the institution submitting this file maintenance request.
56	Payment Account Data	O	C	•	DE 56 is optional when DE 101 (File Name) is MCC111 (PAN-PAR Mapping File).

DE ID	Data Element Name	Org	Sys	Dst	Comments
63	Network Data	•	X	•	Dual Message Authorization System transaction ID code; Mastercard provides this data as a unique identifier for this Issuer File Update Request/ 0302 message.
91	Issuer File Update Code	M	X	•	Issuer File Update function code.
96	Message Security Code	M	•	•	Issuer File Update password or security code used to authenticate Issuer File Update permissions. Data must be provided in EBCDIC hexadecimal format.
101	File Name	M	X	•	Name of file the issuer is maintaining or accessing.
120	Record Data	C	•	•	Contains the specific Issuer File Update detail record data.
127	Private Data	O	X	•	Private use data element, available for message initiator's optional use.

Issuer File Update Request Response/0312

Following is the list of the data elements applicable to this message.

Table 24: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	•	M	M	Constant: 0312 (Issuer File Update Request Response).
-	Bit Map, Primary	•	X	M	Mandatory.
1	Bit Map, Secondary	•	X	M	Mandatory.
2	Primary Account Number (PAN)	•	CE	CE	Must be the same value as in the original Issuer File Update Request/ 0302 message, if present.
7	Transmission Date and Time	•	XE	ME	Must be the same value as in the original Issuer File Update Request/ 0302 message.
11	Systems Trace Audit Number (STAN)	•	XE	ME	Must be the same value as in the original Issuer File Update Request/ 0302 message.

DE ID	Data Element Name	Org	Sys	Dst	Comments
33	Forwarding Institution ID Code	•	XE	ME	Must be the same value as in the original Issuer File Update Request/0302 message, if present.
39	Response Code	•	X	M	Indicates disposition of Issuer File Update Request/0302 message.
44	Additional Response Data	•	X	C	May contain additional response information, based on DE 39 (Response Code).
63	Network Data	•	X	M	Must be the same value as in the original Issuer File Update Request/0302.
91	Issuer File Update Code	•	XE	ME	Must be the same value as in the original Issuer File Update Request/0302 message.
96	Message Security Code	•	XE	M	Must be the same value as in the original Issuer File Update Request/0302 message.
101	File Name	•	ME	M	Must be the same value as in the original Issuer File Update Request/0302 message.
120	Record Data	•	X	C	This data element is used to return file inquiry and rejects records associated with reject responses. Refer to the data element definition for DE 120.
122	Additional Record Data	•	X	C	This data element is used to return additional data resulting from a file inquiry.
127	Private Data	•	XE	C	Must be the same value as in the original Issuer File Update Request/0302 message, if present.

Reversal Request/0400

Following is the list of the data elements applicable to this message.

Table 25: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	•	M	Constant: 0400 (Reversal Request).

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Bit Map, Primary	M	P	M	Mandatory.
1	Bit Map, Secondary	M	P	M	Mandatory.
2	Primary Account Number (PAN)	M	X	M	Must be the same value as in the original Authorization Request/0100 message.
3	Processing Code	M	.	M	Must be the same value as in the original Authorization Request/0100 message.
4	Amount, Transaction	M	.	M	Must be the same value as in the original Authorization Request/0100 message. On a partial approval or purchase only approval, must be the same as the Authorization Request Response/0110 message. Must be a value other than all zeros.
5	Amount, Settlement	.	X	C	The Dual Message Authorization System provides this data element if the issuer chooses to receive settlement amount-related data elements. This value may be different from the Authorization Request/0100 message if the currency conversion rate used by the Dual Message Authorization System has changed between the time the original Authorization Request/0100 message was processed.
6	Amount, Cardholder Billing	.	X	M	The Dual Message Authorization System provides this data element. This value may be different from the Authorization Request/0100 message if the currency conversion rate used by the Dual Message Authorization System has changed between the time the original Authorization Request/0100 message was processed and the time the Reversal Request/0400 message is processed by the Dual Message Authorization System.
7	Transmission Date and Time	M	.	M	Transaction time stamp; UTC date and time that this transaction was entered into the system.

DE ID	Data Element Name	Org	Sys	Dst	Comments
9	Conversion Rate, Settlement	•	X	C	The Dual Message Authorization System provides this data element if the issuer chooses to receive settlement amount-related data elements. This value may be different from the Authorization Request/0100 message if the currency conversion rate used by the Dual Message Authorization System has changed between the time the original Authorization Request/0100 message was processed and the time the Reversal Request/0400 message is processed.
10	Conversion Rate, Cardholder Billing	•	X	M	The Dual Message Authorization System provides this data element. This value may be different from the Authorization Request/0100 message if the currency conversion rate used by the Dual Message Authorization System has changed between the time the original Authorization Request/0100 message was processed and the time the Reversal Request/0400 message is processed.
11	Systems Trace Audit Number (STAN)	M	•	M	Transaction trace number; must be unique value for transaction initiator within each UTC day.
12	Time, Local Transaction	C	•	C	Must be the same value as in the original Authorization Request/0100 message, if present.
13	Date, Local Transaction	C	•	C	Must be the same value as in the original Authorization Request/0100 message, if present.
14	Date, Expiration	C	•	C	Must be the same value as in the original Authorization Request/0100 message, if present.
15	Date, Settlement	•	X	M	The Dual Message Authorization System provides this data element.
16	Date, Conversion	•	X	M	The Dual Message Authorization System provides this data element.
18	Merchant Type	M	•	M	Must be the same value as in the original Authorization Request/0100 message.

DE ID	Data Element Name	Org	Sys	Dst	Comments
19	Acquiring Institution Country Code	•	X	C	The Dual Message Authorization System provides this data element.
20	Primary Account Number (PAN) Country Code	C	•	C	Must be the same value as in the original Authorization Request/0100 message, if present.
22	Point-of-Service (POS) Entry Mode	M	•	M	Must be the same value as in the original Authorization Request/0100 message.
23	Card Sequence Number	C	•	C	Must be the same value as in the original Authorization Request/0100 message, if present.
28	Amount, Transaction Fee	C	•	C	Must be the same value as in the original Authorization Request/0100 message, if present.
32	Acquiring Institution ID Code	M	•	M	Must be the same value as in the original Authorization Request/0100 message.
33	Forwarding Institution ID Code	C	•	C	Must be the same value as in the original Authorization Request/0100 message, if present.
37	Retrieval Reference Number	C	•	C	Must be the same value as in the original Authorization Request/0100 message, if present.
38	Authorization ID Response	C	•	C	Must be the same value as in the original Authorization Request Response/0110 message if present.
39	Response Code	M	•	M	Must be the same value as in the original Authorization Request Response/0110 message or it may contain value 06 (Error [/0400 only]), 17 (Customer cancellation [/0400 only]), 32 (Partial reversal [/0400 only]), 34 (Suspect Fraud [/0400 only]), or 68 (Response received late [/0400 only]).
41	Acceptor Terminal ID	C	•	C	Must be the same value as in the original Authorization Request/0100 message, if present.
42	Acceptor ID	C	•	C	Must be the same value as in the original Authorization Request/0100 message, if present.

DE ID	Data Element Name	Org	Sys	Dst	Comments
43	Acceptor Name and Location	C	•	C	Must be the same value as in the original Authorization Request/0100 message, if present.
48	Additional Data: Private Use	M	X	M	Must contain the transaction category code (TCC) from the original Authorization Request/0100. Subelement 63 (Trace ID) must be present. Subelement 58 (ATM Additional Data) and subelement 77 (Transaction Type Identifier) may be present based on the usage condition defined for each of these subelements in the Data Element Definitions. May also contain the following DE 48 subelements for Visa issuers: • 36 (Visa Defined Data [Visa Only]) • 42 (Electronic Commerce Indicators) • 43 (Universal Cardholder Authentication Field [UCAF]) • 44 (Visa 3-D Secure Electronic Commerce Transaction Identifier [XID]: Visa Only) • 90 (Custom Payment Service Request: Visa Only) • 91 (Custom Payment Service Request Transaction ID: Visa Only) • 94 (Commercial Card Inquiry Request: Visa Only) • 96 (Visa Market-Specific Data Identifier: Visa Only) NOTE: DE 48 subelements not specifically referenced in the Reversal Request/0400 message layout but submitted by an acquirer in the Reversal Request/0400 message will not be forwarded by the Dual Message Authorization System to the issuer or returned to the acquirer.
49	Currency Code, Transaction	M	•	M	Must be the same value as in the original Authorization Request/0100 message.

DE ID	Data Element Name	Org	Sys	Dst	Comments
50	Currency Code, Settlement	•	X	C	The Dual Message Authorization System provides this data element if the issuer chooses to receive settlement amount-related data elements.
51	Currency Code, Cardholder Billing	•	X	M	The Dual Message Authorization System provides this data element.
54	Additional Amounts	C	X	C	Must be the same value as in the original Authorization Request/0100 message. This data element will only be passed to the issuer when DE 54, subfield 2 (Amount Type) contains the value 40 (Amount Cash Back). The Dual Message Authorization System will forward occurrences of this amount type in both the acquirer's currency and the issuer's currency. If the original authorization was a Purchase with Cash Back request (subfield 2, value 40) and the cash back portion of the request was not approved by the issuer (Response Code 87 [Purchase Only Approval]), DE 54 should not be included in the reversal request.
56	Payment Account Data	•	X	C	When Mastercard is the BIN Controller, the PAR value is inserted when available.
61	Point-of-Service (POS) Data	M	•	M	Must be the same value as in the original Authorization Request/0100 message.
62	Intermediate Network Facility (INF) Data	O	•	O	If present in an original Authorization Request/0100 message, may be present in subsequent Reversal Request/0400.
63	Network Data	•	X	M	The Dual Message Authorization System provides this data element.
90	Original Data Elements	M	•	M	Contains certain data elements from the original Authorization Request/0100 message.

DE ID	Data Element Name	Org	Sys	Dst	Comments
95	Replacement Amounts	C	X	C	If this data element is provided by an acquirer in a full reversal message, it must contain all zeros to indicate a full reversal. If this data element is provided by an acquirer in a partial reversal message, it must contain a value other than all zeros to indicate a partial reversal.
105	Multi-Use Transaction Identification Data	C	X	C	
108	Additional Transaction Reference Data	C	•	C	Mandatory for originating institution to submit DE 108 on all Mastercard Send Payment Transactions. Mandatory for originating institution to submit DE 108 on all Funding Transactions. Optional for Mastercard Merchant Presented QR Transactions.
112	Additional Data (National Use)	C	•	C	Contains applicable subelement data.
122	Additional Record Data	C	•	C	
124	Member-defined Data	C	•	C	Member-defined data element, available for message initiator and recipient optional use.
127	Private Data	O	X	•	Private use data element, available for message initiator's optional use.

Reversal Request Response/0410

Following is the list of the data elements applicable to this message.

Table 26: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	•	M	Constant: 0410 (Reversal Request Response)
-	Bit Map, Primary	M	P	M	Mandatory
1	Bit Map, Secondary	M	P	M	Mandatory

DE ID	Data Element Name	Org	Sys	Dst	Comments
2	Primary Account Number (PAN)	ME	X	ME	Must be the same value as in the original Reversal Request/0400 message.
3	Processing Code	ME	.	M	Must be the same value as in the original Reversal Request/0400 message.
4	Amount, Transaction	CE	XE	ME	Must be the same value as in the original Reversal Request/0400 message.
5	Amount, Settlement	CE	X	C	Must be the same value as in the original Reversal Request/0400 message, if present. The Dual Message Authorization System provides this data element if the acquirer chooses to receive settlement amount-related data elements. This data will be present, as defined, except when the Dual Message Authorization System has declined a Reversal Request/0400 message and was unable to complete currency conversion processing.
6	Amount, Cardholder Billing	CE	X	C	Must be the same value as in the original Reversal Request/0400 message. This data will be present, as defined, except when the Dual Message Authorization System has declined a Reversal Request/0400 message and was unable to complete currency conversion processing.
7	Transmission Date and Time	ME	.	ME	Must be the same value as in the original Reversal Request/0400 message.

DE ID	Data Element Name	Org	Sys	Dst	Comments
9	Conversion Rate, Settlement	CE	X	C	Must be the same value as in the original Reversal Request/0400 message, if present. The Dual Message Authorization System provides this data element if the acquirer chooses to receive settlement amount-related data elements. This data will be present, as defined, except when the Dual Message Authorization System has declined a Reversal Request/0400 message and was unable to complete currency conversion processing.
10	Conversion Rate, Cardholder Billing	CE	X	C	Must be the same value as in the original Reversal Request/0400 message. This data will be present, as defined, except when the Dual Message Authorization System has declined a Reversal Request/0400 message and was unable to complete currency conversion processing.
11	Systems Trace Audit Number	ME	.	ME	Must be the same value as in the original Reversal Request/0400 message.
15	Date, Settlement	ME	X	M	Must be the same value as in the original Reversal Request/0400 message.
16	Date, Conversion	CE	X	C	Must be the same value as in the original Reversal Request/0400 message. This data will be present, as defined, except when the Dual Message Authorization System has declined a Reversal Request/0400 message and was unable to complete currency conversion processing.
20	Primary Account Number (PAN) Country Code	CE	.	CE	Must be the same value as in the original Reversal Request/0400 message, if present.
23	Card Sequence Number	CE	.	C	

DE ID	Data Element Name	Org	Sys	Dst	Comments
28	Amount, Transaction Fee	CE	X	CE	Must contain the same value as in the original Reversal Request/0400 message, if present.
32	Acquiring Institution ID Code	ME	•	ME	Must be the same value as in the original Reversal Request/0400 message.
33	Forwarding Institution ID Code	CE	•	CE	Must be the same value as in the original Reversal Request/0400 message, if present.
37	Retrieval Reference Number	CE	•	CE	Must be the same value as in the original Reversal Request/0400 message, if present.
39	Response Code	M	•	M	Must be present in the Reversal Request Response/0410 message. Defines the disposition of a previous message or an action taken as a result of receipt of a previous message.
41	Acceptor Terminal ID	CE	•	CE	Must be the same value as in the original Reversal Request/0400 message, if present.
44	Additional Response Data	C	•	C	Contains additional response data.
48	Additional Data: Private Use	ME	•	ME	Must be the same value as in the original Reversal Request/0400 message.
49	Currency Code, Transaction	ME	•	ME	Must be the same value as in the original Reversal Request/0400 message.
50	Currency Code, Settlement	CE	X	C	Must be the same value as in the original Reversal Request/0400 message, if present. The Dual Message Authorization System provides this data element if the acquirer chooses to receive settlement amount-related data elements. This data will be present, as defined, except when the Dual Message Authorization System has declined a Reversal Request/0400 message and was unable to complete currency conversion processing.

DE ID	Data Element Name	Org	Sys	Dst	Comments
51	Currency Code, Cardholder Billing	CE	X	C	Must be the same value as in the original Reversal Request/0400 message. This data will be present, as defined, except when the Dual Message Authorization System has declined a Reversal Request/0400 message and was unable to complete currency conversion processing.
56	Payment Account Data	C	X	C	When Mastercard is the BIN Controller, issuers are not required to send PAR. Mastercard will insert the PAR in the message before sending to the acquirer, when available. When Mastercard is not the BIN Controller, the issuer may provide the PAR in the response message when available. Mastercard will pass this value to the acquirer.
62	Intermediate Network Facility (INF) Data	CE	•	CE	Must be the same value as in the original Reversal Request/0400 message, if present.
63	Network Data	ME	X	M	Must be the same value as in the original Reversal Request/0400 message.
90	Original Data Elements	ME	•	ME	Must be the same value as in the original Reversal Request/0400 message.
95	Replacement Amounts	CE	X	C	Must be the same value as in the original Reversal Request/0400 message, if present.
105	Multi-Use Transaction Identification Data	CE	X	C	
108	Additional Transaction Reference Data	C	•	C	Mandatory for originating institution to submit DE 108 on all Mastercard Send Payment Transactions, mandatory for originating institution to submit DE 108 on all Funding Transactions. If DE 108 is submitted but the acquirer is not ready to receive, the transaction may fail on the acquirer side.
112	Additional Data (National Use)	C	•	C	Contains applicable subelement data.

DE ID	Data Element Name	Org	Sys	Dst	Comments
121	Authorizing Agent ID Code	C	•	C	Contains the Mastercard member ID of an alternate Stand-In Authorization, that performed Dual Message System processing on-behalf of an issuer or issuer's primary authorizer.
124	Member-defined Data	O	•	C	May contain customer-defined data.
127	Private Data	O	X	CE	Private data for message initiator's use.

Reversal Advice/0420

Following is the list of the data elements applicable to this message.

Table 27: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	•	M	M	Constant: 0420 (Reversal Advice).
-	Bit Map, Primary	•	X	M	Mandatory.
1	Bit Map, Secondary	•	X	M	Mandatory.
2	Primary Account Number (PAN)	•	M	M	Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message.
3	Processing Code	•	M	M	Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message.
4	Amount, Transaction	•	X	M	Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message (except for a partial approval or purchase only approval). For a partial approval or purchase only approval, must be the same value as the Authorization Request Response/0110 message.
5	Amount, Settlement	•	X	C	The Dual Message Authorization System provides this data element if the issuer chooses to receive settlement amount-related data elements.

DE ID	Data Element Name	Org	Sys	Dst	Comments
6	Amount, Cardholder Billing	•	X	M	The Dual Message Authorization System provides this data element.
7	Transmission Date and Time	•	X	M	Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message.
9	Conversion Rate, Settlement	•	X	C	The Dual Message Authorization System provides this data element if the issuer chooses to receive settlement amount-related data elements.
10	Conversion Rate, Cardholder Billing	•	X	M	The Dual Message Authorization System provides this data element.
11	Systems Trace Audit Number (STAN)	•	X	M	Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message.
13	Date, Local Transaction	•	X	C	Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message, if present.
15	Date, Settlement	•	X	M	The Dual Message Authorization System provides this data element.
16	Date, Conversion	•	X	M	The Dual Message Authorization System provides this data element.
19	Acquiring Institution Country Code	•	X	C	The Dual Message Authorization System provides this data element.
20	Primary Account Number (PAN) Country Code	•	X	C	Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message, if present.
22	Point-of-Service (POS) Entry Mode	•	X	M	Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message.
28	Amount, Transaction Fee	•	X	C	Must contain the same value as in the original Authorization Request/0100 or Reversal Request/0400 message, if present.
32	Acquiring Institution ID Code	•	X	M	Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message.

DE ID	Data Element Name	Org	Sys	Dst	Comments
33	Forwarding Institution ID Code	•	X	C	Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message, if present.
37	Retrieval Reference Number	•	X	C	Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message, if present.
38	Authorization ID Response	•	X	C	Must be the same value as in the original Authorization Request Response/0110 or Reversal Request Response/0410 message, if present.
39	Response Code	•	X	C	Must be the same value as in the original Authorization Request Response/0110 message. Otherwise, value 82 (Timeout at issuer), if no Authorization Request Response/0110 or Reversal Request Response/0410 is received. When provided in response to Reversal Request/0400 messages processed by the Dual Message Authorization System on behalf of the issuer, will contain the value from the original Reversal Request/0400 message.
41	Acceptor Terminal ID	•	XE	CE	Must be the same value as in the original Reversal Request/0400 message, if present.
43	Acceptor Name and Location	•	X	C	Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message, if present.

DE ID	Data Element Name	Org	Sys	Dst	Comments
48	Additional Data: Private Use	•	X	C	Must contain the transaction category code (TCC) from the original Authorization Request/0100 or Reversal Request/0400 message. The following subelements may be present in the Reversal Advice if they were present in the original Authorization Request/0100: • 20 (Cardholder Verification Method) • 33 (PAN Mapping File Information) • 38 (Account Category) • 58 (ATM Additional Data) • 63 (Trace ID) • 71 (On-behalf Services) • 77 (Transaction Type Identifier) • 90 (Custom Payment Service Request [Visa Only]) • 91 (Custom Payment Service Request/Transaction ID [Visa Only]) • 94 (Commercial Card Inquiry Request [Visa Only]) • 95 (Mastercard Promotion Code) • 96 (Visa Market-Specific Data Identifier [Visa Only])
49	Currency Code, Transaction	•	X	M	Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message.
50	Currency Code, Settlement	•	X	C	The Dual Message Authorization System provides this data element if the issuer chooses to receive settlement amount-related data elements.
51	Currency Code, Cardholder Billing	•	X	M	The Dual Message Authorization System provides this data element.
54	Additional Amounts	•	X	C	Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message or the original Authorization Request Response/0110 or Reversal Request Response/0410 message.
56	Payment Account Data	•	X	C	When Mastercard is the BIN Controller, the PAR value is inserted when available.

DE ID	Data Element Name	Org	Sys	Dst	Comments
60	Advice Reason Code	•	X	M	Indicates exact purpose of the Reversal Advice/0420 message.
62	Intermediate Network Facility (INF) Data	•	X	C	Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message, if present. Add system-generated advice for Reversal Advice/0420 message.
63	Network Data	•	X	M	The Mastercard Network provides this data element.
90	Original Data Elements	•	X	M	Must be present when the Dual Message Authorization System reverses a Reversal Request/0400 message.
95	Replacement Amounts	•	X	C	If this data element is provided by the acquirer in a partial reversal message, it must be included in the subsequent Reversal Advice/0420 message.
105	Multi-Use Transaction Identification Data	•	X	C	
121	Authorizing Agent ID Code	•	C	C	Must be the same value as in the original Authorization Request Response/0110 or Reversal Request Response/0410 message, if present.
122	Additional Record Data	•	X	C	
124	Member-Defined Data: General Use	•	X	C	
127	Private Data	O	X	•	Must be the same value as in the original Authorization Request/0100 or Reversal Request/0400 message, if present.

Reversal Advice Response/0430

Following is the list of the data elements applicable to this message.

Table 28: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	M	•	Constant: 0430 (Reversal Advice Response)

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Bit Map, Primary	M	X	•	Mandatory.
1	Bit Map, Secondary	M	X	•	Mandatory.
2	Primary Account Number (PAN)	ME	ME	•	Must be the same value as in the original Reversal Advice/0420 message.
3	Processing Code	ME	ME	•	Must be the same value as in the original Reversal Advice/0420 message.
4	Amount, Transaction	ME	XE	•	Must be the same value as in the original Reversal Advice/0420 message.
5	Amount, Settlement	CE	XE	•	Must be the same value as in the original Reversal Advice/0420 message, if present.
6	Amount, Cardholder Billing	ME	XE	•	Must be the same value as in the original Reversal Advice/0420 message.
7	Transmission Date and Time	ME	XE	•	Must be the same value as in the original Reversal Advice/0420 message.
9	Conversion Rate, Settlement	CE	XE	•	Must be the same value as in the original Reversal Advice/0420 message, if present.
10	Conversion Rate, Cardholder Billing	CE	XE	•	Must be the same value as in the original Reversal Advice/0420 message.
11	Systems Trace Audit Number	ME	XE	•	Must be the same value as in the original Reversal Advice/0420 message.
15	Date, Settlement	ME	XE	•	Must be the same value as in the original Reversal Advice/0420 message.
16	Date, Conversion	ME	XE	•	Must be the same value as in the original Reversal Advice/0420 message.
20	Primary Account Number (PAN) Country Code	CE	XE	•	Must be the same value as in the original Reversal Advice/0420 message, if present.
28	Amount, Transaction Fee	CE	XE	•	Must be the same value as in the original Reversal Advice/0420 message, if present.
32	Acquiring Institution ID Code	ME	XE	•	Must be the same value as in the original Reversal Advice/0420 message.
33	Forwarding Institution ID Code	CE	XE	•	Must be the same value as in the original Reversal Advice/0420 message, if present.

DE ID	Data Element Name	Org	Sys	Dst	Comments
37	Retrieval Reference Number	CE	XE	•	Must be the same value as in the original Reversal Advice/0420 message, if present.
39	Response Code	M	X	•	Must be present in the Reversal Advice Response/0430 message.
44	Additional Response Data	C	X	•	Contains additional response data.
49	Currency Code, Transaction	ME	XE	•	Must be the same value as in the original Reversal Advice/0420 message.
50	Currency Code, Settlement	CE	XE	•	Must be the same value as in the original Reversal Advice/0420 message, if present.
51	Currency Code, Cardholder Billing	ME	XE	•	Must be the same value as in the original Reversal Advice/0420 message.
56	Payment Account Data	O	•	•	Issuers may optionally provide DE 56.
62	Intermediate Network Facility (INF) Data	CE	XE	•	Must be the same value as in the original Reversal Advice/0420 message, if present.
63	Network Data	ME	XE	•	Must be the same value as in the original Reversal Advice/0420 message.
90	Original Data Elements	ME	XE	•	Must be the same value as in the original Reversal Advice/0420 message.
95	Replacement Amounts	CE	XE	•	Must be the same value as in the original Reversal Advice/0420 message, if present.
105	Multi-Use Transaction Identification Data	CE	•	•	
127	Private Data	O	P	•	Must be the same value as in the original Reversal Advice/0420 message, if present.

Administrative Request/0600

Following is the list of the data elements applicable to this message.

Table 29: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	P	M	Constant: 0600 (Administrative Request).
-	Bit Map, Primary	M	P	M	Mandatory.
1	Bit Map, Secondary	M	P	M	Mandatory.
2	Primary Account Number (PAN)	M	P	M	Identifies the receiver of the Administrative Request/0600 message.
7	Transmission Date and Time	M	P	M	Transaction time stamp; UTC date and time that this transaction was entered into the interchange.
11	Systems Trace Audit Number (STAN)	M	P	M	Transaction trace number; must be a unique value for the message initiator within each UTC day
32	Acquiring Institution ID Code	M	P	M	Identifies the acquirer of the Administrative Request/0600 message.
33	Forwarding Institution ID Code	C	P	C	Identifies the acquirer processor of the Administrative Request/0600 message, if present.
41	Acceptor Terminal ID	O	P	O	Identifies the merchant terminal ID, if present.
42	Acceptor ID	O	P	O	Identifies the merchant ID, if present.
43	Acceptor Name and Location	O	P	O	Contains the merchant name and address, if present.
60	Advice Reason Code	M	P	M	Indicates specific type of message.
62	Intermediate Network Facility (INF) Data	O	P	O	May contain message initiator network trace information.
63	Network Data	.	X	M	Contains the Banknet Reference Number that the Mastercard Network provides as a unique transaction ID.
113	Reserved for National Use	C	P	C	Contains customer information.
114	Reserved for National Use	C	P	C	Contains customer information.
115	Reserved for National Use	C	P	C	Contains customer information.

DE ID	Data Element Name	Org	Sys	Dst	Comments
116	Reserved for National Use	C	P	C	Contains customer information.
117	Reserved for National Use	C	P	C	Contains customer information.
118	Reserved for National Use	C	P	C	Contains customer information.
119	Reserved for National Use	C	P	C	Contains customer information.
127	Private Data	O	X	•	May contain message initiator information.

Administrative Request Response/0610

Following is the list of the data elements applicable to this message.

Table 30: Message data element layout

DE ID	Date Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	P	M	Constant: 0610 (Administrative Request Response).
-	Bit Map, Primary	M	P	M	Mandatory.
1	Bit Map, Secondary	M	P	M	Mandatory.
2	Primary Account Number (PAN)	ME	P	ME	Must be the same value as in the original Administrative Request/0600 message.
7	Transmission Date and Time	ME	P	ME	Must be the same value as in the original Administrative Request/0600 message.
11	Systems Trace Audit Number (STAN)	ME	P	ME	Must be the same value as in the original Administrative Request/0600 message.
32	Acquiring Institution ID Code	ME	P	ME	Must be the same value as in the original Administrative Request/0600 message.
33	Forwarding Institution ID Code	CE	P	CE	Must be the same value as in the original Administrative Request/0600 message, if present.
39	Response Code	M	P	M	Indicates the disposition of the original Administrative Request/0600 message.

DE ID	Date Element Name	Org	Sys	Dst	Comments
41	Acceptor Terminal ID	CE	P	CE	Must be the same value as in the original Administrative Request/0600 message, if present.
42	Acceptor ID	CE	P	CE	Must be the same value from the original Administrative Request/0600 message, if present.
43	Acceptor Name and Location	CE	P	CE	Must be the same value from the original Administrative Request/0600 message, if present.
44	Additional Response Data	C	P	C	May contain additional error code information.
60	Advice Reason Code	ME	P	ME	Must be the same value from the original Administrative Request/0600 message.
62	Intermediate Network Facility (INF) Data	CE	P	CE	Must be the same value as in the original Administrative Request/0600 message, if present.
63	Network Data	ME	P	ME	Must be the same value as in the original Administrative Request/0600 message.
113	Reserved for National Use	C	P	C	Contains customer information.
114	Reserved for National Use	C	P	C	Contains customer information.
115	Reserved for National Use	C	P	C	Contains customer information.
116	Reserved for National Use	C	P	C	Contains customer information.
117	Reserved for National Use	C	P	C	Contains customer information.
118	Reserved for National Use	C	P	C	Contains customer information.
119	Reserved for National Use	C	P	C	Contains customer information.
127	Private Data	•	X	CE	May contain message initiator information.

Administrative Advice/0620: system-generated

Following is the list of the data elements applicable to this message.

Table 31: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	•	M	M	Constant: 0620 (Administrative Advice).
-	Bit Map, Primary	•	X	M	Mandatory.
1	Bit Map, Secondary	•	X	M	Mandatory.
7	Transmission Date and Time	•	X	M	Transaction time stamp; UTC date and time that this transaction was entered into interchange.
11	Systems Trace Audit Number (STAN)	•	X	M	Transaction trace number; must be unique value for transaction initiator within each UTC day.
33	Forwarding Institution ID Code	•	X	M	Identifies the customer, institution, or Mastercard Network facility originating this Administrative Advice/0620 message.
42	Acceptor ID	•	X	C	Identifies the acceptor that defines the point of the transaction in both local and interchange environments. DE 42 is used as a merchant ID to uniquely identify the merchant in a POS transaction.
48	Additional Data: Private Use	•	X	C	If DE 60 (Advice Reason Code) = 600, DE 48 will not be present in the message.
56	Payment Account Data	•	X	C	When Mastercard is the BIN Controller (as defined by EMVCo) DE 56 will be present and contain the PAR value when one is associated with the PAN. DE 56 is only present when DE 60 (Advice Reason Code) = 0251.
60	Advice Reason Code	•	X	M	Indicates the specific reason for the message.

DE ID	Data Element Name	Org	Sys	Dst	Comments
62	Intermediate Network Facility (INF) Data	•	X	C	If DE 60 (Advice Reason Code) = 600, INF data in the Administrative Advice/0620 message contains MCBN620060000xxx, where xxx is the MIP ID. This data must be returned in an Administrative Advice Response/0630 message. If DE 60 (Advice Reason Code) = 650, INF data in the Administrative Advice/0620 message contains the value RISK, and it must be returned in an Administrative Advice Response/0630 message.
63	Network Data	•	X	M	Contains the Banknet Reference Number that the Mastercard Network provides as a unique transaction ID.
100	Receiving Institution ID Code	•	X	M	Identifies the customer, institution, or Mastercard Network facility that will receive this Administrative Advice/0620.
120	Record Data	•	X	C	If DE 60 (Advice Reason Code) = 600, contains rejected message. DE 120 will also be present if DE 60 (Advice Reason Code) = 0250, 0251, or 0252.
127	Private Data	•	O	O	Private use data element, available for optional use by message initiator.

Administrative Advice/0620: member-generated

Following is the list of the data elements applicable to this message.

Table 32: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	P	M	Constant: 0620 (Administrative Advice).
-	Bit Map, Primary	M	P	M	Mandatory.
1	Bit Map, Secondary	M	P	M	Mandatory.

DE ID	Data Element Name	Org	Sys	Dst	Comments
7	Transmission Date and Time	M	P	M	Transaction time stamp; UTC date and time that this transaction was entered into interchange.
11	Systems Trace Audit Number (STAN)	M	P	M	Transaction trace number; must be unique value for transaction initiator within each UTC day.
33	Forwarding Institution ID Code	M	P	M	Identifies the customer, institution, or Mastercard Network facility originating this Administrative Advice/0620 message.
48	Additional Data: Private Use	C	P	C	This data element is not applicable to this message when used for Administrative textual message transmittal.
60	Advice Reason Code	M	P	M	Must contain value 650 (Administrative textual message transmittal).
62	Intermediate Network Facility (INF) Data	C	P	C	Contains acquiring network trace information that intermediate network facilities (INFs) may require to quickly and accurately route Administrative Advice/0620 messages back to the originating institution.
63	Network Data	•	X	M	Contains the Banknet Reference Number that the Mastercard Network provides as a unique transaction ID.
100	Receiving Institution ID Code	M	P	M	Identifies the customer, institution, or Mastercard Network facility that will receive this Administrative Advice/0620 message.
120	Record Data	C	P	C	Contains textual message. May contain name, address, or interoffice routing information; free text format; used as an aid to insure that administration message is forwarded to a specific individual.
127	Private Data	O	X	•	Private use data element, available for optional use by message initiator.

Administrative Advice Response/0630

Following is the list of the data elements applicable to this message.

Table 33: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	M	•	Constant: 0630 (Administrative Advice Response).
-	Bit Map, Primary	M	X	•	Mandatory.
1	Bit Map, Secondary	M	X	•	Mandatory.
7	Transmission Date and Time	ME	XE	•	Must be the same value from the original Administrative Advice/0620 message.
11	Systems Trace Audit Number (STAN)	ME	X	•	Must be the same value from the original Administrative Advice/0620 message.
33	Forwarding Institution ID	ME	XE	•	Must be the same value from the original Administrative Advice/0620 message.
39	Response Code	M	X	•	Indicates the disposition of the original Administrative Advice/0620 message.
44	Additional Response Data	C	X	•	May contain the additional error code information depending on the value in DE 39.
62	Intermediate Network Facility (INF) Data	CE	XE	•	Must be the same value as in the original Administrative Advice/0620 message, if present.
63	Network Data	ME	XE	•	Must be the same value as in the original Administrative Advice/0620 message.
100	Receiving Institution ID Code	ME	•	•	Must be the same value as in the original Administrative Advice/0620 message.
127	Private Data	O	X	•	Private use data element, available for optional use by message initiator.

Network Management Request/0800: sign-on/sign-off

Following is the list of the data elements applicable to this message.

Table 34: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	M	•	Constant: 0800 (Network Management Request).
-	Bit Map, Primary	M	X	•	Mandatory.
1	Bit Map, Secondary	M	X	•	Mandatory.
2	Primary Account Number (PAN)	M	M	•	Must contain PAN prefix for sign-on/off by prefix functions; must contain Mastercard Group Sign-on for group sign-on/off functions or both the Mastercard Group Sign-on followed by the card prefix for a prefix sign-on/sign-off within a particular group.
7	Transmission Date and Time	M	X	•	Transaction time stamp; contains UTC date and time that this transaction was entered into the system.
11	Systems Trace Audit Number (STAN)	M	X	•	Transaction trace number; must be unique value for transaction indicator within each UTC day.
20	Primary Account Number (PAN) Country Code	C	X	•	Country Code is required if DE 2 (Primary Account Number [PAN]) contains a BIN beginning with 59.
33	Forwarding Institution ID Code	M	X	•	Identifies the customer, institution, or Mastercard Network facility originating this Network Management Request/0800 message.
53	Security-Related Control Information	C	X	•	Issuers performing their own PIN validation and issuers participating in the PIN validation in Stand-In Authorization service must provide the PIN Key Index Number to be used for PIN translation. DE 53 and DE 110 are mutually exclusive.
63	Network Data	•	X	•	Dual Message Authorization System transaction ID code that the Mastercard Network provides as a unique identifier for this transaction.

DE ID	Data Element Name	Org	Sys	Dst	Comments
70	Network Management Information Code	M	X	•	Indicates the specific purpose of this message. Must be the same value as in the original Network Management Request/0800 message.
94	Service Indicator	M	•	•	For values applicable to the Network Management/0800 message, refer to Data element definitions .
96	Message Security Code	M	•	•	Contains the Mastercard customer password security code, allowing access to the Mastercard Network by a customer or Intermediate Network Facilities (INF) processor. Data must be provided in EBCDIC hexadecimal format.
110	Encryption Data	C	•	•	Includes key management encryption data. DE 110 and DE 53 are mutually exclusive.
127	Private Data	O	X	•	Private use data element, available for optional use by message initiator.

Network Management Request/0800: network-connection status, member-generated

Following is the list of the data elements applicable to this message.

Table 35: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	M	•	Constant: 0800 (Network Management Request).
-	Bit Map, Primary	M	X	•	Mandatory.
1	Bit Map, Secondary	M	X	•	Mandatory.
2	Primary Account Number (PAN)	M	M	•	Group Sign-on ID for which the network connection status is being requested.
7	Transmission Date and Time	M	M	•	Transaction time stamp; contains UTC date and time that this transaction was entered into interchange.

DE ID	Data Element Name	Org	Sys	Dst	Comments
11	Systems Trace Audit Number (STAN)	M	X	•	Transaction trace number; must be unique value for transaction indicator within each UTC day.
33	Forwarding Institution ID Code	M	X	•	Identifies the customer, institution, or Mastercard Network facility originating this Network Management Request/0800 message.
63	Network Data	•	X	•	Mastercard Network transaction ID code that the network provides as a unique identifier for this transaction.
70	Network Management Information Code	M	X	•	Indicates the specific purpose of this message. Must be the same value as in the original Network Management Request/0800 message.

Network Management Request/0800: network-connection status, system-generated

Following is the list of the data elements applicable to this message.

Table 36: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	•	M	M	Constant: 0800 (Network Management Request).
-	Bit Map, Primary	•	X	M	Mandatory.
1	Bit Map, Secondary	•	X	M	Mandatory.
2	Primary Account Number (PAN)	•	M	M	Group Sign-on ID for which the Mastercard Network is requesting the network connection status.
7	Transmission Date and Time	•	M	M	Transaction time stamp; contains UTC date and time that this transaction was entered into interchange.
11	Systems Trace Audit Number (STAN)	•	X	M	Transaction trace number; must be unique value for transaction indicator within each UTC day.
33	Forwarding Institution ID Code	•	X	M	Identifies the Mastercard Network facility originating this Network Management Request/0800 message.

DE ID	Data Element Name	Org	Sys	Dst	Comments
63	Network Data	•	X	•	Mastercard Network transaction ID code that the network provides as a unique identifier for this transaction.
70	Network Management Information Code	•	X	M	Indicates the specific purpose of this message. Must be the same value as in the original Network Management Request/0800 message.

Network Management Request/0800: host-session activation/deactivation

Following is the list of the data elements applicable to this message.

Table 37: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	M	•	Constant: 0800 (Network Management Request).
-	Bit Map, Primary	M	X	•	Mandatory.
1	Bit Map, Secondary	M	X	•	Mandatory.
2	Primary Account Number (PAN)	M	M	•	Group Sign-on ID for which the network connection status is being requested.
7	Transmission Date and Time	M	X	•	Transaction time stamp; contains UTC date and time that this transaction was entered into interchange.
11	Systems Trace Audit Number (STAN)	M	X	•	Transaction trace number; must be unique value for transaction indicator within each UTC day.
33	Forwarding Institution ID Code	M	X	•	Identifies the customer, institution, or Mastercard Network facility originating this Network Management Request/0800 message.
70	Network Management Information Code	M	X	•	Indicates the specific purpose of this message. Must be the same value as in the original Network Management Request/0800 message.

Network Management Request/0800: PEK exchange

Following is the list of the data elements applicable to this message.

Table 38: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	•	M	M	Constant: 0800 (Network Management Request).
-	Bit Map, Primary	•	X	M	Mandatory.
1	Bit Map, Secondary	•	X	M	Mandatory.
2	Primary Account Number (PAN)	•	M	M	Member Group ID of the member receiving the working key exchange; the Mastercard Network uses this to route the message to the appropriate issuer or acquirer.
7	Transmission Date and Time	•	X	M	Date and time that this message is transmitted; specified in UTC date and time format.
11	Systems Trace Audit Number (STAN)	•	X	M	Transaction trace number; must be unique value for all messages associated with a given transaction within each UTC day.
33	Forwarding Institution ID Code	•	X	M	Constant: 002202; identifies the Mastercard Network.
48	Additional Data—Private Use	•	O	C	Key Exchange Data Block for variant key change function; the KEK is used to encrypt the PEK.
63	Network Data	•	X	M	Mastercard Network provides this data element; which includes a Banknet Reference Number for this transaction.
70	Network Management Information Code	•	X	M	Indicates the specific purpose of this message. Must be the same value as in the original Network Management Request/0800 message.
110	Encryption Data	•	X	C	Includes key management encryption data.

Network Management Request/0800: PEK exchange on demand

Following is the list of the data elements applicable to this message.

Table 39: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	M	•	Constant: 0800 (Network Management Request).
-	Bit Map, Primary	M	X	•	Mandatory.
1	Bit Map, Secondary	M	X	•	Mandatory.
2	Primary Account Number (PAN)	M	M	•	Member Group ID of the member requesting the working key exchange.
7	Transmission Date and Time	M	X	•	Date and time that this message is transmitted; specified in UTC date and time format.
11	Systems Trace Audit Number (STAN)	M	X	•	Transaction trace number; must be unique value for all messages associated with a given transaction within each UTC day.
33	Forwarding Institution ID Code	M	X	•	ID of the member originating this message; must be the same value as DE 2 (Primary Account Number [PAN]).
63	Network Data	•	X	•	Mastercard Network provides this data element; which includes a Banknet Reference Number for this transaction.
70	Network Management Information Code	M	X	•	Indicates the specific purpose of this message. Must be the same value as in the original Network Management Request/0800 message.
127	Private Data	O	X	•	Private use data element, available for optional use by message initiator; this data is not passed to the Mastercard Network.

Network Management Request Response/0810: network-connection status, member-generated

Following is the list of the data elements applicable to this message.

Table 40: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	M	•	Constant: 0810 (Network Management Request Response).
-	Bit Map, Primary	M	X	•	Mandatory.
1	Bit Map, Secondary	M	X	•	Mandatory.
2	Primary Account Number (PAN)	ME	ME	•	The Member Group ID. Must be the same value as in the original Network Management Request/0800 message.
7	Transmission Date and Time	ME	XE	•	Must be the same value as in the original Network Management Request/0800 message.
11	Systems Trace Audit Number (STAN)	ME	XE	•	Must be the same value as in the original Network Management Request/0800 message.
33	Forwarding Institution ID Code	ME	XE	•	Must be the same value as in the original Network Management Request/0800 message.
39	Response Code	M	X	•	Indicates disposition of original Network Management Request/0800 message.
63	Network Data	ME	XE	•	Must be the same value as in the original Network Management Request/0800 message.
70	Network Management Information Code	ME	XE	•	Indicates the specific purpose of this message. Must be the same value as in the original Network Management Request/0800 message.

Network Management Request Response/0810: network-connection status, system-generated

Following is the list of the data elements applicable to this message.

Table 41: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	•	M	M	Constant: 0810 (Network Management Request Response).
-	Bit Map, Primary	•	X	M	Mandatory.
1	Bit Map, Secondary	•	X	M	Mandatory.
2	Primary Account Number (PAN)	•	ME	ME	The Member Group ID. Must be the same value as in the original Network Management Request/0800 message.
7	Transmission Date and Time	•	XE	ME	Must be the same value as in the original Network Management Request/0800 message.
11	Systems Trace Audit Number (STAN)	•	XE	ME	Must be the same value as in the original Network Management Request/0800 message.
33	Forwarding Institution ID Code	•	XE	ME	Must be the same value as in the original Network Management Request/0800 message.
39	Response Code	•	X	M	Indicates disposition of original Network Management Request/0800 message.
44	Additional Response Data	•	X	C	May be present to provide additional information about message error conditions when the value in DE 39 (Response Code) is 30 (Format error).
63	Network Data	•	X	M	Must be the same value as in the original Network Management Request/0800 message.
70	Network Management Information Code	•	XE	ME	Indicates the specific purpose of this message. Must be the same value as in the original Network Management Request/0800 message.

Network Management Request Response/0810: host-session activation/deactivation

Following is the list of the data elements applicable to this message.

Table 42: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	•	M	M	Constant: 0810 (Network Management Request Response).
-	Bit Map, Primary	•	X	M	Mandatory.
1	Bit Map, Secondary	•	X	M	Mandatory.
2	Primary Account Number (PAN)	•	ME	ME	The Member Group ID. Must be the same value as in the original Network Management Request/0800 message.
7	Transmission Date and Time	•	XE	ME	Must be the same value as in the original Network Management Request/0800 message.
11	Systems Trace Audit Number (STAN)	•	ME	ME	Must be the same value as in the original Network Management Request/0800 message.
33	Forwarding Institution ID Code	•	XE	ME	Must be the same value as in the original Network Management Request/0800 message.
39	Response Code	•	X	M	Indicates disposition of original Network Management Request/0800 message.
44	Additional Response Data	•	X	C	May be present to provide additional information about message error conditions when the value in DE 39 (Response Code) is 30 (Format error).
70	Network Management Information Code	•	XE	ME	Indicates the specific purpose of this message. Must be the same value as in the original Network Management Request/0800 message.

Network Management Request Response/0810: PEK exchange

Following is the list of the data elements applicable to this message.

Table 43: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	M	•	Constant: 0810 (Network Management Request Response).
-	Bit Map, Primary	M	X	•	Mandatory.
1	Bit Map, Secondary	M	X	•	Mandatory.
2	Primary Account Number (PAN)	ME	ME	•	The Member Group ID. Must be the same value as in the original Network Management Request/0800 message.
7	Transmission Date and Time	ME	X	•	Must be the same value as in the original Network Management Request/0800 message.
11	Systems Trace Audit Number (STAN)	ME	ME	•	Must be the same value as in the original Network Management Request/0800 message.
33	Forwarding Institution ID Code	ME	X	•	Must be the same value as in the original Network Management Request/0800 message.
39	Response Code	M	X	•	Indicates disposition of original Network Management Request/0800 message.
44	Additional Response Data	C	X	•	May be present to provide additional information on message error conditions when the value in DE 39 (Response Code) is 30 (Format error).
48	Additional Data: Private Use	M	X	•	Must be the same value as in the original Network Management Request/0800 message.
63	Network Data	ME	XE	•	Must be the same value as in the original Network Management Request/0800 message.
70	Network Management Information Code	ME	X	•	Indicates the specific purpose of this message. Must be the same value as in the original Network Management Request/0800 message.

DE ID	Data Element Name	Org	Sys	Dst	Comments
110	Additional Data-2	O	M	•	Must be the same value as in the original Network Management Request/0800 message.

Network Management Request Response/0810: PEK exchange on demand

Following is the list of the data elements applicable to this message.

Table 44: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	•	M	M	Constant: 0810 (Network Management Request Response).
-	Bit Map, Primary	•	X	M	Mandatory.
1	Bit Map, Secondary	•	X	M	Mandatory.
2	Primary Account Number (PAN)	•	ME	ME	The Member Group ID. Must be the same value as in the original Network Management Request/0800 message.
7	Transmission Date and Time	•	XE	ME	Must be the same value as in the original Network Management Request/0800 message.
11	Systems Trace Audit Number (STAN)	•	ME	ME	Must be the same value as in the original Network Management Request/0800 message.
33	Forwarding Institution ID Code	•	XE	ME	Must be the same value as in the original Network Management Request/0800 message.
39	Response Code	•	X	M	Indicates disposition of original Network Management Request/0800 message.
44	Additional Response Data	•	X	C	May be present to provide additional information on message error conditions when the value in DE 39 (Response Code) is 30 (Format error).
63	Network Data	•	X	M	Must be the same value as in the original Network Management Request/0800 message.

DE ID	Data Element Name	Org	Sys	Dst	Comments
70	Network Management Information Code	•	XE	ME	Indicates the specific purpose of this message. Must be the same value as in the original Network Management Request/0800 message.

Network Management Request Reponse/0810: Sign-On/Sign-Off

Following is the list of the data elements applicable to this message.

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	•	M	M	Constant—0810 (Network Management Request Response).
-	Bit Map, Primary	•	M	M	Mandatory.
1	Bit Map, Secondary	•	M	M	Mandatory.
2	Primary Account Number (PAN)	•	ME	ME	Must be the same value as in the original Network Management Request/0800—Sign-On/Sign-Off.
7	Transmission Date and Time	•	ME	ME	Must be the same value as in the original Network Management Request/0800—Sign-On/Sign-Off.
11	Systems Trace Audit Number (STAN)	•	ME	ME	Must be the same value as in the original Network Management Request/0800—Sign-On/Sign-Off.
33	Forwarding Institution ID Code	•	ME	ME	Must be the same value as in the original Network Management Request/0800—Sign-On/Sign-Off.
39	Response Code	•	M	M	Indicates disposition of original Network Management Request/0800—Sign-On/Sign-Off.
44	Additional Response Data	•	C	C	May be present to provide additional information about message error conditions when the value in DE 39 is 30.
63	Network Data	•	ME	ME	Must be the same value as in the original Network Management Request/0800—Sign-On/Sign-Off.
70	Network Management Information Code	•	ME	ME	Must be the same value as in the original Network Management Request/0800—Sign-On/Sign-Off.

DE ID	Data Element Name	Org	Sys	Dst	Comments
127	Private Data	•	X	CE	Must be the same value as in the original Network Management Request/0800—Sign-On/Sign-Off.

Network Management Advice/0820: PEK exchange

Following is the list of the data elements applicable to this message.

Table 45: Message data element layout

DE ID	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	•	M	M	Constant: 0820 (Network Management Advice).
-	Bit Map, Primary	•	X	M	Mandatory.
1	Bit Map, Secondary	•	X	M	Mandatory.
2	Primary Account Number (PAN)	•	M	M	Member Group ID of the member receiving this working key exchange; the Mastercard Network uses this data to route the transaction to the appropriate issuer or acquirer.
7	Transmission Date and Time	•	X	M	Date and time that this message is transmitted; specified in UTC date and time format.
11	Systems Trace Audit Number (STAN)	•	X	M	Transaction trace number; must be unique value for all messages associated with a given transaction within each UTC day.
33	Forwarding Institution ID Code	•	X	M	Constant: 002202; identifies the Dual Message Authorization System.
48	Additional Data—Private Use	•	X	M	Key Exchange Data Block for key exchange function; the communication key is used to encrypt the working key.
63	Network Data	•	X	M	Mastercard Network provides this data element; which includes a Banknet Reference Number for this transaction.
70	Network Management Information Code	•	X	M	Indicates the specific purpose of this message. Must be the same value as in the original Network Management Request/0800 message.

DE ID	Data Element Name	Org	Sys	Dst	Comments
110	Encryption Data	•	X	M	Includes key management encryption data.

Chapter 4 Data element definitions

This section provides data element (DE) detail definitions of all data elements used in Authorization Platform application messages.

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MCC 104: Local Stoplist File.....	1236
MCC 105: Payment Cancellation File.....	1241
DE 120 (When Blocking Recurring Payments).....	1243
MCC 106: PAN Mapping File.....	1244
MCC 109: Application Transaction Counter File.....	1250
MCC 111: PAN-PAR (Payment Account Reference) Mapping File.....	1252
DE 120 (Error Codes).....	1253
MCC 102: Error Codes.....	1253
MCC 103: Error Codes.....	1254
MCC 104: Error Codes.....	1255
MCC 105: Error Codes.....	1256
MCC 106: Error Codes.....	1258
MCC 109: Error Codes.....	1261
MCC 111: Error Codes.....	1262
DE 121 (Authorizing Agent ID Code).....	1263
DE 122 (Additional Record Data).....	1265
Subelement 001 (Additional Acceptor Data).....	1267
Subfield 001 (Acceptor Street Address).....	1269
Subfield 002 (Acceptor Country Subdivision Code).....	1270
Subfield 003 (Acceptor URL address).....	1270
Subfield 004 (Acceptor Customer Service Phone Number).....	1271
Subfield 005 (Acceptor Phone Number).....	1272
Subfield 006 (Acceptor Additional Contact Information).....	1273
Subfield 007 (Acceptor Tax ID).....	1273
Subfield 008 (Partner ID Code).....	1274
Subfield 009 (Service Location City Name).....	1275

Subfield 010 (Service Location Country Subdivision Code).....	1275
Subfield 011 (Service Location Country Code).....	1276
Subfield 012 (Service Location Postal Code).....	1277
Subfield 013 (Acceptor Geographic Coordinates)	1277
Subfield 014 (Service Location Geographic Coordinates).....	1279
DE 122 Dual Message Authorization System edits.....	1280
DE 123 (Receipt Free Text).....	1282
DE 124 (Member-Defined Data).....	1283
DE 124 (Member-Defined Data: General Use).....	1284
DE 124 (Member-Defined Data: Prepaid Voucher).....	1286
Subfield 1 (Merchant Direct Settlement Indicator).....	1286
DE 124 (Member-Defined Data: Mastercard Send).....	1287
DE 124 (Member-Defined Data: Brazil Maestro Only).....	1287
Subfield 1 (Unique Reference Number).....	1287
Subfield 2 (Sender/Payer/User ID).....	1288
Subfield 3 (Sender/Payer Address).....	1288
Subfield 4 (Additional Sender Information).....	1289
Subfield 6 (Discretionary Message on Sales Slip Supported).....	1290
Subfield 7 (Discretionary Message on Sales Slip Code).....	1290
Subfield 8 (Discretionary Message on Sales Slip Content).....	1291
Subfield 9 (Phoneshop [Phone Company ID]).....	1291
Subfield 10 (Phoneshop [Cell Phone Number]).....	1292
Subfield 11 (Phoneshop [Message Security Code]).....	1292
Subfield 12 (Merchant CNPJ Number).....	1293
Subfield 13 (Total Annual Effective Cost).....	1293
DE 124 (Member-Defined Data [Colombia Domestic Use Only]).....	1294
Subfield 1 (Card Issuer Data).....	1294
Subfield 2 (Tax [IVA]).....	1294
Subfield 3 (Tax Amount Base).....	1295
Subfield 4 (Retailer Data).....	1295
Subfield 5 (Terminal Acquirer Data).....	1296
Subfield 6 (Acquirer Original Processing Code).....	1297
Subfield 7 (Bill Payment/Top up Data).....	1297
Subfield 8 (Local POS Data).....	1298
Subfield 9 (Local Response Codes).....	1298
Subfield 10 (Original Transaction Data).....	1299
Subfield 11 (IAC Tax Amount).....	1300
DE 124 (Member-Defined Data: Ukraine Domestic Transactions).....	1300
DE 125 (New PIN Data).....	1302

DE 126 (Private Data)..... 1303

DE 127 (Private Data)..... 1304

DE 128 (Message Authentication Code)..... 1305

Data element layout

Following is the data element structure for describing data elements. Values and Application Note information is omitted from the data element attributes of those data elements not currently used in Authorization Platform messages.

Attributes

Attribute	Description
Data Representation	Annotation and data length (fixed or variable in length)
Data Element Length	2 or 3 if variable in length
Data Field	Contents of subelement, subfields or N/A
Subfields	Indicates number of subelements, subfields or N/A
Justification	Left, Right, or N/A

Usage

Following is the usage of DE xx (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Message	Org	Sys	Dst
Message types applicable and the entities involved in the message.	.	.	.

Values

Valid values and name of each value listed.

Application Notes

Specific application notes, conditions, and cross-edits where applicable.

Subelement layout

Following is the subelement structure for describing data element subelements.

Attributes

Attribute	Description
Subelement ID	Subelement identifier
Subelement Length	2 or 3 if variable in length
Data Representation	Annotation and data length (fixed or variable in length)
Data Field	Contents of subfields
Subfields	Indicates number of subfields
Justification	See subfields (may be subfield specific)

Usage

Following is the usage of subelement XX (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Message	Org	Sys	Dst
List of message types that use this data element and usage.	•	•	•

Values

The list of subelement values or see subfields.

Subfield layout

Following is the subfield structure for describing data element and subelement subfields. Subfield ID and Subfield Length are omitted from the subfield attributes when subfield ID and length are not included in the data.

Attributes

Attribute	Description
Subfield ID	Subfield identifier (omitted if not part of the data)

Attribute	Description
Subfield Length	2 if variable in length (omitted if not part of the data)
Data Representation	Annotation and data length (fixed or variable in length)
Data Field	Contents of position(s)
Justification	Left, Right, or N/A (justification is not used in describing subelement subfields)

Values

Valid values and name of each value listed.

Position layout

Following is the position structure for describing subelement and subfield positions.

Data Representation	Annotation and data length (fixed or variable in length)
Data Field	Description of Data
Values	List of values

List of data elements (numeric order)

Following is the list of data elements in numeric order.

Number	Name	Data Representation
1	Bit Map, Secondary	b-8
2	Primary Account Number (PAN)	n...19; LLVAR
3	Processing Code	n-6
4	Amount, Transaction	n-12
5	Amount, Settlement	n-12
6	Amount, Cardholder Billing	n-12
7	Transmission Date and Time	n-10
8	Amount, Cardholder Billing Fee ²	n-8

Number	Name	Data Representation
9	Conversion Rate, Settlement	n-8
10	Conversion Rate, Cardholder Billing	n-8
11	Systems Trace Audit Number	n-6
12	Time, Local Transaction	n-6
13	Date, Local Transaction	n-4
14	Date, Expiration	n-4
15	Date, Settlement	n-4
16	Date, Conversion	n-4
17	Date, Capture ²	n-4
18	Merchant Type	n-4
19	Acquiring Institution Country Code ²	n-3
20	Primary Account Number (PAN) Country Code	n-3
21	Forwarding Institution Country Code ²	n-3
22	Point-of-Service (POS) Entry Mode	n-3
23	Card Sequence Number	n-3
24	Network International ID ²	n-3
25	Point-of-Service (POS) Condition Code ²	n-2
26	Point-of-Service (POS) Personal ID Number (PIN) Capture Code	n-2
27	Authorization ID Response Length ²	n-1
28	Amount, Transaction Fee	an-9
29	Amount, Settlement Fee ²	an-9
30	Amount, Transaction Processing Fee ²	an-9
31	Amount, Settlement Processing Fee ²	an-9
32	Acquiring Institution ID Code	n...6; LLVAR
33	Forwarding Institution ID Code	n...6; LLVAR
34	Primary Account Number (PAN), Extended ²	ans...28; LLVAR
35	Track 2 Data	ans...37; LLVAR
36	Track 3 Data ²	ans...104; LLLVAR

² Mastercard currently does not use this data element and it should not be included in an authorization message. A program or service may use it at a later date.

Number	Name	Data Representation
37	Retrieval Reference Number	an-12
38	Authorization ID Response	ans-6
39	Response Code	an-2
40	Service Restriction Code ²	an-3
41	Acceptor Terminal ID	ans-8
42	Acceptor ID	ans-15
43	Acceptor Name and Location	ans-40
44	Additional Response Data	ans...25; LLLVAR
45	Track 1 Data	ans...76; LLLVAR
46	Additional Data—ISO Use ²	ans...999; LLLVAR
47	Additional Data—National Use ³	ans...999; LLLVAR
48	Additional Data—Private Use ⁴	ans...999; LLLVAR
49	Currency Code, Transaction	n-3
50	Currency Code, Settlement	n-3
51	Currency Code, Cardholder Billing	n-3
52	Personal ID Number (PIN) Data	b-8
53	Security-Related Control Information	n-16
54	Additional Amounts	an...240; LLLVAR
55	Integrated Circuit Card (ICC) System-Related Data	b...255; LLLVAR
56	Payment Account Data	an...37; LLLVAR
57–59	Reserved for National Use ³	ans...999; LLLVAR
60	Advice Reason Code ⁵	ans...060; LLLVAR
61	Point-of-Service (POS) Data ⁵	an...026; LLLVAR
62	Intermediate Network Facility (INF) Data ⁵	ans...100; LLLVAR
63	Network Data ⁵	an...050; LLLVAR
64	Message Authentication Code (MAC) ²	b-8

³ This data element is reserved for national use. The Authorization Platform does not perform any processing on this data element. However, if it is included in an authorization message, the network will pass it from the originator to the receiver, provided that both the originator and the receiver are customers of the same national standards organizations.

⁴ This data element is an ISO-designated "private use" data element redefined by Mastercard in accordance with provisions of the ISO 8583–1987 specification.

⁵ Although this data element is designated as binary, it also may contain up to eight bytes of EBCDIC encoded data.

Number	Name	Data Representation
65	Bit Map, Extended ²	b-8
66	Settlement Code ²	n-1
67	Extended Payment Code ²	n-2
68	Receiving Institution Country Code ²	n-3
69	Settlement Institution Country Code ²	n-3
70	Network Management Information Code	n-3
71	Message Number ²	n-4
72	Message Number Last ²	n-4
73	Date, Action	n-6
74	Credits, Number ²	n-10
75	Credits, Reversal Number ²	n-10
76	Debits, Number ²	n-10
77	Debits, Reversal Number ²	n-10
78	Transfers, Number ²	n-10
79	Transfers, Reversal Number ²	n-10
80	Inquiries, Number ²	n-10
81	Authorizations, Number ²	n-10
82	Credits, Processing Fee Amount ²	n-12
83	Credits, Transaction Fee Amount ²	n-12
84	Debits, Processing Fee Amount ²	n-12
85	Debits, Transaction Fee Amount ²	n-12
86	Credits, Amount ²	n-16
87	Credits, Reversal Amount ²	n-16
88	Debits, Amount ²	n-16
89	Debits, Reversal Amount ²	n-16
90	Original Data Elements	n-42
91	Issuer File Update Code	an-1
92	File Security Code ²	an-2
93	Response Indicator ²	an-5
94	Service Indicator	an-7
95	Replacement Amounts	n-42

Number	Name	Data Representation
96	Message Security Code ⁵	b-8
97	Amount, Net Settlement ²	an-17
98	Payee	ans-25
99	Settlement Institution ID Code ²	n...11; LLVAR
100	Receiving Institution ID Code	n...11; LLVAR
101	File Name	ans...17; LLVAR
102	Account ID-1	ans...28; LLVAR
103	Account ID-2	ans...28; LLVAR
104	Digital Payment Data	ans...999; LLLVAR
105–107	Reserved for Mastercard Use ²	ans...999; LLLVAR
108	Additional Transaction Reference Data	ans...999; LLLVAR
109	Reserved for ISO Use ²	ans...999; LLLVAR
110	Additional Data—2	ans...999; LLLVAR
110	Encryption Data	b...999; LLLVAR
111	Reserved for ISO Use ²	ans...999; LLLVAR
112	Additional Data (National Use)	ans...100; LLLVAR
113–119	Reserved for National Use ³	ans...999; LLLVAR
120	Record Data ⁵	ans...999; LLLVAR
121	Authorizing Agent ID Code ⁵	n...6; LLLVAR
122	Additional Record Data ⁵	ans...999; LLLVAR
123	Receipt Free Text	ans...512; LLLVAR
124	Member-defined Data	ans...199; LLLVAR
125	New PIN Data	b-8
126	Private Data	ans...100; LLLVAR
127	Private Data ⁵	ans...100; LLLVAR
128	Message Authentication Code (MAC) ²	b-8

Data elements currently not used by Dual Message Authorization System

This table lists all the data elements not used by the Dual Message Authorization System. For more details on a specific data element, refer to [Data element definitions](#).

Table 46: Data elements currently not used

Data element	Data element title
DE 8	Amount, Cardholder Billing Fee
DE 17	Date, Capture
DE 21	Forwarding Institution Country Code
DE 24	Network International ID
DE 25	Point-of-Service [POS] Condition Code
DE 27	Authorization ID Response Length
DE 29	Amount, Settlement Fee
DE 30	Amount, Transaction Processing Fee
DE 31	Amount, Settlement Processing Fee
DE 34	Primary Account Number [PAN], Extended
DE 36	Track 3 Data
DE 40	Service Restriction Code
DE 46	Expanded Additional Amounts
DE 47	Additional Data: National Use
DE 57- DE 59	Reserved for National Use
DE 66	Settlement Code
DE 67	Extended Payment Code
DE 68	Receiving Institution Country Code
DE 69	Settlement Institution Country Code
DE 71	Message Number
DE 72	Message Number Last
DE 73	Date, Action
DE 74	Credits, Number
DE 75	Credits, Reversal Number

Data element definitions

Data elements currently not used by Dual Message Authorization System

Data element	Data element title
DE 76	Debits, Number
DE 77	Debits, Reversal Number
DE 78	Transfers, Number
DE 79	Transfers, Reversal Number
DE 80	Inquiries, Number
DE 81	Authorizations, Number
DE 82	Credits, Processing Fee Amount
DE 83	Credits, Transaction Fee Amount
DE 84	Debits, Processing Fee Amount
DE 85	Debits, Transaction Fee Amount
DE 86	Credits, Amount
DE 87	Credits, Reversal Amount
DE 88	Debits, Amount
DE 89	Debits, Reversal Amount
DE 92	File Security Code
DE 93	Response Indicator
DE 97	Amount, Net Settlement
DE 98	Payee
DE 99	Settlement Institution ID Code
DE 107	Reserved for Mastercard Use
DE 109	Reserved for ISO Use
DE 111	Reserved for ISO Use

Message Type Identifier

The Message Type Identifier (MTI) is a four-digit numeric data element describing the type of message being interpreted. The MTI is required and must be present as the first data element of each Authorization Platform message.

Attributes

Attribute	Description
Length of Length Field	N/A
Data Representation	n-4
Data Field	Contents of positions 1–4
Subfields	N/A
Justification	N/A

Usage

Message	Org	Sys	Dst
Authorization Request/0100	M	•	M
Authorization Request Response/0110	M	•	M
Authorization Advice/0120—Acquirer-generated	M	•	M
Authorization Advice/0120—Issuer-generated	M	M	•
Authorization Advice/0120—System-generated	•	M	M
Authorization Advice Response/0130—Issuer-generated (Responding to an Acquirer-generated 0120)	M	•	M
Authorization Advice Response/0130—Issuer-generated	M	M	•
Authorization Advice Response/0130—System-generated	•	M	M
Authorization Acknowledgement/0180	M	M	•
Authorization Negative Acknowledgement/0190	•	M	M
Issuer File Update Request/0302	M	M	•
Issuer File Update Request Response/0312	•	M	M
Reversal Request/0400	M	•	M
Reversal Request Response/0410	M	•	M
Reversal Advice/0420	•	M	M

Message	Org	Sys	Dst
Reversal Advice Response/0430	M	M	.
Administrative Request/0600	M	.	M
Administrative Request Response/0610	M	.	M
Administrative Advice/0620	.	M	M
Administrative Advice Response/0630	M	M	.
Network Management Request/0800—Sign-On/Sign-Off	M	M	.
Network Management Request/0800—Host Session Activation/ Deactivation	M	M	.
Network Management Request/0800—Network Connection Status, System-generated	.	M	M
Network Management Request/0800—Network Connection Status, Member-generated	M	M	.
Network Management Request/0800—PEK Exchange	.	M	M
Network Management Request/0800—PEK Exchange—On Demand	M	M	.
Network Management Request Response/0810—Sign-On/Sign-Off	.	M	M
Network Management Request Response/0810—Host Session Activation/Deactivation	.	M	M
Network Management Request Response/0810—Network Connection Status, System-generated	.	M	M
Network Management Request Response/0810—Network Connection Status, Member-generated	M	M	.
Network Management Request Response/0810—PEK Exchange	M	M	.
Network Management Request Response/0810—PEK Exchange—On Demand	.	M	M
Network Management Advice/0820—PEK Exchange	.	M	M

Values

Table 47: Position 1—Version Number

Value	Description
0	= ISO 8583: 1987
1	= ISO 8583: 1992
2–7	= Reserved for ISO use

Value		Description
8	=	Reserved for national use
9	=	Reserved for future use

Table 48: Position 2—Message Class

Value		Description
0	=	Reserved for ISO use
1	=	Authorization
2	=	Financial
3	=	File action
4	=	Reversal/chargeback
5	=	Reconciliation
6	=	Administrative
7	=	Fee collection
8	=	Network management
9	=	Reserved for ISO use

Table 49: Position 3—Message Function

Value		Description
0	=	Request
1	=	Request response
2	=	Advice
3	=	Advice response
4	=	Notification
5–9	=	Reserved for ISO use

Table 50: Position 4—Transaction Originator

Value		Description
0	=	Acquirer

Value		Description
1	=	Acquirer repeat
2	=	Card issuer
3	=	Card issuer repeat
4	=	Other
5	=	Other repeat
6–9	=	Reserved for ISO use

Transaction Data classifications

The following table specifies the category of each data element, subelement, and subfield as Category One Transaction Data or Category Two Transaction Data. Subelements and subfields are assigned the category of the higher level element unless, otherwise specified in the following table.

Data Element	Data Element Description	Transaction Data Category ⁶
DE 1	DE 1: Bit Map, Secondary	One
DE 2	DE 2: Primary Account Number (PAN)	One
DE 3	DE 3: Processing Code	One
DE 3	Subfield 1: Cardholder Transaction Type Code	One
DE 3	Subfield 2: Cardholder Account Type (From)	One
DE 3	Subfield 3: Cardholder Account Type (To)	One
DE 4	DE 4: Amount, Transaction	One
DE 5	DE 5: Amount, Settlement	One
DE 6	DE 6: Amount, Cardholder Billing	One
DE 7	DE 7: Transmission Date and Time	One
DE 8	DE 8: Amount, Cardholder Billing Fee	One

⁶ Transaction Data Category Two applies to any unspecified data element, subelement, or subfield.

Data Element	Data Element Description	Transaction Data Category ⁶
DE 9	DE 9: Conversion Rate, Settlement	One
DE 10	DE 10: Conversion Rate, Cardholder Billing	One
DE 11	DE 11: Systems Trace Audit Number	One
DE 12	DE 12: Time, Local Transaction	One
DE 13	DE 13: Date, Local Transaction	One
DE 14	DE 14: Date, Expiration	One
DE 15	DE 15: Date, Settlement	One
DE 16	DE 16: Date, Conversion	One
DE 17	DE 17: Date, Capture	One
DE 18	DE 18: Merchant Type	One
DE 19	DE 19: Acquiring Institution Country Code	One
DE 20	DE 20: Primary Account Number (PAN) Country Code	One
DE 21	DE 21: Forwarding Institution Country Code	One
DE 22	DE 22: Point of Service Entry Mode	One
DE 22	Subfield 1: POS Terminal PAN Entry Mode	One
DE 22	Subfield 2: POS Terminal PIN Entry Mode	One
DE 23	DE 23: Card Sequence Number	One
DE 24	DE 24: Network International Identifier	Two
DE 25	DE 25: Point of Service Condition Code (ISO)	One
DE 26	DE 26: Point of Service (POS) PIN Capture Code	One
DE 27	DE 27: Authorization Identification Response Length	Two

⁶ Transaction Data Category Two applies to any unspecified data element, subelement, or subfield.

Data Element	Data Element Description	Transaction Data Category ⁶
DE 28	DE 28: Amount, Transaction Fee	One
DE 29	DE 29: Amount, Settlement Fee	One
DE 30	DE 30: Amount, Transaction Processing Fee	Two
DE 32	DE 32: Acquiring Institution Identification Code	One
DE 33	DE 33: Forwarding Institution Identification Code	One
DE 34	DE 34: Primary Account Number, Extended	One
DE 35	DE 35: Track 2 Data	One
DE 35	Subfield 1: Start Sentinel	One
DE 35	Subfield 2: Primary Account Number (PAN)	One
DE 35	Subfield 3: Field Separator	One
DE 35	Subfield 4: Expiration Date	One
DE 35	Subfield 5: Extended Service Code	One
DE 35	Subfield 6: Discretionary Data	One
DE 35	Subfield 7—End Sentinel	One
DE 35	Subfield 8: Longitudinal Redundancy Check (LRC)	One
DE 37	DE 37: Retrieval Reference Number	Two
DE 37	Subfield 1: Transaction Date and Initiator Discretionary Data	Two
DE 37	Subfield 2: Terminal Transaction Number	Two
DE 38	DE 38: Authorization Identification Response	One
DE 39	DE 39: Response Code	One
DE 41	DE 41: Acceptor Terminal ID	One
DE 42	DE 42: Acceptor ID	One

⁶ Transaction Data Category Two applies to any unspecified data element, subelement, or subfield.

Data Element	Data Element Description	Transaction Data Category ⁶
DE 43	DE 43: Acceptor Name and Location	One
DE 43	Subfield 1: Acceptor Name	One
DE 43	Subfield 2: Space	One
DE 43	Subfield 3: Acceptor City	One
DE 43	Subfield 4: Space	One
DE 43	Subfield 5: Acceptor Country Code	One
DE 44	DE 44: Additional Response Data	One
DE 45	DE 45: Track 1 Data	One
DE 45	Subfield 1: Start Sentinel	One
DE 45	Subfield 2: Format Code	One
DE 45	Subfield 3: Primary Account Number (PAN)	One
DE 45	Subfield 4: Field Separator	One
DE 45	Subfield 5: Cardholder Name	One
DE 45	Subfield 6: Field Separator	One
DE 45	Subfield 7: Expiration Date	One
DE 45	Subfield 8: Extended Service Code	One
DE 45	Subfield 9: Discretionary Data	One
DE 45	Subfield 10: End Sentinel	One
DE 45	Subfield 11: Longitudinal Redundancy Check (LRC)	One
DE 48	DE 48: Additional Data	See subelements
DE 48, SE 11	Subelement 11: Key Exchange Data Block (Double- Length Keys)	Two
DE 48, SE 11	Subfield 1: Key Class ID	Two
DE 48, SE 11	Subfield 2: Key Index Number	Two
DE 48, SE 11	Subfield 3: Key Cycle Number	Two

⁶ Transaction Data Category Two applies to any unspecified data element, subelement, or subfield.

Data Element	Data Element Description	Transaction Data Category ⁶
DE 48, SE 11	Subfield 4: PIN Encryption Key (PEK)	Two
DE 48, SE 11	Subfield 5: Key Check Value	Two
DE 48, SE 11	Subelement 11: Key Exchange Data Block (Triple- Length Keys	Two
DE 48, SE 11	Subfield 1: Key Class ID	Two
DE 48, SE 11	Subfield 2: Key Index Number	Two
DE 48, SE 11	Subfield 3: Key Cycle Number	Two
DE 48, SE 11	Subfield 4: PIN Encryption Key (PEK)	Two
DE 48, SE 11	Subfield 5: Key Check Value	Two
DE 48, SE 17	Subelement 17: Authentication Indicator	Two
DE 48, SE 19	Subelement 19: Fraud Notification Information	Two
DE 48, SE 19	Subfield 1: Fraud Notification Service Date	Two
DE 48, SE 19	Subfield 2: Fraud Notification Service Chargeback Counter	Two
DE 48, SE 19	Subfield 3: Fraud Notification Service Second Presentment Rights Indicator	Two
DE 48, SE 21	Subelement 21: Acceptance Data	One
DE 48, SE 21	Subfield 01: mPOS Acceptance Device Type	One
DE 48, SE 21	Subfield 02: Additional Terminal Capability Indicator	One
DE 48, SE 22	Subelement 22: Multi-Purpose Merchant Indicator	Two
DE 48, SE 22	Subfield 01: Low-Risk Merchant Indicator	Two
DE 48, SE 22	Subfield 02: Single Tap Indicator	Two
DE 48, SE 22	Subfield 03: Response to PIN Request	Two

⁶ Transaction Data Category Two applies to any unspecified data element, subelement, or subfield.

Data Element	Data Element Description	Transaction Data Category ⁶
DE 48, SE 22	Subfield 04: Issuer PIN Request in a Single Tap Mode	Two
DE 48, SE 22	Subfield 05: Cardholder/Merchant Initiated Transaction Indicator	Two
DE 48, SE 22	Subfield 06: Message Format Version Code: Acquirer	Two
DE 48, SE 23	Subelement 23: Payment Initiation Channel	One
DE 48, SE 23	Subfield 1: Device Type	One
DE 48, SE 25	Subelement 25: Mastercard Cash Program Data	Two
DE 48, SE 25	Subfield 1: Message Identifier	Two
DE 48, SE 26	Subelement 26: Wallet Program Data	Two
DE 48, SE 26	Subfield 1: Wallet Identifier	Two
DE 48, SE 27	Subelement 27: Transaction Analysis	Two
DE 48, SE 27	Subfield 1: Overview	Two
DE 48, SE 27	Subfield 2: Test Results	Two
DE 48, SE 28	Subelement 28: Cardless ATM Order ID	Two
DE 48, SE 29	Subelement 29: Additional POS Terminal Locations	One
DE 48, SE 30	Subelement 30: Token Transaction Identifier	Two
DE 48, SE 32	Subelement 32: Mastercard Assigned ID	Two
DE 48, SE 32	Subfield 1: Mastercard Assigned ID	Two
DE 48, SE 33	Subelement 33: PAN Mapping File Information	Two
DE 48, SE 33	Subfield 1: Account Number Indicator	Two
DE 48, SE 33	Subfield 2: Account Number PAN	Two

⁶ Transaction Data Category Two applies to any unspecified data element, subelement, or subfield.

Data Element	Data Element Description	Transaction Data Category ⁶
DE 48, SE 33	Subfield 3: Expiration Date	Two
DE 48, SE 33	Subfield 5: Token Assurance Method	Two
DE 48, SE 33	Subfield 6: Token Requestor ID	Two
DE 48, SE 33	Subfield 8: Storage Technology	Two
DE 48, SE 34	Subelement 34: ATC Information	Two
DE 48, SE 34	Subfield 1: ATC Value	Two
DE 48, SE 34	Subfield 2: ATC Discrepancy Value	Two
DE 48, SE 34	Subfield 3: ATC Discrepancy Indicator	Two
DE 48, SE 35	Subelement 35: Contactless Device Request Response	Two
DE 48, SE 35	Subfield 1: Contactless Device Request Response	Two
DE 48, SE 37	Subelement 37: Additional Merchant Data	Two
DE 48, SE 37	Subfield 1: Payment Facilitator ID	Two
DE 48, SE 37	Subfield 2: Independent Sales Organization ID	Two
DE 48, SE 37	Subfield 3: Sponsored Merchant ID	Two
DE 48, SE 38	Subelement 38: Account Category	Two
DE 48, SE 38	Subfield 1: Account Category	Two
DE 48, SE 39	Subelement 39: Account Data Compromise Information	Two
DE 48, SE 39	Positions 1–3 Reserved for Future Use	Two
DE 48, SE 39	Positions 4–6 Reserved for Future Use	Two
DE 48, SE 39	Positions 7–12 Case Key Code #1	Two
DE 48, SE 39	Positions 13–18 Case Key Code #2	Two

⁶ Transaction Data Category Two applies to any unspecified data element, subelement, or subfield.

Data Element	Data Element Description	Transaction Data Category ⁶
DE 48, SE 39	Positions 19–24 Case Key Code #3	Two
DE 48, SE 39	Position 25: ADC Data Type–PAN	Two
DE 48, SE 39	Position 26: ADC Data Type–Expiration Date	Two
DE 48, SE 39	Position 27: ADC Data Type–CVC 2	Two
DE 48, SE 39	Position 28: ADC Data Type–PIN	Two
DE 48, SE 39	Position 29: ADC Data Type–Magnetic Stripe	Two
DE 48, SE 39	Position 30: ADC Data Type–Personal Information	Two
DE 48, SE 40	Subelement 40: Electronic Commerce Merchant/Cardholder Certificate Serial Number	Two
DE 48, SE 40	Subfield 1: Merchant Certificate Serial Number	Two
DE 48, SE 40	Subfield 2: Cardholder Certificate Serial Number	Two
DE 48, SE 41	Subelement 41: Electronic Commerce Certificate Qualifying Information	Two
DE 48, SE 41	Subfield 1: Reserved for future use	Two
DE 48, SE 41	Subfield 2: Reserved for future use	Two
DE 48, SE 41	Subfield 3: Reserved for future use	Two
DE 48, SE 41	Subfield 4: Reserved for future use	Two
DE 48, SE 41	Subfield 5: Reserved for future use	Two
DE 48, SE 41	Subfield 6: Reserved for future use	Two
DE 48, SE 41	Subfield 7: Reserved for future use	Two

⁶ Transaction Data Category Two applies to any unspecified data element, subelement, or subfield.

Data Element	Data Element Description	Transaction Data Category ⁶
DE 48, SE 41	Subfield 8: Reserved for future use	Two
DE 48, SE 41	Subfield 9: Reserved for future use	Two
DE 48, SE 41	Subfield 10: Reserved for future use	Two
DE 48, SE 41	Subfield 11: National ID	Two
DE 48, SE 41	Subfield 12: Reserved for future use	Two
DE 48, SE 41	Subfield 13: Reserved for future use	Two
DE 48, SE 41	Subfield 14: Reserved for future use	Two
DE 48, SE 41	Subfield 15: Reserved for future use	Two
DE 48, SE 41	Subfield 16: Reserved for future use	Two
DE 48, SE 41	Subfield 17: Reserved for future use	Two
DE 48, SE 41	Subfield 18: Reserved for future use	Two
DE 48, SE 41	Subfield 19: Reserved for future use	Two
DE 48, SE 41	Subfield 20: Reserved for future use	Two
DE 48, SE 41	Subfield 21: Reserved for future use	Two
DE 48, SE 42	Subelement 42: Electronic Commerce Indicators	One
DE 48, SE 42	Subfield 1: Electronic Commerce Security Level Indicator (SLI) and UCAF	One
DE 48, SE 42	Collection Indicator	One
DE 48, SE 43	Subelement 43: Universal Cardholder Authentication Field (UCAF)	Two

⁶ Transaction Data Category Two applies to any unspecified data element, subelement, or subfield.

Data Element	Data Element Description	Transaction Data Category ⁶
DE 48, SE 43	Subfield 1: UCAF Data	Two
DE 48, SE 43	Subelement 43: Digital Secure Remote Payment Universal Cardholder Authentication Field (UCAF)	Two
DE 48, SE 48	Digital Commerce Solutions Indicators	Two
DE 48, SE 48	Subfield 1: Digital Commerce Domain Identifier	Two
DE 48, SE 48	Subfield 2: Digital Commerce Program Type	Two
DE 48, SE 48	Subfield 3: Digital Commerce Check-In Entity Type and Name	Two
DE 48, SE 48	Subfield 4: Digital Commerce Check-In Method	Two
DE 48, SE 49	Subelement 49: Time Validation Information	One
DE 48, SE 49	Subfield 1: Time Value	One
DE 48, SE 49	Subfield 2: Time Discrepancy Value	One
DE 48, SE 49	Subfield 3: Time Discrepancy Indicator	One
DE 48, SE 52	Subelement 52: Transaction Integrity Class	Two
DE 48, SE 54	Subelement 54: Ancillary Service Charges	One
DE 48, SE 54	Subfield 1: Ancillary Fee Code	One
DE 48, SE 54	Subfield 2: Ancillary Fee Amount	One
DE 48, SE 56	Subelement 56: Security Services Additional Data for Issuers	Two
DE 48, SE 56	Subfield 1: Security Services Indicator	Two
DE 48, SE 56	Subfield 2: Security Services Data	Two

⁶ Transaction Data Category Two applies to any unspecified data element, subelement, or subfield.

Data Element	Data Element Description	Transaction Data Category ⁶
DE 48, SE 56	Subelement 56: Valid Subfield 1 and Subfield 2 Value Combinations	Two
DE 48, SE 59	Subelement 59: Original Switch Serial Number	Two
DE 48, SE 59	Subfield 1: Original Switch Serial Number	Two
DE 48, SE 61	Subelement 61: POS Data, Extended Condition Codes	One
DE 48, SE 61	Subfield 1: Partial Approval Terminal Support Indicator	One
DE 48, SE 61	Subfield 2—Purchase Amount Only Terminal Support Indicator	One
DE 48, SE 61	Subfield 3—Real-time Substantiation Indicator	One
DE 48, SE 61	Subfield 4: Reserved for Future Use	One
DE 48, SE 61	Subfield 5: Final Authorization Indicator	One
DE 48, SE 63	Subelement 63: Trace ID	One
DE 48, SE 64	Subelement 64: Transit Program	One
DE 48, SE 64	Subfield 1: Transit Transaction Type Indicator	One
DE 48, SE 64	Subfield 2: Transportation Mode Indicator	One
DE 48, SE 65	Subelement 65: Terminal Compliant Indicator	One
DE 48, SE 65	Subfield 1: TLE Compliant	One
DE 48, SE 65	Subfield 2: UKPT/DUKPT Compliant	One
DE 48, SE 66	Subelement 66: Authentication Data	Two
DE 48, SE 66	Subfield 1: Program Protocol	Two
DE 48, SE 66	Subfield 2: Directory Server Transaction ID	Two

⁶ Transaction Data Category Two applies to any unspecified data element, subelement, or subfield.

Data Element	Data Element Description	Transaction Data Category ⁶
DE 48, SE 67	Subelement 67: MoneySend Information	Two
DE 48, SE 67	Subfield 01: Sanctions Score	Two
DE 48, SE 69	Subelement 69: Trace ID of Authorization Advice	Two
DE 48, SE 69	Subfield 1: Network Data	Two
DE 48, SE 69	Subfield 2: Date Settlement	Two
DE 48, SE 70	Subelement 70: Implied Decimal	Two
DE 48, SE 70	Subfield 1: Implied Decimal	Two
DE 48, SE 71	Subelement 71: On-behalf Service (OBS)	Two
DE 48, SE 71	Subfield 1: On-behalf Service Indicator	Two
DE 48, SE 71	Subfield 2: On-behalf Result 1	Two
DE 48, SE 71	Subfield 3: On-behalf Result 2	Two
DE 48, SE 71	Valid Subfield 1 and Subfield 2 Value Combinations	Two
DE 48, SE 72	Subelement 72: Issuer Chip Authentication	Two
DE 48, SE 74	Subelement 74: Additional Processing Information	One
DE 48, SE 74	Subfield 1: Process Indicator	One
DE 48, SE 74	Subfield 2: Processing Information	One
DE 48, SE 74	Valid Subfield 1 and Subfield 2 Value Combinations	One
DE 48, SE 75	Subelement 75: Fraud Scoring Data	Two
DE 48, SE 75	Subfield 1: Assessment Score	Two
DE 48, SE 75	Subfield 2: Score Reason Code	Two
DE 48, SE 75	Subfield 3: Rules Score	Two
DE 48, SE 75	Subfield 4: Rule Reason Code 1	Two
DE 48, SE 75	Subfield 5: Rule Reason Code 2	Two

⁶ Transaction Data Category Two applies to any unspecified data element, subelement, or subfield.

Data Element	Data Element Description	Transaction Data Category ⁶
DE 48, SE 76	Subelement 76: Mastercard Electronic Transaction Indicator	One
	Subfield 1: Mastercard Electronic Program	
DE 48, SE 76	Participation Level	One
DE 48, SE 77	Subelement 77: Transaction Type Identifier	One
DE 48, SE 79	Subelement 79: Chip CVR/TVR Bit Error Results Listing	Two
DE 48, SE 79	Subfield 1: CVR or TVR Identifier	Two
DE 48, SE 79	Subfield 2: Byte ID	Two
DE 48, SE 79	Subfield 3: Bit Identifier	Two
DE 48, SE 79	Subfield 4: Value of Bit in Error	Two
DE 48, SE 80	Subelement 80: PIN Service Code	Two
DE 48, SE 81	Subelement 81: Maestro PIN-less Program Indicator	Two
DE 48, SE 82	Subelement 82: Address Verification Service Request	One
DE 48, SE 82	Subfield 1: AVS Option Code	One
DE 48, SE 83	Subelement 83: Address Verification Service Response	One
DE 48, SE 83	Subfield 1: AVS Result Code	One
DE 48, SE 84	Subelement 84: Merchant Advice Code	One
DE 48, SE 84	Subfield 1: Merchant Advice Code	One
DE 48, SE 87	Subelement 87: Card Validation Code Result	One
DE 48, SE 87	Subfield 1: Code Values	One
DE 48, SE 88	Subelement 88: Magnetic Stripe Compliance Status Indicator	Two
DE 48, SE 88	Subfield 1: Monitoring Status	Two
DE 48, SE 89	Subelement 89: Magnetic Stripe Compliance Error Indicator	Two
DE 48, SE 89	Subfield 1: Data/Code Indicators	Two

⁶ Transaction Data Category Two applies to any unspecified data element, subelement, or subfield.

Data Element	Data Element Description	Transaction Data Category ⁶
DE 48, SE 90	Subelement 90: Lodging and Auto Rental Indicator	One
DE 48, SE 90	Subfield 1: Enrolled Program	One
DE 48, SE 92	Subelement 92: CVC 2 Value	One
DE 48, SE 92	Subfield 1: CVC 2 Value	One
DE 48, SE 93	Subelement 93: Airline Ticket Number	Two
DE 48, SE 93	Subfield 1: Ticket Number	Two
DE 48, SE 94	Subelement 94: Program Participation Indicator	One
DE 48, SE 94	Subfield 1: CVC 2 Validation Program Indicator	One
DE 48, SE 94	Subfield 2: QPS/Contactless Chargeback Eligibility	One
DE 48, SE 94	Subfields 3-20: Reserved for Future Use	One
DE 48, SE 95	Subelement 95: Mastercard Promotion Code	Two
DE 48, SE 95	Subfield 1: Surcharge-Free Alliance	Two
DE 48, SE 98	Subelement 98: Mastercard Corporate Fleet Card ID/Driver Number	Two
DE 48, SE 98	Subfield 1: ID/Driver Number	Two
DE 48, SE 99	Subelement 99: Mastercard Corporate Fleet Card Vehicle Number	Two
DE 48, SE 99	Subfield 1: Vehicle Number	Two
DE 49	DE 49: Currency Code, Transaction	One
DE 50	DE 50: Currency Code, Settlement	One
DE 51	DE 51: Currency Code, Cardholder Billing	One

⁶ Transaction Data Category Two applies to any unspecified data element, subelement, or subfield.

Data Element	Data Element Description	Transaction Data Category ⁶
DE 52	DE 52: Personal Identification Number (PIN) Data	One
DE 53	DE 53: Security Related Control Information	One
DE 54	DE 54: Additional Amounts	One
DE 54	Subfield 1: Account Type	One
DE 54	Subfield 2: Amount Type	One
DE 54	Subfield 3: Currency Code	One
DE 54	Subfield 4: Debit or Credit Indicator	One
DE 54	Subfield 5: Amount	One
DE 55	DE 55: Integrated Circuit Card (ICC) System-Related Data	One
DE 56	DE 56: Payment Account Data	One
DE 56	Subelement 01: Payment Account Data	One
DE 57	DE 57: Reserved for National Use	One
DE 58	DE 58: Authorizing Agent Institution ID	One
DE 59	DE 59: Reserved for National Use	One
DE 60	DE 60: Advice Reason Code	One
DE 60	Subfield 1: Advice Reason Code	One
DE 60	Subfield 2: Advice Reason Detail Code	One
DE 60	Subfield 3: Advice Reason Detail Text	One
DE 60	Subfield 4: Advice Reason Additional Text	One
DE 61	DE 61: Point of Service (POS) Data	One
DE 61	Subfield 1: POS Terminal Attendance	One
DE 61	Subfield 2: Reserved for Future Use	One

⁶ Transaction Data Category Two applies to any unspecified data element, subelement, or subfield.

Data Element	Data Element Description	Transaction Data Category ⁶
DE 61	Subfield 3: POS Terminal Location	One
DE 61	Subfield 4: POS Cardholder Presence	One
DE 61	Subfield 5: POS Card Presence	One
DE 61	Subfield 6: POS Card Retention	One
DE 61	Subfield 7: POS Transaction Status	One
DE 61	Subfield 8: POS Transaction Security	One
DE 61	Subfield 9: Reserved for Future Use	One
DE 61	Subfield 10: Cardholder-Activated Terminal Level	One
DE 61	Subfield 11: POS Card Data Terminal Input Capability	One
DE 61	Subfield 12: POS Authorization Life Cycle	One
DE 61	Subfield 13: POS Country Code (or Sponsored Merchant Information, if applicable)	One
DE 61	Subfield 14: POS Postal Code (or Sponsored Merchant Information, if applicable)	One
DE 62	DE 62: Intermediate Network Facility (INF) Data	Two
DE 63	DE 63: Network Data	Two
DE 63	Subfield 1: Financial Network Code	One
DE 63	Subfield 2: Interchange Rate Indicator	Two
DE 63	Subfield 3: Network Reference Number	Two
DE 63	Subfield 4: Banknet Reference Number	One

⁶ Transaction Data Category Two applies to any unspecified data element, subelement, or subfield.

Data Element	Data Element Description	Transaction Data Category ⁶
DE 63	Subfield 5: Acquirer's Reference Number	One
DE 63	Subfield 6: GCMS Processing Date and Cycle Number	Two
DE 70	DE 70: Network Management Information Code	Two
DE 90	DE 90: Original Data Elements	One
DE 90	Subfield 1: Message Type Identifier	One
DE 90	Subfield 2: System Trace Audit Number	One
DE 90	Subfield 3: Transmission Date and Time	One
DE 90	Subfield 4: Acquiring Institution ID Code	One
DE 90	Subfield 5: Forwarding Institution ID Code	One
DE 95	DE 95: Replacement Amounts	One
DE 95	Subfield 1: Actual Amount, Transaction	One
DE 95	Subfield 2: Actual Amount, Settlement	One
DE 95	Subfield 3: Actual Amount, Cardholder Billing	One
DE 95	Subfield 4: Zero-filled	One
DE 96	DE 96: Message Security Code	Two
DE 100	DE 100: Receiving Institution Identification Code	One
DE 101	DE 101: File Name	Two
DE 102	DE 102: Account Identification-1	One
DE 103	DE 103: Account Identification-2	One
DE 104	DE 104: Digital Payment Data	Two
DE 104	Subelement 1: Digital Payment Cryptogram	Two

⁶ Transaction Data Category Two applies to any unspecified data element, subelement, or subfield.

Data Element	Data Element Description	Transaction Data Category ⁶
DE 105	DE 105: Enhanced Identification Data	Two
DE 105	Subelement 1: AUA Code	Two
DE 105	Subelement 2: Sub-AUA Code	Two
DE 105	Subelement 3: Terminal Device ID	Two
DE 105	Subelement 4: Version	Two
DE 105	Subelement 5: Transaction Identifier	Two
DE 105	Subelement 6: Unique Device Code	Two
DE 105	Subelement 7: Fingerprint Device Code	Two
DE 105	Subelement 8: Reserved for future use	Two
DE 105	Subelement 9: Certificate Identifier	Two
DE 105	Subelement 10: PID Block Type	Two
DE 105	Subelement 11: Primary Biometric Type	Two
DE 105	Subelement 12: Product Code	Two
DE 105	Subelement 13: Return Value	Two
DE 105	Subelement 14: Authenticator Code	Two
DE 105	Subelement 15: Time Stamp	Two
DE 105	Subelement 16: Meta Information	Two
DE 105	Subelement 17: Error Code	Two
DE 105	Subelement 18: Internal Mastercard Use	Two
DE 105	Subelement 71: Internal Mastercard Use	Two
DE 106	DE 106: Private Data	Two
DE 107	DE 107: India On-Soil Data	Two

⁶ Transaction Data Category Two applies to any unspecified data element, subelement, or subfield.

Data Element	Data Element Description	Transaction Data Category ⁶
DE 107	Subelement 1: Private Biometric Data	Two
DE 107	Subelement 1: Accountholder UID Number	Two
DE 107	Subelement 2: License Key	Two
DE 107	Subelement 3: Public IP Address	Two
DE 107	Subelement 4: Key Identifier	Two
DE 107	Subelement 5: HMAC Value	Two
DE 107	Subelement 6: Encryption Key	Two
DE 107	Subelement 7: Location Type	Two
DE 107	Subelement 8: Location Value	Two
DE 108	DE 108: Additional Transaction Reference Data	Two
DE 108	Subelement 01: Receiver/Recipient Data	Two
DE 108	Subelement 02: Sender Data	Two
DE 108	Subelement 03: Transaction Reference Data	Two
DE 108	Subelement 04: Language Description	Two
DE 108	Subelement 05: Digital Account Information	Two
DE 108	Subfield 01: Digital Account Reference Number	Two
DE 108	Subfield 02: Mastercard Merchant Presented QR Receiving Account Number	Two
DE 108	Subelement 06: QR Dynamic Code Data	Two
DE 110	DE 110: Additional Data-2	Two
DE 110	Subelement 1: Merchant ID	Two
DE 110	Subelement 2: Program Registration ID	Two
DE 110	Subelement 3: PLUS Acquirer ID	Two

⁶ Transaction Data Category Two applies to any unspecified data element, subelement, or subfield.

Data Element	Data Element Description	Transaction Data Category ⁶
DE 110	Subelement 4: Visa International Fee Indicator	Two
DE 110	Subelement 5: Additional Card Information	Two
DE 110	Subelement 6: Business Service Arrangement	Two
DE 110	Subelement 7: Settlement Service	Two
DE 110	Subelement 8: Product ID	Two
DE 110	Encryption Data	Two
DE 111	DE 111: Amount, Currency Conversion Assessment	Two
DE 112	DE 112: Additional Data (National Use)	One
DE 112	Subelement 1: Time Payment Options	One
DE 112	Subelement 2: Parcelas Transaction	One
DE 112	Subelement 4: Credit Line Usage Fee (CLUF)	One
DE 112	Subelement 6: Financial Institution ID (FIID)	One
DE 112	Subelement 11: CDC Inquiry and Purchase	One
DE 112	Subelement 12: CDC Inquiry	One
DE 112	Subelement 13-16: CDC Inquiry	One
DE 112	Subelement 18: Post-Dated Transactions (preauthorization and completion)	One
DE 112	Subelement 19: Maestro Installment Transactions (preauthorization and completion)	Two
DE 112	Subelement 22: Positive ID Transactions	One

⁶ Transaction Data Category Two applies to any unspecified data element, subelement, or subfield.

Data Element	Data Element Description	Transaction Data Category ⁶
DE 112	Subelement 23: Positive ID Translation/Validation error code	One
DE 112	Subelement 24: Time-Based Transactions	One
DE 112	Subelement 25: Merchant Generated Installment Payments Transactions (preauthorization and completion)	One
DE 112	Subelement 26: Merchant Generated Installment Payments—Receipt Information	One
DE 112	Subelement 27: ATM Credit Card Cash Advance Installments	One
DE 112	Subelement 29: Brazil Merchant Tax ID	One
DE 112	Subelement 035: Issuer Fee Inquiry Indicator	Two
DE 112	Subelement 036: Issuer Fee Amount	Two
DE 120	DE 120: Record Data	Two
DE 121	DE 121: Authorizing Agent Identification Code	Two
DE 122	DE 122: Additional Record Data	Two
DE 123	DE 123—Mastercard Merchant Data Services	Two
DE 123	Subelement 1: Merchant Data Services	Two
DE 123	Subfield 1: Service Level Indicator	Two
DE 123	Subfield 2: Response Code	Two
DE 123	Subelement 2: Original Merchant Data	Two
DE 123	Subfield 1: POS Merchant Name	Two
DE 123	Subfield 2: Merchant City	Two
DE 123	Subfield 3: Merchant State, Province Code or RegionCode	Two

⁶ Transaction Data Category Two applies to any unspecified data element, subelement, or subfield.

Data Element	Data Element Description	Transaction Data Category ⁶
DE 123	Subelement 3vMastercard Merchant Data Cleansing Service	Two
DE 123	Subfield 1: Cleansed Merchant Street Address	Two
DE 123	Subfield 2: Cleansed Merchant Postal Code	Two
DE 123	Subfield 3: Cleansed Country Code	Two
DE 123	Subfield 4: Cleansed Merchant Telephone Number	Two
DE 123	Subelement 4: Merchant Data Cleansing Plus	Two
DE 123	Subfield 1: Cleansed Legal Corporate Name	Two
DE 123	Subfield 2: Sales Channel—Percentage Brick	Two
DE 123	Subfield 3: Sales Channel—Percentage Online	Two
DE 123	Subfield 4: Sales Channel—Percentage Other	Two
DE 123	Subfield 5: Cleansed Merchant URL	Two
DE 123	Subelement 5: Mastercard Merchant Data Advance	Two
DE 123	Subfield 1: Aggregate Merchant ID	Two
DE 123	Subfield 2: Aggregate Merchant Name	Two
DE 123	Subfield 3: Industry Code	Two
DE 123	Subfield 4: Super Industry Code	Two
DE 123	Subfield 5: Key Aggregate Merchant ID	Two
DE 123	Subfield 6: Channel Distribution ID	Two
DE 123	Subfield 7: NAICS Code	Two

⁶ Transaction Data Category Two applies to any unspecified data element, subelement, or subfield.

Data Element	Data Element Description	Transaction Data Category ⁶
DE 123	Subfield 8: Parent Aggregate Merchant ID	Two
DE 124	DE 124: Member-defined Data	Two
DE 124	Subfield 1: Unique Reference Number	Two
DE 124	Subfield 2: Sender/Payer Name/ User ID	Two
DE 124	Subfield 3: Sender/Payer Address	Two
DE 124	Subfield 4: Reserved Data	Two
DE 125	DE 125: New PIN Data	Two
DE 126	DE 126: Switch Private Data	Two
DE 126	Subfield 1: Settlement Service Data	Two
DE 126	Subfield 2: Single Message System Private Data-1	Two
DE 126	Subfield 3: Cross-Border Transaction Indicator	Two
DE 126	Subfield 4: Currency Indicator	Two
DE 126	Subfield 5: Fraud Score	Two
DE 126	Subfield 6: Fraud Risk Indicator	Two
DE 126	Subfield 7: Single Message System Private Data-2	Two
DE 127	DE 127: Processor Private Data	Two
DE 128	DE 128: Message Authentication Code (MAC)	Two

⁶ Transaction Data Category Two applies to any unspecified data element, subelement, or subfield.

Message types and applicable program or service

The following table specifies the permissible MTI values that may be used within each individual Authorization Platform message. The table key at the end of the table describes the symbols used under program and service columns.

MTI	Description	MC	NP	VI	TE	MS	CI
0100	Authorization Request	√	√	√	√	√	√
0110	Authorization Request Response	√	√	√	√	√	√
0120	Authorization Advice—Acquirer-generated	√	√	•	•	√	√
0120	Authorization Advice—Issuer-generated	√	√	√	√	√	√
0120	Authorization Advice—System-generated	√	√	√	√	√	√
0130	Authorization Advice Response—Issuer-generated	√	√	√	√	√	√
0130	Authorization Advice Response—System-generated	√	√	√	√	√	√
0180	Authorization Acknowledgement	o	o	o	o	o	o
0190	Authorization Negative Acknowledgement	√	√	√	√	√	√
0302	Issuer File Update Request	√	√	•	•	√	√
0312	Issuer File Update Request Response	√	√	•	•	√	√
0400	Reversal Request/0400	√	√	√	•	√	√
0410	Reversal Request Response/0410	√	√	√	•	√	√
0420	Reversal Advice	√	√	√	•	√	√
0430	Reversal Advice Response	√	√	√	•	√	√
0600	Administrative Request/0600	√	√	•	•	•	•
0610	Administrative Request Response/0610	√	√	•	•	•	•
0620	Administrative Advice	√	√	√	•	√	√
0630	Administrative Advice Response	√	√	√	•	√	√
0800	Network Management Request	√	√	√	√	√	√
0810	Network Management Request Response	√	√	√	√	√	√
0820	Network Management Advice	√	√	√	√	√	√

Table Key

√ = The MTI must be provided for the program or service indicated.

O = Optional support (for example, when individual Mastercard customers may elect to support or not support certain message types at their own discretion).

• = The MTI is unavailable for the program or service.

About primary and secondary bit maps

All Authorization Platform messages consist of one or two bit maps, each consisting of 64 bits. Bits are numbered from the left, starting with one. The first bit map contains bits one through 64; the second bit map contains bits 65 through 128.

A bit map is a series of 64 bits used to identify the presence or absence (denoted by 1 or 0) of each data element. The bit map is interpreted from left to right. The left-most bit represents bit number 1 in the Primary Bit Map and bit number 65 in the Secondary Bit Map. The rightmost bit represents bit number 64 in the Primary Bit Map and bit number 128 in the Secondary Bit Map.

Additional bit maps can be accommodated in the ISO 8583–1987 specification by using additional "Bit Maps, Extended" (setting the first bit in any bit map to 1 indicating the presence of a following extended bit map). However, Mastercard currently uses a maximum of two-bit maps (primary and secondary) in Authorization Platform messages, with a maximum number of message data elements ranging from DE 1 through DE 128. Consequently, DE 65 (the first bit in the Secondary Bit Map) must always be 0.

Primary Bit Map

The Primary Bit Map must always be present in a message. The most frequently used data elements are indexed from *DE 1 (Bit Map, Secondary)* through *DE 64 (Message Authentication Code [MAC])*. Infrequently used data elements are indexed from the *DE 66 (Settlement Code)* through *DE 128 (Message Authentication Code [MAC])* message only when data elements in the range *DE 66 (Settlement Code)* through *DE 128 (Message Authentication Code [MAC])* are present in the message.

Table 51: Attributes

Attribute	Description
Data Representation	b-64
Data Length	N/A
Data Field	Fixed length 8 bytes (for each bit map)
Subfields	N/A
Justification	N/A

NOTE: If both bit maps are present, the total length of the bit map data element is 16 bytes.

Usage

Following is the usage of the Primary Bit Map (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 52: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	M	P	M
Authorization Request Response/0110	M	P	M
Authorization Advice/0120: acquirer-generated	M	P	M
Authorization Advice/0120: system-generated	•	X	M
Authorization Advice Response/0130: issuer-generated	M	X	•
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	M	P	M
Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)	M	X	•
Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)	M	X	•
Authorization Response Acknowledgment/0180	M	X	•
Authorization Response Negative Acknowledgment/0190	•	X	M
Issuer File Update Request/0302	M	X	•
Issuer File Update Request Response/0312	•	X	M
Reversal Request/0400	M	P	M
Reversal Request Response/0410	M	P	M
Reversal Advice/0420	•	X	M
Reversal Advice Response/0430	M	X	•
Administrative Request/0600	M	P	M
Administrative Request Response/0610	M	P	M
Administrative Advice/0620: member-generated	M	P	M
Administrative Advice/0620: system-generated	•	X	M
Administrative Advice Response/0630	M	X	•
Network Management Request/0800: sign-on/sign-off	M	X	•

Message	Org	Sys	Dst
Network Management Request/0800: network connection status, member-generated	M	X	•
Network Management Request/0800: network connection status, system-generated	•	X	M
Network Management Request/0800: host session activation/deactivation	•	X	M
Network Management Request/0800: PEK exchange	•	X	M
Network Management Request/0800: PEK exchange, on demand	M	X	•
Network Management Request Response/0810: host session activation/deactivation	•	X	M
Network Management Request Response/0810: network connection status, member-generated	M	X	•
Network Management Request Response/0810: network connection status, system-generated	•	X	M
Network Management Request Response/0810: PEK exchange	M	X	•
Network Management Request Response/0810: PEK exchange-on demand	•	X	M
Network Management Request Response/0810: sign-on/sign-off	•	X	M
Network Management Advice/0820: PEK exchange	•	X	M

Values

Bits corresponding to mandatory data elements for a specific MTI must have a value of 1 to indicate the presence of the data element in the message. Otherwise, the Dual Message Authorization System rejects the message and returns it to the message originator using an Administrative Advice/0620, with the reject reason indicated in *DE 60 (Advice Reason Code)*.

Application notes

This data element is defined and used identically within all Mastercard programs and services.

DE 1 (Bit Map, Secondary)

DE 1 (Bit Map, Secondary) is a series of 64 bits that identifies the presence (1) or absence (0) of each data element in the second segment of a message. This would include *DE 65 (Bit Map, Extended)* through *DE 128 (Message Authentication Code [MAC])*.

Table 53: Attributes

Attribute	Description
Data Representation	b-8
Data Length	N/A
Data Field	Indicates the presence or absence of data elements 65–128
Subfields	N/A
Justification	N/A

Usage

Whether the data element is mandatory, conditional, optional, system-provided, or not required is noted for each applicable message.

Table 54: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	•	C
Authorization Request Response/0110	C	P	C
Authorization Advice/0120: acquirer-generated	C	P	C
Authorization Advice/0120: system-generated	•	X	C
Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)	C	X	•
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	C	P	C
Authorization Response Acknowledgment/0180	C	X	•
Authorization Response Negative Acknowledgment/0190	•	X	C
Issuer File Update Request/0302	M	X	•
Issuer File Update Request Response/0312	•	X	M
Reversal Request/0400	M	P	M

Message	Org	Sys	Dst
Reversal Request Response/0410	M	P	M
Reversal Advice/0420	•	X	M
Reversal Advice Response/0430	M	X	•
Administrative Request/0600	M	P	M
Administrative Request Response/0610	M	P	M
Administrative Advice/0620: system-generated	•	X	M
Administrative Advice/0620: member-generated	M	P	M
Administrative Advice Response/0630	M	X	•
Network Management Request/0800: sign-on/sign-off	M	X	•
Network Management Request/0800: host session activation/ deactivation	M	X	•
Network Management Request/0800: PEK exchange	•	X	M
Network Management Request/0800: PEK exchange, on demand	M	X	•
Network Management Request/0800: network connection status, member-generated	M	X	•
Network Management Request/0800: network connection status, system-generated	•	X	M
Network Management Request Response/0810: host session activation/ deactivation	•	X	M
Network Management Request Response/0810: network connection status, member-generated	M	X	•
Network Management Request Response/0810: network connection status, system-generated	•	X	M
Network Management Request Response/0810: PEK exchange	M	X	•
Network Management Request Response/0810: PEK exchange, on demand	•	X	M
Network Management Request Response/0810: sign-on/sign-off	•	X	M
Network Management Advice/0820: PEK exchange	•	X	M

Values

Bits corresponding to mandatory data elements for a specific MTI must have a value of 1 to indicate the presence of the data element in the message. Otherwise, the Dual Message Authorization System rejects the message and returns it to the message initiator using an Administrative Advice/0620, with the reject reason indicated in *DE 60 (Advice Reason Code)*.

Application notes

This data element is defined and used identically within all Mastercard programs and services.

DE 2 (Primary Account Number [PAN])

DE 2 (Primary Account Number [PAN]) is a series of digits used to identify a customer account or relationship.

Table 55: Attributes

Attribute	Description
Data Representation	n...19; LLVAR
Data Length	2 positions
Data Field	Contents of positions 1–19
Subfields	N/A
Justification	N/A

Usage

Whether the data element is mandatory, conditional, optional, system-provided, or not required is noted for each applicable message.

Table 56: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	M	X	M
Authorization Request Response/0110	ME	X	ME
Authorization Advice/0120: acquirer-generated	M	X	M
Authorization Advice/0120: system-generated	•	M	M
Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)	ME	ME	•
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	ME	X	ME
Issuer File Update Request/0302	C	C	•
Issuer File Update Request Response/0312	•	CE	CE
Reversal Request/0400	M	X	M

Message	Org	Sys	Dst
Reversal Request Response/0410	ME	X	ME
Reversal Advice/0420	.	M	M
Reversal Advice Response/0430	ME	ME	.
Administrative Request/0600	M	.	M
Administrative Request Response/0610	ME	.	ME
Network Management Request/0800: sign-on/sign-off	M	M	.
Network Management Request/0800: network connection status, member-generated	M	M	.
Network Management Request/0800: network connection status, system-generated	.	M	M
Network Management Request/0800: PEK exchange	.	M	M
Network Management Request/0800: PEK exchange, on demand	M	M	.
Network Management Request/0800: host session activation/deactivation	M	M	.
Network Management Request Response/0810: host session activation/deactivation	.	ME	ME
Network Management Request Response/0810: network connection status, member-generated	ME	ME	.
Network Management Request Response/0810: network connection status, system-generated	.	ME	ME
Network Management Request Response/0810: PEK exchange	ME	M	.
Network Management Request Response/0810: PEK exchange on demand	.	ME	ME
Network Management Request Response/0810: sign-on/sign-off	.	ME	ME
Network Management Advice/0820: PEK exchange	.	M	M

Values

Mastercard customer ID numbers, when used in this data element, must be valid values that Mastercard assigns.

DE 2 must be at least 11 digits and contain the variable-length prefix assigned to the specific Issuer ID Number (IIN) for Administrative Request/0600 messages.

Network Management Request/0800 messages may contain the Mastercard Group Sign-on ID only, the card prefix only, or both the Mastercard Group Sign-on ID followed by the card prefix in DE 2.

Application notes

This data element is defined and used identically within all Mastercard programs and services.

About Primary Account Number

DE 2 (Primary Account Number [PAN]) is a series of digits used to identify a customer account or relationship.

Primary Account Number

This data element is used for all numeric PANs up to 19 digits long when PAN is in Authorization/01xx messages, Reversal Advice/04xx messages, and Administrative Advice/06xx messages (when required). *DE 2* may contain a subset of this data within Network Management/08xx messages.

This data element data consists of three primary components:

- Issuing Institution Information (IIN)
- Individual Account ID Number
- PAN check digit

When *DE 2* contains a 59 BIN, *DE 20 (Primary Account Number Country Code)* also must be included in the message to uniquely identify the PAN. Note that 59 numbers are not guaranteed to be unique without an accompanying Country Code.

Specific requirements for PAN composition are described in the ISO 7812 and 7813 specifications. All PANs used in Dual Message Authorization System messages must conform to the ISO PAN encoding requirements, as specified in those specifications.

The Individual Account ID Number encoded or embossed on a card may be a cardholder ID number or a "master account number" related to one or more of the cardholder's accounts.

A card issuer may return the actual number of the cardholder account(s) affected by a transaction in an appropriate response message by using *DE 102 (Account ID-1)*, *DE 103 (Account ID-2)*, or both. However, the PAN used in the original Request message must then remain in *DE 2* for all subsequent messages related to the original request (such as Responses, Advices, Reversals, Chargebacks, Retrieval Requests, and Retrieval Fulfillments).

Issuing Institution Information

PAN may contain Issuing Institution Information (IIN) when required in Authorization/01xx, Issuer File Update/03xx, and Network Management/08xx messages. IIN may consist of either a Mastercard customer ID number, a variable-length issuer card prefix sequence, or a Mastercard assigned Group Sign-on ID as required by a specific message. Where card prefix information is required, the Dual Message Authorization System accommodates variable-length prefix sequences from one to 11 digits.

Where a Group Sign-on ID is required, the Dual Message Authorization System accommodates the five-digit Member Group ID, which Mastercard uses to identify a grouping of issuer account ranges that use the same transaction criteria for routing to the same destination route. More

than one Group Sign-on ID value may be defined for an account range if an issuer chooses to route authorization traffic to different destinations according to transaction criteria.

DE 3 (Processing Code)

DE 3 (Processing Code) describes the effect of a transaction on the customer account and the type of accounts affected.

Table 57: Attributes

Attribute	Description
Data Representation	n-6
Data Length	N/A
Data Field	Contents of subfields 1–3
Subfields	3
Justification	See subfields

Usage

Following is the usage of *DE 3* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 58: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	M	.	M
Authorization Request Response/0110	ME	.	ME
Authorization Advice/0120: acquirer-generated	M	.	M
Authorization Advice/0120: system-generated	.	M	M
Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)	ME	ME	.
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	ME	.	ME
Reversal Request/0400	M	.	M
Reversal Request Response/0410	ME	.	ME
Reversal Advice/0420	.	M	M

Message	Org	Sys	Dst
Reversal Advice Response/0430	ME	ME	.

Values

See subfields.

Application notes

Acquirers and prepaid card issuers in the U.S. region must support POS balance inquiry transaction type and functionality for all prepaid Mastercard credit and Debit Mastercard card account ranges.

The following cross edits occur.

Table 59: Authorization Request/0100 Messages edits

IF...	THEN...
DE 48 (Transaction Category Code) = C	the Cardholder Transaction Type Code must be 01, 17, or 30.
DE 48 (Transaction Category Code) = P	the Cardholder Transaction Type Code must be 28.
DE 48 (Transaction Category Code) = Z	the Cardholder Transaction Type Code must be 01, 30, 91, or 92.

Table 60: Reversal Request/0400 Messages edits

IF...	THEN...
<i>DE 48 (Transaction Category Code) = C</i>	the Cardholder Transaction Type Code must be 01, 17, or 30.
<i>DE 48 (Transaction Category Code) = P</i>	the Cardholder Transaction Type Code must be 28.
<i>DE 48 (Transaction Category Code) = Z</i>	the Cardholder Transaction Type Code must be 01, 30, 91, or 92.

Table 61: Authorization Advice/0120: acquirer-generated messages edits

IF...	THEN...
DE 3, subfield 1 contains one of the following values: 01 = Withdrawal 28 = Payment Transaction 30 = Balance Inquiry 40 = Account Transfer 91 = PIN Unblock 92 = PIN Change or 00 = Purchase and DE 48 (Transaction Category Code) = Z	the Dual Message Authorization System will return to the acquirer an Authorization Advice Response/0130 messages where: DE 39 = 30 DE 44 = 003

Table 62: For Account Status Inquiry Service transactions

WHEN...	THEN the Dual Message Authorization System...
the Authorization Request/0100 message contains DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status), value 8 (Account Status Inquiry Service [ASI]) and DE 3 (Processing Code) contains a value other than 00 (Purchase), 01 (Withdrawal), or 28 (Payment)	sends the acquirer an Authorization Request Response/0110 message where: • DE 39 (Response Code) = 30 • DE 44 (Additional Response Data) = 003

Subfield 1 (Cardholder Transaction Type Code)

DE 3, subfield 1 (Cardholder Transaction Type Code) describes the specific transaction type.

Table 63: Attributes

Attribute	Description
Data Representation	n-2
Data Field	Contents of positions 1–2

Attribute	Description
Justification	N/A

Values

Table 64: Cardholder Account Debits

Value	Description	MC	NP	VI	TE	MS	CI
00	Purchase	√	√	√	√	√	√
01	Withdrawal	√	√	√	√	√	√
02	Debit Adjustment		√				
09	Purchase with Cash Back	√	√			√	
10	Visa Only. Account Funding			√			
17	Cash Disbursement	√	√	√	√		
18	Scrip Issue		√				

Table 65: Cardholder Account Credits

Value	Description	MC	NP	VI	TE	MS	CI
20	Purchase Return/Refund	√	√	√		√	
21	Deposit		√				
22	Credit Adjustment		√				
28	Payment Transaction	√	√	√		√	√

Table 66: Cardholder Account Inquiries

Value	Description	MC	NP	VI	TE	MS	CI
30	Balance Inquiry	√	√	√		√	√

Table 67: Cardholder Account Transfers

Value	Description	MC	NP	VI	TE	MS	CI
40	Account Transfer	√	√				

Table 68: Reserved Values

Value	Description	MC	NP	VI	TE	MS	CI
90	Reserved for Future Use	√					

Table 69: PIN Management Transactions

Value	Description	MC	NP	VI	TE	MS	CI
91	PIN Unblock	√	√			√	√
92	PIN Change	√	√			√	√

Application notes

Value 20 is allowed for MDES related Authorization Request/0100 messages even if the account status is suspended or delete pending.

Value 20 is allowed for Visa gateway transactions to support the authorization of returns or refunds.

Value 20 is allowed for Mastercard and Debit Mastercard transactions and must be supported by Mastercard and Debit Mastercard issuers, to support the authorization of returns or refunds.

Effective with Release 20.Q2, value 20 must be supported by acquirers of Mastercard and Debit Mastercard transactions (support is optional for merchants).

Mastercard requires that issuers receive and respond to refund transactions involving a Mastercard, Debit Mastercard, or Maestro card.

Values 21–24 are not currently supported for use on the Dual Message Authorization System.

Value 30 is only applicable to eligible issuers.

Values 91 and 92 are only applicable to eligible acquirers and issuers.

Subfield 2 (Cardholder From Account Type Code)

DE 3, subfield 2 (Cardholder From Account Type Code) describes the cardholder account type affected for cardholder account debits and inquiries and the from account type for cardholder account transfer transactions.

Table 70: Attributes

Attribute	Description
Data Representation	n-2

Attribute	Description
Data Field	Contents of positions 3–4
Justification	N/A

Values

Subfield 2 must be one of the following values.

Table 71: Cardholder From Account type code

Value	Description	MC	NP	VI	TE	MS	CI
00	Default Account (not specified or not applicable)	✓	✓	✓	✓	✓	✓
10	Savings Account	✓	✓	✓		✓	✓
20	Checking Account	✓	✓	✓		✓	✓
30	Credit Card Account	✓	✓	✓		✓	✓
38	Credit Line Account		✓				
39	Corporate	✓					
40	Universal Account (Customer ID number)		✓				
50	Money Market Investment Account		✓				
60	Stored Value Account	✓					
90	Revolving Loan Account		✓				

Subfield 3 (Cardholder To Account Type Code)

DE 3, subfield 3 (Cardholder To Account Type Code) describes the cardholder account type affected for cardholder account credits and the To Account Type for cardholder account transfer transactions.

Table 72: Attributes

Attribute	Description
Data Representation	n-2
Data Field	Contents of positions 5–6
Justification	N/A

Values

Subfield 3 must be one of the following values.

Table 73: Cardholder To Account type code

Value	Description	MC	NP	VI	TE	MS	CI
00	Default Account (not specified or not applicable)	√	√	√	√	√	√
10	Savings Account		√			√	√
20	Checking Account		√			√	√
30	Credit Card Account	√	√	√			
38	Credit Line Account		√				
40	Universal Account		√				
50	Money Market Investment Account		√				
58	IRA Investment Account		√				
90	Revolving Loan Account		√				
91	Installment Loan Account		√				
92	Real Estate Loan Account		√				

DE 4 (Amount, Transaction)

DE 4 (Amount, Transaction) is the amount of funds the cardholder requested in the local currency of the acquirer or source location of the transaction.

Table 74: Attributes

Attribute	Description
Data Representation	n-12
Data Length	N/A
Data Field	Contents of positions 1–12
Subfields	N/A
Justification	Right with leading zeros

Usage

Following is the usage of *DE 4* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 75: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	M	.	M
Authorization Request Response/0110	CE	X	M
Authorization Advice/0120: acquirer-generated	M	.	M
Authorization Advice/0120: system-generated	.	X	M
Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)	ME	XE	.
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	ME	.	ME
Reversal Request/0400	M	.	M
Reversal Request Response/0410	ME	.	ME
Reversal Advice/0420	.	X	M
Reversal Advice Response/0430	ME	XE	.

Values

Valid numeric data must be present in this data element.

The value must be zero for:

- Authorization Request/0100 and Reversal Request/0400: Balance Inquiry, except where ATM transaction fees are allowed
- Authorization Request/0100: Chargeback Period Extension Request
- Authorization Request/0100 and Reversal Request/0400: PIN Management (PIN Unblock and PIN Change), except where ATM transaction fees are allowed
- Authorization Request/0100: Card Activation Request at POS
- Authorization Request/0100: Account Status Inquiry with Authentication Indicator

DE 28 (Amount, Transaction Fee) must be included in *DE 4* when *DE 28* is present in the message.

Application notes

DE 4 cannot exceed the region or country Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payment Transaction limit.

Table 76: System edits

WHEN...	THEN...
the issuer responds with <i>DE 39 (Response Code)</i> , value 10 (Partial approval) or value 87 (Purchase amount only, no cash back allowed), the issuer is not required to echo <i>DE 4</i>	the Dual Message Authorization System will forward to the acquirer, the Authorization Request Response/0110 message containing the partial approval amount in <i>DE 4</i> .

For Account Status Inquiry Service Transactions

Table 77: System edits

WHEN...	THEN the Dual Message Authorization System...
the Authorization Request/0100 message contains <i>DE 61 (Point-of-Service [POS] Data)</i> , subfield 7 (POS Transaction Status), value 8 (Account Status Inquiry Service [ASI]) and <i>DE 4 (Amount, Transaction)</i> contains a value greater than zero and <i>DE 3 (Processing Code)</i> contains a value of 00 (Purchase)	sends the acquirer an Authorization Request Response/0110 message where: <i>DE 39 (Response Code)</i> = 30 <i>DE 44 (Additional Response Data)</i> = 004

For Automated Fuel Dispenser (AFD) Transactions

The contents of *DE 4* may equal zero when the Authorization Advice/0120 message is sent by a Europe region acquirer for an AFD transaction in Europe and contains *DE 18 (Merchant Type)* = 5542 (Fuel Dispenser, Automated) and *DE 60 (Advice Reason Code)* = 191 (Acquirer Completed Authorization Transaction).

Issuers should be prepared to receive zero amount AFD completion advices (canceling a previously approved authorization) from any acquirer as there is no Dual Message Authorization System edit restricting such advices from Europe region acquirers.

For Authorization Chargeback Protection Period Extension Requests

Table 78: System edits

WHEN...	THEN the Dual Message Authorization System...
the acquirer sends an Authorization Chargeback Protection Period Extension Authorization Request/0100 message that contains the following: • <i>DE 3 (Processing Code)</i>, subfield 1 (Transaction Type), value 00 (Purchase) • <i>DE 4 (Transaction Amount)</i>, value 0 • <i>DE 48 (Additional Data)</i>, subelement 63 (Trace ID) is present • <i>DE 61</i>, subelement 4 (POS Cardholder Presence) is not equal to value 4 (Standing Order/Recurring) • <i>DE 61 (POS Data)</i>, subelement 7 (POS Transaction Status), value 4 (Preauthorized) and the issuer is not available to respond to the request	rejects the transaction where: <i>DE 39 (Response Code)</i> = 91 (Authorization System or issuer system inoperative)
the acquirer sends an Authorization Request/0100 message for an Authorization Chargeback Protection Period Extension request: • <i>DE 3 (Processing Code)</i>, subfield 1 (Transaction Type), value 00 (Purchase) • <i>DE 4 (Transaction Amount)</i>, value 0 • <i>DE 48 (Additional Data)</i>, subelement 63 (Trace ID) is NOT present • <i>DE 61</i>, subelement 4 (POS Cardholder Presence) is not equal to value 4 (Standing Order/Recurring), and • <i>DE 61 (POS Data)</i>, subelement 7 (POS Transaction Status), value 4 (Preauthorized)	rejects the transaction where: • <i>DE 39 (Response Code)</i> = 30 (Format Error) • <i>DE 44 (Additional Response Data)</i> = 0480nn (where nn is the subelement number)

WHEN...	THEN the Dual Message Authorization System...
the acquirer sends an Authorization Request/0100 message for an Authorization Chargeback Protection Period Extension request: • <i>DE 3 (Processing Code)</i>, subfield 1 (Transaction Type), value 00 (Purchase) • <i>DE 4 (Transaction Amount)</i>, value 0 • <i>DE 48 (Additional Data)</i>, subelement 63 (Trace ID) is present • <i>DE 61</i>, subelement 4 (POS Cardholder Presence) is not equal to value 4 (Standing Order/Recurring) • <i>DE 61 (POS Data)</i>, subelement 7 (POS Transaction Status), value 4 (Preauthorized) and if the values for Contactless Transit Aggregated, Transit Debt Recovery Transactions are present: • <i>DE 48 (Additional Data)</i>, subelement 64 (Transit Program), subfield 1 (Transit Transaction Type Individual) = 03 (Post-Authorized Aggregated), 04 (Authorized Aggregated Split Clearing), or 06 (Post-Authorized Aggregated Maestro)	rejects the transaction where: • <i>DE 39 (Response Code)</i> = 30 (Format Error) • <i>DE 44 (Additional Response Data)</i> = 061 (Point-of-Service [POS] Data)

WHEN...	THEN the Dual Message Authorization System...
the acquirer sends an Authorization Request/0100 message for an Authorization Chargeback Protection Period Extension request: • <i>DE 3 (Processing Code)</i>, subfield 1 (Transaction Type), value 00 (Purchase) • <i>DE 4 (Transaction Amount)</i>, value 0 • <i>DE 48 (Additional Data)</i>, subelement 63 (Trace ID) is present • <i>DE 61</i>, subelement 4 (POS Cardholder Presence) is not equal to value 4 (Standing Order/Recurring) • <i>DE 61 (POS Data)</i>, subelement 7 (POS Transaction Status), value 4 (Preauthorized) and if the values for Transit Debt Recovery Transactions are present: • <i>DE 48 (Additional Data)</i>, subelement 64 (Transit Program), subfield 1 (Transit Transaction Type Individual) = 07 (Debt Recovery) or if the values for an installment payment transaction are present: • <i>DE 48 (Additional Data)</i>, subelement 95 (Promotion Code)	rejects the transaction where: • <i>DE 39 (Response Code)</i> = 30 (Format Error) • <i>DE 44 (Additional Response Data)</i> = 004 (Transaction Amount)
the acquirer sends an Acquirer-generated Authorization Advice/0120 or Acquirer-generated Reversal Request/0400 message for an Authorization Chargeback Protection Period Extension request where the message contains the following: • <i>DE 3 (Processing Code)</i>, subfield 1 (Transaction Type), value 00 (Purchase) • <i>DE 4 (Transaction Amount)</i>, value 0 • <i>DE 48 (Additional Data)</i>, subelement 63 (Trace ID) is present • <i>DE 61</i>, subelement 4 (POS Cardholder Presence) is not equal to value 4 (Standing Order/Recurring) • <i>DE 61 (POS Data)</i>, subelement 7 (POS Transaction Status), value 4 (Preauthorized)	rejects the transaction where: • <i>DE 39 (Response Code)</i> = 30 (Format Error) • <i>DE 44 (Additional Response Data)</i> = 004 (Transaction Amount)

DE 5 (Amount, Settlement)

DE 5 (Amount, Settlement) is the amount of funds to be transferred between the acquirer and the issuer equal to *DE 4 (Amount, Transaction)* in the settlement currency. Mastercard programs and services use U.S. dollars as the currency of settlement.

Table 79: Attributes

Attribute	Description
Data Representation	n-12
Data Length	N/A
Data Field	Contents of positions 1–12
Subfields	N/A
Justification	Right with leading zeros

Usage

Following is the usage of *DE 5* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 80: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	•	X	C
Authorization Request Response/0110	CE	X	C
Authorization Advice/0120: acquirer-generated	•	X	C
Authorization Advice/0120: system-generated	•	X	C
Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)	CE	XE	•
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	CE	•	C
Authorization Advice Response/0130: system generated (responding to an acquirer-generated 0120)	CE	XE	•
Reversal Request/0400	•	X	C
Reversal Request Response/0410	CE	X	C
Reversal Advice/0420	•	X	C

Message	Org	Sys	Dst
Reversal Advice Response/0430	CE	XE	.

Values

This data element must contain valid numeric data.

Application notes

All settlement amounts are specified in U.S. dollars. The settlement amount in authorization messages should be interpreted as Amount, Reporting because settlement of funds (for example, accounting posting or memo-posting) does not currently occur online for authorization messages.

The Dual Message Authorization System will provide this data element if the customer chooses to receive settlement amount-related data elements.

When the issuer responds with *DE 39 (Response Code)*, value 10 (Partial approval) or value 87 (Purchase amount only, no cash back allowed), the issuer is not required to echo *DE 5* if it was present in the Authorization Request/0100 message to the issuer. The Dual Message Authorization System will provide the partial approval amount in *DE 5* of the Authorization Request Response/0110 to the acquirer if the acquirer chooses to receive settlement amount-related data elements.

The issuer may receive a Reversal Request/0400 message containing a different value in *DE 5* than was present in *DE 5* of the original Authorization Request/0100 message. This difference occurs when the currency conversion rate used by the Dual Message Authorization System changed between the time the Dual Message Authorization System processed the original Authorization Request/0100 and the time the Dual Message Authorization System processes the Reversal Request/0400 message.

If *DE 4* converts to an amount that is more than 12 digits long in the settlement currency in *DE 5*, the Dual Message Authorization System rejects the transaction with a format error in *DE 4*. When a customer receives this error code, it should verify that the transaction amount is correct.

This data will be present, as defined, except when the Dual Message Authorization System has declined an Authorization Request/0100 message and was unable to complete currency conversion processing.

DE 6 (Amount, Cardholder Billing)

DE 6 (Amount, Cardholder Billing) indicates the transaction amount in the issuer's currency. It is the amount billed to the cardholder in the cardholder account currency, excluding cardholder billing fees.

Table 81: Attributes

Attribute	Description
Data Representation	n-12
Data Length	N/A
Data Field	Contents of positions 1–12
Subfields	N/A
Justification	Right with leading zeros

Usage

Following is the usage of *DE 6* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 82: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	•	X	M
Authorization Request Response/0110	C	•	C
Authorization Advice/0120: acquirer-generated	•	X	M
Authorization Advice/0120: system-generated	•	X	M
Authorization Advice Response/0130: issuer-generated (responding to system-generated 0120)	ME	XE	•
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	CE	•	C
Reversal Request/0400	•	X	M
Reversal Request Response/0410	CE	X	C
Reversal Advice/0420	•	X	M
Reversal Advice Response/0430	ME	XE	•

Values

This data element must contain valid numeric data.

Application notes

The Dual Message Authorization System inserts this data element into all authorization and reversal messages. The Dual Message Authorization System also will insert the following data elements to indicate the:

- Conversion factor used: *DE 10 (Conversion Rate, Cardholder Billing)*
- Conversion date: *DE 16 (Date, Conversion)*
- Issuer's currency: *DE 51 (Currency Code, Cardholder Billing)*

Issuers must adhere to the following when providing a partial approval Authorization Request/0110 message:

- If *DE 39* is 87 (Purchase Amount Only, No Cash Back Allowed), *DE 6* must be less than the requested amount.
- If *DE 39* is 10 (Partial Approval) and transaction is Automated Fuel Dispenser, *DE 6* can be less, equal, or greater than requested amount.
-

Where a minor unit of currency applies, amounts are expressed in the minor unit of currency without a decimal separator (example, value 100 represents USD 1.00).

This data element is defined as a mandatory echo in Authorization Request Response/0110 messages except when the issuer responds with *DE 39 (Response Code)*, value 10 (Partial approval) or value 87 (Purchase amount only, no cash back allowed).

The Dual Message Authorization System forwards the partial approval amount in *DE 6* of the Authorization Request Response/0110 to the acquirer.

The issuer may receive a Reversal Request/0400 message containing a different value in *DE 6* than was present in *DE 6* of the original Authorization Request/0100 message. This difference occurs when the currency conversion rate used by the Dual Message Authorization System changed between the time the Dual Message Authorization System processed the original Authorization Request/0100 and the time the Dual Message Authorization System processes the Reversal Request/0400 message.

The issuer may also receive *DE 6* with a supplemental cardholder fee added for participants in the Issuer Currency Conversion Rate (ICCR) Service or Cross Border Fee Manager (CBFM) Service.

If *DE 4* converts to an amount that is more than 12 digits long in the settlement currency in *DE 5*, the Dual Message Authorization System rejects the transaction with a format error in *DE 4*. When a customer receives this error code, it should verify that the transaction amount is correct.

This data will be present, as defined, except when the Dual Message Authorization System has declined an Authorization Request/0100 message and was unable to complete currency conversion processing.

DE 7 (Transmission Date and Time)

DE 7 (Transmission Date and Time) is the date and time that a message is entered into the Mastercard Network. Date and time must be expressed in Coordinated Universal Time (UTC).

Table 83: Attributes

Attribute	Description
Data Representation	n-10
Data Length	N/A
Data Field	Contents of subfields 1–2
Subfields	2
Justification	See subfields

Usage

Following is the usage of *DE 7* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 84: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	M	•	M
Authorization Request Response/0110	ME	•	ME
Authorization Advice/0120: acquirer-generated	M	•	M
Authorization Advice/0120: system-generated	•	X	M
Authorization Advice Response/0130: issuer-generated (responding to system-generated 0120)	ME	XE	•
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	ME	•	ME
Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)	ME	XE	•
Authorization Response Acknowledgment/0180	ME	XE	•
Authorization Response Negative Acknowledgment/0190	•	XE	ME
Issuer File Update Request/0302	M	X	•
Issuer File Update Request Response/0312	•	XE	ME

Message	Org	Sys	Dst
Reversal Request/0400	M	.	M
Reversal Request Response/0410	ME	.	ME
Reversal Advice/0420	.	X	M
Reversal Advice Response/0430	ME	XE	.
Administrative Request/0600	M	.	M
Administrative Request Response/0610	ME	.	ME
Administrative Advice/0620: member-generated	M	.	M
Administrative Advice/0620: system-generated	.	X	M
Administrative Advice Response/0630	ME	XE	.
Network Management Request/0800: network connection status, member-generated	M	M	.
Network Management Request/0800: network connection status, system-generated	.	M	M
Network Management Request/0800: sign-on/sign-off	M	X	.
Network Management Request/0800: PEK exchange	.	X	M
Network Management Request/0800: PEK exchange, on demand	M	X	.
Network Management Request/0800: host session activation/deactivation	M	X	.
Network Management Request Response/0810 (network connection status, system-generated)	.	XE	ME
Network Management Request Response/0810: network connection status, member-generated	ME	XE	.
Network Management Request Response/0810: PEK exchange	ME	X	.
Network Management Request Response/0810: PEK exchange, on demand	.	XE	ME
Network Management Request Response/0810: host session activation/deactivation	.	XE	ME
Network Management Advice/0820: PEK exchange	.	X	M

Values

See subfields.

Application notes

This data element is defined and used identically within all Mastercard programs and services.

DE 7 must remain unchanged for all messages associated with a given system transaction, which includes all responses and acknowledgments related to an original request message (such as authorization, file update, administrative, network management). For example, the same *DE 7* is used in an Authorization Request/0100 and any related Authorization Request Response/0110, Authorization Advice/0120: acquirer-generated, Authorization Advice/0120: system-generated, or Reversal Advice/0420: system-generated message.

The Reversal Request/0400 message is the exception to this rule. Reversal Request/0400 messages are treated as an originating request and must have a unique *DE 7* value assigned. The *DE 7* value from the original 0100 request is included in *DE 90 (Original Data Elements)* for transaction matching purposes.

Each message initiator must assign a *DE 7* to each originating request message. The transmission time provided in *DE 7* must be within three minutes of the current time recorded by the Mastercard Network.

The combination of a message originator's *DE 11 (Systems Trace Audit Number [STAN])* and *DE 7* must uniquely identify any system transaction the message originator initiates on any given UTC day. These data elements together may be used as key data elements to identify and locate transaction records at some later time for the purpose of error resolution, settlement reconciliation, and retrieval requests.

Subfield 1 (Date)

DE 7, subfield 1 (Date) describes the valid date.

Table 85: Attributes

Attribute	Description
Data Representation	n-4
Data Field	Contents of positions 1–4
Justification	N/A

Values

This subfield must contain a valid date in MMDD format.

Subfield 2 (Time)

DE 7, subfield 2 (Time) describes the valid time.

Table 86: Attributes

Attribute	Description
Data Representation	n-6

Attribute	Description
Data Field	Contents of positions 5–10
Justification	N/A

Values

Time must contain a valid time in hhmmss format.

DE 8 (Amount, Cardholder Billing Fee)

DE 8 (Amount, Cardholder Billing Fee) is the fee the issuer bills to the cardholder in the same currency as *DE 6 (Amount, Cardholder Billing)*.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 9 (Conversion Rate, Settlement)

DE 9 (Conversion Rate, Settlement) is the factor used in the conversion from transaction to settlement amount. *DE 4 (Amount, Transaction)* is multiplied by *DE 9* to determine *DE 5 (Amount, Settlement)*.

Table 87: Attributes

Attribute	Description
Data Representation	n-8
Data Length	N/A
Data Field	Contents of subfields 1–2
Subfields	2
Justification	Right, excluding the decimal indicator that must be the leftmost digit.

Usage

Following is the usage of *DE 9* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 88: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	.	X	C
Authorization Request Response/0110	CE	X	C
Authorization Advice/0120: acquirer-generated	.	X	C
Authorization Advice/0120: system-generated	.	X	C
Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)	CE	XE	.
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	CE	.	CE
Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)	CE	XE	.
Reversal Request/0400	.	X	C
Reversal Request Response/0410	CE	X	C
Reversal Advice/0420	.	X	C
Reversal Advice Response/0430	CE	XE	.

Values

The leftmost digit must be in the range 0–7 and denotes the number of positions that the decimal point shall be moved from the right. Example: For a value of "69972522," conversion rate is 9.972522.

Application notes

The Dual Message Authorization System will insert this data element into a message if the customer chooses to receive settlement amount-related data elements. If the value in *DE 5 (Amount, Settlement)* is equal to the value in *DE 4 (Amount, Transaction)*, the settlement conversion rate is 61000000.

When used in Authorization/01xx messages, this data element should be interpreted as Conversion Rate, Reporting because settlement (for example, account posting or memo posting) of funds does not occur online with Authorization/01xx messages and Reversal Advice/04xx messages.

Note that when this data element is present in a message, *DE 5 (Amount, Settlement)* and *DE 50 (Currency Code, Settlement)* also are present.

The issuer may receive a Reversal Request/0400 message containing a different value in *DE 9* than was present in *DE 9* of the original Authorization Request/0100 message. This difference occurs when the currency conversion rate used by the Dual Message Authorization System

changed between the time the Dual Message Authorization System processed the original Authorization Request/0100 and the time the Dual Message Authorization System processes the Reversal Request/0400 message.

This data will be present, as defined, except when the Dual Message Authorization System has declined an Authorization Request/0100 or Reversal Request/0400 message and was unable to complete currency conversion processing.

Subfield 1 (Decimal Indicator)

DE 9, subfield 1 (Decimal Indicator) indicates the number of positions the decimal point should be moved from the right.

Table 89: Attributes

Attribute	Description
Data Representation	n-1
Data Field	Contents of position 1
Justification	N/A

Values

Must be in the range of 0–7.

Subfield 2 (Conversion Rate)

DE 9, subfield 2 (Conversion Rate) indicates the conversion rate.

Table 90: Attributes

Attribute	Description
Data Representation	n-7
Data Field	Contents of positions 2–8
Justification	N/A

Values

Must be a valid conversion rate.

DE 10 (Conversion Rate, Cardholder Billing)

DE 10 (Conversion Rate, Cardholder Billing) is the factor used in the conversion from transaction to cardholder billing amount. *DE 4 (Amount, Transaction)* is multiplied by *DE 10* to determine *DE 6 (Amount, Cardholder Billing)*.

Table 91: Attributes

Attribute	Description
Data Representation	n-8
Data Length	N/A
Data Field	Contents of subfields 1–2
Subfields	2
Justification	Right, excluding the decimal indicator that must be the leftmost digit.

Usage

Following is the usage of *DE 10* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 92: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	•	X	M
Authorization Request Response/0110	C	•	C
Authorization Advice/0120: acquirer-generated	•	X	M
Authorization Advice/0120: system-generated	•	X	M
Authorization Advice Response/0130: issuer-generated (responding to system-generated 0120)	ME	XE	•
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	ME	•	ME
Reversal Request/0400	•	X	M
Reversal Request Response/0410	CE	X	C
Reversal Advice/0420	•	X	M

Message	Org	Sys	Dst
Reversal Advice Response/0430	CE	XE	.

Values

Fixed length 8 positions; the leftmost decimal indicator denotes the number of positions that the decimal point shall be moved from the right. Example: For data element value "69972522," conversion rate is 9.972522. The leftmost digit must be in the range 0–7.

Application notes

The Dual Message Authorization System inserts this data element into all authorization, advice, and reversal messages. If the value in *DE 6 (Amount, Cardholder Billing)* is equal to the value in *DE 4 (Amount, Transaction)*, the cardholder billing conversion rate is 61000000.

Note that *DE 6 (Amount, Cardholder Billing)* and *DE 51 (Currency Code, Cardholder Billing)* also are present.

The issuer may receive a Reversal Request/0400 message containing a different value in *DE 10* than was present in *DE 10* of the original Authorization Request/0100 message. This occurs when the currency conversion rate used by the Dual Message Authorization System changed between the time the Dual Message Authorization System processed the original Authorization Request/0100 and the time the Dual Message Authorization System processes the Reversal Request/0400 message.

This data will be present, as defined, except when the Dual Message Authorization System has declined an Authorization Request/0100 or Reversal Request/0400 message and was unable to complete currency conversion processing.

Subfield 1 (Decimal Indicator)

DE 10, subfield 1 (Decimal Indicator) indicates the number of positions the decimal point should be moved from the right.

Table 93: Attributes

Attribute	Description
Data Representation	n-1
Data Field	Contents of position 1
Justification	N/A

Values

Must be in the range of 0 through 7.

Subfield 2 (Cardholder Billing Conversion Rate)

DE 10, subfield 2 (Cardholder Billing Conversion Rate) indicates the cardholder billing conversion rate.

Table 94: Attributes

Attribute	Description
Data Representation	n-7
Data Field	Contents of positions 2–8
Justification	N/A

Values

Must be a valid conversion rate.

DE 11 (System Trace Audit Number [STAN])

DE 11 (Systems Trace Audit Number [STAN]) is a number a message initiator assigns to uniquely identify a transaction.

Table 95: Attributes

Attribute	Description
Data Representation	n-6
Data Length	N/A
Data Field	Contents of positions 1–6
Subfields	N/A
Justification	N/A

Usage

Following is the usage of *DE 11* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 96: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	M	.	M
Authorization Request Response/0110	ME	.	ME
Authorization Advice/0120: acquirer-generated	M	.	M
Authorization Advice/0120: system-generated	.	X	M
Authorization Advice Response/0130: issuer-generated (responding to system-generated 0120)	ME	XE	.
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	ME	.	ME
Authorization Advice Response/0130:system-generated (responding to an acquirer-generated 0120)	ME	XE	.
Authorization Response Acknowledgment/0180	ME	XE	.
Authorization Response Negative Acknowledgment/0190	.	XE	ME
Issuer File Update Request/0302	M	X	.
Issuer File Update Request Response/0312	.	XE	ME
Reversal Request/0400	M	.	M
Reversal Request Response/0410	ME	.	ME
Reversal Advice/0420	.	X	M
Reversal Advice Response/0430	ME	XE	.
Administrative Request/0600	M	.	M
Administrative Request Response/0610	ME	.	ME
Administrative Advice/0620: member-generated	M	.	M
Administrative Advice/0620: system-generated	.	X	M
Administrative Advice Response/0630	ME	XE	.
Network Management Request/0800: network connection status, member-generated	M	X	.
Network Management Request/0800: network connection status, system-generated	.	X	M
Network Management Request/0800: sign-on/sign-off	M	X	.
Network Management Request/0800: PEK exchange	.	X	M
Network Management Request/0800: PEK exchange, on demand	M	X	.

Message	Org	Sys	Dst
Network Management Request/0800: host session activation/deactivation	M	X	•
Network Management Request Response/0810: network connection status, member-generated	ME	XE	•
Network Management Request Response/0810: network connection status, system-generated	•	XE	ME
Network Management Request Response/0810: sign-on/sign-off	•	XE	ME
Network Management Request Response/0810: PEK exchange	ME	X	•
Network Management Request Response/0810: PEK exchange, on demand	•	XE	ME
Network Management Request Response/0810: host session activation/deactivation	•	XE	ME
Network Management Advice/0820: PEK exchange	•	X	M

Values

This data element must not be all zeros or blanks.

Application notes

This data element is defined and used identically within all Mastercard programs and services.

DE 11 must remain unchanged for all messages associated with a given system transaction, which includes all responses and acknowledgments related to an original request message (such as authorization, file update, administrative, and network management). For example, the same *DE 11* is used in an Authorization Request/0100 and any related Authorization Request Response/0110, Authorization Advice/0120: acquirer-generated, Authorization Advice/0120: system-generated, or Reversal Advice/0420: system-generated message.

The Reversal Request/0400 message is the exception to this rule. Reversal Request/0400 messages are treated as an originating request and must have a unique *DE 11* value assigned. The *DE 11* value from the original 0100 request is included in *DE 90 (Original Data Elements)* for transaction matching purposes.

Each message initiator must assign *DE 11* to each originating request message.

The combination of a message originator's *DE 11* and *DE 7 (Transmission Date and Time)* must uniquely identify any system transaction the message originator initiates on any given UTC day. These data elements together may be used as key data elements to identify and locate transaction records at some later time for the purpose of error resolution, settlement reconciliation, and retrieval requests.

Acquirers that process more than 999,999 transactions within a UTC day may repeat the same number.

DE 12 (Time, Local Transaction)

DE 12 (Time, Local Transaction) is the local time at which the transaction takes place at the point of acceptor location.

Table 97: Attributes

Attribute	Description
Data Representation	n-6
Data Length	N/A
Data Field	Contents of positions 1–6
Subfields	N/A
Justification	N/A

Usage

Following is the usage of *DE 12* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 98: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	•	C
Authorization Advice/0120: acquirer-generated	C	•	C
Authorization Advice/0120: system-generated	•	C	C
Reversal Request/0400	C	•	C

Values

DE 12 must contain the local time the transaction occurred at the point of interaction, represented in the local time zone and not UTC, in hhmmss format. For reversals and advices, the value must match the original transaction.

If the transaction time is not known at the time of authorization (for example, in the case of a preauthorization request), the acquirer must use the authorization time in this field.

Application notes

This data element is defined and used identically within all Mastercard programs and services.

DE 12 is mandatory for all chip transactions. (If missing, the Dual Message Authorization System will not reject the message; however, a data integrity error will be reported.)

DE 12 is mandatory for all ATM transactions.

DE 12 is mandatory for all other card read transactions. (If missing, the Dual Message Authorization System will not reject the message.)

DE 12 is mandatory for Mastercard Hosted Mobile Phone Top-up transactions.

DE 12 is not required for MDES pre-digitization messages.

For more information, refer to the [MDES Technical Specifications for Dual and Single Message Systems](#).

DE 13 (Date, Local Transaction)

DE 13 (*Date, Local Transaction*) is the local month and day on which the transaction takes place at the point of acceptor location.

Table 99: Attributes

Attribute	Description
Data Representation	n-4
Data Length	N/A
Data Field	Contents of positions 1–4
Subfields	N/A
Justification	N/A

Usage

Following is the usage of DE 13 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 100: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Advice/0120: acquirer-generated	C	.	C
Authorization Advice/0120: system-generated	.	X	C
Reversal Request/0400	C	.	C

Message	Org	Sys	Dst
Reversal Advice/0420	.	X	C

Values

DE 13 must contain the local time the transaction occurred at the point of interaction, represented in the local time zone and not UTC, in hhmmss format. For reversals and advices, the value must match the original transaction.

If the transaction time is not known at the time of authorization (for example, in the case of a preauthorization request), the acquirer must use the authorization time in this field.

Application notes

This data element is defined and used identically within all Mastercard programs and services.

DE 13 is mandatory for all chip transactions (If missing, the Dual Message Authorization System will not reject the message; however, a data integrity error will be reported.)

DE 13 is mandatory for all ATM transactions.

DE 13 is mandatory for all other card read transactions. (If missing, the Dual Message Authorization System will not reject the message.)

DE 13 is mandatory for Mastercard Hosted Mobile Phone Top-up transactions.

DE 13 is not required for MDES pre-digitization messages.

For more information, refer to the [MDES Technical Specifications for Dual and Single Message Systems](#).

DE 14 (Date, Expiration)

DE 14 (Date, Expiration) specifies the year and month after which an issuer designates a cardholder's card to be expired.

Table 101: Attributes

Attribute	Description
Data Representation	n-4
Data Length	N/A
Data Field	Contents of positions 1–4
Subfields	N/A
Justification	N/A

Usage

Following is the usage of *DE 14* (whether it is mandatory, conditional, optional, system-provided, or not required) in applicable messages.

Table 102: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Advice/0120: acquirer-generated	C	.	C
Authorization Advice/0120: system-generated	.	C	C
Reversal Request/0400	C	.	C

Values

This data element must consist of a year and a month in YYMM format.

Application notes

This data element is defined and used identically within all Mastercard programs and services.

Table 103: DE 14 edits

IF...	THEN...
Track 1 or Track 2 (magnetic stripe) data is not present in the Authorization/01xx message	<i>DE 14</i> is conditional.
Track 1 or Track 2 data (magnetic stripe) data is present in the Authorization/01xx message for Mastercard and Visa transactions	<i>DE 14</i> is optional.
Track 1 or Track 2 (magnetic stripe) data is not present in the Authorization/01xx message for transactions other than Mastercard, Maestro, and Visa	<i>DE 14</i> is mandatory.
the expiration date is unavailable	<i>DE 14</i> must not be present (applies to Mastercard and Visa only).
an EMV transaction	<i>DE 14</i> must be populated with mandatory expiration date contained on the chip in TAG 5F24.
an MDES transaction	<i>DE 14</i> must be populated when provided by the merchant.

DE 15 (Date, Settlement)

DE 15 (Date, Settlement) is the date (month and day) that funds will be transferred between an acquirer and an issuer or an appropriate intermediate network facility (INF).

Table 104: Attributes

Attribute	Description
Data Representation	n-4
Data Length	N/A
Data Field	Contents of positions 1–4
Subfields	N/A
Justification	N/A

Usage

Following is the usage of *DE 15* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 105: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	•	X	M
Authorization Request Response/0110	ME	•	M
Authorization Advice/0120: acquirer-generated	•	X	M
Authorization Advice/0120: system-generated	•	X	M
Authorization Advice Response/0130: issuer-generated (responding to system-generated 0120)	ME	XE	•
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	ME	•	M
Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)	•	X	M
Reversal Request/0400	•	X	M
Reversal Request Response/0410	ME	X	M
Reversal Advice/0420	•	X	M

Message	Org	Sys	Dst
Reversal Advice Response/0430	ME	XE	.

Values

This data element must consist of a month and day in MMDD format.

Application notes

This data element is defined and used identically within all Mastercard programs and services.

For Authorization/01xx and Reversal Advice/04xx messages (in which there is no concurrent online settlement of funds), this data element is a date of network reporting. For example, the Dual Message Authorization System provides program- or service-specific financial network business date to inform acquirers and issuers of the reporting date to which a transaction applies. It should be interpreted as Date, Reporting.

In all cases the Dual Message Authorization System determines and inserts DE 15 in all originating Authorization/01xx and Reversal Advice/04xx messages.

The Dual Message Authorization System provides this date. Therefore, this data element must not be present in any originating request or advice messages. Any customer processing system or intermediate network facility (INF) must not change this date in any subsequent response message.

DE 16 (Date, Conversion)

DE 16 (Date, Conversion) indicates the effective date of *DE 9 (Conversion Rate, Settlement)* and also *DE 10 (Conversion Rate, Cardholder Billing)* whenever these data elements are present within a message.

Table 106: Attributes

Attribute	Description
Data Representation	n-4
Data Length	N/A
Data Field	Contents of positions 1-4
Subfields	N/A
Justification	N/A

Usage

Following is the usage of *DE 16* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 107: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	.	X	M
Authorization Request Response/0110	C	.	C
Authorization Advice/0120: acquirer-generated	.	X	M
Authorization Advice/0120: system-generated	.	X	M
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	CE	.	CE
Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)	ME	XE	.
Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)	ME	XE	.
Reversal Request/0400	.	X	M
Reversal Request Response/0410	CE	X	C
Reversal Advice/0420	.	X	M
Reversal Advice Response/0430	ME	XE	.

Values

This data element must consist of a month and day in MMDD format.

Application notes

This data element is defined and used identically within all Mastercard programs and services.

This data will be present, as defined, except when the Dual Message Authorization System has declined an Authorization Request/0100 message and was unable to complete currency conversion processing.

DE 17 (Date, Capture)

DE 17 (Date, Capture) is the month and day the acquirer processed the transaction data.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 18 (Merchant Type)

DE 18 (Merchant Type) is the classification (acceptor business code/merchant category code [MCC]) of the merchant's type of business or service.

Table 108: Attributes

Attribute	Description
Data Representation	n-4
Data Length	N/A
Data Field	Contents of positions 1–4
Subfields	N/A
Justification	N/A

Usage

Following is the usage of *DE 18* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 109: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	M	•	M
Authorization Advice/0120: acquirer-generated	M	•	M
Authorization Advice/0120: system-generated	•	X	M
Reversal Request/0400	M	•	M

Values

This data element must contain a valid MCC. Refer to the [Quick Reference Booklet](#) for valid codes.

Application notes

Table 110: Edits

IF...	THEN...
the transaction is a Mastercard Electronic card Authorization Request/0100 message that occurs at magnetic stripe-reading or chip-reading terminals when the MCC represents a non-face-to-face environment, such as MCC 5542: Fuel Dispenser, Automated when PIN is present or chip is present with CAT 1.	the Dual Message Authorization System sends the acquirer an Authorization Request Response/0100 where <i>DE 39 (Response Code)</i> = 58 (Transaction not permitted to acquirer/terminal).
the transaction is an Authorization Request/0100 or Reversal Request/0400 message and <i>DE 3 (Processing Code)</i> , subfield 1 (Cardholder Transaction Type Code) contains value 01 (Withdrawal) and <i>DE 18</i> is not 6010 (Member Financial Institution: Manual Cash Disbursements) or 6011 (Member Financial Institution: Automated Cash Disbursements)	the Dual Message Authorization System sends an Authorization Request Response/0110 or Reversal Request Response/0410 message where: • <i>DE 39</i> = 30 • <i>DE 44</i> = 018
the OFI is registered for the Mastercard Send Payments Transaction Program and is submitting an Authorization Request/0100 or Reversal Request/0400 message for a Mastercard Send Payment Transaction where: • <i>DE 3</i>, subfield 1 contains value 28 • <i>DE 18</i> does not contain an eligible MCC • <i>DE 48</i>, subelement 77 contains an eligible TTI • <i>DE 61</i>, subfield 10 criteria is met for Mastercard Send Payment Transactions – 0 = Not a CAT transaction – 1 = Authorized Level 1 CAT: Automated dispensing machine with PIN – 2 = Authorized Level 2 CAT: Self-service terminal – 6 = Authorized Level 6 CAT: Electronic commerce • <i>DE 108</i> is present	the Dual Message Authorization System sends the OFI an Authorization Request Response/0110 or Reversal Request Response/0410 message where: • <i>DE 39</i> = 30 (Format error) • <i>DE 44</i> = 018 (Merchant Type)

DE 19 (Acquiring Institution Country Code)

DE 19 (Acquiring Institution Country Code) is based on Mastercard franchise data information.

Table 111: Attributes

Attribute	Description
Data Representation	n-3
Data Length	N/A
Data Field	Contents of positions 1-3
Subfields	N/A
Justification	N/A

Usage

Whether the data element is mandatory, conditional, optional, system-provided, or not required is noted for each applicable message

Table 112: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	•	X	C
Authorization Advice/0120: acquirer-generated	•	X	C
Authorization Advice/0120: system-generated	•	X	C
Reversal Request/0400	•	X	C
Reversal Advice/0420	•	X	C

DE 20 (Primary Account Number [PAN] Country Code)

DE 20 (Primary Account Number [PAN] Country Code) is a code identifying the country where the card issuer is located.

Table 113: Attributes

Attribute	Description
Data Representation	n-3
Data Length	N/A
Data Field	Contents of positions 1–3
Subfields	N/A
Justification	N/A

Usage

Following is the usage of DE 20 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 114: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	•	C
Authorization Request Response/0110	CE	•	CE
Authorization Advice/0120: acquirer-generated	C	•	C
Authorization Advice/0120: system-generated	•	X	C
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	CE	•	CE
Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)	CE	XE	•
Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)	CE	XE	•
Reversal Request/0400	C	•	C
Reversal Request Response/0410	CE	•	CE
Reversal Advice/0420	•	X	C
Reversal Advice Response/0430	CE	XE	•

Message	Org	Sys	Dst
Network Management Request/0800: sign-on/sign-off	C	X	•

Values

Country Codes must be selected from the numeric ISO standard Country Codes. Refer to the [Quick Reference Booklet](#) for valid codes.

Application notes

This data element is defined and used identically within all Mastercard programs and services.

DE 20 is required in any Dual Message Authorization System message when the associated *DE 2 (Primary Account Number [PAN])* is present and begins with a 59 BIN. PANs beginning with a 59 are not guaranteed to be internationally unique without the use of this associated Country Code. When the BIN begins with 59, the country code entered in *DE 20* is identified as that in IPM table 40 issuer account range.

DE 21 (Forwarding Institution Country Code)

DE 21 (Forwarding Institution Country Code) is the code of the country where the forwarding institution is located.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 22 (Point-of-Service [POS] Entry Mode)

DE 22 (Point-of-Service [POS] Entry Mode) indicates the method used for PAN entry to initiate a transaction and the PIN entry capabilities.

Table 115: Attributes

Attribute	Description
Data Representation	n-3
Data Length	N/A
Data Field	Contents of subfields 1–2
Subfields	2
Justification	See subfields

Usage

Following is the usage of *DE 22* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 116: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	M	.	M
Authorization Advice/0120: acquirer-generated	M	.	M
Authorization Advice/0120: system-generated	.	X	M
Reversal Request/0400	M	.	M
Reversal Advice/0420	.	X	M

Values

See subfields.

Application notes

Following *DE 22*, subfield details are the [Dual Message Authorization System edits](#) that describe transaction-specific program and service edits for this data element.

PIN Management Transactions

Chip PIN Management: Only subfield 1, value 05 (PAN auto entry via chip) and subfield 2, value 1 (Terminal has PIN entry capability) are valid.

Magnetic Stripe PIN Management: subfield 1, values 02 and 90 are valid for magnetic stripe PIN change

All ATM Transactions

DE 22, subfield 2, must be value 1 (Terminal has PIN entry capability).

Short Message Service (SMS) Balance Inquiry Service

Issuers participating in the Short Message Service (SMS) Balance Inquiry service will receive *DE 22*, subfield 1 (POS Terminal PAN Entry Mode), value 81 (PAN entry via electronic commerce, including chip).

Subfield 1 (POS Terminal PAN Entry Mode)

DE 22, subfield 1 (POS Terminal PAN Entry Mode) indicates the method used for PAN entry to initiate a transaction.

Table 117: Attributes

Attribute	Description
Data Representation	n-2
Data Field	Contents of positions 1-2
Justification	N/A

Table 118: Values

Value	Description
00	Merchant terminal PAN entry mode unknown
01	Merchant terminal PAN manual entry
02	Merchant terminal PAN auto-entry via magnetic stripe
03	Merchant terminal PAN auto-entry via bar code reader
04	Merchant terminal PAN auto-entry via optical character reader (OCR)
05	Merchant terminal PAN auto-entry via integrated circuit card
07	Merchant terminal PAN auto-entry via contactless M/Chip
09	PAN/Token entry via electronic commerce containing DSRP cryptogram in DE 55 (<i>Integrated Circuit Card [ICC] System-Related Data</i>).
10	Credential on File

Value	Description
79	A hybrid terminal with an online connection to the acquirer failed in sending a chip fallback transaction (in which <i>DE 22</i>, subfield 1 = 80) to the issuer. or A hybrid terminal with no online connection to the acquirer failed to read the chip card. The merchant is prompted to read the magnetic stripe from the card, the magstripe is successfully read and indicates a service code 2XX (or 6XX if card is domestic). To complete the transaction in both cases, a voice transaction takes place during which the merchant communicates the PAN and the expiry date originating from the magstripe track 2 data to the acquirer. The acquirer then sends an online transaction to the issuer in which the value of <i>DE 22</i>, subfield 1 = 79 and in which <i>DE 61</i> subfield 11 indicate that the terminal is chip capable. Refer to the M/Chip Requirements for additional information.
80	Chip card at chip-capable terminal was unable to process transaction using data on the chip; therefore, the terminal defaulted to the magnetic stripe-read PAN. The full track data has been read from the data encoded on the card and transmitted within the Authorization Request/0100 in <i>DE 45</i> (Track 1 Data) or <i>DE 35</i> (Track 2 Data) without alteration or truncation. To use this value, the acquirer must be qualified to use value 90.
81	PAN/Token entry via electronic commerce with optional Identity Check-AAV in <i>DE 48</i> , subelement 43 (UCAF) or DSRP cryptogram in <i>DE 104</i> , subelement 001 (Digital Payment Cryptogram).
82	Electronic Commerce PAN Auto Entry via Server (issuer, acquirer, or third party vendor system).
90	PAN auto-entry via magnetic stripe—the full track data has been read from the data encoded on the card and transmitted within the authorization request in <i>DE 35</i> (Track 2 Data) or <i>DE 45</i> (Track 1 Data) without alteration or truncation.
91	PAN auto-entry via contactless magnetic stripe—the full track data has been transmitted by the acquirer within the authorization request in <i>DE 35</i> (Track 2 Data) or <i>DE 45</i> (Track 1 Data) and forwarded to the issuer without alteration or truncation.
95	Visa only. Chip card with unreliable Card Verification Value (CVV) data.

Subfield 2 (POS Terminal PIN Entry Mode)

DE 22 (Point-of-Service [POS] Entry Mode), subfield 2 (POS Terminal PIN Entry Mode) describes the capability of the terminal device to support/accept PIN entry.

Table 119: Attributes

Attribute	Description
Data Representation	n-1

Attribute	Description
Data Field	Contents of position 3
Justification	N/A

Table 120: Values

Value	Description
0	Unspecified or unknown
1	Terminal has PIN entry capability
2	Terminal does not have PIN entry capability
3	mPOS Software-based PIN entry capability
8	Terminal has PIN entry capability but PIN pad is not currently operative

Dual Message Authorization System edits

The Dual Message Authorization System performs edits on specific programs and services.

Edits are performed on the following programs and services:

- Electronic Commerce Transactions
- Mastercard Consumer Presented QR Transactions
- Chip Transactions
- Magnetic Stripe or Chip-Read Transactions for Mastercard Electronic Card
- Contactless Magnetic Stripe Transactions
- Credential on File Transactions

Electronic commerce transactions

The Dual Message Authorization System performs a cross-edit between *DE 22*, subfield 1 and *DE 48*, subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator), position 3 (UCAF Collection Indicator) and *DE 61* (Point-of-Service [POS] Data).

Table 121: Authorization Request/0100 message

WHEN....	THEN the Dual Message Authorization System...
<i>DE 22</i> , subfield 1 contains the value 81 (PAN entry via electronic commerce, including chip)	<i>DE 48</i> , subelement 42 must be present.

Mastercard Consumer Presented QR Transactions

The Dual Message Authorization System performs the following system edits to help ensure that acquirers supporting merchants that choose to submit tokenized Mastercard Consumer Presented QR transactions are properly using value 03 (PAN auto-entry via barcode reader) in *DE 22 (Point-of-Service [POS] Entry Mode)*, subfield 1 (POS Terminal PAN Entry Mode) in Authorization Request/0100 messages.

Table 122: Authorization Request/0100 message

WHEN...	THEN the Dual Message Authorization System...
an Authorization Request/0100 or Authorization Advice/0120: acquirer-generated message contains <i>DE 22 (Point-of-Service [POS] Entry Mode)</i>, subfield 1 (POS Terminal PAN Entry Mode), value 03 (PAN auto-entry via barcode reader) and <i>DE 48 (Additional Data: Private Use)</i>, subelement 21 (Acceptance Data), subfield 2 (Additional Terminal Capability Indicator) is present and does not contain any of the valid value combinations of 00, 01, 10, or 11	rejects the message and sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where: <i>DE 39 (Response Code)</i> = 30 (Format error) <i>DE 44 (Additional Response Data)</i> = 048021 (indicating the data element in error) NOTE: If <i>DE 48</i>, subelement 21, subfield 2 is not present, the Dual Message Authorization System will continue processing the message.
an Authorization Request/0100 message contains <i>DE 22 (Point-of-Service [POS] Entry Mode)</i>, subfield 1 (POS Terminal PAN Entry Mode), value 03 (PAN auto-entry via barcode reader) and <i>DE 18 (Merchant Type)</i> contains MCC 6011 (Member Financial Institution: Automated Cash Disbursements) or <i>DE 48 (Additional Data: Private Use)</i>, TCC (Transaction Category Code), contains value Z (ATM Cash Disbursement)	rejects the message and sends the acquirer an Authorization Request Response/0110 message where: <i>DE 39 (Response Code)</i> = 58 (Transaction not permitted to acquirer/terminal)
an Authorization Request/0100 message contains both of the following: <i>DE 22 (Point-of-Service [POS] Entry Mode)</i>, subfield 1 (POS Terminal PAN Entry Mode), value 03 (PAN auto-entry via barcode reader); and <i>DE 52 (Personal ID Number [PIN] Data)</i> <i>DE 110 (Encryption Data)</i>, Dataset 01 (PIN Encryption)	rejects the message and sends the acquirer an Authorization Request Response/0110 message where: <i>DE 39 (Response Code)</i> = 58 (Transaction not permitted to acquirer/terminal)

Chip transactions

If Mastercard determines through its Internal Chip Monitoring process that improperly formatted chip transactions are being submitted from acquirers not certified to send chip transactions, Mastercard will notify each acquirer before activating an edit.

For more information, refer to the [M/Chip Processing Services Service Description Guide](#).

Table 123: Authorization Request/0100 message

WHEN...	THEN the Dual Message Authorization System...
the Authorization Request/0100 message contains <i>DE 22</i> , subfield 1 = 05 or 07 or 09 and <i>DE 55</i> is present and Acquirer Chip testing level associated with the acquirer indicates the acquirer is approved for partial grade chip transaction	sends the acquirer an Authorization Request Response/0110 message where: • <i>DE 39 (Response Code)</i> = 30 (Format error) • <i>DE 44 (Additional Response Data)</i> = 055
the Authorization Request/0100 message contains <i>DE 22</i> , subfield 1 contains value 09 and <i>DE 55</i> is not present and Acquirer Chip Testing Level associated with the acquirer indicates the acquirer is approved to send full grade chip transactions	sends the acquirer an Authorization Request Response/0110 message where: • <i>DE 39 (Response Code)</i> = 30 (Format error) • <i>DE 44</i> = 055
the Authorization Request/0100 message contains <i>DE 22</i> , subfield 1 contains value 05 or 07 or 09 and <i>DE 55</i> may or may not be present and Acquirer Chip Testing Level associated with the acquirer indicates the acquirer is not permitted to send chip transactions	sends the acquirer an Authorization Request Response/0110 where: • <i>DE 39 (Response Code)</i> = 30 (Format error) • <i>DE 44 (Additional Response Data)</i> = 022
the Authorization Request/0100 message contains <i>DE 22</i> , subfield 1 contains value 09 and <i>DE 61</i> , subfield 4 does not contain value 5 and <i>DE 61</i> , subfield 10 does not contain value 6	sends the acquirer an Authorization Request Response/0110 where: • <i>DE 39 (Response Code)</i> = 30 (Format error) • <i>DE 44 (Additional Response Data)</i> = 061
the Authorization Request/0100 message contains <i>DE 22</i> , subfield 1 contains value 09 and <i>DE 61</i> , subfield 3 does not contain value 2, 4, or 8	sends the acquirer an Authorization Request Response/0110 where: • <i>DE 39 (Response Code)</i> = 30 (Format error) • <i>DE 44 (Additional Response Data)</i> = 061
<i>DE 55</i> is present in the Authorization Request/0100 message or the Authorization Advice/0120 message and <i>DE 22 (Point-of-Service [POS] Entry Mode)</i> , subfield 1 does not contain value 03, 05, 07, 09, or 81	rejects the Authorization Request/0100 message or Authorization Advice/0120 message where: • <i>DE 39 (Response Code)</i> = 30 (Format error) • <i>DE 44 (Additional Response Data)</i> = 055

WHEN...	THEN the Dual Message Authorization System...
<i>DE 22</i>, subfield 1 contains value 05 or 07 and <i>DE 55</i> may or may not be present and Acquirer Chip Testing Level associated with the acquirer indicates the acquirer is not permitted to send chip transactions	sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where: • <i>DE 39</i> = 30 (Format error) • <i>DE 44</i> = 022
<i>DE 22</i>, subfield 1, contains value 80 and the service code in <i>DE 35</i>—Track 2 Data is not 2xx or 6xx, indicating that it is not a chip card	replaces <i>DE 22</i>, subfield 1, value 80 with value 90 before forwarding the Authorization Request/0100 message to the issuer and Notifies the acquirer of this downgrade by populating <i>DE 48</i>, subelement 74 (Additional Processing Information) in the Authorization Request Response/0110 message where: • Subfield 1 = 90 (Chip Fallback Transaction Downgrade Process) • Subfield 2 = C (Completed Successfully)

Table 124: Authorization Advice/0120: acquirer-generated edit

WHEN...	THEN the Dual Message Authorization System...
<i>DE 22</i> (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) contains value 80 and the service code in <i>DE 35</i>—Track 2 Data is not 2xx or 6xx, indicating that it is not a chip card	Replaces <i>DE 22</i>, subfield 1, value 80 with value 90 before forwarding the Authorization Advice/0120 message to the issuer.

Contactless Magnetic Stripe Transactions

The Dual Message Authorization System will perform the following edit on authorization transactions that contain a contactless magnetic stripe account number that is participating in the Contactless Mapping Service.

Table 125: Authorization Request/0100 message

WHEN...	THEN the Dual Message Authorization System...
<i>DE 22</i>, subfield 1 is value 91 and <i>DE 35</i> or <i>DE 45</i> is not present NOTE: This edit is not performed on an Authorization Advice/0120 message or a Reversal Request/0400 message because track data is not expected for a Reversal Request/0400 message and is optional for an Authorization Advice/0120 message.	rejects the transaction and sends the acquirer an Authorization Request Response/0110 message where: • <i>DE 39</i> = 30 • <i>DE 44</i> = 022

Credential on File Transactions

The Dual Message Authorization System performs the following edits.

Table 126: Authorization Request/0100 message

WHEN...	THEN the Dual Message Authorization System...
an Authorization Request/0100 message contains: • <i>DE 22</i> (<i>Point-of-Service [POS] Entry Mode</i>), subfield 1 (POS Terminal PAN Entry Mode) = 10 (Credential on File), and • <i>DE 61</i> (<i>Point-of-Service [POS] Data</i>), subfield 4 (POS Cardholder Presence) = 5 (Cardholder not present [Electronic order]) and <i>DE 48</i>, subelement 42 is not present	sends an Authorization Request Response/0110 message where: • <i>DE 39</i> (<i>Response Code</i>) = 30 (Format error) • <i>DE 44</i> (<i>Additional Response Data</i>) = 048042

DE 23 (Card Sequence Number)

DE 23 (*Card Sequence Number*) distinguishes among separate cards having the same *DE 2* (*Primary Account Number [PAN]*) or *DE 34* (*Primary Account Number [PAN] Extended*). Issuers

may encode chip cards with Card Sequence Numbers. Acquirers with chip-reading capability may pass this information encoded on the chip in *DE 23* of Authorization Request/0100 messages.

Table 127: Attributes

Attribute	Description
Data Representation	n-3
Data Length	N/A
Data Field	Contents of positions 1–3
Subfields	N/A
Justification	N/A

Usage

Following is the usage of *DE 23* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 128: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	•	C
Authorization Request Response/0110	CE	•	C
Authorization Advice/0120: acquirer-generated	C	•	C
Authorization Advice/0120: system-generated	•	X	C
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	CE	•	CE
Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)	CE	XE	•
Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)	CE	XE	•
Reversal Request/0400	C	•	C
Reversal Request Response/0410	CE	•	C

Values

Acquirers:

DE 23 must be three positions in the numeric range 000–099. For contact chip or contactless chip transactions where *DE 22* subfield 1 is 05 or 07 respectively, *DE 23* must contain the value of the Application PAN Sequence Number (EMV tag 5F34) as personalized on the chip. Considering that the Application PAN Sequence Number (EMV tag 5F34) is defined as a one-byte numeric value in the EMV specification, the terminal or the acquirer software must convert this one-byte numeric value to a three-byte value with leading zeros in *DE 23*. For example, if the value of EMV tag 5F34 on the chip card is 2, then the value of *DE 23* is 002. For chip or contactless chip transactions, if EMV tag 5F34 is not personalized (not present) on the chip, *DE 23* cannot be present in the authorization message.

When *DE 23* (Card Sequence Number) is present in an Authorization Request/0100 message and the value is not in the range of 000–099, then the Dual Message Authorization System forwards the Authorization Request Response/0110 where *DE 39* = 30 (Format Error) and *DE 44* (Additional Response Data) = 023.

If the account range is registered for the Authentication Indicator Type 1 Service, then *DE 23* should be present.

Issuers:

The Dual Message Authorization System does not cross-edit the presence or absence of *DE 23* and the value in *DE 22* (Point-of-Service [POS] Entry Mode) in Authorization Request/0100 messages. However, if *DE 22* has a value 05 (PAN auto-entry via chip) or 07 (PAN auto-entry via contactless M/Chip) and if *DE 23* is present in the Authorization Request/0100, *DE 23* contains the card sequence number from the chip. If *DE 22* has a value other than 05 or 07 and *DE 23* is present, then *DE 23* may contain erroneous information unrelated to the chip transaction.

Because of the potential for *DE 23* to be present in Authorization/01xx messages, chip issuers must be prepared to receive store-and-forward (SAF) records containing *DE 23*.

Issuers must not return *DE 23* in an Authorization Request Response/0110. If an issuer does so, Mastercard will delete *DE 23* before passing the Authorization Request Response/0110 to the acquirer.

Brazil-issued account ranges with Flex Card Service enabled in Brazil, for acquirers

The third position in *DE 23* will determine the routing of a Flex Card service enabled Point of Sale (POS) transaction acquired outside of Brazil, as follows:

- Values 0-4 indicates that the cardholder has selected the credit option. The transaction will be routed and processed as credit.
- Values 5-9 indicates that the cardholder has selected the debit option for processing; debit or credit interchange may apply to the transaction. Acquirers may be required to submit the corresponding interchange if Debit is selected and differentiated from credit for processing and interchange:
 - C = Credit Transaction if *DE 3*, subfield 2 is not equal to 10 or 20.
 - D = Debit Transaction if *DE 3*, subfield 2 is equal to 10 or 20.

Application notes

NOTE: The following feature only applies to tokens that use a secure element or cloud token type.

DE 23 is not provided unless the issuer opts to receive it.

For more information, refer to the [Mastercard Digital Enablement Service Issuer Implementation Guide](#).

DE 24 (Network International ID)

DE 24 (*Network International ID*) identifies a single international network of card issuers.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 25 (Point-of-Service [POS] Condition Code)

DE 25 (*Point-of-Service [POS] Condition Code*) is an ID of the condition under which the transaction takes place at the point of interaction.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 26 (Point-of-Service (POS) Personal ID Number [PIN] Capture Code)

DE 26 (*Point-of-Service [POS] Personal ID Number [PIN] Capture Code*) indicates the technique, maximum number, or both of PIN characters that can be accepted by the POS terminal used to construct the PIN data.

Table 129: Attributes

Attribute	Description
Data Representation	n-2
Data Length	N/A
Data Field	Contents of positions 1–2
Subfields	N/A
Justification	Right

Usage

Following is the usage of *DE 26* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 130: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Advice/0120: system-generated	.	C	C

Values

Table 131: Applicable values

Values	Description
00–03	Invalid
04–12	Indicates the maximum number of PIN characters that the terminal can accept
13–99	Reserved

Application notes

This data element is defined and used identically within all Mastercard programs and services.

DE 26 must be used to indicate the maximum number of PIN characters that the acquiring terminal device (for example; ATM and POS terminal) is capable of accepting.

The Dual Message Authorization System requires that this data element be included in an Authorization Request/0100 message only when *DE 52* (Personal ID Number [PIN] Data) or *DE 110* (Encryption Data), Dataset 01 (PIN Encryption) is present and the maximum PIN character acceptance capability of the terminal is known to be other than 12 digits.

NOTE: This data element is not used to specify the number of PIN characters actually accepted by a POS terminal device.

DE 27 (Authorization ID Response Length)

DE 27 (Authorization ID Response Length) is the maximum length of the authorization response that the acquirer can accommodate. The issuer or its agent is expected to limit response to this length.

NOTE: The Dual Message Authorization System currently does not use this data element.

ISO and Mastercard defines *DE 38 (Authorization ID Response)* to be a six-character data element. All acquirers and issuers are expected to be able to accommodate and use the six-character data element for all authorization ID response codes.

DE 28 (Amount, Transaction Fee)

DE 28 (Amount, Transaction Fee) is the fee charged (for example, by the acquirer) for transaction activity in the currency of DE 4 (Amount, Transaction).

Table 132: Attributes

Attribute	Description
Data Representation	an-9
Data Length	N/A
Data Field	N/A
Subfields	2
Justification	See subfields

Usage

Following is the usage of *DE 28* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 133: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	•	C
Authorization Request Response/0110	CE	X	CE
Authorization Advice/0120: acquirer-generated	C	•	C
Authorization Advice/0120: system-generated	•	X	C

Message	Org	Sys	Dst
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	CE	XE	•
Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)	CE	XE	•
Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)	•	XE	CE
Reversal Request/0400	C	•	C
Reversal Request Response/0410	CE	X	CE
Reversal Advice/0420	•	X	C
Reversal Advice Response/0430	CE	XE	•

Values

This data element may be present in a message whenever an online transaction fee is permitted by the operating rules of a bank card product.

The credit or debit indicator (the first position of the data element) applies to the message recipient. Within acquirer-generated message types, a D (debit) fee amount indicates that the fee is to be applied as a debit to the message recipient, the issuer, (and therefore as a credit to the message originator, the acquirer).

Application notes

The Dual Message Authorization System does not support real-time settlement of transaction fees or processing fees concurrently with each individual transaction. All such fees for each financial network are calculated and settled daily, upon completion of each financial network business day, for all customers.

NOTE: For acquirers that are approved to levy transaction fees, DE 28 must contain the transaction fee amount, and this amount also must be added to the requested amount contained in DE 4 (Amount, Transaction).

Table 134: Errors

WHEN...	THEN the Dual Message Authorization System...
DE 28 is not provided in an Authorization Request Response/0110 or Reversal Request Response/0410 or is provided but it contains a value different from the original request	populates the DE 28 value from the original Authorization Request/0100 or Reversal Request/0400 message in the response message before sending to the acquirer.

WHEN...	THEN the Dual Message Authorization System...
an Authorization Request/0100 message or Reversal Request/0400 message contains <i>DE 28</i> , and subfield 1 is not C or D or if subfield 2 is zeros or is greater than <i>DE 4</i>	forwards to the acquirer an Authorization Request Response/0110 message or Reversal Request Response/0410 where <i>DE 39</i> = 30 (Format Error) and <i>DE 44</i> = 028.
an acquirer sends an Authorization Advice/0120 message and <i>DE 28</i> is not provided in an Authorization Advice Response/0130 message, or is provided, but it contains a value different from the original request	populates the <i>DE 28</i> value from the Authorization Advice/0120 message in the response message before sending to the acquirer.

Subfield 1 (Debit/Credit Indicator)

DE 28, subfield 1 (Debit/Credit Indicator) indicates the program type.

Table 135: Attributes

Attribute	Description
Data Representation	a-1
Data Field	Contents of position 1
Justification	N/A

Table 136: Values

Value	Description
C	Credit
D	Debit

Table 137: Application notes

WHEN an acquirer sends an Authorization Advice/0120 message and...	THEN the Dual Message Authorization System...
DE 28, subfield 1 (Debit/Credit Indicator) contains a value other than C (Credit) or D (Debit)	forwards to the acquirer an Authorization Advice Response/0130 message where: • DE 39 (Response Code) = 30 (Format error). • DE 44 (Additional Response Data) = 028 (indicating the data element in error).

Subfield 2 (Amount)

DE 28, subfield 2 (Amount) indicates the fee amount in the currency of DE 4 (Amount, Transaction).

Table 138: Attributes

Attribute	Description
Data Representation	n-8
Data Field	Contents of positions 2–9
Justification	Right with leading zeros

Values

The fee amount must not contain all zeros or be greater than DE 4 (Amount, Transaction).

DE 29 (Amount, Settlement Fee)

DE 29 (Amount, Settlement Fee) is the fee to be transferred between the acquirer and the issuer equal to DE 28 (Amount, Transaction Fee) in the currency of DE 5 (Amount, Settlement).

NOTE: The Dual Message Authorization System currently does not use this data element.

The Dual Message Authorization System does not support real-time settlement of transaction fees or processing fees concurrently with each individual transaction. All such fees for each financial network are calculated and settled daily, upon completion of each financial network's business day, for all customers.

Subfield 1 (Debit/Credit Indicator)

DE 29, subfield 1 (Debit/Credit Indicator) indicates the program type.

Attributes

Attribute	Description
Data representation	a-1
Data Field	Contents of position 1
Justification	N/A

Values

Value	Description
C	= Credit
D	= Debit

Subfield 2 (Amount)

DE 29, subfield 2 (Amount) indicates the fee amount in the currency of DE 5 (Amount, Settlement).

Attributes

Attribute	Description
Data representation	n-8
Data field	Contents of positions 2–9
Justification	Right with leading zeros

Values

Fee amount in the currency of DE 5 (Amount, Settlement).

DE 30 (Amount, Transaction Processing Fee)

DE 30 (Amount, Transaction Processing Fee) is the fee charged (for example, by the acquirer, issuer, or INF) for the handling and routing of messages in the currency of *DE 4 (Amount, Transaction)*.

NOTE: The Dual Message Authorization System currently does not use this data element.

The Dual Message Authorization System does not support real-time settlement of transaction fees or processing fees concurrently with each individual transaction. All such fees for each financial network are calculated and settled daily, upon completion of each financial network's business day, for all customers.

Subfield 1 (Debit/Credit Indicator)

DE 30, subfield 1 (Debit/Credit Indicator) indicates the program type.

Attributes

Attribute	Description
Data representation	a-1
Data field	Contents of position 1
Justification	N/A

Values

Value	Description
C =	Credit
D =	Debit

Subfield 2 (Amount)

DE 30, subfield 2 (Amount) indicates the fee amount in the currency of DE 4 (Amount, Transaction).

Attributes

Attribute	Description
Data representation	n-8
Data field	Contents of positions 2-9
Justification	Right with leading zeros

Values

Fee amount in the currency of DE 4 (Amount, Transaction).

DE 31 (Amount, Settlement Processing Fee)

DE 31 (*Amount, Settlement Processing Fee*) is the fee charged (for example, by the acquirer, issuer, or INF) for the handling and routing of messages in the currency of DE 5 (*Amount, Settlement*).

NOTE: The Dual Message Authorization System currently does not use this data element.

The Dual Message Authorization System does not support real-time settlement of transaction fees or processing fees concurrently with each individual transaction. All such fees for each financial network are calculated and settled daily, upon completion of each financial network's business day, for all customers.

Subfield 1 (Debit/Credit Indicator)

DE 31, subfield 1 (Debit/Credit Indicator) indicates the program type.

Attributes

Attribute	Description
Data representation	a-1
Data field	Contents of position 1
Justification	N/A

Values

Value	Description
C =	Credit
D =	Debit

Subfield 2 (Amount)

DE 31, subfield 2 (Amount) indicates the fee amount in the currency of DE 5 (*Amount, Settlement*).

Attributes

Attribute	Description
Data representation	n-8
Data field	Contents of positions 2–9

Attribute	Description
Justification	Right with leading zeros

Values

Fee amount in the currency of DE 5 (Amount, Settlement).

DE 32 (Acquiring Institution ID Code)

DE 32 (Acquiring Institution ID Code) identifies the acquiring institution (for example, merchant bank) or its agent.

Table 139: Attributes

Attribute	Description
Data Representation	n-6
Data Length	2 positions
Data Field	Contents of positions 1–6
Subfields	N/A
Justification	N/A

Usage

Following is the usage of *DE 32* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 140: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	M	.	M
Authorization Request Response/0110	ME	.	ME
Authorization Advice/0120: acquirer-generated	M	.	M
Authorization Advice/0120: system-generated	.	X	M
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	ME	.	ME
Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)	ME	XE	.

Message	Org	Sys	Dst
Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)	ME	XE	.
Reversal Request/0400	M	.	M
Reversal Request Response/0410	ME	.	ME
Reversal Advice/0420	.	X	M
Reversal Advice Response/0430	ME	XE	.
Administrative Request/0600	M	.	M
Administrative Request Response/0610	ME	.	ME

Values

A Mastercard customer ID number that Mastercard assigned to the entity acting as the acquiring institution for a transaction.

Application notes

This data element is defined and used identically within all Mastercard programs and services.

DE 32 must contain a six-digit customer ID number assigned by Mastercard that identifies the institution acting as the "acquiring bank" or "merchant bank" for a transaction.

When an institution acts as the customer or intermediate network facility (INF) for an acquirer, the institution must:

- Provide the acquirer's Mastercard-assigned six-digit customer ID in DE 32
- Provide its Mastercard-assigned six-digit customer ID in DE 33 (*Forwarding Institution ID Code*)

The Mastercard customer ID number must be set up in the Dual Message Authorization System for participants to process Administrative Request/06xx messages.

Table 141: Errors

IF...	THEN the Dual Message Authorization System...
in a Mastercard transaction, the acquirer submits a transaction with a value of 90 in DE 22 (<i>Point-of-Service [POS] Entry Mode</i>)	DE 45 (<i>Track 1 Data</i>) or DE 35 (<i>Track 2 Data</i>) must be present, and DE 32 must contain the proper Mastercard customer ID number.
the Authorization Request/0100, Authorization Advice/0120, or Reversal Request/0400 message contains DE 32 and DE 32 is not 6 digits in length	forwards the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 where DE 39 = 30 (Format Error) and DE 44 = 032.

For bridged transactions acquired on the Single Message System, issuers will receive the 6-digit customer ID number assigned by Mastercard that identifies the institution acting as the "acquiring bank" or "merchant bank" for a transaction in *DE 32 (Acquiring Institution Identification Code)* of the authorization message instead of a Mastercard-assigned internal ICA.

NOTE: Transactions that have passed between platforms (Single Message System to Dual Message Authorization System, Dual Message Authorization System to Single Message System) are referred to as bridged transactions.

Visa Transactions

Acquirers must use their Mastercard-assigned customer ID in *DE 32* of Visa-branded authorizations, and must also provide their Visa-assigned acquirer BIN to the Global Customer Service team.

DE 33 (Forwarding Institution ID Code)

DE 33 (Forwarding Institution ID Code) identifies the institution forwarding a Request or Advice message in an interchange system if it is not the same institution as specified in *DE 32 (Acquiring Institution ID Code)*. *DE 33* is used within a message to contain the Mastercard six-digit customer ID number of the customer or INF responsible for directly routing that message to the Dual Message Authorization System.

Table 142: Attributes

Attribute	Description
Data Representation	n-6
Data Length	2 positions
Data Field	Contents of positions 1–6
Subfields	N/A
Justification	N/A

Usage

Following is the usage of *DE 33* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 143: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Request Response/0110	CE	.	CE
Authorization Advice/0120: acquirer-generated	C	.	C
Authorization Advice/0120: system-generated	.	X	C
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	CE	.	CE
Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)	CE	XE	.
Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)	CE	XE	.
Issuer File Update Request/0302	M	X	.
Issuer File Update Request Response/0312	.	XE	ME
Reversal Request/0400	C	.	C
Reversal Request Response/0410	CE	.	CE
Reversal Advice/0420	.	X	C
Reversal Advice Response/0430	CE	XE	.
Administrative Request/0600	C	.	C
Administrative Request Response/0610	CE	.	CE
Administrative Advice/0620: system-generated	.	X	M
Administrative Advice/0620: member-generated	M	.	M
Administrative Advice Response/0630	ME	XE	.
Network Management Request/0800: network connection status, system-generated	.	X	M
Network Management Request/0800: network connection status, member-generated	M	X	.
Network Management Request/0800: PEK exchange	M	X	.
Network Management Request/0800: PEK exchange, on demand	.	X	M
Network Management Request/0800: host session activation/deactivation	M	X	.
Network Management Request/0800: sign-on/sign-off	M	X	.

Message	Org	Sys	Dst
Network Management Request Response/0810: host session activation/deactivation	•	XE	ME
Network Management Request Response/0810: network connection status, member-generated	ME	XE	•
Network Management Request Response/0810: network connection status, system-generated	•	XE	ME
Network Management Request Response/0810: PEK exchange	ME	X	•
Network Management Request Response/0810: PEK exchange, on demand	•	XE	ME
Network Management Advice/0820: PEK exchange	•	X	M

Values

This data element, when used, must contain a valid Mastercard customer ID number.

Application notes

DE 33 must be present in messages together with *DE 32* whenever the Mastercard customer ID number of a customer or INF is different than the Mastercard customer ID number of the actual acquiring institution.

Automated Referral Service (ARS) store-and-forward transactions are included with other Stand-In processing transactions and are identified with a 003850 in *DE 33* of the Authorization Advice/0120 message.

The Mastercard customer ID number must be set up in the Dual Message Authorization System for participants to process Administrative Request/06xx messages.

If an Authorization Request/0100, Authorization Advice/0120, or Reversal Request/0400 message contains *DE 33* and *DE 33* is not 6 digits in length, then the Dual Message Authorization System will forward to the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 where *DE 39* = 30 (Format Error) and *DE 44* = 033.

DE 34 (Primary Account Number [PAN], Extended)

DE 34 (Primary Account Number [PAN], Extended) identifies a customer account or relationship, and is used only when PAN begins with a 59 BIN.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 35 (Track 2 Data)

DE 35 (Track 2 Data) is the information encoded on track 2 of the card magnetic stripe as defined in the ISO 7813 specification, including data element separators but excluding beginning and ending sentinels and longitudinal redundancy check (LRC) characters as defined therein.

Table 144: Attributes

Attribute	Description
Data Representation	ans...37; LLVAR
Data Length	2 positions
Data Field	Contents of positions 1–37
Subfields	N/A
Justification	N/A

Usage

Following is the usage of *DE 35* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 145: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	X	C
Authorization Advice/0120: acquirer-generated	O	.	O
Authorization Advice/0120: system-generated	.	X	C

Values

Whenever *DE 35* is captured automatically at the point of interaction, *DE 35* must contain whatever is encoded on the magnetic stripe (track 2) of the card, regardless of whether the card has been properly encoded with information in accordance with ISO specifications. The ISO 7810, 7811, 7812, and 7813 specifications document the international standards for encoding information on magnetic stripe cards.

For chip transactions, *DE 35* carries data read from the chip as EMV tag 57 (Track 2 Equivalent Data). All ICCs issued by Mastercard customers must support EMV tag 57 (Track 2 Equivalent

Data). Since January 2008, the value of the CVC in EMV tag 57 (Track 2 Equivalent Data) on the chip and the CVC value on the physical magnetic stripe must be different.

DE 35 must contain the hexadecimal digits "0 through 9 and D or = (the equal sign).

Application notes

The account number in *DE 2 (Primary Account Number [PAN])* or *DE 34 (Primary Account Number [PAN], Extended)* must match the account number in *DE 35*.

This data element is defined and used identically within all Mastercard programs and services.

DE 35 is mandatory for all chip transactions (if missing, the Dual Message Authorization System will not reject the message; however, a data integrity error will be reported).

DE 35 is mandatory for all ATM transactions. For Maestro transactions, such as e-commerce, the Dual Message Authorization System creates track 2 data if it is not provided by the acquirer.

DE 35 is optional for 0120 AFD completion advice. Refer to [Automated Fuel Dispenser Completion](#) for more detail.

For MDES tokenized transactions, issuers may opt in through MDES Manager to receive track data in *DE 35 (Track 2 Data)* on Dual Message Authorization System or Single Message System transactions to issuers.

The field separator character (binary 1101) is represented as the EBCDIC character D. However, because many ATM and POS devices perform nonstandard character translation while reading binary coded decimal (BCD) encoded magnetic stripe data, the EBCDIC character "=" may also be used to represent the field separator character in magnetic stripe data.

Following is Track 2 information content and format for Mastercard transactions.

Table 146: Track 2 information content and format for Mastercard transactions

Field ID	Field name	F (Fixed), V (Variable)	Maximum length
1	Start Sentinel These fields are encoded on the card but must be omitted within Track 2 data.	F	n-1
2	Primary Account Number	V	n...19
3	Separator (binary)	F	ans-1
4	Expiration Date	F	ans-4
5	Extended Service Code	F	ans-3
6	Discretionary Data (must include CVC 1)	V	Balance of available digits not to exceed total track length of 40 characters.

Field ID	Field name	F (Fixed), V (Variable)	Maximum length
7	End Sentinel These fields are encoded on the card but must be omitted within Track 2 data.	F	n-1
8	Longitudinal Redundancy Check These fields are encoded on the card but must be omitted within Track 2 data.	F	n-1

Refer to the [Security Rules and Procedures](#) for additional Track 2 data information.

DE 36 (Track 3 Data)

DE 36 (Track 3 Data) is the information encoded on track 3 of the card magnetic stripe as defined in the ISO 4909:1986 specification, including data element separators but excluding beginning and ending sentinels and LRC characters as defined therein.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 37 (Retrieval Reference Number)

DE 37 (Retrieval Reference Number) is a document reference number supplied by the system retaining the original source document of the transaction and assists in locating that source document or a copy thereof. *DE 37* is made available for use by automated merchant POS systems that may be interconnected into the interchange system. Merchant POS systems may assign a unique receipt or sales document ID to be used to satisfy regulatory or legal requirements when the merchant performs source document capture and truncation. *DE 37* may be used to relay source document reference numbers to the issuer at the time each transaction is processed.

Table 147: Attributes

Attribute	Description
Data Representation	ans-12
Data Length	2 positions
Data Field	Contents of positions 1–12
Subfields	2 (for chip transactions)

Attribute	Description
Justification	See subfields

Usage

Following is the usage of *DE 37* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 148: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Request Response/0110	CE	.	CE
Authorization Advice/0120: acquirer-generated	C	.	C
Authorization Advice/0120: system-generated	.	X	C
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	CE	.	CE
Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)	CE	XE	.
Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)	CE	XE	.
Reversal Request/0400	C	.	C
Reversal Request Response/0410	CE	.	CE
Reversal Advice/0420	.	X	C
Reversal Advice Response/0430	CE	XE	.

Values

Number assigned by the acquirer.

When present, *DE 37* must contain a value other than all zeros or all spaces.

DE 37 can have trailing spaces if the transaction is not a Visa or EMV chip transaction.

Application notes

The Authorization System will pass *DE 37*, unaltered, to the receiving entity. The receiving entity must return this data element within any corresponding response.

DE 37 is mandatory for all chip transactions (DE 22 = 05x or 07x), chip fallback transactions (DE 22 = 80x), as well as contactless magnetic stripe transactions (DE 22 = 91x). If DE 37 is not present, the Dual Message Authorization System will not reject the message.

DE 37 is mandatory for all ATM transactions.

DE 37 is mandatory for all other card read transactions. (If missing, the Dual Message Authorization System will not reject the message.)

Visa Transactions: Acquirers that use the Mastercard Network must populate CIS DE 37 with a 12-position, all numeric value containing a valid Julian date, in Visan Authorization Request/0100 and Reversal Request/0400 messages.

The format for CIS DE 37 on Visa transactions is (ydddhnnnnnn):

- yddd (positions 1–4) is the year and day of year, equivalent of the value from DE 7 (*Transmission Date and Time*), y=0–9, ddd=001–366
- hh (positions 5 and 6) is the hours value from the time in DE 7 (*Transmission Date and Time*)
- nnnnnn (positions 7–12) is the value from DE 11 (Systems Trace Audit Number)

Visa will only edit positions 1–4. To avoid problems with system edits that may detect and reject duplicate and reused CIS DE 37 values, Visa recommends that endpoints construct CIS DE 37, positions 5–12 from the data in CIS DE 7 and CIS DE 11.

Acquirers may construct DE 37 in the same manner for Mastercard, as is indicated above for Visa messages.

Subfield 1 (Transaction Date and Initiator Discretionary Data)

DE 37, subfield 1 (Transaction Date and Initiator Discretionary Data) is used to pass chip data. The value information is specific to chip transactions.

Table 149: Attributes

Attribute	Description
Data Representation	ans-7
Data Field	Contents of positions 1–7
Justification	Left

Values

The date (MMDD) the transaction is captured at the point-of-service terminal.

If no discretionary data is included, the remaining three positions of this subfield should be zero-filled.

This subfield is left-justified with trailing zeros.

Subfield 2 (Terminal Transaction Number)

DE 37, subfield 2 (Terminal Transaction Number) is a unique number that identifies the transaction with a specific POS terminal within a specific 24 hour time period. The value information is specific to chip transactions.

Table 150: Attributes

Attribute	Description
Data Representation	ans-5
Data Field	Contents of positions 8–12
Justification	Right

Values

The Terminal Transaction Number: A sequential number, per terminal. Only numeric data may be present in this subfield. This subfield must contain a unique number that identifies the transaction with a specific POS terminal within a specific 24 hour time period.

Mastercard recommends that this subfield contain the value of the Transaction Sequence Counter (EMV tag 9F41), if available.

This subfield is right-justified with leading zeros.

DE 38 (Authorization ID Response)

DE 38 (*Authorization ID Response*) is a transaction response ID code that the authorizing institution assigns. DE 38 is used to transmit a card issuer's "authorization code" for Authorization transactions.

Table 151: Attributes

Attribute	Description
Data Representation	ans-6
Data Length	N/A
Data Field	Contents of positions 1–6
Subfields	N/A
Justification	Left

Usage

Following is the usage of *DE 38* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 152: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	C	.	C
Authorization Advice/0120: acquirer-generated	C	.	C
Authorization Advice/0120: system-generated	.	X	C
Reversal Request/0400	C	.	C
Reversal Advice/0420	.	X	C

Values

When present, *DE 38* must be left-justified and cannot contain all spaces, embedded spaces, all low values, all high values, or all zeros.

Application notes

In general, the authorization code used in *DE 38* may be any combination of alphanumeric characters. However, the following table identifies the standard conventions the major brands use.

Table 153: Authorization codes format

Card type	Length	Authorization ID response code format
Mastercard	6	alphanumeric, with no special characters.
Visa	6	alphanumeric, with no special characters.
American Express	6	alphanumeric, trailing space(s).
Diners Club	6	numeric, trailing space(s).
Discover	6	numeric, trailing space(s).

DE 38 is mandatory in the Authorization Request Response/0110 message when *DE 39* (*Response Code*) contains the value 00 (approved), 08 (Honor with ID), 10 (Partial Approval), 85 (Not Declined), or 87 (Purchase amount only, no cash back allowed). If the transaction is approved by Stand-In processing, a six-digit authorization approval code beginning with "8" appears in this field.

DE 38, position 6 (Account Category Code) may contain an Account Level Management account category code when the account range is participating in an Account Level Management service. For a complete valid value listing, refer to the [Account Level Management Manual](#).

Issuer instructions for ALM

If DE 48, subelement 38 is present in the Authorization Request/0100 message and the issuer approves the request, then the Authorization Request Response/0110 message, DE 38, position 6 should contain the same value as received in DE 48, subelement 38.

Failure to include the category code in DE 38, position 6 on the approved authorization will cause the authorization request to be routed to and processed by the Stand-In System.

Acquirer instructions for ALM

Acquirers should use DE 38, position 6 if the Authorization Request Response/0110 message was approved (DE 39 = 00, 08, 10, 85, or 87). Acquirers should use this value in their clearing process as indicated by IPM Clearing Member Parameter Extract (MPE). If the issuer provides DE 38 in an Authorization Request Response/0110 message and DE 39 is not an approval (00, 08, 10, 85, or 87), the Dual Message Authorization System removes the DE 38 value from the response to the acquirer.

DE 39 (Response Code)

DE 39 (Response Code) defines the disposition of a previous message or an action taken as a result of receipt of a previous message. Response codes also are used to indicate approval or decline of a transaction. In the event an authorization is declined, the response code indicates the reason for rejection and may indicate an action to be taken at the acceptor (for example, capture card).

Table 154: Attributes

Attribute	Description
Data Representation	an-2
Data Length	N/A
Data Field	Contents of positions 1–2
Subfields	N/A
Justification	N/A

Usage

Following is the usage of DE 39 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 155: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	M	•	M
Authorization Advice/0120: acquirer-generated	M	•	M
Authorization Advice/0120: system-generated	•	X	M
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	M	•	M
Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)	M	X	•
Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)	M	X	•
Authorization Response Acknowledgment/0180	M	X	•
Authorization Response Negative Acknowledgment/0190	•	X	M
Issuer File Update Request Response/0312	•	X	M
Reversal Request/0400	M	•	M
Reversal Request Response/0410	M	•	M
Reversal Advice/0420	•	C	C
Reversal Advice Response/0430	M	X	•
Administrative Request Response/0610	M	•	M
Administrative Advice Response/0630	M	X	•
Network Management Request Response/0810: host session activation/deactivation	•	X	M
Network Management Request Response/0810: network connection status, member-generated	M	X	•
Network Management Request Response/0810: network connection status, system-generated	•	X	M
Network Management Request Response/0810: PEK exchange	M	X	•
Network Management Request Response/0810: PEK exchange: on demand	•	X	M
Network Management Request Response/0810: sign-on/sign-off	•	M	M

Values

Valid values are listed by message type.

Application notes

DE 39 must be present in all Request Response, Advice Response, and Acknowledgment messages. In addition, it is also present in Authorization Advice/0120 Stand-In messages to indicate the response code used in the Authorization Request Response/0110 to the original Authorization Request/0100.

The following information applies to purchase only, no cash back allowed responses from an issuer.

Table 156: Associated edits

The issuer must provide...	The acquirer will receive...
the approved amount (purchase amount) in DE 6 (<i>Amount, Cardholder Billing</i>) in the issuer's cardholder billing currency. This amount must be the purchase amount as calculated by subtracting the DE 54 (<i>Additional Amounts</i>) cash back amount and, if applicable, DE 28 (<i>Amount, Transaction Fee</i>) from the DE 6 amount present in the Authorization Request/0100 message in the amount data element that corresponds to the issuer's cardholder billing currency.	the purchase-only approval amount in the acquirer's transaction currency in DE 4
DE 38 (<i>Authorization ID Response</i>)	DE 38
DE 39, value 87 (Purchase only, no cash back allowed)	DE 39, value 87
DE 51 (<i>Currency Code, Cardholder Billing</i>) with the issuer's cardholder billing currency code	an occurrence of the original amount of the transaction in DE 54 (<i>Additional Amounts</i>) in the acquirer's transaction currency. The original amount is identified by DE 54, subfield 2 (Amount Type), value 57 (Original Amount), and subfield 4 (Debit or Credit Indicator), value C plus 12-digit original amount.
	an occurrence of the original amount of the transaction in DE 54 in the issuer's cardholder billing currency. The original amount is identified by DE 54, subfield 2, value 57, and subfield 4, value C plus 12-digit original amount.

Table 157: Associated edits

IF the issuer provides a response code value of...	THEN the Dual Message Authorization System...
87 (Purchase Amount Only, No Cash Back Allowed) in the Authorization Request Response/0110 (applies to Mastercard only)	requires that the amounts in DE 28 (<i>Amount, Transaction Fee</i>) and DE 54 (<i>Additional Amounts</i>) are subtracted from DE 6 (<i>Amount, Cardholder Billing</i>).

Table 158: Associated edits

IF the issuer provides a response code value of...	THEN the Issuer Processing System (IPS)...
30 (Format Error) in the Authorization Request Response/0110 message	also provides <i>DE 44</i> with six positions for data element and subelement format errors (For example, 0480nn for <i>DE48</i> subelement nn), or three positions if no subelements are present (For example, 022 for <i>DE 22</i>).

The Dual Message Authorization System responds as indicated for Mastercard transactions only.

Table 159: Associated edits

IF the issuer provides a response code value of...	THEN the Dual Message Authorization System...
91, 92, or 96 in the Authorization Request Response/0110 (applies to Mastercard only)	automatically invokes Stand-In processing. The Stand-In System processes the transaction according to issuer-specified parameters that generate the issuer-specified response (for example, approve, decline, refer, or capture card). If Stand-In processing not available, invokes Xcode processing according to Product rules.
00, 08, 10, 85, or 87 in the Authorization Request Response/0110	requires that <i>DE 38 (Authorization Identification Response)</i> is present.

Table 160: Associated edits

Invalid values in DE 39 for a Mastercard ATM transaction...	Instead, the issuer should use the following response code...
01, 03, 08, 12, 15, 63, 76, 77, or 87	57
78	14
84 or 94	any valid response code

Table 161: Associated edits

WHEN the acquirer or issuer...	THEN the Dual Message Authorization System...
sends a Network Management/0800 group sign-off message (<i>DE 70 = 062 or 064</i>) within three minutes of the previous 0800 message for the same sign-on Member Group ID (<i>DE 2: Primary Account Number [PAN]</i>)	generates the Network Management Response/0810 message where <i>DE 39 (Response Code) = 96</i> (System error).

Authorization Request Response/0110 response codes

The following *DE 39* response codes are valid in this message.

Table 162: Response codes

Value	Description	Action	MC	NP	VI	TE	MS	CI
00	Approved or completed successfully	Approve	✓	✓	✓	✓	✓	✓
01	Refer to card issuer	Call Issuer	✓	✓	✓	✓	✓	
03	Invalid merchant	Decline	✓	✓	✓	✓	✓	
04	Capture card	Capture	✓	✓	✓	✓	✓	✓
05	Do not honor Issuers should use this response code only when no other reason code applies.	Decline	✓	✓	✓	✓	✓	
08	Honor with ID	Approve	✓	✓	✓	✓	✓	
10	Partial Approval	Approve	✓	✓	✓		✓	
12	Invalid transaction	Decline	✓	✓	✓	✓	✓	
13	Invalid amount	Decline	✓	✓	✓	✓	✓	✓
14	Invalid card number	Decline	✓	✓	✓	✓	✓	✓
15	Invalid issuer	Decline	✓	✓	✓	✓	✓	
30	Format error	Decline	✓	✓	✓	✓	✓	✓
41	Lost card	Capture	✓	✓	✓	✓	✓	✓
43	Stolen card	Capture	✓	✓	✓	✓	✓	✓
51	Insufficient funds/over credit limit	Decline	✓	✓	✓	✓	✓	✓
54	Expired card	Decline	✓	✓	✓	✓	✓	✓

Value	Description	Action	MC	NP	VI	TE	MS	CI
55	Invalid PIN	Decline	√	√	√	√	√	√
57	Transaction not permitted to issuer/cardholder Issuers should use this response code when their host blocks transactions based on specific transaction processing or fraud rules.	Decline	√	√	√		√	√
58	Transaction not permitted to acquirer/terminal	Decline	√	√	√		√	√
61	Exceeds withdrawal amount limit Issuers may also use this response code for transaction types other than ATM.	Decline	√	√	√	√	√	√
62	Restricted card	Decline	√	√	√	√	√	√
63	Security violation	Decline	√	√	√	√	√	

Value	Description	Action	MC	NP	VI	TE	MS	CI
65	Exceeds withdrawal count limit OR Identity Check Soft-Denial of EMV 3DS Authentication (merchant should resubmit authentication for ACS fully authenticated response or due to failed AAV Validation- OBS 05 service) Issuers may also use this response code for transaction types other than ATM transactions. Issuers that opt to decline a contactless or card-not-present transaction and obtain a cardholder verification through a second transaction may use this response code. Refer to M/Chip Requirements for Contact and Contactless for more details on contactless transactions and how to use response code 65. E-commerce merchants should trigger an authentication using Mastercard Identity Check following an authorization decline with response code 65. This could be due to an authentication not completed through an ACS function or due to a failed AAV Validation and a new authentication with new AAV & DS Transaction ID is necessary for the authorization to successfully complete AAV Validation (Dynamic Linking).	Decline	√	√	√	√	√	√
70	Contact Card Issuer	Call Issuer	√	√			√	√
71	PIN Not Changed	Decline	√	√			√	√
75	Allowable number of PIN tries exceeded	Decline	√	√	√	√	√	√
76	Invalid/nonexistent "To Account" specified	Decline	√	√	√		√	√
77	Invalid/nonexistent "From Account" specified	Decline	√	√	√		√	√

Value	Description	Action	MC	NP	VI	TE	MS	CI
78	Invalid/nonexistent account specified (general)	Decline	√	√	√	√	√	
79	Life cycle (Mastercard use only)							
81	Domestic Debit Transaction Not Allowed (Regional use only)	Decline	√					
82	Policy (Mastercard use only)							
83	Fraud/Security (Mastercard use only)							
84	Invalid Authorization Life Cycle	Decline		√				
85	Not declined Valid for all zero amount transactions.	Valid	√	√	√	√	√	√
86	PIN Validation not possible	Decline	√	√	√	√	√	√
87	Purchase Amount Only, No Cash Back Allowed	Approved	√				√	
88	Cryptographic failure This could be due to a failed AAV Validation and a new authentication with new AAV & DS Transaction ID is necessary for the authorization to successfully complete AAV Validation (Dynamic Linking).	Decline	√	√	√	√	√	√
89	Unacceptable PIN—Transaction Declined—Retry	Decline	√	√			√	√
91	Authorization System or issuer system inoperative	See Note	√	√	√	√	√	√
92	Unable to route transaction	See Note	√	√	√	√	√	√
94	Duplicate transmission detected	Decline	√	√	√	√	√	
96	System error	See Note	√	√	√	√	√	√

Value	Description	Action	MC	NP	VI	TE	MS	CI
	Note to the issuer: When a response code of 91, 92, or 96 is received from an issuer, Mastercard will attempt to perform alternate authorization processing (Alternate Issuer, Stand-In or X-Code Processing) according to the rules associated to the Mastercard brand and product. Note to the acquirer: When a response code of 91, 92, or 96 is returned to the acquirer, the transaction should be declined.							

Authorization Advice/0120 response codes

The following *DE 39* response codes are valid in this message.

Table 163: Response codes

Value	Description	Action	MC	NP	VI	TE	MS	CI
00	Approved or completed successfully	Approve	√	√	√	√	√	√
01	Refer to card issuer	Call Issuer	√	√	√	√	√	
03	Invalid merchant	Decline	√	√	√	√	√	
04	Capture card	Capture	√	√	√	√	√	√
05	Do not honor	Decline	√	√	√	√	√	
08	Honor with ID	Approve	√	√	√	√	√	
10	Partial Approval	Approve	√	√	√		√	
12	Invalid transaction	Decline	√	√	√	√	√	
13	Invalid amount	Decline	√	√	√	√	√	√
14	Invalid card number	Decline	√	√	√	√	√	√
15	Invalid issuer	Decline	√	√	√	√	√	
30	Format error	Decline	√	√	√	√	√	√
41	Lost card	Capture	√	√	√	√	√	√
43	Stolen card	Capture	√	√	√	√	√	√
51	Insufficient funds/over credit limit	Decline	√	√	√	√	√	√

Value	Description	Action	MC	NP	VI	TE	MS	CI
54	Expired card	Decline	✓	✓	✓	✓	✓	✓
55	Invalid PIN	Decline	✓	✓	✓	✓	✓	✓
57	Transaction not permitted to issuer/cardholder	Decline	✓	✓	✓		✓	✓
58	Transaction not permitted to acquirer/terminal	Decline	✓	✓	✓		✓	✓
61	Exceeds withdrawal amount limit	Decline	✓	✓	✓	✓	✓	✓
62	Restricted card	Decline	✓	✓	✓	✓	✓	✓
63	Security violation	Decline	✓	✓	✓	✓	✓	
65	Exceeds withdrawal count limit Issuers may also use this response code for transaction types other than ATM transactions. Issuers that opt to decline a contactless or card-not-present transaction and obtain a cardholder verification through a second transaction may use this response code. Refer to M/Chip Requirements for Contact and Contactless for more details on contactless transactions and how to use response code 65. E-commerce merchants should trigger an authentication using Mastercard Identity Check following an authorization decline with response code 65.	Decline	✓	✓	✓	✓	✓	✓
70	Contact Card Issuer	Call Issuer	✓	✓			✓	✓
71	PIN Not Changed	Decline	✓	✓			✓	✓
75	Allowable number of PIN tries exceeded	Decline	✓	✓	✓	✓	✓	✓
76	Invalid/nonexistent "To Account" specified	Decline	✓	✓	✓		✓	✓
77	Invalid/nonexistent "From Account" specified	Decline	✓	✓	✓		✓	✓
78	Invalid/nonexistent account specified (general)	Decline	✓	✓	✓	✓	✓	
84	Invalid Authorization Life Cycle	Decline		✓				

Value	Description	Action	MC	NP	VI	TE	MS	CI
86	PIN Validation not possible	Decline	√	√	√	√	√	√
87	Purchase Amount Only, No Cash Back Allowed	Approved	√				√	
88	Cryptographic failure	Decline	√	√	√	√	√	√
89	Unacceptable PIN—Transaction Declined—Retry	Decline	√	√			√	√
91	Authorization System or issuer system inoperative	Decline	√	√	√	√	√	√
92	Unable to route transaction	Decline	√	√	√	√	√	√
94	Duplicate transmission detected	Decline	√	√	√	√	√	
96	System error	Decline	√	√	√	√	√	√

Authorization Advice Response/0130 response codes

The following *DE 39* response codes are valid in this message.

Table 164: Response codes

Value	Description	Action	MC	NP	VI	TE	MS	CI
00	Approved or completed successfully	—	√	√	√		√	√
12	Invalid transaction	—	√	√	√	√	√	√
14	Invalid card number	—	√				√	√
30	Format error	—	√	√	√		√	√
57	Transaction not permitted to issuer/cardholder. Valid only for Pay with Rewards Service	—	√					
94	Duplicate Transmission detected	—	√	√	√		√	√
96	System error	—	√	√	√		√	√

Authorization Response Acknowledgment/0180 response codes

The following *DE 39* response codes are valid in this message.

Table 165: Response codes

Value	Description	Action	MC	NP	VI	TE	MS	CI
00	Completed successfully	—	✓	✓	✓	✓	✓	✓

Authorization Response Negative Acknowledgment/0190 response codes

The following *DE 39* response codes are valid in this message.

Table 166: Response codes

Value	Description	Action	MC	NP	VI	TE	MS	CI
30	Format error	—	✓	✓	✓	✓	✓	✓
68	Response received late	—	✓	✓	✓	✓	✓	✓
96	System error	—	✓	✓	✓	✓	✓	✓

Issuer File Update Request Response/0312 response codes

The following *DE 39* response codes are valid in this message.

Table 167: Response codes

Value	Description	Action	MC	NP	VI	TE	MS	CI
00	Issuer File Update action completed successfully	—	✓	✓	✓	✓	✓	✓
25	Unable to locate record on file (no action taken)	—	✓	✓	✓	✓	✓	✓
26	Record already in active status	—	✓	✓	✓	✓	✓	✓
27	Issuer File Update field edit error	—	✓	✓	✓	✓	✓	✓
28	Record permanently deleted	—	✓	✓	✓	✓	✓	✓
29	Delete request less than 540 days	—	✓	✓	✓	✓	✓	✓
30	Format error	—	✓	✓	✓	✓	✓	✓
40	Requested function not supported	—	✓	✓	✓	✓	✓	✓
63	Security violation	—	✓	✓	✓	✓	✓	✓

Value	Description	Action	MC	NP	VI	TE	MS	CI
80	Duplicate add, action not performed	—	✓	✓	✓	✓	✓	✓
96	System error	—	✓	✓	✓	✓	✓	✓

Reversal Request/0400 response codes

The following DE 39 response codes are valid in this message.

NOTE: Mastercard recommends that customers use response code 00 for all Stand-In Advice messages.

Table 168: Response codes

Value	Description	Action	MC	NP	VI	TE	MS	CI
00	Approved or completed successfully	—	✓	✓	✓	✓	✓	✓
01	Refer to card issuer	—	✓	✓	✓	✓	✓	
03	Invalid merchant	—	✓	✓	✓	✓	✓	
04	Capture card	—	✓	✓	✓	✓	✓	✓
05	Do not honor	—	✓	✓	✓	✓	✓	
06	Error (/0400 only)	—	✓				✓	✓
08	Honor with ID	—	✓	✓	✓	✓		
10	Partial Approval	—	✓	✓	✓		✓	
12	Invalid transaction	—	✓	✓	✓	✓	✓	
13	Invalid amount	—	✓	✓	✓	✓	✓	✓
14	Invalid card number	—	✓	✓	✓	✓	✓	✓
15	Invalid issuer	—	✓	✓	✓	✓	✓	
17	Customer cancellation (/0400 only)	—	✓				✓	✓
30	Format error	—	✓	✓	✓	✓	✓	✓
32	Partial reversal (/0400 only)	—	✓				✓	✓
34	Suspect Fraud (/0400 only)	—	✓	✓			✓	
41	Lost card	—	✓	✓	✓	✓	✓	✓
43	Stolen card	—	✓	✓	✓	✓	✓	✓

Value	Description	Action	MC	NP	VI	TE	MS	CI
51	Insufficient funds/over credit limit	—	✓	✓	✓	✓	✓	✓
54	Expired card	—	✓	✓	✓	✓	✓	✓
55	Invalid PIN	—	✓	✓	✓	✓	✓	✓
57	Transaction not permitted to issuer/cardholder	—	✓	✓	✓		✓	✓
58	Transaction not permitted to acquirer/terminal	—	✓	✓	✓		✓	✓
61	Exceeds withdrawal amount limit	—	✓	✓	✓	✓	✓	✓
62	Restricted card	—	✓	✓	✓	✓	✓	✓
63	Security violation	—	✓	✓	✓	✓	✓	
65	Exceeds withdrawal count limit	—	✓	✓	✓	✓	✓	✓
68	Response received late (/0400 only)	—	✓				✓	✓
70	Contact Card Issuer	—	✓	✓			✓	✓
71	PIN Not Changed	—	✓	✓			✓	✓
75	Allowable number of PIN tries exceeded	—	✓	✓	✓	✓	✓	✓
76	Invalid/nonexistent "To Account" specified	—	✓	✓	✓		✓	✓
77	Invalid/nonexistent "From Account" specified	—	✓	✓	✓		✓	✓
78	Invalid/nonexistent account specified (general)	—	✓	✓	✓	✓	✓	
84	Invalid Authorization Life Cycle	—		✓				
85	Not declined	—	✓	✓		✓	✓	✓
86	PIN Validation not possible	—	✓	✓	✓	✓	✓	✓
87	Purchase Amount Only, No Cash Back Allowed	—	✓				✓	
88	Cryptographic failure	—	✓	✓	✓	✓	✓	✓
89	Unacceptable PIN—Transaction Declined—Retry	—	✓	✓			✓	✓
91	Authorization System or issuer system inoperative	—	✓	✓	✓	✓	✓	✓
92	Unable to route transaction	—	✓	✓	✓	✓	✓	✓

Value	Description	Action	MC	NP	VI	TE	MS	CI
94	Duplicate transmission detected	—	✓	✓	✓	✓	✓	
96	System error	—	✓	✓	✓	✓	✓	✓

Reversal Request Response/0410 response codes

The following *DE 39* response codes are valid in this message.

Table 169: Response codes

Value	Description	Action	MC	NP	VI	TE	MS	CI
00	Approved or completed successfully	Approve	✓	✓	✓	✓	✓	✓
01	Refer to card issuer	Call Issuer	✓	✓	✓	✓	✓	
12	Invalid transaction	Decline	✓	✓	✓	✓	✓	
13	Invalid amount	Decline	✓	✓	✓	✓	✓	✓
14	Invalid card number	Decline	✓	✓	✓	✓	✓	✓
30	Format error	Decline	✓	✓	✓	✓	✓	✓
41	Lost card	Capture	✓	✓	✓	✓	✓	✓
43	Stolen card	Capture	✓	✓	✓	✓	✓	✓
57	Transaction not permitted to issuer/cardholder	Decline	✓	✓	✓		✓	✓
58	Transaction not permitted to acquirer/terminal	Decline	✓	✓	✓		✓	✓
62	Restricted card	Decline	✓	✓	✓	✓	✓	✓
63	Security violation	Decline	✓	✓	✓	✓	✓	
76	Invalid/nonexistent "To Account" specified	Decline	✓	✓	✓		✓	✓
77	Invalid/nonexistent "From Account" specified	Decline	✓	✓	✓		✓	✓
91	Authorization System or issuer system inoperative	Decline	✓	✓	✓	✓	✓	✓
92	Unable to route transaction	Decline	✓	✓	✓	✓	✓	✓
94	Duplicate transmission detected	Decline	✓	✓	✓	✓	✓	✓
96	System error	Decline	✓	✓	✓	✓	✓	✓

Reversal Advice/0420 response codes

The following *DE 39* response codes are valid in this message.

Table 170: Response codes

Value	Description	Action	MC	NP	VI	TE	MS	CI
00	Approved or completed successfully	Approve	✓	✓	✓	✓	✓	✓
01	Refer to card issuer	Call Issuer	✓	✓	✓	✓	✓	
03	Invalid merchant	Decline	✓	✓	✓	✓	✓	
04	Capture card	Capture	✓	✓	✓	✓	✓	✓
05	Do not honor	Decline	✓	✓	✓	✓	✓	
06	Error	Decline	✓				✓	✓
08	Honor with ID	Approve	✓	✓	✓	✓		
10	Partial Approval	Approve	✓	✓	✓		✓	
12	Invalid transaction	Decline	✓	✓	✓	✓	✓	
13	Invalid amount	Decline	✓	✓	✓	✓	✓	✓
14	Invalid card number	Decline	✓	✓	✓	✓	✓	✓
15	Invalid issuer	Decline	✓	✓	✓	✓	✓	
17	Customer cancellation	Decline	✓				✓	✓
30	Format error	Decline	✓	✓	✓	✓	✓	✓
32	Partial reversal	Decline	✓				✓	✓
34	Suspect Fraud	Decline	✓	✓			✓	
41	Lost card	Capture	✓	✓	✓	✓	✓	✓
43	Stolen card	Capture	✓	✓	✓	✓	✓	✓
51	Insufficient funds/over credit limit	Decline	✓	✓	✓	✓	✓	✓
54	Expired card	Decline	✓	✓	✓	✓	✓	✓
55	Invalid PIN	Decline	✓	✓	✓	✓	✓	✓
57	Transaction not permitted to issuer/cardholder	Decline	✓	✓	✓		✓	✓
58	Transaction not permitted to acquirer/terminal	Decline	✓	✓	✓		✓	✓
61	Exceeds withdrawal amount limit	Decline	✓	✓	✓	✓	✓	✓
62	Restricted card	Decline	✓	✓	✓	✓	✓	✓

Value	Description	Action	MC	NP	VI	TE	MS	CI
63	Security violation	Decline	✓	✓	✓	✓	✓	
65	Exceeds withdrawal count limit	Decline	✓	✓	✓	✓	✓	✓
68	Response received late	Decline	✓				✓	✓
70	Contact Card Issuer	Call Issuer	✓	✓			✓	✓
71	PIN Not Changed	Decline	✓	✓			✓	✓
75	Allowable number of PIN tries exceeded	Decline	✓	✓	✓	✓	✓	✓
76	Invalid/nonexistent "To Account" specified	Decline	✓	✓	✓		✓	✓
77	Invalid/nonexistent "From Account" specified	Decline	✓	✓	✓		✓	✓
78	Invalid/nonexistent account specified (general)	Decline	✓	✓	✓	✓	✓	
82	Timeout at issuer	—	✓	✓	✓		✓	✓
84	Invalid Authorization Life Cycle	Decline		✓				
85	Not declined Valid for zero amount transactions.	Valid	✓	✓		✓	✓	✓
86	PIN Validation not possible	Decline	✓	✓	✓	✓	✓	✓
87	Purchase Amount Only, No Cash Back Allowed	Approved	✓				✓	
88	Cryptographic failure	Decline	✓	✓	✓	✓	✓	✓
89	Unacceptable PIN—Transaction Declined—Retry	Decline	✓	✓			✓	✓
91	Authorization System or issuer system inoperative	Decline	✓	✓	✓	✓	✓	✓
92	Unable to route transaction	Decline	✓	✓	✓	✓	✓	✓
94	Duplicate transmission detected	Decline	✓	✓	✓	✓	✓	
96	System error	Decline	✓	✓	✓	✓	✓	✓

Reversal Advice Response/0430 Message and Administrative Advice Response/0630 response codes

The following *DE 39* response codes are valid in these messages.

Table 171: Response codes

Value	Description	Action	MC	NP	VI	TE	MS	CI
00	Approved or completed successfully	—	✓	✓	✓	✓	✓	✓
30	Format error	—	✓	✓	✓	✓	✓	✓
94	Duplicate Transmission detected	—	✓	✓	✓	✓	✓	✓
96	System error	—	✓	✓	✓	✓	✓	✓

Administrative Request Response/0610 response codes

The following *DE 39* response codes are valid in this message.

Table 172: Response codes

Value	Description	Action	MC	NP	VI	TE	MS	CI
00	Received and processed successfully	None	✓	✓				
30	Format error	Determine error and resubmit request	✓	✓				
57	Transaction not permitted to issuer/cardholder	Decline	✓	✓				
58	Transaction not permitted to acquirer/terminal	Decline	✓	✓				
91	Authorization System or issuer system inoperative	Resubmit request	✓	✓				
92	Unable to route transaction	Determine error and resubmit request	✓	✓				
96	System error	Resubmit request	✓	✓				

Network Management Request Response/0810 response codes

The following *DE 39* response codes are valid in this message.

Table 173: Response codes

Value	Description	Action	MC	NP	VI	TE	MS	CI
00	Approved or completed successfully	—	✓	✓	✓	✓	✓	✓
30	Format error	—	✓	✓	✓	✓	✓	✓
63	Security violation	—	✓	✓	✓	✓	✓	✓
79	Key Exchange Validation failed	—	✓	✓	✓	✓	✓	✓
91	Authorization System or issuer system inoperative	—	✓	✓	✓	✓	✓	✓
94	Duplicate SAF request	—	✓	✓	✓	✓	✓	✓
96	System error	—	✓	✓	✓	✓	✓	✓

DE 40 (Service Restriction Code)

DE 40 (Service Restriction Code) identifies geographic or service availability.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 41 (Acceptor Terminal ID)

DE 41 (Acceptor Terminal ID) is the identifier assigned by the acceptor or the acquirer to the terminal, payment gateway, or other acceptance device used to capture account data to affect a transaction at the acceptor location.

Table 174: Attributes

Attribute	Description
Data Representation	ans-8
Data Length	N/A
Data Field	Contents of positions 1–8
Subfields	N/A

Attribute	Description
Justification	Left with trailing spaces

Usage

Following is the usage of *DE 41* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 175: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Request Response/0110	CE	.	CE
Authorization Advice/0120: acquirer-generated	C	.	C
Authorization Advice/0120: system-generated	.	X	C
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	CE	.	CE
Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)	CE	XE	.
Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)	CE	XE	.
Reversal Request/0400	C	.	C
Reversal Request Response/0410	CE	.	CE
Reversal Advice/0420	CE	XE	.
Administrative Request/0600	O	.	O
Administrative Request Response/0610	CE	.	CE

Processing rules

The Acceptor Terminal ID must be present in all transactions.

Acceptor Terminal ID must uniquely identify each point-of-sale terminal and ATM terminal at the acceptor location. For electronic commerce, phone/mail order, or other card-not-present environments, the value must uniquely identify each payment gateway or other device used to capture account data to affect a transaction at the acceptor location.

DE 42 (Acceptor ID)

DE 42 (Acceptor ID) contains the identifier assigned by the acquirer to the acceptor and its location, or to the payment facilitator.

Table 176: Attributes

Attribute	Description
Data Representation	ans-15
Data Length	N/A
Data Field	Contents of positions 1–15
Subfields	N/A
Justification	Left

Usage

Following is the usage of *DE 42* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 177: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	•	C
Authorization Advice/0120: acquirer-generated	C	•	C
Authorization Advice/0120: system-generated	•	X	C
Reversal Request/0400	C	•	C
Administrative Request/0600	O	•	O
Administrative Request Response/0610	CE	•	CE
Administrative Advice/0620: system-generated	•	X	C

Processing rules

The Acceptor ID must be present in all transactions and must uniquely identify each acceptor and its location.

Payment facilitator-enabled transactions must uniquely identify the payment facilitator. The Acceptor ID of the sponsored merchant must be provided in the sponsored merchant ID data element.

The following Dual Message Authorization System edit will apply:

Table 178: Associated edits

WHEN...	THEN the Authorization System...
DE 42 (Acceptor ID) is not present and DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type code) contains the value 00 (Purchase), 09 (Purchase with Cash Back), or 28 (Payment Transaction) in an Authorization Request/0100, Authorization Advice/0120: acquirer-generated, or a Reversal Request/0400 message	sends the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or a Reversal Request Response/0410 message where: - DE 39 (Response Code) = 30 (Format Error) - DE 44 (Additional Response Data) = 042 (which identifies DE 42 as the source of the error)

American Express Authorizations

An American Express merchant ID consists of the 10-digit numeric value (referred to as "SE number" by American Express) or two-character alphanumeric IATA airline code (which may be followed by the IATA travel agent ID, T + 5-8 digits, separated by a space). Since DE 42 is defined as ans-15, an American Express 10-digit SE number in character format should be left-justified and followed by trailing spaces. This 10-digit number also should pass the Modulus 9 check digit routine.

DE 43 (Acceptor Name and Location)

DE 43 (Acceptor Name and Location) contains the name and location of the acceptor point of interaction.

Table 179: Attributes

Attribute	Description
Data Representation	ans-40 (supports extended character sets)
Data Length	N/A
Data Field	Contents of subfields 1–5
Subfields	5
Justification	See subfields

Usage

Following is the usage of *DE 43* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 180: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Advice/0120: acquirer-generated	C	.	C
Authorization Advice/0120: system-generated	.	X	C
Reversal Request/0400	C	.	C
Reversal Advice/0420	.	X	C
Administrative Request/0600	O	.	O
Administrative Request Response/0610	CE	.	CE

Values

The data element subfields are dependent on the type of message type sent.

Application notes

This data element is required for all Authorization Request/0100 messages for Mastercard and Visa programs and services.

For Authorization Request/0100, use of *DE 43* is conditional (C) based on the program or service being processed. If the program is Mastercard or Visa, use of *DE 43* is mandatory (M). Its usage is optional if the transaction involves Private Label, Travel and Entertainment, or other programs.

This data element satisfies "Regulation-E" requirements. It also is required when the *DE 22* (*Point-of-Service [POS] Entry Mode*) is 05 (PAN auto-entry via integrated circuit card).

As ISO specifies, this data element is an alphanumeric text string of 40 characters. Mastercard requires standardized formatting of this data element as specified in the following table to meet uniform standards for printing this information on customer account statements and billing statements.

This data element is required for all Mastercard programs and services. The Dual Message Authorization System does not perform edits on this data element.

DE 43 is required for Mastercard Hosted Mobile Phone Top-up transactions.

Subfield 1 (Acceptor Name)

DE 43 (Acceptor Name and Location), subfield 1 (Acceptor Name) contains the name of the acceptor that should be recognizable by the cardholder and may include other descriptors, such as a store number.

Table 181: Attributes

Attribute	Description
Data Representation	ans-22
Data Field	Contents of positions 1–22
Justification	Left

Processing rules

The Acceptor Name must be present for all transactions and contain a valid merchant/ATM owner/entity name that is most recognizable to the cardholder.

Unique identifiers, invoice numbers, or additional descriptors must be placed after the merchant/ATM owner/entity name.

When a merchant is a part of a chain or franchise, the name most recognizable to the cardholder must be provided at the beginning of the field, followed by a store number or other unique identifier for each location. For a chain or franchise that has been issued a Mastercard Assigned ID, all submitted transactions must contain the Mastercard Assigned ID data element.

For any MCC in the 3000-3350 range (Airlines), the Abbreviated Airline listed in Chapter 3 of the [Quick Reference Booklet](#) must be provided in positions 1-9 of the field followed by the Airline Ticket Number.

For any MCC in the 3351-3500 range (Automobile/Vehicle Rentals), the first nine characters of the Description listed in Chapter 3 of the [Quick Reference Booklet](#) must be provided in positions 1-9 of the field and then followed by a unique identifier for each location.

For any MCC in the 3501-3999 range (Hotels and Motels), the first nine characters of the Description listed in Chapter 3 of the [Quick Reference Booklet](#) must be provided in positions 1-9 of the field and then followed by a unique identifier for each location.

The (ISO)/IEC 8859-1:1998 (Latin Alphabet No. 1) character set must be used.

Program rules

Payment facilitator-enabled transactions: The sponsored merchant name must be provided. The Payment Facilitator Name or abbreviated name must be provided, followed by an asterisk (*) and the sponsored merchant name. All submitted transactions must contain the Payment Facilitator ID submitted in the Payment Facilitator ID data element.

Staged digital wallet operator: The staged digital wallet operator name must be provided. If issuer authorization for the transaction is occurring at the time of the consumer's purchase of products or services, then the staged digital wallet operator name must be provided, followed by an asterisk (*) and the retailer name. If the retailer is identified with an MCC in the 3000-3999 range, then the appropriate industry MCC value, rather than individual retailer MCC, if assigned, must be provided.

Mastercard® rePower™ transactions must have the value MC repower following the merchant/ATM owner/entity name providing the service.

Payment Transactions must contain the name of the payment provider. Payee information must be provided in the Additional Transaction Reference Data and Mastercard Send Receiver Data data elements.

Country rules

Mexico-acquired transactions must contain only

- uppercase alpha characters (A-Z) (cannot contain accent characters)
- numeric characters (0–9)
- ampersand (&)
- asterisk (*)
- space (cannot be the first character or contain more than one space between words)

Subfield 2 (Space)

DE 43 (Acceptor Name and Location), subfield 2 (Space) indicates a space character.

Table 182: Attributes

Attribute	Description
Data Representation	ans-1
Data Field	Contents of position 23
Justification	N/A

Value

Delimiter (space).

Subfield 3 (Acceptor City)

DE 43 (Acceptor Name and Location), subfield 3 (Acceptor City) indicates the city of the merchant.

Table 183: Attributes

Attribute	Description
Data Representation	ans-13
Data Field	Contents of positions 24–36
Justification	Left

Values

Acceptor City Name must contain the name of the city, town, or village as defined by country postal standards.

For U.S. region transactions, Acceptor City Name must match the U.S. Postal Service list of acceptable cities and spellings by postal code.

Processing rules

The Acceptor City Name must be present for all transactions and contain a valid location for the acceptor. The location of the acceptor will generally be the physical location where the transaction occurred or the business/operations location for transactions that occur virtually.

Program rules

Payment facilitator-enabled transactions: The sponsored merchant city name must be provided.

Subfield 4 (Space)

DE 43 (Acceptor Name and Location), subfield 4 (Space) indicates a space character.

Table 184: Attributes

Attribute	Description
Data Representation	ans-1
Data Field	Contents of position 37
Justification	N/A

Value

Delimiter (space).

Subfield 5 (Acceptor Country Code)

DE 43 (Acceptor Name and Location), subfield 5 (Acceptor Country Code) contains the country code of the acceptor.

Table 185: Attributes

Attribute	Description
Data Representation	a-3
Data Field	Contents of positions 38-40
Justification	Left

Values

Acceptor Country Code must contain a valid three-character alphabetic code as defined in Chapter 2 of the [Quick Reference Booklet](#). When territories have a defined country code in the [Quick Reference Booklet](#), that country code must be used.

Processing rules

The Acceptor Country Code must be present for all transactions.

Program rules

Payment facilitator-enabled transactions: The sponsored merchant country code must be provided.

DE 43, subfield 5 Dual Message Authorization System edits

The Dual Message Authorization System will perform the following system edits.

Authorization Request/0100 message edits for DE 43, subfield 5

Table 186: Associated edits

If...	Then the Dual Message Authorization System...
DE 43, subfield 5 does not contain a valid country code	will return to the acquirer an Authorization Request Response/0110 message where: - DE 39 = 30 - DE 44 = 043

Authorization Advice/0120: acquirer-generated message edits for DE 43, subfield 5

Table 187: Associated edits

If...	Then the Dual Message Authorization System...
DE 43, subfield 5 does not contain a valid country code	will return to the acquirer an Authorization Advice Response/0130: system-generated message where: • DE 39 = 30 • DE 44 = 043

Reversal Request/0400 message edits for DE 43, subfield 5

Table 188: Associated edits

If...	Then the Dual Message Authorization System...
DE 43, subfield 5 does not contain a valid country code	will return to the acquirer a Reversal Request Response/0410 message where: • DE 39 = 30 • DE 44 = 043

DE 44 (Additional Response Data)

DE 44 (Additional Response Data) provides other supplemental data that may be required in response to an authorization or other type of transaction request. This data element may also be present in any response message when DE 39 (Response Code) contains the value 30, indicating that a Format Error condition was detected in the preceding message. In this case, DE 44 will contain either a three- or six-digit value that indicates the data element number (three digits) only or the data element plus subelement numbers (six digits) in which the format error occurred. If the format error occurred in a data element that does not have subelements, DE 44 will contain a three-digit numeric value.

Table 189: Attributes

Attribute	Description
Data Representation	ans...25; LLVAR
Data Length	2 positions

Attribute	Description
Data Field	Contents of positions 1–25
Subfields	N/A
Justification	N/A

Usage

Following is the usage of *DE 44* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 190: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	C	•	C
Authorization Advice/0120: acquirer-generated	C	•	C
Authorization Advice/0120: system-generated	•	X	C
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	CE	•	CE
Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)	CE	XE	•
Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)	C	X	•
Authorization Response Negative Acknowledgment/0190	•	X	C
Issuer File Update Request Response/0312	•	X	C
Reversal Request Response/0410	C	•	C
Reversal Advice Response/0430	C	X	•
Administrative Request Response/0610	C	•	C
Administrative Advice Response/0630	C	X	•
Network Management Request Response/0810: Host Session Activation/Deactivation	•	X	C
Network Management Request Response/0810: PEK exchange	C	X	•
Network Management Request Response/0810: PEK exchange–On Demand	•	X	C
Network Management Request Response/0810: Network Connection Status, System-generated	•	X	C

Message	Org	Sys	Dst
Network Management Request Response/0810: sign-on/sign-off	.	X	C

Values

Contains supplemental data.

DE 44 values by program or service

The following table lists the usage of *DE 44* by program or service. Usage is dependent on values in *DE 39 (Response Code)*. The following table defines the additional response data that is in *DE 44* when the listed values are present in *DE 39*. No data element edits are performed on the variable-length field.

Table 191: DE 44 usage by program or service

IF DE 39 is...	THEN DE 44...	MC	NP	VI	TE	MS	CI
00	N/A						
01	contains the phone number for "call issuer" response codes. Applies to Authorization Request Response/0110 messages.	√	√	√	√	√	
08	contains the ID information for "approve with ID" response codes for example: • B = Blank Name • C = Cardholder's ID number or hologram message • X= Cardholder's last name Applies to Authorization Request Response/0110 messages.	√	√	√	√	√	

IF DE 39 is...	THEN DE 44...	MC	NP	VI	TE	MS	CI
12	should be accessed when the value 12 is present in <i>DE 39</i> for Mastercard Corporate Fleet Card® transactions only as follows: • 01 = Issuer determines that the ID number provided in the Authorization Request/0100 message is invalid causing the transaction to be declined • 02 = Issuer determines that the Driver Number provided in the Authorization Request/0100 message is invalid causing the transaction to be declined • 03 = Issuer determines that the Vehicle Number provided in the Authorization Request/0100 message is invalid causing the transaction to be declined. Applies to Authorization Request Response/0110 messages.	√					
25	contains 120 indicating an error in <i>DE 120 (Record Data)</i> and a three-digit code (such as 120xxx, where xxx indicates the field in the file update request where the error occurred). A listing of these codes can be found in the Error Codes section for <i>DE 120</i> . Applies to Issuer File Update Request Response/0312 messages.	√	√	√	√	√	√
27	contains a 120 indicating an error in <i>DE 120</i> and a three-digit code (such as 120xxx, where xxx indicates the field in the file update request where the error occurred). A listing of these codes can be found in the Error Codes section for <i>DE 120</i> . Applies to Issuer File Update Request Response/0312 messages.	√	√	√	√	√	√
30	contains either the data element/subelement when a format error is detected (for example, 022 for <i>DE 22</i> , or 0480nn for <i>DE 48</i> [subelement nn]). Applies to all response messages.	√	√	√	√	√	√
57	contains 003 indicating an error in <i>DE 3 (Processing Code)</i> when an issuer is unable to process a balance inquiry (30), purchase with cash back (09), purchase return (20), or payment (28) transaction type.	√	√	√		√	√

IF DE 39 is...	THEN DE 44...	MC	NP	VI	TE	MS	CI
85	contains the date and time used in electronic commerce transactions, after which a cardholder may re-apply for a certificate. Format is YYYYMMDDHH. HH is the 24-hour military clock.	√		√		√	

Dual Message Authorization System edits

The Dual Message Authorization System will perform the following system edit.

Table 192: Associated edits

WHEN...	THEN the Dual Message Authorization System...
an Authorization Request/0100, Authorization Advice/0120: acquirer-generated, or Reversal Request/0400 message contains one or both of the following: - DE 48 (Additional Data: Private Use), DE 108 (Additional Transaction Reference Data), or DE 112 (Additional Data [National Use]) with subelements that have incorrect length and/or incorrect format (Data Representations) - Multiple instances of the same subelement (when not permitted) in DE 48	rejects the message and forwards the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 message where DE 44 is 6 positions for subelement format errors: - DE 39 (Response Code) = 30 (Message format error) - DE 44 (Additional Response Data) = 0480nn, or 1080nn, or 1120nn (where nn is the subelement number) (DE 44 is three positions for Dual Message System (Authorization) if no subelements are present, for example, for DE 22 format error: DE 44 = 022)

Examples

- When an edit error occurs on DE 48, the DE 44 data will be populated as these examples below:
 - Error on DE 48 subelement 42, subfield 1: 048042 (no subfield information is provided)
 - Error on DE 48 subelement 61: 048061
- DE 48 TCC format error will be responded with DE 44 = 048000.

For non-DE 48/DE 108/DE 112 format errors, DE 44 will only have DE info and no subelement information. For example, the format error in DE 22 will have DE 44 as 022.

DE 45 (Track 1 Data)

DE 45 (Track 1 Data) is the information encoded on track 1 of the card's magnetic stripe as defined in the ISO 7813 specification, including data element separators but excluding beginning and ending sentinels and LRC characters as defined in this data element definition.

Table 193: Attributes

Attribute	Description
Data Representation	ans...76; LLVAR
Data Length	2 positions
Data Field	Contents of positions 1–76
Subfields	N/A
Justification	N/A

Usage

Following is the usage of *DE 45* (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 194: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Advice/0120: acquirer-generated	O	.	O
Authorization Advice/0120: system-generated	.	X	C

Values

Whenever *DE 45* is captured automatically at the point of interaction, this data element must contain whatever is encoded on the magnetic stripe (track 1) of the card, regardless of whether the card has been properly encoded with information in accordance with ISO specifications.

The ISO 7810, 7811, 7812, and 7813 specifications document the international standards for encoding information on magnetic stripe cards.

Length subelement must not be greater than 76 bytes.

The account number in *DE 2 (Primary Account Number)* or *DE 34 (Primary Account Number [PAN], Extended)* must match the account number in *DE 45*.

Application notes

This data element is defined and used identically within all Mastercard programs and services.

For MDES tokenized transactions, issuers may opt in through MDES Manager to receive track data in *DE 45 (Track 1 Data)* on Dual Message Authorization System or Single Message System transactions to issuers.

DE 45 is optional for 0120 AFD completion advice.

Table 195: Fields in DE 45

Field ID	Field Name	F = Fixed	Maximum Length
		V = Variable	
1	Start Sentinel These fields are encoded on the card but must be omitted within Track 2 data.	F	n-1
2	Format Code-B (encode character B)	F	an-1
3	Primary Account Number	V	n...19
4	Separator (binary)	F	ans-1
5	Cardholder name	V	ans...2-26
6	Separator (binary)	F	ans-1
7	Expiration Date	F	ans-4
8	Extended Service Code	F	ans-3
9	Discretionary Data (must include CVC 1)	V	Balance of available digits not to exceed total track length of 79 characters.
10	End Sentinel These fields are encoded on the card but must be omitted within Track 2 data.	F	n-1
11	Longitudinal Redundancy Check These fields are encoded on the card but must be omitted within Track 2 data.	F	n-1

Refer to the [Security Rules and Procedures](#) for additional Track 1 Data information.

DE 46 (Expanded Additional Amounts)

DE 46 (Additional Data: ISO Use) provides data supplemental to that already conveyed in the specific data elements in the message.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 47 (Additional Data: National Use)

DE 47 (Additional Data—National Use) is reserved for national organizations to define data unique to country applications.

NOTE: The Dual Message Authorization System currently does not use this data element.

This data element should not be present in any Dual Message Authorization System messages. However, if encountered in Dual Message Authorization System messages routed between customers of the same country, it is passed unaltered through the network. The Dual Message Authorization System performs no editing or processing functions on this data element.

DE 48 (Additional Data: Private Use)

DE 48 (Additional Data: Private Use) contains data associated with various Mastercard programs, products, and services. DE 48 is used to convey data that is not related to any other ISO-defined data elements. It is a variable-length data element used for multiple purposes.

Table 196: Attributes

Attribute	Description
Data Representation	ans...999; LLLVAR Although this data element is ans, some subelements deviate from ans and contain binary data. Refer to subelement descriptions of each subelement. The length field must be in the range 001–999. Subelements are identified by valid subelement ID and length.
Data Length	3 positions
Data Field	Contents of subelements
Subelements	99

Attribute	Description
Justification	See subelements

Usage

The following applies to DE 48 usage:

- Subelements do not need to be in any particular order or sequence within DE 48.
- Subelements may contain multiple occurrences of program or service data, up to the maximum limit specified for that subelement. However, the same subelement ID cannot be present within DE 48 more than once. Messages submitted to Mastercard containing multiple occurrences of a subelement within DE 48 will be rejected.
- Message originators and receivers must be able to send and receive any subelement applicable to their business or as mandated by Mastercard.
- Message originators must not use a subelement ID that is not defined by Mastercard. A message containing a subelement not defined by Mastercard will not be rejected by Mastercard. However, the undefined subelement will be removed before being set to the message receiver. The content of such subelements will not be recognized by Mastercard or used in any Mastercard processing.
- For subelements defined with a CE (conditional echo) presence notation for an Org (originator) entity Mastercard will not reject a message from the message originator if the original subelement data is not returned as defined in a response message. Based on the presence notation for the Dst (destination) entity Mastercard will forward the subelement data as received from the message originator if it is present in a response message.

Following is the overall usage of DE 48 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 197: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	M	X	M
Authorization Request Response/0110	C	X	C
Authorization Advice/0120: acquirer-generated	M	X	M
Authorization Advice/0120: system-generated	•	X	M
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	C	X	C
Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)	V		•
Reversal Request/0400	M	X	M
Reversal Request Response/0410	M	X	C

Message	Org	Sys	Dst
Reversal Advice/0420	.	X	C
Administrative Advice/0620: member-generated	C	.	C
Administrative Advice/0620: system-generated	.	X	C
Network Management Request/0800: PEK exchange	.	O	C
Network Management Request Response/0810: PEK exchange	M	X	.
Network Management Advice/0820: PEK exchange	.	X	M

Values

The length field must be in the range 001–999. Subelements are identified by valid subelement ID and length.

Application notes

Mastercard may occasionally introduce new DE 48 subelements and subfields between releases to facilitate special processing within a country, region, or among pilot participants. Mastercard requires customers to be able to successfully process various online messages that may contain new unannounced DE 48 subelements and subfields.

DE 48 and DE 119 contain different subelement definitions. Subelements defined for DE 48 must be sent in DE 48, not DE 119.

This data element's content may vary by program and service. Additional or supplemental information that may be required in a transaction message is specified in Dual Message Authorization System edits for this data element.

For information on other uses of this data element, refer to General message layout.

DE 48 Transaction Category Code

In Authorization/01XX and Reversal/04XX messages, the format is "LLLt," where "t" is the transaction category code (TCC). The message must contain an appropriate TCC as the first byte of data after the length within *DE 48*. The TCC classifies major categories of transactions, such as "Retail Sale," "Cash Disbursement," and "Mail Order."

Table 198: Valid values

TCC	Description	MC	NP	VI	TE	MS	CI
Space	A value of space in place of a valid TCC requests that the Authorization System perform the TCC assignment.	√	√	√	√	√	√
A	Auto/Vehicle Rental	√	√	√	√	√	

TCC	Description	MC	NP	VI	TE	MS	CI
C	Cash Disbursement	✓	✓	✓	✓	✓	✓
F	Restaurant	✓	✓	✓	✓	✓	
H	Hotel/Motel	✓	✓	✓	✓	✓	
O	Hospitalization, College	✓	✓	✓	✓	✓	
P	Payment Transaction	✓	✓			✓	✓
R	Retail Sale	✓	✓	✓	✓	✓	✓
T	Phone, Mail, or Electronic Commerce Order	✓	✓	✓	✓	✓	✓
U	Unique	✓	✓	✓	✓	✓	✓
X	Airline and Other Transportation Services	✓	✓	✓	✓	✓	
Z	ATM Cash Disbursement	✓	✓	✓	✓	✓	✓

NOTE: Mastercard will optionally assign the Transaction Category Code (TCC) in DE 48 (Additional Data: Private Use) of authorization messages on behalf of an acquirer. All acquirers processing through the Mastercard Network may choose to have the Authorization System assign the TCC in authorization messages on their behalf, if TCC is submitted with a <space> value. Refer to the [Quick Reference Booklet](#) for a complete definition of all TCC values.

DE 48 Subelement Encoding Scheme in Authorization Request/0100 Messages

The following table identifies the structure of the DE 48 subelement encoding scheme in the Authorization Request/0100 message.

Table 199: Values

LLL "VAR": 999 maximum bytes (TCC+ Subelement Data)							
3 bytes	1 byte	2 bytes	2 bytes	1-99 bytes	2 bytes	2 bytes	1-99 bytes
		SE ID + Length + Data will not exceed 103 bytes			SE ID + Length + Data will not exceed 103 bytes		
Total Data Element Length	TCC	First Subelement (SE) Data			Second Subelement (SE) Data		
		SE ID	SE Length	SE Variable Length Data	SE ID	SE Length	SE Variable Length data
Mandatory							
1002 maximum bytes (LLL + Subelement Data)							

Table 200: Values

Number of Bytes	Attribute	Description
3	Total Data Element Length	The "LLL" portion of the data element up to 999
1	Transaction Category Code (TCC)	Must be a valid TCC or a space
2	Subelement ID	In the range 00–99
2	Subelement Length	In the range of 01–99
1...99	Subelement Variable Length Data	Contains valid values

DE 48 Subelement Encoding Scheme in Network Management Messages

The following table identifies the structure of the *DE 48* subelement encoding scheme in Network Management messages.

Table 201: Values

LLL	"VAR": 999 maximum bytes (Subelement Data)					
3 bytes	2 bytes	2 bytes	1-96 bytes	2 bytes	2 bytes	1-96 bytes
Total Data Element Length	First Subelement (SE) Data			Second Subelement (SE) Data		
	SE ID	SE Length	SE Variable Length Data	SE ID	SE Length	SE Variable Length data
1002 maximum bytes (LLL + Subelement Data)						

Table 202: Values

Number of Bytes	Attribute	Description
3	Total Data Element Length	The "LLL" portion of LLLVAR
2	Subelement ID	In the range 00–99
2	Subelement Length	In the range of 01–96
1...96	Subelement Variable Length Data	Contains valid values.

List of DE 48 subelements

DE 48 subelements are listed in numeric order. Subelements that are specific to a brand service or program are clearly indicated in the subelement title or description or both.

Table 203: Values

Subelement ID	Subelement name	Data representation
03	Additional Visa Request Data (Visa Only)	an-14
05	Additional Visa Response Data (Visa Only)	ans...19; LLVAR
06	Advanced Digital Transaction Data 1	ans...84; LLVAR
07	Advanced Digital Transaction Data 2	ans...73; LLVAR
08	Advanced Digital Transaction Data 3	ans...98; LLVAR
09	Virtual Card Number	an...99; LLVAR
11	Key Exchange Block Data (Double-length Keys)	an-54
11	Key Exchange Block Data (Triple-length Keys)	an-70
12	Routing Indicator	a-1
13	Mastercard Hosted Mobile Phone Top-up Request Data	ans-47
14	Account Type Indicator	ans-1
15	Dual Message Authorization System Advice Date and Time	n-10
16	Processor Pseudo ICA	n-7
17	Authentication Indicator	n-1
18	Service Parameters	ans...99; LLVAR
19	Reserved for Future Use	N/A
20	Cardholder Verification Method	n-11
21	Acceptance Data	n-5; LLVAR
22	Multi-Purpose Merchant Indicator	ans...34; LLVAR
23	Payment Initiation Channel	an-2
24	Account Level Management [ALM] Service Data	an-30
25	Mastercard Cash Program Data	ans...14; LLVAR
26	Wallet Program Data	an-3
27	Transaction Analysis	an...97; LLVAR
28	Cardless ATM Order ID	an-10
29	Additional POS Terminal Locations	an-1

Subelement ID	Subelement name	Data representation
30	Token Transaction Identifier	ans...64; LLVAR
32	Mastercard Assigned ID	an-6
32	Visa Merchant Identifier (Visa Only)	an-8
33	PAN Mapping File Information	an...93; LLVAR
34	ATC Information	an-11
35	Contactless Non-Card Form Factor Request/Response	an-1
36	Additional Visa Request Data (Visa Only)	ans...94; LLVAR
37	Additional Merchant Data	ans...71; LLVAR
38	Account Category	an-1
39	Account Data Compromise Information	ans-30
40	Electronic Commerce Merchant/Cardholder Certificate Serial Number (Visa Only)	n...40
41	Electronic Commerce Certificate Qualifying Information	ans...95; LLVAR
42	Electronic Commerce Indicators	n-7
43	Universal Cardholder Authentication Field (UCAF)	ans...32; LLVAR
43	3-D Secure for Mastercard Identity Check	ans-28
43	Digital Secure Remote Payment Universal Cardholder Authentication Field [UCAF]	ans-28
43	3-D Secure Electronic Commerce Verification Service [Visa, JCB, Diners Club, and American Express]	n-1
44	Visa 3-D Secure Electronic Commerce Transaction Identifier (XID) (Visa and American Express)	b-20
45	Visa 3-D Secure Electronic Commerce Transaction Response Code (Visa and American Express)	an-1
46	Product ID (Visa Only)	an-2
47	Mastercard Payment Gateway Transaction Indicator	ans...8
48	Digital Commerce Solutions Indicators	ans...73; LLVAR
49	Time Validation Information	n-15
50	Embedded Interchange Data	ans...99; LLVAR
51	Merchant On-behalf Services	ans...99; LLVAR
52	Transaction Integrity Class	an-2
53	E-ID Request Code	ans...99; LLVAR

Subelement ID	Subelement name	Data representation
55	Merchant Fraud Scoring Data	an...32; LLVAR
56	Security Services Additional Data for Issuers	an...99; LLVAR
57	Security Services Additional Data for Acquirers	an...99; LLVAR
58	ATM Additional Data	ans-33
59	Reserved for Future Use	N/A
60	Additional Service Data for Issuers	ans...99; LLVAR
61	POS Data Extended Condition Codes	n-5
62	Real-Time Payment Information	ans...76; LLVAR
63	Trace ID	ans-15
64	Transit Program	n-4
65	Terminal Complaint Indicator	n-2
66	Authentication Data	ans...45; LLVAR
67	Mastercard Send Information	ans...99; LLVAR
68	Financial Account Information	ans...68; LLVAR
69–70	Reserved for Future Use	N/A
71	On-behalf Services	ans...40; LLVAR
72	Issuer Chip Authentication	b...16; LLVAR
73	Mastercard Internal Use Only	n...10
74	Additional Processing Information	an...30; LLVAR
75	Fraud Scoring Data	an...32; LLVAR
76	Mastercard Electronic Acceptance Indicator	a-1
77	Transaction Type Identifier	an-3
78	Payment Service Indicators (Visa Only)	ans-6
79	Chip CVR/TVR Bit Error Result	an...50; LLVAR
80	PIN Service Code	a-2
82	Address Verification Service Request	n-2
83	Address Verification Service Response	a-1
84	American Express Response Codes (American Express Only)	an-3
84	Merchant Advice Code	an-2
84	Visa Response Codes (Visa Only)	an-2

Subelement ID	Subelement name	Data representation
85	Account Status (Visa Only)	a-1
86	Relationship Participant Indicator (Visa Only)	a-1
87	Card Validation Code Result	a-1
87	CVV2 Response (Visa Only)	a-1
88	Magnetic Stripe Compliance Status Indicator	a-1
89	Magnetic Stripe Compliance Error Indicator	a-1
90	Lodging and Auto Rental Indicator	a-1
90	Custom Payment Service Request (Visa Only)	a-1
90	Custom Payment Service Request Response (Visa Only)	a-1
91	Custom Payment Service Request Transaction ID (Visa Only)	an...19; LLVAR
91	Custom Payment Service Response/Transaction ID (Visa Only)	ans...19; LLVAR
91	Acquirer Reference Data (American Express Only)	ans...15; LLVAR
92	CVC 2	n-3
92	CVV2 Data (Visa Only)	ns-6
93	Fleet Card ID Request Data (Visa Only)	ans...19; LLVAR
94	Commercial Card Inquiry Request (Visa Only)	ans-4
94	Commercial Card Inquiry Response (Visa Only)	ans-4
95	Mastercard Promotion Code	an-6
95	American Express CID Card Identifier (American Express Only)	n-4
96	Visa Market-specific Data Identifier (Visa Only)	a-1
97	Prestigious Properties Indicator (Visa Only)	a-1
98	Mastercard Corporate Fleet Card ID/Driver Number	n...6; LLVAR
99	Mastercard Corporate Fleet Card Vehicle Number	n...6; LLVAR

Subelement 03 (Additional Visa Request Data [Visa Only])

DE 48, subelement 03 (Additional Visa Request Data [Visa Only]) will contain additional request data when provided by the acquirer.

Table 204: Attributes

Attribute	Description
Subelement ID	03
Subelement Length	2
Data Representation	an-14
Data Field	Contents of subfields
Subfields	1
Justification	See subfields

Usage shows whether the field is mandatory, conditional, optional, or system-provided in applicable messages.

Table 205: Usage

Message	Org	Sys	Dst
Authorization Request /0100	C	•	C
Authorization Request Response/0110	CE	•	CE
Reversal Request/0400	C	•	C
Reversal Request Response/0410	CE	•	CE

Values

See subfields.

Subfield 01 (Plan Registration System Identifier)

DE 48, subelement 03, subfield 01 (Plan Registration System Identifier) provides a plan identifier for Visa Installment Solutions transactions.

Table 206: Attributes

Attribute	Description
Subfield ID	01

Attribute	Description
Subfield length	2
Data representation	an-10
Data field	Contents of positions 1–10
Justification	N/A

Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

Subelement 04 (Acceptance Environment Response Data [Visa Only])

DE 48, subelement 04 (Acceptance Environment Response Data[Visa Only]) contains the reasons that PSD2 exemptions were not accepted for Visa-branded transactions.

Table 207: Attributes

Attributes	Description
Subelement ID	04
Subelement length	2
Data representation	ans...80; LLVAR
Data field	Contents of positions 1–80
Subfields	N/A
Justification	N/A

Table 208: Usage

Message	Org	Sys	Dst
Authorization Request Response/0110	C	.	C

Values

Refer to Visa's *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

Subelement 05 (Additional Visa Response Data [Visa Only])

DE 48, subelement 05 (Additional Visa Response Data (Visa Only)) will contain additional response data when provided by Visa.

Table 209: Attributes

Attribute	Description
Subelement ID	05
Subelement Data Length	2 positions
Data Representation	ans...19; LLVAR
Data Field	Contents of subfields
Subfields	3
Justification	See subfields

Usage

Whether use is mandatory, conditional, optional, or not required is noted for each applicable message.

Table 210: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	C	•	C
Authorization Reversal Response/0410	C	•	C

Values

See subfields.

Subfield 01 (Commercial Choice Program Data, Agreement ID)

DE 48, subelement 05, subfield 01 (Commercial Choice Program Data, Agreement ID) contains the Commercial Choice Program data agreement ID as received from Visa.

Table 211: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions

Attribute	Description
Data Representation	ans...4; LLVAR
Data Field	Contents of subfield 01
Justification	N/A

Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

Subfield 02 (Digital Authentication Program)

DE 48, subelement 05, subfield 02 (Digital Authentication Program) contains the value to indicate when a transaction qualifies for the Digital Authentication Program as received from Visa.

Table 212: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	ans...2; LLVAR
Data Field	Contents of subfield 02
Justification	N/A

Values

Contains Authentication Program 01 (DAF indicator). Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

Subfield 03 (Authorization Response Source Reason Code)

DE 48, subelement 05, subfield 03 (Authorization Response Source Reason Code) contains the authorization response reason code as received from Visa.

Table 213: Attributes

Attribute	Description
Subfield ID	03
Subfield Data Length	2 positions
Data Representation	ans-1

Attribute	Description
Data Field	Contents of subfield 03
Justification	N/A

Values

Contains the response source reason code that identifies the source of the Visa Field 39 response decision. Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

Subfield 04 (Account Funding Source)

DE 48, subelement 05, subfield 04 (Account Funding Source) contains the Account Funding Source product code value received from Visa.

Table 214: Attributes

Attribute	Description
Subfield ID	04
Subfield Length	2
Data Representation	an-1
Data Field	Contents of subfield 04
Justification	N/A

Subfield 05 (Applied Product Platform Code)

DE 48, subelement 05, subfield 05 (Applied Product Platform Code) contains the Applied Product Platform Code value received from Visa.

Table 215: Attributes

Attribute	Description
Subfield ID	05
Subfield Length	2
Data Representation	an-2
Data Field	Contents of subfield 05
Justification	N/A

Subfield 06 (Applied Cardholder ID Method)

DE 48, subelement 05, subfield 06 (Applied Cardholder ID Method) contains the Cardholder ID Method Indicator value received from Visa.

Table 216: Attributes

Attribute	Description
Subfield ID	06
Subfield Length	2
Data Representation	an-1
Data Field	Contents of subfield 06
Justification	N/A

Subfield 07 (Applied E-Commerce Payment Indicator/ECI Only)

DE 48, subelement 05, subfield 07 (Applied E-Commerce Payment Indicator/ECI Only) contains the Applied E-Commerce Payment Indicator/ECI Only code value received from Visa.

Table 217: Attributes

Attribute	Description
Subfield ID	07
Subfield Length	2
Data Representation	n-2
Data Field	Contents of subfield 07
Justification	N/A

Subelement 06 (Advanced Digital Transaction Data 1)

DE 48, subelement 06 (Advanced Digital Transaction Data 1) provides transaction-related data captured during checkout to the issuer.

Attributes

Attribute	Value	Description
Subelement ID	n-2	
Subelement length	n-2	
Data representation	ans...84; LLVAR	

Attribute	Value	Description
Data field		Contents of subfields
Number of subfields	3	Subfield 01 (Binding ID) Subfield 02 (Phone Number) Subfield 03 (Device Location)
Justification		See subfields

Usage

Usage shows whether the field is mandatory, conditional, optional, or system-provided in applicable messages.

Message	Org	Sys	Dst
Authorization Request/0100	.	X	C

Application notes

The Dual Message System inserts this subelement into Authorization Request/0100 messages when the issuer has opted-in to receive it.

Subfield 01 (Binding ID)

DE 48, subelement 06, subfield 01 (Binding ID) helps to uniquely identify the component of data that is bound to the credential used in the transaction.

Attributes

Attribute	Description
Subfield ID	01
Subfield length	2
Data representation	ans...48; LLVAR
Data field	Contents of subfield 01
Justification	Left

Values

The unique and fixed identifier to a component of data bound to the credential.

Application notes

When DE 104 (Digital Payment Data), subelement 005 (Digital Authentication Data), subfield 001 (Cardholder Verification) indicates a value of...	Then...
1 (Transaction authenticated with a Mastercard-qualified multi-factor authentication [MFA] method) in position 1 NOTE: In Europe region: Transaction authenticated with a Mastercard-qualified MFA method used as Strong Customer Authentication [SCA] delegation.	the component of data is an identifier demonstrating the possession of a consumer device that is bound to the token. Multiple identifiers may be created for the same consumer device bound to the token. It will help establish a pattern of device possession that further confirms the authenticity of the consumer who initiated the transaction.
2 (Transaction authenticated with a Mastercard-operated authentication method) in position 1	
4 (Device wallet transaction authenticated with a Consumer Device Cardholder Verification Method [CDCVM]) in position 1 NOTE: This value may not be present in current device wallet implementations.	the component of data is a unique identifier to the consumer device that is bound to the token. It will help establish a pattern of device usage that further confirms the authenticity of the consumer who initiated the transaction.

Subfield 02 (Phone Number)

DE 48, subelement 06, subfield 02 (Phone Number) contains the phone number that was used to authenticate the consumer during checkout, or the consumer's preferred phone number. The presence of the phone number helps establish the consumer's authenticity when matching the phone number provided during checkout to a list of known phone numbers for the consumer.

Attributes

Attribute	Description
Subfield ID	02
Subfield length	2
Data representation	n...15; LLVAR
Data field	Contents of subfield 02
Justification	Left

Values

The full phone number or partial phone number of the consumer used during checkout.

Subfield 03 (Device Location)

DE 48, subelement 06, subfield 03 (Device Location) contains the latitude and longitude of the device used when the cardholder was authenticated during checkout. This field helps to determine if the transaction was initiated from an unexpected location.

Attributes

Attribute	Description
Subfield ID	03
Subfield length	2
Data representation	ans-9
Data field	Contents of subfield 03
Justification	N/A

Values

Latitude and longitude where the device of the consumer is attempting to check out is located.

Device location latitude an-4, hexadecimal encoded degrees with two decimal places.

Device location longitude an-4, hexadecimal encoded degrees with two decimal places.

Device location lat/long Sector n-1, one of the following values:

Value	Latitude	Longitude
1	N	W
2	N	E
3	S	W
4	S	E

Subelement 07 (Advanced Digital Transaction Data 2)

DE 48, subelement 07 (Advanced Digital Transaction Data 2) provides transaction-related data captured during checkout to the issuer.

Attributes

Attribute	Value	Description
Subelement ID	n-2	
Subelement length	n-2	
Data representation	ans...73; LLVAR	

Attribute	Value	Description
Data field		Contents of subfields
Number of subfields	2	Subfield 01 (Email Address Format) Subfield 02 (Email Address)
Justification		See subfields

Usage

Usage shows whether the field is mandatory, conditional, optional, or system-provided in applicable messages.

Message	Org	Sys	Dst
Authorization Request/0100	.	X	C

Application notes

The Dual Message System inserts this subelement into Authorization Request/0100 messages when the issuer has opted-in to receive it.

Subfield 01 (Email Address Format)

DE 48, subelement 07, subfield 01 (Email Address Format) indicates if an email address was passed in hashed or clear format during checkout.

Attributes

Attribute	Description
Subfield ID	01
Subfield length	2
Data representation	n-1
Data field	Contents of subfield 01
Justification	N/A

Values

Value	Description	Corresponding subfield 02...
0	Hashed email format received during checkout	Contains a hashed email address.

Value	Description	Corresponding subfield 02...
1	Clear email format received during checkout	Contains a clear email address.

Subfield 02 (Email Address)

DE 48, subelement 07, subfield 02 (Email Address) contains the cardholder email address. The presence of the email address helps establish the consumer's authenticity when matching the email address provided during checkout to a list of known email addresses for the consumer.

Attributes

Attribute	Description
Subfield ID	02
Subfield length	2
Data representation	ans...64; LLVAR
Data field	Contents of subfield 02
Justification	Left

Values

The clear email address or hashed email address collected during checkout.

Application notes

Hashing helps the issuer verify the cardholder's email without the merchant or wallet provider providing the full email for privacy and security reasons.

When the merchant or wallet provider provides a hashed email address, Mastercard receives the hash email address and includes it in the data sent to the issuer. The issuer may apply the hash to an email address on file for the cardholder and attempt to match it with the hashed email address provided by the merchant or wallet provider.

When the hash computed by the issuer matches the hash of the one received from the merchant or wallet provider, the comparison with the hash value may be used by the issuer to assist in determining the transaction authorization decision.

The field contains the hash resulting from the following 'emailAddressHash' algorithm, right-padded with spaces. 'emailAddress' is the lower case UTF-8 bytes of the email address.

The following two functions define the hash algorithm:

```
public String emailAddressHash (String emailAddress) {  
    String random8Bytes = 123CCB2F30BA420B  
    return random8Bytes + leastSignificant24bytes(strongerHash(emailAddress +  
    random8Bytes))  
}
```



```
public String strongerHash (String dataToHash) {  
    String currentHash = dataToHash;  
    for (int i = 0; i < 5000; i++) {  
        currentHash = sha256(currentHash);  
    }  
    return sha256(sha256(dataToHash) + currentHash);  
}
```

NOTE: String random8Bytes = 123CCB2F30BA420B is a fixed value for all merchants and wallet providers.

Hash calculation example:

Component	Example
Sample email address	cardholdername@walletprovider.com
emailAddress	63617264686F6C6465726E616D654077616C6C65 7470726F76696465722E636F6D
random8Bytes	123CCB2F30BA420B
Email address hash output	123CCB2F30BA420B2476A8250E0E146164385A34 D34CC9E3A5680D7C5252C481

Subelement 08 (Advanced Digital Transaction Data 3)

DE 48, subelement 08 (Advanced Digital Transaction Data 3) provides transaction-related data captured during checkout to the issuer.

Attributes

Attribute	Value	Description
Subelement ID	n-2	
Subelement length	n-2	
Data representation	ans...98; LLVAR	
Data field		Contents of subfields
Number of subfields	4	Subfield 01 (IP Address) Subfield 02 (Shipping Address Line 1) Subfield 03 (Shipping Address Country) Subfield 04 (Shipping Address Postal Code)

Attribute	Value	Description
Justification		See subfields

Usage

Usage shows whether the field is mandatory, conditional, optional, or system-provided in applicable messages.

Message	Org	Sys	Dst
Authorization Request/0100	.	X	C

Application notes

The Dual Message System inserts this subelement into Authorization Request/0100 messages when the issuer has opted-in to receive it.

Subfield 01 (IP Address)

DE 48, subelement 08, subfield 01 (IP Address) contains a unique string of characters that identifies a device. The presence of the IP address helps determine if the transaction was initiated from an unusual network, helping establish a pattern of safe device usage that further confirms the authenticity of the consumer who initiated the transaction.

Attributes

Attribute	Description
Subfield ID	01
Subfield length	2
Data representation	ans...39; LLVAR
Data field	Contents of subfield 01
Justification	Left

Values

The IP address collected during checkout.

Subfield 02 (Shipping Address Line 1)

DE 48, subelement 08, subfield 02 (Shipping Address Line 1) helps identify the cardholder that initiated the transaction. Shipping address is one component of data that helps establish the

consumer's authenticity when matching the consumer's own billing address or set of commonly used shipping addresses.

Attributes

Attribute	Description
Subfield ID	02
Subfield length	2
Data representation	ans...30; LLVAR
Data field	Contents of subfield 02
Justification	Left

Values

The shipping address line 1 collected during checkout.

Subfield 03 (Shipping Address Country)

DE 48, subelement 08, subfield 03 (Shipping Address Country) contains the three-character alphabetic country code where the shipping address is located. Refer to the *Quick Reference Booklet* for specification format.

Attributes

Attribute	Description
Subfield ID	03
Subfield length	2
Data representation	a-3
Data field	Contents of subfield 03
Justification	N/A

Values

The shipping address country collected during checkout.

Subfield 04 (Shipping Address Postal Code)

DE 48, subelement 08, subfield 04 (Shipping Address Postal Code) contains the postal code of the shipping address.

Attributes

Attribute	Description
Subfield ID	04
Subfield length	2
Data representation	ans...10; LLVAR
Data field	Contents of subfield 04
Justification	Left

Values

The shipping address postal code collected during checkout.

Subelement 09 (Virtual Card Number Data)

DE 48, subelement 09 (Virtual Card Number Data) is used to provide the VCN data required by the issuers for processing MDES token transactions.

Table 218: Attributes

Attribute	Description
Subelement ID	09
Subelement Data Length	02 positions
Data Representation	an...99, LLVAR
Data Field	Contents of subfields 01-03
Subfields	Subfield 01 (Virtual Card Number Indicator) Subfield 02 (Virtual Card Number) Subfield 03 (Virtual Card Number Expiration Date)

Usage

Usage shows whether the field is mandatory, conditional, optional, or system-provided in applicable messages.

Table 219: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	•	X	C
Authorization Advice/0120: system-generated	•	X	C
Authorization Advice/0120: issuer-generated	C	•	•
Authorization Advice/0120: acquirer-generated	•	X	C
Reversal Request/0400	•	X	C
Reversal Advice/0420: system-generated	•	X	C

Subfield 01 (Virtual Card Number Indicator)

DE 48, subelement 09, subfield 01 (Virtual Card Number Indicator) provides the virtual card number indicator information to the issuer.

Table 220: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	an-1
Data Field	Contents of subfield 01
Justification	N/A

Values

V is Virtual card number.

Subfield 02 (Virtual Card Number)

DE 48, subelement 09, subfield 02 (Virtual Card Number) provides the virtual card number associated with the MDES token to the issuer.

Table 221: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions

Attribute	Description
Data Representation	n...19, LLVAR
Data Field	Contents of subfield 02
Justification	N/A

Subfield 03 (Virtual Card Number Expiration Date)

DE 48, subelement 09, subfield 03 contains the VCN expiration date.

Table 222: Attributes

Attribute	Description
Subfield ID	03
Subfield Data Length	2 positions
Data Representation	n-4; format YYMM
Data Field	Contents of subfield 03
Justification	N/A

Subelement 10 (Encrypted PIN Block Key)

DE 48, subelement 10 (Encrypted PIN Block Key) contains 16 hexadecimal characters in the range 0–9 and A–F to represent the 16 bits of a PIN block key encrypted under another key. Individual network, program, or service rules define the other encryption key.

NOTE: The Dual Message Authorization System currently does not use this data element.

Subelement 11 (Key Exchange Block Data [Double-Length Keys])

DE 48, subelement 11 (Key Exchange Block Data [Double-Length Keys]) contains a data block specifically formatted to contain all the control data, encrypted key data, and key check values to complete an encryption key change operation between any two processors (for example, between a customer or and the network).

Table 223: Attributes

Attribute	Description
Subelement ID	11
Subelement Data Length	2 positions
Data Representation	an-54

Attribute	Description
Data Field	Contents of subfields 1–5
Subfields	5
Justification	N/A

Usage

Following is the usage of subelement 11 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 224: Applicable messages

Message	Org	Sys	Dst
Network Management Request/0800: PEK exchange	•	M	M
Network Management Request Response/0810: PEK exchange	M	M	•

Values

Subelement 11 must be encoded for subfields 1–5 defined as follows.

Table 225: Subfield 1 (Key Class ID)

Attribute	Description
Data Representation	an-2
Data Field	Contents of positions 1–2
Values	PK (PIN key)

Table 226: Subfield 2 (Key Index Number)

Attribute	Description
Data Representation	n-2
Data Field	Contents of positions 3–4
Values	00

Table 227: Subfield 3 (Key Cycle Number)

Attribute	Description
Data Representation	n-2
Data Field	Contents of positions 4–6
Values	00–99 (sequential)

Table 228: Subfield 4 (PIN Encryption Key [PEK])

Attribute	Description
Data Representation	an-32
Data Field	Contents of positions 7–38
Values	hex, 0–9, A–F

Table 229: Subfield 5 (Key Check Value)

Attribute	Description
Data Representation	an-16
Data Field	Contents of positions 39–54
Values	hex, 0–9, A–F

Subelement 11 (Key Exchange Block Data [Triple-Length Keys])

DE 48, subelement 11 (Key Exchange Block Data [Triple-Length Keys]) contains a data block specifically formatted to contain all the control data, encrypted key data, and key check values to complete an encryption key change operation between any two processors (for example, between a customer or and the network).

Table 230: Attributes

Attribute	Description
Subelement ID	11
Subelement Data Length	2 positions
Data Representation	an-70

Attribute	Description
Data Field	Contents of subfields 1–5
Subfields	5
Justification	N/A

Usage

Following is the usage of subelement 11 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 231: Applicable messages

Message	Org	Sys	Dst
Network Management Request/0800: PEK exchange	•	M	M
Network Management Request Response/0810: PEK exchange	M	M	•

Values

Subelement 11 must be encoded for subfields 1–5 as defined below.

Table 232: Subfield 1 (Key Class ID)

Attribute	Description
Data Representation	an-2
Data Field	Contents of positions 1–2
Values	PK (PIN key)

Table 233: Subfield 2 (Key Index Number)

Attribute	Description
Data Representation	n-2
Data Field	Contents of positions 3–4
Values	00

Table 234: Subfield 3 (Key Cycle Number)

Attribute	Description
Data Representation	n-2
Data Field	Contents of positions 5–6
Values	00–99 (sequential)

Table 235: Subfield 4 (PIN Encryption Key [PEK])

Attribute	Description
Data Representation	an-48
Data Field	Contents of positions 7–54
Values	hex, 0–9, A–F

Table 236: Subfield 5 (Key Check Value)

Attribute	Description
Data Representation	an-16
Data Field	Contents of positions 55–70
Values	hex, 0–9, A–F

Subelement 12 (Routing Indicator)

DE 48, subelement 12 (Routing Indicator) is defined and derived by the Dual Message Authorization System and passed to the issuer to indicate that alternate issuer host routing has been invoked. Subelement 12 is applicable only to issuers that use an alternate issuer host route instead of the Stand-In System.

Table 237: Attributes

Attribute	Description
Subelement ID	12
Subelement Data Length	2 positions
Data Representation	a-1

Attribute	Description
Data Field	Contents of position 1
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 12 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 238: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	•	X	C

Table 239: Values

Value	Description
A	Alternate issuer host routing
P	Primary issuer host routing

Subelement 13 (Mastercard Hosted Mobile Phone Top-Up Request Data)

DE 48, subelement 13 (Mastercard Hosted Mobile Phone Top-up Request Data) contains the mobile phone number and mobile phone service provider name.

Table 240: Attributes

Attribute	Description
Subelement ID	13
Subelement Data Length	2 positions
Data Representation	ans-47
Data Field	Contents of subfields
Subfields	2
Justification	See subfields

Usage

Following is the usage of subelement 13 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 241: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Request Response/0110	CE	.	CE

Subfield 1 (Mobile Phone Number)

DE 48, subelement 13, subfield 1 (Mobile Phone Number) contains the phone number of the wireless phone for which the customer is purchasing extra service.

Table 242: Attributes

Attribute	Description
Data Representation	ans-17
Data Field	Contents of positions 1–17
Justification	Left

Values

Cannot contain all spaces or all zeros.

Subfield 2 (Mobile Phone Service Provider Name)

DE 48, subelement 13, subfield 2 (Mobile Phone Service Provider Name) contains the name or other identifier of the mobile phone service provider.

Table 243: Attributes

Attribute	Description
Data Representation	ans-30
Data Field	Contents of positions 18–47
Justification	Left

Values

Cannot contain all spaces or all zeros.

Subelement 14 (Account Type Indicator)

DE 48, subelement 14 (Account Type Indicator) contains the value that identifies the cardholder's account type choice. If this subelement is not provided by the acquirer, Mastercard will derive the value. Subelement 14 must only be included on Brazil Flex Card Service-enabled transactions.

Table 244: Attributes

Attribute	Description
Subelement ID	14
Subelement Data Length	2 positions
Data Representation	ans-1
Data Field	Contents of position 1
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 14 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 245: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	X	C
Authorization Request Response/0110	CE	.	C
Authorization Advice/0120	C	X	C
Authorization Advice Response/0130	CE	.	C
Reversal Request/0400	C	X	C
Reversal Request Response/0410	CE	.	C

Acquirers in Brazil must be able to receive DE 48 (*Additional Data: National Use*), subelement 14 (Account Type Indicator) in the following response messages only if they have sent this data

element in an Authorization Request/0100, Authorization Advice/0120, or Reversal Request/0400 message:

- Authorization Request Response/0110 message
- Authorization Advice Response/0130 message
- Reversal Request Response/0410 message

Values

C = Credit Transaction

D = Debit Transaction

Application notes

Subelement 14 (Account Type Indicator) is used in Brazil to identify the transaction as:

- Credit: (Credit/Mastercard). The transaction will be processed by the Dual Message Authorization System with a Mastercard credit brand/product.
- Debit: (Debit/Mastercard). The transaction will be processed by the Dual Message Authorization System as a Debit Mastercard transaction with a Mastercard debit brand/product.

For Brazil issued Flex Card Service enabled POS transactions acquired outside of Brazil, Mastercard will populate *DE 48*, subelement 14 based on the value present in *DE 23*, position 3.

- C = Credit Transaction (values 0, 1, 2, 3, or 4)
- D = Debit Transaction (values 5, 6, 7, 8, or 9)

For Brazil issued Flex Card Service enabled ATM transactions acquired outside of Brazil, Mastercard will populate *DE 48*, subelement 14 based on the value present in *DE 3*, subfield 2:

- C = Credit Transaction if *DE 3*, subfield 2 is not equal to 10 or 20.
- D = Debit Transaction if *DE 3*, subfield 2 is equal to 10 or 20.

For Brazil issued Flex Card Service enabled transactions acquired in Brazil, Mastercard will populate *DE 48*, subelement 14 based on the value present in *DE 3*, subfield 2:

- C = Credit Transaction if *DE 3*, subfield 2 is not equal to 10 or 20.
- D = Debit Transaction if *DE 3*, subfield 2 is equal to 10 or 20.

For Brazil issued Flex Card Service enabled card present transactions acquired outside of Brazil, if the values in *DE 23* is not present then Mastercard will populate *DE 48*, subelement 14 with C = Credit transaction.

For Brazil issued Flex Card Service enabled card not present transactions acquired outside of Brazil, Mastercard will populate *DE 48*, subelement 14 with 'C' = Credit transaction.

Subelement 15 (Authorization System Advice Date and Time)

DE 48, subelement 15 (Authorization System Advice Date and Time), (in UTC units) is a transaction time stamp that the Dual Message Authorization System supplies for each

Authorization Advice/0120: acquirer-generated message. It indicates the date and time that the advice is entered into the network.

Table 246: Attributes

Attribute	Description
Subelement ID	15
Subelement Data Length	2 positions
Data Representation	n-10
Data Field	Contents of subfields 1–2
Subfields	2
Justification	See subfields

Usage

Following is the usage of subelement 15 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 247: Applicable messages

Message	Org	Sys	Dst
Authorization Advice/0120: acquirer-generated	•	X	M
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	CE	•	C

Values

See subfields.

Application notes

Acquirers may use the contents of subelement 15 to aid in the creation of the clearing message.

Subfield 1 (Date)

DE 48, subelement 15, subfield 1 (Date) contains the valid date of the Authorization Advice/0120: acquirer-generated message inserted by the Dual Message Authorization System.

Table 248: Attributes

Attribute	Description
Data Representation	n-4
Data Field	Contents of positions 1-4
Justification	N/A

Values

This subfield contains a valid date in MMDD format.

Subfield 2 (Time)

DE 48, subelement 15, subfield 2 (Time) contains the valid time of the Authorization Advice/0120: acquirer-generated message inserted by the Dual Message Authorization System.

Table 249: Attributes

Attribute	Description
Data Representation	n-6
Data Field	Contents of positions 5-10
Justification	N/A

Values

This subfield contains a valid time in hhmmss format.

Subelement 16 (Processor Pseudo ICA)

DE 48, subelement 16 (Processor Pseudo ICA) identifies the institution submitting a request or advice that is not the same as the institution provided in DE 32 (*Acquiring Institution ID Code*) or DE 33 (*Forwarding Institution ID Code*).

Table 250: Attributes

Attribute	Description
Subelement ID	16

Attribute	Description
Subelement Data Length	2 positions
Data Representation	n-7
Data Field	N/A
Subfields	N/A
Justification	N/A

Usage

Following is the usage of *DE 48*, subelement 16 (whether it is mandatory, conditional, optional, system-provided, or not required) in applicable messages.

Table 251: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	.	X	C
Authorization Advice/0120: acquirer-generated	.	X	C
Authorization Advice/0120: system-generated	.	X	C

Application notes

The Dual Message Authorization System populates this subelement based on the institution submitting the transaction and the issuer's preference for receipt of this data.

Subelement 17 (Authentication Indicator)

DE 48, subelement 17 (Authentication Indicator) is defined by the Dual Message Authorization System and is passed to the issuer to indicate that the transaction qualified for Authentication service.

Table 252: Attributes

Attribute	Description
Subelement ID	17
Subelement Data Length	01 positions
Data Representation	n-1
Subfields	N/A

Usage

Following is the usage of subelement 17 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 253: Applicable messages

Message	Org	Sys	Dst
Authorization request/0100	•	X	C
Authorization request response/0110	•	X	C
Authorization Advice/0120: acquirer-generated	•	X	C
Authorization Advice/0120: system-generated	•	X	C
Authorization advice response/0130: issuer-generated (responding to acquirer-generated 0120)	•	X	C

Table 254: Valid values

Value	Description
1	Transaction qualified for Authentication Service Type 1
2	Transaction qualified for Authentication Service Type 2

Application notes

Cardholder Authentication Service

Acquirers must support receiving subelement 17 in authorization response messages.

Issuers participating in the Cardholder Authentication Service must be prepared to receive the valid cardholder authentication result in subelement 17.

Subelement 18 (Service Parameters)

DE 48, subelement 18 (Service Parameters) defines the service parameters of the transaction.

Table 255: Attributes

Attribute	Value
Subelement ID	18
Subelement Data Length	2 positions

Attribute	Value
Data Representation	ans...99; LLVAR
Data Field	Contents of subfield 01
Subfields	01
Justification	N/A

Usage

Following is the usage of subelement 18 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 256: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	X	.

Values

Contains the Canada Domestic Indicator value in subfield 01.

Subfield 01 (Canada Domestic Indicator)

DE 48, subelement 18, subfield 01 (Canada Domestic Indicator) indicates the merchant accepts Canada domestic Debit Mastercard cards.

Table 257: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	an-1
Data Field	Contents of subfield 01
Justification	N/A

Values

Y = Yes, the merchant accepts Canada domestic Debit Mastercard cards.

Example: 18050101Y—subelement 18 (Service Parameters), length 05, subfield 01 (Canada Domestic Indicator), length 01, value Y.

NOTE: The remaining 94 characters are reserved for future use.

Subelement 20 (Cardholder Verification Method)

Acquirers use *DE 48*, subelement 20 (Cardholder Verification Method) to notify Mastercard that the original transaction is signature/offline PIN-based, no CVM used, or online PIN-based for transaction routing in Authorization Advice/0120: acquirer-generated and Reversal Request/0400 messages.

Table 258: Attributes

Attribute	Value
Subelement ID	20
Subelement Data Length	2 positions
Data Representation	a-1
Data Field	Contents of position 1
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 20 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 259: Applicable messages

Message	Org	Sys	Dst
Authorization Advice/0120: acquirer-generated	M	X	•
Reversal Request/0400	M	X	•
Reversal Advice/0420	•	M	M

Table 260: Values

Value	Description
P	Online PIN verification

Value	Description
S	Can signify signature, "Offline PIN verification" (for chip transactions), "M-PIN" (for Mobile Device with PIN entry capability) or "No CVM used"

If DE 48, subelement 20 is present and the value is not P or S, the transaction will be rejected with format error 30.

Subelement 21 (Acceptance Data)

DE 48 (Additional Data-Private Use), subelement 21 (Acceptance Data) indicates the merchant terminal's capability to support specific programs and services.

Table 261: Attributes

Attribute	Description
Subelement ID	21
Subelement Data Length	2 positions
Data Representation	n-11; LLVAR
Data Field	Contents of subfields
Subfields	Subfield 01: mPOS Acceptance Device Type

Usage

Following is the usage of subelement 21 (whether it is mandatory, conditional, optional, system-provided, or not required) in applicable messages.

Table 262: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Advice/0120: acquirer-generated	C	.	C

Values

See subfields.

Application notes

The subelement should be provided to indicate the merchant terminal's capabilities in supporting specific programs and services.

Subfield 01 (mPOS Acceptance Device Type)

DE 48 (Additional Data: Private Use), subelement 21 (Acceptance Data), subfield 01 (mPOS Acceptance Device Type) identifies the type of device used by the merchant as the terminal for accepting mPOS transactions. This subfield is required for all mPOS terminals.

Table 263: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	n-1
Data Field	Contents of subfield 1
Justification	N/A

Table 264: Values

Value	Description
0	Dedicated mPOS Terminal with PCI compliant dongle (with or without key pad)
1	Off the Shelf Mobile Device
2–9	Reserved for Future Use

Subfield 02 (Additional Terminal Capability Indicator)

DE 48, subelement 21, subfield 02 (Additional Terminal Capability Indicator) identifies Mastercard Consumer Presented QR and barcode terminal capabilities. This subfield is optional. Customers that have opted to submit or process tokenized Mastercard Consumer Presented QR transactions are expected to support this subfield.

Table 265: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	n...2; LLVAR
Data Field	Contents of positions 1–2

Attribute	Description
Justification	N/A
Position 1	Mastercard Consumer Presented QR Capability Support Indicator
Position 2	Mastercard Consumer Presented Barcode Capability Support Indicator

Table 266: Values

Value	Description
0	Not Supported
1	Supported

Subfield 2: Valid Position 1 and Position 2 Value Combinations

Table 267: Valid combinations

Valid Combinations	QR Capability	Barcode Capability
00	Not supported	Not supported
01	Not supported	Supported
10	Supported	Not supported
11	Supported	Supported

Subelement 22 (Multi-Purpose Merchant Indicator)

DE 48, subelement 22 (Multi-Purpose Merchant Indicator) may be used for multiple purposes needed to identify the transaction and may contain data that is unique to programs, services, or other specifics about the transaction.

Table 268: Attributes

Attribute	Description
Subelement ID	22
Subelement Data Length	2 positions
Data Representation	ans...34; LLVAR

Attribute	Description
Data Field	Contents of subfields 01–06
Subfields	6
Justification	See subfields

Usage

Following is the usage of subelement 22 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 269: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Request Response/0110	C	X	C
Authorization Advice/0120: acquirer-generated	O	.	C
Authorization Advice/0120: system-generated	.	X	C
Authorization Advice Response/0130	C	.	C
Reversal Request/0400	O	.	C
Reversal Request Response/0410	C	.	C
Reversal Advice/0420	.	X	C

Application notes

Dual Message Authorization System will remove *DE 48*, subelement 22, subfields 01, 02, 03, 05, and 06 if received in an Authorization Request Response/0110 message from an issuer.

Dual Message Authorization System will remove *DE 48*, subelement 22, subfields 01, 02, 03, 04, 05, and 06 if received in an Reversal Request Response/0410 or Advice Response/0130 message from an issuer.

Refer to the subfields contained within this subelement to obtain the identifier supporting programs, services, or other specifics about the transaction.

Subfield 01 (Low-Risk Merchant Indicator)

DE 48, subelement 22 (Multi-Purpose Merchant Indicator), subfield 01 (Low-Risk Merchant Indicator) is used by acquirers to identify why a low-risk transaction did not use Strong

Customer Authentication (SCA) to prevent an unnecessary decline by an issuer in the European Economic Area (EEA), United Kingdom (UK), and Gibraltar.

For Visa-branded transactions, *DE 48*, subelement 22 (Multi-Purpose Merchant Indicator), subfield 01 (Low-Risk Merchant Indicator) is used to identify the transaction is from a Trusted Merchant, not needing Strong Customer Authentication (SCA) and preventing unnecessary declines by issuers in the European Economic Area (EEA), United Kingdom (UK), and Gibraltar.

Table 270: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	an-2
Data Field	Contents of subfield 01
Justification	N/A

Usage

Following is the usage of subelement 22, subfield 01 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 271: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	O	.	C

Table 272: Values

Value	Description
01	Merchant Initiated Transaction
02	Acquirer Low-Fraud and Transaction Risk Analysis
03	Recurring Payment
04	Low-Value Payment
05	Strong Customer Authentication (SCA) Delegation
06	Secure Corporate Payment

Value		Description
07	=	Authentication Outage Exception
08	=	Trusted Merchant Exemption (Visa Only)

Application notes

Subfield 01 values enable acquirers to provide issuers with an indicator to identify Authorization Request/0100 messages originating from merchants that use an acquirer exemption or other reasons for not applying SCA on transactions. This data will assist issuers in determining whether to approve or decline the transaction and to justify to Local Competent Authorities the reason why the issuers did not apply SCA on the transaction as required under PSD2.

Acquirers should only submit value 08 on Visa-branded transactions. The Dual Message Authorization System drops this value if received on transactions which are not Visa-branded.

Subfield 02 (Single Tap Indicator)

DE 48, subelement 22 (Multi-Purpose Merchant Indicator), subfield 02 (Single Tap Indicator) allows acquirers to indicate that the merchant's point of interaction environment is capable of supporting single tap processing.

Table 273: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	an-1
Data Field	Contents of Subfield 02
Justification	N/A

Usage

Following is the usage of subelement 22, subfield 02 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 274: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C

Table 275: Values

Value	Description
1	= Merchant POI capable of single tap processing

Application notes

Value 1 must be used if the POI supports the service; otherwise, the subfield is not present.

Subfield 03 (Response to PIN Request)

DE 48, subelement 22 (Multi-Purpose Merchant Indicator), subfield 03 (Response to PIN Request) allows acquirers to indicate that the transaction contains an intentionally duplicated (replayed) Application Transaction Counter (ATC) value, within the context of a single tap transaction.

Table 276: Attributes

Attribute	Description
Subfield ID	03
Subfield Data Length	2 positions
Data Representation	an-1
Data Field	Contents of Subfield 03
Justification	N/A

Usage

Following is the usage of subelement 22, subfield 03 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 277: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C

Table 278: Values

Value	Description
1	Transaction contains an intentionally duplicated (replayed) ATC value.

Application notes

Value 1 must be used by acquirers only if the issuer requests a PIN in single tap mode; otherwise, the subfield is not present.

Subfield 04 (Issuer PIN Request in a Single Tap Mode)

DE 48 (Additional Data: Private Use), subelement 22 (Multi-Purpose Merchant Indicator), subfield 04 (Issuer PIN Request in a Single Tap Mode) allows issuers in Europe, Middle East/Africa, Asia/Pacific, and Latin America and the Caribbean regions to request a PIN within the context of a single tap transaction scenario. Does not apply to the United States and Canada regions.

Table 279: Attributes

Attribute	Description
Subfield ID	04
Subfield Data Length	2 positions
Data Representation	an-1
Data Field	Contents of Subfield 04
Justification	N/A

Usage

Following is the usage of subelement 22, subfield 04 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 280: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	C	X	C

Table 281: Valid values

Value	Description
1	Issuer requests PIN in Single Tap mode.

Application notes

Value 1 must be used if the issuer requests a PIN in Single Tap mode; otherwise, the subfield is not present.

When the issuer participates in the On-Behalf Contactless Counters Service, Mastercard will populate subfield 04 in Authorization Request Response/0110 messages and forward to the acquirer when:

- The issuer supports duplicate ATC
- A card present transaction is declined with response code 65 (Decline: Exceeds withdrawal count limits)
- *DE 48*, subelement 22, subfield 02 is value 1 (Merchant capable of single tap processing)

Subfield 05 (Cardholder/Merchant Initiated Transaction Indicator)

DE 48, subelement 22 (Multi-Purpose Merchant Indicator), subfield 05 (Cardholder/Merchant Initiated Transaction Indicator) allows acquirers to identify whether the transaction was initiated by the cardholder or the merchant as well as providing more detail as to the type of stored credential being used for the transaction.

Table 282: Attributes

Attribute	Description
Subfield ID	05
Subfield Data Length	2 positions
Data Representation	an-4
Data Field	Contents of subfield 05
Justification	N/A

Usage

Following is the usage of subelement 22, subfield 05 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 283: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Request Response/0110	CE	.	CE
Authorization Advice/0120: acquirer-generated	C	.	C
Authorization Advice/0120: system-generated	.	X	C
Authorization Advice Response/0130	CE	.	CE
Reversal Request/0400	C	.	C
Reversal Response/0410	CE	.	CE
Reversal Advice/0420	.	X	C

Values

Mastercard will ensure the *DE 48*, subelement 22, subfield 05 values include the ones listed in the CIT and MIT indicator values and examples table.

Table 284: CIT and MIT indicator values and examples

Indicator category	Indicator subcategory	What's included in the definition	Examples
C1 Cardholder-Initiated	01 Credential-on-file (ad-hoc)	Consumer agrees to store the credential on file with the merchant for future cardholder-initiated and/or merchant-initiated transactions that may occur from time to time, or consumer chooses to use a credential previously placed on file with the merchant when making a purchase.	The consumer initiates a purchase and merchant prompts consumer to store the credential-on-file for future cardholder-initiated purchases.
	02 Standing Order (variable amount and fixed frequency)	Consumer agrees to store the credential-on-file and initiates a first transaction in a series intended to be for a variable amount and a fixed frequency.	The initial transaction to store the credential-on-file for a monthly utility payment.
	03 Subscription (fixed amount and frequency)	Consumer agrees to store the credential-on-file and initiates a first transaction in a series intended to be for a fixed amount and a fixed frequency.	The initial transaction to store the credential-on-file for a monthly newspaper subscription.
	04 Installment	Consumer agrees to store the credential-on-file to establish an installment plan and initiates a first transaction in a series. The installment plan should reflect the single purchase of goods/services with a known amount and set frequency over a specified duration.	The initial transaction to store the credential-on-file for a purchase to be paid through installment payments.

Indicator category	Indicator subcategory	What's included in the definition	Examples
M1 Merchant-Initiated Recurring Payment or Installment	01 Unscheduled Credential on File	A transaction made pursuant to an agreement between a cardholder and a merchant, whereby the cardholder authorizes the merchant to store and use the cardholder's account data to initiate one or more future transactions. The transaction can be for a fixed or variable amount but does not occur at regular intervals on a scheduled date.	The consumer agrees to allow a tollway merchant to initiate transactions when the balance in the consumer account with the merchant falls below an established amount (auto top-up).
	02 Standing Order (variable amount and fixed frequency)	A transaction made pursuant to an agreement between a cardholder and a merchant, whereby the cardholder authorizes the merchant to store and use the cardholder's account data for a transaction that must occur at regularly scheduled intervals for a variable amount, as specified in the agreement.	Monthly utility payments.

Indicator category	Indicator subcategory	What's included in the definition	Examples
	03 Subscription (fixed amount and frequency)	A transaction made pursuant to an agreement between a cardholder and a merchant, whereby the cardholder authorizes the merchant to store and use the cardholder's account data for a transaction that must occur at regularly scheduled intervals for a fixed amount, as specified in the agreement. The subscription agreement may include an allowance for introductory pricing or for price changes to occur from time to time.	Monthly subscription or fixed monthly service payment.
	04 Installment	A transaction in a series of transactions made pursuant to an agreement between a cardholder and a merchant, whereby the cardholder authorizes the merchant to store the cardholder's account data for use by the merchant to initiate one or more future transactions for a known amount over a specified duration based on a single purchase.	Purchase a television for \$1,000 USD, pay in four equal installments of \$250 USD (first transaction is CIT, remaining three transactions are MIT).

Indicator category	Indicator subcategory	What's included in the definition	Examples
M2 Merchant-Initiated Industry Practice	05 Partial Shipment	Occurs when an agreed quantity of goods ordered through e-commerce are not all available for shipment at the time of purchase. Each shipment is a separate transaction.	Consumer ordered goods that ship at different times.
	06 Related/Delayed Charge	An additional account charge after initial services have been rendered and payment has been processed.	Hotel mini bar charge after the cardholder has checked out of hotel.
	07 No Show Charge	A penalty charged according to the merchant's cancellation policy.	Cardholder's cancellation of a reservation without providing proper advance notice to the merchant.
	08 Resubmission	Previous attempt to obtain authorization for a transaction has been declined but the issuer's response does not prohibit the merchant from trying again later.	- Insufficient funds/over credit limit response - Transit debt recovery

Subfield 06 (Message Format Version Code: Acquirer)

DE 48, subelement 22 (Multi-Purpose Merchant Indicator), subfield 06 (Message Format Version Code: Acquirer) contains a code that identifies the transaction message standard applied in the formatting and submission of the acceptor-related data elements.

Table 285: Attributes

Attribute	Description
Subfield ID	06
Subfield Data Length	2 positions
Data Representation	an-1
Data Field	Contents of Subfield 06

Attribute	Description
Justification	N/A

Usage

Table 286: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	M	•	•
Authorization Advice/0120: acquirer-generated	M	•	•
Reversal Request/0400	M	•	•
Reversal Advice/0420	M	•	•

Table 287: Valid values

Value	Description
A	Mastercard Pre 23Q2 Release
B	Mastercard Post 23Q2 Release

Application notes

The Message Format Version Code: Acquirer must be present in all transactions and contain a valid value.

AN 6022 Introduction and Standardization of Transaction Data Elements introduced changes to the formatting, alignment, and population criteria of acceptor (Merchant/ATM Owner) name, location and related data elements and introduced additional data elements to fully capture critical Acceptor data. As there may be varying adoption timelines for upstream acquirer, processor and Merchant/ATM Owner systems, acquirers must utilize this code to identify at the transaction level for which standard was applied. Mastercard will rely on this code to modify, realign and append data into the message format standard defined in AN 6022, allowing downstream Mastercard and customer (issuer) systems to operate consistently throughout the transition period and avoid any processing or cardholder disruption.

Value A indicates the transaction does not comply with the new standards as it contains acceptor data in the pre-23Q2 release version format.

Value B indicates the transaction does comply with the new standards as it contains acceptor data in the post-23Q2 release version format.

Subelement 23 (Payment Initiation Channel)

DE 48, subelement 23 (Payment Initiation Channel) provides information about the device type used to identify mobile-initiated (m-commerce) or other EMV Contact, Magnetic stripe Contactless, or M/Chip Contactless transactions.

Table 288: Attributes

Attribute	Description
Subelement ID	23
Subelement Data Length	2 positions
Data Representation	an-2
Data Field	Contents of subfield 1
Subfields	1
Justification	N/A

Usage

Following is the usage of subelement 23 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 289: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Advice/0120	C	.	C
Reversal Request/0400	C	.	C

Values

See subfield 1.

Application notes

Acquirers in regions where it is mandated must ensure that each newly deployed or re-deployed contactless-enabled POS terminal used by their merchants transmits the device type indicator value, if present on the card or non-card form factor, in data element *DE 48 (Additional Data: Private Use)*, subelement 23 (Payment Initiation Channel), subfield 1 (Device Type) of authorization messages.

Subfield 1 (Device Type)

DE 48, subelement 23, subfield 1 (Device Type) indicates the type of device used at the terminal. In the scope of chip transactions it must be populated using the device type field value of chip tag 9F6E.

NOTE: Effective immediately, new values in DE 48, subelement 23, subfield 1 contain an indication of the form factor. Existing values from 00–19 may continue to be used, and their meanings remain unchanged; however, Mastercard will not send these values in future communication as part of DE 48, subelement 23, subfield 1. New values from 20–99 exclusively indicate the form factor only without also indicating the storage technology. Mastercard has introduced new device type values from 20–33 in anticipation of new form factors for payments, with values 34–99 reserved for future use. For information on storage technology of device types (such as Trusted Execution Environment [TEE] or Secure Element), refer to the description of DE 48, subelement 33 (PAN Mapping File Information), subfield 8 (Storage Technology).

Table 290: Attributes

Attribute	Description
Data Representation	an-2
Data Field	Contents of positions 1–2
Justification	N/A

Values

NOTE: Some values from 00–19 may indicate not only the physical form factor but also other attributes such as device technology and payment app specifications.

00 = Card

01 = Mobile Network Operator (MNO) controlled removable secure element (SIM or UICC) personalized for use with a mobile phone or smartphone

02 = Key Fob

03 = Watch using a contactless chip or a fixed (non-removable) secure element not controlled by the MNO

04 = Mobile Tag

05 = Wristband

06 = Mobile Phone Case or Sleeve

07 = Mobile phone or smartphone with a fixed (non-removable) secure element controlled by the MNO, for example, code division multiple access (CDMA)

08 = Removable secure element not controlled by the MNO, for example, memory card personalized for use with a mobile phone or smartphone

- 09 = Mobile Phone or smartphone with a fixed (non-removable) secure element not controlled by the MNO
- 10 = MNO controlled removable secure element (SIM or UICC) personalized for use with a tablet or e-book
- 11 = Tablet or e-book with a fixed (non-removable) secure element controlled by the MNO
- 12 = Removable secure element not controlled by the MNO, for example, memory card personalized for use with a tablet or e-book
- 13 = Tablet or e-book with fixed (non-removable) secure element not controlled by the MNO
- 14 = Mobile phone or smartphone with a payment application running in a host processor
- 15 = Tablet or e-book with a payment application running in a host processor
- 16 = Mobile phone or smartphone with a payment application running in the Trusted Execution Environment (TEE) of a host processor
- 17 = Tablet or e-book with a payment application running in the TEE of a host processor
- 18 = Watch with a payment application running in the TEE of a host processor
- 19 = Watch with a payment application running in a host processor

NOTE: Values from 20–99 exclusively indicate the form factor only without also indicating the storage technology.

Table 291: Values

Value	Meaning	Example
20	Card	
21	Phone	Mobile phone
22	Tablet/e-reader	Tablet computer or e-reader
23	Watch/Wristband	Watch or wristband, including a fitness band, smart strap, disposable band, watch add-on, and security/ID band
24	Sticker	
25	PC	PC or laptop
26	Device Peripheral	Mobile phone case or sleeve
27	Tag	Key fob or mobile tag
28	Jewelry	Ring, bracelet, necklace, and cuff links
29	Fashion Accessory	Handbag, bag charm, and glasses
30	Garment	Dress

Value	Meaning	Example
31	Domestic Appliance	Refrigerator, washing machine
32	Vehicle	Vehicle, including vehicle attached devices
33	Media/Gaming Device	Media or gaming device, including a set top box, media player, and television

Values 34 through 99 equals Reserved for future form factors. Any value in this range may occur within form factor and transaction data without prior notice.

Subelement 24 (Account Level Management [ALM] Service Data)

DE 48, subelement 24 (Account Level Management [ALM] Service Data) supports Regional ALM processing.

Table 292: Attributes

Attribute	Description
Subelement ID	24
Subelement Data Length	2 positions
Data Representation	an-30
Data Field	Contents of subfields 1-4
Subfields	4
Justification	See subfields

Usage

Following is the usage of subelement 24 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 293: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	•	X	C
Authorization Request Response/0110	•	X	C
Authorization Advice/0120: system-generated	•	X	C
Authorization Advice/0120: acquirer-generated	•	X	C
Authorization Advice Response/0130: issuer-generated	•	X	C

Message	Org	Sys	Dst
Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)	•	X	C

Values

See subfields.

Application notes

The following subfields will be populated when applicable to the ALM program indicated by Subfield 1 (ALM Service Code):

- Subfield 2: ALM Product Graduation Plus or Registered Product Code
- Subfield 3: ALM Product Class
- Subfield 4: ALM Rate Type

Subfields that are not applicable to the designated ALM service of the registered PAN will be populated with 'XXX'. The absence of *DE 48*, subelement 24 for opt-in customers means the account range of the PAN does not participate in ALM.

Mastercard reserves the right add new valid codes at any time. Customer must be able to accept and process any value provided within any of the sub elements for authorization and clearing processing. In the event new valid values are added, refer to the [Account Level Management Manual](#).

Subfield 1 (ALM Service Code)

DE 48, subelement 24, subfield 1 (ALM Service Code) indicates the ALM program and service.

ALM Service Code represents the Account Level Management service or combination of ALM services for which the PAN is actively registered. The ALM Service Code is assigned at the time the PAN is registered in Account Level Management and will be populated in the supported authorization messages to ensure ALM processing is applied throughout the lifecycle of the transaction.

ALM processing was not applied to the authorization message when the ALM Service Code is present with one of the following values:

- 00000 = ALM Service Unavailable
 - When the Dual Message Authorization System is unable to make a successful connection to the Account Level Management Platform for PAN registration validation, the customer will receive an ALM Service Unavailable service code value

- 00126 = PAN not registered for ALM, but the issuer's Account Range participates, transaction will process business as usual
 - When the issuer's Account Range is activated for ALM Servicing, identified as having at least one-PAN actively registered, but the PAN as provided in the authorization message is not actively registered in ALM
- All other values provided will represent an individual ALM Service or ALM Service Combinations based on the PAN's active registration

Mastercard reserves the right to add new valid values for ALM Service Code at any time. Customers must be able to accept and process any valid values provided within the authorization messages. If new valid values are added, refer to the [Account Level Management Manual](#).

Table 294: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	an-5
Data Field	Contents of subfield 1
Justification	N/A

Table 295: ALM service names

ALM Service Name (ALM strategies which are enabled by the various Account Level Management Services)	DE 48, SE 24, SF1: ALM Service Code (Represents the ALM service or combination of ALM services for which the PAN is registered. Usage: Account Registration Results, and ALM Service Data in Authorization and Clearing processing.)
AFFLUENT SHORTFALL WORLD ELITE MASTERCARD	01323
AFFLUENT SHORTFALL WORLD ELITE MASTERCARD and PRODUCT GRADUATION PLUS	01425
AFFLUENT SHORTFALL WORLD MASTERCARD	01723
AFFLUENT SHORTFALL WORLD MASTERCARD and PRODUCT GRADUATION PLUS	01825
CONSUMER PRODUCT MONITORING SERVICE: MUSE MASTERCARD QUALIFIED	02217
CONSUMER PRODUCT MONITORING SERVICE: NAME FOR GCMS PRODUCT ID MPE QUALIFIED	02207

ALM Service Name (ALM strategies which are enabled by the various Account Level Management Services)	DE 48, SE 24, SF1: ALM Service Code (Represents the ALM service or combination of ALM services for which the PAN is registered. Usage: Account Registration Results, and ALM Service Data in Authorization and Clearing processing.)
CONSUMER PRODUCT MONITORING SERVICE: REQUALIFICATION 1 (MUSE MASTERCARD REQUALIFIED)	02301
CONSUMER PRODUCT MONITORING SERVICE: REQUALIFICATION 2 (WORLD ELITE REQUALIFIED)	02424
CONSUMER PRODUCT MONITORING SERVICE: WORLD ELITE QUALIFIED	02210
CONSUMER PRODUCT MONITORING SERVICE: WORLD QUALIFIED	02205
ENHANCED VALUE	00402
ENHANCED VALUE and PRODUCT GRADUATION PLUS	00513
PRODUCT GRADUATION PLUS	00616
SMALL BUSINESS SPEND - CREDIT CORE AND PRODUCT GRADUATION PLUS LEVEL 1	00804
SMALL BUSINESS SPEND - CREDIT CORE AND PRODUCT GRADUATION PLUS LEVEL 2	00806
SMALL BUSINESS SPEND - CREDIT CORE AND PRODUCT GRADUATION PLUS LEVEL 3	00808
SMALL BUSINESS SPEND - CREDIT CORE AND PRODUCT GRADUATION PLUS LEVEL 4	00811
SMALL BUSINESS SPEND - CREDIT CORE AND PRODUCT GRADUATION PLUS LEVEL 5	00818
SMALL BUSINESS SPEND - CREDIT CORE LEVEL 1	00703
SMALL BUSINESS SPEND - CREDIT CORE LEVEL 2	00705
SMALL BUSINESS SPEND - CREDIT CORE LEVEL 3	00707
SMALL BUSINESS SPEND - CREDIT CORE LEVEL 4	00710
SMALL BUSINESS SPEND - CREDIT CORE LEVEL 5	00717

ALM Service Name (ALM strategies which are enabled by the various Account Level Management Services)	DE 48, SE 24, SF1: ALM Service Code (Represents the ALM service or combination of ALM services for which the PAN is registered. Usage: Account Registration Results, and ALM Service Data in Authorization and Clearing processing.)
SMALL BUSINESS SPEND - CREDIT WORLD AND PRODUCT GRADUATION PLUS LEVEL 1	01004
SMALL BUSINESS SPEND - CREDIT WORLD AND PRODUCT GRADUATION PLUS LEVEL 2	01006
SMALL BUSINESS SPEND - CREDIT WORLD AND PRODUCT GRADUATION PLUS LEVEL 3	01008
SMALL BUSINESS SPEND - CREDIT WORLD AND PRODUCT GRADUATION PLUS LEVEL 4	01011
SMALL BUSINESS SPEND - CREDIT WORLD AND PRODUCT GRADUATION PLUS LEVEL 5	01018
SMALL BUSINESS SPEND - CREDIT WORLD ELITE AND PRODUCT GRADUATION PLUS LEVEL 1	01204
SMALL BUSINESS SPEND - CREDIT WORLD ELITE AND PRODUCT GRADUATION PLUS LEVEL 2	01206
SMALL BUSINESS SPEND - CREDIT WORLD ELITE AND PRODUCT GRADUATION PLUS LEVEL 3	01208
SMALL BUSINESS SPEND - CREDIT WORLD ELITE AND PRODUCT GRADUATION PLUS LEVEL 4	01211
SMALL BUSINESS SPEND - CREDIT WORLD ELITE AND PRODUCT GRADUATION PLUS LEVEL 5	01218
SMALL BUSINESS SPEND - CREDIT WORLD ELITE LEVEL 1	01103
SMALL BUSINESS SPEND - CREDIT WORLD ELITE LEVEL 2	01105
SMALL BUSINESS SPEND - CREDIT WORLD ELITE LEVEL 3	01107
SMALL BUSINESS SPEND - CREDIT WORLD ELITE LEVEL 4	01110
SMALL BUSINESS SPEND - CREDIT WORLD ELITE LEVEL 5	01117
SMALL BUSINESS SPEND - CREDIT WORLD LEVEL 1	00903
SMALL BUSINESS SPEND - CREDIT WORLD LEVEL 2	00905

ALM Service Name (ALM strategies which are enabled by the various Account Level Management Services)	DE 48, SE 24, SF1: ALM Service Code (Represents the ALM service or combination of ALM services for which the PAN is registered. Usage: Account Registration Results, and ALM Service Data in Authorization and Clearing processing.)
SMALL BUSINESS SPEND - CREDIT WORLD LEVEL 3	00907
SMALL BUSINESS SPEND - CREDIT WORLD LEVEL 4	00910
SMALL BUSINESS SPEND - CREDIT WORLD LEVEL 5	00917
WORLD HIGH VALUE	01519
WORLD HIGH VALUE and PRODUCT GRADUATION PLUS	01620
PAN NOT REGISTERED FOR ALM, BUT ISSUERS ACCOUNT RANGE IS ACTIVE	00126
ALM SERVICE UNAVAILABLE IN AUTHORIZATION	00000

Subfield 2 (ALM Product Graduation Plus or Registered Product Code)

DE 48, subelement 24, subfield 2 (ALM Product Graduation Plus or Registered Product Code) contains the licensed graduated product code or the registered product code for those transactions identified with a PAN registered and qualifying for Product Graduation Plus or Consumer Product Monitoring Service, or a combination of Account Level Management Services.

If the PAN is not actively registered in ALM Product Graduation Plus or Consumer Product Monitoring Service, the field will be populated as 'XXX'.

Mastercard reserves the right to add new product code valid values for eligible ALM Product Graduation Plus or the Consumer Product Monitoring Service at any time and customers must be able to accept, and process any valid values provided within the authorization messages. For new valid values, a complete product code listing, or country availability, refer to the [Account Level Management Manual](#).

Table 296: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	an-3
Data Field	Contents of subfield 2

Attribute	Description
Justification	N/A

Subfield 3 (ALM Product Class)

DE 48, subelement 24, subfield 3 (ALM Product Class) indicates the Product Class Override for the applicable ALM Service. Product Class provides the ability to vary interchange rates by product grouping, while maintaining common interchange structures.

If the PAN is not actively registered in ALM Product Graduation Plus, the field will be populated as 'XXX'.

Mastercard reserves the right to add new valid values for Product Class at any time and customers must be able to accept, and process any valid values provided within the authorization messages. In the event new valid values are added, refer to the [Account Level Management Manual](#).

Table 297: Attributes

Attribute	Description
Subfield ID	03
Subfield Data Length	2 positions
Data Representation	an-3
Data Field	Contents of subfield 3
Justification	N/A

Table 298: Values

IF....DE 48 SE 24 SF 2 is present with one of the following valid values:			THEN....DE 48 SE 24 SF 3 will be present with one of the following values...		
		Product Class associated with the Product Graduation Plus:			
ALM Product Graduation Plus		Valid Values		Product Class Description	
ACS		ACS		ACH FOR CONSUMER	
BPL		BPL		Mastercard® Business Credit Plus	
DBK		DBK		DIGITAL MASTERCARD BLACK	
DCG		DCG		DIGITAL GOLD	
DCS		DCS		DIGITAL STANDARD	

IF....DE 48 SE 24 SF 2 is present with one of the following valid values:			THEN....DE 48 SE 24 SF 3 will be present with one of the following values...		
ALM Product Graduation Plus			Product Class associated with the Product Graduation Plus: Valid Values		
			Product Class Description		
DPL			DPL		
			DIGITAL PLATINUM		
MAB			MAB		
			BUSINESS WORLD ELITE		
MAC			MAC		
			CORPORATE WORLD ELITE		
MBK			MCW		
			WORLD		
MCB			MCB		
			CORPORATE		
MCD			DMC		
			DEBIT MASTERCARD		
MCG			MCG		
			CONSUMER CREDIT GOLD		
MCO			MCO		
			GLOBAL CERTIFIED CORPORATE		
MCS			MCC		
			CONSUMER		
MCT			MCT		
			TITANIUM MASTERCARD		
MCU			MCC		
			CONSUMER		
MCW			MCW		
			WORLD		
MDG			MDG		
			DEBIT MASTERCARD GOLD		
MDH			MDH		
			DEBIT WORLD		
MDI			MDH		
			DEBIT WORLD		
MDJ			MDJ		
			DEBIT MASTERCARD (ENHANCED)		
MDK			MDJ		
			DEBIT MASTERCARD (ENHANCED)		
MDO			DMC		
			DEBIT MASTERCARD		
MDP			MDP		
			DEBIT MASTERCARD PLATINUM		
MDQ			MCO		
			GLOBAL CERTIFIED CORPORATE		
MDR			DMC		
			DEBIT MASTERCARD		
MDS			DMC		
			DEBIT MASTERCARD		
MDU			DMC		
			DEBIT MASTERCARD		
MEB			MEB		
			MASTERCARD EXECUTIVE BUSINESSCARD		

IF.... <i>DE 48 SE 24 SF 2</i> is present with one of the following valid values:			THEN.... <i>DE 48 SE 24 SF 3</i> will be present with one of the following values...		
ALM Product Graduation Plus		Product Class associated with the Product Graduation Plus: Valid Values		Product Class Description	
MEO		MEO		MASTERCARD CORPORATE EXECUTIVE CARD	
MEP		DMC		DEBIT MASTERCARD	
MPC		MCB		CORPORATE	
MPL		MPL		CONSUMER CREDIT PLATINUM	
MUP		DMC		DEBIT MASTERCARD	
MWB		MWB		BUSINESS WORLD	
MWE		MWE		MASTERCARD WORLD ELITE	
MWO		MWO		CORPORATE WORLD	
MXO		DMC		DEBIT MASTERCARD	
MXR		DMC		DEBIT MASTERCARD	
MXS		DMC		DEBIT MASTERCARD	
WDR		MDH		DEBIT WORLD	
XXX		XXX		Product Class Override does not apply, or has not been configured for the ALM Service Code provided in <i>DE 48</i> , subelement 24, subfield 1	

Subfield 4 (ALM Rate Type)

DE 48, subelement 24, subfield 4 (ALM Rate Type) indicates the Rate Type number for the applicable ALM Service. Rate Types provide the ability to vary interchange rates in the event that unique IRDs are not assigned for an applicable business service arrangement. PANs actively registered for ALM Services that do not involve Product Graduation Plus will have an associated ALM Rate Type. Some ALM Services may combine Rate Types with Product Class Overrides (Subfield 3).

If an ALM Rate Type is not supported for the ALM service of the registered PAN, the field will be populated as 'XXX'.

Mastercard reserves the right to add new valid values for Rate Types at any time and customers must be able to accept, and process any valid values provided within the authorization messages. In the event new valid values are added, refer to the [Account Level Management Manual](#).

Table 299: Attributes

Attribute	Description
Subfield ID	04
Subfield Data Length	2 positions
Data Representation	an-3
Data Field	Contents of subfield 4
Justification	N/A

Table 300: Values

Rate type	Rate type description
801	Enhanced Value Rate Adjustment
811	Core Business to Level 1 Adjustment
812	Core Business to Level 2 Adjustment
813	Core Business to Level 3 Adjustment
814	Core Business to Level 4 Adjustment
815	Core Business to Level 5 Adjustment
821	World Business to Level 1 Adjustment
822	World Business to Level 2 Adjustment
823	World Business to Level 3 Adjustment
824	World Business to Level 4 Adjustment
825	World Business to Level 5 Adjustment
831	World Elite Business to Level 1 Adjustment
832	World Elite Business to Level 2 Adjustment
833	World Elite Business to Level 3 Adjustment
834	World Elite Business to Level 4 Adjustment
835	World Elite Business to Level 5 Adjustment
850	World High Value Adjustment
851	World Consumer Shortfall Adjustment
852	World Elite Consumer Shortfall Adjustment

Rate type	Rate type description
881	Canada World Consumer Adjustment *
882	Canada MPE (Mastercard® branding to be announced) *
883	Canada World Elite Adjustment *
884	Canada MUSE Adjustment *
885	Canada MUSE Requalified Down Adjustment *
886	Canada World Elite Requalified Down Adjustment *
XXX	Rate Type does not apply, or has not been configured for the ALM Service Code provided in DE 48, subelement 24, subfield 1

NOTE: *Consumer Product Monitoring Service will process using Rate Types associated with a Registered Product Code in subfield 2 (ALM Product Code).

Subelement 25 (Mastercard Cash Program Data)

DE 48, subelement 25 (Mastercard Cash Program Data) contains information necessary to process Mastercard Cash transactions, which includes message type identifiers.

Table 301: Attributes

Attribute	Description
Subelement ID	25
Subelement Data Length	2 positions
Data Representation	ans...14; LLVAR
Data Field	Contents of subfields
Subfields	1
Justification	See subfields

Usage

Following is the usage of subelement 25 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 302: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C

Message	Org	Sys	Dst
Authorization Advice/0120: system-generated	.	X	C
Reversal Request/0400	C	.	C

Values

See subfields.

Subfield 01 (Message Identifier)

DE 48, subelement 25, subfield 01 (Message Identifier) indicates the type of cash transaction.

Table 303: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	ans...10; LLVAR
Data Field	Contents of positions 1–2, positions 3–10 undefined
Justification	N/A

Table 304: Values

Value	Description
CM	Confirmation message
LP	Linked load request with a purchase
LR	Unlinked load request, or linked load request with no purchase
LU	Linked status update

Subelement 26 (Wallet Program Data)

DE 48 (Additional Data: Private Use), subelement 26 (Wallet Program Data) contains the subfield that will identify transactions submitted using wallets on the Mastercard Digital Enablement Service (MDES) or Masterpass platform.

Table 305: Attributes

Attribute	Description
Subelement ID	26
Subelement Data Length	2 positions
Data Representation	an-3
Data Field	Contents of subfield 1
Subfields	1
Justification	N/A

Usage

Following is the usage of subelement 26 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 306: Applicable

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Advice/0120: acquirer-generated	C	.	C
Authorization Advice/0120: system-generated	.	X	C

Values

See subfield 1.

Subfield 1 (Wallet Identifier)

DE 48, subelement 26, subfield 1 (Wallet Identifier) provides information about transactions initiated through the Masterpass Online platform or through the Mastercard Digital Enablement Service (MDES) and indicates who the Wallet Provider was for the transaction.

Table 307: Attributes

Attribute	Description
Data Representation	an-3
Data Field	Provides the Wallet Identifier
Justification	N/A

NOTE: The following values apply to both Masterpass and MDES transactions.

Table 308: Values

Value	Description
101	Masterpass by Mastercard

NOTE: The following values only apply to MDES transactions.

Table 309: Values

Value	Description
103	Apple Pay
216	Google Pay
217	Samsung Pay
327	Remote commerce programs

Application notes

The Wallet ID is required in Masterpass transactions submitted from certain regions or countries. The Wallet ID cannot contain all zeros, spaces, or special characters.

The Dual Message Authorization System inserts the Wallet Identifier for MDES transactions when available, which identifies the wallet that the transaction came from.

New Wallet Identifier values may be provided in this field without prior written notice.

The following tables indicate the presence of the Wallet Identifier in Masterpass and MDES transactions.

Table 310: Applicable messages

Masterpass Messages	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Advice/0120: acquirer-generated	C	.	C
Authorization Advice/0120: system-generated	.	X	C

NOTE: If the card in Masterpass is an MDES token, then MDES would add the correct Wallet ID to the transaction.

Table 311: Applicable messages

MDES Messages	Org	Sys	Dst
Authorization Request/0100	.	X	C
Authorization Advice/0120: acquirer-generated	.	X	C
Authorization Advice/0120: system-generated	.	X	C
Reversal Request/0400	.	X	C
Reversal Advice/0420	.	X	C

NOTE: For MDES, Mastercard inserts the Wallet Identifier for the wallet that the token was supplied to.

Subelement 27 (Transaction Analysis)

DE 48, subelement 27 (Transaction Analysis) contains information about transaction analysis.

Table 312: Attributes

Attribute	Description
Subelement ID	27
Subelement Data Length	13...97
Data Representation	an...97; LLVAR
Data Field	Contents of subfields 1–2

Attribute	Description
Subfields	2

Usage

Following is the usage of subelement 27 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 313: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	.	X	C
Authorization Advice/0120: system-generated	C	.	C

Application notes

DE 48, subelement 27 and subfields 1 and 2 are set up in Tag, Length, Value format:

27LL01LLVV02LLVVVVVVVVV...

Table 314: Example

Subfield 1	Subfield 2
0102CI	0206SNAOVP

Subfield 1 (Overview)

DE 48, subelement 27, subfield 1 (Overview) provides issuers with a code that represents the overall disposition of the transaction based on the applicable transaction validation results and corresponding decision matrix.

Table 315: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	an-2
Data Field	Contents of subfield 1

Attribute	Description
Justification	N/A

Table 316: Values

Value	Description
CI	Continue processing with information
CW	Continue processing with warning
DI	Decline issuer decision
DS	Decline suspicious

Subfield 2 (Test Results)

DE 48, subelement 27, subfield 2 (Test Results) provides the list of failed validations. If several failed validations generate the same test result code, the code will only be reported one time in subfield 2.

Table 317: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	an...87; LLVAR; The LL length field of LLVAR will be an integral multiple of 3, not to exceed 87.
Data Field	Contents of subfield 2
Justification	N/A

Table 318: Values

Value	Description
CAM	Invalid Card Authentication
CCH	Cross channel
CRN	Consent requirement not fulfilled
CVF	Cardholder verification (on terminal) was not successful

Value	Description
CVU	CVM requirements not fulfilled
CVX	Status CVM unknown
DAF	ODA failed
DAU	ODA was not performed
DMM	Data mismatch
DNC	Data not consistent with application or product
EXP	Token expired
FER	Format error
FUZ	Fuzzing
ICT	Not a valid cryptogram type
NMK	No matching key file/KDI combination
OVE	CDCVM retry exceeded: token suspended
OVF	CDCVM failed
OVP	CDCVM (was possible but) not performed
OVU	CDCVM not performed
PKC	ODA compromised
PPP	PIN Pad Problem
PTB	PIN on terminal bypass
PWE	Possible wedge attack
REP	ATC replay: same UN
SKC	Key compromised
SNA	Request service not allowed for this product
UTP	Unable to process
WOC	Wallet overrule of Mastercard decision on CDCVM

Subelement 28 (Cardless ATM Order ID)

DE 48, subelement 28 (Cardless ATM Order ID) is defined by Mastercard, sent by the acquirer participating in the Cardless ATM program and passed to the issuer participating in the Cardless ATM program.

Table 319: Attributes

Attribute	Description
Subelement ID	28
Subelement Data Length	2 positions
Data Representation	an-10
Data Field	Fixed length, contents of positions 1–10
Subfields	N/A
Justifications	N/A

Usage

Following is the usage of subelement 28 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 320: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	X	C

Values

Cannot contain all spaces or all zeros.

Subelement 29 (Additional POS Terminal Locations)

DE 48, subelement 29 is required when DE 61 (*Point-of-Service [POS] Data*), subfield 3 (POS Terminal Location) is present with the value of 8 (Additional Terminal Operating Environments).

Table 321: Attributes

Attribute	Description
Subelement ID	29
Subelement Data Length	2 positions

Attribute	Description
Data Representation	an-1
Data Field	Contents of position 1
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 29 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 322: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C

Table 323: Values

Value	Description
A	Transaction has been initiated remotely and completed at a physical terminal on premises of the acceptor facility.
B	Transaction has been initiated remotely and completed at a physical terminal off premises of the acceptor facility.

Subelement 30 (Token Transaction Identifier)

DE 48, subelement 30 (Token Transaction Identifier) will contain, when available, the calculated Token Transaction Identifier to identify the transaction. The Token Transaction Identifier is to be retained and used to provide the transaction details associated with an original purchase and subsequent reversal messages. The Token Transaction Identifier is only sent to issuers participating in the Mastercard Digital Enablement Service.

Table 324: Attributes

Attribute	Description
Subelement ID	30
Subelement Data Length	3 positions

Attribute	Description
Data Representation	ans...64; LLVAR
Subfields	N/A

Usage

Following is the usage of subelement 30 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 325: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	•	X	C
Authorization Advice/0120: system-generated	•	X	C
Reversal Request/0400	•	X	C
Reversal Advice/0420	•	X	C

Values

A 44 byte base 64 value will be sent for Mastercard BIN ranges.

A 64 byte value will be sent for Visa BIN ranges.

Application notes

DE 48, subelement 30 will not be present when *DE 3* (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains a value of 20 (Purchase Return/Refund).

Subelement 32 (Mastercard Assigned ID)

DE 48, subelement 32 (Mastercard Assigned ID) contains the merchant ID assigned by Mastercard.

Table 326: Attributes

Attribute	Description
Subelement ID	32
Subelement Data Length	2 positions
Data Representation	an-6
Data Field	Contents of positions 1–6

Attribute	Description
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 32 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 327: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	X	C
Authorization Request Response/0110	OE	X	C
Authorization Advice/0120: acquirer-generated	C	.	C
Authorization Advice/0120: issuer-generated	C	X	.
Authorization Advice/0120: system-generated	.	X	C
Reversal Request/0400	C	.	C
Reversal Advice/0420	.	X	C

Values

The Mastercard Assigned ID must be a value assigned by Mastercard, and is the same value in both the Authorization and Clearing Platforms. Mastercard assigns one ID to the merchant regardless if the merchant is participating in multiple programs. This ID can also be assigned for transaction data integrity purposes. For all acceptors that have been issued a Mastercard Assigned ID, all submitted authorization requests, reversals, and advices must contain the Mastercard Assigned ID in *DE 48*, subelement 32.

This field is required when transactions are submitted for real-time substantiation. Real-time substantiation is indicated by *DE 48*, subelement 61, subfield 3 (Real-time Substantiation Indicator) containing a value of 1 (Merchant terminal verified the purchased items against the Inventory Information Approval System [IIAS]).

Participating merchants must submit the Mastercard Assigned ID for e-commerce transactions that are processed under the Maestro® Recurring Payments Program and the Mastercard® Utility Payment Program.

Application notes for the Mastercard Installments Program with Merchant Participation for qualifying domestic transactions with product code GCS

Optional Echo (OE) indicates that the message initiator may provide the data element in an Authorization Request Response/0110 message. If so, the data element must contain the same value echoed from the original Authorization Request/0100 message.

The Authorization Platform inserts or modifies this subelement in Authorization Request/0100 messages for qualifying installments transactions.

Issuers receive DE 48, subelement 32 in Authorization Request/0100 messages when the acquirer includes the subelement, or when Mastercard has inserted or modified the subelement in the corresponding Authorization Request/0100 message.

Acquirers receive DE 48, subelement 32 in Authorization Request Response/0110 messages when Mastercard has inserted or modified the subelement in the corresponding Authorization Request/0100 message.

Subelement 32 (Visa Merchant Identifier [Visa Only])

DE 48, subelement 32 (Visa Merchant Identifier) contains the merchant ID assigned by Visa.

Table 328: Attributes

Attribute	Description
Subelement ID	32
Subelement Data Length	2 positions
Data Representation	an-8
Data Field	Contents of positions 1–8
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 32 in Visa-branded transactions (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 329: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Advice/0120: system-generated	.	X	C

Message	Org	Sys	Dst
Reversal Advice/0420	.	X	C

Values

The Visa Merchant Identifier (Visa Only) value is assigned by Visa and is only for use with Visa-branded transactions.

Application notes

When the acquirer provides *DE 48*, subelement 32 with a length greater than eight positions on a Visa-branded transaction, the Authorization System rejects the transaction sending the acquirer an Authorization Request Response/0110 message where *DE 39 (Response Code)* contains the value 30 (Format error) and *DE 44 (Additional Response Data)* contains the value 048032.

Subelement 33 (PAN Mapping File Information)

DE 48, subelement 33 (PAN Mapping File Information) supports the mapping between the virtual account data and actual account data.

Table 330: Attributes

Attribute	Description
Subelement ID	33
Subelement Data Length	2 positions
Data Representation	an...93; LLVAR
Data Field	Contents of subfields 1–8
Subfields	8
Justification	See subfields

Usage

Following is the usage of subelement 33 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 331: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	X	C

Message	Org	Sys	Dst
Authorization Request Response/0110	C	X	C
Authorization Advice/0120: acquirer-generated	C	X	C
Authorization Advice/0120: system-generated	.	X	C
Authorization Advice Response/0130: issuer-generated	C	X	C
Reversal Request/0400	C	X	C
Reversal Request Response/0410	C	X	C
Reversal Advice/0420	.	X	C

Values

See subfields.

Application notes

The Dual Message Authorization System inserts this subelement when PAN Mapping Service was performed on the transaction. Subelement 33 may be used for research or audit purposes but need not be used for processing authorization messages.

Acquirers will receive subelement 33 in contact M/Chip, contactless magnetic stripe, or contactless M/Chip transactions when the issuer is participating in the Contactless Mapping Service. Acquirers will also receive subelement 33 in contactless M/Chip, contactless magnetic stripe, magnetic stripe, or Digital Secure Remote Payment transactions when the issuer is participating in the Mastercard Digital Enablement Service. Acquirers do not receive subelement 33 for the Mastercard In Control™ Purchase Control Service or Mastercard In Control Virtual Card Spend Control Service.

Issuers that issue contactless cards or devices that provide a primary account number (PAN) within a transaction different from what is embossed on the card, must ensure that the following is provided in Authorization Request Response/0110 messages when responding to MCC 4111, 4131, 4784, 7523 transactions.

- The value E in DE 48, subelement 33, subfield 1 (Account Number Indicator)
- The embossed number in DE 48, subelement 33, subfield 2 (Account Number)
- The card expiration date of the embossed number in DE 48, subelement 33, subfield 3 (Expiration Date)

Acquirers processing transactions for merchant categories 4111, 4131, 4784, and 7523 must pass the PAN Mapping information back to the merchants.

Subfield 1 (Account Number Indicator)

DE 48, subelement 33, subfield 1 (Account Number Indicator) indicates the type of PAN mapping account.

Table 332: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	an-1
Data Field	Contents of subfield 1
Justification	N/A

Table 333: Values

Value	Description
C	Mastercard Digital Enablement Service Secure Element Token
E	Embossed Account Number Provided by Issuer
F	Mastercard Digital Enablement Service Static Token
H	Mastercard Digital Enablement Service Cloud-based Payments Token
L	Pay with Rewards Loyalty Program Operator [LPO] Card
M	Primary Account Number
P	Contactless Account Number
R	Pay with Rewards Card
V	Virtual Card Number

Subfield 2 (Account Number)

DE 48, subelement 33, subfield 2 (Account Number) indicates the PAN mapping account number.

Table 334: Attributes

Attribute	Description
Subfield ID	02

Attribute	Description
Subfield Data Length	2 positions
Data Representation	n...19, LLVAR
Data Field	Contents of subfield 2
Justification	N/A

Table 335: Values

Value	Description
Acquirer message	Contains primary account number When Mastercard performs token mapping or subfield 2 is provided when using a third party token service provider, subfield 2 contains only the last four digits of the cardholder's PAN. The Authorization Platform will continue to send the full account number for tokenized InControl VCN transactions, and when DE 18 (Merchant Type) contains any of the following MCCs: • MCC 3246: RyanAir • MCC 4111: Transportation-Suburban and Local Commuter Passenger, including Ferries • MCC 4112: Passenger Railways • MCC 4131: Bus Lines • MCC 4511: Air Carries, Airlines- Not Else Where Classified • MCC 4784: Bridge and Road Fees, Tolls • MCC 7523: Automobile Parking Lots and Garages
Issuer message	Contains virtual number or token For Mastercard Digital Enablement Service transactions, subfield 2 is the token rather than the primary account number, if subfield 1 value is C, H, or F.

Subfield 3 (Expiration Date)

DE 48, subelement 33, subfield 3 (Expiration Date) indicates the expiration date of the PAN mapping account.

Table 336: Attributes

Attribute	Description
Subfield ID	03

Attribute	Description
Subfield Data Length	2 positions
Data Representation	n-4; format YYMM
Data Field	Contents of subfield 3
Justification	N/A

Table 337: Values

Value	Description
Acquirer message	Will contain the expiration date when - The issuer provided one for a PAN mapping record added to the MCC106 PAN Mapping File, or - A transit transaction response contains MCC 4111, 4131, 4784, and 7523, or - The Mastercard Digital Enablement Service was applied
Issuer message	May contain virtual card number or token expiration date only if acquirer provided in DE 14 of the authorization message. For contactless transit transactions, subfield 3 is the expiration date of the embossed number rather than the virtual number, if subfield 1 value is E.

Subfield 4 (Product Code)

DE 48, subelement 33, subfield 4 (Product Code) may indicate the product code for subfield 2 account number.

Table 338: Attributes

Attribute	Description
Subfield ID	04
Subfield Data Length	2 positions
Data Representation	an-3
Data Field	Contents of subfield 4
Justification	N/A

Values

Subfield 4 may contain product code for subfield 2 account number.

Subfield 5 (Token Assurance Method)

DE 48, subelement 33, subfield 5 (Token Assurance Method) contains a value indicating the confidence level of the token to PAN/cardholder binding.

Table 339: Attributes

Attribute	Description
Subfield ID	05
Subfield Data Length	2 positions
Data Representation	n-2
Data Field	Contents of subfield 5
Justification	N/A

Values

Contains a value indicating the confidence level of the token to PAN/cardholder relationship.

Table 340: Valid values

TAM Values	TAM Name	Authentication Use Case/MA Services
10	Card Issuer Account Verification	Successful TAR, AS, ASI response with no authentication
11	Card Issuer Interactive Cardholder Authentication – 1 Factor	Token authenticated using MDES Authentication Service (OTP)
14	Card Issuer Asserted Authentication	Push Provisioning into without additional authentication beyond Issuer's own application
20	Push Provisioned Token with Additional Interactive Authentication	Push Provisioned token with MDES Authentication service following token creation
21	Push Provisioned Token with Additional Identity Check Authentication	Push Provisioned token with Identity Check authentication following token creation
22	Identity Check Risk-based Cardholder Authentication	Token authenticated via Identity Check Risk-based authentication (no cardholder challenge)
23	Identity Check Interactive Cardholder Authentication	Token authenticated via Identity Check Interactive step-up (cardholder challenge)

TAM Values	TAM Name	Authentication Use Case/MA Services
24	Token with device-asserted authentication	A Companion Cloud Token or Recurring Payment Cloud Token provisioned from an existing device token, where proof of authentication was provided and validated

Application notes

Subfield 5 is optional for acquirers in the Authorization Request/0100 message. Acquirers may receive the value back in an Authorization Request Response/0110 message.

Subfield 6 (Token Requestor ID)

DE 48, subelement 33, subfield 6 (Token Requestor ID) contains the ID assigned by the Token Service Provider to the Token Requestor.

Table 341: Attributes

Attribute	Description
Subfield ID	06
Subfield Data Length	2 positions
Data Representation	n-11
Data Field	Contents of subfield 6
Justification	N/A

Values

Contains the ID assigned by the Token Service Provider to the Token Requestor. The Token Requestor ID is optional for all token types.

Application notes

Subfield 6 is optional but should be included in messages sent to Mastercard by the acquirer if provided to the acquirer by the merchant. Acquirers may receive the value back in an Authorization Request Response/0110 message.

Subfield 7 (Primary Account Number, Account Range)

DE 48, subelement 33, subfield 7 (Primary Account Number, Account Range) carries either the first nine digits of the cardholder PAN, or the full Visa cardholder PAN in the authorization response for a transaction initiated with a Visa token.

Table 342: Attributes

Attribute	Description
Subfield ID	07
Subfield Data Length	2 positions
Data Representation	n...19; LLVAR
Data Field	Contents of subfield 7
Justification	N/A

Values

The first nine digits of the Visa cardholder PAN, or the full Visa cardholder PAN. Acquirers must be aware that this information must not be forwarded to their merchants.

Subfield 8 (Storage Technology)

DE 48, subelement 33, subfield 8 (Storage Technology) describes the Storage Technology of a requested or created token.

Table 343: Attributes

Attribute	Description
Subfield ID	08
Subfield Data Length	2 positions
Data Representation	an-2
Data Field	Contents of subfield 8
Justification	N/A

Usage

Following is the usage of *DE 48*, subelement 33, subfield 8 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 344: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	•	X	C
Authorization Request Response/0110	•	X	C

Message	Org	Sys	Dst
Authorization Advice/0120: system-generated	•	X	C
Reversal Request/0400	•	X	C
Reversal Advice/0420	•	X	C
Authorization Request/0100: tokenization Eligibility Request (TER)	•	X	C
Authorization Request/0100: tokenization Authorization Request (TAR)	•	X	C
Authorization Request/0100: tokenization Complete Notification (TCN)	•	X	C

Table 345: Values

Value	Description
01	Device Memory
02	Device Memory protected by Trusted Platform Module (TPM)
03	Server
04	Trusted Execution Environment (TEE)
05	Secure Element (SE)
06	Virtual Execution Environment (VEE)

Application notes

Subfield 8 is optional but should be included in messages sent to Mastercard by the acquirer if provided to the acquirer by the merchant. Acquirers may receive the value back in an Authorization Request Response/0110 message.

Subelement 34 (ATC Information)

DE 48, subelement 34 (ATC Information) supports On-behalf Services.

Table 346: Attributes

Attribute	Description
Subelement ID	34
Subelement Data Length	2 positions
Data Representation	an-11
Data Field	Contents of subfields 1–3

Attribute	Description
Subfields	3
Justification	See subfields

Usage

Following is the usage of subelement 34 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 347: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	•	X	C
Authorization Advice/0120: system-generated	•	X	C

Values

See subfields.

Application notes

The Dual Message Authorization System inserts this subelement when the Dynamic CVC 3 Pre-validation Service (OBS 15), Dynamic CVC 3 Validation in Stand-In Processing Service (OBS 16), M/Chip Cryptogram Pre-validation Service (OBS 02), or M/Chip Cryptogram Validation in Stand-In Processing Service (OBS 03) was performed on the transaction and the validation result was V (Valid), A (Valid AC; ATC outside allowed range), E (Valid AC; ATC Replay), T (Valid ARQC/TC and ATC; TVR/CVR invalid), or G (Application Cryptogram is valid but not an ARQC; status of TVR/CVR unknown).

Table 348: Chip Services

Value	Description
V	Valid ARQC/TC and ATC and TVR/CVR
A	Valid Application Cryptogram; ATC out of range
E	Valid Application Cryptogram; ATC replay
T	Valid ARQC/TC and ATC; TVR/CVR invalid
G	Application Cryptogram valid but not an ARQC; ATC valid; status of TVR/CVR unknown

Table 349: CVC Services

Value	Description
V	Valid CVC3 and ATC and TVR/CVR
A	Valid CVC3; ATC out of range
E	Valid CVC; ATC replay

These results are found in *DE 48*, subelement 71, subfield 2 (On-behalf [OB] Result 1).

Subelement 34 should be used for processing authorization messages and maintaining the ATC values on the issuer host. Subfields 1–3 are always present.

The following ATC information will be provided in Dual Message Authorization System and Single Message System financial transaction requests and advices to issuers for MDES transactions initiated with a token, when available. ATC information will not be provided in declined advices.

- *DE 48*, subelement 34, subfield 1 (ATC Value)
- *DE 48*, subelement 34, subfield 2 (ATC Discrepancy Value)
- *DE 48*, subelement 34, subfield 3 (ATC Discrepancy Indicator)

Subfield 1 (ATC Value)

DE 48, subelement 34, subfield 1 (ATC Value) contains the derived full ATC Value used in the validation.

Table 350: Attributes

Attribute	Description
Data Representation	n-5
Data Field	Contents of subfield 1
Justification	Right-justified, leading zeros

Subfield 2 (ATC Discrepancy Value)

DE 48, subelement 34, subfield 2 (ATC Discrepancy Value) is the differential between the transaction ATC and the maximum value allowed by the issuer when the transaction ATC is above the previous ATC, or the differential between the transaction ATC and the minimum value allowed by the issuer when the transaction ATC is below the previous ATC. ATC Discrepancy Value will be zero when the transaction ATC is within the issuer-defined limits.

Table 351: Attributes

Attribute	Description
Data Representation	n-5
Data Field	Contents of subfield 2
Justification	Right-justified, leading zeros

Subfield 3 (ATC Discrepancy Indicator)

DE 48, subelement 34, subfield 3 (ATC Discrepancy Indicator) indicates if the ATC Discrepancy Value is above, below, or within the maximum values allowed by the issuer.

Table 352: Attributes

Attribute	Description
Data Representation	an-1
Data Field	Contents of subfield 3
Justification	N/A

Values

Table 353: Values

Value	Description
G	Indicates that the ATC value is greater than the maximum value allowed
L	Indicates that the ATC value is lower than the minimum value allowed
W	Indicates that the ATC value is within the issuer-defined limits

Subelement 34 subfield data examples

Following are examples of how data may appear in subfield 1, subfield 2, and subfield 3.

Example 1

Last valid ATC value processed by Mastercard was 00011.

Issuer provided a maximum allowed range of 00035.

$00011 + 00035 = 00046$, which represents the maximum ATC value, allowed in the next transaction.

The derived value of the ATC in the next transaction is 00088.

The ATC Discrepancy Value for this transaction is 42 which represents the difference between the maximum ATC value allowed 00046 and the derived ATC value from the transaction 00088.

Subelement 34 will contain the following:

- Subfield 1 = 00088
- Subfield 2 = 00042
- Subfield 3 = G

Example 2

Last valid ATC value processed by Mastercard was 00068.

Issuer provided a minimum allowed range of 00035.

$00068 - 00035 = 00033$, which represents the minimum ATC value allowed in the next transaction.

The derived value of the ATC in the next transaction is 00020.

The ATC Discrepancy Value for this transaction is 13, which represents the difference between the minimum ATC value allowed 00033 and the derived ATC value from the transaction 00020.

Subelement 34 will contain the following:

- Subfield 1 = 00020
- Subfield 2 = 00013
- Subfield 3 = L

Example 3

Last valid ATC value processed by Mastercard was 00037.

Issuer provided a maximum allowed range of 00005.

$00037 + 00005 = 00042$, which represents the maximum ATC value allowed in the next transaction.

The derived value of the ATC in the next transaction is 00040.

The ATC Discrepancy Value for this transaction is 0 because the derived value of the ATC in the next transaction 00040 is greater than the last valid ATC value 00037 and below the maximum ATC value allowed in the next transaction 00042.

Subelement 34 will contain the following:

- Subfield 1 = 00040
- Subfield 2 = 00000
- Subfield 3 = W

Subelement 35 (Contactless Non-Card Form Factor Request/Response)

DE 48, subelement 35 (Contactless Non-Card Form Factor Request/Response) supports Contactless Mapping Service.

Table 354: Attributes

Attribute	Description
Subelement ID	35
Subelement Data Length	2 positions
Data Representation	an-1
Data Field	Contents of position 1
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 35 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 355: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	.	X	C
Authorization Request Response/0110	C	X	.

Values

The contactless issuer will receive the Authorization Request/0100 message containing *DE 48*, subelement 35, value R (Cardholder request for device).

The contactless issuer should respond with an Authorization Request Response/0110 message containing *DE 48*, subelement 35, value A (Approve cardholder request for device) or D (Decline cardholder request for device).

Application notes

Issuers choosing the Contactless Mapping Service "Processing and Issuance of devices" participation option (Passive Participation) will receive an Authorization Request/0100 message when a cardholder requests a contactless device from the issuer-branded, secure website hosted by Mastercard. The issuer may then decide if the cardholder should receive the contactless device.

The Dual Message Authorization System will use the value in *DE 48*, subelement 35 of the Authorization Request Response/0110 message to issue the contactless device or to decline the cardholder's request.

If an issuer chooses not to support subelement 35, the issuer-branded website will evaluate the responses provided for AVS and CVC 2 and approve or decline the cardholder request based on those values.

Subelement 36 (Additional Visa Request Data [Visa Only])

DE 48, subelement 36 (Additional Visa Request Data [Visa Only]) contains additional request data to assist Visa with its processing.

Attributes

Attribute	Description
Subelement ID	36
Subelement Length	2
Data Representation	ans...94; LLVAR
Data Field	Contents of subfields
Subfields	5
Justification	See subfields

Usage

Following is the usage of subelement 36 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Reversal Request/0400	C	.	C

Values

See subfields.

Subfield 01 (Merchant Verification Value [MVV])

DE 48, subelement 36, subfield 01 (Merchant Verification Value) contains the merchant verification value for subelement 36.

Attributes

Attribute	Description
Subfield ID	01
Subfield Length	2
Data Representation	an-10
Data Field	Contents of subfield 1.
Justification	N/A

Values

0–9 and A–F

Subfield 02 (Business Application Identifier [Visa Only])

DE 48, subelement 36, subfield 02 (Business Application Identifier [Visa Only]) identifies the intended use of a Visa-branded transaction.

Table 356: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	a-2
Data Field	Contents of subfield 02
Justification	N/A

Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

Application notes

Subfield 02 (Business Application Identifier [Visa Only]) is only for use with Visa-branded transactions.

Subfield 03 (Acceptor Legal Business Name [Visa Only])

DE 48, subelement 36, subfield 03 (Acceptor Legal Business Name [Visa Only]) identifies the acceptor legal business name.

Table 357: Attributes

Attribute	Description
Subfield ID	03
Subfield Data Length	2 positions
Data Representation	ans...25; LLVAR
Data Field	Contents of subfield 03
Justification	N/A

Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

Application notes

Subfield 03 (Acceptor Legal Business Name [Visa Only]) is only for use with Visa-branded transactions.

Subfield 04 (Payment Facilitator Name [Visa Only])

DE 48, subelement 36, subfield 04 (Payment Facilitator Name [Visa Only]) identifies the payment facilitator name.

Table 358: Attributes

Attribute	Description
Subfield ID	04
Subfield Data Length	2 positions
Data Representation	ans...25; LLVAR
Data Field	Contents of subfield 04
Justification	N/A

Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

Application notes

Subfield 04 (Payment Facilitator Name [Visa Only]) is only for use with Visa-branded transactions.

Subfield 05 (Purpose of Payment [Visa Only])

DE 48, subelement 36, subfield 05 (Purpose of Payment [Visa Only]) identifies the purpose of payment.

Table 359: Attributes

Attribute	Description
Subfield ID	05
Subfield Data Length	2 positions
Data Representation	ans...12; LLVAR
Data Field	Contents of subfield 05
Justification	N/A

Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

Application notes

Subfield 05 (Purpose of Payment [Visa Only]) is only for use with Visa-branded transactions.

Subelement 37 (Additional Merchant Data)

DE 48, subelement 37 (Additional Merchant Data) contains subfields representing the ID of the participating service provider in a transaction and the sub-merchant ID when applicable. The service provider can be a payment facilitator, independent sales organization, or a merchant payment gateway. A transaction may involve more than one service provider.

Table 360: Attributes

Attribute	Description
Subelement ID	37
Subelement Data Length	2 positions
Data Representation	ans...71; LLVAR
Data Field	Contents of subfields 1-5
Subfields	5

Attribute	Description
Justification	See subfields

Usage

Following is the usage of subelement 37 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 361: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Advice/0120: acquirer-generated	C	.	C
Authorization Advice/0120: system-generated	.	C	C

Values

See subfields.

Application notes

Specific application notes, conditions, and cross-edits where applicable.

Valid Combinations of *DE 48*, subelement 37, subfields 1, 2, and 3:

- Subfields 1 and 3 only
- Subfield 2 only
- Subfields 1, 2, and 3 (Mastercard only)

Subfield 1 (Payment Facilitator ID)

DE 48, subelement 37, subfield 1 (Payment Facilitator ID) contains the Payment Facilitator ID assigned by Mastercard or Visa.

Table 362: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	n-11
Data Field	Contents of subfield 1

Attribute	Description
Justification	Right-justified with leading zeros

Values

The Payment Facilitator ID must always be a value assigned by Mastercard and must be the same value in both the Dual Message Authorization and Dual Message Clearing Systems. This ID is the Company ID that will be assigned during the time of registration with Mastercard for a Service Provider as "Payment Facilitator."

The value must be provided by the acquirer when a registered payment facilitator is involved in a transaction and should be right-justified with leading zeros. For example, if the Company ID is 123456, *DE 48*, SE 37, SF 1 should be 00000123456.

Subfield 2 (Independent Sales Organization ID)

DE 48, subelement 37, subfield 2 (Independent Sales Organization ID) contains the Independent Sales Organization ID assigned by Mastercard.

Table 363: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	n-11
Data Field	Contents of subfield 2
Justification	Right-justified with leading zeros

Values

The Independent Sales Organization ID must always be a value assigned by Mastercard and is the same value in both the Dual Message Authorization and Dual Message Clearing Systems. This ID is the "Company ID" that is assigned during the time of registration via Mastercard Connect of a service provider as "Independent Sales Organization."

This value must be provided by the acquirer when a registered independent sales organization is involved in a transaction and should be right-justified with leading zeros. For example, if the Company ID is 123456, *DE 48*, subelement 37, subfield 2 should be 00000123456.

For Visa transactions, the Marketplace ID must always be a value assigned by Visa.

This value must be provided by the acquirer when a Visa-registered Marketplace ID is involved in a transaction and should be right-justified with leading zeros. For example, if the Marketplace ID is 12345678, *DE 48*, subelement 37, subfield 2 should be 00012345678.

Subfield 3 (Sponsored Merchant ID)

DE 48, subelement 37, subfield 3 (Sponsored Merchant ID) contains the identifier assigned by the acquirer or the payment facilitator to the sponsored merchant and its location.

Table 364: Attributes

Attribute	Description
Subfield ID	03
Subfield Data Length	2 positions
Data Representation	ans-15
Data Field	Contents of subfield 3
Justification	Left-justified, trailing spaces

Processing rules

The sponsored merchant ID must be present for all payment facilitator-acquired transactions and must uniquely identify each sponsored merchant and its location.

Subfield 4 (Merchant Country of Origin)

DE 48, subelement 37, subfield 4 (Merchant Country of Origin) contains the ISO numeric country code for the home country of the government that owns or controls the merchant that initiated the transaction.

Table 365: Attributes

Attribute	Description
Subfield ID	04
Subfield Data Length	2 positions
Data Representation	n-3
Data Field	Contents of subfield 4
Justification	n/a

Values

If present, DE 48, subelement 37, subfield 4 must contain a valid ISO numeric country code.

For example, embassies are physically located in countries that are not their "home country" (that is, the embassy of Mexico may be physically located in Singapore; however, the "home country" is Mexico).

Requirements

DE 48, subelement 37, subfield 4 must contain the location information of the government entity, not the location of the merchant.

Usage

Table 366: Applicable messages

Message Type Identifier	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Advice/0120	C	.	C

Subfield 5 (Merchant Payment Gateway ID)

DE 48, subelement 37, subfield 5 (Merchant Payment Gateway ID) identifies the registered Merchant Payment Gateway used by the merchant and is provided by the acquirer when a registered MPG service provider is involved in a transaction.

Table 367: Attributes

Attribute	Description
Subfield ID	05
Subfield Length	2 positions
Data Representation	n-11
Data Field	Contents of subfield 5
Justification	Right-justified with leading zeros

Values

The Merchant Payment Gateway ID is a value assigned by Mastercard. The Merchant Payment Gateway ID is assigned to Merchant Payment Gateway service provider at the time of their registration with Mastercard. The value must be right-justified with leading zeros. For example, if the Merchant Payment Gateway ID is 123456, then *DE 48*, subelement 37, subfield 5 contains 00000123456.

Subelement 38 (Account Category)

DE 48, subelement 38 (Account Category) supports Account Level Management.

Table 368: Attributes

Attribute	Description
Subelement ID	38
Subelement Data Length	2 positions
Data Representation	an-1
Data Field	Contents of position 1
Subfields	NA
Justification	NA

Usage

Following is the usage of subelement 38 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 369: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	.	X	C
Authorization Advice/0120: system-generated	.	X	C
Reversal Advice/0420	.	X	C

For a complete valid values listing, refer to the [Account Level Management Manual](#).

Subelement 39 (Account Data Compromise Information)

DE 48, subelement 39 (Account Data Compromise Information) contains confirmed or suspected account data compromise event information for account ranges.

Table 370: Attributes

Attribute	Description
Subelement ID	39
Subelement Data Length	2 positions

Attribute	Description
Data Representation	ans-30
Data Field	Contents of positions 1–30
Subfields	N/A
Justification	Left

Usage

Following is the usage of subelement 39 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 371: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	•	X	C
Authorization Advice/0120: system-generated	•	X	C

Table 372: Positions 1–3 (Reserved for Future Use)

Attribute	Description
Data Representation	n-3
Data Field	Contents of positions 1–3
Values	000 = Not available

Table 373: Positions 4–6 (Reserved for Future Use)

Attribute	Description
Data Representation	n-3
Data Field	Contents of positions 4–6
Values	N/A

Table 374: Positions 7–12 (Case Key Code number 1)

Attribute	Description
Data Representation	ans-6
Data Field	Contents of positions 7–12
Values	Unique Key to Identify Case

Table 375: Positions 13–18 (Case Key Code number 2)

Attribute	Description
Data Representation	ans-6
Data Field	Contents of positions 13–18
Values	Unique Key to Identify Case

Table 376: Positions 19–24 (Case Key Code number 3)

Attribute	Description
Data Representation	ans-6
Data Field	Contents of positions 19–24
Values	Unique Key to Identify Case

Table 377: Position 25 (Account Number)

Attribute	Description
Data Representation	n-1
Data Field	Contents of position 25
Values	0–9 = Number of events in which the account number has been exposed

Table 378: Position 26 (Expiration Date)

Attribute	Description
Data Representation	n-1

Attribute	Description
Data Field	Contents of position 26
Values	0–9 = Number of events in which expiration date has been exposed

Table 379: Position 27 (CVC 2)

Attribute	Description
Data Representation	n-1
Data Field	Contents of position 27
Values	0–9 = Number of events in which the CVC 2 has been exposed

Table 380: Position 28 (PIN)

Attribute	Description
Data Representation	n-1
Data Field	Contents of position 28
Values	0–9 = Number of events in which the PIN has been exposed

Table 381: Position 29 (Magnetic Stripe)

Attribute	Description
Data Representation	n-1
Data Field	Contents of position 29
Values	0–9 = Number of events in which the Magnetic Stripe has been exposed

Table 382: Position 30 (Personal Information)

Attribute	Description
Data Representation	n-1
Data Field	Contents of position 30

Subelement 40 (Electronic Commerce Merchant/Cardholder Certificate Serial Number [Visa Only])

Attribute	Description
Values	0–9 = Number of events in which the cardholder's personal information has been exposed

Subelement 40 (Electronic Commerce Merchant/Cardholder Certificate Serial Number [Visa Only])

DE 48, subelement 40 (Electronic Commerce Merchant/Cardholder Certificate Serial Number [Visa Only]) contains certificate information on electronic commerce transactions when applicable.

Table 383: Attributes

Attribute	Description
Subelement ID	40
Subelement Data Length	2 positions
Data Representation	n...40 (also contains binary data. See subfields.); LLVAR
Data Field	Contents of subfield 1 or 2 or both
Subfields	2
Justification	N/A

Usage

Following is the usage of subelement 40 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 384: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	•	C
Authorization Request Response/0110	CE	•	CE
Authorization Advice/0120: acquirer-generated	C	X	C
Authorization Advice/0120: system-generated	•	X	C

Values

Contains the contents of subfield 1 or subfield 2 or both.

Subfield 1 (Merchant Certificate Serial Number)

DE 48, subelement 40, subfield 1 (Merchant Certificate Serial Number) is the Merchant Certificate Serial Number in binary data.

Table 385: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	b...16
Data Field	Merchant Certificate Serial Number in binary data.
Justification	N/A
Values	Merchant specific: b...16

Subfield 2 (Cardholder Certificate Serial Number)

DE 48, subelement 40, subfield 2 (Cardholder Certificate Serial Number) is the cardholder certificate serial number in binary format.

Table 386: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	b...16
Data Field	Cardholder Certificate Serial Number in binary data.
Justification	N/A
Values	Cardholder specific: b...16

Subelement 41 (Electronic Commerce Certificate Qualifying Information)

Table 387: Attributes

Attribute	Description
Subelement ID	41

Attribute	Description
Subelement Data Length	2 positions (value in the range of 05–95)
Data Representation	ans...95; LLVAR
Data Field	Contents of subfield(s)
Subfields	18
Justification	N/A

Usage

Following is the usage of subelement 41 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 388: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Request Response/0110	CE	.	CE
Authorization Advice/0120: acquirer-generated	.	C	C
Authorization Advice/0120: system-generated	.	C	C
Reversal Request/0400	C	.	C
Reversal Request Response/0410	CE	.	CE
Reversal Advice/0420	.	C	C

Values

See subfields.

Subfield 1 (Additional Recurring Payments Information)

Subfield 1 (Additional Recurring Payments Information) is used in the Dual Message Authorization System to provide additional data for recurring payments. Currently this is used for domestic transactions in India.

Table 389: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions

Attribute	Description
Data Representation	ans...26

Table 390: Values

Field Name/Positions	Type	Size	Comment
Recurring Payment Mandate Type (Position 1)	Numeric	1	Values are: 1 = fixed amount Recurring Payment 2 = Recurring Payment with maximum amount
Recurring Payment Transaction Type (Position 2)	Numeric	1	Values are: 1 = registration/first transaction 2 = subsequent transaction
Validation Flag (Position 3)	Numeric	1	Values are: 0 = not validated 1 = validated
Maximum Recurring Payment Amount (Positions 4 – 13)	Numeric	10	Maximum amount agreed to by the cardholder. Amount will be specified in DE 49 (Currency Code, Transaction). Example: 1800.00 as 180000, with leading zeros.
Registered Transaction Identifier (Positions 14 – 23)	Alphanumeric	10	Unique identifier for Recurring Payments mandate
Number of Recurring Payments (Positions 24 – 25)	Numeric	2	Indicates the number of recurring payment debits. With leading zeros. Value of 99 means that recurring payments are authorized until canceled or that the Number of Recurring Payments is not indicated.

Field Name/Positions	Type	Size	Comment
Recurring Payment Frequency (Position 26)	Numeric	1	Identifies frequency of the Recurring Payments debit. Values are: 0 – annually 1 – daily 2 – weekly 3 – fortnightly 4 – monthly 5 – quarterly 6 – twice-yearly 8 – ad hoc

Application notes

Subfield 1 facilitates the communication of additional data related to recurring payments between acquirers and issuers. Currently it is only being used for India domestic recurring payments. Mastercard is not enforcing edits to check the presence or content of any data.

DE 48, subelement 41, subfield 1 will include and echo back additional data in Authorization Request Response/0110 message enrollment and first transaction of the standing order instructions, and subsequent recurring transactions within the Authorization Request Response/0110 message.

DE 48, subelement 41, subfield 1 will include and echo back additional data to the acquirer, regardless of issuer interaction.

Subfield 2 (Reserved for Future Use)

DE 48, subelement 41, subfield 2 (Reserved for Future Use) is for future use.

Table 391: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	n-6

Subfield 3 (Reserved for Future Use)

DE 48, subelement 41, subfield 3 (Reserved for Future Use) is for future use.

Table 392: Attributes

Attribute	Description
Subfield ID	03
Subfield Data Length	2 positions
Data Representation	n-3

Subfield 4 (Reserved for Future Use)

DE 48, subelement 41, subfield 4 (Reserved for Future Use) is for future use.

Table 393: Attributes

Attribute	Description
Subfield ID	04
Subfield Data Length	2 positions
Data Representation	an...22

Subfield 5 (Reserved for Future Use)

DE 48, subelement 41, subfield 5 (Reserved for Future Use) is for future use.

Table 394: Attributes

Attribute	Description
Subfield ID	05
Subfield Data Length	2 positions
Data Representation	ans...20

Subfield 6 (Reserved for Future Use)

DE 48, subelement 41, subfield 6 (Reserved for Future Use) is for future use.

Table 395: Attributes

Attribute	Description
Subfield ID	06
Subfield Data Length	2 positions
Data Representation	a...13

Subfield 7 (Reserved for Future Use)

DE 48, subelement 41, subfield 7 (Reserved for Future Use) is for future use.

Table 396: Attributes

Attribute	Description
Subfield ID	07
Subfield Data Length	2 positions
Data Representation	a-3

Subfield 8 (Reserved for Future Use)

DE 48, subelement 41, subfield 8 (Reserved for Future Use) is for future use.

Table 397: Attributes

Attribute	Description
Subfield ID	08
Subfield Data Length	2 positions
Data Representation	an...10

Subfield 9 (Reserved for Future Use)

DE 48, subelement 41, subfield 9 (Reserved for Future Use) is for future use.

Table 398: Attributes

Attribute	Description
Subfield ID	09
Subfield Data Length	2 positions
Data Representation	a...22

Subfield 10 (Reserved for Future Use)

DE 48, subelement 41, subfield 10 (Reserved for Future Use) is for future use.

Table 399: Attributes

Attribute	Description
Subfield ID	10
Subfield Data Length	2 positions
Data Representation	n-9

Subfield 11 (Citizen ID)

DE 48, subelement 41, subfield 11 (Citizen ID) indicates the citizen ID.

Table 400: Attributes

Attribute	Description
Subfield ID	11
Subfield Data Length	2 positions
Data Representation	an...20
Data Field	National ID
Justification	N/A
Values	Cardholder specific

Subfield 12 (Reserved for Future Use)

DE 48, subelement 41, subfield 12 (Reserved for Future Use) is for future use.

Table 401: Attributes

Attribute	Description
Subfield ID	12
Subfield Data Length	2 positions
Data Representation	ans...20

Subfield 13 (Reserved for Future Use)

DE 48, subelement 41, subfield 13 (Reserved for Future Use) is for future use.

Table 402: Attributes

Attribute	Description
Subfield ID	13
Subfield Data Length	2 positions
Data Representation	ans...20

Subfield 14 (Reserved for Future Use)

DE 48, subelement 41, subfield 14 (Reserved for Future Use) is for future use.

Table 403: Attributes

Attribute	Description
Subfield ID	14
Subfield Data Length	2 positions
Data Representation	ans...20

Subfield 15 (Reserved for Future Use)

DE 48, subelement 41, subfield 15 (Reserved for Future Use) is for future use.

Table 404: Attributes

Attribute	Description
Subfield ID	15
Subfield Data Length	2 positions
Data Representation	ans...10

Subfield 16 (Reserved for Future Use)

DE 48, subelement 41, subfield 16 (Reserved for Future Use) is for future use.

Table 405: Attributes

Attribute	Description
Subfield ID	16
Subfield Data Length	2 positions
Data Representation	n-2

Subfield 17 (Reserved for Future Use)

DE 48, subelement 41, subfield 17 (Reserved for Future Use) is for future use.

Table 406: Attributes

Attribute	Description
Subfield ID	17
Subfield Data Length	2 positions
Data Representation	a-1

Subfield 18 (Reserved for Future Use)

DE 48, subelement 41, subfield 18 (Reserved for Future Use) is for future use.

Table 407: Attributes

Attribute	Description
Subfield ID	18
Subfield Data Length	2 positions
Data Representation	a...20

Subelement 42 (Electronic Commerce Indicators)

DE 48, subelement 42 (Electronic Commerce Indicators) contains the electronic commerce indicators representing the security level and cardholder authentication associated with the transaction.

Table 408: Attributes

Attribute	Description
Subelement ID	42
Subelement Data Length	2 positions
Data Representation	n-7 unless an Identity Check downgrade or MDES SLI modification occurs in which case n-19 applies.
Data Field	Contents of subfield 1 (except contents of subfields 1, 2, and 3 for acquirer Authorization Request Response/0110 messages if Identity Check downgrade or MDES SLI modification occurred)
Subfields	3
Justification	N/A

Usage

Subelement 42 must be present in all Authorization Request/0100 messages for electronic commerce transactions. Following is the usage of subelement 42 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 409: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Request Response/0110	CE	X	C
Authorization Advice/0120: acquirer-generated	C	.	C
Authorization Advice/0120: system-generated	.	C	C
Reversal Request/0400	C	.	C

Values

If Identity Check downgrade or MDES SLI modification did not occur, contains the electronic commerce security level indicator and UCAF collection indicator data in subfield 1 that consists of a valid combination of positions 1, 2, and 3.

If Identity Check downgrade or MDES SLI modification occurred, contains contents of subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator), subfield 2 (Original Electronic Commerce Security Level Indicator and UCAF Collection Indicator), and subfield 3 (Reason for UCAF Collection Indicator Downgrade) for acquirer Authorization Request Response/0110 messages.

Application notes

A Mastercard Identity Check transaction is downgraded to a non-Identity Check transaction if an Accountholder Authentication Value (AAV) is not present or is invalid in *DE 48*, subelement 43 (Universal Cardholder Authentication Field [UCAF]).

For transactions initiated with Mastercard Digital Enablement Service (MDES) Tokens, Mastercard provides Dual Message Authorization System edits that validate and when necessary, modify the Security Level Indicator (SLI) in *DE 48*, subelement 42, subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection indicator) in Authorization Request/0100 messages.

Mastercard sends the correct SLI to the issuer in Authorization Request/0100 messages for MDES transactions.

Mastercard notifies acquirers when the SLI has been corrected by inserting two additional *DE 48*, subelement 42 indicators, as follows. All acquirers must be ready to receive *DE 48*, subelement 42, subfield 2 (Original Electronic Commerce Security Level Indicator and UCAF Collection Indicator), and subfield 3 (Reason for UCAF Collection Indicator Downgrade) in the Authorization Request Response/0110 messages.

Subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator)

DE 48, subelement 42, subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator) indicates the electronic commerce security level and UCAF collection in positions 1, 2, and 3.

Table 410: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	n-3
Data Field	Indicates the electronic commerce security level and UCAF collection in positions 1, 2, and 3
Justification	N/A

Values

Valid combination of the security level indicators in positions 1 and 2 and the UCAF collection indicator in position 3.

Position 1 (Security Protocol)

Table 411: Attributes

Attribute	Description
Data Representation	n-1
Data Field	The electronic commerce security level indicator

Table 412: Values

Value	Description
0	Reserved for existing Mastercard Europe/Visa definitions
1	Reserved for future use
2	Channel
3–8	Reserved for future use

Position 2 (Cardholder Authentication)

Table 413: Attributes

Attribute	Description
Data Representation	n-1
Data Field	The cardholder authentication indicator

Table 414: Values

Value	Description
0	Reserved for future use
1	eCommerce / Identity Check
2	Processed through Masterpass
3	Reserved for future use
4	Tokenized payment
5–9	Reserved for future use

Table 415: Valid Combinations of Position 1 and Position 2

Value	Description
21	Channel encryption; cardholder certificate not used (this is the preferred value for Mastercard® Identity Check™)
22	Masterpass-generated transaction
24	Tokenized Payment

Position 3 (UCAF Collection Indicator)

Table 416: Attributes

Attribute	Description
Data Representation	n-1
Data Field	The UCAF collection indicator

Table 417: Values

Value	Description
0	Non-authenticated payment, Identity Check transaction with failed authentication, or Tokenized Payment with Dynamic Token Verification Code (DTVC).
1	UCAF data collection is supported by the merchant, and UCAF data must be present (<i>DE 48</i> , subelement 43 must be present and contain an attempt AAV for Mastercard Identity Check)
2	UCAF data collection is supported by the merchant, and UCAF data must be present (<i>DE 48</i> , subelement 43 must contain a fully authenticated AAV, and DSRP cryptogram shall be optionally present for tokenized transactions). This includes cardholder-initiated transactions (CIT) for authentication.
3	UCAF data collection is supported by the merchant, and UCAF (Mastercard assigned Static Accountholder Authentication Value) data must be present. NOTE: <i>DE 48</i>, subelements 32 and 43 are required for Static AAV transactions. Identifies participation in one of the following programs: • Maestro Recurring Payments Program • Mastercard Utility Payment Program • Maestro Low Risk Merchant Program • Maestro Static AAV for Masterpass
4	Merchant has chosen to share authentication data within authorization; UCAF data collection must be present (<i>DE 48</i> , subelement 43 must contain an Insights AAV for Mastercard Identity Check).
5	Reserved for future use
6	Merchant Risk Based Decisioning
7	Merchant-initiated transactions (<i>DE 48</i> , subelement 43 only required for Identity Check).
8–9	Reserved for future use

Table 418: Valid Combinations of Position 1, Position 2 and Position 3

Value	Description
210	Present in unauthenticated transactions, or in transactions where an Identity Check merchant has chosen not to undertake Identity Check, or where Identity Check failed authentication and the merchant desires to proceed with the transaction. Also present in transactions with a DTVC
211	Mastercard Identity Check transaction. UCAF data contains either an Attempts AAV or a Non-Low Risk AAV for Mastercard Identity Check from Smart Authentication
212	Mastercard Identity Check transaction. UCAF data contains a fully authenticated AAV for Mastercard Identity Check or a Low-Risk AAV for Mastercard Identity Check from Smart Authentication
214	Mastercard Identity Check where merchant has chosen to share data to generate insights within the authorization (used in Identity Check™ Insights or Ekata transactions). Insights or Ekata scores are provided to the Issuer in DE 48, subelement 56
216	Mastercard Identity Check transaction. Merchant Risk Based Decisioning. This includes European PSD2 Acquirer Exemption transactions
217	Mastercard Identity Check transaction. Merchant Initiated Transactions
	NOTE: Mastercard Identity Check split shipments should be managed as 1 authentication, 1 authorization and multiple clearing records.
242	Authenticated tokenized transaction with DSRP cryptogram. Transaction is not eligible for fraud-related chargeback by the issuer
246	Tokenized transaction with DSRP cryptogram
247	Tokenized Merchant Initiated Transaction

Application Notes

For Visa Gateway 3-D Secure transactions the following mapping applies:

- Position 3 "0" maps to Field 60 SLI = 07 (Not Authenticated)

Subfield 2 (Original Electronic Commerce Security Level Indicator and UCAF Collection Indicator)

- Position 3 "1" maps to Field 60 SLI = 06 (3-D Secure merchant, not authenticated)
- Position 3 "2" maps to Field 60 SLI = 05 (Authenticated with CAVV)

For American Express Gateway 3-D Secure transactions the following mapping applies:

- Position 3 "0" maps to Field 61 SLI = 07 (Not Authenticated)
- Position 3 "1" maps to Field 61 SLI = 06 (Attempted with AEVV)
- Position 3 "2" maps to Field 61 SLI = 05 (Authenticated with AEVV)

SLI is the level of security used when a cardmember provides payment information to the merchant during authentication.

JCB and Diners Club use the same Position 3 values (0, 1, and 2) as defined for Mastercard Identity Check.

Subfield 2 (Original Electronic Commerce Security Level Indicator and UCAF Collection Indicator)

DE 48, subelement 42, subfield 2 (Original Electronic Commerce Security Level Indicator and UCAF Collection Indicator) describes the original Security Level Indicators sent by the acquirer in the Authorization Request/0100 message before the Identity Check downgrade or MDES SLI modification occurred.

Table 419: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	n-3
Data Field	Contents of subfield 2
Justification	N/A

Table 420: Positions

Position	Description
1	Security Protocol
2	Cardholder Authentication
3	UCAF Collection Indicator

Subfield 3 (Reason for UCAF Collection Indicator Downgrade)

DE 48, subelement 42, subfield 3 (Reason for UCAF Collection Indicator Downgrade) describes the reason why the Authorization Request/0100 message was downgraded or modified.

Table 421: Attributes

Attribute	Description
Subfield ID	03
Subfield Data Length	2 positions
Data Representation	n-1
Data Field	Contents of subfield 3
Justification	N/A

Table 422: Values

Value	Description
0	Missing Universal Cardholder Authentication Field (UCAF)
1	Invalid Universal Cardholder Authentication Field (UCAF)
2	Invalid Security Level Indicator

Application notes

A Mastercard Identity Check transaction is downgraded to a non-Identity Check transaction if an Accountholder Authentication Value (AAV) is not present or is invalid in DE 48, subelement 43 (Universal Cardholder Authentication Field [UCAF]).

DE 48, subelement 42, subfields 2 and 3 are only present in the Authorization Request Response/0110 message provided by Mastercard to the acquirer if an Identity Check downgrade or MDES SLI modification occurred.

The message format sent in the acquirer Authorization Request Response/0110 message will be in the format 0103xxx0203yyy0301z only if the Identity Check downgrade occurs. Otherwise, the message format will remain as it is today which is 0103xxx.

NOTE: DE 48, subelement 42, subfields 2 and 3 will always be submitted together.

Subelement 43 (Universal Cardholder Authentication Field [UCAF])

DE 48, subelement 43 (Universal Cardholder Authentication Field [UCAF]) contains the encoded Mastercard Identity Check AAV (Accountholder Authentication Value) resulting from all Identity Check fully authenticated or attempts transactions, data for Visa, JCB, Diners Club, or

American Express transactions associated with the 3-D Secure Electronic Commerce Verification Service (if collected), or the Static AAV assigned by Mastercard for Maestro Recurring Payments Program, Mastercard Utility Payment Program, or Maestro Low Risk Merchant Program.

Digital Secure Remote Payments (DSRP) submitted as Electronic Commerce (POS entry mode *DE 22*, subfield 1, value of 10 or 81) transactions should use *DE 104* for communicating the MDES and DSRP cryptogram and other Digital Payment information. Refer to the [Digital Secure Remote Payments: UCAF Formats](#) and [MDES Technical Specifications for Dual and Single Message Systems](#) guides for information on the UCAF formats supported by Mastercard.

Table 423: Attributes

Attribute	Description
Subelement ID	43
Subelement Data Length	2 positions
Data Representation	ans...32; LLVAR
Data Field	Contains UCAF™ data
Subfields	N/A
Justification	N/A

Usage

Subelement 43 must be present whenever UCAF data is collected for electronic commerce transactions. Following is the usage of subelement 43 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 424: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	•	C
Authorization Request Response/0110	CE	•	CE
Authorization Advice/0120: acquirer-generated	C	•	C
Authorization Advice/0120: system-generated	•	X	C
Reversal Request/0400	C	•	C
Reversal Request Response/0410	CE	•	CE

Values

Refer to the specific program or service applications for:

- 3-D Secure for Mastercard Identity Check
- 3-D Secure Electronic Commerce Verification Service (Visa or American Express)
- Digital Secure Remote Payments
- SPA2 AAV and Verification for the Mastercard Identity Check Program
- MDES Technical Specifications for Dual and Single Message Systems

Subelement 43 (3-D Secure for Mastercard Identity Check)

DE 48 (Additional Data: Private Use), subelement 43 (3-D Secure for Mastercard Identity Check) contains UCAF data and is described here for Mastercard Implementation of 3-D Secure for Mastercard Identity Check.

Table 425: Attributes

Attribute	Value
Data Representation	ans-28 (base 64 encoded)
Data Field	The Mastercard 3-D Secure SPA AAV

Values

Values are transaction-specific. Positions 1 and 2 of the Mastercard 3-D Secure SPA AAV are control bytes indicating the Mastercard Identity Check platform that created the field contents. SLI 211 with h and SLI 212 with j are downgraded to SLI 210 (unauthenticated).

Table 426: Mastercard Identity Check Platform

SPA2 Values for Smart Authentication non-low risk and Attempts Only authenticated SLI value 211
kE, kF, kL, kM
SPA2 Values for fully authenticated and CIT (cardholder-initiated transaction) and Smart Authentication low risk authenticated SLI value 212
kA, kB, kC, kD, kG, kH, kJ, kK, kQ, kR, kS, kT

SPA2 Values for Insights authenticated SLI value 214	SPA2 Values for PSD2 Acquirer Exemption authenticated SLI value 216
kX, kY, kZ	kN, kU, kV, kW
SPA2 Values for Recurring MIT (merchant initiated transactions) authenticated SLI value 217	
kO, kP	

NOTE: Control byte kQ is defined for AAV Refresh with SLI = 212, and kF is defined for Attempts Only with SLI = 211. Attempts Only indicates that authentication was attempted by the merchant but was not performed either by the issuer or by Mastercard Identity Check Smart Authentication Stand-In solution.

NOTE: SPA2 AAV leading indicator values i, l, m, and n are reserved for future use.

Application notes

Mastercard will continue to modify the SLI value in Mastercard Digital Enablement Service (MDES) transactions based on the AAV contained in *DE 48*, subelement 43. Acquirers and issuers should expect to see Mastercard Identity Check with SecureCode SLI values in MDES transactions.

NOTE: Issuers must pass the correct security level indicator (SLI) value with the corresponding AAV control byte during cardholder authentication. Mastercard 3DS protocol, Identity Check Program Rules, and DW Integrity Edits do not allow sending Fully Authenticated AAVs for Smart Authentication non-low risk or Attempts Only Authentications nor Smart Authentication non-low risk or Attempts Only AAVs for Fully Authenticated Authentications. Reference: [SPA2 AAV and Verification for the Mastercard Identity Check Program](#).

The following is an example of a properly coded *DE 48*, subelement 43 for Mastercard Identity Check for Fully Authenticated Authorization including AVS request.

T420701032124328kBMAAAAKRIacwR0SdLcB5eBIJjW

Refer to [Mastercard Identity Check Program Guide](#) and [SPA2 AAV and Verification for the Mastercard Identity Check Program](#) for more information.

NOTE: Issuers should not perform Identity Check validation on static AAVs in *DE 48*, subelement 43 nor on Mastercard Identity Check SPA2 AAV generated by Mastercard. Issuers who use the Mastercard AAV Validation Service should refer to the OBS (on-behalf services) field *DE 48*, subelement 71 (On-behalf Services) for services 05 and 06 services. 05 OBS AAV Verification Service is for full-time AAV verification. 06 OBS AAV Verification Service is for Stand-In Authorization when issuers perform self-validation. Issuers enroll in either the 05 OBS AAV Verification Service or the 06 OBS Stand-In AAV Verification Service.

Issuers and their ACS Providers that are sharing IAV (issuer authentication value) data within the SPA2 AAV can always read and verify this data as they have defined for the 4 hexadecimal

bytes shared from their ACS Provider. The IAV can be found in hexadecimal bytes 3-6 after converting the UCAF value from base64 to hexadecimal.

The UCAF field (*DE 48*, subelement 43) is a variable length field up to a maximum of 32 positions. The Mastercard Identity Check AAV is 28 characters in length. There must be no trailing spaces in the UCAF field.

Subelement 43 (Digital Secure Remote Payment Universal Cardholder Authentication Field [UCAF])

DE 48, subelement 43 contains UCAF data and is described here for Mastercard Digital Secure Remote Payments (DSRP).

NOTE: Mastercard requires acquirers to submit the DSRP cryptogram or the new token cryptogram type in *DE 104* (Digital Payment Data), subelement 001 (Digital Payment Cryptogram) instead of submitting it in the existing UCAF *DE 48*, subelement 43 for all tokenized transactions.

Table 427: Attributes

Attribute	Description
Subelement ID	43
Subelement Data Length	2 positions
Data Representation	ans-28; (base 64 encoded)
Data Field	Mastercard Digital Secure Remote Payment UCAF format
Subfields	N/A
Justification	N/A

Values

Transaction specific.

Refer to [MDES Digital Secure Remote Payment Acquirer Implementation Guide](#) for detailed information on the UCAF formats supported by Mastercard.

An MDES transaction might not contain *DE 48*, subelement 43 when the acquirer has submitted a dynamic expiration date (*DE 14*) and a dynamic token verification code (*DE 48*, subelement 92) in order to perform MDES validation services.

A Digital Secure Remote Payment transaction submitted for Mastercard Digital Enablement Service processing that involves an authorization request for a partial shipment or a recurring payment must contain the partial shipment verbiage in subelement 43 as follows: PARTIALbSHIPMENTbbbbbbbbbbbb or PARTIALSHIPMENT00000000000000 unless subelement 42 = 247 in which case subelement 43 is not required. This field contains 28 positions where b represents a space.

Application notes

The format is version dependent.

NOTE: Mastercard requires acquirers to submit the DSRP cryptogram in *DE 104 (Digital Payment Data)*, subelement 001 (Digital Payment Cryptogram) and not in *DE 48*, subelement 43 (UCAF).

Subelement 43 (Static AAV)

DE 48, subelement 43 is the Mastercard implementation of the Static AAV for Maestro Recurring Payments Program, Mastercard Utility Payment Program, Maestro Low Risk Merchant Program, or Maestro Static AAV for Masterpass.

Table 428: Attributes

Attribute	Description
Data Representation	ans-28
Data Field	The Mastercard assigned Static AAV

Values

Position 1 is one of the following values:

- 3 is Transaction processed under the Maestro Recurring Payments Program
- 5 is Transaction processed under the Mastercard Utility Payment Program
- 5 is Maestro Basic Static AAV (Maestro e-commerce transaction initiated through Masterpass)

Position 2–7 is The Mastercard Assigned ID which uniquely identifies the merchant.

Position 8–28 is The merchant name, padded to the right with nines (9).

For the Maestro Low Risk Merchant Program (Europe region only), for subsequent transactions after the first Maestro e-commerce transaction with the same Maestro card, approved merchants will use *DE 48*, subelement 43 in the Authorization Request/0100 message with the Mastercard-assigned static Accountholder Authentication Value (AAV) 5MARPPROGRAM9999999999999999 (28 positions).

Application notes

When subelement 43 contains a Static AAV, subelement 32 (Mastercard Assigned ID) is mandatory. Issuers should not attempt Identity Check validation on a Static AAV.

Subelement 43 (3-D Secure Electronic Commerce Verification Service [Visa, JCB, Diners Club, and American Express])

DE 48, subelement 43 (3-D Secure Electronic Commerce Verification Service [Visa, JCB, Diners Club, and American Express]) is for 3-D Secure Electronic Commerce Verification Service (Visa, JCB, Diners Club, and American Express).

Attributes**Table 429: Position 1 (3-D Secure Electronic Commerce Transaction Indicator)**

Attribute	Description
Subelement ID	43
Subelement Data Length	2 positions
Data Representation	n-1
Data Field	Indicates a 3-D Secure Electronic Commerce transaction
Subfields	N/A
Justification	N/A
Values	8 = Non-Mastercard 3-D Secure Electronic Commerce transaction (Visa, JCB, Diners Club or American Express)

Table 430: Position 2-21 (3-D Secure Electronic Commerce Cardholder Authentication Verification Value [CAVV] Visa Field 126.9 and American Express Verification Value [AEVV] AMEX field 61)

Attribute	Description
Data Representation	b-20
Data Field	The 3-D Secure Electronic Commerce Visa, JCB, Diners Club, CAVV or American Express AEVV value in binary format for usage 2 and 3.
Values	Transaction specific

Application notes

For American Express Gateway 3-D Secure transactions, position 1 is information-only and will not be passed.

For recurring payment transactions, DE 48, subelement 43 will contain the partial shipment verbiage as follows: PARTIALbSHIPMENTbbbbbbbbbbbb or PARTIALSHIPMENT00000000000000 where b represents a space.

NOTE: This does not apply to Visa transactions.

Zeros are literal and must be included as indicated. This field contains 28 positions.

For partial shipment transactions, *DE 48*, subelement 43 will contain the cryptographic data from the original transaction.

Subelement 44 (3-D Secure Electronic Commerce Transaction Identifier [XID][Visa and American Express])

DE 48, subelement 44 (3-D Secure Electronic Commerce Transaction Identifier [XID][Visa and American Express]) contains the 3-D Secure Electronic Commerce Transaction Identifier (XID).

Table 431: Attributes

Attribute	Description
Subelement ID	44
Subelement Data Length	2 positions
Data Representation	b-20
Data Field	The Visa or American Express XID value in binary format
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 44 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 432: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Request Response/0110	CE	.	CE
Reversal Request/0400	C	.	C
Reversal Request Response/0410	CE	.	CE

Values

The Visa XID value in binary. Visa field 126.8 or the American Express XID value in binary—AMEX field 61.

Application notes

The Visa XID value is optional in authorization messages for Visa usage 3 TransStain.
Refer to *Visa Base I Technical Specifications* for the specific formats for Visa 3-D Secure Electronic Commerce Transaction Identifier (XID).

Subelement 45 (3-D Secure Electronic Commerce Transaction Response Code [Visa and American Express])

DE 48, subelement 45 (3-D Secure Electronic Commerce Transaction Response Code) is the 3-D Secure Electronic Commerce Transaction Response Code that contains the Visa Cardholder Authentication Verification Value (CAVV) or the American Express Verification Value (AEVV) results code.

Table 433: Attributes

Attribute	Description
Subelement ID	45
Subelement Data Length	2 positions
Data Representation	an-1
Data Field	Visa 3-D Secure CAVV or American Express 3-D Secure AEVV results code
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 45 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 434: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	C	.	C

Values

Refer to *Visa Dual Message Authorization System* for field 44.13 (CAVV Results Code) for a complete list of values available for this field.

Refer to *American Express Technical Specifications* for AMEX field 61, position 9, American Express Verification Value (AEVV Results Code) for a complete list of values available for this field.

Subelement 46 (Product ID [Visa Only])

DE 48, subelement 46 (Product ID) contains the Visa Product ID value (Visa Only).

Table 435: Attributes

Attribute	Description
Subelement ID	46
Subelement Data Length	2 positions
Data Representation	an-2
Data Field	Contents of positions 1–2
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 46 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 436: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	C	•	C
Reversal Request Response/0410	C	•	C

Values

Refer to *Visa Dual Message Authorization System* for a complete list of values for field 62.23 (Product ID).

Application notes

Acquirers should be prepared to receive subelement 46 in the Authorization Request Response/0110 message when one of the following occurs:

- Subelement 90 (Custom Payment Service Request [Visa]) is included in the Authorization Request/0100, which the Authorization System forwards to the Visa network.
- Subelement 90 is not included in the Authorization Request/0100 message and the Authorization System sends the message non-peer-to-peer to the Visa issuer via the Visa network.

Subelement 47 (Mastercard Payment Gateway Transaction Indicator)

DE 48, subelement 47 (Mastercard Payment Gateway Transaction Indicator) indicates that the transaction is a Mastercard Payment Gateway transaction.

Table 437: Attributes

Attribute	Description
Subelement ID	47
Subelement Data Length	2 positions
Data Representation	ans-8
Data Field	Contents of positions 1–8
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 47 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 438: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Advice/0120: acquirer-generated	.	C	C
Authorization Advice/0120: system-generated	.	C	C
Reversal Request/0400	C	.	.

Values

MC-MPG/W

Table 439: Associated edits

IF...	THEN the Dual Message Authorization System...
the account range does not participate in the MPG Authorization Blocking Service and <i>DE 48</i> , subelement 47 is present but does not contain value MC-MPG/W	sends to the acquirer an Authorization Request Response/0110 message containing: <i>DE 39 (Response Code)</i> = 30 <i>DE 44 (Additional Data)</i> = 048047

Subelement 48 (Digital Commerce Solutions Indicators)

DE 48, subelement 48 (Digital Commerce Solutions Indicators) identifies the domain, program type and optional check-in details of a digital commerce transaction.

Table 440: Attributes

Attribute	Description
Subelement ID	48
Subelement Data Length	2 positions
Data Representation	ans...73; LLVAR
Data Field	Contents of subfields 01-04
Subfields	4
Justification	N/A

Usage

Whether the usage is mandatory, conditional, optional, system-provided, or not required is as noted for each applicable message.

Table 441: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	X	C
Authorization Request Response/0110	C	X	C

Message	Org	Sys	Dst
Authorization Advice/0120: system-generated	.	X	C

Values

See subfields.

Application notes

DE 48, subelement 48 is present in Authorization Request/0100, Authorization Request Response/0110, and Authorization Advice/0120 messages for specific digital commerce transactions.

Subfield 01 (Digital Commerce Domain Identifier)

DE 48, subelement 48 (Digital Commerce Solutions Indicators), subfield 01 (Digital Commerce Domain Identifier) identifies the entity responsible for populating the digital commerce solutions indicators defined in subfield 02 (Digital Commerce Program Type), subfield 03 (Digital Commerce Check-In Entity Type and Name), and subfield 04 (Digital Commerce Check-In Method).

Table 442: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	n-1
Data Field	Contents of subfield 01
Justification	N/A

Table 443: Values

Value	Description
1	Mastercard
2	Acquirer
3	Issuer
4-9	Reserved for future use

Application notes

Mastercard populates the value of 1. Acquirers or issuers provide values 2 and 3.

Subfield 02 (Digital Commerce Program Type)

DE 48, subelement 48 (Digital Commerce Solutions Indicators), subfield 02 (Digital Commerce Program Type) identifies the specific digital commerce program.

Table 444: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	n-1
Data Field	Contents of subfield 02
Justification	N/A

Table 445: Valid values

Value	Description
1	SQR
2	Click to Pay
3	Biometrics Checkout Program
4	Next Gen POI
5-9	Reserved for future use

Application notes

Subfield 2 is present when subfield 1 is present.

When subfield 01 is 1 (Mastercard), subfield 02 must one of the listed values:

- 1 (SQR)
- 2 (Click to Pay)
- 3 (Biometric Checkout Program).

When subfield 01 is 2 (Acquirer) or 3 (Issuer), subfield 02 must be 4 (Next Gen POI).

Subfield 03 (Digital Commerce Check-In Entity Type and Name)

DE 48, subelement 48 (Digital Commerce Solutions Indicators), subfield 03 (Digital Commerce Check-In Entity Type and Name) identifies entity type and entity name. All entities eligible for this subfield must be integrated with the Mastercard Next Gen POI API.

Table 446: Attributes

Attribute	Description
Subfield ID	03
Subfield Data Length	2 positions
Data Representation	ans...15; LLVAR
Data Field	Contents of subfield 03
Justification	N/A

Values

Subfield 03 is an open text field of up to 15 characters.

Application notes

Subfield 03 is:

- Only present when subfield 01 and subfield 02 are present.
- Required for Next Gen Point-of-Interaction transactions.
- Optional for all other transaction types.

Subfield 04 (Digital Commerce Check-In Method)

DE 48, subelement 48 (Digital Commerce Solutions Indicators), subfield 04 (Digital Commerce Check-In Method) identifies the methods a check-in entity applies to validate cardholder presence. This field may also include a reference number for tracking and reporting purpose.

Table 447: Attributes

Attribute	Description
Subfield ID	04
Subfield Data Length	2 positions
Data Representation	ans...40; LLVAR
Data Field	Contents of subfield 04
Justification	N/A

Values

Subfield 04 is an open text field of up to 40 characters.

Application notes

Subfield 04 is:

- Only present when subfield 01 and subfield 02 are present.
- Required for Next Gen Point-of-Interaction transactions.
- Optional for all other transaction types.

Subelement 49 (Time Validation Information)

DE 48, subelement 49 (Time Validation Information) contains time data used to calculate time validation and the results of validation when performed.

Table 448: Attributes

Attribute	Description
Subelement ID	49
Subelement Data Length	15 positions
Data Representation	n-15
Subfields	Subfield 1: Time Value Subfield 2: Time Discrepancy Value Subfield 3: Time Discrepancy Indicator

Usage

Following is the usage of subelement 49 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 449: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	•	X	C
Authorization Advice/0120: system-generated	•	X	C

Subfield 1 (Time Value)

DE 48, subelement 49, subfield 1 (Time Value) contains the time data derived from the acquirer transaction to be used in the time validation.

Table 450: Attributes

Attribute	Description
Data Representation	n-8
Data Field	Contents of subfield 1
Justification	Right-justified, leading zeros

Subfield 2 (Time Discrepancy Value)

DE 48, subelement 49, subfield 2 (Time Discrepancy Value) contains a positive value representing the differential in minutes between the transaction time data and service-calculated time.

Table 451: Attributes

Attribute	Description
Data Representation	n-5
Data Field	Contents of subfield 2
Justification	Right-justified, leading zeros

Subfield 3 (Time Discrepancy Indicator)

DE 48, subelement 49, subfield 3 (Time Discrepancy Indicator) contains a value that indicates if the time discrepancy value is below, above, or within the minimum and maximum values for the time validation window or that indicates time validation was not performed.

Table 452: Attributes

Attribute	Description
Data Representation	n-2
Data Field	Contents of subfield 3
Justification	Left-justified

Table 453: Values

Value	Description
01	Positive value within time validation window
02	Positive value outside time validation window
03	Negative value within time validation window
04	Negative value outside time validation window
05	Unknown (time validation not performed)

Subelement 50 (Embedded Interchange Data)

DE 48, subelement 50 provides data in support of commercial transactions using negotiated interchange rates between acquirers and issuers.

Issuers receive this subelement if they have opted in for the Embedded Flexible Interchange program and if a participating acquirer has included the subelement in the given transaction.

Table 454: Attributes

Attribute	Description
Subelement ID	50
Subelement Data Length	2 positions
Data Representation	ans...99; LLVAR
Data Field	Contents of subfields 01-06
Subfields	6
Justification	See subfields

Following is the usage of *DE 48*, subelement 50 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 455: Usage

	Org	Sys	Dst
Authorization Request/0100 message	C	•	C
Authorization Advice/0120 message: acquirer-generated	C	•	C

Authorization Advice/0120 message: system-generated	•	X	C
Reversal Request/0400 message	C	•	C
Reversal Advice/0420 message	•	X	C

Application notes

DE 48, subelement 50, and its subfields are set up in tag, length, value format.

Registered acquirers provide this subelement to specify the negotiated interchange rate, amount, or both which applies to a commercial transaction.

Acquirers must use one of the valid combination of subfields within this subelement as listed in the Valid subfield combinations table.

Table 456: Valid subfield combinations

Present	Not present
Interchange percentage only	
Subfield 01 (Interchange Indicator), subfield 02 (Interchange Percentage), subfield 03 (Unit Fee Amount), subfield 04 (Flat Interchange Amount)	
- subfield 01 - subfield 02	- subfield 03 - subfield 04
Interchange percentage with unit fee amount	
- subfield 01 - subfield 02 - subfield 03	subfield 04
Flat interchange only	
- subfield 01 - subfield 04	- subfield 02 - subfield 03

The Dual Message Authorization System rejects transactions that do not conform to one of the valid combinations. Subfield 05 (Payment Party) and subfield 06 (Supplemental Interchange Data) are optional subfields and can be included with any of the valid combinations.

Subfields with numeric attributes can take a value of all zeros. For example, an acquirer can use the second combination where the Interchange Percentage is present (and non-zero) and the Unit Fee Amount is all zeros. However, an acquirer cannot include Interchange Percentage, Unit Fee Amount, and Flat Interchange Amount in the same transaction even if one or more of those subfields contains all zeros.

Refer to the subfields contained within this subelement to obtain the supporting programs, services, or other specifics about the transaction.

Subfield 01 (Interchange Indicator)

DE 48, subelement 50, subfield 01 signals that a transaction uses a negotiated interchange rate or amount per the EFI program.

Table 457: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	a-1
Data Field	Contents of subfield 01
Justification	N/A

Table 458: Values

Value	Description
Y	Transaction includes negotiated interchange rate or amount

Application note

This subfield is mandatory for all EFI program transactions.

Subfield 02 (Interchange Percentage)

DE 48, subelement 50, subfield 02 provides the negotiated interchange percentage rate per the EFI program.

Table 459: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	n-6
Data Field	Contents of subfield 02

Attribute	Description
Justification	Right with leading zeros

Values

The interchange percentage carries a percentage value with an implied three-position decimal.

The interchange percentage can take a value of 0.000%, indicating that the negotiated rate is zero. The maximum value for the interchange percentage is 100.000%.

For example, an Interchange Percentage of 1.234% is represented with DE 48, subelement 50, subfield 02, value 001234.

Application notes

Interchange for an EFI program transaction is calculated based on *DE 4 (Amount, Transaction)*, *DE 48 subelement 50 subfield 02 (Interchange Percentage)*, and *DE 48 subelement 50 subfield 03 (Unit Fee Amount)*, if present.

This subfield is conditional for EFI program transactions. It cannot be included in transactions that also contain subfield 04 (Flat Interchange Amount).

Subfield 03 (Unit Fee Amount)

DE 48, subelement 50, subfield 03 provides the negotiated unit fee amount per the EFI program.

Table 460: Attributes

Attribute	Description
Subfield ID	03
Subfield Data Length	2 positions
Data Representation	n-12
Data Field	Contents of subfield 03
Justification	Right with leading zeros

Values

The Unit Fee Amount is expressed in the currency defined by *DE 49 (Currency Code, Transaction)*.

Where a minor unit of currency applies, the Unit Fee Amount is expressed in the minor unit of currency without a decimal separator, such as, value 000000000100 represents USD 1.00.

Application notes

Interchange for an EFI program transaction is calculated based on *DE 4 (Amount, Transaction)*, *DE 48 subelement 50 subfield 02 (Interchange Percentage)*, and *DE 48 subelement 50 subfield 03 (Unit Fee Amount)*.

This subfield is conditional for EFI program transactions. It can only be used in transactions that contain subfield 02. It cannot be included in transactions that also contain subfield 04 (Flat Interchange Amount).

Subfield 04 (Flat Interchange Amount)

DE 48, subelement 50, subfield 04 provides the negotiated flat interchange amount per the EFI program.

Table 461: Attributes

Attribute	Description
Subfield ID	04
Subfield Data Length	2 positions
Data Representation	n-12
Data Field	Contents of subfield 04
Justification	Right with leading zeros

Values

The Flat Interchange Amount is expressed in the currency defined by *DE 49 (Currency Code, Transaction)*.

Where a minor unit of currency applies, the Flat Interchange Amount is expressed in the minor unit of currency without a decimal separator, such as, value 000000000100 represents USD 1.00.

A Flat Interchange Amount of zero indicates that the negotiated interchange is zero.

Application notes

Interchange for an EFI program transaction is determined by subfield 04.

This subfield is conditional for EFI program transactions. It cannot be included in transactions that contain either subfield 02 or subfield 02 and subfield 03.

Subfield 05 (Payment Party)

DE 48, subelement 50, subfield 05 specifies whether the acquirer or issuer pays interchange on this transaction per the EFI program.

Table 462: Attributes

Attribute	Description
Subfield ID	05
Subfield Data Length	2 positions
Data Representation	a-1
Data Field	Contents of subfield 05
Justification	N/A

Values**Table 463: Values**

Value	Description
A	Acquirer pays interchange
I	Issuer pays interchange

Application notes

This subfield is optional for EFI program transactions and must be used only with EFI program transactions.

Subfield 06 (Supplemental Interchange Data)

DE 48, subelement 50, subfield 06 is a free-text field which acquirers can use to send proprietary data to issuers as part of EFI program transactions.

Table 464: Attributes

Attribute	Description
Subfield ID	06
Subfield Data Length	2 positions
Data Representation	ans...59; LLVAR
Data Field	Contents of subfield 06

Attribute	Description
Justification	N/A

Values

Proprietary free-text data.

Application notes

This subfield is optional for EFI program transactions and must be used only with EFI program transactions.

Subelement 51 (Merchant On-behalf Services)

DE 48, subelement 51 (Merchant On-behalf [OB] Services) notifies the acquirer of the On-behalf Service performed on the transaction, and the results. *DE 48* will support multiple occurrences of subelement 51 up to the maximum services for a transaction.

Table 465: Attributes

Attribute	Description
Subelement ID	51
Subelement Data Length	2 positions
Data Representation	ans...99; LLVAR The "LL" length field of LLVAR must be an integral multiple of 4, not to exceed 96.
Data Field	Contents of subfields 1–3
Subfields	3
Justification	N/A

Usage

Following is the usage of subelement 51 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 466: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	.	X	C

Values

See subfields.

Subfield 1 (Merchant On-behalf [OB] Service)

DE 48, subelement 51, subfield 1 (Merchant On-behalf [OB] Service) indicates the service performed on the transaction.

Table 467: Attributes

Attribute	Description
Data Representation	an-2
Data Field	Contents of positions 1–2
Justification	Left-justified

Values

See [Subelement 51 \(Valid Subfield 1 and Subfield 2 Value Combinations\)](#).

Subfield 2 (Merchant On-behalf [OB] Result 1)

DE 48, subelement 51, subfield 2 (Merchant On-behalf [OB] Result 1) indicates the results of the service processing.

Table 468: Attributes

Attribute	Description
Data Representation	an-1
Data Field	Contents of position 3
Justification	N/A

Values

See [Subelement 51 \(Valid Subfield 1 and Subfield 2 Value Combinations\)](#).

Subelement 51 (Valid Subfield 1 and Subfield 2 Value Combinations)

DE 48, subelement 51 (Valid Subfield 1 and Subfield 2 Value Combinations) contains the valid value combination for DE 48, subelement 51, subfield 1 (Merchant On-behalf [OB] Service) and

subfield 2 (Merchant On-behalf [OB] Result 1). The contents of subfield 2 depend on the contents of subfield 1 as described here.

Table 469: Attributes

Attribute	Subfield 1 (Merchant On-behalf [OB] Service)	Subfield 2 (Merchant On-behalf [OB] Result 1)
Data Representation	an-2	an-1
Data Field	Contents of positions 1–2	Contents of position 3
Justification	Left-justified	N/A

Following are the *DE 48*, subelement 51, valid subfield 1 and subfield 2 value combinations.

Table 470: Values

<i>DE 48</i> , Subelement 51, Subfield 1 (Merchant On-behalf [OB] Service) Values (an-2)		<i>DE 48</i> , Subelement 51, Subfield 2 (Merchant On-behalf [OB] Result 1) Values (an-1)	
90	EMS Real-time Fraud Scoring Service for Merchants	C	EMS Real-time Fraud Scoring Service for Merchants was performed successfully.
		I	Invalid; transaction does not qualify for the EMS Real-time Fraud Scoring Service for Merchants due to one of the following: • Transaction is card present, or • Card issuance is outside of the valid issuing region
		U	EMS Real-time Fraud Scoring Service for Merchants was not performed successfully.

Application notes

If the value of subfield 1 and subfield 2 is 90C, then *DE 48*, SE 55, subfield 1 and subfield 2 will be populated.

Subfield 2 will have values that identify the results from the EMS Real-time Fraud Scoring Service for Merchants processing result.

Subfield 3 (Additional Information)

DE 48, subelement 51, subfield 3 (Additional Information) indicates if the acquirer in the transaction is enrolled with the service and if the service was requested.

Table 471: Attributes

Attribute	Description
Data Representation	an-1
Data Field	Contents of position 4
Justification	N/A

Table 472: Values

Value	Description
N	Not qualified for EMS Real-time Fraud Scoring Service for Merchants
blank	No value present

Application notes

Subfield 3 will have an indicator of N if the score was not delivered to the acquirer but the acquirer had requested the transaction be scored. This is based on service participation.

Subelement 52 (Transaction Integrity Class)

DE 48, subelement 52 (Transaction Integrity Class) contains the Mastercard-provided Transaction Integrity Classification for Point of Sale (POS) Purchase and Purchase with Cash Back transactions initiated on the Dual Message Authorization System.

Table 473: Attributes

Attribute	Description
Subelement ID	52
Subelement Data Length	02 positions
Data Representation	an-2
Data Field	Contents of positions 1–2

Attribute	Description
Subfields	N/A

Usage

Following is the usage of subelement 52 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 474: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	•	X	C
Authorization Response/0110	OE	X	C
Authorization Advice/0120: system-generated	•	X	C

Table 475: Values

Classification	Description	Value
Card and Cardholder Present	EMV/Token in a Secure, Trusted Environment	A1
Card and Cardholder Present	EMV/Chip Equivalent	B1
Card and Cardholder Present	Mag Stripe	C1
Card and Cardholder Present	Key Entered	E1
Card and Cardholder Present	Unclassified	U0
Card and/or Cardholder Not Present	Digital Transactions	A2
Card and/or Cardholder Not Present	Authenticated Checkout	B2
Card and/or Cardholder Not Present	Transaction Validation	C2
Card and/or Cardholder Not Present	Enhanced Data	D2
Card and/or Cardholder Not Present	Generic Messaging	E2
Card and/or Cardholder Not Present	Unclassified	U0

Application notes

The Transaction Integrity Class may optionally be provided in the designated clearing messages when provided by Mastercard via the Dual Message Authorization System for Point of Sale

(POS) Purchase and Purchase with Cash Back transactions destined for all issuers in the U.S. region of Mastercard® credit and Debit Mastercard® cards.

Issuers in the U.S. region must support receiving *DE 48 (Additional Data)*, subelement 52 (Transaction Integrity Class) in Authorization Request/0100 messages and optionally provide it in the Authorization Request Response/0110 messages. Issuers in the U.S. region must also support receiving *DE 48 (Additional Data)*, new subelement 52 (Transaction Integrity Class) in Authorization Advice/0120: system-generated messages.

The Transaction Integrity Class is not sent to non-U.S. region issuers in authorization or clearing messages.

Subelement 53 (E-ID Request Code)

DE 48, subelement 53 (E-ID Request Code) enables the issuer to identify that additional cardholder information is being requested by the acquirer. This request code is sent by an acquirer to an issuer in an Account Status Inquiry (ASI) message.

Table 476: Attributes

Attribute	Description
Subelement ID	53
Subelement Length	53 positions
Data Representation	ans...99; LLVAR
Data Field	Contents of subfield 1
Subfields	1

Usage

Following is the usage of subelement 53 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 477: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C

Values

See subfields.

Application notes

Acquirers that support E-ID will include this subelement in an ASI, queuing the issuer to perform further processing outside of the Mastercard Network. This subelement is valid for Europe region activity only.

Subelement 53 (and associated subfield 1) is effective as of 12 June 2018.

For card-not-present transactions, the acquirer should not send subelement 53.

Subfield 1 (E-ID Request Value)

DE 48, subelement 53, subfield 1 (E-ID Request Value) informs the issuer of the proposed action on the cardholder account.

Table 478: Attributes

Attribute	Description
Data Representation	ans-02
Data Field	Contents of positions 1–2
Justification	N/A

Table 479: Values

Value	Description
01	Informs the issuer to pull cardholder personal data for E-ID/Health ID Verification Purposes.
02	Informs the issuer to verify and send the age of the cardholder.
03–99	Reserved for Future Use

Application notes

Issuers must:

- Validate that a valid value is present in the message; and
- Send this cardholder information through a network separate from the Mastercard Network.

Subelement 53 and subfield 1 are effective as of 12 June 2018.

Subelement 55 (Merchant Fraud Scoring Data)

DE 48, subelement 55 (Merchant Fraud Scoring Data) indicates the fraud score on a fraud scoring service transaction when the acquirer requested the transaction to be scored.

Table 480: Attributes

Attribute	Description
Subelement ID	55
Subelement Length	2 positions
Data Representation	an...32; LLVAR
Data Field	Contents of subfields 1–5
Subfields	5
Justification	See subfields

Usage

Following is the usage of subelement 55 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 481: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	.	X	C

Values

Contents of subfield 1–5.

Application notes

The Dual Message Authorization System inserts this subelement when EMS Real-time Fraud Scoring Service for Merchants is performed on the transaction.

Mastercard will require all issuing and acquiring processors to code for, support, and integrate into their systems the data elements, subelements, and values associated with this item.

For additional information about the requirement, refer to the *Global Safety and Security Standards Roadmap*.

NOTE: Mastercard does not require acquirers or issuers to enroll in these optional products and services, but they must code their systems to support these fields to leverage such products and services in a timely manner in the event of a security issue.

Subfield 1 (Merchant Fraud Score/ASI Probability Indicator)

DE 48, subelement 55, subfield 1 (Merchant Fraud Score/ASI Probability Indicator) indicates the transaction risk score or the ASI probability indicator.

Table 482: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	an-3
Data Field	Contents of positions 1–3
Justification	N/A

Values

The EMS Real-time Fraud Scoring Service for Merchants provides the risk score of 001–998, where 001 indicates the least likely fraudulent transaction and 998 indicates the most likely fraudulent transaction. For ASI transactions, the subfield provides a three-digit number from 111 to 999 in response to an ASI request submitted with a future transaction amount value.

Application notes

Acquirers submitting Visa-branded transactions receive the merchant fraud scoring data from Visa in the response messages.

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

Subfield 2 (Reason Code)

DE 48, subelement 55, subfield 2 (Reason Code) contains the reason code.

Table 483: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	an-2
Data Field	Contents of positions 1–2
Justification	N/A

Values

The EMS Real-time Fraud Scoring Service for Merchants provides the score reason code, an alphanumeric code identifying the data used to derive the fraud score. For ASI transactions, the reason code is an alphanumeric code providing additional detail about the value provided for the ASI Probability Indicator. For ASI, the possible values are:

- 00: Successful
- 01: Not Enough Data (Dormant or New Card)

Application notes

Participating acquirers may contact the Risk Solutions Team for a list of the specific score reason codes that apply to their institution.

Subfield 3 (Reserved for Future Use)

DE 48, subelement 55, subfield 3 (Reserved for Future Use) is reserved for future use.

Table 484: Attributes

Attribute	Description
Subfield ID	03
Subfield Data Length	2 positions
Data Representation	an-3
Data Field	Contents of positions 1–3
Justification	N/A

Values

Reserved for Future Use.

Subfield 4 (Reserved for Future Use)

DE 48, subelement 55, subfield 4 (Reserved for Future Use) is reserved for future use.

Table 485: Attributes

Attribute	Description
Subfield ID	04
Subfield Data Length	2 positions
Data Representation	an-2
Data Field	Contents of positions 1–2

Attribute	Description
Justification	N/A

Values

Reserved for Future Use.

Subfield 5 (Reserved for Future Use)

DE 48, subelement 55, subfield 5 (Reserved for Future Use) is reserved for future use.

Table 486: Attributes

Attribute	Description
Subfield ID	05
Subfield Data Length	2 positions
Data Representation	an-2
Data Field	Contents of positions 1–2
Justification	N/A

Values

Reserved for Future Use.

Subelement 56 (Security Services Additional Data for Issuers)

DE 48, subelement 56 (Security Services Additional Data for Issuers) supports Mastercard embedded security services for issuers. Issuers can receive up to 16 instances of subelement 56. Each instance has the same format for each service. Issuers must code to receive the 16 instances that the service supports.

Table 487: Attributes

Attribute	Description
Subelement ID	56
Subelement Data Length	2 positions
Data Representation	an...99; LLVAR The "LL" length field of LLVAR must be an integral multiple of 6, not to exceed 96.
Data Field	Contents of subfields

Attribute	Description
Subfields	2
Justification	N/A

Usage

Following is the usage of subelement 56 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 488: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	•	X	C
Authorization Advice/0120: system-generated	•	C	C

Values

See subfields.

Application notes

Mastercard will require all issuing and acquiring processors to code for, support, and integrate into their systems the data elements, subelements, and values associated with this item.

For additional information about the requirement please refer to the *Global Safety and Security Standards Roadmap*.

NOTE: Mastercard does not require acquirers or issuers to enroll in these optional products and services, but they must code their systems to support these fields to leverage such products and services in a timely manner in the event of a security issue.

Subfield 1 (Security Services Indicator)

DE 48, subelement 56, subfield 1 (Security Services Indicator) provides the security services indicator for issuers.

Table 489: Attributes

Attribute	Description
Data Representation	an-3
Data Field	Contents of positions 1–3

Attribute	Description
Justification	N/A

Values

A three-character code identifying the real-time monitoring service used.

Subfield 2 (Security Services Data)

DE 48, subelement 56, subfield 2 (Security Services Data) contains additional data supporting the issuer fraud score.

Table 490: Attributes

Attribute	Description
Data Representation	an-3
Data Field	Contents of positions 4–6
Justification	N/A

Values

A three-character value will be populated.

Subelement 56 (Valid Subfield 1 and Subfield 2 Value Combinations)

DE 48, subelement 56 (Valid Subfield 1 and Subfield 2 Value Combinations) following is the valid DE 48, subelement 56, subfield 1 (Security Services Indicator) and subfield 2 (Security Services Data) value combinations.

Service Data Content for Decision Intelligence Service: Digital Transaction Insights Feature

Following is the Service Data Content for the Digital Transaction Insights feature of the Decision Intelligence service.

Table 491: Service data content

		Subfield 2 (an-3)																											
Subfield 1 (an-3)	Position 1	Position 2	Position 3																										
Security Services Indicator	Risk Level	Reason Code 1	Reason Code 2																										
AIQ: Digital Transaction Insights	0–9 where higher values indicate higher degree of risk.	The following codes only apply to the Reason Code 1 field. Contact your Mastercard representative for more information about the Reason Code 2 field. • Position 2 = a Mastercard-determined reason code • Position 3 = a Merchant Trust Assessment (merchant's level of trust on EMV 3DS transaction) NOTE: This information is subject to change. <table><tr><th>Reason Code</th><th>Description</th></tr><tr><td>A</td><td>Risk Event - Suspicious Account Activity</td></tr><tr><td>B</td><td>Risk Event - Unknown Device/Account Relationship</td></tr><tr><td>C</td><td>Risk Event - Device or Profile information associated with fraud</td></tr><tr><td>D</td><td>Risk Event - Recent High Risk change to Device or Profile information</td></tr><tr><td>E</td><td>Risk Event - Recent change to Device or Profile Information</td></tr><tr><td>F</td><td>Risk Event - PAN associated with fraud event</td></tr><tr><td>G</td><td>New Account or Insufficient Data</td></tr><tr><td>H</td><td>Merchant/Acquirer: Merchant (fraud) risk high (assessed by Mastercard)</td></tr><tr><td>I</td><td>Merchant/Acquirer: Merchant (fraud) risk low (assessed by Mastercard)</td></tr><tr><td>J</td><td>Environment: Good/Known IP</td></tr><tr><td>K</td><td>Cardholder: Billing address - prior history established</td></tr><tr><td>L</td><td>Cardholder: Email address - prior history established</td></tr></table>		Reason Code	Description	A	Risk Event - Suspicious Account Activity	B	Risk Event - Unknown Device/Account Relationship	C	Risk Event - Device or Profile information associated with fraud	D	Risk Event - Recent High Risk change to Device or Profile information	E	Risk Event - Recent change to Device or Profile Information	F	Risk Event - PAN associated with fraud event	G	New Account or Insufficient Data	H	Merchant/Acquirer: Merchant (fraud) risk high (assessed by Mastercard)	I	Merchant/Acquirer: Merchant (fraud) risk low (assessed by Mastercard)	J	Environment: Good/Known IP	K	Cardholder: Billing address - prior history established	L	Cardholder: Email address - prior history established
Reason Code	Description																												
A	Risk Event - Suspicious Account Activity																												
B	Risk Event - Unknown Device/Account Relationship																												
C	Risk Event - Device or Profile information associated with fraud																												
D	Risk Event - Recent High Risk change to Device or Profile information																												
E	Risk Event - Recent change to Device or Profile Information																												
F	Risk Event - PAN associated with fraud event																												
G	New Account or Insufficient Data																												
H	Merchant/Acquirer: Merchant (fraud) risk high (assessed by Mastercard)																												
I	Merchant/Acquirer: Merchant (fraud) risk low (assessed by Mastercard)																												
J	Environment: Good/Known IP																												
K	Cardholder: Billing address - prior history established																												
L	Cardholder: Email address - prior history established																												

Subfield 1 (an-3)	Subfield 2 (an-3)		
	Position 1	Position 2	Position 3
Security Services Indicator	Risk Level	Reason Code 1	Reason Code 2
		M	Cardholder: Phone Number - prior history established
		N	Cardholder: Shipping address - prior history established
		O	Cardholder: Card number (PAN) behavior established high trust in the current transaction
		P	Environment: Device known
		Q	Environment: Account established on Device
		R	Environment: Session - Trusted/normal/innocent session (no man in the middle attack/no bot, not suspicious account activity)
		S	More than one Cardholder category established
		T	More than one Merchant/Acquirer category established
		U	More than one Environment category established
		V	Co-occurring established link between cardholder and Merchant/Acquirer
		W	Co-occurring established link between cardholder and Environment
		X	Co-occurring established link between Merchant/Acquirer and Environment
		Y	All three categories established
		Z	Most Trusted (future use)

NOTE: The above codes depict an ordering from negative to positive. This list may be further refined. Reason codes A–H reflect riskier, or less information while reason codes I–Z reflect positive information.

NOTE: Identity Check Insights requests can also be submitted with Mastercard Digital Enablement Service (MDES) tokens.

Service Data Content for Decision Intelligence Service: Authorization IQ Feature

Following is the Service Data Content for the Authorization IQ feature of the Decision Intelligence service. Customers that support Decision Intelligence will receive AQV (GDV-based Spend Ranking), AQF (Frequency-based Spend Ranking), and AQS (Segment Qualifier) on all transactions.

NOTE: Positions 2 and 3 use the same reason code definitions.

Table 492: Service data content

Subfield 1 (an-3)	Subfield 2 (an-3)		
	Position 1	Position 2	Position 3
Spending Dimension Insights	Overall Account Spending Insights	Channel Spending Insights	Account Transaction Insights
AQV: GDV-based Spend Ranking Or AQF: Frequency-based Spend Ranking	New (card activity on the network less than 60 days old) 0	New (card activity on the network less than 60 days old) 0 Or None (No network activity in the last 12 months or Dormant Card Number)	0
	High = 1		High = 1
	Med-1 = 4		Med-1 = 4
	Med-2 = 5		Med-2 = 5
	Med-3 = 6		Med-3 = 6
	Low = 9		Low = 9
AQS: Segment Qualifier	Refer to the Segment Definitions table that follows.		

Service Data Content for Decision Intelligence Service: Authentication Risk Analysis

Following is the Service Data Content for the Authentication Risk Analysis (ARA) feature of the Decision Intelligence service.

Table 493: Service data content

Subfield 2 (an-3)				
Subfield 1 (an-3)	Position 1	Position 2	Position 3	Description
Security Services Indicator	Reason code			
ARA: Authentication Risk Analysis	A-Z assessing risk of cardholder	A-Z assessing risk of merchant	A-Z assessing risk of environment	
ARA	A	A	A	Suspicious Account Activity, Merchant (fraud) risk low (assessed by Mastercard)
	A	A	B	Suspicious Account Activity, Merchant (fraud) risk low (assessed by Mastercard), Good/Known IP
	A	A	C	Suspicious Account Activity, Merchant (fraud) risk low (assessed by Mastercard), Recent High Risk Change to Device or Profile Information
	A	A	D	Suspicious Account Activity, Merchant (fraud) risk low (assessed by Mastercard), Recent change to Device or Profile Information
	A	A	E	Suspicious Account Activity, Merchant (fraud) risk low (assessed by Mastercard), Unknown Device/Account Relationship
	A	A	F	Cardholder Billing address prior history established, Merchant (fraud) risk low (assessed by Mastercard), Unknown Device/Account Relationship
	A	A	G	Suspicious Account Activity, Merchant (fraud) risk low (assessed by Mastercard), Environment: Session - Trusted/normal/innocent session (no man in the middle attack/no bot, not suspicious account activity)

Subfield 2 (an-3)				
Subfield 1 (an-3)	Position 1	Position 2	Position 3	Description
Security Services Indicator	Reason code			
	A	A	H	PAN associated with fraud event, Merchant (fraud) risk low (assessed by Mastercard), Unknown Device/Account Relationship
	A	A	I	New Account or Insufficient Data, Merchant (fraud) risk low (assessed by Mastercard), Unknown Device/Account Relationship
	A	A	J	Card number (PAN) behavior established high trust in the current transaction, Merchant (fraud) risk low (assessed by Mastercard), Unknown Device/Account Relationship
	A	A	K	New Account or Insufficient Data, Merchant (fraud) risk low (assessed by Mastercard), Environment: Session - Trusted/normal/innocent session (no man in the middle attack/no bot, not suspicious account activity)
	A	A	L	New Account or Insufficient Data, Merchant (fraud) risk low (assessed by Mastercard)
	A	A	M	New Account or Insufficient Data, Merchant (fraud) risk low (assessed by Mastercard), Recent change to Device or Profile Information
	A	A	N	Cardholder Shipping address prior history established, Merchant (fraud) risk low (assessed by Mastercard), Recent High Risk Change to Device or Profile Information
	A	A	O	New Account or Insufficient Data, Merchant (fraud) risk low (assessed by Mastercard), Recent High Risk Change to Device or Profile Information

Subfield 2 (an-3)				
Subfield 1 (an-3)	Position 1	Position 2	Position 3	Description
Security Services Indicator	Reason code			
	A	A	P	Cardholder Phone number prior history established, Merchant (fraud) risk low (assessed by Mastercard), Unknown Device/Account Relationship
	A	A	Q	Cardholder Card number (PAN) behavior established high trust in the current transaction, Merchant (fraud) risk low (assessed by Mastercard), Recent High Risk Change to Device or Profile Information
	A	A	R	Cardholder Phone number prior history established, Merchant (fraud) risk low (assessed by Mastercard), Recent High Risk Change to Device or Profile Information
	A	A	S	New Account or Insufficient Data, Merchant (fraud) risk high (assessed by Mastercard)
	A	A	T	Cardholder Billing address prior history established, Merchant (fraud) risk low (assessed by Mastercard), Recent High Risk Change to Device or Profile Information
	A	A	U	Cardholder Phone number prior history established, Merchant (fraud) risk low (assessed by Mastercard), Good/Known IP
	A	A	V	Cardholder Billing address prior history established, Merchant (fraud) risk low (assessed by Mastercard), Recent change to Device or Profile Information
	A	A	W	Cardholder Billing address prior history established, Merchant (fraud) risk low (assessed by Mastercard), Good/Known IP

Subfield 2 (an-3)				
Subfield 1 (an-3)	Position 1	Position 2	Position 3	Description
Security Services Indicator	Reason code			
	A	A	X	Cardholder Billing address prior history established, Merchant (fraud) risk low (assessed by Mastercard)
	A	A	Y	PAN associated with fraud event, Merchant (fraud) risk low (assessed by Mastercard), Good/Known IP
	A	A	Z	PAN associated with fraud event, Merchant (fraud) risk low (assessed by Mastercard), Environment: Session - Trusted/normal/innocent session (no man in the middle attack/no bot, not suspicious account activity)
	A	B	A	PAN associated with fraud event, Merchant (fraud) risk low (assessed by Mastercard), Recent change to Device or Profile Information
	A	B	B	Limited information and refer to DTI reason code
	A	B	C	Card number (PAN) behavior established high trust in the current transaction, Merchant/Acquirer: Merchant (fraud) risk low (assessed by Mastercard), Recent change to Device or Profile Information
	A	B	D	Cardholder Billing address prior history established, Merchant/Acquirer: Merchant (fraud) risk low (assessed by Mastercard), Environment: Session - Trusted/normal/innocent session (no man in the middle attack/no bot, not suspicious account activity)
	A	B	E	Cardholder Shipping address prior history established, Merchant/Acquirer: Merchant (fraud) risk low (assessed by Mastercard), Environment: Session - Trusted/normal/innocent session (no man in the middle attack/no bot, not suspicious account activity)

Subfield 2 (an-3)				
Subfield 1 (an-3)	Position 1	Position 2	Position 3	Description
Security Services Indicator	Reason code			
	A	B	F	Cardholder Phone number prior history established, Merchant/Acquirer: Merchant (fraud) risk low (assessed by Mastercard), Environment: Session - Trusted/normal/innocent session (no man in the middle attack/no bot, not suspicious account activity)
	A	B	G	Cardholder Phone number prior history established, Merchant/Acquirer: Merchant (fraud) risk low (assessed by Mastercard)
	A	B	H	Cardholder Shipping address prior history established, Merchant/Acquirer: Merchant (fraud) risk low (assessed by Mastercard), Recent change to Device or Profile Information
	A	B	I	Cardholder Shipping address prior history established, Merchant/Acquirer: Merchant (fraud) risk low (assessed by Mastercard)
	A	B	J	Cardholder Shipping address prior history established, Merchant/Acquirer: Merchant (fraud) risk low (assessed by Mastercard), Good/Known IP
	A	B	K	Cardholder Phone number prior history established, Merchant/Acquirer: Merchant (fraud) risk low (assessed by Mastercard), Recent change to Device or Profile Information

NOTE: The above codes depict the current list of reason codes and an appropriate ordering from negative to positive. This list may be further refined in future.

Segment Definitions for Decision Intelligence Service: Authorization IQ Feature

The Authorization IQ feature of the Decision Intelligence service supports the following series of segments under which each transaction is qualified for dimension analysis.

NOTE: New segments may be added to the Authorization IQ feature of the Decision Intelligence service at any time.

Table 494: Segment Definitions

Segment	Segment Qualifier ID
Channel: Card Present: Domestic Segments	000: Card activity on network less than 60 days old
Travel	001
Retail	002
Gaming	003
Gambling	004
Education/Healthcare	005
Cash (for example: ATM)	006
Fuel	007
Utilities	008
Leisure and Entertainment	009
Digital Goods	010
Professional Services	011
Reserved for Future Use	012–198
Other Card Present: Domestic	199
Channel: Card Present: Cross-Border Segments	200: Card activity on network less than 60 days old
Travel	201
Retail	202
Gaming	203
Gambling	204
Education/Healthcare	205
Cash (for example: ATM)	206
Fuel	207
Utilities	208
Leisure and Entertainment	209
Digital Goods	210
Professional Services	211

Data element definitions
Subelement 56 (Valid Subfield 1 and Subfield 2 Value Combinations)

Segment	Segment Qualifier ID
Reserved for Future Use	212–398
Other Card Present: Cross-Border	399
Channel: Card Not Present: Domestic Segments	400: Card activity on network less than 60 days old
Travel	401
Retail	402
Gaming	403
Gambling	404
Education/Healthcare	405
Cash (for example: ATM)	406
Utilities	407
Leisure and Entertainment	408
Digital Goods	409
Professional Services	410
Reserved for Future Use	411–598
Other Card Not Present: Domestic	599
Channel: Card Not Present: Cross-Border Segments	600: Card activity on network less than 60 days old
Travel	601
Retail	602
Gaming	603
Gambling	604
Education/Healthcare	605
Cash (for example: ATM)	606
Utilities	607
Leisure and Entertainment	608
Digital Goods	609
Professional Services	610
Reserved for Future Use	611–798
Other Card Not Present: Cross-Border	799
Reserved for Future Use	800–899
Dormant Card Number	900: Previously active card with no activity on the network in the last 120 days

Segment	Segment Qualifier ID
Reserved for Future Use	901–999

Service Data Content for Consumer Controls

Following is the Service Data Content for Consumer Controls.

Table 495: Service Data Content

Subfield 1 (an-3)	Subfield 2 (an-3)	
Service Code	Reason Code	Description
INC: Consumer Controls	AAL	Alert All Transactions
	ABD	Alert Budget
	ACB	Alert Cross-border
	ACH	Alert Transaction Channel
	AFL	Alert Filter
	AGE	Alert Geo Location
	AMC	Alert Merchant Category Code
	ATA	Alert Transaction Amount
	DBD	Decline Budget
	DCB	Decline Cross-border
	DCD	Decline Card Disabled
	DCH	Decline Transaction Channel
	DFL	Decline Filter
	DGE	Decline Geo Location
	DMC	Decline Merchant Category Code
	DTA	Decline Transaction Amount
	NAT	No Action Taken

Reason codes for Identity Insights

Identity insight reason codes offer a more specific view about the source of the Mastercard risk analysis, with values from AA–ZZ representing these reasons, with AA as the highest risk to ZZ as the lowest risk.

Table 496: Reason codes

Subfield 1 (an-3)	Subfield 2 (an-3)		
	Position 1	Position 2	Position 3
Security Services Indicator	Risk Level	Reason Code 1	Reason Code 2
DII: Digital Identity Insights	0–9 where higher values indicate higher degree of risk.	The following codes apply to both Reason Code fields 1 and 2, where: - Position 2 = determines the severity of the reason code. A is high, J is medium, and R is low. - Position 3 = determines the physical attribute the risk is associate with. - Use positions 2 and 3 in combination for a holistic view of risk.	

This information is subject to change. AQ-IZ, JG-QZ, and RQ-ZZ are currently not defined and are reserved for future use.

Table 497: Reason codes

Reason code	Description
AA	Consumer identity is suspicious
AB	Increased risk associated with email
AC	Increased risk associated with email and IP
AD	Increased risk associated with email and address
AE	Increased risk associated with email and phone
AF	Increased risk associated with email and name
AG	Increased risk associated with IP
AH	Increased risk associated with IP and phone
AI	Increased risk associated with IP and address
AJ	Increased risk associated with IP and name
AK	Increased risk associated with address
AL	Increased risk associated with address and phone
AM	Increased risk associated with address and name
AN	Increased risk associated with name

Reason code	Description
AO	Increased risk associated with name and phone
AP	Increased risk associated with phone
JA	New consumer or not sufficient information
JB	Medium risk signals associated with email
JC	Medium risk signals associated with address
JD	Medium risk signals associated with phone
JE	Medium risk signals associated with name
JF	Medium risk signals associated with IP
RA	Consumer trust established across all PII elements
RB	Established credibility for email
RC	Established credibility for email and IP
RD	Established credibility for email and address
RE	Established credibility for email and phone
RF	Established credibility for email and name
RG	Established credibility for IP
RH	Established credibility for IP and phone
RI	Established credibility for IP and address
RJ	Established credibility for IP and name
RK	Established credibility for address
RL	Established credibility for address and phone
RM	Established credibility for address and name
RN	Established credibility for name
RO	Established credibility for name and phone
RP	Established credibility for phone

Subelement 57 (Security Services Additional Data for Acquirers)

DE 48, subelement 57 (Security Services Additional Data for Acquirers) supports Mastercard embedded security services for acquirers. Acquirers can receive up to 16 instances of subfields one and two within subelement 57. Each instance has the same format for each service. Acquirers must code to receive the 16 instances that the service supports.

Table 498: Attributes

Attribute	Description
Subelement ID	57
Subelement Data Length	2 positions
Data Representation	an...99; LLVAR The "LL" length field of LLVAR must be an integral multiple of 6, not to exceed 96.
Data Field	Contents of subfields
Subfields	2
Justification	N/A

Usage

Following is the usage of subelement 57 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 499: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	•	X	C

Values

See subfields.

Merchant Risk Predict

Merchant Risk Predict provides predictive, real-time, AI-generated merchant fraud detection scores to acquirers on dual message authorization transactions. The seven scores sit within the

authorization message to acquirers and provide feedback on merchant health, compliance, and chargeback risks.

Subfield 1 (Security Services Indicator)

DE 48, subelement 57, subfield 1 (Security Services Indicator) provides the security services indicator for acquirers.

Table 500: Attributes

Attribute	Description
Data Representation	an-3
Data Field	Contents of positions 1–3
Justification	N/A

Values

A three-character code identifying the Merchant Risk Predict score.

Subfield 2 (Security Services Data)

DE 48, subelement 57, subfield 2 (Security Services Data) contains a likelihood value for each Merchant Risk Predict score.

Table 501: Attributes

Attribute	Description
Data Representation	an-3
Data Field	Contents of positions 4–6
Justification	N/A

Values

A three-character value will be populated for each score indicating the likelihood of occurrence for each fraud type.

Subelement 58 (ATM Additional Data)

DE 48, subelement 58 (ATM Additional Data) only applies to Swedish Domestic Authorization Switching Service (SASS). Subelement 58 contains watermark data captured at an ATM. Watermark data is a card authentication technology supported by Swedish ATMs.

Table 502: Attributes

Attribute	Description
Subelement ID	58
Subelement Data Length	2 positions
Data Representation	ans-33
Data Field	Contents of subfields
Subfields	8
Justification	N/A

Usage

Following is the usage of subelement 58 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 503: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	O	X	C
Reversal Request/0400	O	X	C
Reversal Advice/0420	.	X	C

Values

See subfields.

Subfield 1 (ATM Time)

DE 48, subelement 58, subfield 1 (ATM Time) indicates the ATM time.

Table 504: Attributes

Attribute	Description
Data Representation	n-4
Data Field	Contents of positions 1–4 (hhmm)
Justification	N/A

Subfield 2 (ATM Date)

DE 48, subelement 58, subfield 2 (ATM Date) indicates the ATM date.

Table 505: Attributes

Attribute	Description
Data Representation	n-6
Data Field	Contents of positions 5–10 (YYMMDD)
Justification	N/A

Subfield 3 (Watermark)

DE 48, subelement 58, subfield 3 (Watermark) is the watermark value.

Table 506: Attributes

Attribute	Description
Data Representation	n-12
Data Field	Contents of positions 11–22
Justification	N/A

Subfield 4 (Mark 1)

DE 48, subelement 58, subfield 4 (Mark 1) indicates the card and the watermark status.

Table 507: Attributes

Attribute	Description
Data Representation	ans-2
Data Field	Contents of positions 23–24
Justification	N/A

Table 508: Values

Value	Description
0-	National card

Value	Description
4-	National card, foreign currency
8-	International card
C-	International card, foreign currency
-0	Watermark readable
-B	Watermark unreadable
-C	Watermark missing
-D	Test mode
-E	Test mode

Subfield 5 (Mark 2)

DE 48, subelement 58, subfield 5 (Mark 2) indicates the reason code for reversal from ATM.

Table 509: Attributes

Attribute	Description
Data Representation	an-2
Data Field	Contents of positions 25–26
Justification	N/A

Table 510: Values

Value	Description
00	Dispensing error for bank notes or receipts
02	Error in response
08	Failure to return card, response received from host
48	Failure to return card, single reversal
04	Timeout—card not picked up, response received from host
80	Timeout—card not picked up, response received from host
44	Timeout—card not picked up, single reversal
40	Single reversal, unknown reason
06	Error in response and timeout picking up card

Value	Description
0A	Error in response and failure to return card

Subfield 6 (Mark 3)

DE 48, subelement 58, subfield 6 (Mark 3) indicates stock of bank notes and receipt status.

Table 511: Attributes

Attribute	Description
Data Representation	ans-2
Data Field	Contents of positions 27–28
Justification	N/A

Table 512: Values

Value	Description
0-	Both SEK 100 and SEK 500 notes available
4-	SEK 500 notes not available
8-	SEK 100 notes not available
C-	No money available
-0	Receipt OK
-4	Receipt low
-8	Receipt paper empty
-C	Receipt technical error

Subfield 7 (Card Swallowed Status)

DE 48, subelement 58, subfield 7 (Card Swallowed Status) indicates if the ATM took the card or not.

Table 513: Attributes

Attribute	Description
Data Representation	n-1
Data Field	Contents of position 29

Attribute	Description
Justification	N/A

Table 514: Values

Value	Description
0	Card not swallowed
1	Card swallowed

Subfield 8 (Posting Date)

DE 48, subelement 58, subfield 8 (Posting Date) is the posting date.

Table 515: Attributes

Attribute	Description
Data Representation	n-4
Data Field	Contents of position 30–33
Justification	N/A

Subelement 60 (Additional Service Data For Issuers)

DE 48 subelement 60 (Additional Service Data For Issuers) provides data in support of specific issuer services. Issuers receive this subelement if they have opted in for any of the supported services and if those services were performed on the given transaction.

Table 516: Attributes

Attribute	Description
Subelement ID	60
Subelement Data Length	2 positions
Data Representation	ans...99; LLVAR
Data Field	Contents of subfields 01-04
Subfields	4
Justification	See subfields

Table 517: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	•	X	C
Authorization Advice/0120: acquirer-generated	•	X	C
Authorization Advice/0120: system-generated	•	X	C
Reversal Request/0400	•	X	C

Usage

Following is the usage of *DE 48* subelement 60 in applicable messages, regardless if it is mandatory, conditional, optional, system provided, or not required.

Application notes

DE 48, subelement 60, and subfields 01 through 04 are set up in Tag, Length, Value (TLV) format.

The Dual Message Authorization System inserts this subelement when selected issuer services are performed. The Dual Message Authorization System will remove *DE 48*, subelement 60 if received in Authorization Request/0100, Authorization Advice/0120, or Reversal Request/0400 messages from the acquirer.

Refer to the subfields contained within this subelement to obtain the supporting programs, services, or other specifics about the transaction

Subfield 01 (Reference Conversion Rate)

DE 48, subelement 60, subfield 01 (Reference Conversion Rate) contains the European Central Bank (ECB) reference conversion rate between currencies indicated by *DE 49* (Currency Code, Transaction) and *DE 51* (Currency Code, Cardholder Billing).

Table 518: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	n-8
Data Field	contents of subfield 01
Justification	Right, excluding the decimal indicator which must be the leftmost digit.

Values

The data field for *DE 48*, subelement 60, subfield 01 is formatted similarly as the data fields for *DE 9 (Conversion Rate, Settlement)* and *DE 10 (Conversion Rate, Cardholder Billing)*. Position 1 indicates the number of positions the decimal point should be moved from the right. Must be in the range of 0-7.

Positions 2-8 indicates the conversion rate.

The leftmost digit must be in the range 0–7 and denotes the number of positions that the decimal point shall be moved from the right. Example: For a value of 69972522, reference conversion rate is 9.972522 percent.

Application notes

For European Economic Area (EEA) issuers, the Dual Message Authorization System inserts this subfield for eligible transactions if the account range participates in Cross Border Fee Manager (CBFM) or CBFM ECB Rate Service.

Subfield 02 (Reference Conversion Rate Differential Sign Indicator)

DE 48, subelement 60, subfield 02 (Reference Conversion Rate Differential Sign Indicator) establishes whether the difference of *DE 10 (Conversion Rate, Cardholder Billing)* and *DE 48*, subelement 60, subfield 01 (Reference Conversion Rate) is a positive or negative number.

Table 519: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	a-1
Data Field	contents of subfield 02
Justification:	N/A

Values

DE 48, subelement 60, subfield 02 provides the algebraic sign of the Reference Conversion Rate Differential.

A value of C (for Credit) indicates that *DE 10 (Conversion Rate, Cardholder Billing)* is larger than or equal to *DE 48*, subelement 60, subfield 01 (Reference Conversion Rate).

A value of D (for Debit) indicates that *DE 10 (Conversion Rate, Cardholder Billing)* is smaller than *DE 48*, subelement 60, subfield 01 (Reference Conversion Rate).

Application notes

For EEA issuers, the Dual Message Authorization System inserts this subfield for eligible transactions if the account range participates in CBFM or CBFM ECB Rate Service.

Subfield 03 (Reference Conversion Rate Percent Differential)

DE 48, subelement 60, subfield 03 (Reference Conversion Rate Percent Differential) provides the difference of DE 10 (Conversion Rate, Cardholder Billing) and DE 48, subelement 60, subfield 01 (Reference Conversion Rate), expressed as a percentage of DE 48, subelement 60, subfield 01.

Table 520: Attributes

Attribute	Description
Subfield ID	03
Subfield Data Length	2 positions
Data Representation	n-6
Data Field	contents of subfield 03
Justification:	N/A

Values

DE 48, subelement 60, subfield 03 provides a percentage of DE 48, subelement 60, subfield 01 (Reference Conversion Rate) to communicate the effective currency conversion markup of the transaction.

The Reference Conversion Percent Rate Differential carries a percentage value with an implied two-position decimal.

DE 10 is typically larger than DE 48, subelement 60, subfield 01. Consequently, the difference of these values is typically positive. DE 48, subelement 60, subfield 02 (Reference Conversion Rate Differential) can only represent zero and positive values by definition. The Reference Conversion Rate Differential and Reference Conversion Rate Differential Sign Indicator can be used together to indicate positive and negative differentials.

Example 1: A differential of +123.45 percent is represented with DE 48 SE 60 SF 02 = C and DE 48 SE 60 SF 03 = 012345

Example 2: A differential of 0.00 percent is represented with DE 48 SE 60 SF 02 = C and DE 48 SE 60 SF 03 = 000000

Example 3: A differential of -1.23 percent is represented with DE 48 SE 60 SF 02 = D and DE 48 SE 60 SF 03 = 000123

Application notes

For EEA issuers, the Dual Message Authorization System inserts this subfield for eligible transactions if the account range participates in CBFM or CBFM ECB Rate Service.

Subfield 04 (Currency Rate Unique Identifier)

DE 48, subelement 60, subfield 04 (Currency Rate Unique Identifier) provides a unique identifier for the currency rate dataset used by the Mexico Central Bank (MCB) rate service.

Table 521: Attributes

Attribute	Description
Subfield ID	04
Subfield Data Length	2 positions
Data Representation	an-12
Data Field	Contents of subfield 04
Justification	N/A

The data field for DE 48, subelement 60, subfield 04 contains four pieces of information with construction YYMMDDUUUUUU.

Table 522: Values

Where...	Is...
YY	The two-digit year at the time of conversion, right-justified with leading zeros.
MM	The two-digit month at the time of conversion, right-justified with leading zeros.
DD	The two-digit day at the time of conversion, right-justified with leading zeros.
UUUUUU	A six-digit alphanumeric ID that will be unique each day.

Application notes

The Dual Message Authorization System inserts this subfield if the account range participates in the Mexico Central Bank (MCB) rate service and if the service is applied to the transaction.

Subelement 61 (POS Data Extended Condition Codes)

DE 48, subelement 61 (POS Data Extended Condition Codes) indicates whether the merchant terminal supports a specific program or service.

Table 523: Attributes

Attribute	Description
Subelement ID	61
Subelement Data Length	2 positions
Data Representation	n-5
Data Field	Contents of positions 1–5
Subfields	5
Justification	See subfields

Usage

Following is the usage of subelement 61 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 524: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	X	C
Authorization Advice/0120: acquirer-generated	C	X	C
Authorization Advice/0120: system-generated	•	C	C

Values

See subfields.

Application notes

This subelement should be provided to indicate the merchant terminal's capabilities in supporting specific programs and services.

Subfield 1 (Partial Approval Terminal Support Indicator)

DE 48, subelement 61, subfield 1 (Partial Approval Terminal Support Indicator) indicates if the merchant terminal supports partial approvals.

Table 525: Attributes

Attribute	Description
Data Representation	n-1
Data Field	Contents of position 1
Justification	N/A

Table 526: Values

Value	Description
0	Merchant terminal does not support receipt of partial approvals
1	Merchant terminal supports receipt of partial approvals

Subfield 2 (Purchase Amount Only Terminal Support Indicator)

DE 48, subelement 61, subfield 2 (Purchase Amount Only Terminal Support Indicator) indicates if the merchant terminal supports purchase only approvals.

Table 527: Attributes

Attribute	Description
Data Representation	n-1
Data Field	Contents of position 2
Justification	N/A

Table 528: Values

Value	Description
0	Merchant terminal does not support receipt of purchase only approvals
1	Merchant terminal supports receipt of purchase only approvals

Subfield 3 (Real-time Substantiation Indicator)

DE 48, subelement 61, subfield 3 (Real-time Substantiation Indicator) indicates if the merchant terminal verified the purchased items against the Inventory Information Approval System (IIAS).

Table 529: Attributes

Attribute	Description
Data Representation	n-1
Data Field	Contents of position 3
Justification	N/A

Table 530: Values

Value	Description
0	Merchant terminal did not verify the purchased items against an Inventory Information Approval System (IIAS)
1	Merchant terminal verified the purchased items against an Inventory Information Approval System (IIAS)
2	Merchant claims exemption from using an IIAS based on the IRS 90 percent rule
4	Transaction was submitted as real-time substantiated but from a non-IIAS-certified merchant. Mastercard uses this value to notify the issuer that the merchant could not be substantiated. Acquirers may not use this value.

Subfield 4 (Merchant Transaction Fraud Scoring Indicator)

DE 48, subelement 61, subfield 4 (Merchant Transaction Fraud Scoring Indicator) indicates if the acquirer requested the transaction to be scored by Expert Monitoring for Merchants.

Table 531: Attributes

Attribute	Description
Data Representation	an-1
Data Field	Contents of position 4
Justification	N/A

Table 532: Values

Value	Description
0	No action required
1	Transaction to be scored

Application notes

Only a value of zero is passed to the issuer on the Authorization Request/0100 message.

The Acquirer Generated Authorization Advice/0120 is rejected and will not be sent to EMS for scoring if it is any other value than zero.

Mastercard will require all issuing and acquiring processors to code for, support, and integrate into their systems the data elements, subelements, and values associated with this item.

For additional information about the requirement, refer to the *Global Safety and Security Standards Roadmap*.

NOTE: Mastercard does not require acquirers or issuers to enroll in these optional products and services, but they must code their systems to support these fields to leverage such products and services in a timely manner in the event of a security issue.

Subfield 5 (Final Authorization Indicator)

DE 48, subelement 61, subfield 5 (Final Authorization Indicator) is used to identify final authorization messages.

Table 533: Attributes

Attribute	Description
Data Representation	n-1
Data Field	Contents of position 5
Justification	N/A

Table 534: Values

Value	Description
0	Normal Authorization/Undefined Finality
1	Final Authorization

Application notes

Final authorization is an authorization request for an amount greater than zero, which meets the following conditions:

- Authorization is requested for the final transaction amount.
- The transaction is not expected to be canceled after the authorization request is approved in full (excluding non-completion for technical reasons such as telecommunications failure or terminal failure).

Any transaction corresponding to an authorization identified as a final authorization must be presented within seven calendar days of the authorization approval date. An authorization request is properly identified as a final authorization when *DE 61 (Point-of-Service [POS] Data)*, subfield 7 (POS Transaction Status) contains value 0 and *DE 48 (Additional Data)*, subelement 61 (POS Data Extended Condition Codes), subfield 5 contains value 1.

In the Europe region, all authorization messages must be coded as either preauthorization or final authorization requests; undefined coding is not permitted.

When submitting an Acquirer-generated Authorization Advice/0120, subfield 5 (Final Authorization Indicator) should be populated with the same value as provided in the original Authorization Request/0100 message.

Normal authorization/undefined finality is an authorization request for an amount greater than zero where a final amount may or may not be known and the transaction is not expected to be canceled after the authorization request is approved in full.

An authorization request is identified as normal or undefined finality authorization when *DE 61 (Point-of-Service [POS] Data)*, subfield 7 (POS Transaction Status) contains value 0 and *DE 48 (Additional Data)*, subelement 61 (POS Data Extended Condition Codes), subfield 5 contains value 0 or is not present.

When *DE 61 (Point-of-Service [POS] Data)*, subfield 7 (POS Transaction Status) contains value 4 (Preauthorized request), *DE 48 (Additional Data)*, subelement 61 (POS Data Extended Condition Codes), subfield 5 must either contain value 0 (Normal Authorization/Undefined Finality) or not be present.

Subelement 62 (Real-Time Payment Information)

DE 48, subelement 62 (Real-Time Payment Information) contains settlement-related information associated with specific Mastercard programs that offer the participants real-time settlement options information.

Table 535: Attributes

Attribute	Description
Subelement ID	62
Subelement Length	02 positions

Attribute	Description
Data Representation	ans...76; LLVAR
Data Field	Contents of subfields 1–6
Subfields	See subfields
Justification	N/A

Usage

Following is the usage of subelement 62 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 536: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	C	.	C
Authorization Advice Response/0130	C	.	C
Reversal Request Response/0410	C	.	C

Values

See subfields.

Application notes

This subelement and subfields are provided by the Account Holding Institution (AHI) in the Authorization Response messages and are sent unaltered to the Acquiring Institution by Mastercard.

Subfield 1 (Merchant Direct Participation)

DE 48, subelement 62, subfield 1 (Merchant Direct Participation) indicates whether or not the merchant is a direct participant in the real-time payment network and the party to which their transactions are settled. Transactions processed by a direct merchant are settled directly with the merchant, whereas those processed by an indirect merchant are settled with the Acquiring Institution.

Table 537: Attributes

Attribute	Description
Subfield ID	n-2
Subfield Data Length	2 positions

Attribute	Description
Data Representation	an-1
Data Field	Contents of subfield 1
Justification	N/A

Table 538: Values

Value	Description
Y	Merchant's account used for settlement via a real-time payment network.
N	Acquiring Institution's account used for settlement via a real-time payment network.

Subfield 2 (Type of Transaction)

DE 48, subelement 62, subfield 2 (Type of Transaction) indicates whether or not a transaction will be settled via a real-time payment network and if so, what type of settlement will occur.

Table 539: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	an-1
Data Field	Contents of subfield 2
Justification	N/A

Table 540: Values

Value	Description
D	Intracountry/Domestic real-time payment settlement transaction
C	Intercountry real-time payment settlement transaction
R	Intraregional real-time payment settlement transaction
X	Interregional real-time payment settlement transaction
Z	Other real-time payment settlement not elsewhere defined

Value	Description
N	Acquirer/merchant does not participate in real-time payment settlement

Subfield 3 (Real-Time Payment Rules)

DE 48, subelement 62, subfield 3 (Real-Time Payment Rules) indicates which payment network rules and requirements govern the transaction.

Table 541: Attributes

Attribute	Description
Subfield ID	03
Data Representation	ans...10; LLVAR
Subfield Data Length	2 positions
Data Field	Contents of subfield 3
Justification	N/A

Table 542: Values

Value	Description
PBA	Mastercard Pay by Account

Subfield 4 (Real-Time Payment Settlement Network ID and Name)

DE 48, subelement 62, subfield 4 (Real-Time Payment Settlement Network ID and Name) indicates the real-time payment network through which settlement fulfillment will occur.

Table 543: Attributes

Attribute	Description
Subfield ID	04
Data Representation	n...3; LLVAR
Subfield Data Length	2 positions
Data Field	Contents of subfield 4
Justification	N/A

Table 544: Values

Value	Description
01	On-us Payment
02	Faster Payments
03	UPI
04	IMPS
05	TCH
06	Real-time Payment
07–99	Reserved for future use

NOTE: Customers must be able to support any values between 01–99. New values can be added at any time.

Subfield 5 (Real-Time Payment Funds Transfer Position)

DE 48, subelement 62, subfield 5 (Real-Time Payment Funds Transfer Position) indicates the funds transfer position for a transaction settled through a real-time payment network.

Table 545: Attributes

Attribute	Description
Subfield ID	05
Data Representation	n-2
Subfield Data Length	2 positions
Data Field	Contents of subfield 5
Justification	N/A

Table 546: Values

Value	Description
00	Funds Transfer not initiated
01	Funds transferred to Acquiring Institution or merchant
02	Funds requested from Acquiring Institution or merchant

Value	Description
03–99	Reserved for future use

Subfield 6 (DSS Reference Number)

DE 48, subelement 62, subfield 6 (DSS Reference Number) indicates the unique number for every bank transaction.

Table 547: Attributes

Attribute	Description
Subfield ID	06
Data Representation	ans...35; LLVAR
Subfield Data Length	2 positions
Data Field	Contents of subfield 6
Justification	N/A

Subelement 63 (Trace ID)

DE 48 (Additional Data: Private Use) subelement 63 (Trace ID) contains data from DE 63 (Network Data) subfield 1 (Financial Network Code) and subfield 2 (Banknet Reference Number) and DE 15 (Date, Settlement) that is in the original Authorization Request Response/0110 message.

Table 548: Attributes

Attribute	Description
Subelement ID	63
Subelement Data Length	2 positions
Data Representation	ans-15
Data Field	Contents of positions 1–15
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 63 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 549: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Advice/0120: acquirer-generated	C	.	C
Authorization Advice/0120: system-generated	.	C	C
Reversal Request/0400	M	.	M
Reversal Request Response/0410	ME	.	ME
Reversal Advice/0420	.	M	M

Values

Contains the contents of positions 1–15 as defined below.

Table 550: Values of Positions 1–9 (Network Data)

Attribute	Value
Data Representation	ans-9
Data Field	Contents of positions 1–9

Table 551: Values of Positions 1–9

Contents of *DE 63*, subfield 1 (Financial Network Code) and subfield 2 (Banknet Reference Number) in the original Authorization Request Response/0110 message. The Banknet Reference number is six characters.

Value	Description
Positions 1–3	DE 63, subfield 1
Positions 4–9	DE 63, subfield 2

Table 552: Values of Positions 10–15 (Date Settlement)

Attribute	Description
Data Representation	ans-6
Data Field	Contents of positions 10–15

Values of Positions 10–15

Contents of *DE 15 (Date, Settlement)* in the original Authorization Request Response/0110 message. The four-digit Settlement Date is in MMDD format followed by two spaces.

Application notes

[DE 48 \(Additional Data: Private Use\)](#), subelement 63 must be present in Reversal Request/0400 messages and must contain data in positions 1–15, otherwise the message will be rejected with a format error response where [DE 39 \(Response Code\)](#) is 30 and DE 44 is 048063. [DE 48 \(Additional Data: Private Use\)](#), subelement 63 may contain a value of zeros in Reversal Request/0400 messages or Reversal Advice/0420 messages only when an Authorization Request Response/0110 was not successfully received or processed by the acquirer, or if the customer does not have the original reference number or the settlement date from the authorization.

[DE 48 \(Additional Data: Private Use\)](#), subelement 63 must be provided in Authorization Request/0100 messages to identify an incremental preauthorization and a zero amount Authorization Chargeback Protection Period Extension request.

[DE 48 \(Additional Data: Private Use\)](#), subelement 63 has been added to the acquirer-generated authorization advice/0120 layout to further assist issuers in matching an AFD completion advice to the original preauthorization. For additional details, refer to [Clearing AFD Transactions](#) in the Automated Fuel Dispenser Completion section of the Program and Service Format Requirements chapter.

Application note: European Economic Area (EEA) only

NOTE: The following information applies in the EEA only, in support of Payment Service Directive 2 (PSD2) Regulatory Technical Standards (RTS) regarding Strong Customer Authentication (SCA).

Mastercard uses DE 48, subelement 63 (Trace ID) to provide a reference to the first authorization for recurring payment and merchant-initiated transactions in Dual Message Authorization Request/0100, acquirer-generated advice/0120 and Reversal Request/0400 messages.

Subelement 64 (Transit Program)

DE 48, subelement 64 (Transit Program) is used to identify transit transactions in authorization messages.

NOTE: Transit Program field is not limited to MCC 4111, MCC 4131, and MCC 4784.

Table 553: Attributes

Attribute	Description
Subelement ID	64
Subelement Data Length	2 positions

Attribute	Description
Data Representation	n-4
Data Field	Contents of subfields 1–2
Subfields	2
Justification	N/A

Usage

Following is the usage of subelement 64 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 554: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Advice/0120: acquirer-generated	C	.	C
Authorization Advice/0120: system-generated	.	X	C

Values

Contains the Transit Program values in subfields 1 and 2.

Subfield 1 (Transit Transaction Type Indicator)

DE 48, subelement 64, subfield 1 (Transit Transaction Type Indicator) indicates the transit transaction type identifier.

NOTE: Values 3, 4, and 6 for Transit Transaction Type Indicators subelement 64, subfield 1 are limited to MCC 4111, MCC 4131, and MCC 4784.

Table 555: Attributes

Attribute	Description
Data Representation	n-2
Data Field	Contents of positions 1–2
Justification	N/A

Table 556: Values

Value	Description
01	Prefunded
02	Real-time Authorized
03	Post-Authorized Aggregated
04	Authorized Aggregated Split Clearing
05	Other
06	Post-authorized Aggregated Maestro
07	Debt Recovery
08–99	Reserved for Future Use

Subfield 2 (Transportation Mode Indicator)

DE 48, subelement 64, subfield 2 (Transportation Mode Indicator) indicates the transportation mode for a transit transaction.

Table 557: Attributes

Attribute	Description
Data Representation	n-2
Data Field	Contents of positions 3–4
Justification	N/A

Table 558: Values

Value	Description
00	Unknown
01	Urban Bus
02	Interurban Bus
03	Light Train Mass Transit (Underground Metro, LTR)
04	Train
05	Commuter Train
06	Water Borne Vehicle

Value	Description
07	Toll
08	Parking
09	Taxi
10	High Speed Train
11	Rural Bus
12	Express Commuter Train
13	Para Transit
14	Self Drive Vehicle
15	Coach
16	Locomotive
17	Powered Motor Vehicle
18	Trailer
19	Regional Train
20	Inter City
21	Funicular Train
22	Cable Car
23–99	Reserved for Future Use

Subelement 65 (Terminal Compliant Indicator)

DE 48, subelement 65 (Terminal Compliant Indicator) contains two subfields that identify whether or not a POS terminal from which the transaction originated is Terminal Line Encryption (TLE) and Unique Key Per Terminal (UKPT)/Derived Unique Key Per Terminal (DUKPT) compliant. This field must be present in card-present, POS transactions acquired in India.

Table 559: Attributes

Attribute	Description
Subelement ID	65
Subelement Data Length	2 positions
Data Representation	n-2
Data Field	Contents of subfields 1–2
Subfields	2

Attribute	Description
Justification	See subfields

Usage

Following is the usage of subelement 65 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 560: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	•	C
Authorization Advice/0120: acquirer-generated	C	•	C
Authorization Advice/0120: system-generated	•	C	C

Values

Contains the Terminal Compliant Indicator values in subfields 1 and 2.

Subfield 1 (TLE Compliant)

DE 48, subelement 65, subfield 1 indicates whether or not the POS terminal from which the transaction occurred is Terminal Line Encryption (TLE)-certified.

Table 561: Attributes

Attribute	Description
Data Representation	n-1
Data Field	Contents of subfield 1
Justification	N/A

Table 562: Values

Value	Description
1	Not Certified
2	Certified

Subfield 2 (UKPT/DUKPT Compliant)

DE 48, subelement 65, subfield 2 indicates whether or not the POS terminal from which the transaction occurred is Unique Key Per Terminal/Derived Unique Key Per Terminal (UKPT/ DUKPT)-certified.

Table 563: Attributes

Attribute	Description
Data Representation	n-1
Data Field	Contents of subfield 2
Justification	N/A

Table 564: Values

Value	Description
1	Not Certified
2	Certified

Subelement 66 (Authentication Data)

DE 48 (Additional Data), subelement 66 is populated by the acquirer and passed to the issuer to indicate the Program Protocol being used.

Table 565: Attributes

Attribute	Description
Subelement ID	66
Subelement Data Length	05 or 45 positions
Data Representation	ans...45; LLVAR
Data Field	Contents of subfields
Subfields	Subfield 1: Program Protocol
	Subfield 2: Directory Server Transaction ID

Usage

Following is the usage of subelement 66 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 566: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	X	C
Authorization Request Response/0110	.	CE	C
Authorization Advice/0120: acquirer-generated	C	.	C
Authorization Advice/0120: system-generated	.	C	C

Values

See subfields.

Application notes

This is a conditional field and must be provided by the acquirers' part of the authorization messages if they are participating and have previously authenticated using the Mastercard Identity Check program. This field will not be edited to ensure that these elements were provided on the transaction. If it is provided by the acquirer or processor, it must meet the specifications defined.

Subfield 1 (Program Protocol)

DE 4, subelement 66 (Authentication Data), subfield 1 indicates the Program Protocol.

Table 567: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	an-1
Data Field	Contents of subfield 1
Justification	N/A

Table 568: Values

Value	Description
1	EMV 3-D Secure Version 2.1
2	EMV 3-D Secure Version 2.2
3	EMV 3-D Secure Version 2.3
4	EMV 3-D Secure Version 2.4
5	EMV 3-D Secure Version 2.5
6	EMV 3-D Secure Version 2.6
7	EMV 3-D Secure Version 2.7
8	EMV 3-D Secure Version 2.8
9	EMV 3-D Secure Version 2.9

NOTE: EMV 3DS specification versions are detailed by **major.minor.patch.editorial (2.3.1.1)**. Therefore, all patch and editorial versions beyond 2.3.1.1 should be considered the same as 2.3 for the purposes of this table.

Subfield 2 (Directory Server Transaction ID)

DE 48, subelement 66 (Authentication Data), subfield 2 allows the acquirer to pass the Directory Server Transaction ID during authorization in order to link authentication and authorization data. The Directory Server Transaction ID is generated by the EMV 3DS Mastercard Directory Server during the authentication transaction and passed back to the merchant with the authentication results.

Table 569: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	ans-36
Data Field	Contents of subfield 2
Justification	N/A

Values

The Directory Server Transaction ID is a Universally Unique Transaction ID which can be provided by the processors/acquirers as part of the authentication transaction.

Example of a Directory Server Transaction ID: f38e6948-5388-41a6-bca4-b49723c19437

Application notes

This is a conditional field and must be provided by the acquirer if they are participating and have previously authenticated using the Mastercard EMV 3DS (3DS2) Identity Check program. This field will not be edited to ensure these elements were provided on the transaction. If provided by the acquirer or processor, it must meet the specifications defined.

The Directory Server Transaction ID must not be provided as an all caps value.

Subelement 67 (Mastercard Send Information)

DE 48, subelement 67 (Mastercard Send Information) contains subfield representing the Sanctions Score.

Table 570: Attributes

Attribute	Description
Subelement ID	67
Subelement Data Length	2 positions
Data Representation	ans...99; LLVAR
Data Field	Contents of subfield 1-2
Subfields	2
Justification	See subfields

Usage

Following is the usage of subelement 67 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 571: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	•	X	C
Authorization Request Response/0110	•	X	C
Authorization Advice/0120: system-generated	•	X	C

Values

See subfields.

Subfield 01 (Sanctions Score)

DE 48, subelement 67, subfield 01 (Sanctions Score) contains the Sanctions Score populated by Mastercard.

Table 572: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	n-3
Data Field	Contents of subfield 01
Justification	Right-justified with leading zeros

Values

NOTE: Mastercard has suspended the Mastercard Send Sanctions Scoring Service effective 30 April 2021. Customers will be advised accordingly should Mastercard reinstate the service at a later date

For issuers who opt-in to the Sanctions Scoring service, a Sanctions Score value will be populated by Mastercard on all cross-border Mastercard Send Payment Transactions globally and domestic Mastercard Send Payment Transactions in Egypt, Canada, across Europe and the United States.

The Sanctions Scoring service automatically checks the Sender name (consumer, business, government, and non-government) against select sanctions watchlists such as the OFAC Specially Designated Nationals and Blocked Persons List, UN List, and EU List.

Based on the likelihood of a match of the sender's name to a sanctioned individual or entity, a score of 000 or a score from 070 to 099 is generated and included in DE 48 (Additional Data: Private Use), subelement 67 (Mastercard Send Information), subfield 01 (Sanctions Score) of the authorization request message delivered to the issuer.

A higher score indicates a closer match to names on the applicable screening lists, while lower scores indicate a less likely match. When a score cannot be determined, the value will be 999. A score of 000 indicates that the transaction passes the Sanctions Scoring check.

NOTE: If the Sanctions Score is 100 or if the Sanctions Score produced by the Sanctions Scoring service is greater than or equal to the Blocking Score set up by the issuer in the opt-in service, then the authorization request will be declined and an advice will be sent to the issuer with the value of 33F (Mastercard Send Issuer Blocking—Sanctions Score limit exceeded) in DE 48, subelement 71 (On-behalf [OB] Services), subfield 1 (On-behalf [OB] Service) and subfield 2 (On-behalf [OB] Result 1).

Subfield 02 (Sanctions Score Additional Information)

DE 48, subelement 67, subfield 02 (Sanctions Score Additional Information) allows for communication of additional information regarding the Sanctions Scoring Service.

Table 573: Attributes

Attribute	Description
Subfield ID	02
Subfield length	2
Data representation	ans...58; LLVAR
Data field	Contents of subfield 2
Justification	N/A

Values

Subfield 02 contains a list of watchlist(s) associated with the sanctions score generated by Mastercard, which are separated with a " | " delimiter.

Subelement 68 (Financial Account Information)

DE 48, subelement 68 (Financial Account Information) contains the financial account information associated with the Mastercard Pay by Account token.

Table 574: Attributes

Attribute	Description
Subelement ID	68
Subelement Data Length	02 positions
Data Representation	ans...68; LLVAR
Data Field	Contents of subfield 1
Number of Subfields	See subfields

Usage

Following is the usage of subelement 68 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 575: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	•	X	C
Authorization Request/0100: account status inquiry (merchant-initiated message and MDES pre-digitization message)	•	X	C
Authorization Request/0100: tokenization eligibility	•	X	C
Authorization Request/0100: tokenization authorization	•	X	C
Authorization Request/0100: activation code notification	•	X	C
Authorization Request/0100: tokenization complete notification	•	X	C
Authorization Request/0100: tokenization event notification	•	X	C
Authorization Advice/0120: system-generated	•	X	C
Authorization Advice Response/0130: system-generated	•	X	C
Reversal Request/0400	•	X	C
Reversal Advice/0420	•	X	C
Administrative Advice/0620: issuer token notification advice for activation code notification	•	X	C
Administrative Advice/0620: issuer token notification advice for tokenization complete notification	•	X	C
Administrative Advice/0620: issuer token notification advice for tokenization event notification	•	X	C

Values

See subfields.

Application notes

The *DE 48*, subelement 68 information is added to the message by Mastercard and forwarded to the Account Holding Institution (AHI). Acquiring Institutions will not receive this information.

Subfield 1 (Financial Account Identifier)

DE 48, subelement 68, subfield 1 (Financial Account Identifier) contains the bank account information or other financial account information associated with the Mastercard Pay by Account token at the time the transaction occurred.

Table 576: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	ans...64; LLVAR
Data Field	Contents of subfield 1
Justification	N/A

Values

Financial account identifier must be a minimum of 9 characters and a maximum of 64 characters.

Subelement 71 (On-behalf Services)

DE 48, subelement 71 (On-behalf Services [OBS]) notifies the issuer of the On-behalf Service performed on the transaction and the results. Subelement 71 will support up to ten services for a transaction.

Table 577: Attributes

Attribute	Description
Subelement ID	71
Subelement Data Length	2 positions
Data Representation	ans...40; LLVAR The "LL" length field of LLVAR must be an integral multiple of 4, not to exceed 40.
Data Field	Contents of subfields 1–3
Subfields	3
Justification	N/A

Usage

Following is the usage of subelement 71 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 578: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	.	X	C
Authorization Request Response/0110	CE	XE	.
Authorization Advice/0120: acquirer-generated	.	X	C
Authorization Advice/0120: system-generated	.	X	C
Reversal Request/0400	.	X	C
Reversal Advice/0420	.	X	C

Values

See subfields.

Subfield 1 (On-behalf (OB) Service)

DE 48, subelement 71, subfield 1 (On-behalf [OB] Service) indicates the service performed on the transaction.

Table 579: Attributes

Attribute	Description
Data Representation	an-2
Data Field	Contents of positions 1–2
Justification	Left-justified

Values

Refer to [Subelement 71 \(Valid Subfield 1 and Subfield 2 Value Combinations\)](#) and [Subelement 71 \(Valid Subfield 1 and Subfield 3 Value Combinations\)](#).

Subfield 2 (On-behalf Result 1)

DE 48, subelement 71, subfield 2 (On-behalf [OB] Result 1) indicates the results of the service processing.

Table 580: Attributes

Attribute	Description
Data Representation	an-1
Data Field	Contents of position 3
Justification	N/A

Values

Refer to [Subelement 71 \(Valid Subfield 1 and Subfield 2 Value Combinations\)](#).

Subfield 3 (On-behalf Result 2)

DE 48, subelement 71, subfield 3 (On-behalf Result 2) identifies the results of the service processing.

Table 581: Attributes

Attribute	Description
Data Representation	ans-1
Data Field	Contents of position 4
Justification	N/A

Values

Mastercard use only. May contain a space or a value.

Refer to [Subelement 71 \(Valid Subfield 1 and Subfield 3 Value Combinations\)](#).

Subelement 71 (Valid Subfield 1 and Subfield 2 Value Combinations)

, DE 48 (Additional Data: Private Use) subelement 71 (Valid Subfield 1 and Subfield 2 Value Combinations) are the valid DE 48, subelement 71 (On-behalf Services [OBS]), subfield 1 (On-behalf [OB] Service) and subfield 2 (OB Result 1) value combinations. The contents of subfield 2 depend on the contents of subfield 1 as described here.

Table 582: Attributes

Attribute	Subfield 1 (OB Service)	Subfield 2 (OB Result 1)
Data Representation	an-2	an-1
Data Field	Contents of positions 1–2	Contents of position 3
Justification	Left-justified	N/A

NOTE: DE 48 (Additional Data: Private Use), subelement 71 will contain all applicable, on-behalf service results that were performed on a given transaction; examples include Mastercard Digital Enablement Service (MDES) service results and Mastercard Identity Check service results.

Table 583: Values

DE 48, Subelement 71, Subfield 1 (OB Service) Values (an-2)	DE 48, Subelement 71, Subfield 2 (OB Result 1) Values (an-1)
01 Chip to Magnetic Stripe Conversion Service	C Conversion of the M/Chip transaction to a magnetic stripe transaction was completed from an original POS PAN Entry Mode value of 05 [PAN auto-entry via chip] or 79 [Chip card/PAN entry via manual entry]
	M Conversion of the M/Chip fallback transaction to a magnetic stripe transaction was completed from an original POS PAN Entry Mode value of 80-PAN auto-entry with magnetic stripe
	S Conversion of the M/Chip transaction to a magnetic stripe transaction was completed from an original POS PAN Entry Mode value of 07-PAN auto-entry via contactless M/Chip
02 M/Chip Cryptogram Pre-validation Service	A Valid Application Cryptogram (AC); ATC outside allowed range
	E Valid Application Cryptogram; ATC Replay
	F Format Error
	G Application Cryptogram is valid but not an ARQC nor a TC, status of TVR/CVR unknown
	I Invalid Cryptogram
	K No matching key file for this PAN, PAN expiry date and KDI combination
	T Valid ARQC/TC and ATC; TVR/CVR invalid

DE 48, Subelement 71, Subfield 1 (OB Service) Values (an-2)		DE 48, Subelement 71, Subfield 2 (OB Result 1) Values (an-1)	
03	M/Chip Cryptogram Validation in Stand-In Processing	U	Unable to process
		V	Valid
		X	Security platform time out
		Z	Security platform processing error
		A	Valid Application Cryptogram (AC); ATC outside allowed range
		E	Valid Application Cryptogram; ATC Replay
		F	Format Error
		G	Application Cryptogram is valid but not an ARQC nor a TC, status of TVR/CVR unknown
		I	Invalid Cryptogram
		K	No matching key file for this PAN, PAN expiry date and KDI combination
04	M/Chip Cryptogram Regeneration Service	T	Valid ARQC/TC and ATC; TVR/CVR invalid
		U	Unable to process
		V	Valid
		X	Security platform time out
		Z	Security platform processing error
		C	Regeneration of the M/Chip cryptogram was completed
05	Mastercard® Identity Check™ AAV Verification Service Refer to Subelement 71 (OBS 05 Result Codes) for detailed information on OBS 05 result codes as they relate to SPA2 AAV, DS Transaction ID, and Amount Check (Authorization: Authentication).	A	DS Txn ID not provided; AAV, amount and currency verified
		B	DS Txn ID not provided; AAV and currency verified; amount 0-20% different
		C	DS Txn ID not provided; AAV and currency verified; amount greater than 20% different
		D	DS Transaction ID does not match with PAN and SPA2 AAV within the authentication record
		E	Format Error; AAV is not correctly formatted or failed authentication
		F	Failed MAC key validation; Dynamic Linking validation not provided.
		I	Invalid AAV; Dynamic Linking was not able to find a matching authentication due to expired time, previously used or not authenticated through Directory Server

DE 48, Subelement 71, Subfield 1 (OB Service) Values (an-2)	DE 48, Subelement 71, Subfield 2 (OB Result 1) Values (an-1)
	K No matching key file for this PAN. SPA2 AAV provided is not generated by Mastercard Identity Check (i.e. Leading Indicator & IAV only).
	M Monetary Currency mismatch between authorization and authentication
	P Passed MAC key Validation; Dynamic Linking validation not provided.
	S DS Txn ID present; AAV and currency verified; amount 0-20% different.
	T DS Txn ID present; AAV and currency verified; amount greater than 20% different.
	U Unable to process
	V Valid
	X Security platform time out or Authentication Challenge (Results Request) not received yet; Retry authorization if declined for failed AAV Validation.
	Z Security platform processing error; Authentication Response not received yet; Retry authorization if declined for failed AAV Validation.
06 Mastercard® Identity Check™ AAV Verification in Stand-In Processing	A DS Txn ID not provided; AAV, amount and currency verified.
	B DS Txn ID not provided; AAV and currency verified; amount 0-20% different.
	C DS Txn ID not provided; AAV and currency verified; amount greater than 20% different.
	D DS Transaction ID does not match with PAN and SPA2 AAV within the authentication record
	E Format Error; AAV is not correctly formatted or failed authentication.
	F Failed MAC key Validation; Dynamic Linking validation not provided.
	I Invalid AAV; Dynamic Linking was not able to find a matching authentication due to expired time, previously used or not authenticated through Directory Server.
	K No matching key file for this PAN. SPA2 AAV provided is not generated by Mastercard Identity Check (i.e. Leading Indicator & IAV only).

DE 48, Subelement 71, Subfield 1 (OB Service) Values (an-2)	DE 48, Subelement 71, Subfield 2 (OB Result 1) Values (an-1)
	M Monetary Currency mismatch between authorization and authentication
	P Passed MAC key Validation; Dynamic Linking validation not provided
	S DS Txn ID present; AAV and currency verified; amount 0-20% different.
	T DS Txn ID present; AAV and currency verified; amount greater than 20% different.
	U Unable to process
	V Valid
	X Security platform time out or Authentication Challenge (Results Request) not received yet; Retry authorization if declined for failed AAV Validation.
	Z Security platform processing error or Authentication Response not received yet; Retry authorization if declined for failed AAV Validation.
08 Online PIN Pre-validation (Europe Only)	I Invalid PIN
	P Mandatory PVV not on file
	R PIN retry exceeded (invalid PIN)
	U Unable to process
	V Valid
09 Online PIN Validation in Stand-In (Europe only)	I Invalid PIN
	P Mandatory PVV not on file
	R PIN retry exceeded (invalid PIN)
	U Unable to process
	V Valid
10 CVC 1 Validation Stand-In Service	I Invalid CVC 1
	U Unable to process
	V Valid

DE 48, Subelement 71, Subfield 1 (OB Service) Values (an-2)		DE 48, Subelement 71, Subfield 2 (OB Result 1) Values (an-1)	
11	CVC 1 Pre-Validation Service	I	Invalid CVC 1
		K	No matching key file for this PAN, PAN Expiry date combination
		U	Unable to process
		V	Valid
		NOTE: Issuers that enroll account ranges in the OBS 11 (CVC 1 Pre-Validation Service) that are currently enrolled in OBS 10 (CVC 1 Validation Stand-In Service) must request the removal of those account ranges from the OBS 10 service.	
14	Contactless Mapping Service	C	Conversion of contactless account number to PAN was completed
		I	Invalid
		U	Unable to process
15	Dynamic CVC 3 Pre-validation (with or without Contactless Mapping Service)	A	ATC outside allowed range (applicable when ATC value is dynamic [varying] value)
		E	CVC 3 ATC Replay
		I	Invalid CVC 3
		K	No matching key file for this PAN, PAN expiry date, and KDI combination
		N	Unpredictable Number Mismatch (applicable when the UN is dynamic [varying] value) (Indicates that the number/length in the discretionary data in DE 45 or DE 35 does not match the number/length provided by the issuer during personalization)
		U	Unable to process
		V	Valid
		X	Security platform time out
16	Dynamic CVC 3 Validation in Stand-In Processing	Z	Security platform system error
		A	ATC outside allowed range (applicable when ATC value is dynamic [varying] value)
		E	CVC 3 ATC Replay
		I	Invalid CVC 3
		K	No matching key file for this PAN, PAN expiry date, and KDI combination

DE 48, Subelement 71, Subfield 1 (OB Service) Values (an-2)	DE 48, Subelement 71, Subfield 2 (OB Result 1) Values (an-1)
	N Unpredictable Number Mismatch (applicable when the UN is dynamic [varying] value) (Indicates that the number/length in the discretionary data in <i>DE 45</i> or <i>DE 35</i> does not match the number/length provided by the issuer during personalization)
	U Unable to process
	V Valid
	X Security platform time out
	Z Security platform system error
17 In Control Virtual Card Service	A Virtual Card Number (expiration date does not match) B Virtual Card Number (expiration date expired) C Virtual Card Number Virtual CVC 2 does not match D In Control Validity Period Limit E In Control Transaction Amount Limit Check F In Control Cumulative Amount Limit Check G In Control Transaction Number Usage H In Control Merchant ID Limit I In Control Invalid Virtual Card Number—Real Card Number mapping relationship J In Control MCC Limit K In Control Database Status Bad L In Control Geographic Restriction M In Control Transaction Type Restriction P In Control Transaction Time/Date Restriction U Unable to process V Valid
18 Fraud Scoring Service	C Fraud Scoring Service was performed successfully
	U Fraud Scoring Service was not performed successfully
20 In Control RCN Spend Control Service	D In Control Validity Period Limit
	E In Control Transaction Amount Limit Check
	F In Control Cumulative Amount Limit Check

Data element definitions

Subelement 71 (Valid Subfield 1 and Subfield 2 Value Combinations)

DE 48, Subelement 71, Subfield 1 (OB Service) Values (an-2)	DE 48, Subelement 71, Subfield 2 (OB Result 1) Values (an-1)
	G In Control Transaction Number Usage
	H In Control Merchant ID Limit
	J In Control MCC Limit
	K In Control Database Status Bad
	L In Control Geographic Restriction
	M In Control Transaction Type Restriction
	P In Control Transaction Time/Date Restriction
	U Unable to process
	V Valid
25 Account Data Compromise Information	Y Compromised Event Data Found
	N Compromised Event Data Not Found
	U Unable to process
26 Point-of-Interaction Service	A Amount does not qualify for installment
	C Enrichment completed for Installments
	E Merchant ID does not qualify for installment
	F PAN does not qualify for installments
	I Does not qualify for installment as the installment plan count exceeds the allowed limit
	M MCC does not qualify for Installment
	N Record not found
	P Installment Parameters missing
	U Unable to Process
	V Qualified for Installment
31 Chip CVC to CVC 1 Conversion Service— CVC 1 Key/Decision Matrix Only	C Chip CVC Validated Successfully; Conversion of Chip CVC to CVC 1 Performed Successfully
	F Track Data Formatted Incorrectly
	I Chip CVC Invalid; Conversion of Chip CVC to CVC 1 Not Performed
	K Issuer CVC 1 Key Record not Found for Account Range / Expiry Date combination; Service Not Performed
	U Unable to process

DE 48, Subelement 71, Subfield 1 (OB Service) Values (an-2)	DE 48, Subelement 71, Subfield 2 (OB Result 1) Values (an-1)
32 Chip CVC to CVC 1 Conversion Service— Separate Keys/Decision Matrices	C Chip CVC Validated Successfully; Conversion of Chip CVC to CVC 1 Performed Successfully
	F Track Data Formatted Incorrectly
	I Chip CVC Invalid; Conversion of Chip CVC to CVC 1 Not Performed
	K Issuer Chip CVC Key Record not found and Issuer CVC 1 Key Record not found for Account Range / Expiry Date combination; Service Not Performed
	L Issuer CVC 1 Key Record not Found for Account Range / Expiry Date combination; Service Not Performed
	M Issuer Chip CVC Key Record not Found for Account Range / Expiry Date combination; Service Not Performed
	U Unable to process
33 Send Blocking Service OBS 33 result code values A, B, D, E, F, I, J, K, L, M, N, O, P, Q, R, S, U, V, W, X, Y apply to Send transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payment Transactions). OBS 33 result code value G apply to Mastercard Send Payment Transactions and Gaming and Gambling Payment Transactions. OBS 33 result code value T applies to Funding Transactions. OBS 33 result code value H applies to other transactions, such as coded with legacy TTI or miscoded.	A Send issuer blocking: Transaction limit not allowed for the Transaction Type Identifier (TTI)
	B Send issuer blocking: Merchant not allowed for the Transaction Type Identifier (TTI)
	D Send issuer blocking: Country not allowed for the Transaction Type Identifier (TTI)
	E Send issuer blocking: Domestic activity only allowed for the Transaction Type Identifier (TTI)
	F Send issuer blocking: Sanctions Score limit exceeded for the Transaction Type Identifier (TTI)
	G Send Mastercard blocking: Transaction limit not allowed for the Transaction Type Identifier (TTI) and Country
	H Mastercard blocking: General miscoding
	I Send Mastercard blocking: Reserved for future use
	J Send Mastercard blocking: Transaction Count exceeded for the Transaction Type Identifier (TTI)
	K Send Mastercard blocking: Aggregate transaction amount limit exceeded for the Transaction Type Identifier (TTI)
	L Send issuer blocking: Transaction Count exceeded for the Transaction Type Identifier (TTI)
	M Send issuer blocking: Aggregate transaction amount limit exceeded for the Transaction Type Identifier (TTI)

DE 48, Subelement 71, Subfield 1 (OB Service) Values (an-2)	DE 48, Subelement 71, Subfield 2 (OB Result 1) Values (an-1)
	N Send issuer monitoring: Transaction Count exceeded for the Transaction Type Identifier (TTI)
	O Send issuer monitoring: Aggregate transaction amount limit exceeded for the Transaction Type Identifier (TTI)
	P Send issuer monitoring: Transaction amount limit exceeded for the Transaction Type Identifier (TTI)
	Q Send issuer monitoring: Sanctions Score exceeded
	R Send issuer blocking: Invalid Card
	S Send Mastercard blocking: Product code invalid for the Transaction Type Identifier (TTI) or the use case
	T Send Mastercard blocking: Transaction Amount limit not allowed for the Funding Transaction Type Identifier (TTI)
	U Unable to process
	V Valid
	W Send Mastercard blocking: Geographical restriction
	X Send Mastercard blocking: Reserved for future use
	Y Send Mastercard blocking: Data quality controls
37 Mastercard Merchant Presented QR Blocking Service	D Mastercard Merchant Presented QR Blocking-Transaction Amount Limit Exceeded
	E Mastercard Merchant Presented QR Blocking-Cumulative Transaction Amount Or Count Limit Exceeded
	F Mastercard Merchant Presented QR Blocking-Domestic Activity Only
	U Unable to Process
	V Valid
50 MDES PAN Mapping service	C Conversion of Token to PAN completed successfully
	F Format Error
	I Invalid Token
	U Unable to process
51 MDES Pre-Validation Service for Secure Element tokens (ICC or DSRP data)	A ATC outside allowed range (applicable when ATC value is dynamic [varying] value)
	E ATC Replay
	F Format Error

Data element definitions

Subelement 71 (Valid Subfield 1 and Subfield 2 Value Combinations)

DE 48, Subelement 71, Subfield 1 (OB Service) Values (an-2)	DE 48, Subelement 71, Subfield 2 (OB Result 1) Values (an-1)
	G Application Cryptogram is valid but not an ARQC nor a TC I Invalid Cryptogram K No matching key file for this PAN, PAN expiry date and KDI combination T Valid cryptogram and ATC; TVR/CVR invalid. Cryptogram type (ARQC/AAC/TC) not checked U Unable to process V Valid ARQC/TC and ATC and TVR/CVR X Security platform time out Z Security platform system error
52 MDES Pre-Validation Service for Secure Element tokens (CVC3 data)	A ATC outside allowed range (applicable when ATC value is dynamic [varying] value) E CVC 3 ATC Replay H Invalid Time Validation I Invalid CVC 3 K No matching key file for this PAN, PAN expiry date, and KDI combination N Unpredictable Number Length Indicator Mismatch U Unable to process V Valid X Security platform time out Z Security platform system error
54 Mastercard Digital Enablement Service Digital Payment Data Validation Service Refer to Subelement 71 (Valid Subfield 1 and Subfield 3 Value Combinations) for more information on DE 48, subelement 71, subfield 1, value 54.	A If the transaction amount in DE 4 is less than or equal to the approximate amount in the cryptogram B If the transaction amount in DE 4 is greater than the approximate amount in the cryptogram by 0% to 19.99% C If the transaction amount in DE 4 is greater than the approximate amount in the cryptogram by 20% or more D If the transaction amount in DE 4 is not zero and the approximate amount in the cryptogram is zero
55 Merchant Validation Service	M The submitted merchant data is a match to Mastercard's merchant data

DE 48, Subelement 71, Subfield 1 (OB Service) Values (an-2)		DE 48, Subelement 71, Subfield 2 (OB Result 1) Values (an-1)	
61	MDES Pre-Validation Service for Cloud and Static tokens (ICC, DSRP, or DTVC data)	N	The submitted merchant data is not a match to Mastercard's merchant data
		D	ATC Invalid-Not in list of currently active Single-Use Keys
		E	ATC Replay
		F	Format Error
		I	Invalid MD AC and UMD AC
		K	No matching key file for this PAN, PAN expiry date and KDI combination
		L	Invalid MD AC; Valid UMD AC
		M	Valid MD AC; Invalid UMD AC (Mobile PIN Try Counter Max Limit Reached, Token Suspended)
		P	Valid MD AC; Invalid UMD AC (Invalid Mobile PIN)
		T	Invalid TVR/CVR
		U	Unable to process
		V	Valid
		X	Security platform time out
		Z	Security platform system error
62	MDES Pre-Validation Service for Cloud tokens (CVC3 data)	D	ATC Invalid-Not in list of currently active Single-Use Keys
		E	ATC Replay
		F	Format Error
		H	Invalid Time Validation
		I	Invalid MD AC and UMD AC
		K	No matching key file for this PAN, PAN expiry date and KDI combination
		L	Invalid MD AC; Valid UMD AC
		M	Valid MD AC; Invalid UMD AC (Mobile PIN Try Counter Max Limit Reached, Token Suspended)
		N	Unpredictable Number Length Indicator Mismatch
		P	Valid MD AC; Invalid UMD AC (Invalid Mobile PIN)
		U	Unable to process
		V	Valid
		X	Security platform time out

DE 48, Subelement 71, Subfield 1 (OB Service) Values (an-2)		DE 48, Subelement 71, Subfield 2 (OB Result 1) Values (an-1)	
71	Token Service Provider Cloud Based Payments Chip Validation Service in Stand-In Processing	Z	Security platform system error
		E	ATC Replay
		F	Format Error
		I	Invalid MD AC and UMD AC
		K	No matching key file for this PAN, PAN expiry date and KDI combination
		L	Invalid MD AC; Valid UMD AC
		P	Valid MD AC; Invalid UMD AC (Invalid Mobile PIN)
		T	Invalid TVR/CVR
		U	Unable to process
		V	Valid
		X	Security platform time out
Z	Security platform system error		

Subelement 71 (Valid Subfield 1 and Subfield 3 Value Combinations)

[DE 48 \(Additional Data: Private Use\)](#) DE 48 subelement 71 (Valid Subfield 1 and Subfield 3 Value Combinations) are the valid DE 48, subelement 71 (On-behalf Services [OBS]), subfield 1 (On-behalf [OB] Service) and subfield 3 (OB Result 2) value combinations. The contents of subfield 3 depend on the contents of subfield 1 as described here.

Table 584: Attributes

Attribute	Subfield 1 (OB Service)	Subfield 3 (OB Result 2)
Data Representation	an-2	ans-1
Data Field	Contents of positions 1–2	Contents of position 4
Justification	Left-justified	N/A

Values

Table 585: Values

DE 48, Subelement 71, Subfield 1 (OB Service) Values (an-2)		DE 48, Subelement 71, Subfield 3 (OB Result 2-RCA Identifier Validation) Values (ans-1)	
54 =	Mastercard Digital Enablement Service Digital Payment Data Validation Service	M =	In Europe: Transaction authenticated with Mastercard qualified multifactor authentication method as Strong Customer Authentication, and valid merchant dynamic linking Outside of Europe: Transaction authenticated with Mastercard qualified multifactor authentication method, and valid merchant dynamic linking
		N =	In Europe: Merchant dynamic linking was not performed or was invalid Outside of Europe: Merchant dynamic linking was not performed or was invalid

Subelement 71 (Valid Subfield 1, Subfield 2, and Subfield 3 value combinations)

DE 48 (Additional Data: Private Use), subelement 71 (Valid Subfield 1, Subfield 2, and Subfield 3 value combinations) are the valid DE 48 (Additional Data: Private Use), subelement 71 (On-behalf Service [OBS]), subfield 1 (On-behalf [OB] Service Indicator), subfield 2 (OB Result 1), subfield 3 (On-behalf Result 2) value combinations.

Table 586: Value combinations

Subfield 1: On-behalf (OB) Service	Subfield 2: On-behalf Result 1	Subfield 3: On-behalf Result 2
50 = MDES PAN Mapping service	C = Conversion of Token to PAN completed successfully F = Format Error I = Invalid Token U = Unable to process	SPACE

Table 587: Value combinations

Transaction Description	Subfield 1: On-behalf (OB) Service	Subfield 2: On-behalf Result 1	Subfield 3: On-behalf Result 2
Tokenized transactions that carry Dynamic Token Validation Code	61 - MDES Pre-Validation Service for Cloud and Static tokens (ICC, DSRP, or DTVC data)	V = Valid I = Invalid U = Unable to Process	T = Dynamic Token Validation Code
Tokenized transactions that carry DSRP cryptogram	61 - MDES Pre-Validation Service for Cloud and Static tokens (ICC, DSRP, or DTVC data)	V = Valid I = Invalid U = Unable to Process	SPACE = DSRP Cryptogram

Application notes

Issuers participating in MDES:

- Receive OBS 50 (MDES PAN Mapping service and corresponding result codes in [DE 48 \(Additional Data: Private Use\)](#), subelement 71 indicating the status of PAN mapping for both static and cloud tokenized transactions.
- Receive OBS 61 (MDES Pre-Validation Service for Cloud and Static tokens [ICC, DSRP, or DTVC data] in [DE 48 \(Additional Data: Private Use\)](#), subelement 71, subfield 1 (On-behalf [OB] Service) to indicate that MDES attempted to validate dynamic data submitted by acquirer for both static and cloud tokenized transactions and a valid value in [DE 48 \(Additional Data: Private Use\)](#), subelement 71, subfield 2 (On-behalf Result 1) to indicate the validation outcome of both dynamic data in static and cloud token transactions.
- Receive a new result code T in [DE 48 \(Additional Data: Private Use\)](#), subelement 71, subfield 3 (On-behalf Result 2) to indicate that the acquirer submitted dynamic token verification code (DTVC) and the validation outcome in On-Behalf Result 1 pertains to DTVC.

- Prepare to process static or cloud token transactions without a CVC 2, DTVC validation results, or both, in the authorization request, as in some cases this information might not be provided to the issuer. However, when CVC 2 is provided, issuers should validate it.
- Issuers should be aware that Mastercard attempts to validate the DTVC submitted by acquirers during transaction processing.
 - When DTVC can be validated, Mastercard will send OBS 61 in [DE 48 \(Additional Data: Private Use\)](#), subelement 71 and suppress the DTVC from [DE 48 \(Additional Data: Private Use\)](#), subelement 92 (CVC 2) in Authorization Request/0100 and in Financial Transaction Request/0200 messages.
 - When the validity of the CVC 2 value cannot be determined, Mastercard will send the CVC 2 in [DE 48 \(Additional Data: Private Use\)](#), subelement 92 (CVC 2) to issuers for decisioning. Issuers will not receive OBS 61 in such transactions. In some use cases, the CVC 2 may reflect the value associated with the consumer's primary account number.

Subelement 71 (OBS 05 Result Codes)

DE 48, subelement 71 (OBS 05 Result Codes) provides detailed information on the result codes that can be present in *DE 48 (Additional Data—Private Use)*, subelement 71 (On-behalf Services [OBS]), subfield 2 (OB Result 1) when *DE 48*, subelement 71, subfield 1 (On-behalf [OB] Service) value = 05 (Mastercard Identity Check™ AAV Verification Service).

Table 588: OBS 05 Result Codes

Code	Description	SPA2 AAV	DS Transaction ID	Amount Check (Authorization : Authentication)
A	AAV and Amount Checked Low risk assumption for approval. Applies only to SPA2 AAVs.	Found associated with PAN	Not Present in Authorization	< or = authentication amount
B	Balance to Verify Low risk assumption for approval. Applies only to SPA2 AAVs.	Found associated with PAN	Not Present in Authorization	0%–19.99% of authentication amount
C	Consider the Amount Applies only to SPA2 AAVs. Amount discrepancy greater than 20%; other approval decision factors should be considered.	Found associated with PAN	Not Present in Authorization	20% or > of authentication amount

Code	Description	SPA2 AAV	DS Transaction ID	Amount Check (Authorization : Authentication)
D	DS Transaction ID Failed Applies only to SPA2 AAVs. Service unavailable or discrepancy with critical validation fields; other approval decision factors should be considered.	Found associated with PAN	Not Found associated with PAN and AAV	N/A
F	Failed MAC key Validation; No Dynamic Linking available.	NA	NA	NA
I	Invalid AAV Service unavailable or discrepancy with critical validation fields; other approval decision factors should be considered.	Not Found associated with PAN	N/A	N/A
K	Key Not on File This value usually indicates the issuer is performing Self-Validation on SPA1 AAVs and have not shared their ACS Issuer SPA1 AAV key with Mastercard.			
M	Monetary Currency mismatch between authorization and authentication records.			
P	Passed MAC key Validation; No Dynamic Linking available	NA	NA	NA
S	DS Transaction ID Present – Balance to Verify Low risk assumption for approval. Applies only to SPA2 AAVs.	Found associated with PAN	Found associated with PAN and AAV	0%–19.99% of authentication amount
T	Transaction ID Present – Consider the Amount Applies only to SPA2 AAVs. Amount discrepancy greater than 20%; other approval decision factors should be considered.	Found associated with PAN	Found associated with PAN and AAV	20% or > of authentication amount
U	Service Unavailable Service unavailable or discrepancy with critical validation fields; other approval decision factors should be considered.			

Code	Description	SPA2 AAV	DS Transaction ID	Amount Check (Authorization : Authentication)
V	Valid – All Data Passes Low risk assumption for approval.	Found associated with PAN	Found associated with PAN and AAV	< or = authentication amount
X	Security Platform Timeout Amount discrepancy greater than 20%; other approval decision factors should be considered. Applies only to SPA1 AAVs and 06 service.			
Z	Security Platform Processing Error Amount discrepancy greater than 20%; other approval decision factors should be considered. Applies only to SPA1 AAVs and 06 service.			

Subelement 72 (Issuer Chip Authentication)

DE 48, subelement 72 (Issuer Chip Authentication) carries data used during cryptogram processing.

Table 589: Attributes

Attribute	Description
Subelement ID	72
Subelement Data Length	2 positions
Data Representation	b...16; LLVAR (the "LL" length field of LLVAR must be between 8–16 positions.)
Data Field	Contents of subelement
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 72 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 590: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	•	X	C
Authorization Advice/0120: system-generated	•	X	C

Values

Mastercard generated.

Application notes

Issuers no longer need to echo *DE 48*, subelement 72 in their response messages.

Subelement 74 (Additional Processing Information)

DE 48, subelement 74 (Additional Processing Information) provides additional information about chip transaction processing and results.

Table 591: Attributes

Attribute	Description
Subelement ID	74
Subelement Data Length	2 positions
Data Representation	an...30; LLVAR The "LL" length field of LLVAR must be an integral multiple of 3 not to exceed 30.
Data Field	Contents of subfields
Subfields	2
Justification	See subfields

Usage

Following is the usage of subelement 74 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 592: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	O	X	C

Values

See subfields.

Application notes

DE 48, subelement 74 is not applicable in the 0120/0130, 0400/0410 and 0420/0430 messages. If present, it will be removed before forwarding to its destination.

Chip Cryptogram Validation

Issuers performing chip cryptogram validation may include their validation results in *DE 48*, subelement 74 in the Authorization Request Response/0110 message.

Chip Fallback Transaction Downgrade

The Dual Message Authorization System will include *DE 48*, subelement 74 in the Authorization Request Response/0110 message when the transaction is eligible for the Chip Fallback Transaction Downgrade. In which case *DE 48*, subelement 74, subfield 1 value will be "90" and subfield 2 value will be "C". Refer to the *DE 22: Point-of-Service (POS) Entry Mode* section for additional details.

Table 593: Associated edits

When the issuer sends an Authorization Request Response/0110 message containing DE 48, subelement 74 and...	THEN the Dual Message Authorization System...
Subfield 1 does not contain 50	Will remove <i>DE 48</i> , subelement 74 before forwarding the message to the acquirer.
Subfield 2 contains an invalid value	Will replace the invalid subfield 2 value with an X to indicate an unknown issue before forwarding the message to the acquirer.

When the issuer sends an Authorization Request Response/0110 message containing DE 48, subelement 74 and...	THEN the Dual Message Authorization System...
DE 48 (Additional Data—Private Use), subelement 74 (Additional Processing Information) in the Authorization Advice Response/0130 is received from the issuer contains: • Subfield 1 = value 90 (Chip Fallback Transaction Downgrade Process) • Subfield 2 = value C (Completed Successfully)	Removes this data from transaction before forwarding to the acquirer.

Subfield 1 (Processing Indicator)

DE 48, subelement 74, subfield 1 (Processing Indicator) indicates the transaction processing type.

Table 594: Attributes

Attribute	Description
Data Representation	an-2
Data Field	Identifies the service performed
Justification	N/A

Table 595: Values

Value	Description
02	Mastercard On-behalf Service: M/Chip Cryptogram Pre-validation
03	Mastercard On-behalf Service: M/Chip Cryptogram Validation in Stand-In Processing
50	Issuer Chip Validation
90	Chip Fall-back Transaction Downgrade Process

Subfield 2 (Processing Information)

DE 48, subelement 74, subfield 2 (Processing Information) contains additional information about the issue incurred during chip transaction processing.

Table 596: Attributes

Attribute	Description
Data Representation	an-1
Data Field	Additional information being provided about the service
Justification	N/A

Table 597: Values

Value	Description
A	Valid Application Cryptogram (AC); Application Transaction Counter (ATC) outside allowed range
C	Completed Successfully
E	Valid Application Cryptogram; ATC Replay
F	Format error in DE 55, DE 48, subelement 43, or DE 104, subelement 001
G	Application Cryptogram is valid but is not an ARQC
I	Application Cryptogram invalid
K	No matching key file for this PAN, PAN expiry date and KDI combination
T	Application Cryptogram is valid but TVR/CVR was invalid
U	Application Cryptogram could not be validated due to technical error
X	Issuer provided incorrect subfield 2 value
X	Security platform time out
Z	Security platform processing error

Subelement 74 (Valid Subfield 1 and Subfield 2 Value Combinations)

DE 48, subelement 74 (Valid Subfield 1 and Subfield 2 Value Combinations) is the valid DE 48, subelement 74, subfield 1 (Processing Indicator) and subfield 2 (Processing Information) value combinations. The contents of subfield 2 depend on the contents of subfield 1 as described here.

Table 598: Valid combinations

IF subfield 1 contains...	THEN subfield 2 may contain...
02	A, E, F, G, I, K, T, U, X, or Z
03	A, E, F, G, I, K, T, U, X, or Z
50	A, E, F, G, I, T, U, or X
90	C

Subelement 75 (Fraud Scoring Data)

DE 48, subelement 75 (Fraud Scoring Data) indicates the fraud score on a fraud scoring service transaction.

Mastercard provides two scoring solutions: Expert Monitoring for Issuers or Decision Intelligence. A score will be included in Authorization Request/0100 and Authorization Advice/0120: system-generated messages to the issuer when the Expert Monitoring or Decision Intelligence scoring was performed on a transaction.

Table 599: Attributes

Attribute	Description
Subelement ID	75
Subelement Data Length	2 positions
Data Representation	an...32; LLVAR
Data Field	Contents of subfields 1–5
Subfields	5
Justification	See subfields

Usage

Following is the usage of subelement 75 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 600: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	.	X	C
Authorization Advice/0120: system-generated	.	C	C

Values

Contents of subfields 1–2 and applicable subfields 3–5.

Application notes

The Dual Message Authorization System inserts this subelement when one of the Fraud Scoring Services is performed or when both a Fraud Scoring Service and the Fraud Rule Manager Service are performed on the transaction.

NOTE: When a rule adjusted score is provided in subfield 3, at least one or more rule reason code values will be provided in subfields 4–5. However, rule reason code values may be provided in subfields 4 or 5 with or without a rule adjusted score in subfield 3.

NOTE: Issuers participating in a scoring service will receive subfields 1–2. If those issuers choose to participate in the Fraud Rule Manager Service then those issuers must also prepare to receive subfields 3–5.

NOTE: *DE 48*, subelement 75 will not be included in Authorization Request/0100 messages to the issuer when Mastercard is unable to perform the Fraud Scoring Service.

Expert Monitoring for Issuers and Decision Intelligence services evaluate fraud detection elements to produce a score that indicates a level of risk in the transaction. Both services utilize the same *DE 48*, subelement 75 field with dual message authorization transactions.

For additional details on these services, refer to the [Issuer Security Solutions Utilities Guide](#) and [Mastercard Security Solutions Guide for Issuer Services](#).

Subfield 1 (Fraud Score)

DE 48, subelement 75, subfield 1 (Fraud Score) indicates the transaction risk score. The value in *DE 48*, subelement 75, subfield 1 is the model score as indicated in *DE 48*, subelement 75, subfield 3 (Model Score) unless adjusted by SafetyNet or Rules (Customer Rules or On-behalf of Rules).

Table 601: Attributes

Attribute	Description
Subfield ID	01

Attribute	Description
Subfield Data Length	2 positions
Data Representation	an-3
Data Field	Contents of positions 1–3
Justification	N/A

Values

Fraud Scoring System provides the risk score of 000 through 999, where 000 indicates the least likely fraudulent transaction and 999 indicates the most likely fraudulent transaction.

Subfield 2 (Fraud Reason Code)

DE 48, subelement 75, subfield 2 (Fraud Reason Code) indicates the key factors that influenced the fraud score. This value is the model reason code as indicated in DE 48, subelement 75, subfield 04 (Model Reason Code) unless overwritten by SafetyNet or Rules (Customer Rules or On-behalf of Rules).

Table 602: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	an-2
Data Field	Contents of positions 1–2
Justification	N/A

Values

The Fraud Scoring Service provides the score reason code, an alphanumeric code identifying the data used to derive the fraud score.

NOTE: Participating issuers may contact the Risk Solutions team for a list of the specific score reason codes that apply to their institution.

Subfield 3 (Model Score)

DE 48, subelement 75, subfield 3 (Model Score) contains the model risk score. This score is used in Rule Authoring, Rule Discovery, and Decision Intelligence (DI) Dashboard.

Table 603: Attributes

Attribute	Description
Subfield ID	03
Subfield Data Length	2 positions
Data Representation	an-3
Data Field	Contents of positions 1–3
Justification	N/A

Values

The Fraud Rule Manager Service provides a rules score of 000 through 999, where 000 indicates the least likely fraudulent transaction and 999 indicates the most likely fraudulent transaction.

Subfield 4 (Model Reason Code)

DE 48, subelement 75, subfield 4 (Model Reason Code) indicates the key factors that influenced the model fraud score.

Table 604: Attributes

Attribute	Description
Subfield ID	04
Subfield Data Length	2 positions
Data Representation	an-2
Data Field	Contents of positions 1–2
Justification	N/A

Values

The Fraud Rule Manager Service provides the rule reason code, an alphanumeric code where the information provided gives the data used to derive the rules adjusted score.

Subfield 5 (Reserved for Future Use)

DE 48, subelement 75, subfield 5 (Reserved for Future Use) is reserved for future use.

Subelement 76 (Mastercard Electronic Acceptance Indicator)

DE 48, subelement 76 (Mastercard Electronic Acceptance Indicator) identifies that the transaction is a Mastercard Electronic card transaction. It indicates that the acquirer participates or does not participate in Mastercard Electronic card processing.

Table 605: Attributes

Attribute	Description
Subelement ID	76
Subelement Data Length	2 positions
Data Representation	a-1
Data Field	Contents of position 1
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 76 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 606: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	X	C
Authorization Request Response/0110	O	X	.
Authorization Advice/0120: acquirer-generated	O	X	C
Authorization Advice/0120: system-generated	.	X	C

Table 607: Values

Value	Description
C	Mastercard only participant (not considered a Mastercard Electronic card transaction).

Value	Description
E	Acquirer and its merchant both participate in Mastercard Electronic card processing (considered as a Mastercard Electronic transaction).
M	Acquirer participates in Mastercard Electronic card processing, but the merchant that processed this specific transaction does not participate in Mastercard Electronic (considered not to be a Mastercard Electronic transaction).
U	Unidentified acquirer. It is unknown if the acquirer is a Mastercard Electronic card participant.

Application notes

If acquirers do not populate subelement 76 with a valid value or a value is not present, Mastercard will populate *DE 48*, subelement 76 on behalf of the acquirer based on the acquirer's participation in Mastercard Electronic Card Program and forward the transaction to the issuer.

Acquirers

When the merchant participates in the Mastercard Electronic Card Program, the participating acquirer should send *DE 48*, subelement 76 with a value of E in the Authorization Request/0100 message.

When the merchant does not participate in the Mastercard Electronic Card Program, the participating acquirer should send *DE 48*, subelement 76 with a value of M in the Authorization Request/0100 message.

When participating acquirers do not provide a value or provide an incorrect value in subelement 76, Mastercard will default subelement 76 to the value of E in the Authorization Request/0100 message.

Issuers

Issuers participating in Mastercard Electronic Card must be prepared to receive *DE 48*, subelement 76 with values of C, E, or U in the Authorization Request/0100 and Authorization Advice/0120 messages.

Issuers participating in Mastercard Electronic Card must be prepared to send *DE 48*, subelement 76 with values of C, E, or U in Authorization Request Response/0110 messages.

NOTE: Mastercard will convert the subelement value of M to the value C in the Authorization Request/0100 sent to the issuer. Therefore, the issuer will not receive a value of M.

Subelement 77 (Transaction Type Identifier)

DE 48, subelement 77 (Transaction Type Identifier) indicates the type of additional transaction purpose.

NOTE: All Mastercard Send Transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions) require the presence of this identifier.

Table 608: Attributes

Attribute	Description
Subelement ID	77
Subelement Data Length	2 positions
Data Representation	an-3
Data Field	Contents of positions 1-3
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 77 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

NOTE: This subelement is mandatory when DE 18=MCC is any of the following: MCC 4829, MCC 6536, MCC 6537, MCC 6538, and MCC 6540.

Table 609: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Request Response/0110	CE	.	C
Authorization Advice/0120: system-generated	.	X	C
Reversal Request/0400	C	.	C
Reversal Request Response/0410	CE	.	C
Reversal Advice/0420	.	X	C

Table 610: Values

Value	Description
C04	Gaming Re-pay
C07	General Person-to-Person Transfer
C51	Mastercard Send Indicator (Reserved for Future Use)
C52	General Transfer to Own Account
C53	Agent Cash Out
C54	Payment of Own Credit Card Bill
C55	Business Disbursement
C56	Government/Non-Profit Disbursement
C57	Rapid Merchant Settlement
C58	Cash in at ATM
C59	Cash in at point of Sale
C60	Fast Refund to Original Card
C61	Mastercard Send Indicator (Reserved for Future Use)
C62	Mastercard Send Indicator (Reserved for Future Use)
C63	Mastercard Send Indicator (Reserved for Future Use)
C64	Mastercard Send Indicator (Reserved for Future Use)
C65	General Business to Business Transfer
C66	Mastercard Send Indicator (Reserved for Future Use)
F07	General Person-to-Person Transfer
F08	Person-to-Person Transfer to Card Account
F52	General Transfer to Own Account
F53	Agent Cash Out
F54	Payment of Own Credit Card Bill
F55	Business Disbursement
F61	Transfer to Own Staged Digital Wallet Account
F64	Transfer to Own Debit or Prepaid Account
F65	General Business-to-Business Transfer
P10	Purchase repayment

Value	Description
P11	Business-to-Business Payment

Application notes

If *DE 3*, subfield 1 contains value 28 (Payment Transaction), then *DE 48*, subelement 77 must be present.

If *DE 3*, subfield 1 contains value 00 (Purchase of Goods and Services) and *DE 18* contains values of 4829 (Money Transfer), 6538 (Mastercard Send Funding) or 6540 (Funding Transaction), then *DE 48*, subelement 77 must be present.

Usage of value C04 is limited to eligible acquirers and issuers in eligible countries. Refer to the [Mastercard Gaming and Gambling Payments Program Standards](#).

Usage of value C07 and payment type codes C52–C59, C60–C61, C64–C65, and F07–F65 is limited to eligible countries and eligible acquirers. Refer to the [Mastercard Send and Funding Transactions Program Standards](#).

NOTE: Usage of value C67 is limited to Mastercard Merchant Presented QR transactions. Usage of value C68 is limited to Mastercard Merchant Presented QR refund transactions.

Usage of value P11 (Business-to-Business Payment) is intended to identify transactions between a buyer and a supplier who does not accept card-based payments, via a Business Payment Service Provider who remits the payment to the supplier.

Subelement 78 (Payment Service Indicators [Visa Only])

DE 48, subelement 78 (Payment Service Indicators [Visa Only]) contains subfields which represent various Visa payment service indicators.

Table 611: Attributes

Attribute	Description
Subelement ID	78
Subelement Data Length	2 positions
Data Representation	ans-6
Data Field	Contents of positions 1–6
Subfields	6
Justification	See subfields

Usage

Following is the usage of subelement 78 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 612: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Request Response/0110	C	.	C
Reversal Request/0400	C	.	C
Reversal Request Response/0410	C	.	C

Values

See subfields.

Application notes

Acquirers are not required to populate this subelement in the Authorization Request/0100 message if they do not support any of the Visa payment indicators in subfields 2 through 6 as noted in the following sections.

Subfield 1 (Spend Qualified Indicator)

DE 48, subelement 78, subfield 1 (Spend Qualified Indicator) provides the Visa spend qualification assessment.

Table 613: Attributes

Attribute	Description
Data Representation	ans-1
Data Field	Contents of position 1
Justification	N/A

Table 614: Values

Value	Description
Space	Space will be sent back in the Authorization Request Response/0110 message to the acquirer when Visa field 62.25 is not present in the Visa 0110 message response

Value	Description
N	Spend assessment threshold defined by Visa has not been met
B	Basic spend assessment threshold level met
Q	Highest spend assessment threshold level met

Application notes

The value for this field is mapped from Visa Field 62.25—Spend Qualified Indicator. Refer to Visa's *VisaNet Authorization-Only Online Messages Technical Specifications*. Acquirers will need to populate subfield 1 with a space if any other Payment Service Indicator is submitted in the Authorization Request/0100 and Reversal Request/0400 message. Space will be sent back in subfield 1 to the acquirer if any other Payment Service Indicator is returned back in Authorization Request Response/0110 and Reversal Request Response/0410 messages.

NOTE: The Spend Qualified Indicator is used by acquirers for Visa's Account Level Processing (ALP) program.

Subfield 2 (Dynamic Currency Conversion Indicator)

DE 48, subelement 78, subfield 2 (Dynamic Currency Conversion Indicator) indicates a transaction for which Dynamic Currency Conversion (DCC) is performed by the merchant at the point of sale.

Table 615: Attributes

Attribute	Description
Data Representation	ans-1
Data Field	Contents of position 2
Justification	N/A

Table 616: Values

Value	Description
Y	Dynamic Currency Conversion was performed at the point of sale.
Space	No Dynamic Currency Conversion or not echoed in the Authorization Request Response/0110 or Reversal Request Response/0410 message.

Application notes

If the merchant performs dynamic currency conversion at the point of sale, acquirers must send this value in the Authorization Request/0100 message and in Reversal Request/0400 message in case of Reversal. This value will be mapped to Visa Field 126.19—Dynamic Currency Conversion Indicator. Refer to Visa's *VisaNet Authorization-Only Online Messages Technical Specifications* manual. The acquirer will need to populate subfield 2 with a space if DCC was not performed and any other Payment Service Indicator is submitted in the Authorization Request/0100. Space will be sent back in subfield 2 to the acquirer if any other Payment Service Indicator is returned back in Authorization Request Response/0110 message.

NOTE: Dynamic Currency Conversion (DCC) provides the cardholder the option to pay for goods or services in their own billing currency. DCC occurs when a merchant performs currency conversion locally and submits the transaction in the cardholder's billing currency. Acquirers must ensure that they receive an indicator from their merchants when DCC is performed for a Visa transaction.

Subfield 3 (U.S. Deferred Billing Indicator)

DE 48, subelement 78, subfield 3 (U.S. Deferred Billing Indicator) indicates a transaction for which the billing for merchandise occurred after the merchandise was delivered to the cardholder. This applies to U.S. region acquirers only.

Table 617: Attributes

Attribute	Description
Data Representation	ans-1
Data Field	Contents of position 3
Justification	N/A

Table 618: Values

Value	Description
D	U.S. Deferred Billing Indicator
Space	No Deferred Billing or not echoed in Authorization Request Response/0110 message

Application notes

This code is provided by the merchant through U.S. region acquirers in the Authorization Request/0100 message to indicate that a Visa card is to be billed on a deferred basis, that is, the cardholder is to be billed for merchandise already received. This value will be mapped to Visa Field 126.12—Service Indicators, position 3 (Deferred Billing Indicator). Refer to the Visa's

VisaNet Authorization-Only Online Messages Technical Specifications. Acquirers will need to populate subfield 3 with a "space" if Deferred Billing is not applicable and any other Payment Service Indicator is submitted in the Authorization Request/0100 and Reversal Request /0400 message. "Space" will be sent back in subfield 3 to the acquirer if any other Payment Service Indicator is returned back in Authorization Request Response/0110 and Reversal Request Response/0410 messages.

Subfield 4 (Visa Checkout Indicator)

DE 48, subelement 78, subfield 4 (Visa Checkout Indicator) indicates that the transaction was processed through Visa Checkout.

Table 619: Attributes

Attribute	Description
Data Representation	ans-1
Data Field	Contents of position 4
Justification	N/A

Table 620: Values

Value	Description
Y	Transaction is processed through Visa Checkout
Space	No Visa Checkout or not echoed in the Authorization Request Response/0110 message

Application notes

If the transaction is performed through Visa Checkout, acquirers must send this value in the Authorization Request/0100 message. This value will be mapped to Visa Field 126.18—Agent Unique Account Result. Refer to the Visa's *VisaNet Authorization-Only Online Messages Technical Specifications*. The acquirer will need to populate subfield 4 with a space if Visa Checkout was not performed and any other Payment Service Indicator is submitted in the Authorization Request/0100 and Reversal Request/0400 messages. "Space" will be sent back in subfield 4 to the acquirer if any other Payment Service Indicator is returned back in Authorization Request Response/0110 and Reversal Request Response/0410 messages.

Subfield 5 (Message Reason Code)

DE 48, subelement 78, subfield 5 (Message Reason Code) provides the message reason code of the transaction.

Table 621: Attributes

Attribute	Description
Data Representation	ans-1
Data Field	Contents of position 5
Justification	N/A

Table 622: Values

Value	Description
Space	Default (No message reason code associated with this transaction)
0	Incremental authorization
1	Resubmission
2	Delayed charges
3	Reauthorization
4	No show
5	Account top up
6	Deferred Authorization

Application notes

Acquirers must send the value based on the transaction in the Authorization Request/0100 and Reversal Request/0400 message. This value will be mapped to Visa Field 63.3 Message Reason Code. Refer to Visa's *VisaNet Authorization-Only Online Messages Technical Specifications*.

Subfield 6 (Token Response Information)

DE 48, subelement 78, subfield 6 (Token Response Information) provides the Token Response Information received for Visa-branded authorization transactions from VisaNet (VIP) in the authorization request response.

Table 623: Attributes

Attribute	Description
Data representation	ans-1
Data field	Contents of position 6
Justification	N/A

Table 624: Values

Value	Description
Space	N/A
1	Token program

Application notes

The value for this field is provided by Visa in the Authorization Request Response/0110 message. This field contains a value that can identify transactions eligible for token services. Acquirers will need to retain the value in this field from authorization request response messages and send it in clearing transactions to Visa. Refer to Visa's *VisaNet Authorization-Only Online Messages Technical Specifications*.

Subelement 79 (Chip CVR/TVR Bit Error Results)

DE 48, subelement 79 (Chip CVR/TVR Bit Error Results) provides the Terminal Verification Results (TVR) and Card Verification Results (CVR) bitmask, and expected values registered by the issuer. This serves as notification of bit validation errors detected in the CVR/TVR within the Issuer Application data during M/Chip Cryptogram Validation processing.

Table 625: Attributes

Attribute	Description
Subelement ID	79
Subelement Data Length	2 positions

Attribute	Description
Data Representation	an...50; LLVAR The "LL" length field of LLVAR must be an integral multiple of 5, not to exceed 50.
Data Field	Contents of subfields 1–4
Subfields	4
Justification	N/A

Usage

Following is the usage of subelement 79 (whether it is mandatory, conditional, optional, system provided, or not required) in all applicable messages.

Table 626: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	•	X	C
Authorization Advice/0120: system-generated	•	X	C

Values

See subfields.

Application notes

The Dual Message Authorization System will stop verification of CVR/TVR bits when 10 errors are detected.

Refer to the following example. In the example:

- The binary representation of the TVR Bit Mask for Byte 3 is "10101000"
- The binary representation of the Expected Result for Byte 3 is "00000000"
- The binary representation of the Validation Result for Byte 3 is "10000000"

The Validation Result indicates that the cardholder verification was not successful.

Refer to the *M/Chip Processing Services: Service Description* document for recommended bit mask settings.

Table 627: Binary representations

	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5
TVR Bit Mask: Hex	00	00	A8	00	00
Expected Result	00	00	00	00	00
Validation Result	00	00	80	00	00

Table 628: Subfields

Subfield	Description	Value
Subfield 1	CVR or TVR Identifier	T
Subfield 2	Byte ID	03
Subfield 3	Bit Identifier	8
Subfield 4	Value of Bit in Error	1

Subfield 1 (CVR or TVR Identifier)

DE 48, subelement 79, subfield 1 (CVR or TVR Identifier) indicates whether bit reported in error is part of CVR or TVR.

Table 629: Attributes

Attribute	Description
Data Representation	an-1
Data Field	Contents of position 1
Justification	N/A

Table 630: Values

Value	Description
C	CVR
T	TVR

Subfield 2 (Byte ID)

DE 48, subelement 79, subfield 2 (Byte ID) identifies the byte number of the associated bit reported in error.

Table 631: Attributes

Attribute	Description
Data Representation	an-2
Data Field	Contents of positions 2–3
Justification	N/A

Values

01–99

Subfield 3 (Byte Identifier)

DE 48, subelement 79, subfield 3 (Byte Identifier) identifies the bit number in error within the byte identified in subfield 2.

Table 632: Attributes

Attribute	Description
Data Representation	an-1
Data Field	Contents of position 4
Justification	N/A

Values

1–8

Subfield 4 (Value of Bit in Error)

DE 48, subelement 79, subfield 4 (Value of Bit in Error) identifies the value of the bit in error that was submitted in transaction.

Table 633: Attributes

Attribute	Description
Data Representation	an-1
Data Field	Contents of position 5

Attribute	Description
Justification	N/A

Table 634: Values

Value	Description
0	The bit is turned off
1	The bit is turned on

Subelement 80 (PIN Service Code)

DE 48, subelement 80 (PIN Service Code) indicates the results of PIN processing by the Dual Message Authorization System.

Table 635: Attributes

Attribute	Description
Subelement ID	80
Subelement Data Length	2 positions
Data Representation	a-2
Data Field	Contents of positions 1–2
Subfields	N/A
Justification	N/A

Usage

Subelement 80 is provided by the Dual Message Authorization System in the Authorization Request/0100 message whenever PIN data is present in the Authorization Request/0100 message.

Following is the usage of subelement 80 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 636: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	.	X	C

Message	Org	Sys	Dst
Authorization Request Response/0110	CE	.	C
Authorization Advice/0120: system-generated	C	.	C

Table 637: Values

Value	Description
PD	The Dual Message Authorization System dropped the PIN (Mastercard Use Only for Credit Transactions with PIN)
PV	The Dual Message Authorization System verified the PIN.
TV	The Dual Message Authorization System translated the PIN for issuer verification.
PI	The Dual Message Authorization System was unable to verify the PIN.
TI	The Dual Message Authorization System was unable to translate the PIN.

Subelement 82 (Address Verification Service Request)

DE 48, subelement 82 (Address Verification Service Request) indicates that verification of the cardholder billing address is requested in the authorization message.

Table 638: Attributes

Attribute	Description
Subelement ID	82
Subelement Data Length	2 positions
Data Representation	n-2
Data Field	Contents of positions 1–2
Subfields	N/A
Justification	N/A

Usage

Subelement 82 must be present in the authorization request message whenever cardholder address verification is requested by the acquirer.

Following is the usage of subelement 82 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 639: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Request Response/0110	CE	.	CE
Authorization Advice/0120: system-generated	.	X	C

Table 640: Values

Value	Description
52	AVS and Authorization Request/0100

Application notes

Mastercard all issuing and acquiring processors to code for, support, and integrate into their systems the data elements, subelements, and values associated with this item.

For additional information about the requirement, refer to the *Global Safety and Security Standards Roadmap*.

NOTE: Mastercard does not require acquirers or issuers to enroll in these optional products and services, but they must code their systems to support these fields to leverage such products and services in a timely manner in the event of a security issue.

Subelement 83 (Address Verification Service Response)

DE 48, subelement 83 (Address Verification Service Response) contains the AVS verification response code in the Authorization Request Response/0110 message.

Table 641: Attributes

Attribute	Description
Subelement ID	83
Subelement Data Length	2 positions
Data Representation	a-1
Data Field	Contents of position 1
Subfields	N/A

Attribute	Description
Justification	N/A

Usage

Subelement 83 must be provided by the issuer in the Authorization Request Response/0110 message whenever AVS is requested by the acquirer.

Following is the usage of subelement 83 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 642: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	C	.	C
Authorization Advice/0120: system-generated	.	X	C

Table 643: Values

Value	Description
A	Address matches, postal code does not.
B	Visa only. Street address match. Postal code not verified because of incompatible formats. (Acquirer sent both street address and postal code.)
C	Visa only. Street address and postal code not verified because of incompatible formats. (Acquirer sent both street address and postal code.)
D	Visa only. Street address and postal code match.
F	Visa only. Street address and postal code match. Applies to U.K. only.
G	Visa only. Non-AVS participant outside the U.S.; address not verified for international transaction.
I	Visa only. Address information not verified for international transaction.
M	Visa only. Street addresses and postal code match.
N	Neither address nor postal code matches.
P	Visa only. Postal codes match. Street address not verified because of incompatible formats. (Acquirer sent both street address and postal code.)
R	Retry, system unable to process.

Value	Description
S	AVS currently not supported. U.S. issuers will not be able to send through this response code.
U	No data from issuer/Dual Message Authorization System
W	For U.S. addresses, nine-digit postal code matches, address does not; for address outside the U.S., postal code matches, address does not.
X	For U.S. addresses, nine-digit postal code and address matches; for addresses outside the U.S., postal code and address match.
Y	For U.S. addresses, five-digit postal code and address matches.
Z	For U.S. addresses, five-digit postal code matches, address does not.

Application notes

Mastercard all issuing and acquiring processors to code for, support, and integrate into their systems the data elements, subelements, and values associated with this item.

For additional information about the requirement, refer to the *Global Safety and Security Standards Roadmap*.

NOTE: Mastercard does not require acquirers or issuers to enroll in these optional products and services, but they must code their systems to support these fields to leverage such products and services in a timely manner in the event of a security issue.

Subelement 84 (Merchant Advice Code)

DE 48, subelement 84 (Merchant Advice Code) contains the merchant advice code.

Table 644: Attributes

Attribute	Description
Subelement ID	84
Subelement Data Length	2 positions
Data Representation	an-2
Data Field	Contents of positions 1–2
Subfields	N/A
Justification	N/A

Usage

Subelement 84 is optionally provided by the issuer in the Authorization Request Response/0110 message.

Following is the usage of subelement 84 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 645: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	O	X	O
Authorization Advice/0120: system-generated	.	C	C
Reversal Request Response/0410	O	.	O

Table 646: Values

Value	Description
01	New account information available
02	Cannot approve at this time, try again later
03	Do not try again
04	Token requirements not fulfilled for this token type
05	Negotiated value not approved
21	Payment Cancellation (Mastercard use only)
22	Merchant does not qualify for product code
24	Retry after 1 hour (Mastercard use only)
25	Retry after 24 hours (Mastercard use only)
26	Retry after 2 days (Mastercard use only)
27	Retry after 4 days (Mastercard use only)
28	Retry after 6 days (Mastercard use only)
29	Retry after 8 days (Mastercard use only)
30	Retry after 10 days (Mastercard use only)
40	Consumer non-reloadable prepaid card
41	Consumer single-use virtual card number
42	Sanctions Scoring Service: Score Exceeds Applicable Threshold Value

Value	Description
43	Consumer multi-use virtual card number

Application notes

Merchant Advice Code value of 43 (Consumer Multi-Use Virtual Card Number) may be used on all approval Authorization Request Response/0110 messages and decline Authorization Request Response/0110 messages only when DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status) contains a value of 8 (Account Status Inquiry Service [ASI]).

Subelement 84 (Visa Response Codes (Visa Only))

DE 48, subelement 84 (Visa Response Codes [Visa Only]) contains the Visa Field 39 (Response Codes) as received from Visa.

Table 647: Attributes

Attribute	Description
Subelement ID	84
Subelement Data Length	2 positions
Data Representation	an-2
Data Field	Contents of position 1–2
Subfields	N/A
Justification	N/A

Usage

Whether use is mandatory, conditional, optional, or not required is noted for each applicable message.

Table 648: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	C	.	C
Authorization Reversal Response/0410	C	.	C

Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for the list of valid values and definitions.

Subelement 84 (American Express Response Codes (American Express Only))

DE 48, subelement 84 (American Express Response Codes [American Express only]) contains the American Express Data Field 39 (Action Codes) as received from American Express.

Table 649: Attributes

Attribute	Description
Subelement ID	84
Subelement Data Length	2 positions
Data Representation	an-3
Data Field	Contents of positions 1-3
Subfields	N/A
Justification	N/A

Usage

Whether use is mandatory, conditional, optional, or not required is noted for each applicable message.

Table 650: Applicable messages

Attributes	Org	Sys	Dst
Authorization Request Response/0110	C	.	C

Values

Refer to *American Express Global Credit Authorization Guide (ISO)* for further instructions.

Subelement 85 (Account Status (Visa Only))

DE 48, subelement 85 (Account Status [Visa Only]) identifies the account range as regulated or non-regulated interchange.

Table 651: Attributes

Attribute	Description
Subelement ID	85
Subelement Length	2 positions
Data Representation	a-1
Data Field	Contents of position 1
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 85 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 652: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	C	.	C

Table 653: Values

Value	Description
R	Account is regulated
N	Account is non-regulated

Subelement 86 (Relationship Participant Indicator (Visa Only))

DE 48, subelement 86 (Relationship Participant Indicator) indicates the transaction originates from a cardholder-merchant relationship.

Table 654: Attributes

Attribute	Description
Subelement ID	86
Subelement Length	2 positions
Data Representation	a-1
Data Field	Contents of position 1
Subfields	N/A
Justification	N/A

Usage

Subelement 86 contains an indicator to denote a transaction on a Visa account for a cardholder with a merchant relationship.

Following is the usage of subelement 86 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 655: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	•	C
Authorization Request Response/0110	CE	•	CE
Reversal Request/0400	CE	•	CE

Table 656: Values

Value	Description
C	Credential on File
I	Installment Payment
R	Merchant/Cardholder Periodic Billing

Application notes

Visa Credential-on-File transactions: DE 48, subelement 86 value of C (Credential on File) is conditional when DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) is a value of 10 (Credential on File). Acquirers must submit a value of C when a merchant is initiating the first transaction in a series on behalf of the cardholder using credentials stored on file. For an unscheduled credential-on-file that is merchant-initiated transaction and does not occur at a scheduled interval, acquirers must submit a value of C. Stored credential transactions initiated by cardholders must not submit value C.

Subelement 87 (Card Validation Code Result)

DE 48, subelement 87 (Card Validation Code Result) indicates the CVC 1, CVC 2, or CVC 3 result code.

Table 657: Attributes

Attribute	Description
Subelement ID	87
Subelement Data Length	2 positions
Data Representation	a-1
Data Field	Contents of position 1
Subfields	N/A
Justification	N/A

Usage

Subelement 87 must be provided by the issuer in the Authorization Request Response/0110 message whenever CVC 2 verification is requested by the acquirer.

Subelement 87 is optional whenever DE 45 (Track 1 Data) or DE 35 (Track 2 Data) is present in the Authorization Request/0100 message and CVC 1 is invalid.

Subelement 87 is optionally provided in the Authorization Request Response/0110 message when an issue is encountered during CVC 3 validation.

Following is the usage of subelement 87 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 658: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	.	X	C

Message	Org	Sys	Dst
Authorization Request Response/0110	C	.	C
Authorization Advice/0120: system-generated	.	X	C

Values

Table 659: CVC 1

Value	Description
Y	Invalid CVC 1 (only if DE 35 (Track 2 Data) or DE 45 (Track 1 Data) is present in the Authorization Request/0100 message.) Refer to Application notes section that follows for Stand-In System considerations.

Table 660: CVC 2

Value	Description
M	Valid CVC 2 (match)
N	Invalid CVC 2 (non-match)
P	CVC 2 not processed (issuer temporarily unavailable)
U	CVC 2 Unverified: Mastercard Use Only

Table 661: CVC 3

Value	Description
E	Length of unpredictable number was not a valid length
P	Unable to process
Y	Invalid

Table 662: American Express CID (Card Identifier)

Refer to DE 48 (Additional Data: Private Use), Subelement 95: American Express CID Card Identifier (American Express Only).

Value	Description
Y	Valid CID (match)

Value	Description
N	Invalid CID (non-match)
U	CID not processed

Application notes

If the issuer participates in the CVC 1 validation service and the transaction meets the criteria for the CVC 1 test, the Stand-In System considers the results of the CVC 1 validation in subelement 87 when responding to the Authorization Request/0100 message. The issuer can control the response that the Stand-In System uses when the CVC 1 value is invalid or validation cannot be performed. The default value is 05 (Do Not Honor). To change these values, contact Key Management Services at key_management@Mastercard.com.

If an issuer's BIN is temporarily unable to receive the CVC 2 value, the Dual Message System Authorization inserts the value: U = CVC 2 unverified (Mastercard use only).

Subelement 87 (CVV2 Response [Visa Only])

DE 48, subelement 87 (CVV2 Response) indicates the CVV2 response on a Visa account.

Table 663: Attributes

Attribute	Description
Subelement ID	87
Subelement Data Length	2 positions
Data Representation	a-1
Data Field	Contents of position 1
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 87 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 664: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	C	.	C

Table 665: Values

Value	Description
M	CVV2 match
N	CVV2 no match
P	Not processed
S	CVV2 is on the card, but the merchant has indicated that CVV2 is not present.
U	Issuer is not Visa-certified for CVV2, has not provided Visa encryption keys, or both.

Subelement 88 (Magnetic Stripe Compliance Status Indicator)

DE 48, subelement 88 (Magnetic Stripe Compliance Status Indicator) indicates that the Dual Message Authorization System replaced the DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) value of 90 or 91 with value of 02.

Table 666: Attributes

Attribute	Description
Subelement ID	88
Subelement Data Length	2 positions
Data Representation	a-1
Data Field	Contents of position 1
Subfields	N/A
Justification	N/A

Usage

Subelement 88 is provided by the Dual Message Authorization System whenever the Dual Message Authorization System replaces the DE 22, subfield 1, value of 90 or 91 to a value of 02 due to magnetic stripe compliance downgrade of the acquirer.

Following is the usage of subelement 88 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 667: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	.	X	C
Authorization Request Response/0110	CE	X	C
Authorization Advice/0120: system-generated	.	X	C

Table 668: Values

Value	Description
Y	Dual Message Authorization System replaced DE 22, subfield 1, value 90 or 91 with value 02.

Subelement 89 (Magnetic Stripe Compliance Error Indicator)

DE 48, subelement 89 (Magnetic Stripe Compliance Error Indicator) indicates magnetic stripe compliance errors.

Table 669: Attributes

Attribute	Description
Subelement ID	89
Subelement Data Length	2 positions
Data Representation	a-1
Data Field	Contents of position 1
Subfields	N/A
Justification	N/A

Usage

Subelement 89 is provided by the Dual Message Authorization System whenever errors are detected while editing transactions for magnetic stripe compliance. Following is the usage of subelement 89 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 670: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	•	X	C
Authorization Request Response/0110	CE	X	C
Authorization Advice/0120: system-generated	•	X	C

Table 671: Values

Value	Description
A	Track 1 or Track 2 not present in the message
B	Track 1 and Track 2 present in the message
C	DE 2 (Primary Account Number [PAN]) not equal in track data
D	DE 14 (Expiration Date) not equal in track data
E	Service code invalid in track data
F	Field separator(s) invalid in track data
G	A field within the track data has an invalid length
H	DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) is 80, 90, or 91 when DE 48, Transaction Category Code (TCC) is T
I	DE 61 (Point-of-Service Data), subfield 4 (POS Cardholder Presence) is 1, 2, 3, 4, or 5
J	DE 61 Point-of-Service Data), subfield 5 (POS Card Presence) is 1

Subelement 90 (Lodging and Auto Rental Indicator)

DE 48, subelement 90 (Lodging and Auto Rental Indicator) indicates the presence of Lodging and Auto Rental Service interchange program.

Table 672: Attributes

Attribute	Description
Subelement ID	90
Subelement Data Length	2 positions
Data Representation	α-1
Data Field	Contents of position 1

Attribute	Description
Subfields	N/A
Justification	N/A

Usage

Subelement 90 is provided whenever the cardholder is a preferred customer of the merchant.

Following is the usage of subelement 90 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 673: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Request Response/0110	CE	.	CE
Authorization Advice/0120: system-generated	.	C	C

Table 674: Values

Value	Description
P	Cardholder is enrolled in a merchant preferred customer program and magnetic stripe data may be absent

Subelement 90 (Custom Payment Service Request [Visa Only])

DE 48, subelement 90 (Custom Payment Service Request [Visa Only]) contains the Authorization Characteristics Indicator (Visa Field 62.1).

Table 675: Attributes

Attribute	Description
Subelement ID	90
Subelement Data Length	2 positions
Data Representation	a-1
Data Field	Contents of position 1
Subfields	N/A

Attribute	Description
Justification	N/A

Usage

Subelement 90 is used to indicate a custom payment service transaction on a Visa account whenever applicable.

Following is the usage of subelement 90 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 676: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Reversal Request/0400	C	.	C

Table 677: Values

Value	Description
I	Incremental authorization
P	Preferred customer
R	Recurring payment
Y	Request for Custom Payment Service participation

Application notes

Refer to Visa's *VisaNet Authorization-Only Online Messages Technical Specifications* for a list of all customer request values in Visa field 62.1.

Subelement 90 (Custom Payment Service Request Response [Visa Only])

DE 48, subelement 90 (Custom Payment Service Request Response) contains the Authorization Characteristics Indicator (Visa Field 62.1).

Table 678: Attributes

Attribute	Description
Subelement ID	90

Attribute	Description
Subelement Data Length	2 positions
Data Representation	a-1
Data Field	Contents of position 1
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 90 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 679: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	C	.	C
Reversal Request Response/0410	C	.	C

Values

Acquirer receives Visa customer response code. See Application notes.

Application notes

If the request qualifies for Visa customer and is approved, refer to the Visa's *VisaNet Authorization-Only Online Messages Technical Specifications* for a complete list of customer-qualified codes in Visa Field 62.1. If the original request does not qualify for customer, Visa returns an "N" or "T" in the response.

Subelement 91 (Acquirer Reference Data [American Express Only])

DE 48, subelement 91 (Acquirer Reference Data) contains the 15-digit Transaction Identifier (TID), a unique American Express tracking number. The TID is used to identify and track a card customer transaction throughout its life cycle.

Table 680: Attributes

Attribute	Description
Subelement ID	91
Subelement Data Length	2 positions

Attribute	Description
Data Representation	ans...15; LLVAR
Data Field	Contents of positions 1–15
Subfields	N/A

Usage

Following is the usage of subelement 91 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 681: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	C	.	C

Values

A unique 15 digit TID.

Subelement 91 (Custom Payment Service Request/Transaction ID [Visa Only])

DE 48, subelement 91 (Custom Payment Service Request/Transaction ID) indicates the presence of Custom Payment Service Request response data.

Table 682: Attributes

Attribute	Description
Subelement ID	91
Subelement Data Length	2 positions
Data Representation	an...19; LLVAR
Data Field	Contents of positions 1–19
Subfields	N/A

Usage

Subelement 91 is used to provide transaction ID and/or Validation Code when subelement 90 indicates an incremental authorization or when submitting a reversal on a Visa account.

Following is the usage of subelement 91 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 683: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Reversal Request/0400	C	.	C

Values

Subelement 91 must contain the transaction ID (Visa field 62.2, 15 byte numeric) and/or validation code (Visa field 62.3, 4 bytes alphanumeric) when subelement 90 indicates an incremental authorization or when submitting a reversal on a Visa account.

Subelement 91 (Custom Payment Service Response/Transaction ID [Visa Only])

DE 48, subelement 91 (Custom Payment Service Response/Transaction ID) provides authorization response data when custom payment service (subelement 90) is requested on a Visa account.

Table 684: Attributes

Attribute	Description
Subelement ID	91
Subelement Data Length	2 positions
Data Representation	ans...19; LLVAR
Data Field	Contents of positions 1–4, 1–15, or 1–19 depending on the length field
Subfields	N/A

Usage

Following is the usage of subelement 91 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 685: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	C	.	C
Reversal Request Response/0410	C	.	C

Table 686: Values

Length Field Value	Subelement 91 Value
When length field is 04:	Subelement 91 contains the Visa customer validation code (Visa field 62.3, four bytes alphanumeric). Refer to the Visa's <i>VisaNet Authorization-Only Online Messages Technical Specifications</i> for more information on Visa customer validation codes.
When length field is 15:	Subelement 91 contains the transaction ID (Visa field 62.2, 15 bytes numeric).
When length field is 19:	Subelement 91 contains the transaction ID (Visa field 62.2, 15 byte numeric) and customer validation code (Visa field 62.3, four bytes alphanumeric).

Subelement 92 (CVC 2)

DE 48, subelement 92 (CVC 2) contains the CVC 2 value from the signature panel of the card.

Table 687: Attributes

Attribute	Description
Subelement ID	92
Subelement Data Length	2 positions
Data Representation	n-3
Data Field	Contents of positions 1–3
Subfields	N/A
Justification	N/A

Usage

Subelement 92 contains the CVC 2 value from the signature panel of the card when applicable.

Following is the usage of subelement 92 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 688: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C

Message	Org	Sys	Dst
Authorization Request Response/0110	CE	.	CE
Authorization Advice/0120	.	C	C
Authorization Advice/0120: acquirer-generated	C	.	C

Values

Acquirers must not use 000 as a default when sending this subelement. Acquirers must only provide this subelement when requesting CVC 2 verification.

Application notes

The contents of DE 48, subelement 92—when populated—are passed to the Dual Message System (Authorization) issuer in transactions acquired through the Single Message System.

For Mastercard Digital Enablement Service (MDES) transactions, DE 48, subelement 92 must be populated when provided by the merchant; it might contain a Dynamic Token Verification code.

Subelement 92 (CVV2 Data (Visa Only))

DE 48, subelement 92 (CVV2 Data) consists of the Visa CVV2 presence ID, response type, and CVV2 value.

Table 689: Attributes

Attribute	Description
Subelement ID	92
Subelement Data Length	2 positions
Data Representation	ns-6
Data Field	Contents of positions 1, 2, and 3–6
Subfields	3
Justification	See subfields

Usage

Subelement 92 contains the CVV2 value from the signature panel of the card when applicable.

Following is the usage of subelement 92 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 690: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Request Response/0110	CE	.	CE

Values

Position 1 (CVV2 Presence ID)

Table 691: Attributes

Attribute	Description
Data Representation	n-1
Data Field	Contents of position 1

Table 692: Values

Value	Description
0	Merchant did not provide CVV2 or it was deliberately bypassed
1	CVV2 value present
2	CVV2 is on card, but not legible
9	Cardholder states no CVV2 is on card

Position 2 (CVV2 Response Code)

Table 693: Attributes

Attribute	Description
Data Representation	n-1
Data Field	Contents of position 2

Table 694: Values

Value	Description
0	Only the normal response code in DE 39 should be returned by the issuer
1	The normal response code and CVV2 response code should be returned by the issuer

Positions 3–6 (CVV2 Value)

Table 695: Attributes

Attribute	Description
Data Representation	ns-4
Data Field	Contents of positions 3–6
Justification	Right, blank-filled

Values

CVV2 value from the signature panel of the card. Blank or space-fill as needed. No other special character is allowed.

Subelement 93 (Fleet Card ID Request Data (Visa Only))

DE 48, subelement 93 (Fleet Card ID Request Data) contains the Fleet Card ID information on a Visa account, when applicable.

Table 696: Attributes

Attribute	Description
Subelement ID	93
Subelement Data Length	2 positions
Data Representation	ans...19; LLVAR
Data Field	Contents of subfields
Subfields	2
Justification	N/A

Usage

Following is the usage of subelement 93 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 697: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C

Values

See subfields.

Subfield 1 (Fleet Card ID Request Indicator)

DE 48, subelement 93, subfield 1 (Fleet Card ID Request Indicator) identifies the Fleet Card ID.

Table 698: Attributes

Attribute	Description
Data Representation	ans-2
Data Field	Contents of position 1–2

Values

\$\$ is Fleet Card ID

Subfield 2 (Optional Free-form Informational Text)

DE 48, subelement 93, subfield 2 (Optional Free-form Informational Text) provides additional Point-of-Service (POS) information.

Table 699: Attributes

Attribute	Description
Data Representation	ans...17
Data Field	Contents of positions 3–19

Values

Additional Point-of-Service (POS) information (optional).

Subelement 94 (Commercial Card Inquiry Request (Visa Only))

DE 48, subelement 94 (Commercial Card Inquiry Request) contains an indicator requesting a commercial card inquiry on a Visa account, when applicable.

Table 700: Attributes

Attribute	Description
Subelement ID	94
Subelement Data Length	2 positions
Data Representation	ans-4
Data Field	Contents of positions
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 94 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 701: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C

Values

Table 702: Positions 1–3 (Card Request Indicator)

Attribute	Description
Data Representation	ans-3
Data Field	Contents of positions 1–3 (Hexadecimal value: 5AF0F1)
Values	0 (where 0 is zero) = Commercial Card Inquiry

Table 703: Position 4 (Merchant Request for Commercial Card Type)

Attribute	Description
Data Representation	ans-1
Data Field	Contents of position 4
Values	0 (where 0 is zero) = Request Indicator

Subelement 94 (Commercial Card Inquiry Response (Visa Only))

DE 48, subelement 94 (Commercial Card Inquiry Response) contains the commercial card inquiry response data as a result of a commercial card inquiry on a Visa account.

Table 704: Attributes

Attribute	Description
Subelement ID	94
Subelement Data Length	2 positions
Data Representation	ans-4
Data Field	Contents of positions
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 94 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 705: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	C	.	C

Values

Table 706: Positions 1–3 (Card Request Indicator)

Attribute	Description
Data Representation	ans-3
Data Field	Contents of positions 1–3 (Hexadecimal value: 5AF0F1)
Values	0 (where 0 is zero) = Commercial Card Inquiry

Table 707: Position 4 (Visa Commercial Card Response)

Attribute	Description
Data Representation	a-1
Data Field	Contents of position 4

Table 708: Values

Value	Description
0 (where 0 is zero)	Not a Commercial Card
B	Business Card
R	Corporate Card
S	Purchasing Card

Subelement 95 (Mastercard Promotion Code)

DE 48, subelement 95 (Mastercard Promotion Code) contains the promotion code to identify unique use of a Mastercard product for a specific program or service established between issuers and merchants.

Table 709: Attributes

Attribute	Description
Subelement ID	95
Subelement Data Length	2 positions
Data Representation	an-6

Attribute	Description
Data Field	Contents of positions 1–6
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 95 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 710: Applicable messages

Message	Org	Sys	Dst
Authorization Request /0100	C	.	C
Authorization Request Response/0110	CE	.	CE
Authorization Advice/0120: acquirer-generated	C	.	C
Authorization Advice/0120: system-generated	.	C	C
Reversal Request/0400	C	.	C
Reversal Request Response/0410	CE	.	CE
Reversal Advice/0420	.	C	C

Program or service specific.

Table 711: Values

Value	Description
ARGCTA	Installment payment transaction within Argentina
AGROF1	Mastercard Agro Card
BNDES1	Brazil intracountry transactions using the Mastercard BNDES Card
CHLCTA	Installment payment transaction within Chile
COLCTA	Installment payment transaction within Colombia
GREECE	Installment payment transaction within Greece
HGMINS	Installment payment transaction for Georgia
MCGARS	Identifies Global Automated Service (GARS) Stand-In activity

Value	Description
MCINST	Installment payment transaction
MEXCTA	Installment payment transaction within Mexico
PARCEL	Installment payment transaction within Brazil
PERCTA	Installment payment transaction within Peru
PHINST	Installment payment transaction within Philippines
PRYCTA	Installment payment transaction within Paraguay
URYCTA	Installment payment transaction within Uruguay

Subelement 95 (American Express CID Card Identifier (American Express Only))

DE 48, subelement 95 (American Express CID Card Identifier) contains the American Express Card Identifier (CID) number from the face of the American Express card, when applicable.

Table 712: Attributes

Attribute	Description
Subelement ID	95
Subelement Data Length	2 positions
Data Representation	n-4
Data Field	Contents of positions 1–4
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 95 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 713: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	•	C
Authorization Request Response/0110	CE	•	CE

Values

The four-digit Card Identifier (CID) number on the front of the American Express card.

Subelement 96 (Visa Market-Specific Data Identifier (Visa Only))

DE 48, subelement 96 (Visa Market-Specific Data Identifier) contains the market-specific data identifier.

Table 714: Attributes

Attribute	Description
Subelement ID	96
Subelement Data Length	2 positions
Data Representation	a-1
Data Field	Contents of position 1
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 96 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 715: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Request Response/0110	CE/C	.	CE/C
Reversal Request/0400	C	.	C
Reversal Request Response/0410	CE	.	CE

Table 716: Values

Value	Description
A	Automobile rental
B	Bill Payment Transaction

Value	Description
E	Electronic commerce transaction aggregation
H	Hotel rental
J	B2B invoice payments
M	Healthcare—Medical
N	Failed market-specific data edit
T	Transit (in healthcare transactions only)

Subelement 97 (Prestigious Properties Indicator (Visa Only))

DE 48, subelement 97 (Prestigious Properties Indicator) contains the prestigious property indicator. For participants in the Visa Prestigious Lodging program (conditional).

Table 717: Attributes

Attribute	Description
Subelement ID	97
Subelement Data Length	2 positions
Data Representation	a-1
Data Field	Contents of position 1
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 97 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 718: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	•	C
Authorization Request Response/0110	CE	•	CE

Table 719: Values

Value	Description
D	Visa established limits
B	Visa established limits
S	Visa established limits

Subelement 98 (Mastercard Corporate Fleet Card ID/Driver Number)

DE 48, subelement 98 (Mastercard Corporate Fleet Card ID/Driver Number) allows the corporate customer to verify the user of the card and enables more detailed reporting. It contains the ID of the user of a Fleet card, when applicable.

Table 720: Attributes

Attribute	Description
Subelement ID	98
Subelement Data Length	2 positions
Data Representation	n...6; LLVAR
Data Field	Variable length, contents of subelement
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 98 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 721: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	•	C
Authorization Request Response/0110	CE	•	CE
Authorization Advice/0120: acquirer-generated	C	•	C
Authorization Advice/0120: system-generated	•	C	C

Message	Org	Sys	Dst
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	CE	.	CE

Values

Cardholder unique.

Application notes

If DE 48, subelement 98 (Mastercard Corporate Fleet Card ID/Driver Number) was present in the original 0100 AFD authorization message, subelement 98 must be the same value in acquirer-generated 0120 completion advice.

Subelement 99 (Mastercard Corporate Fleet Card Vehicle Number)

DE 48, subelement 99 (Mastercard Corporate Fleet Card Vehicle Number) allows the corporate customer to verify the user of the card and enables more detailed reporting. It contains the ID of the vehicle used in conjunction with a Fleet card purchase, when applicable.

Table 722: Attributes

Attribute	Description
Subelement ID	99
Subelement Data Length	2 positions
Data Representation	n...6; LLVAR
Data Field	Variable length, contents of subelement
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 99 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 723: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Request Response/0110	CE	.	CE

Message	Org	Sys	Dst
Authorization Advice/0120: acquirer-generated	C	.	C
Authorization Advice/0120: system-generated	.	C	C
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	CE	.	CE

Values

Vehicle-specific.

Application notes

If subelement 99 was present in the original 0100 AFD authorization message, subelement 99 must be the same value in acquirer-generated 0120 completion advice.

DE 48, Dual Message Authorization System Edits

The Dual Message Authorization System performs edits as described in this section.

DE 48, Proper Formatting

The Dual Message Authorization System will perform the following system edit to verify proper formatting of DE 48.

WHEN...	THEN the Dual Message Authorization System...
an Authorization Request/0100, Authorization Advice/0120—Acquirer-generated, or Reversal Request/0400 message contains DE 48 (Additional Data—Private Use), DE 108 (Additional Transaction Reference Data), or DE 112 (Additional Data [National Use]) with subelements that have incorrect length and/or incorrect format (Data Representations), or have multiple instances of the same subelement (when not permitted) within the same data element.	rejects the message and forwards the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 message where DE 44 is 6 positions for subelement format errors: - DE 39 (Response Code) = 30 (Message format error) - DE 44 (Additional Response Data) = 0480nn, or 1080nn, or 1120nn (where nn is the subelement number) (DE 44 is three positions for Dual Message System (Authorization) if no subelements are present, for example, for DE 22 format error: DE 44 = 022)

Examples

- When an edit error occurs on DE 48, the DE 44 data will be populated as these examples below:

- Error on DE 48 subelement 42, subfield 1: 048042 (no subfield information is provided)
- Error on DE 48 subelement 61: 048061
- DE 48 TCC format error will be responded with DE 44 = 048000.

For non-DE 48/DE 108/DE 112 format errors, DE 44 will only have DE info and no subelement information. For example, the format error in DE 22 will have DE 44 as 022 for non-DE48/108/112 data elements.

DE 48, TCC

The Dual Message Authorization System performs the following edit.

Table 724: Edit

WHEN...	THEN the Authorization System...
an Authorization Request/0100, Authorization Advice/0120, or Reversal Request/0400 message contains DE 48, TCC with an invalid value in position 1	rejects the message and forwards the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 message where: • DE 39 = 30 (Format Error) • DE 44 = 048000

DE 48, TCC and DE 3

The Dual Message Authorization System performs the following edits.

Table 725: Associated edits

WHEN TCC contains...	THEN the first two positions of DE 3 (Processing Code) must contain one of the following values...
C = Cash Disbursement	01 = Withdrawal 17 = Cash Disbursement 30 = Balance Inquiry
P = Payment Transaction	28 = Payment Transaction
Z = ATM Cash Disbursement	00 = Purchase 01 = Withdrawal 30 = Balance Inquiry 91 = PIN Unblock 92 = PIN Change Only eligible acquirers may use values 91 or 92.

Table 726: Associated edits

IF this edit...	THEN the Dual Message Authorization System...
fails	rejects the transaction with a format error, indicated by: • DE 39 (Response Code) = 30 • DE 44 (Additional Response Data) = 003
passes	goes to the next edit

Table 727: Associated edits

WHEN...	THEN the Dual Message Authorization System...
TCC contains a space	assigns in an Authorization Request/0100, Authorization Advice/0120: acquirer-generated, and Reversal Request/0400 message, a TCC value based on the acceptor business code/merchant category code (MCC) in DE 18 (Merchant Type) except under the following conditions

Table 728: Associated edits

WHEN...	THEN the Dual Message Authorization System...
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type) contains value 00 (Purchase) and DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) contains value 00 (PAN entry mode unknown) or value 01 (PAN manual entry) and DE 61 (Point-of-Service [POS] Data), subfield 4 (POS Cardholder Presence) does not contain value 0 (Cardholder Present)	assigns TCC value T (Phone, Mail, or Electronic Commerce Order).

DE 48, subelement 14

The Dual Message Authorization System will perform the following system edits.

Table 729: Combo Cards with a Single PAN

WHEN...	THEN the Dual Message Authorization System...
the Authorization Request/0100, the Authorization Advice/0120: acquirer-generated, or the Reversal Request/0400 message contains DE 48 (Additional Data: Private Use), subelement 14 (Account Type Indicator) with values other than C (Credit Transaction) or D (Debit Transaction)	will reject the transaction and forward the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or a Reversal Request Response/0410 with • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 048014 (indicating the data element in error)

DE 48, Subelement 22

The Dual Message Authorization System will perform the following system edits.

Table 730: Associated edits

WHEN...	THEN the Dual Message Authorization System...
DE 48 (Additional Data—Private Use), subelement 22 (Multi-Purpose Merchant Indicator), subfield 01 (Low-Risk Merchant Indicator), subfield 02 (Single Tap Indicator), or subfield 03 (Response to PIN Request) in the Authorization Request/0100 message is not alphanumeric or is an invalid length	rejects the message and sends the acquirer an Authorization Request Response/0110 message where: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 048022 (indicating the data element in error)
the acquirer provides DE 48, subelement 22, subfield 01, subfield 02, or subfield 03 in the Authorization Request/0100 message and the issuer is not in the European Economic Area (EEA)	will not forward the subfield 01, subfield 02, or subfield 03 data to the issuer.
the Dual Message System receives an Authorization Request/0100 message that includes a valid value in DE 48, subelement 22, subfield 02 indicating the merchant terminal supports single tap processing a response, and DE 39 (Response Code) = 65 (Decline: Exceeds withdrawal count limits) from the On-Behalf Contactless Counters Management system	will send the results to the acquirer in an Authorization Request Response/0110 message. When the issuer supports duplicate ATC, the Dual Message Authorization System will add DE 48, subelement 22, subfield 04 (Issuer PIN Request in a Single Tap Mode).

WHEN...	THEN the Dual Message Authorization System...
the Dual Message System receives a response from the issuer with DE 39 (Response Code) = 65 (Decline: Exceeds withdrawal count limits): - And the Authorization Request/0100 message includes a valid value in DE 48, subelement 22, subfield 02 indicating the merchant terminal supports single tap processing - And DE 48, subelement 22, subfield 04 is present in the Authorization Request Response/0110 message	will forward to the acquirer the Authorization Request Response/0110 message with DE 48, subelement 22, subfield 04 to indicate to the acquirer that an issuer supports a PIN in a single tap mode in a subsequent Authorization Request/0100 message.
the Dual Message System receives DE 48, subelement 22, subfield 04 in an Authorization Request Response/0110 message and subfield 04 does not contain alphanumeric data or has an invalid length	will consider the response from the issuer invalid and process consistent with current invalid issuer response processing.
the Dual Message System receives DE 48, subelement 22, subfield 01 in an acquirer-generated advice (Authorization Advice/0120 or Reversal Request/0400 message)	will not forward the subfield 01 data to the issuer.
the Dual Message System receives DE 48, subelement 22, subfield 02 or subfield 03 in a transaction message: - Authorization Advice/0120 - Reversal Request/0400 message	will not forward the subfield 02 or subfield 03 data to the issuer.
the acquirer provides DE 48, subelement 22, subfield 01 in an Authorization Request/0100 message for an ATM transaction	will not forward the subfield 01 data to the issuer.
the Dual Message System receives DE 48, subelement 22, subfield 02 or subfield 03 in an ATM transaction message: Authorization Request/0100	will not forward the subfield 02 or subfield 03 data to the issuer.
the Dual Message System receives DE 48, subelement 22, subfield 01, subfield 02, or subfield 03 in an Authorization Request Response/0110 message	will not forward the subfield 01, subfield 02, or subfield 03 data to the acquirer.

DE 48, subelement 26

The Dual Message Authorization System performs the following edits.

Table 731: Associated edits

WHEN...	THEN the Dual Message Authorization System...
the Authorization Request/0100 or Authorization Advice/0120 message includes DE 48 (Additional Data—Private Use), subelement 26 (Wallet Program Data), subfield 1 (Wallet Identifier) with any combination other than a valid numeric value	sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130—System-generated where: - DE 39 (Response Code) = 30 (Format error) - DE 44 (Additional Response Data) = 048026 (indicating the data element in error)
NOTE: Special characters, spaces, or all zeros not allowed.	

DE 48, subelement 35

The Dual Message Authorization System performs the following edits.

Table 732: Associated edits

Following are Authorization Request/0100 message edits.

WHEN...	THEN the Dual Message Authorization System...
the acquirer sends an Authorization Request/0100 message containing DE 48 (Additional Data—Private Use), subelement 35 (Contactless Non-Card Form Factor Request/Response)	sends the acquirer an Authorization Request Response/0110 message where: - DE 39 = 30 (Format error) - DE 44 = 048035

Table 733: Associated edits

The issuer-branded website will perform the following processing based on the Authorization Request Response/0110 messages.

WHEN...	THEN the issuer-branded website...
the Authorization Request Response/0110 message contains DE 48, subelement 35, value A	arranges for the request contactless card or device to be sent to the cardholder
the Authorization Request Response/0110 message contains DE 48, subelement 35, value D	declines the cardholder's request for the contactless card or device and instructs the cardholder to contact the issuer for more information

WHEN...	THEN the issuer-branded website...
the Authorization Request/0100 message contains DE 48, subelement 35 and the Authorization Request Response/0110 message does not contain DE 48, subelement 35	arranges for the requested contactless card or device to be sent to the cardholder when: Subelement 83 (Address Verification Service Response) is X or Y
or	and
the Authorization Request Response/0110 message containing DE 48, subelement 35 does not contain a valid value of A or D	Subelement 87 (Card Validation Code Result) is M

DE 48, subelement 37

The Dual Message Authorization System performs the following edits.

Table 734: Associated edits

WHEN...	THEN the Dual Message Authorization System...
DE 48 (Additional Data: Private Use), subelement 37 (Additional Merchant Data), subfields 1 (Payment Facilitator ID) or 2 (Independent Sales Organization ID) has values other than 0–9 or contains all zeros in the Authorization Request/0100 or Authorization Advice/0120: acquirer-generated message	sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where: - DE 39 (Response Code) = value 30 (Format error) - DE 44 (Additional Response Data) = value 048037 (indicating the data element in error)
DE 48, subelement 37, subfield 3 (Sponsored Merchant) contains all spaces or all zeros in the Authorization Request/0100 or Authorization Advice/0120: acquirer-generated message	sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where: - DE 39 = value 30 - DE 44 = value 048037

Table 735: Merchant Country of Origin Identifier

WHEN...	THEN the Authorization System...
DE 48 (Additional Data), subelement 37 (Additional Merchant Data), subfield 4 (Merchant Country of Origin) is non-numeric.	will reject the message and forward to the acquirer an Authorization Request Response/0110 message with: - DE 39 (Response Code) = 30 (Format error) - DE 44 (Additional Response Data) = 048037

WHEN...	THEN the Authorization System...
DE 48 (Additional Data), subelement 37 (Additional Merchant Data) subfield 4 (Merchant Country of Origin) is not a valid ISO country code.	will reject the message and forward to the acquirer an Authorization Request Response/0110 message with: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 048037
DE 48 (Additional Data), subelement 37 (Additional Merchant Data), subfield 4 (Merchant Country of Origin) is not 3 numeric digits.	will reject the message and forward to the acquirer an Authorization Request Response/0110 message with: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 048037

Table 736: Associated edits

When...	Then the Authorization System...
an Authorization Request/0100 or Authorization Advice/0120: acquirer-generated message contains DE 48 (Additional Data), subelement 37 (Additional Merchant Data), subfield 5 (Merchant Payment Gateway ID) with a length that is not 11	sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where: • DE 39 = value 30 • DE 44 = value 048037
an Authorization Request/0100 or Authorization Advice/0120: acquirer-generated message contains DE 48, subelement 37, subfield 5 with non-numeric characters	sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where: • DE 39 = value 30 • DE 44 = value 048037

DE 48, subelement 38

The Dual Message Authorization System performs the following edits.

Table 737: Associated edits

WHEN...	THEN...
the Dual Message Authorization System finds that the entire account range no longer participates in Enhanced Value, Product Graduation, or High Value and at least one cardholder account within the account range previously participated in Enhanced Value, Product Graduation, or High Value	the Dual Message Authorization System adds DE 48, subelement 38 (Account Category), value Z to the Authorization Request/0100 message for the following transaction types: • Purchases as indicated by DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) = 00 • Purchase with Cash Back as indicated by DE 3, subfield 1 = 09 and the Dual Message Authorization System forwards the Authorization Request/0100 message to the issuer.
the Dual Message Authorization System provides DE 48, subelement 38 in the Authorization Request/0100 message to the issuer and the issuer populates DE 39 (Response Code) in the Authorization Request Response/0110 message with one of the following values that indicate an authorization approval: • 00 (Approved or completed successfully) • 08 (Honor with ID) • 10 (Partial Approval) • 87 (Purchase Amount Only, No Cash Back Allowed)	DE 38 (Authorization ID Response), position 6 in the Authorization Request Response/0110 message must equal the value of DE 48, subelement 38 of the original Authorization Request/0100 message that was provided to the issuer by the Dual Message Authorization System.
the Dual Message Authorization System provides DE 48, subelement 38 in the Authorization Request/0100 message to the issuer and the issuer populates DE 39 (Response Code) in the Authorization Request Response/0110 message with the value 85 (Not declined) and DE 38 is present	DE 38 (Authorization ID Response), position 6 in the Authorization Request Response/0110 message must equal the value of DE 48, subelement 38 of the original Authorization Request/0100 message that was provided to the issuer by the Dual Message Authorization System. If DE 38 is not present, the Dual Message Authorization System will not perform this validation.

WHEN...	THEN...
DE 38, position 6 in the Authorization Request Response/0110 message does not equal the value of DE 48, subelement 38 of the original Authorization Request/0100 message that was provided to the issuer by the Authorization System	the Dual Message Authorization System rejects the Authorization Request Response/0110 message and sends the Authorization Request/0100 message to the Stand-In System for processing. Stand-In will ensure that DE 38, position 6 of the Authorization Request Response/0110 message matches the value provided by the Dual Message Authorization System in DE 48, subelement 38 of the original Authorization Request/0100 message.
DE 48, subelement 38 is included in the Authorization Request Response/0110 message from the issuer	the Dual Message Authorization System removes DE 48, subelement 38 from the Authorization Request Response/0110 message before providing the 0110 message to the acquirer.

DE 48, Subelement 42 and Subelement 43

The Dual Message Authorization System performs these system edits.

The Dual Message Authorization System performs these edits on Authorization Request/0100 and Authorization Advice/0120 messages.

Table 738: Associated edits

WHEN DE 48, subelement 42, subfield 01, position 3 is...	THEN DE 48, subelement 43...
0 or is not present	cannot contain UCAF data.
1, 2, 3, 4, 5, 6, or 7	must contain UCAF data.

Table 739: Associated edits

IF this edit...	THEN the Authorization System...
fails	rejects the transaction with a format error, indicated by: DE 39 = 30 DE 44 = 048043
is missing or has an invalid AAV leading indicator	downgrades the edit which downgrades missing or invalid AAV leading indicator with associated Identity Check SLI value to SLI = 210.

IF this edit...	THEN the Authorization System...
passes	the Dual Message Authorization System goes to the next edit.

Table 740: Associated edits

WHEN DE 48, subelement 43...	THEN DE 48, subelement 42, subfield 01, position 3 must equal...
contains UCAF data	1, 2, 3, 4, 5, 6, or 7
does not contain UCAF data	0 or not be present

Table 741: Associated edits

IF this edit...	THEN the Authorization System...
fails	rejects the transaction with a format error, indicated by: DE 39 (Response Code) = 30 DE 44 (Additional Response Data) = 048043 (indicating the data element in error)
is missing or has an invalid AAV leading indicator	downgrades the edit which downgrades missing or invalid AAV leading indicator with associated Identity Check SLI value to SLI = 210.
passes	the Dual Message Authorization System goes to the next edit.

The Dual Message Authorization System performs the following edits on Authorization Request/0100 messages.

DSRP transactions initiated with MDES tokens

Table 742: Associated edits

WHEN acquirer submits...		THEN Mastercard will validate/modify Security Level Indicator (SLI) to...
UCAF DE 48, SE 43	Digital Payment Data DE 104, SE 001	SLI DE 48, SE 42, SF 1
DSRP Cryptogram	N/A	SLI based on DSRP cryptogram

WHEN acquirer submits...		THEN Mastercard will validate/ modify Security Level Indicator (SLI) to...
UCAF DE 48, SE 43	Digital Payment Data DE 104, SE 001	SLI DE 48, SE 42, SF 1
N/A	DSRP Cryptogram	SLI based on DSRP cryptogram
Identity Check Fully Authenticated or Merchant Attempt AAV	DSRP Cryptogram	SLI based on AAV
Identity Check Fully Authenticated AAV or Identity Check Merchant Attempt AAV	N/A	SLI based on AAV

DSRP merchant-initiated transactions, where SLI value is not 247

Table 743: Associated edits

WHEN...	THEN the Dual Message Authorization System...
DSRP transaction initiated with a Secure Element or Cloud token, DE 22, subfield 1 = 81 (POS entry mode) and the Authorization Request/0100 contains DE 48, subelement 42, not equal to 247 and DE 48, subelement 43 and DE 104, subelement 001 is not present	rejects the transaction with - DE 39 (Response Code) = 30 (Format error) - DE 44 (Additional Response Data) = 0480nn (where nn is the subelement number)

Non-DSRP transactions initiated with an MDES token and submitted with a merchant attempt Identity Check AAV

Table 744: Associated edits

WHEN...	THEN the Dual Message Authorization System...
a transaction initiated with a Secure Element or Cloud token, DE 22, subfield 1 = 81 (POS entry mode) and the Authorization Request/0100 request contains DE 48, subelement 43, pos1 = h (Merchant Attempt Identity Check AAV)*	if dynamic expiration date and dynamic token verification code was successfully validated by the MDES service, then the Dual Message Authorization System goes to the next edit, otherwise rejects the transaction with - DE 39 (Response Code) = 30 (Format error) - DE 44 (Additional Response Data) = 0480nn (where nn is the subelement number)
*Mastercard Identity Check is no longer generating h SPA1 AAVs	

DE 48, subelement 43 (Static AAV)

The following edits on Authorization Request/0100 and Authorization Advice/0120: acquirer-generated messages are performed on transactions submitted under the Maestro Recurring Payments Program, the Mastercard Utility Payment Program, and the Maestro Low Risk Merchant Program.

Table 745: Associated edits

WHEN the Mastercard assigned static AAV in DE 48, subelement 43, position 1 begins with...	THEN the Dual Message Authorization System...
3 and the acceptance brand is not Maestro	sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where: DE 39 (Response Code) = 30 (Format error) DE 44 = 048043 (identifying the data element in error)
5 and the acceptance brand is not Maestro	sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where: DE 39 = 30 DE 44 = 048043
any value other than 3 or 5 and the acceptance brands are Mastercard, Maestro, or both	sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where: DE 39 = 30 DE 44 = 048043

DE 48, subelement 42 and DE 61

The Dual Message Authorization System performs the following edits.

Table 746: Associated edits

The following edits apply on Electronic Commerce transactions.

WHEN...	THEN...
DE 48, subelement 42, subfield 1, position 3 contains value 1, 2, or 3, and DE 61, subfield 10, contains value 6	DE 61, subfield 4 contains value 4 or 5 and DE 61, subfield 7, must not contain value 2

Table 747: Associated edits

IF this edit...	THEN the Dual Message Authorization System...
fails	generates a format error, indicated by: DE 39 (Response Code) = 30 DE 44 (Additional Response Data) = 061
passes	the Dual Message Authorization System goes to the next edit.

Table 748: Associated edits

The following edits apply on Identity Check Phone Order transactions.

WHEN...	THEN...
DE 48, subelement 42, subfield 1, position 3 contains value 1 or 2 and DE 61, subfield 7 contains value 2	DE 61, subfield 4 must contain a value 3 or 4 and DE 61, subfield 10 must not contain value 6.

Table 749: Associated edits

IF this edit...	THEN the Dual Message Authorization System...
fails	generates a format error indicated by: DE 39 = 30 DE 44 = 061
passes	the Dual Message Authorization System goes to the next edit.

DE 48, subelement 50

Edits for DE 48 (Additional Data, Private Use), subelement 50 (Embedded Interchange Data) are as noted.

Table 750: Associated edits

When the acquirer provides ...	Dual Message Authorization System rejects the transaction sending the acquirer ...	Where...
DE 48, subelement 50, but does not participate in the Embedded Flexible Interchange (EFI) program	• Authorization Request Response/0110 message, or • Reversal Request Response/0410 message	DE 39 (Response Code) contains value 58
DE 48, subelement 50, but DE 02 indicates a Primary Account Number that does not participate in the EFI program	• Authorization Request Response/0110 message, or • Reversal Request Response/0410 message	DE 39 contains value 57
DE 48, subelement 50 but DE 03, subfield 1 is not 00 (Purchase), 20 (Refund)	• Authorization Request Response/0110 message • Authorization Advice Response/0130 message, or • Reversal Request Response/0410 message	DE 39 contains value 30 (Format error) and DE 44 (Additional Response Data) contains value 003
DE 48, subelement 50 with an invalid combination of subfields	• Authorization Request Response/0110 message • Authorization Advice Response/0130 message, or • Reversal Request Response/0410 message	DE 39 contains value 30 and DE 44 (Additional Response Data) contains value 048050

Table 751: DE 48, subelement 50, subfield 01 (Interchange Indicator)

When ...	Then the Dual Message Authorization System...
an Authorization Request/0100, Authorization Advice/0120 message, or Reversal Request/0400 message contains DE 48, subelement 50, without DE 48, subelement 50, subfield 01	rejects the message and forwards to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message, or Reversal Request Response/0410 message with DE 39 = 30 and DE 44 = 048050

When ...	Then the Dual Message Authorization System...
an Authorization Request/0100, Authorization Advice/0120 message, or Reversal Request/0400 message contains DE 48, subelement 50, subfield 01 with a value other than 'Y'	rejects the message and forwards to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message, or Reversal Request Response/0410 message with DE 39 = 30 and DE 44 = 048050

DE 48, subelement 50, subfield 02 (Interchange Percentage)

Table 752: Associated edits

When ...	Then the Dual Message Authorization System...
an Authorization Request/0100, Authorization Advice/0120 message, or Reversal Request/0400 message contains DE 48, subelement 50, subfield 02 with a length that is not six	rejects the message and forwards to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message, or Reversal Request Response/0410 message with DE 39 = 30 and DE 44 = 048050
an Authorization Request/0100, Authorization Advice/0120 message, or Reversal Request/0400 message contains DE 48, subelement 50, subfield 02 containing non-numeric characters	rejects the message and forwards to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message, or Reversal Request Response/0410 message with DE 39 = 30 and DE 44 = 048050
an Authorization Request/0100, Authorization Advice/0120 message, or Reversal Request/0400 message contains DE 48, subelement 50, subfield 02 containing a value greater than 100.000% (100000)	rejects the message and forwards to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message, or Reversal Request Response/0410 message with DE 39 = 30 and DE 44 = 048050

Table 753: DE 48, subelement 50, subfield 03 (Unit Fee Amount)

When ...	Then the Dual Message Authorization System...
an Authorization Request/0100, Authorization Advice/0120 message, or Reversal Request/0400 message contains DE 48, subelement 50, subfield 03 with a length that is not twelve	rejects the message and forwards to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message, or Reversal Request Response/0410 message with DE 39 = 30 and DE 44 = 048050
an Authorization Request/0100, Authorization Advice/0120 message, or Reversal Request/0400 message contains DE 48, subelement 50, subfield 03 containing non-numeric characters	rejects the message and forwards to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message, or Reversal Request Response/0410 message with DE 39 = 30 and DE 44 = 048050

Table 754: DE 48, subelement 50, subfield 04 (Flat Interchange Amount)

When ...	Then the Dual Message Authorization System...
an Authorization Request/0100, Authorization Advice/0120 message, or Reversal Request/0400 message contains DE 48, subelement 50, subfield 04 with a length that is not twelve	rejects the message and forwards to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message, or Reversal Request Response/0410 message with DE 39 = 30 and DE 44 = 048050
an Authorization Request/0100, Authorization Advice/0120 message, or Reversal Request/0400 message contains DE 48, subelement 50, subfield 04 containing non-numeric characters	rejects the message and forwards to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message, or Reversal Request Response/0410 message with DE 39 = 30 and DE 44 = 048050

Table 755: DE 48, subelement 50, subfield 05 (Payment Party)

When ...	Then the Dual Message Authorization System...
an Authorization Request/0100, Authorization Advice/0120 message, or Reversal Request/0400 message contains DE 48, subelement 50, subfield 05 with a value other than 'A' or 'I'	rejects the message and forwards to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message, or Reversal Request Response/0410 message with DE 39 = 30 and DE 44 = 048050

DE 48, Subelement 61

The Dual Message Authorization System performs the following edits.

Table 756: Associated edits

WHEN...	THEN the Dual Message Authorization System...
DE 48, subelement 61 is present and DE 4 contains all zeros	drops DE 48, subelement 61 from the Authorization Request/0100 message to the issuer.

Table 757: Associated edits

WHEN DE 48, subelement 61, subfield 3 is...	THEN the Dual Message Authorization System...
1 (Merchant terminal verified the purchase items against an Inventory Information Approval System [IIAS])	validates that DE 48, subelement 32 (if present) contains a valid Mastercard Assigned ID for IIAS.

Table 758: Associated edits

IF the...	THEN the Dual Message Authorization System...
Mastercard Assigned ID is valid	updates the value in DE 48, subelement 61, subfield 3 as follows: • If the issuer participates in real-time substantiation, sends DE 48, subelement 61, subfield 3, value 1 (Merchant terminal verified the purchased items against an IIAS) • If the issuer does not participate in real-time substantiation sends DE 48, subelement 61, subfield 3, value 0 (Merchant terminal did not verify the purchased items against an IIAS).
Mastercard Assigned ID is not valid or not present in the Authorization Request/0100 message	updates the value in DE 48, subelement 61, subfield 3 as follows: • If the issuer participates in real-time substantiation, sends DE 48, subelement 61, subfield 3, value 4 (Transaction was submitted as real-time substantiated but from a non-IIAS certified merchant). • If the issuer does not participate in real-time substantiation sends DE 48, subelement 61, subfield 3, value 0 (Merchant terminal did not verify the purchased items against an IIAS).

Table 759: Associated edits

The following edits are on Authorization Request/0100 and Authorization Advice/0120—Acquirer-generated messages.

WHEN...	THEN the Dual Message Authorization System...
the acquirer sends DE 48, subelement 61, subfield 3 (Real-time Substantiation Indicator), value 2 (Merchant claims exemption from IAS rules based on the IRS 90 percent rule) in an Authorization Request/0100 message or Authorization Advice/0120: acquirer-generated message and the issuer does not participate in real-time substantiation	forwards the issuer the Authorization Request/0100 or Authorization Advice/0120: acquirer-generated message containing DE 48, subelement 61, subfield 3, value 0 (Merchant terminal did not verify the purchased items against an IAS).
the acquirer sends DE 48, subelement 61, subfield 3 (Real-time Substantiation Indicator), value 2 (Merchant claims exemption from IAS rules based on the IRS 90 percent rule) in an Authorization Request/0100 message or Authorization Advice/0120: acquirer-generated message and the issuer does participate in real-time substantiation	forwards the issuer the Authorization Request/0100 or Authorization Advice/0120—Acquirer-generated message containing DE 48, subelement 61, subfield 3, value 2 (Merchant claims exemption from using the IAS, based on the IRS 90 percent rule).
the acquirer sends DE 48, subelement 61, subfield 3 (Real-time Substantiation Indicator), value other than 0, 1, or 2 in Authorization Request/0100 or Authorization Advice/0120: acquirer-generated message	declines the Authorization: acquirer-generated messages where: • DE 39 (Response Code) is 30 • DE 44 (Response Data) is 048061

DE 48, subelement 66

The Dual Message Authorization System will perform the following system edits.

Table 760: Associated edits

WHEN...	THEN the Dual Message Authorization System...
an Authorization Request/0100 or Authorization Advice/0120 message contains DE 48 (Additional Data: Private Use), subelement 66 (Authentication Data) subfield 1 (Program Protocol) that is the incorrect length or incorrect data representation (alphanumeric)	reject the message and forward to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message with: • DE 39 (Response Code) = 30 (Format Error) • DE 44 (Additional Response Data) = 048066

WHEN...	THEN the Dual Message Authorization System...
an Authorization Request/0100 or Authorization Advice/0120 message contains DE 48, subelement 66 (Authentication Data) subfield 2 (Directory Server Transaction ID) that is not 36 bytes in length or filled with spaces	reject the message and forward to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message with: - DE 39 (Response Code) = 30 (Format Error) - DE 44 (Additional Response Data) = 048066

DE 48, subelement 77

The Dual Message Authorization System will perform the following system edits.

Bill Pay

Table 761: Associated edits

WHEN...	THEN the Dual Message Authorization System...
the Authorization Request/0100 message or the Reversal Request/0400 message contains DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier) with value C91, C92, C93, or C94 and DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) is not value 00 (Purchase)	will reject the transaction with DE 39 (Response Code) = 12 (Invalid Transaction)

High-Risk Purchase

Table 762: Associated edits

WHEN...	THEN the Dual Message Authorization System...
DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier) is present and contains value P70 (Cryptocurrency) but the MCC is not 6051 (Quasi Cash—Merchant) or DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier) is present and contains value P71 (High-risk Securities) but the MCC is not MCC 6211 (Securities—Brokers/Dealers)	rejects the message and sends the acquirer an Authorization Request Response/0110 message where: - DE 39 (Response Code) = 30 (Format error) - DE 44 (Additional Response Data) = 048077 (indicating the data element in error)

Mastercard Send and Funding Transactions

Table 763: Associated edits

WHEN...	THEN the Dual Message Authorization System...
data in DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) and DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier) indicate the transaction is a funding transaction refund and The originating institution identified by the ICA provided in DE 32 (Acquiring Institution ID Code) has not registered to process transactions via the Mastercard Send platform	rejects the transaction and sends the originating institution an Authorization Request Response/0110 or Reversal Request Response/0410 message containing DE 39 (Response Code), value 58 (Transaction not permitted to issuer/cardholder).
data in DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) and DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier) indicate the transaction is a funding transaction refund and the brand product code associated with the transaction does not participate in the Mastercard Funding Program	rejects the transaction and sends the originating institution an Authorization Request Response/0110 or Reversal Request Response/0410 message containing DE 39 (Response Code), value 12 (Invalid Transaction).
data in DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) and DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier) indicate the transaction is a funding transaction refund and the Unique Transaction reference number (DE 108 [Additional Transaction Reference Data], subelement 03 [Transaction Reference Data], subfield 1 [Unique Transaction Reference]) contains all zeros or spaces	sends the originating institution an Authorization Request Response/0110 message to include the following: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 108003 (Additional Data: Private Use)

WHEN...	THEN the Dual Message Authorization System...
data in DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) and DE 48 (Additional Data: Private Use), subelement 77 (Transaction Type Identifier) indicate the transaction is a funding transaction refund and the issuer account range does not participate in Mastercard Send Payment Transactions Program, Mastercard Funding Transaction Program or both.	sends the originating institution an Authorization Request Response/0110 message to include the following: DE 39 (Response Code) = 57 (Transaction not permitted to issuer or cardholder)
the originating institution is registered for the Mastercard Funding Transaction Program and is submitting an Authorization Request/0100 message for funding transactions and funding transaction refunds where: • DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains value 00 or 20 • DE 18 (Merchant Type) does not contain MCC 6538 • DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier) contains value C07, C52, C53, C54, C55, C56, C57, or C65	sends the originating institution an Authorization Request Response/0110 to include the following: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 018 (indicating the data element in error)
the originating institution is registered for the Mastercard Funding Transaction Program and is submitting an Authorization Request/0100 message for funding transactions and funding transaction refunds where: • DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains value 00 or 20 • DE 18 (Merchant Type) does not contain MCC 4829 or MCC 6540 • DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier) contains value F07, F08, F52, F53, F54, F55, F61, F64, or F65	sends the originating institution an Authorization Request Response/0110 to include the following: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 018 (indicating the data element in error)

WHEN...	THEN the Dual Message Authorization System...
the originating institution is registered for the Mastercard Funding Transaction Program and is submitting an Authorization Request/0100 message for funding transactions and funding transaction refunds where: • DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains value 00 or 20 • DE 18 (Merchant Type) contains MCC 6538 • DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier) contains value C07, C52, C53, C54, C55, C56, C57, or C65 • DE 48 (Additional Data: Private Use) Transaction Category Code does not contain value R (Face-to-face retail), T (Non-face-to-face retail), or U (Unique Transaction—Quasi Cash Disbursement).	sends the originating institution an Authorization Request Response/0110 to include the following: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 048000 (Additional Data: Private Use)
the originating institution is registered for the Mastercard Funding Transaction Program and is submitting an Authorization Request/0100 message for funding transactions and funding transaction refunds where: • DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains value 00 or 20 • DE 18 (Merchant Type) contains MCC 4829 or MCC 6540 • DE 48 (Additional Data: Private Use), subelement 77 (Transaction Type Identifier) contains value F07, F08, F52, F53, F54, F55, F61, F64, or F65 • DE 48 (Additional Data: Private Use) Transaction Category Code does not contain value R (Face-to-face retail), T (Non-face-to-face retail), or U (Unique Transaction: Quasi Cash Disbursement).	sends the originating institution an Authorization Request Response/0110 to include the following: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 048000 (Additional Data: Private Use)

DE 48, subelement 78

The Dual Message Authorization System performs the following edits.

Table 764: Associated edits

WHEN DE 48, Subelement 78 is...	THEN DE 2...
D (U.S. Deferred Billing Indicator)	DE 2 (Primary Account Number [PAN]) must be a Visa PAN

Table 765: Associated edits

WHEN...	THEN the Dual Message Authorization System...
the Authorization System receives an Authorization Request/0100 and Reversal Request/0400 for a Visa-branded, Dual Message System (Authorization) transaction with DE 48 (Additional Data—Private Use), subelement 78 (Payment Service Indicators [Visa Only]) where:	sends an Authorization Request Response/0110 and Reversal Request/0400 message where:
- Subfield 1 (Spend Qualified Indicator) contains a value other than Space - Subfield 2 (Dynamic Currency Conversion Indicator) contains a value other than Y or Space - Subfield 3 (U.S. Deferred Billing Indicator) contains a value other than D or Space - Subfield 4 (Visa Checkout Indicator) contains a value other than Y or Space - Subfield 5 (Message Reason Code) contains a value other than Space, 0, 1, 2, 3, 4, 5 - Subfield 6 (Reserved for Future Use) contains a value other than Space	- DE 39 (Response Code) = 30 (Format error) - DE 44 (Additional Response Data) = 048078 (indicating the data element in error)

DE 48, subelement 82

The Dual Message Authorization System performs the following edits.

Table 766: Associated edits

WHEN DE 48, Subelement 82 is...	THEN the Dual Message Authorization System...
51 (AVS-Only)	sends the acquirer an Authorization Request Response/0110 message where: DE 39 = 12 (Invalid Transaction)

WHEN DE 48, Subelement 82 is...	THEN the Dual Message Authorization System...
52 (AVS Request) and DE 39 is 00 (Approved) or 08 (Honor with ID) or 85 (Not declined) and DE 48 SE 83 (AVS Response) is present with a value not equal to X, Y, A, W, Z, N, U, R, S	sends the issuer an Authorization Response Negative Acknowledgment/0190 message where: - DE 39 = 30 (Format Error) - DE 44 = 048083

DE 48, subelement 84

The Dual Message Authorization System performs the following edits.

Table 767: Associated edits

WHEN...	THEN the Dual Message Authorization System...
DE 48, subelement 42 and/or DE 48, subelement 43 are invalid due to token requirements that are not met	returns to the acquirer an Authorization Request Response/0110 message where: - DE 39 (Response Code) = 30 (Format error) - DE 44 (Additional Response Data) = 0480nn (where nn is the subelement number) - DE 48 (Additional Data—Private Use), subelement 84 = 04 NOTE: Issuers do not receive decline advice messages for transactions that are rejected back to acquirers for format errors (DE 39 = 30).

DE 48, subelement 86

The Dual Message Authorization System performs the following edits.

Table 768: Associated edits

WHEN DE 48, Subelement 86 is...	THEN DE 2...
R (Relationship Participant)	DE 2 (Primary Account Number [PAN]) must be a Visa PAN

DE 48, subelement 95

The Dual Message Authorization System performs the following system edit on Authorization Request/0100 and Reversal Request/0400 messages for Brazil installment payment transactions.

Table 769: Associated edits

WHEN...	THEN the Dual Message Authorization System...
DE 48 (Additional Data—Private Use), subelement 95 (Mastercard Promotion Code) contains the value PARCEL,	rejects the transaction sending the acquirer an Authorization Request Response/0110 or Reversal Request Response/0410 where:
And	- DE 39 (Response Code) contains value 30 (Format error) - DE 44 (Additional Response Data) contains value 112001 (indicating the data element in error)
DE 112 (Additional Data [National Use]), subelement 001 (Installment Payment Data) is not present	

DE 48, in Authorization Request Response

The following edit applies to DE 48 and the Authorization Request Response/0110 message.

Table 770: Associated edits

WHEN the issuer...	THEN the Authorization Request Response/0110 message must...
provides response data, such as AVS response, Merchant Advice Code, CVC 2 response, and CVC 1 response, in DE 48 of the Authorization Request Response/0110 message	contain the TCC as the first byte of data within DE 48 and contain all the subelements present in DE 48 of the original Authorization Request/0100 message that also are defined in the Authorization Request Response/0110 message.
does not provide response data	not contain DE 48, including the TCC.

DE 49 (Currency Code, Transaction)

DE 49 (Currency Code, Transaction) is the local currency of the acquirer or source location of the transaction. It specifies the currency used in DE 4 (Amount, Transaction) and DE 95 (Replacement Amounts), subfield 1 (Actual Amount, Transaction).

Table 771: Attributes

Attribute	Description
Data Representation	n-3
Data Length	N/A
Data Field	Contents of positions 1–3
Subfields	N/A
Justification	N/A

Usage

Following is the usage of DE 49 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 772: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	M	•	M
Authorization Request Response/0110	ME	•	ME
Authorization Advice/0120: acquirer-generated	M	•	M
Authorization Advice/0120: system-generated	•	X	M
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	ME	•	ME
Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)	ME	XE	•
Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)	•	XE	ME
Reversal Request/0400	M	•	M
Reversal Request Response/0410	ME	•	ME
Reversal Advice/0420	•	X	M

Message	Org	Sys	Dst
Reversal Advice Response/0430	ME	XE	.

Values

All currency codes must be selected from the numeric ISO standard currency codes. ISO standard currency codes identify DE 49. A list of valid values is available in the [Quick Reference Booklet](#).

Application notes

This data element is defined and used identically within all Mastercard programs and services. Acquirers will receive an Authorization Request Response/0110 message containing a format error in DE 39 when the currency code in DE 49 is a currency code other than those in the ISO Standard Currency Codes table, invalid or pseudo codes will not be accepted.

DE 50 (Currency Code, Settlement)

DE 50 (Currency Code, Settlement) defines the currency of DE 5 (Amount, Settlement), DE 29 (Amount, Settlement Fee), and DE 95 (Replacement Amounts), subfield 2 (Actual Amount, Settlement).

Table 773: Attributes

Attribute	Description
Data Representation	n-3
Data Length	N/A
Data Field	Contents of positions 1–3
Subfields	N/A
Justification	N/A

Usage

Following is the usage of DE 50 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 774: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	.	X	C
Authorization Request Response/0110	CE	X	C
Authorization Advice/0120: acquirer-generated	.	X	C
Authorization Advice/0120: system-generated	.	X	C
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	CE	.	CE
Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)	CE	XE	.
Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)	C	XE	.
Reversal Request/0400	.	X	C
Reversal Request Response/0410	CE	X	C
Reversal Advice/0420	.	X	C
Reversal Advice Response/0430	CE	XE	.

Values

All currency codes must be selected from the numeric ISO standard currency codes.

ISO standard currency codes identify DE 50. A list of valid values is in the [Quick Reference Booklet](#).

Application notes

This data element is defined and used identically within all Mastercard programs and services.

As of the date of this publication, all Mastercard programs and services use U.S. dollars (840) as the currency of settlement for programs and services that the Dual Message Authorization System supports.

The Dual Message Authorization System includes this data element if the customer chooses to receive settlement amount-related data elements.

This data will be present, as defined, except when the Dual Message Authorization System has declined an Authorization Request/0100 message and was unable to complete currency conversion processing.

DE 51 (Currency Code, Cardholder Billing)

DE 51 (Currency Code, Cardholder Billing) defines the currency of DE 6 (Amount, Cardholder Billing), DE 8 (Amount, Cardholder Billing Fee), and DE 95 (Replacement Amounts), subfield 3 (Actual Amount, Cardholder Billing).

Table 775: Attributes

Attribute	Description
Data Representation	n-3
Data Length	N/A
Data Field	Contents of positions 1–3
Subfields	N/A
Justification	N/A

Usage

Following is the usage of DE 51 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 776: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	•	X	M
Authorization Request Response/0110	M	P	M
Authorization Advice/0120: acquirer-generated	•	X	M
Authorization Advice/0120: system-generated	•	X	M
Authorization Advice Response/0130: issuer-generated (Responding to an acquirer-generated 0120)	CE	•	C
Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)	ME	XE	•
Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)	•	X	C
Reversal Request/0400	•	X	M
Reversal Request Response/0410	CE	X	C
Reversal Advice/0420	•	X	M

Message	Org	Sys	Dst
Reversal Advice Response/0430	ME	XE	.

Values

All currency codes must be selected from the numeric ISO standard currency codes.

ISO standard currency codes identify DE 51. A list of valid values is in the [Quick Reference Booklet](#).

Application notes

This data element is defined and used identically within all Mastercard programs and services.

This data will be present, as defined, except when the Dual Message Authorization System has declined an Authorization Request/0100 message and was unable to complete currency conversion processing.

DE 52 (Personal ID Number [PIN] Data)

DE 52 (Personal ID Number [PIN] Data) contains a number assigned to a cardholder intended to uniquely identify that cardholder at the point of interaction. The use of the PIN is subject to bilateral agreement. The data element may contain the PIN itself or a derivative. This data element transmits PIN information from acquirers to issuers (or to the network) for PIN verification or validation.

Table 777: Attributes

Attribute	Description
Data Representation	b-8
Data Length	N/A
Data Field	Contents of bit positions 1–64 (8 bytes)
Subfields	N/A
Justification	N/A

Usage

Following is the usage of DE 52 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 778: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	X	C

Values

The network supports PINs from four to 12 characters long.

The encrypted PIN block always is eight bytes (64 bits) long regardless of the original PIN length.

CIS uses the Data Encryption Standard (DES) algorithm (ISO 9564-2) to translate PIN data. For purposes of PIN translation, DES requires that a 16-digit hexadecimal number and working key be provided. DE 52 houses the 16-digit number in the customer's PIN block format.

Application notes

Because of strict security requirements implemented within the network environment, PINs are never transmitted "in the clear" as character data. In addition, PIN data is never included in Advice or Reversal messages. The primary reason for this is that PIN data is highly sensitive information that is never stored (even in encrypted form) as a permanent component of a transaction, for security reasons. The rules, bylaws, and procedures established for individual programs and services dictate the specific requirements for PIN usage.

This data element is supported for Mastercard transactions.

The Dual Message Application System may perform PIN verification or validation services on behalf of customers that elect to use this optional service. Refer to the appropriate user guide and [Security Rules and Procedures](#) to determine specific PIN verification/validation options that may be selected for individual Mastercard programs and services.

DE 52 is mandatory for all ATM transactions.

Table 779: Associated edits

IF...	THEN the Dual Message Authorization System...
an Authorization Request/0100 message contains DE 52 (PIN Data) and DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) is not 02, 05, 07, 80, 81, 82, 90, or 91	rejects the message and forwards the acquirer an Authorization Request Response/0110 message where: - DE 39 = 30 - DE 44 = 052

NOTE: DE 52, DE 53 and DE 125 are mutually exclusive with DE 110 Dataset01

DE 52 Dual Message Authorization System Edits

The Dual Message Authorization System will perform the following edits for PIN translation failures.

Table 780: Associated edits

When...	THEN the Dual Message Authorization System...
the acquirer uses the Authorization System for PIN translation and DE 52 (Personal ID Number [PIN] Data) is present and DE 53 (Security-Related Control Information), subfield 4 (PIN Key Index Number) contains a value not known to Mastercard, or...	sends the issuer an Authorization Request Response/0100 message where DE 48 (Additional Data: Private Use), subelement 80 (PIN Service Code) has a value of TI (The Authorization System was unable to translate the PIN)
an Authorization System Security Translation Platform fails	

DE 53 (Security-Related Control Information)

DE 53 (Security-Related Control Information) is used with PIN data to provide specific information about PIN block encoding and PIN data encryption to assist the issuer (or its agent) in processing PINs entered at the point of interaction.

Table 781: Attributes

Attribute	Description
Data Representation	n-16
Data Length	N/A
Data Field	Contents of subfields
Subfields	6
Justification	See subfields

Usage

Following is the usage of DE 53 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 782: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	X	C
Network Management Request/0800: sign-on/sign-off	C	C	.

Values

See subfields.

Application notes

Use of DE 53 in online PIN transactions is specific to the PIN translation process performed on the dual message authorization network for Europe Region Acquirers and Issuers. It is not required to be provided in online PIN transactions for messages from non-Europe Region Acquirers.

NOTE: DE 52, DE 53 and DE 125 are mutually exclusive with DE 110 Dataset01.

Subfield 1 (PIN Security Type Code)

DE 53, subfield 1 (PIN Security Type Code) indicates the type of security processing used for the PIN data.

Table 783: Attributes

Attribute	Description
Data Representation	n-2
Data Field	Contents of positions 1–2
Justification	N/A

Table 784: Values

Value	Description
97	Multiple (indexed) keys

Subfield 2 (PIN Encryption Type Code)

DE 53, subfield 2 (PIN Encryption Type Code) indicates the type of security processing used for the PIN data.

Table 785: Attributes

Attribute	Description
Data Representation	n-2
Data Field	Contents of positions 3–4
Justification	N/A

Table 786: Values

Value	Description
01	DES encryption

Subfield 3 (PIN Block Format Code)

DE 53, subfield 3 (PIN Block Format Code) indicates the type of PIN block format used.

Table 787: Attributes

Attribute	Description
Data Representation	n-2
Data Field	Contents of positions 5–6
Justification	N/A

Table 788: Values

Value	Description
01	ANSI 1
10	ISO Format 0

Subfield 4 (PIN Key Index Number)

DE 53, subfield 4 (PIN Key Index Number) indicates the specific PIN key to be used when more than one key is available in a PIN key set.

Table 789: Attributes

Attribute	Description
Data representation	n-4
Data field	Contents of positions 7–10
Justification	N/A

Values

0001–0099

Subfield 5 (Reserved for Future Use)

DE 53, subfield 5 (Reserved) is reserved for future use.

Table 790: Attributes

Attribute	Description
Data Representation	n-2
Data Field	Contents of positions 11–12
Justification	N/A

Values

Not used, default to zero.

Subfield 6 (Reserved for Future Use)

DE 53, subfield 6 (Reserved) is reserved for future use.

Table 791: Attributes

Attribute	Description
Data Representation	n-4
Data Field	Contents of positions 13–16

Attribute	Description
Justification	N/A

Values

Not used, default to zero.

DE 54 (Additional Amounts)

DE 54 (Additional Amounts) provides information on up to three amount types and related account data.

Table 792: Attributes

Attribute	Description
Data Representation	an...240; LLLVAR The "LLL" length field of LLLVAR must be an integral multiple of 20, not to exceed 240.
Data Length	3 positions
Data Field	Contents of subfields 1–5
Subfields	5
Justification	See subfields

Usage

Following is the usage of DE 54 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 793: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	X	C
Authorization Request Response/0110	C	X	C
Authorization Advice/0120: acquirer-generated	C	X	C
Authorization Advice/0120: system-generated	•	X	C
Reversal Request/0400	C	X	C

Message	Org	Sys	Dst
Reversal Advice/0420	.	X	C

Values

See subfields.

Application notes

General

An "occurrence" is defined as one set of the four DE 54 subfields. Depending on the conditions of the message, DE 54 can be sent by the acquirer, issuer, or the Dual Message Authorization System.

Mastercard Network programs and services do not use more than two "additional amounts" amount types within a single Dual Message Authorization System message, except for healthcare amount types which use three.

The message originator can send only two occurrences of the DE 54 subfields in a message. One occurrence is the first amount type in the message originator's currency. The other occurrence is the second amount type in the message originator's currency. The Dual Message Authorization System will then provide an additional occurrence of each amount type to the message recipient in the message recipient's currency. Therefore, the message recipient can receive a maximum of four occurrences of the DE 54 subfields, except for healthcare amount types where the recipient could receive up to six occurrences. Each amount type will have two occurrences, one in the message originator's currency, and one in the message recipient's currency.

When the Authorization Request Response/0110 message contains DE 39, value 10 (Partial Approval) or 87 (Purchase Amount Only, No Cash Back Allowed), the issuer can send only one occurrence of the DE 54 subfields in the issuer's currency that is not equal to DE 54, subfield 2 (Amount Type) value 57 (Original Amount). The Dual Message Authorization System will provide two occurrences of the amount type sent by the issuer to the acquirer, one in the acquirer's currency, and one in the issuer's currency. The Dual Message Authorization System will also provide two occurrences of the DE 54 subfields where DE 54, subfield 2 (Amount Type) is 57 and DE 54, subfield 4 (Debit or Credit Indicator) is C plus the 12-digit original amount. One occurrence will be in the acquirer's currency and one occurrence will be in the issuer's currency. The exception is that if an occurrence of DE 54, subfield 2 value 58 (POI Amount) is sent, then the Dual Message Authorization System will not provide an additional occurrence in the issuer's currency.

Partial approvals

The Authorization Request Response/0110 message contains DE 54, subfield 2, value 57 only if the acquirer can process partial approvals and if the issuer is approving part of the total transaction amount. Issuers may include one additional occurrence of DE 54 subfields in the issuer's currency with an amount type value of 02 to indicate the available balance of the account. If the issuer approves the entire transaction amount, DE 54, subfield 2, value 57 will

not be present in the authorization request response and the issuer may provide more than one occurrence of DE 54 subfields.

Account balance response

The account balance response enables the issuers to include balance information as part of the response to a financial authorization request. Issuers can add account balance response information when responding to authorization requests for prepaid accounts and in ATM transactions. The account balance response is an "unsolicited" transmission of account balance data to the point-of-sale (POS) terminal or ATM. Account balance information returned in the Authorization Request Response/0110 must be provided in the same currency as the Cardholder Billing Currency (DE 51).

Purchase with cash back

Issuers are not required to return DE 54 in Purchase with Cash Back Authorization Request Response/0110 messages. If DE 54 is returned by the issuer, it will be dropped before the response is forwarded to the acquirer. The Authorization Request Response/0110 message only contains DE 54, subfield 2, value 40 when the transaction is processed by X-Code or declined by Mastercard with a format error condition (DE 39 = 30). When reversing a Purchase with Cash Back Authorization where only the purchase amount of the transaction was approved by the issuer (DE 39 equals 87 [Purchase Only Approval]), acquirers are not required to include the cash portion of the original authorization request in DE 54 of the Reversal Request/0400 message. Also, DE 4 (Amount, Transaction) of the reversal request should only contain the actual Purchase Amount of the original authorization request.

Cash back without purchase in participating countries

Cash Back without Purchase is supported for intracountry transactions in participating countries and is identified in the Authorization Request/0100 messages containing DE 3, subfield 1 (Cardholder Transaction Type), value 09 (Purchase with Cash Back) with the presence of DE 54, subfield 2, value 40. For Cash Back without Purchase transactions, DE 54 must be present and equal the same amount in DE 4.

Real-time substantiation

The Authorization Request/0100 or Authorization Advice/0120 messages contain DE 54, subfield 2, values 10, 11, or 12 to indicate the healthcare, prescription, or vision Rx amounts. DE 54, subfield 2, values 10, 11, and 12 are sent only to the issuers that participate in Real-time Substantiation.

Mastercard supports DE 54 as follows for acquirers and issuers:

- Overall size allowed by Mastercard = 240 bytes or 12 occurrences
- Number of occurrences an acquiring customer can send = 4
- Number of occurrences stored in USD currency = 4
- Number of occurrences converted to issuers currency = 4
- Number of occurrences sent to issuer = 8 (4 in acquirer currency, 4 in issuer currency)

The healthcare purchase information passed in DE 54 when these accounts are used at certain U.S. merchant locations is only required in purchase transaction authorization requests. It is not required and should not be expected to be present in refund transaction (purchase return) authorization requests.

Purchase of goods or services with cash back

Refer to [Program and Service Format Requirements, Purchase of Goods or Services with Cash Back, Dual Message Authorization System Edits](#).

POI currency conversion

When POI currency conversion (also known as Dynamic Currency Conversion [DCC]) is performed, Mastercard suggests that acquirers provide the pre-conversion currency and amount in DE 54 with an amount type value of 58 (POI Amount) in subfield 2 for Authorization Request/0100, Authorization Advice/0120, and Reversal Request/0400 messages.

The processing criteria for DCC transactions are as follows:

- The transaction currency code in DE 49 (Currency Code, Transaction) must not be equal to the pre-conversion currency code in DE 54, subfield 3 (Currency Code).
- DE 54 must contain subfield 2 (Amount Type), value 58 (POI Amount), along with all other DE 54 subfields.
- The transaction amount provided in DE 54, subfield 4 (Debit or Credit Indicator) cannot contain all zeros.
- DE 54, subfield 3 must contain a valid, pre-conversion currency code.
- Issuers are not required to return an occurrence of DE 54 with subfield 2, value 58 in Authorization Request Response/0110 message. If it is returned by the issuer, it will be dropped before the response is forwarded to the acquirer.
- For a Purchase with Cash Back transaction to be subjected to DCC, it must contain two occurrences of DE 54 in Authorization Request/0100 or Authorization Advice/0120 messages: one occurrence with a subfield 2 value 58 and another occurrence with a subfield 2 value of 40 (Amount Cash Back).
- Mastercard will allow acquirers to send one occurrence of DE 54 with a subfield 2 value 58 in Reversal Request/0400 messages.
- Mastercard will forward an occurrence of DE 54 with subfield 2, value of 58 in Reversal Advice/0420 messages to the issuers if the acquirer has provided an occurrence of DE 54 with subfield 2, value 58 in Authorization Request/0100 or Reversal Request/0400 message, except when a Reversal Advice/0420 message is sent to the issuer when an Authorization Request Response/0110 message from the issuer is not delivered to the acquirer.

For partial Reversal Request/0400 messages subject to DCC, acquirers may optionally supply the remainder of the transaction amount to be authorized in pre-conversion currency in DE 54, subfield 4 (Debit or Credit Indicator).

Subfield 1 (Account Type)

DE 54, subfield 1 (Account Type) contains the two-digit code as defined in DE 3 (Processing Code), subfield 2 (Cardholder "From Account" Type Code) or subfield 3 (Cardholder "To Account" Type Code).

Table 794: Attributes

Attribute	Description
Data Representation	n-2
Data Field	Contents of positions 1–2
Justification	N/A

Table 795: Values

Value	Description
00	No account specified
10	Savings account
20	Checking account
30	Credit card account

The valid values for this field are the same values as defined for [DE 3 \(Processing Code\)](#), subfield 2 (Cardholder "From Account" Type Code).

Subfield 2 (Amount Type)

DE 54, subfield 2 (Amount Type) indicates the type of amount applied.

Table 796: Attributes

Attribute	Description
Data Representation	n-2
Data Field	Contents of positions 3–4
Justification	N/A
Required as	Conditional

Table 797: Values

Value	Description
01	Ledger balance
02	Available balance
03	Amount owing
04	Amount due
05	Future transaction amount
10	Healthcare eligibility amount
11	Prescription eligibility amount
12	Vision Rx eligibility amount
17	Prepaid online bill pay fee amount
40	Cash back
44	Amount Gratuity
57	Original Amount
58	POI Amount
59	Limit/Balance amount available from Mastercard In Control
80	Copay Amount
90	Available Credit
91	Credit Limit
92	Outstanding hold amounts
95	Foreign exchange fee (Visa only)

Application notes

Amount type 05 is only applicable for Account Status Inquiry (ASI) Authorization transactions. The issuer will not receive an occurrence of DE 54 with amount type 05.

Subfield 3 (Currency Code)

DE 54, subfield 3 (Currency Code) is a three-digit code that must contain a valid numeric code.

Table 798: Attributes

Attribute	Description
Data Representation	n-3
Data Field	Contents of positions 5–7
Justification	Right

Values

Refer to the [Quick Reference Booklet](#).

Subfield 4 (Debit or Credit Indicator)

DE 54, subfield 4 (Debit or Credit Indicator) indicates the amount is a credit or debit indicator.

Table 799: Attributes

Attribute	Description
Data Representation	a-1
Data Field	Contents of position 8
Justification	Right

Table 800: Values

Value	Description
C	Credit amount or positive balance
D	Debit amount or negative balance

Subfield 5 (Amount)

DE 54, subfield 5 (Amount) contains the additional amount.

Table 801: Attributes

Attribute	Description
Data Representation	n-12
Data Field	Contents of positions 9–20
Justification	Right-justified with leading zeros

DE 54 Dual Message Authorization System edits

The Dual Message Authorization System will perform the following system edits.

Table 802: Associated edits

WHEN...	THEN the Dual Message Authorization System...
the acquirer sends an Authorization Request/0100 or an Authorization Advice/0120 message in which: - An occurrence of DE 54 (Additional Amounts) contains a subfield 2 (Amount Type) value of 58 (Additional Amount, POI Amount) and - An occurrence of DE 54 contains a subfield 3 (Currency Code) value that is equal to the DE 49 (Currency Code, Transaction) value or - An occurrence of DE 54 contains an invalid currency code in subfield 3.	will reject the transaction where: - DE 39 (Response Code) = 30 (Format Error) - DE 44 (Additional Response Data) = 054 (indicating the data element in error)
the acquirer sends an Authorization Request/0100 or an Authorization Advice/0120 message in which: - DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains value 09 (Purchase with Cash Back), and - DE 54, subfield 2 contains value 40 (Amount Cash Back), and - DE 54, subfield 3 contains a value that does not match the value in DE 49	will reject the transaction where: - DE 39 = 30 - DE 44 = 054

DE 55 (Integrated Circuit Card [ICC] System-Related Data)

DE 55 (Integrated Circuit Card [ICC] System-Related Data) contains binary data that only the issuer, the issuer agent, or MDES processes; it is used locally by the payment application on the chip at a chip-capable terminal. This data element is present in chip full-grade transactions and can be present in DSRP transactions.

Table 803: Attributes

Attribute	Description
Data Representation	b...255; LLLVAR The "LLL" length field of LLLVAR
Data Length	3 positions
Data Field	Contents of subelements
Subelements	Number of subelements depends on message type
Justification	N/A

Usage

Following is the usage of DE 55 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 804: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Request Response/0110	C	.	C
Authorization Advice/0120: acquirer-generated	C	.	C
Authorization Advice/0120: system-generated	.	X	C

Values

See subelements.

Application notes

For ICC information, refer to [M/Chip Requirements](#).

Refer to [DE 55: Authorization System Edits](#) for a description of the edits performed on this data element.

For DSRP information, refer to Mastercard Cloud-Based Payments Product description.

For MDES tokenized transactions, issuers may opt in through MDES Manager to receive track data in DE 55 (Integrated Circuit Card [ICC] System-Related Data) on Dual Message System (Authorization) or Single Message System transactions to issuers.

DE 55 (Subelement Encoding Scheme)

DE 55 contains chip data formatted in accordance with EMV specifications. Reference Book 3 of the EMV specification for details regarding coding of tag-length-value (TLV) data objects. The chip data in DE 55 consists of a series of subelements in a TLV format.

Table 805: Positions in DE 55

Data Length	Description
1–3	DE 55 Total Length
4 or 4–5	First subelement ID, in binary representation; the length is either one or two positions depending on the definition of the subelement ID in the EMV specification.
5 or 5–6	First subelement length; the value of the length subelement is always one; position depends on the subelement ID length.
6–xxx or 7–xxx	First subelement variable length data; the starting position depends on the subelement ID length.

Positions of the subelement length and variable length data depend on the subelements used.

Subelement ID (Tag), Length, and Variable Length Data may be repeated as needed until all chip data has been presented.

DE 55 Subelements

The following tables list the required and optional subelements in Authorization Request/0100 and Authorization Request Response/0110 messages that contain chip data.

NOTE: Subelements that are not listed as either required or optional must not be included in DE 55.

Required subelements in Authorization Request/0100

For Authorization requests related to Chip full-grade transactions and requests related to e-commerce transactions with EMV compliant ICC data, acquirers must provide the following data in DE 55 subelements.

Table 806: Subelement descriptions

Subelement Description	Tag Value	Component	Each Component Length	Total Subelement Length
Application Cryptogram (AC)	9F26	ID	2	11
		length	1	
		data	8	
Cryptogram Information Data	9F27	ID	2	4
		length	1	
		data	1	
Issuer Application Data (IAD) (Mandatory if the corresponding data object [EMV tag 9F10] is provided by the card to the terminal)	9F10	ID	2	4–35
		length	1	
		data	1–32	
Unpredictable Number	9F37	ID	2	7
		length	1	
		data	4	
Application Transaction Counter	9F36	ID	2	5
		length	1	
		data	2	
Terminal Verification Result (TVR)	95	ID	1	7
		length	1	
		data	5	
Transaction Date	9A	ID	1	5
		length	1	
		data	3	
Transaction Type	9C	ID	1	3
		length	1	
		data	1	
Amount Authorized	9F02	ID	2	9
		length	1	
		data	6	

Subelement Description	Tag Value	Component	Each Component Length	Total Subelement Length
Transaction Currency Code	5F2A	ID	2	5
		length	1	
		data	2	
Application Interchange Profile	82	ID	1	4
		length	1	
		data	2	
Terminal Country Code	9F1A	ID	2	5
		length	1	
		data	2	
Cardholder Verification Method (CVM) Results	9F34	ID	2	6
		length	1	
		data	3	
NOTE: The presence of 9F34 is mandatory for all authorization messages containing DE 55.				
Terminal Capabilities	9F33 (U.S. region only)	ID	2	6
		length	1	
		data	3	
Dedicated File Name	84	ID	1	7–18
		length	1	
		data	5–16	
Amount Other	9F03	ID	2	9
		length	1	
		data	6	

For the tag values the:

- Hexadecimal representation is given here. Every two positions of hexadecimal data is one byte of binary data.
- Lengths are in binary format.
- Total subelement length is the sum of the subelement's ID, length, and data subfields.

When cash back is not permitted by product rules, 9F03 may be absent, or present with a zero value. When cash back is permitted by product rules:

- If there is a cash back amount, 9F03 carries the amount and presence is mandatory
- If there is no cash back amount, the value of 9F03 is zero. 9F03 may be absent, or present with a zero value.

Optional subelements in Authorization Request/0100

When DE 55 is present in the Authorization Request/0100 message, the following subelements are optional in DE 55.

Table 807: Subelement descriptions

Subelement Description	Tag Value	Component	Each Component Length	Total Subelement Length
Application Primary Account Number (PAN) Sequence Number	5F34	ID	2	4
		length	1	
		data	1	
Application Selection Registered Proprietary Data For technical information, refer to M/Chip Requirements for Contact and Contactless .	9F0A	ID	2	3–Var
		length	1	
		data	Var	
Terminal Type	9F35	ID	2	4
		length	1	
		data	1	
Interface Device (IFD) Serial Number	9F1E	ID	2	11
		length	1	
		data	8	
Transaction Category Code	9F53	ID	2	4
		length	1	
		data	1	
Application Version Number	9F09	ID	2	5
		length	1	
		data	2	
Transaction Sequence Counter	9F41	ID	2	5–7
		length	1	
		data	2–4	

Subelement Description	Tag Value	Component	Each Component Length	Total Subelement Length
Terminal Capabilities	9F33 (non-U.S. regions only)	ID	2	6
		length	1	
		data	3	
Third Party Data	9F6E	ID	2	8–35
		length	1	
		data	5–32	
Authenticated Application Data	9F60	ID	2	4–130
		length	1	
		data	var-127	
Kernel Identifier-Terminal	96	ID	1	10
		length	1	
		data	8	

Optional subelements in Authorization Request Response/0110

Issuers may provide the following DE 55 subelements in Authorization Request Response/0110 messages related to:

- Chip full-grade transactions
- E-commerce transactions with EMV-compliant ICC data

If any of these subelements are present in the Authorization Request Response/0110 message, the acquirer must pass the subelements, unaltered, to the IC card.

Table 808: Subelement descriptions

Subelement Description	Tag Value	Component	Each Component Length	Total Subelement Length
Issuer Authentication Data (Provides data to be transmitted to the card for issuer authentication.)	91	ID	1	10–18
		length	1	
		data	8–16	

Subelement Description	Tag Value	Component	Each Component Length	Total Subelement Length
Issuer Script Template 1 (Allows the issuer to provide a post-issuance command for transmission to the card. Present if issuer sends commands to ICC.) Issuers may send more than one instance (maximum of 10 instances or maximum length of DE 55) of subelement 71 in the Authorization Request Response/0110.	71	ID	1	3–129
		length	1	
		data	1–127	
NOTE: Either tag 71 or tag 72 should be included in any single Authorization Request Response/0110 message but not both.				
Issuer Script Template 2 (Allows the issuer to provide a post-issuance command for transmission to the card. Present if issuer sends commands to ICC.) Issuers may send more than one instance (maximum of 10 instances or maximum length of DE 55) of subelement 72 in the Authorization Request Response/0110.	72	ID	1	3–129
		length	1	
		data	1–127	
NOTE: Either tag 71 or tag 72 should be included in any single Authorization Request Response/0110 message but not both.				

DE 55 Dual Message Authorization System Edits

The Dual Message Authorization System performs the following edits.

If Mastercard determines through its Internal Chip Monitoring process that improperly formatted chip transactions are being submitted from acquirers not certified to send chip transactions, Mastercard will notify each acquirer before activating an edit. The Dual Message Authorization System will perform the following edits on the Authorization Request/0100 message.

Table 809: Associated edits

WHEN...	THEN the Dual Message Authorization System...
DE 22, subfield 1 contains value 05 or 07, and DE 55 is present and Acquirer Chip Testing Level associated with the acquirer indicates the acquirer is approved for partial grade chip transactions	sends the acquirer an Authorization Request Response/0110 message where: • DE 39 = 30 • DE 44 = 055
DE 22, subfield 1 contains value 05 and DE 55 is not present and Acquirer Chip Testing Level associated with the acquirer indicates the acquirer is approved to send full grade chip transactions and DE 3, subfield 1 does not contain value 20	sends the acquirer an Authorization Request Response/0110 message where: • DE 39 = 30 • DE 44 = 055
DE 22, subfield 1 contains value 07 and DE 55 is not present and Acquirer Chip Testing Level associated with the acquirer indicates the acquirer is approved to send full grade chip transactions and DE 3, subfield 1 does not contain value 20	sends the acquirer an Authorization Request Response/0110 where: • DE 39 = 30 • DE 44 = 055

Table 810: Associated edits

WHEN...	THEN The Dual Message Authorization System...
DE 55 is present in the Authorization Request/0100 message or the Authorization Advice/0120 message	DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 must be 03, 05, 07, 09, 79, 80, or 81 or the Authorization System rejects the Authorization Request/0100 message or Authorization Advice/0120 message where: • DE 39 = 30 • DE 44 = 055
DE 55 is not present in the Authorization Request/0100 message for a Chip PIN Management transaction	sends the acquirer an Authorization Request Response/0110 message where: • DE 39 = 30 • DE 44 = 055
DE 55 is greater than 255 characters in length in the Authorization Request/0100 message	sends the acquirer an Authorization Request Response/0110 where: • DE 39 = 30 • DE 44 = 055

The Authorization System will perform the following edits on an Authorization Request Response/0110 message.

Table 811: Associated edits

WHEN...	THEN the Dual Message Authorization System...
DE 55 is echoed in the Authorization Request Response/0110 message	sends the issuer an Authorization Response Negative Acknowledgment/0190 message where: • DE 39 = 30 (Format error) • DE 44 = 055

Table 812: Associated edits

WHEN the issuer...	THEN the Dual Message Authorization System...
sends an Authorization Request Response/0110 message where DE 55, subelement 71 is present and exceeds 127 bytes in length	sends an Authorization Response Negative Acknowledgment/0190 message to the issuer where: • DE 39 = 30 • DE 44 = 055

Table 813: Associated edits

IF...	THEN the Dual Message Authorization System...
DE 55 is present in the Authorization Request Response/0110 and DE 55 was not present in the original Authorization Request/0100	sends an Authorization Response Negative Acknowledgment/0190 message to the issuer where: • DE 39 = 30 • DE 44 = 055

Dual Message Authorization System Edits: Cardholder Authentication Service

The Dual Message Authorization System will perform the following system edits related to the Cardholder Authentication Service.

Table 814: Associated edits

WHEN...	THEN the Dual Message Authorization System...
an Account Status Inquiry Transaction for account ranges participating in the Cardholder Authentication Service - the Authorization Request/0100 message contains: • DE 2 (Primary Account Number [PAN]) setup for Type 1 Authentication Service and • DE 4 (Amount, Transaction) has a value of zero and • DE 23 (Card Sequence Number) contains a value of 01X and • DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status), value 8 (Account Status Inquiry Service [ASI]) and • DE 55 (Integrated Circuit Card [ICC] System-Related Data) is present and tag 9F10: – CVR byte1, bit1 is present with a value other than 1 (successful) or – CVR byte2, bit2 is present with a value other than 1 (biometric).	sends the acquirer an Authorization Request Response/0110 message where: • DE 39 (Response Code) = 05 (Do not honor) and Sends the issuer an Authorization Advice/0120 message with DE 60 (Advice Reason Code): • Subfield 1 (Advice Reason Code) value 160 (Authentication Advice to Issuer) and • Subfield 2 (Advice Detail Code) value 0078 (M/Chip Biometric Data not present)

WHEN...	THEN the Dual Message Authorization System...
an Account Status Inquiry Transaction for account ranges participating in the Cardholder Authentication Service - the Authorization Advice/0120: acquirer-generated message contains: • DE 2 (Primary Account Number [PAN]) setup for Type 1 Authentication Service and • DE 4 (Amount, Transaction) has a value of zero and • DE 23 (Card Sequence Number) contains a value of 01X (X can be any number from 0 to 9 inclusive) and • DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status), value 8 (Account Status Inquiry Service [ASI]) and • DE 55 (Integrated Circuit Card [ICC] System-Related Data) is present and tag 9F10: – CVR byte1, bit1 is present with a value other than 1 (successful) or – CVR byte2, bit2 is present with a value other than 1 (biometric).	sends the acquirer an Authorization Advice Response/0130 message where: • DE 39 (Response Code) = 30 (Format Error) and • DE 44 (Additional Response Data) = 055 (indicating the data element in error)
an Account Status Inquiry Transaction for account ranges participating in the Cardholder Authentication Service - the Authorization Request/0100 message contains: • DE 2 (Primary Account Number [PAN]) setup for Type 2 Authentication Service and • DE 4 (Amount, Transaction) has a value of zero and • DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status), value 8 (Account Status Inquiry Service [ASI]) and • DE 55 (Integrated Circuit Card [ICC] System-Related Data) is present and tag 9F10: – CVR byte1, bit1 is present with a value other than 1 (successful) or – CVR byte2, bit2 is present with a value other than 1 (biometric).	sends the acquirer an Authorization Request Response/0110 message where: • DE 39 (Response Code) = 05 (Do not honor) and Sends the issuer an Authorization Advice/0120 message with DE 60 (Advice Reason Code): • Subfield 1 (Advice Reason Code) value 160 (Authentication Advice to Issuer) and • Subfield 2 (Advice Detail Code) value 0078 (M/Chip Biometric Data not present)

WHEN...	THEN the Dual Message Authorization System...
an Account Status Inquiry Transaction for account ranges participating in the Cardholder Authentication Service - the Authorization Advice/0120: acquirer-generated message contains: - DE 2 (Primary Account Number [PAN]) setup for Type 2 Authentication Service and - DE 4 (Amount, Transaction) has a value of zero and - DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status), value 8 (Account Status Inquiry Service [ASI]) and - DE 55 (Integrated Circuit Card [ICC] System-Related Data) is present and tag 9F10: - CVR byte1, bit1 is present with a value other than 1 (successful) or - CVR byte2, bit2 is present with a value other than 1 (biometric).	sends the acquirer an Authorization Advice Response/0130 message where: - DE 39 (Response Code) = 30 (Format Error) and - DE 44 (Additional Response Data) = 055 (indicating the data element in error)

DE 56 (Payment Account Data)

DE 56 (Payment Account Data) contains unique, non-financial reference information associated with the PAN or token used to initiate the transaction.

Table 815: Attributes

Attribute	Description
Data Representation	an...37; LLLVAR
Data Length	3 positions
Data Field	Contents of subelements
Subelements	1
Justification	See subelement

Usage

Following is the usage of DE 56 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 816: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	•	X	C
Authorization Request/0100: Account Status inquiry (merchant initiated message and MDES pre-digitization message)	•	X	C
Authorization Request/0100: Tokenization Complete Notification*	•	X	C
Authorization Request Response/0110	C	X	C
Authorization Request Response/0110: Tokenization Authorization Request Response*	C	X	C
Authorization Request Response/0110: Account Status Inquiry*	C	X	C
Authorization Advice/0120: acquirer-generated	•	X	C
Authorization Advice/0120: system-generated	•	X	C
Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)	C	X	•
Issuer File Update Request/0302	O	•	C
Reversal Request/0400	•	X	C
Reversal Request Response/0410	C	X	C
Reversal Advice/0420	•	X	C
Reversal Advice Response/0430	O	•	•
Administrative Advice/0620: system-generated: issuer token notification advice for tokenization complete notification*	•	X	C

* MDES pre-digitization message

Subelement 01 (Payment Account Data)

DE 56, subelement 01 (Payment Account Data) contains the applicable subfields to carry the unique Payment Account Data specifically associated with the PAN or token used to initiate the transaction.

Table 817: Attributes

	Attribute	Value
Subelement ID	n-2	01
Subelement Data Length	n-2	...33

	Attribute	Value
Data Representation	an...33; LLVAR	
Data Field		Contents of positions 1–2
Number of subfields	1	Payment Account Reference

Subfield 01 (Payment Account Reference [PAR])

DE 56, subelement 01, subfield 01 (Payment Account Reference [PAR]) contains the assigned PAR value. A PAR is a unique value associated with a single PAN and attributed to all tokens associated with that PAN. A PAR can be used to link transactions containing PANs or tokens associated with the same underlying payment account.

Table 818: Attributes

	Attribute	Value
Subfield ID	n-2	01
Subfield Data Length	n-2	...29
Data Representation	an...29; LLVAR	
Justification	N/A	

Table 819: Positions and descriptions

Position	Description
1–4	BIN Controller Identifier: A four-character value assigned by EMVCo
5–29	A unique 25-character alphanumeric uppercase value generated by the BIN Controller and linked to a PAN

Application notes

Mastercard supports PAR for all acquirers and issuers for MDES, MDES Lite, and non-MDES transactions in DE 56, subfield 01, subelement 01.

PAR is not added to Mastercard ATM transactions.

DE 57–DE 59 (Reserved for National Use)

DE 57–DE 59 are reserved for future use.

NOTE: The Dual Message Authorization System currently does not use these data elements.

DE 60 (Advice Reason Code)

DE 60 (Advice Reason Code) indicates to the receiver of an Advice message the specific reason for the transmission of the Advice message.

Table 820: Attributes

Attribute	Description
Data Representation	ans...060; LLLVAR
Data Length	3 positions
Data Field	Contents of subfields
Subfields	3
Justification	See subfields

Usage

Following is the usage of DE 60 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 821: Applicable messages

Message	Org	Sys	Dst
Authorization Advice/0120: acquirer-generated	M	.	M
Authorization Advice/0120: system-generated	.	X	M
Reversal Advice/0420	.	X	M
Administrative Request/0600	M	.	M
Administrative Request Response/0610	ME	.	ME
Administrative Advice/0620: member-generated	M	.	M
Administrative Advice/0620: system-generated	.	X	M

Values

See subfields.

Application notes

Not all Advice Reason Codes may be used within all programs and services.

Subfield 1 (Advice Reason Code)

DE 60, subfield 1 contains the advice reason code.

Table 822: Attributes

Attribute	Description
Data Representation	n-3
Data Field	Contents of positions 1–3
Justification	N/A

Values

This subfield is mandatory for all advice messages and indicates the general purpose of the advice message.

The following tables displays the general values for subfields 1 and 2 based on message type for all products, though each code does not pertain to all products.

DE 60, subfield 1 values, in Authorization Advice/0120

The following values are valid in Authorization Advice/0120 messages.

Table 823: Values

Code	Description	MC	NP	VI	TE	MS	CI
100	Alternate Issuer Route: Issuer selected option This secondary route is to either the Stand-In System or the optional alternate route. This secondary route is to either the Stand-In System or the optional alternate route.	√	√	√		√	√
101	Alternate Issuer Route: Issuer signed out This secondary route is to either the Stand-In System or the optional alternate route.	√	√	√		√	√

Code	Description	MC	NP	VI	TE	MS	CI
102	Alternate Issuer Route: Issuer timed out This secondary route is to either the Stand-In System or the optional alternate route.	√	√	√		√	√
103	Alternate Issuer Route: Issuer unavailable This secondary route is to either the Stand-In System or the optional alternate route.	√	√	√		√	√
105	Transaction processed through X-Code	√	√	√			
107	PIN processing error	√				√	√
108	Alternate Issuer Route: MIP Error This secondary route is to either the Stand-In System or the optional alternate route.	√	√	√		√	√
109	Alternate Issuer Route: Issuer Edit Response Error This secondary route is to either the Stand-In System or the optional alternate route.	√	√	√		√	√
111	Alternate Issuer Route: Issuer Host System Error This secondary route is to either the Stand-In System or the optional alternate route.	√	√	√		√	√
112	Alternate Route: Network Not Dispatched Error This secondary route is to either the Stand-In System or the optional alternate route.	√	√	√		√	√
113	Alternate Route: Issuer Undelivered This secondary route is to either the Stand-In System or the optional alternate route.	√	√	√		√	√
114	Alternate Route: Direct Down Option This secondary route is to either the Stand-In System or the optional alternate route.	√	√	√		√	√
115	Transaction Processed via On-behalf Service Decision	√				√	√
116	Invalid Merchant	√					
120	Transaction Blocking	√				√	√
121	Account Lookup Service	√	√			√	
126	Pay with Rewards Processing Advice to Issuer	√					
140	Unable to convert contactless or virtual account number	√				√	
141	Mastercard Digital Enablement Service Advice to Issuer	√				√	
142	TSP Advice to Issuer	√				√	

Code	Description	MC	NP	VI	TE	MS	CI
150	Send and other transactions Blocking Service Advice to Issuers	√				√	√
151	In Control Processing Advice to Issuer (Mastercard Merchant Presented QR)	√					
160	Authentication Advice to Issuer	√				√	
180	CAT Risk Level 3	√	√				
190	Acquirer Processing System (APS) Approved	√	√	√		√	√
191	Acquirer Processing System (APS) Completed Authorization Transaction	√	√		√	√	
192	M/Chip Offline Advice to Issuer	√				√	
200	In Control Processing Advice to Issuer	√					
300	Transaction processed with On-Demand Decisioning	√				√	√
650	Administrative textual message transmittal	√	√	√		√	√

DE 60, Authorization Advice/0120 edits

The Dual Message Authorization System performs the following system edits on Authorization Advice/0120: acquirer-generated messages.

Table 824: Associated edits

WHEN...	THEN the Dual Message Authorization System...
DE 60 in the Authorization Advice/0120: acquirer-generated messages is not the value 190 (Acquirer Processing System (APS) Approved)	sends the acquirer an Authorization Advice Response/0130 message where:
or	- DE 39 (Response Code) = 30 (Format Error) - DE 44 (Additional Response Data) = 060 (Invalid DE 60 value)
191 Acquirer Completed Authorization Transaction)	
or	
192 (M/Chip Offline Advice to Issuer)	

DE 60, subfield 1 values, in Reversal Advice/0420

The following values are valid in Reversal Advice/0420 messages.

Table 825: Values

Code	Description	MC	NP	VI	TE	MS	CI
400	Banknet advice: Acquirer error; unable to deliver response	√	√	√	√	√	√
401	Banknet advice: Acquirer error; no Acquirer Authorization Response Acknowledgment/0180 or Financial Transaction Acknowledgment/0280	√	√	√		√	√
402	Issuer Time-out	√	√	√		√	√
403	Issuer Sign-out	√	√	√		√	√
409	Issuer Response Error	√	√	√		√	√
410	Reversal message provided by a system other than Banknet	√				√	√
413	Issuer Undelivered	√	√	√		√	√

DE 60, subfield 1 values, in Administrative Request/0600

Following are the valid values in Administrative Request/0600 messages.

Table 826: Values

Code	Description	MC	NP	VI	TE	MS	CI
650	Administrative message containing customer application and account data	√	√				

DE 60, subfield 1 values, in Administrative Request Response/0610

Following are valid values in Administrative Request Response/0610 messages.

Table 827: Values

Code	Description	MC	NP	VI	TE	MS	CI
650	Administrative message containing customer application and account data	√	√				

DE 60, subfield 1 Values, in Administrative Advice/0620

Following are valid values in Administrative Advice/0620 messages.

Table 828: Values

Code	Description	MC	NP	VI	TE	MS	CI
600	Message unreadable or indecipherable or contains invalid data. Subfield 2 (Advice Detail Code) may contain the bit map number of the data element where message scanning was aborted)	√	√	√	√	√	√
650	Administrative textual message transmittal	√	√	√		√	√

DE 60, subfield 1 Value, in Send Blocking Service

These values are valid in Send Blocking Service transactions.

Table 829: Values

Code	Description	MC	NP	VI	TE	MS	CI
150	Send Blocking service processing advice to issuers		X			X	X

Subfield 2 (Advice Detail Code)

DE 60, subfield 2 (Advice Detail Code) is optional, depending on the primary Advice Reason Code; if used, it provides additional (specific) information as to the exact nature of the Advice message. Advice Detail Codes are determined individually for each program and service. The Advice Detail Codes for Mastercard activity appear below. Refer to the appropriate documentation for information on codes for non-Mastercard activity.

Table 830: Attributes

Attribute	Description
Data representation	n-4
Data field	Contents of positions 4–7
Justification	N/A

Table 831: Values

Value	Description
0000	Accept
0001	Reject: negative file
0002	Reject: capture card
0003	Reject: issuer not participating
0004	Reject: invalid PIN
0005	Reject: ATM
0006	Reject: transaction limit test
0009	Reject: Invalid Time Validation
0041	Reject: Payment Cancellation
0078	Reject: M/Chip Biometric Data not Present
0100	Reject: Account Balance Listing, transaction amount exceeds current limit
0101	Reject: Dynamic Decisioning, AI model decline
0216	Reject: DTVC Invalid/Expired
0217	Reject: Channel not Authorized by Cardholder

DE 60, subfield 2 values, in Authorization Advice/0120: issuer-generated

The following values are valid in Authorization Advice/0120: issuer-generated messages.

Table 832: Values

Value	Description
0030	Accept: Member-generated Authorization Advice/0120 sent to the risk scoring server. (Used with Advice Reason Code 650.)

DE 60, subfield 2 values, in Authorization Advice/0120: system-generated

The following values are valid in Authorization Advice/0120—System-generated messages.

Reason codes 0007–1611 apply only to the Authorization Advice/0120: system-generated message.

Table 833: Values

Value	Description
0007	Reject: Premium listing cumulative limit test
0008	Reject: merchant suspicious indicator test
0009	Reject: Invalid Time Validation
0010	Reject: Stand-In Monitoring Rule
0011	Reject: day number 1/number of transactions
0012	Reject: day number 2/number of transactions
0013	Reject: day number 3/number of transactions
0014	Reject: day number 4/number of transactions
0015	Reject: day number 1/amount
0016	Reject: day number 2/amount
0017	Reject: day number 3/amount
0018	Reject: day number 4/amount
0019	Reject: extended Cash Advance cumulative amount
0020	Reject: card number in blocked range
0021	Reject: Premium Listing transaction limit test
0028	Reject: invalid CVC 1
0029	Reject: expired card
0031	Reject: unable to decrypt/encrypt PIN data. (Used with Advice Reason Code 107)
0036	Reject: CVC 1 Unable to process
0047	Reject: CVC 1 No matching key file for this PAN, PAN expiry date combination; status unknown
0216	Reject: DTVC Invalid/Expired
1000	Accept: On-Demand Decisioning Rule
1001	Reject: On-Demand Decisioning Rule

DE 60, subfield 2 values, in Administrative Advice/0620

Following are the valid values in Administrative Advice/0620 messages.

Table 834: Values

Values	Description
0029	MIP-generated risk Advice transactions
0030	Member-generated risk Advice transactions
0250	Activation Code Notification
0251	Tokenization Complete Notification
0252	Tokenization Event Notification

DE 60, subfield 2 values, in Customer Service Messages

Following are the valid values in Customer Service messages.

Table 835: Values

Value	Description
0080	Consumer application request
0081	Consumer application status inquiry
0082	Consumer user lookup inquiry
0083	Consumer account lookup inquiry
0084	Consumer account maintenance request
0085	Consumer counteroffer reply
0086	Consumer preapproved offer inquiry
0090	Business application request
0091	Business application status inquiry
0092	Business user lookup inquiry
0093	Business account lookup inquiry
0094	Business account maintenance request
0095	Business counteroffer reply
0096	Business pre-approved offer inquiry

DE 60, subfield 2 values, in Dynamic CVC 3 Validation

Following are valid values in Dynamic CVC 3 Validation transactions.

Table 836: Values

Value	Description
0037	Reject: No matching key file for this PAN, PAN expiry date, and KDI combination
0038	Reject: Security Platform Time Out
0040	Reject: Security Platform System Error
0042	Reject: CVC 3 Unable to process
0043	Reject: ATC outside allowed range
0044	Reject: CVC 3 Invalid
0045	Reject: CVC 3 Unpredictable number mismatch
0046	Reject: ATC Replay

DE 60, subfield 2 values, in Mastercard In Control Service

The following values are valid in Mastercard In Control™ Service transactions.

Table 837: Values

Value	Description
0060	Reject: Virtual Card Number (expiration date does not match)
0061	Reject: Virtual Card Number (expiration date expired)
0062	Reject: Virtual CVC 2 does not match
0063	Reject: Validity Period Limit: In Control
0064	Reject: Transaction Amount Limit Check
0065	Reject: Cumulative Amount Limit Check
0066	Reject: Transaction Number Usage
0067	Reject: Merchant ID Limit
0068	Reject: Invalid Virtual Card Number–Real Card Number Mapping Relationship
0069	Reject: MCC Limit
0070	Reject: Database Status Bad
0071	Reject: Decline Other
0072	Reject: Geographic Restriction

Value	Description
0073	Reject: Transaction Type Restriction
0075	Reject: Transaction Time/Date Restriction
0076	Reject: Sanctions Score Restriction
0077	Reject: Mastercard Send and Funding Transaction Count Check

DE 60, subfield 2 values, in M/Chip On-Behalf Services

Following are valid values in M/Chip On-Behalf Services transactions.

Table 838: Values

Value	Description
0032	Reject: Chip Data Processing Error
0033	Reject: PIN or CHIP Validation Failed
0034	Reject: Chip validation failed
0035	Reject: TVR/CVR validation failed
0037	Reject: No matching key File for this PAN, PAN expiry date and KDI combination—Validation of ARQC and CVR/TR not performed, status unknown
0038	Reject: Security platform time out
0039	Reject: Cryptogram not ARQC
0040	Reject: Security platform processing error
0043	Reject: ATC outside allowed range
0046	Reject: ATC Replay
0054	Reject: Track Data Format Error
0055	Reject: Chip CVC Invalid
0056	Reject: Chip CVC/CVC 1 Unable to process

DE 60, subfield 2 values, in Mastercard Digital Enablement Service

Following are valid values in Mastercard Digital Enablement Service.

Table 839: Values

Value	Description
0201	Reject: Invalid Token—Primary Account Number mapping relationship

Value	Description
0202	Reject: Token in suspended status
0203	Reject: Token deactivated
0204	Reject: ATC Invalid—Not in List of Currently Active Single Use Keys
0205	Reject: ATC Replay
0206	Reject: Invalid MD AC and UMD AC (Invalid Mobile PIN)
0207	Reject: Valid MD AC; Invalid UMD AC (Mobile PIN Try Counter Max Limit not Reached, Token not Suspended)
0208	Reject: Invalid MD AC; Valid UMD AC
0209	Reject: Valid MD AC; Invalid UMD AC (Mobile PIN Try Counter Max Limit Reached, Token Suspended)
0210	Reject: Unpredictable Number Length Indicator Mismatch
0211	Reject: TVR/CVR validation failed
0212	Reject: Unable to Process
0213	Reject: Invalid Token
0215	Reject: Declined by Transaction Analysis
0032	Reject: Chip Data Processing Error
0034	Reject: Chip validation failed
0035	Reject: TVR/CVR validation failed
0039	Reject: Cryptogram not ARQC
0042	Reject: CVC 3 Unable to process
0043	Reject: CVC 3 ATC outside allowed range
0044	Reject: CVC 3 Invalid
0045	Reject: CVC 3 Unpredictable number mismatch
0046	Reject: CVC 3 ATC Replay

DE 60, subfield 2 values, in Mastercard Merchant Presented QR Service

Following are valid values in Mastercard Merchant Presented QR Service transactions.

Table 840: Values

Value	Description
0064	Reject: Transaction Limit Check
0065	Reject: Cumulative Limit Check

Value	Description
0072	Reject: Geographic Restriction

DE 60, subfield 2 values, in Pay with Rewards

Following are valid values in Mastercard Pay with Rewards service.

Table 841: Values

Value	Description
0120	Reject: Pay with Rewards—Insufficient points balance
0121	Reject: Pay with Rewards—Redemption rule(s) failed
0122	Reject: Pay with Rewards service was not performed successfully
0123	Reject: Pay with Rewards—Account not registered
0124	Reject: Pay with Rewards System error

DE 60, subfield 2 values, in PIN Validation

Following are the valid values in PIN Validation transactions.

Table 842: Values

Value	Description
0050	Reject: Unable to Process
0051	Reject: Invalid PIN
0052	Reject: Mandatory PIN Verification Value (PVV) not on file
0052	Reject: PIN Retry Exceeded (invalid PIN)

DE 60, subfield 2 values, in Private Label Processing

Following are valid values in Private Label Processing transactions.

Table 843: Values

Value	Description
0116	Reject: Merchant not allowed for Private Label transaction

DE 60, subfield 2 values, in TCC, CAT Level, and Promotion Code in Failed Parameter Combinations (Country-Specific)

Following are valid values in TCC, CAT Level, and Promotion Code in Failed Parameter Combinations (country-specific) transactions.

Table 844: Values

Value	Description
0300	Reject: transaction limit test Mastercard default limits caused transaction to fail; card present at point of interaction
0301	Reject: transaction limit test Mastercard default limits caused transaction to fail; card not present at point of interaction
0310	Reject: transaction limit test Customer-defined limits caused transaction to fail; card present at point of interaction
0311	Reject: transaction limit test Customer-defined limits caused transaction to fail; card not present at point of interaction

DE 60, subfield 2 values, in TCC, CAT Level, and Promotion Code in Failed Parameter Combinations (Global)

Following are valid values in TCC, CAT Level, and Promotion Code in Failed Parameter Combinations (global) transactions.

Table 845: Values

Value	Description
0400	Reject: transaction limit test Mastercard default limits caused transaction to fail; card present at point of interaction
0401	Reject: transaction limit test Mastercard default limits caused transaction to fail; card not present at point of interaction

Value	Description
0410	Reject: transaction limit test Customer-defined limits caused transaction to fail; card present at point of interaction
0411	Reject: transaction limit test Customer-defined limits caused transaction to fail; card not present at point of interaction

DE 60, subfield 2 values, in MCC, CAT Level, and Promotion Code in Failed Parameter Combinations (Country-Specific)

Following are valid values for MCC, CAT Level, and Promotion Code in Failed Parameter Combinations (country-specific) transactions.

Table 846: Values

Value	Description
0100	Reject: transaction limit test Mastercard default limits caused transaction to fail; card present at point of interaction
0101	Reject: transaction limit test Mastercard default limits caused transaction to fail; card not present at point of interaction
0110	Reject: transaction limit test Customer-defined limits caused transaction to fail; card present at point of interaction
0111	Reject: transaction limit test Customer-defined limits caused transaction to fail; card not present at point of interaction

DE 60, subfield 2 values, in MCC and Promotion Code in Failed Parameter Combinations (Country-Specific)

Following are valid values in MCC and Promotion Code in Failed Parameter Combinations (country-specific) transactions.

Table 847: Values

Value	Description
0500	Reject: transaction limit test Mastercard default limits caused transaction to fail; card present at point of interaction
0501	Reject: transaction limit test Mastercard default limits caused transaction to fail; card not present at point of interaction
0510	Reject: transaction limit test Customer-defined limits caused transaction to fail; card present at point of interaction
0511	Reject: transaction limit test Customer-defined limits caused transaction to fail; card not present at point of interaction

DE 60, subfield 2 values, in MCC and Promotion Code in Failed Parameter Combinations (Global)

Following are valid values in MCC and Promotion Code in Failed Parameter Combinations (global) transactions.

Table 848: Values

Value	Description
0600	Reject: transaction limit test Mastercard default limits caused transaction to fail; card present at point of interaction
0601	Reject: transaction limit test Mastercard default limits caused transaction to fail; card not present at point of interaction

Value	Description
0610	Reject: transaction limit test Customer-defined limits caused transaction to fail; card present at point of interaction
0611	Reject: transaction limit test Customer-defined limits caused transaction to fail; card not present at point of interaction

DE 60, subfield 2 values, in TCC and Promotion Code in Failed Parameter Combinations (Country-Specific)

Following are valid values in TCC and Promotion Code in Failed Parameter Combinations (country-specific) transactions.

Table 849: Values

Value	Description
0700	Reject: transaction limit test Mastercard default limits caused transaction to fail; card present at point of interaction
0701	Reject: transaction limit test Mastercard default limits caused transaction to fail; card not present at point of interaction
0710	Reject: transaction limit test Customer-defined limits caused transaction to fail; card present at point of interaction
0711	Reject: transaction limit test Customer-defined limits caused transaction to fail; card not present at point of interaction

DE 60, subfield 2 values, in TCC and Promotion Code in Failed Parameter Combinations (Global)

Following are valid values in TCC and Promotion Code in Failed Parameter Combinations (global) transactions.

Table 850: Values

Value	Description
0800	Reject: transaction limit test Mastercard default limits caused transaction to fail; card present at point of interaction
0801	Reject: transaction limit test Mastercard default limits caused transaction to fail; card not present at point of interaction
0810	Reject: transaction limit test Customer-defined limits caused transaction to fail; card present at point of interaction
0811	Reject: transaction limit test Customer-defined limits caused transaction to fail; card not present at point of interaction

DE 60, subfield 2 values, in MCC and CAT Level in Failed Parameter Combinations (Country-Specific)

Following are valid values in MCC and CAT Level in Failed Parameter Combinations (country-specific) transactions.

Table 851: Values

Values	Description
0900	Reject: transaction limit test Mastercard default limits caused transaction to fail; card present at point of interaction
0901	Reject: transaction limit test Mastercard default limits caused transaction to fail; card not present at point of interaction

Values	Description
0910	Reject: transaction limit test Customer-defined limits caused transaction to fail; card present at point of interaction
0911	Reject: transaction limit test Customer-defined limits caused transaction to fail; card not present at point of interaction

DE 60, subfield 2 values, in MCC and CAT Level in Failed Parameter Combinations (Global)

Following are valid values in MCC and CAT Level in Failed Parameter Combinations (global) transactions.

Table 852: Values

Value	Description
1000	Reject: transaction limit test Mastercard default limits caused transaction to fail; card present at point of interaction
1001	Reject: transaction limit test Mastercard default limits caused transaction to fail; card not present at point of interaction
1010	Reject: transaction limit test Customer-defined limits caused transaction to fail; card present at point of interaction
1011	Reject: transaction limit test Customer-defined limits caused transaction to fail; card not present at point of interaction

DE 60, subfield 2 values, in TCC and CAT Level in Failed Parameter Combinations (Country-Specific)

Following are valid values in TCC and CAT Level in Failed Parameter Combinations (country-specific) transactions.

Table 853: Values

Value	Description
1100	Reject: transaction limit test Mastercard default limits caused transaction to fail; card present at point of interaction
1101	Reject: transaction limit test Mastercard default limits caused transaction to fail; card not present at point of interaction
1110	Reject: transaction limit test Customer-defined limits caused transaction to fail; card present at point of interaction
1111	Reject: transaction limit test Customer-defined limits caused transaction to fail; card not present at point of interaction

DE 60, subfield 2 values, in TCC and CAT Level in Failed Parameter Combinations (Global)

Following are valid values in TCC and CAT Level in Failed Parameter Combinations (global) transactions.

Table 854: Values

Value	Description
1200	Reject: transaction limit test Mastercard default limits caused transaction to fail; card present at point of interaction
1201	Reject: transaction limit test Mastercard default limits caused transaction to fail; card not present at point of interaction
1210	Reject: transaction limit test Customer-defined limits caused transaction to fail; card present at point of interaction

Value	Description
1211	Reject: transaction limit test Customer-defined limits caused transaction to fail; card not present at point of interaction

DE 60, subfield 2 values, in MCC in Failed Parameter Combinations (Country-Specific)

Following are valid values in MCC in Failed Parameter Combinations (country-specific) transactions.

Table 855: Values

Value	Description
1300	Reject: transaction limit test Mastercard default limits caused transaction to fail; card present at point of interaction
1301	Reject: transaction limit test Mastercard default limits caused transaction to fail; card not present at point of interaction
1310	Reject: transaction limit test Customer-defined limits caused transaction to fail; card present at point of interaction
1311	Reject: transaction limit test Customer-defined limits caused transaction to fail; card not present at point of interaction

DE 60, subfield 2 values, in MCC in Failed Parameter Combinations (Global)

Following are valid values in MCC in Failed Parameter Combinations (global) transactions.

Table 856: Values

Value	Description
1400	Reject: transaction limit test Mastercard default limits caused transaction to fail; card present at point of interaction

Value	Description
1401	Reject: transaction limit test Mastercard default limits caused transaction to fail; card not present at point of interaction
1410	Reject: transaction limit test Customer-defined limits caused transaction to fail; card present at point of interaction
1411	Reject: transaction limit test Customer-defined limits caused transaction to fail; card not present at point of interaction

DE 60, subfield 2 values, in TCC in Failed Parameter Combinations (Country-Specific)

Following are valid values in TCC in Failed Parameter Combinations (country-specific) transactions.

Table 857: Values

Value	Description
1500	Reject: transaction limit test Mastercard default limits caused transaction to fail; card present at point of interaction
1501	Reject: transaction limit test Mastercard default limits caused transaction to fail; card not present at point of interaction
1510	Reject: transaction limit test Customer-defined limits caused transaction to fail; card present at point of interaction
1511	Reject: transaction limit test Customer-defined limits caused transaction to fail; card not present at point of interaction

DE 60, subfield 2 values, in TCC in Failed Parameter Combinations (Global)

Following are valid values in TCC in Failed Parameter Combinations (global) transactions.

Table 858: Values

Value	Description
1600	Reject: transaction limit test Mastercard default limits caused transaction to fail; card present at point of interaction
1601	Reject: transaction limit test Mastercard default limits caused transaction to fail; card not present at point of interaction
1610	Reject: transaction limit test Customer-defined limits caused transaction to fail; card present at point of interaction
1611	Reject: transaction limit test Customer-defined limits caused transaction to fail; card not present at point of interaction

DE 60, subfield 2 values, in Miscellaneous Processing

Following are valid values in miscellaneous processing transactions.

Table 859: Values

Value	Description
1700	Accept: via X-Code Processing
1701	Reject: via X-Code Processing
1702	Reject: capture card via X-Code Processing
1903	Reject: via MIP Transaction Blocking
2000	Reject: PIN data present in Authorization Request/0100 message (used in conjunction with Reversal Advice/0420 message)
9999	Reject: failed validation test because transaction type or authorization method was invalid

DE 60, Subfield 2 Values in Send and other transactions blocking service

These values are valid in Send and other transactions blocking service transactions.

Table 860: Values

Code	Description
5111	Single txn limit - dom - core txn type
5112	Single txn limit - dom - special txn type
5113	Single txn limit - xb - core txn type
5114	Single txn limit - xb - special txn type
5121	Daily accum limit - dom - core txn type
5122	Daily accum limit - dom - special txn type
5123	Daily accum limit - xb - core txn type
5124	Daily accum limit - xb - special txn type
5125	Daily accum limit - dom+xb - core txn type
5126	Daily accum limit - dom+xb - special txn type
5127	Daily accum limit - dom+xb - group of txn types
5131	Weekly accum limit - dom - core txn type
5132	Weekly accum limit - dom - special txn type
5133	Weekly accum limit - xb - core txn type
5134	Weekly accum limit - xb special txn type
5135	Weekly accum limit - dom+xb - core txn type
5136	Weekly accum limit - dom+xb - special txn type
5137	Weekly accum limit - dom+xb - group of txn types
5141	Monthly accum limit - dom - core txn type
5142	Monthly accum limit - dom - special txn type
5143	Monthly accum limit - xb - core txn type
5144	Monthly accum limit - xb - special txn type
5145	Monthly accum limit - dom+xb - core txn type
5146	Monthly accum limit - dom+xb - special txn type
5147	Monthly accum limit - dom+xb - group of txn types
5150	TTI - Domestic not allowed
5151	TTI - ISS - XB not allowed

Code	Description
5152	TTI - Acceptor - XB not allowed
5153	Special txn type - Domestic not allowed
5154	Special txn type - ISS - XB not allowed
5155	Special txn type - Acceptor - XB not allowed
5156	MCC not allowed for country
5157	MCC-TTI combination not allowed for country
5158	Country code not allowed
5159	Transaction currency not allowed
5160	Other restrictions
5161	Reserved for future use
5162	Reserved for future use
5163	Reserved for future use
5170	Product code is not eligible for TTI
5171	Account not allowed for TTI
5172	Account not allowed for Special txn type
5173	Reserved for future use
5197	Missing TTI
5198	Legacy or restricted use TTI
5199	TC- MCC- TTI combination not allowed

Subfield 3 (Advice Detail Text)

Subfield 3 (Advice Detail Text) is optional and may be used to contain textual information supplementary to the Advice Detail Code. Advice Detail Text data is determined individually for each program and service.

Table 861: Attributes

Attribute	Description
Data Representation	ans...53
Data Field	Contents of positions 8–60
Justification	Left

Values

A two character alphanumeric value that identifies which rule is responsible for ODD. Issuers should be prepared to receive any two character value in DE 60, subfield 3 (Advice Details Text).

For more information on valid values for Send Blocking Service transactions, refer to [Mastercard Send and Funding Transactions Implementation and Operations for MIP Connections](#).

DE 61 (Point-of-Service [POS] Data)

DE 61 (Point-of-Service [POS] Data) indicates the conditions that exist at the point of service at the time of the transaction. Note that DE 61 supersedes and replaces the ISO-specified DE 25 (Point-of-Service [POS] Condition Code) which is not used in the *Mastercard Network Processing Dual Message Authorization System Guide*.

Table 862: Attributes

Attribute	Description
Data Representation	ans...026; LLLVAR
Data Length	3 positions
Data Field	Contents of subfields
Subfields	14
Justification	See subfields

Usage

Following is the usage of DE 61 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 863: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	M	•	M
Authorization Advice/0120: acquirer-generated	M	•	M
Authorization Advice/0120: system-generated	•	X	M
Reversal Request/0400	M	•	M

Values

See subfields.

Three basic categories of data include:

- POS Condition Code: Indicating POS terminal data (mandatory) in subfields 1–11
- POS Authorization Life Cycle: Indicating the transaction is a preauthorization request and the specific amount of days for which the preauthorization will remain valid (the number of days the issuer or its agent will guarantee or hold funds) in subfield 12
- POS geographic reference: Indicating the specific merchant location of the transaction in subfields 13–14

Application notes

This data element will be edited as described in the [Dual Message Authorization System Edits](#).

Subfield 1 (POS Terminal Attendance)

DE 61, subfield 1 (POS Terminal Attendance) indicates if the acceptor is attending the terminal.

Table 864: Attributes

Attribute	Description
Data Representation	n-1
Data Field	Contents of position 1
Justification	N/A

Table 865: Values

Value	Description
0	Attended Terminal
1	Unattended terminal (cardholder-activated terminal [CAT], home PC, mobile phone, PDA)
2	No terminal used (voice/audio response unit [ARU] authorization); server

Subfield 2 (Reserved for Future Use)

DE 61, subfield 2 (Reserved) is reserved.

Table 866: Attributes

Attribute	Description
Data Representation	n-1
Data Field	Contents of position 2
Justification	N/A

Values

Zero-filled.

Subfield 3 (POS Terminal Location)

DE 61, subfield 3 (POS Terminal Location) indicates the terminal location.

Table 867: Attributes

Attribute	Description
Data Representation	n-1
Data Field	Contents of position 3
Justification	N/A

Table 868: Values

Value	Description
0	On premises of acceptor facility
1	Off premises of acceptor facility (merchant terminal—remote location)
2	Off premises of acceptor facility (cardholder terminal including home PC, mobile phone, PDA)
3	No terminal used (voice/ARU authorization); server
4	On premises of acceptor facility (cardholder terminal including home PC, mobile phone, PDA)

Value	Description
8	Additional POS Terminal Locations
NOTE: For more details, refer to DE 48, Subelement 29 (Additional POS Terminal Locations).	

Subfield 4 (POS Cardholder Presence)

DE 61, subfield 4 (POS Cardholder Presence) indicates whether the cardholder is present at the point of service and explains the condition if the cardholder is not present.

Table 869: Attributes

Attribute	Description
Data Representation	n-1
Data Field	Contents of position 4
Justification	N/A

Table 870: Values

Value	Description
0	Cardholder present
1	Cardholder not present, unspecified
2	Cardholder not present (mail/facsimile order)
3	Cardholder not present (phone or Automated Response Unit [ARU])
4	Recurring transactions
5	Cardholder not present (Electronic order [home PC, Internet, mobile phone, PDA])

Subfield 5 (POS Card Presence)

DE 61, subfield 5 (POS Card Presence) indicates if the card is present at the point of service.

Table 871: Attributes

Attribute	Description
Data Representation	n-1
Data Field	Contents of position 5
Justification	N/A

Table 872: Values

Value	Description
0	Card present
1	Card not present

Subfield 6 (POS Card Capture Capabilities)

DE 61, subfield 6 (POS Card Capture Capabilities) indicates whether the terminal has card capture capabilities.

Table 873: Attributes

Attribute	Description
Data Representation	n-1
Data Field	Contents of position 6
Justification	N/A

Table 874: Values

Value	Description
0	Terminal/operator does not have card capture capability
1	Terminal/operator has card capture capability

Subfield 7 (POS Transaction Status)

DE 61, subfield 7 (POS Transaction Status) indicates the purpose or status of the request.

Table 875: Attributes

Attribute	Description
Data Representation	n-1
Data Field	Contents of position 7
Justification	N/A

Table 876: Values

Value	Description
0	Normal request (original presentment)
1	Deferred Authorization
2	Identity Check Phone Order
3	ATM Installment Inquiry
4	Preauthorized request
5	Time Based Payment Authorization Request—Brazil domestic transactions
6	ATC Update
8	Account Status Inquiry Service (ASI)
9	Tokenization Request/Notification

Application notes

For DE 61, subfield 7, value 8, see separate description of Account Status Inquiry Service (both Purchase ASI and Payment ASI) and Product Inquiry Service.

Subfield 8 (POS Transaction Security)

DE 61, subfield 8 (POS Transaction Security) indicates the acceptor's confidence in the transaction security level.

Table 877: Attributes

Attribute	Description
Data Representation	n-1
Data Field	Contents of position 8
Justification	N/A

Table 878: Values

Value	Description
0	No security concern
1	Suspected fraud (merchant suspicious—code 10)
2	ID verified

Subfield 9 (Reserved for Future Use)

DE 61, Subfield 9 (Reserved) is reserved.

Table 879: Attributes

Attribute	Description
Data Representation	n-1
Data Field	Contents of position 9
Justification	N/A

Values

Zero-filled.

Subfield 10 (Cardholder-Activated Terminal Level)

DE 61, subfield 10 (Cardholder-Activated Terminal Level) indicates the type of cardholder activated terminal used by the cardholder to initiate the transaction.

Table 880: Attributes

Attribute	Description
Data Representation	n-1
Data Field	Contents of position 10
Justification	N/A

Table 881: Values

Value	Description
0	Not a CAT transaction
1	Authorized Level 1 CAT: automated dispensing machine with PIN
2	Authorized Level 2 CAT: self-service terminal
3	Authorized Level 3 CAT: limited-amount terminal
4	Authorized Level 4 CAT: in-flight commerce
5	Reserved
6	Authorized Level 6 CAT: electronic commerce
7	Authorized Level 7 CAT: transponder transaction
9	MPOS Acceptance Device
NOTE: Mobile POS are always attended devices and not cardholder activated.	

Application notes

When the recipient of the money transfer (Payment Transaction; DE 3 subfield 1 = 09) is not present, the transaction is not considered a CAT level transaction. In that case the CAT level = 0.

Subfield 11 (POS Card Data Terminal Input Capability Indicator)

DE 61, subfield 11 (POS Card Data Terminal Input Capability Indicator) indicates the set of methods supported by the terminal for the input of account number, card, or mobile device data.

Table 882: Attributes

Attribute	Description
Data Representation	n-1
Data Field	Contents of position 11
Justification	N/A

Table 883: Values

Value	Description
0	Input capability unknown or unspecified.
1	No terminal used (voice/ARU authorization); server.
2	Terminal supports magnetic stripe input only.
3	Terminal supports contactless EMV input. The terminal may also support one or more other card input types, including contactless magstripe input, EMV contact chip input, magnetic stripe input and key entry input.
4	Contactless Mag Stripe (Proximity Chip) Terminal supports contactless magstripe input but not contactless EMV input. The terminal also may support one or more other card input types, including EMV contact chip input, magnetic stripe input, and key entry input.
5	Terminal supports EMV contact chip input and magnetic stripe input.
6	Terminal supports key entry input only.
7	Terminal supports magnetic stripe input and key entry input.
8	Terminal supports EMV contact chip input, magnetic stripe input and key entry input.
9	Terminal supports EMV contact chip input only.

Application notes

DE 61, subfield 11 values 3, 4 (contactless related), 5, 8, and 9 (chip contact related) can ONLY be used if the terminal is chip certified by Mastercard.

DE 61, subfield 11 is used in conjunction with DE 22 (Point of Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode). DE 22, subfield 1 values indicate the method by which the PAN was entered.

DE 61, subfield 11 indicates the terminal data input capabilities and not the specific manner in which the terminal captured the card data for a given transaction.

For example, values 5 and 8 could be used for a magnetic stripe transaction originating from an EMV contact chip capable terminal when the contact magnetic stripe reader was used instead. Similarly, value 3 would be used for a contact chip transaction originating from a contactless EMV capable terminal when the EMV contact chip reader was used.

Subfield 12 (POS Authorization Life Cycle)

DE 61, subfield 12 (POS Authorization Life Cycle) indicates the number of days a preauthorization will stay in effect for Visa Custom Payment Service automobile rentals and hotel reservations—otherwise it must contain zeros.

Table 884: Attributes

Attribute	Description
Data Representation	n-2
Data Field	Contents of positions 12 and 13
Justification	N/A

Values

Indicates the number of days the preauthorization stays in effect; ATM and Maestro POS transactions should use 01, Visa customer transactions use applicable value. Must be zero filled when not applicable.

Subfield 13 (POS Country Code [or Sponsored Merchant Information, if applicable])

DE 61, subfield 13 (POS Country Code) indicates the country of the POS location (not the acquirer location) using ISO-specified codes.

Table 885: Attributes

Attribute	Description
Data Representation	n-3
Data Field	Contents of positions 14–16
Justification	N/A

Values

Refer to the [Quick Reference Booklet](#) for valid country codes.

Subfield 14 (Acceptor Postal Code)

This subfield contains the postal code of the acceptor.

Table 886: Attributes

Attribute	Description
Data representation	ans...10
Data field	Contents of positions 17–26
Justification	Left

Processing rules

The Acceptor Postal Code must be present for all transactions and contain a valid code for countries that utilize postal codes. This information must be populated and formatted as defined by country postal standards.

Program rules

Payment facilitator-enabled transactions: The sponsored merchant postal code must be provided.

Dual Message Authorization System Edits

The Dual Message Authorization System performs the following edits on data element 61 (Point-of-Service [POS] Data).

Authorization Request/0100 Character Edit

Table 887: Associated edits

WHEN the acquirer...	THEN the Dual Message Authorization System...
sends an Authorization Request/0100 message containing less than nine characters	sends an Authorization Request Response/0110 message to the acquirer where: - DE 39 = 30 - DE 44 = 061

WHEN the acquirer...	THEN the Dual Message Authorization System...
sends an Authorization Request/0100 message where DE 61, subfields 1–9 contain a non-numeric character	sends an Authorization Request Response/0110 message to the acquirer where: - DE 39 = 30 - DE 44 = 061

Non-ATM CAT Level 1 Transactions

Table 888: Associated edits

WHEN the Authorization Request/0100 message contains...	THEN the Dual Message Authorization System...
- DE 61, subfield 10, value 1 (Authorized Level 1 CAT: Automated dispensing machine with a PIN) - DE 52 (PIN Data) is not present - DE 55 (Integrated Circuit Card [ICC] System-related Data) is present - DE 110 Dataset01 is not present	forwards the Authorization Request/0100 message to the issuer.
- DE 61, subfield 10, value 1 - DE 52 or Dataset 01 is present - DE 55 is present	forwards the Authorization Request/0100 message to the issuer.
- DE 61, subfield 10, value 1 - DE 52 or Dataset 01 is present - DE 55 is present	forwards the Authorization Request/0100 message to the issuer.
- DE 61, subfield 10, value 1 - DE 52 or Dataset 01 is present - DE 55 is present - DE 110 Dataset 01 is not present	sends an Authorization Request Response/0110 message to the acquirer where: - DE 39 = 30 - DE 44 = 052

Electronic Commerce Transactions

Table 889: Associated edits

IF...	THEN...	Otherwise, the Dual Message Authorization System...
DE 48, subelement 42, position 3 is value 1, 2, 3, 4, 5, 6, or 7 and DE 61, subfield 4 is value 4 (Standing order/recurring transactions) or 5 (Electronic Order)	DE 61, subfield 7 must not contain value 2 (Identity Check Phone Order) and Subfield 10 must contain value 6 (Electronic commerce)	rejects the authorization transaction with a format error indicated by: • DE 39 = 30 • DE 44 = 061

Phone Order Transactions

Table 890: Associated edits

IF...	THEN...	Otherwise, the Dual Message Authorization System...
DE 48, subelement 42, position 3 is value 1 or 2	DE 61, subfield 4 must contain value 3 or 4 and Subfield 7 must contain value 2 (Identity Check Phone Order) and Subfield 10 must not contain value 6	rejects the authorization transaction with a format error indicated by: • DE 39 = 30 • DE 44 = 061

Account Status Inquiry Service

Table 891: Associated edits

WHEN...	THEN the Dual Message Authorization System...
DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status) and DE 4 (Amount, Transaction) do not equal 0 and DE 3 (Processing Code) contains a value other than zero	sends the acquirer an Authorization Request Response/0110 message where: • DE 39 = 30 • DE 44 = 004

WHEN...	THEN the Dual Message Authorization System...
DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status) contains value 8 (Account Status Inquiry Service [ASI]) and DE 4 (Amount, Transaction) contains a value equal to zero and DE 3 (Processing Code) contains a value other than 00 (Purchase), 01 (Withdrawal), and 28 (Payment Transaction)	sends the acquirer an Authorization Request Response/0110 message where: • DE 39 = 30 • DE 44 = 003

Contactless Chip Transaction Edits

Table 892: Associated edits

WHEN the Authorization Request/0100, Authorization Advice/0120, Reversal Request/0400, message contains...	THEN the Dual Message Authorization System...
the PAN entry mode (DE 22 [Point-of-Service Entry Mode], subfield 1 [POS Terminal PAN Entry Mode]) of 07 (PAN auto-entry via contactless M/Chip) and DE 61 (Point-of-Service [POS] Data), subfield 11 (POS Card Data Terminal Input Capability Indicator) is not 3 (Contactless M/Chip [Proximity Chip])	sends the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 message where: • DE 39 = 30 • DE 44 = 061

DE 62 (Intermediate Network Facility [INF] Data)

DE 62 (Intermediate Network Facility [INF] Data), contains "acquiring network trace information" that INFs may require to quickly and accurately route Administrative Advice/0620 messages back to the original acquiring institution. DE 62 assists acquiring INF facilities that connect directly to the Dual Message Authorization System. It allows these INFs to maintain sufficient information within a message to permit immediate online routing of chargebacks and retrieval requests without the requirement of maintaining large online reference databases containing the original transactions.

Table 893: Attributes

Attribute	Description
Data Representation	ans...999; LLLVAR
Data Length	3 positions
Data Field	Contents of positions 1–999
Subfields	N/A
Justification	N/A

Usage

Following is the usage of DE 62 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 894: Applicable messages

Message	Org	Sys	Dst
Authorization request/0100	O	.	O
Authorization request response/0110	CE	.	CE
Authorization advice/0120: acquirer-generated	O	.	O
Authorization advice/0120: system-generated	.	X	C
Authorization advice response/0130: issuer-generated (responding to an acquirer-generated 0120)	CE	.	CE
Authorization advice response/0130: issuer-generated (responding to a system-generated 0120)	CE	XE	.
Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)	CE	XE	.
Reversal request/0400	O	.	O
Reversal request response/0410	CE	.	CE
Reversal advice/0420	.	X	C
Reversal advice response/0430	CE	XE	.
Administrative request/0600	O	.	O
Administrative request response/0610	CE	.	CE
Administrative advice/0620: system generated	.	X	C

Message	Org	Sys	Dst
Administrative advice/0620: member-generated	C	.	C
Administrative advice response/0630	CE	XE	.

Values

DE 62 is a free-format, variable-length alphanumeric data element that may be used to store unique acquiring network ID codes, acquiring network chaining data, or other information useful to INFs in routing online chargebacks and retrieval requests.

Application notes

DE 62 is an optional data element that customers may place in any originating Authorization Request/0100 or Authorization Advice/0120. Subsequently, this data element (if present in an original transaction) is required to be returned without alteration in any Administrative Advice (Retrieval)/0620 pertaining to the original transaction. It contains INF network trace information that allows acquiring INFs to directly route the chargeback or retrieval request to the original acquirer.

When the Dual Message Authorization System generates an Administrative Advice/0620, it places the value MCBN620060000xxx, where xxx is the MIP ID. When a customer generates a corresponding Administrative Advice Response/0630, DE 62 must contain the same value.

When the Dual Message Authorization System generates a System Generated Advice/0420, it sets a value of MCBN620060000. When a customer generates a corresponding System Generated Advice Response/0430, DE 62 must contain the same value.

DE 63 (Network Data)

DE 63 (Network Data) is generated by the Dual Message Authorization System for each originating message routed through the network. The receiver must retain the data element and use it in any response or acknowledgment message associated with the originating message.

Table 895: Attributes

Attribute	Description
Data Representation	an...050; LLLVAR
Data Length	3 positions
Data Field	Contents of subfields
Subfields	2

Attribute	Description
Justification	See subfields

Usage

Following is the usage of DE 63 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 896: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	.	X	M
Authorization Request Response/0110	ME	.	M
Authorization Advice/0120: acquirer-generated	.	X	M
Authorization Advice/0120: system-generated	.	X	M
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	ME	.	M
Authorization Advice Response/0130: issuer-generated (responding to a system-generated 0120)	ME	XE	.
Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)	.	X	M
Authorization Response Acknowledgment/0180	ME	XE	.
Authorization Response Negative Acknowledgment/0190	.	XE	ME
Issuer File Update Request/0302	.	XE	.
Issuer File Update Request Response/0312	.	XE	M
Reversal Request/0400	.	X	M
Reversal Request Response/0410	ME	.	M
Reversal Advice/0420	.	X	M
Reversal Advice Response/0430	ME	XE	.
Administrative Request/0600	.	X	M
Administrative Request Response/0610	ME	.	ME
Administrative Advice/0620: system-generated	.	X	M
Administrative Advice/0620: member-generated	.	X	M
Administrative Advice Response/0630	ME	XE	.
Network Management Request/0800: sign-on/sign-off	.	X	.

Message	Org	Sys	Dst
Network Management Request/0800: Network Connection Status, Member-generated	•	X	•
Network Management Request/0800: Network Connection Status, System-generated	•	X	•
Network Management Request/0800: PEK exchange	•	X	M
Network Management Request/0800: PEK exchange–On Demand	•	X	•
Network Management Request Response/0810: Network Connection Status, Member-generated	ME	XE	•
Network Management Request Response/0810: network connection status, system-generated	•	XE	ME
Network Management Request Response/0810: sign-on/sign-off	•	XE	ME
Network Management Request Response/0810: PEK exchange	ME	X	•
Network Management Request Response/0810: PEK exchange, on demand	•	XE	ME
Network Management Advice/0820: PEK exchange	•	X	M

Values

See subfields.

Application notes

This data element contains blanks if the transaction fails edit checking and is rejected before determining the appropriate Financial Network Code.

Subfield 1 (Financial Network Code)

DE 63, subfield 1 (Financial Network Code) identifies the specific program or service (for example, the financial network, financial program, or card program) with which the transaction is associated. DE 63 will contain the graduated product when the issuer's cardholder account participates in the Product Graduation Account Level Management service.

Table 897: Attributes

Attribute	Description
Data Representation	an-3
Data Field	Contents of positions 1–3
Justification	Left

Table 898: Valid values

Code	Description	MC	NP	VI	TE	MS	CI
ACS	Digital Debit	√					
AMX	American Express This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.				√		
BPC	Bill Pay Commercial	√					
BPD	Business Premium Debit	√					
CBL	Carte Blanche This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.				√		
CIR	Cirrus						√
DAG	Gold Debit Mastercard Salary	√					
DAP	Platinum Debit Mastercard Salary	√					
DAS	Standard Debit Mastercard Salary	√					
DCC	Diners Club This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.				√		
DLG	Debit Mastercard Gold: Delayed Debit	√					
DLH	Debit Mastercard World Embossed: Delayed Debit	√					
DLI	Debit Mastercard Standard ISIC Student Card: Delayed Debit	√					
DLP	Debit Mastercard Platinum: Delayed Debit	√					
DLS	Debit Mastercard Card: Delayed Debit	√					
DLU	Debit Mastercard Unembossed: Delayed Debit	√					
DOS	Standard Debit Mastercard® Social	√					
DSV	Discover This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.				√		
DWF	Debit Mastercard Humanitarian Prepaid This product code is not currently available for issuance in the Canada region.	√					

Code	Description	MC	NP	VI	TE	MS	CI
ETA	Mastercard Installment Payments A	√					
ETB	Mastercard Installment Payments B	√					
ETC	Mastercard Installment Payments C	√					
ETD	Mastercard Installment Payments D	√					
ETE	Mastercard Installment Payments E	√					
ETF	Mastercard Installment Payments F	√					
ETG	Mastercard Installment Payments G	√					
ETH	Prepaid Installment Payments A	√					
ETI	Prepaid Installment Payments B	√					
ETJ	Prepaid Installment Payments C	√					
ETK	Prepaid Installment Payments D	√					
ETL	Prepaid Installment Payments E	√					
ETM	Prepaid Installment Payments F	√					
ETN	Prepaid Installment Payments G	√					
EXC	ADP/Exchange This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.		√				
FIA	Mastercard® B2B VIP 27	√					
FIB	Mastercard® B2B VIP 28	√					
FIC	Mastercard® B2B VIP 29	√					
FID	Mastercard® B2B VIP 30	√					
FIE	Mastercard® B2B VIP 31	√					
FIF	Mastercard® B2B VIP 32	√					
FIG	Mastercard® B2B VIP 33	√					
FIH	Mastercard® B2B VIP 34	√					
GCP	Mastercard Installments Card Premium	√					
GCS	Mastercard Installments Card Standard	√					
GPP	Mastercard Prepaid Installments Card Premium	√					
HNR	Honor This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.		√				

Code	Description	MC	NP	VI	TE	MS	CI
ITT	Instant Teller This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.		√				
JCB	Japanese Credit Bureau This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.		√				
LNC	LINC New York This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.		√				
MAB	World Elite Mastercard® for Business	√					
MAC	Mastercard Corporate World Elite®	√					
MAP	Mastercard Commercial Payments Account	√					
MAQ	Mastercard Prepaid Commercial Payments Account	√					
MBA	Mastercard B2B Product 2 This product code is not currently available for issuance in the Canada region.	√					
MBB	Mastercard Prepaid Consumer	√					
MBC	Mastercard Prepaid Voucher	√					
MBD	Mastercard Professional Debit Business Card	√					
MBE	Mastercard® Electronic Business Card	√					
MBF	Mastercard Alimentação (Food) (MBF–Prepaid Mastercard Food)	√					
MBG	Mastercard B2B Product 3	√					
MBH	Mastercard B2B Product 4	√					
MBI	Mastercard B2B Product 5	√					
MBJ	Mastercard B2B Product 6	√					
MBK	Mastercard Black	√					
MBM	Mastercard® Refeição (Meal) (MBM–Prepaid Mastercard Meal)	√					
MBP	Mastercard Corporate Prepaid	√					
MBS	Mastercard B2B Product 1	√					
MBT	Mastercard Corporate Prepaid Travel	√					
MBW	World Mastercard® Black Edition: Debit	√					

Code	Description	MC	NP	VI	TE	MS	CI
MCA	Mastercard Electronic Basic Card	√					
MCB	Mastercard BusinessCard® Card Mastercard Corporate Card®	√					
MCC	Mastercard Card	√					
MCE	Mastercard Electronic™ Card	√					
MCF	Mastercard Corporate Fleet Card®	√					
MCG	Gold Mastercard® Card	√					
MCH	Mastercard Premium Charge	√					
MCI	Mastercard Select Debit	√					
MCM	Mastercard Corporate Meeting Card This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.	√					
MCO	Mastercard Corporate	√					
MCP	Mastercard Corporate Purchasing Card®	√					
MCS	Mastercard Standard Card	√					
MCT	Titanium Mastercard®	√					
MCU	Mastercard Unembossed	√					
MCV	Merchant-Branded Program	√					
MCW	World Mastercard® Card	√					
MDB	Debit Mastercard BusinessCard Card	√					
MDG	Gold Debit Mastercard®	√					
MDH	World Debit Embossed	√					
MDI	Debit Other Unembossed	√					
MDJ	Debit Mastercard® (Enhanced)	√					
MDK	Debit Mastercard Other 2 Embossed	√					
MDL	Business Debit Other Embossed	√					
MDM	Middle Market Fleet Card	√					
MDN	Middle Market Purchasing Card	√					
MDO	Debit Mastercard Other	√					
MDP	Platinum Debit Mastercard®	√					
MDQ	Middle Market Corporate Card	√					
MDR	Mastercard Debit Brokerage Card	√					

Code	Description	MC	NP	VI	TE	MS	CI
MDS	Debit Mastercard®	√					
MDT	Mastercard Business Debit	√					
MDU	Debit Mastercard Unembossed	√					
MDW	World Elite Debit Mastercard® (Mastercard Black™ Debit LAC region excluding Mexico)	√					
MEB	Mastercard Executive BusinessCard Card	√					
MEC	Mastercard Electronic Commercial	√					
MED	Debit Mastercard Electronic	√					
MEF	Mastercard Electronic Payment Account	√					
MEO	Mastercard Corporate Executive Card® This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.	√					
MEP	Premium Debit Mastercard	√					
MES	Mastercard Enterprise Solution	√					
MET	Titanium Debit Mastercard®	√					
MFB	Flex World Elite	√					
MFD	Flex Platinum	√					
MFE	Flex Charge World Elite	√					
MFH	Flex World	√					
MFL	Flex Charge Platinum	√					
MFW	Flex Charge World	√					
MGF	Mastercard Government Commercial Card	√					
MGP	Mastercard Prepaid Gold Payroll	√					
MGS	Platinum Mastercard® Prepaid General Spend	√					
MHA	Mastercard Healthcare Prepaid Non-tax	√					
MHB	Mastercard HSA Substantiated	√					
MHC	Mastercard Healthcare Credit Non-substantiated	√					
MHH	Mastercard HSA Non-substantiated	√					
MIA	Mastercard Unembossed Prepaid Student Card	√					
MIB	Mastercard Credit Electronic Student Card	√					
MIC	Mastercard Credit Standard Student Card	√					

Code	Description	MC	NP	VI	TE	MS	CI
MID	Mastercard Debit Unembossed Student Card	√					
MIG	Mastercard Credit Unembossed Student Card	√					
MIH	Mastercard Electronic Consumer Non U.S. Student Card	√					
MIJ	Mastercard Debit Unembossed Non U.S. Student Card	√					
MIK	Mastercard Electronic Consumer Prepaid Non U.S. Student Card	√					
MIL	Mastercard Unembossed Prepaid Non U.S. Student Card	√					
MIO	Mastercard Pay & Split	√					
MIP	Mastercard Debit Prepaid Student Card	√					
MIS	Mastercard Debit Standard Student Card	√					
MIU	Debit Mastercard Unembossed Outside US	√					
MKA	Digital Gold Debit	√					
MKB	Digital Platinum Debit	√					
MKC	Digital World Debit	√					
MKD	Digital World Elite Debit	√					
MKE	Digital Platinum	√					
MKF	Digital World	√					
MKG	Digital World Elite	√					
MKH	Digital World Elite Exclusive	√					
MLA	Mastercard Central Travel Solutions Air	√					
MLB	Mastercard Brazil Benefit for Home Improvement	√					
MLC	Mastercard Micro-Business Card	√					
MLD	Mastercard Distribution Card	√					
MLE	Mastercard Pedágio Prepaid Card (MLE–Mastercard Brazil General Benefits)	√					
MLF	Mastercard Agro	√					
MLL	Mastercard Central Travel Solutions Land	√					
MNF	Mastercard Public Sector Commercial Card	√					
MNW	World Mastercard Card (Europe)	√					

Code	Description	MC	NP	VI	TE	MS	CI
MOC	Standard Maestro® Social					√	
MOG	Maestro® Gold This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.					√	
MOP	Maestro® Platinum This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.					√	
MOW	World Maestro					√	
MPA	Prepaid Mastercard Payroll Card	√					
MPB	Mastercard Preferred BusinessCard	√					
MPC	Mastercard Professional Card	√					
MPD	Mastercard Flex Prepaid	√					
MPE	Name for GCMS Product ID MPE	√					
MPF	Prepaid Mastercard Gift Card	√					
MPG	Prepaid Mastercard Consumer Reloadable Card	√					
MPJ	Prepaid Debit Mastercard® Card Gold	√					
MPK	Prepaid Mastercard Government Commercial Card	√					
MPL	Platinum Mastercard® Card	√					
MPM	Prepaid Mastercard Consumer Promotion Card	√					
MPN	Prepaid Mastercard Insurance Card	√					
MPO	Prepaid Mastercard Other Card	√					
MPR	Prepaid Mastercard Travel Card	√					
MPT	Prepaid Mastercard Teen Card	√					
MPV	Prepaid Mastercard Government Benefit Card	√					
MPW	Prepaid Mastercard Corporate Card	√					
MPX	Prepaid Mastercard Flex Benefit Card	√					
MPY	Prepaid Mastercard Employee Incentive Card	√					
MPZ	Prepaid Mastercard Emergency Assistance Card	√					
MRB	Prepaid Mastercard Electronic BusinessCard	√					
MRC	Prepaid Mastercard Electronic Card	√					
MRD	Platinum Debit Mastercard® Prepaid General Spend	√					

Code	Description	MC	NP	VI	TE	MS	CI
MRF	European Regulated Individual Pay	√					
MRG	Prepaid Mastercard Card Outside US	√					
MRH	Mastercard Platinum Prepaid Travel Card	√					
MRJ	Prepaid Mastercard Gold Card	√					
MRK	Prepaid Mastercard Public Sector Commercial Card	√					
MRL	Prepaid Mastercard Electronic Commercial	√					
MRO	Mastercard Rewards Only	√					
MRP	Standard Retailer Centric Payments	√					
MRS	Prepaid Mastercard ISIC Student Card	√					
MRW	Prepaid Mastercard BusinessCard Credit Outside US	√					
MSA	Prepaid Maestro Payroll Card					√	
MSB	Maestro Small Business					√	
MSD	Deferred Debit Mastercard	√					
MSF	Prepaid Maestro Gift Card					√	
MSG	Prepaid Maestro Consumer Reloadable Card					√	
MSI	Maestro					√	
MSJ	Prepaid Maestro Gold					√	
MSM	Prepaid Maestro Consumer Promotion Card					√	
MSN	Prepaid Maestro Insurance Card					√	
MSO	Prepaid Maestro Other Card					√	
MSP	Muse Mastercard	√					
MSQ	Maestro Prepaid (Reserved for Future Use)					√	
MSR	Prepaid Maestro Travel Card					√	
MSS	Maestro Student Card					√	
MST	Prepaid Maestro Teen Card					√	
MSV	Prepaid Maestro Government Benefit Card					√	
MSW	Prepaid Maestro Corporate Card					√	
MSX	Prepaid Maestro Flex Benefit Card					√	
MSY	Prepaid Maestro Employee Incentive Card					√	
MSZ	Prepaid Maestro Emergency Assistance Card					√	
MTA	Mastercard® B2B Product 7	√					

Code	Description	MC	NP	VI	TE	MS	CI
MTB	Mastercard® B2B Product 8	√					
MTC	Mastercard® B2B Product 9	√					
MTD	Mastercard® B2B Product 10	√					
MTE	Mastercard® B2B Product 11	√					
MTF	Mastercard® B2B Product 12	√					
MTG	Mastercard® B2B Product 13	√					
MTH	Mastercard® B2B Product 14	√					
MTI	Mastercard® B2B Product 15	√					
MTJ	Mastercard® B2B Product 16	√					
MTK	Mastercard® B2B Product 17	√					
MTL	Mastercard® B2B Product 18	√					
MTM	Mastercard® B2B Product 19	√					
MTN	Mastercard® B2B Product 20	√					
MTO	Mastercard® B2B Product 21	√					
MTP	Mastercard Platinum Prepaid Travel Card	√					
MTQ	Mastercard® B2B Product 22	√					
MTR	Mastercard® B2B Product 23	√					
MTS	Mastercard® B2B Product 24	√					
MTT	Mastercard® B2B Product 25	√					
MTU	Mastercard® B2B Product 26	√					
MTV	Mastercard® B2B Product 27	√					
MUP	Platinum Debit Mastercard Unembossed	√					
MUS	Prepaid Mastercard Unembossed US	√					
MUW	Mastercard World Domestic Affluent	√					
MVA	Mastercard B2B VIP 1	√					
MVB	Mastercard B2B VIP 2	√					
MVC	Mastercard® B2B VIP 3	√					
MVD	Mastercard B2B VIP 4	√					
MVE	Mastercard B2B VIP 5	√					
MVF	Mastercard B2B VIP 6	√					
MVG	Mastercard B2B VIP 7	√					

Code	Description	MC	NP	VI	TE	MS	CI
MVH	Mastercard B2B VIP 8	√					
MVI	Mastercard B2B VIP 9	√					
MVJ	Mastercard B2B VIP 10	√					
MVK	Mastercard B2B VIP 11	√					
MVL	Mastercard B2B VIP 12	√					
MVM	Mastercard B2B VIP 13	√					
MVN	Mastercard B2B VIP 14	√					
MVO	Mastercard® B2B VIP 15	√					
MVP	Mastercard® B2B VIP 16	√					
MVQ	Mastercard® B2B VIP 17	√					
MVR	Mastercard® B2B VIP 18	√					
MVS	Mastercard® B2B VIP 19	√					
MVT	Mastercard® B2B VIP 20	√					
MVU	Mastercard® B2B VIP 21	√					
MVV	Mastercard® B2B VIP 22	√					
MVW	Mastercard® B2B VIP 23	√					
MVX	Mastercard® B2B VIP 24	√					
MVY	Mastercard® B2B VIP 25	√					
MVZ	Mastercard® B2B VIP 26	√					
MWB	World Mastercard® for Business	√					
MWD	World Deferred	√					
MWE	World Elite™	√					
MWF	Mastercard Humanitarian Prepaid This product code is not currently available for issuance in the Canada region.	√					
MWJ	Consumer Enhanced	√					
MWK	Consumer Plus	√					
MWL	Consumer Plus Flex	√					
MWO	Mastercard Corporate World	√					
MWP	Mastercard World Prepaid	√					
MWR	World Retailer Centric Payment	√					

Code	Description	MC	NP	VI	TE	MS	CI
MXG	Gold Debit Mastercard® Card	√					
NYC	NYCE Network This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.		√				
OLB	Maestro Small Business: Delayed Debit					√	
OLG	Maestro Gold: Delayed Debit					√	
OLI	ISIC Maestro Student Card: Delayed Debit					√	
OLP	Maestro Platinum: Delayed Debit					√	
OLS	Maestro: Delayed Debit					√	
OLW	World Maestro Delayed Debit					√	
PLU	PLUS This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.		√				
PRO	Proprietary Card This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.		√				
PUL	Pulse This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.		√				
PVA	Private Label 1	√					
PVB	Private Label 2	√					
PVC	Private Label 3	√					
PVD	Private Label 4	√					
PVE	Private Label 5	√					
PVF	Private Label 6	√					
PVG	Private Label 7	√					
PVH	Private Label 8	√					
PVI	Private Label 9	√					
PVJ	Private Label 10	√					
PVL	Private label: generic This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.		√				
SAG	Gold Mastercard® Salary-Immediate Debit	√					

Code	Description	MC	NP	VI	TE	MS	CI
SAL	Standard Maestro Salary					√	
SAP	Platinum Mastercard® Salary-Immediate Debit	√					
SAS	Standard Mastercard Salary-Immediate Debit	√					
SBJ	Prepaid Installment Payments H	√					
SBK	Prepaid Installment Payments I	√					
SBP	Small Business Prepaid	√					√
SOS	Standard Mastercard Social-Immediate Debit	√					
SPP	Mastercard Installment Payments P	√					
SPS	Mastercard Installment Payments S	√					
STR	Star Network This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.		√				
SUR	Prepaid Mastercard Unembossed Outside US	√					
TBE	Business: Immediate Debit	√					
TCB	Mastercard Business Card: Immediate Debit	√					
TCC	Mastercard (Mixed BIN): Immediate Debit	√					
TCE	Mastercard Electronic: Immediate Debit	√					
TCF	Mastercard Fleet Card: Immediate Debit	√					
TCG	Gold Mastercard Card: Immediate Debit	√					
TCO	Mastercard Corporate: Immediate Debit	√					
TCP	Mastercard Purchasing Card: Immediate Debit	√					
TCS	Mastercard Standard Card: Immediate Debit	√					
TCW	World Signia Mastercard Card: Immediate Debit	√					
TDN	Middle Market Mastercard Purchasing Card: Immediate Debit	√					
TEB	Mastercard Executive BusinessCard Card: Immediate Debit	√					
TEC	Mastercard Electronic Commercial: Immediate Debit	√					
TEO	Mastercard Corporate Executive Card: Immediate Debit	√					
TIB	ISIC Mastercard Electronic Student Card: Immediate Debit	√					

Code	Description	MC	NP	VI	TE	MS	CI
TIC	ISIC Mastercard Standard Student Card: Immediate Debit	√					
TIU	Mastercard Unembossed: Immediate Debit	√					
TLA	Mastercard Central Travel Solutions Air: Immediate Debit	√					
TNF	Mastercard Public Sector Commercial Card: Immediate Debit	√					
TNW	Mastercard New World: Immediate Debit	√					
TPB	Mastercard Preferred Business Card: Immediate Debit	√					
TPC	Mastercard Professional Card: Immediate Debit	√					
TPL	Platinum Mastercard: Immediate Debit	√					
TWB	World Mastercard® Black Edition: Immediate Debit	√					
VIS	VisaNet This product code is not a valid value in DE 120, MCC103 or MCC104, field 3, Card Program.			√			
WBE	World Mastercard® Black Edition	√					
WDR	World Debit Mastercard Rewards	√					
WMR	World Mastercard Rewards	√					
WPD	World Prepaid Debit	√				√	√

Subfield 2 (Banknet Reference Number)

DE 63, subfield 2 (Banknet Reference Number) is generated by the Authorization System for each originating message it routes. The reference number is guaranteed to be a unique value for any transaction within the specified financial network on any processing day.

Table 899: Attributes

Attribute	Description
Data representation	an...9
Data field	Contents of positions 4–12
Justification	Left

Values

The Banknet reference number is a minimum of six characters and a maximum of nine characters long.

DE 64 (Message Authentication Code)

DE 64 (Message Authentication Code [MAC]) validates the source and the text of the message between the sender and the receiver.

Table 900: Attributes

Attribute	Description
Data Representation	b-8
Data Length	N/A
Data Field	N/A
Subfields	N/A
Justification	N/A

Usage

May contain message authentication code as defined by ISO standards.

Values

The last bit position within any bit map is reserved for DE 64. If authentication is to be used on a message, the MAC information is indicated by the final bit of the final bit map of that message. The final bit of all preceding bit maps shall contain 0; for example, there shall be only one DE 64 per message and that DE 64 must be the last data element of the message.

DE 65 (Bit Map, Extended)

DE 65 (Bit Map, Extended) is a series of eight bytes (64 bits) used to identify the presence (denoted by 1) or the absence (denoted by 0) of each data element in an extended (third) message segment.

Table 901: Attributes

Attribute	Description
Data Representation	b-8

Attribute	Description
Data Length	N/A
Data Field	Contents of bit positions 1–64 (8 bytes)
Subfields	N/A
Justification	N/A

Usage

The Dual Message Authorization System defines only two message segments, the presence or absence of which is indicated by Primary and Secondary Bit Maps. DE 65 would indicate the presence of a third message segment, and must never be present in a Dual Message Authorization System message. The corresponding bit (number 65) must always be 0 in the Secondary Bit Map.

DE 66 (Settlement Code)

DE 66 (Settlement Code) indicates the result of a reconciliation request.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 67 (Extended Payment Code)

DE 67 (Extended Payment Code) indicates the number of months that the cardholder prefers to pay for an item (the item purchased during the course of this transaction) if permitted by the card issuer.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 68 (Receiving Institution Country Code)

DE 68 (Receiving Institution Country Code) is the code of the country where the receiving institution is located.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 69 (Settlement Institution Country Code)

DE 69 (Settlement Institution Country Code) is the code of the country where the settlement institution is located.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 70 (Network Management Information Code)

DE 70 (Network Management Information Code) identifies network status.

Table 902: Attributes

Attribute	Description
Data Representation	n-3
Data Length	N/A
Data Field	Contents of positions 1–3
Subfields	N/A
Justification	N/A

Usage

Following is the usage of DE 70 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 903: Applicable messages

Message	Org	Sys	Dst
Network Management Request/0800: sign-on/sign-off	M	X	•
Network Management Request/0800: host session activation/deactivation	M	X	•
Network Management Request/0800: network connection status, member-generated	M	X	•
Network Management Request/0800: network connection status, system-generated	•	X	M
Network Management Request/0800: PEK exchange: on demand	M	X	•
Network Management Request/0800: PEK exchange	•	X	M

Message	Org	Sys	Dst
Network Management Request Response/0810: host session activation/deactivation	•	XE	ME
Network Management Request Response/0810: network connection status, member-generated	ME	XE	•
Network Management Request Response/0810: network connection status, system-generated	•	XE	ME
Network Management Request Response/0810: PEK exchange	ME	X	•
Network Management Request Response/0810: PEK exchange-on demand	•	XE	ME
Network Management Request Response/0810: sign-on/sign-off	•	XE	ME
Network Management Advice/0820: PEK exchange	•	X	M

Application notes

DE 48 (Additional Data: Private Use) may be used in conjunction with DE 70 to provide complete network status or control information. The Mastercard Network uses this data element in Network Management/08xx messages to convey network control commands and network status information to and from customer information processing systems that interface directly to the Dual Message Authorization System.

This data element is defined and used identically within all Mastercard programs and services.

Network Management Request/0800: sign-on/sign-off

The following values apply to this message.

Table 904: Values

Code	Financial Network	MC	NP	VI	TE	MS	CI
001	Sign-on (by prefix)	✓	✓	✓	✓	✓	✓
002	Sign-off (by prefix)	✓	✓	✓	✓	✓	✓
060	SAF session request	✓	✓	✓	✓	✓	✓
061	Group sign-on (by Mastercard group sign-on)	✓	✓	✓	✓	✓	✓
062	Group sign-off (by Mastercard group sign-on)	✓	✓	✓	✓	✓	✓
063	Group sign-on alternate issuer route	✓	✓			✓	✓
064	Group sign-off alternate issuer route	✓	✓			✓	✓
065	Prefix sign-on (by Group Sign-on ID for primary route)	✓	✓			✓	✓
066	Prefix sign-off (by Group Sign-on ID for primary route)	✓	✓			✓	✓

Code	Financial Network	MC	NP	VI	TE	MS	CI
067	Prefix sign-on (by Group Sign-on ID for alternate issuer route) ¹⁴	√	√			√	√
068	Prefix sign-off (by Group Sign-on ID for alternate issuer route) ¹⁴	√	√			√	√

Network Management Request/0800: Network Connection Status, Member-Generated

The following values apply to this message.

Table 905: Values

Code	Financial Network	MC	NP	VI	TE	MS	CI
270	Network connection status—echo test	√				√	√

Network Management Request/0800: Network Connection Status, System-Generated

The following values apply to this message.

Table 906: Values

Code	Financial Network	MC	NP	VI	TE	MS	CI
270	Network connection status—echo test	√				√	√

Network Management Request/0800: Host Session Activation/Deactivation

The following values apply to this message.

Table 907: Values

Code	Financial Network	MC	NP	VI	TE	MS	CI
081	Host session activation	√	√	√	√	√	√
082	Host session deactivation	√	√	√	√	√	√

Network Management Request/0800: PEK exchange

The following values apply to this message.

Table 908: Values

Code	Financial Network	MC	NP	VI	TE	MS	CI
161	Encryption key exchange	√	√			√	√

Network Management Request/0800: PEK exchange-On Demand

The following values apply to this message.

Table 909: Values

Code	Financial Network	MC	NP	VI	TE	MS	CI
162	Solicitation for Encryption Key Exchange	√	√			√	√
163	Solicitation for Encryption Key Exchange - TR-31 Keyblock	√	√	√	√	√	√
166	Load Comm Key	√	√		√	√	√
167	Load Previous Comm Key	√	√		√	√	√

Network Management Request Response/0810: PEK exchange

The following values apply to this message.

Table 910: Values

Code	Financial Network	MC	NP	VI	TE	MS	CI
163	Solicitation for Encryption Key Exchange - TR-31 Keyblock	√	√	√	√	√	√

Network Management Advice/0820: PEK exchange

The following values apply to this message.

Table 911: Values

Code	Financial Network	MC	NP	VI	TE	MS	CI
164	Encryption TR-31 Block Key Exchange Confirmation of Success	√	√	√	√	√	√

Code	Financial Network	MC	NP	VI	TE	MS	CI
165	Encryption TR-31 Block Key Exchange Advice of Failure	√	√	√	√	√	√

DE 70 Dual Message Authorization System Edits

The Dual Message Authorization System will perform the following system edits.

Table 912: Associated edits

WHEN...	THEN the Dual Message Authorization System...
the acquirer or issuer sends 0800 request with DE 70 = 166/167 and no DE110SF10	will reject with RC 96. No keys will be moved
the acquirer or issuer sends 0800 request with DE 70 = 166 and DE110SF10 present and doesn't match the staged key check value	
the acquirer or issuer sends 0800 request with DE 70 = 167 and DE110SF10 present and doesn't match the previous key check value	

DE 71 (Message Number)

DE 71 (Message Number) is a sequential, cyclic number the message initiator assigns to a message. Message Number is used to monitor the integrity of interchange.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 72 (Message Number Last)

DE 72 (Message Number Last) is a sequential, cyclic number the message initiator assigns to a message, used to monitor the integrity of interchange.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 73 (Date, Action)

DE 73 (Date, Action) specifies the date (year, month, and day) of a future action. In addition, a message originator may use it as a static time such as a birthdate.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 74 (Credits, Number)

DE 74 (Credits, Number) is the numeric sum of credit transactions processed.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 75 (Credits, Reversal Number)

DE 75 (Credits, Reversal Number) is the sum number of reversal credit transactions.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 76 (Debits, Number)

DE 76 (Debits, Number) is the sum number of debit transactions processed.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 77 (Debits, Reversal Number)

DE 77 (Debits, Reversal Number) is the sum number of reversal debit transactions.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 78 (Transfers, Number)

DE 78 (Transfers, Number) is the sum number of all transfer transactions processed.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 79 (Transfers, Reversal Number)

DE 79 (Transfers, Reversal Number) is the sum number of all transfer reversal transactions processed.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 80 (Inquiries, Number)

DE 80 (Inquiries, Number) is the sum number of inquiry transaction requests processed.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 81 (Authorizations, Number)

DE 81 (Authorizations, Number) is the sum number of Authorization Request/0100 and Authorization Advice/0120 messages processed.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 82 (Credits, Processing Fee Amount)

DE 82 (Credits, Processing Fee Amount) is the sum of all processing fees due to an institution or customer for services associated with handling and routing transactions. This Mastercard definition replaces the ISO standard definition.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 83 (Credits, Transaction Fee Amount)

DE 83 (Credits, Transaction Fee Amount) is the sum of all transaction fees due to an institution or customer for processing interchange transactions. This Mastercard definition replaces the ISO standard definition.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 84 (Debits, Processing Fee Amount)

DE 84 (Debits, Processing Fee Amount) is the sum of all processing fees due from an institution or customer for services associated with handling and routing transactions. This Mastercard definition replaces the ISO standard definition.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 85 (Debits, Transaction Fee Amount)

DE 85 (Debits, Transaction Fee Amount) is the sum of all transaction fees due from an institution or customer for processing interchange transactions. This Mastercard definition replaces the ISO standard definition.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 86 (Credits, Amount)

DE 86 (Credits, Amount) is the sum amount of all credit transactions processed exclusive of any fees.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 87 (Credits, Reversal Amount)

DE 87 (Credits, Reversal Amount) is the sum amount of reversal credits processed exclusive of any fees.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 88 (Debits, Amount)

DE 88 (Debits, Amount) is the sum amount of all debit transactions processed exclusive of any fees.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 89 (Debits, Reversal Amount)

DE 89 (Debits, Reversal Amount) is the sum amount of reversal debits processed exclusive of any fees.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 90 (Original Data Elements)

DE 90 (Original Data Elements) is the data elements in the original message, intended to identify a transaction for correction or reversal.

Table 913: Attributes

Attribute	Description
Data Representation	n-42
Data Length	N/A
Data Field	Contents of subfields 1–5
Subfields	5
Justification	see subfields

Usage

Following is the usage of DE 90 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 914: Applicable messages

Message	Org	Sys	Dst
Reversal Request/0400	M	•	M
Reversal Request Response/0410	ME	•	ME
Reversal Advice/0420	•	X	M
Reversal Advice Response/0430	ME	XE	•

Values

See subfields.

Application notes

DE 90 is mandatory in the Reversal Advice/0420 message. DE 90 is always present in the Reversal Advice/0420 message whether the Reversal Advice/0420 message is reversing an Authorization Request/0100 message or advising the issuer that Mastercard processed a Reversal Request/0400 message on behalf of the issuer.

DE 90, subfield 1 must contain the Message Type Identifier (MTI) of the original authorization message. The remaining subfields may contain valid matching values from the original authorization or may be zero-filled if they are not available.

For reversal messages, if DE 7 (Transmission Date and Time) and DE 11 (System Trace Audit Number [STAN]) from the original Authorization 0100/0110 cannot be saved, DE 90, subelements 2 and 3 may be zero-filled within the Reversal Request/0400 message.

Subfield 1 (Original Message Type Identifier)

DE 90, subfield 1 (Original Message Type Identifier) indicates the Message Type Identifier (MTI) of the original message.

Table 915: Attributes

Attribute	Description
Data Representation	n-4
Data Field	Contents of positions 1–4
Justification	N/A

Values

MTI of original message.

Subfield 2 (Original DE 11 [Systems Trace Audit Number])

DE 90, subfield 2 (Original DE 11 [Systems Trace Audit Number]) indicates the Systems Trace Audit Number (STAN) that was in DE 11 of the original message.

Table 916: Attributes

Attribute	Description
Data Representation	n-6
Data Field	Contents of positions 5–10
Justification	N/A

Values

STAN of original message.

Subfield 3 (Original DE 7 [Transmission Date and Time])

DE 90, subfield 3 (Original DE 7 [Transmission Date and Time]) indicates the Transmission Date and Time that was in DE 7 of the original message.

Table 917: Attributes

Attribute	Description
Data Representation	n-10
Data Field	Contents of positions 11–20 (in MMDDhhmmss format)
Justification	N/A

Values

Transmission date and time of original message. This subfield must contain a valid date expressed as month (MM) and day (DD) and a valid time expressed as hours (hh), minutes (mm), and seconds (ss).

Subfield 4 (Original DE 32 [Acquiring Institution ID Code])

DE 90, subfield 4 (Original DE 32 [Acquiring Institution ID Code]) indicates the Acquiring Institution ID Code that was in DE 32 of the original message.

Table 918: Attributes

Attribute	Description
Data Representation	n-11
Data Field	Contents of positions 21–31
Justification	Right with leading zeros

Values

Acquiring Institution ID code of original message.

Subfield 5 (Original DE 33 [Forwarding Institution ID Code])

DE 90, subfield 5 (Original DE 33 [Forwarding Institution ID Code]) indicates the Forwarding Institution ID Code that was in DE 33 of the original message.

Table 919: Attributes

Attribute	Description
Data Representation	n-11
Data Field	Contents of positions 32–42
Justification	Right with leading zeros

Values

Forwarding Institution ID code of original message. If the Forwarding Institution ID code was not present in the original message, then subfield 5 must contain all zeros.

DE 91 (Issuer File Update Code)

DE 91 (Issuer File Update Code) indicates to the system maintaining a file which procedure to follow.

Table 920: Attributes

Attribute	Description
Data Representation	an-1
Data Length	N/A
Data Field	Contents of position 1
Subfields	N/A
Justification	N/A

Usage

Following is the usage of DE 91 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 921: Applicable messages

Message	Org	Sys	Dst
Issuer File Update Request/0302	M	X	•
Issuer File Update Request Response/0312	•	XE	ME

Table 922: Values

Value	Description
1	Add
2	Update
3	Delete
5	Inquiry

Application notes

This data element is defined and used identically within all Mastercard programs and services.

DE 92 (File Security Code)

DE 92 (File Security Code) is an Issuer File Update security code used to indicate that a message originator is authorized to update a file.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 93 (Response Indicator)

DE 93 (Response Indicator) indicates the update action a POS system takes.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 94 (Service Indicator)

DE 94 (Service Indicator) indicates the service a message recipient requires.

Table 923: Attributes

Attribute	Description
Data Representation	ans-7
Data Length	N/A
Data Field	Contents of positions 1–7. Positions 4–7 must contain spaces or zeros.
Subfields	3
Justification	See subfields

Usage

Following is the usage of DE 94 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 924: Applicable messages

Message	Org	Sys	Dst
Network Management Request/0800: sign-on/sign-off	M	•	•

Values

This data element is used in Network Management Request/0800: sign-on/off messages to indicate that a customer is signing on or off for debit card traffic, credit card traffic, or both. It is also used to indicate if the sign-on or sign-off applies to acquirer traffic, issuer traffic, or both.

The Dual Message Authorization System allows credit card transactions and debit card transactions from a single institution to be routed to different processing systems.

Application notes

See subfields.

Subfield 1 (Reserved for Future Use)

DE 94, subfield 1 (Reserved for Future Use) is reserved for future use.

Table 925: Attributes

Attribute	Description
Data Representation	an-1
Data Field	Contents of position 1
Justification	N/A

Table 926: Values

Value	Description
0	Reserved for Future Use

Subfield 2 (Acquirer/Issuer Indicator)

DE 94, subfield 2 (Acquirer/Issuer Indicator) indicates if the sign-on or sign-off message applies to acquirer traffic, issuer traffic, or both.

Table 927: Attributes

Attribute	Description
Data Representation	a-1
Data Field	Contents of position 2
Justification	N/A

Table 928: Values

Value	Description
A	Acquirer only
I	Issuer only
B	Both acquirer and issuer

Subfield 3 (Address Data Indicator)

DE 94, subfield 3 (Address Data Indicator) indicates how the address data is provided.

Table 929: Attributes

Attribute	Description
Data Representation	n-1
Data Field	Contents of position 3
Justification	N/A

Table 930: Values

Value	Description
0	AVS not currently supported
1	Issuer receives complete address data
2	Issuer receives condensed address data. (This supports the algorithm that uses the first five numeric digits in an address [when scanning the address from left to right].)
3	Issuer receives condensed address data. (This supports the algorithm that uses up to the first five numeric digits in an address. This algorithm stops searching for a numeric after it encounters an alphabetic character or space [when scanning the address from left to right].)
4	Issuer receives condensed numeric postal code and condensed numeric address data only. (This supports the algorithm that uses the first five numeric digits in an address [when scanning the address from left to right].)

DE 95 (Replacement Amounts)

DE 95 (Replacement Amounts) contains the actual amount subfields necessary to perform a partial or full reversal of a financial transaction.

Table 931: Attributes

Attribute	Description
Data Representation	n-42
Data Length	N/A

Attribute	Description
Data Field	Contents of subfields 1–4
Subfields	4
Justification	See subfields

Usage

Following is the usage of DE 95 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 932: Applicable messages

Message	Org	Sys	Dst
Reversal Request/0400	C	X	C
Reversal Request Response/0410	CE	X	C
Reversal Advice/0420	•	X	C
Reversal Advice Response/0430	CE	XE	•

Values

See subfields.

Subfield 1 (Actual Amount, Transaction) must contain valid numeric data. All other subfields must be zero-filled; if required, the Authorization System performs currency conversion.

DE 95 is necessary to perform a partial reversal of an authorization transaction. However, the Authorization System will support full reversals via usage of a Reversal Request/0400 message. Therefore, acquirers are not required to include DE 95 in the Reversal Request/0400 message. If DE 95 is included in a Reversal Request/0400 message for full reversals, DE 95 must contain a value of all zeros.

The Dual Message Authorization System requires a message initiator to generate only subfield 1 (Actual Amount, Transaction). The Dual Message Authorization System automatically calculates and inserts subfield 2 (Actual Amount, Settlement) and subfield 3 (Actual Amount, Cardholder Billing).

NOTE: The subfield definitions the Dual Message Authorization System employs differ slightly from the ISO subfield definitions for this data element to accommodate the Mastercard automatic currency conversion service.

Application notes

This data element is defined and used identically within all Mastercard programs and services. Mastercard does not provide currency conversion services for all programs and services.

Mastercard supports DE 95 in Reversal Request/0400 and Reversal Advice/0420 messages. DE 95 must be less than DE 4. If DE 95 contains all zeros, Mastercard will remove DE 95 before forwarding the message to the issuer.

Subfield 1 (Actual Amount, Transaction)

DE 95, subfield 1 (Actual Amount, Transaction) indicates the actual transaction amount.

Table 933: Attributes

Attribute	Description
Data Representation	n-12
Data Field	Contents of positions 1–12
Justification	Right with leading zeros

Values

Full reversal: If present, and the reversal is a full reversal, DE 95, subfield 1 must contain a value of all zeros.

Partial reversal: If the reversal is a partial reversal, DE 95, subfield 1 must contain a value less than the amount in DE 4 (Amount Transaction).

Subfield 2 (Actual Amount, Settlement)

DE 95, subfield 2 (Actual Amount, Settlement) indicates the actual settlement amount in the settlement currency.

Table 934: Attributes

Attribute	Description
Data Representation	n-12
Data Field	Contents of positions 13–24
Justification	Right with leading zeros

Values

Must contain valid numeric data. Absence of data must be indicated with zeros.

Application notes

All settlement amounts are specified in U.S. Dollars. The Dual Message Authorization System will provide this subfield in the settlement currency (U.S. Dollars) if subfield 1 is not all zeros and if the customer chooses to receive settlement amount-related data elements; otherwise, subfield 2 is zero-filled.

Subfield 3 (Actual Amount, Cardholder Billing)

DE 95, subfield 3 (Actual Amount, Cardholder Billing) the actual amount in the issuer currency.

Table 935: Attributes

Attribute	Description
Data Representation	n-12
Data Field	Contents of positions 25–36
Justification	Right with leading zeros

Values

Must contain valid numeric data. Absence of data must be indicated with zeros. The Dual Message Authorization System will provide this subfield in the issuer's cardholder billing currency if subfield 1 is not all zeros.

Subfield 4 (Zero Fill)

DE 95, subfield 4 (Zero Fill) indicates zeros.

Table 936: Attributes

Attribute	Description
Data Representation	n-6
Data Field	Contents of positions 37–42
Justification	N/A

Values

Must contain zeros.

DE 96 (Message Security Code)

DE 96 (Message Security Code) is a verification between an acceptor and a card issuer that a message is authorized to update or modify a special file.

Table 937: Attributes

Attribute	Description
Data Representation	n-8
Data Length	N/A
Data Field	Contents of positions 1–8 (EBCDIC hexadecimal format)
Subfields	N/A
Justification	N/A

Usage

Following is the usage of DE 96 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 938: Applicable messages

Message	Org	Sys	Dst
Issuer File Update Request/0302	M	.	.
Issuer File Update Request Response/0312	M	M	.
Network Management Request/0800: sign-on/sign-off	M	.	.

Values

If an issuer has been assigned a Security Authorization Code that represents the Mastercard customer password, allowing access in the Dual Message Authorization System network by the customer or processor, this data element must contain valid security control data when used in Issuer File Update/0302 and Network Management Request/0800: sign-on/sign-off messages.

DE 97 (Amount, Net Settlement)

DE 97 (Amount, Net Settlement) is the net value of all gross amounts.

NOTE: The Dual Message Authorization System currently does not use this data element.

Subfield 1 (Debit/Credit Indicator)

DE 97, subfield 1 (Debit/Credit Indicator) indicates whether the transaction was credit or debit.

Attributes

Attribute	Description
Data representation	a-1
Data field	Contents of position 1
Justification	N/A

Values

N/A

Subfield 2 (Amount)

DE 97, subfield 2 (Amount) indicates the transaction amount.

Attributes

Attribute	Description
Data representation	n-16
Data field	Contents of positions 2-17
Justification	Right

Values

N/A

DE 98 (Payee)

DE 98 (Payee), is the third-party beneficiary in a payment transaction.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 99 (Settlement Institution ID Code)

DE 99 (Settlement Institution ID Code) identifies the settlement institution or its agent.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 100 (Receiving Institution ID Code)

DE 100 (Receiving Institution ID Code) is the identity of the institution receiving a Request or Advice message in an interchange system if not the same as identified in DE 2 (Primary Account Number [PAN]) or DE 34 (Primary Account Number [PAN], Extended). The Dual Message Authorization System uses DE 100 to determine the destination routing of Administrative/06xx messages. For these messages, DE 33 (Forwarding Institution ID Code) identifies the sender of the message; DE 100 identifies the receiver of the message.

Table 939: Attributes

Attribute	Description
Data Representation	n...11; LLVAR
Data Length	2 positions
Data Field	Contents of positions 1–11
Subfields	N/A
Justification	N/A

Usage

Following is the usage of DE 100 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 940: Applicable messages

Message	Org	Sys	Dst
Administrative Advice/0620: member-generated	M	.	M
Administrative Advice/0620: system-generated	.	X	M
Administrative Advice Response/0630	ME	.	.

Values

Must contain a valid five-digit Mastercard customer ID number. It identifies the destination customer or to receive the message.

Application notes

This data element is defined and used identically within all Mastercard programs and services.

Processing systems responding to originating Administrative Advice/06xx and Network Management/08xx messages must not swap the contents of DE 33 and DE 100 in the Response message to achieve proper routing of the Response to the originator.

DE 101 (File Name)

DE 101 (File Name) is the actual or abbreviated name of the file that the issuer accesses. DE 101 is used in Issuer File Update/03xx messages to identify the specific name of a Dual Message Authorization System data file or program parameter table that is being updated by a customer's Issuer File Update Request/0302.

Table 941: Attributes

Attribute	Description
Data Representation	ans...17; LLVAR
Data Length	2 positions
Data Field	Contents of positions 1–17
Subfields	N/A
Justification	N/A

Usage

Following is the usage of DE 101 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 942: Applicable messages

Message	Org	Sys	Dst
Issuer File Update Request/0302	M	X	•
Issuer File Update Request Response/0312	•	XE	ME

Table 943: Values

Value	Description
MCC102	Stand-In Account File
MCC103	Electronic Warning Bulletin File
MCC104	Local Stoplist File
MCC105	Payment Cancellation File
MCC106	PAN Mapping File
MCC107	Not in use
MCC108	Not in use
MCC109	Contactless Application Transaction Counter (ATC) File
MCC111	PAN-PAR (Payment Account Reference) Mapping File

DE 102 (Account ID 1)

DE 102 (Account ID-1) is a series of digits that identify a customer account or relationship. Customers primarily use it for the from account in a transfer transaction. DE 102 may be used in Authorization Request Response/0110 messages to identify the specific from account that the transaction affected. DE 102 may be used for printing on cardholder transaction receipts.

Table 944: Attributes

Attribute	Description
Data Representation	ans...28; LLVAR
Data Length	2 positions
Data Field	Contents of positions 1–28
Subfields	N/A
Justification	N/A

Usage

Following is the usage of DE 102 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 945: Applicable messages

Messages	Org	Sys	Dst
Authorization Request Response/0110	C	.	C
Authorization Advice/0120: acquirer-generated	C	.	C
Authorization Advice/0120: system-generated	.	C	C

Values

The from account is the account specified in digits three and four of DE 3 (Processing Code).

Application notes

This data element is defined and used identically within all Mastercard programs and services.

DE 103 (Account ID 2)

DE 103 (Account ID-2) is a series of digits that identify a customer account or relationship. Customers primarily use it for the "to" account in a transfer transaction.

Table 946: Attributes

Attribute	Description
Data Representation	ans...28; LLVAR
Data Length	2 positions
Data Field	Contents of positions 1–28
Subfields	N/A
Justification	N/A

Usage

Following is the usage of DE 103 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 947: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	C	.	C
Authorization Advice/0120: acquirer-generated	C	.	C
Authorization Advice/0120: system-generated	.	X	C

Values

The to account is specified in digits five and six of DE 3 (Processing Code).

Application notes

This data element is defined and used identically within all Mastercard programs and services.

DE 104 (Digital Payment Data)

DE 104 (Digital Payment Data) is used in Authorization Request/0100 messages to send digital payment data for e-commerce transactions.

Table 948: Attributes

Attribute	Description
Data Representation	ans...999; LLLVAR
Data Length	3 positions
Data Field	Contents of subelements
Subelements	001: Digital Payment Cryptogram 002: Estimated Amount 003: Remote Commerce Acceptor Identifier 004: Digital Service Provider 005: Digital Authentication Data
Justification	See "Subelements"

Usage

Following is the usage of DE 104 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 949: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	X	C

Values

Transaction specific.

DE 104: Encoding Scheme

Mastercard organizes DE 104 into a group of encoded subelements and subfields. The following table illustrates the structure of DE 104 subelements.

Table 950: Subelement structure

LLL	"VAR" - maximum length varies					
3 bytes	3 bytes	3 bytes	Variable length	3 bytes	3 bytes	Variable length
Total Length	First Subelement (SE) Data			Second Subelement (SE) Data		
	SE ID	SE Length	SE Variable Length Data	SE ID	SE Length	SE Variable Length Data

Table 951: Subelement structure

Number of Bytes	Attribute	Value
3	Total Length field	The "LLL" portion of the data element
3	Subelement ID	In the range 001-099
3	Subelement Length	Variable
1...nnn	Subelement Variable Length Data	Contents of Subfields

The following table illustrates the structure of DE 104 subfields within a subelement.

Table 952: Subfield structure

"VAR" - maximum length varies					
3 bytes	3 bytes	Variable length	3 bytes	3 bytes	Variable length
First subfield (SF) Data			Second subfield (SF) Data		
SF ID	SF Length	SF variable length data	SF ID	SF length	SF Variable length data

Table 953: Subfield structure

Number of bytes	Attribute	Value
3	Subfield ID	In the range 001-099
3	Subfield length	Variable
1...nnn	Subfield variable Length Data	Contents of subfields

Subelement 001 (Digital Payment Cryptogram)

DE 104, subelement 001 (Digital Payment Cryptogram) is where the DSRP Cryptogram will be sent to be processed and verified.

Table 954: Attributes

Attribute	Description
Subelement ID	001
Subelement Data Length	3 positions
Data Representation	ans-28; (base 64 encoded)
Data Field	Contains DSRP Cryptogram
Subfields	N/A
Justification	Left

Usage

Following is the usage of subelement 001 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 955: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	X	C

Values

Transaction specific. Refer to [Digital Secure Remote Payments: UCAF Formats](#) for detailed information on the formats supported by Mastercard, which continue to be applicable to the new Digital Payment Data field.

Application notes

To be included by the acquirer for MDES electronic commerce transactions in Authorization Request/0100 messages.

To be received by the issuer when included by the acquirer in non-MDES transactions in Authorization Request/0100 messages and Reversal Request/0400 messages.

If DE 104, subelement 001 is included on any other message type it will be ignored.

NOTE: Issuers will not receive DE 104, subelement 001 on MDES transactions.

Subelement 002 (Estimated Amount)

DE 104, subelement 002 (Estimated Amount) contains the estimated purchase amount of the transaction when the payment request is presented to the consumer during checkout. This amount may not always be the final amount. The estimated amount is encoded and included within the cryptogram. The estimated amount is in the transaction currency defined in DE 49 (Currency Code, Transaction).

Table 956: Attributes

Attributes	Description
Subelement ID	002
Subelement Data Length	3 positions
Data Representation	n-12
Data Field	Contains estimated amount
Subfields	N/A
Justification	Right with leading zeros

Usage

Following is the usage of subelement 002 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 957: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	.	X	C

Application notes

The Authorization System inserts this subelement into Authorization Request/0100 messages when the issuer has opted-in to receive it.

This subelement is expected to be present in transactions initiated via wallets or remote commerce programs that support DSRP cryptogram format 3.

Subelement 003 (Remote Commerce Acceptor Identifier)

DE 104, subelement 003 (Remote Commerce Acceptor Identifier) contains a merchant identifier such as the merchant business website URL or reverse domain name as presented to the consumer during checkout. The merchant identifier is provided encoded as Base64.

Table 958: Attributes

Attribute	Description
Subelement ID	003
Subelement Data Length	3 positions
Data Representation	ans...150 (Base64 encoded)
Data Field	Contains the Remote Commerce Acceptor Identifier
Subfields	N/A
Justification	Left

Usage

Following is the usage of subelement 003 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 959: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	X	C

Application notes

Acquirers can optionally submit the Remote Commerce Acceptor Identifier in DE 104, subelement 003 within Authorization Request/0100 and Financial Transaction Request/0200 messages encoded as Base64 when provided by the merchant for MDES remote commerce transactions.

Mastercard will forward to the issuer (when opted-in to receive it). Issuers electing to use a Third Party Token Service Provider for device based wallet programs will receive the subelement in the authorization request when included by the acquirer. Applicable to Authorization Request/0100 messages.

This subelement is expected to be present in remote commerce transactions initiated via wallets or remote commerce programs that support DSRP cryptogram format 3. If present, it must be encoded as Base64.

Subelement 004 (Digital Service Provider)

DE 104, subelement 004 (Digital Service Provider) is reserved for specific digital programs and services. It is a free-format, variable-length subelement that may be used for multiple purposes.

Table 960: Attributes

Attribute	Description
Subelement ID	004
Subelement Data Length	3 positions
Data Representation	ans...250; LLLVAR
Data Field	Contents of subfields
Subfields	2
Justification	See subfields

Usage

Following is the usage of subelement 004 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 961: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	.	X	C

Application notes

The Dual Message Authorization System populates this subelement in Authorization Request/0100 messages when the issuer has opted-in to receive it. Mastercard may occasionally introduce new DE 104, subelement 004 values between releases. Mastercard requires customers to code their systems with flexibility to process messages containing new values.

Subfield 001 (Digital Service Provider ID 1)

DE 104, subelement 004 (Digital Service Provider), subfield 001 (Digital Service Provider ID 1) contains the alphanumeric value that identifies the Digital Service Provider type and entity.

This field provides information on either the third-party entity that facilitated a tokenized transaction or the entity that authenticated the cardholder in a remote commerce transaction.

Table 962: Attributes

Attribute	Description
Subfield ID	001
Subfield Data Length	3 positions
Data Representation	an-5
Data Field	Contents of subfield 001
Justification	Left

Values

Table 963: Position 1 (Digital Service Provider Type)

Value	Description
A	Entity that authenticated the cardholder
C	Entity that provided the card on file credential used in the transaction
F	Entity that provided the credential to the merchant during checkout

If position 1 is...	Then position 2-5...
A	The list of available alphanumeric authenticating entity IDs is in <i>AN 6480 Authenticating Entities and Multi-Factor Authentication Methods Values</i> and the <i>MDES Issuer Implementation Guide</i> . Both documents are on the Technical Resource Center (TRC).
C	A list of available IDs is in the <i>MDES Issuer Implementation Guide</i> available on the Technical Resource Center (TRC).
F	A list of available IDs is in the <i>MDES Issuer Implementation Guide</i> available on the Technical Resource Center (TRC).

Application notes

The absence of subfield 001 does not impact the authentication status of a transaction. The subfield serves as additional information and should not be used in isolation to make authorization decisions.

Example: Formatted DE 104, subelement 004, subfield 001

104 225002 012000000000400 003 150d3d3Lm1hc3RlcmNhcmQuY29t... 004 011 001
00511111

Subfield 002 (Digital Service Provider ID 2)

DE 104, subelement 004 (Digital Service Provider), subfield 002 (Digital Service Provider ID 2) contains the alphanumeric value that identifies the Digital Service Provider type and entity.

This field provides information on either the third-party entity that facilitated a tokenized transaction or the entity that authenticated the cardholder in a remote commerce transaction.

Table 964: Attributes

Attribute	Description
Subfield ID	002
Subfield Data Length	3 positions
Data Representation	an-5
Data Field	Contents of subfield 002
Justification	Left

Values

Table 965: Position 1 (Digital Service Provider Type)

Value	Description
A	Entity that authenticated the cardholder
C	Entity that provided the card on file credential used in the transaction
F	Entity that provided the credential to the merchant during checkout

Table 966: Position 2-5 (Digital Service Provider Identifier)

If position 1 is...	Then position 2-5...
A	The list of available alphanumeric authenticating entity IDs is in <i>AN 6480 Authenticating Entities and Multi-Factor Authentication Methods Values</i> and the <i>MDES Issuer Implementation Guide</i> . Both documents are on the Technical Resource Center (TRC).
C	A list of available IDs is in the <i>MDES Issuer Implementation Guide</i> available on the Technical Resource Center (TRC).
F	A list of available IDs is in the <i>MDES Issuer Implementation Guide</i> available on the Technical Resource Center (TRC).

Application notes

The absence of subfield 002 does not impact the authentication status of a transaction. The subfield serves as additional information and should not be used in isolation to make authorization decisions.

Subelement 005 (Digital Authentication Data)

DE 104, subelement 005 (Digital Authentication Data) is reserved for specific digital programs and services. It is a free-format, variable-length sub element that may be used for multiple

purposes. The Attributes table identifies the structure of the subfield encoding scheme in the Authorization Request/0100 message.

Table 967: Attributes

Attribute	Description
Subelement ID	005
Subelement Data Length	3 positions
Data Representation	ans...250; LLLVAR
Data Field	Contents of subfields
Number of subfields	Subfield 001 (Cardholder Verification) Subfield 002 (Authentication Factors)
Justification	See subfields

Usage

Whether the usage is mandatory, conditional, optional, system-provided, or not required is as noted for each applicable message.

Table 968: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	.	X	C

Application notes

The Dual Message Authorization System populates this subelement in Authorization Request/0100 messages when the issuer has opted-in to receive it. Mastercard may occasionally introduce new DE 104, subelement 005 values between releases. Mastercard requires customers to code their systems with flexibility to process messages containing new values.

Subfield 001 (Cardholder Verification)

DE 104, subelement 005 (Digital Authentication Data), subfield 001 (Cardholder Verification) contains an alphanumeric value to indicate the cardholder was authenticated and specifies the

Cardholder Verification Method (CVM) used to authenticate the cardholder during a contactless or remote commerce transaction.

Table 969: Attributes

Attribute	Description
Subfield ID	001
Subfield Data Length	3 positions
Data Representation	an-5
Data Field	Contents of subfield 001
Justification	Left

Values

Table 970: Position 1 (Transaction authenticated with a valid Cardholder Verification Method)

Value	Description
1	Transaction authenticated with a Mastercard-qualified multi-factor authentication (MFA) method. In Europe region: Transaction authenticated with a Mastercard-qualified multi-factor authentication (MFA) method used as Strong Customer Authentication (SCA) delegation.
2	Transaction authenticated with a Mastercard-operated authentication method
3	Transaction authenticated with an Issuer authentication method
4	Device wallet transaction authenticated with a Consumer Device Cardholder Verification Method (CDCVM). This value may not be present in current device wallet implementations.

Table 971: Position 2-5 (Identifier to the Cardholder Verification Method)

If position 1 is...	Then position 2-5 will be...
1	A Mastercard assigned unique identifier for the MFA method. See <i>AN 6480 Authenticating Entities and Multi-Factor Authentication Methods Values</i> for the list of available alphanumeric MFA method IDs.
2	0001: Mastercard authentication method with device binding and device biometrics (FIDO). 0002: Mastercard authentication method with device binding and passive authentication.

If position 1 is...	Then position 2-5 will be...
3	0001: Transaction authenticated by Issuer via Identity Check (EMV 3DS) when Mastercard is 3DS Requestor. 0002: Transaction authenticated by Issuer via MDES.
4	Mastercard assigned unique identifier for the CDCVM method. This value may not be present in current device wallet implementations. TRID: When the value is "4TRID" the Token Requestor ID in DE 48, subelement 33 (PAN Mapping File Information), subfield 006 (Token Requestor ID), identifies the entity that authenticated the cardholder. DDSP: When the value is "4DDSP" the Digital Service Provider ID in DE 104, subelement 004, subfield 001 or subfield 002, identifies the entity that authenticated the cardholder.

The list of available methods is in *AN 6480 Authenticating Entities and Multi-Factor Authentication Methods Values* and the [MDES Issuer Implementation Guide](#). Both documents are on the Technical Resource Center on Mastercard Connect®.

Application notes

The absence of subfield 001 does not impact authentication status of a transaction. The subfield serves as additional information and should not be used in isolation to make authorization decisions.

Subfield 002 (Authentication Factors)

DE 104, subelement 005 (Digital Authentication Data), subfield 002 (Authentication Factors) contains the alphanumeric value that identifies the authentication factors used to authenticate the cardholder during a contactless or remote commerce transaction.

Table 972: Attributes

Attribute	Description
Subfield ID	002
Subfield Data Length	3 positions
Data Representation	an-5
Data Field	Contents of subfield 002
Justification	Left

Table 973: Values

Position	Description	Values
1-2	First authentication factor	value from table below
3-4	Second authentication factor	value from table below
5	Reserved for future use	0

Application notes

The use of authentication factors are subject to any regulatory approvals and compliance with applicable laws and regulations.

Table 974: Authentication factors by value

Value	Authentication factor	Description	Category
00	Not provided	Factor not provided	N/A
01	Consumer device / device token	Token is stored securely on consumer device.	Possession
02	Consumer device / device bound payment credential	Payment credential is bound to consumer device using a device binding process that ensures a unique connection between the device and an application or browser or cryptographic key.	Possession
03	SMS OTP	One Time Password or passcode sent via text message to a phone number	Possession
04	Email OTP	One Time Password or passcode sent via email to an email address only accessible from a registered device	Possession

Value	Authentication factor	Description	Category
05	Email OTP	One Time Password or passcode sent via email to an email address accessible from any device using a knowledge factor	Knowledge
06	Card Validation Code 2 (CVC2)	Static 3-digit number printed on the back of a Mastercard-branded card	Possession
07	Application authentication with unknown factor	Knowledge or inference factor validated by an application (for example, issuer, merchant, or wallet application) exact factor and exact category unknown	Knowledge or Inference
08	Application authentication with knowledge factor	Knowledge factor validated by an application (for example, issuer, merchant, or wallet application) exact factor unknown	Knowledge
09	Application authentication with inference factor	Inference factor validated by an application (for example, issuer, merchant, or wallet application) exact factor unknown	Inference
0A	Device OS authentication with unknown factor	Knowledge or inference factor validated by the operating system on the consumer device - exact factor and category unknown	Knowledge or Inference
0B	Device OS authentication with knowledge factor	Knowledge factor validated by the operating system on the consumer device - exact factor unknown	Knowledge
0C	Device OS authentication with password	Password validated by the operating system on the consumer device	Knowledge

Value	Authentication factor	Description	Category
0D	Device OS authentication with passcode	Passcode validated by the operating system on the consumer device	Knowledge
0E	Device OS authentication with pattern	Pattern validated by the operating system on the consumer device	Knowledge
0F	Device OS authentication with inherence factor	Inherence factor validated by the operating system on the consumer device - exact factor unknown	Inherence
0G	Device OS authentication with fingerprint	Fingerprint validated by the operating system on the consumer device	Inherence
0H	Device OS authentication with facial recognition	Face validated by the operating system on the consumer device	Inherence
0I	Device OS authentication with iris recognition	Iris validated by the operating system on the consumer device	Inherence
0J	Pattern	Pattern validation outside of device OS	Knowledge
0K	Password	Password validation outside of device OS	Knowledge
0L	Passcode	Passcode validation outside of device OS	Knowledge
0M	Facial recognition	Facial recognition outside of device OS	Inherence
0N	Fingerprint	Fingerprint validation outside of device OS	Inherence
0O	Iris recognition	Iris recognition outside of device OS	Inherence
0P	Voice recognition	Voice recognition validated by an application or server	Inherence
0Q	Vein recognition	Vein recognition validated by an application or server	Inherence

Value	Authentication factor	Description	Category
OR	Palm geometry	Palm geometry validated by an application or server	Inherence
OS	Hand geometry	Hand geometry validated by an application or server	Inherence
OT	Behavioral biometrics	Biometrics that relate to behavioral processes created by the human body, for example the way cardholders tap, type, swipe or hold a device	Inherence
OU	Behavioral-based inherence information	Information generated from multiple data points that include elements identifying, cardholder location, spending habits, and transaction history	Inherence

DE 104 Dual Message Authorization System edits

The Dual Message Authorization System will perform the following system edits.

Table 975: Associated edits

WHEN...	THEN the Dual Message Authorization System...
an Authorization Request/0100 message containing DE 104 (Digital Payment Data), subelement 001 (Digital Payment Cryptogram), is of incorrect length or incorrect data representation	will reject the message and forward to the acquirer an Authorization Request Response/0110 message with: - DE 39 (Response Code) = 30 (Format error) - DE 44 (Additional Response Data) = 104001 (indicating the data element in error)
DE 104 (Digital Payment Data), subelement 003 (Remote Commerce Acceptor Identifier) is incorrectly formatted in an Authorization Request/0100 message and/or exceeds 150 characters in length	will reject the message and forward to the acquirer an Authorization Request Response/0110 message with: - DE 39 (Response Code) = 30 (Format error) - DE 44 (Additional Response Data) = 104003 (indicating the data element and subelement in error)

DE 105 (Multi-Use Transaction Identification Data)

DE 105 is alphanumeric, variable length using a tag-length-data encoding scheme to allow for easy addition of new subelements.

Table 976: Attributes

Attribute	Description
Data Representation	ans...999, LLLVAR The length field must be in the range 001-999. Subelements are identified by valid subelement ID and length.
Data Length	3 positions
Data Field	Contents of subelements
Subelements	1
Justification	See subelement

Usage

Whether the data element is mandatory, conditional, optional, system-provided, or not required is noted for each applicable message.

Table 977: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	X	C
Authorization Request Response/0110	CE	X	C
Authorization Advice/0120: acquirer-generated	C	X	C
Authorization Advice/0120: system-generated	•	X	C
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	CE	X	C
Reversal Request/0400	C	X	C
Reversal Request Response/0410	CE	X	C
Reversal Advice/0420	•	X	C
Reversal Advice Response/0430	CE	•	•

Encoding scheme for DE 105

Mastercard organizes DE 105 into a group of encoded subelements and subfields.

Table 978: Structure of DE 105 subelement

LLL	VAR: maximum length varies		
3 bytes	3 bytes	3 bytes	Variable length
Total length	First subelement data		
	Subelement ID	Subelement length	Subelement variable length

Table 979: Structure of DE 105 subelement

Number of Bytes	Attribute	Value
3	Total data element length	The "LLL" portion of the data element
3	Subelement ID	In the range 001–999
3	Subelement length	Variable
1...nnn	Subelement variable length data	Contents of Subfields

Table 980: Structure of DE 105 subfields within a subelement

LLL	VAR: maximum length varies	
3 bytes	3 bytes	Variable length
First subfield data		
Subfield ID	Subfield length	Subfield variable length data

Table 981: Structure of DE 105 subfields

Number of bytes	Attribute	Value
3	Subfield ID	In the range of 001-999
3	Subfield length	Variable
1...nnn	Subfield variable length data	Contains valid values

Subelement 001 (Transaction Link ID [TLID])

DE 105, subelement 001 (Transaction Link ID [TLID]) contains the transaction link ID that is globally unique and can be used as a single unique reference for deterministic matching and linking of all message activity.

Table 982: Attributes

Attribute	Description
Subelement ID	001
Subelement Data Length	3 positions
Data Representation	ans-22
Data Field	N/A
Subfields	N/A
Justification	N/A

Usage

Whether the data element is mandatory, conditional, optional, system-provided, or not required is noted for each applicable message.

Table 983: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	X	C
Authorization Request Response/0110	CE	X	C
Authorization Advice/0120: Acquirer-generated	C	X	C
Authorization Advice/0120: System-generated	.	X	C
Authorization Advice Response/0130: issuer-generated (Responding to an acquirer-generated 0120)	CE	X	C
Authorization Advice Response/0130: system-generated	.	X	C
Reversal Request/0400	C	X	C
Reversal Request Response/0410	CE	X	C
Reversal Advice/0420: System generated	.	X	C
Reversal Advice Response/0430	CE	.	.

NOTE: Authorization Request/0100 pre-digitization messages will not be assigned a TLID. These messages can be identified with DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status) with a value of 9 (Tokenization Request/Notification).

DE 107 (Reserved for Mastercard Use)

DE 107 (Reserved for Mastercard Use) is reserved for Mastercard use.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 106 (Fleet Service Data)

DE 106 (Fleet Service Data) contains information pertaining to Fleet card transactions and the items being purchased.

Table 984: Attributes

Attribute	Description
Data Representation	ans...999; LLLVAR Although this data element is ans, some subelements deviate from ans and contain binary data. Refer to subelement descriptions of each subelement. The length field must be in the range 001-999. Subelements are identified by valid subelement ID and length.
Data Length	Len (3-bytes) – value indicates length of DE 106
Data Field	Contents of subelements
Subelements	5
Justification	See "Subelements"

Usage

The following applies to DE 106 usage:

- For subelements defined with a CE (conditional echo) presence notation for an Org (originator) entity Mastercard will not reject a message from the message originator if the original subelement data is not returned as defined in a response message. Based on the

presence notation for the Dst (destination) entity Mastercard will forward the subelement data as received from the message originator if it is present in a response message.

- Use of DE 106 (Fleet Service Data) for authorization is required only for EMV chip cards personalized using the IFSF/Conexus specification for retrieving supplemental data from acceptors and cardholders.

Following is the overall usage of DE 106 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 985: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	O	•	C
Authorization Request Response/0110	C	•	C
Authorization Advice/0120: acquirer-generated	O	•	C
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	C	•	C

List of DE 106 subelements

DE 106 subelements are listed in numeric order. Subelements that are specific to a brand service or program are clearly indicated in the subelement title or description or both.

Table 986: Subelements

Subelement ID	Subelement name	Data representation
001	Fleet Prompted Data	ans...220; LLLVAR
002	Merchant Fleet Spend, Control Override Capability Indicator	ans...001; LLLVAR
003	Fleet Spend Control Override Items	b...016; LLLVAR
004	Fleet Fuel Information	ans...123; LLLVAR
005	Fleet Non-Fuel Information	ans...552; LLLVAR

Subelement 1 (Fleet Prompted Data)

DE 106, subelement 01 (Fleet Prompted Data) contains up to 5 occurrences of the Prompt Code read from a card chip that indicates prompts that occur at the point of interaction and the data entered in response by the cardholder into the point-of-sale device.

Table 987: Attributes

Attribute	Description
Subelement ID	001
Subelement Data Length	(3-bytes) – value indicates Data field size
Data Representation	ans...220; VAR
Data Field	Contains up to 5 subfields
Subfields	Subfield 01 (Fleet Prompt 1) Subfield 02 (Fleet Prompt 2) Subfield 03 (Fleet Prompt 3) Subfield 04 (Fleet Prompt 4) Subfield 05 (Fleet Prompt 5)

Usage

Usage shows whether the field is mandatory, conditional, optional, or system-provided in applicable messages.

Table 988: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	O	.	C
Authorization Response/0110	C	.	C
Authorization Advice/0120: acquirer-generated	O	.	C
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	C	.	C

Application notes

If subelement 01 was present in the original 0100 AFD authorization message, subelement 01 must be the same values in acquirer-generated 0120 completion advice.

Subfield 1 (Fleet Prompt 1)

DE 106, subelement 01, subfield 01 (Fleet Prompt 1) contains the first occurrence of the Prompt Code and Driver Related Data.

Table 989: Attributes

Attribute	Description
Data Representation	Tag (3-bytes) – value '001' Len (3-bytes) – value indicates Data field size
Data Field	ans...38; VAR Bytes 1-3 (fixed) – Prompt Code 1 (NUM) Bytes 4-38 (varLen) – Driver Related Data 1 (ANS)
Justification	Bytes 1-3 Right-justified Bytes 4-38 Left-justified

Table 990: Values

Position	Description
DE 106, subelement 1, subfield 1 Position 1-3	Prompt Code 1

Table 991: Prompt codes

Prompt code	Prompt
001	User ID
002	Vehicle Number
003	Vehicle Tag
004	Driver ID
005	Odometer
009	Work Order / P.O. Number
010	Invoice Number
011	Trip Number
012	Unit Number
013	Reefer Hours

Prompt code	Prompt
015	ZIP Code / Postal Code
016	Replacement Car
017	Entered Data Alpha
018	Web Portal Data
019	Entered Data Num
021	Job Number
022	Maintenance ID
023	Department Number
024	Trailer Number
025	Delivery Ticket Number
026	Hubometer
033	Sub Fleet Number
035	Transaction Number
036	Control Number
038	Reefer Temperature
040	Driver or Vehicle Card
041	Customer Number
042	Additional Card Data
043	Additional Vehicle Data
044	Engine Hours
045	Tank Level Start
046	Fuel Gauge Level
047	Battery Voltage
048	Coolant Temperature
049	Check Engine Status
050	Fuel Economy
051	Engine RPM
052	Engine Load
053	Engine Oil Temperature
054	Engine Time Total
055	Hard Breaking

Prompt code	Prompt
056	Hard Acceleration
058	Idle Time
065	Total Idle Time
067	Engine Oil Pressure
068	Engine Oil Life
069	Billing ID

Application notes

Prompt Code contains a code that specifies the information requested to be provided by the cardholder at the POI.

Driver Related Data contains the data entered in response to the Prompt Code obtained at the POS.

Subfield 2 (Fleet Prompt 2)

DE 106, subelement 01, subfield 02 (Fleet Prompt 2) contains the second occurrence of the Prompt Code and Driver Related Data.

Table 992: Attributes

Attribute	Description
Data Representation	Tag (3-bytes) – value '002' Len (3-bytes) – value indicates Data field size
Data Field	ans...38; VAR Bytes 1-3 (fixed) – Prompt Code 2 (NUM) Bytes 4-38 (varLen) – Driver Related Data 2 (ANS)
Justification	Left-justified

Table 993: Values

Position	Description
DE 106, subelement 1, subfield 2 Position 1-3	Prompt Code 2

Table 994: Prompt codes

Prompt code	Prompt
001	User ID
002	Vehicle Number
003	Vehicle Tag
004	Driver ID
005	Odometer
009	Work Order / P.O. Number
010	Invoice Number
011	Trip Number
012	Unit Number
013	Reefer Hours
015	ZIP Code / Postal Code
016	Replacement Car
017	Entered Data Alpha
018	Web Portal Data
019	Entered Data Num
021	Job Number
022	Maintenance ID
023	Department Number
024	Trailer Number
025	Delivery Ticket Number
026	Hubometer
033	Sub Fleet Number
035	Transaction Number
036	Control Number
038	Reefer Temperature
040	Driver or Vehicle Card
041	Customer Number
042	Additional Card Data
043	Additional Vehicle Data

Prompt code	Prompt
044	Engine Hours
045	Tank Level Start
046	Fuel Gauge Level
047	Battery Voltage
048	Coolant Temperature
049	Check Engine Status
050	Fuel Economy
051	Engine RPM
052	Engine Load
053	Engine Oil Temperature
054	Engine Time Total
055	Hard Breaking
056	Hard Acceleration
058	Idle Time
065	Total Idle Time
067	Engine Oil Pressure
068	Engine Oil Life
069	Billing ID

Application notes

Prompt Code contains a code that specifies the information requested to be provided by the cardholder at the POI.

Driver Related Data contains the data entered in response to the Prompt Code obtained at the POS.

Subfield 3 (Fleet Prompt 3)

DE 106, subelement 01, subfield 03 (Fleet Prompt 3) contains the third occurrence of the Prompt Code and Driver Related Data.

Table 995: Attributes

Attribute	Description
Data Representation	Tag (3-bytes) – value '003' Len (3-bytes) – value indicates Data field size
Data Field	ans...38; VAR Bytes 1-3 (fixed) – Prompt Code 3 (NUM) Bytes 4-38 (varLen) – Driver Related Data 3 (ANS)
Justification	Left-justified

Table 996: Values

Position	Description
DE 106, subelement 1, subfield 3 Position 1-3	Prompt Code 3

Table 997: Prompt codes

Prompt code	Prompt
001	User ID
002	Vehicle Number
003	Vehicle Tag
004	Driver ID
005	Odometer
009	Work Order / P.O. Number
010	Invoice Number
011	Trip Number
012	Unit Number
013	Reefer Hours
015	ZIP Code / Postal Code

Prompt code	Prompt
016	Replacement Car
017	Entered Data Alpha
018	Web Portal Data
019	Entered Data Num
021	Job Number
022	Maintenance ID
023	Department Number
024	Trailer Number
025	Delivery Ticket Number
026	Hubometer
033	Sub Fleet Number
035	Transaction Number
036	Control Number
038	Reefer Temperature
040	Driver or Vehicle Card
041	Customer Number
042	Additional Card Data
043	Additional Vehicle Data
044	Engine Hours
045	Tank Level Start
046	Fuel Gauge Level
047	Battery Voltage
048	Coolant Temperature
049	Check Engine Status
050	Fuel Economy
051	Engine RPM
052	Engine Load
053	Engine Oil Temperature
054	Engine Time Total
055	Hard Breaking
056	Hard Acceleration

Prompt code	Prompt
058	Idle Time
065	Total Idle Time
067	Engine Oil Pressure
068	Engine Oil Life
069	Billing ID

Application notes

Prompt Code contains a code that specifies the information requested to be provided by the cardholder at the POI.

Driver Related Data contains the data entered in response to the Prompt Code obtained at the POS.

Subfield 4 (Fleet Prompt 4)

DE 106, subelement 01, subfield 04 (Fleet Prompt 4) contains the fourth occurrence of the Prompt Code and Driver Related Data.

Table 998: Attributes

Attribute	Description
Data Representation	Tag (3-bytes) – value '004' Len (3-bytes) – value indicates Data field size
Data Field	ans...38; VAR Bytes 1-3 (fixed) – Prompt Code 4 (NUM) Bytes 4-38 (varLen) – Driver Related Data 4 (ANS)
Justification	Left-justified

Table 999: Values

Position	Description
DE 106, subelement 1, subfield 4 Position 1-3	Prompt Code 4

Table 1000: Prompt codes

Prompt code	Prompt
001	User ID
002	Vehicle Number
003	Vehicle Tag
004	Driver ID
005	Odometer
009	Work Order / P.O. Number
010	Invoice Number
011	Trip Number
012	Unit Number
013	Reefer Hours
015	ZIP Code / Postal Code
016	Replacement Car
017	Entered Data Alpha
018	Web Portal Data
019	Entered Data Num
021	Job Number
022	Maintenance ID
023	Department Number
024	Trailer Number
025	Delivery Ticket Number
026	Hubometer
033	Sub Fleet Number
035	Transaction Number
036	Control Number
038	Reefer Temperature
040	Driver or Vehicle Card
041	Customer Number
042	Additional Card Data
043	Additional Vehicle Data

Prompt code	Prompt
044	Engine Hours
045	Tank Level Start
046	Fuel Gauge Level
047	Battery Voltage
048	Coolant Temperature
049	Check Engine Status
050	Fuel Economy
051	Engine RPM
052	Engine Load
053	Engine Oil Temperature
054	Engine Time Total
055	Hard Breaking
056	Hard Acceleration
058	Idle Time
065	Total Idle Time
067	Engine Oil Pressure
068	Engine Oil Life
069	Billing ID

Application notes

Prompt Code contains a code that specifies the information requested to be provided by the cardholder at the POI.

Driver Related Data contains the data entered in response to the Prompt Code obtained at the POS.

Subfield 5 (Fleet Prompt 5)

DE 106, subelement 01, subfield 05 (Fleet Prompt 5) contains the fifth occurrence of the Prompt Code and Driver Related Data.

Table 1001: Attributes

Attribute	Description
Data Representation	Tag (3-bytes) – value '005' Len (3-bytes) – value indicates Data field size
Data Field	ans...38; VAR Bytes 1-3 (fixed) – Prompt Code 5 (NUM) Bytes 4-38 (varLen) – Driver Related Data 5 (ANS)
Justification	Left-justified

Table 1002: Values

Position	Description
DE 106, subelement 1, subfield 5 Position 1-3	Prompt Code 5

Table 1003: Prompt codes

Prompt code	Prompt
001	User ID
002	Vehicle Number
003	Vehicle Tag
004	Driver ID
005	Odometer
009	Work Order / P.O. Number
010	Invoice Number
011	Trip Number
012	Unit Number
013	Reefer Hours
015	ZIP Code / Postal Code

Prompt code	Prompt
016	Replacement Car
017	Entered Data Alpha
018	Web Portal Data
019	Entered Data Num
021	Job Number
022	Maintenance ID
023	Department Number
024	Trailer Number
025	Delivery Ticket Number
026	Hubometer
033	Sub Fleet Number
035	Transaction Number
036	Control Number
038	Reefer Temperature
040	Driver or Vehicle Card
041	Customer Number
042	Additional Card Data
043	Additional Vehicle Data
044	Engine Hours
045	Tank Level Start
046	Fuel Gauge Level
047	Battery Voltage
048	Coolant Temperature
049	Check Engine Status
050	Fuel Economy
051	Engine RPM
052	Engine Load
053	Engine Oil Temperature
054	Engine Time Total
055	Hard Breaking
056	Hard Acceleration

Prompt code	Prompt
058	Idle Time
065	Total Idle Time
067	Engine Oil Pressure
068	Engine Oil Life
069	Billing ID

Application notes

Prompt Code contains a code that specifies the information requested to be provided by the cardholder at the POI.

Driver Related Data contains the data entered in response to the Prompt Code obtained at the POS.

Subelement 2 (Merchant Fleet Spend Control Override Capability Indicator)

DE 106, subelement 02 (Merchant Fleet Spend Control Override Capability Indicator) indicates if the merchant point-of-sale supports the Fleet Spend Control Override Capability.

Table 1004: Attributes

Attribute	Description
Subelement ID	002
Subelement Data Length	001 positions
Data Representation	ans...001; Fixed
Data Field	One byte indicator
Subfields	0

Usage

Usage shows whether the field is mandatory, conditional, optional, or system-provided in applicable messages.

Table 1005: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	O	.	C

Table 1006: Values

Value	Description
0	Does not support spend control override
1	Supports spend control override

Application notes

Subelement 02 values enable acquirers to provide issuers with an indicator to identify when a merchant can support the Fleet Spend Control Override Capability. This data will assist issuers in determining when they can send revised purchase restriction values in DE 106, subelement 03 (Fleet Spend Control Override Items) in the Authorization Request Response/0110 message to override the purchase restriction values obtained from the card at the POS. This allows issuers to perform real-time adjustments to the types of products allowed to be purchased using that particular Fleet card.

Subelement 3 (Fleet Spend Control Override Items)

DE 106, subelement 03 (Fleet Spend Control Override Items) provides merchants with information, on a transaction basis, to override purchase restrictions to allow products to be purchased that may not have otherwise been allowed.

Table 1007: Attributes

Attribute	Value
Subelement ID	003
Subelement Data Length	016
Data Representation	Binary...16, Fixed
Data Field	N/A
Subfields	0

Usage

Usage shows whether the field is mandatory, conditional, optional, or system-provided in applicable messages.

Table 1008: Applicable messages

Message	Org	Sys	Dst
Authorization Response/0110	C	.	C

Application notes

Subelement 03 enables issuers to provide acquirers with revised purchase restriction values in the Authorization Request Response/0110 message to override the purchase restriction values obtained from Connexus Purchase Restriction Tag DF32 found on the EMV card. This subelement will only be populated when DE 106, subelement 02 contains the value '1' indicating that the merchant point-of-sale supports the Fleet Spend Control Override Capability.

The value in DE 106, subelement 3 must conform to the Connexus format standard defined for Purchase Restriction Tag DF32.

Subelement 4 (Fleet Fuel Information)

DE 106, subelement 04 (Fleet Fuel Information) contains item level detail information of the motor fuel product(s) purchased at the POS.

Table 1009: Attributes

Attribute	Description
Subelement ID	004
Subelement Data Length	3 positions
Data Representation	ans...123; VAR
Data Field	Contains up to 3 subfields
Subfields	Subfield 01 (Fuel Item Purchased 1) Subfield 02 (Fuel Item Purchased 2) Subfield 03 (Fuel Item Purchased 3)

Usage

Usage shows whether the field is mandatory, conditional, optional, or system-provided in applicable messages.

Table 1010: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	O	.	C
Authorization Advice/0120: acquirer-generated	O	.	C

Application notes

DE 106, subelement 04 allows the acquirer to provide issuers with detail information on up to three fuel products purchased.

If the fuel product being purchased and related detail is known at the time of authorization, then this field can be optionally provided in the Authorization Request/0100 message.

If the fuel product being purchased is not known at the time of authorization and not included in the Authorization Request/0100 message, the fuel product and related detail if available can be optionally provided in the Authorization Advice/0120 message.

If subelement 04 was present in the original 0100 AFD authorization message, subelement 04 must be the same values in acquirer-generated 0120 completion advice.

Subfield 1 (Fuel Item Purchased 1)

DE 106, subelement 04, subfield 01 (Fuel Item Purchased 1) contains detail Fleet Fuel Information of the first Fuel Item Purchased.

Table 1011: Attributes

Attribute	Description
Data Representation	Tag (3-bytes) – value '001' Len (3-bytes) – value '035'

Attribute	Description
Data Field	ans...35; Fixed Bytes 1-3 (fixed) – Fleet Fuel Product Code 1 (NUM) Bytes 4-15 (fixed) – Fleet Fuel Unit Price 1 (NUM) Byte 16 (fixed) – Fleet Fuel Unit of Measure 1 (ANS) Bytes 17-22 (fixed) – Fleet Fuel Quantity 1 (NUM) Byte 23 (fixed) – Fleet Fuel Quantity Exponent 1 (NUM) Bytes 24-35 (fixed) – Fleet Fuel Sale Amount 1 (NUM)
Justification	Numeric fields are right-justified with leading zeros ANS fields are left-justified, space filled

Table 1012: Values

Position	Description
DE 106, subelement 4, subfield 1 Positions 1-3	Fleet Fuel Product Code 1

The following list of product code values is based on the Conexus Payment System Product Codes version 4.1.

Table 1013: Product codes

Fleet Fuel product code	Description
001	Regular
002	Mid/Plus
003	Premium/Super
004	Mid/Plus 2
005	Premium/Super 2
011	<Deprecated> Ethanol (5.7% blend)
012	<Deprecated> Mid/Plus Ethanol (5.7% blend)
013	<Deprecated> Premium/Super Ethanol (5.7% blend)

Fleet Fuel product code	Description
014	<Deprecated> Ethanol (7.7% blend)
015	<Deprecated> Mid/Plus Ethanol (7.7% blend)
016	Green Gasoline Regular (Plant based not petroleum based)
017	Green Gasoline Mid/Plus (Plant based not petroleum based)
018	Green Gasoline Premium/Super (Plant based not petroleum based)
019	Regular Diesel #2
020	Premium Diesel #2
021	Regular Diesel #1
022	Compressed Natural Gas
023	Liquid Propane Gas
024	Liquid Natural Gas
026	E-85
027	Reformulated 1
028	Reformulated 2
029	Reformulated 3
030	Reformulated 4
031	Reformulated 5
032	Diesel Off-Road (#1 and #2 Non-Taxable)
033	Diesel Off-Road (Non-Taxable)
034	Biodiesel Blend Off-Road (Non-Taxable)
035	Undefined Fuel - Reserved for Conexxus Future
036	Racing Fuel
037	Mid/Plus 2 (10% Blend)
038	Premium/Super 2 (10% Blend)
039	Mid/Plus Ethanol 2 (15% Blend)
040	Premium/Super Ethanol 2 (15% Blend)
041	<Deprecated> Premium/Super Ethanol (7.7% Blend)
042	Regular Ethanol (10% Blend)

Fleet Fuel product code	Description
043	Mid/Plus Ethanol (10% Blend)
044	Premium/Super Ethanol (10% Blend)
045	B2 Diesel Blend 2% Biodiesel
046	B5 Diesel Blend 5% Biodiesel
047	B10 Diesel Blend 10% Biodiesel
048	B11 Diesel Blend 11% Biodiesel
049	B15 Diesel Blend 15% Biodiesel
050	B20 Diesel Blend 20% Biodiesel
051	B100 Diesel Blend 100% Biodiesel
052	B1 Diesel Blend 1% Biodiesel
053	Additized Diesel #2
054	Additized Diesel #3
060	Renewable Diesel ($\geq$ R95) (meets ASTM D975 in US / EN 590 in Europe)
061	Renewable Diesel (Biodiesel 6%-20%) (meets ASTM D7467 in US)
062	DEF (Diesel Exhaust Fluid)
063	Premium Diesel #1
064	Regular Ethanol (15% Blend)
065	Mid/Plus Ethanol (15% Blend)
066	Premium/Super Ethanol (15% Blend)
067	Premium Diesel Blend $<20\%$ Biodiesel
068	Premium Diesel Blend $\geq 20\%$ Biodiesel
069	B75 Diesel Blend 75% Biodiesel
070	B99 Diesel Blend 99% Biodiesel
099	Miscellaneous Fuel
150	Jet Fuel
151	Aviation Fuel Regular
152	Aviation Fuel Premium
153	Aviation Fuel JP8
154	Aviation Fuel 4

Fleet Fuel product code	Description
155	Aviation Fuel 5
156	Biojet (Diesel)
157	Aviation Biofuel (Gasoline)
174	Miscellaneous Aviation Fuel
225	Marine Fuel 1
226	Marine Fuel 2
227	Marine Fuel 3
228	Marine Fuel 4
229	Marine Fuel 5
230	Marine - Other
231	Marine Diesel
249	Miscellaneous Marine Fuel
300	Kerosene – Low Sulfur
301	White Gas
302	Heating Oil
304	Other Fuel (Non-Taxable)
305	Kerosene – Ultra Low Sulfur
306	Kerosene – Low Sulfur (Non-Taxable)
307	Kerosene – Ultra Low Sulfur (Non-Taxable)
308	EVC-1 – Level 1 charge = 110v 15 amp
309	EVC-2 – Level 2 charge = 240v 15-40 amp
310	EVC-3 – Level 3 charge = 480v 3 phase charge
311	Biodiesel Blend 2% Off-Road (Non-Taxable)
312	Biodiesel Blend 5% Off-Road (Non-Taxable)
313	Biodiesel Blend 10% Off-Road (Non-Taxable)
314	Biodiesel Blend 11% Off-Road (Non-Taxable)
315	Biodiesel Blend 15% Off-Road (Non-Taxable)
316	Biodiesel Blend 20% Off-Road (Non-Taxable)
317	Diesel #1 Off-Road (Non-Taxable)
318	Diesel #2 Off-Road (Non-Taxable)
319	Diesel #1 Premium Off-Road (Non-Taxable)

Fleet Fuel product code	Description
320	Diesel #2 Premium Off-Road (Non-Taxable)
321	Additive Dosage
322	Ethanol Blends E16-E84
323	Low Octane UNL
324	Blended Diesel (#1 and #2)
325	Off-Road Regular (Non-Taxable)
326	Off-Road Mid/Plus (Non-Taxable)
327	Off-Road Premium/Super (Non-Taxable)
328	Off-Road Mid/Plus 2 (Non-Taxable)
329	Off-Road Premium/Super 2 (Non-Taxable)
330	Recreational Fuel (90 Octane)
331	Hydrogen H35
332	Hydrogen H70
333	Renewable Diesel (>=R95) Off-Road (Non-Taxable) (meets ASTM D975 in US / EN 590 in Europe)
334	Biodiesel Blend 1% Off-Road (Non-Taxable)
335	Biodiesel Blend 75% Off-Road (Non-Taxable)
336	Biodiesel Blend 99% Off-Road (Non-Taxable)
337	Biodiesel Blend 100% Off-Road (Non-Taxable)
338	Renewable Diesel (Biodiesel 6%-20%) Off-Road (Non-taxable) (meets ASTM D7467 in US)
399	Miscellaneous Other Fuel

Table 1014: Values

Position	Description
DE 106 subelement 4, subfield 1 Position 16	Fleet Fuel Unit of Measure 1

Table 1015: Measurement values

Value	Unit of measure
1	Gallons

Value	Unit of measure
2	Liters
3	Pounds
4	Kilos
5	Imperial gallons
6	Kilowatt hour
7	Minutes
8	Hours
Space	Not applicable

Application notes

Fleet Fuel Product Code contains a three-digit code that identifies the fuel item purchased.

Fleet Fuel Unit Price contains the price per unit of the item being purchased and must be right-justified with three positions to the right of the decimal.

Fleet Fuel Unit of Measure contains the code indicating the unit of measure for the fuel item purchased.

Fleet Fuel Quantity contains quantity of the fuel item purchased and must be numeric and greater than zero.

Fleet Fuel Quantity Exponent contains the exponent indicating the decimal position of the quantity of the fuel item being purchased. Value must be in the range of 0-5 and is assumed to three positions.

Fleet Fuel Sale Amount contains the sale amount of the fuel item being purchased. Value must be numeric and greater than zero and in the same currency as DE 4 (Amount, Transaction). The currency exponent for this Fleet Fuel Sale Amount is based on the exponent associated with the DE 49 (Currency Code, Transaction) field value.

Subfield 2 (Fuel Item Purchased 2)

DE 106, subelement 04, subfield 02 (Fuel Item Purchased 2) contains detail Fleet Fuel Information of the first Fuel Item Purchased.

Table 1016: Attributes

Attribute	Description
Data Representation	Tag (3-bytes) – value '002' Len (3-bytes) – value '035'

Attribute	Description
Data Field	ans...35; Fixed Bytes 1-3 (fixed) – Fleet Fuel Product Code 2 (NUM) Bytes 4-15 (fixed) – Fleet Fuel Unit Price 2 (NUM) Byte 16 (fixed) – Fleet Fuel Unit of Measure 2 (ANS) Bytes 17-22 (fixed) – Fleet Fuel Quantity 2 (NUM) Byte 23 (fixed) – Fleet Fuel Quantity Exponent 2 (NUM) Bytes 24-35 (fixed) – Fleet Fuel Sale Amount 2 (NUM)
Justification	Numeric fields are right-justified with leading zeros ANS fields are left-justified, space filled

Table 1017: Values

Position	Description
DE 106, subelement 4, subfield 2 Position 1-3	Fleet Fuel Product Code 2

The following list of product code values is based on the Conexus Payment System Product Codes version 4.1.

Table 1018: Product codes

Fleet Fuel product code	Description
001	Regular
002	Mid/Plus
003	Premium/Super
004	Mid/Plus 2
005	Premium/Super 2
011	<Deprecated> Ethanol (5.7% blend)
012	<Deprecated> Mid/Plus Ethanol (5.7% blend)
013	<Deprecated> Premium/Super Ethanol (5.7% blend)

Fleet Fuel product code	Description
014	<Deprecated> Ethanol (7.7% blend)
015	<Deprecated> Mid/Plus Ethanol (7.7% blend)
016	Green Gasoline Regular (Plant based not petroleum based)
017	Green Gasoline Mid/Plus (Plant based not petroleum based)
018	Green Gasoline Premium/Super (Plant based not petroleum based)
019	Regular Diesel #2
020	Premium Diesel #2
021	Regular Diesel #1
022	Compressed Natural Gas
023	Liquid Propane Gas
024	Liquid Natural Gas
026	E-85
027	Reformulated 1
028	Reformulated 2
029	Reformulated 3
030	Reformulated 4
031	Reformulated 5
032	Diesel Off-Road (#1 and #2 Non-Taxable)
033	Diesel Off-Road (Non-Taxable)
034	Biodiesel Blend Off-Road (Non-Taxable)
035	Undefined Fuel - Reserved for Conexxus Future
036	Racing Fuel
037	Mid/Plus 2 (10% Blend)
038	Premium/Super 2 (10% Blend)
039	Mid/Plus Ethanol 2 (15% Blend)
040	Premium/Super Ethanol 2 (15% Blend)
041	<Deprecated> Premium/Super Ethanol (7.7% Blend)
042	Regular Ethanol (10% Blend)

Fleet Fuel product code	Description
043	Mid/Plus Ethanol (10% Blend)
044	Premium/Super Ethanol (10% Blend)
045	B2 Diesel Blend 2% Biodiesel
046	B5 Diesel Blend 5% Biodiesel
047	B10 Diesel Blend 10% Biodiesel
048	B11 Diesel Blend 11% Biodiesel
049	B15 Diesel Blend 15% Biodiesel
050	B20 Diesel Blend 20% Biodiesel
051	B100 Diesel Blend 100% Biodiesel
052	B1 Diesel Blend 1% Biodiesel
053	Additized Diesel #2
054	Additized Diesel #3
060	Renewable Diesel ($\geq$ R95) (meets ASTM D975 in US / EN 590 in Europe)
061	Renewable Diesel (Biodiesel 6%-20%) (meets ASTM D7467 in US)
062	DEF (Diesel Exhaust Fluid)
063	Premium Diesel #1
064	Regular Ethanol (15% Blend)
065	Mid/Plus Ethanol (15% Blend)
066	Premium/Super Ethanol (15% Blend)
067	Premium Diesel Blend $<20\%$ Biodiesel
068	Premium Diesel Blend $\geq 20\%$ Biodiesel
069	B75 Diesel Blend 75% Biodiesel
070	B99 Diesel Blend 99% Biodiesel
099	Miscellaneous Fuel
150	Jet Fuel
151	Aviation Fuel Regular
152	Aviation Fuel Premium
153	Aviation Fuel JP8
154	Aviation Fuel 4

Fleet Fuel product code	Description
155	Aviation Fuel 5
156	Biojet (Diesel)
157	Aviation Biofuel (Gasoline)
174	Miscellaneous Aviation Fuel
225	Marine Fuel 1
226	Marine Fuel 2
227	Marine Fuel 3
228	Marine Fuel 4
229	Marine Fuel 5
230	Marine - Other
231	Marine Diesel
249	Miscellaneous Marine Fuel
300	Kerosene – Low Sulfur
301	White Gas
302	Heating Oil
304	Other Fuel (Non-Taxable)
305	Kerosene – Ultra Low Sulfur
306	Kerosene – Low Sulfur (Non-Taxable)
307	Kerosene – Ultra Low Sulfur (Non-Taxable)
308	EVC-1 – Level 1 charge = 110v 15 amp
309	EVC-2 – Level 2 charge = 240v 15-40 amp
310	EVC-3 – Level 3 charge = 480v 3 phase charge
311	Biodiesel Blend 2% Off-Road (Non-Taxable)
312	Biodiesel Blend 5% Off-Road (Non-Taxable)
313	Biodiesel Blend 10% Off-Road (Non-Taxable)
314	Biodiesel Blend 11% Off-Road (Non-Taxable)
315	Biodiesel Blend 15% Off-Road (Non-Taxable)
316	Biodiesel Blend 20% Off-Road (Non-Taxable)
317	Diesel #1 Off-Road (Non-Taxable)
318	Diesel #2 Off-Road (Non-Taxable)
319	Diesel #1 Premium Off-Road (Non-Taxable)

Fleet Fuel product code	Description
320	Diesel #2 Premium Off-Road (Non-Taxable)
321	Additive Dosage
322	Ethanol Blends E16-E84
323	Low Octane UNL
324	Blended Diesel (#1 and #2)
325	Off-Road Regular (Non-Taxable)
326	Off-Road Mid/Plus (Non-Taxable)
327	Off-Road Premium/Super (Non-Taxable)
328	Off-Road Mid/Plus 2 (Non-Taxable)
329	Off-Road Premium/Super 2 (Non-Taxable)
330	Recreational Fuel (90 Octane)
331	Hydrogen H35
332	Hydrogen H70
333	Renewable Diesel (>=R95) Off-Road (Non-Taxable) (meets ASTM D975 in US / EN 590 in Europe)
334	Biodiesel Blend 1% Off-Road (Non-Taxable)
335	Biodiesel Blend 75% Off-Road (Non-Taxable)
336	Biodiesel Blend 99% Off-Road (Non-Taxable)
337	Biodiesel Blend 100% Off-Road (Non-Taxable)
338	Renewable Diesel (Biodiesel 6%-20%) Off-Road (Non-taxable) (meets ASTM D7467 in US)
399	Miscellaneous Other Fuel

Table 1019: Values

Position	Description
DE 106, subelement 4, subfield 2 Position 16	Fleet Fuel Unit of Measure 2

Table 1020: Measurement values

Value	Unit of measure
1	Gallons

Value	Unit of measure
2	Liters
3	Pounds
4	Kilos
5	Imperial gallons
6	Kilowatt hour
7	Minutes
8	Hours
Space	Not applicable

Application notes

Fleet Fuel Product Code contains a three-digit code that identifies the fuel item purchased.

Fleet Fuel Unit Price contains the price per unit of the item being purchased and must be right-justified with three positions to the right of the decimal.

Fleet Fuel Unit of Measure contains the code indicating the unit of measure for the fuel item purchased.

Fleet Fuel Quantity contains quantity of the fuel item purchased and must be numeric and greater than zero.

Fleet Fuel Quantity Exponent contains the exponent indicating the decimal position of the quantity of the fuel item being purchased. Value must be in the range of 0-5 and is assumed to three positions.

Fleet Fuel Sale Amount contains the sale amount of the fuel item being purchased. Value must be numeric and greater than zero and in the same currency as DE 4 (Amount, Transaction). The currency exponent for this Fleet Fuel Sale Amount is based on the exponent associated with the DE 49 (Currency Code, Transaction) field value.

Subfield 3 (Fuel Item Purchased 3)

DE 106, subelement 04, subfield 03 (Fuel Item Purchased 3) contains detail Fleet Fuel Information of the first Fuel Item Purchased.

Table 1021: Attributes

Attribute	Description
Data Representation	Tag (3-bytes) – value '003' Len (3-bytes) – value '035'

Attribute	Description
Data Field	ans...35; Fixed Bytes 1-3 (fixed) – Fleet Fuel Product Code 3 (NUM) Bytes 4-15 (fixed) – Fleet Fuel Unit Price 3 (NUM) Byte 16 (fixed) – Fleet Fuel Unit of Measure 3 (ANS) Bytes 17-22 (fixed) – Fleet Fuel Quantity 3 (NUM) Byte 23 (fixed) – Fleet Fuel Quantity Exponent 3 (NUM) Bytes 24-35 (fixed) – Fleet Fuel Sale Amount 3 (NUM)
Justification	Numeric fields are right-justified with leading zeros ANS fields are left-justified, space filled

Table 1022: Values

Position	Description
DE 106 SE 4 SF 3 Position 1-3	Fleet Fuel Product Code 3

The following list of product code values is based on the Connexus Payment System Product Codes version 4.1.

Table 1023: Product codes

Fleet Fuel product code	Description
001	Regular
002	Mid/Plus
003	Premium/Super
004	Mid/Plus 2
005	Premium/Super 2
011	<Deprecated> Ethanol (5.7% blend)
012	<Deprecated> Mid/Plus Ethanol (5.7% blend)
013	<Deprecated> Premium/Super Ethanol (5.7% blend)

Fleet Fuel product code	Description
014	<Deprecated> Ethanol (7.7% blend)
015	<Deprecated> Mid/Plus Ethanol (7.7% blend)
016	Green Gasoline Regular (Plant based not petroleum based)
017	Green Gasoline Mid/Plus (Plant based not petroleum based)
018	Green Gasoline Premium/Super (Plant based not petroleum based)
019	Regular Diesel #2
020	Premium Diesel #2
021	Regular Diesel #1
022	Compressed Natural Gas
023	Liquid Propane Gas
024	Liquid Natural Gas
026	E-85
027	Reformulated 1
028	Reformulated 2
029	Reformulated 3
030	Reformulated 4
031	Reformulated 5
032	Diesel Off-Road (#1 and #2 Non-Taxable)
033	Diesel Off-Road (Non-Taxable)
034	Biodiesel Blend Off-Road (Non-Taxable)
035	Undefined Fuel - Reserved for Conexxus Future
036	Racing Fuel
037	Mid/Plus 2 (10% Blend)
038	Premium/Super 2 (10% Blend)
039	Mid/Plus Ethanol 2 (15% Blend)
040	Premium/Super Ethanol 2 (15% Blend)
041	<Deprecated> Premium/Super Ethanol (7.7% Blend)
042	Regular Ethanol (10% Blend)

Fleet Fuel product code	Description
043	Mid/Plus Ethanol (10% Blend)
044	Premium/Super Ethanol (10% Blend)
045	B2 Diesel Blend 2% Biodiesel
046	B5 Diesel Blend 5% Biodiesel
047	B10 Diesel Blend 10% Biodiesel
048	B11 Diesel Blend 11% Biodiesel
049	B15 Diesel Blend 15% Biodiesel
050	B20 Diesel Blend 20% Biodiesel
051	B100 Diesel Blend 100% Biodiesel
052	B1 Diesel Blend 1% Biodiesel
053	Additized Diesel #2
054	Additized Diesel #3
060	Renewable Diesel ($\geq$ R95) (meets ASTM D975 in US / EN 590 in Europe)
061	Renewable Diesel (Biodiesel 6%-20%) (meets ASTM D7467 in US)
062	DEF (Diesel Exhaust Fluid)
063	Premium Diesel #1
064	Regular Ethanol (15% Blend)
065	Mid/Plus Ethanol (15% Blend)
066	Premium/Super Ethanol (15% Blend)
067	Premium Diesel Blend $<20\%$ Biodiesel
068	Premium Diesel Blend $\geq 20\%$ Biodiesel
069	B75 Diesel Blend 75% Biodiesel
070	B99 Diesel Blend 99% Biodiesel
099	Miscellaneous Fuel
150	Jet Fuel
151	Aviation Fuel Regular
152	Aviation Fuel Premium
153	Aviation Fuel JP8
154	Aviation Fuel 4

Fleet Fuel product code	Description
155	Aviation Fuel 5
156	Biojet (Diesel)
157	Aviation Biofuel (Gasoline)
174	Miscellaneous Aviation Fuel
225	Marine Fuel 1
226	Marine Fuel 2
227	Marine Fuel 3
228	Marine Fuel 4
229	Marine Fuel 5
230	Marine - Other
231	Marine Diesel
249	Miscellaneous Marine Fuel
300	Kerosene – Low Sulfur
301	White Gas
302	Heating Oil
304	Other Fuel (Non-Taxable)
305	Kerosene – Ultra Low Sulfur
306	Kerosene – Low Sulfur (Non-Taxable)
307	Kerosene – Ultra Low Sulfur (Non-Taxable)
308	EVC-1 – Level 1 charge = 110v 15 amp
309	EVC-2 – Level 2 charge = 240v 15-40 amp
310	EVC-3 – Level 3 charge = 480v 3 phase charge
311	Biodiesel Blend 2% Off-Road (Non-Taxable)
312	Biodiesel Blend 5% Off-Road (Non-Taxable)
313	Biodiesel Blend 10% Off-Road (Non-Taxable)
314	Biodiesel Blend 11% Off-Road (Non-Taxable)
315	Biodiesel Blend 15% Off-Road (Non-Taxable)
316	Biodiesel Blend 20% Off-Road (Non-Taxable)
317	Diesel #1 Off-Road (Non-Taxable)
318	Diesel #2 Off-Road (Non-Taxable)
319	Diesel #1 Premium Off-Road (Non-Taxable)

Fleet Fuel product code	Description
320	Diesel #2 Premium Off-Road (Non-Taxable)
321	Additive Dosage
322	Ethanol Blends E16-E84
323	Low Octane UNL
324	Blended Diesel (#1 and #2)
325	Off-Road Regular (Non-Taxable)
326	Off-Road Mid/Plus (Non-Taxable)
327	Off-Road Premium/Super (Non-Taxable)
328	Off-Road Mid/Plus 2 (Non-Taxable)
329	Off-Road Premium/Super 2 (Non-Taxable)
330	Recreational Fuel (90 Octane)
331	Hydrogen H35
332	Hydrogen H70
333	Renewable Diesel (>=R95) Off-Road (Non-Taxable) (meets ASTM D975 in US / EN 590 in Europe)
334	Biodiesel Blend 1% Off-Road (Non-Taxable)
335	Biodiesel Blend 75% Off-Road (Non-Taxable)
336	Biodiesel Blend 99% Off-Road (Non-Taxable)
337	Biodiesel Blend 100% Off-Road (Non-Taxable)
338	Renewable Diesel (Biodiesel 6%-20%) Off-Road (Non-taxable) (meets ASTM D7467 in US)
399	Miscellaneous Other Fuel

Table 1024: Values

Position	Description
DE 106, subelement 4, subfield 3 Position 16	Fleet Fuel Unit of Measure 3

Table 1025: Measurement values

Value	Unit of measure
1	Gallons

Value	Unit of measure
2	Liters
3	Pounds
4	Kilos
5	Imperial gallons
6	Kilowatt hour
7	Minutes
8	Hours
Space	Not applicable

Application notes

Fleet Fuel Product Code contains a three-digit code that identifies the fuel item purchased.

Fleet Fuel Unit Price contains the price per unit of the item being purchased and must be right-justified with three positions to the right of the decimal.

Fleet Fuel Unit of Measure contains the code indicating the unit of measure for the fuel item purchased.

Fleet Fuel Quantity contains quantity of the fuel item purchased and must be numeric and greater than zero.

Fleet Fuel Quantity Exponent contains the exponent indicating the decimal position of the quantity of the fuel item being purchased. Value must be in the range of 0-5 and is assumed to three positions.

Fleet Fuel Sale Amount contains the sale amount of the fuel item being purchased. Value must be numeric and greater than zero and in the same currency as DE 4 (Amount, Transaction). The currency exponent for this Fleet Fuel Sale Amount is based on the exponent associated with the DE 49 (Currency Code, Transaction) field value.

Subelement 5 (Fleet Non-Fuel Information)

DE 106, subelement 05 (Fleet Non-Fuel Information) contains item level detail information of the non-fuel product(s) purchased at the POS.

Table 1026: Attributes

Attribute	Description
Subelement ID	005
Subelement Data Length	3 positions
Data Representation	ans...552; VAR

Attribute	Description
Data Field	Contains up to 12 subfields
Subfields	Subfield 01 (Non-Fuel Item Purchased 1) Subfield- 02 (Non-Fuel Item Purchased 2) Subfield 03 (Non-Fuel Item Purchased 3) Subfield 04 (Non-Fuel Item Purchased 4) Subfield 05 (Non-Fuel Item Purchased 5) Subfield 06 (Non-Fuel Item Purchased 6) Subfield 07 (Non-Fuel Item Purchased 7) Subfield 08 (Non-Fuel Item Purchased 8) Subfield 09 (Non-Fuel Item Purchased 9) Subfield 10 (Non-Fuel Item Purchased 10) Subfield 11 (Non-Fuel Item Purchased 11) Subfield 12 (Non-Fuel Item Purchased 12)

Usage

Usage shows whether the field is mandatory, conditional, optional, or system-provided in applicable messages.

Table 1027: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	O	.	C
Authorization Advice/0120: acquirer-generated	O	.	C

Application notes

DE 106, subelement 05 allows the acquirer to provide issuers with detail information on up to 12 non-fuel products purchased.

If subelement 05 was present in the original 0100 AFD authorization message, subelement 05 must be the same values in acquirer-generated 0120 completion advice.

Subfield 1 (Non-Fuel Item Purchased 1)

DE 106, subelement 05, subfield 01 (Non-Fuel Item Purchased 1) contains item level detail information of the non-fuel product(s) purchased.

Table 1028: Attributes

Attribute	Description
Data Representation	Tag (3-bytes) – value '001' Len (3-bytes) – value '040'
Data Field	ans...40; Fixed Bytes 1-3 (fixed) – Fleet Product Code 1 (NUM) Bytes 4-15 (fixed) – Fleet Item Quantity 1 (NUM) Byte 16 (fixed) – Fleet Item Quantity Exponent 1 (NUM) Bytes 17-28 (fixed) – Fleet Item Unit of Measure 1 (ANS) Bytes 29-40 (fixed) – Fleet Extended Item Amount 1 (NUM)
Justification	Numeric fields are right-justified with leading zeros ANS fields are left-justified, space filled

Table 1029: Values

Position	Description
DE 106 subelement 5, subfield 1 Positions 1-3	Fleet Product Code 1

The following list of product code values is based on the Conexus Payment System Product Codes version 4.1.

Table 1030: Product codes

Non-fuel product code	Description
100	General Automotive Merchandise
101	Motor Oil
102	Car Wash

Non-fuel product code	Description
103	Oil Change
104	Oil Filter
105	Work Order
106	Anti-Freeze
107	Washer Fluid
108	Brake Fluid
109	Tires
110	Federal Excise Tax (Tires)
111	Tire Rotation
112	Batteries
113	Lube
114	Inspection
115	Labor
116	Towing
117	Road Service
118	Vehicle Accessories
119	Vehicle Parts
120	Preventative Maintenance
121	Air Conditioning Service
122	Engine Service
123	Transmission Service
124	Brake Service
125	Exhaust Service
126	Body Work
127	Vehicle Glass
128	Synthetic Oil
129	Lamps
130	Wipers
131	Hoses
132	Tire-related (Wheel Balance, Valve Stem)
133	Repairs

Non-fuel product code	Description
134	Service Package
135	Vehicle Parking
136	Truck Tank Cleaning
137	Other Lubricants
138	Vehicle Fuel Additives/Treatment (Injected)
139	Vehicle Rental
140	Air Filter
141	Vehicle Prep
142	Fuel System
149	Miscellaneous Vehicle Products / Services
175	Storage
176	Aircraft Ground Handling
177	Aircraft Ground Power Unit
178	Aircraft Labor
179	Aircraft Work Order
180	Aircraft Maintenance
181	Aircraft Service
182	Transportation
183	De-icing
184	Ramp Fees
185	Catering
186	Hangar Fee
187	Landing Fee
188	Call Out Fee
189	Aircraft Rental
190	Instruction Fee
191	Flight Plans / Weather Brief
192	Charter Fee
193	Communication Fee
194	Aircraft Cleaning
195	Cargo Handling

Non-fuel product code	Description
196	Aircraft Accessories
197	Pilot Supplies
198	Aircraft Parking Fees
199	Aircraft Tie Down Fees
200	Aircraft Sanitation Fees
201	Aircraft Fuel Additive
202	AC Parts
203	Oxygen
204	De-fuel
205	Re-service
206	Static Dissipater Additive
207	Corrosion Inhibitor
208	Airport Fees
209	Overtime Fees
210	IT/Bladder
211	Ground Equipment Service Fees
212	Secure Fees
213	Flow Fee
224	Miscellaneous Aviation Products/Services
250	Marine Service
251	Marine Labor
252	Marine Work Order
253	Launch Fee
254	Slip Rental
299	Miscellaneous Marine Products/Services
400	General Merchandise
401	General Ice
410	General Tobacco
411	Cigarettes
412	Tobacco - Other
420	General Packaged Beverage

Non-fuel product code	Description
421	Packaged Beverages (non-alcoholic)
422	Packaged Juice
423	Other Packaged Beverages
430	General Dispensed Beverage
431	Hot Dispensed Beverages
432	Cold Dispensed Beverages
433	Frozen Dispensed Beverages
434	Other Dispensed Beverages
440	General Snacks
441	Salty Snacks
442	Alternative Snacks
443	Sweet Snacks - Packaged
450	General Candy
460	General Dairy
461	Fluid Milk Products
462	Packaged Ice Cream/Novelties
463	Other Dairy
470	General Grocery
471	Groceries - Edible
472	Groceries - Non-Edible
473	Groceries - Perishable
474	Bread - Packaged
475	Frozen Foods
480	General Alcohol
481	Beer - Alcoholic
482	Beer - Non-Alcoholic
483	Wine
484	Liquor
490	General Deli
491	Packaged Sandwiches/Deli Products
492	Prepared Foods

Non-fuel product code	Description
493	Deli Items
500	General Food service
510	General Lottery
511	Lottery - Instant
512	Lottery - Online
513	Lottery - Other
520	General Money Order
521	Money Order - Vendor Payment
522	Money Order - Payroll Check
523	Money Order - Gift Certificate
524	Money Order - Refund Check
525	Money Order - Official Check
526	Money Order - Rebate Check
527	Money Order - Dividend Check
528	Money Order - Utility Check
530	General Store Service
531	Home Delivery
532	Prepaid Cards - Purchase
533	Prepaid Cards - Activation/Recharge
534	Membership/Loyalty
540	General Health & Beauty Care
550	General Publications
560	PIN Activate Prepaid Card
561	PIN Return Prepaid Card
562	Enable Device/Handset Unlock
563	Disable Device/Handset Lock
564	3rd Party Prepaid Card Activate
565	3rd Party Prepaid Card Reload
566	Financial Prepaid Card Activate
567	Financial Prepaid Card Reload
568	Proprietary Prepaid Card Activate

Non-fuel product code	Description
569	Proprietary Prepaid Card Reload
570	General Purpose Activate
571	General Purpose Reload
572	Real Time Recharge
573	Wireless Real Time Recharge
574	Single Payee Bill Pay
575	Multiple Payee Bill Pay
600	Packaged DEF (Diesel Exhaust Fluid)
601	Packaged B99
602	Packaged B100
603	Packaged Additive
604	Packaged Kerosene
605	Packaged Propane
624	Miscellaneous Packaged Fuels
650	Scales
651	Shower
652	Tire Repair
653	Lodging
654	Wash Out
655	Trailer Wash
656	RV Dump Fee
657	EV Charging Fee
658	EV Battery Exchanges
659	Toll Payments
700	Cannabidiol (CBD)
900	Discount 1
901	Discount 2
902	Discount 3
903	Discount 4
904	Discount 5
905	Coupon 1

Non-fuel product code	Description
906	Coupon 2
907	Coupon 3
908	Coupon 4
909	Coupon 5
910	Lottery Pay Out - Instant
911	Lottery Pay Out - Online
912	Lottery Pay Out - Other
913	Split Tender
914	Tax Discount/Forgiven
915	Local Discount 1
916	Local Discount 2
917	Local Discount 3
918	Local Discount 4
919	Local Discount 5
920	POS/Loyalty Reserved Discount 1
921	POS/Loyalty Reserved Discount 2
922	POS/Loyalty Reserved Discount 3
923	POS/Loyalty Reserved Discount 4
924	POS/Loyalty Reserved Discount 5
949	Miscellaneous Negative Administrative
950	Tax 1
951	Tax 2
952	Tax 3
953	Tax 4
954	Tax 5
955	Cash Back
956	Cash Back Fee
957	Fee 1
958	Fee 2
959	Fee 3
960	Fee 4

Non-fuel product code	Description
961	Fee 5
962	Miscellaneous Aviation Tax
963	GST/HST (Canadian)/VAT 1
964	PST/QST (Canadian) VAT 2
965	SWT Rate (Canadian)
966	Tax 6
967	Tax 7
968	Tax 8
969	Jet Federal Excise Tax
970	AvGas Federal Excise Tax
971	Charity
972	Gratuity
999	Miscellaneous Administrative

Application notes

If the non-fuel product being purchased is not known at the time of authorization and not included in the Authorization Request/0100 message, the non-fuel product and related detail if available can be optionally provided in the Authorization Advice/0120 message.

Fleet Product Code contains the non-fuel related product code of the individual item purchased.

Fleet Item Quantity contains quantity of the non-fuel item(s) purchased and must be numeric and greater than zero.

Fleet Item Quantity Exponent contains the exponents indicating the decimal position of the quantity of the non-fuel item(s) being purchased. Value must be in the range of 0-5.

Fleet Item Unit of Measure contains the code indicating the unit of measure for the non-fuel item purchased.

Fleet Extended Item Amount contains the individual item amount that is normally calculated as price multiplied by quantity. Must contain the amount in the same currency as the Amount, Transaction and be numeric and greater than zero, unless there is zero cost to the customer.

Subfield 2 (Non-Fuel Item Purchased 2)

DE 106, subelement 05, subfield 02 (Non-Fuel Item Purchased 2) contains item level detail information of the non-fuel product(s) purchased.

Table 1031: Attributes

Attribute	Description
Data Representation	Tag (3-bytes) – value '002' Len (3-bytes) – value '040'
Data Field	ans...40; Fixed Bytes 1-3 (fixed) – Fleet Product Code 2 (NUM) Bytes 4-15 (fixed) – Fleet Item Quantity 2 (NUM) Byte 16 (fixed) – Fleet Item Quantity Exponent 2 (NUM) Bytes 17-28 (fixed) – Fleet Item Unit of Measure 2 (ANS) Bytes 29-40 (fixed) – Fleet Extended Item Amount 2 (NUM)
Justification	Numeric fields are right-justified with leading zeros ANS fields are left-justified, space filled

Table 1032: Values

Position	Description
DE 106 subelement 5, subfield 2 Positions 1-3	Fleet Product Code 2

The following list of product code values is based on the Conexus Payment System Product Codes version 4.1.

Table 1033: Product codes

Non-fuel product code	Description
100	General Automotive Merchandise
101	Motor Oil
102	Car Wash

Non-fuel product code	Description
103	Oil Change
104	Oil Filter
105	Work Order
106	Anti-Freeze
107	Washer Fluid
108	Brake Fluid
109	Tires
110	Federal Excise Tax (Tires)
111	Tire Rotation
112	Batteries
113	Lube
114	Inspection
115	Labor
116	Towing
117	Road Service
118	Vehicle Accessories
119	Vehicle Parts
120	Preventative Maintenance
121	Air Conditioning Service
122	Engine Service
123	Transmission Service
124	Brake Service
125	Exhaust Service
126	Body Work
127	Vehicle Glass
128	Synthetic Oil
129	Lamps
130	Wipers
131	Hoses
132	Tire-related (Wheel Balance, Valve Stem)
133	Repairs

Non-fuel product code	Description
134	Service Package
135	Vehicle Parking
136	Truck Tank Cleaning
137	Other Lubricants
138	Vehicle Fuel Additives/Treatment (Injected)
139	Vehicle Rental
140	Air Filter
141	Vehicle Prep
142	Fuel System
149	Miscellaneous Vehicle Products / Services
175	Storage
176	Aircraft Ground Handling
177	Aircraft Ground Power Unit
178	Aircraft Labor
179	Aircraft Work Order
180	Aircraft Maintenance
181	Aircraft Service
182	Transportation
183	De-icing
184	Ramp Fees
185	Catering
186	Hangar Fee
187	Landing Fee
188	Call Out Fee
189	Aircraft Rental
190	Instruction Fee
191	Flight Plans / Weather Brief
192	Charter Fee
193	Communication Fee
194	Aircraft Cleaning
195	Cargo Handling

Non-fuel product code	Description
196	Aircraft Accessories
197	Pilot Supplies
198	Aircraft Parking Fees
199	Aircraft Tie Down Fees
200	Aircraft Sanitation Fees
201	Aircraft Fuel Additive
202	AC Parts
203	Oxygen
204	De-fuel
205	Re-service
206	Static Dissipater Additive
207	Corrosion Inhibitor
208	Airport Fees
209	Overtime Fees
210	IT/Bladder
211	Ground Equipment Service Fees
212	Secure Fees
213	Flow Fee
224	Miscellaneous Aviation Products/Services
250	Marine Service
251	Marine Labor
252	Marine Work Order
253	Launch Fee
254	Slip Rental
299	Miscellaneous Marine Products/Services
400	General Merchandise
401	General Ice
410	General Tobacco
411	Cigarettes
412	Tobacco - Other
420	General Packaged Beverage

Non-fuel product code	Description
421	Packaged Beverages (non-alcoholic)
422	Packaged Juice
423	Other Packaged Beverages
430	General Dispensed Beverage
431	Hot Dispensed Beverages
432	Cold Dispensed Beverages
433	Frozen Dispensed Beverages
434	Other Dispensed Beverages
440	General Snacks
441	Salty Snacks
442	Alternative Snacks
443	Sweet Snacks - Packaged
450	General Candy
460	General Dairy
461	Fluid Milk Products
462	Packaged Ice Cream/Novelties
463	Other Dairy
470	General Grocery
471	Groceries - Edible
472	Groceries - Non-Edible
473	Groceries - Perishable
474	Bread - Packaged
475	Frozen Foods
480	General Alcohol
481	Beer - Alcoholic
482	Beer - Non-Alcoholic
483	Wine
484	Liquor
490	General Deli
491	Packaged Sandwiches/Deli Products
492	Prepared Foods

Non-fuel product code	Description
493	Deli Items
500	General Food service
510	General Lottery
511	Lottery - Instant
512	Lottery - Online
513	Lottery - Other
520	General Money Order
521	Money Order - Vendor Payment
522	Money Order - Payroll Check
523	Money Order - Gift Certificate
524	Money Order - Refund Check
525	Money Order - Official Check
526	Money Order - Rebate Check
527	Money Order - Dividend Check
528	Money Order - Utility Check
530	General Store Service
531	Home Delivery
532	Prepaid Cards - Purchase
533	Prepaid Cards - Activation/Recharge
534	Membership/Loyalty
540	General Health & Beauty Care
550	General Publications
560	PIN Activate Prepaid Card
561	PIN Return Prepaid Card
562	Enable Device/Handset Unlock
563	Disable Device/Handset Lock
564	3rd Party Prepaid Card Activate
565	3rd Party Prepaid Card Reload
566	Financial Prepaid Card Activate
567	Financial Prepaid Card Reload
568	Proprietary Prepaid Card Activate

Non-fuel product code	Description
569	Proprietary Prepaid Card Reload
570	General Purpose Activate
571	General Purpose Reload
572	Real Time Recharge
573	Wireless Real Time Recharge
574	Single Payee Bill Pay
575	Multiple Payee Bill Pay
600	Packaged DEF (Diesel Exhaust Fluid)
601	Packaged B99
602	Packaged B100
603	Packaged Additive
604	Packaged Kerosene
605	Packaged Propane
624	Miscellaneous Packaged Fuels
650	Scales
651	Shower
652	Tire Repair
653	Lodging
654	Wash Out
655	Trailer Wash
656	RV Dump Fee
657	EV Charging Fee
658	EV Battery Exchanges
659	Toll Payments
700	Cannabidiol (CBD)
900	Discount 1
901	Discount 2
902	Discount 3
903	Discount 4
904	Discount 5
905	Coupon 1

Non-fuel product code	Description
906	Coupon 2
907	Coupon 3
908	Coupon 4
909	Coupon 5
910	Lottery Pay Out - Instant
911	Lottery Pay Out - Online
912	Lottery Pay Out - Other
913	Split Tender
914	Tax Discount/Forgiven
915	Local Discount 1
916	Local Discount 2
917	Local Discount 3
918	Local Discount 4
919	Local Discount 5
920	POS/Loyalty Reserved Discount 1
921	POS/Loyalty Reserved Discount 2
922	POS/Loyalty Reserved Discount 3
923	POS/Loyalty Reserved Discount 4
924	POS/Loyalty Reserved Discount 5
949	Miscellaneous Negative Administrative
950	Tax 1
951	Tax 2
952	Tax 3
953	Tax 4
954	Tax 5
955	Cash Back
956	Cash Back Fee
957	Fee 1
958	Fee 2
959	Fee 3
960	Fee 4

Non-fuel product code	Description
961	Fee 5
962	Miscellaneous Aviation Tax
963	GST/HST (Canadian)/VAT 1
964	PST/QST (Canadian) VAT 2
965	SWT Rate (Canadian)
966	Tax 6
967	Tax 7
968	Tax 8
969	Jet Federal Excise Tax
970	AvGas Federal Excise Tax
971	Charity
972	Gratuity
999	Miscellaneous Administrative

Application notes

If the non-fuel product being purchased is not known at the time of authorization and not included in the Authorization Request/0100 message, the non-fuel product and related detail if available can be optionally provided in the Authorization Advice/0120 message.

Fleet Product Code contains the non-fuel related product code of the individual item purchased.

Fleet Item Quantity contains quantity of the non-fuel item(s) purchased and must be numeric and greater than zero.

Fleet Item Quantity Exponent contains the exponents indicating the decimal position of the quantity of the non-fuel item(s) being purchased. Value must be in the range of 0-5.

Fleet Item Unit of Measure contains the code indicating the unit of measure for the non-fuel item purchased.

Fleet Extended Item Amount contains the individual item amount that is normally calculated as price multiplied by quantity. Must contain the amount in the same currency as the Amount, Transaction and be numeric and greater than zero, unless there is zero cost to the customer.

Subfield 3 (Non-Fuel Item Purchased 3)

DE 106, subelement 05, subfield 03 (Non-Fuel Item Purchased 3) contains item level detail information of the non-fuel product(s) purchased.

Table 1034: Attributes

Attribute	Description
Data Representation	Tag (3-bytes) – value '003' Len (3-bytes) – value '040'
Data Field	ans...40; Fixed Bytes 1-3 (fixed) – Fleet Product Code 3 (NUM) Bytes 4-15 (fixed) – Fleet Item Quantity 3 (NUM) Byte 16 (fixed) – Fleet Item Quantity Exponent 3 (NUM) Bytes 17-28 (fixed) – Fleet Item Unit of Measure 3 (ANS) Bytes 29-40 (fixed) – Fleet Extended Item Amount 3 (NUM)
Justification	Numeric fields are right-justified with leading zeros ANS fields are left-justified, space filled.

Table 1035: Values

Position	Description
DE 106 subelement 5, subfield 3 Positions 1-3	Fleet Product Code 3

The following list of product code values is based on the Connexus Payment System Product Codes version 4.1.

Table 1036: Product codes

Non-fuel product code	Description
100	General Automotive Merchandise
101	Motor Oil
102	Car Wash

Non-fuel product code	Description
103	Oil Change
104	Oil Filter
105	Work Order
106	Anti-Freeze
107	Washer Fluid
108	Brake Fluid
109	Tires
110	Federal Excise Tax (Tires)
111	Tire Rotation
112	Batteries
113	Lube
114	Inspection
115	Labor
116	Towing
117	Road Service
118	Vehicle Accessories
119	Vehicle Parts
120	Preventative Maintenance
121	Air Conditioning Service
122	Engine Service
123	Transmission Service
124	Brake Service
125	Exhaust Service
126	Body Work
127	Vehicle Glass
128	Synthetic Oil
129	Lamps
130	Wipers
131	Hoses
132	Tire-related (Wheel Balance, Valve Stem)
133	Repairs

Non-fuel product code	Description
134	Service Package
135	Vehicle Parking
136	Truck Tank Cleaning
137	Other Lubricants
138	Vehicle Fuel Additives/Treatment (Injected)
139	Vehicle Rental
140	Air Filter
141	Vehicle Prep
142	Fuel System
149	Miscellaneous Vehicle Products / Services
175	Storage
176	Aircraft Ground Handling
177	Aircraft Ground Power Unit
178	Aircraft Labor
179	Aircraft Work Order
180	Aircraft Maintenance
181	Aircraft Service
182	Transportation
183	De-icing
184	Ramp Fees
185	Catering
186	Hangar Fee
187	Landing Fee
188	Call Out Fee
189	Aircraft Rental
190	Instruction Fee
191	Flight Plans / Weather Brief
192	Charter Fee
193	Communication Fee
194	Aircraft Cleaning
195	Cargo Handling

Non-fuel product code	Description
196	Aircraft Accessories
197	Pilot Supplies
198	Aircraft Parking Fees
199	Aircraft Tie Down Fees
200	Aircraft Sanitation Fees
201	Aircraft Fuel Additive
202	AC Parts
203	Oxygen
204	De-fuel
205	Re-service
206	Static Dissipater Additive
207	Corrosion Inhibitor
208	Airport Fees
209	Overtime Fees
210	IT/Bladder
211	Ground Equipment Service Fees
212	Secure Fees
213	Flow Fee
224	Miscellaneous Aviation Products/Services
250	Marine Service
251	Marine Labor
252	Marine Work Order
253	Launch Fee
254	Slip Rental
299	Miscellaneous Marine Products/Services
400	General Merchandise
401	General Ice
410	General Tobacco
411	Cigarettes
412	Tobacco - Other
420	General Packaged Beverage

Non-fuel product code	Description
421	Packaged Beverages (non-alcoholic)
422	Packaged Juice
423	Other Packaged Beverages
430	General Dispensed Beverage
431	Hot Dispensed Beverages
432	Cold Dispensed Beverages
433	Frozen Dispensed Beverages
434	Other Dispensed Beverages
440	General Snacks
441	Salty Snacks
442	Alternative Snacks
443	Sweet Snacks - Packaged
450	General Candy
460	General Dairy
461	Fluid Milk Products
462	Packaged Ice Cream/Novelties
463	Other Dairy
470	General Grocery
471	Groceries - Edible
472	Groceries - Non-Edible
473	Groceries - Perishable
474	Bread - Packaged
475	Frozen Foods
480	General Alcohol
481	Beer - Alcoholic
482	Beer - Non-Alcoholic
483	Wine
484	Liquor
490	General Deli
491	Packaged Sandwiches/Deli Products
492	Prepared Foods

Non-fuel product code	Description
493	Deli Items
500	General Food Service
510	General Lottery
511	Lottery - Instant
512	Lottery - Online
513	Lottery - Other
520	General Money Order
521	Money Order - Vendor Payment
522	Money Order - Payroll Check
523	Money Order - Gift Certificate
524	Money Order - Refund Check
525	Money Order - Official Check
526	Money Order - Rebate Check
527	Money Order - Dividend Check
528	Money Order - Utility Check
530	General Store Service
531	Home Delivery
532	Prepaid Cards - Purchase
533	Prepaid Cards - Activation/Recharge
534	Membership/Loyalty
540	General Health & Beauty Care
550	General Publications
560	PIN Activate Prepaid Card
561	PIN Return Prepaid Card
562	Enable Device/Handset Unlock
563	Disable Device/Handset Lock
564	3rd Party Prepaid Card Activate
565	3rd Party Prepaid Card Reload
566	Financial Prepaid Card Activate
567	Financial Prepaid Card Reload
568	Proprietary Prepaid Card Activate

Non-fuel product code	Description
569	Proprietary Prepaid Card Reload
570	General Purpose Activate
571	General Purpose Reload
572	Real Time Recharge
573	Wireless Real Time Recharge
574	Single Payee Bill Pay
575	Multiple Payee Bill Pay
600	Packaged DEF (Diesel Exhaust Fluid)
601	Packaged B99
602	Packaged B100
603	Packaged Additive
604	Packaged Kerosene
605	Packaged Propane
624	Miscellaneous Packaged Fuels
650	Scales
651	Shower
652	Tire Repair
653	Lodging
654	Wash Out
655	Trailer Wash
656	RV Dump Fee
657	EV Charging Fee
658	EV Battery Exchanges
659	Toll Payments
700	Cannabidiol (CBD)
900	Discount 1
901	Discount 2
902	Discount 3
903	Discount 4
904	Discount 5
905	Coupon 1

Non-fuel product code	Description
906	Coupon 2
907	Coupon 3
908	Coupon 4
909	Coupon 5
910	Lottery Pay Out - Instant
911	Lottery Pay Out - Online
912	Lottery Pay Out - Other
913	Split Tender
914	Tax Discount/Forgiven
915	Local Discount 1
916	Local Discount 2
917	Local Discount 3
918	Local Discount 4
919	Local Discount 5
920	POS/Loyalty Reserved Discount 1
921	POS/Loyalty Reserved Discount 2
922	POS/Loyalty Reserved Discount 3
923	POS/Loyalty Reserved Discount 4
924	POS/Loyalty Reserved Discount 5
949	Miscellaneous Negative Administrative
950	Tax 1
951	Tax 2
952	Tax 3
953	Tax 4
954	Tax 5
955	Cash Back
956	Cash Back Fee
957	Fee 1
958	Fee 2
959	Fee 3
960	Fee 4

Non-fuel product code	Description
961	Fee 5
962	Miscellaneous Aviation Tax
963	GST/HST (Canadian)/VAT 1
964	PST/QST (Canadian) VAT 2
965	SWT Rate (Canadian)
966	Tax 6
967	Tax 7
968	Tax 8
969	Jet Federal Excise Tax
970	AvGas Federal Excise Tax
971	Charity
972	Gratuity
999	Miscellaneous Administrative

Application notes

If the non-fuel product being purchased is not known at the time of authorization and not included in the Authorization Request/0100 message, the non-fuel product and related detail if available can be optionally provided in the Authorization Advice/0120 message.

Fleet Product Code contains the non-fuel related product code of the individual item purchased.

Fleet Item Quantity contains quantity of the non-fuel item(s) purchased and must be numeric and greater than zero.

Fleet Item Quantity Exponent contains the exponents indicating the decimal position of the quantity of the non-fuel item(s) being purchased. Value must be in the range of 0-5.

Fleet Item Unit of Measure contains the code indicating the unit of measure for the non-fuel item purchased.

Fleet Extended Item Amount contains the individual item amount that is normally calculated as price multiplied by quantity. Must contain the amount in the same currency as the Amount, Transaction and be numeric and greater than zero, unless there is zero cost to the customer.

Subfield 4 (Non-Fuel Item Purchased 4)

DE 106, subelement 05, subfield 04 (Non-Fuel Item Purchased 4) contains item level detail information of the non-fuel product(s) purchased.

Table 1037: Attributes

Attribute	Description
Data Representation	Tag (3-bytes) – value '004' Len (3-bytes) – value '040'
Data Field	ans...40; Fixed Bytes 1-3 (fixed) – Fleet Product Code 4 (NUM) Bytes 4-15 (fixed) – Fleet Item Quantity 4 (NUM) Byte 16 (fixed) – Fleet Item Quantity Exponent 4 (NUM) Bytes 17-28 (fixed) – Fleet Item Unit of Measure 4 (ANS) Bytes 29-40 (fixed) – Fleet Extended Item Amount 4 (NUM)
Justification	Numeric fields are right-justified with leading zeros ANS fields are left-justified, space filled

Table 1038: Values

Position	Description
DE 106 subelement 5, subfields 4 Positions 1-3	Fleet Product Code 4

The following list of product code values is based on the Conexus Payment System Product Codes version 4.1.

Table 1039: Product codes

Non-fuel product code	Description
100	General Automotive Merchandise
101	Motor Oil
102	Car Wash

Non-fuel product code	Description
103	Oil Change
104	Oil Filter
105	Work Order
106	Anti-Freeze
107	Washer Fluid
108	Brake Fluid
109	Tires
110	Federal Excise Tax (Tires)
111	Tire Rotation
112	Batteries
113	Lube
114	Inspection
115	Labor
116	Towing
117	Road Service
118	Vehicle Accessories
119	Vehicle Parts
120	Preventative Maintenance
121	Air Conditioning Service
122	Engine Service
123	Transmission Service
124	Brake Service
125	Exhaust Service
126	Body Work
127	Vehicle Glass
128	Synthetic Oil
129	Lamps
130	Wipers
131	Hoses
132	Tire-related (Wheel Balance, Valve Stem)
133	Repairs

Non-fuel product code	Description
134	Service Package
135	Vehicle Parking
136	Truck Tank Cleaning
137	Other Lubricants
138	Vehicle Fuel Additives/Treatment (Injected)
139	Vehicle Rental
140	Air Filter
141	Vehicle Prep
142	Fuel System
149	Miscellaneous Vehicle Products / Services
175	Storage
176	Aircraft Ground Handling
177	Aircraft Ground Power Unit
178	Aircraft Labor
179	Aircraft Work Order
180	Aircraft Maintenance
181	Aircraft Service
182	Transportation
183	De-icing
184	Ramp Fees
185	Catering
186	Hangar Fee
187	Landing Fee
188	Call Out Fee
189	Aircraft Rental
190	Instruction Fee
191	Flight Plans / Weather Brief
192	Charter Fee
193	Communication Fee
194	Aircraft Cleaning
195	Cargo Handling

Non-fuel product code	Description
196	Aircraft Accessories
197	Pilot Supplies
198	Aircraft Parking Fees
199	Aircraft Tie Down Fees
200	Aircraft Sanitation Fees
201	Aircraft Fuel Additive
202	AC Parts
203	Oxygen
204	De-fuel
205	Re-service
206	Static Dissipater Additive
207	Corrosion Inhibitor
208	Airport Fees
209	Overtime Fees
210	IT/Bladder
211	Ground Equipment Service Fees
212	Secure Fees
213	Flow Fee
224	Miscellaneous Aviation Products/Services
250	Marine Service
251	Marine Labor
252	Marine Work Order
253	Launch Fee
254	Slip Rental
299	Miscellaneous Marine Products/Services
400	General Merchandise
401	General Ice
410	General Tobacco
411	Cigarettes
412	Tobacco - Other
420	General Packaged Beverage

Non-fuel product code	Description
421	Packaged Beverages (non-alcoholic)
422	Packaged Juice
423	Other Packaged Beverages
430	General Dispensed Beverage
431	Hot Dispensed Beverages
432	Cold Dispensed Beverages
433	Frozen Dispensed Beverages
434	Other Dispensed Beverages
440	General Snacks
441	Salty Snacks
442	Alternative Snacks
443	Sweet Snacks - Packaged
450	General Candy
460	General Dairy
461	Fluid Milk Products
462	Packaged Ice Cream/Novelties
463	Other Dairy
470	General Grocery
471	Groceries - Edible
472	Groceries - Non-Edible
473	Groceries - Perishable
474	Bread - Packaged
475	Frozen Foods
480	General Alcohol
481	Beer - Alcoholic
482	Beer - Non-Alcoholic
483	Wine
484	Liquor
490	General Deli
491	Packaged Sandwiches/Deli Products
492	Prepared Foods

Non-fuel product code	Description
493	Deli Items
500	General Food service
510	General Lottery
511	Lottery - Instant
512	Lottery - Online
513	Lottery - Other
520	General Money Order
521	Money Order - Vendor Payment
522	Money Order - Payroll Check
523	Money Order - Gift Certificate
524	Money Order - Refund Check
525	Money Order - Official Check
526	Money Order - Rebate Check
527	Money Order - Dividend Check
528	Money Order - Utility Check
530	General Store Service
531	Home Delivery
532	Prepaid Cards - Purchase
533	Prepaid Cards - Activation/Recharge
534	Membership/Loyalty
540	General Health & Beauty Care
550	General Publications
560	PIN Activate Prepaid Card
561	PIN Return Prepaid Card
562	Enable Device/Handset Unlock
563	Disable Device/Handset Lock
564	3rd Party Prepaid Card Activate
565	3rd Party Prepaid Card Reload
566	Financial Prepaid Card Activate
567	Financial Prepaid Card Reload
568	Proprietary Prepaid Card Activate

Non-fuel product code	Description
569	Proprietary Prepaid Card Reload
570	General Purpose Activate
571	General Purpose Reload
572	Real Time Recharge
573	Wireless Real Time Recharge
574	Single Payee Bill Pay
575	Multiple Payee Bill Pay
600	Packaged DEF (Diesel Exhaust Fluid)
601	Packaged B99
602	Packaged B100
603	Packaged Additive
604	Packaged Kerosene
605	Packaged Propane
624	Miscellaneous Packaged Fuels
650	Scales
651	Shower
652	Tire Repair
653	Lodging
654	Wash Out
655	Trailer Wash
656	RV Dump Fee
657	EV Charging Fee
658	EV Battery Exchanges
659	Toll Payments
700	Cannabidiol (CBD)
900	Discount 1
901	Discount 2
902	Discount 3
903	Discount 4
904	Discount 5
905	Coupon 1

Non-fuel product code	Description
906	Coupon 2
907	Coupon 3
908	Coupon 4
909	Coupon 5
910	Lottery Pay Out - Instant
911	Lottery Pay Out - Online
912	Lottery Pay Out - Other
913	Split Tender
914	Tax Discount/Forgiven
915	Local Discount 1
916	Local Discount 2
917	Local Discount 3
918	Local Discount 4
919	Local Discount 5
920	POS/Loyalty Reserved Discount 1
921	POS/Loyalty Reserved Discount 2
922	POS/Loyalty Reserved Discount 3
923	POS/Loyalty Reserved Discount 4
924	POS/Loyalty Reserved Discount 5
949	Miscellaneous Negative Administrative
950	Tax 1
951	Tax 2
952	Tax 3
953	Tax 4
954	Tax 5
955	Cash Back
956	Cash Back Fee
957	Fee 1
958	Fee 2
959	Fee 3
960	Fee 4

Non-fuel product code	Description
961	Fee 5
962	Miscellaneous Aviation Tax
963	GST/HST (Canadian)/VAT 1
964	PST/QST (Canadian) VAT 2
965	SWT Rate (Canadian)
966	Tax 6
967	Tax 7
968	Tax 8
969	Jet Federal Excise Tax
970	AvGas Federal Excise Tax
971	Charity
972	Gratuity
999	Miscellaneous Administrative

Application notes

If the non-fuel product being purchased is not known at the time of authorization and not included in the Authorization Request/0100 message, the non-fuel product and related detail if available can be optionally provided in the Authorization Advice/0120 message.

Fleet Product Code contains the non-fuel related product code of the individual item purchased.

Fleet Item Quantity contains quantity of the non-fuel item(s) purchased and must be numeric and greater than zero.

Fleet Item Quantity Exponent contains the exponents indicating the decimal position of the quantity of the non-fuel item(s) being purchased. Value must be in the range of 0-5.

Fleet Item Unit of Measure contains the code indicating the unit of measure for the non-fuel item purchased.

Fleet Extended Item Amount contains the individual item amount that is normally calculated as price multiplied by quantity. Must contain the amount in the same currency as the Amount, Transaction and be numeric and greater than zero, unless there is zero cost to the customer.

Subfield 5 (Non-Fuel Item Purchased 5)

DE 106, subelement 05, subfield 05 (Non-Fuel Item Purchased 5) contains item level detail information of the non-fuel product(s) purchased.

Table 1040: Attributes

Attribute	Description
Data Representation	Tag (3-bytes) – value '005' Len (3-bytes) – value '040'
Data Field	ans...40; Fixed Bytes 1-3 (fixed) – Fleet Product Code 5 (NUM) Bytes 4-15 (fixed) – Fleet Item Quantity 5 (NUM) Byte 16 (fixed) – Fleet Item Quantity Exponent 5 (NUM) Bytes 17-28 (fixed) – Fleet Item Unit of Measure 5 (ANS) Bytes 29-40 (fixed) – Fleet Extended Item Amount 5 (NUM)
Justification	Numeric fields are right-justified with leading zeros ANS fields are left-justified, space filled

Table 1041: Values

Position	Description
DE 106 subelement 5, subfield 5 Positions 1-3	Fleet Product Code 5

The following list of product code values is based on the Conexus Payment System Product Codes version 4.1.

Table 1042: Product codes

Non-fuel product code	Description
100	General Automotive Merchandise
101	Motor Oil
102	Car Wash

Non-fuel product code	Description
103	Oil Change
104	Oil Filter
105	Work Order
106	Anti-Freeze
107	Washer Fluid
108	Brake Fluid
109	Tires
110	Federal Excise Tax (Tires)
111	Tire Rotation
112	Batteries
113	Lube
114	Inspection
115	Labor
116	Towing
117	Road Service
118	Vehicle Accessories
119	Vehicle Parts
120	Preventative Maintenance
121	Air Conditioning Service
122	Engine Service
123	Transmission Service
124	Brake Service
125	Exhaust Service
126	Body Work
127	Vehicle Glass
128	Synthetic Oil
129	Lamps
130	Wipers
131	Hoses
132	Tire-related (Wheel Balance, Valve Stem)
133	Repairs

Non-fuel product code	Description
134	Service Package
135	Vehicle Parking
136	Truck Tank Cleaning
137	Other Lubricants
138	Vehicle Fuel Additives/Treatment (Injected)
139	Vehicle Rental
140	Air Filter
141	Vehicle Prep
142	Fuel System
149	Miscellaneous Vehicle Products / Services
175	Storage
176	Aircraft Ground Handling
177	Aircraft Ground Power Unit
178	Aircraft Labor
179	Aircraft Work Order
180	Aircraft Maintenance
181	Aircraft Service
182	Transportation
183	De-icing
184	Ramp Fees
185	Catering
186	Hangar Fee
187	Landing Fee
188	Call Out Fee
189	Aircraft Rental
190	Instruction Fee
191	Flight Plans / Weather Brief
192	Charter Fee
193	Communication Fee
194	Aircraft Cleaning
195	Cargo Handling

Non-fuel product code	Description
196	Aircraft Accessories
197	Pilot Supplies
198	Aircraft Parking Fees
199	Aircraft Tie Down Fees
200	Aircraft Sanitation Fees
201	Aircraft Fuel Additive
202	AC Parts
203	Oxygen
204	De-fuel
205	Re-service
206	Static Dissipater Additive
207	Corrosion Inhibitor
208	Airport Fees
209	Overtime Fees
210	IT/Bladder
211	Ground Equipment Service Fees
212	Secure Fees
213	Flow Fee
224	Miscellaneous Aviation Products/Services
250	Marine Service
251	Marine Labor
252	Marine Work Order
253	Launch Fee
254	Slip Rental
299	Miscellaneous Marine Products/Services
400	General Merchandise
401	General Ice
410	General Tobacco
411	Cigarettes
412	Tobacco - Other
420	General Packaged Beverage

Non-fuel product code	Description
421	Packaged Beverages (non-alcoholic)
422	Packaged Juice
423	Other Packaged Beverages
430	General Dispensed Beverage
431	Hot Dispensed Beverages
432	Cold Dispensed Beverages
433	Frozen Dispensed Beverages
434	Other Dispensed Beverages
440	General Snacks
441	Salty Snacks
442	Alternative Snacks
443	Sweet Snacks - Packaged
450	General Candy
460	General Dairy
461	Fluid Milk Products
462	Packaged Ice Cream/Novelties
463	Other Dairy
470	General Grocery
471	Groceries - Edible
472	Groceries - Non-Edible
473	Groceries - Perishable
474	Bread - Packaged
475	Frozen Foods
480	General Alcohol
481	Beer - Alcoholic
482	Beer - Non-Alcoholic
483	Wine
484	Liquor
490	General Deli
491	Packaged Sandwiches/Deli Products
492	Prepared Foods

Non-fuel product code	Description
493	Deli Items
500	General Food service
510	General Lottery
511	Lottery - Instant
512	Lottery - Online
513	Lottery - Other
520	General Money Order
521	Money Order - Vendor Payment
522	Money Order - Payroll Check
523	Money Order - Gift Certificate
524	Money Order - Refund Check
525	Money Order - Official Check
526	Money Order - Rebate Check
527	Money Order - Dividend Check
528	Money Order - Utility Check
530	General Store Service
531	Home Delivery
532	Prepaid Cards - Purchase
533	Prepaid Cards - Activation/Recharge
534	Membership/Loyalty
540	General Health & Beauty Care
550	General Publications
560	PIN Activate Prepaid Card
561	PIN Return Prepaid Card
562	Enable Device/Handset Unlock
563	Disable Device/Handset Lock
564	3rd Party Prepaid Card Activate
565	3rd Party Prepaid Card Reload
566	Financial Prepaid Card Activate
567	Financial Prepaid Card Reload
568	Proprietary Prepaid Card Activate

Non-fuel product code	Description
569	Proprietary Prepaid Card Reload
570	General Purpose Activate
571	General Purpose Reload
572	Real Time Recharge
573	Wireless Real Time Recharge
574	Single Payee Bill Pay
575	Multiple Payee Bill Pay
600	Packaged DEF (Diesel Exhaust Fluid)
601	Packaged B99
602	Packaged B100
603	Packaged Additive
604	Packaged Kerosene
605	Packaged Propane
624	Miscellaneous Packaged Fuels
650	Scales
651	Shower
652	Tire Repair
653	Lodging
654	Wash Out
655	Trailer Wash
656	RV Dump Fee
657	EV Charging Fee
658	EV Battery Exchanges
659	Toll Payments
700	Cannabidiol (CBD)
900	Discount 1
901	Discount 2
902	Discount 3
903	Discount 4
904	Discount 5
905	Coupon 1

Non-fuel product code	Description
906	Coupon 2
907	Coupon 3
908	Coupon 4
909	Coupon 5
910	Lottery Pay Out - Instant
911	Lottery Pay Out - Online
912	Lottery Pay Out - Other
913	Split Tender
914	Tax Discount/Forgiven
915	Local Discount 1
916	Local Discount 2
917	Local Discount 3
918	Local Discount 4
919	Local Discount 5
920	POS/Loyalty Reserved Discount 1
921	POS/Loyalty Reserved Discount 2
922	POS/Loyalty Reserved Discount 3
923	POS/Loyalty Reserved Discount 4
924	POS/Loyalty Reserved Discount 5
949	Miscellaneous Negative Administrative
950	Tax 1
951	Tax 2
952	Tax 3
953	Tax 4
954	Tax 5
955	Cash Back
956	Cash Back Fee
957	Fee 1
958	Fee 2
959	Fee 3
960	Fee 4

Non-fuel product code	Description
961	Fee 5
962	Miscellaneous Aviation Tax
963	GST/HST (Canadian)/VAT 1
964	PST/QST (Canadian) VAT 2
965	SWT Rate (Canadian)
966	Tax 6
967	Tax 7
968	Tax 8
969	Jet Federal Excise Tax
970	AvGas Federal Excise Tax
971	Charity
972	Gratuity
999	Miscellaneous Administrative

Application notes

If the non-fuel product being purchased is not known at the time of authorization and not included in the Authorization Request/0100 message, the non-fuel product and related detail if available can be optionally provided in the Authorization Advice/0120 message.

Fleet Product Code contains the non-fuel related product code of the individual item purchased.

Fleet Item Quantity contains quantity of the non-fuel item(s) purchased and must be numeric and greater than zero.

Fleet Item Quantity Exponent contains the exponents indicating the decimal position of the quantity of the non-fuel item(s) being purchased. Value must be in the range of 0-5.

Fleet Item Unit of Measure contains the code indicating the unit of measure for the non-fuel item purchased.

Fleet Extended Item Amount contains the individual item amount that is normally calculated as price multiplied by quantity. Must contain the amount in the same currency as the Amount, Transaction and be numeric and greater than zero, unless there is zero cost to the customer.

Subfield 6 (Non-Fuel Item Purchased 6)

DE 106, subelement 05, subfield 06 (Non-Fuel Item Purchased 6) contains item level detail information of the non-fuel product(s) purchased.

Table 1043: Attributes

Attribute	Description
Data Representation	Tag (3-bytes) – value '006' Len (3-bytes) – value '040'
Data Field	ans...40; Fixed Bytes 1-3 (fixed) – Fleet Product Code 6 (NUM) Bytes 4-15 (fixed) – Fleet Item Quantity 6 (NUM) Byte 16 (fixed) – Fleet Item Quantity Exponent 6 (NUM) Bytes 17-28 (fixed) – Fleet Item Unit of Measure 6 (ANS) Bytes 29-40 (fixed) – Fleet Extended Item Amount 6 (NUM)
Justification	Numeric fields are right-justified with leading zeros ANS fields are left-justified, space filled

Table 1044: Values

Position	Description
DE 106, subfield 5, subfield 6 Positions 1-3	Fleet Product Code 6

The following list of product code values is based on the Connexus Payment System Product Codes version 4.1.

Table 1045: Product codes

Non-fuel product code	Description
100	General Automotive Merchandise
101	Motor Oil
102	Car Wash

Non-fuel product code	Description
103	Oil Change
104	Oil Filter
105	Work Order
106	Anti-Freeze
107	Washer Fluid
108	Brake Fluid
109	Tires
110	Federal Excise Tax (Tires)
111	Tire Rotation
112	Batteries
113	Lube
114	Inspection
115	Labor
116	Towing
117	Road Service
118	Vehicle Accessories
119	Vehicle Parts
120	Preventative Maintenance
121	Air Conditioning Service
122	Engine Service
123	Transmission Service
124	Brake Service
125	Exhaust Service
126	Body Work
127	Vehicle Glass
128	Synthetic Oil
129	Lamps
130	Wipers
131	Hoses
132	Tire-related (Wheel Balance, Valve Stem)
133	Repairs

Non-fuel product code	Description
134	Service Package
135	Vehicle Parking
136	Truck Tank Cleaning
	Other Lubricants
138	Vehicle Fuel Additives/Treatment (Injected)
139	Vehicle Rental
140	Air Filter
141	Vehicle Prep
142	Fuel System
149	Miscellaneous Vehicle Products / Services
175	Storage
176	Aircraft Ground Handling
177	Aircraft Ground Power Unit
178	Aircraft Labor
179	Aircraft Work Order
180	Aircraft Maintenance
181	Aircraft Service
182	Transportation
183	De-icing
184	Ramp Fees
185	Catering
186	Hangar Fee
187	Landing Fee
188	Call Out Fee
189	Aircraft Rental
190	Instruction Fee
191	Flight Plans / Weather Brief
192	Charter Fee
193	Communication Fee
194	Aircraft Cleaning
195	Cargo Handling

Non-fuel product code	Description
196	Aircraft Accessories
197	Pilot Supplies
198	Aircraft Parking Fees
199	Aircraft Tie Down Fees
200	Aircraft Sanitation Fees
201	Aircraft Fuel Additive
202	AC Parts
203	Oxygen
204	De-fuel
205	Re-service
206	Static Dissipater Additive
207	Corrosion Inhibitor
208	Airport Fees
209	Overtime Fees
210	IT/Bladder
211	Ground Equipment Service Fees
212	Secure Fees
213	Flow Fee
224	Miscellaneous Aviation Products/Services
250	Marine Service
251	Marine Labor
252	Marine Work Order
253	Launch Fee
254	Slip Rental
299	Miscellaneous Marine Products/Services
400	General Merchandise
401	General Ice
410	General Tobacco
411	Cigarettes
412	Tobacco - Other
420	General Packaged Beverage

Non-fuel product code	Description
421	Packaged Beverages (non-alcoholic)
422	Packaged Juice
423	Other Packaged Beverages
430	General Dispensed Beverage
431	Hot Dispensed Beverages
432	Cold Dispensed Beverages
433	Frozen Dispensed Beverages
434	Other Dispensed Beverages
440	General Snacks
441	Salty Snacks
442	Alternative Snacks
443	Sweet Snacks - Packaged
450	General Candy
460	General Dairy
461	Fluid Milk Products
462	Packaged Ice Cream/Novelties
463	Other Dairy
470	General Grocery
471	Groceries - Edible
472	Groceries - Non-Edible
473	Groceries - Perishable
474	Bread - Packaged
475	Frozen Foods
480	General Alcohol
481	Beer - Alcoholic
482	Beer - Non-Alcoholic
483	Wine
484	Liquor
490	General Deli
491	Packaged Sandwiches/Deli Products
492	Prepared Foods

Non-fuel product code	Description
493	Deli Items
500	General Food service
510	General Lottery
511	Lottery - Instant
512	Lottery - Online
513	Lottery - Other
520	General Money Order
521	Money Order - Vendor Payment
522	Money Order - Payroll Check
523	Money Order - Gift Certificate
524	Money Order - Refund Check
525	Money Order - Official Check
526	Money Order - Rebate Check
527	Money Order - Dividend Check
528	Money Order - Utility Check
530	General Store Service
531	Home Delivery
532	Prepaid Cards - Purchase
533	Prepaid Cards - Activation/Recharge
534	Membership/Loyalty
540	General Health & Beauty Care
550	General Publications
560	PIN Activate Prepaid Card
561	PIN Return Prepaid Card
562	Enable Device/Handset Unlock
563	Disable Device/Handset Lock
564	3rd Party Prepaid Card Activate
565	3rd Party Prepaid Card Reload
566	Financial Prepaid Card Activate
567	Financial Prepaid Card Reload
568	Proprietary Prepaid Card Activate

Non-fuel product code	Description
569	Proprietary Prepaid Card Reload
570	General Purpose Activate
571	General Purpose Reload
572	Real Time Recharge
573	Wireless Real Time Recharge
574	Single Payee Bill Pay
575	Multiple Payee Bill Pay
600	Packaged DEF (Diesel Exhaust Fluid)
601	Packaged B99
602	Packaged B100
603	Packaged Additive
604	Packaged Kerosene
605	Packaged Propane
624	Miscellaneous Packaged Fuels
650	Scales
651	Shower
652	Tire Repair
653	Lodging
654	Wash Out
655	Trailer Wash
656	RV Dump Fee
657	EV Charging Fee
658	EV Battery Exchanges
659	Toll Payments
700	Cannabidiol (CBD)
900	Discount 1
901	Discount 2
902	Discount 3
903	Discount 4
904	Discount 5
905	Coupon 1

Non-fuel product code	Description
906	Coupon 2
907	Coupon 3
908	Coupon 4
909	Coupon 5
910	Lottery Pay Out - Instant
911	Lottery Pay Out - Online
912	Lottery Pay Out - Other
913	Split Tender
914	Tax Discount/Forgiven
915	Local Discount 1
916	Local Discount 2
917	Local Discount 3
918	Local Discount 4
919	Local Discount 5
920	POS/Loyalty Reserved Discount 1
921	POS/Loyalty Reserved Discount 2
922	POS/Loyalty Reserved Discount 3
923	POS/Loyalty Reserved Discount 4
924	POS/Loyalty Reserved Discount 5
949	Miscellaneous Negative Administrative
950	Tax 1
951	Tax 2
952	Tax 3
953	Tax 4
954	Tax 5
955	Cash Back
956	Cash Back Fee
957	Fee 1
958	Fee 2
959	Fee 3
960	Fee 4

Non-fuel product code	Description
961	Fee 5
962	Miscellaneous Aviation Tax
963	GST/HST (Canadian)/VAT 1
964	PST/QST (Canadian) VAT 2
965	SWT Rate (Canadian)
966	Tax 6
967	Tax 7
968	Tax 8
969	Jet Federal Excise Tax
970	AvGas Federal Excise Tax
971	Charity
972	Gratuity
999	Miscellaneous Administrative

Application notes

If the non-fuel product being purchased is not known at the time of authorization and not included in the Authorization Request/0100 message, the non-fuel product and related detail if available can be optionally provided in the Authorization Advice/0120 message.

Fleet Product Code contains the non-fuel related product code of the individual item purchased.

Fleet Item Quantity contains quantity of the non-fuel item(s) purchased and must be numeric and greater than zero.

Fleet Item Quantity Exponent contains the exponents indicating the decimal position of the quantity of the non-fuel item(s) being purchased. Value must be in the range of 0-5.

Fleet Item Unit of Measure contains the code indicating the unit of measure for the non-fuel item purchased.

Fleet Extended Item Amount contains the individual item amount that is normally calculated as price multiplied by quantity. Must contain the amount in the same currency as the Amount, Transaction and be numeric and greater than zero, unless there is zero cost to the customer.

Subfield 7 (Non-Fuel Item Purchased 7)

DE 106, subelement 05, subfield 07 (Non-Fuel Item Purchased 7) contains item level detail information of the non-fuel product(s) purchased.

Table 1046: Attributes

Attribute	Description
Data Representation	Tag (3-bytes) – value '007' Len (3-bytes) – value '040'
Data Field	ans...40; Fixed Bytes 1-3 (fixed) – Fleet Product Code 7 (NUM) Bytes 4-15 (fixed) – Fleet Item Quantity 7 (NUM) Byte 16 (fixed) – Fleet Item Quantity Exponent 7 (NUM) Bytes 17-28 (fixed) – Fleet Item Unit of Measure 7 (ANS) Bytes 29-40 (fixed) – Fleet Extended Item Amount 7 (NUM)
Justification	Numeric fields are right-justified with leading zeros ANS fields are left-justified, space filled

Table 1047: Values

Position	Description
DE 106, subelement 5, subfield 7 Positions 1-3	Fleet Product Code 7

The following list of product code values is based on the Conexus Payment System Product Codes version 4.1.

Table 1048: Product codes

Non-fuel product code	Description
100	General Automotive Merchandise
101	Motor Oil
102	Car Wash

Non-fuel product code	Description
103	Oil Change
104	Oil Filter
105	Work Order
106	Anti-Freeze
107	Washer Fluid
108	Brake Fluid
109	Tires
110	Federal Excise Tax (Tires)
111	Tire Rotation
112	Batteries
113	Lube
114	Inspection
115	Labor
116	Towing
117	Road Service
118	Vehicle Accessories
119	Vehicle Parts
120	Preventative Maintenance
121	Air Conditioning Service
122	Engine Service
123	Transmission Service
124	Brake Service
125	Exhaust Service
126	Body Work
127	Vehicle Glass
128	Synthetic Oil
129	Lamps
130	Wipers
131	Hoses
132	Tire-related (Wheel Balance, Valve Stem)
133	Repairs

Non-fuel product code	Description
134	Service Package
135	Vehicle Parking
136	Truck Tank Cleaning
137	Other Lubricants
138	Vehicle Fuel Additives/Treatment (Injected)
139	Vehicle Rental
140	Air Filter
141	Vehicle Prep
142	Fuel System
149	Miscellaneous Vehicle Products / Services
175	Storage
176	Aircraft Ground Handling
177	Aircraft Ground Power Unit
178	Aircraft Labor
179	Aircraft Work Order
180	Aircraft Maintenance
181	Aircraft Service
182	Transportation
183	De-icing
184	Ramp Fees
185	Catering
186	Hangar Fee
187	Landing Fee
188	Call Out Fee
189	Aircraft Rental
190	Instruction Fee
191	Flight Plans / Weather Brief
192	Charter Fee
193	Communication Fee
194	Aircraft Cleaning
195	Cargo Handling

Non-fuel product code	Description
196	Aircraft Accessories
197	Pilot Supplies
198	Aircraft Parking Fees
199	Aircraft Tie Down Fees
200	Aircraft Sanitation Fees
201	Aircraft Fuel Additive
202	AC Parts
203	Oxygen
204	De-fuel
205	Re-service
206	Static Dissipater Additive
207	Corrosion Inhibitor
208	Airport Fees
209	Overtime Fees
210	IT/Bladder
211	Ground Equipment Service Fees
212	Secure Fees
213	Flow Fee
224	Miscellaneous Aviation Products/Services
250	Marine Service
251	Marine Labor
252	Marine Work Order
253	Launch Fee
254	Slip Rental
299	Miscellaneous Marine Products/Services
400	General Merchandise
401	General Ice
410	General Tobacco
411	Cigarettes
412	Tobacco - Other
420	General Packaged Beverage

Non-fuel product code	Description
421	Packaged Beverages (non-alcoholic)
422	Packaged Juice
423	Other Packaged Beverages
430	General Dispensed Beverage
431	Hot Dispensed Beverages
432	Cold Dispensed Beverages
433	Frozen Dispensed Beverages
434	Other Dispensed Beverages
440	General Snacks
441	Salty Snacks
442	Alternative Snacks
443	Sweet Snacks - Packaged
450	General Candy
460	General Dairy
461	Fluid Milk Products
462	Packaged Ice Cream/Novelties
463	Other Dairy
470	General Grocery
471	Groceries - Edible
472	Groceries - Non-Edible
473	Groceries - Perishable
474	Bread - Packaged
475	Frozen Foods
480	General Alcohol
481	Beer - Alcoholic
482	Beer - Non-Alcoholic
483	Wine
484	Liquor
490	General Deli
491	Packaged Sandwiches/Deli Products
492	Prepared Foods

Non-fuel product code	Description
493	Deli Items
500	General Food service
510	General Lottery
511	Lottery - Instant
512	Lottery - Online
513	Lottery - Other
520	General Money Order
521	Money Order - Vendor Payment
522	Money Order - Payroll Check
523	Money Order - Gift Certificate
524	Money Order - Refund Check
525	Money Order - Official Check
526	Money Order - Rebate Check
527	Money Order - Dividend Check
528	Money Order - Utility Check
530	General Store Service
531	Home Delivery
532	Prepaid Cards - Purchase
533	Prepaid Cards - Activation/Recharge
534	Membership/Loyalty
540	General Health & Beauty Care
550	General Publications
560	PIN Activate Prepaid Card
561	PIN Return Prepaid Card
562	Enable Device/Handset Unlock
563	Disable Device/Handset Lock
564	3rd Party Prepaid Card Activate
565	3rd Party Prepaid Card Reload
566	Financial Prepaid Card Activate
567	Financial Prepaid Card Reload
568	Proprietary Prepaid Card Activate

Non-fuel product code	Description
569	Proprietary Prepaid Card Reload
570	General Purpose Activate
571	General Purpose Reload
572	Real Time Recharge
573	Wireless Real Time Recharge
574	Single Payee Bill Pay
575	Multiple Payee Bill Pay
600	Packaged DEF (Diesel Exhaust Fluid)
601	Packaged B99
602	Packaged B100
603	Packaged Additive
604	Packaged Kerosene
605	Packaged Propane
624	Miscellaneous Packaged Fuels
650	Scales
651	Shower
652	Tire Repair
653	Lodging
654	Wash Out
655	Trailer Wash
656	RV Dump Fee
657	EV Charging Fee
658	EV Battery Exchanges
659	Toll Payments
700	Cannabidiol (CBD)
900	Discount 1
901	Discount 2
902	Discount 3
903	Discount 4
904	Discount 5
905	Coupon 1

Non-fuel product code	Description
906	Coupon 2
907	Coupon 3
908	Coupon 4
909	Coupon 5
910	Lottery Pay Out - Instant
911	Lottery Pay Out - Online
912	Lottery Pay Out - Other
913	Split Tender
914	Tax Discount/Forgiven
915	Local Discount 1
916	Local Discount 2
917	Local Discount 3
918	Local Discount 4
919	Local Discount 5
920	POS/Loyalty Reserved Discount 1
921	POS/Loyalty Reserved Discount 2
922	POS/Loyalty Reserved Discount 3
923	POS/Loyalty Reserved Discount 4
924	POS/Loyalty Reserved Discount 5
949	Miscellaneous Negative Administrative
950	Tax 1
951	Tax 2
952	Tax 3
953	Tax 4
954	Tax 5
955	Cash Back
956	Cash Back Fee
957	Fee 1
958	Fee 2
959	Fee 3
960	Fee 4

Non-fuel product code	Description
961	Fee 5
962	Miscellaneous Aviation Tax
963	GST/HST (Canadian)/VAT 1
964	PST/QST (Canadian) VAT 2
965	SWT Rate (Canadian)
966	Tax 6
967	Tax 7
968	Tax 8
969	Jet Federal Excise Tax
970	AvGas Federal Excise Tax
971	Charity
972	Gratuity
999	Miscellaneous Administrative

Application notes

If the non-fuel product being purchased is not known at the time of authorization and not included in the Authorization Request/0100 message, the non-fuel product and related detail if available can be optionally provided in the Authorization Advice/0120 message.

Fleet Product Code contains the non-fuel related product code of the individual item purchased.

Fleet Item Quantity contains quantity of the non-fuel item(s) purchased and must be numeric and greater than zero.

Fleet Item Quantity Exponent contains the exponents indicating the decimal position of the quantity of the non-fuel item(s) being purchased. Value must be in the range of 0-5.

Fleet Item Unit of Measure contains the code indicating the unit of measure for the non-fuel item purchased.

Fleet Extended Item Amount contains the individual item amount that is normally calculated as price multiplied by quantity. Must contain the amount in the same currency as the Amount, Transaction and be numeric and greater than zero, unless there is zero cost to the customer.

Subfield 8 (Non-Fuel Item Purchased 8)

DE 106, subelement 05, subfield 08 (Non-Fuel Item Purchased 8) contains item level detail information of the non-fuel product(s) purchased.

Table 1049: Attributes

Attribute	Description
Data Representation	Tag (3-bytes) – value '008' Len (3-bytes) – value '040'
Data Field	ans...40; Fixed Bytes 1-3 (fixed) – Fleet Product Code 8 (NUM) Bytes 4-15 (fixed) – Fleet Item Quantity 8 (NUM) Byte 16 (fixed) – Fleet Item Quantity Exponent 8 (NUM) Bytes 17-28 (fixed) – Fleet Item Unit of Measure 8 (ANS) Bytes 29-40 (fixed) – Fleet Extended Item Amount 8 (NUM)
Justification	Numeric fields are right-justified with leading zeros ANS fields are left-justified, space filled

Table 1050: Values

Position	Description
DE 106, subelement 5, subfield 8 Positions 1-3	Fleet Product Code 8

The following list of product code values is based on the Conexus Payment System Product Codes version 4.1.

Table 1051: Product codes

Non-fuel product code	Description
100	General Automotive Merchandise
101	Motor Oil
102	Car Wash

Non-fuel product code	Description
103	Oil Change
104	Oil Filter
105	Work Order
106	Anti-Freeze
107	Washer Fluid
108	Brake Fluid
109	Tires
110	Federal Excise Tax (Tires)
111	Tire Rotation
112	Batteries
113	Lube
114	Inspection
115	Labor
116	Towing
117	Road Service
118	Vehicle Accessories
119	Vehicle Parts
120	Preventative Maintenance
121	Air Conditioning Service
122	Engine Service
123	Transmission Service
124	Brake Service
125	Exhaust Service
126	Body Work
127	Vehicle Glass
128	Synthetic Oil
129	Lamps
130	Wipers
131	Hoses
132	Tire-related (Wheel Balance, Valve Stem)
133	Repairs

Non-fuel product code	Description
134	Service Package
135	Vehicle Parking
136	Truck Tank Cleaning
137	Other Lubricants
138	Vehicle Fuel Additives/Treatment (Injected)
139	Vehicle Rental
140	Air Filter
141	Vehicle Prep
142	Fuel System
149	Miscellaneous Vehicle Products / Services
175	Storage
176	Aircraft Ground Handling
177	Aircraft Ground Power Unit
178	Aircraft Labor
179	Aircraft Work Order
180	Aircraft Maintenance
181	Aircraft Service
182	Transportation
183	De-icing
184	Ramp Fees
185	Catering
186	Hangar Fee
187	Landing Fee
188	Call Out Fee
189	Aircraft Rental
190	Instruction Fee
191	Flight Plans / Weather Brief
192	Charter Fee
193	Communication Fee
194	Aircraft Cleaning
195	Cargo Handling

Non-fuel product code	Description
196	Aircraft Accessories
197	Pilot Supplies
198	Aircraft Parking Fees
199	Aircraft Tie Down Fees
200	Aircraft Sanitation Fees
201	Aircraft Fuel Additive
202	AC Parts
203	Oxygen
204	De-fuel
205	Re-service
206	Static Dissipater Additive
207	Corrosion Inhibitor
208	Airport Fees
209	Overtime Fees
210	IT/Bladder
211	Ground Equipment Service Fees
212	Secure Fees
213	Flow Fee
224	Miscellaneous Aviation Products/Services
250	Marine Service
251	Marine Labor
252	Marine Work Order
253	Launch Fee
254	Slip Rental
299	Miscellaneous Marine Products/Services
400	General Merchandise
401	General Ice
410	General Tobacco
411	Cigarettes
412	Tobacco - Other
420	General Packaged Beverage

Non-fuel product code	Description
421	Packaged Beverages (non-alcoholic)
422	Packaged Juice
423	Other Packaged Beverages
430	General Dispensed Beverage
431	Hot Dispensed Beverages
432	Cold Dispensed Beverages
433	Frozen Dispensed Beverages
434	Other Dispensed Beverages
440	General Snacks
441	Salty Snacks
442	Alternative Snacks
443	Sweet Snacks - Packaged
450	General Candy
460	General Dairy
461	Fluid Milk Products
462	Packaged Ice Cream/Novelties
463	Other Dairy
470	General Grocery
471	Groceries - Edible
472	Groceries - Non-Edible
473	Groceries - Perishable
474	Bread - Packaged
475	Frozen Foods
480	General Alcohol
481	Beer - Alcoholic
482	Beer - Non-Alcoholic
483	Wine
484	Liquor
490	General Deli
491	Packaged Sandwiches/Deli Products
492	Prepared Foods

Non-fuel product code	Description
493	Deli Items
500	General Food service
510	General Lottery
511	Lottery - Instant
512	Lottery - Online
513	Lottery - Other
520	General Money Order
521	Money Order - Vendor Payment
522	Money Order - Payroll Check
523	Money Order - Gift Certificate
524	Money Order - Refund Check
525	Money Order - Official Check
526	Money Order - Rebate Check
527	Money Order - Dividend Check
528	Money Order - Utility Check
530	General Store Service
531	Home Delivery
532	Prepaid Cards - Purchase
533	Prepaid Cards - Activation/Recharge
534	Membership/Loyalty
540	General Health & Beauty Care
550	General Publications
560	PIN Activate Prepaid Card
561	PIN Return Prepaid Card
562	Enable Device/Handset Unlock
563	Disable Device/Handset Lock
564	3rd Party Prepaid Card Activate
565	3rd Party Prepaid Card Reload
566	Financial Prepaid Card Activate
567	Financial Prepaid Card Reload
568	Proprietary Prepaid Card Activate

Non-fuel product code	Description
569	Proprietary Prepaid Card Reload
570	General Purpose Activate
571	General Purpose Reload
572	Real Time Recharge
573	Wireless Real Time Recharge
574	Single Payee Bill Pay
575	Multiple Payee Bill Pay
600	Packaged DEF (Diesel Exhaust Fluid)
601	Packaged B99
602	Packaged B100
603	Packaged Additive
604	Packaged Kerosene
605	Packaged Propane
624	Miscellaneous Packaged Fuels
650	Scales
651	Shower
652	Tire Repair
653	Lodging
654	Wash Out
655	Trailer Wash
656	RV Dump Fee
657	EV Charging Fee
658	EV Battery Exchanges
659	Toll Payments
700	Cannabidiol (CBD)
900	Discount 1
901	Discount 2
902	Discount 3
903	Discount 4
904	Discount 5
905	Coupon 1

Non-fuel product code	Description
906	Coupon 2
907	Coupon 3
908	Coupon 4
909	Coupon 5
910	Lottery Pay Out - Instant
911	Lottery Pay Out - Online
912	Lottery Pay Out - Other
913	Split Tender
914	Tax Discount/Forgiven
915	Local Discount 1
916	Local Discount 2
917	Local Discount 3
918	Local Discount 4
919	Local Discount 5
920	POS/Loyalty Reserved Discount 1
921	POS/Loyalty Reserved Discount 2
922	POS/Loyalty Reserved Discount 3
923	POS/Loyalty Reserved Discount 4
924	POS/Loyalty Reserved Discount 5
949	Miscellaneous Negative Administrative
950	Tax 1
951	Tax 2
952	Tax 3
953	Tax 4
954	Tax 5
955	Cash Back
956	Cash Back Fee
957	Fee 1
958	Fee 2
959	Fee 3
960	Fee 4

Non-fuel product code	Description
961	Fee 5
962	Miscellaneous Aviation Tax
963	GST/HST (Canadian)/VAT 1
964	PST/QST (Canadian) VAT 2
965	SWT Rate (Canadian)
966	Tax 6
967	Tax 7
968	Tax 8
969	Jet Federal Excise Tax
970	AvGas Federal Excise Tax
971	Charity
972	Gratuity
999	Miscellaneous Administrative

Application notes

If the non-fuel product being purchased is not known at the time of authorization and not included in the Authorization Request/0100 message, the non-fuel product and related detail if available can be optionally provided in the Authorization Advice/0120 message.

Fleet Product Code contains the non-fuel related product code of the individual item purchased.

Fleet Item Quantity contains quantity of the non-fuel item(s) purchased and must be numeric and greater than zero.

Fleet Item Quantity Exponent contains the exponents indicating the decimal position of the quantity of the non-fuel item(s) being purchased. Value must be in the range of 0-5.

Fleet Item Unit of Measure contains the code indicating the unit of measure for the non-fuel item purchased.

Fleet Extended Item Amount contains the individual item amount that is normally calculated as price multiplied by quantity. Must contain the amount in the same currency as the Amount, Transaction and be numeric and greater than zero, unless there is zero cost to the customer.

Subfield 9 (Non-Fuel Item Purchased 9)

DE 106, subelement 05, subfield 09 (Non-Fuel Item Purchased 9) contains item level detail information of the non-fuel product(s) purchased.

Table 1052: Attributes

Attribute	Description
Data Representation	Tag (3-bytes) – value '009' Len (3-bytes) – value '040'
Data Field	ans...40; Fixed Bytes 1-3 (fixed) – Fleet Product Code 9 (NUM) Bytes 4-15 (fixed) – Fleet Item Quantity 9 (NUM) Byte 16 (fixed) – Fleet Item Quantity Exponent 9 (NUM) Bytes 17-28 (fixed) – Fleet Item Unit of Measure 9 (ANS) Bytes 29-40 (fixed) – Fleet Extended Item Amount 9 (NUM)
Justification	Numeric fields are right-justified with leading zeros ANS fields are left-justified, space filled

Table 1053: Values

Position	Description
DE 106, subelement 5, subfield 9 Positions 1-3	Fleet Product Code 9

The following list of product code values is based on the Conexus Payment System Product Codes version 4.1.

Table 1054: Product codes

Non-fuel product code	Description
100	General Automotive Merchandise
101	Motor Oil
102	Car Wash

Non-fuel product code	Description
103	Oil Change
104	Oil Filter
105	Work Order
106	Anti-Freeze
107	Washer Fluid
108	Brake Fluid
109	Tires
110	Federal Excise Tax (Tires)
111	Tire Rotation
112	Batteries
113	Lube
114	Inspection
115	Labor
116	Towing
117	Road Service
118	Vehicle Accessories
119	Vehicle Parts
120	Preventative Maintenance
121	Air Conditioning Service
122	Engine Service
123	Transmission Service
124	Brake Service
125	Exhaust Service
126	Body Work
127	Vehicle Glass
128	Synthetic Oil
129	Lamps
130	Wipers
131	Hoses
132	Tire-related (Wheel Balance, Valve Stem)
133	Repairs

Non-fuel product code	Description
134	Service Package
135	Vehicle Parking
136	Truck Tank Cleaning
137	Other Lubricants
138	Vehicle Fuel Additives/Treatment (Injected)
139	Vehicle Rental
140	Air Filter
141	Vehicle Prep
142	Fuel System
149	Miscellaneous Vehicle Products / Services
175	Storage
176	Aircraft Ground Handling
177	Aircraft Ground Power Unit
178	Aircraft Labor
179	Aircraft Work Order
180	Aircraft Maintenance
181	Aircraft Service
182	Transportation
183	De-icing
184	Ramp Fees
185	Catering
186	Hangar Fee
187	Landing Fee
188	Call Out Fee
189	Aircraft Rental
190	Instruction Fee
191	Flight Plans / Weather Brief
192	Charter Fee
193	Communication Fee
194	Aircraft Cleaning
195	Cargo Handling

Non-fuel product code	Description
196	Aircraft Accessories
197	Pilot Supplies
198	Aircraft Parking Fees
199	Aircraft Tie Down Fees
200	Aircraft Sanitation Fees
201	Aircraft Fuel Additive
202	AC Parts
203	Oxygen
204	De-fuel
205	Re-service
206	Static Dissipater Additive
207	Corrosion Inhibitor
208	Airport Fees
209	Overtime Fees
210	IT/Bladder
211	Ground Equipment Service Fees
212	Secure Fees
213	Flow Fee
224	Miscellaneous Aviation Products/Services
250	Marine Service
251	Marine Labor
252	Marine Work Order
253	Launch Fee
254	Slip Rental
299	Miscellaneous Marine Products/Services
400	General Merchandise
401	General Ice
410	General Tobacco
411	Cigarettes
412	Tobacco - Other
420	General Packaged Beverage

Non-fuel product code	Description
421	Packaged Beverages (non-alcoholic)
422	Packaged Juice
423	Other Packaged Beverages
430	General Dispensed Beverage
431	Hot Dispensed Beverages
432	Cold Dispensed Beverages
433	Frozen Dispensed Beverages
434	Other Dispensed Beverages
440	General Snacks
441	Salty Snacks
442	Alternative Snacks
443	Sweet Snacks - Packaged
450	General Candy
460	General Dairy
461	Fluid Milk Products
462	Packaged Ice Cream/Novelties
463	Other Dairy
470	General Grocery
471	Groceries - Edible
472	Groceries - Non-Edible
473	Groceries - Perishable
474	Bread - Packaged
475	Frozen Foods
480	General Alcohol
481	Beer - Alcoholic
482	Beer - Non-Alcoholic
483	Wine
484	Liquor
490	General Deli
491	Packaged Sandwiches/Deli Products
492	Prepared Foods

Non-fuel product code	Description
493	Deli Items
500	General Food service
510	General Lottery
511	Lottery - Instant
512	Lottery - Online
513	Lottery - Other
520	General Money Order
521	Money Order - Vendor Payment
522	Money Order - Payroll Check
523	Money Order - Gift Certificate
524	Money Order - Refund Check
525	Money Order - Official Check
526	Money Order - Rebate Check
527	Money Order - Dividend Check
528	Money Order - Utility Check
530	General Store Service
531	Home Delivery
532	Prepaid Cards - Purchase
533	Prepaid Cards - Activation/Recharge
534	Membership/Loyalty
540	General Health & Beauty Care
550	General Publications
560	PIN Activate Prepaid Card
561	PIN Return Prepaid Card
562	Enable Device/Handset Unlock
563	Disable Device/Handset Lock
564	3rd Party Prepaid Card Activate
565	3rd Party Prepaid Card Reload
566	Financial Prepaid Card Activate
567	Financial Prepaid Card Reload
568	Proprietary Prepaid Card Activate

Non-fuel product code	Description
569	Proprietary Prepaid Card Reload
570	General Purpose Activate
571	General Purpose Reload
572	Real Time Recharge
573	Wireless Real Time Recharge
574	Single Payee Bill Pay
575	Multiple Payee Bill Pay
600	Packaged DEF (Diesel Exhaust Fluid)
601	Packaged B99
602	Packaged B100
603	Packaged Additive
604	Packaged Kerosene
605	Packaged Propane
624	Miscellaneous Packaged Fuels
650	Scales
651	Shower
652	Tire Repair
653	Lodging
654	Wash Out
655	Trailer Wash
656	RV Dump Fee
657	EV Charging Fee
658	EV Battery Exchanges
659	Toll Payments
700	Cannabidiol (CBD)
900	Discount 1
901	Discount 2
902	Discount 3
903	Discount 4
904	Discount 5
905	Coupon 1

Non-fuel product code	Description
906	Coupon 2
907	Coupon 3
908	Coupon 4
909	Coupon 5
910	Lottery Pay Out - Instant
911	Lottery Pay Out - Online
912	Lottery Pay Out - Other
913	Split Tender
914	Tax Discount/Forgiven
915	Local Discount 1
916	Local Discount 2
917	Local Discount 3
918	Local Discount 4
919	Local Discount 5
920	POS/Loyalty Reserved Discount 1
921	POS/Loyalty Reserved Discount 2
922	POS/Loyalty Reserved Discount 3
923	POS/Loyalty Reserved Discount 4
924	POS/Loyalty Reserved Discount 5
949	Miscellaneous Negative Administrative
950	Tax 1
951	Tax 2
952	Tax 3
953	Tax 4
954	Tax 5
955	Cash Back
956	Cash Back Fee
957	Fee 1
958	Fee 2
959	Fee 3
960	Fee 4

Non-fuel product code	Description
961	Fee 5
962	Miscellaneous Aviation Tax
963	GST/HST (Canadian)/VAT 1
964	PST/QST (Canadian) VAT 2
965	SWT Rate (Canadian)
966	Tax 6
967	Tax 7
968	Tax 8
969	Jet Federal Excise Tax
970	AvGas Federal Excise Tax
971	Charity
972	Gratuity
999	Miscellaneous Administrative

Application notes

If the non-fuel product being purchased is not known at the time of authorization and not included in the Authorization Request/0100 message, the non-fuel product and related detail if available can be optionally provided in the Authorization Advice/0120 message.

Fleet Product Code contains the non-fuel related product code of the individual item purchased.

Fleet Item Quantity contains quantity of the non-fuel item(s) purchased and must be numeric and greater than zero.

Fleet Item Quantity Exponent contains the exponents indicating the decimal position of the quantity of the non-fuel item(s) being purchased. Value must be in the range of 0-5.

Fleet Item Unit of Measure contains the code indicating the unit of measure for the non-fuel item purchased.

Fleet Extended Item Amount contains the individual item amount that is normally calculated as price multiplied by quantity. Must contain the amount in the same currency as the Amount, Transaction and be numeric and greater than zero, unless there is zero cost to the customer.

Subfield 10 (Non-Fuel Item Purchased 10)

DE 106, subelement 05, subfield 10 (Non-Fuel Item Purchased 10) contains item level detail information of the non-fuel product(s) purchased.

Table 1055: Attributes

Attribute	Description
Data Representation	Tag (3-bytes) – value '010' Len (3-bytes) – value '040'
Data Field	ans...40; Fixed Bytes 1-3 (fixed) – Fleet Product Code 10 (NUM) Bytes 4-15 (fixed) – Fleet Item Quantity 10 (NUM) Byte 16 (fixed) – Fleet Item Quantity Exponent 10 (NUM) Bytes 17-28 (fixed) – Fleet Item Unit of Measure 10 (ANS) Bytes 29-40 (fixed) – Fleet Extended Item Amount 10 (NUM)
Justification	Numeric fields are right-justified with leading zeros ANS fields are left-justified, space filled

Table 1056: Values

Position	Description
DE 106, subelement 5, subfield 10 Positions 1-3	Fleet Product Code 10

The following list of product code values is based on the Conexus Payment System Product Codes version 4.1.

Table 1057: Product codes

Non-fuel product code	Description
100	General Automotive Merchandise
101	Motor Oil
102	Car Wash

Non-fuel product code	Description
103	Oil Change
104	Oil Filter
105	Work Order
106	Anti-Freeze
107	Washer Fluid
108	Brake Fluid
109	Tires
110	Federal Excise Tax (Tires)
111	Tire Rotation
112	Batteries
113	Lube
114	Inspection
115	Labor
116	Towing
117	Road Service
118	Vehicle Accessories
119	Vehicle Parts
120	Preventative Maintenance
121	Air Conditioning Service
122	Engine Service
123	Transmission Service
124	Brake Service
125	Exhaust Service
126	Body Work
127	Vehicle Glass
128	Synthetic Oil
129	Lamps
130	Wipers
131	Hoses
132	Tire-related (Wheel Balance, Valve Stem)
133	Repairs

Non-fuel product code	Description
134	Service Package
135	Vehicle Parking
136	Truck Tank Cleaning
137	Other Lubricants
138	Vehicle Fuel Additives/Treatment (Injected)
139	Vehicle Rental
140	Air Filter
141	Vehicle Prep
142	Fuel System
149	Miscellaneous Vehicle Products / Services
175	Storage
176	Aircraft Ground Handling
177	Aircraft Ground Power Unit
178	Aircraft Labor
179	Aircraft Work Order
180	Aircraft Maintenance
181	Aircraft Service
182	Transportation
183	De-icing
184	Ramp Fees
185	Catering
186	Hangar Fee
187	Landing Fee
188	Call Out Fee
189	Aircraft Rental
190	Instruction Fee
191	Flight Plans / Weather Brief
192	Charter Fee
193	Communication Fee
194	Aircraft Cleaning
195	Cargo Handling

Non-fuel product code	Description
196	Aircraft Accessories
197	Pilot Supplies
198	Aircraft Parking Fees
199	Aircraft Tie Down Fees
200	Aircraft Sanitation Fees
201	Aircraft Fuel Additive
202	AC Parts
203	Oxygen
204	De-fuel
205	Re-service
206	Static Dissipater Additive
207	Corrosion Inhibitor
208	Airport Fees
209	Overtime Fees
210	IT/Bladder
211	Ground Equipment Service Fees
212	Secure Fees
213	Flow Fee
224	Miscellaneous Aviation Products/Services
250	Marine Service
251	Marine Labor
252	Marine Work Order
253	Launch Fee
254	Slip Rental
299	Miscellaneous Marine Products/Services
400	General Merchandise
401	General Ice
410	General Tobacco
411	Cigarettes
412	Tobacco - Other
420	General Packaged Beverage

Non-fuel product code	Description
421	Packaged Beverages (non-alcoholic)
422	Packaged Juice
423	Other Packaged Beverages
430	General Dispensed Beverage
431	Hot Dispensed Beverages
432	Cold Dispensed Beverages
433	Frozen Dispensed Beverages
434	Other Dispensed Beverages
440	General Snacks
441	Salty Snacks
442	Alternative Snacks
443	Sweet Snacks - Packaged
450	General Candy
460	General Dairy
461	Fluid Milk Products
462	Packaged Ice Cream/Novelties
463	Other Dairy
470	General Grocery
471	Groceries - Edible
472	Groceries - Non-Edible
473	Groceries - Perishable
474	Bread - Packaged
475	Frozen Foods
480	General Alcohol
481	Beer - Alcoholic
482	Beer - Non-Alcoholic
483	Wine
484	Liquor
490	General Deli
491	Packaged Sandwiches/Deli Products
492	Prepared Foods

Non-fuel product code	Description
493	Deli Items
500	General Food service
510	General Lottery
511	Lottery - Instant
512	Lottery - Online
513	Lottery - Other
520	General Money Order
521	Money Order - Vendor Payment
522	Money Order - Payroll Check
523	Money Order - Gift Certificate
524	Money Order - Refund Check
525	Money Order - Official Check
526	Money Order - Rebate Check
527	Money Order - Dividend Check
528	Money Order - Utility Check
530	General Store Service
531	Home Delivery
532	Prepaid Cards - Purchase
533	Prepaid Cards - Activation/Recharge
534	Membership/Loyalty
540	General Health & Beauty Care
550	General Publications
560	PIN Activate Prepaid Card
561	PIN Return Prepaid Card
562	Enable Device/Handset Unlock
563	Disable Device/Handset Lock
564	3rd Party Prepaid Card Activate
565	3rd Party Prepaid Card Reload
566	Financial Prepaid Card Activate
567	Financial Prepaid Card Reload
568	Proprietary Prepaid Card Activate

Non-fuel product code	Description
569	Proprietary Prepaid Card Reload
570	General Purpose Activate
571	General Purpose Reload
572	Real Time Recharge
573	Wireless Real Time Recharge
574	Single Payee Bill Pay
575	Multiple Payee Bill Pay
600	Packaged DEF (Diesel Exhaust Fluid)
601	Packaged B99
602	Packaged B100
603	Packaged Additive
604	Packaged Kerosene
605	Packaged Propane
624	Miscellaneous Packaged Fuels
650	Scales
651	Shower
652	Tire Repair
653	Lodging
654	Wash Out
655	Trailer Wash
656	RV Dump Fee
657	EV Charging Fee
658	EV Battery Exchanges
659	Toll Payments
700	Cannabidiol (CBD)
900	Discount 1
901	Discount 2
902	Discount 3
903	Discount 4
904	Discount 5
905	Coupon 1

Non-fuel product code	Description
906	Coupon 2
907	Coupon 3
908	Coupon 4
909	Coupon 5
910	Lottery Pay Out - Instant
911	Lottery Pay Out - Online
912	Lottery Pay Out - Other
913	Split Tender
914	Tax Discount/Forgiven
915	Local Discount 1
916	Local Discount 2
917	Local Discount 3
918	Local Discount 4
919	Local Discount 5
920	POS/Loyalty Reserved Discount 1
921	POS/Loyalty Reserved Discount 2
922	POS/Loyalty Reserved Discount 3
923	POS/Loyalty Reserved Discount 4
924	POS/Loyalty Reserved Discount 5
949	Miscellaneous Negative Administrative
950	Tax 1
951	Tax 2
952	Tax 3
953	Tax 4
954	Tax 5
955	Cash Back
956	Cash Back Fee
957	Fee 1
958	Fee 2
959	Fee 3
960	Fee 4

Non-fuel product code	Description
961	Fee 5
962	Miscellaneous Aviation Tax
963	GST/HST (Canadian)/VAT 1
964	PST/QST (Canadian) VAT 2
965	SWT Rate (Canadian)
966	Tax 6
967	Tax 7
968	Tax 8
969	Jet Federal Excise Tax
970	AvGas Federal Excise Tax
971	Charity
972	Gratuity
999	Miscellaneous Administrative

Application notes

If the non-fuel product being purchased is not known at the time of authorization and not included in the Authorization Request/0100 message, the non-fuel product and related detail if available can be optionally provided in the Authorization Advice/0120 message.

Fleet Product Code contains the non-fuel related product code of the individual item purchased.

Fleet Item Quantity contains quantity of the non-fuel item(s) purchased and must be numeric and greater than zero.

Fleet Item Quantity Exponent contains the exponents indicating the decimal position of the quantity of the non-fuel item(s) being purchased. Value must be in the range of 0-5.

Fleet Item Unit of Measure contains the code indicating the unit of measure for the non-fuel item purchased.

Fleet Extended Item Amount contains the individual item amount that is normally calculated as price multiplied by quantity. Must contain the amount in the same currency as the Amount, Transaction and be numeric and greater than zero, unless there is zero cost to the customer.

Subfield 11 (Non-Fuel Item Purchased 11)

DE 106, subelement 05, subfield 11 (Non-Fuel Item Purchased 11) contains item level detail information of the non-fuel product(s) purchased.

Table 1058: Attributes

Attribute	Description
Data Representation	Tag (3-bytes) – value '011' Len (3-bytes) – value '040'
Data Field	ans...40; Fixed Bytes 1-3 (fixed) – Fleet Product Code 11 (NUM) Bytes 4-15 (fixed) – Fleet Item Quantity 11 (NUM) Byte 16 (fixed) – Fleet Item Quantity Exponent 11 (NUM) Bytes 17-28 (fixed) – Fleet Item Unit of Measure 11 (ANS) Bytes 29-40 (fixed) – Fleet Extended Item Amount 11 (NUM)
Justification	Numeric fields are right-justified with leading zeros ANS fields are left-justified, space filled

Table 1059: Values

Position	Description
DE 106, subelement 5, subfield 11 Positions 1-3	Fleet Product Code 11

The following list of product code values is based on the Conexus Payment System Product Codes version 4.1.

Table 1060: Product codes

Non-fuel product code	Description
100	General Automotive Merchandise
101	Motor Oil
102	Car Wash

Non-fuel product code	Description
103	Oil Change
104	Oil Filter
105	Work Order
106	Anti-Freeze
107	Washer Fluid
108	Brake Fluid
109	Tires
110	Federal Excise Tax (Tires)
111	Tire Rotation
112	Batteries
113	Lube
114	Inspection
115	Labor
116	Towing
117	Road Service
118	Vehicle Accessories
119	Vehicle Parts
120	Preventative Maintenance
121	Air Conditioning Service
122	Engine Service
123	Transmission Service
124	Brake Service
125	Exhaust Service
126	Body Work
127	Vehicle Glass
128	Synthetic Oil
129	Lamps
130	Wipers
131	Hoses
132	Tire-related (Wheel Balance, Valve Stem)
133	Repairs

Non-fuel product code	Description
134	Service Package
135	Vehicle Parking
136	Truck Tank Cleaning
137	Other Lubricants
138	Vehicle Fuel Additives/Treatment (Injected)
139	Vehicle Rental
140	Air Filter
141	Vehicle Prep
142	Fuel System
149	Miscellaneous Vehicle Products / Services
175	Storage
176	Aircraft Ground Handling
177	Aircraft Ground Power Unit
178	Aircraft Labor
179	Aircraft Work Order
180	Aircraft Maintenance
181	Aircraft Service
182	Transportation
183	De-icing
184	Ramp Fees
185	Catering
186	Hangar Fee
187	Landing Fee
188	Call Out Fee
189	Aircraft Rental
190	Instruction Fee
191	Flight Plans / Weather Brief
192	Charter Fee
193	Communication Fee
194	Aircraft Cleaning
195	Cargo Handling

Non-fuel product code	Description
196	Aircraft Accessories
197	Pilot Supplies
198	Aircraft Parking Fees
199	Aircraft Tie Down Fees
200	Aircraft Sanitation Fees
201	Aircraft Fuel Additive
202	AC Parts
203	Oxygen
204	De-fuel
205	Re-service
206	Static Dissipater Additive
207	Corrosion Inhibitor
208	Airport Fees
209	Overtime Fees
210	IT/Bladder
211	Ground Equipment Service Fees
212	Secure Fees
213	Flow Fee
224	Miscellaneous Aviation Products/Services
250	Marine Service
251	Marine Labor
252	Marine Work Order
253	Launch Fee
254	Slip Rental
299	Miscellaneous Marine Products/Services
400	General Merchandise
401	General Ice
410	General Tobacco
411	Cigarettes
412	Tobacco - Other
420	General Packaged Beverage

Non-fuel product code	Description
421	Packaged Beverages (non-alcoholic)
422	Packaged Juice
423	Other Packaged Beverages
430	General Dispensed Beverage
431	Hot Dispensed Beverages
432	Cold Dispensed Beverages
433	Frozen Dispensed Beverages
434	Other Dispensed Beverages
440	General Snacks
441	Salty Snacks
442	Alternative Snacks
443	Sweet Snacks - Packaged
450	General Candy
460	General Dairy
461	Fluid Milk Products
462	Packaged Ice Cream/Novelties
463	Other Dairy
470	General Grocery
471	Groceries - Edible
472	Groceries - Non-Edible
473	Groceries - Perishable
474	Bread - Packaged
475	Frozen Foods
480	General Alcohol
481	Beer - Alcoholic
482	Beer - Non-Alcoholic
483	Wine
484	Liquor
490	General Deli
491	Packaged Sandwiches/Deli Products
492	Prepared Foods

Non-fuel product code	Description
493	Deli Items
500	General Food service
510	General Lottery
511	Lottery - Instant
512	Lottery - Online
513	Lottery - Other
520	General Money Order
521	Money Order - Vendor Payment
522	Money Order - Payroll Check
523	Money Order - Gift Certificate
524	Money Order - Refund Check
525	Money Order - Official Check
526	Money Order - Rebate Check
527	Money Order - Dividend Check
528	Money Order - Utility Check
530	General Store Service
531	Home Delivery
532	Prepaid Cards - Purchase
533	Prepaid Cards - Activation/Recharge
534	Membership/Loyalty
540	General Health & Beauty Care
550	General Publications
560	PIN Activate Prepaid Card
561	PIN Return Prepaid Card
562	Enable Device/Handset Unlock
563	Disable Device/Handset Lock
564	3rd Party Prepaid Card Activate
565	3rd Party Prepaid Card Reload
566	Financial Prepaid Card Activate
567	Financial Prepaid Card Reload
568	Proprietary Prepaid Card Activate

Non-fuel product code	Description
569	Proprietary Prepaid Card Reload
570	General Purpose Activate
571	General Purpose Reload
572	Real Time Recharge
573	Wireless Real Time Recharge
574	Single Payee Bill Pay
575	Multiple Payee Bill Pay
600	Packaged DEF (Diesel Exhaust Fluid)
601	Packaged B99
602	Packaged B100
603	Packaged Additive
604	Packaged Kerosene
605	Packaged Propane
624	Miscellaneous Packaged Fuels
650	Scales
651	Shower
652	Tire Repair
653	Lodging
654	Wash Out
655	Trailer Wash
656	RV Dump Fee
657	EV Charging Fee
658	EV Battery Exchanges
659	Toll Payments
700	Cannabidiol (CBD)
900	Discount 1
901	Discount 2
902	Discount 3
903	Discount 4
904	Discount 5
905	Coupon 1

Non-fuel product code	Description
906	Coupon 2
907	Coupon 3
908	Coupon 4
909	Coupon 5
910	Lottery Pay Out - Instant
911	Lottery Pay Out - Online
912	Lottery Pay Out - Other
913	Split Tender
914	Tax Discount/Forgiven
915	Local Discount 1
916	Local Discount 2
917	Local Discount 3
918	Local Discount 4
919	Local Discount 5
920	POS/Loyalty Reserved Discount 1
921	POS/Loyalty Reserved Discount 2
922	POS/Loyalty Reserved Discount 3
923	POS/Loyalty Reserved Discount 4
924	POS/Loyalty Reserved Discount 5
949	Miscellaneous Negative Administrative
950	Tax 1
951	Tax 2
952	Tax 3
953	Tax 4
954	Tax 5
955	Cash Back
956	Cash Back Fee
957	Fee 1
958	Fee 2
959	Fee 3
960	Fee 4

Non-fuel product code	Description
961	Fee 5
962	Miscellaneous Aviation Tax
963	GST/HST (Canadian)/VAT 1
964	PST/QST (Canadian) VAT 2
965	SWT Rate (Canadian)
966	Tax 6
967	Tax 7
968	Tax 8
969	Jet Federal Excise Tax
970	AvGas Federal Excise Tax
971	Charity
972	Gratuity
999	Miscellaneous Administrative

Application notes

If the non-fuel product being purchased is not known at the time of authorization and not included in the Authorization Request/0100 message, the non-fuel product and related detail if available can be optionally provided in the Authorization Advice/0120 message.

Fleet Product Code contains the non-fuel related product code of the individual item purchased.

Fleet Item Quantity contains quantity of the non-fuel item(s) purchased and must be numeric and greater than zero.

Fleet Item Quantity Exponent contains the exponents indicating the decimal position of the quantity of the non-fuel item(s) being purchased. Value must be in the range of 0-5.

Fleet Item Unit of Measure contains the code indicating the unit of measure for the non-fuel item purchased.

Fleet Extended Item Amount contains the individual item amount that is normally calculated as price multiplied by quantity. Must contain the amount in the same currency as the Amount, Transaction and be numeric and greater than zero, unless there is zero cost to the customer.

Subfield 12 (Non-Fuel Item Purchased 12)

DE 106, subelement 05, subfield 12 (Non-Fuel Item Purchased 12) contains item level detail information of the non-fuel product(s) purchased.

Table 1061: Attributes

Attribute	Description
Data Representation	Tag (3-bytes) – value '012' Len (3-bytes) – value '040'
Data Field	ans...40; Fixed Bytes 1-3 (fixed) – Fleet Product Code 12 (NUM) Bytes 4-15 (fixed) – Fleet Item Quantity 12 (NUM) Byte 16 (fixed) – Fleet Item Quantity Exponent 12 (NUM) Bytes 17-28 (fixed) – Fleet Item Unit of Measure 12 (ANS) Bytes 29-40 (fixed) – Fleet Extended Item Amount 12 (NUM)
Justification	Numeric fields are right-justified with leading zeros ANS fields are left-justified, space filled

Table 1062: Values

Position	Description
DE 106, subelement 5, subfield 12 Positions 1-3	Fleet Product Code 12

The following list of product code values is based on the Conexus Payment System Product Codes version 4.1.

Table 1063: Product codes

Non-fuel product code	Description
100	General Automotive Merchandise
101	Motor Oil
102	Car Wash

Non-fuel product code	Description
103	Oil Change
104	Oil Filter
105	Work Order
106	Anti-Freeze
107	Washer Fluid
108	Brake Fluid
109	Tires
110	Federal Excise Tax (Tires)
111	Tire Rotation
112	Batteries
113	Lube
114	Inspection
115	Labor
116	Towing
117	Road Service
118	Vehicle Accessories
119	Vehicle Parts
120	Preventative Maintenance
121	Air Conditioning Service
122	Engine Service
123	Transmission Service
124	Brake Service
125	Exhaust Service
126	Body Work
127	Vehicle Glass
128	Synthetic Oil
129	Lamps
130	Wipers
131	Hoses
132	Tire-related (Wheel Balance, Valve Stem)
133	Repairs

Non-fuel product code	Description
134	Service Package
135	Vehicle Parking
136	Truck Tank Cleaning
137	Other Lubricants
138	Vehicle Fuel Additives/Treatment (Injected)
139	Vehicle Rental
140	Air Filter
141	Vehicle Prep
142	Fuel System
149	Miscellaneous Vehicle Products / Services
175	Storage
176	Aircraft Ground Handling
177	Aircraft Ground Power Unit
178	Aircraft Labor
179	Aircraft Work Order
180	Aircraft Maintenance
181	Aircraft Service
182	Transportation
183	De-icing
184	Ramp Fees
185	Catering
186	Hangar Fee
187	Landing Fee
188	Call Out Fee
189	Aircraft Rental
190	Instruction Fee
191	Flight Plans / Weather Brief
192	Charter Fee
193	Communication Fee
194	Aircraft Cleaning
195	Cargo Handling

Non-fuel product code	Description
196	Aircraft Accessories
197	Pilot Supplies
198	Aircraft Parking Fees
199	Aircraft Tie Down Fees
200	Aircraft Sanitation Fees
201	Aircraft Fuel Additive
202	AC Parts
203	Oxygen
204	De-fuel
205	Re-service
206	Static Dissipater Additive
207	Corrosion Inhibitor
208	Airport Fees
209	Overtime Fees
210	IT/Bladder
211	Ground Equipment Service Fees
212	Secure Fees
213	Flow Fee
224	Miscellaneous Aviation Products/Services
250	Marine Service
251	Marine Labor
252	Marine Work Order
253	Launch Fee
254	Slip Rental
299	Miscellaneous Marine Products/Services
400	General Merchandise
401	General Ice
410	General Tobacco
411	Cigarettes
412	Tobacco - Other
420	General Packaged Beverage

Non-fuel product code	Description
421	Packaged Beverages (non-alcoholic)
422	Packaged Juice
423	Other Packaged Beverages
430	General Dispensed Beverage
431	Hot Dispensed Beverages
432	Cold Dispensed Beverages
433	Frozen Dispensed Beverages
434	Other Dispensed Beverages
440	General Snacks
441	Salty Snacks
442	Alternative Snacks
443	Sweet Snacks - Packaged
450	General Candy
460	General Dairy
461	Fluid Milk Products
462	Packaged Ice Cream/Novelties
463	Other Dairy
470	General Grocery
471	Groceries - Edible
472	Groceries - Non-Edible
473	Groceries - Perishable
474	Bread - Packaged
475	Frozen Foods
480	General Alcohol
481	Beer - Alcoholic
482	Beer - Non-Alcoholic
483	Wine
484	Liquor
490	General Deli
491	Packaged Sandwiches/Deli Products
492	Prepared Foods

Non-fuel product code	Description
493	Deli Items
500	General Food service
510	General Lottery
511	Lottery - Instant
512	Lottery - Online
513	Lottery - Other
520	General Money Order
521	Money Order - Vendor Payment
522	Money Order - Payroll Check
523	Money Order - Gift Certificate
524	Money Order - Refund Check
525	Money Order - Official Check
526	Money Order - Rebate Check
527	Money Order - Dividend Check
528	Money Order - Utility Check
530	General Store Service
531	Home Delivery
532	Prepaid Cards - Purchase
533	Prepaid Cards - Activation/Recharge
534	Membership/Loyalty
540	General Health & Beauty Care
550	General Publications
560	PIN Activate Prepaid Card
561	PIN Return Prepaid Card
562	Enable Device/Handset Unlock
563	Disable Device/Handset Lock
564	3rd Party Prepaid Card Activate
565	3rd Party Prepaid Card Reload
566	Financial Prepaid Card Activate
567	Financial Prepaid Card Reload
568	Proprietary Prepaid Card Activate

Non-fuel product code	Description
569	Proprietary Prepaid Card Reload
570	General Purpose Activate
571	General Purpose Reload
572	Real Time Recharge
573	Wireless Real Time Recharge
574	Single Payee Bill Pay
575	Multiple Payee Bill Pay
600	Packaged DEF (Diesel Exhaust Fluid)
601	Packaged B99
602	Packaged B100
603	Packaged Additive
604	Packaged Kerosene
605	Packaged Propane
624	Miscellaneous Packaged Fuels
650	Scales
651	Shower
652	Tire Repair
653	Lodging
654	Wash Out
655	Trailer Wash
656	RV Dump Fee
657	EV Charging Fee
658	EV Battery Exchanges
659	Toll Payments
700	Cannabidiol (CBD)
900	Discount 1
901	Discount 2
902	Discount 3
903	Discount 4
904	Discount 5
905	Coupon 1

Non-fuel product code	Description
906	Coupon 2
907	Coupon 3
908	Coupon 4
909	Coupon 5
910	Lottery Pay Out - Instant
911	Lottery Pay Out - Online
912	Lottery Pay Out - Other
913	Split Tender
914	Tax Discount/Forgiven
915	Local Discount 1
916	Local Discount 2
917	Local Discount 3
918	Local Discount 4
919	Local Discount 5
920	POS/Loyalty Reserved Discount 1
921	POS/Loyalty Reserved Discount 2
922	POS/Loyalty Reserved Discount 3
923	POS/Loyalty Reserved Discount 4
924	POS/Loyalty Reserved Discount 5
949	Miscellaneous Negative Administrative
950	Tax 1
951	Tax 2
952	Tax 3
953	Tax 4
954	Tax 5
955	Cash Back
956	Cash Back Fee
957	Fee 1
958	Fee 2
959	Fee 3
960	Fee 4

Non-fuel product code	Description
961	Fee 5
962	Miscellaneous Aviation Tax
963	GST/HST (Canadian)/VAT 1
964	PST/QST (Canadian) VAT 2
965	SWT Rate (Canadian)
966	Tax 6
967	Tax 7
968	Tax 8
969	Jet Federal Excise Tax
970	AvGas Federal Excise Tax
971	Charity
972	Gratuity
999	Miscellaneous Administrative

Application notes

If the non-fuel product being purchased is not known at the time of authorization and not included in the Authorization Request/0100 message, the non-fuel product and related detail if available can be optionally provided in the Authorization Advice/0120 message.

Fleet Product Code contains the non-fuel related product code of the individual item purchased.

Fleet Item Quantity contains quantity of the non-fuel item(s) purchased and must be numeric and greater than zero.

Fleet Item Quantity Exponent contains the exponents indicating the decimal position of the quantity of the non-fuel item(s) being purchased. Value must be in the range of 0-5.

Fleet Item Unit of Measure contains the code indicating the unit of measure for the non-fuel item purchased.

Fleet Extended Item Amount contains the individual item amount that is normally calculated as price multiplied by quantity. Must contain the amount in the same currency as the Amount, Transaction and be numeric and greater than zero, unless there is zero cost to the customer.

DE 108 (Additional Transaction Reference Data)

DE 108 (Additional Transaction Reference Data) provides the capability for the acquirers to send sender, receiver, and transaction level data to the issuer in Mastercard Send Transactions (Funding Transactions, Mastercard Send Payment Transactions, Gaming and Gambling

Payments Transactions), Mastercard Merchant Presented QR Transactions, and Visa Account Funding Transactions.

Table 1064: Attributes

Attribute	Description
Data Representation	ans...999; LLLVAR
Data Length	3 positions
Data Field	Contents of subelements
Subelements	11
Justification	See "Subelements"

Usage: Mastercard Send Transactions

Whether the usage is mandatory, conditional, optional, system-provided, or not required is as noted for each applicable message.

NOTE: Refer to the [Mastercard Send and Funding Transactions Program Standards](#) for the full list of mandatory subelements in DE 108.

Table 1065: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Request Response/0110	C	.	C
Authorization Advice/0120: system-generated	.	C	C
Reversal Request/0400	O	.	C
Reversal Request Response/0410	C	.	C

Application notes For Mastercard Send Transactions

For Mastercard Send Payment Transactions and Gaming and Gambling Payments Transactions, acquirers that process transactions must send and receive DE 108 (Additional Transaction Reference Data). Issuers must receive and return DE 108 (Additional Transaction Reference Data) for all Mastercard Send Payment Transactions and Gaming and Gambling Payments Transactions.

For Funding Transactions, acquirers that process transactions must send and be prepared to receive if sending DE 108 (Additional Transaction Reference Data) for all Funding Transactions.

Issuers must receive and return DE 108 (Additional Transaction Reference Data) when submitted by the acquirer on Funding Transactions.

Mastercard Send Transactions, with the exception Mastercard Merchant Presented QR transactions, will not contain the following DE 108 subelements and subfields:

- Subelement 05 (Digital Account Information), subfield 01 (Digital Account Reference Number)
- Subelement 05, subfield 02 (Mastercard Merchant Presented QR Receiving Account Number)
- Subelement 06 (QR Dynamic Code Data)

Application notes For Mastercard Merchant Presented QR Transactions

For Mastercard Merchant Presented QR Funding Transactions, acquirers that process transactions must send DE 108 (Additional Transaction Reference Data). Issuers must receive DE 108 (Additional Transaction Reference Data) for all Mastercard Merchant Presented QR Funding Transactions.

Mastercard Merchant Presented QR Payment Transactions may contain the following DE 108 subelements and subfields.

- Subelement 02 (Sender Data), subfield 11 (Sender Account Number)
- Subelement 03 (Transaction Reference Data), subfield 01 (Unique Transaction Reference Number)
- Subelement 03, subfield 02 (Additional Message) (Payment Transactions only)
- Subelement 03, subfield 03 (Funding Source)
- Subelement 05 (Digital Account Information), subfield 01 (Digital Account Reference Number)
- Subelement 05, subfield 02 (Mastercard Merchant Presented QR Receiving Account Number)
- Subelement 06 (QR Dynamic Code Data)

Mastercard Merchant Presented QR Payment Transactions will not return DE 108 to the acquirer in any response message.

Customers should be aware that QR data for Mastercard Merchant Presented QR Payment Transactions, may be received in either DE 108 subelement 03, subfield 02 or DE 108, subelement 06. Subelement 06 is considered the primary location and should be considered first as the location for QR Data.

Application notes For Mastercard Send and Funding Transactions

Mastercard Send Payment and Funding Transactions will not contain the following DE 108 subelements and subfields:

- Subelement 05 (Digital Account Information), subfield 01 (Digital Account Reference Number)
- Subelement 05, subfield 02 (Mastercard Merchant Presented QR Receiving Account Number)
- Subelement 06 (QR Dynamic Code Data)

Application notes for Account Status Inquiry Transactions related to Mastercard Send and Funding Transactions

For Account Status Inquiry Transactions, acquirers may send and be prepared to receive DE 108 (Additional Transaction Reference Data). Issuers must be prepared to receive and return DE 108 (Additional Transaction Reference Data) with Account Status Inquiry Transactions.

Acquirers that are requesting verification of receiver/recipient or sender data will send DE 108 (Additional Transaction Reference Data) with subelements:

- Subelement 01 (Receiver/Recipient Data) or Subelement 02 (Sender Data)
- Subelement 07 (Data Verification Request)

Issuers must receive DE 108 (Additional Transaction Reference Data) with subelements:

- Subelement 01 (Receiver/Recipient Data) or Subelement 02 (Sender Data)
- Subelement 07 (Data Verification Request)

Issuers must return DE 108 (Additional Transaction Reference Data) with subelement 08 (Data Verification Response).

Application notes for Non-Financial Authorization Transactions

Acquirers and originators requesting validation of account owner information (Name Validation Service) will send DE 108 (Additional Transaction Reference Data) with subelements:

- Subelement 01 (Receiver/Recipient Data) or subelement 02 (Sender Data)
- Subelement 07 (Data Validation Request)

Issuers must respond with DE 108 (Additional Transaction Reference Data) with subelements:

- Subelement 08 (Data Validation Response)

Or, when an issuer participates in the on-behalf Name Match Service the response will include subelements:

- Subelement 09 (Account Information 1)
- Subelement 10 (Account Information 2)
- Subelement 11 (Account Information 3)

Issuers must be enabled for the on-behalf Name Match Service to be able to return DE 108 (Additional Transaction Reference Data) with Subelement 09 (Account Information 1), subelement 10 (Account Information 2), and subelement 11 (Account Information 3).

DE 108 Dual Message Authorization System edits

The Dual Message Authorization System will perform the following system edit to verify proper formatting of DE 108.

Table 1066: Associated edits

WHEN...	THEN the Dual Message Authorization System...
an Authorization Request/0100, Authorization Advice/0120: acquirer-generated, or Reversal Request/0400 message contains DE 48 (Additional Data: Private Use), DE 108 (Additional Transaction Reference Data), or DE 112 (Additional Data [National Use]) with subelements that have incorrect length and/or incorrect format (Data Representations), or have multiple instances of the same subelement (when not permitted) within the same data element.	rejects the message and forwards the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 message where DE 44 is 6 positions for subelement format errors: - DE 39 (Response Code) = 30 (Message format error) - DE 44 (Additional Response Data) = 0480nn, or 1080nn, or 1120nn (where nn is the subelement number) (DE 44 is three positions for Dual Message System (Authorization) if no subelements are present, for example, for DE 22 format error: DE 44 = 022)

Examples

- When an edit error occurs on DE 48, the DE 44 data will be populated as these examples below:
 - Error on DE 48 subelement 42, subfield 1: 048042 (no subfield information is provided)
 - Error on DE 48 subelement 61: 048061
- DE 48 TCC format error will be responded with DE 44 = 048000.

For non-DE 48/DE 108/DE 112 format errors, DE 44 will only have DE info and no subelement information. For example, the format error in DE 22 will have DE 44 as 022 for non-DE48/108/112 data elements.

Subelement 01 (Receiver/Recipient Data)

DE 108 (Additional Transaction Reference Data), subelement 01 (Receiver/Recipient Data) contains the receiver or recipient name, address, phone number, date of birth, and account details.

Table 1067: Attributes

Attribute	Description
Subelement ID	01

Attribute	Description
Subelement Data Length	3 positions
Data Representation	ans...669; LLLVAR
Data Field	Contents of subfields

Attribute	Description
Subfields	Subfield 01 (Receiver/Recipient First Name)
	Subfield 02 (Receiver/Recipient Middle Name)
	Subfield 03 (Receiver/Recipient Last Name)
	Subfield 04 (Receiver/Recipient Street Address)
	Subfield 04 (Receiver/Recipient Street Address Line 1 [Visa Only])
	Subfield 05 (Receiver/Recipient City)
	Subfield 06 (Receiver/Recipient State/Province Code)
	Subfield 07 (Receiver/Recipient Country)
	Subfield 08 (Receiver/Recipient Postal Code)
	Subfield 08 (Receiver/Recipient Postal Code [Visa Only])
	Subfield 09 (Receiver/Recipient Phone Number)
	Subfield 10 (Receiver/Recipient Date of Birth)
	Subfield 11 (Receiver/Recipient Account Number)
	Subfield 12 (Receiver/Recipient Identification Type)
	Subfield 13 (Receiver/Recipient Identification Number)
	Subfield 13 (Receiver/Recipient Identification Number [Visa Only])
	Subfield 14 (Receiver/Recipient Identification Country Code)
	Subfield 15 (Receiver/Recipient Identification Expiration Date)
	Subfield 16 (Receiver/Recipient Nationality)
	Subfield 17 (Receiver/Recipient Country of Birth)
	Subfield 18 (Receiver/Recipient Account Number Type)
	Subfield 19 (Receiver/Recipient Street Address Line 2 [Visa Only])
	Subfield 20 (Receiver/Recipient Street Name [Visa Only])
	Subfield 21 (Receiver/Recipient Building Number [Visa Only])

Attribute	Description
	Subfield 22 (Receiver/Recipient Country Subdivision Code, Minor [Visa Only])
Justification	See subfields

Subfield 01 (Receiver/Recipient First Name)

DE 108, subelement 01, subfield 01 (Receiver/Recipient First Name) contains the Receiver/Recipient first name.

Table 1068: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	ans...35; LLVAR
Data Field	Contents of subfield 01
Justification	N/A

Values

If present, cannot contain all spaces or all numeric values.

Application notes

Subfield 01 must be present for cross-border Mastercard Send Payment Transactions and Gaming and Gambling Payments Transactions, and must be properly formatted.

Subfield 01 is optional for domestic Mastercard Send Payment Transactions, Gaming and Gambling Payments Transactions, and Funding transactions. Mastercard network doesn't enforce system edits on this field and it is optional. However per program rules noted in the [Mastercard Send and Funding Transactions Program Standards](#) guide, it's a required field for domestic and cross-border Funding Transactions and must be provided. Not providing required data in this field may result in compliance assessments.

The first name (Consumer/Business) of the recipient/receiver is included in this subelement.

The table that follows describes the correct submission of business names.

Application notes: Mastercard Merchant Presented QR transactions

Subfield 01 is optional for Mastercard Merchant Presented QR Payment Transactions. It is not edited and will be forward to destination.

Subfield 01 is not applicable for Mastercard Merchant Presented QR Refund Payment or Funding Transactions.

Table 1069: Name example

Business name	First name	Last name
XYZ	XYZ	XYZ
XYZ International	XYZ International	XYZ International
XYZ DBA MA	XYZ DBA MA	XYZ DBA MA

Subfield 02 (Receiver/Recipient Middle Name)

DE 108, subelement 01, subfield 02 (Receiver/Recipient Middle Name) contains the middle name of the Recipient/Receiver.

Table 1070: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	ans-1
Data Field	Contents of subfield 02
Justification	N/A

Values

Valid value will consist of the middle name initial of the Receiver.

Application notes

Subfield 02 is optional for Mastercard Send Transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions) according to the published standards. Refer to the [Mastercard Send and Funding Transactions Program Standards](#) for the full listing of required fields and subelements.

Subfield 03 (Receiver/Recipient Last Name)

DE 108, subelement 01, subfield 03 (Receiver/Recipient Last Name) contains the last name of the Recipient/Receiver.

Table 1071: Attributes

Attribute	Description
Subfield ID	03
Subfield Data Length	2 positions
Data Representation	ans...35; LLVAR
Data Field	Contents of subfield 03
Justification	N/A

Values

If present cannot contain all spaces or all numeric values.

Application notes

NOTE: Refer to [Mastercard Send and Funding Transactions Program Standards](#) for the full listing of required fields and subelements for both funding and payment transactions.

Subfield 03 must be present for cross-border Mastercard Send Payment Transactions and Gaming and Gambling Payments Transactions, and must be properly formatted.

Subfield 01 is optional for domestic Mastercard Send Payment Transactions, Gaming and Gambling Payments Transactions, and Funding transactions. Mastercard network doesn't enforce system edits on this field and it is optional. However per program rules noted in the [Mastercard Send and Funding Transactions Program Standards](#) guide its a required field for domestic and cross-border Funding Transactions and must be provided. Not providing required data in this field may result in compliance assessments.

The last name (Consumer/Business) of the recipient/receiver is included in this subelement.

The table that follows describes the correct submission of business names.

Application notes: Mastercard Merchant Presented QR transactions

Subfield 03 is optional for Mastercard Merchant Presented QR Payment Transactions. It is not edited and will be forwarded to the destination.

Subfield 03 is not applicable for Mastercard Merchant Presented QR Refund Payment or Funding Transactions.

Table 1072: Name example

Business name	First name	Last name
XYZ	XYZ	XYZ
XYZ International	XYZ International	XYZ International
XYZ DBA MA	XYZ DBA MA	XYZ DBA MA

Subfield 04 (Receiver/Recipient Street Address)

DE 108, subelement 01, subfield 04 (Receiver/Recipient Street Address) contains the Street Address of the Recipient/Receiver.

Attributes

Attribute	Description
Subfield ID	04
Subfield length	2
Data representation	ans...50; LLVAR
Data field	Contents of subfield 04
Justification	N/A

Values

Valid value will consist of the street address of the Receiver.

Application notes

Subfield 04 is optional for Mastercard Send Transactions (MoneySend Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

Subfield 04 (Receiver/Recipient Street Address Line 1 [Visa Only])

DE 108, subelement 01, subfield 04 (Receiver/Recipient Street Address Line 1 [Visa Only]) contains the street address of the receiver or recipient.

Table 1073: Attributes

Attribute	Description
Subfield ID	04
Subfield Data Length	2 positions

Attribute	Description
Data Representation	ans...99; LLVAR
Data Field	Contents of subfield 04
Justification	N/A

Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

Application notes

Subfield 04 (Receiver/Recipient Street Address Line 1 [Visa Only]) is only for use with Visa-branded transactions.

Subfield 05 (Receiver/Recipient City)

DE 108, subelement 01, subfield 05 (Receiver/Recipient City) contains the city of the Receiver/Recipient.

Table 1074: Attributes

Attribute	Description
Subfield ID	05
Subfield Data Length	2 positions
Data Representation	ans...25; LLVAR
Data Field	Contents of subfield 05
Justification	N/A

Values

Valid location city name of the Receiver/Recipient.

Application notes

Subfield 05 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

Application notes: Mastercard Merchant Presented QR transactions

Subfield 05 is optional for Mastercard Merchant Presented QR Payment Transactions. It is not edited and will be forwarded to the destination.

Subfield 05 is not applicable for Mastercard Merchant Presented QR Refund Payment or Funding Transactions.

Subfield 06 (Receiver/Recipient State/Province Code)

DE 108, subelement 01, subfield 06 (Receiver/Recipient State/Province Code) contains the state/province code of the Recipient/Receiver.

Table 1075: Attributes

Attribute	Description
Subfield ID	06
Subfield Data Length	2 positions
Data Representation	ans...3; LLVAR
Data Field	Contents of subfield 06
Justification	N/A

Values

Valid location state code of the Receiver/Recipient.

Application notes

Subfield is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions). If submitted it cannot contain spaces or invalid codes when country is U.S. or Canada.

NOTE: Data representation must be alpha (a-3) for Mastercard Send Transactions. For US and Canada, Data representation must be (a-2) for State/Province. Refer to the [Quick Reference Booklet](#) for valid list of State/Province codes.

Subfield 07 (Receiver/Recipient Country)

DE 108, subelement 01, subfield 07 (Receiver/Recipient Country) contains the country of the Recipient/Receiver.

Table 1076: Attributes

Attribute	Description
Subfield ID	07
Subfield Data Length	2 positions
Data Representation	ans-3
Data Field	Contents of subfield 07
Justification	N/A

Values

If present must be a valid ISO country code and must not be part of blocked country list.

Application notes

Subfield 07 is optional for Mastercard Send Transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions).

NOTE: If present for Mastercard Send Transactions, data representation must be alpha (a-3). Refer to the [Quick Reference Booklet](#) for valid list of country codes.

Transaction will be declined if the sender country is subject to comprehensive geographic sanctions published by the Office of Foreign Assets Control (OFAC), <https://ofac.treasury.gov/sanctions-programs-and-country-information>. The current list of countries subject to such sanctions is subject to change.

Subfield 08 (Receiver/Recipient Postal Code)

DE 108, subelement 01, subfield 08 (Receiver/Recipient Postal Code) contains the postal code of the Recipient/Receiver.

Attributes

Attribute	Description
Subfield ID	08
Subfield length	2
Data representation	ans...10; LLVAR
Data field	Contents of subfield 08
Justification	N/A

Values

Valid location Postal Code of the Recipient's/Receiver.

Application notes

Subfield 08 is optional for Mastercard Send Transactions (Funding Transactions, MoneySend Payment Transactions, and Gaming and Gambling Payments Transactions).

Subfield 08 (Receiver/Recipient Postal Code [Visa Only])

DE 108, subelement 01, subfield 08 (Receiver/Recipient Postal Code [Visa Only]) contains the postal code of the receiver or recipient.

Table 1077: Attributes

Attribute	Description
Subfield ID	08
Subfield Data Length	2 positions
Data Representation	ans...16; LLVAR
Data Field	Contents of subfield 08
Justification	N/A

Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

Application notes

Subfield 08 (Receiver/Recipient Postal Code [Visa Only]) is only for use with Visa-branded transactions.

Subfield 09 (Receiver/Recipient Phone Number)

DE 108, subelement 01, subfield 09 (Receiver/Recipient Phone Number) contains the phone number of the Recipient/Receiver.

Table 1078: Attributes

Attribute	Description
Subfield ID	09
Subfield Data Length	2 positions
Data Representation	ans...20; LLVAR
Data Field	Contents of subfield 09
Justification	N/A

Values

Valid phone number of the Recipient/Receiver.

Application notes

Subfield 09 is optional for Mastercard Send Transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions).

Application notes: Mastercard Merchant Presented QR transactions

Subfield 09 is optional for Mastercard Merchant Presented QR Payment Transactions. It is not edited and will be forwarded to the destination.

Subfield 09 is not applicable for Mastercard Merchant Presented QR Refund Payment or Funding Transactions.

Subfield 10 (Receiver/Recipient Date of Birth)

DE 108, subelement 01, subfield 10 (Receiver/Recipient Date of Birth) contains the date of birth of the Receiver/Recipient.

Table 1079: Attributes

Attribute	Description
Subfield ID	10
Subfield Data Length	2 positions
Data Representation	n-8
Data Field	Contents of subfield 10
Justification	N/A

Values

Valid date of birth of the Receiver/Recipient in the format MMDDYYYY.

Application notes

Subfield 10 is optional for Mastercard Send Transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions).

Application notes: Mastercard Merchant Presented QR transactions

Subfield 10 is optional for Mastercard Merchant Presented QR Payment Transactions. It is not edited and will be forwarded to the destination.

Subfield 10 is not applicable for Mastercard Merchant Presented QR Refund Payment or Funding Transactions.

Subfield 11 (Receiver/Recipient Account Number)

DE 108, subelement 01, subfield 11 (Receiver/Recipient Account Number) contains the account number of the Recipient/Receiver.

Table 1080: Attributes

Attribute	Description
Subfield ID	11
Subfield Data Length	2 positions
Data Representation	ans...50; LLVAR
Data Field	Contents of subfield 11
Justification	N/A

Values

Valid Account Number of the Receiver/Recipient.

Application notes

Subfield 11 (Receiver/Recipient Account Number) can have a maximum length of 34 characters for Visa-branded transactions.

Application notes: Mastercard Merchant Presented QR transactions

Subfield 11 is optional for Mastercard Merchant Presented QR Payment Transactions. It is not edited and will be forwarded to the destination. If Mastercard Merchant Presented QR transactions are processed with data in DE 108, subelement 01, subfield 11 (Receiver/Recipient Account Number) exceeding the current supported length, Mastercard Merchant Presented QR edits will decline the transaction.

Subfield 11 is not applicable for Mastercard Merchant Presented QR Refund Payment or Funding Transactions.

Subfield 12 (Receiver/Recipient Identification Type)

DE 108, subelement 01, subfield 12 (Receiver/Recipient Identification Type) contains the identification type of the Recipient/Receiver.

Table 1081: Attributes

Attribute	Description
Subfield ID	12
Subfield Data Length	2 positions

Attribute	Description
Data Representation	n-2
Data Field	Contents of subfield 12
Justification	N/A

Values

If present on Mastercard Send Payment Transactions, Gaming and Gambling Payments Transactions, Visa-branded funding transactions, it must contain one of the following valid values:

Table 1082: Values

Value	Description
00	Passport
01	National Identification Card
02	Driver's License
03	Government Issued
04	Other
05	Date of birth (Visa only)
06	Customer identification (unspecified) (Visa only)
07	Business tax ID (Visa only)
08	Individual tax ID (Visa only)
09	Company registration number (Visa only)
10	Proxy identification (Visa only)
11	Social security number (Visa only)
12	Alien registration number (Visa only)
13	Law enforcement identification (Visa only)
14	Military identification (Visa only)
15	Travel identification (nonpassport) (Visa only)
16	Email (Visa only)
17	Phone number (Visa only)

Application notes

Values 03 and 04 have no corresponding values in Visa format. Acquirers should not send either value 03 or 04 with Visa-branded transactions.

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details on Visa only values.

Subfield 13 (Receiver/Recipient Identification Number)

DE 108, subelement 01, subfield 13 (Receiver/Recipient Identification Number) contains the identification number of the Recipient/Receiver.

Attributes

Attribute	Description
Subfield ID	13
Subfield length	2
Data representation	ans...25; LLVAR
Data field	Contents of subfield 13
Justification	N/A

Values

Valid identification number of the Receiver/Recipient.

Application notes

Subfield 13 is optional for Mastercard Send Transactions (MoneySend Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

Subfield 13 (Receiver/Recipient Identification Number [Visa Only])

DE 108, subelement 01, subfield 13 (Receiver/Recipient Identification Number [Visa Only]) contains the identification number of the receiver or recipient.

Table 1083: Attributes

Attribute	Description
Subfield ID	13
Subfield Data Length	2 positions
Data Representation	ans...35; LLVAR
Data Field	Contents of subfield 13

Attribute	Description
Justification	N/A

Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

Application notes

Subfield 13 (Receiver/Recipient Identification Number [Visa Only]) is only for use with Visa-branded transactions.

Subfield 14 (Receiver/Recipient Identification Country Code)

DE 108, subelement 01, subfield 14 (Receiver/Recipient Identification Country Code) contains the identification country code of the recipient or receiver.

Table 1084: Attributes

Attribute	Description
Subfield ID	14
Subfield Data Length	2 positions
Data Representation	ans-3
Data Field	Contents of subfield 14
Justification	N/A

Values

Valid identification country code of the Receiver/Recipient.

Application notes

Subfield 14 (Receiver/Recipient Identification Country Code) is optional for Mastercard Send Transactions, Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payment Transactions. The field has a three-character ISO country code in Visa-branded transactions.

Subfield 15 (Receiver/Recipient Identification Expiration Date)

DE 108, subelement 01, subfield 15 (Receiver/Recipient Identification Expiration Date) contains the identification expiration date of the Receiver/Recipient.

Table 1085: Attributes

Attribute	Description
Subfield ID	15
Subfield Data Length	2 positions
Data Representation	n-8
Data Field	Contents of subfield 15
Justification	N/A

Values

Valid identification expiration date of the Receiver/Recipient in the format MMDDYYYY.

Application notes

Subfield 15 is optional for Mastercard Send Transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions).

Subfield 16 (Receiver/Recipient Nationality)

DE 108, subelement 01, subfield 16 (Receiver/Recipient Nationality) contains the nationality of the Recipient/Receiver.

Table 1086: Attributes

Attribute	Description
Subfield ID	16
Subfield Data Length	2 positions
Data Representation	ans-3
Data Field	Contents of subfield 16
Justification	N/A

Values

Nationality of the Receiver/Recipient as defined by a valid country code for the country of citizenship.

Application notes

Subfield 16 is optional for Mastercard Send Transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions).

Subfield 17 (Receiver/Recipient Country of Birth)

DE 108, subelement 01, subfield 17 (Receiver/Recipient Country of Birth) contains the country of birth of the Receiver/Receiver.

Table 1087: Attributes

Attribute	Description
Subfield ID	17
Subfield Data Length	2 positions
Data Representation	ans-3
Data Field	Contents of subfield 17
Justification	N/A

Values

Valid Country of birth of the Receiver/Recipient.

Application notes

Subfield 17 is optional for Mastercard Send Transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions).

Subfield 18 (Receiver/Recipient Account Number Type)

DE 108, subelement 01, subfield 18 (Receiver/Recipient Account Number Type) contains the account number type of the Receiver/Recipient.

Table 1088: Attributes

Attribute	Description
Subfield ID	18
Subfield Data Length	2 positions
Data Representation	n-2
Justification	n/a

Values

List of valid values:

- 00: Other
- 01: RTN + Bank Account
- 02: IBAN
- 03: Card Account
- 04: Email
- 05: Phone Number
- 06: Bank account number (BAN) + bank identification code (BIC)
- 07: Wallet ID
- 08: Social Network ID

Application notes

Acquirers are expected per participation requirements to supply Subfield 18 if the related Subfield 11 is provided. Subfield is expected to be one of the valid values for the field.

Subfield 18 is not required for Mastercard QR Transactions and if submitted will be disregarded and not be forwarded to destination.

Subfield 18 is optional for Mastercard Send Transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions).

The Mastercard Network does not enforce system edits on this field. It is therefore optional per specifications. However, per program rules noted in the [Mastercard Send and Funding Transactions Program Standards](#), it is a required field for Funding Transactions and must be provided. Not providing required data in this field as per standards may result in compliance assessments.

Subfield 19 (Receiver/Recipient Street Address Line 2 [Visa Only])

DE 108, subelement 01, subfield 19 (Receiver/Recipient Street Address Line 2 [Visa Only]) contains the street address line 2 of the receiver or recipient.

Table 1089: Attributes

Attribute	Description
Subfield ID	19
Subfield Data Length	2 positions
Data Representation	ans...99; LLVAR
Data Field	Contents of subfield 19
Justification	N/A

Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

Application notes

Subfield 19 (Receiver/Recipient Street Address Line 2 [Visa Only]) is only for use with Visa-branded transactions.

Subfield 20 (Receiver/Recipient Street Name [Visa Only])

DE 108, subelement 01, subfield 20 (Receiver/Recipient Street Name [Visa Only]) contains the street name of the receiver or recipient.

Table 1090: Attributes

Attribute	Description
Subfield ID	20
Subfield Data Length	2 positions
Data Representation	ans...99; LLVAR
Data Field	Contents of subfield 20
Justification	N/A

Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

Application notes

Subfield 20 (Receiver/Recipient Street Name [Visa Only]) is only for use with Visa-branded transactions.

Subfield 21 (Receiver/Recipient Building Number [Visa Only])

DE 108, subelement 01, subfield 21 (Receiver/Recipient Building Number [Visa Only]) contains the building number of the receiver or recipient.

Table 1091: Attributes

Attribute	Description
Subfield ID	21
Subfield Data Length	2 positions
Data Representation	ans...16; LLVAR
Data Field	Contents of subfield 21

Attribute	Description
Justification	N/A

Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

Application notes

Subfield 21 (Receiver/Recipient Building Number [Visa Only]) is only for use with Visa-branded transactions.

Subfield 22 (Receiver/Recipient Country Subdivision Code, Minor [Visa Only])

DE 108, subelement 01, subfield 22 (Receiver/Recipient Country Subdivision Code, Minor [Visa Only]) contains the minor country subdivision code of the receiver or recipient.

Table 1092: Attributes

Attribute	Description
Subfield ID	22
Subfield Data Length	2 positions
Data Representation	ans...16; LLVAR
Data Field	Contents of subfield 22
Justification	N/A

Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

Application notes

Subfield 22 (Receiver/Recipient Country Subdivision Code, Minor [Visa Only]) is only for use with Visa-branded transactions.

Subelement 02 (Sender Data)

DE 108 (Additional Transaction Reference Data), subelement 02 (Sender Data) contains the sender name, address, phone number, date of birth, and account details.

Table 1093: Attributes

Attribute	Description
Subelement ID	02
Subelement Data Length	3 positions
Data Representation	ans...368; LLVAR
Data Field	Contents of subfields
Subfields	Subfield 01 (Sender First Name)
	Subfield 02 (Sender Middle Name)
	Subfield 03 (Sender Last Name)
	Subfield 04 (Sender Street Address)
	Subfield 05 (Sender City)
	Subfield 06 (Sender State/Province Code)
	Subfield 07 (Sender Country)
	Subfield 08 (Sender Postal Code)
	Subfield 09 (Sender Phone Number)
	Subfield 10 (Sender Date of Birth)
	Subfield 11 (Sender Account Number)
	Subfield 12 (Sender Identification Type)
	Subfield 13 (Sender Identification Number)
	Subfield 13 (Sender Identification Number [Visa Only])
	Subfield 14 (Sender Identification Country Code)
	Subfield 15 (Sender Identification Expiration Date)
	Subfield 16 (Sender Nationality)
	Subfield 17 (Sender Country of Birth)
	Subfield 18 (Sender Account Number Type)
Justification	See subfields

Subfield 01 (Sender First Name)

DE 108, subelement 02, subfield 01 (Sender First Name) contains the first name of the Sender.

Table 1094: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	ans...35; LLVAR
Data Field	Contents of subfield 01
Justification	N/A

Values

If present, cannot contain all spaces or all numeric values.

Application notes

Subfield 01 must be present for Mastercard Send Payment Transactions and Gaming and Gambling Payments Transactions and must be properly formatted. Subfield 01 is optional for Funding Transactions.

The first name (Consumer/Business/Government/Non-Government Organization) of the sender is included in this subelement.

The table that follows describes the correct submission of business names.

Application notes: Mastercard Merchant Presented QR transactions

Subfield 01 is optional for Mastercard Merchant Presented QR Payment Transactions. It is not edited and will be forward to destination.

Subfield 01 is not applicable for Mastercard Merchant Presented QR Refund Payment or Funding Transactions.

Table 1095: Name example

Business Name	First Name	Last Name
XYZ	XYZ	XYZ
XYZ International	XYZ	International
XYZ DBA MA	XYZ	DBA MA

Subfield 02 (Sender Middle Name)

DE 108, subelement 02, subfield 02 (Sender Middle Name) contains the middle name initial of the Sender.

Table 1096: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	ans-1
Data Field	Contents of subfield 02
Justification	N/A

Values

Valid value will consist of the middle name initial of the Sender.

Application notes

Subfield 02 is optional for Mastercard Send Transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions).

Subfield 03 (Sender Last Name)

DE 108, subelement 02, subfield 03 (Sender Last Name) contains the last name of the Sender.

Table 1097: Attributes

Attribute	Description
Subfield ID	03
Subfield Data Length	2 positions
Data Representation	ans...35; LLVAR
Data Field	Contents of subfield 03
Justification	N/A

Values

If present, cannot contain all spaces or numeric values.

Application notes

Subfield 03 must be present and properly formatted for Mastercard Send Payment Transactions and Gaming and Gambling Payments Transactions. Subfield 03 is optional for Funding Transactions.

The last name (Consumer/Business/Government/Non-Government Organization) of the sender is included in this subelement.

The table that follows describes the correct submission of business names.

Application notes: Mastercard Merchant Presented QR transactions

Subfield 03 is optional for Mastercard Merchant Presented QR Payment Transactions. It is not edited and will be forwarded to the destination.

Subfield 03 is not applicable for Mastercard Merchant Presented QR Refund Payment or Funding Transactions.

Table 1098: Name example

Business name	First name	Last name
XYZ	XYZ	XYZ
XYZ International	XYZ	International
XYZ DBA MA	XYZ	DBA MA

Subfield 04 (Sender Street Address)

DE 108, subelement 02, subfield 04 (Sender Street Address) contains the street address of the Sender.

Table 1099: Attributes

Attribute	Description
Subfield ID	04
Subfield Data Length	2 positions
Data Representation	ans...50; LLVAR
Data Field	Contents of subfield 04
Justification	N/A

Values

Valid values will consist of street address of the Sender.

Application notes

Subfield 04 is mandatory for Mastercard Send Payment Transactions and Gambling and Gaming Payments Transactions.

Subfield 04 is optional for Funding transactions.

Subfield 05 (Sender City)

DE 108, subelement 02, subfield 05 (Sender City) contains the city of the Sender.

Table 1100: Attributes

Attribute	Description
Subfield ID	05
Subfield Data Length	2 positions
Data Representation	ans...25; LLVAR
Data Field	Contents of subfield 05
Justification	N/A

Values

Valid location city name of the Sender.

Application notes

Subfield 05 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

NOTE: The Mastercard Network does not enforce system edits on this field. It is therefore optional per specifications. However per program rules noted in the [Mastercard Send and Funding Transactions Program Standards](#), it is a required field for Payment Transactions and must be provided. Not providing required data in this field as per standards may result in compliance assessments

Subfield 06 (Sender State/Province Code)

DE 108, subelement 02, subfield 06 (Sender State/Province Code) contains the state/province code of the Sender.

Table 1101: Attributes

Attribute	Description
Subfield ID	06
Subfield Data Length	2 positions

Attribute	Description
Data Representation	ans...3; LLVAR
Data Field	Contents of subfield 06
Justification	N/A

Values

Valid location State code of the Sender. Cannot contain spaces or invalid code when country is U.S. or Canada.

Application notes

Subfield 06 is optional for Mastercard Send Transactions (Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions). If submitted on Mastercard Send Payment Transactions and Gaming and Gambling Payments Transactions, subfield 06 cannot contain spaces or invalid code when country is U.S. or Canada.

NOTE: Data representation must be alpha (a-3) for Mastercard Send Transactions. Refer to [Quick Reference Guide](#) for list of valid country codes.

Subfield 07 (Sender Country)

DE 108, subelement 02, subfield 07 (Sender Country) contains the country code of the Sender.

Table 1102: Attributes

Attribute	Description
Subfield ID	07
Subfield Data Length	2 positions
Data Representation	an-3
Data Field	Contents of subfield 07
Justification	N/A

Values

If present must contain the Mastercard 3-digit alphabetic country code and must not be part of blocked country list.

Application notes

Subfield 07 is mandatory for Mastercard Send Payment Transactions and Gaming and Gambling Payments Transactions and optional for Funding Transactions.

Subfield 07 is optional for Funding Transactions.

Transaction will be declined if the sender country is subject to comprehensive geographic sanctions published by the Office of Foreign Assets Control (OFAC), <https://ofac.treasury.gov/sanctions-programs-and-country-information>. The current list of countries subject to such sanctions is subject to change.

Subfield 08 (Postal Code)

DE 108, subelement 02, subfield 08 (Postal Code) contains the postal code of the Sender.

Table 1103: Attributes

Attribute	Description
Subfield ID	08
Subfield Data Length	2 positions
Data Representation	ans...10; LLVAR
Data Field	Contents of subfield 08
Justification	N/A

Values

Valid location Postal Code of the Sender.

Application notes

Subfield 08 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

Subfield 09 (Sender Phone Number)

DE 108, subelement 02, subfield 09 (Sender Phone Number) contains the phone number of the Sender.

Table 1104: Attributes

Attribute	Description
Subfield ID	09
Subfield Data Length	2 positions
Data Representation	ans...20; LLVAR
Data Field	Contents of subfield 09
Justification	N/A

Values

Valid phone number of the Sender.

Application notes

Subfield 09 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

Subfield 10 (Sender Date of Birth)

DE 108, subelement 02, subfield 10 (Sender Date of Birth) contains the date of birth of the Sender.

Table 1105: Attributes

Attribute	Description
Subfield ID	10
Subfield Data Length	2 positions
Data Representation	n-8
Data Field	Contents of subfield 10
Justification	N/A

Values

Valid values will consist of the date of birth of the Sender in the format MMDDYYYY.

Application notes

Subfield 10 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

Subfield 11 (Sender Account Number)

DE 108, subelement 02, subfield 11 (Sender Account Number) contains the account number of the Sender.

Table 1106: Attributes

Attribute	Description
Subfield ID	11
Subfield Data Length	2 positions
Data Representation	ans...50; LLVAR
Data Field	Contents of subfield 11

Attribute	Description
Justification	N/A

Values

Valid account number of the Sender.

The account number that was used to fund the money transfer (for example, the sender used a credit, debit, or other account to fund the money transfer). When cash is used to fund the money transfer, this field should be populated with a unique transaction reference number or an account number from the OI. This information will assist in identifying the sender if the receiving institution requires additional information from the originating institution for the money transfer.

Application notes

Subfield 11 (Sender Account Number) can have a maximum length of 34 characters for Visa-branded transactions.

Application notes: Mastercard Send Transactions

Subfield 11 must be present for Mastercard Send Payment Transactions and Gaming and Gambling Payments Transactions and must be properly formatted. Subfield 11 is optional for Funding Transactions. When subfield 11 contains a combination of information as specified in subfield 18 [for example, Bank account number (BAN) + Bank Identification Code (BIC)], the data must be separated by '+' and in the order specified for Mastercard Send and Funding Transactions.

Application notes: Mastercard Merchant Presented QR Transactions

Payment Transaction usage:

- Authorization Request/0100: Mandatory. Edit must validate. Forward to destination.
- Reversal Advice/0420: Value from Authorization Request/0100 forwarded to destination.

The enhancement introduced within DE 108, subelement 02, subfield 11 (Sender Account Number) does not apply for MCQR transactions. If MCQR transactions are processed with data in DE 108, subelement 02, subfield 11 (Sender Account Number) exceeding the current supported length, Mastercard MCQR edits will decline the transaction.

Refund payment transaction usage: Reversal Advice/0420—Value from Authorization Request/0100 forwarded to destination

Funding transaction usage: Subfield 11 is not applicable

Subfield 12 (Sender Identification Type)

DE 108, subelement 02, subfield 12 (Sender Identification Type) contains the identification type of the Sender.

Table 1107: Attributes

Attribute	Description
Subfield ID	12
Subfield Data Length	2 positions
Data Representation	n-2
Data Field	Contents of subfield 12
Justification	N/A

Values

If present on Mastercard Send Payment Transactions, Gaming and Gambling Payments Transactions, Visa-branded funding transactions, it must contain one of the following valid values:

Table 1108: Values

Value	Description
00	Passport
01	National Identification Card
02	Driver's License
03	Government Issued
04	Other
05	Date of birth (Visa only)
06	Customer identification (unspecified) (Visa only)
07	Business tax ID (Visa only)
08	Individual tax ID (Visa only)
09	Company registration number (Visa only)
10	Proxy identification (Visa only)
11	Social security number (Visa only)
12	Alien registration number (Visa only)

Value	Description
13	Law enforcement identification (Visa only)
14	Military identification (Visa only)
15	Travel identification (non-passport) (Visa only)
16	Email (Visa only)
17	Phone number (Visa only)

Application notes

Values 03 and 04 have no corresponding values in Visa format. Acquirers should not send either value 03 or 04 with Visa-branded transactions.

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details on Visa-only values.

Subfield 13 (Sender Identification Number)

DE 108, subelement 02, subfield 13 (Sender Identification Number) contains the identification number of the Sender.

Attributes

Attribute	Value
Subfield ID	13
Subfield length	2
Data representation	ans...25; LLVAR
Data field	Contents of subfield 13
Justification	N/A

Values

Valid identification number of the Sender.

Application notes

Subfield 13 is optional for Mastercard Send Transactions (MoneySend Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

Subfield 13 (Sender Identification Number [Visa Only])

DE 108, subelement 02, subfield 13 (Sender Identification Number [Visa Only]) contains the identification number of the Sender.

Table 1109: Attributes

Attribute	Description
Subfield ID	13
Subfield Data Length	2 positions
Data Representation	ans...35; LLVAR
Data Field	Contents of subfield 13
Justification	N/A

Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

Application notes

Subfield 13 (Sender Identification Number [Visa Only]) is only for use with Visa-branded transactions.

Subfield 14 (Sender Identification Country Code)

DE 108, subelement 02, subfield 14 (Sender Identification Country Code) contains the identification country code of the Sender.

Table 1110: Attributes

Attribute	Description
Subfield ID	14
Subfield Data Length	2 positions
Data Representation	ans-3
Data Field	Contents of subfield 14
Justification	N/A

Values

Valid identification country code of the Sender.

Application notes

Subfield 14 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

Subfield 15 (Sender Identification Expiration Date)

DE 108, subelement 02, subfield 15 (Sender Identification Expiration Date) contains the identification expiration date of the Sender.

Table 1111: Attributes

Attribute	Description
Subfield ID	15
Subfield Data Length	2 positions
Data Representation	n-8
Data Field	Contents of subfield 15
Justification	N/A

Values

Valid identification expiration date of the Sender in the format MMDDYYYY.

Application notes

Subfield 15 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

Subfield 16 (Sender Nationality)

DE 108, subelement 02, subfield 16 (Sender Nationality) contains the nationality of the Sender.

Table 1112: Attributes

Attribute	Description
Subfield ID	16
Subfield Data Length	2 positions
Data Representation	ans-3
Data Field	Contents of subfield 16
Justification	N/A

Values

Nationality of the sender as defined by a valid country code for the country of citizenship.

Application notes

Subfield 16 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

Subfield 17 (Sender Country of Birth)

DE 108, subelement 02, subfield 17 (Sender Country of Birth) contains the country of birth of the Sender.

Table 1113: Attributes

Attribute	Description
Subfield ID	17
Subfield Data Length	2 positions
Data Representation	ans-3
Data field	Contents of subfield 17
Justification	N/A

Values

Valid country of birth of the Sender.

Application notes

Subfield 17 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

Subfield 18 (Sender Account Number Type)

DE 108, subelement 02, subfield 18 (Sender Account Number Type) contains the account number type of the Sender.

Table 1114: Attributes

Attribute	Description
Subfield ID	18
Subfield Data Length	2 positions
Data Representation	n-2

Attribute	Description
Justification	N/A

Values

List of valid values:

- 00: Other
- 01: RTN + Bank Account
- 02: IBAN
- 03: Card Account
- 04: Email
- 05: Phone Number
- 06: Bank account number (BAN) + Bank Identification Code (BIC)
- 07: Wallet ID
- 08: Social Network ID

Application notes

Acquirers are expected per participation requirements to supply Subfield 18 if the related Subfield 11 is provided. Subfield is expected to be one of the valid values for the field.

Subfield 18 is not required for Mastercard QR Transactions and if submitted will be disregarded and not be forwarded to destination.

Subfield 18 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payment Transactions).

The Mastercard Network does not enforce system edits on this field. It is therefore optional per specifications. However per program rules noted in the [Mastercard Send and Funding Transactions Program Standards](#), it is a required field for Funding Transactions and must be provided. Not providing required data in this field as per standards may result in compliance assessments

Subelement 03 (Transaction Reference Data)

DE 108, subelement 03 (Transaction Reference Data) contains the transaction reference number, funding score, participation ID, and additional data related to Mastercard Send transactions and Mastercard Merchant Presented QR transactions.

Table 1115: Attributes

Attribute	Description
Subelement ID	03
Subelement Data Length	3 positions

Attribute	Description
Data Representation	ans...138; LLLVAR
Data Field	Contents of subfields
Subfields	Subfield 01 (Unique Transaction Reference) Subfield 01 (Sender Reference Number [Visa Only]) Subfield 02 (Additional Message) Subfield 03 (Funding Score) Subfield 04 (Participant ID) Subfield 05 (Transaction Purpose)
Justification	See subfields

Subfield 01 (Unique Transaction Reference)

DE 108, subelement 03, subfield 01 (Unique Transaction Reference) contains the Unique Transaction Reference number for Mastercard Send Transactions (MoneySend Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions), and for Mastercard Merchant Presented QR funding transactions.

Attributes

Attribute	Value
Subfield ID	01
Subfield length	2
Data representation	ans-19
Data field	Contents of subfield 01
Justification	N/A

Values

Unique Transaction Reference Number (ans-19).

Valid value string will contain a leading zero (0), followed by:

- ICA (n-6)
- Year (n-1)
- Julian Date (n-3)
- Time hhmmss (n-6)
- Transaction Sequence Number (01-99) (n-2)

Example: 0555555801215305401

Application notes

The Mastercard Network does not enforce system edits on this field. However, per program rules noted in the *Mastercard MoneySend and Funding Transactions Program Standards*, it is a required field for Funding Transactions, MoneySend Payment and Gaming and Gambling Payments Transactions must be provided. Not providing required data in this field as per standards may result in compliance assessments

Application Notes: Mastercard Merchant Presented QR transactions

Payment Transaction usage:

- Authorization Request/0100—Optional. If present, edit must validate. Forward to destination if edit validates
- Reversal Advice/0420—Optional. If present, value from Authorization Request/0100 forwarded to destination

Refund Payment Transaction usage:

- Authorization Request/0100—Optional. If present, edit must validate. Forward to destination if edit validates
- Reversal Advice/0420—Value from Authorization Request/0100 forwarded to destination

Funding Transaction usage:

- Authorization Request/0100—Mandatory. If present, edit must validate. Forward to destination if edit validates
- Authorization Advice/0120—Optional. If present, edit must validate. Forward to destination if edit validates
- Reversal Request/0400—Optional. If present, edit must validate. Forward to destination if edit validates
- Reversal Advice/0420—Value from Authorization Request/0100 forwarded to destination

Values for customers that do not need to process beyond 99 transactions per second

Unique Transaction Reference Number (ans-19). Valid value string will contain a leading zero (0), followed by:

- ICA (n-6)
- Year (n-1)
- Julian Date (n-3)
- Time hhmmss (n-6)
- Transaction Sequence Number for customers who do not need to process beyond 99 transactions: (01-99) (n-2)

Example: 0555555801215305401

Values for customers who need to process beyond 99 transactions per second

Unique Transaction Reference Number (ans-19). Valid value string will contain:

- ICA (n-6)
- Year (n-1)
- Julian Date (n-3)
- Time hhmmss (n-6)
- Transaction Sequence Number for customers who need to process beyond 99 transactions: (01-999) (n-3)

Example: 5555558012153054001

Subfield 01 (Sender Reference Number [Visa Only])

DE 108, subelement 03, subfield 01 (Sender Reference Number [Visa Only]) contains the sender reference number.

Table 1116: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	ans...16; LLVAR
Data Field	Contents of subfield 01
Justification	N/A

Values

Refer to *VisaNet Authorization-Only Online Messages Technical Specifications* for further details.

Application notes

Subfield 01 (Sender Reference Number [Visa Only]) is only for use with Visa-branded transactions.

Application notes: Mastercard Merchant Presented QR transactions

Payment Transaction usage:

- Authorization Request/0100: optional. If present, edit must validate. Forward to destination if edit validates
- Reversal Advice/0420: optional. If present, value from Authorization Request/0100 forwarded to destination

Refund Payment Transaction usage:

- Authorization Request/0100: optional. If present, edit must validate. Forward to destination if edit validates
- Reversal Advice/0420: value from Authorization Request/0100 forwarded to destination

Funding Transaction usage:

- Authorization Request/0100: mandatory. If present, edit must validate. Forward to destination if edit validates
- Authorization Advice/0120: optional. If present, edit must validate. Forward to destination if edit validates
- Reversal Request/0400: optional. If present, edit must validate. Forward to destination if edit validates
- Reversal Advice/0420: value from Authorization Request/0100 forwarded to destination

Values for customers that do not need to process beyond 99 transactions per second

Unique Transaction Reference Number (ans-19). Valid value string will contain a leading zero (0), followed by:

- ICA (n-6)
- Year (n-1)
- Julian Date (n-3)
- Time hhmmss (n-6)
- Transaction Sequence Number for customers who do not need to process beyond 99 transactions: (01-99) (n-2)

Example: 0555555801215305401

Values for customers that need to process beyond 99 transactions per second

Unique Transaction Reference Number (ans-19). Valid value string will contain:

- ICA (n-6)
- Year (n-1)
- Julian Date (n-3)
- Time hhmmss (n-6)
- Transaction Sequence Number for customers who need to process beyond 99 transactions: (01-999) (n-3)

Example: 5555558012153054001

Subfield 02 (Additional Message)

DE 108, subelement 03, subfield 02 (Additional Message) contains the additional message.

Table 1117: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions

Attribute	Description
Data Representation	ans...65; LLVAR
Data Field	Contents of subfield 02
Justification	N/A

Application notes

Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

Application notes: Mastercard Merchant Presented QR Transactions

Payment Transaction usage:

- Authorization Request/0100: optional. If present, edit must validate. Forward to destination if edit is validated
- Reversal Advice/0420: optional. If present, value from Authorization Request/0100 forwarded to destination

Refund Payment Transaction usage: subfield 02 is not applicable

Funding Transaction usage:

- Authorization Advice/0120: optional. If present, edit must validate. Forward to destination if edit is validated
- Reversal Request/0400: optional. If present, edit must validate. Forward to destination if edit is validated
- Reversal Advice/0420: value from Authorization Request/0100 forwarded to destination

Subfield 03 (Funding Source)

DE 108, subelement 03, subfield 03 (Funding Source) contains the information representing the Mastercard Send transaction funding source.

Table 1118: Attributes

Attribute	Description
Subfield ID	03
Subfield Data Length	2 positions
Data Representation	n-2
Data Field	Contents of subfield 03
Justification	N/A

Values

Must be present for MPQR payment authorization request messages and, when present, must contain one of the following valid values.

Table 1119: Valid values

Value	Description
01	Credit
02	Debit
03	Prepaid
04	Deposit Account
05	Mobile Money Account
06	Cash
07	Other

Application notes

Subfield 03 is optional for Funding Transactions, Mastercard Send Payment Transactions, and Gaming and Gambling Payments Transactions on Authorization Request/0100 and Reversal Request/0400 messages.

Mastercard will not perform a valid values check on subfield 03 within the Authorization Request/0100 message. Acquirers must send DE 108, subelement 03, subfield 03 on MPQR payment Authorization Request messages.

Table 1120: Application notes for Mastercard Merchant Presented QR Transactions

Transaction	Usage
Payment Transaction: Authorization Request/0100	Mandatory
Payment Transaction: Reversal Advice/0420	Value from Authorization Request/0100 forwarded to destination
Refund Payment Transaction: Reversal Advice/0420	Value from Authorization Request/0100 forwarded to destination
Funding Transaction	Subfield 03 is not applicable

Subfield 04 (Participation ID)

DE 108, subelement 03, subfield 04 (Participation ID) contains participation ID details of the Sender.

Table 1121: Attributes

Attribute	Description
Subfield ID	04
Subfield Data Length	2 positions
Data Representation	ans...30; LLVAR
Data Field	Contents of subfield 04
Justification	N/A

Values

Contains the participation ID of the Sender.

Application notes

Subfield 04 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

Subfield 05 (Transaction Purpose)

DE 108, subelement 03, subfield 05 (Transaction Purpose) contains Mastercard Send transaction purpose details.

Table 1122: Attributes

Attribute	Description
Subfield ID	05
Subfield Data Length	2 positions
Data Representation	n-2
Data Field	Contents of subfield 05
Justification	N/A

Table 1123: Values

Value	Description
00	Family Support
01	Regular Labor Transfers (expatriates)
02	Travel & Tourism
03	Education
04	Hospitalization & Medical Treatment
05	Emergency Need
06	Savings
07	Gifts
08	Other
09	Salary
10	Crowd lending
11	Crypto currency
12	Refund to original card
13	Refund to new card
14	Transfer to another person or organization in the same Staged Digital Wallet (Brazil)
15	Boleto Payment (Brazil)
16	High Risk Securities

Application notes

Subfield 05 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions) and MPQR payment transactions.

Mastercard will not perform a valid values check on subfield 05 within the Authorization Request/0100 message.

Subelement 04 (Language Description)

DE 108 (Additional Transaction Reference Data), subelement 04 (Language Description) contains additional language details supported by the customer.

Table 1124: Attributes

Attribute	Description
Subelement ID	04
Subelement Data Length	3 positions
Data Representation	ans...061; LLLVAR
Data Field	Contents of subfields
Subfields	Subfield 1 (Language Identification)
	Subfield 2 (Language Data)
Justification	See subfields

Subfield 01 (Language Identification)

DE 108, subelement 04, subfield 01 (Language Identification) contains information about the language selected by the customer.

Table 1125: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	ans-3
Data Field	Contents of subfield 01
Justification	N/A

Application notes

Subfield 01 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

Subfield 02 (Language Data)

DE 108, subelement 04, subfield 02 (Language Data) contains additional Mastercard Send information in the language selected by the customer.

Table 1126: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	b...50; LLVAR
Data Field	Contents of subfield 02
Justification	N/A

Application notes

Subfield 02 is optional for Mastercard Send Transactions (Mastercard Send Payment Transactions, Funding Transactions, and Gaming and Gambling Payments Transactions).

Subelement 05 (Digital Account Information)

DE 108 (Additional Transaction Reference Data), subelement 05 (Digital Account Information) contains data specific to Mastercard Merchant Presented QR transactions.

Table 1127: Attributes

Attribute	Description
Subelement ID	05
Subelement Data Length	3 positions
Data Representation	ans...099; LLLVAR
Data Field	Contents of subfields
Subfields	Subfield 01 (Digital Account Reference Number) Subfield 02 (Mastercard Merchant Presented QR Receiving Account Number)
Justification	See subfields

Application notes

Subelement 05 is used only for Mastercard Merchant Presented QR transactions.

Subfield 01 (Digital Account Reference Number)

DE 108 (Additional Transaction Reference Data), subelement 05 (Digital Account Information), subfield 01 (Digital Account Reference Number) is provided in a Mastercard Merchant Presented QR Payment Transaction to allow Receiving Institutions the ability to initiate a refund to the consumer via the Originating Institution.

Table 1128: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	n...19; LLVAR
Data Field	Contents of subfield 01
Justification	N/A

Table 1129: Application notes: Mastercard Merchant Presented QR Transactions

Transaction	Usage
Payment Transaction usage: Authorization Request/0100:	Optional. If present, edit must validate. If not present, Mastercard will populate with value from DE 108, subelement 02, subfield 11 (Sender Account Number); Forward to destination.
Payment Transaction usage: Reversal Advice/0420	Optional. If present, value from Authorization Request/0100 forwarded to destination.
Refund Payment Transaction usage: Reversal Advice/0420	Value from Authorization Request/0100 forwarded to destination.
Funding Transaction usage:	Subfield 01 is not applicable

Subfield 02 (Mastercard Merchant Presented QR Receiving Account Number)

DE 108 (Additional Transaction Reference Data), subelement 05 (Digital Account Information), subfield 02 (Mastercard Merchant Presented QR Receiving Account Number) provided in a

Mastercard Merchant Presented QR Funding Transaction contains the receiving account number.

Table 1130: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	ans...34; LLVAR
Data Field	Contents of subfield 02
Justification	N/A

Values

Valid Account Number of the Receiver.

Table 1131: Application notes: Mastercard Merchant Presented QR Transactions

Transaction	Usage
Payment Transaction usage	Subfield 02 is not applicable
Refund Payment Transaction usage	Subfield 02 is not applicable
Funding Transaction usage: Authorization Request/0100	Optional. If present, edit must validate. Forward to destination if edit validates.
Funding Transaction usage: Authorization Advice/0120	Optional. If present, edit must validate. Forward to destination if edit validates.
Funding Transaction usage: Reversal Request/0400	Optional. If present, edit must validate. Forward to destination if edit validates.
Funding Transaction usage: Reversal Advice/0420	Optional. If present, value from Authorization Request/0100 forwarded to destination.

Subelement 06 (QR Dynamic Code Data)

DE 108 (Additional Transaction Reference Data), subelement 06 (QR Dynamic Code Data) is available for regional data with limited Mastercard editing or data validation.

Table 1132: Attributes

Attribute	Description
Subelement ID	06
Subelement Data Length	3 positions
Data Representation	ans...237; LLLVAR
Data Field	Contents of subelement 06
Subfields	N/A
Justification	Left

Values

If present, must be alphanumeric or special characters and a valid length.

Table 1133: Application notes: Mastercard Merchant Presented QR Transactions

Transaction	Usage
Payment Transaction usage: Authorization Request/0100	Optional. If present, edit must validate. Forward to destination if edit validates.
Payment Transaction usage: Reversal Advice/0420	Optional. If present, value from Authorization Request/0100 forwarded to destination.
Refund Payment Transaction usage	Subelement 06 is not applicable.
Funding Transaction usage: Authorization Request/0100	Optional. If present, edit must validate. Forward to destination if edit validates.
Funding Transaction usage: Authorization Advice/0120	Optional. If present, edit must validate. Forward to destination if edit validates.
Funding Transaction usage: Reversal Request/0400	Optional. If present, edit must validate. Forward to destination if edit validates.

Transaction	Usage
Funding Transaction usage: Reversal Advice/0420	Optional. If present, value from Authorization Request/0100 forwarded to destination.

Subelement 07 (Data Validation Request)

DE 108 (Additional Transaction Reference Data), subelement 07 (Data Validation Request) indicates that account status verification is requested in the Authorization message.

Table 1134: Attributes

Attribute	Description
Subelement ID	07
Subelement Data Length	3 positions
Data Representation	an....6;LLLVAR
Data Field	Contents of subfield 01
Subfields	1
Justification	N/A

Usage

Whether use is mandatory, conditional, optional, or not required is noted for each applicable message.

Table 1135: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	O	.	C
Authorization Request Response/0110	CE	.	CE

Authorization Request/0100 message edits for DE 108, subelement 07

Mastercard will perform the following edits.

Table 1136: Associated edits

When...	Then the Authorization Platform...
an Authorization Request/0100 message contains DE 108 (Additional Transaction Reference Data) with multiple instances of subelement 07 or its subfield.	will reject the message and send an Authorization Request Response/0110 message to the acquirer with: - DE 39 (Response Code) = 30 (Format Error) - DE 44 (Additional Response Data) = 108007 (indicating DE 108 SE07 is in error)
DE 108, subelement 07, subfield 01 (Name Validation Request Type) contains value 10 (Receiver Name Validation Request) and DE 108, subelement 01, subfield 01 (Receiver/Recipient First name), subfield 02 (Receiver/Recipient Middle Name), and subfield 03 (Receiver/Recipient Last Name) are not present.	will not forward DE 108, subelement 07 to the issuer
DE 108, subelement 07, subfield 01 (Name Validation Request Type) contains value 11 (Sender Name Validation Request) and DE 108, subelement 02, subfield 01 (Sender First name), subfield 02 (Sender Middle Name), and subfield 03 (Sender Last Name), are not present.	will not forward DE 108, subelement 07 to the issuer

Subfield 01 (Name Validation Request)

DE 108, subelement 07, subfield 01 (Name Validation Request) indicates the type of validation being requested from the originator.

Table 1137: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	n-2
Data Field	Contents of subfield 01
Justification	N/A

Table 1138: Values

Value	Description
10	Receiver Name Validation Request
11	Sender Name Validation Request

Application notes

Subfield 01 is optional for non-financial authorization transactions. If subfield 01 is present, name information must be provided in DE 108, subelement 01 (Receiver/Recipient Data) or subelement 02 (Sender Data).

Subelement 08 (Data Validation Response)

DE 108 (Additional Transaction Reference Data), subelement 08 (Data Validation Response) contains the Account Status verification response in the Authorization request response message.

Table 1139: Attributes

Attribute	Description
Subelement ID	08
Subelement Data Length	3 positions
Data Representation	an...010; LLLVAR
Data Field	Contents of subfields
Subfields	Subfield 1 (Receiver/Recipient Service Response) Subfield 2 (Sender Service Response)
Justification	See subfields

Usage

Whether use is mandatory, conditional, optional, or not required is noted for each applicable message.

Table 1140: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	C	.	C

If provided, Mastercard will remove subelement 08 from the Authorization Request/0100.

Authorization Response/0110 message edits for DE 108, subelement 08

Mastercard will perform the following edits.

Table 1141: Associated edits

When...	Then the Authorization Platform...
DE 108, subelement 07, subfield 01 (Name Validation Request Type) contains value 11 (Sender Name Validation Request) and DE 108, subelement 08, subfield 01 (Receiver Name Validation Response) includes a value from the issuer	will not forward DE 108, subelement 08, subfield 01 (Receiver Name Validation Response) to the acquirer
DE 108, subelement 07, subfield 01 (Name Validation Request Type) contains value 10 (Receiver Name Validation Request) and DE 108, subelement 08, subfield 02 (Sender Name Validation Response) includes a value from the issuer	will not forward DE 108 subelement 08, subfield 02 (Sender Name Validation Response) to the acquirer

Subfield 01 (Receiver Name Validation Response)

DE 108, subelement 08, subfield 01 (Receiver Name Validation Response) contains the name validation result from the issuer.

Table 1142: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	an-1
Data Field	Contents of subfield 01

Attribute	Description
Justification	N/A

Table 1143: Values

Position	Attribute	Description
1	an-1	Receiver Name Validation Response Contains the results of an attempt to verify that the following data present in the validation request message matched the data on file with the issuer: • Receiver First Name (DE 108, subelement 01, subfield 01) • Receiver Middle Name (DE 108, subelement 01, subfield 02) • Receiver Last Name (DE 108, subelement 01, subfield 03) Valid values include: • A = Match • B = Partial Match • C = No Match • U = Unverified

Application notes

Subfield 01 is conditional for non-financial transactions.

Application notes for Non-Financial Transactions

If DE 108, subelement 07, subfield 01 is received with value '10' in a non-financial Authorization request, and DE 108 subelement 01 (Receiver/Recipient Data), subfield 01 (Receiver/Recipient First name), subfield 02 (Receiver/Recipient Middle Name), and subfield 03 (Receiver/Recipient Last Name) are present, then the issuer should include DE 108, subelement 08, subfield 01 in the response. Issuer provided response is not edited and will be forwarded to the destination. When an issuer opts in for the Name Match service, Mastercard will issue a response in DE 108, subelement 08, subfield 01 on the issuer's behalf. Should a participating issuer provide a value in this subfield, it will be disregarded and replaced with the value provided by Mastercard.

Subfield 02 (Sender Name Validation Response)

DE 108, subelement 08, subfield 02 (Sender Name Validation Response) contains the originator name validation results for the sender.

Table 1144: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	an-1
Data Field	Contents of subfield 02
Justification	N/A

Table 1145: Values

Position	Attribute	Description
1	an-1	Sender Name Validation Response Contains the results of an attempt to verify that the following data present in the validation request message matched the data on file with the issuer: • Sender First Name (DE 108, subelement 02, subfield 01) • Sender Middle Name (DE 108, subelement 02, subfield 02) • Sender Last Name (DE 108, subelement 02, subfield 03) Valid values include: • A = Match • B = Partial Match • C = No Match • U = Unverified

Application notes

Subfield 02 is conditional for non-financial Authorization transactions.

Application notes for Non-Financial Authorization Transactions

If DE 108, subelement 07, subfield 01 is received with value '11' in a non-financial Authorization request, and DE 108 subelement 02 (Sender Data), subfield 01 (Sender First name), subfield 02 (Sender Middle Name), and subfield 03 (Sender Last Name) are present, then the issuer should

include DE 108, subelement 08, subfield 02 in the response. Issuer provided response is not edited and will be forwarded to the destination. When an issuer opts in for the Name Match service, Mastercard will issue a response in Data Element 108, subelement 08, subfield 02 on the issuer's behalf. Should a participating issuer provide a value in this subfield, it will be disregarded and replaced with the value provided by Mastercard.

Subelement 09 (Account Owner Information 1)

DE 108, subelement 09 (Account Owner Information 1) contains the account owner name, sequence number and classification. The account owner information is used for the Name Match Service.

Table 1146: Attributes

Attribute	Description
Subelement ID	09
Data Representation	ans...129; LLLVAR
Subelement Length	3
Data Field	Contents of subfields
Subfields	Subfield 1 (Account Owner First Name) Subfield 2 (Account Owner Middle Name) Subfield 3 (Account Owner Last Name) Subfield 4 (Account Owner Sequence Number) Subfield 5 (Account Owner Classification)
Justification	See subfields

Following is the usage of DE 108, subelement 09 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 1147: Usage

Message	Org	Sys	Dst
Authorization Request Response/0110	C	.	.

Application notes

If provided, Mastercard will remove DE 108, subelement 09 from the Authorization Request/0100.

Authorization Response/0110 message edits for DE 108, subelement 09

Authorization Platform edits for edits for DE 108, subelement 09 are as follows.

When...	Then the Authorization Platform...
an Authorization Request Response/0110 message contains data in DE 108, subelement 09 (Account Owner Information 1) defined with an incorrect length and/or is incorrectly formatted (Data Representations).	will remove DE 108, subelement 09 (Account Owner Information 1) and the message continues processing.
an Authorization Request Response/0110 message contains subfields in DE 108, subelement 09 (Account Owner Information 1) defined with an incorrect length and/or is incorrectly formatted (Data Representations).	will remove the incorrect subfields in DE 108, subelement 09 (Account Owner Information 1) and the message continues processing.
An Authorization Request Response/0110 message contains DE 108, subelement 09 (Account Owner Information 1) with subfields that have multiple instances of the same subelement or subfield within the same data element.	sends an Authorization Response Negative Acknowledgment/0190 message to the issuer with: - DE 39 = 30 - DE 44 = 108

Subfield 01 (Account Owner First Name)

DE 108, subelement 09, subfield 01 (Account Owner First Name) contains the first name of the account owner.

Table 1148: Attributes

Attribute	Description
Subfield ID	01
Data Representation	ans...35; LLVAR
Subfield Length	2
Data Field	Contents of subfield 01
Justification	N/A

Values

If present, cannot contain all spaces or all numeric values.

Application notes

Subfield 01 is optional if the issuer is opted-in for the on-behalf Name Match Service.

Subfield 02 (Account Owner Middle Name)

DE 108, subelement 09, subfield 02 (Account Owner Middle Name) contains the middle name of the account owner.

Table 1149: Attributes

Attribute	Description
Subfield ID	02
Data Representation	ans...35; LLVAR
Subfield Length	2
Data Field	Contents of subfield 02
Justification	N/A

Values

If present, cannot contain all spaces or all numeric values.

Application notes

Subfield 02 is optional if the issuer is opted-in for the on-behalf Name Match Service.

Subfield 03 (Account Owner Last Name)

DE 108, subelement 09, subfield 03 (Account Owner Last Name) contains the last name of the account owner.

Table 1150: Attributes

Attribute	Description
Subfield ID	03
Data Representation	ans...35; LLVAR
Subfield Length	2
Data Field	Contents of subfield 03
Justification	N/A

Values

If present, cannot contain all spaces or all numeric values.

Application notes

Subfield 03 is optional if the issuer is opted-in for the on-behalf Name Match Service.

Subfield 04 (Account Owner Sequence Number)

DE 108, subelement 09, subfield 04 (Account Owner Sequence Number) contains the sequence number of the account owner.

Table 1151: Attributes

Attribute	Description
Subfield ID	04
Data Representation	n-3
Subfield Length	2
Data Field	Contents of subfield 04
Justification	N/A

Values

If present, must be numeric.

Application notes

Subfield 04 is optional if the issuer is opted-in for the on-behalf Name Match Service.

Subfield 05 (Account Owner Classification)

DE 108, subelement 09, subfield 05 (Account Owner Classification) contains the classification category (Personal or Business) of the account owner.

Table 1152: Attributes

Attribute	Description
Subfield ID	05
Data Representation	a-1
Subfield Length	2
Data Field	Contents of subfield 05
Justification	N/A

If present, must contain one of these following valid values.

Table 1153: Values

Value	Description
P	Personal
B	Business

Application notes

Subfield 05 is optional if the issuer is opted-in for the on-behalf Name Match Service.

Subelement 10 (Account Owner Information 2)

DE 108, subelement 10 (Account Owner Information 2) contains the details of the account owner for a given PAN. The account owner information is used for the on-behalf Name Match service.

Table 1154: Attributes

Attribute	Description
Subelement ID	10
Data Representation	ans...129; LLLVAR
Subelement Length	3
Data Field	Contents of subfields
Subfields	Subfield 1 (Account Owner First Name) Subfield 2 (Account Owner Middle Name) Subfield 3 (Account Owner Last Name) Subfield 4 (Account Owner Sequence Number) Subfield 5 (Account Owner Classification)
Justification	See subfields

Whether use is mandatory, conditional, optional, or not required is noted for each applicable message.

Table 1155: Usage

Message	Org	Sys	Dst
Authorization Request Response/0110	C	.	.

Application notes

If provided from originator, Mastercard will remove DE 108, subelement 10 from the Authorization Request/0100 message.

Authorization Response/0110 message edits for DE 108, subelement 10

Authorization Platform edits for DE 108, subelement 10 are as follows.

When...	Then the Authorization Platform...
an Authorization Request Response/0110 message contains data in DE 108, subelement 10 (Account Owner Information 2) defined with an incorrect length and/or is incorrectly formatted (Data Representations).	will remove DE 108, subelement 10 (Account Owner Information 2) and the message continues processing.
an Authorization Request Response/0110 message contains subfields in DE 108, subelement 10 (Account Owner Information 2) defined with an incorrect length and/or is incorrectly formatted (Data Representations)	will remove the incorrect subfields in DE 108, subelement 10 (Account Owner Information 2) and the message continues processing.
An Authorization Request Response/0110 message contains DE 108, subelement 10 (Account Owner Information 2) with subfields that have multiple instances of the same subelement or subfield within the same data element.	sends an Authorization Response Negative Acknowledgment/0190 message to the issuer with: • DE 39 = 30 • DE 44 = 108

Subfield 01 (Account Owner First Name)

DE 108, subelement 10, subfield 01 (Account Owner First Name) contains the first name of the account owner.

Table 1156: Attributes

Attribute	Description
Subfield ID	01
Data Representation	ans...35; LLVAR

Attribute	Description
Subfield Length	2
Data Field	Contents of subfield 01
Justification	N/A

Values

If present, cannot contain all spaces or all numeric values.

Application notes

Subfield 01 is optional if the issuer is opted-in for the on-behalf Name Match Service.

Subfield 02 (Account Owner Middle Name)

DE 108, subelement 10, subfield 02 (Account Owner Middle Name) contains the middle name of the account owner.

Table 1157: Attributes

Attribute	Description
Subfield ID	02
Data Representation	ans...35; LLVAR
Subfield Length	2
Data Field	Contents of subfield 02
Justification	N/A

Values

If present, cannot contain all spaces or all numeric values.

Application notes

Subfield 02 is optional if the issuer is opted-in for the on-behalf Name Match Service.

Subfield 03 (Account Owner Last Name)

DE 108, subelement 10, subfield 03 (Account Owner Last Name) contains the last name of the account owner.

Table 1158: Attributes

Attribute	Description
Subfield ID	03
Data Representation	ans...35; LLVAR
Subfield Length	2
Data Field	Contents of subfield 03
Justification	N/A

Values

If present, cannot contain all spaces or all numeric values.

Application notes

Subfield 03 is optional if the issuer is opted-in for the on-behalf Name Match Service.

Subfield 04 (Account Owner Sequence Number)

DE 108, subelement 10, subfield 04 (Account Owner Sequence Number) contains the sequence number of the account owner.

Table 1159: Attributes

Attribute	Description
Subfield ID	04
Data Representation	n-3
Subfield Length	2
Data Field	Contents of subfield 04
Justification	N/A

Values

If present, must be numeric.

Application notes

Subfield 04 is optional if the issuer is opted-in for the on-behalf Name Match Service.

Subfield 05 (Account Owner Classification)

DE 108, subelement 10, subfield 05 (Account Owner Classification) contains the classification category (Personal or Business) of the account owner.

Table 1160: Attributes

Attribute	Description
Subfield ID	05
Data Representation	a-1
Subfield Length	2
Data Field	Contents of subfield 05
Justification	N/A

If present, must contain one of these valid values.

Table 1161: Values

Value	Description
P	Personal
B	Business

Application notes

Subfield 05 is optional if the issuer is opted-in for the on-behalf Name Match Service.

Subelement 11 (Account Owner Information 3)

DE 108, subelement 11 (Account Owner Information 3) contains the details of the account owner for a given PAN. The account owner information is used for the name validation service.

Table 1162: Attributes

Attribute	Description
Subelement ID	11
Data Representation	ans...129; LLLVAR

Attribute	Description
Subelement Length	3
Data Field	Contents of subfields
Subfields	Subfield 1 (Account Owner First Name) Subfield 2 (Account Owner Middle Name) Subfield 3 (Account Owner Last Name) Subfield 4 (Account Owner Sequence Number) Subfield 5 (Account Owner Classification)
Justification	See subfields

Whether the data element is mandatory, conditional, optional, system-provided, or not required is noted for each applicable message.

Table 1163: Usage

Message	Org	Sys	Dst
Authorization Request Response/0110	C	.	.

Authorization Response/0110 message edits for DE 108, subelement 11

Authorization Platform edits for DE 108, subelement 11 are as follows.

When...	Then the Authorization Platform...
an Authorization Request Response/0110 message contains subfields in DE 108, subelement 11 (Account Owner Information 3) defined with an incorrect length and/or is incorrectly formatted (Data Representations).	will remove DE 108, subelement 11 (Account Owner Information 3) and the message continues processing.
an Authorization Request Response/0110 message contains sunfields in DE 108, subelement 11 (Account Owner Information 1) defined with an incorrect length and/or is incorrectly formatted (Data Representations)	will remove the incorrect subfields in DE 108, subelement 11 (Account Owner Information 3) and the message continues processing.

When...	Then the Authorization Platform...
An Authorization Request Response/0110 message contains DE 108, subelement 11 (Account Owner Information 3) with subfields that have multiple instances of the same subelement or subfield within the same data element.	sends an Authorization Response Negative Acknowledgment/0190 message to the issuer with: - DE 39 = 30 - DE 44 = 108

Subfield 01 (Account Owner First Name)

DE 108, subelement 11, subfield 01 (Account Owner First Name) contains the first name of the account owner.

Table 1164: Attributes

Attribute	Description
Subfield ID	01
Data Representation	ans...35; LLVAR
Subfield Length	2
Data Field	Contents of subfield 01
Justification	N/A

Values

If present, cannot contain all spaces or all numeric values.

Application notes

Subfield 01 is optional if the issuer is opted-in for the on-behalf Name Match Service.

Subfield 02 (Account Owner Middle Name)

DE 108, subelement 11, subfield 02 (Account Owner Middle Name) contains the middle name of the account owner.

Table 1165: Attributes

Attribute	Description
Subfield ID	02
Data Representation	ans...35; LLVAR
Subfield Length	2

Attribute	Description
Data Field	Contents of subfield 02
Justification	N/A

Values

Valid values will consist of the middle name of the Account Owner.

Application notes

Subfield 02 is optional if the issuer is opted-in for the on-behalf Name Match Service.

Subfield 03 (Account Owner Last Name)

DE 108, subelement 11, subfield 03 (Account Owner Last Name) contains the last name of the account owner.

Table 1166: Attributes

Attribute	Description
Subfield ID	03
Data Representation	ans...35; LLVAR
Subfield Length	2
Data Field	Contents of subfield 03
Justification	N/A

Values

If present, cannot contain all spaces or all numeric values.

Application notes

Subfield 03 is optional if the issuer is opted-in for the on-behalf Name Match Service.

Subfield 04 (Account Owner Sequence Number)

DE 108, subelement 11, subfield 04 (Account Owner Sequence Number) contains the sequence number of the account owner.

Table 1167: Attributes

Attribute	Description
Subfield ID	04

Attribute	Description
Data Representation	n-3
Subfield Length	2
Data Field	Contents of subfield 04
Justification	N/A

Values

If present, must be numeric.

Application notes

Subfield 04 is optional if the issuer is opted-in for the on-behalf Name Match Service.

Subfield 05 (Account Owner Classification)

DE 108, subelement 11, subfield 05 (Account Owner Classification) contains the classification category (Personal or Business) of the account owner.

Table 1168: Attributes

Attribute	Description
Subfield ID	05
Data Representation	a-1
Subfield Length	2
Data Field	Contents of subfield 05
Justification	N/A

If present, must contain one of these valid values.

Table 1169: Values

Value	Description
P	Personal
B	Business

Application notes

Subfield 05 is optional if the issuer is opted-in for the on-behalf Name Match Service.

DE 109 (Reserved for ISO Use)

DE 109 (Reserved for ISO Use) is reserved for ISO use.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 110 (Additional Data-2)

DE 110 (Additional Data-2) is reserved for use based on product type.

Table 1170: Attributes

Attribute	Description
Data Representation	ans...999; LLLVAR
Data Length	3 positions, value = ...999
Data Field	Contents of subelements 9–10
Subelements	2
Justification	See Subelements

Usage

Whether the usage is mandatory, conditional, optional, system-provided, or not required is as noted for each applicable message.

Table 1171: Applicable messages

Message	Org	Sys	Dst
Network Management Request/0800: PEK exchange	•	O	C
Network Management Request Response/0810: PEK exchange	O	M	•
Network Management Advice/0820: PEK exchange	•	M	M

Application notes

DE 110 provides supplemental data in a message when a specific ISO-designated data element is not available. It is a free-format, variable-length alphanumeric data element that may be used for multiple purposes. This data element's content may vary by program and service.

The following tables provide formats and descriptions for the subelements in DE 110. Currently, there are two subelements but as subelements are added, the subelement sequence will not have to be in the order of ID value.

Subelement 9 (ANSI X9 TR-31 Key Block Key [128-bit Key Block Protection Key])

DE 110, subelement 9 (ANSI X9 TR-31 Key Block Key [128-bit Key Block Protection Key]) contains the attributes provided in a Network Management Request/0800 message.

Table 1172: Attributes

Attribute	Description
Subelement ID	09
Subelement Data Length	3 positions
Data Representation	an-80
Data Field	Contents of subelement 09
Subfield	N/A
Justification	N/A

Usage

Whether the usage is mandatory, conditional, optional, system-provided, or not required is as noted for each applicable message.

Table 1173: Applicable messages

Message	Org	Sys	Dst
Network Management Request/0800: PEK exchange	•	M	M
Network Management Request Response/0810: PEK exchange	O	C	•
Network Management Request Advice/0820: PEK exchange	•	•	•

Subelement 9 (ANSI X9 TR-31 Key Block Key [192-bit Key Block Protection Key])

DE 110, subelement 9 (ANSI X9 TR-31 Key Block Key [192-bit Key Block Protection Key]) contains the attributes provided in a Network Management Request/0800 message.

Table 1174: Attributes

Attribute	Description
Subelement ID	09
Subelement Data Length	3 positions
Data Representation	an-96
Data Field	Contents of subelement 09
Subfields	N/A
Justification	N/A

Usage

Whether the usage is mandatory, conditional, optional, system-provided, or not required is as noted for each applicable message.

Table 1175: Applicable messages

Message	Org	Sys	Dst
Network Management Request/0800: PEK exchange	•	M	M
Network Management Request Response/0810: PEK exchange	O	C	•
Network Management Request Advice/0820: PEK exchange	•	•	•

Subelement 10 (Key Check Value)

DE 110, subelement 10 (Key Check Value) contains the attributes in a Network Management Request/0800 message.

Table 1176: Attributes

Attribute	Description
Subelement ID	10
Subelement Data Length	3 positions
Data Representation	an-6

Attribute	Description
Data Field	Contents of subelement 10
Subfields	N/A
Justification	N/A

Usage

Whether the usage is mandatory, conditional, optional, system-provided, or not required is as noted for each applicable message.

Table 1177: Applicable messages

Message	Org	Sys	Dst
Network Management Request/0800: PEK exchange	•	M	M
Network Management Request Response/0810: PEK exchange	O	C	•
Network Management Request Advice/0820: PEK exchange	•	M	M

DE 110 (Encryption Data)

DE 110 (Encryption Data) is used in PIN encryption data and key management encryption data.

Table 1178: Attributes

Attribute	Description
Data Representation	b...999; LLLVAR
Data Length	b...999 (EBCIDIC)
Dataset ID	1 byte binary value
Dataset Length	2 byte binary value
Data Field	Dataset Tags
Tag contents	Values
Justification	N/A

Usage

Whether the data element is mandatory, conditional, optional, system-provided, or not required is noted for each applicable message.

Table 1179: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	X	C
Network Management Request/0800: PEK exchange	.	X	C
Network Management Request/0800: sign-on/sign-off	C	X	.
Network Management Advice/0820: PEK exchange	.	.	M

Application notes

The length fields in the Tag Length Value (TLV) are coded according to ISO 8825. If the TLV-coded version of a dataset is used the length of the dataset itself is always coded as a two byte binary value.

- The Dataset ID is the first component in a dataset.
- Dataset ID is a one byte binary identifier given to each dataset within a DE 110 field.
- Dataset length is a two byte binary subfield that contains the total length of the TLV element(s) within the dataset.
- The Tag field contains one byte length hexadecimal code that identifies the content of the Value field.
- The Tag length field follows the ISO 8825 spec. n-byte field that defines the length of the Value field.
- The Value field is a variable-length field that will contain the data specified by the Tag field.

DE 110 layout

DE 110 (Encryption Data) is a usage layout to support the PCI Standards Security Council's mandate for ISO Format 4 PIN Blocks. The layout follows DE 110 specifications in ISO/DIS 13492.

Hierarchy of DE 110 for PIN encryption data and key exchange data

DE 110 contains:

- Dataset ID
- Dataset length
- Tags

Each Tag contains:

- Tag ID

- Tag length
- Tag value

Table 1180: Entity descriptions

Entity	Description
Dataset ID	A one byte hex value specific for each data set Example: 0x01
Dataset length	A two byte hex value. Example: If the length is 24 it will be represented as 0x00 18. This length is the accumulation of all Tags.
Tag ID	A one byte hex specific for each Tag: for example, 0x80

Entity	Description
Tag length	The length field is coded according to ISO 8825. The Dataset length is always coded as a two byte binary value. ISO 8825 defines two forms length octets. - Definite form - Indefinite form (not supported by Mastercard) The definite form has a short and long form. Short form: Tag length consists of a single octet or byte. This form can only be used if the number of octets in the Tag data is less than or equal to 127 bytes. It is encoded as: - Bit 8 is zero - Bits 7 to 1 encode the number of octets or bytes in the contents octets (Tag data) as an unsigned binary integer with bit 7 as the most significant bit. Example: If the Tag data is 38 bytes the Tag length is encoded as: <pre>00100110</pre> Long form: the Tag length consists of an initial octet and one or more additional octets. The initial octet will be encoded as: - Bit 8 is one - Bits 7 to 1 shall encode the number of additional octets in the length octets, as an unsigned binary integer with bit 7 as the most significant bit the value 11111111 must not be used. Coding for additional octets: - First additional octet bits 8 to 1 - Second additional octet bits 8 to 1 - Each additional octet bits 8 to 1 Octets will be encoded as unsigned binary integer equal to the number of octets in the Tag data, with bit 8 of the first additional octet as the most significant bit. Examples: - If the Tag data is 201 bytes the Tag length would be encoded as <pre>10000001 11001001</pre> 10000001: Initial byte which tells how many additional length bytes. 11001001: The actual length byte. - If the tag data is 402 bytes the Tag length would be encoded as <pre>10000010 00000001 10010010</pre>

Entity	Description
Tag value	If the data representation is <i>b</i> it is binary • <i>b</i>-1 is a value of 21 specified by 0x15 • <i>b</i>-2 is a value of 21 specified by 0x0015 • <i>b</i>-8 is a value of 21 specified by 0x00000000000000015 If the data representation is <i>n</i>, only right padded Binary-Coded Decimal (BCD) is allowed and only values 0 to 9. • <i>n</i>-4 is a value of 112 stored in BCD as 0x01, 0x12 • <i>n</i>-8 is a value of 112 stored in BCD as 0x00, 0x00, 0x01, 0x12

Dataset 01 (PIN Encryption)

DE 110, Dataset 01 (PIN Encryption) contains the personal identification number (PIN), a unique identifier used by the cardholder at the point-of-interaction (POI).

Table 1181: Attributes

Attribute	Description
Dataset ID	b-1
Dataset Length	b-2
Data Representation	b...999; LLLVAR
Data Field	cryptogram
Number of Tags	7
	TAG 80 (Control), Mandatory TAG 81 (Key-set Identifier), Mandatory TAG 83 (Algorithm), Conditional TAG 86 (Key Index), Conditional TAG 87 (PIN Block Format), Mandatory TAG 88 (Encrypted PIN Block), Mandatory TAG 89 (Additional Encrypted PIN Block [New PIN Data]), Conditional

TAG 80 (Control)

DE 110, Dataset 01, TAG 80 (Control) field identifies the key management scheme and associated structure of the remainder of the data element.

Table 1182: Attributes

Attribute	Description
Tag ID	80
Tag Data Length	b-1
Data Representation	b-1
Justification	N/A
Required	Mandatory

Table 1183: Values

Value	Description
00	Default value for the Dual Message System

TAG 81 (Key-set Identifier)

DE 110, Dataset 01, TAG 81 (Key-set Identifier) Key-set identifier field is used with DUKPT key management exclusively. The key-set identifier uniquely identifies the base derivation key as defined in the *American National Standard for Financial Services ...Retail Financial Services Symmetric Key Management Part 1: Using Symmetric Techniques (ANSI X9.24 Part 1)*.

Table 1184: Attributes

Attribute	Description
Tag ID	81
Tag Data Length	b-1
Data Representation	b-8
Justification	N/A
Required	Mandatory

Table 1185: Values

Value	Description
000000000000000000	Default value for the Dual Message System

TAG 83 (Algorithm)

DE 110, Dataset 01, TAG 83 (Algorithm) selects the encryption algorithm used to encipher the keys contained in the associated key management data element.

Table 1186: Attributes

Attribute	Description
Tag ID	83
Tag Data Length	b-1
Data Representation	n-2 (Packed binary coded decimal)
Justification	N/A
Required	Conditional

Table 1187: Values

Value	Description
01	Data Encryption Algorithm (DEA)
03	Triple Data Encryption Algorithm (TDEA)
05	Advanced Encryption Standard (AES)

Application notes

TAG 83 is for online PIN transactions performed on the Authorization System for acquirers and issuers in the Europe region. It is not a requirement for acquirers outside of the Europe region.

TAG 86 (Key Index)

DE 110, Dataset 01, TAG 86 (Key Index) indicates the specific PIN key to be used when more than one key is available in a PIN key set.

Table 1188: Attributes

Attribute	Description
Tag ID	86
Tag Data Length	b-1
Data Representation	n-2 (Packed binary coded decimal)
Justification	N/A
Required	Conditional

Table 1189: Values

Value	Description
00-99	PIN key index value for the Dual Message System

Application notes

TAG 86 is for online PIN transactions performed on the Authorization System for acquirers and issuers in the Europe region. It is not a requirement for acquirers outside of the Europe region.

TAG 87 (PIN Block Format)

DE 110, Dataset 01, TAG 87 (PIN Block Format) is the format of the PIN Block defined in ISO 9564-1.

Attributes**Table 1190: Attributes**

Attribute	Description
Tag ID	87
Tag Data Length	b-1
Data Representation	n-2 (Packed binary coded decimal)
Justification	N/A
Required	Mandatory

Table 1191: Values

Value	Description
10	ISO PIN Block Format 0
11	ISO PIN Block Format 1
14	ISO PIN Block Format 4

TAG 88 (Encrypted PIN Block)

DE 110, Dataset 01, TAG 88 (Encrypted PIN Block) contains PIN block encryption as defined in ISO 9564-1.

Table 1192: Attributes

Attribute	Description
Tag ID	88
Tag Data Length	b-1
Data Representation	b-8 or b-16
Justification	N/A
Required	Mandatory

Table 1193: Values

Value	Description
8 byte of PIN Block	Format 0
16 bytes of PIN Block	Format 4

TAG 89 (Additional Encrypted PIN Block)

DE 110, Dataset 01, TAG 89 (Additional Encrypted PIN Block) is used for PIN change transaction.

Table 1194: Attributes

Attribute	Description
Tag ID	89

Attribute	Description
Tag Data Length	b-1
Data Representation	b-8 or b-16
Justification	N/A
Required	Conditional

Table 1195: Values

Value	Description
8 bytes of additional PIN Block	Format 0
16 bytes of additional PIN Block	Format 4

Dataset 04 (Key Exchange)

Dataset 04 (Key Exchange) is a method in cryptography by which cryptographic keys are exchanged between two parties, allowing use of a cryptographic algorithm.

Table 1196: Attributes

Attribute	Description
Dataset ID	b-1
Dataset Length	b-2
Data Representation	b...999; LLLVAR
Data Field Contains	cryptogram
Number of Tags	6
	TAG 80 (Control), Mandatory
	TAG 81 (Key-set Identifier), Mandatory
	TAG 83 (Algorithm), Mandatory
	TAG 86 (Key Index), Mandatory
	TAG 87 (Encrypted Data), Conditional
	TAG 88 (Key Checksum Value), Conditional

Usage

Whether the data element is mandatory, conditional, optional, system-provided, or not required is noted for each applicable message.

Table 1197: Applicable messages

Message	Org	Sys	Dst
Network Management Request/0800: PEK exchange	.	O	C
Network Management Request/0800: sign-on/sign-off	C	C	.
Network Management Advice/0820: PEK exchange	.	M	M

TAG 80 (Control)

DE 110, Dataset 04, TAG 80 (Control) field identifies the key management scheme and associated structure of the remainder of the data element.

Table 1198: Attributes

Attribute	Description
Tag ID	80
Tag Data Length	b-1
Data Representation	b-1
Justification	N/A
Required	Mandatory

Table 1199: Values

Value	Description
00	Default value for the Dual Message System

TAG 81 (Key-set Identifier)

DE 110, Dataset 04, TAG 81 (Key-set Identifier) is a number that uniquely identifies a key-set.

Table 1200: Attributes

Attribute	Description
Tag ID	81
Tag Data Length	b-1
Data Representation	b-8
Justification	N/A
Required	Mandatory

Table 1201: Values

Value	Description
0000000000000000	Default value for the Dual Message System

TAG 83 (Algorithm)

DE 110, Dataset 04, TAG 83 (Algorithm) defines the encryption algorithm used to encipher the keys contained in the associated key management data element.

Table 1202: Attributes

Attribute	Description
Tag ID	83
Tag Data Length	b-1
Data Representation	n-2 (Packed binary coded decimal)
Justification	N/A
Required	Mandatory

Table 1203: Values

Value	Description
01	Data Encryption Algorithm (DEA)

Value	Description
03	Triple Data Encryption Algorithm (TDEA)
05	Advanced Encryption Standard (AES)

TAG 86 (Key Index)

DE 110, Dataset 04, TAG 86 (Key Index) is used when multiple keys are identified with the same key set identifier; for example, key rotation.

Table 1204: Attributes

Attribute	Description
Tag ID	86
Tag Data Length	b-1
Data Representation	n-2 (Packed binary coded decimal)
Justification	N/A
Required	Mandatory

Table 1205: Values

Value	Description
00	Network Management Request/0800: PEK exchange Network Management Request/0820: PEK exchange
00-99	Network Management Request/0800: sign-on/sign-off

TAG 87 (Encrypted Data)

DE 110, Dataset 04, TAG 87 (Encrypted Data) uses a structure within it before encryption to identify the data.

Table 1206: Attributes

Attribute	Value
Tag ID	87

Attribute	Value
Tag length	n-byte
Data representation	b-8 to b-999; LLLVAR
Justification	N/A
Required	Conditional

Values

Hexadecimal values 0-9 and A-F, of the 64 bits of the new encryption key, encrypted under the current communications key.

TAG 88 (Key Checksum Value)

DE 110, Dataset 04, TAG 88 (Key Checksum Value) contains a value used to verify a conveyed key.

Table 1207: Attributes

Attribute	Value
Tag ID	88
Tag Data Length	n-byte
Data Representation	b-2 or b-3; LLLVAR
Justification	N/A
Required	Conditional

Values

The key check value consists of the hexadecimal characters, 0-9 and A-F, of the calculated check value.

Edits

The Dual Message Authorization System will perform the following edits.

Authorization Request/0100 message

Dual Message Authorization System edits for DE 110 (Encryption Data) are listed.

Table 1208: Associated edits

When...	Then the Dual Message Authorization System...
DE 110, Dataset 01 (PIN Encryption), and TAG 80 (Key Management scheme) is not present in Dataset 01	will return to the acquirer an Authorization Request Response/0110 message where: • DE 39 (Response Code) is 30 (Format error) • DE 44 (Additional Response Data) is 110 (indicating DE 110 is in error).
DE 110, Dataset 01, and TAG 81 (Key Set Identifier) is not present in Dataset 01	will return to the acquirer an Authorization Request Response/0110 message where: • DE 39 (Response Code) is 30 (Format error) • DE 44 (Additional Response Data) is 110 (indicating DE 110 is in error).
DE 110, Dataset 01, and TAG 87 (PIN Block Format Code) is not present in Dataset 01	will return to the acquirer an Authorization Request Response/0110 message where: • DE 39 (Response Code) is 30 (Format error) • DE 44 (Additional Response Data) is 110 (indicating DE 110 is in error).
DE 110, Dataset 01, and TAG 88 (Encrypted PIN Block) is not present in Dataset 01	will return to the acquirer an Authorization Request Response/0110 message where: • DE 39 (Response Code) is 30 (Format error) • DE 44 (Additional Response Data) is 110 (indicating DE 110 is in error).
an Authorization Request/0100 message contains DE 110, Dataset 01 and DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) is not 02, 05, 07, 80, 81, 82, 90, or 91	will return to the acquirer an Authorization Request Response/0110 message where: • DE 39 (Response Code) is 30 (Format error) • DE 44 (Additional Response Data) is 110 (indicating DE 110 is in error).
an Authorization Request/0100 message of DE 03 (Processing Code), is 92 (PIN Change) and does not contain DE 110, Dataset 01, TAG 89 (New PIN Data)	will return to the acquirer an Authorization Request Response/0110 message where: • DE 39 (Response Code) is 30 (Format error) • DE 44 (Additional Response Data) is 110 (indicating DE 110 is in error).

When...	Then the Dual Message Authorization System...
an Authorization Request/0100 message with DE110, Dataset 01, Tag 88 is present and acquirer keys managed by Authorization, TAG 83 or TAG 86 not present	will return to the acquirer an Authorization Request Response/0110 message where: - DE 39 (Response Code) is 30 (Format error) - DE 44 (Additional Response Data) is 110 (indicating DE 110 is in error).
an Authorization Request/0100 message DE 110, Dataset 01, TAG 87 is not 10 or 14	will return to the acquirer an Authorization Request Response/0110 message where: - DE 39 (Response Code) is 30 (Format error) - DE 44 (Additional Response Data) is 110 (indicating DE 110 is in error).
an Authorization Request/0100 message contains both: - DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode), value 03 (PAN auto-entry via barcode reader) and - DE 52 (Personal ID Number [PIN] Data) or DE 110, Dataset 01, TAG 88	rejects the message and sends the acquirer an Authorization Request Response/0110 message where DE 39 (Response Code) is 58 (Transaction not permitted to acquirer/terminal).
an Authorization Request/0100 message for a Mastercard Electronic card contains: DE 22, subfield 1, value 02, 03, 05, 07, 09, 80, 90, or 91, and DE 18 (Merchant Type) MCC value is 5542 (Fuel Dispensers, Automated), and DE 52 (Personal ID Number [PIN] Data), and DE 110, Dataset 01, TAG 88 or DE 55 (Integrated Circuit Card [ICC] System-related Data) is not present, and DE 61 (Point-of-Service [POS] Data), subfield 10 (Cardholder Activated Terminal Level) is not a value of 1 (Authorized Level 1 CAT: Automated dispensing machine with a PIN)	sends an Authorization Request Response/0110 message to the acquirer with DE 39 (Response Code), value 58 (Transaction not permitted to acquirer/terminal).
DE 61, subfield 10, value 1 and DE 52 and DE 110, Dataset 01, TAG 88 is not present and DE 55 is not present	sends an Authorization Request Response/0110 message to the acquirer where DE 39 is 30 and DE 44 is 52.

When...	Then the Dual Message Authorization System...
an Authorization Request/0100 message for a Cardless ATM transaction: - Contains DE 18 (Merchant Type), value 6011 (Member Financial Institution: Automated Cash Disbursements), and - DE 48 (Additional Data: Private Data), Transaction Category Code (TCC), value Z (ATM), and - DE 22, subfield 1 (POS Terminal PAN Entry Mode) is present and contains a value other than 09 (PAN/Token entry via electronic commerce containing DSRP cryptogram in DE 55), but DE 52, and - DE 110, Dataset 01, TAG 88 is not present	rejects the message and sends the acquirer an Authorization Request Response/0110 message where: - DE 39 (Response Code) is 30 (Format error) - DE 44 (Additional Response Data) is 52 (indicating the data element in error).
an Authorization Request/0100 message if - DE 3, subfield 1 contains value 91 (PIN unblock) or 92 (PIN Change) and - DE 52 and DE 110, Dataset 01, TAG 88 is not present	sends the acquirer an Authorization Request Response/0110 message where: - DE 39 (Response Code) is 30 (Format error) - DE 44 (Additional Response Data) is 52 (indicating the data element in error).
the Authorization Request/0100 message contains - DE 61, subfield 10, value 1 and - DE 52 and DE 110, Dataset 01, TAG 88 is not present and - DE 55 is present	forwards the Authorization Request/0100 message to the issuer.
the Authorization Request/0100 message contains - DE 61, subfield 10, value 1 and - DE 52 or DE 110, Dataset 01, TAG 88 is present and - DE 55 is present	forwards the Authorization Request/0100 message to the issuer.
the Authorization Request/0100 message contains - DE 61, subfield 10, value 1 and - DE 52 or DE 110, Dataset 01, TAG 88 is present and - DE 55 is not present	forwards the Authorization Request/0100 message to the issuer.

When...	Then the Dual Message Authorization System...
an authorization Request/0100 message, DE 3, subfield 1 contains value 92 (PIN Change) and DE 125 (New PIN Data) and DE 110, Dataset 01, TAG 89 is not present	sends the acquirer an Authorization Request Response/0110 message where: - DE 39 (Response Code) is 30 (Format error) - DE 44 (Additional Response Data) is 125 (indicating the data element in error).

DE 111 (Reserved for ISO Use)

DE 111 (Reserved for ISO Use) is reserved for ISO use.

NOTE: The Dual Message Authorization System currently does not use this data element.

DE 112 (Additional Data [National Use])

DE 112 (Additional Data [National Use]) is reserved for national organizations to define data unique to specific networks or specific programs and services. DE 112 provides other supplemental data in a message when a specific ISO-designated data element is not available. It is a free-format, variable-length, alphanumeric data element used for information on transactions between customers.

Table 1209: Attributes

Attribute	Value
Data Representation	a-1 (for Hungary domestic usage only for processing transactions for Employee Benefits Program).
Data Length	3 positions
Data Field	Contents of subelements
Subelements	Determined by program
Justification	N/A

Usage

Following is the usage of DE 112 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 1210: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Request Response/0110	C	.	C
Authorization Advice/0120: acquirer-generated	C	.	C
Authorization Advice/0120: system-generated	.	C	C
Reversal Request/0400	C	.	C
Reversal Request Response/0410	C	.	C

Values

See subfields.

Application notes

NOTE: Mastercard does not perform currency conversion on any amounts in DE 112.

DE 112 (Encoding Scheme)

Mastercard organizes DE 112 into a group of encoded subelements. The following table illustrates the structure of DE 112.

Table 1211: DE structure

VAR: maximum length varies by program; refer to Data Representation previously for maximum length by program						
LLL						
3 bytes	3 bytes	3 bytes	variable by program	3 bytes	3 bytes	variable by program
Total Length	First Subelement (SE) Data			Second Subelement (SE) Data		
	SE ID	SE Length	SE Variable Length Data	SE ID	SE Length	SE Variable Length Data

Number of Bytes	Attribute	Description
3	Total Length field	The "LLL" portion of the data element up to maximum allowed by program.
3	Subelement ID	In the range 000–099

Number of Bytes	Attribute	Description
		000–069 Defined universally for all programs and services.
		070–099 Defined for individual programs and services.
3	Subelement Length	Varies by program.
1...nnn	Subelement Variable Length Data	Contains valid values.

DE 112 (Dual Message Authorization System Edits)

The Dual Message Authorization System will perform the following system edit to verify proper formatting of DE 112.

Table 1212: Associated edits

WHEN...	THEN the Dual Message Authorization System...
an Authorization Request/0100, Authorization Advice/0120: acquirer-generated, or Reversal Request/0400 message contains DE 48 (Additional Data: Private Use), DE 108 (Additional Transaction Reference Data), or DE 112 (Additional Data [National Use]) with subelements that have incorrect length and/or incorrect format (Data Representations), or have multiple instances of the same subelement (when not permitted) within the same data element.	rejects the message and forwards the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 message where DE 44 is 6 positions for subelement format errors: - DE 39 (Response Code) = 30 (Message format error) - DE 44 (Additional Response Data) = 0480nn, or 1080nn, or 1120nn (where nn is the subelement number) (DE 44 is three positions for Dual Message System (Authorization) if no subelements are present, for example, for DE 22 format error: DE 44 = 022)
DE 112, subelement 045 (Additional Commercial Issuer Data) is not properly formatted	rejects the Authorization Request Response/0110 message and respond back to the Issuer with an Authorization Response Negative Acknowledgment/0190 message with DE 39 (Response Code) = 30 (Format error) and DE 44 (Additional Response Data) = 112.
DE 112, subelement 045 (Additional Commercial Issuer Data) is provided for non-commercial cards	removes the subelement before forwarding the Authorization Request Response/0110 message to the acquirer.

WHEN...	THEN the Dual Message Authorization System...
Stand-In processing occurs on a commercial card B2B transaction	will not provide DE 112, subelement 045.

Examples

- When an edit error occurs on DE 48, the DE 44 data will be populated as these examples below:
 - Error on DE 48 subelement 42, subfield 1: 048042 (no subfield information is provided)
 - Error on DE 48 subelement 61: 048061
- DE 48 TCC format error will be responded with DE 44 = 048000.

For non-DE 48/DE 108/DE 112 format errors, DE 44 will only have DE info and no subelement information. For example, the format error in DE 22 will have DE 44 as 022 for non-DE48/108/112 data elements.

Subelement 042 (Acceptor Tax ID)

Subelement 042 (Acceptor Tax ID) will be used in the Dual Message Authorization System to support DE 112 in any received Authorization Request/0100, Authorization Advice/0120, or Reversal Request/0400 message.

Table 1213: Attributes

Attributes	Value
Subelement ID	042
Data Length	3 positions
Data Length Value	Variable length, 001–012 positions
Data Representation	an...012; LLLVAR
Data Field	Contents of subelement 042
Justification	N/A

Usage

Whether the usage is mandatory, conditional, optional, system-provided, or not required is as noted for each applicable message.

Table 1214: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	X	.
Authorization Advice/0120: acquirer-generated	C	X	.
Reversal Request /0400	C	X	.

Table 1215: Values

Field name	Comment
Description	Tax ID
Base ID	n-11 (no dots or dashes), left filled with zeros
Check digit	an-1

Application notes

This field is required when purchase, purchase with cash back, and refund transactions are submitted by acquirers for merchants in Chile and Paraguay.

The length value is defined as variable length for all markets. However, Chile and Paraguay should use 12 positions exclusively with leading zeros as shown in the examples.

Table 1216: Examples

Tax ID	Format required
18.638.773-K	00018638773K
7.932.677-1	000079326771
763.567-4	000007635674

Subelement 045 (Additional Commercial Issuer Data)

DE 112, subelement 045 is used in the Dual Message Authorization System to provide data required by acquirers and issuers for intracountry processing to be compliant with local regulation for corporate issuer data for B2B transactions.

DE 112, subelement 045 must not be provided in cross-border transactions.

Table 1217: Attributes

Attribute	Description
Subelement ID	045
Subelement Data Length	3 positions
Data Representation	ans...075; LLLVAR
Data Field	Contents of subfields 01-03
Subfields	Subfield 01 (Taxpayer Country Code) Subfield 02 (Taxpayer Identification Number) Subfield 03 (Tax Registration Reason Code)

Usage

This is the usage of DE 112, subelement 045 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 1218: Applicable messages

Message type identifier	Org	Sys	Dst
Authorization Request Response/0110	C	.	C

Subfield 01 (Taxpayer Country Code)

DE 112, subelement 045, subfield 01 is for 3-position alphabetic taxpayer country code of the business buyer, legal entity responsible for the cardholder contract with the issuer of the business or corporate card.

Table 1219: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	a...3; LLVAR
Data Field	Contents of subfield 01
Justification	Left

Subfield 02 (Taxpayer Identification Number)

DE 112, subelement 045, subfield 02 is for unique tax identification code of the business buyer, assigned by the tax authority.

Table 1220: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	ans...30; LLVAR
Data Field	Contents of subfield 02
Justification	N/A

Subfield 03 (Tax Registration Reason Code)

DE 112, subelement 045, subfield 03 is for tax register reason code of the business buyer, assigned by the tax authority.

Table 1221: Attributes

Attribute	Description
Subfield ID	03
Subfield Data Length	2 positions
Data Representation	ans...30; LLVAR
Data Field	Contents of subfield 03
Justification	N/A

Subelement 055 (Brazil Benefits Payment Data)

DE 112, subelement 055 (Brazil Benefits Payment Data) contains data to support processing of intracountry transactions for the Mastercard Benefits Payment Scheme in Brazil.

Table 1222: Attributes

Attributes	Description
Subelement ID	055
Subelement Data Length	3 positions

Attributes	Description
Data Representation	an...11; LLVAR
Data Field	Contents of subfields 01 - 02
Subfields	2
Justification	See subfields

Usage

Following is the usage of subelement 055 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 1223: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Request Response/0110	C	.	C

Values

See subfields.

Subfield 01 (PAT Eligibility)

DE 112, subelement 055, subfield 01 is provided by acquirers to identify merchants that are PAT (Programa de Alimentação do Trabalhador) eligible when a Brazil domestic transaction is processed as part of the Mastercard Benefits Payment Scheme.

Table 1224: Attributes

Attributes	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	a-1
Data Field	Contents of subfield 01
Justification	N/A

Table 1225: Values

Value	Description
Y	Eligible
N	Not Eligible

Subfield 02 (Benefit Program Code)

DE 112, subelement 055, subfield 02 is provided by issuers to identify the specific benefit program applicable to a Brazil domestic transaction when processed within the Mastercard Benefits Payment Scheme.

Table 1226: Attributes

Attributes	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	n-2
Data Field	Contents of subfield 02
Justification	N/A

Table 1227: Values

Value	Description
01	Meal Voucher: Employee Nutrition Program (PAT)
02	Food Voucher: Employee Nutrition Program (PAT)
03	Culture Voucher: Worker's Culture Program (PCT)
04	Meal Voucher: Consolidation of Labor Laws (CLT)
05	Food Voucher: Consolidation of Labor Laws (CLT)

Application notes

Acquirers should use subfield 01 (PAT Eligibility) in the Authorization Request/0100 message to indicate merchant PAT eligibility.

Issuers should use subfield 02 (Benefit Program Code) in the Authorization Request Response/0110 message to indicate the specific benefit program applicable to a transaction.

It is not required for issuers to echo back the subfield 01 in the Authorization Request Response/0110 message, however if provided, it will be passed to the acquirer.

Subelement 60 (Mastercard Network Gateway Services Platform [MNGS] Transport Data) (Mastercard use only)

DE 112, subelement 060 (Mastercard Network Gateway Services Platform [MNGS] Transport Data) (Mastercard use only), is reserved solely for MNGS (Mastercard Network Gateway Services Platform) transport data.

Table 1228: Attributes

Attribute	Description
Subelement ID	060
Subelement Data Length	3 positions
Data Representation	n-399
Data Field	Contents of positions 1–399
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 060 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 1229: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	M	•	C
Authorization Request Response/0110	C	•	C
Authorization Advice/0120: acquirer-generated	C	•	C
Authorization Advice Response/0130: issuer-generated	C	•	C
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	C	•	C
Authorization Advice Response/0130: system-generated	C	•	C
Reversal Request/0400	M	•	C
Reversal Request Response/0410	C	•	C

Values

This subfield contains MNGS specific application transport data.

Application notes

MNGS acquired transactions requiring certain services fulfillment will populate subelement 60 to carry application specific data.

All Regions: Installment Payment Transactions

Following are the technical specifications for Authorization/01xx messages for the global Mastercard Installment Payment Service. Refer to the [Mastercard Installment Payment Service User Guide](#) for additional details.

Subelement 21 (Installment Payment Data 1)

Subelement 21 (Installment Payment Data 1) is used in the Authorization System for installment payment transactions.

Table 1230: Attributes

Attribute	Description
Subelement ID	021
Subelement Data Length	3 positions
Data Representation	an-3
Data Field	Contents of positions 1–3
Subfields	N/A
Justification	Right with leading zeros

Usage

Following is the usage of subelement 021 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 1231: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	.	X	C
Authorization Request Response/0110	C	.	C
Authorization Advice/0120: acquirer-generated	O	.	C
Reversal Request/0400	O	.	O

Message	Org	Sys	Dst
Reversal Request Response/0410	O	.	O

Table 1232: Values

Position	Attribute	Description
1-2	n-2	Installment Type Contains the type of installment. Valid values include: • 20 = Issuer-Financed • 21 = Merchant-Financed • 22 = Acquirer-Financed • 23 = Co-branded Merchant Financed • 24 = Issuer Merchant Co-Financed
3	a-1	Payment Options Payment options provided by issuer. Valid values include: • I (Pay in Installments Only) • F (Pay in Full Only) • B (Pay in Full or Pay in Installments)

Subelement 22 (Installment Payment Data 2)

Subelement 22 (Installment Payment Data 2) is used in the Authorization System for installment payment transactions.

Table 1233: Attributes

Attribute	Description
Subelement ID	022
Subelement Data Length	3 positions
Data Representation	ns...720; LLLVAR The "LLL" length field of LLLVAR must be an integral multiple of 12, not to exceed 720.
Data Field	Contents of subfields 1-7
Subfields	7
Justification	See subfields

Usage

Following is the usage of subelement 022 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 1234: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	.	X	C
Authorization Request Response/0110	C	.	C
Authorization Advice/0120—Acquirer-generated	O	.	C
Reversal Request/0400	O	.	O
Reversal Request Response/0410	O	.	O

Values

See subfields.

Application notes

An “occurrence” is defined as one set of the seven subfields in subelement 022. Depending on the conditions of the message, subelement 022 can be sent by the acquirer, by the issuer, or by Mastercard.

Subfield 1 (Number of Installments)

DE 112, subelement 022, subfield 1 (Number of Installments) contains the number of installment payments provided by the issuer and selected by the cardholder.

Table 1235: Attributes

Attribute	Description
Data Representation	n-2
Data Field	Contents of positions 1–2
Justification	Right with leading zeros

Values

Valid values are 02–99.

Subfield 2 (Interest Rate)

DE 112, subelement 022, subfield 2 (Interest Rate) contains the rate (two decimal places) that the issuer assesses the cardholder for the installment payment.

Table 1236: Attributes

Attribute	Description
Data Representation	ns-5
Data Field	Contents of positions 3–7
Justification	Right with leading zeros

Values

Will be populated as spaces if interest rate is not required to be displayed in the terminal.

Subfield 3 (Installment Fee)

DE 112, subelement 022, subfield 3 (Installment Fee) contains the fee amount in cardholder billing currency that the issuer assesses the cardholder for the installment payments.

Table 1237: Attributes

Attribute	Description
Data Representation	ns-12
Data Field	Contents of positions 8–19
Justification	Right with leading zeros

Values

Will be populated as spaces if installment fee is not required to be displayed in the terminal.

Subfield 4 (Annual Percentage Rate)

DE 112, subelement 022, subfield 4 (Annual Percentage Rate [APR]) contains the rate (two decimal places) that the issuer charges the cardholder for the installment payment.

Table 1238: Attributes

Attribute	Description
Data Representation	ns-5
Data Field	Contents of positions 20–24

Attribute	Description
Justification	Right with leading zeros

Values

Will be populated as spaces if Annual Percentage Rate is not required to be displayed in the terminal.

Subfield 5 (First Installment Amount)

DE 112, subelement 022, subfield 5 (First Installment Amount) contains the amount of the first installment in cardholder billing currency that the issuer will charge the cardholder for the installment payments.

Table 1239: Attributes

Attribute	Description
Data Representation	ns-12
Data Field	Contents of positions 25–36
Justification	Right with leading zeros

Values

Default is all spaces.

Subfield 6 (Subsequent Installment Amount)

DE 112, subelement 022, subfield 6 (Subsequent Installment Amount) contains the amount of the subsequent installments in cardholder billing currency that the issuer will charge the cardholder for the installment payments.

Table 1240: Attributes

Attribute	Description
Data Representation	ns-12
Data Field	Contents of positions 37–48
Justification	Right with leading zeros

Values

Default is all spaces.

Subfield 7 (Total Amount Due)

DE 112, subelement 022, subfield 7 (Total Amount Due) contains the total amount due in cardholder billing currency that the issuer charges the cardholder for the installment payments.

Table 1241: Attributes

Attribute	Description
Data Representation	ns-12
Data Field	Contents of positions 49–60
Justification	Right with leading zeros

Values

Default is all spaces.

Subelement 23 (Installment Payment Data 3)

Subelement 23 (Installment Payment Data 3) is used in the Authorization System for installment payment transactions.

Table 1242: Attributes

Attribute	Description
Subelement ID	023
Subelement Data Length	3 positions
Data Representation	ns-38
Data Field	Contents of positions 1–38
Subfields	N/A
Justification	Right with leading zeros

Usage

Following is the usage of subelement 023 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 1243: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	.	X	C

Message	Org	Sys	Dst
Authorization Request Response/0110	C	.	C
Reversal Request/0400	O	.	O
Reversal Request Response/0410	O	.	O

Table 1244: Values

Position	Description
1–2	Minimum Number of Installments—Contains the number of installment payments provided by the issuer and selected by the cardholder. Valid values are 02–99.
3–4	Maximum number of installments—Contains the number of installment payments provided by the issuer and selected by the cardholder. Valid values are 02–99.
5–9	Interest Rate—Contains the interest rate (2 decimal) that the issuer charges the cardholder for the installment payments. Will be populated as spaces if Interest Rate is not required to be displayed in the terminal.
10–21	Installment Fee—Contains the fee amount that the issuer charges the cardholder for the installment payments in cardholder billing currency. Will be populated as spaces if Installment Fee is not required to be displayed in the terminal. Default is all zeros.
22–26	Annual Percentage Rate (APR)—Contains the annual percentage rate (2 decimal) that the issuer charges the cardholder for the installment payment. Will be populated as spaces if Annual Percentage Rate is not required to be displayed in the terminal.
27–38	Total Amount Due—Contains the total amount due in cardholder billing currency that the issuer charges the cardholder for the installment payment.

Alternate Processing

Installment payments are not eligible for alternate processing (Stand-In System or X-Code System). When an issuer is not available, or offline, the Installment Payment transactions will be

declined using the Authorization Request Response/0110 message containing DE 39 (Response Code), value 91 (Authorization System or issuer system inoperative).

Brazil: Payment Transactions

Following are the technical specifications for Authorization/01xx messages for Brazil payment transactions within Brazil.

Subelement 012 (Brazil Commercial and Financing Data)

DE 112, subelement 012 (Brazil Commercial and Financing Data) contains additional data to support financing data provided by the acquirer, merchant, or issuer.

NOTE: For data requirements for Brazil intracountry transactions using the Mastercard BNDES card, refer to the section that follows. The following table applies to non-BNDES transactions only.

Table 1245: Attributes

Attribute	Description
Subelement ID	012
Subelement Data Length	3 positions
Data Representation	ans-170
Data Field	Contents of positions 1–170
Subfields	N/A
Justification	N/A

Table 1246: Values

Field Name/Positions	Type	Size	Comment
Financing Type (Positions 1–20)	Alpha	20	Type of loan operation, agreed between Merchant and issuer. Examples: AGRO-CUSTEIO, AGRO-INVEST, FINAME, CBN.
Buyer ID Type (Position 21)	Alpha	1	Possible values F or J: F = pessoa Física (person/consumer) J = pessoa Jurídica (company/commercial)

Field Name/Positions	Type	Size	Comment
Buyer ID (Positions 22–35)	Numeric	14	Merchant to fill in with cardholder's CPF (Cadastro de Pessoa Física, a registration number provided by the Brazil government to all persons living in Brazil) or company's CNPJ (Cadastro Nacional da Pessoa Jurídica, registration number provided by the Brazil government to all merchants) number or issuers to supply with company's CNPJ number.
Buyer Phone Number (Positions 36–46)	Numeric	11	Merchant to fill in with buyer phone number.
Purchase Identification (Positions 47–66)	Alphanumeric	20	Invoice data or purchase request data, depending on agreement between merchant and issuer.
Installments Cycle (Positions 67–68)	Numeric	2	Different amortization types than monthly. For future use, it does not affect transaction clearing and settlement.
Interest Rate (Positions 69–86)	Numeric	18 (12.6)	Interest rate on loan. For future use.
Grace Period (Positions 87–89)	Numeric	3	Number of amortization cycles to wait before charging the installments. For future use, it does not affect transaction clearing and settlement.
Grace Period Cycle (Positions 90–92)	Numeric	3	Allow different grace period types than monthly. For future use, it does not affect transaction clearing and settlement.
Grace Period Interest Rate (Positions 93–110)	Numeric	18 (12.6)	Interest rate on grace period. For future use, it does not affect transaction clearing and settlement.
Reference Field 1 (Positions 111–130)	Numeric	20	Acquirer to populate with merchant's CNPJ number.
Reference Field 2 (Positions 131–150)	Alpha	20	Data exchange between merchant and issuer, to identify the transaction.
Reference Field 3 (Positions 151–170)	Alpha	20	Data exchange between merchant and issuer, to identify the transaction.
Filler Data (Positions 171–227)	Alphanumeric	57	Reserved for future Brazil domestic enhancements.

Mastercard Brazil BNDES Card Data Requirements

Mastercard has enhanced the Mastercard Brazilian Development Bank (BNDES) Card to increase card issuance and usage in Brazil and support alternative financing options for small and medium-sized businesses.

Mastercard supports the presence of additional data fields for Brazil intracountry transactions submitted with product code MLD (Mastercard Distribution Card). This addition meets BNDES requirements, in terms of addendum data processed and sent by the BNDES generic module, through the acquirers.

The Mastercard BNDES Card is a solution to access a credit line with subsidized rates by BNDES to finance investments for micro-, small-, and medium-sized businesses established in Brazil.

This product is developed for Brazil domestic use only and transactions are sent to the Mastercard Network using the MLD (Mastercard Distribution Card) product code with a specific promotion code (BNDES1) in all transactions. These are issuer-financed installment (up to 48 installments) transactions, made exclusively through the BNDES Operations Portal.

BNDES has enhanced its Operations Portal's Generic Module, which requires acquirers to send additional data through the Mastercard Network.

NOTE: All Brazil domestic transactions using the Mastercard BNDES card must contain DE 63 (Network Data), subfield 1 (Financial Network Code), value MLD (Mastercard Distribution Card) in the authorization message. The specific promotion code value BNDES1 must be included in DE 48 (Additional Data—Private Use), subelement 95 (Mastercard Promotion Code).

In issuer-financed installment billing, the acquirer submits a single clearing record for the full transaction amount. The issuer then bills the cardholder for the installments in accordance with the terms agreed by the cardholder at the point of sale.

NOTE: A customer's CNPJ number is the registration number provided by the Brazil government to all merchants.

DE 112, subelement 012 (Brazil Commercial and Financing Data) contains additional data to support Brazil intracountry, consumer credit transactions submitted with value MLD (Mastercard Distribution Card) in DE 63 (Network Data), subfield 1 (Financial Network Code).

For Brazil intracountry transactions using the Mastercard BNDES card, the following data must be included.

Table 1247: Attributes

Attribute	Description
Subelement ID	012
Subelement Data Length	3 positions
Data Representation	ans-227

Attribute	Description
Data Field	Contents of positions 1–227
Subfields	N/A
Justification	N/A

Values

- Data Element: DE 112 (Additional Data [National Use])
- Subelement: Subelement 012 (Brazil Commercial and Financing Data)

Table 1248: Positions and value

Positions	Value
Positions 1–20 (Financing Type)	CBN
Position 21 (Buyer ID Type)	J
Positions 22–35 (Buyer ID)	CNPJ number (Buyer Company Tax ID)
Positions 36–110	Spaces
Positions 111–130 (Reference Field 1)	CNPJ number (Merchant Brazilian Tax ID)
Positions 131–227	Spaces

Subelement 013 (Crediário First Simulation)

DE 112 (Additional Data [National Use]), subelement 013 contains the issuer's payment conditions for the first Crediário simulation.

Table 1249: Attributes

Attribute	Description
Subelement ID	013
Subelement Data Length	3 positions
Data Representation	n-131
Data Field	Contents of positions 1–131
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 013 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 1250: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	C	.	C
Authorization Advice/0120: acquirer-generated	C	.	C

Table 1251: Values

Field Name/Positions	Type	Size	Comment
Installment Plan Type (Positions 1–2)	Numeric	2	Identifies whether this is an issuer, a merchant, or a Crediario financed installment plan. Value 25 = Crediario
Total Number of Installments (Positions 3–4)	Numeric	2	Number of installments the cardholder chose to pay. From 02–99
Installment Amount (Positions 5–16)	Numeric	12	Value that the cardholder is going to pay per month. Parcelas (or installment) amount issuer calculates, including calculated interest; two decimal places.
Total Amount (Positions 17–28)	Numeric	12	Total Installment Amount (Monthly installment amount multiplied by the Number of Installments). Transaction total amount issuer calculates, including calculated interest; two decimal places.
Monthly Interest Rate (Positions 29–32)	Numeric	4	Interest Rates the Issuer calculates and charges per month; two decimal places.
Annual Interest Rate (Positions 33–37)	Numeric	5	Interest Rates the issuer calculates and charges per year; two decimal places.
Monthly Total Effective Cost (CET) (Positions 38–49)	Numeric	12	Total amount charged monthly to cardholder including the payments, interest, fees, insurance, and other amounts; two decimal places.

Field Name/Positions	Type	Size	Comment
% Total Effective Cost (Positions 50–61)	Numeric	12	Total Effective Cost, expressed as a percentage; two decimal places.
Date of first installment (Positions 62–67)	Numeric	6	Date of the first installment payment in yymmdd (year, month, day) format.
Taxes (Positions 68–74)	Numeric	7	All taxes, if applicable, that the cardholder is paying, including IOF; two decimal places.
% Taxes of Total Amount (Positions 75–78)	Numeric	4	Percentage of the whole purchase price that the taxes represent.
Fee (Positions 79–85)	Numeric	7	All fees, if applicable, that the cardholder is paying; two decimal places.
% Fee of Total Amount (Positions 86–89)	Numeric	4	Percentage of the whole purchase price that the fees represent.
Insurance (Positions 90–96)	Numeric	7	Any insurance, if applicable, that the cardholder is paying; two decimal places.
% Insurance of Total Amount (Positions 97–100)	Numeric	4	Percentage of the whole purchase price that the insurance represents.
Other (Positions 101–107)	Numeric	7	Any other added values, if applicable, that the cardholder is paying; two decimal places.
% Other of Total Amount (Positions 108–111)	Numeric	4	Percentage of the whole purchase price that any other added values represent.
Total Amount to Merchant (Positions 112–123)	Numeric	12	Total amount that will be paid to the merchant (purchase amount); two decimal places.
% Amount to Merchant of Total Amount (Positions 124–127)	Numeric	4	Percentage of the total amount that will be paid to the merchant.
Reserved for Future Use (Positions 128–131)	Numeric	4	

Application notes

This subelement should be provided to indicate the merchant terminal's capabilities in supporting the first option for the Crediário installment program and services.

Information not populated or optional to a specific field should be zero-filled.

Subelement 014 (Crediário Second Simulation)

DE 112 (Additional Data [National Use]), subelement 014 contains the issuer's payment conditions for the second Crediário simulation.

Table 1252: Attributes

Attribute	Description
Subelement ID	014
Subelement Data Length	3 positions
Data Representation	n-131
Data Field	Contents of positions 1–131
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 014 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 1253: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	C	.	C
Authorization Advice/0120: acquirer-generated	C	.	C

Table 1254: Values

Field Name/Positions	Type	Size	Comment
Installment Plan Type (Positions 1–2)	Numeric	2	Identifies whether this is an issuer, a merchant, or a Crediario financed installment plan. Value 25 = Crediario
Total Number of Installments (Positions 3–4)	Numeric	2	Number of installments the cardholder chose to pay. From 02–99

Field Name/Positions	Type	Size	Comment
Installment Amount (Positions 5–16)	Numeric	12	Value that the cardholder is going to pay per month. Parcelas (or installment) amount issuer calculates, including calculated interest; two decimal places.
Total Amount (Positions 17–28)	Numeric	12	Total Installment Amount (Monthly installment amount multiplied by the Number of Installments). Transaction total amount issuer calculates, including calculated interest; two decimal places.
Monthly Interest Rate (Positions 29–32)	Numeric	4	Interest Rates the Issuer calculates and charges per month; two decimal places.
Annual Interest Rate (Positions 33–37)	Numeric	5	Interest Rates the issuer calculates and charges per year; two decimal places.
Monthly Total Effective Cost (CET) (Positions 38–49)	Numeric	12	Total amount charged monthly to cardholder including the payments, interest, fees, insurance, and other amounts; two decimal places.
% Total Effective Cost (Positions 50–61)	Numeric	12	Total Effective Cost, expressed as a percentage; two decimal places.
Date of first installment (Positions 62–67)	Numeric	6	Date of the first installment payment in yymmdd (year, month, day) format.
Taxes (Positions 68–74)	Numeric	7	All taxes, if applicable, that the cardholder is paying, including IOF; two decimal places.
% Taxes of Total Amount (Positions 75–78)	Numeric	4	Percentage of the whole purchase price that the taxes represent.
Fee (Positions 79–85)	Numeric	7	All fees, if applicable, that the cardholder is paying; two decimal places.
% Fee of Total Amount (Positions 86–89)	Numeric	4	Percentage of the whole purchase price that the fees represent.
Insurance (Positions 90–96)	Numeric	7	Any insurance, if applicable, that the cardholder is paying; two decimal places.
% Insurance of Total Amount (Positions 97–100)	Numeric	4	Percentage of the whole purchase price that the insurance represents.
Other (Positions 101–107)	Numeric	7	Any other added values, if applicable, that the cardholder is paying; two decimal places.
% Other of Total Amount (Positions 108–111)	Numeric	4	Percentage of the whole purchase price that any other added values represent.

Field Name/Positions	Type	Size	Comment
Total Amount to Merchant (Positions 112–123)	Numeric	12	Total amount that will be paid to the merchant (purchase amount); two decimal places.
% Amount to Merchant of Total Amount (Positions 124–127)	Numeric	4	Percentage of the total amount that will be paid to the merchant.
Reserved for Future Use (Positions 128–131)	Numeric	4	

Application notes

This subelement should be provided to indicate the merchant terminal's capabilities in supporting the second option for the Crediário installment program and services.

Information not populated or optional to a specific field should be zero-filled.

Subelement 015 (Crediário Third Simulation)

DE 112 (Additional Data [National Use]), subelement 015 contains the issuer's payment conditions for the third Crediário simulation.

Table 1255: Attributes

Attribute	Description
Subelement ID	015
Subelement Data Length	3 positions
Data Representation	n-131
Data Field	Contents of positions 1–131
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 015 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 1256: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	C	.	C

Message	Org	Sys	Dst
Authorization Advice/0120: acquirer-generated	C	.	C

Table 1257: Values

Field Name/Positions	Type	Size	Comment
Installment Plan Type (Positions 1–2)	Numeric	2	Identifies whether this is an issuer, a merchant, or a Crediario financed installment plan. Value 25 = Crediario
Total Number of Installments (Positions 3–4)	Numeric	2	Number of installments the cardholder chose to pay. From 02–99
Installment Amount (Positions 5–16)	Numeric	12	Value that the cardholder is going to pay per month. Parcelas (or installment) amount issuer calculates, including calculated interest; two decimal places.
Total Amount (Positions 17–28)	Numeric	12	Total Installment Amount (Monthly installment amount multiplied by the Number of Installments). Transaction total amount issuer calculates, including calculated interest; two decimal places.
Monthly Interest Rate (Positions 29–32)	Numeric	4	Interest Rates the Issuer calculates and charges per month; two decimal places.
Annual Interest Rate (Positions 33–37)	Numeric	5	Interest Rates the issuer calculates and charges per year; two decimal places.
Monthly Total Effective Cost (CET) (Positions 38–49)	Numeric	12	Total amount charged monthly to cardholder including the payments, interest, fees, insurance, and other amounts; two decimal places.
% Total Effective Cost (Positions 50–61)	Numeric	12	Total Effective Cost, expressed as a percentage; two decimal places.
Date of first installment (Positions 62–67)	Numeric	6	Date of the first installment payment in yymmdd (year, month, day) format.
Taxes (Positions 68–74)	Numeric	7	All taxes, if applicable, that the cardholder is paying, including IOF; two decimal places.

Field Name/Positions	Type	Size	Comment
% Taxes of Total Amount (Positions 75–78)	Numeric	4	Percentage of the whole purchase price that the taxes represent.
Fee (Positions 79–85)	Numeric	7	All fees, if applicable, that the cardholder is paying; two decimal places.
% Fee of Total Amount (Positions 86–89)	Numeric	4	Percentage of the whole purchase price that the fees represent.
Insurance (Positions 90–96)	Numeric	7	Any insurance, if applicable, that the cardholder is paying; two decimal places.
% Insurance of Total Amount (Positions 97–100)	Numeric	4	Percentage of the whole purchase price that the insurance represents.
Other (Positions 101–107)	Numeric	7	Any other added values, if applicable, that the cardholder is paying; two decimal places.
% Other of Total Amount (Positions 108–111)	Numeric	4	Percentage of the whole purchase price that any other added values represent.
Total Amount to Merchant (Positions 112–123)	Numeric	12	Total amount that will be paid to the merchant (purchase amount); two decimal places.
% Amount to Merchant of Total Amount (Positions 124–127)	Numeric	4	Percentage of the total amount that will be paid to the merchant.
Reserved for Future Use (Positions 128–131)	Numeric	4	

Application notes

This subelement should be provided to indicate the merchant terminal's capabilities in supporting the third option for the Crediário installment program and services.

Information not populated or optional to a specific field should be zero-filled.

Subelement 018 (Brazil Post-Dated Transaction Data)

DE 112, subelement 018 (Brazil Post-Dated Transaction Data) contains additional data to support post-dated transaction data provided by the acquirer.

Table 1258: Attributes

Attribute	Description
Subelement ID	018
Subelement Data Length	3 positions

Attribute	Description
Data Representation	ans-39
Data Field	Contents of positions 1–39
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 018 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 1259: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Request Response/0110	CE	.	C
Authorization Advice/0120	C	.	C
Reversal Request/0400	C	.	C

Table 1260: Values

Field Name	Positions	Attribute	Field Description
Service Code	1–2	n-2	Type of post-dated transaction. Valid Values: 30 = Post-Dated with Guarantee 31 = Post-Dated without Guarantee Must be present and equal to 30 or 31 when DE 61, subfield 7 = 5.
Number of Installments	3	n-1	Valid Values: If present, value will always be 1.
Guarantee	4	a-1	Valid Values: - Y = Yes - N = No

Field Name	Positions	Attribute	Field Description
Guarantee Amount	5–12	n-8	Amount of guarantee to be settled; assumed to be a credit to the issuer. Valid Values: Must be 00000000 if Position 1–2 = 31 (Post-Dated without Guarantee).
Post Settlement Date	13–18	n-6	Proposed settlement date (expected date for completion message arrival). Format: MMDDYY
Original Mastercard Settlement Date	19–24	n-6	Contains zeros on authorization message. Format: MMDDYY
Original Banknet Reference Number	25–33	n-9	Original Banknet reference number assigned by Banknet to original authorization request; contains zeros on authorization request/0100 message.
Authorization Code	34–39	n-6	Contains the online authorization code provided by the issuer on the original authorization response.

Dual Message Authorization System Edits

Table 1261: Post-Dated Transactions

The Dual Message Authorization System performs the following system edits on DE 112, subelement 018 (Brazil Post-Dated Transaction Data), on authorization messages submitted for Post-Dated transactions in Brazil.

WHEN...	THEN the Dual Message Authorization System...
the Authorization Request/0100 message contains DE 112 (Additional Data—National Use), subelement 018 (Brazil Post-Dated Transaction Data) and the issuer account range is not set up for the Post-Dated Payment service or the issuer and acquirer country codes are not the same	will reject the transaction with DE 39 (Response Code) = 57 (Transaction not permitted to issuer/ cardholder)
the Authorization Request/0100 message contains DE 112 (Additional Data—National Use), subelement 018 (Brazil Post-Dated Transaction Data), subfield 1 (Service Code) contains value 30 or 31 and DE 61 (Point of Service Data), subfield 7 (POS Transaction Status) does not contain value 5 (Time Based Payment Authorization Request)	will reject the transaction with - DE 39 (Response Code) = 30 (Format error) - DE 44 (Additional Response Data) = 061 (indicating the data element in error)

WHEN...	THEN the Dual Message Authorization System...
the Authorization Request/0100 message contains DE 61 (Point of Service Data), subfield 7 (POS Transaction Status) does not contain value 5 (Time Based Payment Authorization Request) and DE 112 (Additional Data—National Use), subelement 018 (Brazil Post-Dated Transaction Data), subfield 1 (Service Code) does not contain value 30 or 31	will reject the transaction with • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 112018 (indicating the data element in error)

Alternate Processing

Post-Dated transactions are not eligible for Stand-In or X-Code Processing. If an issuer is not available, or offline, the transactions will be declined using the Authorization Request Response/0110 message containing DE 39 (Response Code), value 91 (Dual Message Authorization System or issuer system inoperative).

Subelement 019 (Original Purchase Amount)

DE 112 (Additional Data [National Use]), subelement 019 contains the amount (two decimal places) that the cardholder wants to finance. Subelement 019 is to be used only for Crediário Account Status Inquiry (ASI) transactions.

Table 1262: Attributes

Attribute	Description
Subelement ID	019
Subelement Data Length	3 positions
Data Representation	n-12
Data Field	Contents of positions 1-12
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 019 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 1263: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	M	•	C

Message	Org	Sys	Dst
Authorization Request Response/0110	C	.	C
Authorization Advice/0120—Acquirer-generated	C	.	C

Values

This subelement must contain valid numeric data.

Application notes

This subelement should be provided to indicate the merchant terminal's capabilities for displaying the original purchase amount for the Crediário installment program.

Information not populated or optional to a specific field should be zero-filled.

Brazil: Merchant Fraud Scoring Data

Following are the technical specifications for Authorization Request/0100 and Authorization Request Response/0110 messages that contain merchant fraud scoring data for card-not-present (CNP) transactions from acquirers in Brazil.

Subelement 028 (Merchant Fraud Score Data)

DE 112, subelement 028 (Merchant Fraud Score Data) should be used by acquirers in Brazil to send merchant fraud score data.

Table 1264: Attributes

Attribute	Description
Subelement ID	028
Subelement Data Length	3 positions
Data Representation	ns-4
Data Field	Contents of positions 1–4
Subfields	N/A
Justification	see values

Usage

Following is the usage of subelement 028 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 1265: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	O	.	C
Authorization Request Response/0110	.	X	C

Values

Acquirers may perform merchant fraud scoring either by themselves or through payment gateways. Acquirers in Brazil can forward the merchant fraud scoring information to issuers using this subelement. Issuers can use this data to know that merchant fraud scoring is done by the acquirer and use the data to make approval decision for CNP transaction.

Merchant Fraud Score data provided by acquirers in Brazil should be in the range -999 to +999. Merchant fraud score should be right-justified with leading zeros if the value is less than three digits.

Application notes

Issuers are not expected to echo DE 112, subelement 028 (Merchant Fraud Score Data) in the Authorization Request Response/0110 message. Mastercard will echo DE 112, subelement 028 (Merchant Fraud Score Data) from Authorization Request/0100 to the acquirers in the Authorization Request Response/0110 message.

Canada domestic: Transactions

Following are the technical specifications for the Authorization/01xx messages for Canada domestic transactions.

Subelement 047 (Payment Consent ID)

DE 112 (Additional Data [National Use]), Subelement 047 (Payment Consent ID), is reserved solely for Interac's KONEK commerce platform payment consent ID.

Table 1266: Attributes

Attribute	Description
Subelement ID	047
Subelement length	3
Data representation	ans-36
Data field	Contents of positions 1-36
Subfields	N/A

Attribute	Description
Justification	N/A

Whether use is mandatory, conditional, optional, or not required is noted for each applicable message.

Table 1267: Usage

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C

Application notes

An acquirer submitting a transaction with the KONEK payment consent ID will include the KONEK payment consent ID in the Authorization Request/0100 message that the issuer will validate before approving the payment transaction in the Canada region.

Chile: Payment Transactions

Following are the technical specifications for Authorization/01xx messages for Chile payment transactions within Chile.

Subelement 010 (Installment Payment Data)

DE 112, Chile, subelement 010 is used in the Dual Message Authorization System for installment payment transactions within Chile.

Subelement 010 must be present if DE 48 (Additional Data: Private Use), subelement 95 (Mastercard Promotion Code) contains the value CHLCTA.

Subelement 010 contains the following subfields used in Chile Domestic Switch (ChIDS) processing. For details on data requirements for these subfields refer to the [Chile Domestic Switch Implementation Guide](#).

- Subfield 1: Total Purchase Amount
- Subfield 2: Number of Installments
- Subfield 3: Installment Type
- Subfield 4: Installment Amount
- Subfield 5: Transaction Interest Rate
- Subfield 6: Deferred Period
- Subfield 7: Deferred Period 1: Installment Amount
- Subfield 8: Deferred Period 1: Interest Rate
- Subfield 9: Deferred Period 1: Period Identifier
- Subfield 10: Deferred Period 2 Installment Amount
- Subfield 11: Deferred Period 2 Interest Rate

- Subfield 12: Deferred Period 2 Period Identifier
- Subfield 13: Deferred Period 3 Installment Amount
- Subfield 14: Deferred Period 3 Interest Rate
- Subfield 15: Deferred Period 3 Period Identifier
- Subfield 16: Simulation Flag
- Subfield 17: Grace Period Flag
- Subfield 18: Grace Period
- Subfield 19: Deferred Period Simulation Flag
- Subfield 20: Reserved for Annual Equivalent Charge Rate (CAE)

Subelement 042 (Acceptor Tax ID)

Subelement 042 (Acceptor Tax ID) will be used in the Dual Message Authorization System to support DE 112 in any received Authorization Request/0100, Authorization Advice/0120, or Reversal Request/0400 message.

Table 1268: Attributes

Tax ID	Description
Subelement ID	042
Subelement Data Length	3 positions
Data Representation	an...012; LLLVAR

Usage

Following is the usage of subelement 042 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages:

Table 1269: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	X	•
Authorization Advice/0120: acquirer-generated	C	X	•
Reversal Request /0400	C	X	•

Application notes

This field is required when purchase, purchase with cash back, and refund transactions are submitted by acquirers for merchants in Chile and Paraguay.

Table 1270: Attributes

Attribute	Description
Data Representation	an
Data Length	3 positions
Length Value	Variable length, 001–012 positions
Data Field	Contents of Subelement 042
Justification	N/A
Description	Tax ID

Colombia: Domestic Transactions

Following are the technical specifications for Authorization/01xx messages for Colombia domestic transactions.

Subelement 035 (Issuer Fee Inquiry Indicator)

DE 112, Colombia, subelement 035 (Issuer Fee Inquiry Indicator) is populated by the acquirer and passed to the issuer to indicate that a transaction qualifies as an Issuer Fee Inquiry.

Table 1271: Attributes

Attribute	Description
Subelement ID	035
Subelement Data Length	3 positions
Data Representation	n-2
Data Field	Contents of positions 1–2
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 035 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 1272: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C

Table 1273: Valid values

Value	Description
01	Indicates Issuer Fee Inquiry Request

Subelement 036 (Issuer Fee Amount)

DE 112, Colombia, subelement 036 (Issuer Fee Amount) is populated by the issuer with the fee amount that would be associated with the intended ATM cash withdrawal.

Table 1274: Attributes

Attribute	Description
Subelement ID	036
Subelement Data Length	3 positions
Data Representation	n-12
Data Field	Contents of positions 1–12
Subfields	N/A
Justification	Right-justified with leading zeros

Usage

Following is the usage of subelement 036 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 1275: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	C	.	C

Table 1276: Values

Value	Description
Valid numeric data	Ex: 000000000123

NOTE: Amounts are expressed without a decimal separator. For currencies that support exponents, users and systems are responsible for placing the decimal separator appropriately. Currently only for use in Colombia, where currency exponent = 2.

Colombia: Payment Transactions

Following are the technical specifications for Authorization/01xx messages for Colombia payment transactions within Colombia.

Subelement 010 (Installment Payment Data)

DE 112, Colombia, subelement 010 (Installment Payment Data) is used in the Dual Message Authorization System for Colombia payment transactions.

Subelement 010 must be present if DE 48, subelement 95 (Mastercard Promotion Code) contains the value COLCTA.

Table 1277: Attributes

Attribute	Description
Subelement ID	010
Subelement Data Length	3 positions
Data Representation	ans-2
Data Field	Contents of positions 1–2
Subfields	N/A
Justification	Left-justified with trailing spaces

Usage

Conditional.

Values

Example Values:

XX = The number of installments (01–XX)

Maximum number of installments accepted in Colombia may vary (99 as maximum value allowed).

Subelement 011 (Customer ID)

DE 112, Colombia, subelement 011 (Customer ID) is the customer associated with a voice, card not present, or manual transaction.

Table 1278: Attributes

Attribute	Description
Subelement ID	011
Subelement Data Length	3 positions
Data Representation	ans-11
Data Field	Contents of positions 1–11
Subfields	N/A
Justification	Left-justified with trailing spaces

Usage

Conditional.

Values

Example Values:

Contains customer ID information.

Cuotas: Payment Transactions

Following are the technical specifications for Authorization/01xx messages for Cuotas payment transactions within Argentina and Uruguay.

Subelement 001 (Installment Payment Data)

DE 112, Cuotas, subelement 001 (Installment Payment Data) is used in the Dual Message Authorization System for Cuotas payment transactions. Subelement 001 must be present if DE 48, subelement 95 (Mastercard Promotion Code) contains the value ARGCTA, or URYCTA.

Table 1279: Attributes

Attribute	Description
Subelement ID	001
Subelement Data Length	3 positions

Attribute	Description
Data Representation	ans...4
Data Field	Contents of positions 1–4 where: XX = positions 1–2 YY = positions 3–4
Subfields	N/A
Justification	N/A

Usage

Whether the usage is mandatory, conditional, optional, system-provided, or not required is as noted for each applicable message.

Table 1280: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	M	•	M
Authorization Request Response/0110		•	M

Values

Table 1281: Cuotas Payment Transactions in the Authorization Request/0100

Used for various Cuotas transactions: From acquirer to issuer in the following format:

XX	Cuotas plan type: 20 = Issuer-financed 21 = Merchant-financed 22 = Acquirer-financed 23 = Average payment financing 24 = Consumer financing (Purchase) 25 = Consumer financing (Manual Cash Advance) 80 = ATM Installment Inquiry 81 = ATM Installment Withdrawal
YY	The total number of Cuotas

Table 1282: Cuotas Payment Transactions in the Authorization Request Response/0110

Used for various Cuotas transactions: From issuer to acquirer in the following format:

XX	Cuotas plan type: 20 = Issuer-financed 21 = Merchant-financed 22 = Acquirer-financed 23 = Average payment financing 24 = Consumer financing (Purchase) 25 = Consumer financing (Manual Cash Advance) 80 = ATM Installment Inquiry 81 = ATM Installment Withdrawal
YY	The total number of Cuotas.

Subelement 003 (Installment Payment Response Data)

DE 112, Cuotas, subelement 003 (Installment Payment Response Data) is used in the Dual Message Authorization System and Cuotas payment transactions to provide issuer response data only for consumer financing plan types 24 and 25.

Table 1283: Attributes

Attribute	Description
Subelement ID	003
Subelement Data Length	3 positions
Data Representation	ans...55
Data Field	Contents of positions 1–55
Subfields	N/A
Justification	N/A

Consumer Financing Cuotas transactions in the Authorization Request Response/0110:

From issuer to acquirer, upon transaction approval; all amounts in transaction currency; use the following format:

Table 1284: Values

Positions	Length	Description
1–12	12	Installment amount including any issuer-calculated interest, insurance, or other charges; two decimal places.
13–17	5	Annual nominal interest percentage rate or all zeros if nominal rate not applicable; two decimal places.
18–22	5	Annual actual interest percentage rate or all zeros if actual rate not applicable; two decimal places.
23–27	5	Insurance percentage rate or all zeros if insurance not applicable; two decimal places.
28–39	12	Insurance amount or all zeros if insurance not applicable; two decimal places.
40–51	12	Issuing charge amount or all zeros if issuing charge not applicable; two decimal places.
52–53	2	Total number of installments, or all zeros if total number of installments not applicable.
54–55	2	Reserved for future use; insert all zeros.

Subelement 027 (ATM Credit Card Cash Advance Installments)

DE 112, Cuotas: Payment Transactions, subelement 027 (ATM Credit Card Cash Advance Installments) is used in the Dual Message Authorization System for credit card cash advance installment transactions performed at the ATM. Subelement 027 contains details of the cash advance installment transaction.

Table 1285: Attributes

Attribute	Description
Subelement ID	027
Subelement Data Length	3 positions
Data Representation	an-4 (Authorization Request/0100) an-137 (Authorization Request Response/0110)
Data Field	Contents of subfields Subfields 1–2 (Authorization Request/0100) Subfields 3–9 (Authorization Request Response/0110)

Attribute	Description
Subfields	9
Justification	See subfields

Usage

Following is the usage of subelement 027 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 1286: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Request Response/0110	C	.	C

Values

See subfields.

Subfield 1 (Transaction Type)

DE 112, Cuotas: Payment Transactions, subelement 027, subfield 1 (Transaction Type) contains the type of ATM credit card cash advance installment transaction.

Table 1287: Attributes

Attribute	Description
Subfield ID	1
Data Representation	an-2
Data Field	Contents of positions 1–2
Justification	N/A

Table 1288: Values

Value	Description
80	ATM Installment Inquiry
81	ATM Installment Withdrawal

Subfield 2 (Requested Number of Installments)

DE 112, Cuotas: Payment Transactions, subelement 027, subfield 2 (Requested Number of Installments) contains the requested number of payments required to payoff the credit card cash advance.

Table 1289: Attributes

Attribute	Description
Subfield ID	2
Data Representation	n-2
Data Field	Contents of positions 3–4
Justification	N/A

Values

Valid values 01–99.

Subfield 3 (Approved Number of Installments)

DE 112, Cuotas: Payment Transactions, subelement 027, subfield 3 (Approved Number of Installments) contains the number of installment payments approved by the issuer.

Table 1290: Attributes

Attribute	Description
Subfield ID	3
Data Representation	n-2
Data Field	Contents of positions 5–6
Justification	Left

Values

Valid values 01–99.

Subfield 4 (Installment Amount)

DE 112, Cuotas: Payment Transactions, subelement 027, subfield 4 (Installment Amount) contains the monthly payment amount.

Table 1291: Attributes

Attribute	Description
Subfield ID	4
Data Representation	n-12
Data Field	Contents of positions 7–8
Justification	Right with leading zeros

Subfield 5 (Total Transaction Amount)

DE 112, Cuotas: Payment Transactions, subelement 027, subfield 5 (Total Transaction Amount) contains the total amount of the credit card cash advance and the interest charged by the issuer.

Table 1292: Attributes

Attribute	Description
Subfield ID	5
Data Representation	n-12
Data Field	Contents of positions 19–30
Justification	Right with leading zeros

Subfield 6 (Yearly Interest Rate)

DE 112, Cuotas: Payment Transactions, subelement 027, subfield 6 (Yearly Interest Rate) contains the yearly interest rate that the issuer charges the cardholder for the installment payment.

Table 1293: Attributes

Attribute	Description
Subfield ID	6
Data Representation	n-4
Data Field	Contents of positions 31–34
Justification	Right with leading zeros

Subfield 7 (Currency Code)

DE 112, Cuotas: Payment Transactions, subelement 027, subfield 7 (Currency Code) contains the currency code the issuer will be charging the cardholder for repayment.

Table 1294: Attributes

Attribute	Description
Subfield ID	7
Data Representation	n-3
Data Field	Contents of positions 35–37
Justification	N/A

Subfield 8 (Member-Defined Data)

DE 112, Cuotas: Payment Transactions, subelement 027, subfield 8 (Member-defined Data) contains member data.

Table 1295: Attributes

Attribute	Description
Subfield ID	8
Data Representation	an-25
Data Field	Contents of positions 38–62
Justification	N/A

Subfield 9 (Member-Defined Data)

DE 112, Cuotas: Payment Transactions, subelement 027, subfield 9 (Member-defined Data) contains member data.

Table 1296: Attributes

Attribute	Description
Subfield ID	9
Data Representation	an-25
Data Field	Contents of positions 63–87
Justification	N/A

Europe Region and Philippines: Payment Transactions

Following are the technical specifications for Authorization/01xx messages for the Europe Region and Philippines payment transactions.

Subelement 009 (Installment Payment Data)

DE 112, subelement 009 (Installment Payment Data) is used in the Dual Message Authorization System for installment payment transactions in Georgia and Philippines.

DE 112, subelement 009 only applies when DE 48 (Additional Data: Private Use), subelement 95 (Mastercard Promotion Code) contains the value HGMINs or PHINST.

NOTE: Customers in the Europe region should refer to global release announcements and the [Mastercard Installment Payment Service User Guide](#) for information about installment payment processing. Mastercard will no longer publish this information in announcements specific to the Europe region as this service is now available globally. Refer to [All Regions: Installment Payment Transactions](#) for more information.

Table 1297: Attributes

Attribute	Description
Subelement ID	009
Subelement Data Length	3 positions
Data Representation	n-33
Data Field	Contents of Positions 1–33
Subfields	N/A
Justification	Right with leading zeros

Usage

Following is the usage of subelement 009 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 1298: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Request Response/0110	C	.	C
Reversal Request/0400	C	.	C
Reversal Request Response/0410	CE	.	CE

Table 1299: Values

Value	Description
Installment type (positions 1–2)	Installment type contains a type of installment. Valid values include 20 (Issuer-financed), 21 (Merchant-financed), and 22 (Acquirer-financed).
Number of installments (positions 3–4)	Number of installments contains the number of installment payments selected by the cardholder. Valid values are 01–99.
Interest rate (positions 5–9)	Interest rate contains the interest rate (2 decimal) that the issuer charges the cardholder for the installment payments. Default rate is 0% ("00000").
First installment amount (positions 10–21)	First installment amount contains the amount of the first installment in transaction currency.
Subsequent installment amount (positions 22–33)	Subsequent installment amount contains the amount of the subsequent installment in transaction currency.

NOTE: DE 112 (Additional Data [National Use]), subelement 009 (Installment Payment Data) is a conditional echo in the Authorization Request Response/0110 message for Philippines installment payment transactions for finance type 21 (merchant-financed).

Dual Message Authorization System Edits

The Dual Message Authorization System performs the following edits on data element 112, subelement 009 (Installment Payment Data), on authorization messages submitted for installment payment transactions in Georgia.

Table 1300: Associated edits

WHEN...	THEN the Dual Message Authorization System...
DE 48 (Additional Data: Private Use), subelement 95 (Mastercard Promotion Code) contains the value HGMINS and DE 112 (Additional Data [National Use]), subelement 009 (Installment Payment Data) is not present in an Authorization Request/0100 message	sends the acquirer an Authorization Request Response/0110 message where: - DE 39 (Response Code) = 30 (Format error) - DE 44 (Additional Response Data) = 112009

WHEN...	THEN the Dual Message Authorization System...
DE 48, subelement 95 contains the value HGMINS and	sends the issuer an Authorization Response Negative Acknowledgment/0190 message where:
DE 112, subelement 009 is present in the Authorization Request/0100 message and	- DE 39 = 30 - DE 44 (Additional Response Data) = 112009 (indicating the data element in error)
DE 112, subelement 009 is not present back in the Authorization Request Response/0110 message	Sends the acquirer an Authorization Request Response/0110 message where: DE 39 = 91 (Authorization System or issuer system inoperative) or 05 (Do not honor)

Subelement 020 (Domestic Acceptor Tax ID)

Subelement 020 (Domestic Acceptor Tax ID) is used in the Croatian Authorization System for payment transactions.

Table 1301: Attributes

Attribute	Description
Subelement ID	020
Subelement Data Length	3 positions
Data Representation	ans-20; LLVAR
Data Field	Contents of Positions 1–20
Subfields	N/A
Justification	Left with trailing spaces

Usage

Following is the usage of subelement 020 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 1302: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Request Response/0110	CE	.	CE
Reversal Request/0400	C	.	C

Message	Org	Sys	Dst
Reversal Request Response/0410	CE	.	CE

Values

Values are assigned by the acquirer.

Application notes

This field is required when installment transactions are submitted by participating acquirers in Croatia.

France: Payment Transactions

The following are technical specifications for payment transactions in France. Acquirer hosts must be able to provide the specific information to issuers through the Mastercard Authorization Request/0100 and Authorization Advice/0120 messages using data element (DE) 112 (Additional Data [National Use]) for any transaction occurring at a merchant in France.

Subelement 020 (Domestic Acceptor Tax ID)

DE 112, France, subelement 020 (Domestic Acceptor Tax ID) is used in the Dual Message Authorization System for French payment transactions. Acquirers send DE 112, subelement 020 to identify the acceptor tax ID (SIRET).

Table 1303: Attributes

Attribute	Description
Subelement ID	020
Subelement Data Length	3 positions
Data Representation	ans...14; LLLVAR
Data Field	Contents of positions 1–14
Subfields	N/A
Justification	N/A

French payment transactions in the Authorization Request/0100 and Authorization Advice/0120, SIRET value:

Table 1304: Values

Positions	Length	Description
1-3	3	To identify subelement 020
4-6	3	Overall subelement length
7-20	14	Acceptor Tax ID

Subelement 096 (Transit Offer ID)

DE 112, France, subelement 096 (Transit Offer ID) is used in the Dual Message Authorization System for French payment transactions. Acquirers send DE 112, subelement 096 to identify the cardholder transit journey.

Table 1305: Attributes

Attribute	Description
Subelement ID	096
Subelement Data Length	3 positions
Data Representation	ans...20; LLLVAR
Data Field	Contents of positions 1-20
Subfields	N/A
Justification	N/A

French payment transactions in the Authorization Request/0100 and Authorization Advice/0120, Transit Offer ID value:

Table 1306: Values

Positions	Length	Description
1-3	3	To identify subelement 096
4-6	3	Overall subelement length
7-27	20	Transit Offer ID

Subelement 097 (Application Selection ID)

DE 112, France, subelement 097 (Application Selection ID) is used in the Dual Message Authorization System for French payment transactions. Acquirers send DE 112, subelement 097 to indicate if the application was selected by the merchant or by the cardholder.

Table 1307: Attributes

Attribute	Description
Subelement ID	097
Subelement Data Length	3 positions
Data Representation	n1, LLL
Data Field	Contents of positions 1
Subfields	N/A
Justification	N/A

Values

0 = Selection by default

1 = Cardholder selection

2-9 = Reserved for future use

French payment transactions in the Authorization Request/0100 and Authorization Advice/0120, Application Selection ID value:

Table 1308: Values

Positions	Length	Description
1-3	3	Identify subelement 097
4-6	2	Subelement length
7	1	Application ID

Hungary: Benefits Program

Following are the technical specifications for transactions in Hungary.

Subelement 013 (Program ID)

DE 112, subelement 013 (Program ID) contains data to support processing of intracountry transactions for Employee Benefits Program in Hungary.

Table 1309: Attributes

Attributes	Description
Subelement ID	013
Settlement Data Length	3 positions
Data Representation	a-1
Data Field	Contents of position 1
Subfields	N/A
Justification	N/A

Table 1310: Values

Value	Description
K	Program identifier

Greece: Payment Transactions

Following are the technical specifications for Authorization/01xx messages for installment payment transactions in Greece.

Subelement 006 (Installment Payment Data)

DE 112, Greece, subelement 006 (Installment Payment Data) is used in the Dual Message Authorization System for Greece payment transactions. It must be present in Authorization Request/0100 and Authorization Request Response/0110 messages when DE 48, subelement 95 (Mastercard Promotion Code) contains the value GREECE.

Table 1311: Attributes

Attribute	Description
Subelement ID	006
Subelement Data Length	3 positions

Attribute	Description
Data Representation	ans...10
Data Field	Contents of positions 1–10
Subfields	N/A
Justification	N/A

Greece payment transactions in the Authorization Request/0100 and Authorization Request Response/0110:

Table 1312: Values

Positions	Length	Description
1–2	2	Type of Credit: 21 = Merchant Financed
3–4	2	Number of Installments • 01 to 99
5–7	3	Grace Period Before First Payment in UZZ format where: • U indicates unit type – N = No grace period – D = Days – W = Weeks – M = Months • ZZ indicates the number of days, weeks or months – 00 to 99
8–10	3	Transaction Currency Code. Must be same value as provided in DE 49 (Currency Code, Transaction). This is the local currency of the acquirer or source location of the transaction. It specifies the currency used in Amount Transaction DE 4. • 978 = Euro (EUR)

Subelement 008 (Installment Payment Response Data)

DE 112, Greece, subelement 008 (Installment Payment Response Data) is used in the Dual Message Authorization System for Greece payment transactions in the Authorization Request Response/0110.

Table 1313: Attributes

Attribute	Description
Subelement ID	008
Subelement Data Length	3 positions
Data Representation	ans...23
Data Field	Contents of positions 1–23
Subfields	N/A
Justification	N/A

Greece payment transactions in the Authorization Request Response/0110:

Table 1314: Values

Positions	Length	Description
1–12	12	Installment amount (with two decimal places) including any issuer-calculated interest, insurance, or other charges.
13–18	6	Due date of first installment (in DDMMYY format).
19–21	3	Financing Currency Code. This is the currency in which the issuer will finance the transaction. It specifies the currency using in Installment Amount above. 978 = Euro (EUR)
22–23	2	Payment Plan. Reserved for future use. Insert all zeros.

India: Payment Transactions

Following are the technical specifications for Authorization/01xx messages for installment payment transactions in India.

Subelement 044 (Installment and Recurring Payment Data: Visa Only)

DE 112, subelement 044 (Installment and Recurring Payment Data: Visa Only) is used by the Dual Message Authorization System for installment and recurring payment transactions for accounts issued in India. This subelement is used only with Visa-branded transactions.

Table 1315: Attributes

Attribute	Description
Subelement ID	044
Subelement Data Length	3 positions
Data Representation	an...117; LLLVAR
Data Field	Contents of subfields 01-12
Subfields	12
Justification	See subfields

Usage

Following is the usage of this subelement and whether it is mandatory, conditional, optional, system-provided, or not required in applicable messages.

Table 1316: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Request Response/0110	CE	.	CE
Reversal Request/0400	C	.	C
Reversal Request Response/0410	CE	.	CE

Application notes

DE 112, subelement 044 and its subfields are set up in Tag, Length, Value format. The Dual Message Authorization System removes this value if received on transactions that are not Visa-branded. The Dual Message Authorization System performs no editing on this subelement's values.

Subfield 01 (Transaction Type)

DE 112, subelement 044, subfield 01 (Transaction Type) defines characteristics based on the installment or recurring payment transaction type.

Table 1317: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	n-2
Data Field	Contents of subfield 01
Justification	N/A

Table 1318: Values

Position	Value
1	1 = Recurring payment and 2 = Installment payment.
2	1 = Registration/first transaction 2 = Subsequent transaction 3 = Modification, and 4 = Cancellation.

Subfield 02 (Recurring Payment Validation)

DE 112, subelement 044, subfield 02 (Recurring Payment Validation) identifies if recurring payment transaction has been validated by the originator.

Table 1319: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	n-1
Data Field	Contents of subfield 02
Justification	N/A

Table 1320: Values

Value	Description
0	Not validated
1	Validated

Subfield 03 (Maximum Recurring Payment Amount)

DE 112, subelement 044, subfield 03 (Maximum Recurring Payment Amount) defines the maximum amount authorized by the cardholder for a recurring payment transaction.

Table 1321: Attributes

Attribute	Description
Subfield ID	03
Subfield Data Length	2 positions
Data Representation	n-12
Data Field	Contents of subfield 03
Justification	Right with leading zeros

Values

The maximum recurring payment amount is expressed in the currency defined by the currency code (subfield 08).

Where a minor unit of currency applies, the maximum recurring payment amount is expressed in the minor unit of currency without a decimal separator (example, value 100 represents INR 1.00).

Subfield 04 (Registered Transaction Identifier)

DE 112, subelement 044, subfield 04 (Registered Transaction Identifier) provides a unique identifier for the installment and recurring payment mandate.

Table 1322: Attributes

Attribute	Description
Subfield ID	04
Subfield Data Length	2 positions
Data Representation	an-10

Attribute	Description
Data Field	Contents of subfield 04
Justification	N/A

Values

Any combination of 10 alphanumeric characters is valid.

Subfield 05 (Number of Payments)

DE 112, subelement 044, subfield 05 (Number of Payments) indicates the number of installment or recurring payment debits.

Table 1323: Attributes

Attribute	Description
Subfield ID	05
Subfield Data Length	2 positions
Data Representation	n-2
Data Field	Contents of subfield 05
Justification	Right with leading zeros

Table 1324: Values

Value	Description
00 to 97	Number of payment debits with leading zeros
98	Number of payments is not indicated
99	Payments are authorized until canceled

Subfield 06 (Payment Frequency)

DE 112, subelement 044, subfield 06 (Payment Frequency) indicates the frequency of payment debits.

Table 1325: Attributes

Attribute	Description
Subfield ID	06

Attribute	Description
Subfield Data Length	2 positions
Data Representation	n-1
Data Field	Contents of subfield 06
Justification	N/A

Table 1326: Values

Value	Description
0	Annually
1	Daily
2	Weekly
3	Biweekly
4	Monthly
5	Quarterly
6	Twice yearly
8	Ad hoc

Subfield 07 (Total Installment Amount)

DE 112, subelement 044, subfield 07 (Total Installment Amount) indicates the total purchase amount corresponding to an installment payment.

Table 1327: Attributes

Attribute	Description
Subfield ID	07
Subfield Data Length	2 positions
Data Representation	n-12
Data Field	Contents of subfield 07
Justification	Right with leading zeros

Values

The total installment amount is expressed in the currency defined by the Currency Code (subfield 08).

Where a minor unit of currency applies, the total installment amount is expressed in the minor unit of currency without a decimal separator (example, value 100 represents INR 1.00).

Subfield 08 (Currency Code)

DE 112, subelement 044, subfield 08 (Currency Code) indicates the ISO currency code associated with the maximum recurring payment amount (subfield 03) or total installment amount (subfield 07).

Table 1328: Attributes

Attribute	Description
Subfield ID	08
Subfield Data Length	2 positions
Data Representation	n-3
Data Field	Contents of subfield 08
Justification	N/A

Values

All currency codes must be selected from the numeric ISO standard currency codes. A list of valid values is available in the [Quick Reference Booklet](#).

Example: A subfield 08 value of 356 indicates the Indian rupee.

Subfield 09 (Installment Interest Rate)

DE 112, subelement 044, subfield 09 (Installment Interest Rate) indicates the interest rate which applies to the installment payment.

Table 1329: Attributes

Attribute	Description
Subfield ID	09
Subfield Data Length	2 positions
Data Representation	n-4
Data Field	Contents of subfield 09
Justification	Right with leading zeros

Values

Carries a percentage value with an implied two-position decimal.

Example: An interest rate of 1.23 percent is represented with DE 112, subelement 044, subfield 09 = 0123

Subfield 10 (Installment Payment Entity)

DE 112, subelement 044, subfield 10 (Installment Payment Entity) identifies the party responsible for managing the splitting and posting of installment payments.

Table 1330: Attributes

Attribute	Description
Subfield ID	10
Subfield Data Length	2 positions
Data Representation	an-1
Data Field	Contents of subfield 10
Justification	N/A

Table 1331: Values

Value	Description
0	Issuer
1	Merchant
2	Acquirer
3	Payment facilitator

Subfield 11 (Installment Interest Charged)

DE 112, subelement 044, subfield 11 (Installment Interest Charged) indicates whether interest is charged to an installment.

Table 1332: Attributes

Attribute	Description
Subfield ID	11
Subfield Data Length	2 positions
Data Representation	an-1
Data Field	Contents of subfield 11

Attribute	Description
Justification	N/A

Table 1333: Values

Value	Description
0	Regular payment
1	Installment payment without interest
2	Installment payment with interest

Subfield 12 (Member-Defined Data)

DE 112, subelement 044, subfield 12 (Member-Defined Data) is used to pass data between the acquirer and issuer as defined by the message originator.

Table 1334: Attributes

Attribute	Description
Subfield ID	12
Subfield Data Length	2 positions
Data Representation	an-20
Data Field	Contents of subfield 12
Justification	N/A

Values

This subfield is optional for transactions that contain installment and recurring payment data.

Japan: Payment Transactions

Following are the technical specifications for Authorization/01xx messages for Japan National Data payment transactions within Japan.

Subelement 030 (Japan Domestic POS Data)

Subelement 030 (Japan Domestic POS Data) is used in the Dual Message Authorization System to provide data required by acquirers and issuers in Japan for intracountry processing.

Table 1335: Attributes

Attribute	Description
Subelement ID	030
Subelement Data Length	3 positions
Data Representation	ans-138
Data Field	Contents of Positions 1–138
Subfields	N/A
Justification	Right with leading zeros

Usage

Following is the usage of subelement 030 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 1336: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	O	.	C
Authorization Request Response/0110	CE	.	C
Authorization Advice/0120: acquirer-generated	O	.	C
Authorization Advice/0120: system-generated	.	C	C
Authorization Advice Response/0130	CE	.	C
Reversal Request/0400	O	.	C
Reversal Request Response/0410	CE	.	.
Reversal Advice/0420: system-generated	.	C	C
Reversal Advice Response/0430	CE	.	.

Table 1337: Values

Value	Positions	Format
Terminal ID	01–13	ans-13
JIS II/ISO	14–82	ans-69 If ISO track data is populated here it must be padded with trailing spaces.
Product Code	83–89	ans-7
Reserved for future use	90	ans-1 space filled
Acquirer Company Code	91–97	an-7
Issuer Company Code	98–104	an-7
Authorization Transmission Mode	105	n-1 1 = Memory 2 = Online
Entry Indicator	106	n-1 1 = Back Stripe ISO 2 = Front Stripe JIS II 3 = Manual 4 = N/A (Domestic Private Label) 5 = IC Chip Data ISO 6 = IC Chip Data JIS II For Reversal Request/0400 messages, if the card is read using the ISO Track (Entry Mode 1), then the track data must be placed in the field JIS II/ISO instead of DE 35 (Track 2 Data) as DE 35 is not present in reversal messages. Note that the ISO track data must be padded with trailing spaces when using the field JIS II/ISO due to its reduced length.
Authorization Type	107	n-1 1 = Normal 2 = Authorization Reservations 3 = Post-Approval Authorization 4 = Card Validity Check 5 = Reversal 6 = Refund 7 = Reversal of Authorization Reservation 8 = Reversal of Post-Approval Authorization

Value	Positions	Format
Approval Number for Post-Approval Authorization	108–113	n-6
Sales Slip Number	114–118	n-5
Sales Slip Number for Reversal/Refund	119–123	n-5 Sales Slip Number for Reversal/Refund is not required when DE 03 (Processing Code), subfield 1 (Cardholder Transaction Type Code) does not equal 20 (Refund/Correction) in an authorization request message. When required, and must be all zeros when a meaningful value is not available.
Tax Amount	124–130	n-7 Format: nnnnnnn
Local Transaction Date	131–138	n-8 Format: YYYYMMDD

Application notes

Because DE 112, subelement 030 contains unique data specific to acquirers and issuers in Japan, Mastercard does not enforce the use of DE 112, subelement 030.

If provided, Mastercard does not validate the contents of DE 112, subelement 030.

If provided, Mastercard will remove subelement 030 from the Authorization Request/0100, Authorization Request Response/0110, Authorization Advice/0120, Authorization Advice Response/0130, Reversal Request/ 0400, and Reversal Request Response/0410 messages when the either acquirer or the issuer country code is not 392 (Japan).

Subelement 031 (Japan Domestic Response Code)

Subelement 031 (Japan Domestic Response Code) is used in the Dual Message Authorization System to provide data required by acquirers and issuers in Japan for intracountry processing.

Table 1338: Attributes

Attribute	Description
Subelement ID	031
Subelement Data Length	3 positions
Data Representation	ans-3

Attribute	Description
Data Field	Contents of Positions 1–3
Subfields	N/A
Justification	Right with leading zeros

Usage

Following is the usage of subelement 031 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 1339: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	O	.	C
Reversal Request Response/0410	O	.	C

Table 1340: Values

Value	Positions	Format
Error Code	01–03	ans-3

Application notes

If present, Mastercard will remove subelement 031 from the Authorization Request Response/0110 and Reversal Request Response/0410 messages when the acquirer or the issuer country code is not 392 (Japan).

Subelement 032 (Japan Payment Options)

Subelement 032 (Japan Payment Options) is used in the Dual Message Authorization System to provide data for Japan domestic transactions.

Table 1341: Attributes

Attribute	Description
Subelement ID	032
Subelement Data Length	3 positions
Data Representation	n-36

Attribute	Description
Data Field	Contents of subfields 1–8
Subfields	8
Justification	See subfields

Usage

Following is the usage of subelement 032 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 1342: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	O	.	C
Authorization Request Response/0110	CE	.	C
Authorization Advice/0120: acquirer-generated	O	.	C
Authorization Advice/0120: system-generated	.	C	C
Authorization Advice Response/0130	CE	.	C
Reversal Request/0400	O	.	C
Reversal Request Response/0410	CE	.	C
Reversal Advice/0420: system-generated	.	C	C
Reversal Advice Response/0430	CE	.	.

Values

See subfields.

Application notes

If subelement 032 is present then all subfields must also be present.

Mastercard provides subelement 032 to facilitate communication of data between the acquirer and issuer. Mastercard does not enforce edits to check the presence of any data. Transactions that do not include DE 112, subelement 032 data will not be rejected.

Mastercard will remove subelement 032 from the Authorization Request/0100, Authorization Request Response/0110, Authorization Advice/0120, Authorization Advice Response/0130, Reversal Request/0400, and Reversal Request Response/0410 messages when either the acquirer or the issuer country code is not 392 (Japan).

Subfield 1 (Payment Option Code)

DE 112, subelement 032, subfield 1 (Payment Option Code) indicates how the cardholder wants to pay for the transaction.

Table 1343: Attributes

Attribute	Description
Data Representation	n-2
Data Field	Contents of positions 1–2
Justification	Left

Table 1344: Values

Value	Description
10	Single Payment
21	Bonus Pay
22	Bonus Pay (with number of bonuses)
23	Bonus Pay (with bonus month)
24	Bonus Pay (with number of bonuses and bonus month)
31	Bonus Pay with Installments
32	Bonus Pay with Installments (with bonus amount)
33	Bonus Pay with Installments (with number of bonuses, bonus month)
34	Bonus Pay with Installments (with number of bonuses, bonus month, bonus amount)
61	Installments (Twice Pay/Three or More)
80	Revolving

Application notes

This field is required and must contain one of the listed values.

Subfield 2 (Bonuses Per Year)

DE 112, subelement 032, subfield 2 (Bonuses Per Year) indicates if the cardholder receives one or two bonuses per year.

Table 1345: Attributes

Attribute	Description
Data Representation	n-2
Data Field	Contents of positions 3–4
Justification	Right with leading zeros

Table 1346: Values

Value	Description
01	One bonus per year
02	Two bonuses per year
00	Not Applicable

Application notes

This field must contain values 01 or 02 for Payment Option Codes 22, 24, 33, or 34; otherwise zero filled.

Subfield 3 (First Bonus Month)

DE 112, subelement 032, subfield 3 (First Bonus Month) identifies month of the next bonus payment received by the cardholder following the purchase date.

Table 1347: Attributes

Attribute	Description
Data Representation	n-2
Data Field	Contents of positions 5–6
Justification	Right with leading zeros

Table 1348: Values

Value	Description
01–12	Month
00	Not Applicable

Application notes

This field must contain a value 01 to 12 for Payment Option Codes 23, 24, 33, or 34; otherwise zero filled.

Subfield 4 (First Bonus Amount)

DE 112, subelement 032, subfield 4 (First Bonus Amount) indicates the amount of first bonus payment. The currency is the same as the transaction currency.

Table 1349: Attributes

Attribute	Description
Data Representation	n-12
Data Field	Contents of positions 7–18
Justification	Right with leading zeros

Table 1350: Values

Value	Description
000000000000–999999999999	Amount of first bonus payment. The currency is the same as the transaction currency. Default is all zeros.

Application notes

This field must contain a Bonus Amount for Payment Option Codes 32 and 34; otherwise zero filled.

Subfield 5 (Second Bonus Month)

DE 112, subelement 032, subfield 5 (Second Bonus Month) identifies the month of the second bonus payment received by the cardholder following the purchase date.

Table 1351: Attributes

Attribute	Description
Data Representation	n-2
Data Field	Contents of positions 19–20
Justification	Right with leading zeros

Table 1352: Values

Value	Description
01–12	Month
00	Not Applicable

Application notes

This field must contain a value of 01 to 12 when Payment Option Code is 24, 33, or 34 and subfield 2 (Bonuses Per Year) indicates 2; otherwise it should be zero filled.

Subfield 6 (Second Bonus Amount)

DE 112, subelement 032, subfield 6 (Second Bonus Amount) indicates the amount of second bonus payment. The currency is the same as the transaction currency.

Table 1353: Attributes

Attribute	Description
Data Representation	n-12
Data Field	Contents of positions 21–32
Justification	Right with leading zeros

Table 1354: Values

Value	Description
000000000000–999999999999	Amount of second bonus payment. The currency is the same as the transaction currency. Default is all zeros.

Application notes

This field should contain a Bonus Amount when Payment Option Code is 34 and subfield 2 (Bonuses Per Year) indicates 2; otherwise it should be zero filled.

Subfield 7 (Total Number of Installments)

DE 112, subelement 032, subfield 7 (Total Number of Installments) indicates the number of installments.

Table 1355: Attributes

Attribute	Description
Data Representation	n-2
Data Field	Contents of positions 33–34
Justification	Right with leading zeros

Table 1356: Values

Value	Description
01–99	Number of installments
00	Not Applicable

Application notes

This field must contain a value of 01 to 99 when Payment Option Code is 31, 32, 33, 34, or 61; otherwise zero filled.

Subfield 8 (First Installment Month)

DE 112, subelement 032, subfield 8 (First Installment Month) indicates the month in which the first installment will be paid.

Table 1357: Attributes

Attribute	Description
Data Representation	n-2
Data Field	Contents of positions 35–36
Justification	Right with leading zeros

Table 1358: Values

Value	Description
01–12	Month
00	Not Applicable

Application notes

This field must contain a value of 01 to 12 when Payment Option Code is 31, 32, 33, 34, or 61; otherwise zero filled.

Mexcta: Payment Transactions

Following are the technical specifications for Authorization/01xx messages for Mexcta payment transactions within Mexico.

Subelement 004 (Credit Line Usage Fee (CLUF))

DE 112, Mexcta, subelement 004 (Credit Line Usage Fee [CLUF]) contains the currency code and fee amount associated with the CLUF for domestic credit card cash advance (CCCA) ATM transactions in Authorization Request Response/0110 and Authorization Reversal Request/0400 Messages.

Table 1359: Attributes

Attribute	Description
Subelement ID	004
Subelement Data Length	3 positions
Data Representation	n-11

Attribute	Description
Data Field	Contents of positions 1–11
Subfields	N/A
Justification	Right-justified

Values

Numeric currency code (positions 1–3).

Fee amount (positions 4–11).

Subelement 005 (Issuing Bank Name (AKA Doing Business As [DBA]))

DE 112, Mexcta, subelement 005 (Issuing Bank Name [AKA Doing Business As (DBA)]) contains the issuing bank name (DBA) for ATM transactions where the Credit Line Usage Fee (CLUF) applies in Authorization Request Response/0110 and Authorization Reversal Request/0400 Messages.

Table 1360: Attributes

Attribute	Description
Subelement ID	005
Subelement Data Length	3 positions
Data Representation	ans-20
Data Field	Contents of positions 1–20
Subfields	N/A
Justification	Left-justified, trailing spaces

Subelement 006 (Financial Institution ID (FIID))

DE 112, Mexcta, subelement 006 (Financial Institution ID [FIID]) is used in Credit Line Usage Fee (CLUF) transactions on domestic ATM credit card cash advance transactions in the Dual Message Authorization System in Authorization Request Response/0110 and Authorization Reversal Request/0400 Messages.

Table 1361: Attributes

Attribute	Description
Subelement ID	006
Subelement Data Length	3 positions

Attribute	Description
Data Representation	ans-10
Data Field	Contents of positions 1–10
Subfields	N/A
Justification	N/A

Values

Valid FIID.

Subelement 007 (Installment Payment Data)

DE 112, Mexcta, subelement 007 (Installment Payment Data) is used in the Dual Message Authorization System for Mexcta payment transactions.

Table 1362: Attributes

Attribute	Description
Subelement ID	007
Subelement Data Length	3 positions
Data Representation	ans-9
Data Field	Contents of positions 1–9
Subfields	N/A
Justification	N/A

Values

Table 1363: Mexcta Payment Transactions in the Authorization Request/0100

Value	Description
XX	Installment pay plan types (positions 1–2). Valid values include: • 00 (No promotion) • 03 (Without interest for the cardholder) • 05 (With interest for the cardholder) • 07 (Buy today, pay later)
YY	The total number of installments (positions 3–4). Valid value is 01–99.
ZZ	Grace period before first payment (positions 5–6). Valid values is 00–99.

Value	Description
NNN	Transaction Currency Code (positions 7–9). Same value as DE 49 (Currency Code, Transaction). This is the local currency of the acquirer or source location of the transaction. It specifies the currency used in DE 4 (Amount, Transaction). 484 = Mexican Peso

Table 1364: Mexcta Payment Transactions in the Authorization Request Response/0110

Value	Description
XX	Installment pay plan types (positions 1–2). Valid values include: • 00 (No promotion) • 03 (Without interest for the cardholder) • 05 (With interest for the cardholder) • 07 (Buy today, pay later)
YY	The total number of installments (positions 3–4). Valid value is 01–99.
ZZ	Grace period before first payment (positions 5–6). Valid value is 00–99.
NNN	Transaction Currency Code (positions 7–9). Same value as DE 49 (Currency Code, Transaction). This is the local currency of the acquirer or source location of the transaction. It specifies the currency used in DE 4 (Amount, Transaction). 484 = Mexican Peso

Subelement 008 (Installment Payment Response Data)

DE 112, Mexcta, subelement 008 (Installment Payment Response Data) is used in the Dual Message Authorization System and Mexcta payment transactions in the Authorization Request Response/0110.

Table 1365: Attributes

Attribute	Description
Subelement ID	008
Subelement Data Length	23 positions
Data Representation	ans-23
Data Field	Contents of positions 1–23

Attribute	Description
Subfields	N/A
Justification	N/A

Table 1366: Values

Value	Description
Installment amount (positions 1–12)	Installment amount (with two decimal places) including any issuer-calculated interest, insurance, or other charges.
Due Date of First Installment (positions 13–18)	Due Date of First Installment (in binary format: ddmmyyyy)
Finance currency code (positions 19–21)	This is the currency in which the issuer will finance the transaction. This specifies the currency used in installment amount, above 484 (Mexican Peso)
Payment Plan (positions 22–23)	Reserved for future use. Insert all zeros.

Netherlands: IBAN (Account Inquiry)

Following are the technical specifications for Account Status Inquiry (ASI) Authorization Request Response/0110 messages that enable customers in the Netherlands to share customer data such as the International Bank Account Number (IBAN).

Subelement 037 (Additional Cardholder Information)

DE 112 (Additional Data [National Use]), subelement 037 contains cardholder information to help track debit card activity.

Table 1367: Attributes

Attribute	Description
Subelement ID	037
Subelement Data Length	003 positions
Data Representation	an...076; LLLVAR
Data Field	Contents of subfields 1–2
Subfields	Subfield 01: Primary Cardholder Identifier Subfield 02: Secondary Cardholder Identifier

Usage

Following is the usage of subelement 37 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 1368: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	C	.	C

Values

See subfields.

Application notes

Issuers will add DE 112, SE 037 as part of the ASI Authorization Request Response/0110 messages for the Netherlands and add the IBAN and the four-digit card number on the respective subfields. This will be done only after successfully validating that the ASI Authorization Request/0100 message is received by the issuer is a domestic transit ASI request.

Subfield 01 (Primary Cardholder Identifier)

DE 112, subelement 037 (Additional Cardholder Information), subfield 01 (Primary Cardholder Identifier) is used by customers to share the unique cardholder identifier.

Table 1369: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	an...34; LLVAR
Data Field	Contents of subfield 01
Justification	Left

Values

For Netherlands intracountry transit activity issuers will populate this subfield with the 18-digit International Bank Account Number (IBAN) within the ASI Authorization Request Response/0110 message to the acquirer.

Issuers will populate this field only upon successful validation of ASI Authorization Request/0100 message and then respond back with an ASI Authorization Request Response/0110 message with this additional DE 112, SE 037, SF 01.

Example

The sample below is an example of an International Bank Account Number (IBAN) for the Netherlands.

Table 1370: Sample

IBAN	NL91ABAB0XX71XX300
ISO Country Code	NL (The Netherlands)
IBAN Check Digits	91
BBAN	ABAB0XX71XX300
Bank Identifier	ABAB
Account Number	0XX71XX300

The IBAN consists of a two-letter country code, two check digits and a Basic Bank Account Number (BBAN). A BBAN includes information about the domestic bank and account number. The IBAN print format adds one space after every four characters whereas the electronic format contains no spaces.

Subfield 02 (Secondary Cardholder Identifier)

DE 112, subelement 037 (Additional Cardholder Information), subfield 02 (Secondary Cardholder Identifier) is used by customers to share the unique cardholder identifier.

Table 1371: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	an...34; LLVAR
Data Field	Contents of subfield 02
Justification	Left

Values

For Netherlands intracountry transit-related ASI Authorization Request Response/0110 messages will be populated with the four-digit card number information of the cardholder, provided by the issuer to the acquirer, which is then provided to the merchant.

Paraguay: Payment Transactions

Following are the technical specifications for Authorization/01xx messages for Paraguay payment transactions within Paraguay.

Subelement 042 (Acceptor Tax ID)

Subelement 042 (Acceptor Tax ID) will be used in the Dual Message Authorization System to support DE 112 in any received Authorization Request/0100, Authorization Advice/0120, or Reversal Request/0400 message.

Table 1372: Attributes

Attribute	Description
Subelement ID	042
Subelement Data Length	3 positions
Data Representation	ans...12; LLLVAR

Usage

Following is the usage of subelement 042 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages:

Table 1373: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	X	•
Authorization Advice/0120: acquirer-generated	C	X	•
Reversal Request /0400	C	X	•

Application notes

This field is required when purchase, purchase with cash back, and refund transactions are submitted by acquirers for merchants in Chile and Paraguay.

Parcelas: Payment Transactions

Following are the technical specifications for Authorization/01xx messages for Parcelas payment transactions within Brazil.

Subelement 001 (Installment Payment Data)

DE 112, Parcelas, subelement 001 (Installment Payment Data) is used in the Dual Message Authorization System for Parcelas payment transactions.

Table 1374: Attributes

Attribute	Description
Subelement ID	001
Subelement Data Length	3 positions
Data Representation	ans-4
Data Field	Contents of positions 1–4
Subfields	N/A
Justification	N/A

Values

Table 1375: Parcelas Payment Transactions in the Authorization Request/0100

Used for both issuer- or merchant-financed Parcelas transactions: From acquirer to issuer requiring approval; use the following format:

XX	Parcelas plan type:
	20 = Issuer-financed
	21 = Merchant-financed
	25 = Credário
YY	The total number of Parcelas

Table 1376: Parcelas Payment Transactions in the Authorization Request Response/0110

Used for both issuer- or merchant-financed Parcelas transactions: From acquirer to issuer requiring approval. Echoing this subelement in the Authorization Request Response is unique to Brazil processing. Use the following format:

XX	Parcelas plan type:
	20 = Issuer-financed

	21 = Merchant-financed
	25 = Crediário
YY	The total number of Parcelas

Subelement 002 (Installment Payment Response Data)

DE 112, Parcelas, subelement 002 (Installment Payment Response Data) is used in the Dual Message Authorization System and Parcelas payment transactions.

Table 1377: Attributes

Attribute	Description
Subelement ID	002
Subelement Data Length	3 positions
Data Representation	ans-32 if format 1 is used and ans-4 if format 2 is used
Data Field	Contents of positions 1–32 or 1–4
Subfields	N/A
Justification	N/A

Values

Table 1378: Format 1: Issuer-Financed Parcelas transactions in the Authorization Request Response/0110

Issuer acknowledgment format 1, from issuer to acquirer, upon transaction approval; all amounts are express in transaction currency; contents are as follows:

Value	Description
Positions 1–4	Parcelas information, same as subelement 001.
Positions 5–16	Parcelas (or installment) amount issuer calculates, including calculated interest; two decimal places.
Positions 17–28	Transaction total amount issuer calculates, including calculated interest; two decimal places.
Positions 29–32	Monthly interest rate issuer calculates; two decimal places.

Table 1379: Format 2: Merchant-Financed Parcelas transactions in the Authorization Request Response/0110

Issuer acknowledgment format 2, from issuer to acquirer, upon transaction approval; Parcelas information copied from subelement 001 as follows:

Value	Description
21	Merchant-financed
YY	The total number of Parcelas

Subelement 016 (Additional Installment Payment Response Data)

DE 112, Parcelas, subelement 016 (Additional Installment Payment Response Data) is used in the Dual Message Authorization System and Parcelas payment transactions.

Table 1380: Attributes

Attribute	Description
Subelement ID	016
Subelement Data Length	3 positions
Data Representation	ans-74
Data Field	Contents of positions 1–74
Subfields	N/A
Justification	N/A

All Authorization Request Response/0110 messages in response to Parcelas Authorization Request/0100 messages must include additional information to identify the total effective cost of installment transactions.

Table 1381: Values

Value	Description
Positions 1–12	Total Effective Cost (TEC) (N12)
Positions 13–19	Taxes (N7)
Positions 20–23	% Taxes of Total Amount (N4)
Positions 24–30	Fee (N7)
Positions 31–34	% Fee of Total Amount (N4)

Value	Description
Positions 35–41	Insurance (N7)
Positions 42–45	% Insurance of Total Amount (N4)
Positions 46–52	Other (N7)
Positions 53–56	% Other of Total Amount (N4)
Positions 57–63	Total Amount to Merchant (N7)
Positions 64–67	% Amount to Merchant of Total Amount (N4)
Positions 68–74	% Total Effect Cost (N7)

Percta: Payment Transactions

Following are the technical specifications for Authorization/01xx messages for Percta payment transactions within Peru.

Subelement 007 (Installment Payment Data)

DE 112, Percta, Subelement 007 (Installment Payment Data) is used in the Dual Message Authorization System for Percta payment transactions.

Table 1382: Attributes

Attribute	Description
Subelement ID	007
Subelement Data Length	3 positions
Data Representation	ans-8
Data Field	Contents of positions 1–8
Subfields	N/A
Justification	N/A

Values

Table 1383: Percta Payment Transactions in the Authorization Request/0100

Value	Description
XX	Type of credit (positions 1–2). Valid value is 20 (Issuer-financed) In the case of Peru, issuer finance is the only finance model.
YY	The total number of installments (positions 3–4). Valid value is 01–99.

Value	Description
Z	Grace period before first payment (position 5). Valid values are: • 0 = No grace period • 1 = One month • 2 = Two months
NNN	Transaction Currency Code (positions 6–8). Echo of DE 49 (Currency Code, Transaction) This is the local currency of the acquirer or source location of the transaction. It specifies the currency used in DE 4 (Amount, Transaction). 604 = Peru Nuevo Sol 840 = United States dollar

Table 1384: Percta Payment Transactions in the Authorization Request Response/0110

Value	Description
XX	Type of credit (positions 1–2). Valid value is 20 (Issuer-financed) In the case of Peru, issuer finance is the only finance model.
YY	The total number of installments (positions 3–4). Valid value is 01–99.
Z	Grace period before first payment (position 5). Valid values are: • 0 = No grace period • 1 = One month • 2 = Two months
NNN	Transaction Currency Code (positions 6–8). Echo of DE 49 (Currency Code, Transaction) This is the local currency of the acquirer or source location of the transaction. It specifies the currency used in DE 4 (Amount, Transaction). 604 = Peru Nuevo Sol 840 = United States dollar

Subelement 008 (Installment Payment Response Data)

DE 112, Percata, subelement 008 (Installment Payment Response Data) is used in the Dual Message Authorization System and Percata payment transactions.

Table 1385: Attributes

Attribute	Description
Subelement ID	008
Subelement Data Length	3 positions
Data Representation	ans-23
Data Field	Contents of positions 1-23
Subfields	N/A
Justification	N/A

Table 1386: Values

Value	Description
Installment amount (positions 1-12)	Installment amount (with two decimal places) including any issuer-calculated interest, insurance, or other charges.
Due Date of First Installment (positions 13-18)	Due Date of First Installment (in binary format: ddmmyyyy)
Finance currency code (positions 19-21)	This is the currency in which the issuer will finance the transaction. This specifies the currency used in installment amount, above 604 (Peru Nuevo Sol) or 840 (United States dollar)
Payment Plan (positions 22-23)	Reserved for future use. Insert all zeros.

Spain: Domestic ATM Transactions

Following are the technical specifications for Authorization/01xx messages for domestic ATM transactions within Spain.

Subelement 017 (ATM Domestic Fee)

DE 112, subelement 017 (ATM Domestic Fee) contains fees charged for ATM domestic transactions in Spain.

Table 1387: Attributes

Attribute	Description
Subelement ID	017
Subelement Data Length	3 positions
Data Representation	n...48; LLLVAR
Data Field	Contents of subfields 1–3
Subfields	3
Justification	See subfields.

Usage

Following is the usage of subelement 017 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 1388: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	O	.	O
Authorization Request Response/0110	O	.	O
Reversal Request/0400	O	.	O

Values

See subfields.

Subfield 1 (ATM Service Fee)

DE 112, subelement 017, subfield 1 (ATM Service Fee) contains fee amount in the currency of DE 49 (Currency Code, Transaction) charged to the issuer to compensate the ATM acquirer for the ATM cost incurred for ATM domestic transactions in Spain.

Table 1389: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	n-12
Data Field	Contents of subfield 1
Justification	Right with leading zeros

Values

ATM Service Fee amount in transaction currency.

Subfield 2 (ATM Disloyalty Fee)

DE 112, subelement 017, subfield 2 (ATM Disloyalty Fee) contains fee amount in the currency of DE 51 (Currency Code, Cardholder Billing), charged by the issuer to the cardholder for using another bank's ATM for ATM domestic transactions in Spain.

Table 1390: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	n-12
Data Field	Contents of subfield 2
Justification	Right with leading zeros

Values

ATM Disloyalty Fee amount in Cardholder Billing currency.

Subfield 3 (Credit Card Fee)

DE 112, subelement 017, subfield 3 (Credit Card Fee) contains fee amount in the currency of DE 51 (Currency Code, Cardholder Billing) charged by the issuer to the cardholder for the grant of credit for ATM domestic transactions in Spain.

Table 1391: Attributes

Attribute	Description
Subfield ID	03
Subfield Data Length	2 positions
Data Representation	n-12
Data Field	Contents of subfield 3
Justification	Right with leading zeros

Values

Credit Card Fee amount in Cardholder Billing currency.

Taiwan: National Data

Following are the technical specifications specific to Taiwan domestic transactions.

Subelement 038 (Taiwan National Data)

DE 112, subelement 038 (Taiwan National Data) is used in the authorization message to pass national data between acquirers and issuers for intracountry processing.

Table 1392: Attributes

Attribute	Description
Subelement ID	038
Subelement Data Length	3 positions
Data Representation	ans..256; LLLVAR
Data Field	Contents of positions 1–256
Subfields	N/A
Justification	N/A

Usage

Following is the usage of subelement 038 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 1393: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	O	.	C
Authorization Request Response/0110	O	.	C
Reversal Request/0400	O	.	C
Reversal Request Response/0410	O	.	C

Values

Market agreed values for installment data and other national specific data fields.

United Kingdom: Debt Repayment Transactions

Following are the technical specifications for Authorization/01xx messages for United Kingdom debt repayment transactions within the U.K.

Subelement 033 (UK Recipient Details)

DE 112, subelement 033 (UK Recipient Details) supports the transmission of recipient details in an authorization request from acquirer to issuer.

Table 1394: Attributes

Attribute	Description
Subelement ID	033
Subelement Data Length	3 positions
Data Representation	ans...89; LLLVAR
Data Field	Contents of subfields 1–4
Subfields	4
Justification	See subfields

Usage

Following is the usage of subelement 033 (whether it is mandatory, conditional, optional, or system provided) in applicable messages.

Table 1395: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C

Values

See subfields.

Subfield 1 (Recipient Last Name)

DE 112, subelement 033 (UK Recipient Details), subfield 1 (Recipient Last Name) contains the last name of the recipient.

Table 1396: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	ans...35; LLVAR
Data Field	Contents of subfield 1
Justification	N/A

Values

Contains valid last name of the recipient (only the first six letters of the last name are required).

Subfield 2 (Recipient Postal Code)

DE 112, subelement 033 (UK Recipient Details), subfield 2 (Recipient Postal Code) contains the postal code of the recipient.

Table 1397: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	10 positions
Data Representation	ans-10
Data Field	Contents of subfield 2

Attribute	Description
Justification	left-justified with trailing spaces

Values

Contains valid location postal code of the recipient. First part represents the district, then space-filled.

Subfield 3 (Recipient Date of Birth)

DE 112, subelement 033 (UK Recipient Details), subfield 3 (Recipient Date of Birth) contains the date of birth of the recipient.

Table 1398: Attributes

Attribute	Description
Subfield ID	03
Subfield Data Length	08 positions
Data Representation	n-8
Data Field	Contents of subfield 3
Justification	N/A

Values

Contains valid date of birth of the recipient in the format YYYYMMDD.

Subfield 4 (Recipient Account Number)

DE 112, subelement 033 (UK Recipient Details), subfield 4 (Recipient Account Number) contains valid account number of the recipient.

Table 1399: Attributes

Attribute	Description
Subfield ID	04
Subfield Data Length	2 positions
Data Representation	ns...20; LLVAR
Data Field	Contents of subfield 4
Justification	N/A

Values

Contains the recipient account number.

DE 113 (Reserved for National Use)

Mastercard recommends that DE 113 contain Application Generic Data and Application Banking Data. This data element is typically present for consumer and business application requests, counteroffer replies, and pre-approved offer inquiries.

Table 1400: Attributes

Attribute	Description
Data Representation	ans...999; LLLVAR
Data Length	3 positions
Data Field	Contents of positions 1–999
Subfields	N/A
Justification	N/A

Usage

Following is the usage of DE 113 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 1401: Applicable messages

Message	Org	Sys	Dst
Administrative Request/0600	C	.	C
Administrative Request Response/0610	C	.	C

Application notes

Mastercard edits DE 113 for valid attributes but does not edit or log data contents.

At least one of DE 113–119 is mandatory within Administrative 0600/0610 messages.

Recommended format construction

113LLL<field_name>data<field_name>data...<field_name>data

113 is data element number

LLL is total length of all field names, separators and data

< indicates start of field_name

field_name is unique name for each field

> indicates end of field_name

Generic Data, Administrative Request/0600 Message

DE 113, Generic Data, may contain the following fields.

Table 1402: Field details

Field name	Maximum length	Description
client_id	16	Unique identifier of credit client
store_code	16	Unique identifier of each store location
credit_unit	5	Credit unit designation
appl_source_type	3	Origin of application
source_code	19	Application source code
appl_source_empid	11	Employee ID that obtained application request
appl_type	1	• C = Consumer • B = Business
account_type	1	For Business applications use: • R = Revolving • N = Non-revolving • I = Installment • P = Invoice/Net Pay • M = Co-Brand For Consumer applications use: • 1 = Individual • 2 = Joint • 3 = Authorized Buyer
language_pref	3	• ENG = English • SPA = Spanish • FRE = French • and other supported ISO 639 codes
sale_pending_amt	12	Amount of initial sale
credit_limit_req_amt	12	Amount of requested credit limit

Field name	Maximum length	Description
single_purch_limit	12	Amount limit per purchase
bill_addr1	25	Billing address line 1
bill_addr2	25	Billing address line 2
bill_city	20	Billing city
bill_sub_ctype	2	Billing subnational entity alpha abbreviation; for example U.S. state code abbreviations
bill_post_code	10	Billing postal code
bill_ctype	3	Billing country alpha abbreviation
bill_phone_nbr	15	Billing phone number
bill_email_addr	54	Billing e-mail address
membership_nbr	16	Unique identifier of member or customer
correlation_id	16	Correlation identifier may be assigned when response to application response is status O
reference_nbr	13	Reference number assigned to original application
offer_accept_ind	1	Y or N indicates acceptance of counteroffer
temp_pass_days	2	Number of days requested for a temporary charge pass
pre_approval_nbr	13	Preapproval reference number

NOTE: The above fields are not required to be present. Additional fields may be present as needed. Fields need not be in any particular order. Fields may be less than specified maximum length.

Usage

Application Generic Data recommended usage when:

- DE 60 = 6500080 (Consumer Application Request)
- DE 60 = 6500085 (Consumer Counteroffer Reply)
- DE 60 = 6500090 (Business Application Request)
- DE 60 = 6500095 (Business Counteroffer Reply)

Banking Data, Administrative Request/0600 Message

DE 113, Banking Data, may contain the following fields.

Table 1403: Field details

Field name	Maximum length	Description
bank_name	35	Name of bank
bank_addr1	25	Bank address line 1
bank_addr2	25	Bank address line 2
bank_city	20	Bank city
bank_sub_ctype	2	Bank subnational entity alpha abbreviation; for example U.S. state code abbreviations
bank_post_code	10	Bank postal code
bank_ctype	3	Bank country alpha abbreviation
bank_phone_nbr	15	Bank phone number
checking_acct_nbr	17	Checking account number
savings_acct_nbr	17	Savings account number

NOTE: The above fields are not required to be present. Additional fields may be present as needed. Fields need not be in any particular order. Fields may be less than specified maximum length.

Usage

Application Banking Data recommended usage when:

- DE 60 = 6500080 (Consumer Application Request)
- DE 60 = 6500090 (Business Application Request)

DE 114 (Reserved for National Use)

Mastercard recommends that DE 114 contain Consumer Application Data or Consumer Maintenance Data. This data element is typically present for consumer application requests, application status inquiries, preapproved offer inquiries, or consumer maintenance requests as well as consumer application or consumer maintenance responses. DE 114 also may be present for business application requests that require a personal guarantee.

Table 1404: Attributes

Attribute	Description
Data Representation	ans...999; LLLVAR
Data Length	3 positions
Data Field	Contents of positions 1–999
Subfields	N/A
Justification	N/A

Usage

Following is the usage of DE 114 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 1405: Applicable messages

Message	Org	Sys	Dst
Administrative Request/0600	C	.	C
Administrative Request Response/0610	C	.	C

Application notes

Mastercard edits DE 114 for valid attributes but does not edit or log data contents.

At least one of DE 113–119 is mandatory within Administrative 0600/0610 messages.

Recommended format construction

114LLL<field_name>data<field_name>data...<field_name>data

114 is data element number

LLL is total length of all field names, separators and data

< indicates start of field_name

field_name is unique name for each field

> indicates end of field_name

Consumer Application Request Data Administrative Request/0600 Message

DE 114, Consumer Application Request Data, may contain the following fields.

Table 1406: Field details

Field name	Maximum length	Description
cons_first_name	15	Consumer's first name
cons_middle_initial	1	Consumer's middle initial
cons_last_name	20	Consumer's last name
cons_suffix	6	Consumer's suffix; SR, JR, III, PHD
birth_date	10	Consumer's date of birth; CCYY/MM/DD
national_id	20	Consumer's national identification number United States—use Social Security number
local_id_type	1	Consumer's local ID type
		D = Driver's license
		I = Identification card
		O = Other
local_id_location	2	Consumer's local ID location United States—use alpha state codes
local_id_nbr	24	Consumer's local ID number
home_addr1	25	Consumer's home address line 1
home_addr2	25	Consumer's home address line 2
home_city	20	Consumer's home city
home_sub_ctype	2	Consumer's home subnational entity alpha abbreviation; for example U.S. state code abbreviations
home_post_code	10	Consumer's home postal code
home_ctype	3	Consumer's home country alpha abbreviation
home_phone_nbr	15	Consumer's home phone number
rent_own_ind	1	Consumer's home status
		B = Board or other
		M = Military
		O = Own

Field name	Maximum length	Description
		P = Live with parents or other relatives
		R = Rent
residence_time	2	Number of years lived in current residence
prev_residence_time	2	Number of years lived in previous residence
prev_home_addr1	25	Consumer's previous home address line 1
prev_home_addr2	25	Consumer's previous home address line 2
prev_home_city	20	Consumer's previous home city
prev_home_sub_ctype	2	Consumer's previous home subnational entity alpha abbreviation; for example U.S. state code abbreviations
prev_home_post_code	10	Consumer's previous home postal code
prev_home_ctype	3	Consumer's previous home country alpha abbreviation
employment_time	2	Number of years employed by current employer
work_phone_nbr	15	Consumer's work phone number
annual_income_amt	12	Consumer's total annual income amount
credit_ins_ind	1	Y or N—Consumer's acceptance of credit insurance
card_qty	3	Quantity of cards requested
client_employee	1	Y or N—Is applicant an employee of client

NOTE: The above fields are not required to be present. Additional fields may be present as needed. Fields need not be in any particular order. Fields may be less than specified maximum length.

Usage

Consumer Application Request Data recommended usage when:

- DE 60 = 6500080 (Consumer Application Request)
- DE 60 = 6500090 (Business Application Request)

Consumer Status Inquiry or Preapproved Offer Inquiry Data Administrative Request/0600 Message

DE 114, Consumer Status Inquiry or Preapproved Offer Inquiry Data may contain the following fields.

Table 1407: Field details

Field name	Maximum length	Description
client_id	16	Unique identifier of credit client
store_code	16	Unique identifier of each store location
reference_nbr	13	Application reference number provided if available
cons_first_name	15	Consumer's first name
cons_middle_initial	1	Consumer's middle initial
cons_last_name	20	Consumer's last name
cons_suffix	6	Consumer's suffix; SR, JR, III, PHD
birth_date	10	Consumer's date of birth; CCYY/MM/DD
national_id	20	Consumer's national identification number United States—use Social Security number
local_id_nbr	24	Consumer's local ID number United States—use driver's license or other ID
home_phone_nbr	15	Consumer's home phone number
pre_approval_nbr	13	Preapproved reference number
temp_pass_days	2	Number of days requested for a temporary charge pass

NOTE: The above fields are not required to be present. Additional fields may be present as needed. Fields need not be in any particular order. Fields may be less than specified maximum length.

Usage

Consumer Application Inquiry Data recommended usage when:

- DE 60 = 6500081 (Consumer Application Status Inquiry)
- DE 60 = 6500086 (Consumer Preapproved Offer Inquiry)

Consumer Account Maintenance Data Administrative Request/0600 Message

DE 114, Consumer Account Maintenance Data may contain the following fields.

Table 1408: Field details

Field name	Maximum length	Description
client_id	16	Unique identifier of credit client
store_code	16	Unique identifier of each store location
account_nbr	19	Consumer account number
account_status	1	Update account status only to I = Inactive
cons_first_name	15	Update consumer's first name
cons_middle_initial	1	Update consumer's middle initial
cons_last_name	20	Update consumer's last name
cons_suffix	6	Update consumer's suffix; SR, JR, III, PHD
local_id_type	1	Update consumer's local ID type
		D = Driver's license
		I = Identification card
local_id_location	2	O = Other
		Update consumer's local ID location
		United States: use alpha State codes
local_id_nbr	24	Update consumer's local ID number
home_addr1	25	Update consumer's home address line 1
home_addr2	25	Update consumer's home address line 2
home_city	20	Update consumer's home city
home_sub_ctr	2	Update consumer's home subnational entity alpha abbreviation; for example U.S. state code abbreviations
home_post_code	10	Update consumer's home postal code
home_ctr	3	Update consumer's home country alpha abbreviation
home_phone_nbr	15	Update consumer's home phone number
bill_addr1	25	Update billing address line 1
bill_addr2	25	Update billing address line 2
bill_city	20	Update billing city

Field name	Maximum length	Description
bill_sub_ctype	2	Update billing subnational entity alpha abbreviation; for example U.S. state code abbreviations
bill_post_code	10	Update billing postal code
bill_ctype	3	Update billing country alpha abbreviation
bill_phone_nbr	15	Update billing phone number
bill_email_addr	54	Update billing e-mail address
work_phone_nbr	15	Update consumer's work phone number
bank_name	35	Update name of bank
bank_addr1	25	Update bank address line 1
bank_addr2	25	Update bank address line 2
bank_city	20	Update bank city
bank_sub_ctype	2	Update bank subnational entity alpha abbreviation; for example U.S. state code abbreviations
bank_post_code	10	Update bank postal code
bank_ctype	3	Update bank country alpha abbreviation
checking_acct_nbr	17	Update checking account number
savings_acct_nbr	17	Update savings account number
credit_ins_ind	1	Update Y or N consumer's acceptance of credit insurance
annual_income_amt	12	Update consumer's total annual income amount
language_pref	3	ENG = English, SPA = Spanish, FRE = French, and other supported ISO 639 codes
NOTE: Repeat following User information as necessary for multiple users.		
user_function	1	A = Add, D = Delete
user_first_name	15	User's first name
user_middle_initial	1	User's middle initial
user_last_name	20	User's last name
user_suffix	6	User's suffix; SR, JR, III, PHD

NOTE: The above fields are not required to be present. Additional fields may be present as needed. Fields need not be in any particular order. Fields may be less than specified maximum length.

Table 1409: Field details

Field name	Maximum length	Description
bank_phone_nbr	15	Update bank phone number
reference_nbr	13	Reference number assigned to each application
appl_source_type	3	Origin of application
cons_first_name	15	Consumer's first name
cons_middle_initial	1	Consumer's middle initial
cons_last_name	20	Consumer's last name
cons_suffix	6	Consumer's suffix; SR, JR, III, PHD
home_addr1	25	Consumer's home address line 1
home_addr2	25	Consumer's home address line 2
home_city	20	Consumer's home city
home_sub_ctype	2	Consumer's home subnational entity alpha abbreviation; for example U.S. state code abbreviations
home_post_code	10	Consumer's home postal code
home_ctype	3	Consumer's home country alpha abbreviation
home_phone_nbr	15	Consumer's home phone number
language_pref	3	ENG = English, SPA = Spanish, FRE = French, and other supported ISO 639 codes
temp_pass_exp_date	10	Expiration date of the temporary charge pass; format CCYY/MM/DD
appl_status	1	A = Approved
		D = Declined
		C = Call
		O = Counteroffer
		P = Pending
		M = Mail-based Offer Approved
account_nbr	19	Account number present if status = A

Field name	Maximum length	Description
credit_limit_amt	12	Credit limit amount present if status = A or O

Usage

Consumer Account Maintenance Data recommended usage when:

DE 60 = 6500084 (Consumer Account Maintenance Request)

Consumer Application Response Data Administrative Request Response/0610 Message

DE 114, Consumer Application Response Data may contain the following fields.

Table 1410: Field details

Field name	Maximum length	Description
reference_nbr	13	Reference number assigned to each application
appl_source_type	3	Origin of application
cons_first_name	15	Consumer's first name
cons_middle_initial	1	Consumer's middle initial
cons_last_name	20	Consumer's last name
cons_suffix	6	Consumer's suffix; SR, JR, III, PHD
home_addr1	25	Consumer's home address line 1
home_addr2	25	Consumer's home address line 2
home_city	20	Consumer's home city
home_sub_ctype	2	Consumer's home subnational entity alpha abbreviation; for example U.S. state code abbreviations
home_post_code	10	Consumer's home postal code
home_ctype	3	Consumer's home country alpha abbreviation
home_phone_nbr	15	Consumer's home phone number
language_pref	3	ENG = English, SPA = Spanish, FRE = French, and other supported ISO 639 codes
temp_pass_exp_date	10	Expiration date of the temporary charge pass; format CCYY/MM/DD
appl_status	1	A = Approved
		D = Declined

Field name	Maximum length	Description
		C = Call
		O = Counteroffer
		P = Pending
		M = Mail-based Offer Approved
account_nbr	19	Account number present if status = A
credit_limit_amt	12	Credit limit amount present if status = A or O
card_expiry_date	4	Card expiration date may be present if status = A; format YYMM
credit_phone_nbr	15	Phone number may be present for application inquiries by customer or store
credit_terms	256	Credit terms may be present if applicable. Credit terms may include relevant information such as for revolving, non-revolving, or installment credit account.
correlation_id	16	Correlation identifier may be assigned when response to application request is status O
pre_approval_nbr	13	Preapproval reference number
pre_appr_end_date	10	Expiration date of the pre approval offer; format CCYY/MM/DD

NOTE: The above fields are not required to be present. Additional fields may be present as needed. Fields need not be in any particular order. Fields may be less than specified maximum length.

Usage

Consumer Application Response Data recommended usage when:

- DE 60 = 6500080 (Consumer Application Request)
- DE 60 = 6500081 (Consumer Application Status Inquiry)
- DE 60 = 6500085 (Consumer Counteroffer Reply)
- DE 60 = 6500086 (Consumer Preapproved Offer Inquiry)

Consumer Account Maintenance Data Administrative Request Response/0610 Message

DE 114, Consumer Account Maintenance Data may contain the following fields.

Table 1411: Field details

Field name	Maximum length	Description
account_nbr	19	Consumer account number
account_status_maint	1	S = successful, U = unsuccessful
account_status	1	Update account status only to I = Inactive
cons_first_name_maint	1	S = successful, U = unsuccessful
cons_first_name	15	Update consumer's first name
cons_middle_initial_maint	1	S = successful, U = unsuccessful
cons_middle_initial	1	Update consumer's middle initial
cons_last_name_maint	1	S = successful, U = unsuccessful
cons_last_name	20	Update consumer's last name
cons_suffix_maint	1	S = successful, U = unsuccessful
cons_suffix	6	Update consumer's suffix; SR, JR, III, PHD
local_id_type_maint	1	S = successful, U = unsuccessful
local_id_type	1	Update consumer's local ID type
local_id_location_maint	1	S = successful, U = unsuccessful
local_id_location	2	Update consumer's local ID location
local_id_nbr_maint	1	S = successful, U = unsuccessful
local_id_nbr	24	Update consumer's local ID number
home_addr1_maint	1	S = successful, U = unsuccessful
home_addr1	25	Update consumer's home address line 1
home_addr2_maint	1	S = successful, U = unsuccessful
home_addr2	25	Update consumer's home address line 2
home_city_maint	1	S = successful, U = unsuccessful
home_city	20	Update consumer's home city
home_sub_ctype_maint	1	S = successful, U = unsuccessful
home_sub_ctype	2	Update home subnational entity alpha abbreviation; for example U.S. state code abbreviations

Field name	Maximum length	Description
home_post_code_maint	1	S = successful, U = unsuccessful
home_post_code	10	Update consumer's home postal code
home_ctype_maint	1	S = successful, U = unsuccessful
home_ctype	3	Update consumer's home country alpha abbreviation
home_phone_nbr_maint	1	S = successful, U = unsuccessful
home_phone_nbr	15	Update consumer's home phone number
bill_addr1_maint	1	S = successful, U = unsuccessful
bill_addr1	25	Update billing address line 1
bill_addr2_maint	1	S = successful, U = unsuccessful
bill_addr2	25	Update billing address line 2
bill_city_maint	1	S = successful, U = unsuccessful
bill_city	20	Update billing city
bill_sub_ctype_maint	1	S = successful, U = unsuccessful
bill_sub_ctype	2	Update billing subnational entity alpha abbreviation; for example U.S. state code abbreviations
bill_post_code_maint	1	S = successful, U = unsuccessful
bill_post_code	10	Update billing postal code
bill_ctype_maint	1	S = successful, U = unsuccessful
bill_ctype	3	Update billing country alpha abbreviation
bill_phone_nbr_maint	1	S = successful, U = unsuccessful
bill_phone_nbr	15	Update billing phone number
bill_email_addr_maint	1	S = successful, U = unsuccessful
bill_email_addr	54	Update billing e-mail address
work_phone_nbr_maint	1	S = successful, U = unsuccessful
work_phone_nbr	15	Update consumer's work phone number
bank_name_maint	1	S = successful, U = unsuccessful
bank_name	35	Update name of bank
bank_addr1_maint	1	S = successful, U = unsuccessful
bank_addr1	25	Update bank address line 1
bank_addr2_maint	1	S = successful, U = unsuccessful

Field name	Maximum length	Description
bank_addr2	25	Update bank address line 2
bank_city_maint	1	S = successful, U = unsuccessful
bank_city	20	Update bank city
bank_sub_ctype_maint	1	S = successful, U = unsuccessful
bank_sub_ctype	2	Update bank subnational entity alpha abbreviation; for example U.S. state code abbreviations
bank_post_code_maint	1	S = successful, U = unsuccessful
bank_post_code	10	Update bank postal code
bank_ctype_maint	1	S = successful, U = unsuccessful
bank_ctype	3	Update bank country alpha abbreviation
bank_phone_nbr_maint	1	S = successful, U = unsuccessful
bank_phone_nbr	15	Update bank phone number
checking_acct_nbr_maint	1	S = successful, U = unsuccessful
checking_acct_nbr	17	Update checking account number
savings_acct_nbr_maint	1	S = successful, U = unsuccessful
savings_acct_nbr	17	Update savings account number
credit_ins_ind_maint	1	S = successful, U = unsuccessful
credit_ins_ind	1	Update Y or N consumer's acceptance of credit insurance
annual_income_amt_maint	1	S = successful, U = unsuccessful
annual_income_amt	12	Update consumer's total annual income amount
language_pref_maint	1	S = successful, U = unsuccessful
language_pref	3	ENG = English, SPA = Spanish, FRE = French, and other supported ISO 639 codes
NOTE: Repeat following User information as necessary for multiple users.		
user_function_maint	1	S = successful, U = unsuccessful
user_function	1	A=Add, D=Delete
user_first_name	15	User's first name

Field name	Maximum length	Description
user_middle_initial	1	User's middle initial
user_last_name	20	User's last name
user_suffix	6	User's suffix; SR, JR, III, PHD

NOTE: The above fields are not required to be present. Additional fields may be present as needed. Fields need not be in any particular order. Fields may be less than specified maximum length.

Usage

Consumer Account Maintenance Data recommended usage when:

DE 60 = 6500084 (Consumer Account Maintenance Request)

DE 115 (Reserved for National Use)

DE 115 (Reserved for National Use) contains either the Business Application Data or Business Maintenance Data. This data element is typically present for business application requests, application status inquiries, preapproved offer inquiries, or business maintenance requests as well as business application or business maintenance responses.

Table 1412: Attributes

Attribute	Description
Data Representation	ans...999; LLLVAR
Data Length	3 positions
Data Field	Contents of positions 1–999
Subfields	N/A
Justification	N/A

Usage

Following is the usage of DE 115 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 1413: Applicable messages

Message	Org	Sys	Dst
Administrative Request/0600	C	.	C
Administrative Request Response/0610	C	.	C

Application notes

Mastercard edits DE 115 for valid attributes but does not edit or log data contents. At least one DE 113–119 is mandatory within Administrative 0600/0610 messages.

Recommended format construction

115LLL<field_name>data<field_name>data...<field_name>data

115 is data element number

LLL is total length of all field names, separators and data

< indicates start of field_name

field_name is unique name for each field

> indicates end of field_name

Business Application Request Data Administrative Request/0600 Message

DE 115, Business Application Request Data may contain the following fields.

Table 1414: Field details

Field name	Maximum length	Description
bus_lgl_name	24	Business legal name
doing_bus_as_name	40	Doing business as name
bus_addr1	25	Business address line 1
bus_addr2	25	Business address line 2
bus_city	20	Business city
bus_sub_ctype	2	Business subnational entity alpha abbreviation; for example U.S. state code abbreviations
bus_post_code	10	Business postal code
bus_ctype	3	Business country alpha abbreviation

Field name	Maximum length	Description
bus_phone_nbr	15	Business phone number
bill_contact_name	19	Business billing contact name such as Accounts Payable or individual name
annual_rev_amt	12	Business total annual revenue amount
employee_qty	6	Number of business employees
sic	8	Standard Industry Code
lgl_structure	1	C = Corporation
		P = Partnership
		S = Sole Proprietorship
		L = Limited Liability Corp
		D = Limited Partnership
		blank = Unknown or all other
corp_structure	1	F = Fortune 1000
		G = Government (national and local)
		P = Professional
		N = Non-profit
		R = Religious
		blank = Unknown or all other
bus_structure	1	B = Branch of parent organization
		P = Parent organization
		I = Independent organization
in_business_since	4	Year that business started
tax_exempt	1	Y if exempt from local taxes; N if not exempt
tax_id	30	National government tax identifier
po_req	1	Y if purchase order required; N if not required
card_qty	3	Quantity of cards requested
purchase_instructions	40	Contains special instructions applicable to purchases
signature_first_name	15	First name of person submitting business application
signature_mid_initial	1	Middle initial of person submitting business application

Field name	Maximum length	Description
signature_last_name	20	Last name of person submitting business application
signature_title	15	Title of person submitting business application
signature_present	1	Y or N: Signature of person submitting business application present on physical application
guar_signature_ind	1	Y or N: Personal guarantor signature present
guar_signature_date	10	Expiration date of the temporary charge pass; format CCYY/MM/DD
fax_info_sent	1	Y or N: Additional information sent via fax, such as list of authorized buyers, financial statement

NOTE: The above fields are not required to be present. Additional fields may be present as needed. Fields need not be in any particular order. Fields may be less than specified maximum length.

Usage

Business Application Data recommended usage when:

DE 60 = 6500090 (Business Application Request)

Business Application Status Inquiry Data or Business Preapproved Offer Inquiry Data Administrative Request/0600 Message

DE 115, Business Application Status Inquiry Data or Business Preapproved Offer Inquiry Data may contain the following fields.

Table 1415: Field details

Field name	Maximum length	Description
client_id	16	Unique identifier of credit client
store_code	16	Unique identifier of each store location
reference_nbr	13	Application reference number if available
bus_lgl_name	24	Business legal name
doing_bus_as_name	40	Doing business as name
tax_id	30	National government tax identifier
bus_phone_nbr	15	Business phone number

Field name	Maximum length	Description
pre_approval_nbr	13	Preapproval reference number
temp_pass_days	2	Number of days requested for a temporary charge pass

NOTE: The above fields are not required to be present. Additional fields may be present as needed. Fields need not be in any particular order. Fields may be less than specified maximum length.

Usage

Business Application Inquiry Data recommended usage when DE 60 = 6500091 (Business Application Status Inquiry)

DE 60 = 6500096 (Business Preapproved Offer Inquiry)

Business Account Maintenance Data Administrative Request/0600 Message

DE 115, Business Account Maintenance Data may contain the following fields.

Table 1416: Field details

Field name	Maximum length	Description
client_id	16	Unique identifier of credit client
store_code	16	Unique identifier of each store location
account_nbr	19	Business account number
account_status	1	Update account status only to C closed or canceled
bus_lgl_name	24	Update business legal name
doing_bus_as_name	40	Update doing business as name
bus_addr1	25	Update business address line 1
bus_addr2	25	Update business address line 2
bus_city	20	Update business city
bus_sub_ctype	2	Update business subnational entity alpha abbreviation; for example U.S. state code abbreviations
bus_post_code	10	Update business postal code
bus_ctype	3	Update business country alpha abbreviation
bus_phone_nbr	15	Update business phone number

Field name	Maximum length	Description
annual_rev_amt	12	Update business total annual revenue amount
sic	8	Update standard industry code
lgl_structure	1	C = Corporation
		P = Partnership
		L = Limited Liability Corp
		D = Limited Partnership
		S = Sole Proprietorship
corp_structure	1	blank = Unknown or all other
		F = Fortune 1000
		G = Government (national and local)
		P = Professional
		N = Non-profit
		R = Religious
bus_structure	1	blank = Unknown or all other
		B = Branch of parent organization
		P = Parent organization
tax_exempt	1	I = Independent organization
tax_exempt	1	Update—Y if exempt from local taxes; N if not exempt
tax_id	30	Update national government tax identifier
po_req	1	Update—Y if purchase order required; N if not required
card_qty	3	Update quantity of cards requested (additional cards)
bill_addr1	25	Update billing address line 1
bill_addr2	25	Update billing address line 2
bill_city	20	Update billing city
bill_sub_ctype	2	Update billing subnational entity alpha abbreviation; for example U.S. state code abbreviations
bill_post_code	10	Update billing postal code
bill_ctype	3	Update billing country alpha abbreviation

Field name	Maximum length	Description
bill_phone_nbr	15	Update billing phone number
bill_contact_name	19	Update billing contact name such as Accounts Payable or individual name
bill_email_addr	54	Update billing e-mail address
bank_name	35	Update name of bank
bank_addr1	25	Update bank address line 1
bank_addr2	25	Update bank address line 2
bank_city	20	Update bank city
bank_sub_ctype	2	Update bank subnational entity alpha abbreviation; for example U.S. state code abbreviations
bank_post_code	10	Update bank postal code
bank_ctype	3	Update bank country alpha abbreviation
bank_phone_nbr	15	Update bank phone number
checking_acct_nbr	17	Update checking account number
savings_acct_nbr	17	Update savings account number
credit_ins_ind	1	Update Y or N business's acceptance of credit insurance
annual_income_amt	12	Update business's total annual income amount
single_purch_limit	12	Update amount limit per purchase
purchase_instructions	40	Update special instructions applicable to purchases

NOTE: Repeat following User information as necessary for multiple users.

user_function	1	A=Add, D=Delete
user_first_name	15	User's first name
user_middle_initial	1	User's middle initial
user_last_name	20	User's last name
user_suffix	6	User's suffix; SR, JR, III, PHD

NOTE: The above fields are not required to be present. Additional fields may be present as needed. Fields need not be in any particular order. Fields may be less than specified maximum length.

Usage

Business Account Maintenance Data recommended usage when:

DE 60 = 6500094 (Business Account Maintenance Request)

Business Application Response Data Administrative Request Response/0610 Message

DE 115, Business Application Response Data may contain the following fields.

Table 1417: Field details

Field name	Maximum length	Description
reference_nbr	13	Reference number assigned to each application
appl_source_type	3	Origin of application
account_type	1	R = Revolving N = Non-revolving I = Installment P = Invoice/Net Pay M = Co-Brand
bus_lgl_name	24	Business legal name
doing_bus_as_name	40	Doing business as name
bus_addr1	25	Business address line 1
bus_addr2	25	Business address line 2
bus_city	20	Business city
bus_sub_ctype	2	Business subnational entity alpha abbreviation; for example U.S. state code abbreviations
bus_post_code	10	Business postal code
bus_ctype	3	Business country alpha abbreviation
bus_phone_nbr	15	Business phone number
bill_contact_name	19	Business billing contact name such as Accounts Payable or individual name
po_req	1	Y if purchase order required; N if not required

Field name	Maximum length	Description
purchase_instructions	40	Contains special instructions applicable to purchases
language_pref	3	ENG = English SPA = Spanish FRE = French and other supported ISO 639 codes
temp_pass_exp_date	10	Expiration date of the temporary charge pass; format CCYY/MM/DD
fax_info_sent	1	Y or N: Additional information sent via fax, such as list of authorized buyers, financial statement
fax_info_rcvd	1	Y or N: Additional information received via fax, such as list of authorized buyers, financial statement
appl_status	1	A = Approved
		D = Declined
		C = Call
		O = Counteroffer
		P = Pending
account_nbr	19	Account number present if status = A
credit_limit_amt	12	Credit limit amount present if status = A or O
card_expiry_date	4	Card expiration date may be present if status = A; format YYMM
credit_phone_nbr	15	Phone number may be present for application inquiries by customer or store.
credit_terms	256	Credit terms may be present if applicable. Credit terms may include relevant information such as for revolving, non-revolving, or installment credit account.
correlation_id	16	Correlation identifier may be assigned when response to application request is status O
pre_approval_nbr	13	Preapproval reference number
pre_appr_end_date	10	Expiration date of the pre-approval offer; format CCYY/MM/DD

NOTE: The above fields are not required to be present. Additional fields may be present as needed. Fields need not be in any particular order. Fields may be less than specified maximum length.

Usage

Business Application Response Data recommended usage when:

- DE 60 = 6500090 (Business Application Request)
- DE 60 = 6500091 (Business Application Status Inquiry)
- DE 60 = 6500095 (Business Counteroffer Reply)
- DE 60 = 6500096 (Business Preapproved Offer Inquiry)

Business Account Maintenance Data Administrative Request Response/0610 Message

DE 115, Business Account Maintenance Data may contain the following fields.

Table 1418: Field details

Field name	Maximum length	Description
account_nbr	19	Consumer account number
account_status_maint	1	S = successful, U = unsuccessful
account_status	1	Update account status only to C closed or canceled
bus_lgl_name_maint	1	S = successful, U = unsuccessful
bus_lgl_name	24	Business legal name
doing_bus_as_name_maint	1	S = successful, U = unsuccessful
doing_bus_as_name	40	Doing business as name
bus_addr1_maint	1	S = successful, U = unsuccessful
bus_addr1	25	Business address line 1
bus_addr2_maint	1	S = successful, U = unsuccessful
bus_addr2	25	Business address line 2
bus_city_maint	1	S = successful, U = unsuccessful
bus_city	20	Business city
bus_sub_ctry_maint	1	S = successful, U = unsuccessful
bus_sub_ctry	2	Business subnational entity alpha abbreviation; for example U.S. state code abbreviations
bus_post_code_maint	1	S = successful, U = unsuccessful
bus_post_code	10	Business postal code

Field name	Maximum length	Description
bus_ctype_maint	1	S = successful, U = unsuccessful
bus_ctype	3	Business country alpha abbreviation
bus_phone_nbr_maint	1	S = successful, U = unsuccessful
bus_phone_nbr	15	Business phone number
annual_rev_amt_maint	1	S = successful, U = unsuccessful
annual_rev_amt	12	Business total annual revenue amount
sic_maint	1	S = successful, U = unsuccessful
sic	8	Standard Industry Code
lgl_structure_maint	1	S = successful, U = unsuccessful
lgl_structure	1	C = Corporation
		P = Partnership
		L = Limited Liability Corp
		D = Limited Partnership
		S = Sole Proprietorship
		blank = Unknown or all other
corp_structure_maint	1	S = successful, U = unsuccessful
corp_structure	1	F = Fortune 1000
		G = Government (national and local)
		P = Professional
		N = Non-profit
		R = Religious
		blank = Unknown or all other
bus_structure_maint	1	S = successful, U = unsuccessful
bus_structure	1	B = Branch of parent organization
		P = Parent organization
		I = Independent organization
tax_exempt_maint	1	S = successful, U = unsuccessful
tax_exempt	1	Y if exempt from local taxes; N if not exempt
tax_id_maint	1	S = successful, U = unsuccessful
tax_id	30	National government tax identifier
po_req_maint	1	S = successful, U = unsuccessful

Field name	Maximum length	Description
po_req	1	Y if purchase order required; N if not required
card_qty_maint	1	S = successful, U = unsuccessful
card_qty	3	Quantity of cards requested (additional cards)
bill_addr1_maint	1	S = successful, U = unsuccessful
bill_addr1	25	Update billing address line 1
bill_addr2_maint	1	S = successful, U = unsuccessful
bill_addr2	25	Update billing address line 2
bill_city_maint	1	S = successful, U = unsuccessful
bill_city	20	Update billing city
bill_sub_ctype_maint	1	S = successful, U = unsuccessful
bill_sub_ctype	2	Update billing subnational entity alpha abbreviation; for example U.S. state code abbreviations
bill_post_code_maint	1	S = successful, U = unsuccessful
bill_post_code	10	Update billing postal code
bill_ctype_maint	1	S = successful, U = unsuccessful
bill_ctype	3	Update billing country alpha abbreviation
bill_phone_nbr_maint	1	S = successful, U = unsuccessful
bill_phone_nbr	15	Update billing phone number
bill_contact_name_maint	1	S = successful, U = unsuccessful
bill_contact_name	19	Update billing contact name such as Accounts Payable or individual name
bill_email_addr_maint	1	S = successful, U = unsuccessful
bill_email_addr	54	Update billing e-mail address
bank_name_maint	1	S = successful, U = unsuccessful
bank_name	35	Update name of bank
bank_addr1_maint	1	S = successful, U = unsuccessful
bank_addr1	25	Update bank address line 1
bank_addr2_maint	1	S = successful, U = unsuccessful
bank_addr2	25	Update bank address line 2
bank_city_maint	1	S = successful, U = unsuccessful

Field name	Maximum length	Description
bank_city	20	Update bank city
bank_sub_ctype_maint	1	S = successful, U = unsuccessful
bank_sub_ctype	2	Update bank subnational entity alpha abbreviation; for example U.S. state code abbreviations
bank_post_code_maint	1	S = successful, U = unsuccessful
bank_post_code	10	Update bank postal code
bank_ctype_maint	1	S = successful, U = unsuccessful
bank_ctype	3	Update bank country alpha abbreviation
bank_phone_nbr_maint	1	S = successful, U = unsuccessful
bank_phone_nbr	15	Update bank phone number
checking_acct_nbr_maint	1	S = successful, U = unsuccessful
checking_acct_nbr	17	Update checking account number
savings_acct_nbr_maint	1	S = successful, U = unsuccessful
savings_acct_nbr	17	Update savings account number
credit_ins_ind_maint	1	S = successful, U = unsuccessful
credit_ins_ind	1	Update Y or N business's acceptance of credit insurance
annual_income_amt_maint	1	S = successful, U = unsuccessful
annual_income_amt	12	Update business's total annual income amount
single_purch_limit_maint	1	S = successful, U = unsuccessful
single_purch_limit	12	Update amount limit per purchase
purchase_instructions_maint	1	S = successful, U = unsuccessful
purchase_instructions	40	Contains special instructions applicable to purchases

NOTE: Repeat following User information as necessary for multiple users.

user_function_maint	1	S = successful, U = unsuccessful
user_function	1	A=Add, D=Delete
user_first_name	15	User's first name

Field name	Maximum length	Description
user_middle_initial	1	User's middle initial
user_last_name	20	User's last name
user_suffix	6	User's suffix; SR, JR, III, PHD

NOTE: The above fields are not required to be present. Additional fields may be present as needed. Fields need not be in any particular order. Fields may be less than specified maximum length.

Usage

Business Account Maintenance Data recommended usage when:

DE 60 = 6500094 (Business Account Maintenance Request)

DE 116 (Reserved For National Use)

Mastercard recommends that DE 116 contain Consumer User Lookup Data and Consumer Account Lookup Data. This data element is typically present to request consumer user and account information and provide consumer user account information.

Table 1419: Attributes

Attribute	Description
Data Representation	ans...999; LLLVAR
Data Length	3 positions
Data Field	Contents of positions 1–999
Subfields	N/A
Justification	N/A

Usage

Following is the usage of DE 116 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 1420: Applicable messages

Message	Org	Sys	Dst
Administrative Request/0600	C	.	C

Message	Org	Sys	Dst
Administrative Request Response/0610	C	.	C

Application notes

Mastercard edits DE 116 for valid attributes but does not edit or log data contents. At least one of DE 113–119 is mandatory within Administrative 0600/0610 messages.

Recommended format construction

116LLL<field_name>data<field_name>data...<field_name>data

116 is data element number

LLL is total length of all field names, separators and data

< indicates start of field_name

field_name is unique name for each field

> indicates end of field_name

Consumer User Lookup Inquiry Data Administrative Request/0600

DE 116, Consumer User Lookup Inquiry may contain the following fields.

Table 1421: Field details

Field name	Maximum length	Description
client_id	16	Unique identifier of credit client
store_code	16	Unique identifier of each store location
account_nbr	19	Account number
national_id	20	Consumer's national identification number United States—use Social Security number
cons_first_name	15	Consumer's first name
cons_middle_initial	1	Consumer's middle initial
cons_last_name	20	Consumer's last name
home_phone_nbr	15	Consumer's home phone number
user_list	1	Y = Request list of consumer account users (The maximum number of users to be provided will depend upon the client system as well as the 0610 maximum message length)

NOTE: The above fields are not required to be present. Additional fields may be present as needed. Fields need not be in any particular order. Fields may be less than specified maximum length.

Usage

Customer User Lookup Inquiry Data recommended usage when:

DE 60 = 6500082 (Consumer User Lookup Inquiry)

NOTE: DE 118 in 0610 response message contains the Authorized Users.

Consumer Account Lookup Inquiry Data Administrative Request/0600 Message

DE 116, Consumer Account Lookup Inquiry Data may contain the following fields.

Table 1422: Field details

Field name	Maximum length	Description
client_id	16	Unique identifier of credit client
store_code	16	Unique identifier of each store location
account_nbr	19	Account number
national_id	20	Consumer's national identification number United States—use Social Security number
cons_first_name	15	Consumer's first name
cons_middle_initial	1	Consumer's middle initial
cons_last_name	20	Consumer's last name
home_phone_nbr	15	Consumer's home phone number
user_first_name	15	User's first name
user_middle_initial	1	User's middle initial
user_last_name	20	User's last name
user_suffix	6	User's suffix; SR, JR, III, PHD

NOTE: The above fields are not required to be present. Additional fields may be present as needed. Fields need not be in any particular order. Fields may be less than specified maximum length.

Usage

Customer Account Lookup Inquiry Data recommended usage when:

DE 60 = 6500083 (Consumer Account Lookup Inquiry)

Consumer Account Lookup Response Data Administrative Request Response/0610 Message

DE 116, Consumer Account Lookup Response Data may contain the following fields.

Table 1423: Field details

Field name	Maximum length	Description
account_nbr	19	Account number
cons_first_name	15	Consumer's first name
cons_middle_initial	1	Consumer's middle initial
cons_last_name	20	Consumer's last name
home_addr1	25	Consumer's home address line 1
home_addr2	25	Consumer's home address line 2
home_city	20	Consumer's home city
home_sub_ctype	2	Consumer's home subnational entity alpha abbreviation; for example U.S. state code abbreviations
home_post_code	10	Consumer's home postal code
home_ctype	3	Consumer's home country alpha abbreviation
home_phone_nbr	15	Consumer's home phone number
work_phone_nbr	15	Consumer's work phone number
account_status	1	A = Active
		B = Blocked
		I = Inactive
credit_limit_amt	12	Credit limit amount
available_credit_amt	12	Available credit amount
bal_owe_amt	12	Balance owed amount
pay_owe_amt	12	Next payment owed amount
pay_due_date	10	Next payment due date; format CCYY/MM/DD
last_pay_amt	12	Last payment received amount
last_pay_date	10	Last payment received data CCYY/MM/DD
acct_open_date	10	Account open date; format CCYY/MM/DD

Field name	Maximum length	Description
custsvc_phone_nbr	15	Customer service phone number may be present for account lookup inquiries by customer or store.
language_pref	3	ENG = English, SPA = Spanish, FRE = French, and other supported ISO 639 codes
credit_terms	256	Credit terms may be present if applicable. Credit terms may include relevant information such as for revolving, non-revolving, or installment credit account.

NOTE: The above fields are not required to be present. Additional fields may be present as needed. Fields need not be in any particular order. Fields may be less than specified maximum length.

Usage

Customer Account Lookup Response Data recommended usage when:

DE 60 = 65–83 (Consumer Account Lookup Inquiry).

DE 117 (Reserved for National Use)

Mastercard recommends that DE 117 contain Business User Lookup Data and Business Account Lookup Data. This data element is typically present to request business user and account information and provide business user account information.

Table 1424: Attributes

Attribute	Description
Data Representation	ans...999; LLLVAR
Data Length	3 positions
Data Field	Contents of positions 1–999
Subfields	N/A
Justification	N/A

Usage

Following is the usage of DE 117 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 1425: Applicable messages

Message	Org	Sys	Dst
Administrative Request/0600	C	.	C
Administrative Request Response/0610	C	.	C

Application notes

Mastercard edits DE 117 for valid attributes but does not edit or log data contents.
At least one of DE 113–119 is mandatory within Administrative 0600/0610 messages.

Recommended format construction

117LLL<field_name>data<field_name>data...<field_name>data

117 is data element number

LLL is total length of all field names, separators and data

< indicates start of field_name

field_name is unique name for each field

> indicates end of field_name

Business User Lookup Inquiry Data Administrative Request/0600 Message

DE 117, Business User Lookup Inquiry Data may contain the following fields.

Table 1426: Fields

Field name	Maximum length	Description
client_id	16	Unique identifier of credit client
store_code	16	Unique identifier of each store location
account_nbr	19	Account number
tax_id	30	National government tax identifier
bus_lgl_name	24	Business legal name
doing_bus_as_name	40	Doing business as name
bus_phone_nbr	15	Business phone number
auth_user_id	5	ID assigned to authorized user

Field name	Maximum length	Description
user_list	1	Y = Request list of consumer account users (the maximum number of users to be provided will depend upon the client system as well as the 0610 maximum message length)

NOTE: The above fields are not required to be present. Additional fields may be present as needed. Fields need not be in any particular order. Fields may be less than specified maximum length.

Usage

Business User Lookup Inquiry Data recommended usage when:

DE 60 = 6500092 (Business User Lookup Inquiry)

NOTE: DE 118 in 0610 response message contains the Authorized Users.

Business Account Lookup Inquiry Data Administrative Request/0600 Message

DE 117, Business Account Lookup Inquiry Data may contain the following fields.

Table 1427: Field details

Field name	Maximum length	Description
client_id	16	Unique identifier of credit client
store_code	16	Unique identifier of each store location
account_nbr	19	Account number
tax_id	30	National government tax identifier
bus_lgl_name	24	Business legal name
doing_bus_as_name	40	Doing business as name
bus_phone_nbr	15	Business phone number
auth_user_id	5	ID assigned to authorized user
user_first_name	15	User's first name
user_middle_initial	1	User's middle initial
user_last_name	20	User's last name
user_suffix	6	User's suffix; SR, JR, III, PHD

Field name	Maximum length	Description
user_list	1	Y = Request list of consumer account users (the maximum number of users to be provided will depend upon the client system as well as the 0610 maximum message length)

NOTE: The above fields are not required to be present. Additional fields may be present as needed. Fields need not be in any particular order. Fields may be less than specified maximum length.

Usage

Business Account Lookup Request Data recommended usage when:

DE 60 = 6500093 (Business Account Lookup Inquiry)

Business Account Lookup Response Data Administrative Request Response/0610 Message

DE 117, Business Account Lookup Response Data may contain the following fields.

Table 1428: Field details

Field name	Maximum length	Description
account_nbr	19	Account number
tax_id	30	National government tax identifier
bus_lgl_name	24	Business legal name
doing_bus_as_name	40	Doing business as name
bus_addr1	25	Business address line 1
bus_addr2	25	Business address line 2
bus_city	20	Business city
bus_sub_ctype	2	Business subnational entity alpha abbreviation; for example U.S. state code abbreviations
bus_post_code	10	Business postal code
bus_ctype	3	Business country alpha abbreviation
bus_phone_nbr	15	Business phone number
account_status	1	A = Active B = Blocked

Field name	Maximum length	Description
		I = Inactive
credit_limit_amt	12	Credit limit amount
available_credit_amt	12	Available credit amount
bal_owe_amt	12	Balance owed amount
pay_owe_amt	12	Next payment owed amount
pay_due_date	10	Next payment due date; format CCYY/MM/DD
last_pay_amt	12	Last payment received amount
last_pay_date	10	Last payment received data CCYY/MM/DD
acct_open_date	10	Account open date; format CCYY/MM/DD
custsvc_phone_nbr	15	Customer service phone number may be present for account lookup inquiries by customer or store.
language_pref	3	ENG = English, SPA = Spanish, FRE = French, and other supported ISO 639 codes
credit_terms	256	Credit terms may be present if applicable. Credit terms may include relevant information such as for revolving, non-revolving, or installment credit account.
po_req	1	Y if purchase order required; N if not required
tax_exempt	1	Y if exempt from local taxes; N if not exempt
single_purch_limit	12	Amount limit per purchase
purchase_instructions	40	Contains special instructions applicable to purchases

NOTE: The above fields are not required to be present. Additional fields may be present as needed. Fields need not be in any particular order. Fields may be less than specified maximum length.

Usage

Business Account Lookup Response Data recommended usage when:

DE 60 = 6500093 (Business Account Lookup Inquiry)

DE 118 (Reserved for National Use)

Mastercard recommends that DE 118 contain Authorized Users. This data element may be present for consumer and business application requests and lookup responses.

Table 1429: Attributes

Attribute	Description
Data Representation	ans...999; LLLVAR
Data Length	3 positions
Data Field	Contents of positions 1–999
Subfields	N/A
Justification	N/A

Usage

Following is the usage of DE 118 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 1430: Applicable messages

Message	Org	Sys	Dst
Administrative Request/0600	C	.	C
Administrative Request Response/0610	C	.	C

Application notes

Mastercard edits DE 118 for valid attributes but does not edit or log data contents.

At least one of DE 113–119 is mandatory within Administrative 0600/0610 messages.

Recommended format construction

118LLL<field_name>data<field_name>data...<field_name>data

118 is data element number

LLL is total length of all field names, separators and data

< indicates start of field_name

field_name is unique name for each field

> indicates end of field_name

Authorized User Data Administrative Request/0600 Message

DE 118, Authorized User Data may contain the following fields.

Table 1431: Field details

Field name	Max length	Description
auth_user	0	The first field for each authorized user occurrence
user_first_name	15	User's first name
user_middle_initial	1	User's middle initial
user_last_name	20	User's last name
user_suffix	6	User's suffix; SR, JR, III, PHD
auth_user_id	5	ID assigned to authorized user

NOTE: The above fields are not required to be present. Additional fields may be present as needed. Fields need not be in any particular order. Fields may be less than specified maximum length. The above fields may be repeated as necessary within this DE 118.

Usage

Authorized User Data recommended usage when:

DE 60 = 6500080 (Consumer Application Request)

DE 60 = 6500090 (Business Application Request)

The number of users depends upon the client system as well as the 0610 maximum message length.

Field usage example

Assume for two authorized users and no assigned auth user id:

<auth_user> <user_first_name>John <user_middle_initial> Q <user_last_name> Public
<user_suffix>Mr <auth_user> <user_first_name> Mary <user_middle_initial>J <user_last_name>
Public <user_suffix> Mrs

Trade Reference Data Administrative Request/0600 Message

DE 118, Trade Reference may contain the following fields.

Table 1432: Field details

Field name	Maximum length	Description
trade_ref	0	The first field for each trade reference occurrence
trade_ref_name	40	Trade reference name
trade_ref_phone_nbr	15	Trade reference phone number
trade_ref_acct_nbr	28	Applicant's account number with trade reference

NOTE: The above fields are not required to be present. Additional fields may be present as needed. Fields need not be in any particular order. Fields may be less than specified maximum length. The above fields may be repeated as necessary within this DE 118.

Usage

Authorized User Data recommended usage when:

DE 60 = 6500090 (Business Application Request)

The number of users depends upon the client system as well as the 0610 maximum message length.

Field usage example

Assume for two trade references:

```
<trade_ref> <trade_ref_name>BigCityHardware <trade_ref_phone_nbr> 18885555555  
<trade_ref_acct_nbr>00000000 <trade_ref> <trade_ref_name> MidCityHardware  
<trade_ref_phone_nbr>18005555555 <trade_ref_acct_nbr> 000000000
```

Authorized User Response Data Administrative Request/0610 Message

DE 118, Authorized User Response Data may contain the following fields.

Table 1433: Field details

Field name	Maximum length	Description
auth_user	0	The header field for each occurrence
user_first_name	15	User's first name

Field name	Maximum length	Description
user_middle_initial	1	User's middle initial
user_last_name	20	User's last name
user_suffix	6	User's suffix; SR, JR, III, PHD
auth_user_id	5	ID assigned to authorized user

NOTE: The above fields are not required to be present. Additional fields may be present as needed. Fields need not be in any particular order. Fields may be less than specified maximum length. The above fields may be repeated as necessary within this DE 118.

Usage

Authorized User Data recommended usage when:

- DE 60 = 6500081 (Consumer Application Status Inquiry)
- DE 60 = 6500082 (Consumer User Lookup Inquiry)
- DE 60 = 6500083 (Consumer Account Lookup Inquiry)
- DE 60 = 6500091 (Business Application Status Inquiry)
- DE 60 = 6500092 (Business User Lookup Inquiry)
- DE 60 = 6500093 (Business Account Lookup Inquiry).
- The number of users depends upon the client system, as well as the 0610 maximum message length

Field usage example

Assume for two authorized users and no assigned auth user id:

<auth_user> <user_first_name>John <user_middle_initial> Q <user_last_name> Public
<user_suffix>Mr <auth_user> <user_first_name> Mary <user_middle_initial>J <user_last_name>
Public <user_suffix> Mrs

DE 119 (Additional Data: Private Use 2)

DE 119 (Additional Data: Private Use 2) contains data associated with various Mastercard programs, products, and services. DE 119 is used to convey data that is not related to any other ISO-defined data elements. It is a variable-length data element used for multiple purposes.

Table 1434: Attributes

Attribute	Description
Data representation	ans...999; LLLVAR
Data element length	3

Attribute	Description
Data field	Contents of subelements
Subelements	001 to 999 (future use)
Justification	Not applicable

Usage

The following applies to DE 119 usage:

- Subelements do not need to be in any particular order or sequence within DE 119.
- Customers must be able to send and receive all subelements available within DE 119.
- For subelements defined with a CE (conditional echo) presence notation for an Org (originator) entity, Mastercard will not reject a message from the message originator if the original subelement data is not returned as defined in a response message. Based on the presence notation for the Dst (destination) entity, Mastercard will forward the subelement data as received from the message originator if it is present in a response message.
- Subelements 900-999 are reserved for Mastercard internal use.

Values

The length field must be in the range 001–999. Subelements are identified by valid subelement ID and length.

Application notes

Mastercard may introduce new DE 119 subelements and subfields between releases to facilitate special processing within a country, region, or among pilot participants. Mastercard requires customers to be able to successfully process various online messages that may contain new unannounced DE 119 subelements and subfields.

DE 119 and DE 48 contain different subelement definitions. Subelements defined for DE 119 must be sent in DE 119, not DE 48.

This data element's content may vary by program and service. Additional or supplemental information that may be required in a transaction message is specified in Authorization Platform Edits for this data element.

For information on other uses of this data element, refer to the message layouts.

DE 119 encoding scheme

Mastercard organizes DE 119 into a group of encoded subelements and subfields. The following tables illustrate the structure of DE 119 subelements and subfields within a subelement.

Table 1435: Structure of DE 119 subelements

LLL	VAR (maximum length varies)					
3 bytes	3 bytes	3 bytes	Variable length	3 bytes	3 bytes	Variable length
Total data element length	First subelement data			Second subelement data		
	Subelement ID	Subelement length	Subelement variable length data	Subelement ID	Subelement length	Subelement variable length data

Table 1436: Attributes of DE 119 subelements

Number of bytes	Attribute	Value
3	Total data element length	The "LLL" portion of the data element
3	Subelement ID	In the range 001-999
3	Subelement length	Variable
1...nnn	Subelement variable length data	Contents of subfields

DE 119 subelement encoding scheme

The following tables illustrate the structure of DE 119 subfields within a subelement.

Table 1437: Structure of DE 119 subfields within a subelement

VAR (maximum length varies)					
3 bytes	3 bytes	Variable length	3 bytes	3 bytes	Variable length
First subfield data			Second subfield data		
Subfield ID	Subfield length	Subfield variable length data	Subfield ID	Subfield length	Subfield variable length data

Table 1438: Attributes of DE 119 subfields within a subelement

Number of bytes	Attribute	Value
3	Subfield ID	In the range 001-999
3	Subfield length	Variable
1...nnn	Subfield variable length data	Contents of subfields

Using DE 113–119 in Administrative 06xx Messages

The following information explains how to use DE 113–119 in Administrative 0600/0610 messages. Mastercard recommends that the contents of DE 113–119 from the 0600 message not be returned in the 0610 message to avoid the maximum message length constraint of 8k bytes.

Table 1439: DE 60 = 6500080 Consumer Application Request

MTI	DE	Presence	Data description
0600	113	Recommended	Application Generic Data
0600	113	Recommended	Application Banking Data
0600	114	Recommended	Consumer Application Request Data
0600	118	Optional	Authorized User Data
0610	114	Recommended	Consumer Application Response Data

Table 1440: DE 60 = 6500081 Consumer Application Status Inquiry

MTI	DE	Presence	Data description
0600	114	Recommended	Consumer Application Status Inquiry Data
0610	114	Recommended	Consumer Application Response Data
0610	118	Optional	Authorized User Response Data

Table 1441: DE 60 = 6500082 Consumer User Lookup Inquiry

MTI	DE	Presence	Data description
0600	116	Recommended	Consumer User Lookup Inquiry Data
0610	118	Recommended	Authorized User Response Data

Table 1442: DE 60 = 6500083 Consumer Account Lookup Inquiry

MTI	DE	Presence	Data description
0600	116	Recommended	Consumer Account Lookup Inquiry Data
0610	116	Recommended	Consumer Account Lookup Response Data
0610	118	Optional	Authorized User Response Data

Table 1443: DE 60 = 6500084 Consumer Account Maintenance Request

MTI	DE	Presence	Data description
0600	114	Recommended	Consumer Account Maintenance Data
0610	114	Recommended	Consumer Account Maintenance Data

Table 1444: DE 60 = 6500085 Consumer Counteroffer Reply

MTI	DE	Presence	Data Description
0600	113	Recommended	Application Generic Data
0610	114	Recommended	Consumer Application Response Data

Table 1445: DE 60 = 6500086 Consumer Preapproved Offer Inquiry

MTI	DE	Presence	Data Description
0600	114	Recommended	Consumer Application Status Inquiry Data
0610	114	Recommended	Consumer Application Response Data

Table 1446: DE 60 = 6500090 Business Application Request

MTI	DE	Presence	Data Description
0600	113	Recommended	Application Generic Data
0600	113	Recommended	Application Banking Data
0600	114	Optional	Consumer Application Request
0600	115	Recommended	Business Application Request Data
0600	118	Optional	Authorized User Data
0600	118	Optional	Trade Reference Data
0610	115	Recommended	Business Application Response Data

Table 1447: DE 60 = 6500091 Business Application Status Inquiry

MTI	DE	Presence	Data Description
0600	115	Recommended	Business Application Status Inquiry Data
0610	115	Recommended	Business Application Response Data
0610	118	Optional	Authorized User Response Data

Table 1448: DE 60 = 6500092 Business User Lookup Inquiry

MTI	DE	Presence	Data Description
0600	117	Recommended	Business User Lookup Inquiry Data
0610	118	Recommended	Authorized User Response Data

Table 1449: DE 60 = 6500093 Business Account Lookup Inquiry

MTI	DE	Presence	Data Description
0600	117	Recommended	Business Account Lookup Inquiry Data
0610	117	Recommended	Business Account Lookup Response Data
0610	118	Optional	Authorized User Response Data

Table 1450: DE 60 = 6500094 Business Account Maintenance Request

MTI	DE	Presence	Data Description
0600	115	Recommended	Business Account Maintenance Data
0610	115	Recommended	Business Account Maintenance Data

Table 1451: DE 60 = 6500095 Business Counteroffer Reply

MTI	DE	Presence	Data Description
0600	113	Recommended	Application Generic Data
0610	115	Recommended	Business Application Response Data

Table 1452: DE 60 = 6500096 Business Preapproved Offer Inquiry

MTI	DE	Presence	Data Description
0600	115	Recommended	Business Application Status Inquiry Data
0610	115	Recommended	Business Application Response Data

DE 120 (Record Data)

DE 120 (Record Data) is a variable-length data element used for transmitting file record data or textual character string data in various message types.

Table 1453: Attributes

Attribute	Description
Data Representation	ans...999; LLLVAR (supports extended character sets)
Data Length	3 positions
Data Field	Contents of subfields 1, or 2, or 3, or 4
Subfields	4
Justification	See subfields

Usage

Following is the usage of DE 120 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 1454: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	X	C
Authorization Request Response/0110	C	X	C
Issuer File Update Request/0302	C	.	.
Issuer File Update Request Response/0312	.	X	C
Administrative Advice/0620: system-generated	.	X	C
Administrative Advice/0620: member-generated	C	.	C

Values

See subfields.

Application notes

The following applies to DE 120.

Authorization Request/0100: AVS messages

When DE 120 is present in Authorization Request/0100 messages, DE 120 contains Address Verification Service data. Only one of the four possible subfields will be present in a message from an acquirer or to an issuer. In Authorization Request Response/0110: AVS messages, DE 120 contains the AVS data originally provided in the Authorization Request/0100 message.

Issuer File Update Request/0302 messages

When DE 120 is present in Issuer File Update Request/0302 messages, DE 120 contains the new, actual file record data used in "add" or "change" file maintenance requests. In Issuer File Update Request Response/0312 messages, DE 120 contains actual record data for file maintenance requests.

Administrative Advice/0620 messages

When DE 120 is present in Administrative Advice/0620 messages, DE 120 contains the following:

- In Administrative Advice/0620 messages, DE 120 includes the new subfields Token Requestor ID, Wallet ID, and Device Type.

- In Administrative Advice/0620: System-generated message where DE 60 is 600, DE 120 contains the first 200 bytes of an indecipherable message.
- In Administrative Advice/0620: Member-generated message where DE 60 is 650, DE 120 contains member-provided free format textual data.

Subfield 01 (AVS Service Indicator 1)

DE 120, subfield 01 (AVS Service Indicator 1) contains AVS data in this format for an issuer whose AVS Service Indicator status is 1.

Table 1455: Attributes

Attribute	Description
Subfield ID	01
Subfield Data Length	2 positions
Data Representation	ans...29
Data Field	Contents of positions 1–29
Justification	Left, trailing spaces

Table 1456: Values

Value	Description
Positions 1–9: Postal Code	
Postal Code	Cardholder postal/ZIP code
Positions 10–29: Address (Mastercard)	
Address	Cardholder address
Positions 10–29: Address (Visa, American Express)	
Address	Cardholder address

Application notes

Some merchants or acquirers are currently limited to supporting only numeric data.

The Authorization System will not truncate the address data passed from the acquirer if greater than 29 positions. DE 120 is an LLLVAR field; therefore, it will indicate the length of the data being passed. The issuer should code to accept the data if it is more than 29 positions.

Subfield 02 (AVS Service Indicator 2)

DE 120, subfield 02 (AVS Service Indicator 2) contains AVS data in this format for an issuer whose AVS Service Indicator status is 2.

Table 1457: Attributes

Attribute	Description
Subfield ID	02
Subfield Data Length	2 positions
Data Representation	an-14
Data Field	Contents of positions 1–14
Justification	Left, blank-filled

Table 1458: Values

Value	Description
Positions 1–9: Postal Code	
Postal Code	Cardholder postal/ZIP code
Positions 10–14: Address (Mastercard)	
Address	Cardholder address

Application notes

Issuer receives condensed address data. (This supports the algorithm that uses the first five numeric digits in an address [when scanning the address from left to right].)

Subfield 03 (AVS Service Indicator 3)

DE 120, subfield 03 (AVS Service Indicator 3) contains AVS data in this format for an issuer whose AVS Service Indicator status is 3.

Table 1459: Attributes

Attribute	Description
Subfield ID	03
Subfield Data Length	2 positions
Data Representation	an-14

Attribute	Description
Data Field	Contents of positions 1–14
Justification	Left, blank-filled

Table 1460: Values

Value	Description
Positions 1–9: Postal Code	
Postal Code	Cardholder postal/ZIP code
Positions 10–14: Address (Mastercard)	
Address	Cardholder address

Application notes

The issuer receives condensed address data. (This subfield supports the algorithm that uses up to the first five numeric digits that appear before the first alphabetic character or space in the address when scanning the address from left to right.)

Subfield 04 (AVS Service Indicator 4)

DE 120, subfield 04 (AVS Service Indicator 4) contains AVS data in this format for an issuer whose AVS Service Indicator status is 4.

Table 1461: Attributes

Attribute	Description
Subfield ID	04
Subfield Data Length	2 positions
Data Representation	an-14
Data Field	Contents of positions 1–14
Justification	Left, blank-filled

Table 1462: Values

Value	Description
Positions 1–9: Postal Code	

Value	Description
Postal Code	Cardholder postal/ZIP code
Positions 10–14: Address (Mastercard)	
Address	Cardholder address

Application notes

Issuer supports AVS, receives condensed numeric postal code and condensed numeric address only. (This supports the algorithm that uses only the numeric digits in a postal code and the first five numeric digits in an address [when scanning the address from left to right].)

Online file maintenance

DE 120 (Record Data), contains the record detail for the file type identified in DE 101 (File Name). The record detail contained in DE 120 depends on the value provided in DE 91 (Issuer File Update Code). The following table lists the allowed DE 91 values for each DE 101 file type.

Table 1463: Applicable MCCs

DE 101 (File Name)	DE 91 (Issuer File Update Code)			
	1 Add	2 Update	3 Delete	5 Inquiry
MCC102 (Stand-In Account File)	✓	✓	✓	✓
MCC103 (Electronic Warning Bulletin File)	✓		✓	✓
MCC104 (Local Stoplist File)	✓		✓	✓
MCC105 (Payment Cancellation File)	✓	✓	✓	✓
MCC106 (PAN Mapping File)	✓	✓	✓	
MCC109 (Application Transaction Counter [ATC] File)	✓			✓
MCC111 (PAN-PAR [Payment Account Reference] Mapping File)		✓		

MCC 102: Stand-In Account File

DE 120 (Record Data) contains Stand-In Account file data when DE 101 (File Name) contains MCC102.

Table 1464: Associated edits

Following are the MCC102 data fields that should be included in DE 120 depending on the value in DE 91 (Issuer File Update Code).

When DE 91 Contains...	Include These Fields in DE 120...
1 = Add	use all appropriate fields
2 = Update	Use all appropriate fields
3 = Delete	Use field 1
5 = Inquiry	Use field 1

Table 1465: Fields

Following is the DE 120 Layout for MCC102.

Field ID	Field Name	Attributes	Comments/Valid Values
1	Primary Account Number	an-19	Number that is embossed or encoded or both on the card. Customers may only input account numbers for BINs assigned to the associated Mastercard-assigned customer ID. Format: Sixteen digit Mastercard account number, followed by three spaces.
2	Entry Reason	an-1	Reason for listing this card. Valid values: C = Credit F = Fraud G = ATM Premium Listing L = Lost O = Other P = Capture Card S = Stolen U = Unauthorized Use V = Premium Listing X = Counterfeit

Field ID	Field Name	Attributes	Comments/Valid Values
3	Date Last Update Activity	n-6	Inquiry only and is ignored on add, change, and delete. (This field is returned in the 0312 response message only for inquiry requests.) Date of the last maintenance activity occurring on this account. Format: MMDDYY MM = month DD = day YY = year
4	Time Last Update Activity	n-4	Inquiry only and is ignored on add, change, and delete. (This field returned in the 0312 response message only for inquiry requests.) Time of the last maintenance activity occurring on this account. Format: hhmm hh = hour mm = minute
5	PIN Length	n-2	No longer applicable. Valid values: zeros.
6	Premium Listing Accumulative Limit	n-12	Valid only for entry reason V. For other reason codes, use 12 zeros.
7	Premium Listing Limit Currency Code	n-3	Valid only for Entry Reason V. For other reason codes, use three zeros.
8	Issuer-defined Purge Date	n-8	The issuer-defined purge date must be at least 180 days after the account list date in YYYYMMDD format. This field is optional.
9	Card Sequence Number	n-3	The card sequence number of the listed card. Only required for card-level support.
10	Card Expiration Date	n-6	The expiration date of the listed card in YYYYMM format. Only required for card-level support.

MCC 103: Electronic Warning Bulletin File

DE 120 (Record Data) contains Electronic Warning Bulletin file data when DE 101 (File Name) contains MCC103.

Table 1466: Fields

Following is the DE 120 Layout for MCC103.

Field ID	Field Name	Attributes	Comments/Valid Values
1	Primary Account Number	an-19	Number that is embossed or encoded on the card. Customers can only input account numbers for BINs assigned to the associated Mastercard-assigned customer ID. Format: Sixteen digit Mastercard account number followed by three spaces.
2	Customer ID	n-6	Mastercard customer ID assigned to the BIN. Format: right-justified, zero-filled
3	Card Program	a-3	Type of card bearing the account number. Refer to DE 63 (Network Data), for the list of card program values allowed. Card programs not allowed for MCC103 are footnoted in DE 63, subfield 1.
4	Response Code	n-2	Response Code this account listing should prompt for an authorization request. Valid value: 04 capture card
5	Entry Reason	a-1	Reason for listing this card. Valid values: C = Credit F = Fraud O = Other X = Counterfeit
6	Filler	an-25	Reserved for future AMS enhancements.

Field ID	Field Name	Attributes	Comments/Valid Values
7	Regional Indicator/ Purge Date	an-7	Issuer-requested region and purge date(s) associated with this account number listing. Field is non-positional (excludes regions and purge dates that do not apply). Field may occur up to six times for that many regions and purge dates. Must be entered in ascending order (1, A,..., E). Format: RYYMMDD R = Region Valid region values: 1 = United States A = Canada B = Latin America/Caribbean C = Asia/Pacific D = Europe E = Middle East/Africa YY = purge date year MM = purge date month DD = purge date day NOTE: To purge an account on the card expiration date, enter that date as the purge date. For additional information on purge dates, refer to the Account Management System User Manual.

MCC 104: Local Stoplist File

DE 120 (Record Data) contains Local Stoplist file data when DE 101 (File Name) contains MCC104.

Table 1467: Associated edits

Following are the MCC104 data fields that should be included in DE 120 depending on the value in DE 91 (Issuer File Update Code).

When DE 91 Contains...	Include These Fields in DE 120...
1 = Add	Use all applicable fields.
NOTE: Add will update an existing record if the system finds a match on the master file.	
1 = Update	Use all applicable fields that require update.
1 = Delete (deletes account from one region)	Use fields one.
1 = Delete (deletes account from one country)	Use fields one, seven, eight, and ten.
1 = Delete (deletes account from one subcountry)	Use fields one, seven, eight, nine, and ten.
3 = Delete (deletes account from all regions, countries, or subcountries)	Use field one.
5 = Inquiry	Use field one.

Table 1468: Fields

Following is the DE 120 Layout for MCC104.

Field ID	Field Name	Attributes	Comments/Valid Values
1	Primary Account Number	an-19	Number that is embossed, encoded, or both on the card. Customers may input only those account numbers that contain their assigned prefixes. Format: 16-digit Mastercard account number followed by three spaces.
2	Customer ID	n-6	Mastercard-assigned customer ID, a Mastercard customer may input only an account number with a BIN assigned to it. Format: Right-justified, zero-filled
3	Card Program	a-3	Type of card bearing the account number. Refer to DE 63 (Network Data), for the list of card program values allowed. Card programs not allowed for MCC104 are footnoted in DE 63, subfield 1.
4	Response Code	n-2	Response Code this account listing should prompt for an authorization request. Valid value: 04 capture card

Field ID	Field Name	Attributes	Comments/Valid Values
5	Entry Reason	a-1	Reason for listing this card. Valid values: C = Credit (Prompts an Authorization response of capture card at regional level) OR (Prompts an Authorization response of decline at the country/subcountry level) F = Fraud O = Other X = Counterfeit
6	Filler	an-25	Reserved for future AMS enhancements.
7	Region	an-1	The region associated with this account listing. If a country or subcountry listing is entered, this field must contain the issuer's region as follows: Valid region values: 1 = United States A = Canada B = Latin America/Caribbean C = Asia/Pacific D = Europe E = Middle East/Africa Region values must be entered in ascending order (1, A, B, C, D, E).
8	Country	an-3	If applicable, contains the country associated with this account listing. Must be a participating country within the issuer's region; values are three-digit country code or spaces. Refer to the Quick Reference Booklet for participating countries. Enter spaces if no country is specified.

Field ID	Field Name	Attributes	Comments/Valid Values
9	Subcountry	an-2	If applicable, contains the subcountry associated with this account listing. Must be a participating subcountry within the issuer's country; values are two-digit subcountry code or spaces. Refer to the Account Management System User Manual. Enter spaces if no subcountry is specified.
10	Purge Date	n-6	Purge date (format YYMMDD) associated with this account number listing as follows: YY = purge date year MM = purge date month DD = purge date day
11	Region	an-1	If applicable, second region associated with this account listing. If a second country or subcountry listing is entered, this field must contain the issuer's region. Region values must be entered in ascending order (1, A, B, C, D, E).
12	Country	an-3	If applicable, second country associated with this account listing; must be a participating country within the issuer's region.
13	Subcountry	an-2	If applicable, second subcountry associated with this account listing; must be a participating subcountry within the issuer's country.
14	Purge Date	n-6	If applicable, second purge date associated with this account number listing.
15	Region	an-1	If applicable, third region associated with this account listing; if a third country or subcountry listing is entered, this field must contain the issuer's region; region values must be entered in ascending order (1, A, B, C, D, E).
16	Country	an-3	If applicable, third country associated with this account listing; must be a participating country within the issuer's region.

Field ID	Field Name	Attributes	Comments/Valid Values
17	Subcountry	an-2	If applicable, third subcountry associated with this account listing; must be a participating subcountry within the issuer's country.
18	Purge Date	n-6	If applicable, third purge date associated with this account number listing.
19	Region	an-1	If applicable, fourth region associated with this account listing; if a fourth country or subcountry listing is entered, this field must contain the issuer's region; region values must be entered in ascending order (1, A, B, C, D, E).
20	Country	an-3	If applicable, fourth country associated with this account listing; must be a participating country within the issuer's region.
21	Subcountry	an-2	If applicable, fourth subcountry associated with this account listing; must be a participating subcountry within the issuer's country.
22	Purge Date	n-6	If applicable, fourth purge date associated with this account number listing.
23	Region	an-1	If applicable, fifth region associated with this account listing; if a fifth country or subcountry listing is entered, this field must contain the issuer's region; region values must be entered in ascending order (1, A, B, C, D, E).
24	Country	an-3	If applicable, fifth country associated with this account listing; must be a participating country within the issuer's region.
25	Subcountry	an-2	If applicable, fifth subcountry associated with this account listing; must be a participating subcountry within the issuer's country.
26	Purge Date	n-6	If applicable, fifth purge date associated with this account number listing.
27	Region	an-1	If applicable, sixth region associated with this account listing; if a sixth country or subcountry listing is entered, this field must contain the issuer's region; region values must be entered in ascending order (1, A, B, C, D, E).

Field ID	Field Name	Attributes	Comments/Valid Values
28	Country	an-3	If applicable, sixth country associated with this account listing; must be a participating country within the issuer's region.
29	Subcountry	an-2	If applicable, sixth subcountry associated with this account listing; must be a participating subcountry within the issuer's country.
30	Purge Date	n-6	If applicable, sixth purge date associated with this account number listing.

MCC 105: Payment Cancellation File

DE 120 (Record Data) contains Payment Cancellation file data when DE 101 (File Name) contains MCC105.

Table 1469: Associated edits

Following are the MCC105 data fields that should be included in DE 120 depending on the value in DE 91 (Issuer File Update Code).

When DE 91 Contains...	Include These Fields in DE 120...
1 = Add	Use all applicable fields; fields 1–5 are mandatory, fields 6–13 are optional
2 = Update	Use all applicable fields; fields 1–5 are mandatory, fields 6–7 are optional. Fields 8–13 not applicable.
3 = Delete	Use fields 1, 4, and 5; fields 2 and 3 must be filled with spaces. Fields 6–13 not applicable.
5 = Inquiry	Use fields 1, 4, and 5; fields 2 and 3 must be filled with spaces. Fields 6–13 not applicable.

Table 1470: Fields

Following is the modified DE 120 Layout for MCC105. Amount fields 6 and 7, when provided, must include a minor unit of currency. For example, USD 100.50 entered as 000000010050. Fields 8–9 are sent in the 0312 response message.

Field ID	Field Name	Attributes	Comments/Valid Values
1	Primary Account Number	an-19	Number that is embossed, encoded, or both on the card. Customers may only input account numbers for BINs assigned to the associated customer ID that Mastercard assigned Format: Left-justified, with trailing spaces
2	Entry Reason	an-1	Reason for listing this card. Valid value: A = Payment Cancellation
3	Issuer-defined Purge Date	n-8	The issuer-defined purge date must be no more than 15 months beyond the account list date in YYYYMMDD format.
4	Acquirer ID	n-6	Valid acquirer ICA.
5	Acceptor ID	ans-15	Must be unique per acquirer ICA. Must be a value other than spaces. NOTE: This field is case sensitive.
6	Transaction Amount (low)	n-12	Optional field. Right-justified with leading zeros. Indicates a single transaction amount or the start of a transaction amount range. Transaction amount must be stated in the transaction currency and not the cardholder billing currency.
7	Transaction Amount (high)	n-12	Optional field. Right-justified with leading zeros. Indicates the end of a transaction amount range. Do not use this field if specifying a single amount. If used, must be greater than Transaction Amount (low). Transaction amount must be stated in the transaction currency and not the cardholder billing currency.
8	Activity Date	n-6	Format is MMDDYY (Month, Day, Year); Provide spaces in request message if fields beyond Field ID 9 are present. System populates data in the response message.

Field ID	Field Name	Attributes	Comments/Valid Values
9	Activity Time	n-4	Format is hhmm (hour, minute); Provide spaces in request message if fields beyond Field ID 9 are present. System populates data in the response message.
10	Acceptor ID Code 2	ans-15	Must be unique per acquirer ICA.
11	Acceptor ID Code 3	ans-15	Must be unique per acquirer ICA.
12	Acceptor ID Code 4	ans-15	Must be unique per acquirer ICA.
13	Acceptor ID Code 5	ans-15	Must be unique per acquirer ICA.

DE 120 (When Blocking Recurring Payments)

Following are various uses of DE 120 for MCC105:

- Blocking all recurring payment arrangements with all merchants

A block of all recurring payments arrangements can only be accomplished by setting up blocks for each merchant and acquirer combination.

- Blocking all recurring payment arrangements with one merchant, or blocking a cardholder's only recurring payment with one merchant
 - Fields 1–5 are required for Add
 - Fields 6–7 should not be provided
- Blocking one of many recurring payment arrangements with one merchant

The one recurring payment arrangement can only be blocked if its transaction amount is different than any of the other recurring payment arrangements with a particular merchant.

The recurring payment arrangement to be blocked has a fixed (same amount each billing) transaction amount, such as an insurance premium billing

- Fields 1–5 are required for Add
- Field 6 should be provided to specify the fixed transaction amount
- Field 7 should not be provided

The recurring payment arrangement to be blocked has a variable (different amount each billing) transaction amount, such as a long distance telephone billing. For example, if the variable billing previous amounts were between USD 15.00 and USD 50.00, field 6 should contain USD 15.00 and field 7 should contain USD 50.00. This results in transactions being blocked within this specified amount range.

- Fields 1–5 are required for Add
- Field 6 is provided to specify the lowest transaction amount billable
- Field 7 is provided to specify the highest transaction amount billable

Usage of the amount range should only be used if the billing amount of the recurring payment to be blocked is unique and does not overlap with other recurring payments from the same merchant that should not be blocked. If one recurring payment arrangement billing

amount was typically between USD 15.00 and USD 50.00 but another recurring payment from the same merchant either had a variable amount that sometimes fell within USD 15.00 to USD 50.00 or had a fixed amount within USD 15.00 to USD 50.00, both recurring payments are blocked.

- The following guidelines apply for the multiple Acceptor IDs:
 - Issuers will only be able to provide multiple Acceptor IDs when PAN (field 1) and Acquirer ICA (field 4) are provided in DE 120.
 - The additional Acceptor IDs must be added in sequence. For example, if codes are entered in Acceptor ID fields 5, 10, and 12, skipping field 11, the 0302 Payment Cancellation MCC105 add request will be rejected with DE 39 = 27 (Edit Error) and DE 44 will contain field number 120011.
 - If issuers provide transaction amount or transaction amount range in the 0302 Payment Cancellation add list request, the transaction amount and transaction amount range will be applicable to all the given Acceptor IDs in the message.

MCC 106: PAN Mapping File

DE 120 (Record Data) contains Token to PAN Mapping File data when DE 101 (File Name) contains MCC106.

Table 1471: Associated edits

Following are the MCC106 data fields that should be included in DE 120 depending on the value in DE 91 (Issuer File Update Code).

When DE 91 Contains...	Include These Fields in DE 120...
1 = Add	Fields 1–2 mandatory, field 3 optional
2 = Update	Fields 1–2 mandatory, fields 3–5 optional Valid combinations of fields 1–5 are: • Fields 1–3 • Fields 1–3 and field 5; field 4 must contain spaces • Fields 1–2 and field 4; field 3 must contain spaces • Fields 1–2 and field 5; fields 3 and 4 must contain spaces
3 = Delete	Use field 1 only. All records associated to contactless account number will be deleted.

Table 1472: Fields

Following is the DE 120 Layout for MCC106 in the Issuer File Update Request/0302 Message.

Field ID	Field Name	Attributes	Comments/Valid Values
1	Contactless Account Number	an-19	Number that is assigned to the contactless card or device and transmitted from the card or device to the X terminal. Format: Left-justified, with trailing spaces
2	PAN	an-19	Number that is embossed, encoded, or both on the card. Customers may only input account numbers for BINs assigned to the associated customer ID that Mastercard assigned. Format: Left-justified, with trailing spaces
3	Card Expiration Date (PAN)	n-4	Expiration date that is embossed, encoded, or both on the card that represents the cardholder primary account number. Format: YYMM
4	Contactless account number (replacement)	an-19	Replacement account number that is assigned to the contactless card or device and transmitted from card or device to the contactless terminal. Format: Left-justified, with trailing spaces
5	PAN (replacement)	an-19	Number that is embossed, encoded, or both on the replacement card. Customers may only input account numbers for BINs assigned to the associated customer ID that Mastercard assigned Format: Left-justified, with trailing spaces

Table 1473: Fields

The following DE 120 layout applies to Issuer File Update Request/0302 Messages requesting the replacement of a PAN associated with an individual Mastercard Digital Enablement Service (MDES) token, or with all tokens associated with current PAN, in the MCC106 PAN Mapping File.

Field ID	Field Name	Attributes	Comments/Valid Values
1	Mapping File Indicator	an-1	M = Mastercard Digital Enablement Service Tokens excluding Static Tokens A = All Mastercard Digital Enablement Service Tokens including Static Tokens Best practice for the issuer to use value A

Field ID	Field Name	Attributes	Comments/Valid Values
2	Replacement PAN	an-19	Number replacing the number that is embossed, encoded, or both on the card (the primary account number). Customers may input only account numbers for BINs assigned to the associated customer ID assigned by Mastercard. The issuer has the option to provide a replacement primary account number for association to the token. Format: Left-justified, with trailing spaces
3	Replacement PAN Expiration Date	n-4	Expiration date that is associated with the number replacing the number that is embossed, encoded, or both on the card that represents the cardholder primary account number. If the issuer has provided a replacement primary account number, this field contains the expiration date that is associated with the replacement primary account number. If the issuer has not provided a replacement primary account number, this field contains the expiration date of the existing primary account number. Format: YYMM
4	Primary Account Card Sequence Number	ans-3	If the issuer has provided a replacement primary account number, this field will contain the card sequence number associated with the replacement primary account number. If the issuer has not provided a replacement primary account number, this field will contain the card sequence number associated with the original primary account number.

Field ID	Field Name	Attributes	Comments/Valid Values
5	Notify Wallet Service Provider Indicator	n-1	If the issuer has provided a replacement primary account number and replacement primary account number expiration date, this field indicates whether the Wallet Provider should be notified of the change. Values: 0 = Update token mapping information and notify the Wallet Provider with the primary account number information 1 = Update token mapping information, but do not notify the Wallet Provider with the primary account number information 2 = Do not update token mapping information, but do update the Wallet Provider with the primary account number information
6	Token	an-19	Surrogate value for a PAN that is consistent with ISO message requirements and is a 13 to 19-digit numeric value that passes basic validation rules of an account number, including the LUHN check. Format: Left-justified, with trailing spaces If field not present, update shall apply to all token to PAN mappings associated with the PAN as follows: This field is optional. If present, the token to PAN mapping update will only be applied to the specified token. If not present, the token to PAN mapping update will be applied to all tokens associated with the current PAN.

Table 1474: Fields

The following DE 120 layout applies to Issuer File Update Request/0302 Messages requesting the suspension, deactivation, or resumption of an MDES token in the MCC106 PAN Mapping File.

Field ID	Field Name	Attributes	Comments/Valid Values
1	Mapping File Indicator	an-1	M = Mastercard Digital Enablement Service Tokens excluding Static Tokens A = All Mastercard Digital Enablement Service Tokens including Static Tokens Best practice for the issuer to use value A
2	Action Required	an-1	S = Suspend token D = Deactivate token C = Resume token
3	Notify Wallet Service Provider Indicator	n-1	This field indicates whether the Wallet Provider should be notified of the change. Values: 0 = Update token mapping information and notify the Wallet Provider with the primary account number information.
4	Token—If Updating a Specific Token	an-19	Surrogate value for a PAN that is consistent with ISO message requirements and is a 13 to 19-digit numeric value that passes basic validation rules of an account number, including the LUHN check. Format: Left-justified, with trailing spaces. If field not present, update shall apply to all token to PAN mappings associated with the PAN.

Table 1475: Fields

The following DE 120 layout applies to Issuer File Update Request Response/0312 Messages created by Mastercard in response to an MCC106 PAN Mapping File update request.

Field ID	Field Name	Attributes	Comments/Valid Values
1	Contactless account number	an-19	Number that is assigned to the contactless card or device and transmitted from the card or device to contactless terminal. Format: Left-justified, with trailing spaces

Field ID	Field Name	Attributes	Comments/Valid Values
2	PAN	an-19	Number that is embossed, encoded, or both on the card. Customers may only input account numbers for BINs assigned to the associated customer ID that Mastercard assigned. Format: Left-justified, with trailing spaces
3	Card Expiration Date (PAN)	n-4	Expiration date that is embossed, encoded, or both on the actual card. Format: YYMM
4	Contactless account number (replacement)	an-19	Replacement number that is assigned to the contactless card or device and transmitted from the card or device to contactless terminal. Format: Left-justified, with trailing spaces
5	PAN (replacement)	an-19	Number that is embossed, encoded, or both on the replacement card. Customers may only input account numbers for BINs assigned to the associated customer ID that Mastercard assigned. Format: Left-justified, with trailing spaces
6	Date-Last-Update Activity	n-6	Format: MMDDYY (Month, Day, Year)
7	Time-Last-Update Activity	n-4	Format: hhmm (hour, minute)

MCC106—PAN Mapping File – Token Service Provider Maintenance

Table 1476: Fields

The following table provides the technical details of DE 120 layout for MCC106 in the Issuer File Update Request/0302 message.

Field ID	Field Name	Attributes	Comments/Valid Values
15	KDI Value	an-3	Conditional Required for Cloud Processing Type 2D. Optional for other Cloud Types.
16	AIP List ID	an-5	Conditional Required for Cloud Processing Type 2D. Optional for other Cloud Types.

Field ID	Field Name	Attributes	Comments/Valid Values
17	CVM List ID Primary	an-3	Conditional Required for Cloud Processing Type 2D. Optional for other Cloud Types.
18	CVM List ID Secondary	an-3	Conditional Required for Cloud Processing Type 2D. Optional for other Cloud Types.
19	CVN	an-2	Conditional Required for Cloud Processing Type 2D. Optional for other Cloud Types.

MCC 109: Application Transaction Counter File

DE 120 (Record Data) contains Application Transaction Counter file data when DE 101 (File Name) contains MCC109.

Table 1477: Associated edits

Following are the MCC109 data fields that should be included in DE 120 depending on the value in DE 91 (Issuer File Update Code).

When DE 91 Contains...	Include These Fields in DE 120...
1 = Add	Fields 1–4 mandatory
3 = Delete	Fields 1–4 mandatory
NOTE: Applicable only when submitting an R311 bulk file deletion request.	
5 = Inquiry	Fields 1–3 mandatory

Table 1478: Associated edits

Following is the DE 120 Layout for MCC109 in the Issuer File Update Request/0302 message. Fields 1 through 3 must be considered contiguous to identify if an entry exists when trying to delete a record.

Field ID	Field Name	Attributes	Comments/Valid Values
1	Contactless Account Number	an-19	Number that is assigned to the contactless card or device and transmitted from the card or device to the contactless terminal Format: Left-justified, with trailing spaces
2	Card Sequence Number	n-1	May be zero
3	Contactless Account Expiration Date	n-4	Expiration date Format: YYMM
4	Contactless Application Transaction Counter (ATC) Value	n-5	Right-justified, leading zeros

Table 1479: Associated edits

Following is the DE 120 Layout for MCC109 in the Issuer File Update Request Response/0312 message.

Field ID	Field Name	Attributes	Comments/Valid Values
1	Contactless Account Number	an-19	Number that is assigned to the contactless card or device and transmitted from the card or device to contactless terminal Format: Left-justified, with trailing spaces
2	Card Sequence Number	n-1	May be zero
3	Contactless Account Expiration Date	n-4	Expiration date Format: YYMM
4	Contactless Application Transaction Counter (ATC) Value	n-5	Right-justified, leading zeros
5	Filler	an-1	
6	Issuer Customer ID	n-6	
7	Date/Time-Last-Update Activity	n-12	Format: MMDDYYHHMMSS
8	Entry Change Indicator	an-1	Indicates how the ATC Value was last changed B = batch entry O = online entry T = transaction

Field ID	Field Name	Attributes	Comments/Valid Values
9	Creation Date/Time Stamp	n-14	Format: MMDDCCYYHHMMSS
10	Last Transaction Time	n-4	Last transaction time for valid ATC. Format: hhmm

MCC 111: PAN-PAR (Payment Account Reference) Mapping File

DE 120 (Record Data) contains PAN to PAR Mapping File data when DE 101 (File Name) contains MCC111.

Only PAN replacement and PAR record deletion actions are allowed. Both are handled as an "update".

Table 1480: Associated edits

Following are the MCC111 data fields that should be included in DE 120 depending on the combination of values in DE 91 (Issuer File Update Code) and DE 120 subfield 1 (Action Required).

When DE 91 Contains...	Include These Subfields in DE 120...
2 = Update	Subfield 1 is mandatory
This is the only valid Issuer File Update Code value for MCC111	- When subfield 1 = U (Update), subfield 2 must contain a replacement PAN - When subfield 1 = D (Delete), all other DE 120 subfields must contain spaces

Table 1481: Subfields

The following DE 120 layout applies to Issuer File Update Request/0302 messages requesting the replacement of a PAN associated with a PAR value in the MCC111 PAN-PAR Mapping File. This action should be performed when the issuer is replacing the consumer's current PAN with a new PAN.

Subfield ID	Subfield Name	Attributes	Comments/Valid Values
1	Action Required	an-1	U = Update the PAN-PAR mapping record
2	Replacement PAN	an-19	Number that is embossed, encoded, or both on the replacement card. Customers may only input account numbers for BINs assigned to the associated customer ID assigned by Mastercard. Format: Left-justified, with trailing spaces

Table 1482: Subfields

The following DE 120 layout applies to Issuer File Update Request/0302 messages requesting the deletion of a PAN-PAR mapping record in the MCC111 PAN-PAR Mapping File. This action should be performed only when the issuer has permanently closed/cancelled the consumer's PAN.

Subfield ID	Subfield Name	Attributes	Comments/Valid Values
1	Action Required	an-1	D = Delete the PAN-PAR mapping record associated with the PAN in DE 2 (Primary Account Number)

DE 120 (Error Codes)

Following are the DE 120 Error Codes.

MCC 102: Error Codes

The Dual Message Authorization System returns an Issuer File Update Request Response/0312 message where DE 39 contains the value 27 and DE 44 contains the value 120xxx (xxx indicates the DE 120 field in which the error occurred).

Table 1483: Related errors and error messages

Error Code	Error Message
120001	Primary Account Number (PAN), extended not numeric -Or- BIN in Primary Account Number (PAN), extended not on FIM -Or- Origin of message invalid for BIN in Primary Account Number (PAN), extended -Or- Check digit in Primary Account Number (PAN), extended not correct -Or- Primary Account Number (PAN), extended not on Account File (for delete or inquiry).
120002	Entry Reason not one of the following: (C, F, G, L, O, P, S, U, V, X).
120005	PIN length not numeric or spaces.
120006	Entry-Reason = V and Premium Listing Accumulative Limit Amount not numeric.
120007	Invalid Currency Code
120008	Invalid Issuer-defined Purge Date
120009	Invalid Card Sequence Number

Error Code	Error Message
120010	Invalid Card Expiration Date

MCC 103: Error Codes

The Dual Message Authorization System returns an Issuer File Update Request Response/0312 message where DE 39 contains the value 25 or 27 and DE 44 contains the value 120xxx (xxx indicates the DE 120 field in which the error occurred).

When DE 39 contains the value 25 (Unable to locate record on file [no action taken]), DE 44 contains the following error code:

Table 1484: Related errors and error messages

Error Code	Error Message
120001	Primary account number not on file.

When DE 39 contains the value 27, DE 44 contains one of the following error codes:

Table 1485: Related errors and error messages

Error Code	Error Message
120001	Primary account number not on file -Or- Primary Account Number (PAN), extended is not numeric -Or- BIN in Primary Account Number (PAN), extended is invalid -Or- Check digit in cardholder number is not correct and product code is not equal to MNS (Non-standard) -Or- Origin of message invalid for BIN in Primary Account Number (PAN), Extended.
120002	Issuing ICA (customer ID) is not associated with Primary Account Number (PAN), Extended BIN.
120003	Product code is not equal to a valid card program value.
120004	Response code not equal to 04 (capture card).
120005	Entry reason is not C, F, O, or X.

Error Code	Error Message
120006	Regional indicator is not 1, A, B, C, D, or E. -Or- Regional indicator is not in ascending order.
120007	Region-requested purge date invalid. -Or- Region-requested purge date not equal to current or future date. -Or- Requested purge date must be at least one day beyond current date if account is not already listed.

MCC 104: Error Codes

The Dual Message Authorization System returns an Issuer File Update Request Response/0312 message where DE 39 contains the value 25, 27, or 30 and DE 44 contains the value 120xxx (xxx indicates the DE 120 field in which the error occurred.)

When DE 39 contains the value 25, DE 44 contains the following error code:

Table 1486: Related errors and error messages

Error Code	Error Message
120001	Account not on file.

When DE 39 contains the value 27, DE 44 contains one of the following error codes:

Table 1487: Related errors and error messages

Error Code	Error Message
120001	Primary Account Number (PAN), Extended is not numeric -Or- BIN in Primary Account Number (PAN), Extended is invalid -Or- Check digit in cardholder number is not correct and product code is not equal to PNS (Non-standard) -Or- Origin of message invalid for BIN in Primary Account Number (PAN), Extended.

Error Code	Error Message
120002	Issuing ICA (customer ID) is not associated with Primary Account Number (PAN)
120003	Card Program is not equal to a valid value -Or- BIN is not set up for proprietary card program, and card program in the request equals PRO or PNS.
120004	Response code is not 04.
120005	Entry reason is not C, F, O, or X.
120006	Regional indicator is not equal to 1, A, B, C, D, or E -Or- Regional indicator is not in ascending order.
120007	Country invalid or outside issuer region.
120008	Subcountry invalid or outside issuer country.
120009	Purge date invalid -Or- Purge date not equal to current or future date -Or- Requested purge date must be at least one day beyond current date if account is not already listed.

MCC 105: Error Codes

If the Dual Message Authorization System detects an error in the Issuer File Update Request/0302 message, it returns an Issuer File Update Request Response/0312 message containing error codes specific to each file name. DE 39 (Response Code) contains value 27 (Issuer File Update field edit error) and DE 44 (Additional Response Data) contains the value 120xxx (where xxx indicates the DE 120 field in which the error occurred).

Table 1488: Related errors and error messages

Error Code	Error Message
120001	Primary Account Number (PAN) not within account range that is a Payment Cancellation participant -Or- Primary Account Number (PAN) not numeric -Or- BIN in Primary Account Number (PAN) not valid -Or- Check digit in Primary Account Number (PAN) not correct -Or- Primary Account Number (PAN) not on Payment Cancellation File (for update, delete, or inquiry)
120002	Entry Reason is not (A)
120003	Invalid Issuer-defined Purge Date format -Or- Issuer-defined Purge Date not equal to or greater than system date -Or- Issuer-defined Purge Date not greater than system date by 15 months
120004	Acquirer ICA not numeric
120005	Acceptor ID not present
120006	Transaction Amount (low limit) not numeric
120007	Transaction Amount (high limit) not numeric -Or- Transaction Amount (high limit) not greater than Transaction Amount (low limit) -Or- Transaction Amount (high limit) is present but Transaction Amount (low limit) is not present

Error Code	Error Message
120010–120013	Invalid Acceptor ID If multiple Acceptor ID occurrences are not provided in sequence, then Add request will be rejected with edit error on the respective field that has spaces in-between two valid Acceptor IDs. For example: If fields 5, 10, 12 are provided with valid values and field 11 is provided with spaces, then Add request will be rejected with edit error on field 11, DE44=120011.

MCC 106: Error Codes

DE 39 contains the appropriate value and DE 44 contains value 120xxx (where xxx indicates the DE 120 field in which the error occurred).

When DE 39 contains value 25 (Unable to locate record on file), DE 44 contains the following error code:

Table 1489: Related errors and error messages

Error Code	Error Message
120001	Contactless account number not on MCC106 (PAN Mapping File) (for update or delete)
120002	PAN not on MCC106 (PAN Mapping File) (for update)

When DE 39 contains value 27 (Issuer File Update Field Edit Error), DE 44 contains one of the following error codes:

Table 1490: Related errors and error messages

Error Code	Error Message
120001	Contactless account number not numeric -Or- Prefix in contactless account number not valid -Or- Check digit in contactless account number not correct -Or- Contactless account number not within account range that is a Contactless Mapping Service participant

Error Code	Error Message
120002	PAN not numeric -Or- Prefix in PAN not valid -Or- Check digit in PAN not correct -Or- PAN and contactless account number prefixes do not have same country and product -Or- PAN prefix should not be participating in the Contactless Mapping Service
120003	Invalid PAN card expiration date
120004	Replacement contactless account number not numeric -Or- Prefix in Replacement contactless account number not valid -Or- Check digit in Replacement contactless account number not correct -Or- Replacement contactless account number not within account range that is a Contactless Mapping Service participant
120005	Replacement PAN not numeric -Or- Prefix in Replacement PAN not valid -Or- Check digit in Replacement PAN not correct -Or- Replacement PAN and contactless account number prefixes do not have same country and product -Or- Replacement PAN prefix should not be participating in the Contactless Mapping Service

Error Code	Error Message
120015	Invalid Token Type -Or- Invalid Cloud Processing Type -Or- Invalid Key Derivation Index
120016	Invalid Token Type -Or- Invalid Cloud Processing Type -Or- Invalid AIP List ID
120017	Invalid Token Type -Or- Invalid Cloud Processing Type -Or- Invalid CVM List ID Primary
120018	Invalid Token Type -Or- Invalid Cloud Processing Type -Or- Invalid CVM List ID Secondary
120019	Invalid Token Type -Or- Invalid Cloud Processing Type -Or- Invalid CVN

When DE 39 contains value 80 (Duplicate add, action not performed), DE 44 contains the following error code:

Table 1491: Related errors and error messages

Error Code	Error Message
120001	Contactless account number is a duplicate
120004	Replacement contactless account number is a duplicate

MCC 109: Error Codes

DE 39 contains the appropriate value and DE 44 contains value 120xxx (where xxx indicates the DE 120 field in which the error occurred).

When DE 39 contains value 25 (Unable to locate record on file), DE 44 contains the following error code:

Table 1492: Related errors and error messages

Error Code	Error Message
120001	Contactless Account Number/Card Sequence Number/Contactless Account Expiration Date not on Contactless Application Transaction Counter (ATC) File (MCC109) (for add or inquiry)

When DE 39 contains value 27 (Issuer File Update Field Edit Error), DE 44 contains one of the following error codes:

Table 1493: Related errors and error messages

Error Code	Error Message
120001	Contactless Account Number not numeric -Or- Prefix in contactless account not valid -Or- Check digit in Contactless Account Number not correct
120002	Card Sequence Number not numeric
120003	Contactless Account Expiration Date invalid
120004	ATC not numeric

When DE 39 contains value 80 (Duplicate add, action not performed), DE 44 contains the following error code:

Table 1494: Related errors and error messages

Error Code	Error Message
120001	Contactless Account Number/Card Sequence Number/Contactless Account Expiration Date already exists on file (applies to Add)

MCC 111: Error Codes

DE 39 contains the appropriate value and DE 44 contains value 120xxx (where xxx indicates the DE 120 subfield in which the error occurred).

When DE 39 contains value 25 (Unable to locate record on file [no action taken]), DE 44 contains the following error code:

Table 1495: Related errors and error messages

Error Code	Error Message
120001	The PAN in DE 2 is not associated with a PAR value in the MCC111 PAN-PAR Mapping File and therefore the action requested in subfield 1 cannot be performed.

When DE 39 contains value 27 (Issuer File Update field edit error), DE 44 contains the following error code:

Table 1496: Related errors and error messages

Error Code	Error Message
120001	The Action Required value in DE 120 subfield 1 is not D (Delete) or U (Update). Value is space or other than D or U. -Or- The Action Required value in DE 120 subfield 1 is D (Delete) but the PAN in DE 2 is associated with a token in the MCC106 Token-PAN Mapping File that is in active or suspended status. Therefore the PAN-PAR mapping record cannot be deleted.

Error Code	Error Message
120002	The content of DE 120 subfield 2 is not properly formatted • Not numeric (alpha, special, or spaces) • Incorrect length • Invalid check digit -Or- The Action Required value in DE 120 subfield 1 is U (Update) but no replacement PAN is present in subfield 2. -Or- The Action Required value in DE 120 subfield 1 is U (Update) but the account range of the replacement PAN in subfield 2 is not eligible for association with a PAR value. -Or- The Action Required value in DE 120 subfield 1 is U (Update) but the replacement PAN in subfield 2 is already associated with another PAR value.

When DE 39 contains value 80 (Duplicate Add, action not performed), DE 44 contains the following error code:

Table 1497: Related errors and error messages

Error Code	Error Message
120002	The PAN value in DE 2 is a duplicate of the PAN value in DE 120 subfield 2. The replacement PAN in subfield 2 must be different than the current PAN in DE 2.

DE 121 (Authorizing Agent ID Code)

DE 121 (Authorizing Agent ID Code), when used, must contain the appropriate Mastercard-assigned customer ID number that uniquely identifies the Dual Message Authorization System Stand-In processing facility or alternate routing customer responsible for performing Stand-In processing on-behalf of the issuer.

Table 1498: Attributes

Attribute	Description
Data Representation	n...6; LLLVAR

Attribute	Description
Data Length	3 positions
Data Field	Contents of positions 1–6
Subfields	N/A
Justification	N/A

Usage

Following is the usage of DE 121 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 1499: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	C	.	C
Authorization Advice/0120: acquirer-generated	C	.	C
Authorization Advice/0120: system-generated	.	C	C
Reversal Request Response/0410	C	.	C
Reversal Advice/0420	.	C	C

Values

Must contain a valid Mastercard customer ID number. Any values reserved for Mastercard use will be in the range 000000-000999.

The following values may be present in DE 121:

- If the 0110 message was responded to by the issuer, the value may contain the issuer's Mastercard ID.
- If the 0110 message was responded to by the MIP X-Code System, the value will be 000000.
- If the 0110 message was responded to by the Stand-In System, the value will be 000001.
- If the 0110 message was responded to by an alternate issuer host route, the value will be 000002.
- If a decline occurred due to an on-behalf service, the value will be 000003.
- If the 0110 message was responded to by the Mastercard Rewards System, the value will be 000004 (Mastercard Rewards System).

Application notes

This data element is defined and used identically within all Mastercard programs and services.

When Stand-In processing or “alternate authorizer” performs an Authorization Request/0100 on behalf of an issuer, it must insert this data element into the appropriate response message (for example, Authorization Request Response/0110) and into the appropriate Authorization Advice/0120—System-generated message so that the entire transaction audit trail clearly identifies the specific authorizing agent that actually approved the transaction.

The acquirer must insert this value in the Authorization Advice/0120 message when DE 121 is present in the Authorization Request Response/0110 message.

DE 121 is not required in Authorization Request/0100 or Authorization Advice/0120 messages the original issuer or its designated “primary” authorizing agent initiates.

DE 121, value 000002 is applicable only to issuers that use an alternate issuer host route instead of the Stand-In System. The issuer will provide in the Authorization Request Response/0110 message DE 121, value 000002 when an alternate issuer processed the original request.

DE 121 will have the value 000003 when a decline occurred due to an on-behalf service.

DE 121, value 000004 is applicable only to issuers participating in the Mastercard Pay with Rewards service and will only be present in the Authorization Advice/0120: system-generated messages notifying issuers of a declined request. The specific reason for the decline will be specified in DE 60 (Advice Reason Code).

DE 122 (Additional Record Data)

DE 122 is a free-format, variable-length data element used for transmitting file record data in various message types.

Table 1500: Attributes

Attribute	Description
Data Representation	ans...999; LLLVAR
Data Length	3 positions
Data Field	Contents of subelements
Subfields	1
Justification	See “Subelements”

Usage

Following is the usage of DE 122 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 1501: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	•	C
Authorization Advice/0120: acquirer-generated	C	•	C
Authorization Advice/0120: system-generated	•	X	C
Issuer File Update Request Response/0312	•	X	C
Reversal Request/0400	C	•	C
Reversal Advice/0420	•	X	C

Application notes

Subelement 001 contains the Acceptor Street Address, the Acceptor Country Subdivision Code, Acceptor URL Address, Acceptor Customer Service Phone Number, Acceptor Phone Number, Acceptor Additional Contact Information, Acceptor Tax ID, Partner ID Code, Service Location, City Name, Service Location Country Subdivision Code, Service Location Country Code, and Service Location Postal Code, Acceptor Geographic Coordinates, Service Location Geographic Coordinates and should be used for the specified message types.

Encoding scheme for DE 122

Mastercard organizes DE 122 into a group of encoded subelements and subfields.

Table 1502: Structure of DE 122 subelement

LLL	VAR: maximum length varies		
3 bytes	3 bytes	3 bytes	Variable length
Total length	First subelement data		
	Subelement ID	Subelement length	Subelement variable length

Table 1503: Values

Number of Bytes	Attribute	Value
3	Total data element length	The "LLL" portion of the data element

Number of Bytes	Attribute	Value
3	Subelement ID	In the range 001–099
3	Subelement length	Variable
1...nnn	Subelement variable length data	Contents of Subfields

Table 1504: Structure of DE 122 subfields within a subelement

LLL	VAR: maximum length varies	
3 bytes	3 bytes	Variable length
First subfield data		
Subfield ID	Subfield length	Subfield variable length data

Table 1505: Values

Number of bytes	Attribute	Value
3	Subfield ID	In the range of 001-099
3	Subfield length	Variable
1...nnn	Subfield variable length data	Contains valid values

Subelement 001 (Additional Acceptor Data)

DE 122 (Additional Record Data), subelement 001 (Additional Acceptor Data) contains data for identifying, locating, and contacting acceptors.

Table 1506: Attributes

Attribute	Description
Subelement ID	001
Subelement Data Length	3 positions
Data Representation	ans...543; LLLVAR
Data Field	Contents of subfields

Attribute	Description
Subfields	Subfield 001 (Acceptor Street Address) Subfield 002 (Acceptor Country Subdivision Code) Subfield 003 (Acceptor URL address) Subfield 004 (Acceptor Customer Service Phone Number) Subfield 005 (Acceptor Phone Number) Subfield 006 (Acceptor Additional Contact Information) Subfield 007 (Acceptor Tax ID) Subfield 008 (Partner ID Code) Subfield 009 (Service Location City Name) Subfield 010 (Service Location Country Subdivision Code) Subfield 011 (Service Location Country Code) Subfield 012 (Service Location Postal Code) Subfield 013 (Acceptor Geographic Coordinates) Subfield 014 (Service Location Geographic Coordinates)
Justification	See subfields

Table 1507: Usage

Message	Org	Sys	Dst
Authorization Request/0100	C	•	C
Authorization Advice/0120: acquirer-generated	C	•	C
Authorization Advice/0120: system-generated	•	X	C
Reversal Request/0400	C	•	C
Reversal Advice/0420	•	X	C

Subfield 001 (Acceptor Street Address)

DE 122, subelement 001, subfield 001 (Acceptor Street Address) contains the street address of the acceptor, which should include the street number, street name, and other identifiers of the precise location, such as a building or unit number.

Table 1508: Attributes

Attribute	Description
Subfield ID	001
Subfield Data Length	3 positions
Data Representation	ans...048; LLLVAR
Data Field	Contents of subfield 001
Justification	Left

Values

Acceptor Street Address must contain all Address Line 1 and Address Line 2 information, including

- the name of the street or throughfare
- the locality or community
- the building or house number; and
- any apartment, unit, suite, or floor number, if applicable.

This information must be populated and formatted as defined by country postal standards.

Processing rules

The Acceptor Street Address must be present for all transactions and contain a valid location for the acceptor. The location of the acceptor will generally be the physical location where the transaction occurred or business location for transactions that occur virtually.

If the information for Acceptor Street Address does not exist per country postal standards, then the value of "Unavailable" must be provided.

Program rules

Payment facilitator-enabled transactions: The sponsored merchant street address must be provided.

Subfield 002 (Acceptor Country Subdivision Code)

DE 122, subelement 001, subfield 002 (Acceptor Country Subdivision Code) contains the ISO 3166-2 code identifying the state, province, or other country subdivision of the acceptor.

Table 1509: Attributes

Attribute	Description
Subfield ID	002
Subfield Data Length	3 positions
Data Representation	ans-003
Data Field	Contents of subfield 002
Justification	Left with trailing spaces

Values

Acceptor Country Subdivision Code must contain a valid code as defined in Chapter 2 of the [Quick Reference Booklet](#) and must be followed by trailing spaces for codes that are less than three positions.

Processing rules

The Acceptor Country Subdivision Code must be present for all transactions when defined in Chapter 2 of the [Quick Reference Booklet](#).

Program rules

Payment facilitator-enabled transaction: The sponsored merchant country subdivision code must be provided.

Subfield 003 (Acceptor URL address)

DE 122, subelement 001, subfield 003 (Acceptor URL Address) contains the web address of the acceptor.

Table 1510: Attributes

Attribute	Description
Subfield ID	003
Subfield Data Length	3 positions
Data Representation	ans...255; LLLVAR

Attribute	Description
Data Field	Contents of subfield 003
Justification	Left

Processing rules

The Acceptor URL Address must be present for all electronic commerce transactions and contain a valid website address. For all other transactions, it must be provided unless a customer service phone is provided for contact.

Program rules

Payment facilitator-enabled transaction: The sponsored merchant acceptor URL should be provided.

Subfield 004 (Acceptor Customer Service Phone Number)

DE 122, subelement 001, subfield 004 (Acceptor Customer Service Phone Number) contains the phone number of the acceptor that can be used for transaction inquiries.

Table 1511: Attributes

Attribute	Description
Subfield ID	004
Subfield Data Length	3 positions
Data Representation	ans...016; LLLVAR
Data Field	Contents of subfield 004
Justification	Left

Values

Acceptor Customer Service Phone Number must contain a valid value defined and formatted as per local telecom or the ITU Telecommunication Standardization Sector (ITU-T) standards.

For U.S. region and Canada region transactions, if present, Acceptor Customer Service Phone Number must be 10 positions with the beginning three positions numeric.

Application notes

The Acceptor Customer Service Phone Number should be present for all transactions and contain a valid phone number, unless a URL is provided for contact.

For U.S. region and Canada transactions Acceptor Customer Service Phone Number must be present for mail order, phone order, and standing/recurring transactions.

Program rules

Payment facilitator-enabled transactions: The sponsored merchant customer service phone number should be provided.

Subfield 005 (Acceptor Phone Number)

DE 122, subelement 001, subfield 005 (Acceptor Phone Number) contains the business phone number of the acceptor, if different from the acceptor customer service phone number.

Table 1512: Attributes

Attribute	Description
Subfield ID	005
Subfield Data Length	3 positions
Data Representation	ans...016; LLLVAR
Data Field	Contents of subfield 005
Justification	Left

Values

Acceptor Phone Number must contain a valid value defined and formatted as per local telecom or the ITU Telecommunication Standardization Sector (ITU-T) standards.

For U.S region and Canada region transactions, if present, Acceptor Phone Number must be 10 positions with the beginning three positions numeric.

Processing rules

The acceptor phone number can be provided when different from the acceptor customer service phone number.

Subfield 006 (Acceptor Additional Contact Information)

DE 122, subelement 001, subfield 006 (Acceptor Additional Contact Information) contains additional information for contacting the acceptor, such as an additional phone number or a contact name.

Table 1513: Attributes

Attribute	Description
Subfield ID	006
Subfield Data Length	3 positions
Data Representation	ans...025; LLLVAR
Data Field	Contents of subfield 006
Justification	Left

Processing rules

Acceptor Additional Contact Information may contain additional information, such as secondary phone numbers, customer service operation hours, and contact name.

Subfield 007 (Acceptor Tax ID)

DE 122, subelement 001, subfield 007 (Acceptor Tax ID) contains the identifier of the acceptor issued by a taxation authority.

Table 1514: Attributes

Attribute	Description
Subfield ID	007
Subfield Data Length	3 positions
Data Representation	ans-021
Data Field	Contents of subfield 007
Justification	Left

Values

The subfield must contain a valid value formatted per issuing taxation authority standards in positions 001 through 020 with trailing spaces and position 021 with a value from the following table.

Table 1515: Values

Value	Description
Y	Acceptor Tax ID provided
N	Acceptor Tax ID not provided
R	Refused to provide Acceptor Tax ID

Processing rules

Acceptor Tax ID may be optionally provided.

Subfield 008 (Partner ID Code)

DE 122, subelement 001, subfield 008 (Partner ID Code) identifies a specific partnership agreement, generally between specific acceptors and issuers.

Table 1516: Attributes

Attribute	Description
Subfield ID	008
Subfield Data Length	3 positions
Data Representation	ans...006; LLLVAR
Data Field	Contents of subfield 008
Justification	Left

Values

Acceptors and issuers may establish Partner ID codes. If present, this subfield cannot contain all low values, all high values, all zeros, or all spaces.

Processing rules

This subfield may be used for transactions on accounts issued under co-branding agreements when one of the co-branding partners originates the transaction. It also may be used when the agreement is between the acceptor and the clearing system, such as in certain promotional discounts.

Subfield 009 (Service Location City Name)

DE 122, subelement 001, subfield 009 (Service Location City Name) contains the name of the city where the cardholder received the services, if it is different from the location identified by the acceptor city name.

Table 1517: Attributes

Attribute	Description
Subfield ID	009
Subfield Data Length	3 positions
Data Representation	ans...013; LLLVAR
Data Field	Contents of subfield 009
Justification	Left

Values

Service Location City Name must contain the name of the city, town, or village as defined by country postal standards.

Processing rules

Service Location City Name must be present for all transactions when different from the location identified in the Acceptor City Name. Alternatively, DE 122, subelement 001, subfield 014 (Service Location Geographic Coordinates) can be provided instead of populating this subfield.

For the U.S. region, this field must match the U.S. Postal Service list of acceptable cities and spellings.

Subfield 010 (Service Location Country Subdivision Code)

DE 122, subelement 001, subfield 010 (Service Location Country Subdivision Code) contains the code of the country subdivision where the cardholder received the services, if it is different from the location identified by the acceptor country subdivision code.

Table 1518: Attributes

Attribute	Description
Subfield ID	010
Subfield Data Length	3 positions
Data Representation	ans-003

Attribute	Description
Data Field	Contents of subfield 010
Justification	Left with trailing spaces

Values

Service Location Country Subdivision Code must contain a valid code as defined in Chapter 2 of the [Quick Reference Booklet](#) and must be followed by trailing spaces for codes that are less than three positions.

Processing rules

The Service Location Subdivision Code must be present for all transactions when different from the location identified in the Acceptor Subdivision Code. Alternatively, DE 122, subelement 001, subfield 014 (Service Location Geographic Coordinates) can be provided instead of populating this subfield.

Subfield 011 (Service Location Country Code)

DE 122, subelement 001, subfield 011 (Service Location Country Code) contains the code of the country where the cardholder received the services, if it is different from the location identified by the acceptor country code.

Table 1519: Attributes

Attribute	Description
Subfield ID	011
Subfield Data Length	3 positions
Data Representation	ans-003
Data Field	Contents of subfield 011
Justification	Left

Values

Acceptor Country Code must contain a valid three-character alphabetic code as defined in Chapter 2 of the [Quick Reference Booklet](#). When territories have a defined country code in the [Quick Reference Booklet](#), that country code must be used.

Processing rules

The Service Location Country Code must be present for all transactions when different from the location identified in the Acceptor Country Code. Alternatively, DE 122, subelement 001,

subfield 014 (Service Location Geographic Coordinates) can be provided instead of populating this subfield.

Subfield 012 (Service Location Postal Code)

DE 122, subelement 001, subfield 012 (Service Location Postal Code) contains the postal code of the country where the cardholder received the services, if it is different from the location identified by the acceptor postal code.

Table 1520: Attributes

Attribute	Description
Subfield ID	012
Subfield Data Length	3 positions
Data Representation	ans...010; LLLVAR
Data Field	Contents of subfield 012
Justification	Left

Values

Service Location Postal Code must contain a valid code as defined by country postal standards.

Processing rules

The Service Location Postal Code must be present for all transactions when different from the Acceptor Postal Code and contain a valid code for countries that utilize postal codes. This information must be populated and formatted as defined by country postal standards. Alternatively, DE 122, subelement 001, subfield 014 (Service Location Geographic Coordinates) can be provided instead of populating this subfield.

Subfield 013 (Acceptor Geographic Coordinates)

DE 122, subelement 001, subfield 013 (Acceptor Geographic Coordinates) contains the latitude and longitude of the acceptor's location.

Table 1521: Attributes

Attribute	Description
Subfield ID	013
Subfield Length	3
Data Representation	ns...020; LLLVAR
Data Field	Contents of subfield 013

Attribute	Description
Justification	N/A

Values

Acceptor Geographic Coordinates, when present, must contain valid latitude and longitude values, comma-separated, represented in decimal degrees to five decimal places in positions 001 through 020 with these format standards:

- Length of geographic coordinates is a range of 15 - 20 positions
- Subfield should not contain alpha characters
- Subfield contains up to two '-'
- Subfield contains precisely two ':'
- Subfield contains precisely one ','
- At least one numeric position is found before the first ':'
- Five numeric positions are found between first ':' and ','
- At least one numeric position is found between ',' and the second ':'
- Five numeric positions are found after second ':'
- Latitude is greater than or equal to -90.00000 and less than or equal to 90.00000
- Longitude is greater than or equal to -180.00000 and less than or equal to 180.00000
- No space character follows the comma separator

Table 1522: Latitude and longitude examples

Location	Latitude and longitude
Anloga, Ghana	5.78381,0.72175
Lagos, Portugal	37.11000,-8.67880
Anchorage, AK	61.21630,-149.89500
Pago Pago, Eastern District, American Samoa	-14.27451,-170.70616

Processing rules

Acceptor Geographic Coordinates can optionally be provided for all transactions and contain valid latitude and longitude values representing the acceptor location.

If the data present in subfield 013 is not formatted according to the "Values" specification, Mastercard will remove the subfield and continue processing the transaction.

Subfield 014 (Service Location Geographic Coordinates)

DE 122, subelement 001, subfield 014 (Service Location Geographic Coordinates) contains the latitude and longitude where the cardholder received the service, if it is different from the acceptor's location.

Table 1523: Attributes

Attribute	Description
Subfield ID	014
Subfield Length	3
Data Representation	ns...020; LLLVAR
Data Field	Contents of subfield 014
Justification	N/A

Values

Service location geographic coordinates, when present, must contain valid latitude and longitude values, comma-separated, represented in decimal degrees to five decimal places in positions 001 through 020 with these format standards:

- Length of geographic coordinates is a range of 15 - 20 positions
- Subfield should not contain alpha characters
- Subfield contains up to two '-'
- The subfield contains precisely two '.'
- Subfield contains precisely one ','
- At least one numeric position is before the first '.'
- Five numeric positions are found between the first '.' and ','
- At least one numeric position is found between ',' and the second '.'
- Five numeric positions are found after second '.'
- Latitude is greater than or equal to -90.00000 and less than or equal to 90.00000
- Longitude is greater than or equal to -180.00000 and less than or equal to 180.00000
- No space character follows the comma separator

Table 1524: Latitude and longitude examples

Location	Latitude and longitude
Anloga, Ghana	5.78381,0.72175
Lagos, Portugal	37.11000,-8.67880

Location	Latitude and longitude
Anchorage, AK	61.21630,-149.89500
Pago Pago, Eastern District, American Samoa	-14.27451,-170.70616

Processing rules

Service location geographic coordinates can optionally be provided instead of, or in addition to, subfields 009 through 012 (Service Location - City Name, Country Subdivision Code, Country Code, Country Postal Code) when those subfields are required.

If the data present in subfield 014 is not formatted according to the "Values" specification, Mastercard will remove the subfield and continue processing the transaction.

DE 122 Dual Message Authorization System edits

The Dual Message Authorization System will perform the following system edits.

Authorization Request/0100 message edits for DE 122, subelement 1, subfields 1 through 12

Table 1525: Associated edits

If...	Then the Dual Message Authorization System...
DE 122, subelement 1, subfields 1-12 extend past the valid field lengths	will return to the acquirer an Authorization Request Response/0110 message where: - DE 39 = 30 - DE 44 = 122001

Authorization Advice/0120 message edits for DE 122, subelement 1, subfields 1 through 12

Table 1526: Associated edits

If...	Then the Dual Message Authorization System...
DE 122, subelement 1, subfields 1-12 extend past the valid field lengths	will return to the acquirer an Authorization Advice Response/0130: system-generated message where: - DE 39 = 30 - DE 44 = 122001

Reversal Request/0400 message edits for DE 122, subelement 1, subfields 1 through 12

Table 1527: Associated edits

If...	Then the Authorization System...
DE 122, subelement 1, subfields 1-12 extend past the valid field lengths	will return to the acquirer a Reversal Request Response/0410 message where: • DE 39 = 30 • DE 44 = 122001

Table 1528: DE 122, subelement 001, subfield 013 (Acceptor Geographic Coordinates)

If...	Then the Authorization Platform...
DE 122, subelement 001, subfield 013 length is not within expected limits	will return to the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 message where: • DE 39 = 30 • DE 44 = 122001
DE 122, subelement 001, subfield 013 includes an alpha character	will return to the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 message where: • DE 39 = 30 • DE 44 = 122001

Table 1529: DE 122, subelement 001, subfield 014 (Service Location Geographic Coordinates)

If...	Then the Authorization Platform...
DE 122, subelement 001, subfield 014 length is not within expected limits	will return to the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 message where: • DE 39 = 30 • DE 44 = 122001

If...	Then the Authorization Platform...
DE 122, subelement 001, subfield 014 includes an alpha character	will return to the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 message where: • DE 39 = 30 • DE 44 = 122001

DE 123 (Receipt Free Text)

DE 123 (Receipt Free Text) only applies to the Swedish Domestic Authorization Switching Service (SASS), Peru, and the Mastercard Installment Payment Service. For SASS, DE 123 contains information to be printed on a receipt (not displayed on the terminal screen) for balance inquiry and ATM transactions (where DE 3 [Processing Code] is value 01 [Withdrawal] or value 30 [Balance Inquiry]). For Peru and Mastercard Installment Payment Service, DE 123 contains a text message to be printed on point-of-sale (POS) sales receipts.

Table 1530: Attributes

Attribute	Description
Data Representation	ans...512; LLLVAR
Data Length	3 positions
Subfields	N/A
Justification	N/A

Usage

Whether the usage is mandatory, conditional, optional, system-provided, or not required is as noted for each applicable message.

Table 1531: Applicable messages

Message	Org	Sys	Dst
Authorization Request Response/0110	O	X	C

Values

Proprietary receipt free data up to 512 characters.

Application notes

Issuers may provide DE 123 for balance inquiry transactions, combined authorization response and account balance transactions, and ATM withdrawal transactions.

Table 1532: Associated edits

WHEN...	THEN the Dual Message Authorization System...
for SASS, DE 123 is present in an Authorization Request Response/0110 message where DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) does not contain value 01 (Withdrawal) or 30 (Balance Inquiry)	sends the issuer an Authorization Response Negative Acknowledgment/0190 where: • DE 39 = 30 • DE 44 = 123

Table 1533: Associated edits

WHEN...	THEN the Dual Message Authorization System...
DE 123 contains a length greater than 512 characters	sends the issuer an Authorization Response Negative Acknowledgment/0190 where: • DE 39 = 30 • DE 44 = 123 The Dual Message Authorization System will not send DE 123 in an Authorization Request Response/0110 message to an acquirer that is not registered to receive this data.

DE 124 (Member-Defined Data)

DE 124 (Member-defined Data) is described for general use and Prepaid Voucher, Brazil Maestro only, Bill Payment at the ATM only, and Mastercard Digital Enablement Service use.

NOTE: For detailed data layout requirements when using DE 124 with the Mastercard Digital Enablement Service, refer to the [MDES Technical Specifications for Dual and Single Message Systems](#).

DE 124 (Member-Defined Data: General Use)

DE 124 (Member-Defined Data: General Use) may be used to submit up to 299 bytes of customer-defined data. DE 124 can contain program-specific data as defined by the DE 124 subelements.

Table 1534: Attributes

Attribute	Description
Data Representation	ans...299; LLLVAR
Data Length	3 positions
Data Field	Contents of positions 1–299
Subelements	Determined by program
Justification	N/A

Usage

Whether the usage is mandatory, conditional, optional, system-provided, or not required is as noted for each applicable message.

Table 1535: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	.	C
Authorization Request Response/0110	C	.	C
Authorization Advice/0120: acquirer-generated	O	.	C
Authorization Advice/0120: system-generated	.	X	C
Reversal Request/0400	O	.	C
Reversal Request Response/0410	O	.	C
Reversal Advice/0420	.	X	C

Application notes

DE 124 may be required in support of programs to convey information between customers in an authorization message.

Acquirers

Table 1536: Associated edits

IF DE 124 is...	THEN...
up to 299 bytes in acquirer-generated authorization messages	the Dual Message Authorization System will not edit the content of DE 124 and will forward the entire length of the data to the issuer.
more than 299 bytes of data in acquirer-generated authorization messages	the Dual Message Authorization System will only send the first 299 bytes of data to the issuer.

Absolute positioning of data in DE 124 subfields is required; padding will be necessary. Customers must select all State, Province, and Country Codes from the [Quick Reference Booklet](#). If a country code is used, it must be the ISO 3-character alphabetic (not numeric) Country Code. If used, a State or Province Code should be right-justified in this subfield with one leading blank space. Customers must not use all zeros, all low values, or all high values when formatting DE 124.

Subfields 6–13: are used for Maestro credit usage in Brazil.

Issuers

Table 1537: Associated edits

IF DE 124 is...	THEN...
Up to 299 bytes in issuer-generated response messages	The Dual Message Authorization System will not edit the content of DE 124 and will forward the entire length of the data to the acquirer.
More than 299 bytes of data in issuer-generated response messages	The Dual Message Authorization System will truncate to 299 bytes the length of the data sent to the acquirer.

The Authorization Request/0100 and Authorization Request Response/0110 messages will be able to contain DE 124 data independently from one another. Therefore, issuers may:

- Send back the same value in DE 124 in the Authorization Request Response/0110 message as was present in DE 124 of the Authorization Request/0100 message.
- Send a different value in DE 124 in the Authorization Request Response/0110 message as was present in DE 124 of the Authorization Request/0100 message.
- Not send DE 124 in the Authorization Request Response/0110 message. In this case, the acquirer will not receive DE 124 in the Authorization Request Response/0110 message.
- Include DE 124 in the Authorization Request Response/0110 message even when the acquirer did not send it in the Authorization Request/0100 message.

DE 124 (Member-Defined Data: Prepaid Voucher)

DE 124 (Member-Defined Data: Prepaid Voucher) is for use only for Mastercard prepaid meal and food voucher programs in Europe.

Subfield 1 (Merchant Direct Settlement Indicator)

DE 124 (Member-Defined Data: Prepaid Voucher), subfield 1 (Merchant Direct Settlement Indicator) is for use only for Mastercard prepaid meal and food voucher programs in Europe.

Table 1538: Attributes

Attribute	Description
Subfield Data Length	3 positions
Data Representation	ans-3
Data Field	Contents of positions 1–3
Justification	N/A

Values

B00 equals voucher transactions will be settled bilaterally by the issuer directly with the merchant.

Value B00 is valid for use only in voucher transactions in the Europe region.

An acquirer must clear the transaction via the Mastercard Clearing Platform by using the bilateral settlement indicator (IPM Private Data Subelement [PDS] 0165 [Settlement Indicator], subfield 1 [Settlement Indicator] = B) in the First Presentment/1240 message.

Application notes

Issuers must include this indicator in DE 124 Authorization Request Response/0110 messages for voucher transactions they will settle directly with the merchant.

When responding to an Authorization Request/0100 message for a transaction the issuer will settle directly with the merchant, the issuer must include the Prepaid Voucher format of DE 124, with subfield 1 containing Merchant Direct Settlement Indicator value B00.

The issuer must not include this indicator in response messages for meal voucher transactions that require settlement by Mastercard via the Mastercard settlement process.

The presence of the settlement indicator value B00 in DE 124, subfield 1, will serve as indication to the acquirer that the transaction will be settled by the issuer directly with the merchant. Receipt of this indicator value in an authorization response message by the acquirer indicates the acquirer must include PDS 0165 (Settlement Indicator), subfield 1 (Settlement Indicator) with value B (Bilateral settlement agreement applies; the Mastercard Clearing Platform clears the transaction but does not settle it) in the related clearing First Presentment/1240 message.

This value allows the acquirer to clear the message without triggering the Mastercard settlement process.

DE 124 (Member-Defined Data: Mastercard Send)

Acquirers and issuers are able to exchange DE 124 data in Mastercard Send (funding, refund, and payment), Gaming and Gambling, and MPQR payment transaction Authorization Request/Response messages.

Acquirers and issuers must use DE 108 (Additional Transaction Reference Data) when sending and receiving Receiver and Sender information.

DE 124 (Member-Defined Data: Brazil Maestro Only)

DE 124 (Member-Defined Data: Brazil Maestro Only) is used only for Brazil Maestro transactions.

Subfields 1 through 8: are used for Maestro credit usage in Brazil.

Subfield 1 (Unique Reference Number)

DE 124, subfield 1 (Unique Reference Number) contains a unique reference number.

Table 1539: Attributes

Attribute	Description
Data Representation	ans-19
Data Field	Contents of positions 1–19
Justification	Left-justified with trailing spaces

Values

Valid value string will contain a leading zero (0), followed by:

ICA (n-6)

Year (n-1)

Julian Date (n-3)

Time hhmmss (n-6)

Transaction Sequence number (01–99) (n-2)

Example: 0555555801215305401

Subfield 2 (Sender/Payer/User ID)

DE 124, subfield 2 (Sender/Payer/User ID) contains sender consumer, business, government, and non-government names, payer name, or user ID data.

Table 1540: Attributes

Attribute	Description
Data Representation	ans-24
Data Field	Contents of positions 20–43
Justification	Left-justified with trailing spaces

Values

Sender consumer, business, government, and non-government names, payer name, or user ID value up to 24 character spaces. If the consumer name is populated, the format must be (last name, first name).

Subfield 3 (Sender/Payer Address)

DE 124, subfield 3 (Sender/Payer Address) contains sender/payer address data.

Table 1541: Attributes

Attribute	Description
Data Representation	ans-91
Data Field	Contents of positions 44–134
Justification	Left-justified with trailing spaces

Table 1542: Values

Field	Data	Edit
Street address	ans-50	Must be present Must not be all blanks or all zeros
City	ans-25	Must be present if applicable

Field	Data	Edit
State/province code	ans-3	If country is U.S. or Canada, state must be a valid state code for U.S. or a valid province code for Canada Must not be all blanks or zeros For all other countries, may contain all spaces or an applicable state code
Country code	ans-3	Must be present Must not be all blanks or all zeros Must be a valid country code Must not be a blocked country
Postal code	ans-10	Must be present if applicable

NOTE: Transaction will be declined if the sender Country is subject to comprehensive geographic sanctions published by the Office of Foreign Assets Control (OFAC), <https://ofac.treasury.gov/sanctions-programs-and-country-information>. The current list of countries subject to such sanctions is subject to change.

Subfield 4 (Additional Sender Information)

DE 124, subfield 4 (Additional Sender Information) contains telephone number, date of birth, or optional message.

Table 1543: Attributes

Attribute	Description
Data Representation	ans-65
Data Field	Contents of positions 135–199
Justification	Left-justified with trailing spaces

Values

Additional sender information is formatted as follows:

Telephone number (n-20)

Date of Birth (n-8) (MMDDYYYY)

Optional message (ans-37)

Subfield 6 (Discretionary Message on Sales Slip Supported)

DE 124, subfield 6 (Discretionary Message on Sales Slip Supported), used in association with Maestro credit in Brazil, is sent by the acquirer in the Authorization Request/0100 message, to indicate whether the POS terminal supports the discretionary message on the sales slip.

Table 1544: Attributes

Attribute	Description
Data Representation	an-1
Data Field	Contents of position 1–6
Justification	N/A

Table 1545: Values

Value	Description
0	No Merchant terminal does not support the printing of messages sent by the issuer
1	Yes Merchant terminal supports the printing of messages sent by the issuer

Subfield 7 (Discretionary Message on Sales Slip Code)

DE 124, subfield 7 (Discretionary Message on Sales Slip Code) is sent by the issuer in the Authorization Request Response/0110 message, to indicate the number of the message to be printed on the sales slip.

Table 1546: Attributes

Attribute	Description
Data Representation	an-3
Data Field	Contents of positions 7–9
Justification	Left

Values

Valid value will contain the number of the message to be printed on the sales slip.

Subfield 8 (Discretionary Message on Sales Slip Content)

DE 124, subfield 8 (Discretionary Message on Sales Slip Content) contains the variable part of the message to be printed on the sales slip.

Table 1547: Attributes

Attribute	Description
Data Representation	an-10
Data Field	Contents of positions 10–19
Justification	Left

Values

Valid value will consist of the variable part of the message to be printed on the sales slip.

Subfield 9 (Phoneshop [Phone Company ID])

DE 124, subfield 9 (Phoneshop [Phone Company ID]) is sent by the acquirer in the Authorization Request/0100 message, to identify the telephone company that provides the service for the Phoneshop product (for example, Telefonica = 01).

Table 1548: Attributes

Attribute	Description
Data Representation	an-2
Data Field	Contents of positions 20–21
Justification	Left

Values

Valid value will consist of the phone company ID.

Subfield 10 (Phoneshop [Cell Phone Number])

DE 124, subfield 10 (Phoneshop [Cell Phone Number]) is sent by the acquirer in the Authorization Request/0100 message, to indicate the cardholder cell phone number for the Phoneshop product.

Table 1549: Attributes

Attribute	Description
Data Representation	n-10
Data Field	Contents of positions 22–31
Justification	Right

Values

Valid value will consist of the cell phone number.

Subfield 11 (Phoneshop [Message Security Code])

DE 124, subfield 11 (Phoneshop [Message Security Code]) is sent by the issuer in the Authorization Request Response/0110 message, to indicate the security code that the merchant sent for validation.

Table 1550: Attributes

Attribute	Description
Data Representation	an-4
Data Field	Contents of positions 32–35
Justification	Left

Values

Valid value will consist of the message security code.

Subfield 12 (Merchant CNPJ Number)

DE 124, subfield 12 (Merchant CNPJ Number) is sent by the acquirer in the Authorization Request/0100 message, to indicate the Merchant CNPJ number.

Table 1551: Attributes

Attribute	Description
Data Representation	n-14
Data Field	Contents of positions 36–49
Justification	Right

Values

Valid value will consist of the Merchant CNPJ Number (a registration number provided by the government to all merchants).

Subfield 13 (Total Annual Effective Cost)

DE 124, subfield 13 (Total Annual Effective Cost) is sent by the issuer in the Authorization Request Response/0110 message, to indicate the total annual effective cost in a financing transaction. A financing transaction includes purchase or cash withdrawal using a credit card.

Table 1552: Attributes

Attribute	Description
Data Representation	n-6
Data Field	Contents of positions 50–55
Justification	Right

Values

NOTE: Optionally, issuers may submit spaces preceding the positional start of subfield 13. Data submitted for this subfield should begin in position 50.

Valid value will consist of the total annual effective cost in a financing transaction (including interest amount, taxes, and fees charged to the cardholder). This amount is mandated by local law number 3517 (implemented to ensure the cardholder is enforced about the total effective cost of the transaction). This applies to purchase or cash withdrawal at an ATM using a credit card.

DE 124 (Member-Defined Data [Colombia Domestic Use Only])

DE 124 (Member-defined Data [Colombia Domestic Use Only]) is used only for Colombia domestic activity transactions.

Subfields 1 through 11: are used for Colombia domestic activity.

Subfield 1 (Card Issuer Data)

DE 124 (Member-defined Data [Colombia Domestic Use Only]), subfield 1 (Card Issuer Data) defines the FIID and Logical Network associated with the card issuer.

Table 1553: Attributes

Attribute	Description
Data Representation	ans-19
Data Field	Contents of positions 1–19
Justification	Left-justified with trailing spaces

Usage

Optional.

Example values

FIID (1–4)

Logical Network ID (5–8) = Examples are as follow: PRO1, MDS, BNET, ASCR, PRO2

- PRO1 = Redeban Multicolor
- MDS or BNET = Mastercard
- ASCR = AScredibanco
- PRO2 = Credibanco

Constant Value (9–19) = zeros

Subfield 2 (Tax [IVA])

DE 124 (Member-defined Data [Colombia Domestic Use Only]), subfield 2 (Tax [IVA]) defines the tax for Colombia domestic activity.

Table 1554: Attributes

Attribute	Description
Data Representation	n-12
Data Field	Contents of positions 20–31

Attribute	Description
Justification	Right-justified with leading zeros

Usage

Required. If no value, it should contain zeroes.

Example values

Amount is in Colombian pesos and carries an implied two position decimal.

Subfield 3 (Tax Amount Base)

DE 124 (Member-defined Data [Colombia Domestic Use Only]), subfield 3 (Tax Amount Base) defines the tax amount base for Colombia domestic activity.

Table 1555: Attributes

Attribute	Description
Data Representation	n-12
Data Field	Contents of positions 32–43
Justification	Right-justified with leading zeros

Usage

Required.

Values

Amount is in Colombian pesos and carries an implied two position decimal.

Subfield 4 (Retailer Data)

DE 124 (Member-defined Data [Colombia Domestic Use Only]), subfield 4 (Retailer Data) defines the local merchant ID for Colombia domestic activity.

Table 1556: Attributes

Attribute	Description
Data Representation	ans-27
Data Field	Contents of positions 44–70
Justification	Left-justified with trailing spaces

Usage

Optional.

Example values

Each acquirer may elect to provide the merchant ID in their original format. Example to follow:

Unique merchant ID (44–53)

Spaces (54–62)

Terminal Group (63–66)

Retailer Region (67–70)

Subfield 5 (Terminal Acquirer Data)

DE 124 (Member-defined Data [Colombia Domestic Use Only]), subfield 5 (Terminal Acquirer Data) identifies information about the terminal owner for Colombia domestic activity.

Table 1557: Attributes

Attribute	Description
Data Representation	ans-16
Data Field	Contents of positions 71–86
Justification	Right-justified

Usage

Optional.

Example values

FIID (71–74)

Logical Network (75–78)

Constant Value = zeros

Subfield 6 (Acquirer Original Processing Code)

DE 124 (Member-defined Data [Colombia Domestic Use Only]), subfield 6 (Original Processing Code) contains the original processing code for Colombia domestic activity.

Table 1558: Attributes

Attribute	Description
Data Representation	n-6
Data Field	Contents of positions 87–92
Justification	Right-justified

Subfield 7 (Bill Payment/Top up Data)

DE 124 (Member-defined Data [Colombia Domestic Use Only]), subfield 7 (Bill Payment/Top up Data) provides information related to bill payment for Colombia domestic activity.

Table 1559: Attributes

Attribute	Description
Data Representation	ans-36
Data Field	Contents of positions 93–128
Justification	Left-justified with trailing spaces

Table 1560: Values

Value	Description
Service Code (93–96)	Number used to identify the provider company (for example: 0001 = Bogota Power Light Company)
Service Description (97–126)	Service Provider Bill ID (for example: Invoice number, additional billing information)
Originator Device (127)	Alphanumeric (for example, P)
Filler (128)	Space

Subfield 8 (Local POS Data)

DE 124 (Member-defined Data [Colombia Domestic Use Only]), subfield 8 (Local POS Data) is present when transaction is a web payment or recurring payment. This field is optional.

Table 1561: Attributes

Attribute	Description
Data Representation	n-1
Data Field	Contents of positions 129
Justification	N/A

Usage

Optional.

Table 1562: Values

Values	Description
6	Preauthorization/recurring payment
7	Web payment, Electronic orders

Subfield 9 (Local Response Codes)

DE 124 (Member-defined Data [Colombia Domestic Use Only]), subfield 9 (Local Response Codes) identifies unique response codes for Colombia domestic activity.

Table 1563: Attributes

Attribute	Description
Data Representation	ans-2
Data Field	Contents of positions 130–131
Justification	Left-justified with trailing spaces

Usage

Optional.

Example values

M2 = Invalid national customer identifier

Subfield 10 (Original Transaction Data)

DE 124 (Member-defined Data [Colombia Domestic Use Only]), subfield 10 (Original Transaction Data) is present for reversal messages to help uniquely identify the original message.

Table 1564: Attributes

Attribute	Description
Data Representation	ans-42
Data Field	Contents of positions 132–173
Justification	Left-justified with trailing spaces

Usage

Optional.

Table 1565: Example values

Example value	Example description
Original Message Type (132–135)	100 or 0200
Sequence Number (136–147)	
Original Transaction Date (148–151)	MMDD
Original Transaction Time (152–157)	HHMMSS
Capture Date (158–161)	MMDD
Filler (162–173)	All zeros

Subfield 11 (IAC Tax Amount)

DE 124 (Member-defined Data [Colombia Domestic Use Only]), subfield 11 (IAC Tax Amount) defines the IAC Tax for Colombia domestic activity.

Table 1566: Attributes

Attribute	Description
Data Representation	n-12
Data Field	Contents of positions 174–185
Justification	Right-justified with leading zeros

DE 124 (Member-Defined Data: Ukraine Domestic Transactions)

DE 124 (Member-Defined Data: Ukraine Domestic Transactions) enables acquirers to pass merchant identification data to issuers for Ukraine domestic transactions.

- The first three positions contain the length of the remaining positions.
- The next five positions contain the identifier UKR01.
- The next two positions contain one of the tags identifying the type of ID being transmitted.
- The next two positions contain the length of the ID value being transmitted.
- The final set of positions contain the ID value (with the length as indicated by the preceding two positions).

Values

Table 1567: Identifier details for Ukraine domestic transactions

Identifier	TAG	Length (LL)	Value	Description
UKR01	01	08	ABCD0001	Unified State Register of Enterprises and Organizations of Ukraine (USREOU). The following eight characters include a valid identity number assigned to corporate entities by the Unified State Register of Enterprises and Organizations of Ukraine.

Identifier	TAG	Length (LL)	Value	Description
UKR01	02	10	ABCDXYZ123	Registration number of the taxpayer's record card (RNOTRC). The following 10 characters include a valid identity number assigned to private entrepreneurs, known as the registration number of the taxpayer's record card.
UKR01	03	08-13	ABCDXYZ1 or ABCDXYZ123456	Passport. The following eight or 13 characters (depending on the type of passport) include a valid passport letter series (if applicable) and number for private entrepreneurs for whom the RNOTRC is not available.

Examples of Ukraine domestic transactions:

- 017UKR010108ABCD0001
- 019UKR010210ABCDXYZ123
- 017UKR010308ABCDXYZ1
- 022UKR010313ABCDXYZ123456

Application notes

Acquirers must provide DE 124 in Authorization Request/0100 messages for transactions that may fall into the scope of the National Bank of Ukraine requirement.

Acquirers must transmit identification data for all transaction types, which may fall into the scope of the relevant National Bank of Ukraine requirements. Identification of the relevant use cases is solely the acquirer's responsibility.

DE 125 (New PIN Data)

DE 125 (New PIN Data) consists of a binary block containing a derived encrypted value calculated from the new PIN introduced by the cardholder at the ATM offering the PIN change service.

Table 1568: Attributes

Attribute	Description
Data Representation	b-8
Data Length	N/A
Data Field	Contents of bit positions 1-64 (8 bytes)
Subfields	N/A
Justification	N/A

Usage

Following is the usage of DE 125 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 1569: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	C	X	C

Values

DE 125 contains the new PIN, which is formatted into one of the supported PIN block formats and is then encrypted. The PIN block format and encryption method used must be the same as the one used for the existing PIN that is stored in DE 52. DE 125 is only required in Authorization Request/0100: PIN Change messages. Otherwise, it must not be present.

DE 126 (Private Data)

DE 126 (Private Data) is reserved for future use.

Table 1570: Attributes

Attribute	Description
Data Representation	ans...100; LLLVAR
Data Length	3 positions
Data Field	Contents of positions 1–100
Subfields	N/A
Justification	N/A

Usage

By provision of the ISO 8583:1987 specification, Mastercard defined this data element for use as Private Data available for Mastercard's optional use for additional acquirer data.

The Dual Message Authorization System does not pass data placed in this data element through to the message receiver; rather, the Dual Message Authorization System temporarily stores the data. The Dual Message Authorization System does not return this data to the message originator in any subsequent response to an original Request, Advice, Response, or Acknowledgment message.

Values

The length subelement must be in the range 001–100.

Application notes

This data element is defined and used identically within all Mastercard programs and services.

DE 127 (Private Data)

DE 127 (Private Data) may contain any private-use data that the customer may want to include in a message. Any Authorization System message originator may use DE 127.

Table 1571: Attributes

Attribute	Description
Data Representation	ans...100; LLLVAR
Data Length	3 positions
Data Field	Contents of positions 1–100
Subfields	N/A
Justification	N/A

Usage

Following is the usage of DE 127 (whether it is mandatory, conditional, optional, system provided, or not required) in applicable messages.

Table 1572: Applicable messages

Message	Org	Sys	Dst
Authorization Request/0100	O	X	.
Authorization Request Response/0110	O	X	CE
Authorization Advice/0120: acquirer-generated	O	X	.
Authorization Advice Response/0130: system-generated (responding to an acquirer-generated 0120)	.	X	CE
Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120)	CE	X	CE
Authorization Response Acknowledgment/0180	CE	.	.
Authorization Response Negative Acknowledgment/0190	.	X	CE
Issuer File Update Request/0302	O	X	.
Issuer File Update Request Response/0312	.	X	CE
Reversal Request/0400	O	X	.
Reversal Request Response/0410	O	X	CE

Message	Org	Sys	Dst
Reversal Advice/0420	.	XE	CE
Reversal Advice Response/0430	CE	.	.
Administrative Request/0600	O	X	.
Administrative Request Response/0610	.	X	CE
Administrative Advice/0620: system-generated	.	O	O
Administrative Advice/0620:member-generated	O	X	.
Administrative Advice Response/0630	.	X	.
Network Management Request/0800: sign-on/sign-off	O	X	.
Network Management Request/0800: PEK exchange, on demand	O	X	.
Network Management Request Response/0810: sign-on/sign-off	.	X	CE

Values

The length must be in the range 001–100.

Application notes

Data placed in DE 127 is not passed through to the message receiver; rather, the Authorization System temporarily stores the data and returns it to the message originator in any subsequent response to an original Request, Advice, Response, or Acknowledgment message.

This data element is defined and used identically within all Mastercard programs and services.

Acquirers that provide DE 127 in the Authorization Request/0100 message will receive the first twenty bytes of the value provided in DE 072 (Data Record) of IPM Fee Collection/1740–783 for interregional non-financial ATM transactions.

DE 128 (Message Authentication Code)

DE 128 (Message Authentication Code [MAC]) validates the source and the text of the message between the sender and the receiver.

Table 1573: Attributes

Attribute	Description
Data Representation	b-8
Data Length	N/A
Data Field	Contents of bit positions 1–64 (8 bytes)

Attribute	Description
Subfields	N/A
Justification	N/A

Usage

May contain message authentication code as defined by ISO standards.

Values

Not applicable.

Application notes

This data element should not be present in any Dual Message Authorization System message.

The last bit position within any bit map is reserved for DE 128. If authentication is to be used on a message, the MAC information is indicated by the final bit of the final bit map of that message. The final bit of all preceding bit maps shall contain zero; for example, there shall be only one DE 128 per message and that DE 128 must be the last data element of the message.

Chapter 5 Program and service format requirements

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Product Value Constraints

Product value constraints list types of transactions permitted by card program, and the interrelated value constraints for certain data elements carried in the Dual Message Authorization System messages.

The business rules related to the card programs Mastercard supports differentiate between the nature of the transactions permitted for:

- Mastercard (MCC)
- Debit Mastercard (DMC)
- Maestro (MSI)
- Cirrus (CIR)
- Private Label (PVL)

Permitted Transactions by card program

Following are the types of transactions permitted by card program and the interrelated value constraints for certain data elements in Dual Message Authorization System messages.

The interrelated data elements are:

- DE 3 (Processing Code)
- DE 22 (Point-of-Service [POS] Entry Mode)
- DE 52 (Personal ID Number [PIN] Data)
- DE 61 (Point-of-Service [POS] Data)

Permitted Transactions by Card Program

Use the reference number in the Ref column in the following tables with the specific value constraints in the corresponding [Value Constraints by Transaction Type](#) table.

Table 1574: ATM Withdrawal

Ref	Transaction Type	PAN Entry Mode	CVM	MCC	DMC	MSI	CIR	PVL	Comments
1.1	ATM	Magnetic Stripe	Online PIN	X	X	X	X		For this message, DE 18 must contain value 6011 (Automated Cash Disbursements). For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.
1.2	ATM	Chip	Online PIN	X	X	X	X		For this message, DE 18 must contain value 6011 (Automated Cash Disbursements). For MCC, DMC, MSI, contactless are transactions permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.

Table 1575: Cash Advance

Ref	Transaction Type	PAN Entry Mode	CVM	MC C	DM C	MSI	CIR	PVL	Comments
2.1	Cash Advance	Manual	Signature	X	X				For this message, DE 18 must contain value 6010 (Manual Cash Disbursements)

Ref	Transaction Type	PAN Entry Mode	CVM	MC C	DM C	MSI	CIR	PVL	Comments
2.2	Cash Advance	Magnetic Stripe	Signature	X	X				For this message, DE 18 must contain value 6010 (Manual Cash Disbursements) For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.
2.3	Cash Advance	Magnetic Stripe	Online PIN	X	X				For this message, DE 18 must contain value 6010 (Manual Cash Disbursements) For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.
2.4	Cash Advance	Chip	Offline PIN	X	X				For this message, DE 18 must contain value 6010 (Manual Cash Disbursements)

Program and service format requirements
Permitted Transactions by card program

Ref	Transaction Type	PAN Entry Mode	CVM	MC C	DM C	MSI	CIR	PVL	Comments
2.5	Cash Advance	Chip	Online PIN	X	X				For this message, DE 18 must contain value 6010 (Manual Cash Disbursements) For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.
2.6	Cash Advance	Chip	Signature	X	X				For this message, DE 18 must contain value 6010 (Manual Cash Disbursements) For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.
2.7	Electronic Cash Advance	Magnetic Stripe	Online PIN			X	X		For this message, DE 18 must contain value 6010 (Manual Cash Disbursements)

Ref	Transaction Type	PAN Entry Mode	CVM	MC C	DM C	MSI	CIR	PVL	Comments
2.8	Electronic Cash Advance	Chip	Offline PIN			X	X		For this message, DE 18 must contain value 6010 (Manual Cash Disbursements) A waiver is required for this item. Issuers must register.
2.9	Electronic Cash Advance	Chip	Online PIN			X	X		For this message, DE 18 must contain value 6010 (Manual Cash Disbursements)

Table 1576: Purchase of Goods and Services

Ref	Transaction Type	PAN Entry Mode	CVM	MCC	DMC	MSI	CIR	PVL	Comments
3.1	Purchase	Manual	Signature	X	X			X	
3.2	Purchase	Magnetic Stripe	Signature	X	X	X		X	For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits. For MSI, country restrictions apply.

Ref	Transaction Type	PAN Entry Mode	CVM	MCC	DMC	MSI	CIR	PVL	Comments
3.3	Purchase	Magnetic Stripe	Online PIN	X	X	X			For MCC, DMC, MSI contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits. For MSI, country restrictions apply.
3.4	Purchase	Chip	Offline PIN	X	X	X			For MCC, DMC, MSI contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits. For MSI, country restrictions apply.
3.5	Purchase	Chip	Signature	X	X			X	
3.6	Purchase	Chip	Online PIN	X	X	X			For MCC, DMC, MSI contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits. For MSI, country restrictions apply.
3.7	Purchase	Chip	No CVM	X	X	X			For MCC, DMC, MSI contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.

Ref	Transaction Type	PAN Entry Mode	CVM	MCC	DMC	MSI	CIR	PVL	Comments
3.8	Purchase	Credential on File	No CVM	X	X	X			
3.9	MO/TO	Manual	No CVM	X	X	X		X	
3.10	MO/TO	Credential on File	No CVM	X	X	X			
3.11	Recurring Payment	Credential on File	N/A	X	X	X			
3.12	CAT Level 1	Magnetic Stripe	Online PIN	X	X				For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.
3.13	CAT Level 1	Chip	Online PIN	X	X				For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.
3.14	CAT Level 1	Chip	Offline PIN						
3.15	CAT Level 2	Magnetic Stripe	No CVM	X	X			X	For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.

Ref	Transaction Type	PAN Entry Mode	CVM	MCC	DMC	MSI	CIR	PVL	Comments
3.16	CAT Level 2	Chip	No CVM	X	X			X	For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.
3.17	CAT Level 3	Magnetic Stripe	No CVM	X	X			X	For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.
	CAT Level 3	Chip	Offline PIN						
3.18	CAT Level 3	Chip	No CVM	X	X			X	For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.
3.19	CAT Level 4	Magnetic Stripe	No CVM	X	X			X	For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.

Ref	Transaction Type	PAN Entry Mode	CVM	MCC	DMC	MSI	CIR	PVL	Comments
3.20	CAT Level 4	Chip	No CVM	X	X			X	For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.
3.21	CAT Level 6	Credential on File	No CVM	X	X	X			
3.22	CAT Level 6	Electronic Commerce	No CVM	X	X	X		X	
3.23	CAT Level 6	Electronic Commerce	Offline PIN	X	X	X			
3.24	CAT Level 6	Electronic Commerce	Online PIN	X	X	X			
3.25	CAT Level 7	N/A	N/A	X	X			X	No specific constraints defined.

Table 1577: Payments

Ref	Transaction Type	PAN Entry Mode	CVM	MCC	DMC	MSI	CIR	PVL	Comments
4.1	Payment Transactions	N/A	No CVM	X	X	X		X	For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits. No specific constraints defined.

Table 1578: PIN Change Management

Ref	Transaction Type	PAN Entry Mode	CVM	MCC	DMC	MSI	CIR	PVL	Comments
5.1	ATM PIN unblock	Chip	Online PIN	X	X	X	X		For this message, DE 18 must contain value 6010 (Manual Cash Disbursements)
5.2	ATM PIN change	Chip	Online PIN	X	X	X	X		For this message, DE 18 must contain value 6010 (Manual Cash Disbursements)
5.3	ATM PIN change	Magnetic Stripe	Online PIN	X	X	X	X		For this message, DE 18 must contain value 6010 (Manual Cash Disbursements)

Table 1579: Balance Inquiry

Ref	Transaction Type	PAN Entry Mode	CVM	MCC	DMC	MSI	CIR	PVL	Comments
6.1	ATM Balance Inquiry	Magnetic Stripe	Online PIN	X	X	X	X		For this message, DE 18 must contain value 6010 (Manual Cash Disbursements) For MCC, DMC, MSI contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.

Ref	Transaction Type	PAN Entry Mode	CVM	MCC	DMC	MSI	CIR	PVL	Comments
6.2	ATM Balance Inquiry	Chip	Online PIN	X	X	X	X		For this message, DE 18 must contain value 6010 (Manual Cash Disbursements) For MCC, DMC, MSI contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits. Applies to prepaid cards only.
6.3	POS Balance Inquiry	Magnetic Stripe	Signature	X	X	X		X	For MCC, DMC, MSI contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.
6.4	POS Balance Inquiry	Magnetic Stripe	Online PIN	X	X	X	X	X	For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.
6.5	POS Balance Inquiry	Chip	Offline PIN	X	X	X		X	

Ref	Transaction Type	PAN Entry Mode	CVM	MCC	DMC	MSI	CIR	PVL	Comments
6.6	POS Balance Inquiry	Chip	Signature	X	X			X	For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.
6.7	POS Balance Inquiry	Chip	Online PIN	X	X			X	For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.
6.8	SMS Balance Inquiry	eCommerce	No CVM	X	X				

Table 1580: Purchase with Cash Back (PWCB)

Ref	Transaction Type	PAN Entry Mode	CVM	MCC	DMC	MSI	CIR	PVL	Comments
7.1	PWCB	Magnetic Stripe	Signature		X	X			Approved intracountry countries, as indicated in Mastercard Rules . For MSI, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.

Ref	Transaction Type	PAN Entry Mode	CVM	MCC	DMC	MSI	CIR	PVL	Comments
7.2	PWCB	Magnetic Stripe	Online PIN		X	X			Approved intracountry countries, as indicated in Mastercard Rules .
7.3	PWCB	Chip	Offline PIN		X	X			Approved intracountry countries, as indicated in Mastercard Rules .
7.4	PWCB	Chip	Online PIN		X	X			For DMC, MSI, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.
7.5	PWCB	Manual	Signature			X			

Table 1581: Refund

Ref	Transaction Type	PAN Entry Mode	CVM	MCC	DMC	MSI	CIR	PVL	Comments
8.1	Refund	Manual	Signature	X	X			X	
8.2	Refund	Magnetic Stripe	Signature	X	X	X		X	For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.
8.3	Refund	Magnetic Stripe	Online PIN			X			

Ref	Transaction Type	PAN Entry Mode	CVM	MCC	DMC	MSI	CIR	PVL	Comments
8.4	Refund	Chip	Signature	X	X				For MCC, DMC, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.
8.5	Refund	Chip	Online PIN			X			For MSI, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.
8.6	Refund	Chip	Offline PIN	X	X				
8.7	Refund	Chip	No CVM			X			For MSI, contactless transactions are permitted. No CVM is required for contactless non-ATM transactions under contactless CVM limits.

Value Constraints by Transaction Type

Following are value constraints by transaction type.

Value Constraints by Transaction Type

Table 1582: ATM Withdrawal

Ref	Transaction Type, PAN Entry Mode, and Cardholder Verification Method	DE 3 values	DE 22 values	CVM Method	DE 61, SF 10 values	Comments
1.1	ATM, Magnetic Stripe, Online PIN	010000	901, 91x (or 801)	P	0 or 1	For this message, DE 18 must contain value 6011 (Automated Cash Disbursements).
1.2	ATM, Chip, Online PIN	010000	051, 071	P	0 or 1	For this message, DE 18 must contain value 6011 (Automated Cash Disbursements).

Table 1583: Cash Advance

Ref	Transaction Type, PAN Entry Mode, and Cardholder Verification Method	DE 3 values	DE 22 values	CVM Method	DE 61, SF 10 values	Comments
2.1	Cash Advance, Manual, Signature	170000	00x, 01x (or 79x)	S	0	For this message, DE 18 must contain value 6010 (Manual Cash Disbursements).
2.2	Cash Advance, Magnetic Stripe, Signature	170000	90x, 91x (or 80x)	S	0	For this message, DE 18 must contain value 6010 (Manual Cash Disbursements).
2.3	Cash Advance, Magnetic Strip, Online PIN	170000	90x, 91x (or 80x)	P	0	For this message, DE 18 must contain value 6010 (Manual Cash Disbursements).

Ref	Transaction Type, PAN Entry Mode, and Cardholder Verification Method	DE 3 values	DE 22 values	CVM Method	DE 61, SF 10 values	Comments
2.4	Cash Advance, Chip, Offline PIN	170000	051	S	0	For this message, DE 18 must contain value 6010 (Manual Cash Disbursements).
2.5	Cash Advance, Chip Online PIN	170000	051, 071	P	0	For this message, DE 18 must contain value 6010 (Manual Cash Disbursements).
2.6	Cash Advance, Chip, Signature	170000	05x, 07x	S	0	For this message, DE 18 must contain value 6010 (Manual Cash Disbursements).
2.7	Electronic Cash Advance, Magnetic Stripe, Online PIN	010000	901	P	0	For this message, DE 18 must contain value 6010 (Manual Cash Disbursements).
2.8	Electronic Cash Advance, Chip, Offline PIN	010000	051	S	0	For this message, DE 18 must contain value 6010 (Manual Cash Disbursements).
2.9	Electronic Cash Advance, Chip Online PIN	010000	051	P	0	For this message, DE 18 must contain value 6010 (Manual Cash Disbursements).

Table 1584: Purchase of Goods and Services

Ref	Transaction Type, PAN Entry Mode, and Cardholder Verification Method	DE 3 values	DE 22 values	CVM Method	DE 61, SF 10 values
3.1	Purchase, Manual, Signature	00xx00	00x, 01x (or 79x)	S	0
3.2	Purchase, Magnetic Stripe, Signature	00xx00	90x, 91x (or 80x)	S	0

Ref	Transaction Type, PAN Entry Mode, and Cardholder Verification Method	DE 3 values	DE 22 values	CVM Method	DE 61, SF 10 values
3.3	Purchase, Magnetic Stripe, Online PIN	00xx00	901, 911 (or 80x)	P	0
3.4	Purchase, Chip, Offline PIN	00xx00	051	S	0
3.5	Purchase, Chip, Signature	00xx00	05x, 07x	S	0
3.6	Purchase, Chip, Online PIN	00xx00	051, 071	P	0
3.7	Purchase, Chip, No CVM	00xx00	05x, 07x	None	0
3.8	Purchase, Credential on File, No CVM	00xx00	10x	None	0
3.9	MO/TO, Manual, No CVM	00xx00	01x, 10x	None	0
3.10	MO/TO, Credential on File, No CVM	00xx00	10x	None	0
3.11	Recurring Payment, Credential on File, No CVM	00xx00	10x	No CVM	0
3.12	CAT Level 1, Magnetic Stripe, Online PIN	00xx00	901, 911 (or 801)	P	1
3.13	CAT Level 1, Chip, Online PIN	00xx00	051, 071	P	1
3.14	CAT Level 1, Chip, Offline PIN	00xx00	051	S	1
3.15	CAT Level 2, Magnetic Stripe, No CVM	00xx00	90x, 91x (or 80x)	None	2
3.16	CAT Level 2, Chip, No CVM	00xx00	05x, 07x	None	2
3.17	CAT Level 3, Magnetic Stripe, No CVM	00xx00	90x, 91x (or 80x)	None	3
3.18	CAT Level 3, Chip, No CVM	00xx00	05x, 07x	None	3
3.19	CAT Level 4, Magnetic Stripe, No CVM	00xx00	90x (or 80x)	None	4
3.20	CAT Level 4, Chip, No CVM	00xx00	05x, 07x	None	4
3.21	CAT Level 6, Credential on File, No CVM	00xx00	10x	None	6
3.22	CAT Level 6, Electronic Commerce, No CVM	00xx00	10x, 81x, 82x	None	6
3.23	CAT Level 6, Electronic Commerce, Offline PIN	00xx00	10x, 81x	S	6
3.24	CAT Level 6, Electronic Commerce, Online PIN	00xx00	10x, 81x, 82x	P	6
3.25	CAT Level 7	00xx00	No constraints	Any constraints	7

Table 1585: Payments

Ref	Transaction Type, PAN Entry Mode, and Cardholder Verification Method	DE 3 values	DE 22 values	CVM Method	DE 61, SF 10 values
4.1	Payment Transactions, No CVM	280000	No constraints	None	No constraints

Table 1586: PIN Change Management

Ref	Transaction Type, PAN Entry Mode, and Cardholder Verification Method	DE 3 values	DE 22 values	CVM Method	DE 61, SF 10 values	Comments
5.1	ATM PIN unblock, Chip, Online PIN	910000	051	P	0 or 1	For this message, DE 18 must contain value 6011 (Automated Cash Disbursements).
5.2	ATM PIN change, Chip, Online PIN	920000	051	P	0 or 1	For this message, DE 18 must contain value 6011 (Automated Cash Disbursements).
5.3	ATM PIN change Magnetic Stripe, Online PIN	920000	021, 901	P	0 or 1	

Table 1587: Balance Inquiry

Ref	Transaction Type, PAN Entry Mode, and Cardholder Verification Method	DE 3 values	DE 22 values	CVM Method	DE 61, SF 10 values	Comments
6.1	ATM Balance Inquiry, Magnetic Stripe, Online PIN	30xx00	901, 911	P	0 or 1	For this message, DE 18 must contain value 6011 (Automated Cash Disbursements).

Ref	Transaction Type, PAN Entry Mode, and Cardholder Verification Method	DE 3 values	DE 22 values	CVM Method	DE 61, SF 10 values	Comments
6.2	ATM Balance Inquiry, Chip, Online PIN	30xx00	051, 071	P	0 or 1	
6.3	POS Balance Inquiry, Magnetic Stripe, Signature	300000 or 303000	90x, 91x (or 80x)	S	0	
6.4	POS Balance Inquiry, Magnetic Stripe, Online PIN	300000 or 303000	901, 911 (or 801)	P	0	
6.5	POS Balance Inquiry, Chip, Offline PIN	300000 or 303000	051	S	0	
6.6	POS Balance Inquiry, Chip, Signature	300000 or 303000	05x, 07x	S	0	
6.7	POS Balance Inquiry, Chip, Online PIN	300000 or 303000	051, 071	P	0	
6.8	SMS Balance Inquiry					

Table 1588: Purchase with Cash Back (PWCB)

Ref	Transaction Type, PAN Entry Mode, and Cardholder Verification Method	DE 3 values	DE 22 values	CVM Method	DE 61, SF 10 values
7.1	PWCB, Magnetic Stripe, Signature	09xx00	90x (or 80x)	S	0
7.2	PWCB, Magnetic Stripe, Online PIN	09xx00	901 (or 801)	P	0
7.3	PWCB, Chip Offline PIN	09xx00	051	S	0
7.4	PWCB, Chip Online PIN	09xx00	051	P	0
7.5	PWCB, Manual, Signature	09xx00	01x	S	0

Table 1589: Refund

Ref	Transaction Type, PAN Entry Mode, and Cardholder Verification Method	DE 3 values	DE 22 values	CVM Method	DE 61, SF 10 values
8.1	Refund, Manual, Signature	20xx00	00x, 01x	S	0
8.2	Refund, Magnetic Stripe, Signature	20xx00	90x, 91x (or 80x)	S	0
8.3	Refund, Magnetic Stripe, Online PIN	20xx00	901 (or 801)	P	0
8.4	Refund, Chip, Signature	20xx00	05x, 07x	S	0
8.5	Refund, Chip, Online PIN	20xx00	051	P	0
8.6	Refund, Chip, Offline PIN	20xx00	051	S	0

Account Status Inquiry Service

Account Status Inquiry Service allows acquirers to send Account Status Inquiry transactions to validate aspects of a cardholder account.

The Account Status Inquiry Service supports purchase account status inquiry transactions and payment account status inquiry transactions for Mastercard® credit, Debit Mastercard®, and Maestro® acceptance brands.

In addition to Mastercard products, Account Status Inquiry Service transaction processing supports the following types of acceptance brands:

- American Express
- Diners
- Discover
- JCB
- Private label cards
- Visa

The Account Status Inquiry Service does not provide any chargeback rights in the event of a dispute. The Account Status Inquiry service (the "Service") is an Activity (as defined in the Mastercard Transaction Processing Rules), subject to the Mastercard Transaction Processing Rules. Customer shall comply with all applicable provisions of the Mastercard Transaction Processing Rules. The Service shall include the production and delivery of deliverables, including the ASI Probability Indicator ("Deliverables").

The Deliverables are only an input to and should not be used in isolation or as the sole method for Customer's and merchant's retry or other related decisioning. In particular, Customer acknowledges and agrees that its acceptance and use of the Deliverables is only one aspect of

the Customer's banking activities. Customer will have sole responsibility for ensuring that Customer's and merchant's use of the Deliverables is compliant with applicable law or governing authority. In addition, Mastercard is not providing the Service as investment, legal or financial advice, and is not making any representation or warranty about Mastercard's business operations. Customer agrees to indemnify Mastercard for any loss to Mastercard caused by Customer's failure to comply with applicable law or regulation when using the Service.

Purchase Account Status Inquiry / Recurring Payment Account Status Inquiry

A Purchase Account Status Inquiry (ASI) or a Recurring Payment ASI is an optional service that allows merchants to validate that a cardholder account is open and the account is not listed in the Electronic Warning Bulletin—without negatively affecting a cardholder's funds availability when establishing a recurring or bill payment relationship, validating a card-not-present purchase before fulfillment, or before submitting an authorization request for the full amount of a recurring payment. A Recurring Payment ASI must comply with all existing recurring payment transaction identification requirements.

Account Status Inquiry Transaction Process

1. The acquirer submits an Account Status Inquiry transaction request in the Authorization Request/0100 message containing:
 - DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code), value 00 (Purchase)
 - DE 4 (Amount, Transaction) with a transaction amount of zero
 - DE 48 (Additional Data—Private Use), subelement 82 (Address Verification Service Request), value 52 (AVS and Authorization Request/0100) for AVS requests (optional)
 - DE 48, subelement 92 (CVC 2 Value), CVC 2 value (optional)
 - DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status), value 8 (Account Status Inquiry Service [ASI])

If Recurring Payment ASI (for an ongoing recurring payment arrangement):

 - DE 61 (Point-of-Service [POS] Data), subfield 4 (POS Cardholder Presence), value 4 (Standing order/recurring transactions)
2. The issuer sends the acquirer an Authorization Request Response/0110 message where DE 39 (Response Code) may contain either value 00 (Approved or completed successfully), 85 (Not declined), or a valid business decline response. If the issuer is unable to reply, the acquirer will receive a response code of 91 (Authorization System or issuer system inoperative).
3. If applicable, the issuer provides the applicable AVS response in DE 48, subelement 83 (Address Verification Service Response), and provides a valid CVC 2 response in DE 48, subelement 87 (Card Validation Code Result) in the Authorization Request Response/0110 message.

NOTE: Product Inquiry messages do not include address verification and/or CVC 2 validation requests. See separate description of Product Inquiry Service.

Acquirers are prohibited from placing a value of one major unit of currency or any other nominal test amount (including equivalent units of local currency such as EUR 1 or JPY 1) that does not represent an actual purchase amount in DE 4 of the Authorization Request/0100 message.

Transactions occurring at an automated fuel dispenser in the U.S. region identified with MCC 5542 may submit USD 1 (or local currency equivalent) authorization requests when value 4 (Preauthorized request) is used in DE 61, subfield 7.

Contactless transit aggregated transactions generated by the transit authority and held for a period of time before being cleared may continue to submit an authorization request for USD 1 or an amount up to the cardholder verification method (CVM) limit amount published in the *Chargeback Guide* on the day of the transaction (or local currency equivalent) authorization requests. The transaction should contain value 0 (Normal request) or 4 (Preauthorized request) in DE 61, subfield 7.

MDES Account Status Inquiries

Account Status Inquiry messages can also be initiated with MDES tokens. In this case the applicable MDES services will be performed when MDES data is provided.

For more information on Tokenized ASI messages, see the *MDES Technical Specifications* manual.

Country- and Currency-Related Data in MDES Pre-Digitization and ASI Messages

Issuers that use the country-level authorization service for cards designated for local use only and have chosen to receive MDES pre-digitization and ASI messages must be aware that when a funding account range is configured for local use only, Mastercard will modify the country- and currency-related data fields in the MDES pre-digitization and ASI messages to match the issuer country code and primary currency code (as applicable) of the funding account range, according to the configurations provided by the issuer during the onboarding process. The country- and currency-related data fields include the following:

- DE 43 (Acceptor Name and Location), subfield 5 (Acceptor Country Code)
- DE 49 (Currency Code, Transaction)
- DE 61 (Additional POS Data), subfield 13 (POS Country Code)

Issuers that are not configured for local use only will continue to see the MDES pre-digitization and ASI messages with the United States country and currency codes.

Payment Account Status Inquiry

A Payment Account Status Inquiry (Payment ASI) enables payers to verify that issuers will accept a Payment Transaction on behalf of a payee (recipient) for the actual transaction amount for a particular recipient card account before collecting funds from the sender and initiating the Payment Transaction authorization to credit an account.

Benefits

Issuers can provide a response to an inquiry sent by the payer, indicating whether the issuer will accept a subsequent Payment Transaction authorization for a particular recipient account and

amount. Obtaining a successful Payment ASI helps to eliminate the need for a payer to return funds to the sender and can be useful in cases where returning funds to the sender is impractical or difficult.

The Payment ASI helps improve the payer's experience when confirming a recipient's card account for future use. Issuers can clearly identify:

- The difference between a Payment ASI transaction and a Payment Transaction authorization, so that they can respond and act accordingly.
- The difference between a Payment ASI transaction and a Purchase ASI transaction.

How It Works

Acquirers can submit an ASI transaction for a payment to determine whether an issuer will post the specified amount to a particular card account before collecting funds from the sender.

Payment ASI transactions are non-financial transactions. Issuers must not credit the receiving cardholder account based on the amount provided in the Payment ASI request.

- When an issuer receives a Payment ASI transaction of zero, the issuer should confirm that the recipient's card account exists and can be used in the future for receiving funds.
- When an issuer receives a Payment ASI transaction for greater than zero, the issuer should confirm that the amount specified does not exceed a prepaid load limit or other types of limits they may have for Payment Transactions.
- When the issuer provides an approval response to the Payment ASI for the receiving account, the issuer is indicating that it is expecting to approve a subsequent Payment Transaction request message when it is received for the account.

NOTE: The receiving issuer may reject the subsequent Payment Transaction if circumstances for the account have changed.

WHEN...	THEN the Authorization Platform...
A Payment ASI qualifies for a Mastercard® MoneySend™ Payment Transaction	Applies regular MoneySend Payment edits.
An acquirer advice Payment ASI is initiated	Sends the acquirer an Authorization Advice/0130 message where DE 39 = 12 (Invalid transaction).
An acquirer reversal of a Payment ASI is initiated	Sends the acquirer a Reversal Request Response/0410 message where DE 39 = 12 (Invalid transaction).

Payment Account Status Inquiry Transaction Messages

Following is a list of the data elements and values applicable to Payment ASI Authorization Request/0100 messages.

Data Element	Value	Comment
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)	28	Payment
DE 4 (Amount, Transaction)	Equal to or greater than 0	Equal to or greater than zero
DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier)	C02, C03, C04	
DE 48 (Additional Data—Private Use), subelement 82 (Address Verification Service Request)	52	Optional AVS
DE 48 (Additional Data—Private Use), subelement 92 (CVC 2)	CVC 2	Optional CVC 2
DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status)	8	Account Status Inquiry Service (ASI)

MoneySend Payment ASI Transaction Messages

Following is a list of the data elements and values applicable to MoneySend Payment ASI Authorization Request/0100 messages.

Data Element	Value	Comment
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)	28 (Payment Transaction)	
DE 4 (Amount, Transaction)	Equal to or greater than 0	For transaction-value limits, refer to the <i>Mastercard MoneySend and Funding Transactions Program Standards</i> .
DE 18 (Merchant Type)	MCC 6536 (MoneySend Intracountry) MCC 6537 (MoneySend Intercountry)	

Data Element	Value	Comment
DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier)	C07 (General Person-to-Person)	MoneySend Funding/ Payment Transaction Type Indicators
	C52 (General Transfer to Own Account)	
	C53 (Agent Cash Out)	
	C54 (Payment of Own Credit Card Bill)	
	C55 (Business Disbursement)	
	C56 (Government/Non-profit Disbursement)	
	C57 (Rapid Merchant Settlement)	
	C58 (Cash in at ATM)	
	C59 (Cash in at point of sale)	
	C65 (General Business to Business Transfer)	
DE 48 (Additional Data—Private Use), subelement 82 (Address Verification Service Request)	52 (AVS and Authorization Request/0100)	Optional AVS
DE 48 (Additional Data—Private Use), subelement 92 (CVC 2)	CVC 2	Optional CVC 2 If the originating institution does not have CVC data for push payment transactions on the Mastercard Network, this field should be left blank. The receiving institution must not decline MoneySend payment transactions on the basis that the CVC data is not populated.
DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status)	8 (Account Status Inquiry Service [ASI])	
DE 61 (Point-of-Service [POS] Data), subfield 10 (Cardholder-Activated Terminal Level)	0 (Not a CAT transaction)	

Data Element	Value	Comment
DE 108 (Additional Transaction Reference Data)		DE 108 fields are required for MoneySend Payment ASI Transactions. For technical requirements, refer to the <i>MoneySend and Funding Transactions Program Standards</i> .

Authorization Platform Edits

The Authorization Platform performs the following edits on Account Status Inquiry transactions.

WHEN...	THEN the Authorization Platform...
DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status) contains value 8 (Account Status Inquiry Service [ASI]) and DE 4 (Amount, Transaction) contains a value other than zero	Sends the acquirer an Authorization Request Response/0110 message where: - DE 39 (Response Code) = 30 (Format error) - DE 44 (Additional Response Data) = 004 (indicating the data element in error)
DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status) contains value 8 (Account Status Inquiry Service [ASI]) and DE 4 (Amount, Transaction) contains a value equal to zero and DE 3 (Processing Code) contains a value other than 00 (Purchase)	Sends the acquirer an Authorization Request Response/0110 message where: - DE 39 = 30 - DE 44 = 003
The acquirer sends an Authorization Advice/0120 or acquirer-generated Reversal Request/0400 message with the value combinations above	Sends the acquirer an Authorization Advice/0130 or Reversal Request Response/0410 message where DE 39 = 12 (Invalid transaction).

Name Validation Service

The Name Validation Service validates that the name provided by a registered user matches the name associated to the card at the issuer. Customers can validate cardholder name information by including a name validation request in an Account Status Inquiry (ASI) message.

Participation requirements

Acquirers should be aware that name validation can be included in an ASI message.

Issuers will receive ASI service messages with a request for cardholder name validation. Issuers must validate the name information provided in the request message and respond with a result indicating the likelihood of a match.

The availability and effectiveness of name validation may vary, based on the issuer's capability to perform the validation.

Name Validation Process

1. The acquirer creates an ASI Authorization Request/0100 message, when submitting name validation requests, the message will contain:
 - a. Data Element (DE) 108 (Additional Transaction Reference Data), subelement 02 (Sender Data), subfield 01 (Sender First name), subfield 02 (Sender Middle Name), and subfield 03 (Sender Last Name), DE 108 subelement 07 (Data Verification Type), subfield 01 (Data Verification Type) = 11 (Sender Data Verification Request)
 - b. DE 108, subelement 01 (Receiver/Recipient Data), subfield 01 (Receiver/Recipient First name), subfield 02 (Receiver/Recipient Middle Name), and subfield 03 (Receiver/Recipient Last Name), DE 108, subelement 07, subfield 01 = 10 (Receiver/Recipient Data Verification Request)
2. The issuer (or issuer processor) compares the name information provided in DE 108 to the cardholder name(s) on file
3. The issuer generates an ASI Authorization Response/0110 message with the name validation response included:
 - a. When sender name validation was requested, DE 108, subelement 08, subfield 02 (Sender Service Response) is populated with one of the following validation result values: A (match), B (partial match), C (no match), or U (unverified)
 - b. When receiver name validation was requested, DE 108, subelement 08 (Data Verification Response), subfield 01 (Receiver/Recipient Service Response) is populated with one of the following validation result values: A (match), B (partial match), C (no match), or U (unverified)

For more information

For more information on the Name Validation Service, refer to [Name Validation Service Issuer Name Matching Guide](#).

Address Verification Service

Address Verification Service (AVS) is a fraud deterrent service that provides greater security to merchants and cardholders. It helps to protect against fraudulent use of cards by verifying the cardholder's billing address.

How AVS Works

The acquirer requests the cardholder billing address verification as part of the Authorization Request/0100. The issuer performs the verification and returns the appropriate information to the acquirer in the Authorization Request Response/0110.

Alternate Processing

The Stand-In and X-Code systems do not perform AVS. Transactions that contain an AVS request in an Account Status Inquiry Service request will receive a response indicating the service is not available. Transactions that contain an authorization request and an AVS request will receive the appropriate authorization response in addition to a response indicating the AVS service is not available.

NOTE:

- For more information about issuer and acquirer participation in AVS, requirements, and service options, refer to the *Customer Interface Specification and Authorization Manual*.
- The Global Safety and Security Standards mandate that issuers, acquirers, and processors must code for the Authorization security fields that are required to support the Address Verification service. The service themselves are not mandate, but issuers, acquirers, and processors must code for the authorization message field that support this service.
- Issuers in North America are required to ensure all U.S. and Canada issued cards support and respond to all AVS requests from merchants.

It is the issuer's responsibility to ensure that all their account ranges for U.S. and Canada issued cards are enabled for AVS. Sign-on value 0 (AVS not currently supported) must not be used.

AVS response code U (No response from issuer/authorization platform) may be utilized where there is a decline authorization and the issuer cannot disclose the AVS response.

Authorization Request/0100: AVS and authorization request

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Table 1590: Applicable data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 4 (Amount, Transaction)	M	.	M	Must be a valid amount for authorization request with AVS requests.

Data Element	Org	Sys	Dst	Values/Comments
DE 48 (Additional Data—Private Use), TCC (Transaction Category Code)	M	•	M	Must be a valid TCC.
DE 48, subelement 82 (Address Verification Request)	M	•	M	52 = AVS and Authorization Request
DE 120 (Record Data)				The acquirer must provide the non-condensed cardholder billing address. NOTE: Note: Some merchants/acquirers are currently limited to supporting only numeric data. The issuer will receive only one occurrence of DE 120 subfields depending on the AVS Service Indicator identified in DE 94 of the Network Management Request/0800 message. The AVS Service Indicator identifies how the issuer expects to receive address data.
DE 120, subfield 1 (AVS Service Indicator 1)	M	•	C	Postal code (an-9) Address (ans...20 Mastercard) Address (ans...20 Visa) Note: At a minimum the Postal Code must be provided and cannot be less than nine bytes long (Postal code left-justified and blank filled if necessary up to nine bytes).
DE 120, subfield 2 (AVS Service Indicator 2)	•	X	C	Postal Code (an-9) Address (an-5)
DE 120, subfield 3 (AVS Service Indicator 3)	•	X	C	Postal Code (an-9) Address (an-5)
DE 120, subfield 4 (AVS Service Indicator 4)	•	X	C	Postal Code (an-9) Address (an-5)

Authorization Request Response/0110: AVS and Authorization Request Response

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0110 data elements apply.

Table 1591: Applicable data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 4 (Transaction Amount)	ME	P	ME	Must be the same value as in the original Authorization Request/0100 message.
DE 39 (Response Code)	M	P	M	Contains one of the response codes listed for this data element in the Data Element Definitions chapter of this manual.
DE 48 (Additional Data—Private Use), TCC (Transaction Category Code)	ME	P	ME	Must be the same value as in the original Authorization Request/0100 message.
DE 48, subelement 82 (Address Verification Request)	ME	P	ME	Must be the same value as in the original Authorization Request/0100 message.
DE 48, subelement 83 (Address Verification Response)	M	P	M	Contains one of the response codes listed for this data element in the Data Element Definitions chapter of this manual.
DE 120 (Record Data)	ME	P	ME	Must be the same value as in the original Authorization Request/0100 message.

NOTE: Acquirers always receive the original information in DE 120 of the Authorization Request Response/0110 (the non-condensed subfield 01 contents) that they provided in the Authorization Request/0100.

Network Management Request/0800: sign-on/sign-off

Following is a list of the data elements and values applicable to this message type. All mandatory Network Management Request/0800: AVS sign-on data elements apply.

Table 1592: Applicable data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 94 (Service Indicator), subfield 3 (Address Data Indicator)	M	.	M	0 = AVS not supported 1 = Issuer supports AVS, receives all the address data (non-condensed) 2 = Issuer supports AVS, receives condensed address data 3 = Issuer supports AVS, receives condensed address data 4 = Issuer supports AVS, receives condensed numeric postal code and condensed numeric address data only.

Alternate Processing

When the issuer is unavailable the transaction is processed by Stand-In Authorization or X-Code as follows.

Table 1593: Authorization Request/0100: Request and Address Verification Service

WHEN...	THEN the Dual Message Authorization System....
the Authorization Request/0100 contains DE 48, subelement 82 value 52	sends an Authorization Request Response/0110 message containing DE 39, value (based on the decision for the authorization request portion of the transaction). if the issuer supports AVS then DE 48, subelement 83 contains value R if the issuer does not support AVS then DE 48, subelement 83 contains value S

DE 48 and DE 120 structure in AVS transactions

Following is the structure of DE 48 and DE 120 in AVS transactions.

Table 1594: Data element structure

The following table illustrates DE 48, subelements 82 (Address Verification Service Request) option code and subelement 83 (Address Verification Service Response) result code in an AVS transaction.

LLL	VAR: 999 maximum bytes (TCC + AVS Data)						
3 bytes	1 byte	2 bytes	2 bytes	2 bytes	2 bytes	2 bytes	1 byte
Total Data Element Length	TCC	AVS Request Data			AVS Response Data		
		Subelement ID 82	Subelement Length 02	AVS Request Code	Subelement ID 83	Subelement Length 01	AVS Result Code
Mandatory							
1002 maximum bytes (LLL + TCC + AVS Data)							

Table 1595: Data element structure

The following table illustrates DE 120 contents for subfield 01 (AVS Service Indicator 1).

LLL	VAR: 33 maximum bytes (Mastercard) – VAR—53 maximum bytes (Visa)			
3 bytes	2 bytes	2 bytes	9 bytes	1–20 bytes (Mastercard) 1–40 bytes (Visa)
Total Data Element Length	Subfield ID 01	Subfield Length Variable	Cardholder postal/ZIP code	Cardholder Address

Table 1596: Data element structure

The following table illustrates DE 120 contents for subfield 02–04 (AVS Service Indicator 2–4).

LLL	Fixed 18 bytes			
3 bytes	2 bytes	2 bytes	9 bytes	5 bytes
Total Data Element Length	Subfield ID 02, or 03, or 04	Subfield Length	Cardholder postal/ZIP code	Cardholder Address

For more information

For details about data requirements, refer to [Data element definitions](#).

Authorization Platform Edits

The Authorization Platform performs the following edits for an AVS-only transaction.

NOTE: The AVS Only request was discontinued 28 June 2011.

WHEN the acquirer...	THEN the Authorization Platform...
DE 48, subelement 82 (AVS Request Data) = 51 (AVS-Only)	Sends the acquirer an Authorization Request Response/0110 message where: DE 39 = 12 (Invalid Transaction)

The Authorization Platform performs the following edit for AVS and Authorization Request/0100 messages.

WHEN the acquirer...	THEN the Authorization Platform...
Sends an Authorization Request/0100 message with DE 48, subelement 82 (AVS Request Data) = 52 (AVS and Authorization Request) and DE 120 (Record Data), subfield 01 (AVS Service Indicator 1) is less than nine bytes long	Sends the acquirer an Authorization Request Response/0110 message with: • DE 39 = 30 (Format Error) • DE 44 = 120

The Authorization Platform performs the following edits for AVS and Authorization Request Response/0110 messages.

WHEN the issuer...	THEN the Authorization Platform...
Sends an Authorization Request Response/0110 message where DE 39 is 00 (Approved) or 08 (Honor with ID) or 85 (Not declined) and DE 48 subelement 82 is 52 (AVS Request) DE 48, subelement 83 (AVS Response) is present with a value not equal to X, Y, A, W, Z, N, U, R, S.	Sends the issuer an Authorization Response Negative Acknowledgement/0190 message where: • DE 39 = 30 (Format Error) • DE 44 = 048083 (indicating the data element in error)

WHEN the issuer...	THEN the Authorization Platform...
Sends an Authorization Request Response/0110 message with either an Approved or Declined response where: - DE 48 (Additional Data—Private Use), subelement 82 (Address Verification Service Request) contains value 52 (AVS and Authorization Request/0100) and - DE 48, subelement 83 (Address Verification Service Response) is present with a value not equal to A (Address matches, postal code does not), N (Neither address nor postal code matches), R (Retry, system unable to process), S (AVS currently not supported), U (No data from issuer/Authorization Platform), W (For U.S. addresses, nine-digit postal code matches, address does not; for address outside the U.S., postal code matches, address does not), X (For U.S. addresses, nine-digit postal code and address matches; for addresses outside the U.S., postal code and address match), Y (For U.S. addresses, five-digit postal code and address matches), and Z (For U.S. addresses, five-digit postal code matches, address does not)	Will set the default value as U and will forward the Authorization Request Response/0110 message to the acquirer. and Will not send the issuer an Authorization Response Negative Acknowledgement/0190 message where: - DE 39 (Response Code) = 30 (Format Error) - DE 44 (Additional Response Data) = 048083 (indicating the data element in error)
Sends an Authorization Request Response/0110 message where: - DE 48, subelement 82 contains value 52 and - DE 48, subelement 83 is not present	Will set the default value as U and will forward the Authorization Request Response/0110 message to the acquirer.

Automated Fuel Dispenser Completion

Acquirers use the Automated Fuel Dispenser (AFD) Completion messages to advise the issuer of the total amount of the Automated Fuel Dispenser (AFD) transaction.

An AFD transaction is identified with MCC 5542 and may occur at a cardholder-activated terminal (CAT) or as an e-commerce transaction.

The acquirer sends an Authorization Request/0100 message to verify a minimum availability of funds in the cardholder's account for an Automated Fuel Dispenser (AFD) transaction. If the request is not declined, the acquirer sends an Authorization Advice/0120: acquirer-generated message to the issuer, specifying the total amount of the AFD transaction. The value in DE 60,

subfield 1 (Advice Reason Code) is 191 (Acquirer Processing System [APS] Completed Authorization Transaction).

The Dual Message Authorization System forwards all Authorization Advice/0120: acquirer-generated messages directly to the issuer. The issuer must respond with an Authorization Advice Response/0130: responding to an acquirer-generated 0120 message acknowledging receipt of the Authorization Advice/0120: acquirer-generated message. If the issuer is not available or does not respond within timer limits, the Authorization Advice/0120: acquirer-generated message will be sent to the Stand-In Authorization. The Stand-In Authorization responds with an Authorization Advice Response/0130: system-generated message to the acquirer to acknowledge receipt of the Authorization Advice/0120: acquirer-generated message. The advice message will be added to the SAF for later distribution to the issuer.

NOTE: The Data Integrity Automated Fuel Dispenser (AFD) edit for acquirers of AFD merchants that are located in the U.S. region is described in *Data Integrity Monitoring Program*.

AFD message scenarios

Mastercard supports the following usage scenarios for AFD authorization messages.

Table 1597: AFD message scenarios

WHEN the AFD Authorization Request Response/0110 contains...	THEN the AFD acquirer...
DE 39 (Response Code), value 00 (Approved or completed successfully)	submits the Authorization Advice/0120: acquirer-generated message containing the final fuel sale amount.
DE 39, value 10 (Partial Approval)	submits the Authorization Advice/0120: acquirer-generated message containing the final fuel sale amount that is less than or equal to the partial approval amount.
DE 39, value 00 (Approved or completed successfully) or value 10 (Partial Approval) and the transaction is canceled at the pump	submits a Reversal Request/0400 message, or an Authorization Advice/0120 message with a DE 04 zero amount, to cancel the AFD transaction.

NOTE: The AFD Authorization Advice/0120: acquirer-generated message is not a replacement for the First Presentment/1240 message to the clearing system on a credit transaction.

Authorization Request/0100: Automated Fuel Dispenser Preauthorization

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Table 1598: Applicable data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 4 (Amount, Transaction)	M	•	M	Must be at least an amount no less than the equivalent of USD 1.
DE 18 (Merchant Type)	M	•	M	5542 = Fuel Dispenser, Automated
DE 61 (Point of Service Data), subfield 7 (POS Transaction Status)	M	•	M	May contain one of the following values: 0 = Normal request (original presentment) 4 = Preauthorized request
DE 61, subfield 10 (Cardholder Activated Terminal Level)	M	•	M	2 = Authorized Level 2 CAT: Self-service terminal, or 1 = Authorized Level 1 CAT if PIN used

Authorization Advice/0120: acquirer-generated (Automated Fuel Dispenser Completion)

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Advice/0120 data elements apply.

Table 1599: Applicable data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 4 (Amount, Transaction)	M	•	M	Contains the completed transaction amount and not the requested amount.
DE 18 (Merchant Type)	M	•	M	5542 = Fuel Dispenser, Automated
DE 39 (Response Code)	M	•	M	Valid values: 00 = Approved or completed successfully 10 = Partial Approval
DE 48 (Additional Data: Private Use), subelement 15 (Dual Message Authorization System Advice Date and Time), subfield 1 (Date)	•	X	M	Contains the valid date of the Authorization Advice/0120: acquirer-generated inserted by the Dual Message Authorization System in MMDD format.

Data Element	Org	Sys	Dst	Values/Comments
DE 48, subelement 15, subfield 2 (Time)	•	X	M	Contains the valid time of the Authorization Advice/0120: acquirer-generated inserted by the Dual Message Authorization System in hhmmss format.
DE 48, subelement 20 (Cardholder Verification Method)	M	X	•	
DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code)	M	•	M	191 = Acquirer Processing System (APS) Completed Authorization Transaction

NOTE: Acquirers must not submit DE 48, subelement 82 (Address Verification Service Request) or subelement 92 (CVC 2) as these services are not performed on the Authorization Advice/0120: acquirer-generated message and will cause the 0120 message to be rejected by the Dual Message Authorization System.

Authorization Advice/0120: acquirer-generated (Automated Fuel Dispenser Completion)

Acquirers of Automated Fuel Dispenser (AFD) merchants located in the U.S. and Canada regions must send an Authorization Advice/0120 message to the issuer providing the actual transaction amount for each approved AFD transaction within 60 minutes after the original Authorization Request/0100 message was submitted. Acquirers of AFD merchants located in the Europe region must send an Authorization Advice/0120 or Reversal Request/0400 message within 20 minutes of receiving the Authorization Request Response/0110 message. In other regions, acquirers may optionally support this message for Mastercard and Debit Mastercard AFD transactions.

Customers should be aware of the critical requirements for proper processing of acceptor business code (MCC) 5542 (Fuel Dispenser, Automated) transactions.

Critical AFD Advice Message Data

The following information provides a summary of the critical acquirer requirements for processing AFD transactions:

- Authorization Advice/0120 (Automated Fuel Dispenser Completion) messages must contain DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code), value 191 (completed authorization) for AFD transactions with an original Authorization Request/0100 message.
- The following Authorization Advice/0120 message data elements must match the value submitted within the original Authorization Request/0100 message for issuer transaction matching purposes:
 - DE 2 (Primary Account Number)
 - DE 7 (Transmission Date and Time)
 - DE 11 (System Trace Audit Number [STAN])
 - DE 32 (Acquiring Institution ID Code)

- DE 33 (Forwarding Institution ID Code), if present in the Authorization Request/0100 message
- DE 38 (Authorization ID Response) and DE 39 (Response Code) with the same value as received in the original Authorization Request Response/0110 message
- DE 48 (Additional Data), subelement 63 (Trace ID) to further assist issuers in matching an AFD completion advice to the original preauthorization
- DE 48 (Additional Data), subelement 98 (Mastercard Corporate Fleet Card® ID/Driver Number) and/or subelement 99 (Mastercard Corporate Fleet Card® Vehicle Number), with the same values as submitted in the original Authorization Request/0100 message, if present
- DE 121 (Authorizing Agent ID Code), with same value as was received in the original Authorization Request Response/0110 message, if present

Track Data in AFD Advice Message: acquirers

The Authorization Advice/0120 message layout shows DE 35 (Track 2 Data) and DE 45 (Track 1 Data) as optional. Acquirers of AFD merchants are reminded that the presence of track data in the Authorization Request/0100 message for a card-present transaction does not necessitate inclusion within the AFD Advice message. This data is not needed by issuers for matching an AFD advice to an original authorization, and storage of track data for submission of the AFD Advice may not be PCI compliant.

Track Data in AFD Advice Message: issuers

The Authorization Advice/0120: acquirer-generated message for an AFD transaction contains the completion amount and other reference data from the original Authorization Request/0100 message data. Issuers are reminded that DE 35 and DE 45 track data is optional in an AFD Advice message and may not be present, regardless of the Point-of-Service (POS) entry mode value in DE 22. As such, the absence or presence of track data in the AFD Advice message should not result in a format error from issuers for card-present fuel purchases.

For details about data requirements, refer to [Data element definitions](#) section.

Authorization Advice Response/0130: Issuer-generated (responding to an acquirer-generated 0120)

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Advice Response/0130: issuer-generated (responding to an acquirer-generated 0120) data elements apply.

Table 1600: Applicable data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 48 (Additional Data: Private Use), subelement 15 (Dual Message Authorization System Advice Date and Time), subfield 1 (Date)	ME	•	ME	This should match what was received by the issuer in the Authorization Advice/0120 message, which was provided by the Dual Message Authorization System.
DE 48 (Additional Data: Private Use), subelement 15 (Dual Message Authorization System Advice Date and Time), subfield 2 (Time)	ME	•	ME	This should match what was received by the issuer in the Authorization Advice/0120 message, which was provided by the Dual Message Authorization System.

Alternate Processing

Alternate processing is provided if an issuer does not respond or is unable or unavailable to receive the Authorization Advice/0120: acquirer-generated AFD completion messages.

The Dual Message Authorization System responds to the acquirer with an Authorization Advice Response/0130 message containing DE 39 (Response Code), value 00 (Approved or completed successfully) and sends the Authorization Advice/0120: acquirer-generated message to SAF. SAF messages will be sent to the issuer when the issuer is available. DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code) keeps value 191 (Acquirer Processing System [APS] Completed Authorization Transaction) when delivered from SAF.

Clearing AFD Transactions

Mastercard recommends that the Trace ID (DE 15 [Settlement Date] and DE 63 [Network Data]) of the completion advice response be used within the clearing presentment.

As an alternative, if the Trace ID of the original preauthorization is used for clearing presentment, that same Trace ID must be included in the AFD completion advice message within DE 48, subelement 63 (Additional Data, Trace ID).

Account Level Management

Mastercard Account Level Management is a platform that enables specialized processing so that Mastercard can manage capabilities at the individual card account level.

Mastercard Account Level Management provides issuers the flexibility of qualifying cardholder accounts for competitive interchange as well as upgrading cardholder accounts to a different card product. The following are the key capabilities of Account Level Management functionality. Availability of, and distinctions of, are based on location:

- Enhanced Value (United States and select countries in the Europe region – Moldova and Ukraine)
- Product Graduation Plus (United States and select countries in the Europe region – Moldova and Ukraine)
- Product Graduation Select (select countries in the Latin America and Caribbean and Europe regions). Refer to the *Account Level Management User Manual* for a complete listing of all participating Countries
- Consumer Product Monitoring Service (Canada)
- High Value (United States)
- Spend Shortfall (World Mastercard and World Elite Mastercard) (United States)
- Small Business Spend Processing (United States)

Refer to the *Account Level Management User Manual* for detailed information.

Alternate Processing

Both Stand-In Authorization and X-Code verify that DE 38, position 6 in the Authorization Request Response/0110 message contains the value provided by the Dual Message Authorization System in DE 48, subelement 38 of the Authorization Request/0100 message.

Stand-In Authorization for Product Graduation

Stand-In Authorization supports Product Graduation. For product graduated accounts, Stand-In Authorization uses the product code in DE 63, subfield 1 instead of the product code associated with the authorization account range when applying Stand-In Authorization limits at the product code level.

Graduated Product Code Limits

Mastercard offers issuers graduated product code limits as an option for defining Stand-In Authorization parameters for graduated accounts. Issuers can establish limits at the authorization account range level for each product code supported for Product Graduation. These limits will apply specifically to accounts within that account range that are graduated to that product and can be different from the limits that the issuer establishes for an account range that is associated to the same product code. If the issuer takes no action to establish limits for product codes supported for Product Graduation, the Mastercard default limits for the graduated product will apply. Mastercard provides the *Stand-In Processing Transaction Category*

Code *Global Parameters* (Form 041g) available on Mastercard Connect™ for issuer defined limits.

X-Code Processing

If the product code in DE 63, subfield 1 is different than the product code assigned to the authorization account range, X-Code applies product code limits using the product code in DE 63, subfield 1.

ATM Bill Payment Service

Bill payment transactions at the ATM help increase the acquirer's transaction volume by providing a convenient method for cardholders to initiate bill pay request transactions.

ATM Bill Payment service supports:

- Europe acquired ATM Bill Payment transactions
- Non-Europe acquired ATM Bill Payment transactions

Authorization Request/0100: ATM Bill Payment, Europe Acquired

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Table 1601: Applicable data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)	M	•	M	00 = Purchase
DE 18 (Merchant Type)	M	•	M	Contains value 6050 = Quasi Cash: Member Financial Institution or a more precise MCC related to the nature of the bill that is being paid, for example, MCC 4900 (Utilities: Electric, Gas, Heating Oil, Sanitary, Water) for utilities bills. MCC 6011 (Member Financial Institution: Automated Cash Disbursements) must not be used.
DE 48 (Additional Data—Private Use), TCC (Transaction Category Code)	M	•	M	U = Unique
DE 61 (Point-of-Service [POS] Data), subfield 10 (Cardholder-Activated Terminal Level Indicator)	M	•	M	0 = Not a CAT transaction 1 = Authorized Level 1 CAT: automated dispensing machine with PIN

Data Element	Org	Sys	Dst	Values/Comments
DE 124 (Member-defined Data)	O	•	C	Contains details relating to the bill being paid.

Authorization Request/0100: ATM Bill Payment, Non-Europe Acquired

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Table 1602: Applicable data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)	M	•	M	00 = Purchase
DE 18 (Merchant Type)	M	•	M	Must contain value 6539 = Funding Transaction, Excluding Mastercard Send
DE 48 (Additional Data—Private Use), TCC (Transaction Category Code)	M	•	M	U = Unique
DE 61 (Point-of-Service [POS] Data), subfield 10 (Cardholder-Activated Terminal Level Indicator)	M	•	M	One of the following values: 1 = Authorized Level 1 CAT: automated dispensing machine with PIN 2 = Authorized Level 1 CAT: self-service terminal
DE 124 (Member-defined Data)	O	•	C	Contains details relating to the bill being paid.

Dual Message Authorization System Edits

The Dual Message Authorization System applies the following edits on ATM Bill Payment Service transactions.

Table 1603: Applicable data elements

WHEN the acquirer...	THEN the Dual Message Authorization System...
sends an Authorization Request/0100, Authorization Advice/0120, or Reversal Request/0400 message containing DE 18 (Merchant Type), value 6539 (Funding Transaction, Excluding Mastercard Send)	sends the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 message with DE 39 (Response Code), value 58 (Transaction not permitted to acquirer/terminal).

ATM Credit Card Cash Advance in Installments

This service allows cardholders to initiate an inquiry at the ATM requesting a credit card cash advance to be repaid in monthly installments (for example, 6, 12, or 18 months).

With this enhancement, a cardholder can initiate an inquiry at the ATM requesting a credit card cash advance to be repaid in monthly installments (for example, 6, 12, or 18 months). The issuer can accept the installment terms requested by the cardholder, decline the request, or decline the request and offer an alternate installment schedule. The cardholder has the option to accept or reject the issuer's terms and conditions for the installment payments. If the cardholder accepts the issuer's terms and conditions for repayment, the issuer will then approve or decline the cash advance transaction. If approved, the cardholder will receive the funds, along with the installment payment details printed on the ATM receipt. The issuer will bill the cardholder for the amount of the transaction in the agreed-upon installments.

Acquirers entering this market must be able to:

- Provide ATM screens that offer an installment payment option.
- Provide the ability to print the issuer's installment payment details on the ATM receipt.

The ATM Credit Card Cash Advance in Installments service is currently available to single message transactions (Single Message System-acquired activity).

The ATM Credit Card Cash Advance in Installments service is not available to ATM acquirers in the Europe region.

Authorization Request/0100: ATM Installment Inquiry

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Table 1604: Applicable data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)	M	•	M	01 (Withdrawal)
DE 4 (Amount, Transaction)	M	•	M	Contains the amount of the cash advance
DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status)	M	•	M	Identifies that the cardholder has initiated an ATM installment inquiry transaction 3 = ATM Installment Inquiry
DE 112 (Additional Data [National Use]), subelement 001 (Installment Payment Data)	C	•	C	Identifies the transaction type. 80xx = ATM Installment Inquiry, Number of Installments 81xx = ATM Installment Withdrawal, Number of Installments
DE 112, subelement 027 (ATM Credit Card Cash Advance Installments), subfield 1 (Transaction Type)	C	•	C	80 = ATM Installment Inquiry
DE 112, subelement 027 (ATM Credit Card Cash Advance Installments), subfield 2 (Requested Number of Installments)	C		C	01–99 = the number of installments requested by the cardholder

Authorization Request Response/0110: ATM Installment Inquiry

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0110 data elements apply.

Table 1605: Applicable data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)	ME	•	ME	01 (Withdrawal)
DE 4 (Amount, Transaction)	CE	X	CE	Contains the amount of the cash advance
DE 112 (Additional Data [National Use]), subelement 001 (Installment Payment Data)	C	•	C	Identifies the transaction type. 80xx = ATM Installment Inquiry, Number of Installments 81xx = ATM Installment Withdrawal, Number of Installments
DE 112, subelement 027 (ATM Credit Card Cash Advance Installments), subfield 1 (Transaction Type)	C	•	C	80 = ATM Installment Inquiry
DE 112, subelement 027, subfield 2 (Requested Number of Installments)	C	•	C	01–99 = the number of installments requested by the cardholder
DE 112, subelement 027, subfield 3 (Approved Number of Installments)	C	•	C	The number of installment payments approved by the issuer
DE 112, subelement 027, subfield 4 (Installment Amount)	C	•	C	The monthly payment amount
DE 112, subelement 027, subfield 5 (Total Transaction Amount)	C	•	C	The amount of the cash advance and the interest for the transaction
DE 112, subelement 027, subfield 6 (Yearly Interest Rate)	C	•	C	The interest rate that will be charged to the cardholder by the issuer
DE 112, subelement 027, subfield 7 (Currency Code)	C	•	C	The currency code the issuer is charging the cardholder for repayment
DE 112, subelement 027, subfields 8–11 (Member-defined Data)	C	•	C	Member-defined data

Authorization Request/0100: ATM Installment Withdrawal

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Table 1606: Applicable data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)	M	•	M	01 = Withdrawal
DE 4 (Amount, Transaction)	M	•	M	Contains the amount of the cash advance.
DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status)	M	•	M	Identifies that the cardholder has initiated an ATM installment inquiry transaction. 0 = Normal request (original presentment)
DE 112 (Additional Data [National Use]), subelement 001 (Installment Payment Data)	C	•	C	Identifies the transaction type. 81xx = ATM Installment Withdrawal, Number of Installments
DE 112, subelement 027 (ATM Credit Card Cash Advance Installments), subfield 1 (Transaction Type)	C	•	C	81 = ATM Installment Withdrawal
DE 112, subelement 027 (ATM Credit Card Cash Advance Installments), subfield 2 (Requested Number of Installments)	C	•	C	01–99 = The number of installments requested by the cardholder

Authorization Request Response/0110: ATM Installment Withdrawal

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0110 data elements apply.

Table 1607: Applicable data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)	ME	•	ME	01 = Withdrawal

Data Element	Org	Sys	Dst	Values/Comments
DE 4 (Amount, Transaction)	CE	X	CE	Contains the amount of the cash advance.
DE 112 (Additional Data [National Use]), subelement 001 (Installment Payment Data)	C	•	C	Identifies the transaction type. 81xx = ATM Installment Withdrawal, Number of Installments
DE 112, subelement 027 (ATM Credit Card Cash Advance Installments), subfield 1 (Transaction Type)	C	•	C	81 = ATM Installment Withdrawal
DE 112, subelement 027, subfield 2 (Requested Number of Installments)	C	•	C	01–99 = The number of installments requested by the cardholder.
DE 112, subelement 027, subfield 3 (Approved Number of Installments)	C	•	C	01–99 = The number of installment payments approved by the issuer.
DE 112, subelement 027, subfield 4 (Installment Amount)	C	•	C	The cardholder's monthly payment amount.
DE 112, subelement 027, subfield 5 (Total Transaction Amount)	C	•	C	The amount of the cash advance and the interest for the transaction.
DE 112, subelement 027, subfield 6 (Yearly Interest Rate)	C	•	C	The interest rate that will be charged to the cardholder by the issuer.
DE 112, subelement 027, subfield 7 (Currency Code)	C	•	C	The currency code the issuer is charging the cardholder for repayment.
DE 112, subelement 027, subfields 8–11	C	•	C	Member-defined data.

Authorization and Preauthorization Processing Standards

All authorizations initiated by acceptors in the Asia/Pacific, Europe, and Middle East/Africa regions must clearly be distinguished as a final authorization or a preauthorization. Authorizations initiated by acceptors in all other regions must be distinguished as a preauthorization, final authorization, or undefined authorization.

For additional detail on Authorization and Preauthorization Processing Standards refer to the [Transaction Processing Rules](#).

Balance Inquiry: ATM

Issuers that participate in this service must test with the Customer Implementation Services team. Cirrus is the controlling gateway. It forwards balance inquiry requests only to issuers who are certified for receipt. Issuers must then reply with a balance inquiry response.

Authorization Request/0100: ATM Balance Inquiry

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Table 1608: Applicable data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code)	M	•	M	Must be 30xx00
DE 4 (Amount, Transaction)	M	•	M	Must be zero unless an acquirer applied an ATM transaction fee for an ATM transaction in a country where application of an ATM transaction fee is allowed.
DE 18 (Merchant Type)	M	•	M	MCC must be one of the following values: 6010 (Member Financial Institution—Manual Cash Disbursements) 6011 (Member Financial Institution—Automated Cash Disbursements)
DE 28 (Amount, Transaction Fee)	C	•	C	May contain an ATM transaction fee, if applicable for transactions in a country where application of an ATM transaction fee is allowed.
DE 35 (Track 2 Data)	C	•	C	Must omit if DE 45 present
DE 41 (Acceptor Terminal ID)	M	•	M	
DE 43 (Acceptor Name and Location)	M	•	M	
DE 45 (Track 1 Data)	C	•	C	Must omit if DE 35 present
DE 52 (Personal ID Number [PIN] Data)	M	•	M	

Authorization Request/0100: ATM Balance Inquiry Edits

In addition to the standard Dual Message Authorization System edits on Authorization Request/0100 messages, the Dual Message Authorization System will apply the following edits on an Authorization Request/0100: ATM balance inquiry message.

Table 1609: Edits

IF...	THEN the Dual Message Authorization System...
the issuer does not participate in ATM balance inquiry	returns the Authorization Request Response/0110 message containing: DE 39 = 57 (Transaction not permitted to issuer/ cardholder).
DE 18 (Merchant Type) is not 6010 or 6011	returns the Authorization Request Response/0110 message containing: DE 39 = 30 DE 44 = 018
if neither DE 45 (Track 1 Data) nor DE 35 (Track 2 Data) is present in the Authorization Request/0100 message	returns the Authorization Request Response/0110 message containing: DE 39 = 30 DE 44 = 035
DE 54 (Additional Amounts) is present	returns the Authorization Request Response/0110 message containing: DE 39 = 30 DE 44 = 054

Authorization Request Response/0110: ATM Balance Inquiry

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0100 data elements apply.

Table 1610: Applicable data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 28 (Amount, Transaction Fee)	CE	X	CE	Must be the same value as in the original Authorization Request/0100 message, if present.

Data Element	Org	Sys	Dst	Values/Comments
DE 39 (Response Code)	M	•	M	00 = Approved or completed successfully or 85 = Not declined
DE 54 (Additional Amounts), subfield 1 (Account Type)	M	•	M	Must be the same value as DE 3, subfield 2 in the original Authorization Request/0100 message
DE 54, subfield 2 (Amount Type)	M	•	M	02 = Available Balance
DE 54, subfield 3 (Currency Code)	M	•	M	Must be a valid three-digit currency code that matches the issuer's currency code.
DE 54, subfield 4 (Debit or Credit Indicator)	M	•	M	C = Credit amount plus 12 digits or D = Debit amount plus 12 digits

Authorization Request Response/0110: ATM Balance Inquiry Edits

In addition to the standard Dual Message Authorization System edits on Authorization Request Response/0110 messages, the Dual Message Authorization System will apply the following edits on an Authorization Request Response (ATM balance inquiry)/0110 message.

Table 1611: Edits

IF...	THEN...
DE 39 (Response Code) contains the value 00 or 85 and DE 54 (Additional Amounts) is not present	the Dual Message Authorization System will send to the issuer an Authorization Negative Acknowledgment/0190 message containing: DE 39 = 30 DE 44 = 054
DE 54, subfield 3 (Currency Code) is not a valid currency code	the Dual Message Authorization System will generate an Authorization Response Negative Acknowledgment/0190 message containing: DE 39 = 30 DE 44 = 054

IF...	THEN...
DE 54, subfield 4 (Debit or Credit Indicator) is not C plus 12 digits or D plus 12 digits	the Dual Message Authorization System will generate an Authorization Response Negative Acknowledgment/0190 message containing: DE 39 = 30 DE 44 = 054

Authorization Advice/0120: Acquirer-Generated (ATM Balance Inquiry Edits)

The Dual Message Authorization System will perform the following edit on Authorization Advice/0120: acquirer-generated messages for ATM Balance Inquiry transactions.

Table 1612: Edits

IF...	THEN...
the acquirer submits an Authorization Advice/0120: acquirer-generated message containing DE 3 (Processing Code), value 30 (Balance Inquiry)	the Dual Message Authorization System will reject the message and forward the acquirer an Authorization Advice Response/0130 message containing: DE 39 = 30 DE 44 = 003

Alternate Processing

ATM Balance inquiry transactions are not eligible for alternate (Stand-In Authorization or alternate issuer host routing) or X-Code Processing.

If the primary issuer is not available to the ATM balance inquiry request, an Authorization Request Response/0110 is returned to the acquirer with DE 39 (Response Code) value 91 (Authorization System or issuer system inoperative).

Balance Inquiry: Point-of-Sale

When attempting to make a purchase at the point of sale, cardholders uncertain of the remaining balance on a Mastercard prepaid card or a private label card can initiate a balance inquiry at the point of sale to make a more informed decision about how to use the card's funds for the purchase. POS balance inquiry helps the cardholder to completely redeem the funds on the prepaid card, reduces the potential of a declined authorization request when the purchase amount exceeds the funds available on the card, and helps to avoid extended checkout times and lost sales.

The acquirer will receive the cardholder account balance in both the issuer's currency and the transaction currency. The acquirer has the option to provide the merchant with the cardholder account balance in the issuer's currency, transaction currency, or both. The merchant will then display the appropriate account balance on the customer's printed receipt.

Acquirer participation in POS balance inquiry is optional. To request a POS balance inquiry, the acquirer provides the appropriate data elements as defined below.

Acquirers in the U.S. region must support POS balance inquiry transaction type and functionality for all prepaid Mastercard credit and Debit Mastercard card account ranges.

Prepaid card issuers in the U.S. region must support online POS balance inquiries for all prepaid Mastercard credit and Debit Mastercard account ranges.

Issuer participation in POS balance inquiry is optional. To request participation, issuers must submit the *Point-of-Sale (POS) Balance Inquiry Participation Request Form* (Form 771) to Global Customer Service.

Authorization Request/0100: POS Balance Inquiry

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Table 1613: Applicable data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)	M	•	M	30 (Balance Inquiry)
DE 3, subfield 2 (Cardholder "From Account" Type Code)	M	•	M	00 (Default Account) or 30 (Credit Card Account)
DE 3, subfield 3 (Cardholder "To Account" Type Code)	M	•	M	00 (Default Account)
DE 4 (Amount, Transaction)	M	•	M	Must be zero.
DE 18 (Merchant Type)	M	•	M	For POS balance inquiries, the MCC must be a value other than: 6010 = Member Financial Institution: Manual Cash Disbursements or 6011 = Member Financial Institution: Automated Cash Disbursements

Data Element	Org	Sys	Dst	Values/Comments
DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)	M	•	M	Must contain one of the following values: 02 = PAN auto-entry via magnetic stripe 05 = PAN auto-entry via chip 07 = PAN auto-entry via contactless M/Chip 80 = Chip card at chip-capable terminal was unable to process transaction using data on the chip; therefore, the terminal defaulted to the magnetic stripe-read PAN. 90 = PAN auto-entry via magnetic stripe 91 = PAN auto-entry via contactless magnetic stripe
DE 35 (Track 2 Data)	C	•	C	Track information encoded on the magnetic stripe must be presented from either Track 2 or Track 1.
DE 41 (Acceptor Terminal ID)	M	•	M	Must contain terminal ID at the acceptor location.
DE 43 (Acceptor Name and Location)	M	•	M	Must contain name and location of the acceptor as known by the cardholder.
DE 45 (Track 1 Data)	C	•	C	Track information encoded on the magnetic stripe must be presented from either Track 1 or Track 2.

Authorization Request Response/0110: POS Balance Inquiry

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0110 data elements apply.

Issuers participating in POS balance inquiry and responding to an Authorization Request/0100—POS Balance Inquiry message must send an Authorization Request Response/0110 message containing balance information in DE 54 (Additional Amounts) when DE 39 contains the value 00 (Approved or completed successfully) or 85 (Not declined).

Table 1614: Applicable data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 39 (Response Code)	M	•	M	00 = Approved or completed successfully or 85 = Not declined
DE 54 (Additional Amounts), subfield 1 (Account Type)	M	•	M	Must be the same value as DE 3, subfield 2 in the original Authorization Request/ 0100 message 00 = Default Account or 30 = Credit Card Account
DE 54, subfield 2 (Amount Type)	M	•	M	02 = Available Balance
DE 54, subfield 3 (Currency Code)	M	•	M	Must be a valid three-digit currency code that matches the issuer's currency code.
DE 54, subfield 4 (Debit or Credit Indicator)	M	•	M	C = Credit amount plus 12 digits or D = Debit amount plus 12 digits

Authorization Request/0100: POS Balance Inquiry Edits

In addition to the standard Dual Message Authorization System edits on Authorization Request/0100 messages, the Dual Message Authorization System will apply the following edits on an Authorization Request/0100: POS balance inquiry message.

Table 1615: Edits

IF...	THEN the Dual Message Authorization System...
the issuer participates in POS balance inquiry	forwards the POS balance inquiry to the issuer.
the issuer does not participate in POS balance inquiry	returns the Authorization Request Response/0110 message containing: DE 39 = 57 (Transaction not permitted to issuer/ cardholder).

IF...	THEN the Dual Message Authorization System...
DE 4 (Transaction Amount) is not zero	returns the Authorization Request Response/0110 message containing: DE 39 = 30 DE 44 = 004
DE 18 (Merchant Type) is 6010 or 6011	returns the Authorization Request Response/0110 message containing: DE 39 = 30 DE 44 = 018
DE 22 (Point-of-Service (POS) Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) is not 02, 05, 07, 10, 80, 82, 90, or 91	returns the Authorization Request Response/0110 message containing: DE 39 = 30 DE 44 = 022
DE 41 (Acceptor Terminal ID) is not present	returns the Authorization Request Response/0110 message containing: DE 39 = 30 DE 44 = 041
DE 43 (Acceptor Name and Location) is not present	returns the Authorization Request Response/0110 message containing: DE 39 = 30 DE 44 = 043
if neither DE 45 (Track 1 Data) nor DE 35 (Track 2 Data) is present in the Authorization Request/0100: POS Balance Inquiry message except if DE 22, subfield 1 = 10 or 81	returns the Authorization Request Response/0110 message containing: DE 39 = 30 DE 44 = 035
NOTE: This edit is exempt when DE 22, subfield 1 contains values 10 or 81.	
DE 54 (Additional Amounts) is present	returns the Authorization Request Response/0110 message containing: DE 39 = 30 DE 44 = 054

Authorization Request Response/0110: POS Balance Inquiry Edits

In addition to the standard Dual Message Authorization System edits on Authorization Request Response/0110 messages, the Dual Message Authorization System will apply the following edits on an Authorization Request Response/0110: POS balance inquiry message.

Table 1616: Edits

IF...	THEN...
DE 39 (Response Code) contains the value 00 or 85 and DE 54 (Additional Amounts) is not present	the Dual Message Authorization System will send to the issuer an Authorization Negative Acknowledgment/0190 message containing: DE 39 = 30 DE 44 = 054
DE 39 contains a value other than 00 or 85 and DE 54 is present	the Dual Message Authorization System will remove DE 54 from the Authorization Request Response/0110 message before sending the message to the acquirer
DE 54, subfield 2 (Amount Type) is not 02 (Available Balance)	the Dual Message Authorization System will generate an Authorization Response Negative Acknowledgment/0190 message containing: DE 39 = 30 DE 44 = 054
DE 54, subfield 3 (Currency Code) is not a valid currency code	the Dual Message Authorization System will generate an Authorization Response Negative Acknowledgment/0190 message containing: DE 39 = 30 DE 44 = 054
DE 54, subfield 4 (Debit or Credit Indicator) is not C plus 12 digits or D plus 12 digits	the Dual Message Authorization System will generate an Authorization Response Negative Acknowledgment/0190 message containing: DE 39 = 30 DE 44 = 054

Authorization Advice/0120: acquirer-generated (POS Balance Inquiry Edits)

The Dual Message Authorization System will perform the following edits on Authorization Advice/0120: POS balance inquiry acquirer-generated messages.

Table 1617: Edits

IF...	THEN...
The acquirer submits an Authorization Advice/0120: acquirer-generated message containing DE 3 (Processing Code), value 30 (Balance Inquiry)	The Dual Message Authorization System will reject the message and forward the acquirer an Authorization Advice Response/0130 message containing: DE 39 = 30 DE 44 = 003

Alternate Processing

POS Balance inquiry transactions are not eligible for alternate (Stand-In Authorization or alternate issuer host routing) or X-Code Processing.

If the primary issuer is not available to respond to the POS balance inquiry request, an Authorization Request Response/0110 is returned to the acquirer with DE 39 (Response Code) value 91 (Authorization System or issuer system inoperative).

Balance Inquiry: Short Message Service

The Balance Inquiry Short Message Service (SMS) provides cardholders with real-time access to their account balance information via their mobile devices.

Issuers that want to participate in the SMS Balance Service Program must complete a contract and Issuer SMS Balance Inquiry Service information form.

Issuers participating in the SMS Balance Inquiry service will receive these balance inquiries with DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode), value 81 PAN/Token entry via electronic commerce with optional Identity Check-AAV in DE 48, subelement 43 (UCAF) or DSRP cryptogram in DE 104, subelement 001 (Digital Payment Cryptogram).

Authorization Request/0100: Short Message Service Balance Inquiry

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Table 1618: Applicable data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)	M	•	M	30 = Balance Inquiry
DE 18 (Merchant Type)	M	•	M	5969 = Mail Order/Telephone Order Providers
DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)	M	•	M	81 = PAN/Token entry via electronic commerce with optional Identity Check-AAV in DE 48, subelement 43 (UCAF) or DSRP cryptogram in DE 104, subelement 001 (Digital Payment Cryptogram)
DE 48 (Additional Data: Private Use), subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator), position 3 (UCAF Collection Indicator)	M	•	M	Zero (0)

Chip-Specific Value Constraints

The Dual Message Authorization System supports the authorization of chip transactions generated using an integrated circuit card (ICC).

Acquirers may choose to acquire chip-related data in one of the two modes:

- Chip Partial Grade mode
- Chip Full Grade mode

The content of chip-related Authorization Request/0100 messages will vary, depending on the mode used. Issuers of ICCs must be able to accept authorization requests in either mode.

Chip Partial Grade Value Constraints

Following is a list of specific data elements and data element values required in the Authorization Request/0100 message for a chip partial grade transaction.

Table 1619: Data elements and values

DE 22, subfield 1	DE 22, subfield 1 value	DE 35	DE 52 and DE 23
Offline PIN	05	ICC data as EMV tag 57 (Track 2 Equivalent Data)	Not provided
Online PIN	05	ICC data as EMV tag 57 (Track 2 Equivalent Data)	Must be provided
Signature	05	ICC data as EMV tag 57 (Track 2 Equivalent Data)	Not provided
No CVM	05	ICC data as EMV tag 57 (Track 2 Equivalent Data)	Not provided

Table 1620: Data element requirements

Data Element	Requirements
DE 22	For chip transactions, acquirers must provide value 05x. For a chip transaction where the magnetic stripe was used as a fallback technology, acquirers must provide the value 80x. In this case, acquirers must provide the full, unaltered track data. For a chip transaction where manual PAN entry was used as a fallback technology, acquirers may provide the value 79x.
DE 23	The card sequence number may now be provided in chip partial grade transactions where DE 55 is not present. DE 23 must be provided if EMV tag 5F34 (Application PAN Sequence Number) is present on the ICC.
DE 35	The ICC data as EMV tag 57 (Track 2 Equivalent Data). This data corresponds with the data stored in Track 2 of the magnetic stripe. For chip transactions, acquirers must provide DE 35 EMV tag 57 if the data object was present on the ICC. For a chip transaction where the magnetic stripe was used as a fallback technology, DE 35 must contain the actual Track 1 or Track 2 data from the magnetic stripe, or in the case of Maestro the actual Track 2 data because only Track 2 is present on the card.
DE 52	DE 52 must be provided if the Cardholder verification method was Online PIN.

Chip Full Grade Value Constraints

This mode is for acquirers who accept chip transactions and whose infrastructure allows them to provide chip-specific data. Acquirers operating in this mode must comply with all chip partial grade requirements and the chip full grade requirements.

Table 1621: Data element requirements

Data Element	Requirements
DE 23	DE 23 must be provided if EMV tag 5F34 is present on the ICC.
DE 35	The ICC data as EMV tag 57 (Track 2 Equivalent Data). This data corresponds with the data stored in Track 2 of the magnetic stripe. For chip transactions, acquirers must provide DE 35 EMV tag 57 if the data object was present on the ICC. For a chip transaction where the magnetic stripe was used as a fallback technology, DE 35 must contain the actual Track 1 or Track 2 data from the magnetic stripe, or in the case of Maestro the actual Track 2 data because only Track 2 is present on the card.
DE 37	DE 37, subfield 2 contains the value of Transaction Sequence Counter (EMV Tag 9F41), right-justified, left-padded with zeros.
DE 52	DE 52 must be provided if the CVM is Online PIN.
DE 55	DE 55 must be provided as specified in Message layouts . DE 55 must be tag-length-value (TLV)-encoded and must contain the information (mandatory and optional) as specified in Data element definitions .
DE 125	Acquirers must provide DE 125 for PIN change transactions where DE 52 also must be present.

Contact and Contactless Chip Specific Value Constraints

Following is the usage of DE 61, subfield 11 in conjunction with DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode). DE 22, subfield 1 values indicate the method by which the PAN was entered.

Table 1622: Value constraints

DE 61, subfield 11 value		Description
3	=	Merchant terminal PAN auto-entry via bar code reader Value 3 indicates that the terminal supports contactless M/Chip and contactless magnetic stripe transactions. The terminal may also support other card input types, including contact transactions. DE 22, subfield 1 applicable values: • 07 (PAN auto-entry via contactless M/Chip) • 91 (PAN auto-entry via contactless magnetic stripe)
4	=	Merchant terminal PAN auto-entry via optical character reader (OCR) Value 4 indicates that the terminal supports contactless magnetic stripe transactions. The terminal may also support other card input types, including contact transactions. DE 22, subfield 1 applicable value: • 91 (PAN auto-entry via contactless magnetic stripe)
5	=	Merchant terminal PAN auto-entry via integrated circuit card DE 22, subfield 1 applicable values: • 05 (PAN auto-entry via chip) • 79 (Hybrid terminal with an online connection to acquirer failed in sending chip to magnetic stripe fallback or reading the chip card) • 80 (Chip card at chip-capable terminal was unable to process transaction; therefore, terminal defaulted to the magstripe-read PAN) • 90 (PAN auto-entry via magnetic stripe-track data is required)
8	=	EMV specification (compatible chip reader), magnetic stripe reader and key entry. DE 22, subfield 1 applicable values: • 01 (PAN manual entry) • 05 (PAN auto-entry via chip) • 79 (Hybrid terminal with an online connection to acquirer failed in sending chip to magstripe fallback or reading the chip card) • 80 (Chip card at chip-capable terminal was unable to process transaction using data on the chip) • 90 (PAN auto-entry via magnetic stripe-track data is required)

DE 61, subfield 11 value		Description
9	=	PAN/Token entry via electronic commerce containing DSRP cryptogram in DE 55 (Integrated Circuit Card [ICC] System-Related Data). DE 22, subfield 1 applicable value: 05 (PAN auto-entry via chip)

Canada Region Debit Mastercard Merchant Acceptance

The Dual Message Authorization System allows the issuance of the Debit Mastercard® brand by Canada region issuers and allows Canada region merchants to accept domestically issued Debit Mastercard cards.

The Dual Message Authorization System prevents the accidental acceptance of Canada region-issued Debit Mastercard cards and allows issuers to issue Debit Mastercard cards either as stand-alone cards or as cards bearing other acceptance marks.

Acquirers

Acquirers in the Canada region that have established an agreement with their merchants to process Debit Mastercard transactions must support DE 48 (Additional Data—Private Use), subelement 18 (Service Parameters), subfield 01 (Canada Domestic Indicator) and DE 39 (Response Code), value 81 (Domestic Debit Transaction Not Allowed).

For merchants that accept Canada region-issued Debit Mastercard cards, acquirers in Canada must:

- Include DE 48, subelement 18 (Service Parameters) in the Authorization Request/0100 message indicating the merchant has agreed to process Debit Mastercard.
- Support DE 39, response code value 81 (Domestic Debit Transaction Not Allowed) in the Authorization Request Response/0110 message indicating a transaction has been declined.

Issuers

Issuers in the Canada region that choose to issue Debit Mastercard cards must notify Mastercard Customer Implementation Services (CIS) if the account ranges bear other acceptance marks.

Issuers in Canada must:

- Notify Customer Implementation Services (CIS):
 - When the Debit Mastercard BIN being implemented will bear other acceptance marks.
 - The point-of-interaction (POI) environments (face-to-face or non-face-to-face) in which other acceptance marks will exist.

- Ensure Debit Mastercard cards comply with the Canada region chapter in *Mastercard Rules*.
- Immediately notify Mastercard through CIS of any changes in the account ranges affecting compliance with Mastercard Rules.

Dual Message Authorization System Edits

The Dual Message Authorization System will perform the following edits on transactions that are acquired in Canada, on a card issued in Canada when the acceptance brand of that card is Debit Mastercard.

Table 1623: Associated edits

For transactions with a Canada region-issued Debit Mastercard card that bears other acceptance marks with international only Debit Mastercard routing.

WHEN...	THEN the Dual Message Authorization System...
the country code of the merchant in DE 61 (Point-of-Service [POS] Data), subfield 13 (POS Country Code) is the same as the issuer	rejects the transaction where DE 39 (Response Code) = 81 (Domestic Debit Transaction Not Allowed).

Table 1624: Associated edits

For transactions with a Canada region-issued Debit Mastercard card bearing other acceptance marks with only domestic face-to-face functionality for the other acceptance marks on the card.

WHEN...	THEN the Dual Message Authorization System...
• the country code of the merchant in DE 61 (Point-of-Service [POS] Data), subfield 13 (POS Country Code) is the same as the issuer • DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) is value 00 (Purchase), 09 (Purchase with Cash Back), or 28 (Payment Transaction) • DE 61, subfield 4 (POS Cardholder Presence) is value 0 (Cardholder present) or value 1 (Cardholder not present, unspecified)	rejects the transaction where DE 39 (Response Code) = 81 (Domestic Debit Transaction Not Allowed).

Table 1625: Associated edits

For transactions with a Canada region-issued Debit Mastercard card bearing other acceptance marks with domestic non-face-to-face functionality for the other acceptance marks on the card.

WHEN...	THEN the Dual Message Authorization System...
- the country code of the merchant in DE 61 (Point-of-Service [POS] Data), subfield 13 (POS Country Code) is the same as the issuer - DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) is value 00 (Purchase), 09 (Purchase with Cash Back), or 28 (Payment Transaction) - DE 61, subfield 4 (POS Cardholder Presence) is value 2 (Mail/facsimile order), 3 (Phone/ARU order), 4 (Standing order/recurring transactions), or 5 (Electronic order [home PC, Internet, mobile phone, PDA]) - DE 48 (Additional Data: Private Use), subelement 18 (Service Parameters), subfield 1 (Canada Domestic Indicator) is not present or is value other than Y	rejects the transaction where DE 39 (Response Code) = 81 (Domestic Debit Transaction Not Allowed).

Table 1626: Associated edits

For transactions with a Canada region-issued Debit Mastercard card not containing other acceptance marks.

WHEN...	THEN the Dual Message Authorization System...
- the country code of the merchant in DE 61 (Point-of-Service [POS] Data), subfield 13 (POS Country Code) is the same as the issuer - DE 48 (Additional Data: Private Use), subelement 18 (Service Parameters), subfield 1 (Canada Domestic Indicator) is not present or is a value other than Y	rejects the transaction where DE 39 (Response Code) = 81 (Domestic Debit Transaction Not Allowed).

Cardholder Authentication Service

Mastercard offers the capability to indicate the result of biometric authentication to global customers. The Cardholder Authentication Service is an optional service for issuers.

Overview

To help reduce identity fraud, Mastercard has expanded the capability to indicate the result of biometric verification to global issuers participating in the Cardholder Authentication Service.

The authentication indicator (in association with Biometric Card solution) indicates that the cardholder:

- Is the user of the card; and
- Was present at the time of the transaction.

The Cardholder Authentication Service is an optional service that helps participating issuers reduce fraud and increase approvals through biometric verification of the cardholder's identity. With the Cardholder Authentication Service, participating issuers are notified when a transaction is successfully authenticated through the verification of a cardholder's biological trait—such as a thumbprint—with a biometric-enabled card.

Data Requirements

Mastercard will populate the indicator in DE 48 (Additional Data—Private Use), subelement 17 (Authentication Indicator) based on the presence of the biometric authentication result values in DE 55 (Integrated Circuit Card [ICC] System-Related Data), subelement 9F10, CVR byte1, bit1 and byte2, bit2.

The biometric authentication result values available for DE 55, subelement 9F10, CVR byte1, bit1 are as follows:

- 0—Not successful
- 1—Successful

Mastercard will forward the biometric verification result indicators to the issuer in the following messages:

- Authorization Request/0100
- Authorization Advice/0120 (Acquirer-generated and System-generated)

Mastercard will only populate DE 48, subelement 17 with an authentication indicator if the biometric authentication is successful.

Acquirers

Acquirers globally must support receiving DE 48, subelement 17 in the following authorization response messages:

- Authorization Request Response/0110
- Authorization Advice Response/0130—Issuer-generated (Responding to Acquirer-generated 0120)

Issuers

In addition to the standard chip data validation processing for chip transactions, issuers that have registered for or are already participating in the Cardholder Authentication Service must verify DE 55, subelement 9F10, CVR byte1, bit1 and byte2, bit2 to use the final cardholder authentication result in DE 48, subelement 17 in the following messages:

- Authorization Request/0100
- Authorization Advice/0120 (Acquirer-generated and System-generated)

Dual Message Authorization System Edits: Cardholder Authentication Service

The Dual Message Authorization System will perform the following system edits related to the Cardholder Authentication Service.

Table 1627: Associated edits

WHEN...	THEN the Dual Message Authorization System...
an Account Status Inquiry Transaction for account ranges participating in the Cardholder Authentication Service - the Authorization Request/0100 message contains: - DE 2 (Primary Account Number [PAN]) setup for Type 1 Authentication Service and - DE 4 (Amount, Transaction) has a value of zero and - DE 23 (Card Sequence Number) contains a value of 01X and - DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status), value 8 (Account Status Inquiry Service [ASI]) and - DE 55 (Integrated Circuit Card [ICC] System-Related Data) is present and tag 9F10: - CVR byte1, bit1 is present with a value other than 1 (successful) or - CVR byte2, bit2 is present with a value other than 1 (biometric).	sends the acquirer an Authorization Request Response/0110 message where: - DE 39 (Response Code) = 05 (Do not honor) and Sends the issuer an Authorization Advice/0120 message with DE 60 (Advice Reason Code): - Subfield 1 (Advice Reason Code) value 160 (Authentication Advice to Issuer) and - Subfield 2 (Advice Detail Code) value 0078 (M/Chip Biometric Data not present)

WHEN...	THEN the Dual Message Authorization System...
an Account Status Inquiry Transaction for account ranges participating in the Cardholder Authentication Service - the Authorization Advice/0120: acquirer-generated message contains: • DE 2 (Primary Account Number [PAN]) setup for Type 1 Authentication Service and • DE 4 (Amount, Transaction) has a value of zero and • DE 23 (Card Sequence Number) contains a value of 01X (X can be any number from 0 to 9 inclusive) and • DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status), value 8 (Account Status Inquiry Service [ASI]) and • DE 55 (Integrated Circuit Card [ICC] System-Related Data) is present and tag 9F10: – CVR byte1, bit1 is present with a value other than 1 (successful) or – CVR byte2, bit2 is present with a value other than 1 (biometric).	sends the acquirer an Authorization Advice Response/0130 message where: • DE 39 (Response Code) = 30 (Format Error) and • DE 44 (Additional Response Data) = 055 (indicating the data element in error)
an Account Status Inquiry Transaction for account ranges participating in the Cardholder Authentication Service - the Authorization Request/0100 message contains: • DE 2 (Primary Account Number [PAN]) setup for Type 2 Authentication Service and • DE 4 (Amount, Transaction) has a value of zero and • DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status), value 8 (Account Status Inquiry Service [ASI]) and • DE 55 (Integrated Circuit Card [ICC] System-Related Data) is present and tag 9F10: – CVR byte1, bit1 is present with a value other than 1 (successful) or – CVR byte2, bit2 is present with a value other than 1 (biometric).	sends the acquirer an Authorization Request Response/0110 message where: • DE 39 (Response Code) = 05 (Do not honor) and Sends the issuer an Authorization Advice/0120 message with DE 60 (Advice Reason Code): • Subfield 1 (Advice Reason Code) value 160 (Authentication Advice to Issuer) and • Subfield 2 (Advice Detail Code) value 0078 (M/Chip Biometric Data not present)

WHEN...	THEN the Dual Message Authorization System...
an Account Status Inquiry Transaction for account ranges participating in the Cardholder Authentication Service - the Authorization Advice/0120: acquirer-generated message contains: • DE 2 (Primary Account Number [PAN]) setup for Type 2 Authentication Service and • DE 4 (Amount, Transaction) has a value of zero and • DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status), value 8 (Account Status Inquiry Service [ASI]) and • DE 55 (Integrated Circuit Card [ICC] System-Related Data) is present and tag 9F10: – CVR byte1, bit1 is present with a value other than 1 (successful) - or – CVR byte2, bit2 is present with a value other than 1 (biometric).	sends the acquirer an Authorization Advice Response/0130 message where: • DE 39 (Response Code) = 30 (Format Error) and • DE 44 (Additional Response Data) = 055 (indicating the data element in error)

Card Validation Code 2

To comply with card validation code 2 (CVC 2) requirements, acquirers and issuers must process transactions according to the guidelines in this topic.

The following message flows describe authorization message processing for CVC 2 verification transaction in these scenarios:

- When the CVC 2 is verified
- When the CVC 2 is unverified (because the issuer was temporarily unable to receive the CVC 2 value)
- When the CVC 2 is processed by Stand-In
- When the CVC 2 is processed by X-Code

Authorization Request/0100: CVC 2 Verified

Following are the stages of an Authorization Request/0100: CVC2 transaction when the value is verified.

1. The merchant provides the CVC 2 value, and the acquirer generates an Authorization Request/0100 message with the CVC 2 value in DE 48 (Additional Data: Private Use), subelement 92 (CVC 2).

2. The Dual Message Authorization System forwards the Authorization Request/0100 message to the issuer.
3. After the issuer transmits the Authorization Request Response/0110 message, the Dual Message Authorization System verifies that one of the following values appears in DE 48, subelement 87 (Card Validation Code Result):
 - M (Valid CVC 2: match)
 - N (Invalid CVC 2: non-match)
 - P (CVC 2 not processed: issuer temporarily unavailable)

Table 1628: Associated edits

IF DE 48, subelement 87 contains...	AND IF the transaction was...	THEN the Dual Message Authorization System...
no response or an invalid response	not approved	forwards the Authorization Request Response/0110 message to the acquirer with DE 48, subelement 87 = P and the issuer will not receive an Authorization Response Negative Acknowledgment/0190 message
no response or an invalid response	approved	forwards the Authorization Request/0100 message to the alternate authorization service provider for processing and the issuer will receive an Authorization Response Negative Acknowledgment/0190 message

Authorization Request/0100: CVC 2 Unverified

Following are the stages of an Authorization Request/0100: CVC 2 unverified transaction (issuer temporarily unable to receive the CVC 2 value).

1. The merchant provides the CVC 2 value, and the acquirer generates an Authorization Request/0100 message with the CVC 2 value in DE 48, subelement 92.
2. The Dual Message Authorization System forwards the Authorization Request/0100 message to the issuer, but the issuer is temporarily unable to receive the CVC value. The MIP forwards the Authorization Request/0100 message to the issuer without DE 48, subelement 92.

3. After the issuer transmits the Authorization Request Response/0110 message, the Dual Message Authorization System places value U (CVC 2 unverified: Mastercard use only) in DE 48, subelement 87 and forwards it to the acquirer.
4. The acquirer transmits the CVC 2 response code, provided by Mastercard in DE 48, subelement 87 of the Authorization Request Response/0110 message, to the merchant. The acquirer is responsible for ensuring that the merchant receives the CVC 2 response code.

Authorization Request/0100: CVC 2 Processed by Stand-In

Following are the stages of an Authorization Request/0100: CVC 2 transaction processed by Stand-In Authorization.

1. The merchant provides the CVC 2 value, and the acquirer generates an Authorization Request/0100 message with the CVC 2 value in DE 48, subelement 92.
2. Stand-In Authorization responds in the following conditions:
 - The issuer is not signed in.
 - The transaction cannot be delivered to the issuer.
 - The issuer is not responding.
 - The issuer's Authorization Request Response/0110 message fails an edit check and is rejected.
3. Stand-In Authorization generates the Authorization Request Response/0110 message.

Table 1629: Associated edits

WHEN the issuer is...	THEN the Dual Message Authorization System...
temporarily unable to receive the CVC 2 value	places value U (CVC 2 unverified: Mastercard use only) in DE 48, subelement 87 and forwards the Authorization Request Response/0110 message to the acquirer.
temporarily unable to process the CVC 2 value	places value P (CVC 2 not processed, issuer temporarily unavailable) in DE 48, subelement 87 and forwards the Authorization Request Response/0110 message to the acquirer.

4. The acquirer transmits the CVC 2 response code, provided by the issuer or Mastercard in DE 48, subelement 87 of the Authorization Request Response/0110 message, to the merchant. The acquirer is responsible for ensuring that the merchant receives the CVC 2 response code.
5. Stand-In Authorization also generates Authorization Advice/0120 messages as appropriate and stores the advice in the Store-and-Forward (SAF) queue. Stand-In Authorization will include value U or P in DE 48, subelement 87 for issuers that are temporarily unavailable or otherwise unable to process the transaction.

Authorization Request/0100: CVC 2 Processed by X-Code

Following are the stages of an Authorization Request/0100: CVC 2 transaction processed by X-Code Processing.

1. The merchant supplies the CVC 2 value to acquirer.
2. The acquirer creates an Authorization Request/0100 message with the CVC 2 value, but the transmission is not successful.
3. The X-Code Processing by the acquirer's MIP responds based on current X-Code Processing rules.

Table 1630: Associated edits

WHEN the issuer is...	THEN the Dual Message Authorization System...
Temporarily unable to receive the CVC 2 value	Places value U (CVC 2 unverified: Mastercard use only) in DE 48, subelement 87 and forwards the Authorization Request Response/0110 message to the acquirer.
Temporarily unable to process the CVC 2 value	Places a P (CVC 2 not processed, issuer temporarily unavailable) in DE 48, subelement 87 and forwards the Authorization Request Response/0110 message to the acquirer.

4. The acquirer transmits the CVC 2 response code, provided by Mastercard in DE 48, subelement 87 of the Authorization Request Response/0110 message, to the merchant. The acquirer is responsible for ensuring that the merchant receives the CVC 2 response code.
5. The Dual Message Authorization System sends an Authorization Advice/0120 message to the issuer.

CVC 2 DE 48 Structure

The following diagram illustrates DE 48 subelements related to CVC 2 transactions.

Table 1631: CVC 2 DE 48 Structure

LLL	"VAR"						
3 bytes	1 byte	2 bytes	2 bytes	3 bytes	2 bytes	2 bytes	1 byte
Total Data Element Length	TCC	SE ID 92	SE Length 03	CVC 2 Value	SE ID 87	SE Length 01	Card Validation Code Result

Authorization Request/0100: CVC 2

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Table 1632: Applicable data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 14 (Date, Expiration)	C	•	C	Required for issuers to validate CVC 2 value
DE 48 (Additional Data: Private Use), TCC (Transaction Category Code)	M	•	M	Must contain the appropriate TCC code.
DE 48, subelement 87 (Card Validation Code Result)	•	X	C	If an issuer's BIN is temporarily unable to receive the CVC 2 value, the Dual Message Authorization System inserts the following value: U = CVC 2 unverified (Mastercard use only)
DE 48, subelement 92 (CVC 2)	C	•	C	Must be the three-digit CVC 2 value sent by the merchant to the acquirer.

Authorization Request Response/0110: CVC 2

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0110 data elements apply.

Table 1633: Applicable data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 48 (Additional Data: Private Use), TCC (Transaction Category Code)	ME	•	ME	Must be present for registered CVC 2 issuers. Must also be present for unregistered CVC 2 issuers if the issuer is providing additional subelements (such as AVS response.) Must be present for all issuers.

Data Element	Org	Sys	Dst	Values/Comments
DE 48, subelement 87 (Card Validation Code Result), CVC 1 and CVC 2	M	.	M	Contains one of the following CVC 2 response codes: M = Valid CVC 2 (match) N = Invalid CVC 2 (non-match) P = CVC 2 not processed (issuer temporarily unavailable) U = CVC 2 Unverified: Mastercard Use Only
DE 48, subelement 92 (CVC 2)	CE	X	CE	Contains the CVC 2 code provided by the merchant to the acquirer.

Dual Message Authorization System Edits

The Dual Message Authorization System performs the following edit for CVC 2.

Table 1634: Applicable edits

IF DE 48, subelement 87 (Card Validation Code Result) contains...	AND if the transaction was...	THEN the Dual Message Authorization System...
no response or an invalid response	approved	Mastercard forwards the Authorization Request Response/0110 message to the acquirer with DE 48, subelement 87, value P (CVC 2 not processed, issuer temporarily unavailable) the issuer will not receive an Authorization Response Negative Acknowledgment/0190 message

Card Validation Code 3

Mastercard supports card validation code 3 (CVC 3) result data in Authorization Request Response/0110 messages using DE 48, subelement 87 (Card Validation Code Result). Issuers performing CVC 3 validation in-house on contactless transactions may use these CVC 3 values. When the CVC 3 is valid, the Authorization Request Response/0110 message should not contain DE 48, subelement 87.

Authorization Request Response/0110: CVC 3 Result

Mastercard encourages issuers to include DE 48, subelement 87 when they perform CVC 3 validation and there is an issue with the CVC 3. When the CVC 3 is not valid, the issuer should send the acquirer the Authorization Request Response/0110 message containing CVC 3 result data. Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0110 data elements apply.

Table 1635: Applicable data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 48 (Additional Data: Private use), subelement 87 (Card Validation Code Result)	C	.	C	E = Length of unpredictable number was not a valid length P = Unable to process Y = Invalid

Contactless CVC 3 Processing Service

Mastercard provides contactless processing services to assist issuers processing contactless transactions with the validation of the Card Validation Code 3 (CVC 3) to ensure the values provided by the acquirer match the issuer's expected values.

Mastercard offers the following on-behalf services:

- **Dynamic CVC 3 Pre-validation Service:** This is an optional, stand-alone service for issuers with contactless-enabled authorization systems that do not support dynamic CVC 3 validation. The Contactless Mapping Service may be used with the Dynamic CVC 3 Pre-validation Service.

Mastercard will perform the dynamic CVC 3 validation on behalf of the issuer before forwarding the Authorization Request/0100 message. If the issuer is unavailable, Mastercard will consider the results of the dynamic CVC 3 validation when the transaction is processed by the Stand-In Authorization.

- **Dynamic CVC 3 Validation in Stand-In Authorization Service:** This is a mandated service for issuers with contactless-enabled authorization systems that support dynamic CVC 3 validation in-house that would like Mastercard to provide dynamic CVC 3 validation when the issuer is not available and the transaction is processed by the Stand-In Authorization.

The Contactless Mapping Service is for issuers with non-contactless-enabled authorization systems that are not able to support the processing of a separate contactless account number and the validation of dynamic CVC 3 value.

Participation in any CVC 3 on-behalf service or services is defined by account range. Validation will be facilitated by information the issuer provides to Mastercard. Issuers will be requested to

provide confidential key data to Mastercard that will be critical in the validation process (unless Mastercard arranges for devices to be sent to cardholders on behalf of issuers).

Authorization Request/0100: CVC 3

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

The Authorization Request/0100 message will contain CVC 3 values in DE 45 (Track 1 Data) or DE 35 (Track 2 Data) following the configuration provided by the issuer. The Mastercard contactless account number is entered via contactless magnetic stripe where DE 22 (Point of Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) is 91.

Mastercard will apply the appropriate algorithm to DE 45 or DE 35 using the parameter data provided by the issuer to determine the validity of the CVC 3.

Table 1636: Data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 22 (Point of Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)	M	•	M	91 = PAN auto-entry via contactless magnetic stripe—the full track data has been read from the data on the card and transmitted within the authorization request in DE 35 (Track 2 Data) or DE 45 (Track 1 Data) without alteration or truncation. The Mastercard contactless account number is entered via contactless magnetic stripe.
DE 35 (Track 2 Data)	C	•	C	Contains CVC 3 values following the configuration by the issuer.
DE 45 (Track 1 Data)	C	•	C	Contains CVC 3 values following the configuration by the issuer.
DE 48 (Additional Data—Private Use), subelement 71 (On-behalf Services), subfield 1 (On-behalf [OB] Service)	M	•	M	15 = Dynamic CVC 3 Pre-validation (with or without Contactless Mapping Service) 16 = Dynamic CVC 3 Validation in Stand-In Authorization

Data Element	Org	Sys	Dst	Values/Comments
DE 48, subelement 71, subfield 2 (On-behalf [OB] Result 1) when subfield 1 contains value 15 or 16	M	•	M	A = ATC outside allowed range (applicable when ATC value is dynamic [varying value]) E = CVC 3 ATC Replay I = Invalid CVC 3 N = Unpredictable Number Mismatch (applicable when the UN is dynamic [varying value]). (Indicates that the number/length in the discretionary data in DE 45 or DE 35 does not match the number/length provided by the issuer during personalization.) U = Unable to process V = Valid
DE 48, subelement 71, subfield 3 (On-behalf [OB] Result 2)	M	•	M	This subfield is for Mastercard use only.

Dynamic CVC 3 Application Transaction Counter (ATC) Processing

The ATC is used in the calculation of the dynamic CVC 3 cryptogram. The ATC counter value is managed by the chip on a contactless card that increments with every authorization request that occurs at a contactless-enabled terminal.

The Dual Message Authorization System supports DE 48, subelement 71 (On-behalf Service), subfield 2 (On-behalf [OB] Result 1), value E (CVC 3 ATC Replay) in the Authorization Request/0100 message to indicate to the issuer that the ATC value has been used in a previous transaction.

When the result of Dynamic CVC 3 validation in DE 48, subelement 71, subfield 2 is V (Valid) or A (ATC outside allowed range), the Dual Message Authorization System compares the transaction's ATC value to the stored list of ATC values for each contactless account number/sequence number/expiration date combination.

If the ATC used in validating the CVC 3 value is not in the stored list of ATC values, the Dual Message Authorization System adds the transaction's five-digit ATC value to the stored list of ATC values.

If the ATC used in validating the CVC 3 value is in the stored list of ATC values, the Dual Message Authorization System replaces DE 48, subelement 71, subfield 2, value V or A with value E in the Authorization Request/0100 message to the issuer. The Dual Message Authorization System does not add the transaction's ATC again to the stored list of ATC values.

Replacing the DE 48, subelement 71, subfield 2 value V or A, with value E does not affect any data contained in the message that supports value V or A. For example, DE 48, subelement 34

supports value A, but remains unchanged if value E replaces DE 48, subelement 71, subfield 2, value A.

The Dual Message Authorization System populates DE 48, subelement 87 (Card Validation Code Result) value Y (Invalid) in the Authorization Request Response/0110 message to the acquirer.

NOTE: The Dual Message Authorization System only adds to the list of stored ATC values for transactions that have been processed online through the Mastercard Dual Message Authorization System. ATC values for transactions processed offline are not available for the Dual Message Authorization System to use in CVC 3 ATC replay processing.

Dynamic CVC 3 Application Transaction Counter (ATC) Information

DE 48, subelement 34 (Dynamic CVC 3 ATC Information) provides issuers with information about the ATC value derived for use in dynamic CVC 3 validation processing.

DE 48, subelement 34 will be conditionally present in Authorization Request/0100 messages and Authorization Advice/012: system-generated messages when the issuer participates in the Dynamic CVC 3 Pre-validation service or the Dynamic CVC 3 Validation in Stand-In Authorization service.

The Dual Message Authorization System will provide DE 48, subelement 34 when the Dynamic CVC Pre-validation service or the Dynamic CVC 3 Validation in Stand-In Authorization service is performed and validation results in DE 48, subelement 71 (On-behalf Services), subfield 2 (On-behalf [OB] Result 1) contain the value A (ATC outside allowed range), E (CVC 3 ATC Replay), or V (Valid). DE 48, subelement 34 also may be sent as part of the Contactless Mapping service if implemented with Dynamic CVC 3 Pre-validation.

When responding to the Authorization Request/0100 message, issuers should consider the results of the CVC 3 validation in DE 48, subelement 71, subfield 2 and DE 48, subelement 34 (if present).

MCC109 (Application Transaction Counter File)

The MCC109 (Application Transaction Counter [ATC] file) contains a range of ATC values for a contactless account number/card sequence number/expiration date combination for contactless cards that were personalized using dynamic values in the ATC and unpredictable number (UN). Issuers participating in the Dynamic CVC 3 Pre-validation service or the Dynamic CVC 3 Validation in Stand-In Authorization service may provide this file to Mastercard. Issuers may only add to or inquire about this file.

Mastercard supports entry of the ATC data in DE 101 (File Name), value (MCC109) via the following methods:

- Issuer File Update/0302 messages
- Bulk file request (R311)
- Mastercard eService

MCC109 (Application Transaction Counter [ATC] File) maintenance requests submitted via the Issuer File Update Request/0302 message or Mastercard eService are applied immediately.

Maintenance requests submitted by bulk file are applied one time per day at 18:00 (Central Time).

NOTE: This functionality does not apply to contactless cards and devices that were personalized to send zeros in the ATC and UN.

Dual Message Authorization System Edits

The Dual Message Authorization System performs the following edits on Issuer File Update Request/0302 messages when DE 101 (File Name) contains value MCC109 and DE 120 (Record Data) contains the layout for the contactless ATC File.

Table 1637: Edits

Field ID	Field Name	Authorization Edit
1	Contactless Account Number	- Contactless account number must be present - Contactless account number is numeric - Contactless account number prefix is valid - Contactless account number check digit is correct - Contactless account number required in the Application Transaction Counter (ATC) File (MCC109) for additions and inquiries
2	Card Sequence Number	- Card sequence number must be present - Card sequence number is numeric - Value may be zero - Card sequence number is required in the ATC File (MCC109) for additions and inquiries
3	Contactless Account Expiration Date	- Contactless Account Expiration Date is valid YYMM format - Contactless Account Expiration Date is required in the ATC File (MCC109) for additions and inquiries
4	Contactless Application Transaction Counter (ATC) Value	- Contactless ATC is numeric (may be zero) - Contactless ATC is required in the ATC File (MCC109) for a additions and inquiries

NOTE: Fields 1 through 3 are mandatory for an inquiry. Fields 1 through 4 are mandatory for an add.

Card Validation Code Result

The acquirer will receive DE 48, subelement 87 when the CVC 3 validation result was not valid.

Acquirers may receive the following:

- Y = Indicates the ATC was determined to be outside the allowed range specified by the issuer, the ATC was determined to be a replay, or the CVC 3 value was determined to be invalid.

- E = Indicates the length of the unpredictable number was not a valid length resulting in an Unpredictable Number (UN) that was not valid.
- P = Indicates Mastercard was unable to process the CVC 3 validation.

If the CVC 3 data is valid, the Dual Message Authorization System will not include DE 48, subelement 87 in the Authorization Request Response/0110 message to the acquirer.

The issuer will receive the Authorization Advice/0120 message, DE 48, subelement 87 when CVC 3 was invalid in the Authorization Request/0100 message.

Optional Non-Valid CVC 3 Processing

Issuers may elect to have the Dual Message Authorization System respond to Authorization Request/0100 messages on their behalf when the CVC 3 value is not valid. The issuer will receive an Authorization Advice/0120: system-generated message when the Dual Message Authorization System responds.

For information about how to participate, refer to the [Contactless On-behalf Services Guide](#).

The following table describes the process that occurs when the issuer wants the Dual Message Authorization System to respond to Authorization Request/0100 messages on its behalf during CVC 3 validation.

Table 1638: Associated edits

WHEN the CVC 3 validation results contain a value of...	THEN the Dual Message Authorization System...
A, E, I, N, or U	sends an Authorization Request Response/0110 message to the acquirer where: • DE 39 = 05 (Do not honor) • DE 48, subelement 87 = E, P, or Y generates an Authorization Advice/0120 message to the issuer containing: • DE 39, value 05 (Do not honor) • DE 48, subelement 34 (if created) • DE 48, subelement 71 with the appropriate values • DE 48, subelement 87, value E, P, or Y • DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code), value 115 (Transaction processed via MIP On-behalf Service Decision) • DE 60, subfield 2 (Advice Detail Code), one of the following values: – 0042 (CVC 3 Unable to process) – 0043 (CVC 3 ATC outside allowed range) – 0044 (CVC 3 Invalid) – 0045 (CVC 3 Unpredictable number mismatch) – 0046 (CVC 3 ATC Replay)
V	forwards the Authorization Request/0100 to the issuer for processing

Alternate Processing

Mastercard supports issuers subscribing to the Dynamic CVC 3 Pre-validation services that are not available to respond to the Authorization Request/0100 message and issuers subscribing to the Dynamic CVC 3 Validation in Stand-In Authorization service by responding to the authorization message on behalf of the issuer. Mastercard will consider the results of the CVC 3 validation in DE 48, subelement 71 when responding to the Authorization Request/0100.

The values in DE 60 (Advice Reason Code), subfield 2 (Advice Detail Code) are dependent upon the values contained in DE 48, subelement 71, subfield 2 (OB Result 1) as described below.

Table 1639: Associated edits

IF DE 48, subelement 71, subfield 2 contains the value...	THEN DE 60, subfield 2 will contain...
A	0043 (CVC 3 [ATC outside allowed range])
E	0046 (CVC 3 ATC Replay)
I	0044 (CVC 3 [Invalid])
K	0037 (Reject: No matching key file for this PAN, PAN expiry date, and KDI combination)
N	0045 (CVC 3 [Unpredictable number mismatch])
U	0042 (CVC 3 [Unable to process])
X	0038 (Reject: Security Platform Time Out)
Z	0040 (Reject: Security Platform System Error)

MIP X-Code Processing does not perform CVC 3 validation. MIP X-Code Processing will respond based on the type of transaction and the MIP X-Code limits defined.

Contactless Mapping Service for Contactless M/Chip and Contact M/Chip Transactions

The Contactless Mapping Service will be performed for participating account ranges when DE 22, subfield 1 contains value 05 (PAN auto-entry via chip) or value 07 (PAN auto-entry via contactless M/Chip).

Processing

The following process describes how the Dual Message Authorization System processes Authorization Request/0100, Authorization Advice/0120: acquirer-generated, and Reversal Request/0400 messages for Contactless Mapping Service transactions.

Table 1640: Associated edits

WHEN the message contains...	THEN the Dual Message Authorization System...
DE 22, subfield 1, value 05 (PAN auto-entry via chip) and The Contactless Mapping Service was performed on the transaction	changes the value in DE 22, subfield 1 from 05 to 06, indicating that the Contactless Mapping Service occurred.

WHEN the message contains...	THEN the Dual Message Authorization System...
DE 22, subfield 1, value 07 (PAN auto-entry via contactless M/Chip) and The Contactless Mapping Service was performed on the transaction	changes the value in DE 22, subfield 1 from 07 to 08, indicating that the Contactless Mapping Service occurred.

The Contactless Mapping Service occurs on Authorization Request/0100, Authorization Advice/0120: acquirer-generated, and Reversal Request/0400 messages. The following example describes the Contactless Mapping Service being performed on an Authorization Request/0100 message.

Stages describing the process that occurs when an Authorization Request/0100 transaction is generated from a contactless M/Chip or contact M/Chip card eligible for the Contactless Mapping Service.

1. When a cardholder initiates an eligible contactless M/Chip or contact M/Chip transaction on a contactless M/Chip or contact M/Chip terminal, the contactless M/Chip PAN or contact M/Chip PAN is passed to the terminal.
2. The acquirer sends an Authorization Request/0100 message containing the contactless M/Chip PAN or contact M/Chip account number to the Dual Message Authorization System.
3. The Dual Message Authorization System maps the contactless M/Chip or contact M/Chip account number to the cardholder's PAN (or funding account), and then forwards the authorization message containing the following values to the issuer:
 - DE 2 (Primary Account Number [PAN]) containing the cardholder's PAN
 - DE 14 (Date, Expiration) containing the cardholder's expiry date, if provided
 - DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)
 - DE 23 (Card Sequence Number) containing the contactless M/Chip or contact M/Chip card sequence number, if provided
 - DE 35 (Track 2 Data), if provided, is removed
 - DE 45 (Track 1 Data), if provided, is removed
 - DE 48 (Additional Data: Private Use), subelement 33 (Contactless Mapping Information) containing contactless M/Chip or contact M/Chip PAN information
 - DE 48, subelement 71 (On-behalf Services), identifying the Contactless Mapping Service and any other on-behalf services performed
 - DE 55 (Integrated Circuit Card [ICC] System-related Data), as originally sent by the terminal
4. The issuer responds with an Authorization Request Response/0110 message containing the cardholder's PAN in DE 2.
5. The Dual Message Authorization System maps the PAN back to the contactless M/Chip or contact M/Chip account number and places the cardholder's PAN in DE 48, subelement 33, and then sends the contactless M/Chip or contact M/Chip account number in DE 2 to the acquirer.

6. The acquirer forwards the contactless M/Chip or contact M/Chip account number and other authorization response information to the contactless terminal.

The Contactless Mapping Service for contactless M/Chip and contact M/Chip transactions (DE 22, subfield 1, value 05 or 07) requires participation in the Dynamic CVC 3 Pre-validation Service unless the account range comprises M/Chip cards that have not had the magnetic stripe profile programmed.

For more information about the Mastercard Contactless Mapping Service, refer to the [Contactless On-behalf Services Guide](#).

Dual Message Authorization System Edits

The following edit will be performed on Authorization Request/0100, Authorization Advice/0120, and Reversal Request/0400 messages to enforce the current valid expiration date edit that exists for both contactless M/Chip and contact M/Chip transactions that are eligible for the Contactless Mapping Service.

Table 1641: Associated edits

WHEN...	THEN the Dual Message Authorization System...
the expiration date in DE 14, DE 35, or DE 45 (whichever data element is first to contain the expiration date) is expired	rejects the transaction and sends the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 message where DE 39 (Response Code) is 54 (Expired Card).
and	
DE 22, subfield 1 is value 05 or 07	
and	
the account is part of an account range that participates in the Contactless Mapping Service	

Cross-Border Fee Manager Service

The Cross-Border Fee Manager service is an optional service that gives issuers the flexibility to set markup rates for both currency-conversion and cross-border transactions, taking into account a variety of parameters.

Through this service, Mastercard offers two different methods, Currency Conversion Method and Cross Border Flex Method. Issuers may use these methods to add a markup.

Participating issuers may implement any combination of the following methods and add the associated fees to the total cardholder billing amount.

- **Currency Conversion Method:** Allows issuers to charge cardholders a percentage rate, fixed amount, or both by applying a markup to currency-converted transactions that may be further filtered by setup parameters.
- **Cross-Border Flex Method:** Allows issuers to charge cardholders an additional percentage rate, fixed amount, or both by applying a markup to cross-border transactions that may be further filtered by setup parameters.

NOTE: Issuers may only participate in the Currency Conversion Method and the Cross-Border Flex Method where permitted by local regulations.

For more information, refer to the [Cross-Border Fee Manager Service Program Guide](#).

Currency conversion

The Dual Message Authorization System automatically provides a currency conversion service to acquirers and issuers to allow processing of Authorization/01xx and Reversal/04xx messages in the customer's preferred currency.

Acquirers may submit online messages in any valid Mastercard supported transaction currency. Acquirers and issuers will receive amount-related data elements in the acquirer's transaction currency and issuer's cardholder billing currency even if they use the same currency. Acquirers and issuers have the option to receive amount-related data elements in the settlement currency (always U.S. dollars).

Acquirers and issuers that want to change the option to receive settlement amount-related data elements must complete the *Currency Conversion Parameters: Acquirer and Issuer Usage* (Form 391). Please contact a Global Customer Service Representative.

Amount-related data elements in Authorization and Reversal messages

The following table lists how acquirers and issuers will send and receive amount-related data elements in authorization and reversal messages.

Table 1642: Data elements

DE	Acquirer Sends	Issuer Receives	Issuer Returns	Acquirer Receives
4	In acquirer's transaction currency	In acquirer's transaction currency	In acquirer's transaction currency (echo except when responding with partial approval or purchase amount only—no cash back allowed)	In acquirer's transaction currency

Program and service format requirements
Amount-related data elements in Authorization and Reversal messages

DE	Acquirer Sends	Issuer Receives	Issuer Returns	Acquirer Receives
5	N/A	In U.S. dollars if issuer receives settlement amount-related data elements Not present if issuer does not receive settlement amount-related data elements	In U.S. dollars if issuer receives settlement amounts (echo except when responding with partial approval or purchase amount only—no cash back allowed) Not present if issuer does not receive settlement amount-related data elements	In U.S. dollars if acquirer receives settlement amount-related data elements. Not present if acquirer does not receive settlement amount-related data elements
6	N/A	In issuer's cardholder billing currency	In issuer's cardholder billing currency (echo except when responding with partial approval or purchase amount only—no cash back allowed)	In issuer's cardholder billing currency
9	N/A	Rate used to convert DE 4 amount from acquirer's transaction currency to U.S. dollars, if issuer receives settlement amount-related data elements Not present if issuer does not receive settlement amount-related data elements	Rate used to convert DE 4 amount from acquirer's transaction currency to U.S. dollars, if issuer receives amount-related data elements (echo) Not present if issuer does not receive settlement amount-related data elements	Rate used to convert DE 4 amount from the acquirer's transaction currency to U.S. dollars, if acquirer receives settlement amount-related data elements Not present if acquirer does not receive settlement amount-related data elements
10	N/A	Rate used to convert DE 4 amount from acquirer's transaction currency to issuer's cardholder billing currency	Rate used to convert DE 4 amount from acquirer's transaction currency to issuer's cardholder billing currency (echo)	Rate used to convert DE 4 amount from acquirer's transaction currency to issuer's cardholder billing currency
16	N/A	Month and day conversion rate is effective	Month and day conversion rate is effective (echo)	Month and day conversion rate is effective
28	Acquirer's transaction currency	Acquirer's transaction currency	Acquirer's transaction currency (echo)	Acquirer's transaction currency
49	Acquirer's transaction currency code.	Acquirer's transaction currency code	Acquirer's transaction currency code (echo)	Acquirer's transaction currency code

Program and service format requirements
Amount-related data elements in Authorization and Reversal messages

DE	Acquirer Sends	Issuer Receives	Issuer Returns	Acquirer Receives
50	N/A	Settlement currency code (840) if issuer receives settlement amount-related data elements Not present if issuer does not receive settlement amount-related data elements	Settlement currency code (840) if issuer receives settlement amount-related data elements (echo) Not present if issuer does not receive settlement amount-related data elements	Settlement currency code (840) if acquirer receives settlement amount-related data elements Not present if acquirer does not receive settlement amount-related data elements
51	N/A	Issuer's cardholder billing currency code	Issuer's cardholder billing currency code (echo)	Issuer's cardholder billing currency code
54	If applicable to the transaction, one occurrence of each amount type in acquirer's transaction currency	One occurrence of each amount type in acquirer's transaction currency One occurrence of each amount type in issuer's cardholder billing currency	If applicable to the transaction, one occurrence of each amount type in issuer's cardholder billing currency (not an echo of what the acquirer sent)	One occurrence of each amount type in acquirer's transaction currency One occurrence of each amount type in issuer's cardholder billing currency
95	If applicable, DE 95, subfield 1 in acquirer's transaction currency. DE 95, subfields 2–4 contain zeros	DE 95, subfield 1 in acquirer's transaction currency DE 95, subfield 2 in U.S. dollars if issuer receives settlement amount-related data elements DE 95, subfield 2 zero-filled if issuer does not receive settlement amount-related data elements DE 95, subfield 3 in issuer's cardholder billing currency DE 95, subfield 4 zero-filled	Same values as received (echo)	DE 95, subfield 1 in acquirer's transaction currency DE 95, subfield 2 in U.S. dollars if acquirer receives settlement amount-related data elements DE 95, subfield 2 zero-filled if acquirer does not receive settlement amount-related data elements DE 95, subfield 3 in issuer's cardholder billing currency DE 95, subfield 4 zero-filled

Dual Message System Processing

The following Dual Message System processing notes apply to currency conversion processing.

If the acquirer provides DE 5, DE 6, DE 9, DE 10, DE 16, DE 50, or DE 51, the Dual Message Authorization System will overwrite with the appropriate values in the message sent to the issuer. DE 5, DE 6, DE 9, DE 10, DE 16, DE 50, or DE 51 will be present, as defined, except when the Dual Message Authorization System has declined an Authorization Request/0100 message and was unable to complete currency conversion processing.

If the settlement currency (U.S. dollars) is the same as the acquirer's currency, DE 5 will be the same value as in DE 4 and DE 9 will contain the value 61000000.

If the acquirer and issuer are the same currency, DE 6 will be the same value as in DE 4 and DE 10 will contain the value 61000000.

DE 54 will not contain additional occurrences of amount types in the settlement currency. DE 54 will only contain occurrences of amount types in U.S. dollars if it is the currency of the acquirer or issuer.

Data Elements Acquirers Send in Request, Advice, and Reversal Messages

Acquirers are required to send the following currency conversion related data elements in authorization request, advice, and reversal messages.

Table 1643: Data elements

DE	Note	0100	0120	0400
4	Acquirer's transaction currency	P	P	P
5	Always U.S. dollars			
6	Issuer's cardholder billing currency			
9	Factor used in conversion from transaction amount to settlement amount			
10	Factor used in conversion from transaction amount to cardholder billing amount			
16	Always present			
28	Acquirer's transaction currency	C	C	C
49	Acquirer's transaction currency code	P	P	P
50	Always 840			
51	Issuer's cardholder billing currency code			
54	Occurrence of each amount type in acquirer's transaction currency			
54	Occurrence of each amount type in issuer's cardholder billing currency	C	C	C

DE	Note	0100	0120	0400
95, sf 1	Acquirer's transaction currency			P
95, sf 2	Always U.S. dollars if acquirer chooses to receive settlement amounts, zero fill if not selected			Z
95, sf 3	Issuer's cardholder billing currency			Z
95, sf 4	Zero fill			Z

Table Key

C under MTI = indicates that the data element may be present in that message .

X under MTI = indicates that the data element will be present if the acquirer chooses to receive settlement amount-related data elements.

Z under 0400 = indicates that if DE 95, subfield 1 is present, subfields 2–4 are zero-filled.

Data Elements Acquirers Receive in Request, Advice, and Reversal Messages

Acquirers are expected to receive the following currency conversion related data elements in Request, Advice, and Reversal responses.

Table 1644: Data elements

DE	Note	0110	0130	0410
4	Acquirer's transaction currency	P	P	P
5	Always U.S. dollars	X	X	X
6	Issuer's cardholder billing currency	P	P	P
9	Factor used in conversion from transaction amount to settlement amount	X	X	X
10	Factor used in conversion from transaction amount to cardholder billing amount	C	C	C
16	Always present	C	C	C
28	Acquirer's transaction currency	C	C	C
49	Acquirer's transaction currency code	P	P	P
50	Always 840	X	X	X
51	Issuer's cardholder billing currency code	C	C	C
54	Occurrence of each amount type in acquirer's transaction currency	C		

DE	Note	0110	0130	0410
54	Occurrence of each amount type in issuer's cardholder billing currency	C		
95, sf 1	Acquirer's transaction currency			P
95, sf 2	Always U.S. dollars if acquirer chooses to receive settlement amounts, zero fill if not selected			X
95, sf 3	Issuer's cardholder billing currency			P
95, sf 4	Zero fill			P

Table Key

P under MTI = indicates that the data element may be present in that message.

X under MTI = indicates that the data element will be present if the acquirer chooses to receive settlement amount-related data elements.

Data Elements Issuers Receive in Request, Advice, and Reversal messages

Issuers are required to receive the following data elements in authorization and reversal messages.

Table 1645: Data elements

DE	Note	0100	0120	0400	0420
4	Acquirer's transaction currency	P	P	P	P
5	Always U.S. dollars	X	X	X	X
6	Issuer's cardholder billing currency	P	P	P	P
9	Factor used in conversion from transaction amount to settlement amount	X	X	X	X
10	Factor used in conversion from transaction amount to cardholder billing amount	P	P	P	P
16	Always present	P	P	P	P
28	Acquirer's transaction currency	P	P	P	P
49	Acquirer's transaction currency code	P	P	P	P
50	Always 840	X	X	X	X
51	Issuer's cardholder billing currency code	P	P	P	P
54	Occurrence of each amount type in acquirer's transaction currency	P	P	P	P

DE	Note	0100	0120	0400	0420
54	Occurrence of each amount type in issuer's cardholder billing currency	P	P	P	P
95, sf 1	Acquirer's transaction currency			P	P
95, sf 2	Always U.S. dollars if selected, zero fill if not selected			X	X
95, sf 3	Issuer's cardholder billing currency			P	P
95, sf 4	Zero fill			P	P

Table Key

P under MTI = indicates that the data element will be present in that message

X under MTI = indicates that the data element will be present if the issuer chooses to receive settlement amount-related data elements

Data Elements Issuers Send in Request, Advice, and Reversal Responses

Issuers are expected to send the following currency conversion related data elements in Request, Advice, and Reversal responses.

Table 1646: Data elements

DE	Note	0110	0130	0410	0430
4	Acquirer's transaction currency	P	P	P	P
5	Always U.S. dollars	X	X	X	X
6	Issuer's cardholder billing currency	P	P	P	P
9	Factor used in conversion from transaction amount to settlement amount	X	X	X	X
10	Factor used in conversion from transaction amount to cardholder billing amount	P	P	P	P
16	Always present	P	P	P	P
28	Acquirer's transaction currency	C	C	C	P
49	Acquirer's transaction currency code	P	P	P	P
50	Always 840	X	X	X	X
51	Issuer's cardholder billing currency code	P	P	P	P
54	Occurrence of each amount type in acquirer's transaction currency				

DE	Note	0110	0130	0410	0430
54	Occurrence of each amount type in issuer's cardholder billing currency	P			
95, sf 1	Acquirer's transaction currency				P
95, sf 2	Always U.S. dollars if selected, zero fill if not selected			X	X
95, sf 3	Issuer's cardholder billing currency			P	P
95, sf 4	Zero fill			P	P

Table Key

P under MTI = indicates that the data element will be present in that message

X under MTI = indicates that the issuer will return the data element in the response message if the issuer chooses to receive settlement amount-related data elements in the original request or advice message

Alternate Processing

Following are details regarding currency conversion in alternate processing.

Authorization Advice/0120 and Reversal Advice/0420 messages that the issuer receives through Store-and-forward (SAF) processing will always contain amount-related data elements in the acquirer's transaction currency and the issuer's cardholder billing currency.

SAF messages will contain amount-related data elements in the settlement currency (U.S. dollars) according to the issuer's preference for receiving these data elements.

Dual Message Authorization System edits

The Dual Message Authorization System ensures that amount-related data elements, if present in the following response messages from the issuer, are the same values as provided in the request messages sent to the issuer.

Issuers should echo amount-related data elements in response messages as they were received in the request messages. The exception is when an issuer provides DE 39 (Response Code), value 10 (Partial approval) or 87 (Purchase amount only, no cash back allowed) in an Authorization Request Response/0110 message. When providing either of these responses, the issuer is required to provide DE 6 and DE 51 according to the specifications for partial approvals and purchase of goods or services with cash back.

Table 1647: Authorization Request Response/0110

IF...	THEN the Dual Message Authorization System...
the value in DE 39 is not 10 (Partial approval) or 87 (Purchase amount only, no cash back allowed), and the amount-related data elements returned by the issuer are different from the values received by the issuer	generates and forwards to the issuer an Authorization Response Negative Acknowledgment/0190 message where: DE 39 = 30 DE 44 = the data element in error

Table 1648: Reversal Request Response/0410

IF...	THEN the Dual Message Authorization System...
the amount-related data elements returned by the issuer are different from the values received by the issuer	generates and forwards to the issuer an Authorization Response Negative Acknowledgment/0190 where: DE 39 = 30 DE 44 = the data element in error

Table 1649: Authorization Request/0100 or Authorization Advice/0120

IF...	THEN the Dual Message Authorization System...
no occurrence of DE 54, subfield 3 (Currency Code) is equal to the currency code in DE 49	generates and forwards to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where: DE 39 = 30 DE 44 = 054
DE 54 contains more than two occurrences of the DE 54 subfields	generates and forwards to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where: DE 39 = 30 DE 44 = 054

IF...	THEN the Dual Message Authorization System...
DE 54 contains more than one occurrence of the DE 54 subfields for a given amount type (subfield 2)	generates and forwards to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where: DE 39 = 30 DE 44 = 054
The combined length of one occurrence of DE 54 subfields 1–4 is not equal to 20 bytes	generates and forwards to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where: DE 39 = 30 DE 44 = 054
DE 54, subfields 1–3 are not numeric	generates and forwards to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where: DE 39 = 30 DE 44 = 054
The first position in DE 54, subfield 4 is not C (Credit amount or positive balance) or D (Debit amount or negative balance), followed by 12 numeric digits	generates and forwards to the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where: DE 39 = 30 DE 44 = 054

Table 1650: Authorization Request Response/0110

IF...	THEN the Dual Message Authorization System...
no occurrence of DE 54, subfield 3 (Currency Code) is equal to the currency code in DE 51	generates and forwards to the issuer an Authorization Response Negative Acknowledgment/0190 message where: DE 39 = 30 DE 44 = 054

IF...	THEN the Dual Message Authorization System...
the value in DE 39 is not 10 or 87 and DE 54 contains more than two occurrences of the DE 54 subfields	generates and forwards to the issuer an Authorization Response Negative Acknowledgment/0190 message where: DE 39 = 30 DE 44 = 054
the value in DE 39 is 10 or 87 and DE 54 contains more than one occurrence of DE 54, subfield 2 (Amount Type) that is not equal to 57 (Original Amount)	generates and forwards to the issuer an Authorization Response Negative Acknowledgment/0190 message where: DE 39 = 30 DE 44 = 054
DE 54 contains more than one occurrence of DE 54 subfields for a given amount type (subfield 2)	generates and forwards to the issuer an Authorization Response Negative Acknowledgment/0190 message where: DE 39 = 30 DE 44 = 054
the combined length of one occurrence of DE 54 subfields 1–4 is not equal to 20 bytes	generates and forwards to the issuer an Authorization Response Negative Acknowledgment/0190 message where: DE 39 = 30 DE 44 = 054
DE 54, subfields 1–3 are not numeric	generates and forwards to the issuer an Authorization Response Negative Acknowledgment/0190 message where: DE 39 = 30 DE 44 = 054
the first position in DE 54, subfield 4 is not C or D, followed by 12 numeric digits	generates and forwards to the issuer an Authorization Response Negative Acknowledgment/0190 message where: DE 39 = 30 DE 44 = 054

Electronic Commerce Processing

Electronic commerce transactions are non-face-to-face online transactions using electronic media over any public network, such as the Internet, or private network, such as an extranet. E-

commerce processing allows transactions to be initiated from a cardholder-controlled device, such as a PC or mobile phone, for purchasing goods and services on the Internet.

All e-commerce transactions must be uniquely identified by the acquirer in the authorization. This permits the issuer to assess the degree of risk associated with the transaction and to support any processing requirements associated with interchange compliance.

Customer requirements

- All issuers must be able to receive and process all e-commerce data present in the Authorization Request/0100 messages.
- All acquirers must properly identify e-commerce messages within the Authorization Request/0100 message. They must be able to receive and to process e-commerce Authorization Request Response/0110 messages.

The Authorization Request/0100 includes the security level indicator (in DE 48, subelement 42 [Electronic Commerce Indicators] subfield 1 [Electronic Commerce Security Level Indicator and UCAF Collection Indicator]) that indicates the following security attributes of electronic commerce transactions:

- Security protocol, which indicates the security protocol that was used to facilitate the transaction (for example, channel encryption).
- Cardholder authentication, which indicates the method of cardholder authentication used to facilitate the transactions.
- UCAF collection indicator, which indicates both merchant UCAF readiness and inclusion of any specific authentication data in the UCAF field (DE 48, subelement 43).

Mastercard messaging requirements regarding the security of e-commerce transactions may vary depending upon the security protocol involved in the transaction. Methods may include:

- No security protocol
- Channel encryption, which is a security layer that resides between an application and its transport layers, for example, SSL (Secure Sockets Layer) and HTTPS (Hypertext Transfer Protocol Secure)
- Mastercard® Identity Check™ Transaction using the Universal Cardholder Authentication Field (UCAF™)

If an e-commerce item that ships late is re-authorized for message reason code 4808 (Authorization-Related Chargeback) chargeback protection, this transaction will take on the security characteristics of the original authorization within a dispute resolution.

Best Practices for E-Commerce Transactions

Mastercard provides best practices for the management of electronic commerce (e-commerce) transactions.

The following sections provide recommended processing for authorization, authorization reversal, and clearing when dealing with estimated amounts and multi-item purchases where all items may or may not be delivered or are not delivered at the same time. These best practices

are intended to help guide dual message acquirers, issuers, and processors in the usage of these transactions.

Background

E-commerce authorizations are intended to reserve funds for subsequent clearing presentments once online purchases are dispatched (for example, physical items shipped or electronic content delivered). The following information may also be applied to mail order/telephone order (MOTO) (non-Travel and Entertainment [T&E]) transactions.

Guiding Principles

The following is provided as guidance for e-commerce processing. Note that unless otherwise specified, these are best practices and not mandatory:

- An approved e-commerce authorization will have only one first presentment, unless multi-clearing processing is utilized with the proper message reason codes indicating that the issuer should maintain hold of funds for subsequent presentments.

NOTE: As specified in the *Chargeback Guide*, airline ticket and installment purchases are allowed multiple first presentments against one approved authorization.

- Mastercard recommends that merchants submit reversals as soon as an adjustment to the original authorization amount is known. For additional detail about reversals, refer to the separate sections on Authorization and Preauthorization Processing Standards, Incremental Preauthorization Standards, and Reversal Processing.

NOTE: Merchants and acquirers must submit a full or partial reversal (as applicable) within seven calendar days of an original undefined authorization or final authorization request, and within 30 calendar days of an original preauthorization request. Merchants and acquirers must submit a full or partial reversal within 24 hours of transaction cancellation or of the transaction completing for an amount different from the authorized amount. Refer to Authorization and Preauthorization Processing Standards, and Incremental Preauthorization Standards for more information about revised Standards for authorizations and preauthorizations.

- Issuers must release any hold of funds once a clearing presentment has been matched (using the Trace ID in addition to other data elements) to the original e-commerce authorization, unless multi-clearing processing is utilized.
- If an e-commerce item ships late (beyond the authorization expiration date), merchants may submit a chargeback extension request message to avoid chargeback for message reason code 4808 (Authorization-Related Chargeback). If the issuer approves the extension request, the merchant will be protected from chargeback reason 4808 as long as the item ships prior to the new authorization expiration date.
- If an e-commerce item ships late and the merchant has not requested or did not receive approval of a chargeback period extension request and the authorization chargeback protection period had expired, the merchant must submit a new authorization for the item to be shipped to avoid chargeback message reason code 4808 (Authorization-Related Chargeback). The new authorization will take on the security characteristics of the original authorization within a dispute resolution.

NOTE: The best practice for extending the payment guarantee and avoiding a chargeback is to submit an incremental preauthorization to refresh the authorization date. A zero amount (USD 0) can be used in the incremental preauthorization. Without Universal Cardholder Authentication Field (UCAF™) data present, a re-authorized transaction must be presented within clearing for non-UCAF interchange (as applicable by region).

Specific Scenarios for E-Commerce Transactions

Depending on the merchant's inventory system and the nature of the item being purchased online, an e-commerce transaction may be submitted as either a preauthorization or a final authorization by acceptors globally.

For more information about preauthorization and final authorization usage requirements, refer to Authorization and Preauthorization Processing Standards, Incremental Preauthorization Standards, and the *Transaction Processing Rules*.

The following describes the recommended processing when dealing with estimated amounts and multi-item purchases where all items may or may not be delivered or are not delivered at the same time.

IF...	THEN...
The entire e-commerce purchase is canceled (for any reason)	Submit a full reversal.
The initial preauthorization is for an estimated amount based on anticipated sales tax, or estimated shipping weight	Submit partial reversal and present clearing if the effective amount is less than the preauthorization, or submit incremental preauthorization and present clearing if the effective amount is greater. (Merchant could also authorize and clear the difference separately, or just present clearing for the greater amount and risk possible issuer chargeback.)
One item within a multi-item purchase is canceled (customer cancellation or out-of-stock)	Submit a partial reversal and present clearing when the remaining items ship. Refer to the following section Alternate Processing—Canceling a Single Item.
The customer requests split shipment based on inventory availability, or e-commerce aggregator processes order through multiple suppliers	Refer to the following section E-Commerce Split or Partial Shipments.

Alternate Processing—Canceling a Single Item

Issuers will release any hold of funds once a clearing presentment has been matched (using the Trace ID in addition to other data elements) to the original authorization. Accordingly, a merchant is not required to submit a partial reversal if the lower amount is processed by

Mastercard clearing within 24 hours of finalization of the transaction—assuming multi-clearing processing is not utilized.

E-Commerce Split or Partial Shipments

The following are options for managing e-commerce split or partial shipments.

Available Option	Comments
Utilize multi-clearing processing with the proper partial/final presentment message reason code.	This option is the best practice for processing split shipments due to fulfillment delays or multiple suppliers.
Submit partial reversal for unshipped items. Present clearing for shipped items. Separately re-authorize and clear as late items ship.	Merchant risks decline on re-authorization. (Refer to previous note about UCAF security/interchange for re-authorized transactions.)
Do not present clearing on a multi-item purchase until all items ship.	Merchant cash flow may be impacted due to item inventory delays.
Secure consumer permission to bill up front, ahead of the order shipment. Present clearing for entire order upon authorization.	Issuer inquiry or chargebacks could result if merchant billing terms and conditions are not understood.
Submit separate authorizations and clear as each item ships.	Each item is processed as a separate transaction.

No Security Protocol

This section describes an Authorization Request/0100—Electronic Commerce transaction with no security protocol.

1. The cardholder browses the Internet until the cardholder is ready to make a purchase from a merchant. In this example, the cardholder does not have authentication of the merchant at any time.
2. The cardholder's browser sends the purchase and payment information over the Internet to the merchant. In this example, no security protocol protects the request.
3. The merchant receives the purchase information and has access to all of the cardholder account data that the cardholder provided (including payment information).
4. The merchant requests authorization from the acquirer.
5. The acquirer generates an Authorization Request/0100 message, including both of the following:
 - Cardholder payment data
 - Appropriate data element values that identify this as an e-commerce transaction with no security protocol and no cardholder authentication
6. The issuer receives the authorization request and generates an Authorization Request Response/0110 message.
7. The acquirer receives the Authorization Request Response/0110 message and sends the response back to the merchant.

8. The merchant provides acknowledgment to the cardholder (still unprotected as it travels over the Internet).

Channel Encryption

This section describes an Authorization Request/0100: Electronic Commerce transaction that uses channel encryption protocol between the cardholder and the merchant.

1. The cardholder browses the Internet until the cardholder is ready to make a purchase from a merchant.
2. The cardholder's browser, which supports channel encryption (that is, SSL), sends the purchase and payment information to the merchant. Channel encryption protects the information over the Internet.
3. The merchant receives the purchase information and has access to all the cardholder account data that the cardholder provided.
4. The merchant requests authorization from the acquirer.
5. The acquirer generates an Authorization Request/0100 message, including both of the following:
 - The cardholder payment data
 - The appropriate data element values that identify this as an e-commerce transaction with channel encryption protocol and no cardholder authentication.
6. The issuer receives the authorization request and generates an Authorization Request Response/0110 message.
7. The acquirer receives the Authorization Request Response/0110 and sends the response back to the merchant.
8. The merchant provides acknowledgment to the cardholder (protected by channel encryption as it travels over the Internet).

NOTE: An approved e-commerce authorization must only have one first presentment submitted for clearing unless the authorization is for airline tickets or installment purchases. In the case of airline tickets and installment purchases multiple first presentments against one approve authorization is allowed.

Authorization Request/0100: Electronic Commerce Purchase

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Table 1651: Data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 22 (Point-of-Service (POS) Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)	M	.	M	81 = PAN entry via electronic commerce, including chip

Data Element	Org	Sys	Dst	Values/Comments
DE 48 (Additional Data—Private Use), TCC (Transaction Category Code)	M	•	M	Contains one of the following values: P = Payment Transaction T = Phone, Mail, or Electronic Commerce Order U = Unique X = Airline and Other Transportation Services (irrespective of whether the transaction origin is face to face or not)
DE 48, subelement 40 (Electronic Commerce Merchant/Cardholder Certificate Serial Number [Visa Only]), subfield 1 (Merchant/Certificate Serial Number)	C	•	C	DE 48 subelement 40 is not required for any Mastercard e-commerce programs. Merchant certificate serial number in binary format
DE 48, subelement 40, subfield 2 (Cardholder Certificate Serial Number)	C	•	C	Cardholder certificate serial number in binary format
DE 48, subelement 42 (Electronic Commerce Indicators) subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator)	M	•	M	Contains security level in positions 1 and 2 and UCAF collection indicator in position 3. Position 1 = Security Protocol Position 2 = Cardholder Authentication Position 3 = UCAF Collection Indicator
DE 48, subelement 43 (Universal Cardholder Authentication) for Mastercard Identity Check issuer or cardholder generated authentication data	C	•	C	Authentication data generated by Identity Check-compliant solution.
DE 48, subelement 43 (Universal Cardholder Authentication) for static AAV program.	C	•	C	A static Accountholder Authentication Value (AAV) assigned by Mastercard for use with this Program.
DE 48, subelement 43 (Universal Cardholder Authentication) for 3-D Secure Electronic Commerce Verification Service (Visa and American Express)	C	•	C	Position 1 (3-D Secure Electronic Commerce Transaction Indicator) 8 = Indicates Secure Electronic Commerce Transaction Position 2–21 (3-D Secure Electronic Commerce Cardholder Authentication Verification Value [CAVV]) = Cardholder Authentication Verification Value (CAVV)

Data Element	Org	Sys	Dst	Values/Comments
DE 48, subelement 44 (Electronic Commerce Transaction Identifier [XID] [Visa and American Express])	C	•	C	Contains the 3-D Secure Electronic Commerce Transaction Identifier (XID) value in binary format.
DE 61 (Point-of-Service (POS) Data), subfield 3 (POS Terminal Location)	M	•	M	2 = Off premises of acceptor facility (cardholder terminal including home PC, mobile phone, PDA)
DE 61, subfield 4 (POS Cardholder Presence)	M	•	M	When the first transaction in a recurring payment arrangement is an electronic commerce transaction: • 4 = Standing order/recurring transactions For each subsequent transaction in the same recurring payment arrangement use: • 4 = Standing order/recurring transactions For all other e-commerce transactions use: • 5 = Electronic order (home PC, Internet, mobile phone, PDA) NOTE: For Intra-EEA Transactions and to Intracountry Transactions in the EEA, recurring payment arrangements must be identified as specified by the registered switch of the Customer's choice.
DE 61, subfield 7 (POS Transaction Status)	M	•	M	Must not contain value 2 (Identity Check Phone Order). NOTE: An e-commerce transaction may be submitted as either a preauthorization (DE 61, subfield 7 value 4) or final authorization (DE 61, subfield 7, value 0 and DE 48, subelement 61, subfield 5, value 1) by acceptors in the Europe region.
DE 61, subfield 10 (Cardholder-Activated Terminal Level)	M	•	M	6 = Authorized Level 6 CAT: Electronic commerce

Authorization Request Response/0110: Electronic Commerce Purchase

When any response data is present in DE 48 (such as AVS response, CVC 2 response), the issuer uses a normal Authorization Request Response/0110, with all subelements present in the original Authorization Request/0100 message echoed back. Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Table 1652: Data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 48 (Additional Data—Private Use), TCC (Transaction Category Code)	CE	•	CE	Must be the same value as in the original Authorization Request/0100 message.
DE 48, subelement 40 (Electronic Commerce Merchant/Cardholder Certificate Serial Number [Visa Only])	CE	•	CE	Must be the same value as in the original Authorization Request/0100 message, if present.
DE 48, subelement 42 (Electronic Commerce Security Level Indicator)	ME	•	ME	Must be the same value as in the original Authorization Request/0100 message.
DE 48, subelement 43 (Universal Cardholder Authentication Field [UCAF])	CE	•	CE	Must be the same value as in the original Authorization Request/0100 message, if present.
DE 48, subelement 44 Electronic Commerce Transaction Identifier [XID] [Visa and American Express])	CE	•	CE	Must be the same value as in the original Authorization Request/0100 message, if present. (Visa and American Express.)
DE 48, subelement 45 (Three-Domain (3-D) Secure Electronic Commerce Transaction Response Code [Visa and American Express])	C	•	C	Contains the Visa 3-D Secure CAVV Results Code, if present. (Visa and American Express).

Mastercard Identity Check

The Mastercard Identity Check™ program encompasses varying solution-oriented protocols that build upon the infrastructure requirements for channel encryption with the additional benefit of cardholder authentication. When used with components of the Mastercard Dual Message Authorization System, this program provides a mechanism for online merchants to potentially receive an enhanced payment guarantee similar to what retailers receive with qualifying physical point-of-sale transactions.

When participating in the Mastercard Identity Check program, the SLI and UCAF data must be included in the authorization to the issuer when it is available in DE 48 (Additional Data: Private Use), subelement 42 (Electronic Commerce Indicators) and 43 (Universal Cardholder Authentication Field [UCAF]). It is a variable length (maximum 32 characters) field with a flexible data structure that can be tailored to support the needs of issuer security and

authentication approaches. Mastercard AAV UCAF data is always a base64 value with a length of 28 characters.

For more information on the Mastercard Identity Check program, refer to the [Mastercard Identity Check Program Guide](#).

Static Accountholder Authentication Value

Several Mastercard programs enable enrolled merchants that have met specific qualification criteria, to accept Mastercard or Maestro cards for electronic commerce (e-commerce) transactions without using Mastercard Identity Check to authenticate every transaction. Merchants are required to perform full Mastercard Identity Check authentication on the first transaction they perform for any individual cardholder.

These Mastercard programs are as follows:

- Maestro Recurring Payments Program
- Mastercard Utility Payment Program
- Maestro Low Risk Merchant Program

As part of these programs, Mastercard:

- Assigns participating merchants a static Accountholder Authentication Value (AAV) for use with transactions that are processed without Mastercard Identity Check authentication.
- Allocates participating merchants a Mastercard Assigned ID.

Mastercard static AAV UCAF data is identified in Authorization Request/0100 messages by DE 48 (Additional Data: Private Use), subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator), position 3 (UCAF Collection Indicator), value 3 (UCAF data collection is supported by the merchant, and UCAF [Mastercard assigned Static Accountholder Authentication Value] data must be present).

Acquirers must provide the Mastercard assigned merchant ID in DE 48, subelement 32 (Mastercard Assigned ID) of Authorization Request/0100 and Authorization Advice/0120—Acquirer-generated messages.

The combination of static AAV data submitted in the Universal Cardholder Authentication Field (DE 48, subelement 43) and the Mastercard assigned merchant ID (DE 48, subelement 32) is unique and is verified by the Dual Message Authorization System.

NOTE: The Mastercard Identity Check AAV Verification Service and Mastercard Identity Check AAV Verification in Stand-In Authorization will not be performed on Mastercard or Maestro e-commerce transactions that are processed under the Maestro Recurring Payments Program, the Mastercard Utility Payment Program, or the Maestro Low Risk Merchant Program. These on-behalf services will continue to be performed on Mastercard and Maestro e-commerce transactions that are authenticated using Mastercard Identity Check.

For more information

For information on the Identity Check program, refer to [Mastercard Identity Check](#).

Forgotten Card at ATM

Forgotten card at ATM service invokes when a cardholder leaves the terminal without taking his or her card or when the cardholder cannot retrieve the card from the terminal for a technical reason. This service is available only to customers that participate in Swedish Domestic Authorization Switching Service (SASS).

Reversal Request/0400: Forgotten Card

Following is a list of the data elements and values applicable to this message type. All mandatory Reversal Request/0400 data elements apply.

Table 1653: Applicable data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)	M	•	M	Contains one of the following values: 01 = Withdrawal 30 = Balance Inquiry
DE 4 (Amount, Transaction)	M	•	M	Contains value zero
DE 48 (Additional Data—Private Use), subelement 58 (ATM Additional Data)	O	X	C	Subfield data may be present for Swedish ATMs

NOTE: If the issuer does not support DE 48, subelement 58, the Dual Message Authorization System will send the acquirer a Reversal Request Response/0410 message containing DE 39 (Response Code), value 57 (Transaction not permitted to issuer/cardholder).

Mastercard Gaming and Gambling Payment Transactions

This section describes Mastercard Gaming and Gambling Payment Transaction processing in the Europe and Middle East/Africa (MEA) regions and in the United States region.

The Dual Message System (Authorization) supports Gaming and Gambling Payment Transaction processing for the Mastercard and Maestro card brands in Europe and Mastercard card brands in Middle East/Africa region countries where the crediting of gambling winnings is permitted by law.

The Dual Message System (Authorization) also supports Gaming and Gambling Payment Transaction processing for the Mastercard and Maestro brands in the United States Region to enable fast receipt of gaming payments.

Mastercard Gaming and Gambling Payment Transaction Processing in the Europe and Middle East/Africa Regions

The Dual Message System (Authorization) supports Gaming Payment Transactions in Authorization Request/0100 and Reversal Request/0400 messages for the Mastercard® and Maestro® card brands in the Europe and Mastercard card brands in Middle East/Africa (MEA) regions where online gaming and the crediting of gaming winnings on cards is permitted by law.

Acquirers submit their online Gaming and Gambling Payment Transaction Authorization Request/0100 and Reversal Request/0400 messages with the following values:

- DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type), value 28 (Payment Transaction)
- DE 4 (Amount, Transaction)
- DE 48 (Additional Data: Private Use), TCC value P (Payment Transaction)
- DE 48, subelement 77, value C04 (Gaming Re-pay)
- DE 108 (Additional Transaction Reference Data) information must be present to capture the sender and receiver information needed as defined per the *Mastercard Gaming and Gambling Payments Program Standards*

If an acquirer has not completed an implementation project to submit Gaming and Gambling Payment Transactions and sends a Gaming and Gambling Payment Transaction authorization request, the Authorization Platform declines the request with DE 39 (Response Code), value 58 (Transaction not permitted to acquirer or terminal).

Issuers may receive Gaming and Gambling Payment Transactions if they are located in Europe and Middle East/Africa region countries where online gaming and crediting of gaming winnings on cards is permitted by law. For a list of these Europe region countries that processes and receives Gaming and Gambling payment transactions, refer to *Mastercard Gaming and Gambling Payments Program Standards*. Issuers in a country where the Mastercard Gaming and Gambling Payments Program is available are automatically enrolled into the program. If the issuer does not participate in gaming and gambling, the Authorization Platform declines the Gaming and Gambling Payment Transaction request with DE 39 (Response Code), value 57 (Transaction not permitted to issuer/cardholder).

Alternate Processing

Gaming and Gambling Payment Transactions are routed to an alternate issuer host. Gaming and Gambling Payment Transactions are not routed to the Stand-In System or X-Code System.

When there is either no alternate issuer host or no response from an alternate issuer host, the Authorization Platform will decline the transaction with DE 39 (Response Code), value 91 (Authorization System or issuer system inoperative).

For More Information

For additional details about data requirements, refer to the *Mastercard Rules* and *Mastercard Gaming and Gambling Payments Program Standards*.

Authorization Request/0100: Gaming Payment

Following is a list of the data elements and values applicable to this message type for Gaming and Gambling Payment Transactions in the Europe and Middle East/Africa (MEA) regions. All mandatory Authorization Request/0100 data elements apply.

Data Element	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)	M	•	M	28 = Payment Transaction
DE 4 (Amount, Transaction)	M	•	M	
DE 18 (Merchant Type)	M	•	M	Must contain value 7995 = Gambling Transaction
DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)	M	•	M	81 = PAN entry via electronic commerce, including chip
DE 48 (Additional Data—Private Use), TCC	M	•	M	P = Payment Transaction
DE 48, subelement 77 (Transaction Type Identifier)	M	•	M	C04 = Gaming Re-pay

Reversal Request/0400: Gaming Payment

Following is a list of the data elements and values applicable to this message type for Gaming and Gambling Payment Transactions in the Europe and Middle East/Africa (MEA) regions. All mandatory Reversal Request/0400 data elements apply.

Data Element	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)	M	•	M	28 = Payment Transaction
DE 4 (Amount, Transaction)	M	•	M	
DE 18 (Merchant Type)	M	•	M	Must contain value 7995 = Gambling Transaction
DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)	M	•	M	81 = PAN entry via electronic commerce, including chip
DE 48 (Additional Data—Private Use), TCC	M	•	M	P = Payment Transaction
DE 48, subelement 77 (Transaction Type Identifier)	M	•	M	C04 = Gaming Re-pay

Mastercard Gaming and Gambling Payment Transaction Processing in the United States Region

The Authorization Platform supports Gaming Payment Transactions in Authorization Request/0100 and Reversal Request/0400 messages for the Mastercard and Maestro brands in the United States region to help enable faster receipt of gaming payments.

An acquirer in the United States region submits a Gaming Payment Transaction Authorization Request/0100 and Reversal Request/0400 message with the following values:

- DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type), value 28 (Payment Transaction)
- DE 4 (Amount, Transaction)
- DE 18 (Merchant Type)
 - MCC 7800 (Government-owned Lottery [U.S. Region Only])
 - MCC 7801 (Internet Gambling [U.S. Region Only])
 - MCC 7802 (Government Licensed Horse/Dog Racing [U.S. Region Only])
 - MCC 7994 (Video Game Arcades/Establishments)
- DE 48 (Additional Data—Private Use), TCC value P (Payment Transaction)
- DE 48, subelement 77, value C04 (Gaming Re-pay)
- DE 108 (Additional Transaction Reference Data) information must be present to capture the sender and receiver information needed as defined per the *Mastercard Gaming and Gambling Payments Program Standards*

An issuer in the United States region may receive Gaming Payment Transactions identified as described above along with DE 108, subelement 01 (Receiver/Recipient Data), DE 108, subelement 02 (Sender Data), and DE 108, subelement 03 (Transaction Reference Data).

Alternate Processing

The Gaming Payment Transaction must not be routed to the Stand-In System or X-Code System. When a Gaming Payment Transaction identified using MCC 7800 is routed to:

- The Stand-In System, the Authorization Request/0100 message will be declined with DE 39 (Response Code), value 05 (Do Not Honor); or
- The X-Code System, the Authorization Request/0100 message will be declined with DE 39 (Response Code), value 01 (Refer to Card Issuer).

Authorization Platform Edits

The following edits apply to Gaming and Gambling Payment Transactions.

WHEN...	THEN the Authorization Platform...
- DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) equals 28 (Payment) and - DE 48 (Additional Data: Private Use), subelement 77 (Transaction Type Identifier) equals C04 (Gaming Re-pay) and the originating institution identified by the institution provided in DE 32 (Acquiring Institution ID Code) has not registered to process gaming payment transactions	declines the transaction and sends the acquirer an Authorization Request Response/0110 or Reversal Request Response/0410 message containing DE 39 (Response Code), value 58 (Transaction not permitted to acquirer/terminal).
- DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) equals 28 (Payment) and DE 48 (Additional Data: Private Use), subelement 77 (Transaction Type Identifier) equals C04 (Gaming Re-pay) and - The account range is not registered to do Gaming Payment Transactions	declines the transaction and sends the acquirer an Authorization Request Response/0110 or Reversal Request Response/0410 message containing DE 39 (Response Code), value 57 (Transaction not permitted to issuer/cardholder).
- DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) equals 28 (Payment) and DE 48 (Additional Data: Private Use), subelement 77 (Transaction Type Identifier) equals C04 (Gaming Re-pay) and - The account range is not registered to do Gaming Payment Transactions	declines the transaction and sends the acquirer an Authorization Request Response/0110 containing DE 39 (Response Code), value 12 (Invalid Transaction).
- DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) equals 28 (Payment) - DE 48 (Additional Data: Private Use), subelement 77 (Transaction Type Identifier) equals C04 (Gaming Re-pay), and - authorization is sent to X-Code	declines the transaction and sends the acquirer an Authorization Request Response/0110 containing DE 39 (Response Code), value 91 (Unable to route transaction).

Issuer Currency Conversion Rate (ICCR) service

The ICCR is a parameter-based service that charges cardholders for currency conversion by applying a mark-up in clearing.

This supplemental fee is added to the transaction amount when the transaction currency is different from the cardholder billing currency. This fee is included in DE 6 (Amount, Cardholder Billing).

Issuers set the additional fee for the ICCR by account range, which applies to First Presentment messages.

Issuers can receive daily audit report files of the clearing transactions where the ICCR was applied. Bulk File TN70 (On Request Report Data File Transfer Raw Format) supports the daily audit report files.

ICCR applies to POS and ATM transactions acquired worldwide.

In the Dual Message Authorization System, all payment transactions (processing code = 28) will be bypassed for ICCR.

ICCR Service Overview

The Issuer Currency Conversion Rate (ICCR) is a parameter-based system that charges cardholders by applying a mark-up called ICCR in clearing.

Issuers in the Europe region can add this supplemental fee to a transaction amount when the transaction currency is different from the cardholder billing currency. This fee is listed under DE 6 (Amount, Cardholder Billing).

Issuers will be able to set their issuer currency conversion rate by account range, which will apply to First Presentment messages.

Issuers can receive daily audit report files of the clearing transactions where the ICCR was applied. A new bulk file type, Bulk File TN70 (On Request Report Data File Transfer Raw Format), which replaces bulk file type TAV8, supports the daily audit report files.

ICCR applies to POS and ATM transactions acquired worldwide.

In the dual message system, all payment transactions (processing code = 28) will be bypassed for ICCR.

ICCR Enrollment

All issuers in the Europe region that choose to enroll in the ICCR service must submit the correct enrollment form.

The *ICCR Service Enrollment Form*, available on Mastercard Connect™, must be completed and submitted before issuers can participate in the ICCR service.

Forms should be submitted to the issuer's local customer delivery manager.

Incremental Preauthorization

Incremental Preauthorization provides a means to associate incremental, or additional, preauthorizations to a single clearing presentment. It is a common practice for merchant categories to submit incremental preauthorizations and settle those through a single clearing presentment. These standards provide a method for issuers, with certainty, to associate incremental preauthorizations to a single presentment providing them the means to more effectively manage cardholder open-to-buy balances.

Incremental Preauthorization Transaction Processing

Each incremental preauthorization transaction associated with a single cardholder event must reference the original preauthorization. When incremental Authorization Request/0100 messages are sent for the same transaction, acquirers must include DE 63 (Network Data) and DE 15 (Date, Settlement) information from the original Authorization Request Response/0110 message in DE 48 (Additional Data: Private Use), subelement 63 (Trace ID) as follows:

- Positions 1–3 = value from DE 63, subfield 1 (Financial Network Code)
- Positions 4–9 = value from DE 63, subfield 2 (Banknet Reference Number)
- Positions 10–13 = value from DE 15 (Date, Settlement)
- Positions 14–15 = blank filled

The First Presentment/1240 message submitted by the acquirer must include information from the original Authorization Request Response/0110 message in Integrated Product Message DE 63 (Transaction Life Cycle ID), subfield 2 (Trace ID). Upon receipt of the clearing record, the issuer must release holds created by all authorizations properly related to the original preauthorization.

Guiding Principles

This section provides insight into the intent of the Incremental Preauthorization process.

- The primary intent of an incremental preauthorization is to increase the amount of an original preauthorization. The authorized amount is equal to the cumulative amount of all authorizations (plus tolerance if applicable). The allowable tolerances are described in the *Chargeback Guide*.
- The original preauthorization is the reference. All attributes associated with the original are inherited by the incremental preauthorizations. Transactions that would have incremental preauthorizations will likely be submitted as preauthorizations.
- The usage of incremental preauthorizations is optional. Acquirers can, as an alternative, submit clearing records (presentments) for each individual authorization.
- Incremental preauthorizations are recognized for all MCCs.
- Transactions that may have incremental preauthorizations are recommended to be submitted as preauthorizations. Incremental preauthorizations must be coded as preauthorizations.

Specific Scenarios

Table 1654: Typical Usage

The following table depicts the process of a typical incremental preauthorization. The merchant successfully completes an original preauthorization for 100, followed by an incremental preauthorization of 50. This total amount is then processed as a first presentment.

Message Type	DE 63 (Banknet Reference Number)	DE 15 (Settlement Date)	Trace ID	
			(Authorization: DE 48, subelement 63)	(Clearing: DE 63, subfield 2)
DE 4 (Amount, Transaction)				
Original Preauthorization (0100/0110) The Authorization Request/0100 message to issuer, the Authorization Request Response/0110 message to acquirer.	MCC123ABC	1105	N/A	100
Incremental Preauthorization (0100/0110) The Authorization Request/0100 message to issuer, the Authorization Request Response/0110 message to acquirer.	MCC456ABC	1106	MCC123ABC1105	50
First Presentment (1240–200)	N/A	N/A	MCC123ABC1105	150

Incremental Preauthorization Reversals

Acquirers are encouraged to recognize the original preauthorization and all related incremental preauthorizations in aggregate in the event they need to reverse part or the entire transaction amount. The recommended method of submitting reversals is illustrated in the following table.

Table 1655: Incremental Preauthorization Reversals

In the following scenario, the merchant successfully completes an original preauthorization for 100, followed by an incremental preauthorization of 50, but then wants to cancel the transaction.

Message Type	DE 63 (Banknet Reference Number)	DE 15 (Settlement Date)	Trace ID		
			(Authorization: DE 48, subelement 63)	(Clearing: DE 63, subfield 2)	DE 4 (Amount, Transaction)
Original Preauthorization (0100/0110)	MCC123ABC	1105	N/A	100	N/A
The Authorization Request/0100 message to issuer, the Authorization Request Response/0110 message to acquirer.					
Incremental Preauthorization (0100/0110)	MCC456ABC	1106	MCC123ABC1105	50	N/A
The Authorization Request/0100 message to issuer, the Authorization Request Response/0110 message to acquirer.					

Message Type	DE 63 (Banknet Reference Number)	DE 15 (Settlement Date)	Trace ID		
			(Authorization: DE 48, subelement 63)	(Clearing: DE 63, subfield 2)	DE 4 (Amount, Transaction)
Reversal (0400/0410)	MCC789ABC	1106	MCC123ABC1105	150	0
In the Reversal transaction, DE 90 (Original Data Elements) will contain DE 11 (System Trace Audit Number [STAN]), DE 7 (Transmission Date and Time), DE 32 (Acquiring Institution ID Code), and DE 33 (Forwarding Institution ID Code) from the Authorization Request Response/0110 message that corresponds to the original preauthorization request. The Transaction Amount (DE 4) is recommended to contain 150, while the Replacement Amount (DE 95) is recommended to contain 0, indicating a full reversal.					

Table 1656: Incremental Preauthorization Reversals

In the next scenario, the merchant successfully completes an original preauthorization for 100, followed by an incremental preauthorization of 50. The merchant then reduces the preauthorization by 10 using a partial reversal.

Message Type	DE 63 (Banknet Reference Number)	DE 15 (Settlement Date)	Trace ID		
			(Authorization: DE 48, subelement 63)	(Clearing: DE 63, subfield 2)	DE 95 (Replacement Amount)
Original Preauthorization (0100/0110) The Authorization Request/0100 message to issuer, the Authorization Request Response/0110 message to acquirer.	MCC123ABC	1105	N/A	100	N/A
Incremental Preauthorization (0100/0110) The Authorization Request/0100 message to issuer, the Authorization Request Response/0110 message to acquirer.	MCC456ABC	1106	MCC123ABC1105	50	N/A

Message Type	DE 63 (Banknet Reference Number)	DE 15 (Settlement Date)	Trace ID		
			(Authorization: DE 48, subelement 63)	(Clearing: DE 63, subfield 2)	DE 4 (Amount, Transaction)
Reversal (0400/0410)	MCC789ABC	1106	MCC123ABC1105	150	140
In the Reversal transaction, DE 90 (Original Data Elements) will contain DE 11 (System Trace Audit Number [STAN]), DE 7 (Transmission Date and Time), DE 32 (Acquiring Institution ID Code), and DE 33 (Forwarding Institution ID Code) from the Authorization Request Response/0110 message that corresponds to the original preauthorization request. The Transaction Amount (DE 4) is recommended to contain 150, while the Replacement Amount (DE 95) is recommended to contain 0, indicating a full reversal.					

Message Type	DE 63 (Banknet Reference Number)	DE 15 (Settlement Date)	Trace ID		
			(Authorization: DE 48, subelement 63)	(Clearing: DE 63, subfield 2)	DE 4 (Amount, Transaction)
Clearing (1240–200)	N/A	N/A	MCC123ABC1105	140	N/A

Acquirers are expected to send reversals once they become aware that an adjustment is necessary. An exception to this is if the acquirer will be processing the first presentment within 24 hours of knowing a reversal is necessary.

Guaranteed Reservations

Guaranteed reservations are common in the hotel, motel, vehicle rental, and cruise line industries. Guaranteed reservations are managed uniquely for each of these segments. The recommended practice for each of these is as follows:

- Vehicle Rental: An account status inquiry is issued at the time of the reservation to confirm the account is in good standing. No advance authorization is issued nor is any amount held or liability assumed for no-shows.
- Cruise Lines: Typically, cruise lines charge the entire amount of the cruise at the time of the reservation. Any charges incurred during the cruise are submitted as a new authorization with incremental preauthorizations as needed.
- Hotel Reservation: A hotel, motel, resort, or other lodging merchant participating in the Guaranteed Reservations service has the option of utilizing the rules in regard to no-show policies. The cardholder is obligated to cancel a confirmed reservation before 18:00 at the hotel, motel, or resort (merchant's local time). If the cardholder fails to cancel the confirmed reservation, the merchant can charge the cardholder a no-show charge equal to one night's lodging. In this case, it is appropriate for the merchant to authorize and clear the amount that would be charged for one night's lodging if they so choose. Any amount above that would be a violation of Mastercard rules and be at risk for a chargeback.

Authorization Chargeback Protection Period Extension Request

In support of the time limit for chargeback protection related to Dual Message System authorizations, Mastercard provides an Authorization Chargeback Protection Period Extension Request for use by Dual Message System acquirers, which is a non-financial (zero amount) incremental preauthorization Authorization Request/0100 message.

Authorizations originally coded as a preauthorization may require a longer chargeback protection period. To increase the effective duration of the chargeback protection period, the merchant may submit incremental preauthorization requests for either an additional amount or zero amount for the same transaction on later dates. Incremental preauthorizations for an additional amount may be used to increase the authorized amount held against the card

account and to extend the chargeback protection period associated with the original preauthorization. Incremental preauthorizations for a zero amount may be used to extend only the chargeback protection period associated with the original preauthorization.

If the issuer declines the chargeback extension request and the original preauthorization subsequently expires, the acquirer must request a new authorization.

NOTE: Each approved preauthorization has an extended payment guarantee period of 30 calendar days from the authorization approval date. The issuer payment guarantee period is limited to a maximum period counting from the authorization or preauthorization date. This maximum period is 30 calendar days for Mastercard authorizations properly coded as preauthorizations and is seven calendar days for all other Mastercard authorizations and for all Maestro and Cirrus authorizations and preauthorizations.

Issuers must be prepared to receive and process a chargeback protection period extension Authorization Request/0100 message.

To increase only the effective duration of the chargeback protection period of an original preauthorization, the merchant may submit an incremental preauthorization request for a zero amount. The acquirer must submit the Authorization Request/0100 as follows.

- DE 3 (Processing Code), subfield 1 (Transaction Type) = 00 (Purchase)
- DE 4 (Transaction Amount) = Zero (0) (Additional Amount)
- DE 61 (POS Data), subelement 7 (POS Transaction Status) = 4 (Preauthorized request)
- DE 61, subfield 4 (POS Cardholder Presence) is not equal to 4 (Recurring transactions)
- DE 48 (Additional Data), subfield 63 (Trace ID) must be present and contain the Trace ID data (DE 15 and DE 63) from the original preauthorization request in positions 1–15

NOTE: Authorization Chargeback Protection Period Extension Requests are applicable to all types of purchase preauthorization transactions, except Mastercard contactless transit aggregated or transit debt recovery transactions nor to merchant-financed or acquirer-financed installment payment transactions properly coded as preauthorizations. If submitted, the message will be declined.

Authorization Advice/0120: acquirer-generated and Reversal Request/0400 messages are not applicable for Authorization Chargeback Protection Period Extension Requests. If submitted, the message will be declined.

The issuer, at its discretion, will send the acquirer an Authorization Request Response/0110 message where DE 39 may contain one of the following values.

- 00 (Approved)
- 05 (Do not honor)
- 85 (Not declined)
- Other valid business declines
Examples of other acceptable business declines include values 41 (Lost Card), 43 (Stolen Card), and 54 (Expired Card)

If the issuer is unable to reply, the acquirer will receive a response code of 91 (Authorization System or issuer system inoperative).

When the chargeback protection period of a preauthorization is extended because of an incremental preauthorization, it is extended for 30 calendar days from the date of the latest approved incremental preauthorization. Such requests, if approved by the issuer and when properly coded as an incremental preauthorization, will give rise to an extended chargeback protection period and optionally additional approved amounts that may be cleared under the conditions that would apply to the security parameters when applied to the original authorization. In other words:

- To the extent that interchange levels are determined by the security parameters of the authorization, then the parameters of the original authorization will be considered.
- To the extent that chargebacks consider the security parameters of the authorization, then the parameters of the original authorization will be taken into account.

When the chargeback protection period expires, issuers must release any block they may have placed on the cardholder's account in relation to the authorization.

Several other options may be used when the authorization lifecycle expires before the transaction is finalized:

- Submit a new preauthorization within 30 calendar days of the planned date of the stay. This creates the risk of a declined transaction. If the original preauthorization has not expired, Mastercard recommends that a reversal is sent of the first transaction to instruct the issuer to release the funds in advance of receiving the second transaction.
- Periodically use the Account Status Inquiry Service to confirm that the card is in good standing prior to the stay. An authorization for the anticipated amount may be submitted in advance of the stay (but no earlier than 30 calendar days before).
- Use the Advance Resort Deposit process as described in the [Chargeback Guide](#) to authorize and clear the transaction at the time of the reservation.

Long Running Transactions

Occasionally, transactions have an extended life. Examples of these scenarios are vehicle rentals for multiple weeks, extended resort stays, or long cruises. Mastercard expects merchants to authorize the estimated amount at the initiation of the service. Incremental preauthorizations would be requested if the permitted tolerance (if applicable) is exceeded. For more information about transactions with permissible tolerances, refer to [Transaction Processing Rules](#), Merchant Acceptance section.

With the specific intent of extending the authorization lifecycle, an incremental preauthorization is recommended. As noted earlier, the authorization lifecycle is based on the date of the latest incremental preauthorization.

Extending Security Features

Mastercard® Identity Check™ protocol authentication relates only to the transaction (and any related incremental preauthorizations if applicable) for which they were originally provided. An acquirer may retain these features by submitting an incremental preauthorization transaction with an amount greater than zero.

Authorization Platform Edits

The Authorization Platform will perform the following system edits related to Authorization Chargeback Protection Period Extension requests.

WHEN...	THEN the Authorization Platform...
The acquirer sends an Authorization Chargeback Protection Period Extension Authorization Request/0100 message that contains the following: • DE 3 (Processing Code), subfield 1 (Transaction Type), value 00 (Purchase) • DE 4 (Transaction Amount), value 0 • DE 48 (Additional Data), subelement 63 (Trace ID) is present • DE 61, subelement 4 (POS Cardholder Presence) is not equal to value 4 (Standing Order/Recurring) • DE 61 (POS Data), subelement 7 (POS Transaction Status), value 4 (Preauthorized) and the issuer is not available to respond to the request	Rejects the transaction where: • DE 39 (Response Code) = 91 (Authorization System or issuer system inoperative)
The acquirer sends an Authorization Request/0100 message for an Authorization Chargeback Protection Period Extension request: • DE 3, subfield 1, value 00 • DE 4, value 0 • DE 48, subelement 63 is NOT present • DE 61, subelement 4 is not equal to value 4, and • DE 61, subelement 7, value 4	Rejects the transaction where: • DE 39 = 30 (Format Error) • DE 44 (Additional Response Data) = 0480nn (where nn is the subelement number)

WHEN...	THEN the Authorization Platform...
If the acquirer sends an Authorization Request/0100 message for an Authorization Chargeback Protection Period Extension request: - DE 3, subfield 1, value 00 - DE 4, value 0 - DE 48, subelement 63 is present - DE 61, subelement 4 is not equal to value 4 - DE 61, subelement 7, value 4 And if the values for Contactless Transit Aggregated, Transit Debt Recovery Transactions are present: - DE 48, subelement 64 (Transit Program), subfield 1 (Transit Transaction Type Individual) = 03 (Post-Authorized Aggregated), 04 (Authorized Aggregated Split Clearing), or 06 (Post-Authorized Aggregated Maestro)	Rejects the transaction where: - DE 39 = 30 - DE 44 = 061 (Point-of-Service [POS] Data)
If the acquirer sends an Authorization Request/0100 message for an Authorization Chargeback Protection Period Extension request: - DE 3, subfield 1, value 00 - DE 4, value 0 - DE 48, subelement 63 is present - DE 61, subelement 4 is not equal to value 4 - DE 61, subelement 7, value 4 And if the values for Transit Debt Recovery Transactions are present: - DE 48, subelement 64, subfield 1 = 07 (Debt Recovery) Or if the values for an installment payment transaction are present: - DE 48, subelement 95 (Promotion Code)	Rejects the transaction where: - DE 39 = 30 - DE 44 = 004 (Transaction Amount)

WHEN...	THEN the Authorization Platform...
The acquirer sends an Acquirer-Generated Authorization Advice/0120 or Acquirer-Generated Reversal Request/0400 message for an Authorization Chargeback Protection Period Extension request where the message contains the following: • DE 3, subfield 1, value 00 • DE 4, value 0 • DE 48, subelement 63 is present • DE 61, subelement 4 is not equal to value 4 • DE 61, subelement 7, value 4	Rejects the transaction where: • DE 39 = 30 • DE 44 = 004

Maestro Pre-authorized Transactions

The Authorization Platform allows acquirers to request pre-authorization on transactions for which the amount is not yet determined. Once the issuer approves the pre-authorization request and after the transaction has taken place and the final amount is determined, the acquirer must then send an Authorization Advice/0120 message to the issuer within 20 minutes of the authorization response message completion. This service is available only for Maestro Petrol transactions.

Authorization Request/0100: Maestro Preauthorization

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Table 1657: Data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 4 (Amount, Transaction)	M	•	M	Maximum amount determined by the acquirer or merchant.
DE 18 (Merchant Type)	M	•	M	Must contain value 5542 = Fuel Dispenser, Automated
DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status)	M	•	M	Must contain value 4 = Preauthorized request.
DE 61, subfield 10 (Card-Activated Terminal Level)	M	•	M	1 = Authorized Level 1 CAT: Automated dispensing machine with PIN

Authorization Advice/0120: Maestro Preauthorization Completion

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Advice/0120 data elements apply.

Table 1658: Data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 4 (Amount, Transaction)	M	•	M	Must contain the final transaction amount
DE 48 (Additional Data—Private Use), subelement 15 (Dual Message Authorization System Advice Date and Time)	•	X	M	Dual Message Authorization System supplies date and time.
DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code)	M	•	M	191 = Acquirer Processing System (APS) completed Note: Issuers must also be prepared to receive value 190 (Acquirer Processing System (APS) approved)

Matching Request and Advice Messages

To match the original Authorization Request/0100 with the Authorization Advice/0120, the following DEs should be used:

- DE 2 (Primary Account Number [PAN])
- DE 7 (Transmission Date and Time)
- DE 11 (System Trace Audit Number [STAN])
- DE 32 (Acquiring Institution ID Code)
- DE 33 (Forwarding Institution ID Code)

Maestro Recurring Payments Program

The Maestro® Recurring Payments Program enables the Maestro brand to accept recurring payments for electronic commerce (e-commerce) transactions. This program enables Maestro-branded products to provide enhanced value to customers.

Acquirer requirements

- Acquirers in the Europe region that process Maestro e-commerce transactions may participate in the Maestro Recurring Payments Program.
- Participating acquirers must submit the following values in Authorization Request/0100, Authorization Advice/0120: acquirer-generated, and Reversal Request/0400: acquirer-

generated messages to identify Maestro e-commerce recurring payment transactions received from enrolled merchants:

- DE 22 (Point of Service Data Code), subfield 1 (Terminal Data: Card Data Input Capability), value 81 (PAN/Token entry via electronic commerce with optional Identity Check-AAV in DE 48, subelement 43 [UCAF] or DSRP cryptogram in DE 104, subelement 001 [Digital Payment Cryptogram])
- DE 48 (Additional Data: Private Use), subelement 32 (Mastercard Assigned ID)
- DE 48, subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator), position 3 (UCAF Collection Indicator), value 3 (UCAF data collection is supported by the merchant, and UCAF (Mastercard Assigned Static AAV) data must be present)
- DE 48, subelement 43 (Static AAV)
- DE 61 (Point-of-Service [POS] Data), subfield 4 (POS Cardholder Presence), value 4 (Recurring transactions)
- DE 61 (Point-of-Service [POS] Data), subfield 10 (Cardholder-Activated Terminal Level), value 6 (Authorized Level 6 CAT: Electronic commerce)

Issuer requirements

- Maestro issuers must support receipt and processing of recurring payment e-commerce transactions, as identified by the presence of the following data elements in Authorization Request/0100, Authorization Advice/0120: acquirer-generated, Authorization Advice/0120: system-generated, Reversal Request/0400: acquirer-generated, and Reversal Request/0400: system-generated messages:
 - DE 22, subfield 1, value 81
 - DE 48, subelement 32
 - DE 48, subelement 42, subfield 1, position 3, value 3
 - DE 48, subelement 43
 - DE 61, subfield 4, value 4
 - DE 61, subfield 10, value 6
- Maestro issuers also must provide the following data elements in Authorization Request Response/0110, Authorization Advice Response/0130: system-generated, and Reversal Request Response/0410 messages:
 - DE 48, subelement 42
 - DE 48, subelement 43

Authorization Request/0100: Maestro Recurring Payment

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Table 1659: Data elements

Data Element ID and Name	Org	Sys	Dst	Comments
DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)	M	•	M	81 = PAN/Token entry via electronic commerce with optional Identity Check-AAV in DE 48, subelement 43 (UCAF) or DSRP cryptogram in DE 104, subelement 001 (Digital Payment Cryptogram).
DE 48 (Additional Data—Private Use), subelement 32 (Mastercard Assigned ID)	C	•	C	The value assigned by Mastercard
DE 48, subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator), position 3 (UCAF Collection Indicator)	C	•	C	3 = UCAF data collection is supported by the merchant, and UCAF (Mastercard assigned static Accountholder Authentication Value) data must be present. NOTE: DE 48, subelements 32 and 43 are required for Static AAV transactions.
DE 48, subelement 43 (Static AAV)	C	•	C	Contains the contents of positions 1–28. When DE 48, subelement 43 contains a static AAV, DE 48, subelement 32 is mandatory.
DE 61 (Point-of-Service [POS] Data), subfield 4 (POS Cardholder Presence)	M	•	M	4 = Recurring transactions
DE 61, subfield 10 (Cardholder-Activated Terminal Level)	M	•	M	6 = Authorized Level 6 CAT: Electronic Commerce

Dual Message Authorization System Edits

The following edits are performed on Authorization Request/0100 messages for Maestro Recurring Payment Program transactions.

Mastercard uses the following edit to verify participation in the Maestro Recurring Payments Program. Dual Message Authorization System uses this edit to validate the value in DE 48, subelement 32 for Authorization Request/0100 and Authorization Advice/0120: acquirer-generated messages.

NOTE: The same Static AAV value can be used for the Maestro Recurring Payments Program, the Mastercard Utility Payment Program, and the Maestro Low Risk Merchant Program.

Table 1660: Applicable edits

WHEN...	THEN the Dual Message Authorization System...
an Authorization Request/0100 or Authorization Advice/0120: system-generated message with DE 48, subelement 42, subfield 1, position 3, value 3	validates that DE 48, subelement 32 and subelement 43 are present and contain valid values.

Existing Edits

The following existing edits will be applied to the Maestro Recurring Payments Program in the Dual Message Authorization System where the DE 48, subelement 42, subfield 1, position 3, value 3 (UCAF data collection is supported by the merchant, and UCAF [Mastercard assigned Static Accountholder Authentication Value] data must be present) data must be present for Authorization Request/0100 and Authorization Advice/0120: acquirer-generated messages submitted under the Maestro Recurring Payments Program for e-commerce initiated transactions.

Table 1661: Applicable edits

WHEN...	THEN...
DE 48, subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator), position 3 (UCAF Collection Indicator) has the following value: 3 (UCAF data collection supported by merchant, and UCAF [Mastercard Assigned Static AAV Value] data is present)	DE 48, subelement 43 (Static AAV) must contain the static AAV assigned by Mastercard for the Maestro Recurring Payments Program for e-commerce transactions.
DE 48, subelement 43 (Static AAV) contains the static AAV assigned by Mastercard for the Maestro Recurring Payments Program for e-commerce transactions	DE 48, subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator), position 3 (UCAF Collection Indicator) must contain the value 3 (UCAF data collection supported by merchant, and UCAF [Mastercard-assigned Static AAV Value] data is present).

WHEN...	THEN...
DE 48, subelement 42, subfield 1, position 3 (UCAF Collection Indicator) has the value 3 (UCAF data collection supported by merchant, and UCAF [Mastercard Assigned Static AAV Value] data is present)	for recurring payment e-commerce transactions: - DE 61 (Point-of-Service Data) subfield 4 (POS Cardholder Presence) must contain a value of 4 (Recurring transactions) and - DE 61, subfield 10 (Cardholder-Activated Terminal Level) must contain a value of 6 (Authorized Level 6 CAT: Electronic commerce)

Magnetic Stripe Compliance

The Authorization Platform provides for positive identification of card-read transactions and transactions in which software, hardware, or card failures prevent complete and accurate data capture. A card-read transaction is one in which the entire unaltered track 1 or track 2 is read and captured by the POS device, then transmitted without truncation in an Authorization Request/0100. As a function of this process, the Authorization Platform performs online edits of track data on Mastercard transactions. If the system finds deficiencies, the findings are forwarded to the issuer and acquirer.

Acquirers and issuers must process transactions according to the guidelines in this subsection to comply with the Mastercard magnetic stripe program.

Acquirer Requirements

When the entire unaltered magnetic stripe from track 1 or track 2 encoded on the card is present in DE 45 (Track 1 Data) or DE 35 (Track 2 Data), acquirers must provide DE 22 (Point-of-Service [POS] Entry Mode, subfield 1 (POS Terminal PAN Entry Mode), with the value 80, 90, or 91 in the Authorization Request/0100.

Acquirers also must provide the proper customer ID that Mastercard assigned directly to the entity acting on the acquiring institution's behalf in DE 32 (Acquiring Institution ID Code). However, if the Authorization Request/0100 is routed to the Mastercard Network via a customer processing system (CPS) or intermediate network facility (INF), the proper customer ID that Mastercard assigned directly to the processor entity must be provided in DE 33 (Forwarding Institution ID Code).

The Authorization Platform provides track data validation and point of interaction validation. If the track data, DE 61 (Point-of-Service [POS] Data), or TCC in DE 48 (Additional Data—Private Use) contains an edit error, the Authorization Platform will provide DE 48, subelement 89 (Magnetic Stripe Compliance Error Indicator) in the Authorization Request/0100 and Authorization Request Response/0110 messages.

If a valid acquirer customer ID is not provided in DE 32, the Authorization Platform:

- Changes the value in DE 22, subfield 01 from 80, 90, or 91 to 02.
- Provides DE 48, subelement 88, with a value of Y.

When applicable, this information is provided in the Authorization Request/0100 message sent to the issuer and the Authorization Request Response/0110 message sent to the acquirer to indicate this condition.

Issuer Requirements

Issuers must encode the CVC 1 value on both track 1 and track 2. In addition, issuers must indent print the CVC 2 value into the signature panel after the account number.

Issuers can receive DE 22, subfield 1, value 02 if:

- An acquirer does not comply; therefore, Mastercard changed the 80, 90, or 91 to 02.
- An acquirer did not submit a 80, 90, or 91 but rather a 02. Issuers must be able to process a 80, 90, or 91 in DE 22 and optionally process error codes in DE 48.

Issuers may indicate a CVC 1 error by providing a Y in DE 48, subelement 87.

Use the Authorization Request/0100 for transmitting magnetic stripe-read transactions. Refer to the *Security Rules and Procedures* manual for subelement information for DE 45 (Track 1 Data) and DE 35 (Track 2 Data).

Authorization Request/0100: Magnetic Stripe-read

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Data Element	Org	Sys	Dst	Values/Comments
DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)	M	•	M	Must be one of the following values: 80 = Chip card at chip-capable terminal was unable to process transaction using data on the chip 90 = PAN auto-entry via magnetic stripe 91 = PAN auto-entry via contactless magnetic stripe
DE 48 (Additional Data—Private Use), TCC (Transaction Category Code)	M	•	M	Contains a valid transaction category code.
DE 48, subelement 88 (Magnetic Stripe Compliance Status Indicator)	•	X	C	Y = Authorization Platform replaced DE 22 value 90 or 91 with value 02.
DE 48, subelement 89 (Magnetic Stripe Compliance Error Indicator)	•	X	C	Authorization Platform provides this subelement, when applicable. Indicates Track data, POS data, or TCC errors.

Authorization Request Response/0110: Magnetic Stripe-read

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0110 data elements apply.

Data Element	Org	Sys	Dst	Values/Comments
DE 48 (Additional Data—Private Use), TCC (Transaction Category Code)	CE	•	CE	Must contain the same value as in the Authorization Request/0100.
DE 48, subelement 87 (Card Validation Code Result)	C	•	C	Issuer provides this subelement, when applicable.
DE 48, subelement 88 (Magnetic Stripe Compliance Status Indicator)	CE	•	CE	Must contain the same value as in the Authorization Request/0100.
DE 48, subelement 89 (Magnetic Stripe Compliance Error Indicator)	CE	•	CE	Must contain the same value as in the Authorization Request/0100.

Mastercard Cardless ATM Program

The Mastercard® Cardless ATM Program provides cardholders with the ability to initiate cash withdrawals using an issuer's mobile banking application (app) and to complete the transaction at an enabled ATM without the use of their physical card.

How It Works

Following is a high-level overview of the Mastercard Cardless ATM transaction process:

1. Cardholders can view the fee of an ATM cash withdrawal in advance of conducting the transaction from an enabled ATM using their mobile banking app.
2. Cardholders initiate the withdrawal using their mobile banking app after their arrival at the ATM or any time in advance of the withdrawal.
3. At the ATM, cardholders scan a Quick Response (QR) code or manually enter a code displayed on the ATM.
4. After viewing any transaction fees in their mobile banking app, cardholders authenticate the withdrawal using their mobile device.
5. Mastercard passes the token and details of the withdrawal to the ATM operator about the authorization request.
6. If the transaction is approved by the issuer, the ATM dispenses cash to cardholders.

The Mastercard Cardless ATM program supports cardless cash withdrawals initiated on the Mastercard Network or those received from alternate networks via the Mastercard Shared Service Platform (MSSP) or Alternative Network Routing Solution (ANRS) protocols.

Participation in the Cardless ATM program is on an opt-in basis.

NOTE: Mastercard will not enable Visa tokenized accounts carrying a Mastercard ATM brand with this enhancement.

For More Information

For more information on the data specifications of the Mastercard Cardless ATM program, refer to the following sections of this manual:

- DE 48 (Additional Data—Private Use), [Subelement 28 \(Cardless ATM Order ID\)](#)
- DE 48, [Subelement 29 \(Additional POS Terminal Locations\)](#)
- DE 61 (Point-of-Service [POS] Data), [Subfield 3 \(POS Terminal Location\)](#), value 8 (Additional POS Terminal Locations)
- [Dual Message Authorization System Edits](#)

To Participate

Issuers and acquirers must register with Mastercard to participate in the Mastercard Cardless ATM Program. Issuers and acquirers interested in registering for this program should contact their Mastercard Account Representative.

Dual Message Authorization System Edits

The Dual Message Authorization System will perform the following system edits.

Table 1662: Associated edits

WHEN...	THEN the Dual Message Authorization System...
an Authorization Request/0100 message for a Cardless ATM transaction contains DE 18 (Merchant Type), value 6011 (Member Financial Institution—Automated Cash Disbursements) and DE 48 (Additional Data), Transaction Category Code (TCC), value Z (ATM) and DE 22, subfield 2 (POS Terminal PIN Entry Mode) value 1 (Terminal has PIN entry capability) is present, but DE 22, subfield 1 (POS Terminal PAN Entry Mode) contains a value other than 09 (PAN/Token entry via electronic commerce containing DSRP cryptogram in DE 55)	rejects the message and sends the acquirer an Authorization Request Response/0110 message where: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 022 (indicating the data element in error)

WHEN...	THEN the Dual Message Authorization System...
an Authorization Request/0100 message for a Cardless ATM transaction contains DE 18 (Merchant Type), value 6011 (Member Financial Institution, Automated Cash Disbursements) and DE 48 (Additional Data), Transaction Category Code (TCC), value Z (ATM) and DE 22, subfield 1 (POS Terminal PAN Entry Mode) is present and contains a value other than 09 (PAN/Token entry via electronic commerce containing DSRP cryptogram in DE 55), but DE 52 (Personal ID Number [PIN] Data) is not present	rejects the message and sends the acquirer an Authorization Request Response/0110 message where: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 052 (indicating the data element in error)
an Authorization Request/0100 message for a Cardless ATM transaction does not contain DE 22 (Point-of-Service [POS] Entry Mode), subfield 2 (POS Terminal PIN Entry Mode), value 1 (Terminal has PIN entry capability) or value 8 (Terminal has PIN entry capability but PIN pad is not currently operative)	rejects the message and sends the acquirer an Authorization Request Response/0110 message where: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 022 (indicating the data element in error)
an Authorization Request/0100 message for a Cardless ATM transaction does not contain DE 48 (Additional Data), subelement 29 (Additional POS Terminal Locations), value A (Transaction has been initiated remotely and completed at a physical terminal on premises of the acceptor facility) or value B (Transaction has been initiated remotely and completed at a physical terminal off premises of the acceptor facility)	rejects the message and sends the acquirer an Authorization Request Response/0110 message where: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 048029 (indicating the data element and subelement in error)
an Authorization Request/0100 message for a Cardless ATM transaction does not contain DE 55 (Integrated Circuit Card [ICC] System-Related Data)	rejects the message and sends the acquirer an Authorization Request Response/0110 message where: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 055 (indicating the data element in error)

WHEN...	THEN the Dual Message Authorization System...
an Authorization Request/0100 message for a Cardless ATM transaction does not contain the following: • DE 61 (Point-of-Service [POS] Data) – Subfield 1 (POS Terminal Attendance), value 1 (Unattended terminal [cardholder-activated terminal (CAT), home PC, mobile phone, PDA]) – Subfield 2 (Reserved for Future Use), value 0 (zero-filled) – Subfield 3 (POS Terminal Location), value 8 (Additional POS Terminal Location) – Subfield 4 (POS Cardholder Presence), value 5 (Electronic order [home PC, Internet, mobile phone, PDA]) – Subfield 5 (POS Card Presence), value 1 (Card not present) – Subfield 6 (POS Card Capture Capabilities), value 0 (Terminal/operator does not have card capture capability) – Subfield 7 (POS Transaction Status), value 0 (Normal request [original presentment]) – Subfield 8 (POS Transaction Security), value 0 (No security concern) – Subfield 9 (Reserved for Future Use), value 0 (zero-filled) – Subfield 10 (Cardholder-Activated Terminal Level), value 6 (Authorized Level 6 CAT: Electronic commerce)	rejects the message and sends the acquirer an Authorization Request Response/0110 message where: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 061 (indicating the data element in error)
an Authorization Request/0100 message for a Cardless ATM transaction does not contain DE 48 (Additional Data), subelement 29 (Additional POS Terminal Locations) and DE 61 (Point-of-Service [POS] Data), subfield 3 (POS Terminal Location) contains value 8 (Additional POS Terminal Locations)	rejects the message and sends the acquirer an Authorization Request Response/0110 message where: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 048029 (indicating the data element and subelement in error)

WHEN...	THEN the Dual Message Authorization System...
an Authorization Request/0100 message for a Cardless ATM transaction contains DE 48 (Additional Data), subelement 29 (Additional POS Terminal Locations) and DE 61 (Point-of-Service [POS] Data), subfield 3 (POS Terminal Location) contains a value other than 8 (Additional POS Terminal Locations)	rejects the message and sends the acquirer an Authorization Request Response/0110 message where: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 048029 (indicating the data element and subelement in error)
an Authorization Request/0100 message for a Cardless ATM transaction contains DE 48 (Additional Data), subelement 29 (Additional POS Terminal Locations) with values other than A (Transaction has been initiated remotely and completed at a physical terminal on premises of the acceptor facility) or B (Transaction has been initiated remotely and completed at a physical terminal off premises of the acceptor facility)	rejects the message and sends the acquirer an Authorization Request Response/0110 message where: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 048029 (indicating the data element and subelement in error)
an Authorization Request/0100 message for a Cardless ATM transaction contains DE 48 (Additional Data), subelement 29 (Additional POS Terminal Locations) and the length is not 01	rejects the message and sends the acquirer an Authorization Request Response/0110 message where: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 048029 (indicating the data element and subelement in error)
an Authorization Request/0100 message for a Cardless ATM transaction contains DE 48 (Additional Data), subelement 28 (Cardless ATM Order ID) where the length is not 10 characters and there are all spaces or all zeros	rejects the message and sends the acquirer an Authorization Request Response/0110 message where: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 048028 (indicating the data element and subelement in error)
an Authorization Request/0100 message for a Cardless ATM transaction does not contain DE 48 (Additional Data), subelement 28 (Cardless ATM Order ID)	rejects the message and sends the acquirer an Authorization Request Response/0110 message where: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 048028 (indicating the data element and subelement in error)

WHEN...	THEN the Dual Message Authorization System...
a Cardless ATM transaction is submitted and is routed to X-Code for processing	rejects the message and sends the acquirer an Authorization Request Response/0110 message where: DE 39 (Response Code) = 05 (Do not honor)

Mastercard Commercial Payments Account

This section describes Dual Message Authorization System edits supporting the introduction of the MAP (Mastercard Commercial Payments Account) product code in Brazil and the Mastercard European Economic Area subregion.

Dual Message Authorization System Edit to Support MAP in Brazil

Mastercard offers a domestic card-not-present (CNP) commercial payment solution in Brazil using the product code MAP.

The Dual Message Authorization System performs the following system edit to ensure product code MAP is used only in CNP transactions in Brazil.

Table 1663: Edits

When...	Then the Dual Message Authorization System...
PAN in DE 2 contains the issuer account range associated with brand product MAP and the merchant country code is NOT EQUAL to Brazil (BRA) and DE 61, subfield 5 is NOT EQUAL to value 1	rejects Authorization Request/0100 and Authorization Advice/0120: acquirer-generated messages with: DE 39 (Response Code) = 12 (Invalid Transactions) is returned in Authorization Request Response/0110 and Authorization Advice Response/0130 messages.

Dual Message Authorization System Edits to Support MAP in Mastercard European Economic Area Subregion

The Dual Message Authorization System will perform the following system edits.

Table 1664: Edits

WHEN...	THEN the Dual Message Authorization System...
the product code is MAP and merchant country code is not in the below country list: • Austria (040) • Belgium (056) • Bulgaria (100) • Croatia (191) • Czech Republic (203) • France (250) • Germany (280) • Greece (300) • Iceland (352) • Ireland (372) • Italy (380) • Liechtenstein (438) • Luxembourg (442) • Malta (470) • Netherlands (528) • Norway (578) • Portugal (620) • Slovakia (703) • Slovenia (705) • Sweden (752) • United Kingdom (826)	will reject Authorization Request/0100 and Authorization Advice/0120: acquirer-generated messages with: • DE 39 (Response Code) = 12 (Invalid Transaction) returned in Authorization Request Response/0110 and Authorization Advice Response/0130 messages.

Table 1665: Edits

WHEN...	THEN the Dual Message Authorization System...
PAN in DE 2 contains issuer account range associated with brand product MAP and Card Present Indicator DE 61 (Point-of-Service [POS] Data), subfield 5 (POS Card Presence) is not equal to value 1 (Card not present), and merchant country code is not in the below country list: • Austria (040) • Belgium (056) • Bulgaria (100) • Croatia (191) • Czech Republic (203) • France (250) • Germany (280) • Greece (300) • Iceland (352) • Ireland (372) • Italy (380) • Liechtenstein (438) • Luxembourg (442) • Malta (470) • Netherlands (528) • Norway (578) • Portugal (620) • Slovakia (703) • Slovenia (705) • Sweden (752) • United Kingdom (826)	will reject Authorization Request/0100 and Authorization Advice/0120: acquirer-generated messages with: • DE 39 (Response Code) = 12 (Invalid Transaction) returned in Authorization Request Response/0110 and Authorization Advice Response/0130 messages. NOTE: This edit is also applicable with Region C to Region C messages.

MDES Register for Issuer Pre-digitization Network Message Values

The following Dual Message Authorization System message layouts support real-time messages for tokenization.

MDES Pre-Digitization API documentation that describes the API messages and parameters is available on the Mastercard Developers site.

Tokenization Eligibility Request (TER)

Following is a list of the data elements, values, and subfields applicable for Tokenization Eligibility Request (TER).

- [Authorization Request/0100 for TER](#)
- [DE 124 Subfields in Authorization Request/0100 for TER](#)
- [Authorization Request Response/0110 for TER](#)
- [DE 124 Subfields for Authorization Request Response/0110 for TER](#)

Authorization Request/0100 for TER

All mandatory Authorization Request/0100 data elements apply.

Table 1666: Authorization Request/0100 for TER

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 2 (Primary Account Number)	•	X	M	Cardholder's primary account number
DE 3 (Processing Code)	•	X	M	00 = Purchase
DE 4 (Amount, Transaction)	•	X	M	Will be zero
DE 14 (Date, Expiration)	•	X	C	Cardholder's primary account expiration date
DE 22 (POS Entry Mode)	•	X	M	Subfield 1 (POS Terminal PAN Entry Mode) = 01 (PAN manual entry) Subfield 2 (POS Terminal PIN Entry Mode) = 0 (Unspecified or unknown)
DE 35 (Track 2 Data)	•	X	C	The Authorization Platform creates and provides Track 2 data for Maestro tokenization messages.
DE 43 (Card Acceptor Name/ Location for All Transactions), subfield 5 (Card Acceptor State or Country Code)	•	X	M	The acquirer in these messages is the Consumer Facing Entity Name (CFEN), with a three alphanumeric Country Code corresponding to "USA". For Local-Use-Only account ranges, Mastercard overrides this value to provide the three alphanumeric Country Code of the issuer.
DE 48 Transaction Category Code	•	X	C	T (Phone, Mail, or Electronic Commerce Order)
DE 48 (Additional Data—Private Use), subelement 23 (Payment Initiation Channel)	•	X	C	Value indicating the type of device for which the consumer is requesting tokenization of a primary account number.

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 48 (Additional Data—Private Use), subelement 26 (Wallet Program Data), subfield 1 (Wallet Identifier)	•	X	C	Contains the identifier associated with the Wallet Provider
DE 48 (Additional Data—Private Use), subelement 33 (PAN Mapping File Information), subfield 1 (Account Number Indicator)	•	X	C	Contains the type of PAN mapping account. • C = When MDES Secure Element Device Token • H = When MDES Cloud-based Payments Device Token • F = When MDES Static Token
DE 48 (Additional Data—Private Use), subelement 33 (PAN Mapping File Information), subfield 2 (Account Number)	•	X	C	Indicates the PAN mapping account
DE 48 (Additional Data—Private Use), subelement 33 (PAN Mapping File Information), subfield 6 (Token Requestor ID)	•	X	C	Contains the ID assigned by the Token Service Provider to the Token Requestor.
DE 48 (Additional Data—Private Use), subelement 33 (PAN Mapping File Information), subfield 8 (Storage Technology)	•	X	C	Contains a value indicating the Storage Technology of the requested token.
DE 49 (Currency Code, Transaction)	•	X	M	Value indicating the local currency of the acquirer or source location of the transaction. The acquirer in these messages is Mastercard, with a three digits Currency Code of "840" corresponding to USD as Currency Code used in USA. For Local-Use-Only account ranges, Mastercard overrides this value to provide the issuer's three digits Currency Code. This corresponds to the local currency used in that country.
DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status)	•	X	M	9 = Tokenization Request/Notification

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 61 (Additional POS Data), subfield 13 (POS Country Code)	•	X	M	Indicates the country of the POS location (not the acquirer location) using ISO-specified codes. The POS Country Code is set to "840", corresponding to USA. For Local-Use-Only account ranges, Mastercard overrides this value to provide the issuer's three digits Country Code.
DE 124 (Member-defined Data)	•	X	M	See layout for DE 124.

DE 124 Subfields in Authorization Request/0100 for TER

Absolute positioning of data in DE 124 subfields is required. Subfields not containing values will be populated with spaces when they are followed by subfields that contain values. If one or more subfields at the end of the data element do not contain a value, the subfield(s) will not be present and the total length of DE 124 will be reduced by the subfield length(s).

Table 1667: DE 124 Subfields in Authorization Request/0100 for TER

Subfield Name	Attributes	Values/Comments
Message Type	ans-2	TE = Tokenization Eligibility Request (TER)
Correlation ID	an-14	Identifier assigned by Mastercard to associate related tokenization request/notification messages.
Primary Account Number Source	an-1	Identifies the method by which the cardholder is attempting to tokenize a primary account number. 1 = Card on file 2 = Card added manually 3 = Card added via application 4 = Existing token credential 5 = Account added via browser
Payment Application Instance ID	ans-48, left-justified, padded with spaces	Identifier associated with the payment application instance installed onto a device.

Subfield Name	Attributes	Values/Comments
Number of Active Tokens for the Primary Account Number	ans-2, leading zeros	Number of active or suspended tokens for the primary account number provisioned to a consumer. Space-filled when requested token digitized to a server. NOTE: Space-filled means if data is not available then length of that field is filled with space; this is specific to ISO message not for API.

Subfield Name	Attributes	Values/Comments
Wallet Provider Account ID Hash	ans-64	When provided by the Wallet Provider, the issuer may use this hash value to match against known identifiers for the cardholder; for example, their email addresses on file. If the hash values match, this may aid an issuer's digitization decision by providing additional factors to help verify that the Wallet Provider account holder is indeed their cardholder, or to differentiate between primary and secondary cardholders. The Wallet Provider computes the hash over an email address; MDES receives this and includes it in the data sent to the issuer. The issuer computes the hash of the email address on file for the cardholder; if it matches the hash of the one received from the Wallet Provider, the comparison with the hash value received from the Wallet Provider may be used by the issuer to assist determining the digitization decision. Hashing is used to help the issuer verify the cardholder email (or other relevant identifiers that the issuer may have for the cardholder) without the Wallet Provider providing the full email for privacy and security reasons. When the Wallet Provider is Apple Pay, the hash is generated using the PBKDF2 algorithm (PKCS #5). PBKDF2 is performed using 10 iterations, a salt, and the lowercase account ID as the password. The salt is calculated by taking the lower case UTF-8 bytes from the account ID and performing a SHA-256. Hash calculation example: Example: Input (Password): csharp@walletprovider.com Salt: 41404d1bca85ddb59ab21466e277ac1ac5f61470be120c82a21b1e45b5248123 Count: 10 Output: 7098014b646d44c6f3b454c5d54f7a32b3b46e2b0c8e2367f3e5307e3036dfe6 For all other Wallet Providers, the field contains the hash resulting from the following "accountIdHash" algorithm, rightpadded with spaces. "accountId" is the lower case UTF-8 bytes of the account ID: <pre>public String accountIdHash (String accountId) {String random8Bytes = 123CCB2F30BA420B return random8Bytes + lessSignificant24bytes(strongerHash(accountId + random8Bytes))} public String strongerHash(String dataToHash) {String currentHash = dataToHash; for (int i = 0; i < 5000; i++) {currentHash = sha256(currentHash);} return sha256(sha256(dataToHash) + currentHash);}</pre>

Subfield Name	Attributes	Values/Comments
NOTE: String random8Bytes = 123CCB2F30BA420B" is a fixed value for all Wallet Providers. Hash calculation example: - email address: cardholdername@walletprovider.com - accountId: 63617264686F6C6465726E616D654077616C6C657470726F 7669646 5722E636F6D - random8Bytes: 123CCB2F30BA420B - output: 123CCB2F30BA420B17F837DF60E2FC9D6965A74476849FC D43A640F 792A2B358		
Cardholder Name	ans-27, left-justified, padded with spaces	This field may be present and contain the name of the cardholder. The format is either LASTNAME/FIRSTNAME with the names delimited by a slash "/" (Example: SMITH/JOE) or the format is FIRSTNAME LASTNAME (Example: JOE SMITH). If the cardholder's name is longer than 27 positions, the data will be truncated to the maximum length of 27.
Token Type	an-1	Contains a value indicating the type of requested token. C = Mastercard Cloud-based Payments F = Static Token S = Embedded Secure Element

Authorization Request Response/0110 for TER

All mandatory Authorization Request Response/0110 data elements apply.

Table 1668: Authorization Request Response/0110 for TER

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code)	ME	ME	•	Must be the same value as the original Authorization Request/0100
DE 4 (Amount, Transaction)	ME	ME	•	Must be the same value as the original Authorization Request/0100

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 39 (Response Code)	M	•	•	One of the following values indicating how the issuer wants Mastercard to proceed with the tokenization process: • 00 = Continue • 05 = Decline • 85 = Continue, but require additional authentication
DE 124 (Member-defined Data)	C	•	•	See layout for DE 124. If the issuer responds with DE 39 = 00 or 85, the issuer may optionally provide DE 124. If the issuer responds with DE 39 = 05, DE 124 must not be present.

DE 124 Subfields for Authorization Request Response/0110 for TER

Absolute positioning of data in DE 124 subfields is required. Each value must be padded with spaces if the length does not fit exactly the length of the subfield. Subfields not containing values must be populated with spaces when they are followed by subfields that contain values. If one or more subfields at the end of the data element do not contain a value, the subfield(s) will not be present and the total length of DE 124 will be reduced by the subfield length(s).

Table 1669: DE 124 Subfields for Authorization Request Response/0110 for TER

Subfield Name	Attributes	Values/Comments
Issuer Product Configuration ID	ans-10	The unique product configuration identifier provided by the issuer that identifies a particular set of card art, texts and other product related data, provided during the issuer enablement or maintenance process. This ID is case-sensitive. By specifying the Issuer Product Configuration ID on the Token Eligibility Request Response, an issuer may override the usual Product Configuration assigned as the default at Account Range level. This allows greater control over exactly which card art and related card product data gets applied as part of a specific tokenization process. However, this does not provide a mechanism to override the Issuer Terms and Conditions file which always comes from the default Product Configuration(s) assigned at Account Range level.
Primary Account Number Card Sequence Number	ans-3, leading zeros	The card sequence number associated with the primary account number.

Subfield Name	Attributes	Values/Comments												
Token Data and Activation Method(s)	ans...186	When the issuer responds with DE 39 = 85 (Continue, but require additional authentication), the issuer may provide one or more methods by which the consumer can retrieve the activation code. If the issuer responds with DE 39 = 00 (Continue) or 05 (Decline), or if the issuer responds with DE 39 = 85 but uses an issuer-specified default for the activation method, this field will not be present. Each method will consist of:												
<table> <tr> <th>Name</th><th>Attributes</th><th>Values/Comments</th></tr> <tr> <td>Activation Method Type</td><td>n-1</td><td> 1 = Masked mobile phone number 2 = Masked email address 3 = Automated call center phone number 4 = Call center phone number 5 = Website 6 = Mobile application 7 = Masked voice call phone number 9 = Identity check EMV 3DS authentication </td></tr> <tr> <td>Activation Method Value</td><td>ans...183</td><td>Refer to the following examples.</td></tr> <tr> <td>Delimiter</td><td>ans...2</td><td> If multiple activation methods are provided by the issuer, the issuer must separate each method with the " " delimiter. Following the last character of the final method in the field, the issuer should include two " " delimiters. Values: " " indicates another Activation Method follows. </td></tr> </table>			Name	Attributes	Values/Comments	Activation Method Type	n-1	1 = Masked mobile phone number 2 = Masked email address 3 = Automated call center phone number 4 = Call center phone number 5 = Website 6 = Mobile application 7 = Masked voice call phone number 9 = Identity check EMV 3DS authentication	Activation Method Value	ans...183	Refer to the following examples.	Delimiter	ans...2	If multiple activation methods are provided by the issuer, the issuer must separate each method with the " " delimiter. Following the last character of the final method in the field, the issuer should include two " " delimiters. Values: " " indicates another Activation Method follows.
Name	Attributes	Values/Comments												
Activation Method Type	n-1	1 = Masked mobile phone number 2 = Masked email address 3 = Automated call center phone number 4 = Call center phone number 5 = Website 6 = Mobile application 7 = Masked voice call phone number 9 = Identity check EMV 3DS authentication												
Activation Method Value	ans...183	Refer to the following examples.												
Delimiter	ans...2	If multiple activation methods are provided by the issuer, the issuer must separate each method with the " " delimiter. Following the last character of the final method in the field, the issuer should include two " " delimiters. Values: " " indicates another Activation Method follows.												

Subfield Name	Attributes	Values/Comments
		" " indicates the last Activation Method.

Table 1670: Examples

1(###) ### 4567 |

1 = Masked mobile phone number

The 1 will be followed by the masked mobile phone number, then the delimiter.

2a***d@anymail.com |

2 = Masked email address

The 2 will be followed by the consumer's masked email address (the issuer will mask according to their own format), then the delimiter.

3(555) 123 4567 |

3 = Automated call center phone number

The 3 will be followed by the phone number, then the delimiter. This phone number is not masked.

4(555) 123 8901 |

4 = Call center phone number

The 4 will be followed by the phone number, then the delimiter. This phone number is not masked.

5http://www.anybank.com |

5 = Website

The 5 will be followed by the issuer's website URL, then the delimiter.

6com.anybank.mobileapp |

6 = Mobile app

The 6 will be followed by the issuer's mobile app information, the content of which depends upon the mobile device operating system, then the delimiter.

7(###) ### 2345 ||

7 = Masked voice call phone number

The 7 will be followed by the masked voice call phone number. If this were the last Activation Method provided, the value is followed by two delimiters.

9 = Identity check - EMV 3DS

Tokenization Authorization Request (TAR)

Following is a list of the data elements, values, and subfields applicable for Tokenization Authorization Request (TAR).

- [Authorization Request/0100 for TAR](#)
- [DE 124 Subfields in Authorization Request/0100 for TAR](#)
- [Authorization Request Response/0110 for TAR](#)
- [DE 124 Subfields in Authorization Request Response/0110 for TAR](#)

Authorization Request/0100 for TAR

All mandatory Authorization Request/0100 data elements apply.

Table 1671: Authorization Request/0100 for TAR

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 2 (Primary Account Number)	•	X	M	Cardholder's primary account number
DE 3 (Processing Code)	•	X	M	00 = Purchase
DE 4 (Amount, Transaction)	•	X	M	Will be zero
DE 14 (Date, Expiration)	•	X	C	Cardholder's primary account expiration date
DE 22 (POS Entry Mode)	•	X	M	Subfield 1 (POS Terminal PAN Entry Mode) Values: • 01 = PAN manual entry • 05 = Merchant terminal PAN auto-entry through integrated circuit card (reserved for Card DIP) • 07 = Merchant terminal PAN auto-entry through Contactless M/Chip Subfield 2 (POS Terminal PIN Entry Mode) = 0 (Unspecified or unknown)
DE 23 (Card Sequence Number)	•	X	C	Distinguishes among separate cards having the same DE 2 (Primary Account Number [PAN]) This is present only when issuer has opted to validate chip data for tap to add card provisioning.

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 35 (Track 2 Data)	•	X	C	The Authorization Platform creates and provides Track 2 data for Maestro tokenization messages. This is present only when issuer has opted to validate chip data for tap to add card provisioning.
DE 43 (Card Acceptor Name/ Location for All Transactions), subfield 5 (Card Acceptor State or Country Code)	•	X	M	The acquirer in these messages is the Consumer Facing Entity Name (CFEN), with a three alphanumeric Country Code corresponding to "USA". For Local-Use-Only account ranges, Mastercard overrides this value to provide the three alphanumeric Country Code of the issuer.
DE 48 Transaction Category Code	•	X	C	T (Phone, Mail, or Electronic Commerce Order)
DE 48 (Additional Data - Private Use), subelement 23 (Payment Initiation Channel)	•	X	C	Value indicating the type of device for which the consumer is requesting tokenization of a primary account number.
DE 48 (Additional Data - Private Use), subelement 26 (Wallet Program Data), subfield 1 (Wallet Identifier)	•	X	C	Contains the identifier associated with the Wallet Provider.
DE 48 (Additional Data - Private Use), subelement 33 (PAN Mapping File Information), subfield 1 (Account Number Indicator)	•	X	C	Contains the type of PAN mapping account. Possible values: • C = when MDES Secure Element Device Token • H = when MDES Cloud-based Payments Device Token • F = when MDES Static Token
DE 48 (Additional Data - Private Use), subelement 33 (PAN Mapping File Information), subfield 2 (Account Number)	•	X	C	Indicates the PAN mapping account.
DE 48 (Additional Data - Private Use), subelement 33 (PAN Mapping File Information), subfield 6 (Token Requestor ID)	•	X	C	Contains the ID assigned by the Token Service Provider to the Token Requestor.

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 48 (Additional Data - Private Use), subelement 33 (PAN Mapping File Information), subfield 8 (Storage Technology)	•	X	C	Contains a value indicating the storage technology of the requested token.
DE 48 (Additional Data - Private Use), subelement 71 (On-behalf Services [OBS]) subfield 1 (On-behalf [OB] Service)	•	X	C	This value notifies the issuer of the OBS performed by Mastercard and the result of chip data validation. This is present only when issuer has opted to use OBS for tap to add card provisioning.
DE 48 (Additional Data - Private Use), subelement 71 (On-behalf Services [OBS]) subfield 2 (On-behalf [OB] Result 1)	•	X	C	Indicates the results of the service processing. This is present only when issuer has opted to use OBS for tap to add card provisioning.
DE 48 (Additional Data - Private Use), subelement 82 (Address Verification Service)	•	X	C	52 = AVS and Authorization Request/0100
DE 48 (Additional Data - Private Use), subelement 92 (CVC 2)	•	X	C	CVC 2 value from the signature panel of the card when applicable
DE 49 (Currency Code, Transaction)	•	X	M	Value indicating the local currency of the acquirer or source location of the transaction. The acquirer in these messages is Mastercard, with a three digits Currency Code of "840" corresponding to USD as Currency Code used in USA. For Local-Use-Only account ranges, Mastercard overrides this value to provide the issuer's three digits Currency Code. This corresponds to the local currency used in that country.
DE 55 (Integrated Circuit Card [ICC] System-Related Data)	•	X	C	Integrated Circuit Card [ICC] System-Related Data, binary data captured from chip data. This is present only when issuer has opted to validate chip data for tap to add card provisioning.
DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status)	•	X	M	9 = Tokenization Request/Notification

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 61 (Additional POS Data), subfield 13 (POS Country Code)	•	X	M	Indicates the country of the POS location (not the acquirer location) using ISO-specified codes. The POS Country Code is set to "840", corresponding to USA. For Local-Use-Only account ranges, Mastercard overrides this value to provide the issuer's three digits Country Code.
DE 120 (Record Data)	•	X	C	AVS postal code and address data in the format specified by the issuer
DE 124 (Member-defined Data)	•	X	M	See layout for DE 124.

DE 124 Subfields in Authorization Request/0100 for TAR

Absolute positioning of data in DE 124 subfields is required. Subfields not containing values will be populated with spaces when they are followed by subfields that contain values. If one or more subfields at the end of the data element do not contain a value, the subfield(s) will not be present and the total length of DE 124 will be reduced by the subfield length(s).

Table 1672: DE 124 Subfields in Authorization Request/0100 for TAR

Subfield Name	Attributes	Values/Comments
Message Type	an-2	TA = Tokenization Authorization Request (TAR)
Correlation ID	an-14	Identifier assigned by Mastercard that can be used by the issuer to associate related tokenization request/notification messages.
Primary Account Number Source	an-1	Identifies the method which the cardholder is attempting to tokenize a primary account number. 1 = Card on file 2 = Card added manually 3 = Card added via application 4 = Existing token credential 5 = Account added via browser 6 = Account added via chip data
Payment Application Instance ID	ans-48, left-justified, padded with spaces	Identifier associated with the payment application instance installed onto a device.

Subfield Name	Attributes	Values/Comments
Device Source IP Address	ans-12, left-justified, padded with spaces	Variable length IP address. Each octet of the IP address is converted to hex, and joined into one string, with the order maintained.
Wallet Service Provider Account ID Hash	ans-64	This field may be present and contain PBKDF2 hash of the consumer's account ID with the Wallet Provider. The Account ID is typically expected to be an email address. PBKDF2 is performed using 10 iterations, a salt, and the lowercase account ID as the password. The salt is calculated by taking the lower case UTF-8 bytes from the account ID and performing a SHA-256. Example: Input (Password): csharp@walletprovider.com Salt: 41404d1bca85ddb59ab21466e277ac1ac5f61470be120c82a21b1e45b5248123 Count: 10 Output: 7098014b646d44c6f3b454c5d54f7a32b3b46e2b0c8e2367f3e5307e3036dfe6
Cardholder Name	ans-27, left-justified, trailing spaces	This field may be present and contain the name of the cardholder. The format is either LASTNAME/FIRSTNAME with the names delimited by a slash "/" (Example: SMITH/JOE) or the format is FIRSTNAME LASTNAME (Example: JOE SMITH). If the cardholder's name is longer than 27 positions, the data will be truncated to the maximum length of 27.
Wallet Provider Tokenization Recommendation	an-1	Tokenization decision suggested by the Wallet Provider. One of the following values: 0 = Decline 1 = Approve 2 = Require additional authentication
Wallet Provider Tokenization Recommendation Standard Version	an-2, left-justified, padded with spaces	The version of the standards the Wallet Provider is using to determine the suggested tokenization recommendation.
Wallet Provider Device Score	n-1	Score assigned by Wallet Provider for the device. Value between 1 and 5.
Wallet Provider Account Score	n-1	Score assigned by Wallet Provider for the primary account number. Value between 1 and 5.

Subfield Name	Attributes	Values/Comments
Number of Active Tokens for the Primary Account Number	ans-2, leading zeros	Number of active or suspended tokens for the primary account number provisioned to a consumer. Space-filled when requested token digitized to a server. NOTE: Space-filled means if data is not available then length of that field is filled with space; this is specific to ISO message not for API.
Wallet Service Provider Tokenization Recommendation Reason Codes	ans-6	Code indicating the specific reason the Wallet Provider is suggesting the tokenization recommendation. The data of this field is a hex-encoded bitmap, whereby each bit corresponds to a specific Reason Code. The bitmap is big-endian with the least significant bit corresponding to Reason Code 01, with the next least significant bit corresponding to Reason Code 02 and so on. For example, if Reason Codes 01, 05, and 16 were encoded, the bitmap would be 0000000010000000000010001 and the hex value of this field would be 008011. If the Wallet Provider has no reason, this field will contain spaces.

Subfield Name	Attributes	Values/Comments
Device Location	ans-9	Latitude and longitude where the device the consumer is attempting to tokenize a card onto is located. Device Location Latitude - an-4; hexadecimal encoded degrees with 2 decimal places. The first two characters represent the latitude value before the decimal place, and the last two characters represent the latitude after the decimal place. Device Location Longitude - an-4; hexadecimal encoded degrees with 2 decimal places. The first two characters represent the longitude value before the decimal place, and the last two characters represent the longitude after the decimal place. Device Location Lat/Long Sector - n-1; one of the following values: 1 = Latitude = N, Longitude = W (positive latitude, negative longitude) 2 = Latitude = N, Longitude = E (positive latitude, positive longitude) 3 = Latitude = S, Longitude = W (negative latitude, negative longitude) 4 = Latitude = S, Longitude = E (negative latitude, positive longitude) For example, a value of 264B5A4B1 would be interpreted as: - Latitude (first four characters) 264B - First hex digit (for value before decimal place) = 0x26 = 38 (decimal) - Second hex digit (for value after decimal place) = 0x4B = 75 (decimal) - Longitude (next four characters) 5A4B - First hex digit (for value before decimal place) = 0x5A = 90 (decimal) - Second hex digit (for value after decimal place) = 0x4B = 75 (decimal) - Device Location Sector (final digit) 1 = NW - Final location = 38.75N, 90.75W This field will contain spaces if the Wallet Provider has not provided this information.

Subfield Name	Attributes	Values/Comments
Mobile Number Last 4 Digits	ans-4	Last four digits of the consumer's mobile phone number This field will contain spaces if the Wallet Provider has not provided this information.
Token Type	an-1	Contains a value indicating the type of requested token. C = Mastercard Cloud-based Payments F = Static S = Embedded Secure Element
Consumer Identifier	an-88	Unique identifier for each consumer, which financial institutions will then need to verify before a consumer can add their credentials to the payment-enabled devices. Position 212–299 in the DE 124 of the Authorization Request/0100 - Tokenization Authorization message only. Issuers and processors will not receive the new Consumer Identifier in DE 124 unless they explicitly opt to receive it.

Authorization Request Response/0110 for TAR

All mandatory Authorization Request Response/0110 data elements apply.

Table 1673: Authorization Request Response/0110 for TAR

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code)	ME	ME	•	Must be the same value as the original Authorization Request/0100
DE 4 (Amount, Transaction)	ME	ME	•	Must be the same value as the original Authorization Request/0100
DE 39 (Response Code)	M	•	•	One of the following values indicating how the issuer wants Mastercard to proceed with the tokenization process: • 00 = Approve • 05 = Decline • 85 = Approve, but require additional authentication
DE 48 Transaction Category Code	CE	CE	•	Must be the same value as the original Authorization Request/0100

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 48 (Additional Data—Private Use), subelement 82 (Address Verification Service)	CE	CE	•	52 = AVS and Authorization Request/0100
DE 48 (Additional Data—Private Use), subelement 83 (Address Verification Service Response)	C	•	•	The AVS response code
DE 48 (Additional Data—Private Use), subelement 87 (Card Validation Code Result)	C	•	•	The CVC 2 result code
DE 48 (Additional Data—Private Use), subelement 92 (CVC 2)	CE	CE	•	Must be the same value as the original Authorization Request/0100
DE 56 (Payment Account Data)	C	X	C	PAR value provided by the Issuer when Issuer or its designated service provider is BIN Controller (Mastercard is not BIN Controller).
DE 124 (Member-defined Data)	C	•	•	See layout for DE 124. If the issuer responds with DE 39 = 00 or 85, the issuer may optionally provide DE 124. If the issuer responds with DE 39 = 05, DE 124 must not be present.

DE 124 Subfields in Authorization Request Response/0110 for TAR

Absolute positioning of data in DE 124 subfields is required. Subfields not containing values must be populated with spaces when they are followed by subfields that contain values. If one or more subfields at the end of the data element do not contain a value, the subfield(s) will not be present and the total length of DE 124 will be reduced by the subfield length(s).

Table 1674: DE 124 Subfields in Authorization Request Response/0110 for TAR

Subfield Name	Attributes	Values/Comments
Issuer Product Configuration ID	ans-10	The unique product configuration identifier provided by the issuer that identifies a particular set of card art, texts and other product related data, provided during the issuer enablement or maintenance process. This ID is case-sensitive. By specifying the Issuer Product Configuration ID on the Tokenization Authorization Request Response, an issuer may override the usual Product Configuration assigned as the default at Account Range level. This allows greater control over exactly which card art and related card product data gets applied as part of a specific tokenization process. However, this does not provide a mechanism to override the Issuer Terms and Conditions file which always comes from the default Product Configuration(s) assigned at Account Range level.
Primary Account Number Card Sequence Number	ans-3, leading zeros	The card sequence number associated with the primary account number.
Token Data and Activation Method(s)	ans...286	The subfield can have zero to multiple groupings of three Token Data and Activation Method Type, Token Data and Activation Method Value, and Delimiter. When the issuer responds with DE 39 = 85 (Approve, but require additional authentication), the issuer may provide one or more activation methods by which the consumer can retrieve the activation code. If the issuer responds with DE 39 = 00 (Approve) or 05 (Decline) activation methods will not be present. If the issuer responds with DE 39 = 85 but uses an issuer-specified default for the activation method, activation methods may not be present. When the issuer responds with DE 39 = 00 (Approve) or DE 39 = 85 (Approved, but require additional authentication), the issuer may provide zero to multiple token personalization data.

Subfield Name	Attributes	Values/Comments
Token Data and Activation Method Type	an-1	1 = Masked mobile phone number (activation method) 2 = Masked email address (activation method) 3 = Automated call center phone number (activation method) 4 = Call center phone number (activation method) 5 = Website (activation method) 6 = Mobile application (activation method) 7 = Masked voice call phone number (activation method) 9 = Identity check EMV 3DS (activation method) a = Alternate account identifier t = One or multiple token personalization data items

Subfield Name	Attributes	Values/Comments
Token Data and Activation Method Value	ans...283	When the Token Data and Activation Method Type is 1, 2, 3, 4, 5, 6, 7 or 9 the issuer must provide values used to specify an activation method. See Examples 1, 2, 3, and 4 for examples of Token Data and Activation Method(s) containing activation method values. Issuers have the option of providing an alternate account identifier in Tokenization Authorization Response messages (Token Data and Activation Method Type = a). An Alternate Account Identifier is a cardholder-friendly reference to a bank account, for example an IBAN (International Bank Account Number). It is typically useful for a Wallet Provider to display a suffix of this identifier to the cardholder to help them identify their tokenized card when they are not aware of their Account PAN. This will enable wallets to display alternate account identifier suffixes depending on the wallet functionality. Issuers must refer to each wallet's documentation to determine if/how the alternate account identifier suffixes are displayed to cardholders. An alternate account identifier is in the format of a string with minimum 9 and maximum 64 characters, with no spaces allowed. An IBAN is an example of an alternate account identifier, as for example, GB82WEST12345698765432. When the Token Data and Activation Method Type is t, the issuer must provide one or multiple proprietary token personalization data items that MDES will include in the token profile. See Example 8 for an example of Token Data and Activation Method(s) containing multiple token personalization data items. Each token personalization data item can have one to multiple instances of couples Token Personalization Data and Separator. Token Personalization Data: Base64 encoded data. Structure and content of data is defined separately per <i>Mastercard Digital Enablement Service (MDES) Issuer Implementation Guide</i>. See Examples 6 and 7. Separator ans...1 Each separate token personalization data (including after the last) will be delimited by a "~" character.

Subfield Name	Attributes	Values/Comments
		See Example 5 for an example of Token Data and Activation Method(s) containing multiple activation method values and multiple token personalization data items.
Delimiter	ans-2	If multiple Token Data and Activation Methods are provided by the issuer, the issuer must separate each method with the " " delimiter. NOTE: When at least another grouping of three Token Data and Activation Method Type, Token Data and Activation Method Value, and Delimiter follows, then the Delimiter has attributes of ans-1. Delimiter Values: " " indicates that another Token Data or Activation Method follows. Following the last character of the final Token Data and Activation Method in the field, the issuer should include two " " delimiters. " " indicates the last Token Data and Activation Method.

Table 1675: Examples

Example	Description
1	An example of Token Data and Activation Method(s) subfield containing only a masked mobile phone number activation method: 1(###)###4567
2	An example of Token Data and Activation Method(s) subfield containing the masked email address activation method and the automated call center phone number activation method: 2a***d@anymail.com 3(555)1234567
3	An example of Token Data and Activation Method(s) subfield containing the call center phone number activation method, the website activation method and the masked voice call phone number activation method: 4(555)1238901 5http://www.anybank.com 6com.anybank.mobileapp
4	An example of Token Data and Activation Method(s) subfield containing only a masked voice call phone number activation method: 7(###)###4567

Example	Description
5	An example of Token Data and Activation Method(s) subfield containing mobile application activation method, masked email address activation method and some personalization data for the token profile: 6com.anybank.mobileapp 2a***d@anymail.com tBJ9uCwJQAAEwOGECAwQFX1UCR1I=~CwECAwQFBgc=~
6	An example of Token Personalization Data: "BJ9uCwJQAAEwOGECAwQFX1UCR1I=" that is "049F6E 0B 0250 0001 3039 0102030405 5F55 02 4652" base64 encoded. The purpose of Token Personalization Data is to replace default Third Party Data (9F6E) and add Issuer Country Code (5F55) in Contactless FCI Issuer Discretionary Data.
7	An example of Token Personalization Data: "CwECAwQFBgc=" that is "0B01020304050607" base64 encoded. The purpose of Token Personalization Data is to create a new tag "9F70" (Protected Data Envelope) readable through GET DATA on both Management and Contactless Payment interfaces.
8	An example of Token Data and Activation Method(s) subfield containing multiple token personalization data items: tBJ9uCwJQAAEwOGECAwQFX1UCR1I=~CwECAwQFBgc=~

Authentication Methods Request (AMR)

Following is a list of the data elements, values, and subfields applicable for Authentication Methods Request (AMR).

- [Authorization Request/0100 for AMR](#)
- [DE 124 Subfields in Authorization Request/0100 for AMR](#)
- [Authorization Request Response/0110 for AMR](#)
- [DE 124 Subfields in Authorization Request Response/0110 for AMR](#)

Authorization Request/0100 for AMR

All mandatory Authorization Request/0100 data elements apply.

Table 1676: Authorization Request/0100 for AMR

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 2 (Primary Account Number)	•	X	M	Cardholder's primary account number
DE 3 (Processing Code)	•	X	M	00 = Purchase
DE 4 (Amount, Transaction)	•	X	M	Will be zero
DE 14 (Date, Expiration)	•	X	C	Cardholder's primary account expiration date

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 22 (POS Entry Mode)	•	X	M	Subfield 1 (POS Terminal PAN Entry Mode) = 01 PAN manual entry Subfield 2 (POS Terminal PIN Entry Mode) = 0 (Unspecified or unknown)
DE 35 (Track 2 Data)	•	X	C	The Authorization Platform creates and provides Track 2 data for Maestro tokenization messages.
DE 43 (Card Acceptor Name/ Location for All Transactions), subfield 5 (Card Acceptor State or Country Code)	•	X	M	The acquirer in these messages is the Consumer Facing Entity Name (CFEN), with a three alphanumeric Country Code corresponding to "USA". For Local-Use-Only account ranges, Mastercard overrides this value to provide the three alphanumeric Country Code of the issuer.
DE 48 Transaction Category Code	•	X	C	T (Phone, Mail, or Electronic Commerce Order)

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 48 (Additional Data - Private Use) subelement 06 (Advanced Digital Transaction Data 1) Subfield 1 (Binding ID) Subfield 2 (Phone Number) Subfield 3 (Device Location)	•	X	C	Subfield 1 contains the Bind ID - Helps to uniquely identify the component of data that is bound to the credential. Subfield 2 contains the Phone Number - The full phone number or partial phone number of the consumer. Subfield 3 contains the Device Location - Latitude and longitude where the device the cardholder is using is located for authentication. Device Location Latitude - an-4; hexadecimal encoded degrees with 2 decimal places. Device Location Longitude - an-4; hexadecimal encoded degrees with 2 decimal places. Device Location Lat/Long Sector - n-1; one of the following values: 1 = Latitude = N, Longitude = W 2 = Latitude = N, Longitude = E 3 = Latitude = S, Longitude = W 4 = Latitude = S, Longitude = E This field will contain spaces if the Wallet Provider has not provided this information.
DE 48 (Additional Data - Private Use), subelement 07 (Advanced Digital Transaction Data 2) Subfield 1 (Email Address Format) Subfield 2 (Email Address)	•	X	C	Subfield 1 Email Address Format - Value 0 indicates - Contains a hashed email address. Value 1 indicates - Contains a clear email address. Subfield 1 Email Address - The clear email address or hashed email address.
DE 48 (Additional Data - Private Use), subelement 08 (Advanced Digital Transaction Data 2), subfield 1 (IP Address)	•	X	C	Subfield 1 IP Address
DE 48 (Additional Data - Private Use), subelement 23 (Payment Initiation Channel)	•	X	C	Value indicating the type of device for which the consumer is requesting tokenization of a primary account number.

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 48 (Additional Data - Private Use), subelement 26 (Wallet Program Data), subfield 1 (Wallet Identifier)	•	X	C	Contains the identifier associated with the Wallet Provider.
DE 48 (Additional Data - Private Use), subelement 33 (PAN Mapping File Information) Subfield 1 (Account Number Indicator) Subfield 2 (Account Number) Subfield 3 (Expiration Date) Subfield 5 (Token Assurance Method) Subfield 6 (Token Requestor ID) Subfield 8 (Storage Technology)	•	X	C	Subfield 1 Indicates the type of PAN mapping account. Subfield 2 Indicates the PAN mapping account. Subfield 3 Indicates the expiration date of the PAN mapping account. Subfield 5 Indicates Token Assurance Method Subfield 6 Contains the ID assigned by the Token Service Provider to the Token Requestor. Subfield 8 Indicates Storage Technology of the token.
DE 49 (Currency Code, Transaction)	•	X	M	Value indicating the local currency of the acquirer or source location of the transaction. The acquirer in these messages is Mastercard, with a three digits Currency Code of "840" corresponding to USD as Currency Code used in USA. For Local-Use-Only account ranges, Mastercard overrides this value to provide the issuer's three digits Currency Code. This corresponds to the local currency used in that country.
DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status)	•	X	M	9 = Tokenization Request/Notification
DE 61 (Additional POS Data), subfield 13 (POS Country Code)	•	X	M	Indicates the country of the POS location (not the acquirer location) using ISO-specified codes. The POS Country Code is set to "840", corresponding to USA. For Local-Use-Only account ranges, Mastercard overrides this value to provide the issuer's three digits Country Code.

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 104 (Digital Payment Data) Subelement 004 (Digital Service Provider)	•	X	C	Subfield 001 (Digital Service Provider ID 1) Subfield 002 (Digital Service Provider ID 2)
DE 104 (Digital Payment Data) Subelement 005 (Digital Authentication Data)	•	X	C	Subfield 001 (Cardholder Verification) Subfield 002 (Authentication Factors)
DE 124 (Member-defined Data)	•	X	M	See layout for DE 124.

DE 124 Subfields in Authorization Request/0100 for AMR

Absolute positioning of data in DE 124 subfields is required. Subfields not containing values will be populated with spaces when they are followed by subfields that contain values. If one or more subfields at the end of the data element do not contain a value, the subfield(s) will not be present and the total length of DE 124 will be reduced by the subfield length(s).

Table 1677: DE 124 Subfields in Authorization Request/0100 for AMR

Subfield Name	Attributes	Values/Comments
Message Type	an-2	AM = Authentication Methods Request (AMR)
Correlation ID	an-14	Identifier assigned by Mastercard that can be used by the issuer to associate related tokenization request/notification messages.
Token Unique Reference	ans-48, left-justified, padded with spaces	Service-allocated unique reference to the token.
Token Type	an-1	Contains a value indicating the type of requested token. C = Mastercard Cloud-based Payments F = Static Token S = Embedded Secure Element

Subfield Name	Attributes	Values/Comments
Reason Codes	ans-6	The reason for the cardholder authentication. The reason could impact the generation and validation of the authentication code in a number of ways including but not limited to the time period of validity of the code and the number of attempts allowed to validate the code. New reason codes can be added at any time and should not result in a failure. 01 = ADD_CARD – The account holder is being authenticated to verify account ownership at the time of adding a card in merchant system. 02 = VERIFY_ACCOUNT - The account holder is being authenticated to verify account ownership to use a service 03 = OTHER 04 = PAYMENT_TRANSACTION - The account holder is being authenticated to initiate payment transaction 05 = TOKEN_BINDING - The account holder is being authenticated to bind a token to bind id. 06 = ACCOUNT_BINDING - The account holder is being authenticated to bind associated account with bind id (device). The data of this field is a hex-encoded bitmap, whereby each bit corresponds to a specific Reason Code. The bitmap is big-endian with the least significant bit corresponding to Reason Code 01, with the next least significant bit corresponding to Reason Code 02 and so on. For example, if Reason Codes 01 and 02 were encoded, the bitmap would be 00000000000000000000000011 and the hex value of this field would be 000003.
Token Request Account Score	n-1	Score assigned by Token Requestor for the primary account number/Token. Value between 1 and 5.
Token Requestor Device Score	n-1	Score assigned by Token Requestor for the device. Value between 1 and 5. Device used at the time of authentication.
Account Lifetime	n-2	The lifetime (in months) of the account with the Token Requestor.
authRequestCorrelationId	ans-64	Value linking different authentication requests/messages.

Subfield Name	Attributes	Values/Comments
Device OS Name	an-1	Operating System of the device that is used for cardholder authentication. • 0 = Android • 1 = iOS • 2 = Windows • 3 = Tizen • 4 = PAGER_EMBEDDED • 5 = ANDROID_WEAR • 6 = EMBEDDED_OS • z = None of the above
Device Name	ans-20 left-justified padded with space	Device Name set by the consumer or manufacturer.
Recently authenticated by trusted authenticating entity ID	an-5	Identification of Mastercard trusted authenticator name who recently authenticated cardholder.
Recent authentication proof authenticated by trusted authenticating entity ID	ans-35	Authentication Proof shared by the trusted authenticator.
Recent authentication time, authenticated by trusted authenticating entity ID	n-10	Date and time when this authentication happened. Time is in UTC in the following format: YYMMDDhhmm.

Authorization Request Response/0110 for AMR

All mandatory Authorization Request Response/0110 data elements apply.

Table 1678: Authorization Request Response/0110 for AMR

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code)	ME	ME	•	Must be the same value as the original Authorization Request/0100
DE 4 (Amount, Transaction)	ME	ME	•	Must be the same value as the original Authorization Request/0100

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 39 (Response Code)	M	•	•	00 - Requested use case is approved and Cardholder Authentication is not required. 85 - Cardholder Authentication is required. All other: Requested use case is not approved.
DE 48 Transaction Category Code	CE	CE	•	Must be the same value as the original Authorization Request/0100
DE 124 (Member-defined Data)	M	•	•	See layout for DE 124.

DE 124 Subfields in Authorization Request Response/0110 for AMR

Absolute positioning of data in DE 124 subfields is required. Subfields not containing values must be populated with spaces when they are followed by subfields that contain values. If one or more subfields at the end of the data element do not contain a value, the subfield(s) will not be present and the total length of DE 124 will be reduced by the subfield length(s).

Table 1679: DE 124 Subfields in Authorization Request Response/0110 for AMR

Subfield Name	Attributes	Values/Comments
Token Data and Authentication Method(s)	ans...199	The subfield can have zero to multiple groupings of three Token Data and Authentication Method Type, Token Data and Authentication Method Value, and Delimiter.
Token Data and Authentication Method Type	an-1	1 = Masked mobile phone number 2 = Masked email address 3 = Automated call center phone number 4 = Call center phone number 5 = Website 6 = Mobile application 7 = Masked voice call phone number 8 = Push Notification (means issuer supports push notification to deliver Authentication Code in issuer app) 9 = EMV 3DS (Identity Check)

Subfield Name	Attributes	Values/Comments
Token Data and Authentication Method Value	ans...196	Authentication Value is required for authentication type - 1, 2, 5. Authentication Value is not required for authentication type - 6, 8, 9. Authentication Value guidance for authentication type 6: value will be PUSH_NOTIFICATION if an issuer wants to their application to open through push notification using Android intent and iOS deep link URL. Issuer will make sure that they configure iOS deep link URL and Android intent information through MDES Manager. PUSH_NOTIFICATION is not applicable to activate a device token, it is only applicable for post-tokenization authentication.
Delimiter	ans-2	If multiple Token Data and Activation Methods are provided by the issuer, the issuer must separate each method with the " " delimiter. NOTE: When at least another grouping of three Token Data and Activation Method Type, Token Data and Activation Method Value, and Delimiter follows, then the Delimiter has attributes of ans-1. Delimiter Values: " " indicates that another Token Data or Activation Method follows. Following the last character of the final Token Data and Activation Method in the field, the issuer should include two " " delimiters. " " indicates the last Token Data and Activation Method.

Activation Code Notification (ACN)

Following is a list of the data elements, values, and subfields applicable for Activation Code Notification (ACN).

- [Authorization Request/0100 for ACN](#)
- [DE 124 Subfields in Authorization Request/0100 for ACN](#)
- [Authorization Request Response/0110 for ACN](#)

Authorization Request/0100 for ACN

All mandatory Authorization Request/0100 data elements apply.

Table 1680: Authorization Request/0100 for ACN

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 2 (Primary Account Number)	•	X	M	Cardholder's primary account number
DE 3 (Processing Code)	•	X	M	00 = Purchase
DE 4 (Amount, Transaction)	•	X	M	Will be zero
DE 14 (Date, Expiration)	•	X	C	Cardholder's primary account expiration date
DE 22 (POS Entry Mode)	•	X	M	Subfield 1 (POS Terminal PAN Entry Mode) = 01 (PAN manual entry) Subfield 2 (POS Terminal PIN Entry Mode) = 0 (Unspecified or unknown)
DE 35 (Track 2 Data)	•	X	C	The Authorization Platform creates and provides Track 2 data for Maestro tokenization messages.
DE 43 (Card Acceptor Name/ Location for All Transactions), subfield 5 (Card Acceptor State or Country Code)	•	X	M	The acquirer in these messages is the Consumer Facing Entity Name (CFEN), with a three alphanumeric Country Code corresponding to "USA". For Local-Use-Only account ranges, Mastercard overrides this value to provide the three alphanumeric Country Code of the issuer.
DE 48 Transaction Category Code	•	X	C	T (Phone, Mail, or Electronic Commerce Order)
DE 48 (Additional Data—Private Use), subelement 23 (Payment Initiation Channel)	•	X	C	Value indicating the type of device for which the consumer is requesting tokenization of a primary account number.
DE 48, subelement 26 (Wallet Program Data), subfield 1 (Wallet Identifier)	•	X	C	Contains the identifier associated with the Wallet Provider

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 48 (Additional Data—Private Use), subelement 33 (PAN Mapping File Information), subfield 1 (Account Number Indicator)	•	X	C	Contains the type of PAN mapping account. Possible values: • C = When MDES Secure Element Device Token • H = When MDES Cloud-based Payments Device Token • F = When MDES Static Token
DE 48 (Additional Data—Private Use), subelement 33 (PAN Mapping File Information), subfield 2 (Account Number)	•	X	C	Indicates the PAN mapping account
DE 48 (Additional Data—Private Use), subelement 33 (PAN Mapping File Information), subfield 6 (Token Requestor ID)	•	X	C	Contains the ID assigned by the Token Service Provider to the Token Requestor.
DE 49 (Currency Code, Transaction)	•	X	M	Value indicating the local currency of the acquirer or source location of the transaction. The acquirer in these messages is Mastercard, with a three digits Currency Code of "840" corresponding to USD as Currency Code used in USA. For Local-Use-Only account ranges, Mastercard overrides this value to provide the issuer's three digits Currency Code. This corresponds to the local currency used in that country.
DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status)	•	X	M	9 = Tokenization Request/Notification
DE 61 (Additional POS Data), subfield 13 (POS Country Code)	•	X	M	Indicates the country of the POS location (not the acquirer location) using ISO-specified codes. The POS Country Code is set to "840", corresponding to USA. For Local-Use-Only account ranges, Mastercard overrides this value to provide the issuer's three digits Country Code.
DE 124 (Member-defined Data)	•	X	M	See layout for DE 124.

DE 124 Subfields in Authorization Request/0100 for ACN

Absolute positioning of data in DE 124 subfields is required. Subfields not containing values will be populated with spaces when they are followed by subfields that contain values. If one or more subfields at the end of the data element do not contain a value, the subfields will not be present and the total length of DE 124 will be reduced by the subfield lengths.

Table 1681: DE 124 Subfields in Authorization Request/0100 for ACN

Subfield Name	Attributes	Values/Comments
Message Type	an-2	AC = Activation Code Notification (ACN)
Correlation ID	an-14	The Identifier assigned by Mastercard to associate related tokenization request/notification messages.
Activation Code	ans-8, left-justified, padded with spaces	Activation code assigned by Mastercard that will be provided to the issuer for delivery to the consumer to complete the tokenization process.
Activation Code Expiration Date and Time	n-10	The date and time that the Activation Code expires specified in UTC units. Format: YYMMDDhhmm.
Consumer's Activation Method Preference	ans-165, left-justified, padded with spaces NOTE: For post-tokenization authentication flow type: ans - 150, left justified, padded with space	This field contains the activation method selected by the consumer, if only one was offered by the issuer, then that activation method. There will be only one method contained within this field. This field will only be present if the cardholder provides a choice. If the reason code is not present, then the additional space will not be padded with space.

Name	Attributes	Values/Comments
Activation Method Type	n-1	1 = Masked mobile phone number 2 = Masked email address 3 = Automated call center phone number 4 = Call center phone number 5 = Website

Subfield Name	Attributes	Values/Comments
		6 = Mobile application. If the value is PUSH_NOTIFICATION, then the issuer should send a push notification to the cardholder to open the application and complete authentication. In this case, the issuer must use the Customer Service API to share the authentication result.
		7 = Masked voice call phone number
		8 = Push Notification. The issuer must share the activation code to the cardholder using a push notification.
		9 = EMV 3DS (Identity Check)
	Activation Method Value	ans...164 Refer to the following examples

Subfield Name	Attributes	Values/Comments
Reason Code	ans-6	The reason the account holder is being authenticated. The reason could impact the generation and validation of the code in various ways including but not limited to the time period of validity of the code and the number of attempts allowed to validate the code. New reason codes can be added at any time and should not result in a failure. Bit Position Meaning 01 "ADD_CARD" - The account holder is being authenticated to verify account ownership at the time of adding a card in the consent requestor system Note: In the context of MDES - token might be in an active state. 02 "VERIFY_ACCOUNT" - The account holder is being authenticated to verify account ownership to use a service 03 OTHER - Requires Token Binding 04 - PAYMENT_TRANSACTION – The account holder is being authenticated to initiate a payment transaction 05 - TOKEN_BINDING – The account holder is being authenticated to bind a token to bind ID. 06 - ACCOUNT_BINDING – The account holder is being authenticated to bind associated account with bind id (device) The data of this field is a hex-encoded bitmap, whereby each bit corresponds to a specific Reason Code. The bitmap is big-endian with the least significant bit corresponding to Reason Code 01, with the next least significant bit corresponding to Reason Code 02 and so on. For example, if Reason Codes 01 and 02 were encoded, the bitmap would be 0000000000000000000011 and the hex value of this field would be 000003. NOTE: This parameter is present only if an issuer has opted for Authentication Service for Click to Pay and Secure Card on File Tokens.
Token Unique Reference	ans-48, left-justified, padded with spaces	Service-allocated unique reference to the token

Table 1682: Examples

1(###) ### 4567

1 = Masked mobile phone number

The 1 will be followed by the masked mobile phone number.

2a***d@anymail.com

2 = Masked email address

The 2 will be followed by the consumer's masked email address (the issuer will mask according to their own format).

3(555) 123 4567

3 = Automated call center phone number

The 3 will be followed by the phone number. This phone number is not masked.

4(555) 123 8901

4 = Call center phone number

The 4 will be followed by the phone number. This phone number is not masked.

5http://www.anybank.com

5 = Website

The 5 will be followed by the issuer's website URL.

6PUSH_NOTIFICATION

6 = Mobile app

The 6 will be followed by the PUSH_NOTIFICATION, that informs issuer that send a push notification to cardholder mobile application instance.

7(###) ### 2345

7 = Masked voice call phone number

The 7 will be followed by the masked voice call phone number.

8

8 = Push notification

No value is require for 8. Issuer must send a push notification to cardholder mobile banking application with authentication code.

Authorization Request Response/0110 for ACN

All mandatory Authorization Request Response/0110 data elements apply.

Table 1683: Authorization Request Response/0110 for ACN

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code)	ME	ME	•	Must be the same value as the original Authorization Request/0100
DE 4 (Amount, Transaction)	ME	ME	•	Must be the same value as the original Authorization Request/0100
DE 39 (Response Code)	M	•	•	00 = Approved

NOTE: DE 124 is not present.

Tokenization Complete Notification (TCN)

Following is a list of the data elements, values, and subfields applicable for Tokenization Complete Notification (TCN).

- [Authorization Request/0100 for TCN](#)
- [DE 124 Subfields in Authorization Request/0100 for TCN](#)
- [Authorization Request Response/0110 for TCN](#)

Authorization Request/0100 for TCN

All mandatory Authorization Request/0100 data elements apply.

Table 1684: Authorization Request/0100 for TCN

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 2 (Primary Account Number)	•	X	M	Cardholder's primary account number
DE 3 (Processing Code)	•	X	M	00 = Purchase
DE 4 (Amount, Transaction)	•	X	M	Will be zero
DE 14 (Date, Expiration)	•	X	C	Cardholder's primary account expiration date
DE 22 (POS Entry Mode)	•	X	M	Subfield 1 (POS Terminal PAN Entry Mode) = 01 (PAN manual entry) Subfield 2 (POS Terminal PIN Entry Mode) = 0 (Unspecified or unknown)

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 35 (Track 2 Data)	•	X	C	The Authorization Platform creates and provides Track 2 data for Maestro tokenization messages.
DE 43 (Card Acceptor Name/Location for All Transactions), subfield 5 (Card Acceptor State or Country Code)	•	X	M	The acquirer in these messages is the Consumer Facing Entity Name (CFEN), with a three alphanumeric Country Code corresponding to "USA". For Local-Use-Only account ranges, Mastercard overrides this value to provide the three alphanumeric Country Code of the issuer.
DE 48 Transaction Category Code	•	X	C	T (Phone, Mail, or Electronic Commerce Order)
DE 48 (Additional Data—Private Use), subelement 23 (Payment Initiation Channel)	•	X	C	Value indicating the type of device for which the consumer is requesting tokenization of a primary account number.
DE 48, subelement 26 (Wallet Program Data), subfield 1 (Wallet Identifier)	•	X	C	Contains the identifier associated with the Wallet Provider.
DE 48 (Additional Data—Private Use), subelement 33 (PAN Mapping File Information), subfield 1 (Account Number Indicator), subfield 2 (Account Number), subfield 3 (Expiration Date), subfield 5 (Token Assurance Method), subfield 6 (Token Requestor ID), and subfield 8 (Storage Technology)	•	X	C	Subfield 1 indicates the type of PAN mapping account. Subfield 2 indicates the PAN mapping account number. Subfield 3 indicates the expiration date of the PAN mapping account. Subfield 5 indicates Token Assurance Method. Subfield 6 contains the ID assigned by the Token Service Provider to the Token Requestor. Subfield 8 indicates Storage Technology of the token.

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 49 (Currency Code, Transaction)	•	X	M	Value indicating the local currency of the acquirer or source location of the transaction. The acquirer in these messages is Mastercard, with a three digits Currency Code of "840" corresponding to USD as Currency Code used in USA. For Local-Use-Only account ranges, Mastercard overrides this value to provide the issuer's three digits Currency Code. This corresponds to the local currency used in that country.
DE 56 (Payment Account Data)	•	X	C	This is only for MDES Tokenization Complete Notifications.
DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status)	•	X	M	9 = Tokenization Request/Notification
DE 61 (Additional POS Data), subfield 13 (POS Country Code)	•	X	M	Indicates the country of the POS location (not the acquirer location) using ISO-specified codes. The POS Country Code is set to "840", corresponding to USA. For Local-Use-Only account ranges, Mastercard overrides this value to provide the issuer's three digits Country Code.
DE 124 (Member-defined Data)	•	X	M	See layout for DE 124.

DE 124 Subfields in Authorization Request/0100 for TCN

Absolute positioning of data in DE 124 subfields is required. Subfields not containing values will be populated with spaces when they are followed by subfields that contain values. If one or more subfields at the end of the data element do not contain a value, the subfield(s) will not be present and the total length of DE 124 will be reduced by the subfield length(s).

Table 1685: DE 124 Subfields in Authorization Request/0100 for TCN

Subfield Name	Attributes	Values/Comments
Message Type	an-2	TC = Tokenization Complete Notification (TCN)
Correlation ID	an-14	Identifier assigned by Mastercard to associate related tokenization request/notification messages.

Subfield Name	Attributes	Values/Comments
Number of Active Tokens for the Primary Account Number	ans-2, leading zeros	Number of active or suspended tokens for the primary account number provisioned to a consumer. Space-filled when requested token digitized to a server. NOTE: Space-filled means if data is not available then length of that field is filled with space; this is specific to ISO message not for API.
Issuer Product Configuration ID	ans-10	The unique product configuration identifier provided by the issuer that identifies a particular set of card art, texts and other product related data, provided during the issuer enablement or maintenance process. This ID is case-sensitive.
Consumer Language	a-2	Language preference selected by the consumer.
Device Name	ans-20, left-justified, padded with spaces	Name that the consumer has associated to the device with the Wallet Provider.
Final Tokenization Decision	ans-1	The final tokenization decision that was used in the tokenization of the card: 1 = Approve 2 = Approve, but require additional authentication
Final Tokenization Decision Indicator	ans-1	The element of the Service that was responsible for determining the final tokenization decision: 1 = Tokenization Eligibility Response 2 = Tokenization Authorization Response 3 = Issuer predefined tokenization rules 4 = Mobile Application
T&C Identifier	ans-32, left-justified, padded with spaces	Identifier associated with the version of terms and conditions accepted by the consumer.
T&C Date and Time	ans-10	Date and time that the consumer accepted the terms and conditions of the Service specified in UTC units. Format: YYMMDDhhmm
Number of Activation Attempts	ans-1	Number of activation code entry attempts by the cardholder. Space-filled when DE 124, subfield 14 (Token Type) value is F.
Token Unique Reference	ans-48, left-justified, padded with spaces	Service-allocated unique reference to the token.
Primary Account Number Unique Reference	ans-48, left-justified, padded with spaces	Service-allocated unique reference to the tokenized Primary Account Number at the wallet level.

Subfield Name	Attributes	Values/Comments
Token Type	an-1	Contains a value indicating the type of token present in DE 48, subelement 33, subfield 2 (Account Number) in an 0100 Tokenization Complete Notification message. C = Mastercard Cloud-based Payments F = Static Token S = Embedded Secure Element

Authorization Request Response/0110 for TCN

All mandatory Authorization Request Response/0110 data elements apply.

Table 1686: Authorization Request Response/0110 for TCN

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code)	ME	ME	•	Must be the same value as the original Authorization Request/0100.
DE 4 (Amount, Transaction)	ME	ME	•	Must be the same value as the original Authorization Request/0100.
DE 39 (Response Code)	M	•	•	00 = Approved

NOTE: DE 124 is not present.

Tokenization Event Notification (TVN)

Following is a list of the data elements, values, and subfields applicable for Tokenization Event Notification (TVN).

- [Authorization Request/0100 for TVN](#)
- [DE 124 Subfields in Authorization Request/0100 for TVN](#)
- [Authorization Request Response/0110 for TVN](#)

Authorization Request/0100 for TVN

All mandatory Authorization Request/0100 data elements apply.

Table 1687: Authorization Request/0100 for TVN

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 2 (Primary Account Number)	•	X	M	Cardholder's primary account number
DE 3 (Processing Code)	•	X	M	00 = Purchase

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 4 (Amount, Transaction)	•	X	M	Will be zero
DE 14 (Date, Expiration)	•	X	C	Cardholder's primary account expiration date
DE 22 (POS Entry Mode)	•	X	M	Subfield 1 (POS Terminal PAN Entry Mode) = 01 (PAN manual entry) Subfield 2 (POS Terminal PIN Entry Mode) = 0 (Unspecified or unknown)
DE 35 (Track 2 Data)	•	X	C	The Authorization Platform creates and provides Track 2 data for Maestro tokenization messages.
DE 43 (Card Acceptor Name/ Location for All Transactions), subfield 5 (Card Acceptor State or Country Code)	•	X	M	The acquirer in these messages is the Consumer Facing Entity Name (CFEN), with a three alphanumeric Country Code corresponding to "USA". For Local-Use-Only account ranges, Mastercard overrides this value to provide the three alphanumeric Country Code of the issuer.
DE 48 Transaction Category Code	•	X	C	T (Phone, Mail, or Electronic Commerce Order)
DE 48 (Additional Data - Private Use) subelement 06 (Advanced Digital Transaction Data 1) subfield 1 (Binding ID)	•	X	C	Subfield 1 contains the Binding Id - The unique and fixed identifier to a component of data bound to the credential.
DE 48 (Additional Data - Private Use), subelement 23 (Payment Initiation Channel)	•	X	C	Value indicating the type of device for which the consumer is requesting tokenization of a primary account number.
DE 48, subelement 26 (Wallet Program Data), subfield 1 (Wallet Identifier)	•	X	C	Contains the identifier associated with the Wallet Provider.

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 48 (Additional Data - Private Use), subelement 33 (PAN Mapping File Information), subfield 1 (Account Number Indicator), subfield 2 (Account Number), subfield 3 (Expiration Date), subfield 5 (Token Assurance Method), subfield 6 (Token Requestor ID), and subfield 8 (Storage Technology)	•	X	C	Subfield 1 indicates the type of PAN mapping account. Subfield 2 indicates the PAN mapping account number. Subfield 3 indicates the expiration date of the PAN mapping account. Subfield 5 indicates Token Assurance Method. Subfield 6 contains the ID assigned by the Token Service Provider to the Token Requestor. Subfield 8 indicates Storage Technology of the token.
DE 49 (Currency Code, Transaction)	•	X	M	Value indicating the local currency of the acquirer or source location of the transaction. The acquirer in these messages is Mastercard, with a three digits Currency Code of "840" corresponding to USD as Currency Code used in USA. For Local-Use-Only account ranges, Mastercard overrides this value to provide the issuer's three digits Currency Code. This corresponds to the local currency used in that country.
DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status)	•	X	M	9 = Tokenization Request/Notification
DE 61 (Additional POS Data), subfield 13 (POS Country Code)	•	X	M	Indicates the country of the POS location (not the acquirer location) using ISO-specified codes. The POS Country Code is set to "840", corresponding to USA. For Local-Use-Only account ranges, Mastercard overrides this value to provide the issuer's three digits Country Code.
DE 104 (Digital Payment Data), subelement 004 (Digital Service Provider)	•	X	C	Subfield 001 (Digital Service Provider ID 1) Subfield 002 (Digital Service Provider ID 2)

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 104 (Digital Payment Data), subelement 005 ((Digital Authentication Data))	•	X	C	Subfield 001 (Cardholder Verification) Subfield 002 (Authentication Factors)
DE 124 (Member-defined Data)	•	X	M	See layout for DE 124.

DE 124 Subfields in Authorization Request/0100 for TVN

Absolute positioning of data in DE 124 subfields is required. Subfields not containing values will be populated with spaces when they are followed by subfields that contain values. If one or more subfields at the end of the data element do not contain a value, the subfield(s) will not be present and the total length of DE 124 will be reduced by the subfield length(s).

Table 1688: DE 124 Subfields in Authorization Request/0100 for TVN

Subfield Name	Attributes	Values/Comments
Message Type	an-2	TV = Tokenization Event Notification (TVN)
Correlation ID	an-14	Identifier assigned by Mastercard to associate related tokenization request/notification messages.
Tokenization Event Indicator	an-1	Value indicating the event that has occurred on the Mastercard Digital Enablement Service for the token. 1 = Funding Account Update 2 = Payment Channel Preference Updated 3 = Deactivate 4 = Deleted from consumer device 5 = Authentication Performed NOTE: Only applicable for wallets supporting the enrollment using a physical card in a physical store. 6 = Suspend 7 = Resume 8 = Tokenization Exception Event 9 = Replacement (token re-digitization) - a - Token Binding - b - Account Binding

Subfield Name	Attributes	Values/Comments
Tokenization Event Reason Code	an-2	If Tokenization Event Indicator contains value 8 (Tokenization Exception Event), this field contains a value indicating the event reason. If the Tokenization Event Indicator contains a value of 3 (Deactivate), 6 (Suspend), or 7 (Resume), this field will not be present. 00 = Activation code retries exceeded 01 = Activation code expired or invalidated 02 = Activation code entered incorrectly by cardholder
Event Requestor	ans-1	If the Tokenization Event Indicator contains a value of 3 (Deactivate), 6 (Suspend), or 7 (Resume), this field will contain a value indicating the party that requested the event. If the Tokenization Event Indicator contains a value of 8 (Tokenization Exception Event) this field will be space filled. 0 = Indicates the Tokenization Event was requested by the Wallet Provider or Token Requestor 1 = Indicates the Tokenization Event was requested by the Funding Account issuer 2 = Indicates the Tokenization Event was requested by the Cardholder 3 = Indicates the Tokenization Event was requested in relation to a systematic event triggered by Mobile PIN Validation security (applicable to Tokenization Event Indicator value of 6 (Suspend), or 7 (Resume) only) 4 = Indicates the Tokenization Event was requested in relation to a systematic event triggered by the deletion of tokens, which did not have a transaction in the set (default or issuer-defined) number of days and were deleted by consumer app. 5 = Reserved for future use 6 = Reserved for future use 7 = Reserved for future use 8 = Reserved for future use 9 = Reserved for future use
Token Unique Reference (TUR)	ans-48, left justified, padded with spaces	Service-allocated unique reference to the token.

Subfield Name	Attributes	Values/Comments
Cardholder Disabled Payment Channel (India only)	an-3	This field is present only if Tokenization Event Indicator contains value 2 (Payment Channel Preference Updated). The data of this field is a hex-encoded bitmap, whereby each bit corresponds to a specific payment channel disabled by the cardholder. The bitmap is big-endian with the least significant bit corresponding to Payment Channel 01, with the next least significant bit corresponding to Payment Channel 02 and so on. For example, if Payment Channels 01, 03, and 05 were encoded, the bitmap would be 0000 0001 0101 and the hex value of this field would be 15. MDES will pass 0 if all payment channels are enabled. Bit Position and Payment Channel 1 = Contactless 2 = Merchant Presented QR (MPQR) 3 = Customer Presented QR (CPQR) 4 = Digital Secure Remote Payment (DSRP) In-App 5 = Person to Person (P2P) 6 = Dynamic Magnetic Stripe Data (DMSD) 7 to 12 = Reserved for future use
Binding Status	n-1	If the Tokenization Event Indicator contains a value of a (Token Binding), b (Account Binding) this field will contain a value indicating the status of Binding 1 = Active 2 = Deactivated (deleted)

Authorization Request Response/0110 for TVN

All mandatory Authorization Request Response/0110 data elements apply.

Table 1689: Authorization Request Response/0110 for TVN

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code)	ME	ME	•	Must be the same value as the original Authorization Request/0100
DE 4 (Amount, Transaction)	ME	ME	•	Must be the same value as the original Authorization Request/0100

Data Element ID and Name	Org	Sys	Dst	Values/Comments
DE 39 (Response Code)	M	.	.	00 = Approved

NOTE: DE 124 will not be present.

DE 120 Layout for MCC106 Mastercard Digital Enablement Service

The following tables provides the technical details of DE 120 (Record Data) for Token Update and PAN Update - Deactivate/Suspend/Resume Token.

- [Issuer File Update Request/0302 - Maintenance \(Token/PAN Update\)](#)
- [Issuer File Update Request Response/0312 - Issuer Token Maintenance Response \(Token/PAN Update\)](#)
- [Token Update](#)
- [PAN Update - Deactivate/Suspend/Resume Token](#)

NOTE: For In Control customers, check the status of the In Control VCN before updating a token status. Refer to the [Mastercard In Control for Commercial Payments User and Company Guide](#).

Issuer File Update Request/0302 - Maintenance (Token/PAN Update)

Following is a list of the data elements and values applicable to this message type. All mandatory Issuer File Update Request/0302 data elements apply.

Table 1690: Issuer File Update Request/0302 - Maintenance (Token/PAN Update)

Data Element and Name	Org	Sys	Dst	Values/Comments
DE 2 (Primary Account Number)	M	M	.	Contains the cardholder's primary account number
DE 91 (Issuer File Update Code)	M	M	.	Contains the Issuer File Update function code. Valid value will be 2 (Update).
DE 101 (File Name)	M	M	.	MCC106 (PAN Mapping File)
DE 120 (Record Data)	M	M	.	Contains token and PAN information. Refer to the layout of DE 120.

Token Update

The following table provides the technical details of DE 120 (Record Data) when DE 101 (File Name) contains the value MCC106 (Mastercard Digital Enablement Service Mapping File PAN Update) in the Issuer File Update Request/0302 message.

Table 1691: Token Update

Field ID	Field Name	Attributes	Values/Comments
1	Mapping File Indicator	an-1	M = Mastercard Digital Enablement Service Tokens excluding Static Tokens A = All Mastercard Digital Enablement Service Tokens including Static Tokens Best practice for the issuer to use value A.
2	Replacement PAN	ans-19	Number replacing the number that is embossed, encoded, or both on the card (the primary account number). Customers may input only account numbers for BINs assigned to the associated customer ID assigned by Mastercard. The issuer has the option to provide a replacement primary account number for association to the token. Format: Left-justified, with trailing spaces
3	Replacement PAN Expiration Date	n-4	Expiration date that is associated with the number replacing the number that is embossed, encoded, or both on the card that represents the cardholder primary account number. If the issuer has provided a replacement primary account number, this field contains the expiration date that is associated with the replacement primary account number. If the issuer has not provided a replacement primary account number, this field contains the expiration date of the existing primary account number. Format: YYMM

Field ID	Field Name	Attributes	Values/Comments
4	Primary Account Card Sequence Number	ans-3	If the issuer has provided a replacement primary account number, this field will contain the card sequence number associated with the replacement primary account number. If the issuer has not provided a replacement primary account number, this field will contain the card sequence number associated with the original primary account number. Format: Left-justified, padded with leading zeros.
5	Notify Wallet Service Provider Indicator	n-1	If the issuer has provided a replacement primary account number and replacement primary account number expiration date, this field indicates whether the Wallet Provider should be notified of the change. Values: • 0 = Update token mapping information and notify the Wallet Provider with the primary account number information • 1 = Update token mapping information, but do not notify the Wallet Provider with the primary account number information • 2 = Do not update token mapping information, but do update the Wallet Provider with the primary account number information
6	Token—If replacing a specific token	ans-19	Surrogate value for a PAN that is consistent with ISO message requirements and is a 13 to 19-digit numeric value that passes basic validation rules of an account number, including the LUHN check. Format: Left-justified, with trailing spaces If field not present, update shall apply to all token to PAN mappings associated with the PAN.

PAN Update - Deactivate/Suspend/Resume Token

The following table provides the technical details of DE 120 (Record Data) when DE 101 (File Name) contains the value MCC106 (Mastercard Digital Enablement Service Mapping File PAN Update) in the Issuer File Update Request/0302 message.

Table 1692: PAN Update - Deactivate/Suspend/Resume Token

Field ID	Field Name	Attributes	Values/Comments
1	Mapping File Indicator	an-1	M = Mastercard Digital Enablement Service tokens excluding Static Tokens. A = All Mastercard Digital Enablement Service tokens including Static Tokens. Best practice for the issuer to use value A.
2	Action Required	an-1	S = Suspend token D = Deactivate token C = Resume token
3	Notify Wallet Service Provider indicator	n-1	This field indicates whether the Wallet Provider should be notified of the change. Value: 0 = Update token mapping information and notify the Wallet Provider.
4	Token - If updating a specific token	ans-19	Surrogate value for a PAN that is consistent with ISO message requirements and is a 13-19-digit numeric value that passes basic validation rules of an account number, including the LUHN check. Format: Left-justified, with trailing spaces. If not present, do across the board switch - based on Field 1. If field not present, update shall apply to all token to PAN mappings associated with the PAN.
5	Date-Last-Update-Activity	n-6	Format: MMDDYY (Month, Day, Year) Format: Spaces must be provided if there are fields present after subfield 6.
6	Time-Last-Update-Activity	n-4	Format: hhmm (hour, minute) Format: Spaces must be provided if there are fields present after subfield 7.

Issuer File Update Request Response/0312 - Issuer Token Maintenance Response (Token/PAN Update)

Following is a list of the data elements and values applicable to this message type. All mandatory Issuer File Update Request Response/0312 data elements apply.

Table 1693: Issuer File Update Request Response/0312 - Issuer Token Maintenance Response (Token/PAN Update)

Data Element and Name	Org	Sys	Dst	Values/Comments
DE 2 (Primary Account Number)	•	ME	ME	Must be the same value as in the original Issuer File Update Request/0302.
DE 39 (Response Code)	•	M	M	Indicates disposition of Issuer File Update Request/0302.
DE 44 (Additional Response Data)	•	C	C	May contain additional response information, based on DE 39. Error messages should spell out field in error (ex: 120XXX)
DE 91 (Issuer File Update Code)	•	ME	ME	Must be the same value as in the original Issuer File Update Request/0302.
DE 101 (File Name)	•	ME	ME	Must be the same value as in the original Issuer File Update Request/0302.
DE 120 (Record Data)	•	ME	ME	Must be the same value as in the original Issuer File Update Request/0302.

DE 120 Layout for Administrative Advice/0620 Issuer Token Notification Advice

The following tables provides the technical details of DE 120 (Record Data) for Issuer Token Notification Advice for Activation Code Notification, Tokenization Complete Notification and Tokenization Event Notification.

- [Activation Code Notification](#)
- [Tokenization Complete Notification](#)
- [Tokenization Event Notification](#)

Activation Code Notification

The following table provides the technical details of DE 120 (Record Data) when DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code) contains the value 141 (Mastercard Digital Enablement Service Advice to Issuer) and DE 60, subfield 2 (Advice Detail Code) contains the value 0250 (Activation Code Notification) in the Administrative Advice/0620 message.

Absolute positioning of data in DE 120 subfields is required. Subfields not containing values will be populated with spaces when they are followed by subfields that contain values. If one or more subfields at the end of the data element do not contain a value the subfields will not be present and the total length of DE 120 will be reduced by the subfield lengths.

Table 1694: Activation Code Notification

Field ID	Field Name	Attributes	Values/Comments
1	Primary Account Number	an-19	Cardholder's primary account number
2	Primary Account Number Expiry Date	n-4	Cardholder's primary account number expiration date
3	Correlation ID	an-14	The identifier assigned by Mastercard to associate related tokenization request/notification messages.
4	Activation Code	ans-8, left-justified, padded with spaces	Activation code assigned by Mastercard that will be provided to the issuer for delivery to the consumer to complete the tokenization process.
5	Activation Code Expiration Date and Time	n-10	The date and time that the Activation Code expires specified in UTC units. Format: YYMMDDhhmm.
6	Customer's Activation Method Preference	ans-165, left-justified, padded with spaces	This field contains the activation method selected by the consumer, if only one was offered by the issuer, then that activation method will be present. There will be only one method contained within this field. This field will only be present if the cardholder made a selection.
	Name	Attributes	Values/Comments
	Activation Method Type	n-1	1 = Masked mobile phone number 2 = Masked email address 3 = Automated call center phone number 4 = Call center phone number 5 = Website

Field ID	Field Name	Attributes	Values/Comments
			6 = Mobile application. If the value is PUSH_NOTIFICATION, then the issuer should send a push notification to the cardholder to open the application and complete authentication. In this case, the issuer must use the Customer Service API to share the authentication result. 7 = Masked voice call phone number 8 = Push Notification. The issuer must share the activation code with the cardholder using a push notification. 9 = EMV 3DS (Identity Check)
			Activation Method Value ans...164 See examples below.
7	Token Requestor ID	n-11	Contains the identifier assigned by the Token Service Provider to the Token Requestor.
8	Wallet ID	ans-3	Contains the identifier associated with the Wallet Provider. The presence of this field is conditional.
9	Device Type	ans-2	Indicates the type of device used. The presence of this field is conditional.

Field ID	Field Name	Attributes	Values/Comments
10	Reason Code	ans-6	The reason the account holder is being authenticated. The reason could impact the generation and validation of the code in various ways including but not limited to the time period of validity of the code and the number of attempts allowed to validate the code. New reason codes can be added at any time and should not result in a failure. Bit Position Meaning 01 ADD_CARD - The account holder is being authenticated to verify account ownership at the time of adding a card in the consent requestor system NOTE: In the context of MDES - token might be in an active state. 02 VERIFY_ACCOUNT - The account holder is being authenticated to verify account ownership to use a service 03 OTHER - 04 PAYMENT_TRANSACTION - The account holder is being authenticated to initiate a payment transaction 05 TOKEN_BINDING - The account holder is being authenticated to bind a token to bind ID. 06 ACCOUNT_BINDING - The account holder is being authenticated to bind associated account with bind ID (device). The data of this field is a hex-encoded bitmap, whereby each bit corresponds to a specific Reason Code. The bitmap is big-endian with the least significant bit corresponding to Reason Code 01, with the next least significant bit corresponding to Reason Code 02 and so on. For example, if Reason Codes 01, and 02 the bitmap would be 00000000000000000000000011 and the hex value of this field would be 000003. NOTE: This parameter is present only if an issuer has opted for Authentication Service for Click to Pay and Secure Card on File Tokens.
11	Token Unique Reference	ans-48, left-justified, padded with spaces	Service-allocated unique reference to the token.

Field ID	Field Name	Attributes	Values/Comments
12	authRequestCorrelationId	ans-64	Value linking different authentication messages.

Table 1695: Examples

"1(###) ### 4567"

1 = Masked mobile phone number

The "1" will be followed by the masked mobile phone number.

"2a***d@anymail.com"

2 = Masked email address

The "2" will be followed by the consumer's masked email address (the issuer will mask according to their own format).

"3(555) 123 4567"

3 = Automated call center phone number

The "3" will be followed by the phone number. This phone number is not masked.

"4(555) 123 8901"

4 = Call center phone number

The "4" will be followed by the phone number. This phone number is not masked.

"5http://www.anybank.com"

5 = Website

The "5" will be followed by the issuer's website URL.

"6com.anybank.mobileapp"

6 = Mobile app

The "6" will be followed by the issuer's mobile app information, the content of which depends upon the mobile device operating system.

"7(###) ### 2345"

7 = Masked voice call phone number

The "7" will be followed by the masked voice call phone number.

Tokenization Complete Notification

The following table provides the technical details of DE 120 (Record Data) when DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code) contains the value 141 (Mastercard Digital Enablement Service Advice to Issuer) and DE 60, subfield 2 (Advice Detail Code) contains the value 0251 (Tokenization Complete Notification) in the Administrative Advice/0620 message.

Absolute positioning of data in DE 120 subfields is required. Subfields not containing values will be populated with spaces when they are followed by subfields that contain values. If one or more subfields at the end of the data element do not contain a value the subfield(s) will not be present and the total length of DE 120 will be reduced by the subfield length(s).

Table 1696: Tokenization Complete Notification

Field ID	Field Name	Attributes	Values/Comments
1	Token	an-19	Surrogate value for a PAN that is consistent with ISO message requirements and is a 13 to 19-digit numeric value that passes basic validation rules of an account number, including the LUHN check. Format: Left-justified, with trailing spaces
2	Token Expiration Date	n-4	Expiration date associated with the token.
3	PAN	an-19	Number that is embossed, encoded, or both on the card (also known as the primary account number). Customers may input only account numbers for BINs assigned to the associated customer ID assigned by Mastercard. Format: Left-justified, with trailing spaces
4	PAN Expiration Date	n-4	Expiration date that is embossed, encoded, or both on the card that represents the cardholder primary account number (primary account number). Format: YYMM
5	Token Service Provider Identification	a-1	M = Mastercard Digital Enablement Service
6	Token Assurance Method	n-2	Assurance method assigned to the token (value between 00 and 99).
7	Token Requestor ID	n-11	The ID assigned by Mastercard to the token requestor.

Field ID	Field Name	Attributes	Values/Comments
8	Contactless Usage	n-1	Contains value indicating if the token is permitted for use in contactless transactions. Values: 0 = Token is not permitted for use in contactless transactions 1 = Token is permitted for use in contactless transactions
9	Card on File Electronic Commerce Usage	n-1	Contains value indicating if the token is permitted for use in card on file electronic commerce transactions. Values: 0 = Token is not permitted for use in Card on File electronic commerce transactions 1 = Token is permitted for use in card on file electronic commerce transactions
10	Mobile/Digital Wallet Electronic Commerce Usage	n-1	Contains value indicating if the token is permitted for use in mobile/digital wallet electronic commerce transactions. Values: 0 = Token is not permitted for use in mobile/digital wallet electronic commerce transactions 1 = Token is permitted for use in mobile/digital wallet electronic commerce transactions
11	Correlation ID	an-14	Identifier assigned by Mastercard to associate related tokenization request/notification messages.
12	Number of Active Tokens for the Primary Account Number	ans-2, leading zeros	Number of active or suspended tokens for the primary account number digitized to consumer devices. Space-filled when token present in DE 48, subelement 33, subfield 2 (Account Number) in an 0100 Tokenization Complete Notification message is provisioned to a server. Presence of this field is conditional.

Field ID	Field Name	Attributes	Values/Comments
13	Issuer Product Configuration ID	ans-10	The unique product configuration identifier applied to the token, as provided by the issuer, identifying a particular set of card art, texts, and other product related data, that were provided during the issuer enablement or maintenance process. Presence of this field is conditional. This ID is case-sensitive.
14	Consumer Language	a-2	Language preference selected by the consumer. Presence of this field is conditional.
15	Device Name	ans-20, left-justified, padded with spaces	Name that the consumer has associated to the device with the Wallet Provider. Presence of this field is conditional.
16	Final Tokenization Decision	ans-1	The final tokenization decision that was used in the tokenization of the card. • 1 = Approve • 2 = Approve but requires additional authentication Presence of this field is conditional.
17	Final Tokenization Decision Indicator	ans-1	The element of the Service that was responsible for determining the final tokenization decision: • 1 = Tokenization Eligibility Response • 2 = Tokenization Authorization Response • 3 = Issuer pre-defined tokenization rules • 4 = Mobile Application Presence of this field is conditional.
18	T&C Identifier	ans-32, left justified, padded with spaces	Identifier associated with the version of terms and conditions accepted by the consumer. Presence of this field is conditional.
19	T&C Date and Time	ans-10	Date and time that the consumer accepted the terms and conditions of the Service, specified in UTC units. Format: YYMMDDhhmm

Field ID	Field Name	Attributes	Values/Comments
20	Number of Activation Attempts	ans-1	Number of activation code entry attempts by the cardholder. Space-filled when DE124, SF14 (Token Type) value is F. Presence of this field is conditional.
21	Token Unique Reference	ans-48, left justified, padded with spaces	Service-allocated unique reference to the token.
22	Primary Account Number Unique Reference	ans-48, left justified, padded with spaces	Service-allocated unique reference to the tokenized Primary Account Number at the wallet level.
23	Token Type	an-1	Contains a value indicating the type of token present in DE 120, field 1 (Token) in an 0620 Tokenization Complete Notification message. • C = Mastercard Cloud-Based Payments • F = Static Token • S = Embedded Secure Element
24	Wallet ID	ans-3	Contains the identifier associated with the Wallet Provider. Presence of this field is conditional.
25	Device Type	ans-2	Indicates the type of device used. Presence of this field is conditional.
26	Storage Technology	an-2	Contains a value indicating the storage technology of the token. Presence of this field is conditional.

NOTE: The Token Requestor ID is already being included in DE 120, field ID 7 of the Administrative Advice/0620—Issuer Token Notification Advice for Tokenization Complete Notification message.

Tokenization Event Notification

The following table provides the technical details of DE 120 (Record Data) when DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code) contains the value 141 (Mastercard Digital Enablement Service Advice to Issuer) and DE 60, subfield 2 (Advice Detail Code) contains the value 0252 (Tokenization Event Notification) in the Administrative Advice/0620 message.

Absolute positioning of data in DE 120 subfields is required. Subfields not containing values will be populated with spaces when they are followed by subfields that contain values. If one or more subfields at the end of the data element do not contain a value the subfield(s) will not be present and the total length of DE 120 will be reduced by the subfield length(s).

Table 1697: Tokenization Event Notification

Field ID	Field Name	Attributes	Values/Comments
1	Primary Account Number	an-19	Cardholder's primary account number
2	Primary Account Number Expiry Date	n-4	Cardholder's primary account expiration date
3	Token (PAN mapping file information)	an-19	Token
4	Token Expiration Date (PAN mapping file information)	an-4	Token expiration date
5	Token Service Provider Identification	an-1	M = Mastercard Digital Enablement Service
6	Correlation ID	an-14	Identifier assigned by Mastercard to associate related tokenization request/notification messages
7	Tokenization Event Indicator	n-1	Value indicating the event that has occurred on the Mastercard Digital Enablement Service for the token 1 = Funding Account Update 2 = Payment Channel Preference Updated (India only) 3 = Deactivate 4 = Deleted from consumer device 5 = Authentication Performed NOTE: Only applicable for wallets supporting the enrollment using a physical card in a physical store. 6 = Suspend 7 = Resume 8 = Tokenization Exception Event 9 = Replacement - a = Token Binding - b = Account Binding

Field ID	Field Name	Attributes	Values/Comments
8	Tokenization Event Reason Code	an-2	If the Tokenization Event Indicator contains value 8 (Tokenization Exception Event), this field contains a value indicating the event reason. If the Tokenization Event Indicator contains a value of 3 (Deactivate), 6 (Suspend), or 7 (Resume), this field will not be present. 00 = Activation code retries exceeded 01 = Activation code expired or invalidated 02 = Activation code entered incorrectly by cardholder
9	Contactless Usage	n-1	Contains value indicating if the token is permitted for use in contactless transactions. Values: 0 = Token is not permitted for use in contactless transactions 1 = Token is permitted for use in contactless transactions
10	Card on File Electronic Commerce Usage	n-1	Contains value indicating if the token is permitted for use in card on file electronic commerce transactions. Values: 0 = Token is not permitted for use in Card on File electronic commerce transactions 1 = Token is permitted for use in card on file electronic commerce transactions
11	Mobile/Digital Wallet Electronic Commerce Usage	n-1	Contains value indicating if the token is permitted for use in mobile/digital wallet electronic commerce transactions. Values: 0 = Token is not permitted for use in mobile/digital wallet electronic commerce transactions 1 = Token is permitted for use in mobile/digital wallet electronic commerce transactions

Field ID	Field Name	Attributes	Values/Comments
12	Event Requestor	ans-1	If the Tokenization Event Indicator contains a value of 3 (Deactivate), 6 (Suspend), or 7 (Resume), this field will contain a value indicating the party that requested the event. If the Tokenization Event Indicator contains a value of 8 (Tokenization Exception Event) this field will be space filled. 0 = Indicates the Tokenization Event was requested by the Wallet Provider or Token Requestor 1 = Indicates the Tokenization Event was requested by the Funding Account issuer 2 = Indicates the Tokenization Event was requested by the Cardholder 3 = Indicates the Tokenization Event was requested in relation to a systematic event triggered by Mobile PIN Validation security (applicable to Tokenization Event Indicator value of 6 (Suspend), or 7 (Resume) only) 4 = Indicates the Tokenization Event was requested in relation to a systematic event triggered by the deletion of tokens, which did not have a transaction in the set (default or issuer-defined) number of days and were deleted by consumer app. 5 = Reserved for future use 6 = Reserved for future use 7 = Reserved for future use 8 = Reserved for future use 9 = Reserved for future use
13	Token Requestor ID	n-11	Contains the identifier assigned by the Token Service Provider to the Token Requestor.
14	Wallet ID	ans-3	Contains the identifier associated with the Wallet Provider. Presence of this field is conditional.
15	Device Type	ans-2	Indicates the type of device used. Presence of this field is conditional.
16	Token Unique Reference (TUR)	ans-48, left justified, padded with spaces	Service-allocated unique reference to the token.

Field ID	Field Name	Attributes	Values/Comments
17	Cardholder Disabled Payment Channel (India only)	an-3	This field is present only if Tokenization Event Indicator contains value 2 (Payment Channel Preference Updated). The data of this field is a hex-encoded bitmap, whereby each bit corresponds to a specific payment channel disabled by the cardholder. The bitmap is big-endian with the least significant bit corresponding to Payment Channel 01, with the next least significant bit corresponding to Payment Channel 02 and so on. For example, if Payment Channels 01, 03, and 05 were encoded, the bitmap would be 0000 0001 0101 and the hex value of this field would be 15. MDES will pass 0 if all payment channels are enabled. Bit Position and Payment Channel 1 = Contactless 2 = Merchant Presented QR (MPQR) 3 = Customer Presented QR (CPQR) 4 = Digital Secure Remote Payment (DSRP) In-App 5 = Person to Person (P2P) 6 = Dynamic Magnetic Stripe Data (DMSD) 7 to 12 = Reserved for future use
18	Binding Status	n-1	If the Tokenization Event Indicator contains a value of a (Token Binding), b (Account Binding) this field will contain a value indicating the status of Binding 1= Active, 2 = Deactivated (deleted)
19	Digital Service Provider ID 1	an-5	Contains the alphanumeric value that identifies the Digital Service Provider type and entity. This field provides information on either the third-party entity that facilitated a tokenized transaction or the entity that authenticated the cardholder in a remote commerce transaction.

Field ID	Field Name	Attributes	Values/Comments
20	Digital Service Provider ID 2	an-5	Contains the alphanumeric value that identifies the Digital Service Provider type and entity. This field provides information on either the third-party entity that facilitated a tokenized transaction or the entity that authenticated.
21	Cardholder Verification	an-5	Contains an alphanumeric value to indicate the cardholder will be authenticated and specifies the Cardholder Verification Method (CVM) will be used to authenticate the cardholder during a contactless or remote commerce transaction.
22	Authentication Factors	an-5	Contains the alphanumeric value that identifies the authentication factors will be used to authenticate the cardholder during a contactless or remote commerce transaction.
23	Binding ID	ans-48	The unique and fixed identifier to a component of data bound to the credential.

Administrative Advice/0620 - Issuer Token Notification Advice

Mastercard will support new layouts for Administrative Advice/0620 messages to communicate activation codes to issuers in support of token issuance and to notify issuers that Mastercard has assigned a token on their behalf. Following is a list of the data elements and values applicable to this message type. All mandatory Administrative Advice/0620 - System-generated data elements apply.

Table 1698: Administrative Advice/0620 - Issuer Token Notification Advice

Data Element and Name	Org	Sys	Dst	Values/Comments
Message Type Identifier (MTI)	•	X	M	Constant—0620 (Administrative Advice)
Bit Map, Primary	•	X	M	Mandatory
Bit Map, Secondary	•	X	M	Mandatory
DE 7 (Transmission Date and Time)	•	X	M	Transaction time stamp; UTC date and time that this transaction was entered into interchange
DE 11 (Systems Trace Audit Number [STAN])	•	X	M	Transaction trace number; must be unique value for transaction initiator within each UTC day

Data Element and Name	Org	Sys	Dst	Values/Comments
DE 33 (Forwarding Institution ID Code)	•	X	M	Identifies the customer, institution, or Authorization Platform facility originating the Administrative Advice/0620 message
DE 42 (Card Acceptor Identification Code)	•	X	M	Identifies the card acceptor.
DE 56 (Payment Account Data)	•	X	C	This is only for MDES Tokenization Complete Notifications.
DE 60 (Advice Reason Code)	•	X	M	Indicates the specific reason for the transmission of the Advice message Subfield 1 (Advice Reason Code) = 141 (Mastercard Digital Enablement Service Advice to Issuer) Subfield 2 (Advice Detail Code) = one of the following values: • 0250 = Activation Code Notification (ACN) • 0251 = Tokenization Complete Notification • 0252 = Tokenization Event Notification
DE 63 (Network Data)	•	X	M	Contains the Banknet Reference Number that the Authorization Platform provides as a unique transaction ID.
DE 100 (Receiving Institution ID Code)	•	X	M	Identifies the customer, institution, or Authorization Platform facility that will receive this Administrative Advice/0620.
DE 120 (Record Data)	•	X	M	Contains activation code information. Refer to the associated DE 120 layout.

Administrative Advice Response/0630 - Issuer Token Notification Advice Response

Following is a list of the data elements and values applicable to this message type. All mandatory Administrative Advice Response/0630 data elements apply.

Table 1699: Administrative Advice Response/0630 - Issuer Token Notification Advice Response

Data Element and Name	Org	Sys	Dst	Values/Comments
Message Type Identifier (MTI)	M	M	•	Constant—0630 (Administrative Advice Response)

Data Element and Name	Org	Sys	Dst	Values/Comments
Bit Map, Primary	M	M	•	Mandatory
Bit Map, Secondary	M	M	•	Mandatory
DE 7 (Transmission Date and Time)	ME	ME	•	Must be the same value as in the original Administrative Advice/0620
DE 11 (Systems Trace Audit Number [STAN])	ME	ME	•	Must be the same value as in the original Administrative Advice/0620
DE 33 (Forwarding Institution ID Code)	ME	ME	•	Must be the same value as in the original Administrative Advice/0620
DE 39 (Response Code)	M	M	•	Indicates the disposition of the original Administrative Advice/0620
DE 44 (Additional Response Data)	C	C	•	May contain the additional error code information depending on the value in DE 39
DE 63 (Network Data)	ME	ME	•	Must be the same value as in the original Administrative Advice/0620
DE 100 (Receiving Institution ID Code)	ME	ME	•	Must be the same value as in the original Administrative Advice/0620

MDES for Merchants

This section describes enhancements to the Mastercard Digital Enablement Service (MDES) for merchants program that provides a network tokenization service for merchants. The section provides the technical requirements for issuers choosing to participate in the program and for acquirers supporting merchants that are enrolled in the program.

Mastercard offers specific values to existing data elements in the Dual Message Authorization System to help acquirers and issuers identify the MDES for merchants program and participants throughout the authorization flow.

The following process explains an authorization transaction request when a token transaction is generated by an MDES for merchant participant.

1. When a merchant initiates the token transaction, the tokenized transaction data is transmitted to the acquirer.
2. The acquirer initiates an Authorization Request/0100 message containing the following data and sends it to the Mastercard Network for initial purchase:
 - DE 2 (Primary Account Number [PAN]) with the token
 - DE 22 (POS Entry Mode), subfield 1 (POS Terminal PAN Entry Mode), value 10 (Credential on File)

- DE 48 (Additional Data: Private Use), subelement 42 containing value 246 indicating Merchant Risk Management Decisioning
- DE 48 (Additional Data: Private Use), subelement 43 (Universal Cardholder Authentication Field [UCAF]) with cryptographic data
- DE 61 (Point of Service [POS] Data), subfield 3 (POS Terminal Location)
 - Value 2 (Off premises), or
 - Value 4 (On premises of acceptor facility [cardholder terminal including home PC, mobile phone, PDA])

Acquirers submit a merchant-initiated Authorization Request/0100 message containing the following data:

- DE 2 (Primary Account Number [PAN]) with the token
 - DE 22 (POS Entry Mode), subfield 1 (POS Terminal PAN Entry Mode), value 10 (Credential on File)
 - DE 48 (Additional Data: Private Use), subelement 42 (Electronic Commerce Indicators), subfield 1 containing values:
 - Position 1, value 2 (Channel)
 - Position 2, value 4 (Digital Secure Remote Payment [DSRP] with UCAF data)
 - Position 3, value 7 (Merchant-initiated transaction)
 - DE 48 (Additional Data: Private Use), subelement 43 not present
 - DE 61 (Point-of-Service [POS] Data), subfield 3 = 2 (Off premises of acceptor facility [cardholder terminal including home PC, mobile phone, PDA])
 - DE 61 (Point-of-Service [POS] Data), subfield 4 = 4 (Standing order/recurring transactions) for recurring payment transactions or = 5 (Electronic order [home PC, Internet, mobile phone, PDA]) for partial shipments.
3. The Dual Message Authorization System attempts to identify a PAN mapping relationship between the token sent by the acquirer, and a PAN in the PAN mapping database.
- If the mapping or validation fails, MDES rejects the transaction back to the acquirer and merchant with the appropriate response, and then generates an Authorization Advice/0120 response message to inform the issuer of the action.
 - If the MDES services are successful, the Dual Message Authorization System sends the issuer the authorization transaction request containing:
 - DE 2 (Primary Account Number [PAN]) with the cardholder's primary account number
 - DE 14 (Date, Expiration) with the primary account number's expiration date
 - DE 22 (Point-of-Service [POS] Entry Mode), subfield 1, value 10 (Credential on File)
 - DE 48 (Additional Data: Private Use), subelement 26 (Wallet Program Data), subfield 1 (Wallet Identifier) value = 327
 - DE 48 (Additional Data: Private Use), subelement 30 (Token Transaction Identifier)
 - DE 48 (Additional Data: Private Use), subelement 33 containing:
 - Subfield 1 (Account Number Indicator), value H (Cloud-Based Token)
 - Subfield 2 (Account Number) containing the token
 - Subfield 3 (Expiration Date) containing the token expiration date
 - Subfield 5 (Token Assurance Method) static value 00 to 99

- Subfield 6 (Token Requestor ID)
- Subfield 8 (Storage Technology)
- DE 48 (Additional Data: Private Use), subelement 42 (Electronic Commerce Indicators), subfield 1 containing value 246 indicating Merchant Risk Management Decisioning
- DE 48 (Additional Data: Private Use), subelement 42 (Electronic Commerce Indicators), subfield 1 containing value 247 for merchant-initiated transactions

NOTE: The Dual Message Authorization System will validate or modify to the correct security level indicator (SLI) if needed. See the following for more details.

- DE 48 (Additional Data: Private Use), subelement 43 (Universal Cardholder Authentication Field [UCAF]) may be present if the issuer has opted-in to receive cryptographic data
 - DE 48 (Additional Data: Private Use), subelement 71 (On-behalf Services) including:
 - 50C = MDES PAN Mapping service
 - 61V = MDES Pre-Validation Service for Cloud and Static tokens (ICC, DSRP, or DTVC data)
4. The issuer responds with Authorization Request Response/0110 message. The Dual Message Authorization System provides the response to the acquirer containing the following:
- DE 2 (Primary Account Number [PAN]) contains the token
 - DE 48 (Additional Data: Private Use), subelement 33, containing:
 - Subfield 1 (Account Number Indicator), value M (Primary Account Number)
 - Subfield 2 (Account Number)
 - Subfield 3 (Expiration Date) containing the expiration date
 - Subfield 5 (Token Assurance Method) static value 00 to 99
 - Subfield 6 (Token Requestor ID)
5. The acquirer forwards the Authorization Request Response/0110 including the token to the merchant.

Mastercard Fraud Scoring Services

This section describes the Mastercard Fraud Scoring Services, which includes Fraud Rule Manager.

Fraud Rule Manager

Fraud Rule Manager is an optional service that allows participating issuers to publish and enact business rules specific to their portfolio of accounts. Fraud Rule Manager enables issuers to automate precise, highly variable fraud decisions during authorization to respond to evolving fraud trends quickly and easily.

For more information on the Mastercard Fraud Scoring Services, refer to [Mastercard Security Solutions Guide for Issuer Services](#).

Fraud Rule Manager

Fraud Rule Manager is an optional service that allows participating issuers to publish and enact business rules specific to their portfolio of accounts. The outcome of the rules may be to introduce a Rule Adjusted Score, Rule Reason Codes, or both into the authorization message when rule criteria authorized by the issuer are met.

To Participate

Issuers that want to participate in this service must complete a contract and registration form. Issuers interested in registering for this service should contact their Mastercard representative or the Mastercard Risk Solutions Team at riskolutions@Mastercard.com.

For More Information

For more information about this service, refer to the *Authorization Manual*.

Decision Intelligence

This section describes Decision Intelligence, one of the two Mastercard Fraud Scoring Services.

Overview

This section provides an overview of Decision Intelligence.

Decision Intelligence is a real-time authorization decisioning solution that applies thousands of data points and sophisticated modeling techniques to each transaction, simplifying these insights into a single transaction decision score that helps issuers to fine-tune their authorization decisions with the goal of approving genuine transactions and declining fraudulent ones.

Decision Intelligence ultimately shifts transaction decisioning from the basis of a risk score to a decision score that is based on both the fraud risk and rewards of approving a transaction. With a more comprehensive analysis of the risk and reward factors, Decision Intelligence assesses both negative and positive data points to calculate a single score that shows where the transaction falls on a scale from decline to approve—thus helping issuers authorize more transactions with confidence, increase their profitability, and improve the cardholder experience.

Decision Intelligence supports all Mastercard brands (Mastercard®, Maestro®, Cirrus®), segments (consumer, commercial), and products (credit, debit, prepaid) for transactions processed via the Mastercard Network.

How It Works

This section briefly describes how Decision Intelligence works.

The value of Decision Intelligence lies in its ability to evaluate multiple factors throughout the shopping experience to decide whether or not a transaction makes sense for a particular consumer. Decision Intelligence evaluates information about consumers, merchants, and issuers during the shopping experience to help issuers decide whether a transaction should be approved or declined.

For each transaction, Decision Intelligence assesses:

- Consumer transaction attributes based on consumer account and device information

- Transaction security based on widespread fraud monitoring, fraud rules, transaction fraud models, and profiles
- Cardholder segmentation based on insights into account spending that help define the value and engagement of the cardholder with their issuer

The resulting decision score shows the transaction on a risk-reward continuum (with a value from 0–999). A lower score signifies an excellent approvability quotient, while a higher score indicates more reasons for declining the transaction. Issuers can easily integrate the decision score into their authorization decision processes and cardholder strategies.

Issuers' Potential Benefit from Decision Intelligence

Decision Intelligence enriches and simplifies the decision management process with insights across multiple key dimensions that can potentially help issuers gain incremental business in revenue by:

- Helping increase approvals of genuine transactions without increasing their risk exposure
- Deepening consumer relationships with a more consistent, satisfying experience across shopping choices

Decision Intelligence potentially helps issuers to save money from:

- Fewer false positive or erroneous declines of genuine transactions
- Less revenue lost from fraud and chargebacks
- Improved productivity and lower costs (operational, IT, customer service, reputational, and so on) of managing fraud
- Decreased customer service costs due to more approved transactions
- Lower operational costs in capital, information technology (IT) development, and customer service by outsourcing decisioning through Mastercard's established, scalable platform

Message Specification Requirements

To limit the development impact on its customers, Decision Intelligence utilizes the preexisting Expert Monitoring for Issuers data fields. Because many issuers or their processors already process these fields, there may not be additional development effort required to support the fields for this service.

Issuers participating in the Decision Intelligence service must support the following existing fields.

On-behalf Service (OBS)

DE 48 (Additional Data—Private Use)

- Subelement 71 (On-behalf Services)
 - Subfield 1 (On-behalf Service) with a value of 18C (Fraud Scoring Service was performed successfully)

Fraud Scoring Data

DE 48 (Additional Data—Private Use)

- Subelement 75 (Fraud Scoring Data)
 - Subfield 1 (Fraud Score) with a default value of 001
 - Subfield 2 (Score Reason Code) with a default value of 00
 - Subfield 3 (Rules Score)
 - Subfield 4 (Rules Reason Code 1)
 - Subfield 5 (Rules Reason Code 2)

Security Services Additional Data for Issuers

DE 48 (Additional Data—Private Use)

- Subelement 56 (Security Services Additional Data for Issuers)
 - Subfield 1 (Security Services Indicator)
 - Subfield 2 (Security Services Data)

To Participate

Issuers interested in participating in Decision Intelligence should contact their Mastercard account representative. To participate, issuers must complete a Service Agreement to indicate the appropriate bank identification numbers (BINs)/account ranges to be configured for the service.

For More Information

For details on the Authorization IQ and Digital Transaction Insights features of Decision Intelligence, refer to DE 48 (Additional Data—Private Use), Subelement 56 (Security Services Additional Data for Issuers)—Valid Subfield 1 (Security Services Indicator) and Subfield 2 (Security Services Data) Value Combinations in the Data Element Definitions chapter of this manual.

Expert Monitoring for Merchants

Expert Monitoring for Merchants provides participating acquirers with a real-time fraud score on U.S.-issued, dual message, card-not-present (CNP) authorization transactions.

Benefits

Expert Monitoring for Merchants provides the following benefits:

- Provides CNP merchants with a real-time behavior-based fraud score at the time of transaction authorization
- Delivers a predictive fraud score on Mastercard transactions to merchants within the authorization response message
- Uniquely scores transactions using a targeted fraud model that evaluates the current transaction against a comprehensive view of the cardholder's transaction history
- Enables merchants to use the fraud score as a highly predictive data point in their current fraud detection solution to increase fraud detection while reducing chargebacks, fraud losses, and manual review costs

To Participate

Acquirers that want to participate in Expert Monitoring for Merchants must complete a contract and registration form. Acquirers interested in registering for this service should contact their Mastercard representative.

For More Information

For more information about this service, refer to the *Authorization Manual*.

Authorization Request/0100: Fraud Scoring

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Data Element	Org	Sys	Dst	Values/Comments
DE 48, subelement 71 (On-behalf Services), subfield 1 (On-behalf [OB] Service)	•	X	C	18 = Fraud Scoring Service
DE 48, subelement 71 (On-behalf Services), subfield 2 (On-behalf Result 1)	•	X	C	Contains one of the following values: C = Fraud Scoring Service was performed successfully U = Fraud Scoring Service was not performed successfully
DE 48, subelement 75 (Fraud Scoring Data), subfield 1 (Fraud Score Data)	•	X	C	The Authorization Platform inserts this subelement when the Fraud Scoring Service is performed on the transaction. Fraud Scoring System provides the risk score of 000–999 in subfield 1, where 999 is more likely to be fraudulent than a score of 000.
DE 48, subelement 75 (Fraud Scoring Data), subfield 2 (Score Reason Code)	•	X	C	Fraud Scoring System provides the score reason code in subfield 2, which indicates the key factors that influenced the fraud score. Subfield 2 is provided whenever a fraud score is provided in subfield 1 (Fraud Score Data).

Alternate Processing

The following table indicates the Authorization Advice/0120 content when the transaction is qualified for fraud scoring, but the issuer is unavailable and Stand-In processing is invoked. This information applies to both Expert Monitoring for Issuers and Decision Intelligence.

Stand-In and X-Code processing do not consider the fraud assessment score when performing an authorization decision for an issuer.

IF...	THEN the Authorization Advice/0120—System-generated message will contain...
If the original Authorization Request/0100 message was successfully scored	DE 48, subelement 75 and DE 48, subelement 71, subfield 1, value 18 and subfield 2, value C indicating Fraud Scoring Service was performed on the transaction
If the original Authorization Request/0100 message was not successfully scored	DE 48, subelement 71, subfield 1, value 18 and Subfield 2, value U indicating Fraud Scoring Service was not performed on the transaction

Mastercard Hosted Mobile Phone Top-Up ATM Transactions

Mastercard Hosted Mobile Phone Top-up supports Maestro®, Debit Mastercard®, and Mastercard® card transactions performed at ATMs in the Europe region. Mobile phone top-up functionality enables customers to top up (that is, add credit to) their prepaid mobile phone service. Mobile Phone Top-up enables acquirers to provide cardholders the ability to pay-as-you-go at the ATM for additional mobile phone minutes.

For more information on the Mastercard Hosted Mobile Phone Top-Up ATM Transaction, refer to the [Transaction Processing Rules](#).

Authorization Request/0100: Mastercard Hosted Mobile Phone Top-Up

The following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Table 1700: Applicable data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)	M	.	M	00 = Purchase
DE 4 (Amount, Transaction)	M	.	M	The top-up amount.
DE 12 (Time, Local Transaction)	C	.	C	The local time at which the transaction takes place at the point of acceptor location.
DE 13 (Date, Local Transaction)	C	.	C	The local month and day on which the transaction takes place at the point of acceptor location.
DE 18 (Merchant Type)	M	.	M	Acceptor business code (MCC) 4814 (Telecommunication Services including but not limited to prepaid phone services and recurring phone services)
DE 43 (Acceptor Name and Location)	C	.	C	
DE 48 (Additional Data: Private Use), TCC (Transaction Category Code)	M	.	M	Z = ATM Cash Disbursement
DE 48 (Additional Data: Private Use), subelement 13 (Mastercard Hosted Mobile Phone Top-up Request Data)	C	.	C	Subfield 1 = Mobile Phone Number (Must be left-justified and cannot contain all spaces or all zeros.) Subfield 2 = Mobile Phone Service Provider Name (Must be left-justified and cannot contain all spaces or all zeros.)
DE 52 (Personal ID Number [PIN] Data)	M	X	M	DE 52 is mandatory for all ATM transactions.
DE 61 (Point-of-Service [POS] Data), subfield 10 (Cardholder-Activated Terminal Level)	C	.	C	1 = Authorized Level 1 CAT: Automated dispensing machine with PIN

NOTE: Alternate processing is not applicable for Mastercard Hosted Mobile Phone Top-up transactions. If the primary issuer is not available or does not provide a timely response, the Dual Message Authorization System will send the acquirer an Authorization Request Response/0110 message with DE 39, value 91 (Authorization System or issuer system inoperative).

Dual Message Authorization System Edits

The following edits are performed on Authorization Request/0100 messages for Mastercard Hosted Mobile Phone Top-up transactions.

Table 1701: Authorization Request/0100 Edits

When the message contains...	Then the Dual Message Authorization System...
DE 48 (Additional Data: Private Use), subelement 13 and One of the following data elements is not present:	sends the acquirer an Authorization Request Response/0110 message where: DE 39 (Response Code) = 30 (Format error) DE 44 (Additional Response Data) = the data element in error
• DE 12 (Time, Local Transaction) • DE 13 (Date, Local Transaction) • DE 43 (Acceptor Name and Location)	
DE 48(Additional Data: Private Use), subelement 13 is incorrectly formatted (for example, incorrect length, subfield 1 or 2 is not present, or subfield 1 or 2 contains all zeros or spaces)	sends the acquirer an Authorization Request Response/0110 message where: DE 39 (Response Code) = 30 DE 44 (Additional Response Data) = 048013

Table 1702: Authorization Request Response/0110 Edits

When the value in...	Then the Dual Message Authorization System...
DE 48(Additional Data: Private Use), subelement 13 is different from the original Authorization Request/0100 message	sends the issuer an Authorization Response Negative Acknowledgment/0190 message where: DE 39 (Response Code) = 30 DE 44 (Additional Response Data) = 048013

Table 1703: Authorization Advice/0120: acquirer-generated edits

When the message contains...	Then the Dual Message Authorization System...
DE 48 (Additional Data: Private Use), subelement 13	declines the request with an Authorization Advice Response/0130 message where: DE 39 (Response Code) = 30 DE 44 (Additional Response Data) = 048013

Table 1704: Reversal Request/0400 Edits

When the message contains...	Then the Dual Message Authorization System...
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) is 00 (Purchase) and DE 48 position 1 (TCC) is Z and DE 48, subelement 13 (Mastercard Hosted Mobile Phone Top-up Request Data) is present	declines the request with a Reversal Request Response/0410 message where: DE 39 (Response Code) = 30 DE 44 (Additional Response Data) = 048013

Mastercard In Control Service

The Mastercard In Control™ platform provides for a number of advanced authorization, transaction routing, and alert controls designed to assist issuers in creating new and enhanced payment products.

Features

The Mastercard In Control platform allows issuers to leverage "off-the-shelf" solutions and to create customized offerings depending on the needs of their customers.

Among the advanced new features issuers can leverage to support their commercial card portfolios are:

- Enhanced authorization controls that direct how, when, and where cards may be used to a greater level of specificity than previously supported
- Robust alert functionality that provides personalized real-time communication about transaction activities
- A limited use number feature that allows authorization, spending limits, and usability controls to be set on a transaction-by-transaction basis, providing enhanced levels of security, control, data capture, and traceability on every purchase

Authorization Request/0100: In Control Purchase Control

The following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply. This layout applies to participating issuers only.

Table 1705: Data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 2 (Primary Account Number)	M	.	M	Contains the cardholder's real account number
DE 14 (Date, Expiration)	C	.	C	The cardholder's real account number expiration date
DE 48 (Additional Data—Private Use), subelement 33 (PAN Mapping File Information), subfield 1 (Account Number Indicator)	.	X	C	Indicates Mastercard In Control virtual card number V = Virtual Card Number
DE 48, subelement 33, subfield 2 (Account Number)	.	X	C	Virtual card number
DE 48, subelement 33, subfield 3 (Expiration Date)	.	X	C	VCN Expiration date
DE 48, subelement 71 (On-Behalf Services), subfield 1 (On-Behalf [OB] Service)	.	X	C	17 = In Control Virtual Card Service
DE 48, subelement 71 (On-Behalf Services), subfield 2 (On-Behalf [OB] Result 1)	.	X	C	V = Valid NOTE: If the Purchase Control service is unsuccessful, refer to Subfield 2 (On-behalf Result 1) for a list of other valid values.

Dual Message System Processing

This message describes Dual Message System processing of a Mastercard In Control Purchase Control transaction.

1. The acquirer sends an Authorization Request/0100, Authorization Advice/0120—Acquirer-generated, or a Reversal Request/0400 message to the Mastercard Network containing the In Control virtual card number in DE 2 (Primary Account Number [PAN]).
2. The Authorization Platform applies unique controls for the In Control virtual card number and performs mapping to the cardholder's primary account number.
3. The Authorization Platform sends an Authorization Request/0100, Authorization Advice/0120—Acquirer-generated, or a Reversal Request/0400 message to the issuer containing:

- DE 2 (Primary Account Number [PAN]) and DE 14 (Date, Expiration) with the cardholder's real account number and associated expiration date
 - DE 48 (Additional Data—Private Use), subelement 33 (PAN Mapping File Information) where subfield 1 (Account Number Indicator) = value V (Virtual Card Number), subfield 2 (Account Number) = VCN (Virtual card number), and subfield 3 (Expiration Date) = VCN expiration date
 - DE 48, subelement 71 (On-Behalf Services) where subfield 1 (On-Behalf Services) = value 17 (In Control Virtual Card Service), subfield 2 (On-Behalf [OB] Result 1)—value V (Valid)
4. The issuer approves or declines the authorization request by sending an Authorization Request Response/0110, Authorization Advice Response/0130—Issuer-generated, or a Reversal Request Response/0410 message.
 5. The Authorization Platform maps the cardholder's primary account number back to the In Control virtual card number, places it in DE 2 (Primary Account number [PAN]), and then forwards the Authorization Request Response/0110, Authorization Advice Response/0130, and Reversal Request Response/0410 messages to the acquirer.
 6. The acquirer forwards the virtual card number and the authorization response information to the merchant.
 7. If In Control processing cannot be completed, the Authorization Platform sends the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Request Response/0410 containing DE 39 (Response Code), value 96 (System error).

Mastercard In Control Virtual Card Service

The Mastercard In Control Virtual Card Service allows issuers to process partial approvals.

Issuers must opt-in to the Mastercard In Control Virtual Card Service when choosing to process partial approvals by registering with Mastercard.

Mastercard In Control Real Card Spend Control

The Mastercard In Control Real Card Spend Control service allows cardholders to establish spend control rules for their preexisting payment cards to be enforced during the authorization process.

These rules can be defined for both card-present and card-not-present transaction environments, and depending on the cardholder's predefined response rules, may result in an alert notification being generated to the cardholder or the transaction being declined on their behalf.

NOTE: Spend control rules can only be applied to transactions that flow through the Mastercard Network. Spend control rules are not applied to Authorization Advice/0120: acquirer-generated and Reversal Request/0400 messages.

Processing a Mastercard In Control service-eligible transaction

This process describes the stages of the Dual Message Authorization System processing for a Mastercard In Control Service-eligible transaction.

The acquirer sends the Authorization Request/0100 message containing the Mastercard In Control real card number in DE 2 (Primary Account Number [PAN]) to the Mastercard Network.

Table 1706: Associated edits

When...	Then the Dual Message Authorization System...
the transaction complies with the control rules established by the cardholder	sends the issuer the Authorization Request/0100 message where DE 48 (Additional Data: Private Use), subelement 71 (On-behalf Services), subfield 1 (On-behalf [OB] Service) contains value 20 (In Control—RCN Spend Control) and subfield 2 (On-behalf [OB] Result 1) contains value V (Valid).
the transaction does not comply with the control rules established by the cardholder and the cardholder has registered an action response of decline and alert notification upon rule failure	declines the transaction and sends the acquirer an Authorization Request Response/0110 message where DE 39 (Response Code) contains a decline response. and sends the issuer an Authorization Advice/0120: system-generated message where: - DE 48 (Additional Data: Private Use), subelement 71 (On-behalf Services): - Subfield 1 (On-behalf [OB] Service) contains value 20 - Subfield 2 (On-behalf Result 1) contains the spend control rule that failed - DE 60 (Advice Reason Code): - Subfield 1 (Advice Reason Code) contains value 200 (In Control Processing Advice to Issuer) - Subfield 2 (Advice Detail Code) contains a valid advice reason code and sends the cardholder an alert notification indicating rule failure.

When...	Then the Dual Message Authorization System...
the transaction does not comply with the spend control rules established by the cardholder and	sends the issuer an Authorization Request/0100 message where DE 48, subelement 71, subfield 1 is value 20 and subfield 2 indicates the spend control rule that failed.
the cardholder has registered an action response of "alert notification only" upon rule failure, but no decline action	and sends the cardholder an alert notification indicating rule failure.

NOTE: If the transaction is declined (by the issuer, alternate issuer, Stand-In Authorization, or X-Code Processing), or the issuer/alternate issuer performs a partial approval or purchase amount only approval, the Dual Message Authorization System sends an Authorization Advice/0120: system-generated message to Mastercard In Control to update the disposition of the transaction.

Table 1707: Associated edits

When...	Then the Dual Message Authorization System...
Mastercard In Control is unavailable or responds late or the spend control rules have recently been removed	populates DE 48 (Additional Data: Private Use), subelement 71 (On-behalf Services), subfield 1 (On-behalf [OB] Service) contains value 20 (In Control—RCN Spend Control) and Subfield 2 (On-behalf Result 1), value U (Unable to process) and forwards the transaction to issuer.

Authorization Request/0100: In Control Real Card Spend Control

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Table 1708: Data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 2 (Primary Account Number)	M	.	M	Account range must participate in Mastercard In Control Real Card Spend Control Service; DE 2 contains the Mastercard In Control real card number
DE 48 (Additional Data—Private Use), subelement 71 (On-behalf Services), subfield 1 (On-behalf [OB] Service)	.	X	C	20 = In Control RCN Spend Control Service

Data Element	Org	Sys	Dst	Values/Comments
DE 48 (Additional Data—Private Use), subelement 71 (On-behalf Services), subfield 2 (On-behalf [OB] Result 1)	•	X	C	Indicates the results of the service processing. Valid subfield 2 values: D, E, F, G, H, J, K, L, M, P, U, or V

Authorization Advice/0120: In Control Real Card Spend Control

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Advice/0120 data elements apply.

Table 1709: Data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 2 (Primary Account Number)	•	M	M	Account range must participate in Mastercard In Control Real Card Spend Control Service
DE 48 (Additional Data—Private Use), subelement 71 (On-behalf Services), subfield 1 (On-behalf [OB] Service)	•	M	M	20 = In Control RCN Spend Control Service
DE 48 (Additional Data—Private Use), subelement 71 (On-behalf Services), subfield 2 (On-behalf [OB] Result 1)	•	M	M	Indicates the results of the service processing. Valid subfield 2 values: D, E, F, G, H, J, K, L, M, P, U, or V
DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code)	•	M	M	200 = In Control Processing Advice to Issuer
DE 60 (Advice Reason Code), subfield 2 (Advice Detail Code)	•	M	M	Reject reason
DE 121 (Authorizing Agent ID Code)	•	C	C	000003 = Applicable only to issuers participating in the In Control services and will only be present in Authorization Advice/0120: system-generated messages notifying issuers of a declined request due to a failed spend control rule defined by the cardholder

Mastercard In Control Virtual Card Mapping and Spend Control Service

The Mastercard In Control platform leverages the same authorization and clearing data elements for mapping virtual card numbers to their real card account numbers as in the Mastercard In Control Purchase Control service and the Mastercard Contactless Mapping service. Issuers currently supporting the Mastercard Contactless Mapping service need only to recognize new data values in existing data elements to support the Mastercard In Control Virtual Card Mapping and Spend Control service.

Authorization Request/0100: In Control Virtual Card Mapping and Spend Control Service

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Table 1710: Data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 2 (Primary Account Number [PAN])	M	.	M	The real card number
DE 48 (Additional Data—Private Use), Subelement 33 (PAN Mapping File Information)	.	X	C	The virtual account data
DE 48, Subelement 71	.	X	C	The on-behalf service performed on the transaction

Exception Processing

The following message flow describes the Dual Message Authorization System processing for a Mastercard In Control Virtual Card Mapping and Spend Control service authorization transaction that was declined by In Control processing.

1. The acquirer sends an Authorization Request/0100 message to the Dual Message Authorization System containing the In Control virtual card number in DE 2 (Primary Account Number [PAN]).
2. The Dual Message Authorization System applies unique controls for the In Control virtual card number and performs mapping to the cardholder's primary account number.
3. If the transaction fails mapping:
 - The Dual Message Authorization System declines the Authorization Request/0100, Authorization Advice/0120, or Reversal Request/400 message and sends the acquirer an

- Authorization Request Response/0110, Authorization Advice Response/0130: system-generated, or a Reversal Request Response/0410 message.
- If the transaction fails spend controls, the Dual Message Authorization System declines the Authorization Request/0100, Authorization Advice/0120, or Reversal Request/400 message and sends the acquirer an Authorization Request Response/0110 message.
4. If the transaction fails spend controls but PAN mapping is successful, the Dual Message Authorization System sends an Authorization Advice/0120: system-generated message to the issuer containing:
- DE 2 (Primary Account Number [PAN])
 - DE 48 (Additional Data: Private Use), subelement 33 (PAN Mapping File Information)
 - DE 48 (Additional Data: Private Use), subelement 71 (On-Behalf Services), subfield 1 (On-Behalf Services), value 17 (In Control Virtual Card Service) and subfield 2 (On-Behalf [OB] Result 1) containing the appropriate result value
 - DE 60 (Advice Reason code), subfield 1 (Advice Reason Code) = 200 (In Control Processing Advice to Issuer), subfield 2 (Advice Detail Code) = appropriate advice reason code

Mastercard Installment Payment Service

The Mastercard Installment Payment Service is a service offered by Mastercard to Dual Message Authorization System-connected issuers to support installment payments for authorization and clearing processing. This service enables cardholders to choose to pay for a purchase in installment payments and select the installment payment terms from choices provided at the time of purchase.

For additional guidance on the Mastercard Installment Payment Service, refer to the [Mastercard Installments Payment Services Program Guide](#).

Mastercard Merchant Presented QR

This service offers a mobile payment option for person-to-merchant transactions in markets where conventional transactions rely heavily on cash for retail purchases.

Mastercard Merchant Presented QR is a consumer-initiated, push payment transaction initiated with a mobile device. Using a smartphone, the consumer makes a cashless payment by scanning a Mastercard Merchant Presented QR code at any merchant accepting Mastercard Merchant Presented QR transactions. This product fulfills the need to make payments instantly to merchants in a direct, convenient, and secure method for both the merchant and cardholder.

This service operates in the four-party business model, which enables funds to be transferred from a licensed Mastercard Originating Institution (OI), the consumer's bank, to a Receiving Institution (RI), the merchant's bank. The transaction originates at the OI and routes to the merchant after successfully authenticating and authorizing the debit from the consumer's account. The RI will process the payment transaction and notify the merchant that the payment

has been received successfully. The RI will credit the merchant account after processing the payment transaction.

For more information about Mastercard Merchant Presented QR, refer to the [Mastercard QR: Send Program Guide](#).

Dual Message Authorization System Edits

The Dual Message Authorization System performs the following system edits.

Mastercard Merchant Presented QR Funding Transaction Processing

The Dual Message Authorization System validates Mastercard Merchant Presented QR transactions that are identified as:

- DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code), value 00 (Purchase)
- DE 48 (Additional Data), subelement 77 (Transaction Type Identifier), value C67 (Mastercard Merchant Presented QR)

Mastercard Merchant Presented QR Payment Transaction

The Dual Message Authorization System validates Merchant Presented QR Payment transactions that are identified as:

- DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code), value 28 (Payment Transaction)
- DE 48 (Additional Data), subelement 77 (Funding/Payment Transaction Type Indicator), value C67 (Masterpass QR)

The Dual Message Authorization System validates Merchant Presented QR Refund Payment transactions that are identified as:

- DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code), value 28 (Payment Transaction)
- DE 48 (Additional Data), subelement 77 (Funding/Payment Transaction Type Indicator), new value C68 (Masterpass QR Refund)

Table 1711: Validate Merchant Code edits

When...	Then the Dual Message Authorization System...
DE 18 (Merchant Code) contains any one of the following MCCs in an Authorization Request/0100 message for a Mastercard Merchant Presented QR Payment Transaction, Mastercard Merchant Presented QR Refund Payment Transaction, or for a Mastercard Merchant Presented QR Funding Transaction: • 4829 = Money Transfer • 6010 = Manual Cash • 6011 = ATM • 6532 = Payment Transaction—Customer Financial Institution • 6533 = Payment Transaction: Merchant • 6538 = Mastercard Send Funding • 6540 = Funding (exclusive of Mastercard Send) • 6555 = MA Initiated Rewards/Rebate • 6050 = Quasi Cash • 6051 = Quasi Cash—merchant	rejects the transaction and forwards the Originating Institution an Authorization Request Response/0110 with: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 018 (indicating the data element in error)

Table 1712: Validate the presence of mandatory data edits

When...	Then the Dual Message Authorization System...
DE 108 (Additional Transaction Reference Data), subelement 02 (Sender Data), subfield 11 (Sender Account Number) is not present, not numeric, contains all zeros, or exceeds 20 characters	rejects the transaction and forwards the Originating Institution an Authorization Request Response/0110 with • DE 39 = 30 (format error) • DE 44 (Additional Response Data) = 108002 (indicating the data element in error)

Table 1713: Validate the presence of mandatory data edits

When...	Then the Dual Message Authorization System...
DE 108 (Additional Transaction Reference Data), subelement 02 (Transaction Reference Data), subfield 3 (Funding Source) is not present or does not contain one of the following values: 01 (Credit), 02 (Debit), 03 (Prepaid), 04 (Deposit Account), 05 (Mobile Money Account), 06 (Cash), or 07 (Reserved)	Rejects the transaction and forwards the Originating Institution an Authorization Request Response/0110 with • DE 39 = 30 (format error) • DE 44 (Additional Response Data) = 108002 (indicating the data element in error)

Table 1714: Validate transaction amount limit edits

When...	Then the Dual Message Authorization System...
the transaction amount limit is exceeded	rejects the transaction and forwards the Originating Institution an Authorization Request Response/0110 with DE 39 = 05 (Do not honor) an Authorization Advice/0120 message is sent to the Receiving Institution with: • DE 60 (Advice Reason Code), subfield 1 (Values, in Authorization Advice/0120), value 151 (Blocking Processing Advice to Issuer [Mastercard Merchant Presented QR]). • DE 60, subfield 2 (Values, in Mastercard In Control Service), code 0064 (Reject: Transaction Limit Check). • DE 48 (Additional Data: Private Use), subelement 71 (Valid Subfield 1 and Subfield 2 Value Combinations), value 37D (Mastercard Merchant Presented QR Blocking: Transaction Amount Limit Exceeded).

Table 1715: Validate cumulative transaction amount limit edits

When...	Then the Dual Message Authorization System...
the cumulative transaction amount limit is exceeded	rejects the transaction and forwards the Originating Institution an Authorization Request Response/0110 with DE 39 = 05 (Do not honor) an Authorization Advice/0120 message is sent to the Receiving Institution with: • DE 60 (Advice Reason Code), subfield 1 (Values, in Authorization Advice/0120), value 151 (Blocking Processing Advice to Issuer [Mastercard Merchant Presented QR]). • DE 60, subfield 2 (Values, in Mastercard In Control Service), code 0065 (Reject: Cumulative Limit Check). • DE 48 (Additional Data: Private Use), subelement 71 (Valid Subfield 1 and Subfield 2 Value Combinations), value 37E (Mastercard Merchant Presented QR Blocking: Cumulative Transaction Amount Limit Exceeded).

Table 1716: Validate the Originating Institution edits

When...	Then the Dual Message Authorization System...
the Originating Institution, not registered for participation in the Mastercard Merchant Presented QR service, submits a transaction containing DE 48 (Additional Data), subelement 77 (Transaction Type Identifier), value C67	rejects the transaction and forwards the Originating Institution an Authorization Request Response/0110 with DE 39 = 58 (Transaction not permitted to acquirer/terminal)

Table 1717: Domestic activity edits

When...	Then the Dual Message Authorization System...
if the Mastercard Merchant Presented QR Receiving Institution has chosen to limit Mastercard Merchant Presented QR to domestic activity, then if the PAN country code does not equal the country code value in DE 32 (Acquiring Institution Identification Code)	rejects the transaction and forwards the OI an Authorization Request Response/0110 with DE 39 = 05 (Do not honor) an Authorization Advice/0120 message is sent to the Receiving Institution with • DE 60 (Advice Reason Code), subfield 1 (Values, in Authorization Advice/0120), code 151 (In Control Processing Advice to Issuer [Mastercard Merchant Presented QR]). • DE 60, subfield 2 (Values, in Mastercard In Control Service) code 0072 (Reject: Geographic Restriction). • DE 48 (Additional Data: Private Use), subelement 71 (Valid Subfield 1 and Subfield 2 Value Combinations), value 37F (Mastercard Merchant Presented QR Blocking—Domestic Activity Only).

Table 1718: Valid authorization messages edits

When...	Then the Dual Message Authorization System...
the Originating Institution submits an Authorization Request/0100 message for a valid Mastercard Merchant Presented QR transaction	forwards the Authorization Request/0100 message to the Receiving Institution with DE 48, subelement 71, subfield 1, subfield 2, containing value 37V (Valid).
the Originating Institution submits an Authorization Request/0100 message for any other system issue that prevents successful Mastercard Merchant Presented QR service processing	forwards the Authorization Request/0100 message to the Receiving Institution with DE 48, subelement 71, subfield 1, subfield 2, containing value 37U (Unable to process).

Table 1719: Reject reversal messages

When...	Then the Dual Message Authorization System...
the Originating Institution submits a Reversal Request/0400 message for a Mastercard Merchant Presented QR transaction	rejects the transaction and forwards the Originating Institution a Reversal Request Response/0410 with DE 39 = 58 (Transaction not permitted to acquirer/terminal)

Table 1720: Edits on payment transactions

When...	Then the Dual Message Authorization System...
DE 108 (Additional Transaction Reference Data), subelement 05 (Digital Account Information), subfield 01 (Digital Account Reference Number) in an Authorization Request/0100 message does not contain a valid PAN	rejects the message and responds to the Originating Institution with an Authorization Request Response/0110 message that contains DE 39 (Response Code) = 12 (Invalid)

Table 1721: Edits on payment transactions

When...	Then the Dual Message Authorization System...
DE 108 (Additional Transaction Reference Data), subelement 05 (Digital Account Information), subfield 01 (Digital Account Reference Number) in an Authorization Request/0100 message is not properly formatted	rejects the message and responds to the Originating Institution with an Authorization Request Response/0110 message that contains DE 39 (Response Code) = 30 (Format error)
- DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code), value 28 (Payment Transaction) - DE 48 (Additional Data), subelement 77 (Transaction Type Identifier), value C67 (Mastercard Merchant Presented QR)	

Table 1722: Edits on payment transactions

When...	Then the Dual Message Authorization System...
DE 108 (Additional Transaction Reference Data), subelement 05 (Digital Account Information), subfield 01 (Digital Account Reference Number) in an Authorization Request/0100 message contains a Digital Account Reference Number that is assigned already or is mapped to a different funding account	rejects the message and responds to the Originating Institution with an Authorization Request Response/0110 message that contains DE 39 (Response Code) = 12 (Invalid)

Table 1723: Edits on payment transactions

When...	Then the Dual Message Authorization System...
DE 108 (Additional Transaction Reference Data), subelement 05 (Digital Account Information), subfield 01 (Digital Account Reference Number) is not contained in an Authorization Request/0100 message, and the value in DE 108 (Additional Transaction Reference Data), subelement 02 (Sender Data), subfield 11 (Sender Account Number) is a valid Mastercard PAN	populates DE 108, subelement 05 (Digital Account Information), subfield 01 (Digital Account Reference Number) in the Authorization Request/0100 message and forwards the message to the Receiving Institution/merchant bank.

Table 1724: Edits on payment transactions

When...	Then the Dual Message Authorization System...
DE 108 (Additional Transaction Reference Data), subelement 05 (Digital Account Information), subfield 01 (Digital Account Reference Number) is not contained in an Authorization Request/0100 message, the value in DE 108, subelement 02, subfield 11 is not a valid Mastercard PAN	forwards the message to the Receiving Institution without the PAN field populated. The merchant handles the refund as a manual cash distribution.

Table 1725: Edits on payment transactions

When...	Then the Dual Message Authorization System...
the Originating Institution submits a Reversal Request/0400 message for a Mastercard Merchant Presented QR Refund transaction	rejects the message and responds to the Originating Institution with a Reversal Request Response/0410 message that contains DE 39 = 58 (Transaction not permitted to acquirer/terminal)

Table 1726: Edit on funding transactions

When...	Then the Dual Message Authorization System...
the Originating Institution is not registered for participation in the Mastercard Merchant Presented QR service and submits a funding transaction in an Authorization Request/0100 message containing DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code), value 00 (Purchase) and DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier), value C67 (Mastercard Merchant Presented QR)	rejects the message and responds to the Originating Institution with an Authorization Request Response/0110 message that contains DE 39 = 58 (Transaction not permitted to acquirer/terminal)

Table 1727: Edit on refund payment transactions

When...	Then the Dual Message Authorization System...
the Receiving Institution is not registered for participation in the Mastercard Merchant Presented QR service and submits a refund transaction in an Authorization Request/0100 message containing DE 48 (Additional Data: Private Use), subelement 77 (Transaction Type Identifier), value C68 (Mastercard Merchant Presented QR Refund)	rejects the message and responds to the Originating Institution with an Authorization Request Response/0110 message that contains DE 39 = 58 (Transaction not permitted to acquirer/terminal)

Table 1728: Mastercard Merchant Presented QR Payment Transaction edits

When...	Then the Dual Message Authorization System...
the individual transaction limit or the cumulative transaction amount or count limit is exceeded in an Authorization Request/0100 message for a Mastercard Merchant Presented QR Payment Transaction	rejects the message and forwards the Originating Institution an Authorization Request Response/0110 message with: DE 39 (Response Code) = 05 (Do Not Honor) rejects the message and forwards to the Receiving Institution an Authorization Advice/0120: system-generated message with: • DE 48 (Additional Data), subelement 71 (On-behalf Service [OBS]), subfield 1 (On-behalf Service Indicator), value 37 (Mastercard Merchant Presented QR Blocking Service) • DE 48, subelement 71, subfield 2 (On-behalf Result 1), value E (Mastercard Merchant Presented QR Blocking: Cumulative Transaction Amount or Count Limit Exceeded) • DE 60 (Advice Reason Code), in the table Subfield 1 Values, in Authorization Advice/0120, value 151 (In Control Processing Advice to Issuer [Mastercard Merchant Presented QR]) • DE 60, in the table Subfield 2 Values, in Mastercard Merchant Presented QR Service, value 0064 (Reject: Transaction Limit Check) when the individual transaction limit is exceeded or • DE 60, in the table Subfield 2 Values, in Mastercard Merchant Presented QR Service, value 0065 (Reject: Cumulative Limit Check) when the cumulative counts or amounts limit is exceeded

Table 1729: Mastercard Merchant Presented QR Funding Transaction edits

When...	Then the Dual Message Authorization System...
an Authorization Request/0100 message for a Mastercard Merchant Presented QR Funding Transaction does not contain DE 108 (Additional Transaction Reference Data), subelement 03 (Transaction Reference Data), subfield 01 (Unique Transaction Reference)	rejects the transaction and forwards the Originating Institution an Authorization Request Response/0110 with: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 108003 (indicating the data element)
an Authorization Request/0100 message for a Mastercard Merchant Presented QR Funding Transaction contains DE 108 (Additional Transaction Reference Data), subelement 05 (Digital Account Information), subfield 02 (Mastercard Merchant Presented QR Receiving Account Number): • That is not a valid length • That is not alphanumeric special characters • That contains leading spaces • That contains all zeros	rejects the transaction and forwards the OI an Authorization Request Response/0110 with: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 108005 (indicating the data element in error)
an Authorization Request/0100 message for a Mastercard Merchant Presented QR Funding Transaction contains DE 108 (Additional Transaction Reference Data), subelement 06 (Dynamic QR Data): • That is not a valid length • That is not alphanumeric special characters	rejects the transaction and forwards the OI an Authorization Request Response/0110 with: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 108006 (indicating the data element in error)

Table 1730: Mastercard Merchant Presented QR Refund Transaction edits

When...	Then the Dual Message Authorization System...
the individual transaction limit or the cumulative transaction amount or count limit is exceeded in an Authorization Request/0100 message for a Mastercard Merchant Presented QR Refund Transaction	rejects the message and forwards to the Receiving Institution an Authorization Request Response/0110 message with: DE 39 (Response Code) = 05 (Do Not Honor) rejects the message and forwards to the Originating Institution an Authorization Advice/0120: system-generated message with: • DE 48 (Additional Data), subelement 71 (On-behalf Service [OBS]), subfield 1 (On-behalf Service Indicator), value 37 (Mastercard Merchant Presented QR Blocking Service) • DE 48, subelement 71, subfield 2 (On-behalf Result 1), value E (Mastercard Merchant Presented QR Blocking—Cumulative Transaction Amount or Count Limit Exceeded) • DE 60 (Advice Reason Code), in the table Subfield 1 Values, in Authorization Advice/0120, value 151 (In Control Processing Advice to Issuer [Mastercard Merchant Presented QR]) • DE 60, in the table Subfield 2 Values, in Mastercard Merchant Presented QR Service, value 0064 (Reject: Transaction Limit Check) when the individual transaction limit is exceeded or • DE 60, in the table Subfield 2 Values, in Mastercard Merchant Presented QR Service, value 0065 (Reject: Cumulative Limit Check) when the cumulative counts or amounts limit is exceeded

Table 1731: Mastercard Merchant Presented QR Refund Payment Processing edits

When...	Then the Dual Message Authorization System...
the Receiving Institution submits an Authorization Request/0100 message for a Mastercard Merchant Presented QR refund transaction it will contain the Digital Account Reference Number in DE 2 (Primary Account Number [PAN])	includes this value in DE 2 of the Authorization Request/0100 message sent to the Originating Institution and the following data is provided in DE 108 (Additional Transaction Reference Data): • Subelement 02 (Sender Data), subfield 11 (Sender Account Number) • Subelement 03 (Transaction Reference Data), subfield 03 (Funding Source)

Alternate Processing

If the receiving institution is unable to respond, Mastercard Merchant Presented QR transactions are declined.

If the Mastercard Merchant Presented QR Blocking Service is not available:

- The platform that called the service receives notice that the service is not available. The following information is sent to the Receiving Institution in the Authorization Request/0100 message for an authorization decision.
 - DE 48 (Additional Data: Private Use), subelement 71 (On-behalf Service), subfield 1 (On-behalf Service Indicator), value 37 (Mastercard Merchant Presented QR Blocking Service)
 - DE 48, subelement 71, subfield 2 (On-behalf Result 1), value U (Unable to process)

Alternate Processing for Refund Payment Transactions

If the consumer's bank is unable to respond to a payment or refund payment request, the Mastercard Merchant Presented QR refund transaction is declined.

Stand-In limits are set to zero.

X-Code Processing

Mastercard Merchant Presented QR transactions is declined by the Dual Message Authorization System in X-Code Processing with DE 39 (Response Code), value 91 (Authorization System or issuer system inoperative).

For more information about Mastercard Merchant Presented QR, refer to the [Mastercard QR: Send Program Guide](#).

Mastercard Send programs

Mastercard Send includes the Mastercard Send Payments Transaction program, the Mastercard Funding Transaction program, and the Mastercard Gaming and Gambling Payments program.

The Mastercard Send programs allow processing of funding and payments transactions on the Dual Message Authorization System to facilitate various money transfer and remittance flows globally for person to person, person to merchant, business to consumer and business to business processing. Consumers can leverage their Mastercard®, Debit® Mastercard®, Cirrus®, or Maestro® card to send and access funds for both domestic and cross-border transactions. Funding and payment transactions can be enabled through multiple channels such as an ATM, a bank branch, a stand-alone kiosk, mobile, or remote.

Mastercard Funding Transaction program

The funding transaction program facilitates both intracountry and cross-border funding transactions from Mastercard, Maestro, and Cirrus-branded cards.

A funding transaction is a point-of-sale (POS) transaction for the purchase of funds transfer services that involves the transfer of funds from an eligible account (by an acquirer or an originating institution on behalf of the cardholder or the sender) for the purpose of either: (a) funding a subsequent transactions and linking funds transfer from the Sender to another person or entity; or (b) transferring funds into another eligible financial account held by that sender.

Mastercard Send Payment program

The Mastercard Send Payment program enables person-to-person transfers, account-to-account transfers, agent cash out, credit card bill payment, business to consumer disbursement, government/non-government organization to consumer disbursement, and business money transfers by allowing consumers to use their Mastercard, Debit Mastercard, Cirrus, or Maestro card to send and access funds. The Mastercard Send Payment program also allows the use of multiple channels to initiate transactions such as an ATM, a bank branch, a stand-alone kiosk, mobile phone, or over the Internet.

Mastercard Gaming and Gambling Payments program

A gaming payment transaction is a type of payment transaction that transfers winnings or value usable for gambling or gaming to a Mastercard or Maestro account. It is affected pursuant to, and subject to, the Gaming and Gambling Payments Standards. The funds are transferred by an Originating Institution, or an acquirer, to the recipient's receiving account at the Receiving Institution, or an issuer. Gaming payment transactions serve the growing demand by originating institutions and receiving institutions for fast, secure, and convenient ways for customers to send and receive gaming payments, respectively.

For a list of countries that process and receive Gaming and Gambling payment transactions, refer to [Mastercard Gaming and Gambling Payments Program Standards](#).

For a list of MCCs supported under the Gaming and Gambling program, refer to [Mastercard Gaming and Gambling Payments Program Standards](#).

Table 1732: Gaming and gambling payment transaction identification

DE 3, Subfield 1 (Cardholder Transaction Type Code)	DE 48, Subelement 77 (Transaction Type Identifier)	DE 18 (Merchant Type)
28	C04	MCC Supported under the Gaming and Gambling program

For more information about implementing the Mastercard Send Payment Transaction Program and Gaming and Gambling Payments Program, refer to the [Mastercard Send and Funding Transactions Program Standards](#) or contact the Global Customer Service team.

Authorization Request/0100: Mastercard Funding Transactions

Originating Institutions (OIs) that are registered for the Mastercard Funding Transaction program will be required to send an Authorization Request/0100 and Reversal Request/0400 message with the required data elements in addition to the data elements that are specific to this service.

Table 1733: Applicable data elements

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Data Element	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)	M	.	M	00 = Funding Transaction
DE 18 (Merchant Type)	M	.	M	MCC 4829 (Money Transfer) MCC 6538 (Mastercard Send Funding) MCC 6540 (Funding Transaction)
DE 48 (Additional Data—Private Use), TCC	M	.	M	R = Face-to-face retail (applicable to MCCs 6538 and 6540) T = Non-face-to-face retail (applicable to MCCs 6538 and 6540) U = Unique Transaction—Quasi Cash Disbursement (applicable to MCC 4829)

Data Element	Org	Sys	Dst	Values/Comments
DE 48, subelement 77 (Transaction Type Identifier)	M	.	M	C07 = General Person-to-Person Transfer C52 = General Transfer to Own Account C53 = Agent Cash Out C54 = Payment of Own Credit Card Bill C55 = Business Disbursement C56 = Government/Non-profit Disbursement C57 = Rapid Merchant Settlement C65 = General Business to Business Transfer F07 = General Person-to-Person Transfer F08 = Person-to-Person Transfer to Card Account F52 = General Transfer to Own Account F53 = Agent Cash Out F54 = Payment of Own Credit Card Bill F55 = Business Disbursement F61 = Transfer to Own Staged Digital Wallet Account F64 = Transfer to Own Debit or Prepaid Account F65 = General Business-to-Business Transfer
DE 108, Additional Transaction Reference Data	O	.	C	Required for all Mastercard Funding Transactions

Reversal Request/0400: Mastercard Funding Transactions

Following is a list of the data elements and values applicable to this message type. All mandatory Reversal Request/0400 data elements apply.

Table 1734: Applicable data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)	M	•	M	00 = Funding Transaction
DE 18 (Merchant Type)	M	•	M	MCC 4829 (Money Transfer) MCC 6538 (Mastercard Send Funding) MCC 6540 (Funding Transaction)
DE 48 (Additional Data—Private Use), TCC	M	•	M	R = Face-to-face retail (applicable to MCCs 6538 and 6540) T = Non-face-to-face retail (applicable to MCCs 6538 and 6540) U = Unique Transaction—Quasi Cash Disbursement (applicable to MCC 4829)
DE 108 (Additional Transaction Reference Data)	O	•	C	Required for all Mastercard Funding Transactions.

Dual Message Authorization System edits

The Dual Message Authorization System performs the following edits on the Authorization Request/0100 and Reversal Request/0400 messages for funding transactions.

NOTE: Refer to the [Mastercard Gaming and Gambling Payments Program Standards](#) for the full list of mandatory fields and subelements.

Table 1735: Associated edits

When...	Then the Dual Message Authorization System...
data in DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) and DE 48 (Additional Data: Private Use), subelement 77 (Transaction Type Identifier) indicate the transaction is a funding transaction refund and The originating institution identified by the ICA provided in DE 32 (Acquiring Institution ID Code) has not registered to process transactions via the Mastercard Send platform	rejects the transaction and sends the originating institution an Authorization Request Response/0110 or Reversal Request Response/0410 message containing DE 39 (Response Code), value 58 (Transaction not permitted to issuer/cardholder).
data in DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) and DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier) indicate the transaction is a funding transaction refund and the brand product code associated with the transaction does not participate in the Mastercard Send transaction	rejects the transaction and sends the originating institution an Authorization Request Response/0110 or Reversal Request Response/0410 message containing DE 39 (Response Code), value 12 (Invalid Transaction).
data in DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) and DE 48 (Additional Data: Private Use), subelement 77 (Transaction Type Identifier) indicate the transaction is a funding transaction refund and the Unique Transaction reference number (DE 108 [Additional Transaction Reference Data], subelement 03 [Transaction Reference Data], subfield 1 [Unique Transaction Reference]) contains all zeros or spaces	sends the originating institution an Authorization Request Response/0110 message to include the following: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 108003 (Additional Data—Private Use)

When...	Then the Dual Message Authorization System...
data in DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) and DE 48 (Additional Data: Private Use), subelement 77 (Transaction Type Identifier) indicate the transaction is a funding transaction refund and The issuer account range does not participate in the Mastercard Send Payment Transaction Program, Mastercard Funding Transaction Program, or both.	sends the originating institution an Authorization Request Response/0110 message to include DE 39 (Response Code) = 57 (Transaction not permitted to issuer or cardholder)
the originating institution is registered for the Mastercard Send service and is submitting Authorization Request/0100 messages for funding transactions and funding transaction refunds where: • DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains value 00 or 20 • DE 18 (Merchant Type) does not contain MCC 6538 • DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier) contains value C07, C52, C53, C54, C55, C56, C57, or C65	sends the originating institution an Authorization Request Response/0110 to include the following: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 018 (indicating the data element in error)
the originating institution is registered for the Mastercard Funding Transaction Program and is submitting an Authorization Request/0100 message for funding transactions and funding transaction refunds where: • DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains value 00 or 20 • DE 18 (Merchant Type) does not contain MCC 4829 or MCC 6540 • DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier) contains value F07, F08, F52, F53, F54, F55, F61, F64, or F65	sends the originating institution an Authorization Request Response/0110 to include the following: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 018 (indicating the data element in error)

When...	Then the Dual Message Authorization System...
the originating institution is registered for the Mastercard Send service and is submitting an Authorization Request/0100 message for funding transactions and funding transaction refunds where: - DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains value 00 or 20 - DE 18 (Merchant Type) contains MCC 6538 - DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier) contains value C07, C52, C53, C54, C55, C56, C57, or C65 - DE 48 (Additional Data: Private Use) Transaction Category Code does not contain value R (Face-to-face retail), T (Non-face-to-face retail), or U (Unique Transaction—Quasi Cash Disbursement).	sends the originating institution an Authorization Request Response/0110 to include the following: - DE 39 (Response Code) = 30 (Format error) - DE 44 (Additional Response Data) = 048000 (Additional Data: Private Use)
the originating institution is registered for the Mastercard Send service and is submitting an Authorization Request/0100 message for funding transactions and funding transaction refunds where: - DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains value 00 or 20 - DE 18 (Merchant Type) contains MCC 4829 or MCC 6540 - DE 48 (Additional Data: Private Use), subelement 77 (Transaction Type Identifier) contains value F07, F08, F52, F53, F54, F55, F61, F64, or F65 - DE 48 (Additional Data: Private Use) Transaction Category Code does not contain value R (Face-to-face retail), T (Non-face-to-face retail), or U (Unique Transaction: Quasi Cash Disbursement).	sends the originating institution an Authorization Request Response/0110 to include the following: - DE 39 (Response Code) = 30 (Format error) - DE 44 (Additional Response Data) = 048000 (Additional Data: Private Use)

Authorization Request/0100: Mastercard Send payment transactions

Originating Institutions (OIs) that are registered for the Mastercard Send Payments Transaction program are required to send Authorization Request/0100 and Reversal Request/0400

messages with the required data elements in addition to the data elements that are specific to this program.

Table 1736: Applicable edits

The following table is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Data Element	Org	Sys	Dst	Values/Comments						
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)	M	•	M	28 = Payment Transaction						
DE 18 (Merchant Type)	M	•	M	Must be one of the following values: MCC 6536 (Mastercard Send Intracountry) MCC 6537 (Mastercard Send Intercountry)						
DE 48 (Additional Data: Private Use), TCC	M	•	M	P = Payment Transaction NOTE: If TCC is not supplied by the acquirer, Mastercard will insert a "P".						
DE 48, subelement 77 (Transaction Type Identifier)	C	•	C	C07 = General Person-to-Person Transfer C52 = General Transfer to Own Account C53 = Agent Cash Out C54 = Payment of Own Credit Card Bill C55 = Business Disbursement C56 = Government/Non-profit Disbursement C57 = Rapid Merchant Settlement C58 = Cash in at ATM C59 = Cash in at point of sale C65 = General Business to Business Transfer						
DE 61 (Point-of-Service [POS] Data), subfield 10 (Cardholder-Activated Terminal Level)	M	•	M	The Dual Message Authorization System validates the POI to ensure DE 61, subfield 10 is one of the following values: <table><tr><th>IF...</th><th>THEN...</th></tr><tr><td>at a bank branch</td><td>0 = Not a CAT transaction</td></tr><tr><td>at an ATM</td><td>1 = Authorized Level 1 CAT: Automated dispensing machine with PIN</td></tr></table>	IF...	THEN...	at a bank branch	0 = Not a CAT transaction	at an ATM	1 = Authorized Level 1 CAT: Automated dispensing machine with PIN
IF...	THEN...									
at a bank branch	0 = Not a CAT transaction									
at an ATM	1 = Authorized Level 1 CAT: Automated dispensing machine with PIN									

Data Element	Org	Sys	Dst	Values/Comments
				at an unmanned kiosk 2 = Authorized Level 2 CAT: Self-service terminal
				on the Internet 6 = Authorized Level 6 CAT: Electronic commerce
				8 = Reserved for future use
DE 108 (Additional Transaction Reference Data)	M	X	C	Edits are applied to specific subelements and subfields within DE 108. These edits are described later in this section.

Reversal Request/0400: Mastercard Send Payment Transaction

Following is a list of the data elements and values applicable to this message type. All mandatory Reversal Request/0400 data elements apply.

Table 1737: Applicable edits

Data Element	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)	M	•	M	28 = Payment Transaction
DE 18 (Merchant Type)	M	•	M	Must be one of the following values: MCC 6536 (Mastercard Send Intracountry) MCC 6537 (Mastercard Send Intercountry)
DE 48 (Additional Data: Private Use), TCC	M	•	M	P = Payment Transaction
DE 48, subelement 77 (Transaction Type Identifier)	C	•	C	C07 = General Person-to-Person Transfer C52 = General Transfer to Own Account C53 = Agent Cash Out C54 = Payment of Own Credit Card Bill C55 = Business Disbursement C56 = Government/Non-profit Disbursement C57 = Rapid Merchant Settlement C58 = Cash in at ATM C59 = Cash in at point of sale C65 = General Business to Business Transfer

Data Element	Org	Sys	Dst	Values/Comments
DE 61 (Point-of-Service [POS] Data), subfield 10 (Cardholder-Activated Terminal Level)	M	•	M	The Dual Message Authorization System will validate the POI to ensure DE 61, subfield 10 is one of the following values: C07, C52–C59, C65, C67
				IF... THEN...
				at a bank branch 0 = Not a CAT transaction
				at an ATM 1 = Authorized Level 1 CAT: Automated dispensing machine with PIN
				at an unmanned kiosk 2 = Authorized Level 2 CAT: Self-service terminal
				on the Internet 6 = Authorized Level 6 CAT: Electronic commerce
DE 108 (Additional Transaction Reference Data)	C	X	C	Mandatory for acquirers to send for all Mastercard Send Payment and Gaming and Gambling Payment Transactions and optional for all Funding Transactions. Issuers are required to send DE 108 in the response message; issuers must echo, unedited, the information sent by the originator/acquirer.

NOTE: A Mastercard Send Payment transaction may only be reversed by the acquirer for reason of a documented clerical error and upon agreement with the issuer. In such an event, the error must be reversed within 60 seconds of the time the Mastercard Send Payment transaction was authorized.

Dual Message Authorization System edits

The Dual Message Authorization System performs the following edits on the Authorization Request/0100 and Reversal Request/0400 messages for Mastercard Send Payment and Gaming and Gambling Payment transactions.

Table 1738: Associated edits

When...	Then the Authorization Platform...
an originating institution identified by the ICA number provided in DE 32 (Acquiring Institution ID Code) attempts to submit a Mastercard Send Payment Transaction or Gaming and Gambling Payment transactions but has not registered for the Mastercard Send Payment Transaction Program or Gaming and Gambling payment program	rejects the transactions and sends the originating institution an Authorization Request Response/0110 or Reversal Request Response/0410 message containing DE 39 (Response Code), value 58 (Transaction not permitted to acquirer/terminal).
the originating institution is registered for the Mastercard Send service and is submitting a Mastercard Send Payment Transaction where: • DE 3, subfield 1 contains value 28 • DE 18 does not contain MCC 6536 or MCC 6537 • DE 48, subelement 77 contains value C07, C52, C53, C54, C55, C56, C57, C58, C59, C60, or C65 • DE 61, subfield 10 criteria are met for Moneysend Payment transaction – 0 = Not a CAT transaction – 1 = Authorized Level 1 CAT: Automated dispensing machine with PIN – 2 = Authorized Level 2 CAT: Self-service terminal – 6 = Authorized Level 6 CAT: Electronic commerce • DE 108 is present	sends the originating institution an Authorization Request Response/0110 or Reversal Response/0410 message where: • DE 39 = 30 • DE 44 = 018 (Merchant Type)

When...	Then the Authorization Platform...
the originating institution is registered for the Mastercard Send Payment Transaction Program or Mastercard Gaming and Gambling Payments Program and is submitting a Mastercard Send Payment Transaction or Mastercard Gaming and Gambling Payments Program where: • DE 3, subfield 1 contains value 28 • DE 18 contains value MCC associated with Mastercard Send Payment or Gaming and Gambling payment program • DE 48, subelement 77 contains the Transactions Type Identifier associated with Mastercard Send Payment or Gaming and Gambling payment program • DE 61, subfield 10 criteria is not met • DE 108 is present	sends the originating institution an Authorization Request Response/0110 or Reversal Request Response/0410 message where: • DE 39 = 30 • DE 44 = 061 (Point-of-Service [POS] Data)
the originating institution is registered for the Mastercard Send Payment Transaction Program or Mastercard Gaming and Gambling Payments Program and is submitting a Mastercard Send Payment Transaction or Gaming and Gambling Payment transactions where: • DE 3, subfield 1 contains value 28 • DE 18 contains associated with Mastercard Send Payment or Gaming and Gambling payment program • DE 48, subelement 77 contains value C07, C52, C53, C54, C55, C56, C57, C58, C59, C60, or C65 • DE 61, subfield 10 criteria are met on Moneysend Payment transaction • DE 108 is not present	sends the originating institution an Authorization Request Response/0110 or Reversal Response/0410 message where: • DE 39 = 30 • DE 44 = 108000
• DE 3, subfield 1 contains value 28 • DE 48, subelement 77 contains the Transactions Type Identifier associated with Mastercard Send Payment or Gaming and Gambling payment program • DE 61, subfield 13 (POS Country Code), and the licensed country of DE 2 (Primary Account Number) do not match (indicating a cross-border transaction) • DE 108, subelement 01, subfield 01 (Receiver/Recipient First Name) is missing, all spaces, all numbers, or any combination of spaces and numbers	sends the originating institution an Authorization Request Response/0110 or Reversal Response/0410 message where: • DE 39 = 30 • DE 44 = 108001 (Receiver/Recipient Data)

When...	Then the Authorization Platform...
DE 3, subfield 1 contains value 28 and DE 61, subfield 10 contains value 1 and DE 52 and DE 55 are not present	forwards the Authorization Request/0100 message to the issuer.
the Mastercard Send platform does not support an acquiring country	declines the request and sends the originating institution an Authorization Request Response/0110 message where DE 39 = 58 (Transaction not permitted to acquirer/terminal)
- DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) equals 28 (Payment) - DE 48 (Additional Data: Private Use), subelement 77 (Transaction Type Identifier) contains Transactions Type Identifier associated with Mastercard Send Payment or Gaming and Gambling payment program and The account range is not registered for the Mastercard Send Payment Transaction Program Mastercard Gaming and Gambling Payments Program	declines the request and sends the originating institution an Authorization Request Response/0110 message where DE 39 = 57 (Transaction not permitted to issuer/cardholder)
the Originating Institution is registered for the Mastercard Send payment transaction Program or Mastercard Gaming and Gambling Payments Program and is submitting an Authorization Request/0100 message for a Mastercard Send Payment or Gaming and Gambling Payment transaction where: - DE 3, subfield 1 contains value 28 - DE 18 contains MCC associated with Mastercard Send Payment or Gaming and Gambling Payments Program - DE 48, subelement 77 contains Transactions Type Identifier associated with Mastercard Send Payment or Gaming and Gambling Payments program - DE108, subfield 02, subelement 04 (Sender Street Address) is not present or contains all zeros, all spaces, or any combination of zeros and spaces	sends the Originating Institution an Authorization Request Response/0110 message where: - DE 39 = 30 (Format Error) - DE 44 = 108002 (Sender Data)

When...	Then the Authorization Platform...
the Originating Institution is registered for the Mastercard Send payment transaction Program or Mastercard Gaming and Gambling Payments Program and is submitting an Authorization Request/0100 message for a Mastercard Send Payment or Gaming and Gambling Payment transaction where: • DE 3, subfield 1 contains value 28 • DE 18 contains MCC values associated with Mastercard Send Payment or Gaming and Gambling Payments program • DE 48, subelement 77 contains Transactions Type Identifier associated with Mastercard Send Payment or Gaming and Gambling Payments Program • DE 108, subelement 02, subfield 07 Sender Country code is present but not a valid ISO recognized alpha Country code (Refer to the Quick Reference Booklet for valid Alpha [3 Characters] Country code)	sends the Originating Institution an Authorization Request Response/0110 message where: • DE 39 = 30 • DE 44 = 108002 (Sender Data)
the Originating Institution is registered for the Mastercard Send payment transaction Program or Mastercard Gaming and Gambling Payments Program and is submitting an Authorization Request/0100 message for a Mastercard Send payment transaction or Gaming and Gambling payment transaction where: • DE 3, subfield 1 contains value 28 • DE 18 contains MCC associated with Mastercard Send Payment or Gaming and Gambling Payments Program • DE 48, subelement 77 contains Transactions Type Identifier associated with Mastercard Send Payment or Gaming and Gambling Payments Program • DE 108, subelement 01, subfield 07 (Receiver/Recipient Country) is valid but blocked (part of the Mastercard Send Blocking Country List)	sends the Originating Institution an Authorization Request Response/0110 to include the following: DE 39 (Response Code) = 57 (Transaction not permitted to issuer/cardholder) and sends the RFI an Authorization Advice Message/0120 where: • DE 48, subelement 71, subfield 1, value 33 (Send Blocking Service) • DE 48, subelement 71, subfield 2, value W (Send Mastercard Blocking—Country not allowed for the Send Transaction) • DE 60, subfield 1, value 200 (In Control Processing Advice to Issuer) • DE 60, subfield 2, value 0072 (Reject: Geographic Restriction)

When...	Then the Authorization Platform...
the Originating Institution is registered for the Mastercard Send payment transaction Program or Mastercard Gaming and Gambling Payments Program and is submitting an Authorization Request/0100 message for a Mastercard Send payment transaction or Gaming and Gambling payment transaction where: • DE 3, subfield 1 contains value 28 • DE 18 contains MCC value associated with Mastercard Send Payment or Gaming and Gambling Payments Program • DE 48, subelement 77 contains Transactions Type Identifier associated with Mastercard Send Payment or Gaming and Gambling Payments Program • DE 108, subelement 02, subfield 07 (Sender Country) is USA or CAN • DE 108, subelement 02, subfield 06 (Sender State/Province Code) is not present or contains all zeros or spaces or invalid code for USA and CAN	sends the Originating Institution an Authorization Request Response/0110 message where: • DE 39 = 30 • DE 44 = 108002 (Sender Data)
the Originating Institution is registered for the Mastercard Send payment transaction Program or Mastercard Gaming and Gambling Payments Program and is submitting an Authorization Request/0100 message for a Mastercard Send payment transaction or Gaming and Gambling payment transaction where: • DE 3, subfield 1 contains value 28 • DE 18 contains MCC associated with Mastercard Send Payment or Gaming and Gambling Payments Program • DE 48, subelement 77 contains Transactions Type Identifier associated with Mastercard Send Payment or Gaming and Gambling Payments Program • DE 61, subfield 10 criteria are met • DE 108 is present and valid	sends an Authorization Request Response/0110 message to the acquirer where: DE 39 (Response Code) = 12 (Invalid Transaction)

When...	Then the Authorization Platform...
an Originating Institution that participates in Payment Transfer Activity (PTA) submits a non-Mastercard Send or non-Mastercard Merchant Presented QR transaction that does not contain DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type), value 28 (Payment Transaction) in the Authorization Request/0100 message	rejects the transaction and sends the Authorization Request Response/0110 message to the acquirer where: DE 39 (Response Code) = 12 (Decline: Invalid Transaction)
Mastercard Send or Gaming and Gambling payment transactions containing DE 108 (Additional Transaction Reference Data), subelement 01 (Receiver/Recipient Data), subfield 07 (Receiver/Recipient Country) is USA or CAN and DE 108, subelement 01 (Receiver/Recipient Data), subfield 06 (Receiver/Recipient State/Province Code) contains all spaces or all zeros in the Authorization Request/0100 message	rejects the transaction and sends the Authorization Request Response/0110 message to the acquirer where: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 108001 (Receiver/Recipient Data)
DE 108 (Additional Transaction Reference Data), subelement 02 (Sender Data), subfield 04 (Sender Street Address) for Mastercard Send payment transactions is not present or contains all zeros, all spaces, or any combination of these two values in the Authorization Request/0100 message	rejects the transaction and sends the Authorization Request Response/0110 message to the acquirer where: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 108002
DE 108 (Additional Transaction Reference Data), subelement 02 (Sender Data), subfield 07 (Sender Country) for Mastercard Send payment transactions or Gaming and Gambling Payment transactions is not present or contains all zeros, all spaces, or any combination of these two values in the Authorization Request/0100 message	rejects the transaction and sends the Authorization Request Response/0110 message to the acquirer where: • DE 39 (Response Code) = 30 (Format error) • DE 44 (Additional Response Data) = 108002
Gaming and Gambling Payment transactions where: DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) equals 28 (Payment) and DE 48 (Additional Data: Private Use), subelement 77 (Transaction Type Identifier) equals C04 (Gaming Re-pay) and DE 61, subfield 13 (POS Country Code) and the licensed country of DE 2 (Primary Account Number) does not match and cross-border is not permitted as per Mastercard Gaming and Gambling Payments Program Standards	rejects the transaction and sends the Authorization Request Response/0110 message to the acquirer where: DE 39 (Response Code) = 12 (Invalid Transaction)

Mastercard MoneySend Issuer Transaction Controls

The Mastercard® MoneySend™ Issuer Transaction Controls is an On-behalf Service that Receiving Institutions can use to manage their risk. Monitoring and blocking options include velocity by number of transactions and cumulative dollar amount, and sanctions score.

Network Blocking

Mastercard sets single transaction amounts and 30-day accumulative amount limits to receiving accounts for Mastercard Send Payment, Gaming and Gambling payment transactions. The maximum amount limit for single transactions and 30-day accumulative amount limits vary by program type. Mastercard declines transactions over the set limits with DE 39, value 05 (Do not honor).

Sanctions Scoring

Mastercard provides a Sanctions Scoring service to assist with meeting Anti-Money Laundering (AML) obligations.

NOTE: Mastercard has suspended the Mastercard Send Sanctions Scoring Service effective 30 April 2021. Customers will be advised accordingly should Mastercard reinstate the service at a later date.

The sanctions score is provided to acquirers and issuers and is available for all cross-border Mastercard Send Payment transactions and intracountry Mastercard Send payment transactions in Egypt, Canada, across Europe, and the U.S. This service provides a score based on screening the individual sender name (consumer, business, government, and non-government) in the authorization message to support real-time decision making.

Stand-In Authorization and X-Code Processing

Mastercard Send payment and Gaming and Gambling payment transactions are declined by Stand-In Authorization and X-Code Processing. Mastercard Send funding transactions are evaluated against limits and processing in accordance with the issuer Stand-In limits or X-Code controls pertinent to the applicable information in the message (such as Brand, MCC, Purchase (DE 3 = 00), and so forth).

For More Information About Implementing the MoneySend Transaction Program, Gaming and Gambling

For more information about implementing the MoneySend Payment Transaction Program and Gaming and Gambling Payments Program, refer to the Mastercard *Mastercard MoneySend and Funding Transactions Program, Gaming and Gambling Standards* or contact the Global Customer Service team.

Mastercard Pay By Account Program

The Mastercard® Pay by Account for retail payments (referred to as Pay by Account Program) helps issuers, referred to as Account Holding Institutions (AHIs) such as banks, credit unions, and digital players, enrolled in the Pay by Account Program, accelerate the digital transformation of

everyday payments by offering consumers additional payment choices, using a secure retail payment solution.

The Pay by Account Program is designed to function as a traditional Debit Mastercard to leverage the Mastercard global acceptance network across in-store (contactless and QR) and e-commerce channels.

The Pay by Account Program enables:

- Account holders to use a digital account-based payment solution through an AHI's online and mobile platforms to process retail transactions. The solution enables an account holder to select an eligible consumer account to make a payment to a merchant across a range of use cases including online, in-app, and in-store purchases; and
- A merchant, through its acquirer, which is referred to as an Acquiring Institution, enrolled in the Pay by Account Program to (1) opt-in and receive real-time settlement of the merchant's proceeds from the Acquiring Institution or (2) process the transaction as a Debit Mastercard transaction and receive settlement of the merchant's proceeds from the Acquiring Institution through its usual processes for the Dual Message Authorization System.

To Participate

Acquiring Institutions and AHIs that choose to participate in the Mastercard Pay by Account Program must register by contacting their Mastercard account representative or Global Customer Service.

MCC 106: MDES PAN Mapping File

Account Holding Institution (AHI) can request two distinct types of maintenance for Mastercard Pay by Account tokens stored in the MDES PAN Mapping File: Token status changes and financial account information replacement.

Requests for Mastercard Pay by Account token record maintenance are made using the Issuer File Update Request/0302 message. This message meets all the general message requirements and the following Mastercard Pay by Account requirements:

- DE 2 (Primary Account Number) must not be present
- DE 91 (Issuer File Update Code) must contain value 2 (Update)
- DE 101 (File Name) must contain the value MCC106 (MDES PAN Mapping File)
- DE 120 must be present and formatted as described in the following table

Table 1739: Field values

Field ID	Field Name	Attributes	Comments/Valid Values
1	Mapping File Indicator	an-1	Required F = Financial Account Maintenance

Field ID	Field Name	Attributes	Comments/Valid Values
2	ICA	n-6	Required ICA of the Account Holding Institution (AHI) at which the funding account resides. Cannot be all zeros; must contain a valid ICA. Must be the same as the ICA currently associated with the account. Format: Right-justified with leading zeros for ICAs less than six-digits (for example, 001234).
3	Country Code	an-3	Required Numeric three-digit country code of the bank at which the funding account resides. Cannot be all spaces nor all zeros; must be a valid three-digit Mastercard-supported country code. Alpha country codes are not supported in this field. Must be the same as the country code currently associated with the account. Customers can reference the <i>Mastercard Quick Reference Booklet</i> for the listing of valid country codes.
4	Action Required	an-1	Required C = Resume Token D = Deactivate Token S = Suspend Token U = Update (Replace) Financial Account Information
5	Current Financial Account Information	ans-64	Required The financial account information currently associated with one or more Mastercard Pay by Account tokens. The value provided must be a minimum of 9 characters, and a maximum of 64 characters.

Field ID	Field Name	Attributes	Comments/Valid Values
6	Token	an-19	Conditional A token associated with the financial account information. Contains all spaces when requesting an update action to replace the current financial account information. Subfield 6 must contain: - A valid token or all spaces when subfield 4 (Action Required) contains C (Resume Token), D (Deactivate Token), or S (Suspend Token). The presence of a token indicates a request to change the status of only the token provided. The presence of all spaces indicates a request to change the status of all tokens associated with the financial account information in subfield 5. - All spaces when subfield 4 contains U (Update Financial Account Information).
7	Replacement Financial Account Information	ans-64	Conditional The new financial account information that is associated with all Mastercard Pay by Account tokens with the same ICA, country code, and related current financial account information. When requesting an update to the financial account information (subfield 4, value U), all tokens associated with the current financial account information are updated with the new financial account information; financial account information for a specific token (in a multi-token scenario) cannot be replaced. A specific token cannot be provided when subfield 4 contains U. The value provided must be a minimum of 9 characters, and a maximum of 64 characters. NOTE: The update to Financial Account Information will proceed regardless of the Replacement Financial Account Information already being associated to existing tokens. Those tokens that are already associated with new financial accounts are not impacted by this operation.

Table 1740: Field values

The following DE 120 layout applies to Issuer File Update Request Response/0312 messages created by Mastercard in response to MCC106 maintenance for Mastercard Pay by Account Token-PAN Financial Account information mapping records.

Field ID	Field Name	Attributes	Comments/Valid Values
1	Mapping File Indicator	an-1	F = Financial Account Maintenance The value is echoed from the Issuer File Maintenance Request/0302 Message.
2	ICA	n-6	ICA of the Account Holding Institution (AHI) at which the funding account resides. The value is echoed from the Issuer File Maintenance Request/0302 Message.
3	Country Code	an-3	Numeric three-digit country code of the bank at which the funding account resides. The value is echoed from the Issuer File Maintenance Request/0302 Message.
4	Action Required	an-1	C = Resume Token D = Deactivate Token S = Suspend Token U = Update (Replace) Financial Account Information The value is echoed from the Issuer File Maintenance Request/0302 Message.
5	Current Financial Account Information	ans-64	The financial account information provided in the original request message. The value is echoed from the Issuer File Maintenance Request/0302 Message.
6	Token	an-19	The token or space value provided in the original request message. The value is echoed from the Issuer File Maintenance Request/0302 Message.
7	Replacement Financial Account Information	ans-64	The financial account information provided in the original request message. The value is echoed from the Issuer File Maintenance Request/0302 Message.

Field ID	Field Name	Attributes	Comments/Valid Values
8	Date of Last Update Activity	n-6	Contains the date on which the record was most recently maintained. Format: MMDDYY (Month, Day, Year) Present only in the Issuer File Update Request Response/0312 Message sent by Mastercard. It must not be present in the Issuer File Maintenance Request/0302 Message sent to Mastercard.
9	Time of Last Update Activity	n-4	Contains the time at which the record was updated on the most recent maintenance date. Format: HHMM (hour, minute) Present only in the Issuer File Update Request Response/0312 Message sent by Mastercard. It must not be present in the Issuer File Maintenance Request/0302 Message sent to Mastercard.

Dual Message Authorization System Edits

The Dual Message Authorization System performs the following system edits related to Mastercard Pay By Account transactions.

Table 1741: Associated edits

When...	Then the Dual Message Authorization System
DE 2 (Primary Account Number [PAN]) is present in the Issuer File Update Request/0302 Message and contains a Mastercard Pay by Account PAN And DE 120 (Record Data), field 1 contains value M (Mastercard Digital Enablement Service Tokens excluding Static Tokens) or A (All Mastercard Digital Enablement Service Tokens including Static Tokens) And DE 120 (Record Data), field 2 (Action Required) does not contain value C (Resume Token), D (Deactivate Token), or S (Suspend Token)	rejects the message with: - DE 39 (Response Code) = 30 (Format error) - DE 44 (Additional Response Data) = 002 (indicating the data element in error)

When...	Then the Dual Message Authorization System
DE 91 (Issuer File Update Code) contains value 1 (Add), 3 (Delete), or 4 (Inquiry) And DE 101 (File Name) = MCC106 (MDES PAN Mapping File) and DE 120 (Record Data), field 1 (Mapping File Indicator) contains value F (Financial Account Maintenance)	rejects the message with: - DE 39 (Response Code) = 40 (Requested function not supported) - DE 44 (Additional Response Data) = 101 (indicating the data element in error)
DE 120 (Record Data), field 1 (Mapping File Indicator) contains value F (Financial Account Maintenance) and the PAN stored in the Token-PAN mapping record is not a Mastercard Pay by Account PAN	rejects the message with: - DE 39 (Response Code) = 27 (Issuer File Update field edit error) - DE 44 (Additional Response Data) = 120001 (indicating the data element in error)
DE 120 (Record Data), field 1 (Mapping File Indicator), contains value F (Financial Account Maintenance) And DE 2 (Primary Account Number [PAN]) is present in the Issuer File Update Request/0302 Message containing a Mastercard Pay by Account PAN	rejects the message with: - DE 39 (Response Code) = 30 (Format Error) - DE 44 (Additional Response Data) = 002 (indicating the data element in error)
DE 120 (Record Data), field 1 (Mapping File Indicator) contains value F (Financial Account Maintenance) and DE 120, field 2 (ICA) is zero-filled or does not contain a valid ICA or DE 120, field 2 does not match the ICA present in the existing mapping record	rejects the message with: - DE 39 (Response Code) = 27 (Issuer File Update field edit error) - DE 44 (Additional Response Data) = 120002 (indicating the data element in error)

When...	Then the Dual Message Authorization System
DE 120 (Record Data), field 1 (Mapping File Indicator) contains value F (Financial Account Maintenance) and DE 120, field 3 (Country Code) contains any non-numeric characters, or is zero-filled, or does not contain a valid Mastercard-supported three-digit country code value or DE 120, field 3 does not match the country code present in the existing mapping record	rejects the message with: - DE 39 (Response Code) = 27 (Issuer File Update field edit error) - DE 44 (Additional Response Data) = 120003 (indicating the data element in error)
DE 120 (Record Data), field 1 (Mapping File Indicator) contains value F (Financial Account Maintenance) and DE 120 (Record Data), field 4 (Action Required) contains a space or blank, or is other than value C (Resume Token), D (Deactivate Token), S (Suspend Token), or U (Update Financial Account Information) or DE 120 (Record Data), field 4 (Action Required) contains value C, D, or S and subfield 7 (Replacement Financial Account Information) is not space or blank filled	rejects the message with: - DE 39 (Response Code) = 27 (Issuer File Update field edit error) - DE 44 (Additional Response Data) = 120004 (indicating the data element in error)
DE 120 (Record Data), field 1 (Mapping File Indicator) contains value F (Financial Account Maintenance) and DE 120 (Record Data), field 4 (Action Required) contains value U (Update Financial Account Information) and subfield 7 (Replacement Financial Account Information) is space or blank filled	rejects the message with: - DE 39 (Response Code) = 27 (Issuer File Update field edit error) - DE 44 (Additional Response Data) = 120004 (indicating the data element in error)

When...	Then the Dual Message Authorization System
DE 120 (Record Data), field 1 (Mapping File Indicator) contains value F (Financial Account Maintenance) and DE 120 (Record Data), field 4 (Action Required) contains value C (Resume Token), D (Deactivate Token), or S (Suspend Token) and subfield 6 (Token) contains a value other than spaces or blanks or a valid numeric Mastercard Pay by Account token or DE 120 (Record Data), field 4 (Action Required) contains value U (Update Financial Account Information) and field 6 (Token) is not space or blank filled	rejects the message with: - DE 39 (Response Code) = 27 (Issuer File Update field edit error) - DE 44 (Additional Response Data) = 120004 (indicating the data element in error)
DE 120 (Record Data), field 1 (Mapping File Indicator) contains value F (Financial Account Maintenance) and DE 120 (Record Data), field 5 (Current Financial Account Information) contains a space or blank or DE 120 (Record Data), field 5 (Current Financial Account Information) cannot be located in the Token-PAN mapping file	rejects the message with: - DE 39 (Response Code) = 27 (Issuer File Update field edit error) - DE 44 (Additional Response Data) = 120005 (indicating the data element in error)
DE 120 (Record Data), field 1 (Mapping File Indicator) contains value F (Financial Account Maintenance) and DE 120 (Record Data), field 6 (Token) cannot be located in the Token-PAN mapping file	rejects the message with: - DE 39 (Response Code) = 25 (Unable to locate record on file [no action taken]) - DE 44 (Additional Response Data) = 120006 (indicating the data element in error)
DE 120 (Record Data), field 1 (Mapping File Indicator) contains value F (Financial Account Maintenance) and DE 120 (Record Data), field 5 (Current Financial Account Information) contains the same value as field 7 (Replacement Financial Account Information)	rejects the message with: - DE 39 (Response Code) = 80 (Duplicate Add, action not performed) - DE 44 (Additional Response Data) = 120007 (indicating the data element in error)

When...	Then the Dual Message Authorization System
DE 120 (Record Data), field 1 (Mapping File Indicator) contains value F (Financial Account Maintenance) and DE 120 (Record Data), field 4 (Action Required) contains value U (Update Financial Account Information) and field 7 (Replacement Financial Account Information) value provided is fewer than 9 characters or more than 64 characters	rejects the message with: - DE 39 (Response Code) = 27 (Issuer File Update field edit error) - DE 44 (Additional Response Data) = 120007 (indicating the data element in error)
DE 120 (Record Data), field 1 (Mapping File Indicator) contains value F (Financial Account Maintenance) and DE 120 (Record Data), field 7 (Replacement Financial Account Information) contains the same financial account information as the Token-PAN Mapping File	rejects the message with: - DE 39 (Response Code) = 80 (Duplicate Add, action not performed) - DE 44 (Additional Response Data) = 120007 (indicating the data element in error)

Mastercard Safety Net

This section provides a brief overview of the Mastercard Safety Net service, part of the Mastercard Global Safety and Security Standards suite of services.

Summary and Background

Mastercard Safety Net provides issuers with a second line of defense to limit the impact of a large-scale fraud attack on one or more of their payment channels (for example, in ATM or e-commerce) when their payment systems are breached. The service also identifies when large-scale fraud attacks are occurring, using insights from the Mastercard Network, so that appropriate action can be taken by the issuer. It does not replace an issuer's primary fraud prevention system.

As applicable, Mastercard requires issuers to participate in Mastercard Safety Net across their entire consumer and commercial credit and debit portfolios, without the availability of an opt-out.

Issuers may enroll in the Mastercard Safety Net Alert Only option if desired through **Manage My Accounts** in Mastercard Connect™.

Data Requirements

If a transaction is declined by the Safety Net service, issuers will receive an Authorization Advice/0120 message with the following values:

- DE 48, subelement 71, subfield 1 = 18 (Fraud Scoring Service)
- DE 48, subelement 71, subfield 2 = C (Fraud Scoring Service was performed successfully)
- DE 48, subelement 75, subfield 1 = 998 or 000
- DE 48, subelement 75, subfield 2 = NM (Network Monitor)
- DE 60, subfield 1 = 120 (Transaction Blocking)
- DE 121 = 000003 (Decline occurred due to an on-behalf service)

For more information about these data elements and values, refer to [Data element definitions](#).

For More Information

For more information, refer to the [Safety Net User Guide](#).

Masterpass Transactions

This section provides a brief overview of the Masterpass™ by Mastercard® service.

Masterpass is a secure and convenient method for consumers to conduct e-commerce wallet transactions or transactions originated from other wallets. Masterpass enables e-commerce merchants to convert browsing customers into buyers by providing a fast, convenient, and secure checkout experience.

How It Works

The following information provides a high-level overview of the Masterpass wallet transaction process:

1. The consumer clicks the **Masterpass** button (or the **BUY WITH Masterpass** button) on a merchant's website to go the Masterpass sign-in page.
2. The consumer chooses the integrated Masterpass digital wallet that he or she wants to use, and then successfully completes authentication.
3. The consumer selects the preferred payment card and shipping address.
4. Masterpass securely transfers the consumer's payment and shipping information to the merchant's website confirmation page, where the merchant completes the checkout process.
5. The consumer's payment information is submitted to the merchant's acquirer for processing.

Authorization Processing

Masterpass transactions submitted to the Masterpass platform for processing are identified by DE 48 (Additional Data—Private Use), subelement 26 (Wallet Program Data), subfield 1 (Wallet Identifier) containing a three-digit value.

Masterpass transactions are identified in the following messages:

- Authorization Request/0100

- Authorization Advice/0120—Acquirer-generated
- Authorization Advice/0120—System-generated

NOTE: Acquirers in the Europe region must support the Masterpass identifiers in these authorization messages.

The data values are generated by the Masterpass platform and passed to the merchant along with the consumer's checkout information (for example, card credentials, shipping address, and email address).

There will not be an edit to determine if these values are present; however, if a value exists, it will be validated and rejected if it consists of special characters, all zeros, or spaces.

For More Information

More information on Masterpass can be found at www.masterpass.com.

Authorization Request/0100: Masterpass Online Wallet

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply and all other transaction conditions that identify specific transactions must be provided. For e-commerce transactions, the transaction details and values must be provided.

Data Element	Org	Sys	Dst	Values/Comments
DE 48 Additional Data—Private Use	M	X	M	Contains applicable subelement data.
DE 48 Additional Data—Private Use, subelement 26 (Wallet Program Data)	C	.	C	Contains the subfield representing the wallet information. Conditional for Masterpass Online transactions.

Authorization Advice/0120: Acquirer-Generated

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Advice/0120 data elements apply.

Data Element	Org	Sys	Dst	Values/Comments
DE 48 Additional Data—Private Use	M	X	M	Contains applicable subelement data.
DE 48 Additional Data—Private Use, subelement 26 (Wallet Program Data)	C	.	C	Contains the subfield representing the wallet information. Conditional for Masterpass Online transactions.

Authorization Advice/0120: System-Generated

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Advice/0120 data elements apply.

Data Element	Org	Sys	Dst	Values/Comments
DE 48 Additional Data—Private Use	.	M	M	Contains applicable subelement data.
DE 48 Additional Data—Private Use, subelement 26 (Wallet Program Data)	.	C	C	Contains the subfield representing the wallet information. Conditional for Masterpass Online transactions.

Authorization Platform Edits

The Authorization Platform performs the following edits on Masterpass Transactions.

WHEN the message contains...	THEN the Authorization Platform...
The Authorization Request/0100 or Authorization Advice/0120 message includes DE 48 (Additional Data—Private Use), subelement 26 (Wallet Program Data), subfield 1 (Wallet Identifier) with a value other than the following: 0–9 - A–Z or a–z Subelement 26 cannot contain all zeros. If all zeros is present, transaction will be rejected.	Sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130—System-generated where: - DE 39 (Response Code) = 30 (Format error) - DE 44 (Additional Response Data) = 048026 (indicating the data element in error)

Merchant Advice Codes

Mastercard supports the use of Merchant Advice codes for issuers to communicate clearly with merchants:

- The reason for approving or declining a transaction
- The actions merchants can take to continue to serve their customers

Issuers can use Merchant Advice codes to provide specific direction to acquirers.

Merchant Advice Codes Used with Response Codes

Issuers can use the Merchant Advice Codes in conjunction or association with Response Codes to indicate to merchants how to respond to various transaction scenarios.

DE 48, Subelement 84 (Merchant Advice Codes)

Issuers can use the following values to indicate Merchant Advice codes in Authorization Request Response/0110 messages, DE 48, subelement 84.

Value	Description
01	New Account Information Available
02	Try Again Later
03	Do Not Try Again
04	Token requirements not fulfilled for this token type
05	Negotiated value not approved
21	Payment Cancellation
22	Merchant does not qualify for product code
24	Retry after 1 hour (Mastercard use only)
25	Retry after 24 hours (Mastercard use only)
26	Retry after 2 days (Mastercard use only)
27	Retry after 4 days (Mastercard use only)
28	Retry after 6 days (Mastercard use only)
29	Retry after 8 days (Mastercard use only)
30	Retry after 10 days (Mastercard use only)
40	Consumer non-reloadable prepaid card
41	Consumer single-use virtual card number

Authorization Request Response/0110, DE 39 Response Codes

The following table lists the most common DE 39 values that issuers send in conjunction with Merchant Advice codes in DE 48, subelement 84.

Value	Description
00	Approved
05	Do not honor
14	Invalid card number

Value	Description
51	Insufficient funds/over credit limit
54	Expired card

Examples of Combined DE 48, Subelement 84 and DE 39 Values

The following table provides examples of how issuers and acquirers should use the combination of DE 48, subelement 84, and DE 39.

DE 39	DE 48, subelement 84	Merchant Advice Description	Examples of Reason for Decline	Suggested Merchant Action
00 05 14 51 54	01	New account information available	- Expired card - Account upgrade - Portfolio sale - Conversion	Obtain new account information before next billing cycle
51	02	Try again later	- Over credit limit - Insufficient funds	Recycle transaction 72 hours later
05 14 51 54	03	Do not try again	- Account closed - Fraudulent	Obtain another type of payment from customer
30	04	Token requirements not fulfilled for this token type	Invalid token to PAN mapping	Obtain another type of payment from customer
05	21	Payment Cancellation	Cardholder cancelled recurring agreement	Do not resubmit transaction

Merchant actions

As part of the Mastercard Authorization Optimizer service, these are the actions merchants should take using the combination of DE 48, subelement 84, and DE 39 to make better decisions on card-not-present authorizations.

Table 1742: DE 48, subelement 84, and DE 39

DE 39	DE 48, subelement 84	Merchant advice description	Merchant action
79 or 82	01	Updated information needed	Updated information may be available in Mastercard ABU database, secure new information before reattempting.
79 or 82	03	Do not try again	Updated credentials not found in the Mastercard ABU database, do not retry.
83	01	Additional information needed	Ensure that card information is correct. Authentication may improve likelihood of an approval, retry using authentication such as EMV 3DS.
83	03	Do not try again	Suspected fraud do not retry.
79 or 82 or 83	02	Try again later	Retry transaction later.

M/Chip Processing Services

Mastercard provides M/Chip Processing services to assist issuers that choose to migrate to Mastercard M/Chip technology while minimizing changes to their host systems. The Stand-In System provides chip security features for customers currently using M/Chip technology. The security features provide additional verification for chip transactions when the issuer's host is unavailable.

The Mastercard M/Chip Processing services include:

- Chip to Magnetic Stripe Conversion
- M/Chip Cryptogram Pre-validation

- M/Chip Cryptogram Validation in Stand-In processing
- Combined Service Option

Refer to the *Authorization Manual* for M/Chip Processing service programs and service information.

The Mastercard M/Chip Processing services support issuers using M/Chip technology in their authorization processing by performing all, or part of the M/Chip-related authorization processing on-behalf of the issuer in a designated account range.

The Chip to Magnetic Stripe Conversion and the M/Chip Cryptogram Pre-validation services are provided on a permanent basis. The M/Chip Cryptogram Validation in Stand-In processing service is provided on a dynamic basis if and when the issuer is not able to respond to an authorization request.

Mastercard provides the following DE 48 (Additional Data–Private Use) subelements that support M/Chip Processing services:

- Subelement 71 (On-behalf Services)
- Subelement 72 (Issuer Chip Authentication)
- Subelement 79 (Chip CVR/TVR Bit Error Results)

Program Use of M/Chip Processing Service Data Elements

The following table defines the M/Chip Processing Services use of DE 48, subelement 71, subelement 72, subelement 74, and subelement 79.

DE 48 Subelement:	Description
Subelement 71 (On-behalf Services)	Subelement 71 notifies the issuer of the M/Chip Processing service performed on the transaction. Transactions that meet the criteria for the service contain subelement 71 data in DE 48 of the Authorization Request/0100 to notify the issuer of the service performed and the results. Issuers must return subelement 71 in the Authorization Request Response/0110 message when subelement 71 is present in the Authorization Request/0100 message. The acquiring MIP removes subelement 71 before sending the Authorization Request Response/0110 message to the acquirer host.

DE 48 Subelement:	Description
Subelement 72 (Issuer Chip Authentication)	Mastercard uses subelement 72 to carry data used during cryptogram processing. Issuers subscribing to the M/Chip Cryptogram Pre-validation service will have subelement 72 in the Authorization Request/0100 message when the ARQC is valid as determined by Mastercard. Issuers that use the M/Chip Cryptogram Validation in Stand-In processing service or the M/Chip Cryptogram Pre-validation service will have subelement 72 in the Authorization Advice/0120 message. In this situation, subelement 72 contains the ARPC (Authorization Response Cryptogram) present in DE 55, subelement ID 91 (Issuer Authentication Data) in the Authorization Request Response/0110 based on the approval or decline decision. NOTE: Subelement 72 should not be returned in the Authorization Request Response/0110. The Authorization Platform returns subelement 72 in the Authorization Request Response/0110 message when subelement 72 is present in the Authorization Request/0100 message. The acquiring MIP removes subelement 72 before sending the Authorization Request Response/0110 message to the acquirer host.
Subelement 74 (Additional Processing Information)	For M/Chip Processing services, subelement 74 notifies the acquirer that there was an issue with the cryptogram validation. Mastercard provides the acquirer an Authorization Request Response/0110 message containing subelement 74 when there is an issue with cryptogram validation during M/Chip Processing service processing. Issuers may provide the acquirer an Authorization Request Response/0110 message containing subelement 74 when there is an issue with cryptogram validation when they perform their own validation.

DE 48 Subelement:	Description
Subelement 79 (Chip CVR/TVR Bit Error Results)	Issuers that participate in the M/Chip Cryptogram Pre-validation Service or the M/Chip Cryptogram Validation in Stand-In Processing Service receive subelement 79 in the Authorization Request/0100 and the Authorization Advice/0120—System-generated when errors are detected in the CVR/TVR within the Issuer Application data during M/Chip Cryptogram Validation processing. Subelement 79 is present when subelement 71, subfield 2 (OB Result 1) contains the value T (TVR/CVR was invalid). Subelement 79 should not be returned in the Authorization Request Response/0110.

Chip To Magnetic Stripe Conversion

The Chip to Magnetic Stripe Conversion service is an optional service that removes M/Chip-related data element (DE) 55 (Integrated Circuit Card [ICC] System-related Data), if present and DE 23 (Card Sequence Number), if present and changes the value in DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) before sending the Authorization Request/0100 message to the issuer for processing.

Chip to Magnetic Stripe Conversion affects issuers only. Acquirers are not affected by this service. Issuers subscribing to the Chip to Magnetic Stripe Conversion service must ensure that the cards are not personalized to expect Issuer Authentication Data in DE 55 in the Authorization Request Response/0110 message.

Authorization Advice/0120 messages contain the converted magnetic stripe transaction and include DE 48, subelement 71.

Issuers should be aware that the acquirer submits the original M/Chip transaction into clearing.

The Global Clearing Management System (GCMS) offers the Chip to Magnetic Stripe Conversion service. Issuers should refer to the *GCMS Reference Manual* for information about this optional service.

To request this service, issuers must contact Mastercard and identify the account range that the service supports.

Authorization and Stand-In Processing

The following Dual Message System and Stand-In processing applies for M/Chip Processing Services.

Dual Message System Processing

The issuer receives the Authorization Request/0100 message containing DE 48, subelement 71 with the value 01C, 01M, or 01S indicating that the Chip to Magnetic Stripe Conversion On-behalf service was performed on the transaction.

IF the transaction is Full-Grade and...	THEN the Authorization Platform will...
DE 55 is present, and	Remove DE 55
DE 23 is present, and	Remove DE 23
DE 22, subfield 1 contains:	
05 = PAN auto-entry via chip	Change DE 22, subfield 1 to 90 = PAN auto-entry via magnetic stripe and DE 48, subelement 71, subfield 2, value C (Conversion of the M/Chip transaction to a magnetic stripe transaction was completed from an original POS PAN Entry Mode value of 05 [PAN auto-entry via chip] or 79 [Chip card/PAN entry via manual entry])
07 = PAN auto-entry via contactless M/Chip	Change DE 22, subfield 1 to 90 = PAN auto-entry via magnetic stripe and DE 48, subelement 71, subfield 2, value S (Conversion of the M/Chip transaction to a magnetic stripe transaction was completed from an original POS PAN Entry Mode value of 07 [PAN auto-entry via contactless M/Chip])
79 = Chip card/PAN entry via manual entry	Change DE 22, subfield 1 to 01 = PAN manual entry and DE 48, subelement 71, subfield 2, value C (Conversion of the M/Chip transaction to a magnetic stripe transaction was completed from an original POS PAN Entry Mode value of 05 [PAN auto-entry via chip] or 79 [Chip card/PAN entry via manual entry])
80 = Chip card/PAN via magnetic stripe	Change DE 22, subfield 1 to 90 = PAN auto-entry via magnetic stripe and DE 48, subelement 71, subfield 2, value M (Conversion of the M/Chip fallback transaction to a magnetic stripe transaction was completed from an original POS PAN Entry Mode value of 80 [PAN auto-entry with magnetic stripe])

IF the transaction is Partial-Grade and...	THEN the Authorization Platform will...
DE 55 is not present, and DE 23 is present	Remove DE 23
DE 22, subfield 1 contains: 05 = PAN auto-entry via chip	Change DE 22, subfield 1 to 90 = PAN auto-entry via magnetic stripe and DE 48, subelement 71, subfield 2, value C (Conversion of the M/Chip transaction to a magnetic stripe transaction was completed from an original POS PAN Entry Mode value of 05 [PAN auto-entry via chip] or 79 [Chip card/PAN entry via manual entry])
07 = PAN auto-entry via contactless M/Chip	Change DE 22, subfield 1 to 90 = PAN auto-entry via magnetic stripe and DE 48, subelement 71, subfield 2, value S (Conversion of the M/Chip transaction to a magnetic stripe transaction was completed from an original POS PAN Entry Mode value of 07 [PAN auto-entry via contactless M/Chip])
79 = Chip card/PAN entry via manual entry	Change DE 22, subfield 1 to 01 = PAN manual entry and DE 48, subelement 71, subfield 2, value C (Conversion of the M/Chip transaction to a magnetic stripe transaction was completed from an original POS PAN Entry Mode value of 05 [PAN auto-entry via chip] or 79 [Chip card/PAN entry via manual entry])
80 = Chip card/PAN via magnetic stripe	Change DE 22, subfield 1 to 90 = PAN auto-entry via magnetic stripe and DE 48, subelement 71, subfield 2, value M (Conversion of the M/Chip fallback transaction to a magnetic stripe transaction was completed from an original POS PAN Entry Mode value of 80 [PAN auto-entry with magnetic stripe])

IF DE 55 is present and...	THEN the Authorization Platform will...
DE 22 is not 03, 05, 07, 09, 79, or 80	Reject the transaction for a format error: DE 39 = 30 DE 44 = 055

Stand-In and X-Code Processing

Stand-In processes the Authorization Request/0100 message as a magnetic stripe transaction if the issuer is not available. X-Code processes the Authorization Request/0100 message as a magnetic stripe transaction if Stand-In processing is not available.

The Authorization Advice/0120 message contains the converted magnetic stripe transaction and includes DE 48, subelement 71.

Subelement 71 (On-behalf Services)...	Contains...		
Subfield 1 (OB Service)	01	=	Chip to Magnetic Strip Conversion
Subfield 2 (OB Result 1)	C	=	Conversion of the M/Chip transaction to a magnetic stripe transaction was completed from an original POS PAN Entry Mode value of 05 [PAN auto-entry via chip] or 79 [Chip card/PAN entry via manual entry])
	M	=	Conversion of the M/Chip fallback transaction to a magnetic stripe transaction was completed from an original POS PAN Entry Mode value of 80 [PAN auto-entry with magnetic stripe]
	S	=	Conversion of the M/Chip transaction to a magnetic stripe transaction was completed from an original POS PAN Entry Mode value of 07 [PAN auto-entry via contactless M/Chip])
Subfield 3 (OB Result 2)	Blank	=	No value present

M/Chip Cryptogram Pre-validation

The M/Chip Cryptogram Pre-validation service validates the Authorization Request Cryptogram (ARQC) and generates the Authorization Response Cryptogram (ARPC) for issuers subscribing to this service. Mastercard supports cryptogram validation for M/Chip Select 2.0, M/Chip Lite

2.1, M/Chip 4.0 (EMV96 and EMV2000), or Common Core Definition (EMV-CCD) session key derivations.

Transactions that qualify for this service contain the following data elements:

- DE 55 (Integrated Circuit Card [ICC] System Related Data)
- DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) must be one of the following values:
 - 05 = PAN auto-entry via chip
 - 07 = PAN auto-entry via contactless M/Chip
 - 09 = PAN entry via electronic commerce, including remote chip

The Authorization Platform performs the following edits:

IF...	THEN the Authorization Platform will...
DE 55 format is not valid or the required subfields are not present	Reject the transaction for a format error with: DE 39 = 30 DE 44 = 055
DE 48, subelement 71 or subelement 72 are present in the Authorization Request/0100 message from the acquirer host	Reject the transaction for a format error with: DE 39 = 30 DE 44 = 0480nn (where nn is the subelement number)

Decline Option for Issuers

Mastercard has enhanced the M/Chip Cryptogram Pre-validation service with a decline option indicator to decline transactions on-behalf of participating issuers when the Application Request Cryptogram or the chip data validation for the transaction is not successful.

For the decline option indicator, Mastercard will forward the transaction to the issuer or decline the transaction on-behalf of the issuer on the basis of the information included in the decision matrix. For each value of the On-behalf Service (OBS) results, the decision matrix includes which action to take (forward or decline the transaction).

Mastercard supports the following four options for issuers registering to the OBS 02 service.

- Option 1: Do not use the decline option for neither contact nor contactless transactions.
- Option 2: Use the decline option for contact and contactless transactions.
- Option 3: Use the decline option for contact transactions and do not use the decline option for contactless transactions.
- Option 4: Use the decline option for contactless transactions and do not use the decline option for contact transactions.

Validation of the Application Cryptogram

Validation of the ARQC in the Application Cryptogram subfield supports parameters linked to each key, based on the PAN (account range), Floor Expiry date, and DMK index (also known as the Key Derivation Index-KDI).

Issuers must provide Mastercard with the keys and parameters according to the On-behalf key management (OBKM) documentation set. Mastercard generates the ARPC unless there was a technical failure. The session key derivations supported include:

- M/Chip2 (Lite and Select) for the ARQC and the ARPC
- M/Chip4 (EMV96 for Lite and Select) for the ARQC and the ARPC
- M/Chip4 (EMV2000 for Lite and Select) for the ARQC and the ARPC
- Common Core Definition (EMV-CCD) for the ARQC and ARPC

Subelement 71—On-behalf Service notifies the issuer that the M/Chip Cryptogram Pre-validation service was performed on the transaction. This service is indicated in subfield 1 OB Service with the value 02. Subfield 2 OB Result 1 provides information that can be used by the issuer in the authorization decision process.

Several tests are performed on DE 55 during the ARQC validation process. The following identifies the tests and the values in subfield 2 (OB Result 1). If a test produces a negative result, no additional tests are performed.

IF starting with the application cryptogram validation...	THEN the value in subelement 71, subfield 2 (OB Result 1) is...
The Application Cryptogram could not be validated due to a technical error	U
A format error is detected in DE 55	F
The Application Cryptogram is invalid	I
The cryptogram in the Application Cryptogram is valid but the AC is not an ARQC nor a TC	G
The Application Cryptogram was a valid ARQC but the TVR/CVR was invalid	T
There is no matching key file for this PAN, PAN expiry date and KDI combination	K
The security platform has timed out	X
The security platform has experienced a processing error	Z
No errors were detected in the previous tests, the ARQC is valid	V

Issuers will receive additional detail on the specific Card Verification Results (CVR) and Terminal Verification Results (TVR) found to be in error during the cryptogram validation process.

Information about the specific bits in error submitted in the transaction is forwarded to the issuer in DE 48, subelement 79 (Chip CVR/TVR Bit Error Results) of the Authorization Request/0100 message and the Authorization Advice/0120—System-generated message.

The Authorization Request Response/0110 message is returned to the acquirer when a format error occurs. The issuer receives the Authorization Request/0100 message in all other transactions.

When DE 48, subelement 71, subfield 2 contains U, F, I, G, or T, Mastercard sends the acquirer an Authorization Request Response/0110 message where DE 48, subelement 74, subfield 1 is value 02 and subfield 2 is subelement 71, subfield 2 value.

Generation of the Issuer Chip Authentication Data

Mastercard generates the ARPC assuming an approval response from the issuer (DE 39 = 00). This ARPC is placed in DE 48, subelement 72 based on key information provided by the issuer. The ARPC is generated when the ARQC is valid.

Mastercard uses subelement 72 to carry data used during cryptogram processing. The issuer must return this data in the Authorization Request Response/0110 message. If the issuer declines the transaction, Mastercard re-generates the ARPC before sending the Authorization Request Response/0110 message to the acquirer host. Mastercard removes subelements 71 and 72 from the Authorization Request Response/0110 message before sending the message to the acquirer host.

If Stand-In responds to the Authorization Request/0100 message, subelement 72 is also present in the Authorization Advice/0120 message sent by Stand-In processing.

Stand-In processing provides authorization support when the issuer is not available. Normal Stand-In processing occurs when the Application Cryptogram is determined to be valid based on the value in subelement 71, subfield 2 (OB Result 1).

Stand-In processing uses the decision matrix values from the OBKM interface when subelement 71, subfield 2 (OB Result 1) contains a value of G, I, T, or U indicating the Application Cryptogram is invalid.

The Authorization Response Cryptogram is generated based on the Stand-In response. The Authorization Request Response/0110 message created includes DE 55, subfield ID 91 Issuer Authentication Data.

The Authorization Advice/0120 message includes:

- DE 48, subelement 71 that identifies the service performed on the transaction
- DE 48, subelement 72 that contains the Authorization Response Cryptogram generated based on the approved or declined response
- DE 48, subelement 79 that identifies the CVR/TVR bits found to be in error
- DE 55 containing the ARQC from the Authorization Request/0100 message

DE 60 (Advice Reason Code)

DE 60, subfield 2 contains values depending on values contained in DE 48, subelement 71, subfield 2 as shown in the following table.

IF subfield 2 (OB Result 1) contains...	THEN DE 60 (Advice Reason Code), subfield 2 (Advice Detail Code) contains...
F	0059 = Format Error
G	0039 = Application Cryptogram is valid but not an ARQC nor a TC; Status of TVR/CVR unknown
I	0034 = Invalid Chip Cryptogram
K	0037 = No matching key file for this PAN, PAN expiry date and KDI combination—Validation of ARQC and CVR/TVR not performed, status unknown
T	0035 = TVR/CVR validation failed
U	0032 = Reject: Chip Data Processing Error -Or- Another valid advice reason code
V	0000 = Accept -Or- Another valid advice reason code
X	0038 = Security platform time out
Z	0040 = Security platform processing error

Acquirers and issuers are both affected by M/Chip Cryptogram Pre-validation service.

Acquirers should be aware that Issuer Chip Authentication Data may not be present in the Authorization Request Response/0110 message if the ARPC was not generated. The card may decline the transaction.

The M/Chip Cryptogram Pre-validation service is mandated. Issuers must contact Mastercard and identify the account range that the service supports. Issuers are also required to provide the keys and parameters according to the OBKM documentation set.

Mastercard notifies the issuer that the M/Chip Cryptogram Pre-validation service was performed on the transaction by the presence of DE 48, subelement 71.

The Authorization Advice/0120 message provides the ARQC from the Authorization Request/0100 message in DE 55. The ARPC generated based on an approved or declined response is in DE 48, subelement 72.

If Stand-In is not available, the Authorization Request/0100 message is processed by X-Code. No ARQC validation/ARPC generation occurs on X-Code processing.

Issuers should contact their Customer Implementation Services specialist to sign-up and test the M/Chip Processing services.

Registration forms Chip Project Request Workbook (CPRW) and Parameter Worksheet will be updated to include new options for M/Chip Cryptogram Pre-Validation Service processing.

Issuers opting for the M/Chip Cryptogram Pre-Validation Service with the decline option indicator should work with their business and Customer Implementation Services representatives to enable the M/Chip Cryptogram Pre-Validation Service with the decline option indicator by specifying which existing or new account ranges they want to support using the CPRW or *Parameter Worksheet* if issuers are in the Europe region.

Alternate Processing

If a transaction is routed to the Stand-In System, the Stand-In System will respond to the acquirer based on the service results and on the instructions established by the issuer.

Authorization Platform Edits

The Authorization Platform will perform the following system edits.

WHEN...	THEN the Authorization Platform...
EMV tag values other than 91, 71, and 72 are present and sent in the Authorization Request Response/0110 message or if the tag-length-value (TLV) structure of DE 55 (Integrated Circuit Card System Related Data) is incorrect	Sends an Authorization Negative Acknowledgement/0190 message to the issuer as a format edit error where: • DE 39 (Response Code) = 30 (Format Error) • DE 44 (Additional Response Data) = 055 (indicating the data element in error)

Combined Service Option

Issuers can choose an option that combines the Chip to Magnetic Stripe Conversion and M/Chip Cryptogram Pre-validation services.

This option allows two M/Chip processing services to be performed on a single transaction, providing issuers with a bridge to maximize the benefits of chip card processing capabilities while minimizing impacts on their authorization systems. These two services also are available individually.

The Combined Service Option is optional.

M/Chip Cryptogram Validation in Stand-In Processing

The M/Chip Cryptogram Validation in Stand-In processing service supports issuers that process chip transactions on an on-going basis. M/Chip Cryptogram Validation in Stand-In processing

supports chip cryptogram validation and generation of the ARPC when the issuer is signed-out, the transaction cannot be delivered to the issuer, or the issuer timed-out.

Mastercard performs the cryptographic support and provides Dual Message System processing for issuers using the M/Chip Select 2.0, M/Chip Lite 2.1, M/Chip 4.0 (EMV 96 and EMV 2000), and Common Core Definition (EMV-CCD) session key derivations.

Transaction Qualifiers

The processing performed by Mastercard for the M/Chip Cryptogram Validation in Stand-In processing service is very similar to the processing performed for the M/Chip Cryptogram Pre-validation service.

Transactions containing the following data elements qualify for this service:

- DE 55 must be present
- DE 22, subfield 1 containing one of the following values:
 - 05 = PAN auto-entry via chip
 - 07 = PAN auto-entry via contactless M/Chip
 - 09 = PAN entry via electronic commerce, including remote chip

For transactions that qualify for this service:

- The ARQC is validated and the TVR/CVR fields are validated according to a default pattern
- Stand-In processing approves or declines the transaction
- The ARPC is generated
- The Authorization Request Response/0110 message is created
- The Authorization Advice/0120 message is created

Authorization Platform Edits

The Authorization Platform performs the following edits.

IF....	THEN the Authorization Platform will...
DE 55 format is not valid or the required subfields are not present	Reject the transaction with format error in DE 39 = 30 DE 44 = 055
DE 48 subelement 71 or subelement 72 are present in the Authorization Request/0100 from the acquirer	Reject the transaction with format error in DE 39 = 30 DE 44 = 0480nn (where nn is the subelement number)

Validation of the Application Cryptogram

Validation of the ARQC in the Application Cryptogram subfield supports parameters linked to each key, based on the PAN (account range), Floor Expiry date, and DMK index (also known as the Key Derivation Index–KDI).

Mastercard generates the ARPC unless there was a technical failure or the ARQC was invalid.

The session key derivations supported include:

- M/Chip2 (Lite and Select) for the ARQC and the ARPC
- M/Chip4 (EMV 96 for Lite and Select) for the ARQC and the ARPC
- M/Chip4 (EMV 2000 for Lite and Select) for the ARQC and the ARPC
- Common Core Definition (EMV-CCD) for the ARQC and ARPC

Subelement 71 notifies the issuer that the Dynamic M/Chip Stand-In service was performed on the transaction. This is indicated in subfield 1 (OB Service) with the value 03. Subfield 2 (OB Result 1) provides information regarding the results of the ARQC validation.

Several tests are performed on DE 55 during the ARQC validation process. The following identifies the tests and the values in subfield 2. If a test produces a negative result, no additional tests are performed.

IF starting with the application cryptogram validation...	THEN the value in subelement 71, subfield 2 (OB Result 1) is...
The Application Cryptogram could not be validated due to a technical error	U
A format error is detected in DE 55	F (Mastercard Internal only)
The Application Cryptogram is invalid	I
The cryptogram in the Application Cryptogram is valid but the AC is not an ARQC nor a TC	G
The Application Cryptogram was a valid ARQC but the TVR/CVR was invalid	T
There is no matching key file for this PAN, PAN expiry date and KDI combination	K
The security platform has timed out	X
The security platform has experienced a processing error	Z
No errors were detected in the previous tests, the ARQC is valid	V

Issuers will receive additional detail on the specific Card Verification Results (CVR) and Terminal Verification Results (TVR) found to be in error during the cryptogram validation process. Information regarding the specific bits in error submitted in the transaction will be forwarded to

the issuer in DE 48, subelement 79 (Chip CVR/TVR Bit Error Results) of the Authorization Request/0100 message and the Authorization Advice/0120—System-generated message.

The Authorization Request Response/0110 message is returned to the acquirer when a format error occurs. Stand-In processing will use the issuer defined values from the OBKM interface in all other transactions.

When DE 48, subelement 71, subfield 2 contains U, F, I, G, or T, Mastercard will send the acquirer an Authorization Request Response/0110 message where DE 48, subelement 74, subfield 1 is value 03 and subfield 2 is subelement 71, subfield 2 value.

Generation of the Issuer Chip Authentication Data

Mastercard generates the ARPC in DE 48, subelement 72 based on key information provided by the issuer.

Mastercard uses subelement 72 to carry data used during cryptogram processing. Subelement 72 will be present in the Authorization Advice/0120 message sent by Stand-In.

Stand-In processing provides the authorization support when the issuer is signed out, the transaction cannot be delivered to the issuer, or the issuer timed out. Normal Stand-In processing occurs when the ARQC was determined to be valid based on the value in subelement 71, subfield 2.

Stand-In uses the decision matrix values from the OBKM interface when subelement 71, subfield 2 contains the value G, I, T, or U.

The Authorization Response Cryptogram is generated based on the Stand-In response. The Authorization Request Response/0110 message created includes DE 55, subfield ID 91 Issuer Authentication Data.

The Authorization Advice/0120 message includes:

- DE 48, subelement 71 that identifies the service performed and results
- DE 48, subelement 72 that contains the Authorization Response Cryptogram generated based on approved or declined response
- DE 48, subelement 79 that identifies the CVR/TVR bits found to be in error
- DE 55 containing the ARQC from the Authorization Request/0100 message

DE 60: Advice Reason Code

DE 60, subfield 2 contains values depending on values contained in DE 48, subelement 71, subfield 2 as shown in the following table.

IF subfield 2 (OB Result 1) contains...	THEN DE 60 (Advice Reason Code), subfield 2 (Advice Detail Code) contains...
F	0059 = Format Error
G	0039 = Application Cryptogram is valid but not an ARQC nor a TC; Status of TVR/CVR unknown
I	0034 = Invalid Chip Cryptogram

IF subfield 2 (OB Result 1) contains...	THEN DE 60 (Advice Reason Code), subfield 2 (Advice Detail Code) contains...
K	0037 = No matching key file for this PAN, PAN expiry date and KDI combination—Validation of ARQC and CVR/TVR not performed, status unknown
T	0035 = TVR/CVR validation failed
U	0032 = Reject: Chip Data Processing Error -Or- Another valid advice reason code
V	0000 = Accept -Or- Another valid advice reason code
X	0038 = Security platform time out
Z	0040 = Security platform processing error

Acquirers and issuers are both affected by M/Chip Cryptogram Validation in Stand-In processing service.

Acquirers should be aware that Issuer Chip Authentication Data may not be present in the Authorization Request Response/0110 message if the ARPC was not generated. The card may decline the transaction.

The Authorization Advice/0120 message provides the ARQC from the Authorization Request/0100 in DE 55. The ARPC generated is in DE 48, subelement 72 when the transaction was approved.

If Stand-In processing is not available, the Authorization Request/0100 message is processed by X-Code processing. No ARQC validation/ARPC generation occurs in X-Code processing.

The M/Chip Cryptogram Validation in Stand-In processing service is mandated.

Issuers should contact their Customer Implementation Services specialist for information regarding signing up and testing for the M/Chip Processing services.

Issuers must contact Mastercard and identify the account range that the service will support.

Alternate Processing

If a transaction is routed to the Stand-In System, the Stand-In System will respond to the acquirer based on the service results and on the instructions established by the issuer.

Authorization Platform Edits

The Authorization Platform will perform the following system edits.

WHEN...	THEN the Authorization Platform...
EMV tag values other than 91, 71, and 72 are present and sent in the Authorization Request Response/0110 message or if the tag-length-value (TLV) structure of DE 55 (Integrated Circuit Card System Related Data) is incorrect	Sends an Authorization Negative Acknowledgement/0190 message to the issuer as a format edit error where: - DE 39 (Response Code) = 30 (Format Error) - DE 44 (Additional Response Data) = 055 (indicating the data element in error)

MIP Transaction Blocking

MIP Transaction Blocking provides issuers with easy-to-manage, flexible controls that supplement their authorization strategy.

MIP Transaction Blocking extends an issuer's authorization capabilities, augmenting their defenses to protect portfolios from fraud attacks, as well as manage specialized payment programs with ease. An issuer's participation in the MIP Transaction Blocking Service is optional.

This service may not have the effect of causing the acquirer to be in noncompliance with any Mastercard Standard, including Rule 6.4, Selective Authorization.

Issuers may want to apply MIP Transaction Blocking to:

- Prevent the authorization of fraudulent transactions for compromised account ranges
- Prevent the authorization of transactions due to operational emergencies (for example, the issuer encounters problems supporting certain processing codes or POS PAN entry modes)
- Decline authorizations for an account range in combination with the point-of-interaction (POI) country in the event of local issues that necessitate preventing transactions from being authorized until those issues are resolved
- Protect assigned BIN ranges before they go into production or account ranges in production that are not actively used

MIP Transaction Blocking provides issuers the ability to decline authorizations in the Dual Message System based upon the issuing ICA number or account range, and one or more of the following transaction parameters:

- DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)
- DE 18 (Merchant Type)
- DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)

- DE 32 (Acquiring Institution ID Code)
- DE 61 (Point-of-Service [POS] Data), subfield 10 (Card Activated Terminal Level [CAT])
- DE 61 (Point-of-Service [POS] Data), subfield 13 (POS Country Code [or Sponsored Merchant Information, if applicable])

MIP Transaction Blocking can also provide Full BIN Blocking on authorization requests for an entire BIN range or a segment of a BIN range, defined up to 11 digits, and is available globally on both the Dual and Single Message Systems.

MIP Transaction Block Setup

Issuers must provide the following information for each transaction block setup.

The following information is required for each transaction block setup:

- ICA or Routing and Transit Number and account range
- The data elements identifying the type of transaction to be blocked
- Effective date
- The response code that will be returned in the Authorization Request Response/0110 message to the acquirer when the transaction matches the blocking criteria. Issuers may choose a response code from the following list:
 - 03 (Invalid merchant)
 - 04 (Capture card)
 - 05 (Do not honor)
 - 12 (Invalid transaction)
 - 57 (Transaction not permitted to issuer/cardholder)
 - 58 (Transaction not permitted to acquirer/terminal)

NOTE: If issuers do not specify a decline response code, the default response code is 05 (Do not honor).

- Registration for the optional MIP Transaction Blocking ICA Level Block Summary (SI738010-AA) and delivery endpoint.

MIP Transaction Blocking ICA Level Block Summary (SI738010-AA)

The MIP Transaction Blocking ICA Level Block Summary (SI738010-AA) is an optional report showing each ICA level block for the issuer and the account ranges associated to that block. The ICA Blocking Summary report will assist issuers in determining the account ranges that are blocked via ICA level MIP Transaction Blocking records.

To Participate

Issuers may choose to participate in and customize their MIP Transaction Blocking settings by completing the *MIP Transaction Blocking Service Request Form* (Form 810) and submitting the form to the Global Customer Service team.

For questions about participating in MIP Transaction Blocking, contact the Global Customer Service team.

Dual Message Authorization System Edits

The Dual Message Authorization System performs the following edits on Authorization Request/0100: MIP Transaction Blocking transactions.

Table 1743: Associated edits

WHEN...	THEN the Dual Message Authorization System...
the acquirer sends an Authorization Request/0100 message and the transaction matches a MIP transaction block record	sends the acquirer an Authorization Request Response/0110 message where: DE 39 = The response code identified in the MIP transaction block setup and may send the issuer an Authorization Advice/0120 message where • DE 39 = The response code sent in the Authorization Request Response/0110 • DE 60 (Advice Reason Code), subfield 1 (Advice Detail Code) will contain 120 • DE 121 (Authorizing Agent ID Code) will contain 000003 (Decline occurred due to an on-behalf service)
the transaction does not match any MIP transaction block	forwards the Authorization Request/0100 message to the issuer or to the Stand-In Authorization if the issuer is unavailable.

Full BIN Block

The Full BIN Block service blocks authorization requests for an entire BIN range or a segment of a BIN range, defined up to 11 digits.

Through this service, issuers can protect assigned BIN ranges before they go into production. Issuers also can protect account ranges in production that are not actively used. The Full BIN Block service will decline any authorization request in the identified account range with response code 14 (Invalid card number).

The Full BIN Block service is available globally on both the Dual Message Authorization and Single Message Systems. This service is recommended when a BIN has been assigned by Mastercard for use by the issuer, but that is not yet supported by the issuer's authorization system.

Participation in the Full BIN Block service is optional.

Benefits

Issuers may want to apply authorization blocking for inactive BINs to:

- Block an entire BIN range or a segment of a BIN range.
- Combat fraud attacks on inactive BIN ranges, defined as “live” on Mastercard systems but not in production on the issuer host system.

To Participate

To sign up for the Full BIN Block service, issuers must complete the *MIP Transaction Blocking Service Request Form* (Form 810).

Dual Message Authorization System Edits

The Dual Message Authorization System performs the following edits on Authorization Request/0100—Full BIN Block transactions.

Table 1744: Associated edits

WHEN...	THEN the Dual Message Authorization System...
the acquirer sends an Authorization Request/0100 message and the transaction falls within a Full BIN Block	sends the acquirer an Authorization Request Response/0110 message where: DE 39 = 14 (Invalid card number. The transaction is declined.)

Partial Approvals

The Authorization Platform allows acquirers to indicate whether the merchant terminal supports receipt of partial approvals in an authorization request message.

If the acquirer has indicated that the merchant terminal supports receipt of partial approvals, issuers can approve a portion of the requested transaction amount by responding with the approved amount and a partial approval response code in the authorization message.

All Debit Mastercard® card issuers in the U.S. region must support partial approvals and updates to the cardholder’s open-to-buy balance upon receipt of a reversal (full or partial).

The United Kingdom must support partial approvals for debit/prepaid cards and balance responses for prepaid cards. For MCC 5542 only partial approval support is required.

Authorization Request/0100: Partial Approval

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Table 1745: Data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 48 (Additional Data: Private Use), subelement 61 (POS Data, Extended Condition Codes), subfield 1 (Partial Approval Terminal Indicator)	C	.	C	1 = Merchant terminal supports receipt of partial approvals

Authorization Request Response/0110: Partial Approval

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0110 data elements apply.

Table 1746: Data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 4 (Amount, Transaction)	.	X	M	Partial approval amount in acquirer transaction currency.
DE 5 (Amount, Settlement)	.	X	C	Partial approval amount in the settlement currency.
DE 6 (Amount, Cardholder Billing)	M	.	M	Partial approval amount in issuer cardholder billing currency. This amount can be less than or equal to the original amount present in the Authorization Request/0100 message in the amount data element that corresponds to the issuer cardholder billing currency. This amount may be less than, greater than, or equal to the amount present in the Authorization Request/0100 message when the DE 18 (Merchant Type) is value 5542 (Fuel Dispenser, Automated)
DE 38 (Authorization ID Response)	M	.	M	Authorization Code.
DE 39 (Response Code)	M	.	M	10 = Partial Approval
DE 51 (Currency Code, Cardholder Billing)	M	.	M	Issuer cardholder billing currency code.

Data Element	Org	Sys	Dst	Values/Comments
DE 54 (Additional Amounts), subfield 1 (Account Type)	•	X	M	00 = Default Account (not specified or not applicable)
DE 54 (Additional Amounts), subfield 2 (Amount Type)	•	X	M	57 = Original Amount
DE 54 (Additional Amounts), subfield 3 (Currency Code)	•	X	M	Refer to the Quick Reference Booklet
DE 54, subfield 4 (Debit or Credit Indicator)	•	X	M	C plus 12 digits

NOTE: Issuers responding with DE 39, value 10 will not be required to echo DE 4 (Amount, Transaction) in the Authorization Request Response/0110 message. Likewise, if DE 5 (Amount, Settlement) was present in the Authorization Request/0100 message to the issuer, the issuer will not be required to echo DE 5 in the Authorization Request Response/0110 message when responding with DE 39, value 10. The issuer will provide the partial approval amount in DE 6 (Amount, Cardholder Billing).

The Dual Message Authorization System additionally provides the partial approval amount to the acquirer in the following data elements of the Authorization Request Response/0110 message:

- DE 4 in acquirer's transaction currency
- DE 5 in U.S. dollars if the acquirer receives settlement amount-related data elements
- DE 6 in the issuer's cardholder billing currency

The Dual Message Authorization System will provide two additional occurrences of the original amount in DE 54 of the Authorization Request Response/0110 message to the acquirer; one in the acquirer's currency and one in the issuer cardholder billing currency.

Reversal Request/0400: Partial Approval

In some cases, the cardholder or merchant may elect not to complete the transaction after receiving the partial approval response from the issuer. Mastercard supports full acquirer-generated reversal messages to allow the merchant to cancel the partial approval.

In addition to all other applicable data elements for the Reversal Request/0400 message, acquirers should submit Reversal Request/0400 messages with the following data elements for a partial approval:

- The partial approval amount in DE 4 that was present in the Authorization Request Response/0110 message to the acquirer, not the original amount present in DE 4 of the Authorization Request/0100 message from the acquirer
- DE 39, value 10

When processing a Reversal Request/0400 message for a partial approval (DE 39, value 10), the issuer should increase the cardholder open-to-buy using the partial approval amount present in the amount data element of the Reversal Request/0400 message that corresponds to the issuer cardholder billing currency.

The Reversal Request/0400 message will not contain DE 48, subelement 61 or DE 54, subelement 2, value 57.

Reversal Advice/0420: Partial Approval

Following are data elements applicable to this message in addition to the required data elements.

If the Dual Message Authorization System generates a Reversal Advice/0420 message after the issuer has responded to the Authorization Request Response/0110 message with DE 39, value 10, the Dual Message Authorization System will provide the following data elements:

- Partial approval amount in DE 4 in the acquirer transaction currency
- Partial approval amount in DE 5 if the issuer has opted to receive amounts in the settlement currency (U.S. dollars)
- Partial approval amount in DE 6 in the issuer cardholder billing currency
- DE 39, value 10
- Original amount in DE 54 in the issuer cardholder billing currency (and acquirer transaction currency if different from issuer cardholder billing currency)

When processing an Reversal Advice/0420 message for a partial approval (DE 39, value 10), the issuer should increase the cardholder open-to-buy using the partial approval amount present in the amount data element of the Reversal Advice/0420 message that corresponds to the issuer's cardholder billing currency.

Authorization Advice/0120: Acquirer-Generated

The following data elements apply to this message in addition to the required data elements.

When an acquirer creates an Authorization Advice/0120: acquirer-generated message to advise the issuer of an approved authorization performed by the acquirer, it may include DE 48, subelement 61 in the Authorization Advice/0120: acquirer-generated message if it was present in the original Authorization Request/0100 message.

DE 39, value 10 is not a valid value for Authorization Advice/0120: acquirer-generated messages. If an Authorization Advice/0120: acquirer-generated message contains DE 39, value 10 and the advice is not for an Automated Fuel Dispenser Completion (DE 18 = 5542), the Dual Message Authorization System will generate an Authorization Advice Response/0130 message containing DE 39, value 30 and DE 44 (Additional Data), value 039.

If an Authorization Advice/0120: acquirer-generated message contains DE 39, value 10 and the advice is for an Automated Fuel Dispenser Completion, the Dual Message Authorization System will change the DE 39 value to 00 (Approved or completed successfully) before forwarding the advice to the issuer.

Alternate Processing

The Stand-In Authorization and the X-Code Processing will not provide a partial approval response to an authorization request because these systems do not maintain card balances.

However, if an Authorization Request/0100 message containing DE 48, subelement 61 is processed by the Stand-In Authorization or the X-Code Processing, the corresponding Authorization Advice/0120: system-generated message will contain DE 48, subelement 61.

Authorization Platform Edits

The Authorization Platform will perform the following edits.

Authorization Request/0100 Edits

The Authorization Platform performs the following edits on the Authorization Request/0100 message.

WHEN...	THEN the Authorization Platform...
DE 48, subelement 61, subfield 1 contains a value other than 0 or 1	Generates an Authorization Request Response/0110 message containing: DE 39 = 30 DE 44 = 048061
DE 48, subelement 61, subfields 4–5 contain values other than 0	Generates an Authorization Request Response/0110 message containing: DE 39 = 30 DE 44 = 048061
DE 54 is present in the Authorization Request/0100 message where DE 54, subfield 2 contains value 57 (Original Amount)	Generates an Authorization Request Response/0110 message containing: DE 39 = 30 DE 44 = 054

Authorization Request Response/0110 Edits

The Authorization Platform performs the following edits on the Authorization Request Response/0110 message when the issuer has provided DE 39 with value 10.

WHEN...	THEN the Authorization Platform...
DE 48, subelement 61 is not present or DE 48, subelement 61, subfield 1 does not contain value 1 or DE 3, subfield 1 is 09 in the Authorization Request/0100 message sent to the issuer	Generates an Authorization Response Negative Acknowledgement/0190 message to the issuer where: DE 39 = 30 DE 44 = 039
DE 38 is not present	Generates an Authorization Response Negative Acknowledgement/0190 message to the issuer where: DE 39 = 30 DE 44 = 038
DE 6 is not present	Generates an Authorization Response Negative Acknowledgement/0190 message to the issuer where: DE 39 = 30 DE 44 = 006
The issuer responds with DE 39, value 10 (Partial Approval) and DE 6 is a partial approval amount equal to the original requested amount in the Authorization Request/0100 message	Allows the partial approval amount to be equal to the requested amount.
DE 6 (Amount, Cardholder Billing) is an amount greater than or equal to the original amount provided in the Authorization Request/0100 message And DE 39 (Response Code) is 10 (Partial Approval) And DE 18 (Merchant Type) is not 5542 (Fuel Dispenser, Automated)	Rejects the Authorization Request Response/0110 message with an Authorization Response Negative Acknowledgement/0190 where: DE 39 = 30 DE 44 = 006
DE 51 is not present or is not the issuer's correct cardholder billing currency code	Generates an Authorization Response Negative Acknowledgement/0190 message to the issuer where: DE 39 = 30 DE 44 = 051

Authorization Advice (Acquirer-generated)/0120 Edits

The Authorization Platform performs the following edit on the Authorization Advice/0120—Acquirer-generated message.

WHEN...	THEN the Authorization Platform...
DE 39 contains value 10 (Partial Approval) and the advice message is not for an Automated Fuel Dispenser Completion message (DE 18 = 5542)	Generates an Authorization Advice Response (System-generated)/0130 message containing: DE 39 = 30 DE 44 = 039

Payment Transactions

A Payment Transaction facilitates the movement of funds between two parties—a payer (sender) and a payee (recipient). This transaction can be used to support several business opportunities, such as person-to-person payments, merchant rebates and rewards, loading value to a debit or prepaid account, issuer rebates and rewards.

A Payment Transaction also may be used to initiate a Private Label prepaid card activation request. For card activation requests there is no movement of funds. Refer to the Private Label Processing section in this chapter of this manual for more information.

Issuers must be able to receive and support Payment Transactions unless they have regulatory restrictions that would prevent their participation in Payment Transactions.

Payment Transactions are identified in the following messages:

- Authorization Request/0100
- Authorization Advice/0120—System-generated
- Reversal Request/0400
- Reversal Advice/0420

Authorization Request/0100: Payment Transaction Message

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Table 1747: Data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) (Cardholder account credits)	M	.	M	28 = Payment Transaction

Data Element	Org	Sys	Dst	Values/Comments
DE 3, subfield 2 (Cardholder "From Account" Type Code)	M	.	M	00 = Default account (account not specified or not applicable)
DE 3, subfield 3 (Cardholder "To Account" Type Code)	M	.	M	Any valid value for subfield 3.
DE 18 (Merchant Type)	M	.	M	Valid merchant type value for Payment Transaction.
DE 48 (Additional Data—Private Use), TCC (Transaction Category Code)	M	.	M	P = Payment Transaction
DE 48, subelement 77 (Transaction Type Identifier)	M	.	M	Refer to Subelement 77 (Transaction Type Identifier) for more information.

Authorization Request Response/0110: Payment Transaction

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0110 data elements apply.

Table 1748: Data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code)	ME	.	ME	Processing Code value in the original Authorization Request/0100 message.
DE 48 (Additional Data—Private Use), TCC (Transaction Category Code)	CE	.	CE	P = Payment Transaction

Dual Message Authorization System Edits

Issuers must process transaction amounts in a Payment Transaction as a credit to the cardholder account. Following are Authorization Request/0100 and Reversal Request/0400 edits.

Table 1749: Applicable edits

WHEN...	THEN the Dual Message Authorization System...
DE 3, subfield 1, value 28 and TCC value P are not both present	sends the acquirer a response where: DE 39 = 30 (Format error) DE 44 = 048000

WHEN...	THEN the Dual Message Authorization System...
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) does not contain value 28 (Payment Transaction) and DE 48, subelement 77 is present	removes DE 48, subelement 77 from the following messages: • Authorization Request/0100 • Reversal Request/0400 • Reversal Advice/0420
DE 3, subelement 1 is value 28 and DE 48, subelement 77 is not included	sends the acquirer a response where: DE 39 = 30 (Format error) DE 44 = 048077
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains value 28 (Payment Transaction), and DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier), contains value C09 (Card Activation) and DE 61 (Point of Sale [POS] Data) subfields does not contain the following values: • Subfield 1 (POS Terminal Attendance) = 0 (Attended Terminal) • Subfield 3 (POS Terminal Location) = 0 (On premises of acceptor facility) • Subfield 4 (POS Cardholder Presence) = 0 (Cardholder present) • Subfield 5 (POS Card Presence) = 0 (Card present) • Subfield 10 (Cardholder-Activated Terminal [CAT] Level) = 0 (Not a CAT transaction)	the Dual Message Authorization System declines the request with a format error response where: DE 39 (Response Code) = 30 DE 44 (Response Data) = 061

WHEN...	THEN the Dual Message Authorization System...
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains value 28 (Payment Transaction), and DE 48 (Additional Data: Private Use), subelement 77 (Transaction Type Identifier), contains value C09 (Card Activation) and DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) does not contain one of the following values: • 02 = Merchant terminal PAN auto-entry via magnetic stripe • 05 = Merchant terminal PAN auto-entry via integrated circuit card • 07 = Merchant terminal PAN auto-entry via contactless M/Chip • 80 = Chip card at chip-capable terminal was unable to process transaction using data on the chip; therefore, the terminal defaulted to the magnetic stripe-read PAN. The full track data has been read from the data encoded on the card and transmitted within the Authorization Request/0100 in DE 45 (Track 1 Data) or DE 35 (Track 2 Data) without alteration or truncation. To use this value, the acquirer must be qualified to use value 90. • 90 = PAN auto-entry via magnetic stripe, the full track data has been read from the data encoded on the card and transmitted within the authorization request in DE 35 (Track 2 Data) or DE 45 (Track 1 Data) without alteration or truncation. • 91 = PAN auto-entry via contactless magnetic stripe, the full track data has been read from the data on the card and transmitted within the authorization request in DE 35 (Track 2 Data) or DE 45 (Track 1 Data) without alteration or truncation.	the Dual Message Authorization System declines the request with a format error response where: DE 39 (Response Code) = 30 DE 44 (Response Data) = 022

NOTE: The Dual Message Authorization System will reject the transaction with a format error on DE 3 if the acquirer sends an Authorization Advice/0120: acquirer-generated message where DE 3, subfield 1 contains value 28.

PIN Management Service

Mastercard supports chip and magnetic stripe PIN management services.

Chip PIN Management Service

The Chip PIN Management Service is an optional service that enables cardholders to perform the transactions at ATMs that support Mastercard®, Maestro®, or Cirrus® chip cards.

Transactions supported are:

- PIN change—Allows cardholders to change the PIN code on their chip card.
- PIN unblock—Allows cardholders to unblock the PIN code on their chip card by resetting the PIN try counter.

The Chip PIN Management Service is available only to full grade acquirers and chip grade issuers. For example, the Chip PIN Management Service is *not* available for:

- Transactions initiated with a chip card using magnetic stripe technology
- Issuers that use the Chip to Magnetic Stripe Conversion on-behalf service
- Issuers that use the M/Chip Cryptogram Pre-validation

Authorization Request/0100: PIN Change or PIN Unblock (Chip Card)

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Table 1750: Data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 2 (Primary Account Number [PAN])	M	•	M	
DE 3 (Processing Code), subfield 1	M	•	M	91 = PIN Unblock or 92 = PIN Change
DE 3, subfield 2 (Cardholder Transaction Type Code)	M	•	M	00 = Default Account
DE 3, subfield 3 (Cardholder "To Account" Type Code)	M	•	M	00 = Default Account
DE 4 (Amount, Transaction)	M	•	M	Must be zero unless an ATM transaction fee has been applied by an acquirer for an ATM transaction in a country where application of an ATM transaction fee is allowed.

Program and service format requirements
Authorization Request/0100: PIN Change or PIN Unblock (Chip Card)

Data Element	Org	Sys	Dst	Values/Comments
DE 18 (Merchant Type)	M	•	M	6011 = Member Financial Institution—Automated Cash Disbursements
DE 22 (Point-of-Service (POS) Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)	M	•	M	05 = Merchant terminal PAN auto-entry via integrated circuit card
DE 22, subfield 2 (POS Terminal PIN Entry Mode)	M	•	M	1 = Terminal has PIN entry capability
DE 28 (Amount, Transaction Fee)	C	•	C	May contain an ATM transaction fee, if applicable for transactions in a country where application of an ATM transaction fee is allowed.
DE 32 (Acquiring Institution ID Code)	M	•	M	
DE 48 (Additional Data—Private Use), TCC (Transaction Category Code)	M	•	M	Space or Z = ATM Cash Disbursement
DE 52 (Personal ID Number [PIN] Data)	C	X	C	Required for ATM transactions
DE 55 (Integrated Circuit Card (ICC) System-Related Data)	C	•	C	Conditional data for chip-based transactions
DE 125 (New PIN Data)	C	X	C	Must be present for all online chip card PIN change transactions; otherwise not present DE 125 contains the new PIN, which is formatted into one of the supported PIN block formats and is then encrypted. The PIN block format and encryption method used must be the same as the one used for the existing PIN that is stored in DE 52.

Authorization Request Response/0110: PIN Change or PIN Unblock (Chip Card)

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0110 data elements apply.

Table 1751: Data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 39 (Response Code)	M	.	M	70 = Contact Card Issuer 71 = PIN Not Changed 85 = Not Declined (Successful PIN Change or PIN unblock) (recommended for chip cards) 89 = Unacceptable PIN—Transaction Declined—Retry
DE 55 (Integrated Circuit Card [ICC] System-Related Data)	C	.	C	Contains system-generated Authorization Response Cryptogram (ARPC). Depending on the card application, DE 55 may contain an issuer script. The issuer script message contains instructions to the chip card: • If the issuer approves the PIN unblock request, the script message instructs the chip card to unblock the PIN on the card. • If the issuer declines the PIN unblock request, the issuer may optionally provide additional instructions to block the chip card (for example, block the card or the card application) using the related issuer script.

Reversal Request/0400: PIN Change (Chip Card)

Following is a list of the data elements and values applicable to this message type. All mandatory Reversal Request/0400 data elements apply.

Table 1752: Data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 2 (Primary Account Number [PAN])	M	.	M	

Data Element	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code), subfield 1	M	.	M	92 = PIN Change
DE 3, subfield 2 (Cardholder Transaction Type Code)	M	.	M	00 = Default Account
DE 3, subfield 3 (Cardholder "To Account" Type Code)	M	.	M	00 = Default Account
DE 4 (Amount, Transaction)	M	.	M	Must be zero unless an ATM transaction fee has been applied by an acquirer for an ATM transaction in a country where application of an ATM transaction fee is allowed.
DE 18 (Merchant Type)	M	.	M	6011 = Member Financial Institution: Automated Cash Disbursements
DE 22 (Point-of-Service (POS) Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)	M	.	M	05 = Merchant terminal PAN auto-entry via integrated circuit card
DE 22, subfield 2 (POS Terminal PIN Entry Mode)	M	.	M	1 = Terminal has PIN entry capability
DE 28 (Amount, Transaction Fee)	C	.	C	May contain an ATM transaction fee, if applicable for transactions in a country where application of an ATM transaction fee is allowed.
DE 32 (Acquiring Institution ID Code)	M	.	M	
DE 48 (Additional Data—Private Use), TCC (Transaction Category Code)	M	X	M	Space or Z = ATM Cash Disbursement
DE 48, subelement 20 (Cardholder Verification Method)	M	X	M	P = Online PIN Verification

Magnetic Stripe PIN Management Service

Mastercard has expanded the optional Chip PIN Management service to allow cardholders to change the PIN associated with their non-chip (magnetic stripe) Mastercard credit, Debit Mastercard, Maestro, or Cirrus card at any ATM that supports this functionality.

This expansion allows issuers globally to provide their cardholders with additional conveniences for their magnetic stripe cards.

Participating acquirers in the Europe region that process transactions through the Dual Message System can indicate a PIN change transaction performed on a magnetic stripe card in the Authorization Request/0100—PIN Change (Magnetic Stripe Card) message.

Authorization Request/0100: PIN Change (Magnetic Stripe Card)

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Table 1753: Data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 2 (Primary Account Number [PAN])	M	•	M	
DE 3 (Processing Code), subfield 1	M	•	M	92 = PIN Change
DE 3, subfield 2 (Cardholder Transaction Type Code)	M	•	M	00 = Default Account
DE 3, subfield 3 (Cardholder "To Account" Type Code)	M	•	M	00 = Default Account
DE 4 (Amount, Transaction)	M	•	M	Must be zero unless an ATM transaction fee has been applied by an acquirer for an ATM transaction in a country where application of an ATM transaction fee is allowed.
DE 18 (Merchant Type)	M	•	M	6011 = Member Financial Institution—Automated Cash Disbursements
DE 22 (Point-of-Service (POS) Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)	M	•	M	02 = Merchant terminal PAN auto-entry via magnetic stripe—track data is not required or 90 = PAN auto-entry via magnetic stripe—the full track data has been read
DE 22, subfield 2 (POS Terminal PIN Entry Mode)	M	•	M	1 = Terminal has PIN entry capability
DE 28 (Amount, Transaction Fee)	C	•	C	May contain an ATM transaction fee, if applicable for transactions in a country where application of an ATM transaction fee is allowed.
DE 32 (Acquiring Institution ID Code)	M	•	M	
DE 48 (Additional Data—Private Use), TCC (Transaction Category Code)	M	•	M	Space or Z = ATM Cash Disbursement

Data Element	Org	Sys	Dst	Values/Comments
DE 52 (Personal ID Number [PIN] Data)	C	X	C	Required for ATM transactions.
DE 125 (New PIN Data)	C	X	C	Must be present for all online chip card PIN change transactions; otherwise not present. DE 125 contains the new PIN, which is formatted into one of the supported PIN block formats and is then encrypted. The PIN block format and encryption method used must be the same as the one used for the existing PIN that is stored in DE 52.

NOTE: In the event of a transaction failure or time out, the acquirer must reverse the PIN change transaction so that the issuer is aware that the PIN change was not completed at the ATM. Consequently, the new PIN should not be considered active.

Authorization Request Response/0110: PIN Change (Magnetic Stripe Card)

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0110 data elements apply.

Table 1754: Data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 39 (Response Code)	M	.	M	00 = Approved or completed successfully (Successful PIN Change) 70 = Contact Card Issuer 71 = PIN Not Changed 85 = Not Declined (Successful PIN Change) (recommended for chip cards) 89 = Unacceptable PIN—Transaction Declined—Retry

Reversal Request/0400: PIN Change (Magnetic Stripe Card)

Following is a list of the data elements and values applicable to this message type. All mandatory Reversal Request/0400 data elements apply.

Table 1755: Data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code), subfield 1	M	•	M	92 = PIN Change
DE 3, subfield 2 (Cardholder Transaction Type Code)	M	•	M	00 = Default Account
DE 3, subfield 3 (Cardholder "To Account" Type Code)	M	•	M	00 = Default Account
DE 4 (Amount, Transaction)	M	•	M	Must be zero unless an ATM transaction fee has been applied by an acquirer for an ATM transaction in a country where application of an ATM transaction fee is allowed.
DE 18 (Merchant Type)	M	•	M	6011 = Member Financial Institution: Automated Cash Disbursements
DE 22 (Point-of-Service (POS) Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)	M	•	M	02 = Merchant terminal PAN auto-entry via magnetic stripe: track data is not required or 90 = PAN auto-entry via magnetic stripe: the full track data has been read from the data encoded on the card and transmitted within the authorization request in DE 35 (Track 2 Data) or DE 45 (Track 1 Data) without alteration or truncation.
DE 22, subfield 2 (POS Terminal PIN Entry Mode)	M	•	M	1 = Terminal has PIN entry capability
DE 28 (Amount, Transaction Fee)	C	•	C	May contain an ATM transaction fee, if applicable for transactions in a country where application of an ATM transaction fee is allowed.
DE 48 (Additional Data—Private Use), TCC (Transaction Category Code)	M	X	M	Space or Z = ATM Cash Disbursement

Data Element	Org	Sys	Dst	Values/Comments
DE 48, subelement 20 (Cardholder Verification Method)	M	X	M	P = Online PIN Verification

Authorization Request/0100: Edits (Magnetic Stripe Card)

The following edits apply for PIN Management magnetic stripe transactions.

Table 1756: Applicable edits

WHEN...	THEN the Dual Message Authorization System...
DE 3, subfield 1 contains value 91 (PIN Unblock) and DE 22, subfield 1 does not contain value 05 (PAN auto-entry via chip)	sends the acquirer an Authorization Request Response/0110 message where: DE 39 (Response Code) = 57 (Transaction not permitted to issuer/cardholder)
DE 3, subfield 1 contains value 92 (PIN Change) and DE 125 is not present	sends the acquirer an Authorization Request Response/0110 message where: DE 39 = 30 (Format error) DE 44 (Additional Response Data) = 125 (indicating the data element in error)
DE 3, subfield 1 contains value 92 and DE 22, subfield 1, does not contain one of the following values: 02 (PAN auto-entry via magnetic stripe—track data is not required) 90 (PAN auto-entry via magnetic stripe—the full track data has been read)	sends the acquirer an Authorization Request Response/0110 message where: DE 39 = 30 DE 44 = 022
DE 3, subfield 1 contains value 92 and DE 22, subfield 2 does not contain value 1 (Terminal has PIN entry capability)	sends the acquirer an Authorization Request Response/0110 message where: DE 39 = 30 DE 44 = 022
DE 3, subfield 1 contains value 92 and DE 52 is not present	sends the acquirer an Authorization Request Response/0110 message where: DE 39 = 30 DE 44 = 052

WHEN...	THEN the Dual Message Authorization System...
DE 3, subfield 1 contains value 92 and DE 4 does not contain a value of all zeros, indicating a zero-amount transaction and DE 28 is not present	sends the acquirer an Authorization Request Response/0110 message where: DE 39 = 30 DE 44 = 004
DE 3, subfield 1 contains value 92 and DE 48, TCC contains a value other than Z (ATM Cash Disbursement) or space	sends the acquirer an Authorization Request Response/0110 message where: DE 39 = 30 DE 44 = 048000
the acquirer does not participate in the PIN Management service	sends the acquirer an Authorization Request Response/0110 message where: DE 39 = 58 (Transaction not permitted to acquirer/ terminal)
the issuer does not participate in the PIN Management service	sends the acquirer an Authorization Request Response/0110 message where: DE 39 = 57 (Transaction not permitted to issuer/ cardholder)
the issuer is not able to respond to the PIN Management transaction on time	sends the acquirer an Authorization Request Response/0110 message where: DE 39 = 91 (Authorization System or issuer system inoperative)

Authorization Advice/0120: Acquirer-Generated Edits (Magnetic Stripe Card)

The following edits apply to PIN Management magnetic stripe card transactions.

Table 1757: Applicable edits

WHEN...	THEN the Dual Message Authorization System...
DE 3, subfield 1 contains value 92	sends the acquirer an Authorization Advice Response/0130 message where: DE 39 = 30 DE 44 = 003

Reversal Request/0400: Edits (Magnetic Stripe Card)

The following edits apply to PIN Management, magnetic stripe card transactions.

Table 1758: Applicable edits

WHEN...	THEN the Dual Message Authorization System...
DE 3, subfield 1 contains value 92 and DE 22, subfield 1, does not contain one of the following values: • 02 • 90	sends the acquirer a Reversal Request Response/0410 message where: DE 39 = 30 DE 44 = 022
DE 3, subfield 1 contains value 92 and DE 22, subfield 2 does not contain value 1	sends the acquirer a Reversal Request Response/0410 message where: DE 39 = 30 DE 44 = 022
DE 3 (Processing Code) subfield 1 (Cardholder Transaction Type Code) contains value 92 and DE 4 does not contain a value of all zeros, indicating a zero-amount transaction and DE 28 is not present	sends the acquirer a Reversal Request Response/0410 message where: DE 39 = 30 DE 44 = 004
DE 3, subfield 1 contains value 92 and DE 48, TCC contains a value other than Z or space	sends the acquirer a Reversal Request Response/0410 message where: DE 39 = 30 DE 44 = 048000
the acquirer does not participate in the PIN Management service	sends the acquirer a Reversal Request Response/0410 message where DE 39 = 58 (Transaction not permitted to acquirer/terminal)
the issuer does not participate in the PIN Management service	sends the acquirer a Reversal Request Response/0410 message where DE 39 = 57 (Transaction not permitted to issuer/cardholder)

Issuer Response Options to a Magnetic Stripe PIN Change Request

For PIN change transactions, the issuer may respond using one of the following methods.

- If the card is a non-chip (magnetic stripe) card or if the card is a chip card not supporting offline PIN and the issuer approves, the issuer sends an Authorization Request Response/0110 message containing DE 39 (Response Code), value 00 (Approved or completed successfully) or DE 39, value 85 (Not declined).

NOTE: When approving a PIN change transaction, Mastercard recommends that issuers of chip cards send DE 39, value 85 in the Authorization Request Response/0110 message.

- If the card is a non-chip (magnetic stripe) card or if the card is a chip card not supporting offline PIN and the issuer declines, the issuer sends an Authorization Request Response/0110 message containing DE 39, value 70 (Contact Card Issuer), DE 39, value 71 (PIN Not Changed), or DE 39, value 89 (Unacceptable PIN—Transaction Declined—Retry).
- If the card is a chip card supporting offline PIN, the issuer must decline and send an Authorization Request Response/0110 message containing DE 39, value 57 (Transaction not permitted to issuer/cardholder).

Issuers that have chip cards personalized with both online and offline PIN must not approve PIN change transactions when DE 22 (Point of Service [POS] Entry Mode, subfield 1 (POS Terminal PAN Entry Mode) contains value 02 or value 90. If these transactions are approved, the offline PIN will be out-of-sync with the online PIN, and subsequent offline transactions may be declined due to invalid PIN.

If an issuer's account range participates in Chip to Magnetic Stripe Conversion or M/Chip Cryptogram Pre-validation on-behalf services, that account range cannot participate in the Chip PIN Management service, even if the account range also contains magnetic stripe cards.

PIN Processing for Europe Region Customers

Mastercard supports PIN translation for acquirers and issuers and issuer PIN validation for Track 1 and Track 2 on the Mastercard Network for customers in the Europe Region.

For information about PIN processing capabilities, refer to [Transaction Processing Rules](#).

PIN Translation

The Dual Message Authorization System performs PIN translation on DE 52 (Personal ID Number [PIN] Data) and DE 125 (New PIN Data) for acquirers and issuers that use the Mastercard Network.

When issuers perform PIN processing in-house or when they participate in the Online PIN Validation in Stand-In Authorization, issuers must specify DE 53, subfield 4 (PIN key index number) in the Network Management Request/0800: sign-on/sign-off message, all remaining DE 53, subfields may be zero filled.

Issuers participating in the Online PIN Pre-validation Service are not required to send DE 53 in the Network Management Request/0800: sign-on/sign-off message.

Acquirers specify the key used to encrypt the PIN using the Authorization Request/0100 message.

The Key Management Services (KMS) group provides customers a Member Key ID (MKID) for each service. The MKID is used to notify the KMS group of new security keys. Each customer will associate the PIN key index number to identify a specific security key. Customers may define a maximum of 99 security keys for use during PIN translation.

DE 53 (Security-related Control Information)

Acquirers will specify DE 53, subfield 3 (PIN Block Format Code) when sending the Authorization Request/0100 message.

Issuers that use the PIN translation service to translate DE 52 and DE 125 (New PIN Data) will receive DE 53, subfield 4 (PIN Key Index Number) that identifies the PIN key used for translation.

PIN translation also will be performed on DE 125, if present, when the PIN translation was successfully completed for DE 52.

PIN Translation Edits

The following edits apply to Authorization Request/0100 messages (for acquirers only).

Table 1759: Associated edits

WHEN...	THEN the Dual Message Authorization System...
the acquirer uses the Dual Message Authorization System for PIN translation and DE 52 (Personal ID Number [PIN] Data) is present and DE 53 (Security-related Control Information) is not present	sends the acquirer an Authorization Request Response/0110 message where: - DE 39 = 30 (Format error) - DE 44 = 053 (DE 53 is the data element in error)
the acquirer uses the Dual Message Authorization System for PIN translation and DE 52 is present and DE 53, subfield 3 (PIN Block Format Code) is not valid. (Valid values are 01 and 10.)	sends the acquirer an Authorization Request Response/0110 message where: - DE 39 = 30 - DE 44 = 053
the acquirer uses the Dual Message Authorization System for PIN translation and DE 52 is present and DE 53, subfield 4 (PIN Key Index Number) contains a PIN Key Index Number not known to Mastercard	sends the acquirer an Authorization Request Response/0110 message where: - DE 39 = 88 (Cryptographic failure) - DE 48 (Additional Data: Private Use), subelement 80 (PIN Service Code) = TI (The Dual Message Authorization System was unable to translate the PIN)
the acquirer does not use the Dual Message Authorization System for PIN translation and DE 53 is present	removes DE 53 and forwards the Authorization Request/0100 message to the Single Message System

WHEN...	THEN the Dual Message Authorization System...
the Dual Message Authorization System Security Translation Platform fails	sends the acquirer an Authorization Request Response/0110 message where: • DE 39 = 88 • DE 48, subelement 80 = TI

The following edits apply to Network Management Request/0800 sign-on messages.

Table 1760: Associated edits

WHEN...	THEN the Dual Message Authorization System...
The issuer uses the Dual Message Authorization System for PIN translation and DE 53 (Security-Related Control Information), subfield 4 (PIN Key Index Number) contains a PIN Key Index Number not known to Mastercard	Sends the issuer a Network Management Request Response/0810 message where: • DE 39 = 30 • DE 44 = 053

PIN Validation

For information about PIN processing capabilities, refer to [Transaction Processing Rules](#).

PIN validation results will be provided in DE 48, subelement 71 (On-behalf Services) in Authorization Request/0100 and Authorization Advice/0120: system-generated messages when PIN validation has been performed on behalf of the issuer. DE 48, subelement 71 will contain the following values related to PIN validation:

- Subfield 1 (On-behalf [OB] Service) will contain the following values:
 - 08 (Online PIN Pre-validation)
 - 09 (Online PIN Validation in Stand-In)
- Subfield 2 (On-behalf [OB] Result 1) when subfield 1 is value 08 or 09:
 - I (Invalid)
 - P (Mandatory PVV not on file)
 - R (PIN retry exceeded [invalid PIN])
 - U (Unable to process)
 - V (Valid)
- Subfield 3 (On-behalf [OB] Result 2) is blank when it is sent to the issuer

The Dual Message Authorization System will manage tracking the number of PIN failed attempts at the card level (for example, PAN, card sequence number from the track and expiration date).

DE 52 (Personal ID Number [PIN] Data) is not included in the Authorization Request/0100 message when PIN validation is performed.

Table 1761: Associated edits

WHEN the issuer subscribes to the...	THEN the issuer...
Online PIN Validation in Stand-In Authorization	chooses the PIN Failed Attempts limit, which must be less than or equal to five. Stand-In Authorization will reset the PIN Failed Attempts counter to zero when a valid PIN is entered and the count has not exceeded the issuer defined maximum. Stand-In Authorization will reset the PIN Failed Attempts counter to zero after the Stand-In Authorization maintenance.
Online PIN Pre-validation	does not choose the PIN Failed Attempts. Mastercard will allow five PIN Failed Attempts. The Dual Message Authorization System will reset the PIN Failed Attempts counter to zero when a valid PIN is entered and the count has not exceeded five. The Dual Message Authorization System will reset the PIN Failed Attempts counter to zero after 24 hours has passed.

PIN Validation edits

The following edits apply to Authorization Request/0100 messages (for issuers only).

Table 1762: Associated edits

WHEN...	THEN the Dual Message Authorization System...
the issuer uses the Dual Message Authorization System for PIN pre-validation and the PIN is valid and the number of PIN failed attempts were not yet exceeded	forwards the Authorization Request/0100 message where: - DE 48, subelement 71, subfield 1 = 08 (Online PIN Pre-validation) - DE 48, subelement 71, subfield 2 = V (Valid)
the issuer uses the Dual Message Authorization System for PIN pre-validation and the PIN validation cannot be performed	forwards the Authorization Request/0100 message where: - DE 48, subelement 71, subfield 1 = 08 - DE 48, subelement 71, subfield 2 = U (Unable to process)

WHEN...	THEN the Dual Message Authorization System...
the issuer uses the Dual Message Authorization System for PIN pre-validation and the number of PIN retries were exceeded	forwards the Authorization Request/0100 message where: • DE 48, subelement 71, subfield 1 = 08 • DE 48, subelement 71, subfield 2 = R (PIN retry exceeded)
the issuer uses the Dual Message Authorization System for PIN pre-validation and the PIN validation fails	forwards the Authorization Request/0100 message where: • DE 48, subelement 71, subfield 1 = 08 • DE 48, subelement 71, subfield 2 = I (Invalid)
the issuer uses the Dual Message Authorization System for PIN pre-validation and has specified Mandatory PVV On File and the PVV was not found	forwards the Authorization Request/0100 message where: • DE 48, subelement 71, subfield 1 = 08 • DE 48, subelement 71, subfield 2 = P (Mandatory PVV not on file)
the issuer uses the Dual Message Authorization System for PIN Validation in Stand-In Authorization, PIN validation fails, and the number of PIN retries were not yet exceeded	sends the acquirer an Authorization Request Response/0110 message where: • DE 39 = issuer defined • DE 48, subelement 80 = PI (Dual Message Authorization System unable to process PIN) sends the issuer an Authorization Advice/0120: system-generated message where: • DE 48, subelement 71, subfield 1 = 09 (PIN Validation in Stand-In) • DE 48, subelement 71, subfield 2 = I (Invalid) • DE 60, subfield 2 = 0051 (invalid PIN)
the issuer uses the Dual Message Authorization System for PIN Validation in Stand-In and the PIN validation fails	sends the acquirer an Authorization Request Response/0110 message where: • DE 39 = issuer defined • DE 48, subelement 80 = PI sends the issuer an Authorization Advice/0120 (System-generated) message where: • DE 48, subelement 71, subfield 1 = 09 • DE 48, subelement 71, subfield 2 = U (Unable to process) • DE 60, subfield 2 = 0050 (Unable to process)

WHEN...	THEN the Dual Message Authorization System...
the issuer uses the Dual Message Authorization System for PIN Validation in Stand-In and the PIN is valid and the number of PIN retries were not yet exceeded	sends the acquirer an Authorization Request Response/0110 message where: • DE 39 = determined by remaining Stand-In Authorization tests • DE 48, subelement 80 = PV (Valid PIN) sends the issuer an Authorization Advice/0120 (System-generated) message where: • DE 48, subelement 71, subfield 1 = 09 • DE 48, subelement 71, subfield 2 = V (Valid)
the issuer uses the Dual Message Authorization System for PIN Validation in Stand-In and the number of PIN retries were exceeded	sends the acquirer an Authorization Request Response/0110 message where: • DE 39 = issuer defined response code • DE 48, subelement 80 = PI sends the issuer an Authorization Advice/0120: system-generated message where: • DE 48, subelement 71, subfield 1 = 09 • DE 48, subelement 71, subfield 2 = R (PIN retry exceeded) • DE 60, subfield 2 = 0052 (PIN Retry Exceeded-invalid PIN)
the issuer uses the Dual Message Authorization System for PIN Validation in Stand-In and has specified Mandatory PVV On File and the PVV was not found	sends the acquirer an Authorization Request Response/0110 message where: • DE 39 = issuer defined value • DE 48, subelement 80 = PI sends the issuer an Authorization Advice/0120: system-generated message where: • DE 48, subelement 71, subfield 1 = 09 • DE 48, subelement 71, subfield 2 = P (Mandatory PVV not on file) • DE 60, subfield 2 = 0052 (Mandatory PVV not on file)

WHEN...	THEN the Dual Message Authorization System...
the issuer uses the Dual Message Authorization System for PIN Validation in Stand-In and the PIN translation fails	sends the acquirer an Authorization Request Response/0110 message where: • DE 39 = 88 • DE 48, subelement 80 = TI sends the issuer an Authorization Advice/0120: system-generated message where: • DE 48, subelement 71, subfield 1 = 09 • DE 48, subelement 71, subfield 2 = U (Unable to process) • DE 60, subfield 2 = 0050 (Unable to process)

PIN Key Management

The Key Management Services (KMS) group will manage the security keys for customers using the PIN translation and PIN validation services supported by the Dual Message Authorization System on the Mastercard Network.

PIN Verification Value/PIN Offset on File Service

The PIN Verification Value (PVV)/PIN Offset used in PIN validation is usually located on Track 1 or Track 2 of a card's magnetic stripe. Mastercard offers a service by which the issuer may optionally send the PVV/PIN offsets, in a file to Mastercard in order to override the information encoded in the track of a card's magnetic stripe.

Mastercard activates the PVV/PIN Offset data from an issuer upon receipt of the PVV/PIN Offset File. Issuers may send a full file replacements weekly.

Processing Transactions Using PVV/PIN Offset

Issuers that use the optional PVV/PIN Offset on File processing must request participation in the service for each card range and (optionally) expiry date range.

To process a transaction, the Dual Message Authorization System checks that PVV/PIN Offset details are held on file for the relevant card range and expiry date range. If details are on file for the card range and expiry date range, the Dual Message Authorization System checks for the individual card's details within the issuer's stored file.

The Dual Message Authorization System checks the entry in the PVV/PIN Offset file to ensure that the card's PAN, card sequence number, and expiry date match. If they do match, the Dual Message Authorization System can use the PVV value on file. If the PAN, card sequence number, and expiry date do not match, the Dual Message Authorization System processes the transaction according to the processing parameters that the issuer specifies for the relevant card range and expiry date.

Processing Parameters

Mastercard stores the issuer PVV/PIN Offset files for use in the Online PIN Pre-validation and the Online PIN Validation in Stand-In on-behalf services. When the Dual Message Authorization

System cannot find a matching entry (with the correct PAN, card sequence number, and expiry date values) in the issuer PVV/PIN Offset file, it needs further instructions to process the transaction properly.

The issuer must select one of the following processing options for each card range and expiry date:

- Optional on file: With this option, if no matching PVV/PIN Offset entry is found in the PVV/PIN Offset file, the Dual Message Authorization System retrieves the PVV value from Track 1 or Track 2 of the relevant card.
- Mandatory on file: With this option, the Dual Message Authorization System, having checked the Track 1 or Track 2 information for the issuer card against the entries held in the issuer PVV/PIN Offset file, performs one of the following actions:
 - For Stand-In services, the Dual Message Authorization System returns an Authorization Request Response/0110 message to the acquirer and an Authorization Advice/0120: system-generated message to the issuer with the appropriate response code, detailing the result of the processing.
 - For Pre-validation services, the Dual Message Authorization System forwards the results of the PIN validation to the issuer in an Authorization Request/0100 message and to the acquirer in an Authorization Request Response/0110 message.

The Global Parameters section provides an indicator to identify issuer participation.

PVV/PIN Offset File Format

Following is the PVV/PIN Offset File format that has a fixed length of 64 characters. Issuers can use Bulk ID RA85 or CONNECT:Direct to send the PVV/PIN Offset file to Mastercard.

Table 1763: PVV/PIN Offset File Header Record

Field Name	Attribute	Length	Comments/Values
Record type ID	Alphanumeric	3	HDR—header record
File version number	Numeric	3	100 (initial version)
Customer ID	Alphanumeric	11	First six digits must contain the issuers ICA with leading zeros Seventh digit and beyond contains trailing spaces
Transmission code	Alphanumeric	10	Member assigned transmission code. May contain all spaces or zeros.
Transmission sequence	Numeric	4	0000–9999 Wraps at 9999
Transmission date	Numeric	6	YYMMDD in UTC

Field Name	Attribute	Length	Comments/Values
Transmission time	Numeric	6	hhmmss in UTC
Input file type	Alphanumeric	1	F: Full file replace
Filler	Alphanumeric	20	Spaces

Table 1764: PVV/PIN Offset File Detail Record

Field Name	Attribute	Length	Comments/Values
Record type ID	Alphanumeric	3	DTL—detail record
Update code	Alphanumeric	1	A = Add/Update
PAN	Numeric	19	With trailing spaces
Card expiry date	Numeric	4	YYMM (must contain a valid year and month)
Card sequence number	Numeric	1	Must contain the Sequence Number associated with the PAN or 0. A value of 0 indicates that the card sequence number should not be used as criteria when matching to the PVV file.
PVV/PIN Offset	Numeric	6	Trailing spaces
Filler	Alphanumeric	30	Spaces

Table 1765: PVV/PIN Offset File Trailer Record

Field Name	Attribute	Length	Comments/Values
Record type ID	Alphanumeric	3	TRL—trailer record
Number of detail records	Numeric	11	Detail record count
Filler	Alphanumeric	50	Spaces

Alternate Processing

Mastercard will support issuers subscribing to the Online PIN Pre-validation service (OB service code value 08) that are not available to respond to the Authorization Request/0100 message and issuers subscribing to the PIN Validation in Stand-In service (OB service code value 09) by responding to the Authorization Request/0100 message on behalf of the issuer. Mastercard will

consider the results of the PIN validation in DE 48, subelement 71 when responding to the Authorization Request/0100 message.

The values contained in DE 60 (Advice Reason Code), subfield 2 (Advice Detail Code) are dependent on the values contained in DE 48, subelement 71, subfield 2 (OB Result 1) as described in the following table.

Table 1766: Values

WHEN DE 48, subelement 71, subfield 2 contains the value...	THEN DE 60, subfield 2 will contain...
U (Unable to Process)	0050
I (Invalid)	0051
P (Mandatory PVV not on file)	0052
R (PIN Retry Exceeded [invalid PIN])	0052

PIN Processing for Non-Europe Customers

Personal identification number (PIN) is a proven technology that is not easily circumvented. It is secure within actual market conditions. Mastercard permits acquirers to use PIN technology at the point of sale that prompts Mastercard cardholders for a PIN.

This section contains PIN processing procedures for non-Europe customers. For information about PIN processing procedures for Europe region acquirers and issuers that process through the Mastercard Network, refer to [PIN Processing for Europe Region Customers](#).

Acquirer Requirements

Acquirers must comply with the following before processing Mastercard purchase transactions that contain a PIN (such as CAT Level 1 transactions) in Authorization/01xx messages.

- Support either static or dynamic PIN Encryption Key (PEK) exchanges
- Comply with the Mastercard Magnetic Stripe Compliance Program
- Correctly format Authorization Request/0100—PIN messages
- Correctly format Authorization Request Response/0110—PIN messages

Support either Static or Dynamic PIN Encryption Key (PEK) exchanges

The Dual Message Authorization System and customers' system use PEK to encrypt or decrypt PINs. PEKs provide a secure means of passing PIN information in Authorization/01xx messages. Acquirers must support either static or dynamic PEK exchanges.

NOTE: For increased security, Mastercard strongly recommends using dynamic PEK exchanges.

Following are the differences between the two options.

Table 1767: PEK options

PEK	Description
Static	• Customers and the Dual Message Authorization System establish these keys offline. • Customers and the Dual Message Authorization System must establish a single PEK. This PEK must be associated with the acquirer's customer ID.
Dynamic	• Customers and the Dual Message Authorization System exchange these keys automatically online every 24 hours or every 2,500 transactions, whichever occurs first. • Customers and the Dual Message Authorization System use Key Encryption Keys (KEKs) to encrypt and decrypt the PEKs during PEK exchanges between the acquirer and the Dual Message Authorization System. Each acquirer must establish a KEK with Mastercard to exchange the PEKs dynamically with the Dual Message Authorization System. This KEK must be associated with the acquirer's customer ID.

Mastercard Magnetic Stripe Compliance Program Compliance

PIN verification requires valid track data, only acquirers that comply with the Mastercard magnetic stripe compliance program and provide complete and unaltered track data in their Authorization Request/0100 messages are able to support PIN processing.

Authorization Request/0100: PIN Transactions

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Table 1768: Data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)	M	•	M	Must contain one of the following values: • 05 = Merchant terminal PAN auto-entry via integrated circuit card • 07 = Merchant terminal PAN auto-entry via contactless M/Chip • 80 = Chip card at chip-capable terminal was unable to process transaction using data on them chip; therefore, the terminal defaulted to the magnetic stripe-read PAN • 90 = PAN auto-entry via magnetic stripe: the full track data has been read from the data encoded on the card and transmitted within the authorization request in DE 35 (Track 2 Data) or DE 45 (Track 1 Data) without alteration or truncation
DE 26 (POS PIN Capture Code)	C	•	C	Must be present when DE 52 (PIN Data) is present and the maximum PIN characters that the terminal accepts is something other (more or less) than 12 characters.
DE 32 (Acquiring Institution ID Code)	M	•	M	If present, must be the same ID as the Member Group ID for the Dual Message Authorization System to translate the PIN data that the acquirer provides.
DE 33 (Forwarding Institution ID Code)	C	•	C	If present, must be the same ID as the Member Group ID for the Dual Message Authorization System to translate the PIN data that the acquirer provides.

Data Element	Org	Sys	Dst	Values/Comments
DE 35 (Track-2 Data)	C	•	C	Must be present and represent the information as encoded on the magnetic stripe of the card (or the equivalent data if a chip card and chip-capable terminal are used at the point of interaction) or DE 45 (Track-1 Data)
DE 45 (Track-1 Data)	C	•	C	Must be present and represent the information as encoded on the magnetic stripe of the card (or the equivalent data if a chip card and chip-capable terminal are used at the point of interaction)
DE 52 (PIN Data)	C	X	C	The 16-digit number Data Encryption Standard (DES) hexadecimal number in the ANSI PIN block format (also referred to as "ISO Format-0")

Authorization Request Response/0110: PIN Transactions

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Table 1769: Data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 39 (Response Code)	M	•	M	Contains response code indicating the Dual Message Authorization System or the issuer could or could not verify the PIN entered at the point of interaction. Refer to the following table that identifies possible responses based on PIN processing conditions.
DE 48 (Additional Data—Private Use), TCC	C	•	C	Must contain the appropriate TCC code.

Data Element	Org	Sys	Dst	Values/Comments
DE 48, subelement 80 (PIN Service Code)	C	.	C	Identifies whether the Dual Message Authorization System attempted to verify or translate the PIN. Valid values: - PV = The Dual Message Authorization System verified the PIN - TV = The Dual Message Authorization System translated the PIN for issuer verification - PI = The Dual Message Authorization System was unable to verify the PIN - TI = The Dual Message Authorization System was unable to translate the PIN

Acquirer-related PIN processing errors, indicated by DE 39 (Response Code), value 30 (Format Error) and DE 44 (Additional Response Data) value 052 (DE 52 PIN Data), occur if the Dual Message Authorization System is unable to decrypt the PIN data. Possible causes are:

- The PEK that the Dual Message Authorization System and customer share is not synchronized. It should be the same key.
- The Dual Message Authorization System is unable to determine the proper PEK because DE 32 or DE 33 is incorrect.
- The acquirer is not qualified to be in the Mastercard Magnetic Stripe Compliance Program.
- The acquirer did not correctly establish the static PEK or KEK with Mastercard.
- The DE 22, subfield 1 value is not valid for transactions that contain a PIN.

Following are possible Authorization Request Response/0110 message responses:

Table 1770: Associated edits

IF...	THEN...
The Dual Message Authorization System on behalf of the issuer was unable to verify the PIN.	DE 39 = 55 (Invalid PIN) DE 48, subelement 80 (PIN Service Code) = PI
The Dual Message Authorization System translated the PIN, but the issuer was unable to verify the PIN.	DE 39 = 55 DE 48, subelement 80 = TV
The Dual Message Authorization System was unable to decrypt the PIN for the acquirer.	DE 39 = 30 (Format Error) DE 44 = 52 DE 48, subelement 80 = TI

IF...	THEN...
PIN processing was successful (verification or translation).	DE 39 = any valid code DE 48, subelement 80 = TV or PV
The Dual Message Authorization System was unable to decrypt the PIN for the issuer.	DE 39 = 91 (Authorization System or issuer system inoperative) DE 48, subelement 80 = TI

Issuer Requirements

Issuers must comply with the following to process Mastercard transactions that contain a PIN in Authorization Request/0100 messages.

- Receive purchase or ATM transactions that contain a PIN
- Support Static or Dynamic PEK exchanges
- Process applicable data elements in Authorization Request/0100 messages
- Process applicable data elements in Authorization Request Response/0110 messages
- Process applicable data elements in Authorization Advice/0120: system-generated messages
- Process applicable data elements in Reversal Advice/0420: system-generated messages

For information, refer to [Transaction Processing Rules](#).

Receive Purchase Transactions that Contain a PIN

Issuers must be able to receive purchase transactions that contain a PIN using one of the following network options.

- Through a Mastercard Network connection using Authorization Request/0100 messages
- The default method for issuers that currently receive their Mastercard ATM cash disbursement and ATM balance inquiry activity using this interface. Static PEKs that are already active and operational at the Single Message System will be used for each of the issuer's bank identification number (BIN)/card ranges.

Issuers may choose to support dynamic PEK exchanges instead.

- Through a Single Message System connection using Financial Transaction Request/0200 messages

The default method for issuers that currently receive their Debit Mastercard activity, ATM activity, or both using this interface. These transactions will be identified as preauthorization requests with a DE 61, subfield 7, value 4 because they must be cleared through the Mastercard Global Clearing Management System, with all other purchase activity. PEKs that already are active and operational at the Single Message System will be used for each of the issuer's BIN/card ranges.

Support Static or Dynamic PEK exchanges

For dynamic PEK exchanges, the Dual Message Authorization System and the issuer must establish a new single KEK for all BINs that the issuer will process. This new KEK must be

associated with the issuer's Group ID that the issuer uses for sign-on, sign-off, and store-and-forward (SAF) sessions for the associated BINs. The issuer must complete the *Customer-Initiated Key Part Exchange Form* (Form 536).

NOTE: For increased security, Mastercard strongly recommends using dynamic PEKs.

Authorization Request/0100: PIN Messages

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Table 1771: Data elements

Data Element	Org	Sys	Dst	Values/Comments	
DE 48 (Additional Data—Private Use), TCC	M	.	M	Must contain the appropriate TCC code.	
DE 48, subelement 80 (PIN Service Code)	.	X	C	This subelement contains the results of PIN processing as follows:	
				IF the issuer chooses to...	THEN the Dual Message Authorization System...
				Use the Mastercard PIN Verification Service	• Omits DE 52 from the Authorization Request/0100 message sent to the issuer. • Sends DE 48, subelement 80, value PV (The Dual Message Authorization System verified the PIN). • If the PIN is invalid, the Dual Message Authorization System sends the acquirer an Authorization Request Response/0110 message where DE 39 contains the value 55.

Data Element	Org	Sys	Dst	Values/Comments
DE 52 (Personal ID Number (PIN) Data)	C	X	C	Will be present if the Dual Message Authorization System does not perform PIN verification. The issuer verifies the PIN data in DE 52. Either DE 45 or DE 35 will be present in the message to provide the PIN verification value (PVV) or offset, depending on the issuer's verification method.

Table 1772: Values

IF...	THEN...
DE 52 is present	DE 48, subelement 80 is value TV
DE 52 is not present	DE 48, subelement 80 is value PV

Authorization Advice/0120: PIN Messages

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Advice/0120 data elements apply.

Table 1773: Data elements

Data Element	Org	Sys	Dst	Values/Comments				
DE 48 (Additional Data—Private Use), TCC	•	M	M	Must contain the appropriate TCC code.				
DE 48, subelement 80 (PIN Service Code)				This subelement contains the results of PIN processing as follows: <table><tr><th>IF the issuer chooses to...</th><th>THEN the Dual Message Authorization System...</th></tr><tr><td>Have the Dual Message Authorization System verify the PIN</td><td> • Omits DE 52 from the Authorization Advice/0120 message sent to the issuer. • Sends DE 48, subelement 80, value TI (The Dual Message Authorization System was unable to translate the PIN). </td></tr></table>	IF the issuer chooses to...	THEN the Dual Message Authorization System...	Have the Dual Message Authorization System verify the PIN	• Omits DE 52 from the Authorization Advice/0120 message sent to the issuer. • Sends DE 48, subelement 80, value TI (The Dual Message Authorization System was unable to translate the PIN).
IF the issuer chooses to...	THEN the Dual Message Authorization System...							
Have the Dual Message Authorization System verify the PIN	• Omits DE 52 from the Authorization Advice/0120 message sent to the issuer. • Sends DE 48, subelement 80, value TI (The Dual Message Authorization System was unable to translate the PIN).							

Data Element	Org	Sys	Dst	Values/Comments
DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code)	•	M	M	107 (PIN Processing Error)
DE 60, subfield 2 (Advice Detail Code)	•	M	M	Provides additional details as follows: 0030 = Reject: Unable to verify PIN data 0031 = Reject: Unable to decrypt/encrypt PIN data

Reversal Advice/0420: PIN Messages

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

The issuer must be able to receive Reversal Advice/0420 messages with DE 60 values in addition to all other applicable data elements.

The Reversal Advice/0420 message identifies the corresponding Authorization Request Response/0110 message being reversed that contained PIN data.

Table 1774: Data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code)	•	M	M	400 = Banknet advice: APS error; unable to deliver response
DE 60, subfield 2 (Advice Detail Code)	•	M	M	2000 = Reject: PIN data present in Authorization Request/0100 message

Alternate Processing

The issuer may choose to have the Dual Message Authorization System verify PIN data on their behalf. If the Dual Message Authorization System performs PIN verification, it will perform Stand-In Authorization and X-Code Processing using the issuer or Mastercard existing authorization-qualifying criteria, when applicable.

If the Dual Message Authorization System does not verify the PIN data and the issuer is unavailable or unable to process the Authorization Request/0100 message, the Dual Message Authorization System will respond to the acquirer with an Authorization Request Response/0110 message indicating the issuer could not process the Authorization Request/0100 message, except in situations where an issuer chooses to allow transactions with unverified PINs in Stand-In Authorization.

Support for Both Acquiring and Issuing Processing

Customers that support both acquiring and issuing authorization processing may use the same PEK for all purchase transactions that contain a PIN. To accomplish this, the customer must use the same customer ID as follows:

- As the Group ID for establishing the static PEK or the KEK.
- In DE 32 (Acquiring Institution ID Code) or DE 33 (Forwarding Institution ID Code) in all acquired transactions.
- In DE 2 and DE 33 of the Network Management Request/0800: PEK exchange–on demand message.
- In DE 2 of the Network Management Request/0800: group sign-on message.

Cleartext Use Prohibited

If there is a major problem with security equipment (for example, a faulty tamper-resistant security module [TRSM] or DES circuit board), the Dual Message Authorization System suspends all transaction processing with that customer. The customer must not send unencrypted (cleartext) PIN data.

For more information on PIN encipherment, refer to the [Security Rules and Procedures Guide](#).

Emergency Static PEK or Emergency KEK Process

Following are the stages of the emergency static PEK or KEK process.

1. Mastercard considers the faulty customer as down.
2. Authorized Mastercard personnel randomly generate both parts of an emergency static PEK or KEK.
3. Mastercard personnel call the security or operations staff of the customer. Mastercard gives the emergency static PEK or KEK verbally to the customer.
4. Both the Mastercard personnel and the customer insert the new emergency static PEK or KEK in their TRSMs.
5. The Dual Message Authorization System initiates dynamic PEK exchange with a Network Management Request/0800: PEK exchange, using the new emergency static PEK or KEK.

Mastercard and the customer are to use the emergency static PEK or KEK process only as an interim measure to get the customer up as quickly as possible following a key exchange failure. Mastercard limits the use of the emergency static PEK or KEK in any one occurrence to six business days. The customer must notify their security officers responsible for key management immediately of the security failure situation and must conduct a secure key exchange at the earliest possible time.

Previous PEKs

After exchanging new PEKs statically or dynamically, the Dual Message Authorization System and the customer are responsible for preserving the previous PEK for five minutes. They do this in the event that the current PEK becomes inoperative. If the current PEK has a Sanity Check error during this five-minute interval, the Dual Message Authorization System or the customer should attempt to use the previous PEK. If the previous PEK is also inoperative, then refer to the steps previously discussed to determine action.

PIN Verification Value on File Service

For issuers that use the Single Message System for PIN-processing services and elect to use the PIN Validation in Stand-In service, the Single Message System will perform the PIN validation service using the PVV for a card based on the PVV value provided in a file by the issuer.

Issuers that choose to register for this service must send a PVV/PIN Offset file to the Single Message System to update cardholder PINs and must use Single Message System-managed security keys. A Customer Implementation Services specialist will assist issuers with setting up a file or adding new files for testing and production.

Participating issuers must send a PVV/PIN Offset file to the Single Message System to provide and update cardholder PIN information. This file must be sent using the secure Global File Transfer (GFT) methods approved by Mastercard. The following bulk file IDs have been established for use when submitting PVV/PIN Offset files: RM29 (Production), RM31 (MTF).

Issuers also must ensure that they have exchanged the PIN verification keys (PVKs) with the Single Message System before using this service.

PVV/PIN Offset File Format

The PVV/PIN Offset file has a fixed length of 64 characters. Issuers may use Bulk ID RM29 (Production), RM31 (MTF) or CONNECT:Direct to send the PVV/PIN Offset file to Mastercard. Refer to the following tables for information about the file header, detail, and trailer records.

Table 1775: PVV/PIN Offset File Header Record

Field Name	Attribute	Length	Comments and Values
Record Type ID	Alphanumeric	3	HDR—header record
File Version Number	Numeric	3	100 (initial version)
Customer ID	Alphanumeric	11	The first six digits must contain the issuers ICA with leading zeros. The seventh digit and beyond contains trailing spaces.
Transmission code	Alphanumeric	10	Member-assigned transmission code. May contain all spaces or zeros.
Transmission sequence	Numeric	4	0000–9999 Wraps at 9999
Transmission date	Numeric	6	YYMMDD in UTC
Transmission time	Numeric	6	hhmmss in UTC
Input file type	Alphanumeric	1	F = Full file replacement
Filler	Alphanumeric	20	Spaces

Table 1776: PVV/PIN Offset File Detail Record

Field Name	Attribute	Length	Comments and Values
Record Type ID	Alphanumeric	3	DTL = detail record
Update code	Alphanumeric	1	A = add/update
PAN	Alphanumeric	19	With trailing spaces
Card expiry date	Numeric	4	YYMM (must contain a valid year and month)
Card sequence number	Numeric	1	Values can be in the range of 0–9 Must contain the Sequence Number associated with the PAN or 0. A value of 0 indicates that the card sequence number should not be used as criteria when matching to the PVV file.
PVV/PIN Offset	Numeric	6	Trailing spaces
Filler	Alphanumeric	30	Spaces

Table 1777: PVV/PIN Offset File Trailer Record

Field Name	Attribute	Length	Comments and Values
Record type ID	Alphanumeric	3	TRL = trailer record
Number of detail records	Numeric	11	Detail record count
Filler	Alphanumeric	50	Spaces

Alternate Processing

In instances where Stand-In Authorization is not accessible, X-Code Processing is initiated. X-Code Processing does not perform PIN verification. Therefore, authorization requests received with an unverified PIN will be declined.

PIN Translation and Verification Process

The following describes the key exchanges during the PIN translation and verification process.

1. The cardholder enters the PIN at the point of interaction. The PIN is encrypted by the terminal's hardware under a PIN encryption key and is then sent to the acquirer.

2. The acquirer receives the encrypted PIN, which the acquirer then decrypts using the terminal PEK stored in a TRSM. The acquirer then creates the ANSI PIN block and encrypts it using the DES algorithm as follows:

First Block: PIN Data: The first digit of this block contains the control character 0, followed by a number representing the PIN length (maximum 12), and then the PIN itself. The acquirer then fills the remaining digits of the block on the right with hexadecimal F's to complete the 16-digit account number.

Second Block: PAN Data: The first four digits of the second block contain 0000, followed by the 12 right-most digits of the PAN, excluding the check digit. The acquirer then pads zeros to the left to complete the 16-digit data element.

In formatting an ANSI block, the acquirer performs an XOR function on the two 16-digit blocks.

After creating the PIN block, the acquirer sends it through the DES algorithm along with the 16-digit PEK that the acquirer and the Dual Message Authorization System share, producing the translated PIN block as follows. The acquirer may encrypt the PIN using:

- a. Single DES algorithm

Encrypt PIN information using a single DES key algorithm with single-length PEKs.

- b. Triple DES algorithm with double length PEKs

When a customer chooses to support triple DES, the customer must support both sending and receiving double-length (16-byte) PEKs.

- c. Triple DES algorithm with triple length PEKs

3. The acquirer sends the translated PIN block to the Dual Message Authorization System. The Dual Message Authorization System translates it from the acquirer PEK to the issuer PEK using the procedures described in step 2. The Dual Message Authorization System then sends the translated PIN block to the issuer for verification.

DE 52 (PIN Data) is not in the Authorization Request/0100 message going to the issuer if the Dual Message Authorization System is performing PIN verification on the issuer's behalf.

4. The issuer then translates the PIN block and verifies the PIN using:

- a. Single DES algorithm

- b. Triple DES algorithm with double length PEKs

- c. Triple DES algorithm with triple length PEKs

Detection of PEK Corruption Using Sanity Checks

Issuers and the Dual Message Authorization System perform Sanity Checks on PIN blocks as part of the PIN encryption/decryption process. This subsection describes Sanity Check errors that indicate PEK out-of-synchronization conditions that may occur during PIN decryption. It also explains ensuing steps that the Dual Message Authorization System and customer take.

All Tamper Resistant Security Modules (TRSMs) must be able to detect possible corruption of PEKs by performing Sanity Checks on the PIN block as part of decryption. When Sanity Check errors occur, customers should use the following procedures to resolve the problem.

Dual Message Authorization System Sanity Check Error

The following describes the process when the Dual Message Authorization System TRSM discovers a PEK or KEK error.

1. The Dual Message Authorization System's Sanity Check reveals an out-of-synchronization condition with the PEK that it shares with the acquirer.
2. If five consecutive transactions fail the Sanity Check, and the Dual Message Authorization System has performed steps 1–3 with each of the previous five transactions from the acquirer, the Dual Message Authorization System takes the following action:
 - For a static PEK, Mastercard starts emergency procedures to create a new static PEK.
 - For a dynamic PEK, the Dual Message Authorization System sends the acquirer a Network Management Request/0800: PEK exchange message with a new PEK. If there are problems with the new PEK, the Dual Message Authorization System will initiate procedures to establish a new KEK.
3. For a dynamic PEK, the acquirer sends a Network Management Request Response/0810: PEK exchange to the Dual Message Authorization System acknowledging receipt of the Network Management Request/0800: PEK exchange message exchanging the PEK. If the message does not contain a DE 39 value of 00, completed successfully, the Dual Message Authorization System sends another Network Management Request/0800: PEK exchange message.
4. For a dynamic PEK, the Dual Message Authorization System sends a Network Management Advice/0820: PEK exchange message to the acquirer that notifies the customer that the new PEK is active and operational. Subsequent Authorization Request/0100 messages must use the new PEK to encrypt the PIN in DE 52.

Issuer Sanity Check Error

The issuer TRSMs must be able to detect possible corruption of PEKs. The following describes the process when the issuer TRSM discovers a PEK error using a Sanity Check.

1. The issuer's Sanity Check reveals an out-of-synchronization condition with the PEK that it shares with the Dual Message Authorization System.
2. The issuer attempts to decrypt the PIN block using the prior PEK, if available.
3. If the Sanity Check fails using the prior PEK, the issuer sends an Authorization Request Response/0110 message.
4. If five consecutive transactions fail the Sanity Check, and the issuer has performed steps 1–3 with each of the five transactions, the issuer takes the following appropriate action:
 - For a static PEK, the issuer contacts Mastercard to start emergency procedures to create a new static PEK.
 - For a dynamic PEK, the issuer sends the Dual Message Authorization System a Network Management Request/0800: PEK exchange–On Demand message to create a new PEK. If there are problems with the new PEK, the issuer contacts Mastercard to start emergency procedures to establish a new KEK.
5. For a dynamic PEK, the Dual Message Authorization System acknowledges receipt of the request by sending a Network Management Request Response/0810:PEK exchange message.

6. For a dynamic PEK, the Dual Message Authorization System sends to the issuer a Network Management Request/0800: PEK exchange message with a new PEK.
7. For a dynamic PEK, the issuer sends a Network Management Request Response/0810: PEK exchange message to the Dual Message Authorization System acknowledging receipt of the Network Management Request/0800: PEK exchange message exchanging the PEK. If the message does not contain a DE 39 value of 00, completed successfully, the Dual Message Authorization System sends another Network Management Request/0800: PEK exchange message.
8. For a dynamic PEK, the issuer receives a Network Management Advice/0820: PEK exchange message from the Dual Message Authorization System that notifies the customer that the new PEK is active and operational. Subsequent Authorization Request/0100 messages must use the new PEK to encrypt the PIN in DE 52.

Private Label Processing

Under the Mastercard Private Label Program, private label issuers use Mastercard account ranges. The use of a Mastercard account range on an approved private label program facilitates the seamless switching of private label transactions via the four-party model (issuer, acquirer, merchant, Mastercard).

Mastercard Private Label card programs will use Mastercard or Maestro BIN ranges.

In addition to the following information, refer to the *Private Label Rules* manual for more details.

Authorization Request/0100: Private Label Processing

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Data Element	Org	Sys	Dst	Values/Comments
DE 63 (Network Data), subfield 1 (Financial Network Code)	.	X	M	A valid private label financial network code.

Card Activation for Private Label Processing

Private Label enables issuers to allow consumers buying their Private Label prepaid card to activate them when purchased at the merchant location. This limits risks for the merchants as

their Private Label cards (on display for sale and already loaded with a predefined amount) will be activated only when purchased.

Authorization Request/0100 and Reversal Request/0400: Card Activation at Point of Sale

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 and Reversal Request/0400 data elements apply.

Data Element	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code, subfield 1 (Transaction Type)	M	.	M	Must be value 28 (Payment Transaction)
DE 4 (Amount, Transaction)	M	.	M	Must be zero
DE 18 (Merchant Type)	M	.	M	Must contain a value other than 6010 (Member Financial Institution—Manual Cash Disbursements) or 6011 (Member Financial Institution—Automated Cash Disbursements)

Data Element	Org	Sys	Dst	Values/Comments
DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)	M	.	M	Must contain one of the following values: • 02 = PAN entry mode unknown • 05 = PAN auto-entry via chip • 07 = PAN auto-entry via contactless M/Chip • 80 = Chip card at chip-capable terminal was unable to process transaction using data on the chip; therefore, the terminal defaulted to the magnetic stripe-read PAN. The full track data has been read from the data encoded on the card and transmitted within the Authorization Request/0100 in DE 45 (Track 1 Data) or DE 35 (Track 2 Data) without alteration or truncation. To use this value, the acquirer must be qualified to use value 90. • 90 = PAN auto-entry via magnetic stripe—The full track data has been read from the data encoded on the card and transmitted within the authorization request in DE 35 (Track 2 Data) or DE 45 (Track 1 Data) without alteration or truncation. • 91 = PAN auto-entry via contactless magnetic stripe—The full track data has been read from the data on the card and transmitted within the authorization request in DE 35 (Track 2 Data) or DE 45 (Track 1 Data) without alteration or truncation.
DE 48 (Additional Data—Private Use), TCC (Transaction Category Code)	M	.	M	Must contain value P
DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier)	C	.	C	Must contain value C09 (Card Activation)
DE 61 (Point of Sale [POS] Data), subfield 1 (POS Terminal Attendance)	M	.	M	Must contain value 0 (Attended Terminal)
DE 61 (Point of Sale [POS] Data), subfield 3 (POS Terminal Location)	M	.	M	Must contain value 0 (On premises of acceptor facility)

Data Element	Org	Sys	Dst	Values/Comments
DE 61 (Point of Sale [POS] Data), subfield 4 (POS Cardholder Presence)	M	.	M	Must contain value 0 (Cardholder present)
DE 61 (Point of Sale [POS] Data), subfield 5 (POS Card Presence)	M	.	M	Must contain value 0 (Card present)
DE 61 (Point of Sale [POS] Data), subfield 10 (Cardholder-Activated Terminal [CAT] Level)	M	.	M	Must contain value 0 (Not a CAT transaction)

Alternate Processing

Private Label prepaid card activation transactions are not eligible for alternate (Stand-In or alternate issuer host routing) or X-Code processing. If the primary issuer is not available to respond to a card activation request, an Authorization Request Response/0110 is returned to the acquirer with DE 39 (Response Code) value 91 (Authorization System or issuer system inoperative).

If the issuer is not available to respond to a reversal of a card activation request, a Reversal Request Response/0410 is returned to the acquirer with DE 39 (Response Code) value 00 (Approval) and the issuer receives notification of the response the Authorization Platform provided on their behalf in a Reversal Advice/0420 message.

Authorization Platform Edits

The Authorization Platform will perform the following edits for Authorization Request/0100 and Reversal Request/0400 messages.

WHEN...	THEN the Authorization Platform...
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains value 28 (Payment Transaction), and DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier), contains value C09 (Card Activation) and DE 61 (Point of Sale [POS] Data) subfields does not contain the following values:	The Authorization Platform declines the request with a format error response where: DE 39 (Response Code) = 30 DE 44 (Response Data) = 061
- Subfield 1 (POS Terminal Attendance) = 0 (Attended Terminal) - Subfield 3 (POS Terminal Location) = 0 (On premises of acceptor facility) - Subfield 4 (POS Cardholder Presence) = 0 (Cardholder present) - Subfield 5 (POS Card Presence) = 0 (Card present) - Subfield 10 (Cardholder-Activated Terminal [CAT] Level) = 0 (Not a CAT transaction)	

WHEN...	THEN the Authorization Platform...
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains value 28 (Payment Transaction), and DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier), contains value C09 (Card Activation) and DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode) does not contain one of the following values: • 02 = PAN entry mode unknown • 05 = PAN auto-entry via chip • 07 = PAN auto-entry via contactless M/Chip • 80 = Chip card at chip-capable terminal was unable to process transaction using data on the chip; therefore, the terminal defaulted to the magnetic stripe-read PAN. The full track data has been read from the data encoded on the card and transmitted within the Authorization Request/0100 in DE 45 (Track 1 Data) or DE 35 (Track 2 Data) without alteration or truncation. To use this value, the acquirer must be qualified to use value 90. • 90 = PAN auto-entry via magnetic stripe—the full track data has been read from the data encoded on the card and transmitted within the authorization request in DE 35 (Track 2 Data) or DE 45 (Track 1 Data) without alteration or truncation. • 91 = PAN auto-entry via contactless magnetic stripe—the full track data has been read from the data on the card and transmitted within the authorization request in DE 35 (Track 2 Data) or DE 45 (Track 1 Data) without alteration or truncation.	The Authorization Platform declines the request with a format error response where: DE 39 (Response Code) = 30 DE 44 (Response Data) = 022

Card Activation Plus Initial Load for Private Label Processing

Private label acquirer and issuers can use the Card Activation Plus Initial Load service to provide their cardholders the ability to set the amount to load on their private label prepaid card at the moment of purchase and activation at the point-of-sale (POS) terminal.

For private label prepaid cards, the Authorization Platform can process one transaction for both activating and initially loading a card.

The Authorization Platform will allow a valid amount in DE 4 (Transaction Amount) for private label prepaid card activation authorization messages.

Acquirer Processing

- Acquirers that want to offer the private label card activation plus initial load functionality must support Authorization Request/0100 and Reversal Request/0400 messages containing DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier), value C09 (Card Activation).
- Acquirers already supporting private label card activation at POS functionality and wanting to offer card activation plus initial load must send Authorization Request/0100 and Reversal Request/0400 messages containing DE 4 (Amount, Transaction) with a valid amount and DE 48, subelement 77, value C09.
- Acquirers must submit Reversal Request/0400 messages for the full amount when reversing card activation plus initial load messages. Acquirers must ensure that DE 95 (Replacement Amounts) is not present in Reversal Request/0400 messages or, if present, that DE 95, subfield 1 (Actual Amount, Transaction) contains an amount equal to zero.

Issuer Processing

- Issuers that want to offer the private label card activation plus initial load functionality must support Authorization Request/0100 and Reversal Request/0400 messages containing DE 48 (Additional Data—Private Use), subelement 77 (Transaction Type Identifier), value C09 (Card Activation).
- Private label card issuers already supporting private label card activation at POS requests and wanting to offer card activation plus initial load must be able to accept Authorization Request/0100 and Reversal Request/0400 messages where DE 4 contains a valid amount, DE 6 (Amount, Cardholder Billing), and optionally DE 5 (Amount, Settlement), and DE 48, subelement 77 is C09.

Product Inquiry Service

The Product Inquiry Service allows an acquirer to send a product inquiry authorization request message to Mastercard.

NOTE: Applies only to the U.S. region.

As part of the authorization response to an acquirer, Mastercard will provide an acquirer with the product code associated with the particular Mastercard card number. Additionally, because product codes for various card programs potentially fall under different interchange rate structures, Mastercard will also populate the authorization response message with the applicable account category code as defined by the Account Level Management Service.

The information received by the acquirer through a Product Inquiry Service request, along with the published Mastercard interchange rate schedule and rate criteria, can be used by an acquirer and merchant to determine the product and associated interchange rate that may be applied to a purchase transaction for that card.

Acquirers

Acquirers that choose to support Product Inquiry Service transactions must send Authorization Request/0100 messages containing DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status), value 8 (Account Status Inquiry Service [ASI]) and DE 4 (Amount, Transaction) with a transaction amount of zero when submitting Product Inquiry Service transaction requests.

NOTE: Product Inquiry Service messages that include address verification and/or CVC 2 validation requests will be deemed to be an attempt to mitigate fraud. As such, they will be designated as an Account Status Inquiry Service transaction and billed accordingly.

Issuers

Issuers will receive Authorization Request/0100 messages containing DE 61, subfield 7, value 8, and DE 4 with a transaction amount of zero. Issuers must respond to these transactions with value 00 (Approved or completed successfully), 85 (Not Declined), 05 (Do Not Honor) or other valid business decline responses in DE 39 (Response Code). Invalid business declines include values 03 (Invalid merchant), 12 (Invalid transaction), 13 (Invalid amount), 30 (Format error), 51 (Insufficient funds/over credit limit), 57 (Transaction not permitted to issuer/cardholder), and 58 (Transaction not permitted to acquirer/terminal).

For More Information

For more information about the Product Inquiry Service, contact the Global Customer Service team.

Authorization Request/0100: Product Inquiry Service

Following is the list of the data elements applicable to this message. All mandatory Authorization Request/0100 data elements apply.

Table 1778: Data elements

DE ID	Data Element Name	Org	Sys	Dst	Comments
4	Amount, Transaction	M	.	M	Transaction amount of zero, in the acquirer's currency, at the point of interaction.
61	Point-of-Service (POS) Data, subfield 7 (POS Transaction Status), value 8 (Account Status Inquiry Service [ASI])	M	.	M	

Proximity Payments

The Mastercard Proximity Payments solution, which includes Mastercard contactless magnetic stripe and Mastercard contactless M/Chip, is part of the global Proximity Payments Program and is designed to enrich the traditional card with a contactless interface.

The contactless interface provides cardholder and merchant benefits that are particularly relevant in environments such as:

- Unattended point-of-service (POS) devices (for example, gas pumps and vending machines)
- High-traffic environments (for example, quick service and drive-through restaurants)

Proximity payments do not require cardholders holding a contactless Mastercard chip card to swipe or insert the card into a card reader or terminal. Instead, cardholders place the contactless card in proximity of a specially equipped merchant terminal to make a payment.

Authorization Request/0100: Proximity Payments

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Table 1779: Data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 22 (Point-of-Service [POS] Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)	M	•	M	Contains one of the following values: • 07 = Merchant terminal PAN auto-entry via contactless M/Chip • 91 = PAN auto-entry via contactless magnetic stripe—the full track data has been read from the data on the card and transmitted within the Authorization Request/0100 message in DE 35 (Track 2 Data) or DE 45 (Track 1 Data) without alteration or truncation.

Data Element	Org	Sys	Dst	Values/Comments
DE 61 (Point-of-Service [POS] Data), subfield 11 (POS Card Data Terminal Input)	M	•	M	Contains one of the following values: • 3 = Terminal supports contactless EMV input. The terminal may also support one or more other card input types, including contactless magstripe input, EMV contact chip input, magnetic stripe input and key entry input. • 4 = Contactless Mag Stripe (Proximity Chip) Terminal supports contactless magstripe input but not contactless EMV input. The terminal also may support one or more other card input types, including EMV contact chip input, magnetic stripe input, and key entry input.

Purchase of Goods or Services with Cash Back

Mastercard allows the use of Purchase of Goods or Services with Cash Back in Authorization/01xx and Reversal/04xx messages for Debit Mastercard and Maestro cards.

Participation Mandate

All issuers of Debit Mastercard and Maestro cards are required to support the receipt of authorization and reversal requests for Purchase of Goods or Services with Cash Back transactions. The Purchase of Goods or Services with Cash Back service is automatically associated with all Debit Mastercard and Maestro account ranges.

Issuers will continue to approve or decline Purchase of Goods or Services with Cash Back transactions at their discretion.

Terminal Support Indicator

The Dual Message Authorization System allows acquirers to indicate whether the merchant terminal supports receipt of the purchase amount only approval response code in an authorization message.

The Authorization Request/0100 message must contain DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code), value 09 (Purchase with Cash Back) and DE 48 (Additional Data: Private Use), subelement 61 (POS Data, Extended Condition Codes), subfield 2 (Purchase Amount Only Terminal Support Indicator), value 1 (Merchant terminal supports receipt of purchase-only approval).

For more information, refer to the [Transaction Processing Rules](#).

Authorization Request/0100: Purchase of Goods or Services with Cash Back

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Table 1780: Data elements

Data Element	Org	Sys	Dst	Values/Comments
DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)	M	.	M	Must contain value 09 = Purchase of Goods or Services with Cash Back
DE 4 (Amount, Transaction)	M	.	M	Must contain the transaction amount, inclusive of the amount cash back.
DE 48 (Additional Data—Private Use), TCC (Transaction Category Code)	M	.	M	
DE 48, subelement 61 (POS Data, Extended Condition Codes), subfield 2 (Purchase Amount Only Terminal Support Indicator)	C	.	C	1 = Merchant terminal supports receipt of purchase only approvals
DE 54 (Additional Amounts), subfield 1 (Account Type)	M	X	M	The Dual Message Authorization System provides a second occurrence of DE 54 to the issuer if billing currency is different than transaction currency. Must contain one of the following values: 00 = Default account (not specified or not applicable) 10 = Savings Account 20 = Checking Account
DE 54, subfield 2 (Amount Type)	M	X	M	40 = Amount Cash Back
DE 54, subfield 3 (Currency Code)	M	X	M	Must contain the valid three-digit numeric value present in DE 49 (Currency Code, Transaction)
DE 54, subfield 4 (Debit or Credit Indicator)	M	X	M	D = (Debit amount) plus 12 digits right-justified with leading zeros

Issuer Response Options

Issuers have several options in how they respond to a Purchase of Goods or Services with Cash Back request.

When the issuer receives an Authorization Request 0100/message containing DE 48 (Additional Data: Private Use), subelement 61 (POS Data, Extended Condition Codes), subfield 2 (Purchase

Amount Only Terminal Support Indicator), value 1 (Merchant terminal supports receipt of purchase only approvals), the issuer has the option to:

- Approve the entire transaction amount
- Decline the entire transaction amount
- Respond with a purchase amount only approval

The issuer must approve the entire purchase amount. Partial approval (DE 39, value 10) of the purchase amount is not allowed.

The following table describes the information that the issuer must provide in the Authorization Request Response/0110 message when responding with a purchase-only approval and the information that the acquirer will receive for a purchase-only approval.

Table 1781: Issuer Response Options

The Issuer must provide...	The Acquirer will receive...
the approved amount (purchase amount) in DE 6 (Amount, Cardholder Billing) in the issuer's cardholder billing currency. This amount must be the purchase amount as calculated by subtracting the DE 54 (Additional Amounts) cash back amount and, if applicable, the DE 28 (Amount, Transaction Fee) from the DE 6 amount present in the Authorization Request/0100 message in the amount data element that corresponds to the issuer's cardholder billing currency.	the purchase-only approval amount in the acquirer's transaction currency in DE 4
DE 38 (Authorization ID Response)	DE 38
DE 39 value 87 (Purchase only, no cash back allowed)	DE 39, value 87
DE 51 (Currency Code, Cardholder Billing) with the issuer's cardholder billing currency code	an occurrence of the original amount of the transaction in DE 54 (Additional Amounts) in the acquirer's transaction currency. The original amount is identified by DE 54, subfield 2 (Amount Type), value 57 (Original Amount), and subfield 4 (Debit or Credit Indicator), value C plus 12-digit original amount.
	an occurrence of the original amount of the transaction in DE 54 in the issuer's cardholder billing currency. The original amount is identified by DE 54, subfield 2, value 57, and subfield 4, value C plus 12-digit original amount.

NOTE: Issuers responding with DE 39, value 87 will not be required to echo DE 4 (Amount, Transaction) in the Authorization Request Response/0110. Likewise, if DE 5 (Amount, Settlement) was present in the Authorization Request/0100 message to the issuer, the issuer will not be required to echo DE 5 in the Authorization Request Response/0110 when responding with DE 39, value 87. The issuer will provide the purchase-only approval amount in DE 6 and the issuer currency code in DE 51.

Reversal Request/0400

In some cases, the cardholder or merchant may elect not to complete the transaction after receiving the purchase-only approval response from the issuer. Mastercard supports full reversal messages to allow the merchant to cancel the transaction.

In addition to all other applicable data elements for the Reversal Request/0400 message, acquirers should submit Reversal Request/0400 messages with the following data elements for a reversal of a purchase-only approval:

- Purchase-only approval amount in DE 4 that was present in the Authorization Request Response/0110 message to the acquirer, not the original amount present in DE 4 of the Authorization Request/0100 message from the acquirer
- DE 39, value 87

The Dual Message Authorization System will perform currency conversion if appropriate and will provide the following data elements in the Reversal Request/0400 message to the issuer:

- Purchase-only approval amount in DE 4 in the acquirer's transaction currency
- Purchase-only approval amount in DE 5 in U.S. dollars if the issuer has opted to receive this data element in the message
- Purchase-only approval amount in DE 6 in the issuer's cardholder billing currency
- DE 39, value 87

When processing a Reversal Request/0400 for a purchase-only approval (DE 39, value 87), the issuer should increase the cardholder's open-to-buy.

Reversal Advice/0420

If the Dual Message Authorization System generates a Reversal Advice/0420 message after the issuer has responded to the Authorization Request Response/0110 message with DE 39, value 87, the Dual Message Authorization System will provide the following data elements.

- Purchase-only approval amount in the acquirer's transaction currency in DE 4
- Purchase-only approval amount in DE 5 in U.S. dollars, if the issuer has opted to receive this data element in the message
- Purchase-only approval amount in DE 6 in the issuer's cardholder billing currency
- DE 39, value 87
- Original amount in DE 54 in the issuer's cardholder billing currency and acquirer's transaction currency

When processing a Reversal Advice/0420 for a purchase-only approval (DE 39, value 87), the issuer should increase the cardholder's open-to-buy.

Authorization Advice/0120

In addition to the standard data elements that are part of the system-generated Authorization Advice/0120 message, these messages should include the following data elements for purchase-only approvals.

- Purchase-only approval amount in DE 4 in the acquirer's transaction currency

- Purchase-only approval amount in DE 5 in the settlement currency (US dollars)
- Purchase only-approval amount in DE 6 in the issuer's cardholder billing currency
- DE 39, value 87 as provided by the issuer or Stand-In in the Authorization Request Response/0110
- Original amount in DE 54 in the issuer's cardholder billing currency and acquirer's transaction currency

Authorization Advice/0120: acquirer-generated

DE 39, value 87 is not a valid value for Authorization Advice/0120: acquirer-generated messages. If an Authorization Advice/0120: acquirer-generated message contains DE 39, value 87, the Dual Message Authorization System will generate an Authorization Advice Response/0130 message where DE 39 contains value 30 and DE 44 contains value 039.

Alternate Processing

The Mastercard Network will invoke Stand-In Authorization or X-code Processing in accordance with issuer parameters and Mastercard rules and may subsequently approve the transaction for the following situations:

- If a format error is detected on incoming Authorization Request Response/0110 messages for all brands, regardless of the response code provided by the issuer in DE 39 or response code is invalid.
- If the issuer responds to an Authorization Request/0100 message with an Authorization Request Response/0110 message with one of the following values in DE 39 (Response Code) that indicates a system error, the transaction was unable to route, or the issuer is unavailable:
 - DE 39, value of 91 (Destination processor [customer or] not available)
 - DE 39, value of 92 (Authorization system or issuer system inoperative)
 - DE 39, value of 96 (System error)
- The Authorization Request/0100 message to the issuer times out or the issuer is marked down.

Issuers have the option to choose whether Purchase with Cash Back (PWCB) transactions are eligible for processing by Stand-In Authorization. An issuer may choose to allow PWCB for only PIN-based transactions, only Signature-based transactions, or allow both PIN and signature-based transactions be processed by Stand-In Authorization.

Mastercard provides issuers with parameters to define whether or not a Purchase With Cash Back transaction should be forwarded to Stand-In Authorization for processing. For a transaction to qualify for the PIN-based category, PIN Data will have to be present in the Authorization Request/0100 message. If PIN Data was not present in the incoming message, the transactions will be categorized as a signature-based transaction.

If the issuer chooses to have PIN, Signature or both PIN and Signature Purchase With Cash Back transactions excluded from Stand-In Authorization, the Dual Message Authorization System will provide the acquirer with an Authorization Request Response/0110 message with a DE 39 Response of 91 if the issuer is not able to respond to the Authorization Request/0100 message.

If the issuer chooses to process PIN, Signature or both PIN and Signature Purchase with Cash Back transactions in Stand-In, Stand-In Authorization may provide the purchase-only amount response DE 39, value 87 when the Authorization Request/0100 message contains DE 3, subfield 1, value 09, and DE 48, subelement 61, subfield 2 contains value 1 and the cash back transaction limit or the cash accumulation limits have been exceeded but the purchase limits have not.

Purchase with Cash Back transactions will not be processed with X-Code Processing.

For more information, refer to the [Stand-In Authorization Reference Guide](#).

Dual Message Authorization System Edits

The Dual Message Authorization System will perform the following edits on Purchase of Goods or Services with Cash Back transactions.

Authorization Request/0100

Table 1782: Associated edits

WHEN...	THEN the Dual Message Authorization System...
DE 3, subfield 1 with the value 09 and DE 54 is not present	DE 39 (Response Code) = 30 DE 44 (Additional Response Data) = 003
DE 3, subfield 1 with the value 09 and DE 54, subfield 1 is not a valid two-digit numeric value	DE 39 = 30 DE 44 = 054
DE 3, subfield 1 with the value 09 and DE 54, subfield 2 is not 40	DE 39 = 30 DE 44 = 003
DE 3, subfield 1 with the value 09 and DE 54, subfield 3 is not the same value as in DE 49	DE 39 = 30 DE 44 = 054
DE 3, subfield 1 with the value 09 and DE 54, subfield 4 is not D followed by 12 numeric digits	DE 39 = 30 DE 44 = 054
DE 3, subfield 1 with the value 09 and the cash back amount in DE 54 is greater than the amount in DE 4 (Amount, Transaction)	DE 39 = 30 DE 44 = 054
DE 3, subfield 1 with the value 09 and the Primary Account Number (PAN) is not within an account range that supports Purchase With Cash Back	DE 39 = 57

WHEN...	THEN the Dual Message Authorization System...
DE 48, subelement 61, subfield 2 contains a value other than 0 or 1	Generates an Authorization Request Response/0110 message containing: DE 39 = 30 DE 44 = 048061
DE 54 is present in the Authorization Request/0100 message where DE 54, subfield 2 contains value 57 (Original Amount)	Generates an Authorization Request Response/0110 message containing: DE 39 = 30 DE 44 = 054
DE 54, subfield 2 with the value 40 and DE 3, subfield 1 is not 09	DE 39 = 30 DE 44 = 003

If the Authorization Request/0100 message passes all Dual Message Authorization System edits, the Dual Message Authorization System will forward the Authorization Request/0100 message to the issuer.

NOTE: Acquirers should send only one occurrence of DE 54, subfields 1–4 in Authorization Request/0100 and Reversal Request/0400 messages.

Authorization Request Response/0110

The Dual Message Authorization System will perform the following edits on the Authorization Request Response/0110 message when the issuer has provided DE 39 with value 87.

The Dual Message Authorization System will provide two additional occurrences of DE 54, subfields 1–4 in the Authorization Request/0100 message to the issuer; one in the acquirer's transaction currency and one in the issuer's cardholder billing currency. This additional occurrence will be appended to the end of DE 54 before sending to the issuer.

If the issuer provides DE 54, subfield 2 with the value 40 in the Authorization Request Response/0110 message, Mastercard will not forward the cash back amount to the acquirer in the Authorization Request Response/0110 message.

Table 1783: Associated edits

WHEN...	THEN the Dual Message Authorization System...
DE 48, subelement 61 is not present or DE 48, subelement 61, subfield 2 does not contain value 1 or DE 3, subfield 1 is not 09 in the Authorization Request/0100 message sent to the issuer	Sends to the issuer an Authorization Response Negative Acknowledgment/0190 message where: DE 39 = 30 DE 44 = 039

WHEN...	THEN the Dual Message Authorization System...
DE 38 is not present	Sends to the issuer an Authorization Response Negative Acknowledgment/0190 message where: DE 39 = 30 DE 44 = 038
DE 6 is not present	Sends to the issuer an Authorization Response Negative Acknowledgment/0190 message where: DE 39 = 30 DE 44 = 006
The amount in DE 6 of the Authorization Request Response/0110 message does not equal the purchase amount based on the original amounts present in the Authorization Request/0100 message.	Sends to the issuer an Authorization Response Negative Acknowledgment/0190 message where: DE 39 = 30 DE 44 = 039
The amount in DE 6 of the Authorization Request Response/0110 message is greater than the original amount in the Authorization Request/0100 message.	Sends to the issuer an Authorization Response Negative Acknowledgment/0190 message where: DE 39 = 30 DE 44 = 006
DE 51 is not present or is not the issuer's correct cardholder billing currency code	Sends to the issuer an Authorization Response Negative Acknowledgment/0190 message where: DE 39 = 30 DE 44 = 051

Authorization Advice/0120: acquirer-generated

Table 1784: Associated edits

WHEN...	THEN the Dual Message Authorization System...
DE 39 contains value 87	Creates an Authorization Advice Response/0130—System-generated message containing: DE 39 = 30 DE 44 = 039

Reversal Request/0400

Table 1785: Associated edits

WHEN...	THEN the Dual Message Authorization System...
DE 54 is not formatted correctly in relation to alphanumeric specifications for the subfields	Returns the acquirer a Reversal Request Response/0410 message where: DE 39 = 30 DE 44 = 054
DE 3, subfield 1, contains value 09 and DE 54, subfield 2, value 40 is not present and DE 39 does not contain value 87	Returns the acquirer a Reversal Request Response/0410 message where: DE 39 = 30 DE 44 = 003
DE 3, subfield 1 does not contain value 09 and DE 54, subfield 2, value 40 is present	Returns the acquirer a Reversal Request Response/0410 message where: DE 39 = 30 DE 44 = 003

Real-Time Substantiation

The Real-time Substantiation (formerly referred to as Auto Substantiation) service supports substantiation at the point of sale (POS) for qualified expenses incurred on a Flexible Spending Account (FSA) and Healthcare Reimbursement Arrangement (HRA) cards when used at a merchant with a qualifying Inventory Information Approval System (IIAS).

Participation in Real-Time Substantiation

Issuers must notify Mastercard if they want to support real-time substantiation processing by completing the *Real-time Substantiation Participation Request Form* and providing the form to their Global Customer Service representative.

Participation authorizes Mastercard to provide the issuer with real-time substantiation information in the Authorization Request/0100 or the Authorization Advice/0120 message. The Authorization Platform will remove healthcare related amounts from the Authorization Request/0100 or the Authorization Advice/0120 message for non-participating issuers.

Mastercard supports the issuance of Mastercard Assigned IDs for merchants.

To indicate an IIAS-compliant merchant, acquirers must provide DE 48 (Additional Data—Private Use), subelement 32 (Mastercard Assigned ID) in Authorization Request/0100, Authorization Advice/0120, and Reversal Request/0400 messages when the transaction is indicated as real-time substantiated (DE 48, subelement 61 (POS Data Extended Condition Codes), subfield 3

(Real-time Substantiation Indicator), value 1 (Merchant terminal verified the purchased items against an Inventory Information Approval System [IIAS]).

To obtain Mastercard Assigned IDs for IIAS merchant validation, acquirers should send an e-mail message to sigis_merchant_setup@mastercard.com. The request should specify whether it is an addition or an update of the Mastercard Assigned ID, and at a minimum should include the acquirer name, telephone number, e-mail address, acquirer ID, processor ID, merchant parent/owner name (if applicable), Mastercard Assigned ID (if the request is for an update to the existing Mastercard Assigned ID), and merchant contact information.

Acquirers with existing Mastercard Assigned IDs for their merchants should continue to use those values but must notify Mastercard that those merchants are Special Interest Group for IIAS Standards (SIGIS)-compliant.

NOTE: Acquirers with health care IIAS merchant that have received a Visa-assigned Merchant Verification Value from Visa, should include it in DE 48, subelement 36 (Visa Defined Data) for gateway mapping to Visa Field 62.20.

Merchant Terminal Verification

Following are the details on how a merchant terminal is IIAS-compliant.

Acquirers must provide DE 48 (Additional Data—Private Use), subelement 32 (Mastercard Assigned ID) in Authorization Request/0100, Authorization Advice/0120, and Reversal Request/0400 messages when the transaction is indicated as real-time substantiated (DE 48, subelement 61 (POS Data Extended Condition Codes), subfield 3 (Real-time Substantiation Indicator), value 1 (Merchant terminal verified the purchased items against an Inventory Information Approval System [IIAS])).

When the merchant terminal has verified the purchased items against an IIAS, the acquirer should populate the Authorization Request/0100 message with DE 48, subelement 61 and the following subfield values:

- Subfield 1 (Partial Approval Terminal Support Indicator) and subfield 2 (Purchase Amount Only Terminal Support Indicator) must contain values of zero or 1.
- Subfield 3 must contain a value of 1 (Merchant terminal verified the purchased items against the IIAS).
 - If the issuer is not participating in real-time substantiation, this value will be changed by the Authorization Platform to 0 (Merchant terminal did not verify the purchased items against the IIAS).
 - To indicate to issuers participating in real-time substantiation that the transaction was submitted as IIAS but from a non-IIAS compliant merchant, Mastercard will populate subfield 3 with the value 4 (Transaction was submitted as real-time substantiated, but from a non-IIAS certified merchant).
- Subfields 4 and 5 are reserved for future use and must contain values of zero.

Acquirers will not receive DE 48, subelement 61 in the Authorization Request Response/0110 message.

When an acquirer creates an Authorization Advice/0120 message to advise the issuer of an approved authorization performed by the acquirer, DE 48, subelement 61 should be present if it was present in the original Authorization Request/0100 message.

Real-Time Substantiation Amounts

Mastercard defines DE 54 (Additional Amounts), subfield 2 (Amount Type) with a redefined value of 10 (Healthcare Eligibility Amount). When the acquirer is providing the real-time substantiation indicator in DE 48, subelement 61, subfield 3, this redefined amount type allows the acquirer to indicate the portion of DE 4 (Amount, Transaction) that is eligible for real-time substantiation.

In addition to the Healthcare Eligibility Amount (value 10), Mastercard supports amount type DE 54, subfield 2, value 11 (Prescription Eligibility Amount), which allows the acquirer to indicate the portion of the healthcare eligibility amount that includes the amount spent for prescriptions.

DE 54, subfield 2, value 11 must only be present when the acquirer provides subfield 2, value 10 in the Authorization Request/0100 or the Authorization Advice/0120 message. In addition, the amount in subfield 2, value 11 must be less than or equal to the amount of subfield 2, value 10.

Mastercard also supports DE 54, subfield 2, value 12 (Vision Rx Eligibility Amount), which allows the acquirer to indicate the portion of the amount spent on vision Rx or vision products/services. When a prescription is issued and dispensed by a licensed vision center for vision products/services (for example, tests/exams, lenses, glass frames, and so on), the patient portion of the cost is considered an eligible item under IRS regulations for processing as per the Special Interest Group for IAS Standards (SIGIS). Acquirers may provide DE 54, subfield 2, value 12 by itself in the Authorization Request/0100 or the Authorization Advice/0120 messages or in combination with value 10 or values 10 and 11.

Values 10, 11, or 12 will not be returned to the acquirer in the Authorization Request Response/0110 or the Authorization Advice Response/0130 message.

Transaction Processing Examples

Following are examples of Real-time Substantiation transaction processing. The examples only show one occurrence of DE 54 amounts in USD. However, standard currency conversion rules will apply. Therefore, acquirers and issuers will always receive amount-related data elements in the acquirer's transaction currency and the issuer's cardholder billing currency for DE 54.

Because Real-time Substantiation transactions can be used in combination with partial approvals, when the issuer receives an Authorization Request/0100 message containing DE 48, subelement 61, subfield 1, value 1, the issuer has the option to:

- Approve the entire transaction amount.
- Decline the entire transaction amount.
- Respond with a partial approval.

For the issuer to respond with a partial approval, DE 48, subelement 61, subfield 1 must contain a value of 1; otherwise, the issuer must approve or decline the entire transaction amount.

Partial approvals are not valid for Authorization Advice/0120 messages and will be rejected with a format error.

Example 1—Partial Approval: Entire Healthcare Eligibility Amount

This example illustrates that the issuer approved the entire Healthcare Eligibility Amount, which in this case is a partial approval of the total transaction amount. In this scenario, the merchant would ask the cardholder to use another form of payment for the remaining USD 60 or remove some items from the purchase.

Authorization Request/0100	Authorization Request Response/0110	Authorization Request Response/0110
	From Issuer	To Acquirer
- DE 48, subelement 61 = 10100 (indicates terminal can handle partial approvals and has verified against IIAS) - DE 4 = USD 100 - DE 54, subfield 2, value 10 = USD 40	- DE 6 = USD 40 - DE 39 = 10	- DE 4 = USD 40 100 - DE 54, subfield 2, value 57 (Original Amount) = USD - DE 39 = 10

Example 2—Partial Approval: Entire Vision Prescription Amount

This example illustrates that the issuer approved the entire Vision Prescription Amount, which in this case is a partial approval of the total transaction amount. In this scenario, the merchant would ask the cardholder to use another form of payment for the remaining USD 60 or remove some items from the purchase.

Authorization Request/0100	Authorization Request Response/0110	Authorization Request Response/0110
	From Issuer	To Acquirer
- DE 48, subelement 61 = 10100 (indicates terminal can handle partial approvals and has verified against IIAS) - DE 4 = USD 100 - DE 54, subfield 2, value 12 = USD 40	- DE 6 = USD 40 - DE 39 = 10	- DE 4 = USD 40 - DE 54, subfield 2, value 57 (Original Amount) = USD 100 - DE 39 = 10

Example 3—Partial Approval: Partial Healthcare Eligibility Amount

This example illustrates that the issuer approved only a portion of the Healthcare Eligibility Amount, which in this case is a partial approval of the total transaction amount. In this scenario, the merchant would ask the cardholder to use another form of payment for the remaining USD 80 or remove some items from the purchase.

Authorization Request/0100	Authorization Request Response/0110	Authorization Request Response/0110
	From Issuer	To Acquirer
- DE 48, subelement 61 = 10100 - DE 4 = USD 100 - DE 54, subfield 2, value 10 = USD 40	- DE 6 = USD 20 - DE 39 = 10	- DE 4 = USD 20 - DE 54, subfield 2, value 57 = USD 100 - DE 39 = 10

Example 4—Partial Approval: Partial Vision Prescription Amount

This example illustrates that the issuer approved only a portion of the Vision Prescription Amount, which in this case is a partial approval of the total transaction amount. In this scenario, the merchant would ask the cardholder to use another form of payment for the remaining USD 80 or remove some items from the purchase.

Authorization Request/0100	Authorization Request Response/0110	Authorization Request Response/0110
	From Issuer	To Acquirer
- DE 48, subelement 61 = 10100 - DE 4 = USD 100 - DE 54, subfield 2, value 12 = USD 40	- DE 6 = USD 20 - DE 39 = 10	- DE 4 = USD 20 - DE 54, subfield 2, value 57 = USD 100 - DE 39 = 10

Example 5—Full Approval: Entire Healthcare Eligibility Amount

In this example, the merchant terminal is indicating that they do not support partial approvals; therefore, a full approval or decline is required. The issuer approves the full transaction; thereby approving the entire Healthcare Eligibility Amount.

Authorization Request/0100	Authorization Request Response/0110 From Issuer	Authorization Request Response/0110 To Acquirer
• DE 48, subelement 61 = 00100 • DE 4 = USD 100 • DE 54, subfield 2, value 10 = USD 100	• DE 4 = USD 100 • DE 39 = 00	• DE 4 = USD 100 • DE 39 = 00 • No DE 54, subfield 2, value 57 (Original Amount) is provided in this scenario because the issuer responded with DE 39 = 00

Example 6—Full Approval: Entire Vision Prescription Amount

In this example, the merchant terminal is indicating that they do not support partial approvals; therefore, a full approval or decline is required. The issuer approves the full transaction; thereby approving the entire Vision Prescription Amount.

Authorization Request/0100	Authorization Request Response/0110 From Issuer	Authorization Request Response/0110 To Acquirer
• DE 48, subelement 61 = 00100 • DE 4 = USD 100 • DE 54, subfield 2, value 12 = USD 100	• DE 4 = USD 100 • DE 39 = 10	• DE 4 = USD 100 • DE 39 = 00 • No DE 54, subfield 2, value 57 (Original Amount) is provided in this scenario since the issuer responded with DE 39 = 00

Example 7—Partial Approval: Partial Healthcare Eligibility Amount, including Prescriptions

This example illustrates that the issuer approved only a portion of the Healthcare and Prescription Eligibility amounts. In this scenario, the merchant would ask the cardholder to use another form of payment for the remaining USD 80 or remove some items from the purchase.

Authorization Request/0100	Authorization Request Response/0110 From Issuer	Authorization Request Response/0110 To Acquirer
• DE 48, subelement 61 = 10100 • DE 4 = USD 100 • DE 54, subfield 2, value 10 = USD 40 • DE 54, subfield 2, value 11 = USD 30	• DE 6 = USD 20 • DE 39 = 10	• DE 4 = USD 20 • DE 54, subfield 2, value 57 = USD 100 • DE 39 = 10

Example 8—Partial Approval: Partial Healthcare Eligibility Amount and Vision Prescription Amount , including Prescriptions and Vision Prescriptions

This example illustrates that the issuer approved only a portion of the Healthcare, Prescription Eligibility, and Vision Prescription amounts. In this scenario, the merchant would ask the cardholder to use another form of payment for the remaining USD 80 or remove some items from the purchase.

Authorization Request/0100	Authorization Request Response/0110 From Issuer	Authorization Request Response/0110 To Acquirer
• DE 48, subelement 61 = 10100 • DE 4 = USD 100 • DE 54, subfield 2, value 10 = USD 20 • DE 54, subfield 2, value 11 = USD 10 • DE 54, subfield 2, value 12 = USD 10	• DE 6 = USD 20 • DE 39 = 10	• DE 4 = USD 20 • DE 54, subfield 2, value 57 = USD 100 • DE 39 = 10

Example 9—Full Approval: Entire Healthcare Eligibility Amount, including Prescriptions

In this example, the merchant terminal is indicating that they do not support partial approvals; therefore, a full approval or decline is required. The issuer approves the full transaction; thereby approving the entire Healthcare Eligibility Amount, including the Prescription Eligibility Amount.

Authorization Request/0100	Authorization Request Response/0110 From Issuer	Authorization Request Response/0110 To Acquirer
- DE 48, subelement 61 = 00100 - DE 4 = USD 100 - DE 54, subfield 2, value 10 = USD 100 - DE 54, subfield 2, value 11 = USD 60	- DE 4 = USD 100 - DE 39 = 00	- DE 4 = USD 100 - DE 39 = 00 - No DE 54, subfield 2, value 57 (Original Amount) is provided in this scenario since the issuer responded with DE 39 = 00

Example 10—Full Approval: Entire Healthcare Eligibility Amount and Vision Prescription Amount, including Prescriptions and Vision Prescriptions

In this example, the merchant terminal is indicating that they do not support partial approvals; therefore, a full approval or decline is required. The issuer approves the full transaction; thereby approving the entire Healthcare Eligibility Amount including the Prescription Eligibility Amount and Entire Vision Prescription Amount.

Authorization Request/0100	Authorization Request Response/0110 From Issuer	Authorization Request Response/0110 To Acquirer
- DE 48, subelement 61 = 00100 - DE 4 = USD 100 - DE 54, subfield 2, value 10 = USD 60 - DE 54, subfield 2, value 11 = USD 10 - DE 54, subfield 2, value 12 = USD 40	- DE 4 = USD 100 - DE 39 = 10	- DE 4 = USD 100 - DE 39 = 00 - No DE 54, subfield 2, value 57 (Original Amount) is provided in this scenario since the issuer responded with DE 39 = 00

Authorization Platform Edits

The Authorization Platform performs the following edits on Real-time Substantiation transactions.

Authorization Request/0100 and Authorization Advice/0120

NOTE: DE 54 (Additional Amounts), subfield 2 (Amount Type), value 12 (Vision Rx Eligibility Amount) is an independent field and can be used without value 10 (Healthcare Eligibility Amount).

NOTE: When DE 54 contains multiple occurrences, the DE 54, subfield 2, value 10 (Healthcare Eligibility Amount) can co-exist with amount types other than the value 11 (Prescription Eligibility Amount).

WHEN...	THEN the Authorization Platform...
DE 54, subfield 2, values 10, 11, or 12 are present and DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) is not 00 (Purchase of goods or services)	Sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where: - DE 39 = 30 (Format error) - DE 44 = 054
DE 54 (Additional Amounts), subfield 2 (Amount Type) is greater than transaction amount in DE 4 (Amount, Transaction) and the healthcare amount types (10 or 12) are populated separately or The sum of healthcare and vision amount types (10 or 12) are populated together and is greater than request amount in DE 4	Sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where: - DE 39 (Response Code) = 30 (Format error) - DE 44 (Additional Response Data) = 054
DE 54 subfield 2, values 10, 11, or 12 are present and the issuer does not support healthcare substantiation transaction processing	Removes DE 54 subfield 2, value 10 and 11 occurrences from the Authorization Request/0100 or Authorization Advice/0120 message to the issuer.
DE 54, subfield 2, value 11 is present and DE 54, subfield 2, value 10 is not present	Sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where - DE 39 = 30 - DE 44 = 054
The amount in DE 54, subfield 2 value 11 is greater than the amount in DE 54, subfield 2, value 10	Sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where - DE 39 = 30 - DE 44 = 054
DE 54 contains more than one occurrence of subfield 2, values 10, 11, or 12	Sends the acquirer an Authorization Request Response/0110 or Authorization Advice Response/0130 message where: - DE 39 = 30 - DE 44 = 054

WHEN DE 48, Subelement 32 is present and...	THEN the Authorization Platform...
The length of the Mastercard Assigned ID is less than six digits in the Authorization Request/0100, Authorization Advice/0120—System-generated, and Reversal Request/0400 message	Sends the acquirer an Authorization Request Response/0110, Authorization Advice Response/0130, or Reversal Response/0410 message where: • DE 39 = 30 • DE 44 = 048032
WHEN DE 48, subelement 61, subfield 3 is...	THEN the Authorization Platform...
1 (Merchant terminal verified the purchase items against an Inventory Information Approval System [IIAS])	Validates that DE 48, subelement 32 (if present) contains a valid Mastercard Assigned ID for IIAS.
If the Mastercard Assigned ID is valid	Forwards the Authorization Request/0100 message to the issuer.
If the Mastercard Assigned ID is not valid or not present in the Authorization Request/0100 message	Updates the value in DE 48, subelement 61, subfield 3 as follows: • If the issuer participates in real-time substantiation, sends DE 48, subelement 61, subfield 3, value 4 (Transaction was submitted as real-time substantiated but from a non-IIAS certified merchant). • If the issuer does not participate in real-time substantiation sends DE 48, subelement 61, subfield 3, value 0 (Merchant terminal did not verify the purchased items against an Inventory Information Approval System [IIAS]).
WHEN DE 48, subelement 61, subfield 3 is...	THEN the Authorization Platform...
4 (Transaction was submitted as real-time substantiated but from a non-IIAS certified merchant) in the Authorization Request/0100 message	Sends the acquirer an Authorization Request Response/0110 message where: • DE 39 = 30 • DE 44 = 048061

Reversal Request/0400

WHEN...	THEN the Authorization Platform...
DE 54, subfield 2, values 10, 11, or 12 are present in the Reversal Request/0400 message	Removes DE 54, subfield 2, values 10, 11, and 12 from the Reversal Request/0400 message to the issuer.

Refund Transactions

A refund transaction allows a cardholder to return goods or services previously purchased from a merchant. The Authorization Platform accepts refund transactions for Mastercard® and Maestro® transactions, in addition to Private Label transactions.

An acquirer sends an Authorization Request/0100 message with DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) containing value 20xx00 for the refund (where xx indicates the type of account to be credited). The issuer approves or declines the refund transaction using an Authorization Request Response/0110 message.

The refund Authorization Request/0100 message can be reversed using a Reversal Request/0400 message, if the merchant or cardholder decides not to complete the refund.

All refund transactions are routed to the same issuer host as for point-of-service (POS) transactions.

Refund transactions are not processed by the Stand-In System or X-Code alternate processing.

- If the issuer is not available to respond to a refund Authorization Request/0100 message, the Authorization Platform sends an Authorization Request Response/0110 message to the acquirer with DE 39 (Response Code), value 91 (Authorization System or Issuer System Inoperative).
- If the issuer is not available to respond to a refund Reversal Request/0400 message, the Authorization Platform sends a Reversal Request Response/0410 message to the acquirer with DE 39, value 00 (Approved or completed successfully). The Authorization Platform also sends a Reversal Advice/0420 message to the issuer as notification of the reversal that was responded to on their behalf.

Refund Transaction Support Requirements

All Mastercard and Debit Mastercard issuers must be able to approve or decline refund transaction authorization request messages.

Effective as of 17 April 2020 with Release 20.Q2, all acquirers of Mastercard and Debit Mastercard transactions must be able to send refund transaction authorization request messages and forward the issuer's response to the merchant, for any merchant that chooses to initiate refund transaction authorization requests. Support of refund transaction authorization request messages is optional for merchants.

The Swedish Domestic Authorization Switching Service includes support of refund transactions. For more information about refund transactions, refer to the *Transaction Processing Rules*.

Reversal Processing

Mastercard supports the reversal of a full transaction amount or a partial transaction amount using reversal processing.

Guiding principles for Authorization Reversal Processing

Mastercard provides the following update to its best practices for the management of authorization reversals. These best practices are intended to guide dual message acquirers, issuers, and processors in the usage of these transactions.

Background

Merchant-initiated authorization reversal messages are intended to assist issuers in managing "open-to-buy." This processing function is especially critical to debit and prepaid card issuers. Merchants and acquirers submit authorization reversal messages when:

- The final transaction amount was less than the authorized amount
- The cardholder cancels or chooses not to complete all or part of the transaction
- All or a portion of the goods or services could not be provided (for example, out-of-stock items)
- The authorization was submitted in error (for example, duplicate request or technical failures)

Guiding principles

The following is provided as guidance for reversal processing. Note that unless otherwise specified, these are best practices and are not mandatory:

- Authorization reversal messages are intended to fully or partially release a hold of funds prior to clearing the transaction.
- Purchase returns (or refunds) may be submitted by the merchant for authorization. If the merchant or cardholder wishes to cancel the transaction, the merchant should initiate a full reversal.
- Mastercard recommends that merchants submit reversals as soon as an adjustment to the original authorization amount is known. Only approved authorization requests or transactions that have timed out are required to be reversed.
 - Merchants and acquirers must submit a full or partial reversal (as applicable) within seven calendar days of an original undefined authorization or final authorization request and within 30 calendar days of an original preauthorization request.
 - Merchants and acquirers must submit a full or partial reversal within 24 hours of transaction cancellation or of the transaction completing for an amount different from the authorized amount.

NOTE: Refer to the [Incremental Preauthorization](#) section in this chapter of this manual for more information about revised Standards for authorizations and preauthorizations.

- Reversals must contain the mandatory matching data elements from the original authorization for an issuer to release the appropriate funds, subject to the issuer's risk or fraud control policies.

Specific Scenarios

Specific Scenarios are described as follows:

- If an incremental authorization has been submitted, a reversal should reference original transaction data, including the original Trace ID, and DE 4 (Amount, Transaction) should be the sum of the original authorized amount plus any incremental amounts.
- Issuers will release any hold of funds once a clearing presentment has been matched (using the Trace ID in addition to other data elements) to the original authorization. Accordingly, a merchant is not required to submit a partial reversal if the lower amount is processed by Mastercard clearing within 24 hours of finalization of the transaction—assuming multi-clearing processing is not utilized.
- Partial reversals must be used when multi-clearing processing is utilized, since issuers will maintain a hold of funds for subsequent presentments.

Reversal Message Data Element Details

As follows are details about reversal messaging data elements:

- Mandatory Reversal Request/0400 data elements used for issuer matching, which must contain the same value from the original Authorization 0100/0110 message:
 - DE 2 (Primary Account Number [PAN])
 - DE 3 (Processing Code)
 - DE 4 (Amount, Transaction)
 - DE 38 (Authorization ID Response)
 - DE 48, subelement 63 (Trace ID) (from original DE 15 [Date, Settlement] and DE 63 [Network Data])
 - DE 49 (Currency Code, Transaction)
 - DE 90 (Original Data Elements) (from original DE 7 [Transmission Date and Time], DE 11 [System Trace Audit Number (STAN)], and DE 32 [Acquiring Institution ID Code] and DE 33 [Forwarding Institution ID Code])

Reversals submitted for technical failures (DE 39 reversal reason code 06—Error) will not contain DE 38 (Authorization ID) or DE 48, subelement 63 (Trace ID), because no Authorization Request Response/0110 message was received. If Trace ID data from the original authorization is not available due to technical failures, DE 48, subelement 63 in the reversal must contain zeros.

DE 90, subfield 1 must contain the Message Type Identifier (MTI) of the original authorization message. The remaining subfields may contain valid matching values from the original authorization or may be zero-filled if they are not available.

- Mandatory Reversal Request/0400 message data elements that may use default values consistent with the merchant initiating the reversal, unless supplied directly from the merchant point-of-interaction (POI) (for example, merchants with multiple acceptor business codes [MCCs]):
 - DE 18 (Merchant Type)
 - DE 22 (Point-of-Service [POS] Entry Mode)
 - DE 32 (Acquiring Institution ID Code) and DE 33 (Forwarding Institution ID Code) by using default values consistent with the merchant, DE 32 and DE 33 will be the same values as in the original 0100 message
 - DE 39 (Response Code)
 - DE 41 (Acceptor Terminal ID)
 - DE 42 (Acceptor ID)
 - DE 43 (Acceptor Name and Location)
 - DE 61 (Point-of-Service [POS] Data)

Mastercard recommends that DE 39 (Response Code) be populated with one of the following Reversal Request/0400 reason codes:

- 06 is Error
- 17 is Customer Cancellation
- 32 is Partial Reversal
- 34 is Suspect Fraud
- 68 is Response Received Late
- Minimum list of original Authorization 0100/0110 data elements to be saved for submitting a Reversal Request/0400 message:
 - DE 2 (Primary Account Number [PAN])
 - DE 3 (Processing Code)
 - DE 4 (Amount, Transaction)
 - DE 7 (Transmission Date and Time)
 - DE 11 (System Trace Audit Number [STAN])
 - DE 15 (Date, Settlement)
 - DE 38 (Authorization ID Response)
 - DE 49 (Currency Code, Transaction)
 - DE 63 (Network Data)

NOTE:

DE 15, DE 38, and DE 63 will only be available if an approved authorization response was received.

If DE 7 and DE 11 from the original authorization cannot be saved, DE 90, subelements 2 and 3 may be zero-filled within the Reversal Request/0400 message.

- Optional/Conditional data elements that must contain the same value from the original Authorization 0100 if included in the Reversal Request/0400 message:
 - DE 12 (Time, Local Transaction)
 - DE 13 (Date, Local Transaction)

- DE 14 (Date, Expiration)
- DE 23 (Card Sequence Number)
- DE 28 (Amount, Transaction Fee)
- DE 37 (Retrieval Reference Number)
- DE 54 (Additional Amounts)

Full Reversals

The Dual Message Authorization System supports full reversal functionality using the Reversal 04xx messages.

Full reversal functionality may be used for, but is not limited to, the following scenarios:

- Acquirer cannot deliver a response message 0110 or 0410 to the merchant
- Acquirer cannot match the response message 0110 or 0410 to the original request
- The authorization response message received contains errors
- The authorization response message was received too late

Partial Reversals

The Dual Message Authorization System supports partial reversal functionality using the Reversal/04xx messages.

Partial reversal functionality is useful in adjusting a portion of the original authorization amount in the following scenarios:

- A merchant ships only a portion of merchandise.
- A cardholder returns a rental vehicle earlier than originally reserved.
- A cardholder checks out of a hotel earlier than originally reserved.
- A cardholder cancels a portion of the transaction.

Mastercard supports Reversal Request/0400 and Reversal Advice/0420 messages where DE 95 (Replacement Amounts), subfield 1 (Actual Amount, Transaction) contains a value other than all zeros. DE 95, subfield 1, in the originating reversal, must be a lesser amount than the amount in DE 4 (Amount, Transaction).

Refer to the [Chargeback Guide](#) for rules related to partial reversal functionality.

Reversals of Balance Inquiry Transactions

Mastercard supports Reversal Request/0400 and Reversal Advice/0420 messages containing DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code), value 30 (Balance Inquiry). In inquiry transactions, DE 4 must contain a value of all zeros unless an ATM transaction fee DE 28 (Amount, Transaction Fee) has been applied by an acquirer for an ATM transaction in a country where of an ATM transaction fee is allowed.

Reversals of Purchase of Goods or Services with Cash Back Transactions

Mastercard provides DE 54 (Additional Amounts), subfield 2 (Amount Type), value 40 (Amount Cash Back) to issuers in the Reversal Request/0400 message if DE 54, subfield 2, value 40 was contained in the Reversal Request/0400 message from the acquirer. Acquirers provide DE 4

(Amount, Transaction) in the Reversal Request/0400 message for the full, original amount of the Authorization Request/0100 message. With Purchase of Goods and Services with Cash Back transactions, DE 4 contains the cash back amount, as defined in DE 54.

As with Authorization Request/0100 message processing, the Dual Message Authorization System edits Reversal Request/0400 messages when DE 54, subfield 2, value 40 is present. In this case, the Dual Message Authorization System checks to ensure that DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) contains value 09 (Purchase with Cash Back).

If DE 54, subfield 2, value 40 is not present, the Dual Message Authorization System performs an additional check to ensure that the Reversal Request/0400 message contains DE 39 (Response Code), value 87 (Purchase only, no cash back allowed).

When an acquirer submits a full or partial reversal for a Purchase Amount Only approval where DE 39 (Response Code) of the Authorization Request Response/0110 message contains value 87, Mastercard recommends that the acquirer not send DE 54 in the reversal message. If the acquirer intends the Reversal Request/0400 message as a partial reversal, DE 95 (Replacement Amounts) will also be present, containing the adjusted amount of the original authorization.

DE 54, subfield 2, value 40 is the only instance of DE 54 that Mastercard will forward to issuers in a Reversal Request/0400 message. All other instances of DE 54 will be removed from the Reversal Request/0400 message before forwarding the message to the issuer.

DE 54, subfield 2, value 40 also may be included in the Reversal Advice/0420 message.

Alternate Processing

While no Stand-In Authorization processing tests will be applied to the Reversal Request/0400 message, the Dual Message Authorization System supports alternate processing of Reversal Request/0400 and Reversal Request Response/0410 messages when the issuer is unable or unavailable to respond to the Reversal Request/0400 message.

The Dual Message Authorization System will perform the following processing when an issuer is unable or is unavailable to respond to the Reversal Request/0400 message:

Table 1786: Associated edits

WHEN...	THEN the Dual Message Authorization System...
the Dual Message Authorization System times out or The issuer is signed-out or the issuer responded with a Reversal Request Response/0410 message with an error or the Dual Message Authorization System cannot deliver the Reversal Request/0400 message to the issuer	sends the acquirer a Reversal Request Response/0410 message where: DE 39 = 00 (Approved or Completed successfully) sends a Reversal Advice/0420 message to SAF for immediate availability to the issuer where: • DE 39 is The value from the acquirer's Reversal Request/0400 message • DE 60 contains one of the following values as applicable: – 402 = Issuer Time-out – 403 = Issuer Sign-out – 409 = Issuer Response Error – 413 = Issuer Undelivered
the issuer responds with an error in the Reversal Request Response/0410 message	responds to the issuer with a Negative Acknowledgment/0190 message where: • DE 39 = 30 • DE 44 = Data element in error provides a Reversal Advice/0420 message to the issuer from SAF for immediate availability to the issuer (as defined above)
the Dual Message Authorization System cannot deliver the Reversal Request Response/0410 message to the acquirer	will not send a Reversal Advice/0420 message to the issuer. in this case, acquirers have the responsibility to resend the reversal to the issuer.

NOTE: Customers in the Europe region that route to an alternate issuer host for alternate processing instead of Stand-In, will still receive an Authorization Advice/0120: acquirer-generated as described here. Alternate issuer host processing does not send Reversal Request/0400 or Authorization Advice/0120 messages to the alternate host.

Dual Message Authorization System Edits

The Dual Message Authorization System performs the following edits on reversal transaction processing.

Reversal Request/0400

The Dual Message Authorization System will perform the following edits on the Reversal Request/0400 message as it relates to DE 95 (Replacement Amounts) and partial reversals.

Table 1787: Associated edits

WHEN...	THEN the Dual Message Authorization System...
DE 95 (Replacement Amounts) is equal to or greater than DE 4 (Amount, Transaction)	sends the acquirer a Reversal Request Response/0410 message where: DE 39 = 30 DE 44 = 095
DE 95 (Replacement Amounts) is equal to zero	removes DE 95 from the Reversal Request/0400 message before providing the Reversal Request/0400 message to the issuer.

The Dual Message Authorization System will perform the following edits on the Reversal Request/0400 message as it relates to DE 54 (Additional Amounts) and reversals of Purchase of Goods or Services with Cash Back transactions.

Table 1788: Associated edits

WHEN...	THEN the Dual Message Authorization System...
DE 54 is not formatted correctly in relation to alphanumeric specifications for the subfields	returns the acquirer a Reversal Request Response/0410 message where: DE 39 = 30 DE 44 = 054
DE 3, subfield 1, contains value 09 and DE 54, subfield 2, value 40 is not present and DE 39 does not contain value 87	returns the acquirer a Reversal Request Response/0410 message where: DE 39 = 30 DE 44 = 003
DE 3, subfield 1 does not contain value 09 and DE 54, subfield 2, value 40 is present	returns the acquirer a Reversal Request Response/0410 message where: DE 39 = 30 DE 44 = 003

The Dual Message Authorization System will perform the following edits on the Reversal Request/0400 message as it relates to reversals of Balance Inquiry transactions.

Table 1789: Associated edits

WHEN the Reversal Request/0400 message...	THEN the Dual Message Authorization System...
contains a value of all zeros in DE 4 (Amount, Transaction) and DE 3, subfield 1, does not contain a value of 30 (Balance Inquiry)	returns the acquirer a Reversal Request Response/ 0410 message where: DE 39 = 30 DE 44 = 004
contains a value of 30 (Balance Inquiry) in DE 3, subfield 1 then DE 4 (Amount, Transaction) must contain a value of all zeros unless the transaction contains DE 28 (Amount, Transaction Fee), otherwise	returns the acquirer a Reversal Request Response/ 0410 message where: DE 39 = 30 DE 44 = 004

Visa Transaction Processing

Mastercard supports Visa transaction processing as described in this section.

Visa Custom Payment Service

Mastercard supports Visa Custom Payment Service Authorization Request/0100 messages to accommodate customers. Mastercard accepts and forwards all Visa Authorization Request/0100 messages containing Visa Custom Payment Service information to the Visa network. If the Visa network is unavailable to authorize a transaction through the Mastercard gateway, Mastercard processes the authorization through X-Code and will decline the Visa transaction.

NOTE: This topic applies only to Dual Message System processing of Visa authorization transactions.

Authorization Request/0100: Visa Custom Payment Service

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Data Element	Org	Sys	Dst	Values/Comments
DE 2 (Primary Account Number [PAN])	M	.	M	PAN begins with 4 indicating a Visa card.
DE 14 (Date, Expiration)	C	.	C	Must be present for magnetic stripe transactions and contain same expiration date as in DE 35 or DE 45.
DE 18 (Merchant Type)	M	.	M	Must contain a valid Custom Payment Service MCC.

Data Element	Org	Sys	Dst	Values/Comments
DE 22 (Point-of-Service (POS) Entry Mode, subfield 1, POS Terminal PAN Entry Mode)	M	•	M	Must contain 90 to indicate that the track 1 or track 2 data is unaltered; where card presence is not required or the transaction is key entered, 01 is valid.
DE 35 (Track 2 Data)	C	•	C	Track 2 or Track 1 must be present and passed unaltered if the DE 22 value is 90.
DE 37 (Retrieval Reference Number)	C	•	C	Must be present for Custom Payment Service incremental authorizations and reversals.
DE 42 (Acceptor ID)	M	•	M	Must be present for Custom Payment Service transactions.
DE 43 (Acceptor Name and Location)	M	•	M	Must be present for Custom Payment Service transactions.
DE 45 (Track 1 Data)	C	•	C	Track 1 or Track 2 must be present and passed unaltered if the DE 22 value is 90.
DE 48, subelement 90 (Custom Payment Service Request [Visa field 62.1])	M	•	M	Indicates a request for a Visa Custom Payment Service qualified transaction and must contain one of the following values: I = Incremental authorization P = Preferred customer R = Recurring payment Y = Custom Payment Service participation request. Refer to Visa Base I Technical Specifications manual for a list of all CPS request values in Visa field 62.1.
DE 48, subelement 91 (Visa Custom Payment Service Request/ Transaction ID, [Visa field 62.2 and 62.3])	C	•	C	Must be present for incremental authorization and reversal transactions and contain the Transaction ID and/or the CPS validation code of the original transaction.

Data Element	Org	Sys	Dst	Values/Comments
DE 48, subelement 96 (Visa Market-Specific Data [Visa field 62.4])	C	.	C	Must be present for hotel or automobile transactions and contain one of the following values: A = Automobile rental B = Bill payment transaction E = Electronic commerce transaction aggregation H = Hotel rental J = B2B invoice payments M = Healthcare—Medical N = Failed market-specific data edit T = Transit (in healthcare transactions only)
DE 48, subelement 97 (Visa Prestigious Property Indicator [Visa field 62.6])	C	.	C	May be present for participants in the Visa Prestigious Lodging program and contains one of the following values: D = Visa established limits B = Visa established limits S = Visa established limits
DE 61 (Point-of-Service [POS] Data), subfield 12 (POS Authorization Life Cycle)	M	.	M	Must contain the authorization life cycle if DE 48, subelement 96 (Market-specific Data) is present.

Authorization Request Response/0110: Visa Custom Payment Service

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0110 data elements apply.

Data Element	Org	Sys	Dst	Values/Comments
DE 48, subelement 85 (Account Status [Visa Only])	C	.	C	Identifies the account range as regulated or non-regulated interchange. Must be one of the following values: R = Account is regulated N = Account is non-regulated

Program and service format requirements
DE 48 (Structure in a Visa Custom Payment Service Transaction)

Data Element	Org	Sys	Dst	Values/Comments
DE 48, subelement 90 (Custom Payment Service Request [Visa field 62.1])	M	•	M	If the request qualifies for Visa CPS and is approved, refer to the Visa Base I Technical Specifications manual for a complete list of CPS-qualified codes in Visa field 62.1. If the original request does not qualify for CPS, Visa returns an "N" or "T" in the response.
DE 48, subelement 91 (Visa Custom Payment Service Request/Transaction ID)	C	•	C	Must be present if DE 48 subelement 90 is present. Contains one of the following response lengths: When length field is 04: Subelement 91 contains the Visa CPS validation code (Visa field 62.3, four bytes alphanumeric). Refer to the Visa Base I Technical Specifications manual for more information on Visa CPS validation codes. When length field is 15: Subelement 91 contains the transaction ID (Visa field 62.2, 15 bytes numeric). When length field is 19: Subelement 91 contains the transaction ID (Visa field 62.2, 15 byte numeric) and CPS validation code (Visa field 62.3, four bytes alphanumeric).
DE 48, subelement 96 (Visa Market-Specific Data [Visa field 62.4])	C	•	C	Must be present if present in the original Authorization Request/0100 message. Contains response data for Visa market-specific data and must be one of the following values: A = Automobile rental H = Hotel rental N = Failed market-specific data edit

DE 48 (Structure in a Visa Custom Payment Service Transaction)

Following is the structure of DE 48 (Additional Data—Private Use) in a Visa Custom Payment Service transaction.

LLL	"VAR"—999 maximum bytes (TCC + Subelement (SE) data)...
-----	---

3 bytes	1 byte	2 bytes	2 bytes	1 byte	2 bytes	2 bytes	Variable
Total Data Element Length	TCC	First Subelement Data		Second Subelement Data			
mandatory		SE ID 90	SE 90 Length	Authorization Characteristics Indicator	SE ID 91	SE 91 Length	Transaction ID and/or CPS Validation Code
1002 maximum bytes (LLL +TCC + Subelement Data)...							
..."VAR"—999 maximum bytes (TCC + Subelement (SE) data)							
2 bytes	2 bytes	1 byte		2 bytes	2 bytes	1 byte	
Third Subelement Data			Fourth Subelement Data				
SE ID 96	SE 96 Length	Market-specific Data ID	SE ID 97	SE 97 Length	Prestigious Property Indicator		
...1002 maximum bytes (LLL +TCC + Subelement Data)							

Visa Programs

Mastercard supports Visa transactions as described in this section.

Visa CVV2

Following are Visa CVV2 Authorization/01xx message layouts.

Authorization Request/0100: Visa CVV2

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Data Element	Org	Sys	Dst	Values/Comments
DE 2 (Primary Account Number)	M	.	M	PAN begins with 4 indicating a Visa card.
DE 48, subelement 92 (CVV2 Data), subfield 1 (CVV2 Presence ID)	C	.	C	0 = Merchant did not provide CVV2 or it was deliberately bypassed 1 = CVV2 value present 2 = CVV2 is on card, but not legible 9 = Cardholder states no CVV2 is on card

Data Element	Org	Sys	Dst	Values/Comments
DE 48, subelement 92, subfield 2 (CVV2 Response Code)	C	.	C	0 = Only the normal response code in DE 39 should be returned by the issuer 1 = The normal response code and CVV2 response code should be returned by the issuer
DE 48, subelement 92, subfield 3 (CVV2 Value)	C	.	C	CVV2 value; right-justified and blank-filled

Authorization Request Response/0110: Visa CVV2

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0110 data elements apply.

Data Element	Org	Sys	Dst	Values/Comments
DE 48, subelement 87 (CVV2 Response)	C	.	C	Contains one of the following CVV2 response codes: M = CVV2 match N = CVV2 no match P = Not processed S = CVV2 is on the card, but the Merchant has indicated that CVV2 is not present U = Issuer is not Visa-certified for CVV2, has not provided Visa encryption keys, or both
DE 48, subelement 92 (CVV2 Data)	CE	.	CE	Must be the same value as in the original Authorization/0100 message.

Visa Fleet Card ID

Following are Visa Fleet Card ID Authorization/01xx message layouts.

Authorization Request/0100: Visa Fleet Card ID

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Data Element	Org	Sys	Dst	Values/Comments
DE 2 (Primary Account Number [PAN])	M	.	M	PAN begins with 4 indicating a Visa card.

Data Element	Org	Sys	Dst	Values/Comments
DE 48 (Additional Data—Private Use), subelement 93 (Fleet Card ID Request Data), subfield 1 (Fleet Card ID Request Indicator)	C	.	C	Contains request indicator value "\$" for Fleet Card.
DE 48, subelement 93, subfield 2 (Optional Free-form Informational Text)	C	.	C	Contains free-form information text. Additional Point-of-Service (POS) information.

NOTE: The Authorization Platform does not forward DE 48, subelement 93 back to the acquirer in Authorization Request Response/0110 messages.

Visa Commercial Card Inquiry

Following are Visa Commercial Card Inquiry Authorization/01xx message layouts.

Authorization Request/0100: Visa Commercial Card Inquiry

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request/0100 data elements apply.

Data Element	Org	Sys	Dst	Values/Comments
DE 2 (Primary Account Number [PAN])	M	.	M	PAN begins with 4 indicating a Visa card.
DE 48 (Additional Data—Private Use), subelement 94 (Commercial Card Inquiry Request), subfield 1 (Card Request Indicator)	C	.	C	!01 = Request indicator for Commercial Card
DE 48, subelement 94, subfield 2 (Merchant Request for Commercial Card Type)	C	.	C	0 = Merchant request for Commercial Card type

Authorization Request Response/0110: Visa Commercial Card Inquiry

Following is a list of the data elements and values applicable to this message type. All mandatory Authorization Request Response/0110 data elements apply.

Data Element	Org	Sys	Dst	Values/Comments
DE 48 (Additional Data—Private Use), subelement 94 (Commercial Card Inquiry Request), subfield 1 (Card Request Indicator)	C	.	C	!01 = Request indicator for Commercial Card

Data Element	Org	Sys	Dst	Values/Comments
DE 48, subelement 94, subfield 2 (Merchant Request for Commercial Card Type)	C	.	C	Contains one of the following values: 0 = Decline or not a Commercial Card B = Business Card R = Corporate Card S = Purchase Card

Visa Token Processing

Mastercard has expanded the functionality of the Visa Gateway to support tokenization processing for partial shipments, recurring payments, and e-commerce mobile transactions for Visa-branded, dual-message authorization transactions submitted to the Mastercard Network.

These enhancements affect acquirers that use the Mastercard Network to submit Visa-branded dual-message authorization transactions.

Tokenized Recurring Payment Messages

Acquirers must be prepared to identify tokenized recurring payment transactions in Authorization Request/0100 and Reversal Request/0400 messages.

- Existing value 4 (Standing order/recurring transactions) in DE 61 (Point-of-Service [POS] Data), subfield 4 (POS Cardholder Presence)
- Presence of DE 48 (Additional Data—Private Use), subelement 43 (Secure Electronic Commerce Verification Service) containing the partial shipment verbiage as follows: PARTIALbSHIPMENTbbbbbbbbbbbb or PARTIALSHIPMENT00000000000000 where b represents a space. This field contains 28 positions.
- Existing values 2 (Channel) in position 1 (Security Protocol), 1 (Cardholder certificate not used) in position 2 (Cardholder Authentication), and 0 (UCAF data collection is not supported by the merchant or an Identity Check merchant has chosen not to undertake Identity Check on this transaction) in position 3 (UCAF Collection Indicator) in DE 48 (Additional Data—Private Use), subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator)
- Existing value 81 (PAN manual entry via e-commerce) in DE 22 (Point-of-Service Entry Mode), subfield 1 (POS Terminal PAN Entry Mode)

Tokenized Partial Shipment Messages

Acquirers must be prepared to identify tokenized partial shipment transactions in Authorization Request/0100 and Reversal Request/0400 messages whose original transaction was token-based with a valid cryptogram. Cryptographic data from the original transaction must be resent in Authorization Request/0100 messages for partial shipments, but is not required in Reversal Request/0400 messages.

- DE 61 (Point-of-Service [POS] Data), subfield 4 (POS Cardholder Presence), value 5 (Electronic order [home PC, Internet, mobile phone, PDA])

- DE 48 (Additional Data—Private Use), subelement 43 containing the cryptographic data from the original transaction
- DE 48, subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator):
 - Position 1 (Security Protocol), value 2 (Channel)
 - Position 2 (Cardholder Authentication), value 1 (Cardholder certificate not used)
 - Position 3 (UCAF Collection Indicator), value 2 (UCAF data collection is supported by the merchant, and UCAF data must be present [DE 48, subelement 43 must contain a fully authenticated AAV])
- DE 22 (Point-of-Service Entry Mode), subfield 1 (POS Terminal PAN Entry Mode), values 10 (Credential on File) or 81 (PAN manual entry via e-commerce)

NOTE: DE 48, subelement 43 is not required in Reversal Request/0400 messages for partial shipments.

Tokenized E-Commerce with Mobile Device Messages

Acquirers must be prepared to receive the tokenized e-commerce with mobile device transaction identifier in Authorization Request Response/0110 and Reversal Request Response/0410 messages.

- Value B (Tokenized e-commerce with mobile device) in DE 48, subelement 90 (Custom Payment Service Request Response [Visa Only])
- Value Y (Transaction is processed through Visa Checkout) in DE 48, subelement 78 (Payment Service Indicators [Visa Only]), subfield 4 (Visa Checkout Indicator)

Authorization Request/0100: Visa Token Request

Following is a list of the data elements and values applicable to the Authorization Request/0100 message type for token requests for Visa primary account numbers. All mandatory Authorization Request/0100 data elements apply except where noted.

DE ID	Data Element Name	Org	Sys	Dst	Values/Comments
2	Primary Account Number	M	•	M	Cardholder's primary account number
3	Processing Code	M	•	M	00 = Purchase
4	Amount, Transaction	M	•	M	Must be zero
14	Date, Expiration	M	•	M	Cardholder's primary account expiration date
48	Transaction Category Code	M	•	M	Contains the appropriate Transaction Category Code (TCC).

DE ID	Data Element Name	Org	Sys	Dst	Values/Comments
48	Additional Data—Private Use, subelement 33 (PAN Mapping File Information)	M	•	M	This is a Tag-Length-Value field. The tags in the subelement will indicate which subfields are present. Only subfield 6 is required. Subfield 6 (Token Requestor ID) = The ID assigned by the Visa Token Vault to the Token Requestor.
48	Additional Data—Private Use, subelement 42 (Electronic Commerce Indicators)	C	•	C	Contains the electronic commerce security level indicator and UCAF collection indicator data in subfield 1 that consists of a valid combination of positions 1, 2, and 3.
48	Additional Data—Private Use, subelement 43—3-D Secure Electronic Commerce Verification Service	C	•	C	Position 1 = 8 (Non-Mastercard 3-D Secure Electronic Commerce transaction [Visa, JCB, Diners Club, or American Express]) Position 2–21 (3-D Secure Electronic Commerce Cardholder Authentication Verification Value [CAVV])
48	Additional Data—Private Use, subelement 82 (Address Verification Service Request)	M	•	M	52 = AVS and Authorization Request/0100
48	Additional Data—Private Use, subelement 92 (CVV2 Data [Visa Only])	C	•	C	CVV2 value from the signature panel of the card when applicable.
61	Point-of-Service (POS) Data, subfield 7 (POS Transaction Status)	M	•	M	8 = Account Status Inquiry
120	Record Data	M	•	M	The cardholder's billing address

Authorization Request Response/0110: Visa Token Request Response

Following is a list of the data elements and values applicable to the Authorization Request Response/0110 message for responses to Visa token requests. All mandatory Authorization Request Response/0110 data elements apply except where noted.

DE ID	Data Element Name	Org	Sys	Dst	Values/Comments
2	Primary Account Number	ME	•	ME	Cardholder's primary account number.
3	Processing Code	ME	•	ME	00 = Purchase

DE ID	Data Element Name	Org	Sys	Dst	Values/Comments
4	Amount, Transaction	CE	X	M	Must be zero.
39	Response Code	M	.	M	Contains the applicable response code.
48	Additional Data—Private Use, subelement 33 (PAN Mapping File Information)	C	.	C	This is a Tag-Length-Value field. The tags in the subelement will indicate which subfields are present. • Subfield 1 (Account Number Indicator) = C (Mastercard Digital Enablement Service Device Account Number) • Subfield 2 (Account Number) = Token • Subfield 5 (Token Assurance Method) = Value between 00 and 99 indicating the Token Assurance Method assigned by the Visa Token Vault. • Subfield 6 (Token Requestor ID) NOTE: The Visa token expiration date will be the same as the expiration date of the primary account number.
48	Additional Data—Private Use, subelement 45 (3-D Secure Electronic Commerce Transactions Response Code)	C	.	C	The Visa Cardholder Authentication Verification Value (CAVV) results code.
48	Additional Data—Private Use, subelement 82 (Address Verification Service)	CE	.	CE	52 = AVS and Authorization Request/0100
48	Additional Data—Private Use, subelement 83 (Address Verification Service Response)	C	.	C	The AVS verification response code.
48	Additional Data—Private Use, subelement 87 (CVV2 Response [Visa Only])	C	.	C	The CVV2 response.
120	Record Data, subfield 01 (AVS Service Indicator 1)	C	X	C	Must be the same value as in the original Authorization Request/0100 message, if present.

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